

ISC Class XII Mathematics

Part I Complete Solved Solutions – Chapters 1–6

Step-by-Step ISC Board Solutions – Full Master Volume

Chapters Covered in This Volume

- I. Chapter 1: Relations and Functions
- II. Chapter 2: Inverse Trigonometric Functions
- III. Chapter 3: Matrices
- IV. Chapter 4: Determinants
- V. Chapter 5: Continuity and Differentiability
- VI. Chapter 6: Applications of Derivatives

Contents

- Chapter 1: Relations and Functions
- Chapter 2: Inverse Trigonometric Functions
- Chapter 3: Matrices
- Chapter 4: Determinants
- Chapter 5: Continuity and Differentiability
- Chapter 6: Applications of Derivatives

Index of Problems

Chapter 1: Relations and Functions

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
--------	---------------------------	----------	------	---------	------

Exercise 1.1 – Types of Relations: Reflexive, Symmetric, Transitive & Equivalence Relations

1	If $A = \{-1, 1, 3\}$, then what is the number of relations on A ?	1	Numerical	1.1	87
2	(i) If $R = \{(x, y) : x + 2y = 8\}$ is a relation on $\mathbb{N}$, write the range of ...	2	Numerical	1.1	87
3	Let A be any non-empty set. State true or false:	3	Numerical	1.1	88
(i)	Identity relation on A is reflexive.	3	Numerical	1.1	88
(ii)	Every reflexive relation on A is identity relation on A .	3	Numerical	1.1	88
(iii)	Identity relation on A is symmetric.	3	Numerical	1.1	88
(iv)	Identity relation on A is an equivalence relation.	3	Numerical	1.1	88
(v)	Universal relation on A is an equivalence relation.	3	Numerical	1.1	88
4	State the reason for the relation R in the set $\{1, 2, 3\}$ given by ...	4	Numerical	1.1	89
5	The relation R on $A = \{1, 2, 3\}$ is defined by $R = \{(1, 2), (2, 1)\}$. What kind of ...	4	Numerical	1.1	90
6	Write the smallest equivalence relation on A where $A = \{1, 2, 3\}$.	3	Numerical	1.1	90
7	If $A = \{0, 1, 2, \dots, 9\}$ and the relation R on A is defined by ...	4	Numerical	1.1	91
8	Is the relation R on the set $\mathbb{R}$ of real numbers defined by ...	5	Numerical	1.1	92
9	Is the relation R on the set $\mathbb{Q}$ of rational numbers defined by ...	5	Numerical	1.1	93
10	If a relation R on the set $\{a, b, c\}$ is defined by $R = \{(b, b)\}$, then classify ...	4	Numerical	1.1	93
11	If the relation R on the set $A = \{1, 2, 3\}$ is defined by ...	4	Numerical	1.1	94
12	How many reflexive relations are possible on a set A of order 3?	6	Numerical	1.1	94

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
13	If R be the relation in the set $\{1, 2, 3, 4\}$ given by ...	4	Numerical	1.1	95
(i)	R is reflexive and symmetric but not transitive.	4	Numerical	1.1	95
(ii)	R is reflexive and transitive but not symmetric.	4	Numerical	1.1	95
(iii)	R is symmetric and transitive but not reflexive.	4	Numerical	1.1	95
14	Determine whether each of the following relations are reflexive, symmetric and ...	5	Numerical	1.1	96
(i)	Relation R in the set $A = \{1, 2, 3, \dots, 10\}$ defined by ...	5	Numerical	1.1	96
(ii)	Relation R in the set $\mathbb{Z}$ of all integers defined by ...	5	Numerical	1.1	96
(iii)	Relation R on the set $A = \{1, 2, 3, 4, 5, 6\}$ defined by ...	5	Numerical	1.1	96
15	If the relation R in the set A , where $A = \{1, 2, 3, 4, 5, 6\}$, is defined by ...	4	Numerical	1.1	97
16	If R is the relation defined on the set of natural numbers $\mathbb{N}$ as follows: ...	2	Numerical	1.1	97
17	Determine whether each of the following relations in the set A of human beings in a ...	5	Numerical	1.1	98
(i)	$R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$	5	Numerical	1.1	98
(ii)	$R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$	5	Numerical	1.1	98
(iii)	$R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$	5	Numerical	1.1	98
(iv)	$R = \{(x, y) : x \text{ and } y \text{ live within 2 kilometres}\}$	5	Numerical	1.1	98
18	Determine whether each of the following relations in the set A of students at a ...	5	Numerical	1.1	99
(i)	$R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are honest}\}$	5	Numerical	1.1	99
(ii)	$R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are obedient}\}$	5	Numerical	1.1	99
(iii)	$R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are hardworking}\}$	5	Numerical	1.1	99
19	Show that the relation R in the set $A = \{1, 2, 3\}$ given by ...	4	Proof	1.1	100

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
20	Explain why $x + 4y = 12$, $x, y \in \mathbb{N}$ is NOT a symmetric relation.	4	Numerical	1.1	101
21	Let R be a relation defined over $\mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers, defined ...	4	Numerical	1.1	101
22	Let R be the relation on $\mathbb{W}$ (set of whole numbers) defined by ...	2	Numerical	1.1	102
23	Let R be the relation defined on $\mathbb{Q}_0$ (set of non-zero rational numbers) ...	5	Numerical	1.1	103
24	A relation R is defined on a set of real numbers $\mathbb{R}$ as ...	5	Numerical	1.1	104
25	Check whether the relation R on the set $\mathbb{N}$ of natural numbers given by ...	5	Numerical	1.1	105
26	Let R be the relation on $\mathbb{R}$ (set of real numbers) defined by ...	5	Proof	1.1	105
27	Determine whether each of the following relations are reflexive, symmetric and ...	5	Numerical	1.1	106
(i)	$R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$	5	Numerical	1.1	106
(ii)	$R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$	5	Numerical	1.1	106
28	Let T be the set of all triangles drawn in a plane with R as a relation in T given ...	7	Proof	1.1	107
29	Show that the relation R in the set A of all books in a library of a school, given by ...	7	Proof	1.1	108
30	Show that the relation R on the set $\mathbb{Z}$ of all integers defined by ...	7	Proof	1.1	108
31	Show that the relation R on the set $\mathbb{Z}$ of all integers defined by ...	8	Numerical	1.1	109
32	Prove that the relation R in the set $A = \{5, 6, 7, 8, 9\}$, given by ...	8	Numerical	1.1	110
33	Give examples of relations which are	9	Numerical	1.1	111
(i)	symmetric but neither reflexive nor transitive.	9	Numerical	1.1	111
(ii)	transitive but neither reflexive nor symmetric.	9	Numerical	1.1	111
(iii)	symmetric and transitive but not reflexive.	9	Numerical	1.1	111
(iv)	symmetric and reflexive but not transitive.	9	Numerical	1.1	111

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(v)	reflexive and transitive but not symmetric.	9	Numerical	1.1	111
34	Give an example of a relation R on $A = \{a, b, c\}$ which is	9	Numerical	1.1	112
(i)	neither reflexive nor symmetric but transitive.	9	Numerical	1.1	112
(ii)	neither symmetric nor transitive but reflexive.	9	Numerical	1.1	112
(iii)	neither transitive nor reflexive but symmetric.	9	Numerical	1.1	112
35	Show that the relation S in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by ...	8	Numerical	1.1	113
36	Show that the relation R in the set $A = \{x \in \mathbb{W}, 0 \leq x \leq 17\}$...	8	Numerical	1.1	114
(i)	$R = \{(a, b) : a - b \text{ is a multiple of } 5\}$	8	Numerical	1.1	114
(ii)	$R = \{(a, b) : a = b\}$	8	Numerical	1.1	114
37	Show that the relation R defined by $(a, b) R (c, d) \iff a + d = b + c$ on ...	10	Numerical	1.1	115
38	Show that the relation R in the set $\mathbb{N} \times \mathbb{N}$ defined by ...	10	Proof	1.1	116
39	The Earth has 24 time zones, defined by dividing the Earth into 24 equal longitudinal ...	5	Numerical	1.1	117
(i)	Check whether the relation R is reflexive, symmetric and transitive.	5	Numerical	1.1	117
(ii)	Is relation R an equivalence relation?	5	Numerical	1.1	117

Exercise 1.2 – Functions: Domain and Range

1	A function is defined by $f(x) = \frac{3x^2 + 2x - 1}{x + 1}$...	11	Numerical	1.2	118
2	(i) Is the relation f defined by $f(x) = (\text{piecewise definition})$ a function? ...	12	Numerical	1.2	119
3	A real function f is defined by $f(x) = 3x - 2, x \in \mathbb{R}$, find the value of ...	11	Numerical	1.2	120
4	If a real function f is defined by $f(x) = 2x^2 - 3$, find x such that $f(x) = 15$.	13	Numerical	1.2	120
5	Given $f(x) = x^3 - 1$, find x if $f(x)$ is 215.	13	Numerical	1.2	121

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
6	A function is given by the formula $f(x) = 144 - 16x^2$. Calculate $f(2)$. Also find the ...	13	Numerical	1.2	121
7	A function 'f' is defined by $f(x) = 2x^2 + 3$, for all $x \in \mathbb{R}$. Find (i) ...	13	Numerical	1.2	122
8	If $f(x) = 2x^2 + 1$ and domain of $f = \{-2, -1, 0, 1, 3\}$, find (i) range of f (ii) ...	14	Numerical	1.2	123
9	A function 'f' is defined by $f(x) = x^2 + 3$, $x \in \mathbb{N}$ and $x \leq 5$. (i) Find ...	14	Numerical	1.2	124
10	Find the domain of the following real functions: (i) $f(x) = x + 2 $ (ii) ...	15	Numerical	1.2	124
11	Find the range of the following functions: (i) $f(x) = x^2 + 2$, $x \in \mathbb{R}$ (ii) ...	16	Numerical	1.2	125
12	Find the domain and the range of the function $f(x) = 2x^2 + 1$. Also find $f(-2)$ and the ...	16	Numerical	1.2	126
13	Find the domain and the range of the following functions: (i) $f(x) = \sqrt{x + 2}$ (ii) ...	17	Numerical	1.2	127
14	Find the domain and the range of the following functions: (i) ...	18	Numerical	1.2	128
15	Find the domain and the range of the following functions: (i) $f(x) = \sqrt{16 - x^2}$...	17	Numerical	1.2	129
16	Find the domain and the range of the following functions: (i) $f(x) = x - 3 $ (ii) ...	15	Numerical	1.2	130
17	If a real function f is defined by $f(x) = \frac{ x - x}{2x}$, then find its domain ...	15	Numerical	1.2	131
18	Find the domain and the range of the function f defined by ...	15	Numerical	1.2	132
19	If $f(x) = \log \left(\frac{1+x}{1-x} \right)$, then show that ...	19	Proof	1.2	133
20	Find the domain of the following functions: (i) $\log(4x - 3)$ (ii) ...	20	Numerical	1.2	134
21	Find the domain of the following functions: (i) $f(x) = \frac{1}{3 - 2 \sin x}$ (ii) ...	21	Numerical	1.2	135

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
22	Find the range of the following functions: (i) $f(x) = 2 - 3 \cos x$ (ii) ...	21	Numerical	1.2	136

Exercise 1.3 – One-One, Onto and Bijective Functions

1	Show that the function $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(x) = 3x - 2$ is ...	22	Proof	1.3	137
2	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 2x - 1$ is not ...	22	Proof	1.3	137
3	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n^2$ is ...	22	Proof	1.3	138
4	Is the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 2$ one-one? ...	22	Numerical	1.3	138
5	What is the range of the function $f(x) = \frac{ x-1 }{x-1}$?	15	Numerical	1.3	139
6	Find the domain of the function $f(x) = \frac{1}{5 - 3 \sin x}$.	21	Numerical	1.3	140
7	Find the range of the function $f(x) = \frac{1}{2 - \cos x}$.	21	Numerical	1.3	141
8	Find the range of the following functions:	21	Numerical	1.3	141
(i)	$f(x) = \sin x + \cos x$	21	Numerical	1.3	141
(ii)	$f(x) = 3 \sin x + 4 \cos x$	21	Numerical	1.3	141
9	The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 2x - 3 $.	22	Numerical	1.3	142
(i)	Is f one-one?	22	Numerical	1.3	142
(ii)	Is f onto?	22	Numerical	1.3	142
10	If $A = \{1, 2, 3\}$ and $B = \{3, 5\}$, then write the number of functions from A to ...	23	Numerical	1.3	143
11	If $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$, then write the number of functions from A ...	23	Numerical	1.3	143
12	If $A = \{a, b\}$ and $B = \{1, 3, 5\}$, then write the number of one-one functions from ...	23	Numerical	1.3	144
13	If $A = \{a, b, c\}$ and $B = \{p, q\}$, then write the number of onto functions from A ...	23	Numerical	1.3	145
14	If $A = \{1, 2, 3, 4\}$ and $B = \{-1, 3\}$, then write the number of onto functions from ...	23	Numerical	1.3	145

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
15	If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, then write the number of	23	Numerical	1.3	146
(i)	one-one onto functions from A to B	23	Numerical	1.3	146
(ii)	bijections from A to B .	23	Numerical	1.3	146
16	The given function $f : \mathbb{R} \rightarrow \mathbb{R}$ is not 'onto' function. Give reason.	22	Proof	1.3	146
17	Let $A = \{1, 2\}$, $B = \{3, 6\}$, function $f : A \rightarrow B$ be defined by $f(x) = 3x$ and ...	24	Proof	1.3	147
18	Check the injectivity and surjectivity of the following functions:	22	Numerical	1.3	148
(i)	$f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2$	22	Numerical	1.3	148
(ii)	$f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^2$	22	Numerical	1.3	148
(iii)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$	22	Numerical	1.3	148
(iv)	$f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^3$	22	Numerical	1.3	148
(v)	$f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^3$	22	Numerical	1.3	148
(vi)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$	22	Numerical	1.3	148
19	Let $\mathbb{I}$ be the set of all integers. Show that the function ...	22	Proof	1.3	149
20	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is ...	22	Proof	1.3	150
21	Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and $f = \{(1, 4), (2, 5), (3, 6)\}$ be a ...	22	Proof	1.3	150
22	In each of the following cases, state whether the function is bijective or not ...	22	Numerical	1.3	151
(i)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 1$	22	Numerical	1.3	151
(ii)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3 - 4x$	22	Numerical	1.3	151
(iii)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1 + x^2$	22	Numerical	1.3	151
23	Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^4$. Show that f is neither ...	22	Proof	1.3	152
24	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$ is ...	22	Proof	1.3	152
25	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by ...	22	Proof	1.3	153
26	Show that the function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is ...	22	Proof	1.3	154
27	If the function $f : \mathbb{I} \rightarrow \mathbb{I}$ is defined by ...	25	Numerical	1.3	154
(i)	Find $f(3)$.	25	Numerical	1.3	154

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	Find $f(4)$.	25	Numerical	1.3	154
(iii)	Find x , if $f(x) = -9$.	25	Numerical	1.3	154
28	See parts below:	25	Proof	1.3	156
(i)	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by ...	25	Proof	1.3	156
(ii)	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by ...	25	Proof	1.3	156
29	Find the number of all one-one functions from the set $\{1, 2, 3, 4\}$ to itself.	23	Numerical	1.3	157
30	Show that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{5x-3}{4}$...	22	Proof	1.3	157
31	Let $f : \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ be a function ...	22	Numerical	1.3	158
32	If $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$, then show that the function ...	22	Proof	1.3	160
33	(a) If a real-valued function is given by $f(x) = \sqrt{25-x^2}$ is an onto function ...	22	Numerical	1.3	161

Exercise 1.4 – Composition of Functions

1	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = (3-x^3)^{1/3}$, then find ...	26	Numerical	1.4	162
2	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = (5-x^5)^{1/5}$, then find ...	26	Numerical	1.4	163
3	If $f(x) = x $ and $g(x) = [x]$ for all $x \in \mathbb{R}$, find	27	Numerical	1.4	163
(i)	$(g \circ f)\left(-\frac{5}{3}\right)$	27	Numerical	1.4	163
(ii)	$(f \circ g)\left(-\frac{5}{3}\right)$	27	Numerical	1.4	163
4	If $f(x) = x+7$ and $g(x) = x-7$, find $(f \circ g)(7)$.	27	Numerical	1.4	165
5	If $f : \{2, 3, 4, 5\} \rightarrow \{3, 4, 5, 9\}$ and $g : \{3, 4, 5, 9\} \rightarrow \{7, 11, 15\}$ are ...	28	Numerical	1.4	165
6	Let $A = \{3, 4, 5, 6\}$, $B = \{13, 14, 15, 16\}$ and $C = \{23, 24, 25\}$. If ...	28	Numerical	1.4	166
7	Let $A = \{1, 2, 3, 4\}$. If the functions $f : A \rightarrow A$ and $g : A \rightarrow A$ are defined ...	28	Numerical	1.4	166
(i)	$g \circ f$	28	Numerical	1.4	166
(ii)	$f \circ g$	28	Numerical	1.4	166
(iii)	$f \circ f$	28	Numerical	1.4	166

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
8	If the functions f and g are given by $f = \{(1, 2), (3, 5), (4, 1)\}$ and ...	28	Numerical	1.4	167
(i)	find the domain of f and g	28	Numerical	1.4	167
(ii)	find the range of f and g	28	Numerical	1.4	167
(iii)	write $f \circ g$ and $g \circ f$ as sets of ordered pairs	28	Numerical	1.4	167
9	Let $\mathbb{R}$ be the set of all real numbers. Find $g \circ f$ and $f \circ g$...	26	Numerical	1.4	168
(i)	$f(x) = 2x - 1$ and $g(x) = x^2 + 3$ for all $x \in \mathbb{R}$	26	Numerical	1.4	168
(ii)	$f(x) = x + 1$ and $g(x) = x $ for all $x \in \mathbb{R}$	26	Numerical	1.4	168
(iii)	$f(x) = x^3$ and $g(x) = 2x^2 + 1$ for all $x \in \mathbb{R}$	26	Numerical	1.4	168
(iv)	$f(x) = \sin x$ and $g(x) = 5x^2$ for all $x \in \mathbb{R}$	26	Numerical	1.4	168
10	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^2 - 3x + 2$ for all ...	26	Numerical	1.4	169
11	Let $\mathbb{R}$ be the set of all real numbers. If the functions ...	26	Numerical	1.4	170
(i)	$f \circ g$	26	Numerical	1.4	170
(ii)	$g \circ f$	26	Numerical	1.4	170
(iii)	$f \circ f$	26	Numerical	1.4	170
(iv)	$g \circ g$	26	Numerical	1.4	170
12	If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by ...	26	Proof	1.4	171
13	If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by ...	26	Proof	1.4	172
14	If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by ...	26	Numerical	1.4	173
(i)	$g \circ f$	26	Numerical	1.4	173
(ii)	$f \circ g$	26	Numerical	1.4	173
(iii)	$f \circ f$	26	Numerical	1.4	173
15	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by ...	22	Numerical	1.4	174

Exercise 1.5 – Invertible Functions and Inverse of a Function

1	If $f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$, find f^{-1} .	29	Numerical	1.5	175
2	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = 3x + 5$, then find ...	30	Numerical	1.5	176
3	If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{3x-2}{5}$ is invertible ...	30	Numerical	1.5	176
4	If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-7}{4}$ is invertible ...	30	Numerical	1.5	177

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
5	If f is an invertible function defined as $f(x) = \frac{3x-4}{5}$, write $f^{-1}(x)$.	30	Numerical	1.5	178
6	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = ax - b \dots$	30	Numerical	1.5	178
7	If $f : \mathbb{R} - \left\{-\frac{3}{5}\right\} \rightarrow \mathbb{R}$ is a function defined by $\dots$	30	Numerical	1.5	179
8	If $f : \mathbb{R} - \left\{-\frac{4}{3}\right\} \rightarrow \mathbb{R}$ is a function defined by $\dots$	30	Numerical	1.5	180
9	If $f(x) = [4 - (x - 7)^3]^{1/5}$ is a real invertible function, then find $f^{-1}(x)$.	30	Numerical	1.5	181
10	The function $f : X \rightarrow Y$ is not invertible. State the reason.	31	Numerical	1.5	181
11	Which one of the following graphs is a function of x ?	32	Numerical	1.5	182
12	Let $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$ be a function defined by $f(1) = a$, $f(2) = b$ and $\dots$	33a	Proof	1.5	183
13	Let $S = \{1, 2, 3\}$. Determine whether the functions $f : S \rightarrow S$ defined as below $\dots$	33a	Numerical	1.5	184
(i)	$f = \{(1, 1), (2, 2), (3, 3)\}$	33a	Numerical	1.5	184
(ii)	$f = \{(1, 2), (2, 1), (3, 1)\}$	33a	Numerical	1.5	184
(iii)	$f = \{(1, 3), (3, 2), (2, 1)\}$	33a	Numerical	1.5	184
(iv)	$f = \{(1, 3), (2, 2), (3, 1)\}$	33a	Numerical	1.5	184
14	Explain why for the following functions $f : X \rightarrow Y$, we cannot define f^{-1} :	31	Proof	1.5	185
(i)	$X = \mathbb{R} - \{0\}, Y = \mathbb{R}$ and $f(x) = \frac{1}{x}$ for all $\dots$	31	Proof	1.5	185
(ii)	$X = \{-1, 0, 1, 2\}, Y = \{2, 3\}$ and $f(x) = 2$ for $x = -1, 0, 1$ and $\dots$	31	Proof	1.5	185
(iii)	$X = \mathbb{R} = Y$ and $f(x) = x^2 + 1$ for all $x \in \mathbb{R}$	31	Proof	1.5	185
15	Let $S = \{a, b, c\}$ and $T = \{1, 2, 3\}$. Find f^{-1} of the following functions $\dots$	29	Numerical	1.5	185
(i)	$f = \{(a, 3), (b, 2), (c, 1)\}$	29	Numerical	1.5	185
(ii)	$f = \{(a, 2), (b, 1), (c, 1)\}$	29	Numerical	1.5	185
16	Let $\mathbb{R}$ be the set of all real numbers. Show that the function $\dots$	33b	Proof	1.5	186
17	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $\dots$	33b	Proof	1.5	187
18	Consider a bijective function $f : \mathbb{R}_0^+ \rightarrow (7, \infty)$ given by $\dots$	33b	Numerical	1.5	188

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
19	If $A = \mathbb{R} - \{-3\}$ and $B = \mathbb{R} - \{2\}$ and a function $f : A \rightarrow B$ be ...	33b	Proof	1.5	189
20	If $f : A \rightarrow A$ and $A = \mathbb{R} - \{\frac{8}{5}\}$, show that the function ...	33b	Proof	1.5	190
21	Consider the function $f : \mathbb{R}_+ \rightarrow [9, \infty)$ defined by $f(x) = x^2 + 9$...	33b	Proof	1.5	191
22	If $\mathbb{R}_0$ is the set of all non-zero real numbers, show that the function ...	33b	Proof	1.5	192
23	Prove that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2 + x + 1$ is ...	33b	Proof	1.5	193
24	Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 4x^2 + 12x + 15$, where S is ...	33b	Proof	1.5	194
25	Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 9x^2 + 6x - 5$, where S is ...	33b	Numerical	1.5	195
26	If function $f : \mathbf{R} \rightarrow \mathbf{R}$ is defined by $f(x) = 5x - 3$, then find ...	34	Numerical	1.5	197
27	A teacher wrote $f(x) = x^9$ on the board and asked her student to examine whether the ...	35	Numerical	1.5	197
28	A part of the graph of the function $f(x) = 2x^3 - 3x^2 - 12x + 8, x \in \mathbf{R}$...	36	Numerical	1.5	199
(i)	Explain why 'f' does not have an inverse.	36	Numerical	1.5	199
(ii)	The domain of 'f' is now restricted to $a \leq x \leq b$ where $a < 0$ and ...	36	Numerical	1.5	199
29	"A function f is called a self-inverse function if $f^{-1}(x) = f(x)$ for all values of ...	37	Proof	1.5	200

Chapter 2: Inverse Trigonometric Functions

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
--------	---------------------------	----------	------	---------	------

Exercise 2.1 – Principal Values, Domain and Range of Inverse Trigonometric Functions

1	Find the principal values of the following:	1	Numerical	2.1	202
---	---	---	-----------	-----	-----

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	$\sin^{-1}(\frac{1}{\sqrt{2}})$	1	Numerical	2.1	202
(ii)	$\cos^{-1}(\frac{1}{2})$	1	Numerical	2.1	202
(iii)	$\tan^{-1}(\sqrt{3})$	1	Numerical	2.1	202
2	Find the principal values of the following:	1	Numerical	2.1	203
(i)	$\sec^{-1}(\frac{2}{\sqrt{3}})$	1	Numerical	2.1	203
(ii)	$\cot^{-1}(\sqrt{3})$	1	Numerical	2.1	203
(iii)	$\operatorname{cosec}^{-1}(\sqrt{2})$	1	Numerical	2.1	203
3	Find the principal values of the following:	1	Numerical	2.1	204
(i)	$\cos^{-1}(-\frac{1}{2})$	1	Numerical	2.1	204
(ii)	$\tan^{-1}(-1)$	1	Numerical	2.1	204
(iii)	$\operatorname{cosec}^{-1}(-\sqrt{2})$	1	Numerical	2.1	204
4	Find the principal values of the following:	1	Numerical	2.1	205
(i)	$\cot^{-1}(-\frac{1}{\sqrt{3}})$	1	Numerical	2.1	205
(ii)	$\sec^{-1}(-\frac{2}{\sqrt{3}})$	1	Numerical	2.1	205
(iii)	$\sin^{-1}(-1)$	1	Numerical	2.1	205
5	Evaluate the following:	2	Numerical	2.1	206
(i)	$\cos^{-1}(\cos \frac{4\pi}{3})$	2	Numerical	2.1	206
(ii)	$\sin^{-1}(\sin \frac{4\pi}{3})$	2	Numerical	2.1	206
(iii)	$\tan^{-1}(\tan \frac{3\pi}{4})$	2	Numerical	2.1	206
6	Evaluate the following:	2	Numerical	2.1	207
(i)	$\cos^{-1}(\cos \frac{5\pi}{3})$	2	Numerical	2.1	207
(ii)	$\tan^{-1}(\tan \frac{9\pi}{8})$	2	Numerical	2.1	207
(iii)	$\sec^{-1}(\sec \frac{9\pi}{5})$	2	Numerical	2.1	207
7	Show that:	2	Numerical	2.1	209
(i)	$\tan^{-1}(\tan \frac{5\pi}{6}) \neq \frac{5\pi}{6}$, what is its value?	2	Numerical	2.1	209
(ii)	$\cos^{-1}(\cos(-\frac{\pi}{6})) \neq -\frac{\pi}{6}$, what is its value?	2	Numerical	2.1	209
(iii)	$\sin^{-1}(\sin \frac{5\pi}{3}) \neq \frac{5\pi}{3}$, what is its value?	2	Numerical	2.1	209
8	Find the domain of the following functions:	3	Numerical	2.1	210
(i)	$\sin^{-1}(1-x)$	3	Numerical	2.1	210
(ii)	$\sec^{-1}(2x-3)$	3	Numerical	2.1	210
(iii)	$\cos^{-1} x + \sin x$	3	Numerical	2.1	210
9	See parts below:	4	Numerical	2.1	211
(i)	Write the range of one branch of $\sin^{-1} x$, other than the principal branch.	4	Numerical	2.1	211
(ii)	Write the range of one branch of $\cos^{-1} x$, other than the principal branch.	4	Numerical	2.1	211
10	Find the maximum value of $(\cos^{-1} x)^2$.	5	Numerical	2.1	212

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
11	If $\cot^{-1}(\frac{3}{4}) = x$, find the value of $\cos x$.	6	Numerical	2.1	212
12	See parts below:	6	Numerical	2.1	213
(i)	If $\tan^{-1}(\frac{3}{4}) = x$, find the values of $\cos x$ and $\sin x$.	6	Numerical	2.1	213
(ii)	If $\cot^{-1}(-\frac{1}{7}) = x$, find the values of $\sin x$ and $\cos x$.	6	Numerical	2.1	213
(iii)	If $\tan^{-1} x = \sin^{-1}(\frac{1}{2})$, find the value of x .	6	Numerical	2.1	213
13	Evaluate the following:	7	Numerical	2.1	214
(i)	$\cot(\tan^{-1} \sqrt{3})$	7	Numerical	2.1	214
(ii)	$\sin(\frac{\pi}{6} - \sin^{-1}(-\frac{\sqrt{3}}{2}))$	7	Numerical	2.1	214
(iii)	$\cos(\cos^{-1}(-\frac{\sqrt{3}}{2}) + \frac{\pi}{6})$	7	Numerical	2.1	214
14	Prove the following:	7	Proof	2.1	216
(i)	$\sin^{-1}(-\frac{1}{2}) + \cos^{-1}(-\frac{\sqrt{3}}{2}) = \frac{2\pi}{3}$	7	Proof	2.1	216
(ii)	$\tan^{-1}(-1) + \cos^{-1}(-\frac{1}{\sqrt{2}}) = \frac{\pi}{2}$	7	Proof	2.1	216
15	Using principal values, find the values of:	7	Numerical	2.1	217
(i)	$\cos^{-1}(\frac{1}{2}) - 2 \sin^{-1}(-\frac{1}{2})$	7	Numerical	2.1	217
(ii)	$\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3})$	7	Numerical	2.1	217
(iii)	$\tan^{-1}(1) + \cos^{-1}(-\frac{1}{2})$	7	Numerical	2.1	217
(iv)	$\operatorname{cosec}^{-1}(-1) + \cot^{-1}(-\frac{1}{\sqrt{3}})$	7	Numerical	2.1	217
16	Find the values of the following:	8	Numerical	2.1	218
(i)	$\tan^{-1}(2 \cos(2 \sin^{-1} \frac{1}{2}))$	8	Numerical	2.1	218
(ii)	$\tan^{-1}(2 \sin(2 \cos^{-1} \frac{\sqrt{3}}{2}))$	8	Numerical	2.1	218
(iii)	$\sin^{-1}(\frac{1}{2} \sin(2 \cot^{-1}(-1)))$	8	Numerical	2.1	218
(iv)	$\cot(\sin^{-1}(\cos(\tan^{-1} 1)))$	8	Numerical	2.1	218
17	Find the value of:	2	Numerical	2.1	220
(i)	$\tan^{-1}(\tan \frac{5\pi}{6}) + \cos^{-1}(\cos \frac{13\pi}{6})$	2	Numerical	2.1	220
(ii)	$\tan^{-1}(-\frac{1}{\sqrt{3}}) + \cot^{-1}(\frac{1}{\sqrt{3}}) + \tan^{-1}(\sin(-\frac{\pi}{2})) \dots$	2	Numerical	2.1	220
18	(see parts below)	7	Numerical	2.1	221
(i)	Evaluate: $\sin^{-1}(\sin \frac{3\pi}{4}) + \cos^{-1}(\cos \pi) + \tan^{-1}(1)$	7	Numerical	2.1	221
(ii)	Evaluate: ...	7	Numerical	2.1	222
(iii)	Evaluate: ...	7	Numerical	2.1	223
19	(see parts below)	9	Numerical	2.1	224
(i)	Evaluate: $\cos(\sin^{-1}(-\frac{3}{5}))$	9	Numerical	2.1	224
(ii)	Evaluate: $\operatorname{cosec}(\cos^{-1}(-\frac{12}{13}))$	9	Numerical	2.1	225
(iii)	Evaluate: $\sin(2 \cot^{-1} \frac{1}{5})$	9	Numerical	2.1	226
20	Show that: $\tan(\frac{1}{2} \sin^{-1} \frac{3}{4}) = \frac{4-\sqrt{7}}{2}$	10	Proof	2.1	227

Exercise 2.2 – Properties and Simplification of Inverse Trigonometric Functions

1	Evaluate the following:	2	Numerical	2.2	229
(i)	$\sin^{-1}(\sin 2)$	2	Numerical	2.2	229

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	$\sin^{-1}(\sin 5)$	2	Numerical	2.2	229
(iii)	$\tan^{-1}(\tan 4)$	2	Numerical	2.2	229
(iv)	$\cos^{-1}(\cos 10)$	2	Numerical	2.2	229
2	Find the values of:	7	Numerical	2.2	230
(i)	$\sin(\sin^{-1} x + \cos^{-1} x), \quad x \leq 1$	7	Numerical	2.2	230
(ii)	$\cot(\tan^{-1} x + \cot^{-1} x), \quad x \in \mathbb{R}$	7	Numerical	2.2	230
3	Find the values of:	7	Numerical	2.2	231
(i)	$\csc(\sin^{-1} x + \cos^{-1} x), \quad x \leq 1$	7	Numerical	2.2	231
(ii)	$\cos(\sec^{-1} x + \csc^{-1} x), \quad x \geq 1$	7	Numerical	2.2	231
4	Find the values of:	9	Numerical	2.2	231
(i)	$\tan^{-1}(\sin(-\frac{\pi}{2}))$	9	Numerical	2.2	231
(ii)	$\cos(\tan^{-1} \frac{3}{4})$	9	Numerical	2.2	231
(iii)	$\tan(\cos^{-1} \frac{8}{17})$	9	Numerical	2.2	231
5	Simplify the following:	11	Numerical	2.2	232
(i)	$\tan^{-1}(\frac{\sin x}{1+\cos x})$	11	Numerical	2.2	232
(ii)	$\tan^{-1}(\frac{x}{\sqrt{1-x^2}}), \quad x < 1$	11	Numerical	2.2	232
(iii)	$\sin^{-1}(\frac{x}{\sqrt{1+x^2}})$	11	Numerical	2.2	232
(iv)	$\csc^{-1}(\frac{\sqrt{1+x^2}}{x})$	11	Numerical	2.2	232
(v)	$\cot^{-1}(\frac{x}{\sqrt{1-x^2}}), \quad x < 1$	11	Numerical	2.2	232
(vi)	$\csc^{-1}(\frac{1}{\sqrt{1-x^2}}), \quad x < 1$	11	Numerical	2.2	232
6	Solve the following equations for x :	12	Numerical	2.2	235
(i)	$\tan^{-1} x = \sin^{-1}(\frac{1}{\sqrt{2}})$	12	Numerical	2.2	235
(ii)	$\sin^{-1} x = \cos^{-1}(\frac{\sqrt{3}}{2})$	12	Numerical	2.2	235
(iii)	$\cos^{-1} x = \sin^{-1}(-\frac{1}{2})$	12	Numerical	2.2	235
(iv)	$\cos(\sin^{-1} x) = \frac{1}{2}$	12	Numerical	2.2	235
(v)	$4 \sin^{-1} x + \cos^{-1} x = \pi$	12	Numerical	2.2	235
(vi)	$\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$	12	Numerical	2.2	235
(vii)	$5 \tan^{-1} x + 3 \cot^{-1} x = 2\pi$	12	Numerical	2.2	235
(viii)	$\tan^{-1} x^{-1} = \cot^{-1}(\frac{4}{x}), \quad x > 0$	12	Numerical	2.2	235
7	If $\sin(\sin^{-1} \frac{2}{3} + \cos^{-1} x) = 1$, then find the value of x .	12	Numerical	2.2	237
8	Find the values of:	13	Numerical	2.2	238
(i)	$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$	13	Numerical	2.2	238
(ii)	$\cot^{-1} 2 + \cot^{-1} 3$	13	Numerical	2.2	238
9	If $\tan^{-1} x - \tan^{-1} y = \frac{\pi}{4}, \quad xy > -1$, then find the value of ...	13	Numerical	2.2	239
10	See parts below:	7	Numerical	2.2	240
(i)	If $\tan^{-1} x = \frac{\pi}{10}$ for some $x \in \mathbb{R}$, find the value ...	7	Numerical	2.2	240
(ii)	If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then find the value of ...	7	Numerical	2.2	240
(iii)	If $\sec^{-1} x + \sec^{-1} y = \frac{2\pi}{3}$, then find the value of ...	7	Numerical	2.2	240
11	If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then find the value of ...	5	Numerical	2.2	241

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
12	Prove the following:	13	Proof	2.2	242
(i)	$\tan^{-1}\left(\frac{1+x}{1-x}\right) = \frac{\pi}{4} + \tan^{-1} x, \quad x < 1$	13	Proof	2.2	242
(ii)	$\tan^{-1} x + \cot^{-1}(x+1) = \tan^{-1}(x^2 + x + 1)$	13	Proof	2.2	242
(iii)	$\cos[\tan^{-1}\{\cot(\sin^{-1} x)\}] = x$	13	Proof	2.2	242
13	Prove the following (for suitable values of x and y):	13	Proof	2.2	243
(i)	$\tan^{-1} \sqrt{x} + \tan^{-1} \sqrt{y} = \tan^{-1} \frac{\sqrt{x} + \sqrt{y}}{1 + \sqrt{xy}}$	13	Proof	2.2	243
(ii)	$\tan^{-1} \frac{x + \sqrt{x}}{1 - x^{3/2}} = \tan^{-1} x + \tan^{-1} \sqrt{x}$	13	Proof	2.2	243
14	Write the following functions in simplest form:	11	Numerical	2.2	244
(i)	$\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$	11	Numerical	2.2	244
(ii)	$\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$	11	Numerical	2.2	244
(iii)	$\tan^{-1}\left(\frac{2\sqrt{x}}{1-x}\right)$	11	Numerical	2.2	244
(iv)	$\sin^{-1}\left(\sqrt{\frac{x}{1+x}}\right)$	11	Numerical	2.2	244
(v)	$\tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)$	11	Numerical	2.2	244
(vi)	$\cos^{-1}(1 - 2x^2)$	11	Numerical	2.2	244
(vii)	$\sec^{-1}\left(\frac{1}{2x^2-1}\right)$	11	Numerical	2.2	244
(viii)	$\tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right)$	11	Numerical	2.2	244
15	Prove that: $\sin(\cot^{-1}(\cos(\tan^{-1} x))) = \sqrt{\frac{x^2+1}{x^2+2}}$	8	Proof	2.2	246
16	Find the value of ...	13	Numerical	2.2	247
17	Prove the following:	13	Proof	2.2	248
(i)	$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{2}{11} = \tan^{-1} \frac{3}{4}$	13	Proof	2.2	248
(ii)	$\tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} = \tan^{-1} \frac{2}{9}$	13	Proof	2.2	248
(iii)	$\tan^{-1} 2 - \tan^{-1} 1 = \tan^{-1} \frac{1}{3}$	13	Proof	2.2	248
(iv)	$\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{2}$	13	Proof	2.2	248
(v)	Prove that $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$.	13	Proof	2.2	250
(vi)	Prove that ...	13	Proof	2.2	251
(vii)	Prove that $\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3 = \frac{\pi}{2}$.	13	Proof	2.2	252
(viii)	Prove that $\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$.	13	Proof	2.2	253
18	If ...	13	Proof	2.2	254
19	(see parts below)	13	Proof	2.2	255
(i)	Prove that $\sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{2}{\sqrt{5}} = \frac{\pi}{2}$.	13	Proof	2.2	255
(ii)	Prove that $\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$.	13	Proof	2.2	256
(iii)	Prove that $\cos^{-1} \frac{3}{5} + \sin^{-1} \frac{12}{13} = \sin^{-1} \frac{56}{65}$.	13	Proof	2.2	257
(iv)	Prove that $\tan^{-1} \frac{1}{3} + \sec^{-1} \frac{\sqrt{5}}{2} = \frac{\pi}{4}$.	13	Proof	2.2	258
(v)	Prove that $\cos^{-1} \frac{4}{5} + \tan^{-1} \frac{3}{5} = \tan^{-1} \frac{27}{11}$.	13	Proof	2.2	259
(vi)	Prove that $\cos^{-1} \frac{5}{\sqrt{41}} + \cot^{-1} \frac{4}{5} = \frac{\pi}{2}$.	13	Proof	2.2	259

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(vii)	Prove that ...	13	Proof	2.2	260
(viii)	Prove that $\sin^{-1} \frac{8}{17} + \cos^{-1} \frac{4}{5} = \cot^{-1} \frac{36}{77}$.	13	Proof	2.2	261
(ix)	Prove that ...	13	Proof	2.2	262
(x)	Prove that ...	13	Proof	2.2	263
20	If $\cos^{-1}(\frac{x}{a}) + \cos^{-1}(\frac{y}{b}) = \alpha$, prove that ...	13	Proof	2.2	264
21	(see parts below)	9	Numerical	2.2	265
(i)	Evaluate $\tan(2 \tan^{-1} \frac{1}{2} - \cot^{-1} 3)$.	9	Numerical	2.2	265
(ii)	Evaluate $\tan(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4})$.	9	Numerical	2.2	266
22	(see parts below)	9	Numerical	2.2	267
(i)	Find the value of $\tan(\cos^{-1} \frac{3}{5} + \tan^{-1} \frac{1}{4})$.	9	Numerical	2.2	267
(ii)	Find the value of $\cos(\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{5}{13})$.	9	Numerical	2.2	268
(iii)	Find the value of $\cot(\cos^{-1} \frac{4}{5} + \sin^{-1} \frac{2}{\sqrt{13}})$.	9	Numerical	2.2	268
(iv)	Find the value of $\sin(2 \tan^{-1} \frac{2}{3}) + \cos(\tan^{-1} \sqrt{3})$.	9	Numerical	2.2	269
23	(see parts below)	12	Numerical	2.2	270
(i)	Solve the equation for x : ...	12	Numerical	2.2	270
(ii)	Solve the equation for x : $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{4}$.	12	Numerical	2.2	271
(iii)	Solve the equation for x : $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1} \frac{6}{17}$.	12	Numerical	2.2	272
(iv)	Solve the equation for x : ...	12	Numerical	2.2	273
(v)	Solve the equation for x : ...	12	Numerical	2.2	274
(vi)	Solve the equation for x : ...	12	Numerical	2.2	275
(vii)	Solve the equation for x : $\cos(\sin^{-1} x) = \frac{1}{9}$.	12	Numerical	2.2	276
(viii)	Solve the equation for x : $\cos(2 \sin^{-1} x) = \frac{1}{9}$.	12	Numerical	2.2	277
(ix)	Solve the equation for x : $\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$.	12	Numerical	2.2	278
(x)	Solve the equation for x : $\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$.	12	Numerical	2.2	279
(xi)	Solve the equation for x : ...	12	Numerical	2.2	279
(xii)	Solve the equation for x : ...	12	Numerical	2.2	280

Chapter 3: Matrices

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
--------	---------------------------	----------	------	---------	------

Exercise 3.1 – Order, Elements and Construction of Matrices

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
1	If A is a matrix of type $p \times q$ and R is a row of A , then what is the type of ...	1	Numerical	3.1	282
2	If A is a column matrix with 5 rows, then what type of matrix is a row of A ?	1	Numerical	3.1	282
3	If a matrix has 18 elements, what the possible orders it can have? What, if it has 11 ...	2	Numerical	3.1	282
4	(i) If A is a 3×3 matrix whose elements are given by ...	3	Numerical	3.1	283
5	(i) For a 2×2 matrix, $A = [a_{ij}]$, whose elements are given by ...	3	Numerical	3.1	284
6	Construct a 2×2 matrix $A = [a_{ij}]$ whose elements a_{ij} are given by:	4	Numerical	3.1	285
(i)	$a_{ij} = 2i - j$	4	Numerical	3.1	285
(ii)	$a_{ij} = 2i - 3j $	4	Numerical	3.1	285
7	Construct a 2×3 matrix $B = [b_{ij}]$ whose elements b_{ij} are given by:	4	Numerical	3.1	286
(i)	$b_{ij} = i - 3j$	4	Numerical	3.1	286
(ii)	$b_{ij} = (i + 2j)^2$	4	Numerical	3.1	286
8	See parts below:	5	Numerical	3.1	287
(i)	If ...	5	Numerical	3.1	287
(ii)	If ...	5	Numerical	3.1	287
(iii)	Find the value of y , if ...	5	Numerical	3.1	287
(iv)	Find the value of x , if ...	5	Numerical	3.1	287
(v)	If ...	5	Numerical	3.1	287
(vi)	If ...	5	Numerical	3.1	287
(vii)	If ...	5	Numerical	3.1	287
9	Write the value of $x - y + z$ from the following equation: ...	5	Numerical	3.1	289
10	If ...	5	Numerical	3.1	290
11	Find the values of a, b, c and d from the following equations: ...	5	Numerical	3.1	290
12	Find x, y, a and b if ...	5	Numerical	3.1	292
13	See parts below:	6	Numerical	3.1	293
(i)	Write the number of all possible matrices of order 2×3 with each entry ...	6	Numerical	3.1	293
(ii)	Write the number of all possible matrices of order 2×2 with each entry ...	6	Numerical	3.1	293

Exercise 3.2 – Addition, Subtraction and Scalar Multiplication of Matrices

1	If A and B are matrices of order $3 \times n$ and $m \times 5$ respectively, then find ...	1	Numerical	3.2	294
---	--	---	-----------	-----	-----

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
2	If $A = \begin{bmatrix} 2 & 43 & 2 \end{bmatrix} \dots$	7	Numerical	3.2	294
(i)	$A + B$	7	Numerical	3.2	294
(ii)	$A - B$	7	Numerical	3.2	294
(iii)	$3A - C$	7	Numerical	3.2	294
3	Compute the following:	7	Numerical	3.2	296
(i)	$\begin{bmatrix} -1 & 4 & -68 & 5 & 162 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 12 & 7 & 68 & 0 & 53 & 2 & 4 \end{bmatrix} \dots$	7	Numerical	3.2	296
(ii)	$\begin{bmatrix} \cos^2 x & \sin^2 x & \sin^2 x & \cos^2 x \end{bmatrix} + \begin{bmatrix} \sin^2 x & \cos^2 x & \cos^2 x & \sin^2 x \end{bmatrix} \dots$	7	Numerical	3.2	296
4	If $A = \begin{bmatrix} 1 & 5 & -2 & 3 \end{bmatrix}$ and ...	7	Numerical	3.2	297
5	If $A = \begin{bmatrix} 1 & 2 & 32 & 3 & 1 \end{bmatrix}$ and ...	7	Numerical	3.2	298
6	If ...	8	Numerical	3.2	298
7	If $A = \text{diagonal } [1, -2, 5]$, $B = \text{diagonal } [3, 0, -4]$ and ...	7	Numerical	3.2	299
8	If $A = \begin{bmatrix} 2 & -35 & 0 \end{bmatrix}$ and ...	5	Numerical	3.2	300
9	If ...	5	Numerical	3.2	300
10	If ...	5	Numerical	3.2	301
11	Given ...	5	Numerical	3.2	302
12	If ...	5	Numerical	3.2	303
13	Find X if $Y = \begin{bmatrix} 3 & 21 & 4 \end{bmatrix}$ and ...	8	Numerical	3.2	304
14	Find matrices X and Y , if $X + Y = \begin{bmatrix} 7 & 02 & 5 \end{bmatrix}$ and ...	8	Numerical	3.2	305
15	Find matrices X and Y if ...	8	Numerical	3.2	306
16	Find a matrix X such that $3A - 2B + X = O$, where ...	8	Numerical	3.2	307
17	If $A = \begin{bmatrix} 9 & 17 & 8 \end{bmatrix}$ and ...	8	Numerical	3.2	308

Exercise 3.3 – Multiplication of Matrices

1	If $A = \begin{bmatrix} -1 & 2 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -17 \end{bmatrix}$, write the ...	9	Numerical	3.3	309
2	If $A = \begin{bmatrix} 2 & 1 & 44 & 1 & 5 \end{bmatrix}$ and ...	9	Numerical	3.3	310
3	If A is a matrix of order 3×3 and B is a matrix of order 2×3 , can ...	9	Numerical	3.3	311
4	If $A = \begin{bmatrix} 1 & 2 & -1 & 12 & 3 \end{bmatrix} \dots$	10	Numerical	3.3	311
5	If $A = \begin{bmatrix} 2 & 43 & 1 & -1 & 2 \end{bmatrix} \dots$	10	Numerical	3.3	312
6	If $\begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 & 0 \end{bmatrix} = O \dots$	11	Numerical	3.3	313
7	If ...	11	Numerical	3.3	314
8	Write the value of $x + y + z$ if ...	11	Numerical	3.3	314
9	Compute the indicated products:	12	Numerical	3.3	315
(i)	$\begin{bmatrix} 1 & -22 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 32 & 3 & 1 \end{bmatrix} \dots$	12	Numerical	3.3	315
(ii)	$\begin{bmatrix} 3 & -1 & 3 & -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & -31 & 03 & 1 \end{bmatrix}$	12	Numerical	3.3	315
	...				
10	If $A = \begin{bmatrix} \alpha & \beta\gamma & -\alpha \end{bmatrix}$ and $A^2 = I$, find ...	11	Numerical	3.3	316
11	If $A = \begin{bmatrix} 1 & 00 & -1 \end{bmatrix}$ and ...	12	Numerical	3.3	317
12	See parts below:	13	Proof	3.3	318
(i)	If $A = \begin{bmatrix} 1 & 11 & 1 \end{bmatrix}$ and ...	13	Proof	3.3	318

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	If $A = \begin{bmatrix} 1 & 23 & 4 \end{bmatrix} \dots$	13	Proof	3.3	318
13	If $A = \begin{bmatrix} \alpha & 01 & 1 \end{bmatrix} \dots$	11	Numerical	3.3	320
14	If A is a square matrix such that $A^2 = I$, then find the simplified value of $\dots$	14	Numerical	3.3	320
15	Write the following as a single matrix:	15	Numerical	3.3	321
(i)	$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -5 & 7 \end{bmatrix} \begin{bmatrix} 2 & -1 & 50 & 2 & 47 & 5 & 0 \end{bmatrix} - \dots$	15	Numerical	3.3	321
(ii)	$\begin{bmatrix} 3 & 2 & 57 & -4 & 0 \\ 7 & -85 & 9 \end{bmatrix} \begin{bmatrix} 2 & 22 & -13 & 5 \end{bmatrix} - \dots$	15	Numerical	3.3	321
16	See parts below:	16	Proof	3.3	322
(i)	If $\dots$	16	Proof	3.3	322
(ii)	If $\dots$	16	Proof	3.3	322
17	If $A = \begin{bmatrix} 2 & -13 & 2 \end{bmatrix}$ and $\dots$	15	Numerical	3.3	324
18	If $A = \begin{bmatrix} 2 & 1 \end{bmatrix} \dots$	13	Proof	3.3	325
19	See parts below:	11	Numerical	3.3	326
(i)	Solve for x and y , given that $\dots$	11	Numerical	3.3	326
(ii)	Solve for x and y , given that $\dots$	11	Numerical	3.3	326
20	See parts below:	17	Proof	3.3	327
(i)	If $A = \begin{bmatrix} 3 & -5 & -4 & 2 \end{bmatrix}$, show that $\dots$	17	Proof	3.3	327
(ii)	If $A = \begin{bmatrix} 1 & 24 & 1 \end{bmatrix}$, find the value of $\dots$	17	Proof	3.3	327
21	If $A = \begin{bmatrix} 1 & 00 & 1 \end{bmatrix}$ and $\dots$	13	Proof	3.3	329
22	If $A = \begin{bmatrix} 4 & 2 & -1 & 1 \end{bmatrix}$, prove that $(A - 2I)(A - 3I) = O$.	17	Proof	3.3	330
23	If $A = \begin{bmatrix} 3 & -24 & -2 \end{bmatrix}$, then find k so that $\dots$	11	Numerical	3.3	331
24	(see parts below)	17	Proof	3.3	332
(i)	If $A = \begin{bmatrix} 1 & 22 & 1 \end{bmatrix}$, show that $f(A) = O$, where $\dots$	17	Proof	3.3	332
(ii)	If $A = \begin{bmatrix} -1 & 23 & 1 \end{bmatrix}$, find $f(A)$, where $\dots$	17	Numerical	3.3	333
25	If $A = \begin{bmatrix} 2 & 31 & 2 \end{bmatrix}$ and $\dots$	17	Numerical	3.3	334
(i)	find λ, μ so that $A^2 = \lambda A + \mu I$	17	Numerical	3.3	334
(ii)	prove that $A^3 - 4A^2 + A = O$	17	Numerical	3.3	334
26	If $A = \begin{bmatrix} 1 & 2 & 22 & 1 & 22 & 2 & 1 \end{bmatrix}$, verify that $\dots$	17	Proof	3.3	336
27	See parts below:	13	Proof	3.3	337
(i)	Let A and B be square matrices of same order. Does $\dots$	13	Proof	3.3	337
(ii)	Let A and B be square matrices of order 3×3 . Is $(AB)^2 = A^2B^2$? $\dots$	13	Proof	3.3	337

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
28	If A and B are square matrices of the same order such that $AB = BA$, prove that:	13	Proof	3.3	338
(i)	$(A + B)(A - B) = A^2 - B^2$	13	Proof	3.3	338
(ii)	$(A + B)^2 = A^2 + 2AB + B^2$	13	Proof	3.3	338
(iii)	$(A - B)^2 = A^2 - 2AB + B^2$	13	Proof	3.3	338
(iv)	$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$	13	Proof	3.3	338
(v)	$(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$	13	Proof	3.3	338
29	If $P = \begin{bmatrix} p & q & -q & p \end{bmatrix}$ and ...	13	Proof	3.3	339
30	Find x , if:	11	Numerical	3.3	340
(i)	$\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 22 & 5 & 11 & 5 & 3 & 2 \end{bmatrix} \begin{bmatrix} 12x \\ 1 \\ 0 \end{bmatrix} = 1$	11	Numerical	3.3	340
(ii)	$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 02 & 0 & 11 & 0 & 2 \end{bmatrix} \begin{bmatrix} 02x \\ 11 \\ 0 \end{bmatrix} = 11$	11	Numerical	3.3	340
31	If $A = \begin{bmatrix} 2 & 3 & -1 & 2 \end{bmatrix}$ and $f(x) = x^2 - 4x + 7$, show that ...	18	Numerical	3.3	341
32	If $A = \begin{bmatrix} 1 & 11 & 1 \end{bmatrix}$, prove that ...	19	Proof	3.3	343
33	See parts below:	20	Numerical	3.3	344
(i)	Find a 2×2 matrix B such that ...	20	Numerical	3.3	344
(ii)	If $A = \begin{bmatrix} 3 & -4 & -1 & 2 \end{bmatrix}$, find matrix B such that ...	20	Numerical	3.3	344
34	Find the matrix A such that ...	20	Numerical	3.3	345
35	Three students Ram, Mohan and Ankit go to a shop to buy stationary. Ram purchases 2 dozen ...	21	Numerical	3.3	347

Exercise 3.4 – Transpose, Symmetric and Skew-Symmetric Matrices

1	If $A = \begin{bmatrix} 2 & -3 & 0 & -1 & 4 & 5 \end{bmatrix}$, then find $(3A)'$.	22	Numerical	3.4	348
2	If $A = \begin{bmatrix} 2 & -1 & 54 & 0 & 3 \end{bmatrix}$ and ...	22	Numerical	3.4	349
3	If ...	22	Numerical	3.4	350
4	If $A^T = \begin{bmatrix} 3 & 4 & -1 & 20 & 1 \end{bmatrix}$ and ...	22	Numerical	3.4	351
5	For what value of k the matrix $\begin{bmatrix} 0 & k & -6 & 0 \end{bmatrix}$ is a ...	23	Numerical	3.4	352
6	(see parts below)	23	Proof	3.4	353
(i)	Show that $(A + A')$ is a symmetric matrix, if ...	23	Proof	3.4	353
(ii)	If $A = \begin{bmatrix} 5 & -1 & -2 & 6 \end{bmatrix}$, determine whether $A - A^T$ is ...	23	Numerical	3.4	353
7	(see parts below)	23	Proof	3.4	354
(i)	If $A = \begin{bmatrix} 5 & ab & 0 \end{bmatrix}$ and A is symmetric matrix, show that ...	23	Proof	3.4	354
(ii)	If $A = \begin{bmatrix} 5 & xy & 0 \end{bmatrix}$ and A is symmetric, then write the ...	23	Numerical	3.4	355
(iii)	If the matrix $\begin{bmatrix} 6 & -x^2 & 2x & -15 & 10 \end{bmatrix}$ is symmetric, find the ...	23	Numerical	3.4	356

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
8	If A is a square matrix, prove that $A'A$ is symmetric.	24	Proof	3.4	356
9	If $A = \begin{bmatrix} 3 & 57 & 9 \end{bmatrix}$ is written as $A = P + Q$, where P is ...	25	Numerical	3.4	357
10	If A, B are symmetric matrices and $AB = BA$, then show that AB is symmetric.	24	Proof	3.4	358
11	If A, B and AB are all symmetric matrices, then show that $AB = BA$.	24	Proof	3.4	358
12	If A, B are skew-symmetric matrices and $AB = BA$, then show that AB is symmetric.	24	Proof	3.4	359
13	If A, B are skew-symmetric matrices of same order and AB is symmetric, then show that ...	24	Proof	3.4	359
14	Let $A = \begin{bmatrix} 5 & -16 & 7 \end{bmatrix}$ and ...	22	Proof	3.4	360
15	If $A = \begin{bmatrix} 3 & \sqrt{3} & 24 & 2 & 0 \end{bmatrix}$ and ...	22	Proof	3.4	361
16	If $A' = \begin{bmatrix} -2 & 31 & 2 \end{bmatrix}$ and ...	22	Numerical	3.4	362
17	If $A = \begin{bmatrix} 1 & -43 \end{bmatrix}$ and ...	22	Proof	3.4	363
18	If $A = \begin{bmatrix} 1 & 24 & 15 & 6 \end{bmatrix}$ and ...	22	Proof	3.4	364
19	If $A = \begin{bmatrix} 2 & 4 & 03 & 9 & 6 \end{bmatrix}$ and ...	22	Proof	3.4	365
20	If $A = \begin{bmatrix} 3 & 2 & -1 & 1 \end{bmatrix}$ and ...	22	Numerical	3.4	367
21	If $A = \begin{bmatrix} \sin \alpha & \cos \alpha & -\cos \alpha & \sin \alpha \end{bmatrix}$	26	Proof	3.4	367
22	... If $A = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 & -2 & 1 & 2 & -2 & -2 & -1 \end{bmatrix}$...	26	Proof	3.4	368
23	If $A = \begin{bmatrix} 1 & 2 & 22 & 1 & x-2 & 2 & -1 \end{bmatrix}$ is a matrix ...	26	Numerical	3.4	369
24	Find the integral value of x if ...	22	Numerical	3.4	370
25	Show that the matrix A , where ...	23	Proof	3.4	371
26	If the matrix $\begin{bmatrix} 0 & 2b & -23 & 1 & 33a & 3 & -1 \end{bmatrix}$ is ...	23	Numerical	3.4	372
27	If A and B are symmetric matrices of the same order, prove that $AB + BA$ is ...	24	Proof	3.4	373
28	If A and B are symmetric matrices such that AB and BA are both defined, then prove ...	24	Proof	3.4	373
29	Express each of the following matrices as the sum of a symmetric and a skew-symmetric ...	25	Numerical	3.4	374
(i)	$\begin{bmatrix} 3 & 51 & -1 \end{bmatrix}$	25	Numerical	3.4	374
(ii)	$\begin{bmatrix} 3 & -41 & -1 \end{bmatrix}$	25	Numerical	3.4	374
30	Express the following matrices as the sum of symmetric and skew-symmetric matrices:	25	Numerical	3.4	376
(i)	$\begin{bmatrix} 3 & 2 & 54 & 1 & 30 & 6 & 7 \end{bmatrix}$	25	Numerical	3.4	376

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	$\begin{bmatrix} 2 & 4 & -67 & 3 & 51 & -2 & 4 \end{bmatrix}$	25	Numerical	3.4	376
31	For what value of x , is ...	23	Numerical	3.4	377
32	If the matrix $A = \begin{bmatrix} a & 6 & b2c & x+1 & 8-1 & -8 & y-3 \end{bmatrix}$ is ...	23	Numerical	3.4	378

Chapter 4: Determinants

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
--------	---------------------------	----------	------	---------	------

Exercise 4.1 – Evaluation of Determinants: Minors and Cofactors

1	(see parts below)	1	Numerical	4.1	380
(i)	Evaluate the determinant: ...	1	Numerical	4.1	380
(ii)	Evaluate the determinant: ...	1	Numerical	4.1	380
(iii)	Evaluate the determinant: ...	2	Numerical	4.1	381
2	(see parts below)	1	Numerical	4.1	381
(i)	Evaluate the determinant: ...	1	Numerical	4.1	381
(ii)	Evaluate the determinant: ...	3	Numerical	4.1	382
(iii)	Evaluate the determinant: ...	3	Numerical	4.1	383
3	(see parts below)	2	Numerical	4.1	383
(i)	Find the value of $ \sin A \quad -\sin B \cos A \quad \cos B $, where ...	2	Numerical	4.1	383
(ii)	Find the value of $ \cos A \quad \sin A - \sin B \quad \cos B $, where ...	2	Numerical	4.1	384
4	(see parts below)	4	Numerical	4.1	385
(i)	If $A = \begin{bmatrix} 5 & 6 & -3 & -4 & 3 & 2 & -4 & -7 & 3 \end{bmatrix}$, then write the ...	4	Numerical	4.1	385
(ii)	If A_{ij} is the cofactor of the element a_{ij} of the determinant ...	4	Numerical	4.1	385
5	(see parts below)	5	Numerical	4.1	386
(i)	Find the value of x if: $ 2x+5 \quad 35x+2 \quad 9 = 0$	5	Numerical	4.1	386
(ii)	Find the value of x if: $ x+2 \quad 3x+5 \quad 4 = 3$	5	Numerical	4.1	387
(iii)	Find the value of x if: ...	5	Numerical	4.1	388
6	(see parts below)	5	Numerical	4.1	388
(i)	If ...	5	Numerical	4.1	388
(ii)	If ...	5	Numerical	4.1	389
7	(see parts below)	5	Numerical	4.1	390
(i)	If $x \in \mathbf{N}$ and ...	5	Numerical	4.1	390
(ii)	If $x \in \mathbf{I}$ and ...	5	Numerical	4.1	391

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
8	If $A = \begin{bmatrix} 1 & 24 & 2 \end{bmatrix}$, then show that $ 2A = 4 A $.	6	Proof	4.1	391
9	(see parts below)	3	Numerical	4.1	392
(i)	Evaluate the determinant: $\begin{vmatrix} 0 & -1 & 21 & 0 & 3 & -2 & -3 & 0 \end{vmatrix}$	3	Numerical	4.1	392
(ii)	Evaluate the determinant: $\begin{vmatrix} 3 & -4 & 51 & 1 & -22 & 3 & 1 \end{vmatrix}$	3	Numerical	4.1	393
(iii)	Evaluate the determinant: ...	3	Numerical	4.1	394
10	Prove that: ...	3	Proof	4.1	395
11	Show that the value of the determinant ...	2	Proof	4.1	396
12	Using cofactors of elements of second row, evaluate ...	7	Numerical	4.1	397

Exercise 4.2 – Properties of Determinants and Proving Identities

1	(see parts below)	8	Numerical	4.2	398
(i)	Without expanding, find the value of: ...	8	Numerical	4.2	398
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2	399
(iii)	Without expanding, find the value of: ...	8	Numerical	4.2	399
2	(see parts below)	8	Numerical	4.2	400
(i)	Without expanding, find the value of: ...	8	Numerical	4.2	400
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2	401
3	(see parts below)	8	Numerical	4.2	402
(i)	Without expanding, find the value of: ...	8	Numerical	4.2	402
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2	403
4	(see parts below)	8	Numerical	4.2	404
(i)	Without expanding, find the value of: ...	8	Numerical	4.2	404
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2	405
5	If A is a square matrix of order 2 and $ A = -5$, find the value of $ 3A $.	6	Numerical	4.2	405
6	If A is a square matrix of order 3 and $ A = 4$, then write the value of $ 2A $.	6	Numerical	4.2	406
7	If A is a square matrix of order 3 and $ A = 5$, find the value of $ 3A $.	6	Numerical	4.2	406
8	If A is a square matrix of order 3 and $ A = -2$, find the value of $ -5A $.	6	Numerical	4.2	407
9	If A is a matrix of order 3×3 and its determinant is 4, then find $ 3A $.	6	Numerical	4.2	407

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
10	If A is a square matrix of order 3 and $ 3A = k A $, then find the value of k .	6	Numerical	4.2	408
11	(see parts below)	9	Numerical	4.2	408
(i)	If A is a square matrix such that $ A = 5$, write the value of $ AA^T $.	9	Numerical	4.2	408
(ii)	If A is a square matrix satisfying $AA' = I$, write the value of $ A $.	9	Numerical	4.2	409
12	If A is a square matrix of order 3 such that $A^2 = 2A$, then find the value of $ A $.	9	Numerical	4.2	410
13	If $A = \begin{bmatrix} 1 & 23 & -1 \end{bmatrix}$ and ...	9	Numerical	4.2	410
14	If $A = \begin{bmatrix} x & 22 & x \end{bmatrix}$ and $ A^3 = 125$, find the value(s) of ...	9	Numerical	4.2	411
15	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2	412
16	Show that one root of the equation ...	5	Proof	4.2	413
17	Without expanding, prove that ...	10	Proof	4.2	414
18	Without expanding, prove that ...	10	Proof	4.2	415
19	(see parts below)	5	Numerical	4.2	416
(i)	Use properties of determinants to solve for x : ...	5	Numerical	4.2	416
(ii)	Use properties of determinants to solve for x : ...	5	Numerical	4.2	417
20	If $s = a + b + c$, then prove that ...	10	Proof	4.2	418
21	If a, b and c are in A.P., then find the value of ...	8	Numerical	4.2	419
22	Using the properties of determinant, prove that ...	10	Proof	4.2	420
23	Using properties of determinants, prove that: ...	10	Proof	4.2	421
24	(see parts below)	10	Proof	4.2	423
(i)	Prove that $\begin{vmatrix} 1+a & b & ca & 1+b & ca & b & 1+c \end{vmatrix} = 1+a+b+c$	10	Proof	4.2	423
(ii)	Prove that ...	10	Proof	4.2	424
(iii)	Prove that $\begin{vmatrix} y+k & y & yy & y+k & yy & y & y+k \end{vmatrix} = k^2(3y+k)$	10	Proof	4.2	425
(iv)	Prove that ...	10	Proof	4.2	426
25	Show that: ...	10	Proof	4.2	427
26	(see parts below)	10	Proof	4.2	428
(i)	Prove that: ...	10	Proof	4.2	428
(ii)	Prove that: ...	10	Proof	4.2	430
27	(see parts below)	10	Proof	4.2	431
(i)	Prove that: ...	10	Proof	4.2	431

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	Prove that: $\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix} = xy$	10	Proof	4.2	431
(iii)	Prove that: ...	10	Proof	4.2	432
28	Prove that: ...	10	Proof	4.2	433
29	Prove that: ...	10	Proof	4.2	434
30	Prove that: ...	10	Proof	4.2	436
31	Prove that: ...	10	Proof	4.2	437
32	Prove that: ...	10	Proof	4.2	438
33	Prove that: ...	10	Proof	4.2	440
34	(see parts below)	10	Proof	4.2	441
(i)	Prove that: ...	10	Proof	4.2	441
(ii)	Prove that: ...	10	Proof	4.2	442
35	Prove that: ...	10	Proof	4.2	443
36	Prove that: ...	10	Proof	4.2	444
37	Prove that: ...	10	Proof	4.2	445
38	If a, b, c are all non-zero and: ...	10	Proof	4.2	446
39	Without expanding any of the determinants given below, prove that: ...	10	Proof	4.2	447
40	(see parts below)	5	Numerical	4.2	448
(i)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2	448
(ii)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2	449
(iii)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2	450
(iv)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2	452
41	If a, b, c are all different and: ...	10	Proof	4.2	452
42	In a triangle ABC , if: ...	10	Proof	4.2	454

Exercise 4.3 – Applications of Determinants: Area, Collinearity and Equation of a Line

1	(see parts below)	11	Numerical	4.3	455
(i)	Using determinants, find the area of the triangle whose vertices are $(3, 8)$, $(-4, 2)$...	11	Numerical	4.3	455
(ii)	Using determinants, find the area of the triangle whose vertices are $(2, 7)$, $(1, 1)$...	11	Numerical	4.3	457
2	Show that the points $(1, 0)$, $(6, 0)$, $(0, 0)$ are collinear.	12	Proof	4.3	458

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
3	Using determinants, determine whether the points $P(a, b + c)$, $Q(b, c + a)$ and ...	12	Numerical	4.3	459
4	Using determinants, find the value of λ so that the points $(\lambda, 7)$...	12	Numerical	4.3	460
5	Using determinants, find the equation of the line passing through the points $(1, 2)$ and ...	13	Numerical	4.3	461
6	Find the value(s) of k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$ and ...	11	Numerical	4.3	462
7	Find the value(s) of x if the area of the triangle with vertices $(x, 4)$, $(2, -6)$ and ...	11	Numerical	4.3	463
8	If the points $(a, 0)$, $(0, b)$ and $(1, 1)$ are collinear, prove that $a + b = ab$.	12	Proof	4.3	465

Exercise 4.4 – Adjoint and Inverse of a Matrix

1	(see parts below)	14	Numerical	4.4	466
(i)	For what value of x , is the following matrix singular? ...	14	Numerical	4.4	466
(ii)	If $0 < x < \pi$ and the matrix $\begin{bmatrix} \cos x & 13 & 4 \cos x \end{bmatrix}$ is ...	14	Numerical	4.4	467
2	(see parts below)	14	Numerical	4.4	468
(i)	Determine if the following matrix is non-singular: ...	14	Numerical	4.4	468
(ii)	Determine if the following matrix is non-singular: ...	14	Numerical	4.4	468
3	(see parts below)	15	Numerical	4.4	469
(i)	Write the adjoint of the matrix: $\begin{bmatrix} 2 & -14 & 3 \end{bmatrix}$	15	Numerical	4.4	469
(ii)	Write the adjoint of the matrix: $\begin{bmatrix} 3 & 0 & -2 & -5 \end{bmatrix}$	15	Numerical	4.4	470
4	(see parts below)	14	Numerical	4.4	471
(i)	For what value of k , the matrix $\begin{bmatrix} 2 & k3 & 5 \end{bmatrix}$ has no ...	14	Numerical	4.4	471
(ii)	If $A = \begin{bmatrix} 3 & 102 & 7 \end{bmatrix}$, then write A^{-1} .	16	Numerical	4.4	471
(iii)	Write the inverse of the matrix $\begin{bmatrix} 3 & -11 & 2 \end{bmatrix}$.	16	Numerical	4.4	472
5	(see parts below)	17	Numerical	4.4	473
(i)	If $A = \begin{bmatrix} 8 & 23 & 2 \end{bmatrix}$, then find $ \text{adj } A $.	17	Numerical	4.4	473
(ii)	If A is a square matrix of order 3 and $ A = 5$, then find $ \text{adj } A $.	17	Numerical	4.4	474
(iii)	If A is a non-singular square matrix of order 3 such that $ \text{adj } A = 64$, find ...	17	Numerical	4.4	474

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(iv)	If A is a square matrix of order 3 such that $ A = -2$, then find $ 3\text{adj } A $.	17	Numerical	4.4	475
(v)	If $ A^{-1} = 5$, then find the value of $ A $.	17	Numerical	4.4	475
(vi)	If $ A = -4$ and ...	17	Numerical	4.4	476
(vii)	If $A = \begin{bmatrix} 2 & 35 & -2 \end{bmatrix}$ be such that $A^{-1} = kA$, then find ...	16	Numerical	4.4	476
(viii)	If $A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 00 & 3 & 00 & 0 & 3 \end{bmatrix}$, then ...	17	Numerical	4.4	477
(ix)	If the value of a determinant of third order is 12, then what is the value of the ...	17	Numerical	4.4	478
6	(see parts below)	17	Numerical	4.4	479
(i)	Without computing $\text{adj } A$, find $ \text{adj } A $ if: ...	17	Numerical	4.4	479
(ii)	Without computing $\text{adj } A$, find $ \text{adj } A $ if: ...	17	Numerical	4.4	479
(iii)	Without computing $\text{adj } A$, find $ \text{adj } A $ if: ...	17	Numerical	4.4	480
7	(see parts below)	17	Numerical	4.4	481
(i)	If $A = \begin{bmatrix} 2 & 17 & 5 \end{bmatrix}$, find $ A\text{adj } A $.	17	Numerical	4.4	481
(ii)	If $A = \begin{bmatrix} a & 0 & 00 & a & 00 & 0 & a \end{bmatrix}, a \neq 0$, find ...	17	Numerical	4.4	481
(iii)	If $A = \begin{bmatrix} 3 & -14 & 5 \end{bmatrix}$, find $\text{adj}(\text{adj } A)$.	17	Numerical	4.4	482
8	(see parts below)	18	Proof	4.4	483
(i)	If A is a non-singular matrix, then show that $ A^{-1} = \frac{1}{ A }$.	18	Proof	4.4	483
(ii)	If A is a matrix such that $A^3 = I$, then show that A is invertible.	18	Proof	4.4	483
(iii)	If A is a matrix such that $A^4 = I$, then show that $A^{-1} = A^3$.	18	Proof	4.4	484
9	If $A = \begin{bmatrix} 3 & -54 & 2 \end{bmatrix}$, find $A(\text{adj } A)$.	19	Numerical	4.4	485
10	If the matrix $\begin{bmatrix} x+4 & x & xx & x+4 & xx & x & x+4 \end{bmatrix}$ is ...	14	Numerical	4.4	485
11	(see parts below)	19	Numerical	4.4	486
(i)	Find the adjoint of the matrix: $A = \begin{bmatrix} -2 & 13 & 4 \end{bmatrix}$ Also ...	19	Numerical	4.4	486
(ii)	Find the adjoint of the matrix: $A = \begin{bmatrix} 2 & 3 & -4 & -6 \end{bmatrix}$ Also ...	19	Numerical	4.4	488
12	If $A = \begin{bmatrix} 3 & 1 & 22 & -3 & -11 & 2 & 1 \end{bmatrix}$, find ...	19	Numerical	4.4	490
13	Find the adjoint of the following matrices:	19	Numerical	4.4	492

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	$A = \begin{bmatrix} 1 & -1 & 22 & 3 & 5 & -2 & 0 & 1 \end{bmatrix}$	19	Numerical	4.4	492
(ii)	$A = \begin{bmatrix} 1 & -1 & 23 & 0 & -21 & 0 & 3 \end{bmatrix}$	19	Numerical	4.4	492
14	If ...	19	Numerical	4.4	496
15	For the matrix $A = \begin{bmatrix} 2 & -1 & 34 & 2 & 50 & 4 & -1 \end{bmatrix}$, verify ...	19	Proof	4.4	498
16	For the matrix $A = \begin{bmatrix} 5 & 2 & -3 & -1 \end{bmatrix}$, verify that ...	19	Proof	4.4	500
17	See parts below:	18	Proof	4.4	501
(i)	Given $A = \begin{bmatrix} 2 & -3 & -4 & 7 \end{bmatrix}$, compute A^{-1} and ...	18	Proof	4.4	501
(ii)	If $A = \begin{bmatrix} 4 & 52 & 1 \end{bmatrix}$, show that ...	18	Proof	4.4	501
18	Find the inverse of the matrix ...	18	Proof	4.4	503
19	If $A = \begin{bmatrix} 0 & 1 & 31 & 2 & x2 & 3 & 1 \end{bmatrix}$ and ...	18	Numerical	4.4	504
20	If $A = \begin{bmatrix} 2 & -35 & 1 \end{bmatrix}$ and ...	21	Proof	4.4	506
21	Find the inverse of each of the following matrices:	16	Numerical	4.4	508
(i)	$A = \begin{bmatrix} 1 & 0 & 03 & 3 & 05 & 2 & -1 \end{bmatrix}$	16	Numerical	4.4	508
(ii)	$A = \begin{bmatrix} -1 & 1 & 23 & -1 & 1 & -1 & 3 & 4 \end{bmatrix}$	16	Numerical	4.4	508
22	Given $A = \begin{bmatrix} 5 & 0 & 42 & 3 & 21 & 2 & 1 \end{bmatrix}$...	21	Numerical	4.4	512
23	See parts below:	20	Numerical	4.4	514
(i)	Show that $A = \begin{bmatrix} 5 & 3 & -1 & -2 \end{bmatrix}$ satisfies the ...	20	Numerical	4.4	514
(ii)	If $A = \begin{bmatrix} 2 & -1 & -1 & 2 \end{bmatrix}$ and I is the identity ...	20	Numerical	4.4	514
(iii)	If $A = \begin{bmatrix} 3 & -24 & -2 \end{bmatrix}$, then find the value of ...	20	Numerical	4.4	514
(iv)	If $A = \begin{bmatrix} 1 & 3 & 22 & 0 & -11 & 2 & 3 \end{bmatrix}$, then show ...	20	Numerical	4.4	514
24	See parts below:	20	Numerical	4.4	518
(i)	If $A = \begin{bmatrix} 1 & 2 & 22 & 1 & 22 & 2 & 1 \end{bmatrix}$, prove that ...	20	Numerical	4.4	518
(ii)	Find A^{-1} if ...	20	Numerical	4.4	518
25	See parts below:	18	Proof	4.4	521
(i)	If ...	18	Proof	4.4	521
(ii)	If $A = \begin{bmatrix} 1 & -1 & 12 & -1 & 01 & 0 & 0 \end{bmatrix}$, prove ...	18	Proof	4.4	521
(iii)	If ...	18	Proof	4.4	521
26	Show that for the matrix ...	20	Numerical	4.4	523
27	See parts below:	20	Numerical	4.4	525
(i)	Find a 2×2 matrix B such that ...	20	Numerical	4.4	525
(ii)	Solve the matrix equation ...	20	Numerical	4.4	525
28	See parts below:	20	Numerical	4.4	527

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	Find the matrix A satisfying the matrix equation ...	20	Numerical	4.4	527
(ii)	Find the matrix P satisfying the matrix equation ...	20	Numerical	4.4	527

Exercise 4.5 – Solving Systems of Linear Equations Using Matrices

1	(see parts below)	22	Numerical	4.5	529
(i)	Use matrix method to solve the following system of equations: $5x + 2y = 4$ $7x + 3y = 5$	22	Numerical	4.5	529
(ii)	Use matrix method to solve the following system of equations: $4x - 3y = 3$ $3x - 5y = 7$	22	Numerical	4.5	529
2	(see parts below)	23	Numerical	4.5	531
(i)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5	531
(ii)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5	532
(iii)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5	532
(iv)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5	534
3	Solve the following system of equations by using matrix method: $x + y = 1$ $7x + 7y = 7$	24	Numerical	4.5	535
4	(see parts below)	22	Numerical	4.5	536
(i)	Using matrix method, solve the following system of equations: $x - 2y + 3z = 6$...	22	Numerical	4.5	536
(ii)	Using matrix method, solve the following system of equations: $x + y + z = 6$...	22	Numerical	4.5	537
(iii)	Using matrix method, solve the following system of equations: $x + y + z = 9$...	22	Numerical	4.5	538
(iv)	Using matrix method, solve the following system of equations: $x - 2y = 10$...	22	Numerical	4.5	539
(v)	Using matrix method, solve the following system of equations: $3x + y + z = 1$...	22	Numerical	4.5	541
(vi)	Using matrix method, solve the following system of equations: $2x - 3y + 5z = 11$...	22	Numerical	4.5	542

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
5	Solve the following system of linear equations using matrix method: ...	25	Numerical	4.5	543
6	State the condition under which the following equations have a unique solution: ...	22	Numerical	4.5	545
7	The sum of three numbers is 6. If we multiply third number by 3 and add second to it we ...	26	Numerical	4.5	547
8	Garima owns a cafe that serves burgers, samosas, and juice bottles. On a particular ...	26	Numerical	4.5	548
(i)	Form a set of simultaneous equations for the above information.	26	Numerical	4.5	548
(ii)	Solve the set of equations formed in (i) by matrix method.	26	Numerical	4.5	548
9	If $A = \begin{bmatrix} 1 & 2 & -3 & 2 & 23 & -3 & -4 \end{bmatrix}$, find A^{-1} ...	27	Numerical	4.5	550
10	If $A = \begin{bmatrix} 2 & -3 & 5 & 3 & 2 & -4 & 1 & -2 \end{bmatrix}$, find A^{-1} ...	27	Numerical	4.5	551
11	If $A = \begin{bmatrix} 3 & 5 & -2 & 3 \end{bmatrix}$, find A^{-1} and use it to solve the ...	28	Numerical	4.5	553
12	If $A = \begin{bmatrix} 1 & -1 & 12 & 1 & -3 & 1 & 1 \end{bmatrix}$, find A^{-1} and ...	28	Numerical	4.5	554
13	Given two matrices A and B where ...	29	Numerical	4.5	556
14	If $A = \begin{bmatrix} -3 & -2 & -4 & 2 & 1 & 2 & 1 & 3 \end{bmatrix}$ and ...	29	Numerical	4.5	557
15	(see parts below)	30	Numerical	4.5	559
(i)	Using matrices, solve the following system of homogeneous equations: $5x + 5y + 2z = 0$...	30	Numerical	4.5	559
(ii)	Using matrices, solve the following system of homogeneous equations: $3x + y - 2z = 0$...	30	Numerical	4.5	560
16	(see parts below)	30	Numerical	4.5	561
(i)	Using matrices, solve the following system of homogeneous equations: $2x - 3y - z = 0$...	30	Numerical	4.5	561
(ii)	Using matrices, solve the following system of homogeneous equations: $x + y - z = 0$...	30	Numerical	4.5	562

Chapter 5: Continuity and Differentiability

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
Exercise 5.1 – Continuity: Checking Continuity at a Point					
1	(see parts below)	1	Numerical	5.1	565
(i)	Examine the following function for continuity at the indicated point: ...	1	Numerical	5.1	565
(ii)	Examine the following function for continuity at the indicated point: ...	1	Numerical	5.1	565
2	If $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$ and $f(x) = 2$ when $x = 1$, show that ...	1	Proof	5.1	566
3	A function f is defined as $f(x) =$ (piecewise definition). Show that f is ...	1	Proof	5.1	567
4	Is the function f defined by $f(x) =$ (piecewise definition) continuous at (i) ...	2	Numerical	5.1	567
5	Is the function f defined by $f(x) =$ (piecewise definition) continuous at ...	1	Numerical	5.1	568
6	Is the function f defined by $f(x) =$ (piecewise definition) continuous at ...	1	Numerical	5.1	569
7	If $f(x) =$ (piecewise definition), find k so that f may be continuous at ...	3	Numerical	5.1	570
8	If $f(x) =$ (piecewise definition), find k so that f may be continuous at ...	3	Numerical	5.1	571
9	If $f(x) =$ (piecewise definition), find λ so that f may be continuous at ...	3	Numerical	5.1	571
10	(see parts below)	4	Numerical	5.1	572
(i)	$f(x) = \frac{x^2-9}{x-3}$ is not defined at $x = 3$. What type of removable ...	4	Numerical	5.1	572
(ii)	If $f(x) =$ (piecewise definition), find k so that the function f may be ...	3	Numerical	5.1	573
(iii)	Determine the value of ' k ' for which the following function is continuous at $x = 3$: ...	3	Numerical	5.1	574
(iv)	If $f(x) =$ (piecewise definition), find k so that the function f may be ...	3	Numerical	5.1	574
11	If $f(x) =$ (piecewise definition), find k so that the function f may be ...	3	Numerical	5.1	575
12	Is the function f defined by $f(x) = \tan x$ continuous at $x = \frac{\pi}{2}$?	1	Numerical	5.1	576
13	Is the function f defined by $f(x) = x $ continuous at $x = 0$?	5	Numerical	5.1	577
14	Is the function f defined by $f(x) = x - 1 $ continuous at $x = 1$?	5	Numerical	5.1	577
15	Is the function f defined by $f(x) = x - x $ continuous at $x = 0$?	5	Numerical	5.1	578

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
16	(see parts below)	2	Numerical	5.1	579
(i)	Examine the following function for continuity at the indicated point: ...	2	Numerical	5.1	579
(ii)	Examine the following function for continuity at the indicated point: ...	2	Numerical	5.1	580
17	Is the function f defined by $f(x) =$ (piecewise definition) continuous at ...	2	Numerical	5.1	581
18	(see parts below)	5	Numerical	5.1	582
(i)	Examine the following function for continuity at $x = 0$: ...	5	Numerical	5.1	582
(ii)	Examine the following function for continuity at $x = 4$: ...	5	Numerical	5.1	583
19	(see parts below)	6	Numerical	5.1	584
(i)	Examine the following function for continuity at $x = 0$: ...	6	Numerical	5.1	584
(ii)	Examine the following function for continuity at $x = 0$: ...	6	Numerical	5.1	585
(iii)	Examine the following function for continuity at $x = 0$: ...	6	Numerical	5.1	586
20	(see parts below)	3	Numerical	5.1	586
(i)	If $f(x) =$ (piecewise definition) is continuous at $x = 5$, find the value of k .	3	Numerical	5.1	586
(ii)	Find the value of k so that the function $f(x) =$ (piecewise definition) is ...	3	Numerical	5.1	587
(iii)	For what value of k is the following function continuous at $x = 2$? ...	3	Numerical	5.1	588
(iv)	Find the value of k so that the function f defined by ...	3	Numerical	5.1	589
(v)	For what value of k is the function $f(x) =$ (piecewise definition) continuous at ...	3	Numerical	5.1	590
(vi)	Find the value of k for which the function f given as ...	3	Numerical	5.1	591
(vii)	Find the value(s) of ' λ ' for which the function f given as ...	3	Numerical	5.1	592
21	Find the relationship between a and b so that the function f defined by ...	3	Numerical	5.1	593
22	If the function f defined by $f(x) =$ (piecewise definition) is continuous at ...	3	Numerical	5.1	593
23	If $f(x) =$ (piecewise definition) is continuous at $x = 0$, then find the values ...	3	Numerical	5.1	595
24	(see parts below)	4	Numerical	5.1	596
(i)	Consider the function defined as follows, for $x \neq 0$: ...	4	Numerical	5.1	596

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	Consider the function defined as follows, for $x \neq 0$: ...	4	Numerical	5.1	597
25	Examine the function $f(x) =$ (piecewise definition) for continuity at $x = 0$.	7	Numerical	5.1	598
26	Show that the function $f(x) = x + x - 1 , x \in \mathbb{R}$, is continuous both at ...	5	Proof	5.1	599
27	Examine the function $f(x) =$ (piecewise definition) for continuity at $x = 0$.	8	Numerical	5.1	600

Exercise 5.2 – Continuity: Piecewise Functions and Finding Constants

1	Is the function f defined by $f(x) = [x]$ continuous at:	1	Numerical	5.2	601
(i)	$x = 0$	1	Numerical	5.2	601
(ii)	$x = 1$	1	Numerical	5.2	601
(iii)	$x = \frac{1}{2}$?	1	Numerical	5.2	601
2	(see parts below)	2	Numerical	5.2	603
(i)	Examine the following function for continuity: $f(x) =$ (piecewise definition)	2	Numerical	5.2	603
(ii)	Examine the following function for continuity: $f(x) =$ (piecewise definition)	2	Numerical	5.2	604
3	(see parts below)	2	Numerical	5.2	605
(i)	Examine the following function for continuity: $f(x) =$ (piecewise definition)	2	Numerical	5.2	605
(ii)	Examine the following function for continuity: $f(x) =$ (piecewise definition)	2	Numerical	5.2	606
4	(see parts below)	5	Numerical	5.2	607
(i)	Examine the continuity of $f(x) = x $.	5	Numerical	5.2	607
(ii)	Examine the continuity of $f(x) = x - 5 $.	5	Numerical	5.2	608
5	(see parts below)	1	Numerical	5.2	609
(i)	Examine the continuity of $f(x) = \frac{x^2 - 25}{x + 5}$.	1	Numerical	5.2	609
(ii)	Examine the continuity of $f(x) = \frac{1}{x - 5}$.	1	Numerical	5.2	609
6	(see parts below)	6	Numerical	5.2	610
(i)	Examine the following function for continuity: $f(x) =$ (piecewise definition)	6	Numerical	5.2	610
(ii)	Examine the following function for continuity: $f(x) =$ (piecewise definition)	5	Numerical	5.2	611
7	(see parts below)	2	Numerical	5.2	612
(i)	Examine the following function for continuity: $f(x) =$ (piecewise definition)	2	Numerical	5.2	612
(ii)	Examine the continuity of $f(x) = \tan x$ for $0 \leq x \leq \frac{\pi}{4}$.	1	Numerical	5.2	613

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
8	(see parts below)	2	Numerical	5.2	614
(i)	Locate the points of discontinuity (if any) of the function: ...	2	Numerical	5.2	614
(ii)	Locate the points of discontinuity (if any) of the function: ...	2	Numerical	5.2	615
9	(see parts below)	1	Numerical	5.2	616
(i)	Locate the points of discontinuity (if any) of the function: ...	1	Numerical	5.2	616
(ii)	Locate the points of discontinuity (if any) of the function: ...	6	Numerical	5.2	617
10	(see parts below)	2	Numerical	5.2	618
(i)	Locate the points of discontinuity (if any) of the function: ...	2	Numerical	5.2	618
(ii)	Locate the points of discontinuity (if any) of the function: ...	2	Numerical	5.2	619
11	Find the value of the constant k so that the function f defined by ...	3	Numerical	5.2	620
12	Determine the value of k for which the following function is continuous at $x = 3$: ...	3	Numerical	5.2	621
13	(see parts below)	3	Numerical	5.2	622
(i)	For what choice of a and b is the following function continuous: ...	3	Numerical	5.2	622
(ii)	For what choice of a and b is the following function continuous: ...	3	Numerical	5.2	624
14	Consider $f(x) =$ (piecewise definition)	4	Numerical	5.2	625
(i)	Determine if there is a removable discontinuity at $x = 3$.	4	Numerical	5.2	625
(ii)	Classify it as missing point or isolated point.	4	Numerical	5.2	625
(iii)	Redefine the function to make it continuous.	4	Numerical	5.2	625

Exercise 5.3 – Continuity: Algebra and Composition of Continuous Functions

1	Is $\tan x$ a continuous function?	9	Numerical	5.3	626
2	State which of the following functions are continuous?	9	Numerical	5.3	627
(i)	$\sin x + \cos x$	9	Numerical	5.3	627
(ii)	$\sin x - \cos x$	9	Numerical	5.3	627
(iii)	$\sin x \cdot \cos x$	9	Numerical	5.3	627
3	Is the function f defined by $f(x) = 2 - 3x + x $ a continuous function?	5	Numerical	5.3	627
4	Is the function f defined by $f(x) = \sin x $ a continuous function?	10	Numerical	5.3	628

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
5	Is the function f defined by $f(x) = \cos x $ a continuous function?	10	Numerical	5.3	629
6	Prove that the following functions are continuous:	9	Proof	5.3	629
(i)	$\frac{2x^3-7x^2+3}{(x-1)(x+3)}$	9	Proof	5.3	629
(ii)	$ \sec x + \tan x $	9	Proof	5.3	629
7	Examine the following functions for continuity:	10	Numerical	5.3	630
(i)	$\cos(x^2)$	10	Numerical	5.3	630
(ii)	$\sin(3x^2 - 5)$	10	Numerical	5.3	630
(iii)	$\tan^{-1}(2x^2 + 3)$	10	Numerical	5.3	630

Exercise 5.4 – Differentiability: Definition and Relation to Continuity

1	If a function is derivable at a point, is it necessary that it must be continuous at that ...	11	Numerical	5.4	631
2	If a function is continuous at a point, is it necessary that it must be derivable at that ...	11	Numerical	5.4	632
3	Is the function $f(x) = x $ derivable at $x = 0$?	12	Numerical	5.4	633
4	Is the function $f(x) = \cot x$ derivable at $x = 0$?	13	Numerical	5.4	634
5	Is the function $f(x) = \sec x$ derivable at $x = \frac{\pi}{2}$?	13	Numerical	5.4	634
6	When a function is called derivable?	14	Numerical	5.4	635
7	If a function is derivable at $x = c$, then write the value of $\lim_{x \rightarrow c} f(x)$.	11	Numerical	5.4	635
8	Which of the following functions are derivable?	13	Numerical	5.4	636
(i)	$\sin x$	13	Numerical	5.4	636
(ii)	$\cos x$	13	Numerical	5.4	636
(iii)	$\tan x$	13	Numerical	5.4	636
(iv)	$\cot x$	13	Numerical	5.4	636
(v)	$\sec x$	13	Numerical	5.4	636
(vi)	$\csc x$	13	Numerical	5.4	636
9	Examine the function $f(x) =$ (piecewise definition) for differentiability at ...	15	Numerical	5.4	636
10	If $f(2) = 4$ and $f'(2) = 4$, then evaluate ...	16	Numerical	5.4	638
11	Examine the following function for continuity at $x = 1$ and differentiability at $x = 2$: ...	15	Numerical	5.4	638
12	Show that the function $f(x) = 2x - x $ is continuous at $x = 0$ but not differentiable ...	12	Proof	5.4	640

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
13	Show that $f(x) = x - 4 $ is continuous but not differentiable at $x = 4$.	12	Proof	5.4	641
14	If the function $f(x) = x - 3 + x - 4 $, then show that f is not differentiable at ...	12	Proof	5.4	642
15	Show that the function f is continuous at $x = 1$ for all values of a where ...	17	Numerical	5.4	643
16	Prove that the function $f(x) =$ (piecewise definition) is not differentiable at ...	11	Proof	5.4	644
17	If $f(x) =$ (piecewise definition), then show that f is not differentiable at ...	15	Proof	5.4	645
18	If $f(x) =$ (piecewise definition) is a differentiable function in $(0, 2)$, then ...	17	Numerical	5.4	646
19	If the function $f(x) =$ (piecewise definition) is differentiable at $x = 3$, find ...	17	Numerical	5.4	647
20	(see parts below)	18	Numerical	5.4	648
(i)	Examine for continuity and differentiability: ...	18	Numerical	5.4	648
(ii)	Examine for continuity and differentiability: ...	18	Numerical	5.4	649
(iii)	Examine for continuity and differentiability: ...	18	Numerical	5.4	650
21	Find the values of a and b so that the following function is differentiable for all ...	17	Numerical	5.4	651
22	Check the differentiability of $f(x) = \cos x $ at $x = \frac{\pi}{2}$.	12	Numerical	5.4	652

Exercise 5.5 – Derivatives of Standard and Absolute Value Functions

1	(see parts below)	19	Numerical	5.5	653
(i)	If $f(x) = \sqrt{4 - x^2}$, $x \in (-2, 2)$, find $f'(x)$.	19	Numerical	5.5	653
(ii)	If $f(x) = \sqrt{1 - x^2}$, $x \in (0, 1)$, find $f'(x)$.	19	Numerical	5.5	654
2	(see parts below)	19	Numerical	5.5	654
(i)	Differentiate $\sqrt{2x - 1}$ w.r.t. x .	19	Numerical	5.5	654
(ii)	Differentiate $(3x^2 - 9x + 5)^9$ w.r.t. x .	19	Numerical	5.5	655
3	(see parts below)	20	Numerical	5.5	655
(i)	Differentiate $\sqrt{3x + 2} + \frac{1}{\sqrt{2x^2 + 4}}$ w.r.t. x .	20	Numerical	5.5	655
(ii)	Differentiate $\sin^3 x + \cos^6 x$ w.r.t. x .	20	Numerical	5.5	656
4	(see parts below)	21	Numerical	5.5	656
(i)	Differentiate $\sin(x^2)$ w.r.t. x .	21	Numerical	5.5	656
(ii)	Differentiate $\cos \sqrt{x}$ w.r.t. x .	21	Numerical	5.5	657
5	(see parts below)	19	Numerical	5.5	657
(i)	Differentiate $\sin(ax + b)$ w.r.t. x .	19	Numerical	5.5	657

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	Differentiate $\tan(2x + 3)$ w.r.t. x .	19	Numerical	5.5	658
6	(see parts below)	21	Numerical	5.5	658
(i)	Differentiate $\cos(\sin x)$ w.r.t. x .	21	Numerical	5.5	658
(ii)	Differentiate $\sin(\cos(x^2))$ w.r.t. x .	22	Numerical	5.5	659
7	(see parts below)	22	Numerical	5.5	659
(i)	Differentiate $\cos(\tan \sqrt{x+1})$ w.r.t. x .	22	Numerical	5.5	659
(ii)	Differentiate $\sqrt{\tan \sqrt{x}}$ w.r.t. x .	22	Numerical	5.5	660
8	(see parts below)	23	Numerical	5.5	660
(i)	If $f(x) = x - 1$, find $\frac{d}{dx}((f \circ g)(x))$.	23	Numerical	5.5	660
(ii)	If $f(x) = x + 7$ and $g(x) = x - 7$, find $\frac{d}{dx}((f \circ f)(x))$.	23	Numerical	5.5	661
9	If ...	24	Proof	5.5	662
10	Find the derivative of $ 2x^2 - 3 $ with respect to x .	25	Numerical	5.5	662
11	If $f(x) = \cos x $, find $f'(\frac{3\pi}{4})$.	26	Numerical	5.5	663
12	The table below shows some of the values of differentiable functions u and v and their ...	27	Numerical	5.5	664
13	(see parts below)	28	Numerical	5.5	665
(i)	Differentiate $\sqrt{\frac{a^2-x^2}{a^2+x^2}}$ w.r.t. x .	28	Numerical	5.5	665
(ii)	Differentiate $\sin \sqrt{x} + \cos^2 \sqrt{x}$ w.r.t. x .	20	Numerical	5.5	665
14	(see parts below)	28	Numerical	5.5	666
(i)	Differentiate $\frac{\sin(ax+b)}{\cos(cx+d)}$ w.r.t. x .	28	Numerical	5.5	666
(ii)	Differentiate $\frac{\sin x + x^2}{\cot 2x}$ w.r.t. x .	28	Numerical	5.5	667
15	(see parts below)	20	Numerical	5.5	667
(i)	Differentiate $\sin^m x \cos^n x$ w.r.t. x .	20	Numerical	5.5	667
(ii)	Differentiate $\sin^n(ax^2 + bx + c)$ w.r.t. x .	22	Numerical	5.5	668
16	(see parts below)	29	Numerical	5.5	668
(i)	Differentiate $\frac{\sqrt{a+x}-\sqrt{a-x}}{\sqrt{a+x}+\sqrt{a-x}}$ w.r.t. x .	29	Numerical	5.5	668
(ii)	Differentiate $\frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{\sqrt{x^2+1}-\sqrt{x^2-1}}$ w.r.t. ...	29	Numerical	5.5	669
17	If $f(x) = \sqrt{\frac{\sec x - 1}{\sec x + 1}}$, find $f'(x)$. Also find ...	30	Numerical	5.5	670
18	(see parts below)	31	Numerical	5.5	671
(i)	Find the derivative of $\cos(2x + \frac{\pi}{3})$ at $x = \frac{\pi}{3}$.	31	Numerical	5.5	671
(ii)	Find the derivative of $\frac{1-\sin x}{1+\cos x}$ at $x = \frac{\pi}{2}$.	28	Numerical	5.5	672
19	If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, then show that ...	32	Proof	5.5	672
20	If $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, prove that ...	30	Proof	5.5	673
21	Differentiate $ \cos x $ w.r.t. x . Is this function differentiable? What can you say ...	25	Numerical	5.5	674
22	Prove that the derivative of an odd function is always an even function.	33	Proof	5.5	675

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
23	If $2f(x) + 3f(-x) = x^2 + x + 1$, find $f'(1)$.	34	Numerical	5.5	676
24	If $f(x) = \tan 2x $, then find the value of $f'(x)$ at $x = \frac{\pi}{3}$.	26	Numerical	5.5	677
25	If $y = \cos^3(\sec^2 2t)$, find $\frac{dy}{dt}$.	22	Numerical	5.5	677
26	The figure given below shows the graph of a function $y = f(x)$. What is the derivative of ...	25	Numerical	5.5	678

Exercise 5.6 – Derivatives of Inverse Trigonometric Functions

1	Find the domain of differentiability of the following functions:	35	Numerical	5.6	680
(i)	$\sin^{-1} x$	35	Numerical	5.6	680
(ii)	$\cos^{-1} x$	35	Numerical	5.6	680
(iii)	$\sec^{-1} x$	35	Numerical	5.6	680
(iv)	$\csc^{-1} x$	35	Numerical	5.6	680
(v)	$\tan^{-1} x$	35	Numerical	5.6	680
(vi)	$\cot^{-1} x$	35	Numerical	5.6	680
2	Find the derivatives of the following functions:	36	Numerical	5.6	681
(i)	$\tan^{-1}(x\sqrt{x})$	36	Numerical	5.6	681
(ii)	$\cos^{-1}(3x)$	36	Numerical	5.6	681
(iii)	$\sqrt{\sin^{-1}(2x)}$	36	Numerical	5.6	681
(iv)	$\sin(\tan^{-1} x)$	36	Numerical	5.6	681
(v)	$\cot^{-1} \sqrt{x}$	36	Numerical	5.6	681
(vi)	$(\tan^{-1} x)^2$	36	Numerical	5.6	681
3	Find the derivatives of the following functions:	37	Numerical	5.6	683
(i)	$\sec^{-1}(\csc x)$	37	Numerical	5.6	683
(ii)	$\tan^{-1}(\cot x)$	37	Numerical	5.6	683
4	If $f(x) = \cot^{-1}(\tan x)$, show that $f'(x) = -1$.	37	Proof	5.6	684
5	If $y = \sin^{-1} x + \cos^{-1} x$, find $\frac{dy}{dx}$.	37	Numerical	5.6	685
6	If $f(x) = \sin^{-1} x + \sec^{-1} \frac{1}{x}$, find $f'(x)$.	37	Numerical	5.6	685
7	If $y = \tan^{-1} x + \cot^{-1} \frac{1}{x}$, $x > 0$, find $\frac{dy}{dx}$.	37	Numerical	5.6	686
8	If $y = \sin^{-1}(\cos x) + 3 \csc^{-1}(\sec x)$, find $\frac{dy}{dx}$.	37	Numerical	5.6	686
9	If $y = 2 \cos^{-1}(\sin x) + 3 \cot^{-1}(\tan x)$, find $\frac{dy}{dx}$.	37	Numerical	5.6	687
10	Differentiate the following functions w.r.t. x :	38	Numerical	5.6	688
(i)	$x \tan^{-1} x$	38	Numerical	5.6	688
(ii)	$x \cos^{-1} \sqrt{x}$	38	Numerical	5.6	688
(iii)	$\frac{\sin^{-1} x}{x}$	38	Numerical	5.6	688
11	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	689
(i)	$\cos^{-1}(\frac{1-x}{1+x})$	39	Numerical	5.6	689

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	$\sec^{-1}\left(\frac{1}{x-1}\right)$	39	Numerical	5.6	689
(iii)	$\sin^{-1}\left(\frac{1}{\sqrt{x+1}}\right)$	39	Numerical	5.6	689
12	Differentiate the following functions w.r.t. x :	37	Numerical	5.6	691
(i)	$\sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$	37	Numerical	5.6	691
(ii)	$x(\sin^{-1} x)^2 + 2\sqrt{1-x^2} \sin^{-1} x - 2x$	37	Numerical	5.6	691
13	Differentiate the following functions w.r.t. x :	37	Numerical	5.6	692
(i)	$\tan^{-1}(\sin x + \cos x)$	37	Numerical	5.6	692
(ii)	$\tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$	37	Numerical	5.6	692
(iii)	$\tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$	37	Numerical	5.6	692
(iv)	$\tan^{-1}\left(\frac{\sin x}{1+\sin x}\right)$	37	Numerical	5.6	692
14	Differentiate the following functions w.r.t. x :	37	Numerical	5.6	694
(i)	$\cot^{-1}(\csc x + \cot x)$	37	Numerical	5.6	694
(ii)	$\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$	37	Numerical	5.6	694
(iii)	$\cot^{-1}\left(\sqrt{\frac{1-\sin x}{1+\sin x}}\right)$	37	Numerical	5.6	694
(iv)	$\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$	37	Numerical	5.6	694
15	See parts below:	32	Proof	5.6	695
(i)	If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, prove that ...	32	Proof	5.6	695
(ii)	If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, prove that $(1+x^2)\frac{dy}{dx} = 2$.	32	Proof	5.6	695
16	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	696
(i)	$\tan^{-1}\left(\frac{2x}{1-x^2}\right)$	39	Numerical	5.6	696
(ii)	$\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$	39	Numerical	5.6	696
(iii)	$\cos^{-1}(1-2x^2)$	39	Numerical	5.6	696
(iv)	$\sin^{-1}(2x\sqrt{1-x^2})$	39	Numerical	5.6	696
17	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	698
(i)	$\sin^{-1}(3x-4x^3)$	39	Numerical	5.6	698
(ii)	$\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$	39	Numerical	5.6	698
(iii)	$\cos^{-1}\left(\frac{2x}{1+x^2}\right)$	39	Numerical	5.6	698
(iv)	$\sec^{-1}\left(\frac{1}{4x^3-3x}\right)$	39	Numerical	5.6	698
18	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	699
(i)	$\sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$	39	Numerical	5.6	699
(ii)	$\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$	39	Numerical	5.6	699
(iii)	$\tan^{-1}(\sqrt{1+x^2} - x)$	39	Numerical	5.6	699
(iv)	$\cot^{-1}(\sqrt{1+x^2} + x)$	39	Numerical	5.6	699
19	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	701
(i)	$\tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right)$	39	Numerical	5.6	701

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	$\sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$	39	Numerical	5.6	701
(iii)	$\cos^{-1}\left(\sqrt{\frac{1+x}{2}}\right)$	39	Numerical	5.6	701
(iv)	$\sin^{-1}\left(\frac{x}{\sqrt{x^2+a^2}}\right)$	39	Numerical	5.6	701
20	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	702
(i)	$\sin^{-1}\left(\sqrt{\frac{1+x^2}{2}}\right)$	39	Numerical	5.6	702
(ii)	$\tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$	39	Numerical	5.6	702
(iii)	$\sin^2(\cot^{-1}\sqrt{\frac{1+x}{1-x}})$	39	Numerical	5.6	702
(iv)	$\sin^{-1}\left(\frac{2\sec x}{1+\sec^2 x}\right)$	39	Numerical	5.6	702
21	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	704
(i)	$\cos^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right), -\frac{\pi}{4} < x < \frac{\pi}{4} \dots$	39	Numerical	5.6	704
(ii)	$\cos^{-1}\left(\frac{3\cos x - 4\sin x}{5}\right)$	39	Numerical	5.6	704
(iii)	$\sin^{-1}\left(\frac{x+\sqrt{1-x^2}}{\sqrt{2}}\right)$	39	Numerical	5.6	704
(iv)	$\cos^{-1}\left(\frac{3x+4\sqrt{1-x^2}}{5}\right)$	39	Numerical	5.6	704
22	Differentiate the following functions w.r.t. x :	39	Numerical	5.6	705
(i)	$\tan^{-1}\left(\frac{4\sqrt{x}}{1-4x}\right)$	39	Numerical	5.6	705
(ii)	$\tan^{-1}\left(\frac{x^{1/3}+a^{1/3}}{1-x^{1/3}a^{1/3}}\right)$	39	Numerical	5.6	705
(iii)	$\sin(2\sin^{-1}x)$	39	Numerical	5.6	705
(iv)	$\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right) \dots$	39	Numerical	5.6	705
23	If $y = \sin(\tan^{-1} e^x)$, then find $\frac{dy}{dx}$ at $x = 0$.	31	Numerical	5.6	707
24	If $y = \csc(\cot^{-1} x)$, then prove that $\sqrt{1+x^2}\frac{dy}{dx} - x = 0$.	32	Proof	5.6	708

Exercise 5.7 – Implicit Differentiation

1	(see parts below)	40	Numerical	5.7	709
(i)	Find $\frac{dy}{dx}$ in $x - y = \pi$.	40	Numerical	5.7	709
(ii)	Find $\frac{dy}{dx}$ in $x^2 + y^2 = 25$.	40	Numerical	5.7	710
2	(see parts below)	40	Numerical	5.7	710
(i)	Find $\frac{dy}{dx}$ in $xy = c^2$.	40	Numerical	5.7	710
(ii)	Find $\frac{dy}{dx}$ in $\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$.	40	Numerical	5.7	711
3	(see parts below)	40	Numerical	5.7	712
(i)	Find $\frac{dy}{dx}$ in $\sqrt{x} + \sqrt{y} = \sqrt{c}$.	40	Numerical	5.7	712
(ii)	Find $\frac{dy}{dx}$ in $x^3 + y^3 = 3axy$.	40	Numerical	5.7	713
4	(see parts below)	40	Numerical	5.7	714
(i)	Find $\frac{dy}{dx}$ in $2x + 3y = \sin x$.	40	Numerical	5.7	714
(ii)	Find $\frac{dy}{dx}$ in $2x + 3y = \sin y$.	40	Numerical	5.7	714
5	(see parts below)	40	Numerical	5.7	715
(i)	Find $\frac{dy}{dx}$ when $(x^2 + y^2)^2 = xy$.	40	Numerical	5.7	715
(ii)	Find $\frac{dy}{dx}$ when $\sin(x + y) = \frac{2}{3}$.	40	Numerical	5.7	716

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
6	If $x^{2/3} + y^{2/3} = 2$, find $\frac{dy}{dx}$ at $(1, 1)$.	41	Numerical	5.7	716
7	(see parts below)	42	Proof	5.7	717
(i)	Use implicit differentiation to verify that $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \dots$	42	Proof	5.7	717
(ii)	Use implicit differentiation to verify that $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \dots$	42	Proof	5.7	718
8	(see parts below)	40	Numerical	5.7	720
(i)	Find $\frac{dy}{dx}$, when $y + \sin y = \cos x$.	40	Numerical	5.7	720
(ii)	Find $\frac{dy}{dx}$, when $ax + by^2 = \cos y$.	40	Numerical	5.7	720
(iii)	Find $\frac{dy}{dx}$, when $y \sec x + \tan x + x^2 y = 0$.	40	Numerical	5.7	721
(iv)	Find $\frac{dy}{dx}$, when $\sin^2 x + \cos^2 y = 1$.	40	Numerical	5.7	722
(v)	Find $\frac{dy}{dx}$, when $xy + y^2 = \tan x + y$.	40	Numerical	5.7	722
(vi)	Find $\frac{dy}{dx}$, when $y = \tan(x + y)$.	40	Numerical	5.7	723
(vii)	Find $\frac{dy}{dx}$, when $\sin(xy) + \frac{x}{y} = x^2 - y$.	40	Numerical	5.7	724
(viii)	Find $\frac{dy}{dx}$, when $\sec(x + y) = xy$.	40	Numerical	5.7	725
9	If $\sec\left(\frac{x+y}{x-y}\right) = a$, prove that $\frac{dy}{dx} = \frac{y}{x}$.	43	Proof	5.7	726
10	(see parts below)	44	Proof	5.7	727
(i)	If $y = x \sin(a + y)$, prove that ...	44	Proof	5.7	727
(ii)	If $\sin x = y \sin(x + a)$, prove that $\frac{dy}{dx} = \frac{\sin a}{\sin^2(x+a)}$.	44	Proof	5.7	728
(iii)	If $\sin y = x \sin(a + y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$.	44	Proof	5.7	729
11	If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$, prove that ...	39	Proof	5.7	729
12	(see parts below)	45	Proof	5.7	731
(i)	If $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$, prove that ...	45	Proof	5.7	731
(ii)	If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$, prove that ...	45	Proof	5.7	731
(iii)	If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots \infty}}}$, prove that ...	45	Proof	5.7	732
13	If $y = x + \frac{1}{x + \frac{1}{x + \dots \text{to } \infty}}$, prove that ...	45	Proof	5.7	733

Exercise 5.8 – Differentiation of Exponential and Logarithmic Functions

1	Differentiate the following functions w.r.t. x :	20	Numerical	5.8	734
(i)	$\log x + \frac{1}{x}$	20	Numerical	5.8	734
(ii)	$10^x + \frac{1}{3}e^x - 2 \log x$	20	Numerical	5.8	734
2	Differentiate the following functions w.r.t. x :	19	Numerical	5.8	735
(i)	e^{-x}	19	Numerical	5.8	735
(ii)	$\sin(\log x), \quad x > 0$	19	Numerical	5.8	735

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
3	Differentiate the following functions w.r.t. x :	19	Numerical	5.8	736
(i)	$e^{\cos x}$	19	Numerical	5.8	736
(ii)	$e^{\sin^{-1} x}$	19	Numerical	5.8	736
4	Differentiate the following functions w.r.t. x :	22	Numerical	5.8	737
(i)	e^{x^3}	22	Numerical	5.8	737
(ii)	$\sqrt{e^{\sqrt{x}}}$, $x > 0$	22	Numerical	5.8	737
5	Differentiate the following functions w.r.t. x :	19	Numerical	5.8	737
(i)	$\log(\log x)$, $x > 1$	19	Numerical	5.8	737
(ii)	$\log_7(\log x)$, $x > 1$	19	Numerical	5.8	737
6	Differentiate the following functions w.r.t. x :	22	Numerical	5.8	738
(i)	$\log(\cos e^x)$	22	Numerical	5.8	738
(ii)	$\tan^{-1}(e^{\sin x})$	22	Numerical	5.8	738
7	Is the function $f(x) = \log(2x - 1)$ derivable at $x = 0$?	35	Numerical	5.8	739
8	If $f(x) = \log(\log x)$, find $f'(e)$.	31	Numerical	5.8	740
9	If $y = \log(\tan x)$, show that $\frac{dy}{dx} = 2 \csc 2x$.	33	Proof	5.8	740
10	If $f(1) = 4$ and $f'(1) = 2$, find the value of the derivative of $\log f(e^x)$ w.r.t. ...	31	Numerical	5.8	741
11	If $f(x) = e^{xg(x)}$, $g(0) = 2$ and $g'(0) = 1$, then find $f'(0)$.	31	Numerical	5.8	742
12	Differentiate the following functions w.r.t. x :	28	Numerical	5.8	743
(i)	$\frac{e^x}{\sin x}$	28	Numerical	5.8	743
(ii)	$\frac{e^x}{1 + \sin x}$	28	Numerical	5.8	743
13	Differentiate the following functions w.r.t. x :	38	Numerical	5.8	744
(i)	$2x \tan^{-1} x - \log(1 + x^2)$	38	Numerical	5.8	744
(ii)	$\frac{\cos x}{\log x}$, $x > 0$	38	Numerical	5.8	744
14	Differentiate the following functions w.r.t. x :	20	Numerical	5.8	745
(i)	$e^{\sec^2 x} + 3 \cos^{-1} x$	20	Numerical	5.8	745
(ii)	$e^x + e^{x^2} + \dots + e^{x^5}$	20	Numerical	5.8	745
15	Differentiate the following functions w.r.t. x :	28	Numerical	5.8	746
(i)	$\frac{5^x \log x}{x^2 + 1}$	28	Numerical	5.8	746
(ii)	$\cos(\log x + e^x)$	28	Numerical	5.8	746
16	Differentiate the following functions w.r.t. x :	30	Numerical	5.8	747
(i)	$\log\left(\frac{x + \sqrt{x^2 - a^2}}{x - \sqrt{x^2 - a^2}}\right)$	30	Numerical	5.8	747

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	$\log \sqrt{\frac{1-\cos x}{1+\cos x}}$	30	Numerical	5.8	747
17	Differentiate the following functions w.r.t. x :	36	Numerical	5.8	749
(i)	$e^{\cot^{-1} x^2}$	36	Numerical	5.8	749
(ii)	$x\sqrt{1+x^2} + \log(x + \sqrt{x^2+1})$	36	Numerical	5.8	749
18	See parts below:	32	Proof	5.8	750
(i)	If $y = \frac{\log(x+\sqrt{x^2+1})}{\sqrt{x^2+1}}$, prove that ...	32	Proof	5.8	750
(ii)	If $y = e^{2\log x + 3x}$, prove that $\frac{dy}{dx} = x(2+3x)e^{3x}$.	32	Proof	5.8	750
19	Differentiate $\tan^{-1}(\frac{2^{x+1}}{1-4^x})$ w.r.t. x .	39	Numerical	5.8	751
20	Find $\frac{dy}{dx}$ when	40	Numerical	5.8	752
(i)	$xy + xe^{-y} + ye^x = x^2$	40	Numerical	5.8	752
(ii)	$e^{x-y} = \log(\frac{x}{y})$	40	Numerical	5.8	752
(iii)	$5^x + 5^y = 5^{x+y}$	40	Numerical	5.8	752
21	If $\log(x^2 + y^2) = 2 \tan^{-1}(\frac{y}{x})$, show that ...	40	Proof	5.8	754
22	If $y \log x = x - y$, prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.	44	Proof	5.8	755
23	Differentiate the following functions w.r.t. x :	28	Numerical	5.8	756
(i)	$\log_x(2x-3)$	28	Numerical	5.8	756
(ii)	$\log_{\cos x} \sin x$	28	Numerical	5.8	756

Exercise 5.9 – Logarithmic Differentiation

1	(see parts below)	46	Numerical	5.9	758
(i)	Differentiate $(x+3)^2(x+4)^3(x+5)^4$ w.r.t. x .	46	Numerical	5.9	758
(ii)	Differentiate $\cos x \cos 2x \cos 3x$ w.r.t. x .	46	Numerical	5.9	758
2	(see parts below)	46	Numerical	5.9	759
(i)	Differentiate $e^x \cos^3 x \sin^2 x$ w.r.t. x .	46	Numerical	5.9	759
(ii)	Differentiate $\frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$ w.r.t. x .	46	Numerical	5.9	759
3	(see parts below)	47	Numerical	5.9	760
(i)	Differentiate $(\sin x)^{\cos x}, 0 < x < \pi$ w.r.t. x .	47	Numerical	5.9	760
(ii)	Differentiate $(\sin x)^{\sin x}, 0 < x < \pi$ w.r.t. x .	47	Numerical	5.9	761
(iii)	Differentiate $(\tan x)^{\tan x}, 0 < x < \frac{\pi}{2}$ w.r.t. x .	47	Numerical	5.9	762
4	(see parts below)	47	Numerical	5.9	762
(i)	Differentiate $x^{\sin x}, x > 0$ w.r.t. x .	47	Numerical	5.9	762
(ii)	Differentiate $(2x+3)^{x-5}, x > -\frac{3}{2}$ w.r.t. x .	47	Numerical	5.9	763
5	(see parts below)	47	Numerical	5.9	764

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	Differentiate $(\log x)^{\cos x}$, $x > 1$ w.r.t. x .	47	Numerical	5.9	764
(ii)	Differentiate $(\log x)^{\log x}$, $x > 1$ w.r.t. x .	47	Numerical	5.9	764
6	(see parts below)	48	Proof	5.9	765
(i)	If $y = x^y$, prove that $x \frac{dy}{dx} = \frac{y^2}{1-y \log x}$.	48	Proof	5.9	765
(ii)	If $x = e^{x/y}$, prove that $\frac{dy}{dx} = \frac{x-y}{x \log x}$.	48	Proof	5.9	766
(iii)	If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\log x}{(\log x e)^2}$.	48	Proof	5.9	767
(iv)	If $x^{16}y^9 = (x^2 + y)^{17}$, prove that $\frac{dy}{dx} = \frac{2y}{x}$.	48	Proof	5.9	768
7	Find the derivative of $x^x + a^x + x^a + a^a$ for some fixed $a > 0, x > 0$.	49	Numerical	5.9	769
8	(see parts below)	49	Numerical	5.9	770
(i)	Differentiate $x^{\log x} + (\log x)^x$ w.r.t. x .	49	Numerical	5.9	770
(ii)	Differentiate $(\sin x)^{\cos x} + x^{\sin x}$ w.r.t. x .	49	Numerical	5.9	771
(iii)	Differentiate $x^{\cos x} + (\cos x)^x$ w.r.t. x .	49	Numerical	5.9	772
(iv)	Differentiate $x^{\cos x} + (\sin x)^{\tan x}$ w.r.t. x .	49	Numerical	5.9	772
(v)	Differentiate $(\sin 2x)^x + \sin^{-1} \sqrt{3x}$ w.r.t. x .	49	Numerical	5.9	773
(vi)	Differentiate $e^{\sin x} + (\tan x)^x$ w.r.t. x .	49	Numerical	5.9	774
(vii)	Differentiate $x^x - 2^{\sin x}$ w.r.t. x .	49	Numerical	5.9	775
(viii)	Differentiate $x^x \cos x + \frac{x^2+1}{x^2-1}$ w.r.t. x .	49	Numerical	5.9	775
9	(see parts below)	49	Numerical	5.9	776
(i)	If $y = (\log x)^{\cos x} + \frac{x^2+1}{x^2-1}$, find $\frac{dy}{dx}$.	49	Numerical	5.9	776
(ii)	If $y = e^{x^2 \cos x} + (\cos x)^x$, find $\frac{dy}{dx}$.	49	Numerical	5.9	777
10	(see parts below)	48	Numerical	5.9	778
(i)	Find $\frac{dy}{dx}$ when $x^y y^x = a^b$.	48	Numerical	5.9	778
(ii)	Find $\frac{dy}{dx}$ when $x^y + y^x = a^b$.	49	Numerical	5.9	779
(iii)	Find $\frac{dy}{dx}$ when $x^x + y^x = 1$.	49	Numerical	5.9	780
(iv)	Find $\frac{dy}{dx}$ when $(\sin x)^y = x + y$.	48	Numerical	5.9	781
(v)	Find $\frac{dy}{dx}$ when $xy = e^{x-y}$.	48	Numerical	5.9	782
(vi)	Find $\frac{dy}{dx}$ when $(\cos x)^y = (\cos y)^x$.	48	Numerical	5.9	782
11	If $y = x^{x^{\dots \infty}}$, prove that ...	45	Proof	5.9	783
12	If $y = x^x$, then find $\frac{dy}{dx}$ at $x = 1$.	50	Numerical	5.9	784
(i)	Find $\frac{dy}{dx}$ if $y = (\log x)^{\cos x}$.	61	Numerical	5.9	785
(ii)	Find $\frac{dy}{dx}$ if $y = x^{e^{x^2 \cos x}}$.	61	Numerical	5.9	786
(i)	Find $\frac{dy}{dx}$ if $x^y + y^x = a^b$. (Implicit differentiation / Logarithmic ...	62	Numerical	5.9	787
(ii)	Find $\frac{dy}{dx}$ if $x^{y-1} + y^{x-1} = 1$.	62	Numerical	5.9	788
(iii)	Find $\frac{dy}{dx}$ if $x^x + y^x = xy$.	62	Numerical	5.9	789
(iv)	Find $\frac{dy}{dx}$ if $(x+y)^y = \log(\sin x)$.	61	Numerical	5.9	790
(v)	Find $\frac{dy}{dx}$ if $y^x = x^y$.	61	Numerical	5.9	791
(vi)	Find $\frac{dy}{dx}$ if $y = \frac{y \tan x + \log \cos y}{x \tan y + \log \cos x}$...	62	Numerical	5.9	792

Exercise 5.10 – Parametric Differentiation

1	(see parts below)	51	Numerical	5.10	793
---	-------------------	----	-----------	------	-----

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	Find $\frac{dy}{dx}$ when $x = at^2$ and $y = 2at$.	51	Numerical	5.10	793
(ii)	Find $\frac{dy}{dx}$ when $x = 2at^2$ and $y = at^4$.	51	Numerical	5.10	794
2	(see parts below)	51	Numerical	5.10	795
(i)	Find $\frac{dy}{dx}$ when $x = a \cos \theta$ and $y = b \cos \theta$.	51	Numerical	5.10	795
(ii)	Find $\frac{dy}{dx}$ when $x = a \cos \theta$ and $y = a \sin \theta$.	51	Numerical	5.10	796
3	(see parts below)	51	Numerical	5.10	797
(i)	Find $\frac{dy}{dx}$ when $x = a \sec \theta$ and $y = b \tan \theta$.	51	Numerical	5.10	797
(ii)	Find $\frac{dy}{dx}$ when $x = \sin t$ and $y = \cos 2t$.	51	Numerical	5.10	798
4	(see parts below)	51	Numerical	5.10	799
(i)	Find $\frac{dy}{dx}$ when $x = 4t$ and $y = \frac{4}{t}$.	51	Numerical	5.10	799
(ii)	Find $\frac{dy}{dx}$ when $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$.	51	Numerical	5.10	800
(iii)	Find $\frac{dy}{dx}$ when $x = a(t - \sin t)$ and $y = a(1 + \cos t)$.	51	Numerical	5.10	801
5	(see parts below)	51	Numerical	5.10	802
(i)	Find $\frac{dy}{dx}$ when $x = a \cos^3 t$ and $y = b \sin^3 t$.	51	Numerical	5.10	802
(ii)	Find $\frac{dy}{dx}$ when $x = a(1 - \sin t)$ and $y = a(1 + \cos t)$.	51	Numerical	5.10	803
6	(see parts below)	51	Numerical	5.10	804
(i)	Find $\frac{dy}{dx}$ when $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$.	51	Numerical	5.10	804
(ii)	Find $\frac{dy}{dx}$ when $x = a(1 - \cos \theta)$ and $y = a(\theta + \sin \theta)$.	51	Numerical	5.10	805
7	(see parts below)	51	Numerical	5.10	806
(i)	Find $\frac{dy}{dx}$ when $x = e^t(\sin t + \cos t)$ and $y = e^t(\sin t - \cos t)$.	51	Numerical	5.10	806
(ii)	Find $\frac{dy}{dx}$ when $x = \cos \theta - \cos 2\theta$ and ...	51	Numerical	5.10	808
8	(see parts below)	51	Numerical	5.10	808
(i)	Find $\frac{dy}{dx}$ when $x = a(\cos \theta + \cos 2\theta)$ and ...	51	Numerical	5.10	808
(ii)	Find $\frac{dy}{dx}$ when $x = \frac{1+\log t}{t^2}$ and $y = \frac{3+2\log t}{t}$.	51	Numerical	5.10	809
9	Find $\frac{dy}{dx}$ when $x = 3 \cos \theta - 2 \cos^3 \theta$ and ...	51	Numerical	5.10	811
10	(see parts below)	52	Numerical	5.10	812
(i)	If $x = 2 \cos t - \cos 2t$ and $y = 2 \sin t - \sin 2t$, find $\frac{dy}{dx}$ at ...	52	Numerical	5.10	812
(ii)	If $x = 3 \sin t - \sin 3t$ and $y = 3 \cos t - \cos 3t$, find $\frac{dy}{dx}$ at ...	52	Numerical	5.10	813
(iii)	If $x = a \sin 2t$ and $y = a(\cos 2t + \log \tan t)$, find $\frac{dy}{dx}$.	51	Numerical	5.10	814

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
11	(see parts below)	52	Numerical	5.10	816
(i)	If $x = \frac{2bt}{1+t^2}$ and $y = \frac{a(1-t^2)}{1+t^2}$, find $\frac{dy}{dx}$ at ...	52	Numerical	5.10	816
(ii)	If $x = ae^\theta(\sin \theta - \cos \theta)$ and ...	52	Numerical	5.10	817
12	(see parts below)	53	Numerical	5.10	818
(i)	Differentiate $\frac{x^2}{1-x^2}$ with respect to x^2 .	53	Numerical	5.10	818
(ii)	Differentiate $\sin x^2$ with respect to x^3 .	53	Numerical	5.10	819
(iii)	Differentiate $\cot^3(2x+1)$ with respect to x^2+1 .	53	Numerical	5.10	820
(iv)	Differentiate $\sin^2 x$ with respect to $e^{\cos x}$.	53	Numerical	5.10	821
13	(see parts below)	54	Numerical	5.10	822
(i)	Differentiate $\sin^{-1}(\frac{2x}{1+x^2})$ with respect to $\tan^{-1} x$.	54	Numerical	5.10	822
(ii)	Differentiate $\tan^{-1}(\frac{2x}{1-x^2})$ with respect to $\tan^{-1} x$.	54	Numerical	5.10	823
14	Differentiate $\cos^{-1}(\frac{1}{\sqrt{1+t^2}})$ with respect to ...	53	Numerical	5.10	824
15	(see parts below)	54	Numerical	5.10	825
(i)	Differentiate $\sin^{-1}(\frac{2x}{1+x^2})$ with respect to ...	54	Numerical	5.10	825
(ii)	Differentiate $\tan^{-1}(\frac{3x-x^3}{1-3x^2})$ with respect to ...	54	Numerical	5.10	826
16	(see parts below)	54	Numerical	5.10	827
(i)	Differentiate $\tan^{-1}(\frac{\sqrt{1+x^2}-1}{x})$ with respect to $\tan^{-1} x$.	54	Numerical	5.10	827
(ii)	Differentiate $\tan^{-1}(\frac{\sqrt{1-x^2}}{x})$ with respect to ...	54	Numerical	5.10	828
(iii)	Differentiate $\sec^{-1}(\frac{1}{\sqrt{1-x^2}})$ with respect to ...	54	Numerical	5.10	829
17	Prove that the derivative of $\tan^{-1}(\frac{x}{1+\sqrt{1-x^2}})$ with respect to ...	54	Proof	5.10	830
18	Differentiate x^x with respect to $x \log x$.	53	Numerical	5.10	831
19	If $x = \sqrt{a^{\tan^{-1} t}}$ and $y = \sqrt{a^{\cot^{-1} t}}$, show that ...	43	Proof	5.10	832

Exercise 5.11 – Second-Order Derivatives (Miscellaneous)

1	(see parts below)	63	Numerical	5.11	833
(i)	Find the second order derivative of $x^3 + \tan x$.	63	Numerical	5.11	833
(ii)	Find the second order derivative of $\log x$.	63	Numerical	5.11	834
(iii)	Find the second order derivative of $\tan^{-1} x$.	63	Numerical	5.11	834
2	(see parts below)	63	Numerical	5.11	835
(i)	If $y = \log(x-2)$, $x > 2$, find $\frac{d^2y}{dx^2}$.	63	Numerical	5.11	835

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	If $y = \cot x$, find $\frac{d^2y}{dx^2}$ at $x = \frac{\pi}{4}$.	63	Numerical	5.11	836
(iii)	If $x = at^2$ and $y = 2at$, then find $\frac{d^2y}{dx^2}$.	64a	Numerical	5.11	837
3	(see parts below)	63	Numerical	5.11	837
(i)	Find the second order derivatives of $\sin^{-1} x$.	63	Numerical	5.11	837
(ii)	Find the second order derivatives of $x \cos x$.	65	Numerical	5.11	838
(iii)	Find the second order derivatives of $x \sin 2x$.	65	Numerical	5.11	839
(iv)	Find the second order derivatives of $e^x \sin 5x$.	65	Numerical	5.11	840
(v)	Find the second order derivatives of $e^{2x} \sin 3x$.	65	Numerical	5.11	841
(vi)	Find the second order derivatives of $\log(\log x)$.	63	Numerical	5.11	842
(vii)	Find the second order derivatives of $\frac{\log x}{x}$.	66	Numerical	5.11	842
(viii)	Find the second order derivatives of $x^2 \log \cos x $.	65	Numerical	5.11	843
(ix)	Find the second order derivatives of $\frac{2x+1}{2x+3}$.	66	Numerical	5.11	844
4	(see parts below)	63	Numerical	5.11	845
(i)	Find the second order derivative of $\sec ax$.	63	Numerical	5.11	845
(ii)	Find the second order derivative of $\cot(1 - 2x)$.	63	Numerical	5.11	846
(iii)	Find the second order derivative of $\sin 3x \cos 5x$.	65	Numerical	5.11	847
(iv)	Find the second order derivative of $\sin^3 x$.	63	Numerical	5.11	848
(v)	Find the second order derivative of $\cos(2x^2 - 1)$.	63	Numerical	5.11	848
(vi)	Find the second order derivative of $\sqrt{1 - x^2}$.	63	Numerical	5.11	849
5	If $y = \cos^{-1} x$, find $\frac{d^2y}{dx^2}$ in terms of y alone.	67	Numerical	5.11	850
6	(see parts below)	68a	Proof	5.11	851
(i)	If $y = \cot x$, prove that $\frac{d^2y}{dx^2} + 2y \frac{dy}{dx} = 0$.	68a	Proof	5.11	851
(ii)	If $y = 5 \cos x - 3 \sin x$, prove that $\frac{d^2y}{dx^2} + y = 0$.	68a	Proof	5.11	851
7	(see parts below)	68a	Proof	5.11	852
(i)	If $y = x + \tan x$, prove that $\cos^2 x \frac{d^2y}{dx^2} - 2y + 2x = 0$.	68a	Proof	5.11	852
(ii)	If $y = \tan x + \sec x$, prove that ...	68a	Proof	5.11	853
8	(see parts below)	68a	Proof	5.11	854

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	If $y = \sec x - \tan x$, prove that $\cos x \frac{d^2y}{dx^2} = y^2$.	68a	Proof	5.11	854
(ii)	If $y = x + \cot x$, prove that $\sin^2 x \frac{d^2y}{dx^2} - 2y + 2x = 0$.	68a	Proof	5.11	854
9	(see parts below)	68a	Proof	5.11	855
(i)	If $y = ae^{mx} + be^{-mx}$, prove that $y_2 - m^2y = 0$.	68a	Proof	5.11	855
(ii)	If $y = ae^{2x} + be^{-x}$, prove that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$.	68a	Proof	5.11	856
10	(see parts below)	68a	Proof	5.11	856
(i)	If $y = A \cos nx + B \sin nx$, show that $\frac{d^2y}{dx^2} + n^2y = 0$.	68a	Proof	5.11	856
(ii)	If $y = A \sin 2x + B \cos 2x$ and $\frac{d^2y}{dx^2} - ky = 0$, find the value of k .	69	Numerical	5.11	857
11	(see parts below)	68a	Proof	5.11	858
(i)	If $y = \tan^{-1} x$, prove that $(1 + x^2) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = 0$.	68a	Proof	5.11	858
(ii)	If $y = \sin^{-1} x$, prove that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 0$.	68a	Proof	5.11	858
12	(see parts below)	68a	Proof	5.11	859
(i)	If $y = \sin(\log x)$, prove that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$.	68a	Proof	5.11	859
(ii)	If $y = 3 \cos(\log x) + 4 \sin(\log x)$, prove that ...	68a	Proof	5.11	860
13	If $y = x \cos x$, prove that ...	68a	Proof	5.11	860
14	(see parts below)	68a	Proof	5.11	861
(i)	If $y = (\cos^{-1} x)^2$, prove that $(1 - x^2)y_2 - xy_1 = 2$.	68a	Proof	5.11	861
(ii)	If $y = (\tan^{-1} x)^2$, prove that ...	68a	Proof	5.11	862
(iii)	If $y = (\sec^{-1} x)^2, x > 1$, show that ...	68a	Proof	5.11	862
15	(see parts below)	68a	Proof	5.11	863
(i)	If $y = \log(x + \sqrt{x^2 + 1})$, prove that ...	68a	Proof	5.11	863
(ii)	If $y = \log(x + \sqrt{x^2 + a^2})$, prove that $(x^2 + a^2)y_2 + xy_1 = 0$.	68a	Proof	5.11	864
(iii)	If $y = (\log x)^2$, prove that $x^2y'' + xy' = 2$.	68a	Proof	5.11	865
16	If $y = \cos(\sin x)$, prove that ...	68b	Proof	5.11	865
17	(see parts below)	68c	Proof	5.11	866
(i)	If $y = e^{\tan^{-1} x}$, prove that ...	68c	Proof	5.11	866
(ii)	If $x = \tan(\frac{1}{a} \log y)$, then prove that ...	68d	Proof	5.11	867
(iii)	If $y = e^{a \cos^{-1} x}, -1 < x < 1$, prove that ...	68e	Proof	5.11	868
18	If $y = \log(\frac{x}{a+bx})^x$, prove that ...	68f	Proof	5.11	869
19	If $\sqrt{x} + \sqrt{y} = \sqrt{a}$, find $\frac{d^2y}{dx^2}$ at $x = a$.	75	Numerical	5.11	870

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
20	If $x^m y^n = (x + y)^{m+n}$, then prove that: (i) $\frac{dy}{dx} = \frac{y}{x}$ (ii) ...	68g	Proof	5.11	871
21	If $x = a(\theta - \sin \theta)$, $y = a(1 + \cos \theta)$, find $\frac{d^2y}{dx^2}$.	64b	Numerical	5.11	873
22	If $x = \log t$ and $y = \frac{1}{t}$, prove that ...	64c	Proof	5.11	874
23	If $x = a(\frac{1-t^2}{1+t^2})$ and $y = \frac{2at}{1+t^2}$, prove that ...	64d	Proof	5.11	874
24	If $x = a \sin pt$ and $y = b \cos pt$, find the value of $\frac{d^2y}{dx^2}$ at $t = 0$.	64e	Numerical	5.11	876
25	If $x = a \sec^3 \theta$ and $y = a \tan^3 \theta$, find $\frac{d^2y}{dx^2}$ at ...	64f	Numerical	5.11	876
26	If $x = a(1 + \cos t)$, $y = a(t + \sin t)$, find $\frac{d^2y}{dx^2}$ at ...	64g	Numerical	5.11	877
27	If $x = \cos t + \log(\tan \frac{t}{2})$, $y = \sin t$, then find $\frac{d^2y}{dx^2}$ at ...	64h	Numerical	5.11	878

Chapter 6: Applications of Derivatives

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
--------	---------------------------	----------	------	---------	------

Exercise 6.1 – Rate of Change of Quantities

1	Find the rate of change of the area of a circle with respect to its radius when the radius ...	1a	Numerical	6.1	880
2	If the radius of a circle is increasing at the rate of 0.5 cm/sec, at what rate ...	1b	Numerical	6.1	880
3	If the radius of a circle is increasing at the rate of 3 cm/sec, at what rate is ...	1c	Numerical	6.1	881
4	If the sides of a square are decreasing at the rate of 1.5 cm/sec, at what rate ...	1d	Numerical	6.1	882
5	If an edge of a variable cube is increasing at the rate of 10 cm/sec, at what ...	1e	Numerical	6.1	883
6	A balloon which always remains spherical has a variable radius. Find the rate at which its ...	1f	Numerical	6.1	883
7	If the radius of a balloon which always remains spherical is increasing at the rate of ...	1g	Numerical	6.1	884
8	If the radius of a soap bubble is increasing at the rate of $\frac{1}{2}$ cm/sec ...	1h	Numerical	6.1	885

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
9	A cylindrical tank of radius 10 m is being filled with wheat at the rate of ...	1i	Numerical	6.1	886
10	(i) The total revenue in rupees received from the sale of x units of a product is given ...	10	Numerical	6.1	887
11	Find the rate of change of area of a circle of radius r with regard to its radius. How ...	1a	Numerical	6.1	888
12	Find the rate of change of the whole surface of a closed circular cylinder of radius r ...	1j	Numerical	6.1	889
13	A stone is dropped into a quiet lake and waves move in circles at the speed of ...	1c	Numerical	6.1	890
14	A stone is dropped into a quiet lake and waves move in circles at a speed of ...	1c	Numerical	6.1	890
15	If the circumference of circle is increasing at the constant rate, prove that rate of ...	12a	Proof	6.1	891
16	Find the point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the ...	13	Numerical	6.1	892
17	The median of an equilateral triangle is increasing at the rate of ...	1k	Numerical	6.1	893
18	The length x of a rectangle is decreasing at the rate of 5 cm/minute and the ...	1l	Numerical	6.1	894
19	The volume of a cube is increasing at the rate of 9 cubic centimetres per second ...	1m	Numerical	6.1	895
20	A particle moves along the curve $y = \frac{2}{3}x^3 + 1$. Find the points on the curve ...	12c	Numerical	6.1	897
21	A man of height 2 metres walks at a uniform speed of 5 km/h away from a lamp post which is ...	18a	Numerical	6.1	898
22	A girl of height 1.6 m walks at the rate of 50 metres per minute away from a lamp which is ...	18a	Numerical	6.1	899
23	A kite is 120 m high and 130 m of string is out. If the kite is moving horizontally at the ...	18b	Numerical	6.1	901
24	A circular cone, with semi-vertical angle 45° , is fixed with its axis vertical and ...	18c	Numerical	6.1	902

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
25	Sand is pouring from a pipe at the rate of $12 \text{ cm}^3/\text{sec}$. The falling sand ...	18d	Numerical	6.1	904
26	A conical vessel whose height is 4 metres and of base radius 2 metres is being filled with ...	18e	Numerical	6.1	905
27	Water is dripping out at a steady rate of 1 cu cm/sec through a tiny hole at the ...	18f	Numerical	6.1	907
28	The population of rabbits in a forest is modelled by the function below: ...	24	Proof	6.1	908

Exercise 6.2 – Tangents and Normals

1	What is the slope of the tangent to the following curves:	25a	Numerical	6.2	910
(i)	$y = x^3 - x$ at $x = 2$?	25a	Numerical	6.2	910
(ii)	$y = 3x^4 - 4x$ at $x = 4$?	25a	Numerical	6.2	910
(iii)	$x^2 + 3y + y^2 = 5$ at the point $(1, 1)$?	25a	Numerical	6.2	910
(iv)	$y = x^3 - 3x + 2$ at the point whose x -coordinate is 3?	25a	Numerical	6.2	910
(v)	$y = 3x^2 + 4x$ at the point whose x -coordinate is -2 ?	25a	Numerical	6.2	910
(vi)	$y = 2 \sin^2(3x)$ at $x = \frac{\pi}{6}$?	25a	Numerical	6.2	910
2	What is the slope of the normal to the following curves:	25b	Numerical	6.2	912
(i)	$y = x^3 - 5x^2 - x + 1$ at the point $(1, -4)$?	25b	Numerical	6.2	912
(ii)	$y = 2x^2 + 3 \sin x$ at $x = 0$?	25b	Numerical	6.2	912
3	Find the point on the curve $y = 3x^2 - 2x + 1$ at which the slope of the tangent is 4.	27a	Numerical	6.2	914
4	Determine the point on the curve $y = 3x^2 - 1$ at which the slope of the tangent is 3.	27a	Numerical	6.2	914
5	At what point on the curve $y = x^2$ does the tangent make an angle of 45° with the ...	27b	Numerical	6.2	915
6	See parts below:	27c	Numerical	6.2	916
(i)	Find the point on the curve $y = x^2 - 2x + 3$ at which the tangent is parallel ...	27c	Numerical	6.2	916
(ii)	Find the point on the curve $y = x^2 + 2x - 1$, where the tangent is parallel to ...	27c	Numerical	6.2	916
7	Find the slope of the normal to the curve $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ at ...	25c	Numerical	6.2	917

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
8	Find the slope of the normal to the curve $x = 1 - a \sin \theta$, $y = b \cos^2 \theta$...	25c	Numerical	6.2	918
9	Write the equation of the normal to the curve $y^2 = 8x$ at the origin.	37a	Numerical	6.2	919
10	Find the slopes of the tangents at the three points where the curve $y = x(x^2 - 3x - 4)$...	25d	Numerical	6.2	920
11	Show that the tangents to the curve $y = 7x^3 + 11$ at the points where $x = 2$ and ...	27d	Proof	6.2	921
12	If the tangent to the curve $y = x^3 + ax + b$ at $(1, -6)$ is parallel to the line ...	27e	Numerical	6.2	922
13	Determine the point of the curve $y = 3x^2 - 5$ at which the tangent is perpendicular to a ...	27f	Numerical	6.2	922
14	Determine the points on the curve $y = x^3 - 3x^2 - 9x + 7$ at which the tangents are ...	27c	Numerical	6.2	923
15	Find the points on the curve $x^2 + y^2 - 2x - 3 = 0$ at which the tangents are parallel ...	27c	Numerical	6.2	924
16	Find the points on the curve $\frac{x^2}{4} + \frac{y^2}{25} = 1$ at which the ...	27c	Numerical	6.2	925
(i)	parallel to x -axis	27c	Numerical	6.2	925
(ii)	parallel to y -axis.	27c	Numerical	6.2	925
17	Find the points on the curve $x^2 + y^2 = 13$, the tangent at each one of which is ...	27g	Numerical	6.2	926
18	Find the points on the curve $y = x^3 - 3x^2 - 4x$ at which the tangent lines are parallel ...	27g	Numerical	6.2	927
19	Find the equations of the tangent and the normal to each of the following curves at ...	37b	Numerical	6.2	928
(i)	$y = x^3$ at $(1, 1)$	37b	Numerical	6.2	928
(ii)	$y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(1, 3)$	37b	Numerical	6.2	928
(iii)	$y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$	37b	Numerical	6.2	928
(iv)	$y = x^2$ at $(0, 0)$	37b	Numerical	6.2	928
(v)	$y = \sin^2 x$ at $x = \frac{\pi}{2}$	37b	Numerical	6.2	928
20	Find the equation of the normal to the curve $ay^2 = x^3$ at the point (am^2, am^3) .	37c	Numerical	6.2	930
21	See parts below:	37d	Numerical	6.2	931

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	Find the equation of the tangent to the curve $\sqrt{x} + \sqrt{y} = a$ at the ...	37d	Numerical	6.2	931
(ii)	Find the equations of the tangent and the normal to the curve ...	37d	Numerical	6.2	931
22	See parts below:	37e	Numerical	6.2	932
(i)	Find the equation of tangent to the curve given by $x = a \sin^3 t$...	37e	Numerical	6.2	932
(ii)	Find the equation of the tangent to the curve $x = \sin 3t$, $y = \cos 2t$ at ...	37e	Numerical	6.2	932
23	See parts below:	37f	Numerical	6.2	934
(i)	Find the equations of all lines having slope 2 and that are tangents to the ...	37f	Numerical	6.2	934
(ii)	Find the equation of the tangent to the curve $y = x + \frac{4}{x^2}$ which is ...	37f	Numerical	6.2	934
24	Find the equation of the tangent to the curve $y = x^2 - 2x + 7$ which is parallel to the line $2x - y + 9 = 0$	42a	Numerical	6.2	935
(i)	perpendicular to the line	42a	Numerical	6.2	935
(ii)	$5y - 15x = 13$.	42a	Numerical	6.2	935
25	See parts below:	42a	Numerical	6.2	936
(i)	Find the equations of the tangents to the curve $y = 4x^3 - 3x + 5$ which are ...	42a	Numerical	6.2	936
(ii)	Find the equations of the tangents to the curve $3x^2 - y^2 = 2$ which are ...	42a	Numerical	6.2	936
26	Find the equation of the tangent line to the curve $y = \sqrt{5x - 3} - 5$, which is ...	42a	Numerical	6.2	938
27	Find the equations of the normals to the curve $3x^2 - y^2 = 8$ which are parallel to the ...	42b	Numerical	6.2	940
28	Find the equation of the normal to the curve $y = x + \frac{1}{x}$, $x > 0$...	42b	Numerical	6.2	942
29	Find the equation of the normal to the curve $x^2 = 4y$ which passes through the point ...	37g	Numerical	6.2	944
30	(see parts below)	45a	Numerical	6.2	945
(i)	Find the angle of intersection between the curves $xy = 6$ and $x^2y = 12$.	45a	Numerical	6.2	945
(ii)	Find the angle of intersection between the curves $y = 4 - x^2$ and $y = x^2$.	45a	Numerical	6.2	947

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
31	Find the angle of intersection of the curves $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$ at ...	45a	Numerical	6.2	948
32	Prove that the curves $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ intersect at right angles at ...	45b	Proof	6.2	950
33	Find the condition that the curves $2x = y^2$ and $2xy = k$ intersect orthogonally.	45b	Numerical	6.2	951
34	Prove that the curves $xy = 4$ and $x^2 + y^2 = 8$ touch each other.	45c	Proof	6.2	953
35	Find the angle of intersection between the curves $y^2 = 4x$ and $x^2 = 4y$.	45a	Numerical	6.2	954

Exercise 6.3 – Solving Inequalities via Sign-Scheme Method

1	Solve the following inequalities for $x \in \mathbb{R}$:	48a	Numerical	6.3	956
(i)	$x(x - 2)(x - 5)(x + 3) > 0$	48a	Numerical	6.3	956
(ii)	$x^4 - 5x^2 + 4 \geq 0$	48a	Numerical	6.3	956
2	Find all real values of x which satisfy:	48b	Numerical	6.3	958
(i)	$x^3(x - 1)(x - 2) > 0$	48b	Numerical	6.3	958
(ii)	$x^2(x - 1)(x - 2) \leq 0$	48b	Numerical	6.3	958
3	Solve the following rational inequalities for $x \in \mathbb{R}$:	48c	Numerical	6.3	959
(i)	$\frac{1}{x-2} \leq 1$	48c	Numerical	6.3	959
(ii)	$\frac{(x+1)(x-3)}{x+2} \geq 0$	48c	Numerical	6.3	959

Exercise 6.4 – Increasing and Decreasing Functions

1	Show that the following functions are strictly increasing on $\mathbb{R}$:	51a	Proof	6.4	961
(i)	$f(x) = 3x + 17$	51a	Proof	6.4	961
(ii)	$f(x) = 7x - 3$	51a	Proof	6.4	961
2	See parts below:	51b	Proof	6.4	962
(i)	Show that the function f given by $f(x) = 3 - 7x$ is strictly decreasing.	51b	Proof	6.4	962
(ii)	Show that the function $f(x) = 1 - \frac{1}{x}$ is strictly increasing.	51b	Proof	6.4	962
3	See parts below:	51c	Proof	6.4	963
(i)	Prove that the function $f(x) = 3 + \frac{1}{x}$ is strictly decreasing.	51c	Proof	6.4	963
(ii)	Show that the function $f(x) = x - \frac{1}{x}$ is strictly increasing.	51c	Proof	6.4	963
4	See parts below:	51d	Proof	6.4	964

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	Show that the function $f(x) = e^{-2x}$ is strictly decreasing on $\mathbb{R}$.	51d	Proof	6.4	964
(ii)	Show that the function $f(x) = 5^x$ is strictly increasing.	51d	Proof	6.4	964
(iii)	Show that the function $f(x) = \log x$ is strictly increasing.	51d	Proof	6.4	964
(iv)	Show that the function $f(x) = -x + \cos x$ is decreasing.	51d	Proof	6.4	964
5	Find the values of 'a' for which the function $f(x) = b - ax + \sin x$ is increasing on ...	55a	Numerical	6.4	965
6	See parts below:	56a	Numerical	6.4	966
(i)	Find the interval in which the function f given by $f(x) = 7 - 4x - x^2$ is ...	56a	Numerical	6.4	966
(ii)	If $f(x) = x^3 - 48x$, find the interval where $f : \mathbb{R} \rightarrow \mathbb{R}$ is ...	56a	Numerical	6.4	966
(iii)	Find the interval in which the function $f(x) = 2x^3 - 3x$ is strictly ...	56a	Numerical	6.4	966
7	Show that the function f defined by $f(x) = (x - 1)e^x + 1$ is an increasing function ...	51e	Proof	6.4	968
8	For what value of k , the function $f(x) = kx^3 + 5$ is decreasing?	55b	Numerical	6.4	969
9	See parts below:	51f	Proof	6.4	970
(i)	Prove that the function $f(x) = ax + b$ is strictly decreasing iff $a < 0$.	51f	Proof	6.4	970
(ii)	Prove that the function $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.	51f	Proof	6.4	970
10	Prove that the function $f(x) = x^3 - 6x^2 + 12x + 5$ is increasing on $\mathbb{R}$.	51g	Proof	6.4	971
11	Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ is strictly increasing on ...	51g	Proof	6.4	971
12	Prove that the function $f(x) = 5 - 3x + 3x^2 - x^3$ is decreasing on $\mathbb{R}$.	51g	Proof	6.4	972
13	See parts below:	51h	Proof	6.4	973
(i)	Show that $f(x) = 2x - \sin 2x$ is an increasing function.	51h	Proof	6.4	973
(ii)	Show that the function $f(x) = \frac{16 \sin x}{4 + \cos x} - x$ is strictly ...	51h	Proof	6.4	973
14	See parts below:	62a	Numerical	6.4	974

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(i)	Prove that the function f given by $f(x) = \log \cos x$ is strictly decreasing ...	62a	Numerical	6.4	974
(ii)	If $f(x) = a(\tan x - \cot x)$, where $a > 0$, then find whether $f(x)$ is ...	62a	Numerical	6.4	974
15	Determine whether the following functions are increasing or decreasing for the stated ...	62b	Numerical	6.4	975
(i)	$f(x) = \frac{1}{x}, \quad x < 0$	62b	Numerical	6.4	975
(ii)	$f(x) = \frac{1}{1+x^2}, \quad x \geq 0$	62b	Numerical	6.4	975
(iii)	$f(x) = \cos^2 x$ in $[0, \frac{\pi}{2}]$	62b	Numerical	6.4	975
(iv)	$f(x) = -\frac{x}{2} + \sin x, \quad -\frac{\pi}{3} < x < \frac{\pi}{3}$	62b	Numerical	6.4	975
16	Which of the following functions are strictly decreasing on $(0, \frac{\pi}{2})$?	62c	Numerical	6.4	977
(i)	$\cos x$	62c	Numerical	6.4	977
(ii)	$\cos 2x$	62c	Numerical	6.4	977
(iii)	$\sin^2 x$	62c	Numerical	6.4	977
(iv)	$\tan x$	62c	Numerical	6.4	977
17	Consider the statement "There exists at least one value of $b \in \mathbb{R}$ for which ...	55c	Numerical	6.4	977
18	Prove that the function $f(x) = \sin(2x + \frac{\pi}{4})$ is decreasing for ...	51i	Proof	6.4	978
19	Prove that the function $f(x) = \sin x$ is:	51j	Proof	6.4	979
(i)	strictly increasing in $(0, \frac{\pi}{2})$	51j	Proof	6.4	979
(ii)	strictly decreasing in $(\frac{\pi}{2}, \pi)$	51j	Proof	6.4	979
(iii)	neither increasing nor decreasing in $(0, \pi)$	51j	Proof	6.4	979
20	Prove that the function f defined by $f(x) = \cos x$ is:	51k	Proof	6.4	980
(i)	strictly increasing in $(\pi, 2\pi)$	51k	Proof	6.4	980
(ii)	strictly decreasing in $(0, \pi)$	51k	Proof	6.4	980
(iii)	neither increasing nor decreasing in $(0, 2\pi)$	51k	Proof	6.4	980
21	Prove that the following functions are neither increasing nor decreasing:	69	Proof	6.4	981
(i)	$f(x) = x^2 - x + 1$ on $(-1, 1)$	69	Proof	6.4	981
(ii)	$f(x) = \frac{1}{x^2+1}$ on $\mathbb{R}$	69	Proof	6.4	981
22	Prove that the function $f(x) = x $ is:	51l	Proof	6.4	982
(i)	strictly increasing in $(0, \infty)$	51l	Proof	6.4	982
(ii)	strictly decreasing in $(-\infty, 0)$	51l	Proof	6.4	982

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
23	Prove that the function f given by $f(x) = \frac{4\sin x}{2+\cos x} - x$ is ...	51m	Proof	6.4	983
24	(see parts below)	56b	Numerical	6.4	984
(i)	Find the intervals in which the function $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly ...	56b	Numerical	6.4	984
(ii)	Find the intervals in which the function $f(x) = 4x^3 - 6x^2 - 72x + 30$ is strictly ...	56b	Numerical	6.4	985
(iii)	Find the intervals in which the function $f(x) = 2x^3 - 9x^2 + 12x + 15$ is strictly ...	56b	Numerical	6.4	985
(iv)	Find the intervals in which the function $f(x) = -2x^3 - 9x^2 - 12x + 1$ is strictly ...	56b	Numerical	6.4	986
(v)	Find the intervals in which the function $f(x) = x + \frac{1}{x}, x \neq 0$ is strictly ...	56c	Numerical	6.4	987
(vi)	Find the intervals in which the function $f(x) = \frac{4x^2+1}{x}, x \neq 0$ is ...	56c	Numerical	6.4	988
(vii)	Find the intervals in which the function $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$ is ...	56d	Numerical	6.4	989
(viii)	Find the intervals in which the function $f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$...	56d	Numerical	6.4	990
(ix)	Find the intervals in which the function $f(x) = x^4 - 4x$ is strictly increasing or ...	56d	Numerical	6.4	990
(x)	Find the intervals in which the function $f(x) = (x-1)^3(x-2)^2$ is strictly ...	56e	Numerical	6.4	991
25	(see parts below)	56f	Numerical	6.4	992
(i)	Find the interval in which the function $f(x) = x^2e^{-x}$ is increasing.	56f	Numerical	6.4	992
(ii)	Find the intervals in which the function given by $f(x) = x^2e^x$ is strictly increasing ...	56f	Numerical	6.4	993
26	Find the intervals in which the function $f(x) = 2\log(x-2) - x^2 + 4x - 5$ is strictly ...	56g	Numerical	6.4	993

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
27	Find the intervals in which the function f given by ...	56h	Numerical	6.4	994
(i)	strictly increasing	56h	Numerical	6.4	994
(ii)	strictly decreasing	56h	Numerical	6.4	994

Exercise 6.5 – Maxima and Minima Without Derivative Tests

1	Find the maximum and the minimum values (if any) of the following functions:	79a	Numerical	6.5	995
(i)	$f(x) = x^2$	79a	Numerical	6.5	995
(ii)	$f(x) = (2x - 1)^2 + 3$	79a	Numerical	6.5	995
2	Find the maximum and the minimum values (if any) of the following functions:	79a	Numerical	6.5	997
(i)	$f(x) = -(x - 1)^2 + 10$	79a	Numerical	6.5	997
(ii)	$f(x) = 9x^2 + 12x + 2$	79a	Numerical	6.5	997
3	Find the maximum and the minimum values (if any) of the following functions:	79b	Numerical	6.5	998
(i)	$f(x) = x $	79b	Numerical	6.5	998
(ii)	$f(x) = x^3 + 1$	79b	Numerical	6.5	998
4	Find the maximum and the minimum values (if any) of the following functions:	79b	Numerical	6.5	999
(i)	$f(x) = x + 2 - 1$	79b	Numerical	6.5	999
(ii)	$f(x) = 3 + x $	79b	Numerical	6.5	999
5	Find the maximum and the minimum values (if any) of the following functions:	79c	Numerical	6.5	1000
(i)	$f(x) = - x + 1 + 3$	79c	Numerical	6.5	1000
(ii)	$f(x) = \sin 2x$	79c	Numerical	6.5	1000
6	Find the maximum and the minimum values (if any) of the following functions:	79c	Numerical	6.5	1001
(i)	$f(x) = \sin 2x + 5$	79c	Numerical	6.5	1001
(ii)	$f(x) = \sin 4x + 3 $	79c	Numerical	6.5	1001
7	Find the maximum and the minimum values (if any) of the following functions:	79c	Numerical	6.5	1002
(i)	$f(x) = \sin 3x - 3$	79c	Numerical	6.5	1002
(ii)	$f(x) = 4 - \cos 3x $	79c	Numerical	6.5	1002
8	See parts below:	79c	Numerical	6.5	1003
(i)	Is the minimum value of $\cos x$ zero?	79c	Numerical	6.5	1003
(ii)	What are the minimum and maximum values of $2 - 3 \cos x$?	79c	Numerical	6.5	1003

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
9	What are the maximum and minimum values of $3 \sin x + 4 \cos x$?	79d	Numerical	6.5	1003
10	Find the maximum and minimum values, if any, of the following functions:	79e	Numerical	6.5	1005
(i)	$f(x) = x$ in $(0, 1)$	79e	Numerical	6.5	1005
(ii)	$f(x) = x$ in $[0, 1]$	79e	Numerical	6.5	1005
11	Prove that the function $f(x) = x^3 + x^2 + x + 1$ does not have maxima or minima.	84	Proof	6.5	1006
12	Find the (absolute) maximum and the (absolute) minimum values of the following ...	85a	Numerical	6.5	1007
(i)	$f(x) = (x - 1)^2 + 3$ in $[-3, 1]$	85a	Numerical	6.5	1007
(ii)	$f(x) = 2x^3 - 15x^2 + 36x + 1$ on the interval $[1, 5]$	85a	Numerical	6.5	1007
(iii)	$f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ in $[0, 3]$	85a	Numerical	6.5	1007
(iv)	$f(x) = x + \sin x$ in $[0, 2\pi]$	85a	Numerical	6.5	1007
(v)	$f(x) = \sin x + \sqrt{3} \cos x$ in $[0, \pi]$	85a	Numerical	6.5	1007
(vi)	$f(x) = 3 + x + 1 $ in $[-2, 3]$	85a	Numerical	6.5	1007
13	Find the maximum and minimum values of $x + \sin 2x$ on $[0, 2\pi]$. Also find points of ...	85b	Numerical	6.5	1009
14	Find the absolute maximum and absolute minimum values of the function ...	85c	Numerical	6.5	1011

Exercise 6.6 – Maxima and Minima Using Derivative Tests

1	Find the turning points of the following functions and distinguish between them. Also ...	88a	Numerical	6.6	1013
(i)	$f(x) = 2x^3 - 21x^2 + 36x - 20$	88a	Numerical	6.6	1013
(ii)	$f(x) = x^3 - 3x^2 + 3x$	88a	Numerical	6.6	1013
2	Find the coordinates of stationary points on the curve $y = x^3 - 3x^2 - 9x + 7$, and ...	88a	Numerical	6.6	1015
3	Find the points of local maxima and minima (if any) of each of the following ...	88b	Numerical	6.6	1017
(i)	$f(x) = x^2$	88b	Numerical	6.6	1017
(ii)	$f(x) = x^3 - 3x$	88b	Numerical	6.6	1017
(iii)	$f(x) = x^3 - 12x$	88b	Numerical	6.6	1017
(iv)	$f(x) = x^3 - 6x^2 + 9x + 15$	88b	Numerical	6.6	1017
(v)	$f(x) = x^3 - 3x + 3$	88b	Numerical	6.6	1017
(vi)	$f(x) = 2x^3 - 6x^2 + 6x + 5$	88b	Numerical	6.6	1017
(vii)	$f(x) = \frac{x}{2} + \frac{2}{x}, \quad x > 0$	88b	Numerical	6.6	1017

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(viii)	$f(x) = \frac{1}{x^2+2}$	88b	Numerical	6.6	1017
(ix)	$f(x) = 2 \sin x - x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$	88b	Numerical	6.6	1017
(x)	$f(x) = \sin x + \cos x, \quad 0 < x < \frac{\pi}{2}$	88b	Numerical	6.6	1017
(xi)	$f(x) = 3x^4 + 4x^3 - 12x^2 + 12$	88b	Numerical	6.6	1017
4	For what values of a and b , the function $f(x) = x^3 + ax^2 + bx - 3$ has local ...	90a	Numerical	6.6	1020
5	Given $2f(x) = \log x ^a - bx^2 + x$, if $f(x)$ has extreme values at $x = -1$ and ...	90b	Numerical	6.6	1021
6	Aditya hires a speed boat in an island to do water sports, which is moving along a curve ...	92	Numerical	6.6	1022
7	Anu enjoys a roller coaster ride by first going downwards and then upwards to the ...	93	Numerical	6.6	1024
(i)	What is the rate of change of displacement with respect to the time?	93	Numerical	6.6	1024
(ii)	How many seconds will it take her to go to its maximum height?	93	Numerical	6.6	1024

Exercise 6.7 – Applications of Maxima and Minima

1	Find two numbers whose sum is 24 and whose product is as large as possible.	94	Numerical	6.7	1025
2	The sum of two positive numbers is 20. Find the numbers	95	Numerical	6.7	1026
(i)	if the sum of their squares is minimum.	95	Numerical	6.7	1026
(ii)	if the product of the square of one and the cube of the other is maximum.	95	Numerical	6.7	1026
3	The product of two positive numbers is 16. Find the numbers	96	Numerical	6.7	1028
(i)	if their sum is least.	96	Numerical	6.7	1028
(ii)	if the sum of one and the square of the other is least.	96	Numerical	6.7	1028
4	Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.	97	Numerical	6.7	1030
5	Find two positive numbers x and y such that their sum is 35 and the product x^2y^5 ...	97	Numerical	6.7	1032
6	Sum of two numbers is 5. If the sum of the cubes of these numbers is least, then find the ...	98	Numerical	6.7	1033

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
7	Three numbers are given whose sum is 180 and the ratio of first two of them is 1 : 2. If ...	99	Numerical	6.7	1034
8	Find the maximum profit that a company can make, if the profit function is given by ...	100	Numerical	6.7	1035
9	The cost (in Rs.) per article C , of manufacturing a certain article is given by the ...	101	Numerical	6.7	1036
10	Prove that of all rectangles with given area, the square has the smallest perimeter.	102	Proof	6.7	1037
11	Of all rectangles each of which has perimeter 24 cm, find the one having maximum area ...	103	Numerical	6.7	1039
12	Prove that of all rectangles with given perimeter, the square has the largest area.	103	Proof	6.7	1040
13	Prove that the perimeter of a right-angled triangle of given hypotenuse is maximum when ...	104	Proof	6.7	1041
14	Two sides of a triangle have lengths a and b , and the angle between them is θ ...	105	Numerical	6.7	1043
15	A right-angled triangle with constant area is given. Prove that the hypotenuse of the ...	106	Proof	6.7	1045
16	The lengths of the sides of a triangle are $9 + x^2$, $9 + x^2$ and $18 - 2x^2$...	107	Numerical	6.7	1046
(i)	the area of the triangle in terms of x .	107	Numerical	6.7	1046
(ii)	the value of x for which this area is maximum.	107	Numerical	6.7	1046
17	The perimeter of a triangle is 8 cm. If one of the sides is 3 cm, what are the lengths of ...	108	Numerical	6.7	1048
18	A sheet of paper is to contain 18 cm^2 of printed matter. The margins at the top ...	109	Numerical	6.7	1049
19	Show that of all rectangles inscribed in a given fixed circle the square has the maximum ...	110	Proof	6.7	1051
20	See parts below:	111	Proof	6.7	1053
(i)	The sum of perimeters of a circle and a square is k , where k is some ...	111	Proof	6.7	1053

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
(ii)	Given the sum of perimeters of a circle and a square, show that the sum of areas ...	111	Proof	6.7	1053
21	A wire 10 metres long is cut into two parts. One part is bent into the shape of a circle ...	112	Numerical	6.7	1055
22	A wire of length 36 cm is cut into two pieces. One of the piece is turned into the form of ...	113	Numerical	6.7	1057
23	A window is in the form of a rectangle above which there is a semicircle. If the perimeter ...	114	Proof	6.7	1058
24	A rectangular area of 9000 m^2 is to be surrounded by a fence with two opposite ...	116	Numerical	6.7	1060
25	See parts below:	117	Numerical	6.7	1062
(i)	Find the point on the parabola $y^2 = 4x$ which is nearest to the point ...	117	Numerical	6.7	1062
(ii)	Find the point on the curve $x^2 = 8y$ which is nearest to the point (2, 4).	117	Numerical	6.7	1062
26	A running track of 440 m is to be laid out enclosing a football field. The ...	118	Numerical	6.7	1064
27	Find the maximum area of an isosceles triangle inscribed in the ellipse ...	119	Numerical	6.7	1066
28	Show that the surface area of a closed cuboid with square base and given volume is minimum ...	120	Proof	6.7	1068
29	An open box with a square base is to be made of given iron sheet of area ...	121	Proof	6.7	1069
30	The volume of a closed metal box with a square base is 4096 cm^3 . The cost of ...	122	Numerical	6.7	1071
31	An open tank with a square base and vertical sides is to be constructed from a metal sheet ...	123	Proof	6.7	1072
32	A tank with rectangular base and rectangular sides, open at the top, is to be constructed ...	124	Numerical	6.7	1073
33	A rectangular sheet of tin 45 cm by 24 cm is to be made into a box ...	125	Numerical	6.7	1075
34	Show that the semi-vertical angle of a cone of the maximum volume and of given slant ...	126	Proof	6.7	1076

Q. No.	Topic / Statement Summary	Type No.	Kind	Section	Page
35	Show that a right circular cylinder of given volume, which is open at the top, has minimum ...	127	Proof	6.7	1078
36	See parts below:	128	Numerical	6.7	1079
(i)	Show that a closed cylindrical vessel of given volume has the least (total) ...	128	Numerical	6.7	1079
(ii)	A closed right circular cylinder has a volume of 2156 cu. cm. What will ...	128	Numerical	6.7	1079
37	Show that the radius of a closed right circular cylinder of given surface area and maximum ...	129	Proof	6.7	1081
38	Find the maximum volume of the cylinder which can be inscribed in a sphere of radius ...	130	Numerical	6.7	1082
39	A cone is inscribed in a sphere of radius 12 cm. If the volume of the cone is ...	131	Numerical	6.7	1084
40	Show that the altitude of a right circular cone of maximum curved surface area which can ...	132	Proof	6.7	1085
41	Show that the height of a right circular cylinder of greatest volume which can be ...	133	Proof	6.7	1086
42	The sum of the surface areas of a rectangular parallelopiped with sides x , $2x$ and ...	115	Numerical	6.7	1088
43	A manufacturer plans to construct a cylindrical can to hold one cubic metre of oil. If the ...	134	Numerical	6.7	1091

Formula & Standard Results Sheet

Chapter 1: Relations and Functions

I. Relations – Types and Equivalence Relations (Ex 1.1, 1.2)

Number of Relations

$$n(A \times B) = n(A) \cdot n(B), \quad \text{Total relations from } A \text{ to } B = 2^{n(A) \cdot n(B)}$$

On a set A alone, total relations on $A = 2^{n(A)^2}$

Domain and Range of a Relation

$$\text{Domain}(R) = \{x : (x, y) \in R\}, \quad \text{Range}(R) = \{y : (x, y) \in R\}$$

Domain is the set of first components; Range is the set of second components

Reflexive Relation

$$(a, a) \in R \quad \forall a \in A$$

Every element of A must be related to itself

Symmetric Relation

$$(a, b) \in R \implies (b, a) \in R$$

Must hold for every pair in R , not just some

Transitive Relation

$$(a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$$

Check every chain of two related pairs

Equivalence Relation

R is reflexive + symmetric + transitive

A relation is an equivalence relation only if all three properties hold simultaneously

Identity and Universal Relations

$$I_A = \{(a, a) : a \in A\}, \quad \text{Universal relation} = A \times A$$

I_A is always reflexive, symmetric, transitive; the universal relation is always an equivalence relation

Equivalence Class

$$[a] = \{x \in A : (x, a) \in R\}$$

The equivalence class of a collects every element related to a ; equivalence classes partition A

II. Functions – Domain and Range (Ex 1.2)

Value of a Function

$$f(a) = f(x)|_{x=a}$$

Substitute the given input directly into the rule for f

Range of a Function

$$\text{Range}(f) = \{f(x) : x \in \text{Domain}(f)\}$$

Found by expressing x in terms of $y = f(x)$ and finding the valid values of y

Modulus Function

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Domain is all of $\mathbb{R}$; always defined and non-negative

Square-Root Type Functions

$$\sqrt{g(x)} \text{ needs } g(x) \geq 0, \quad \frac{1}{\sqrt{g(x)}} \text{ needs } g(x) > 0$$

A root in the denominator forces the strict inequality since division by zero is undefined

Rational Function

$$\frac{P(x)}{Q(x)}, \quad Q(x) \neq 0$$

Domain excludes every root of the denominator $Q(x)$

Logarithmic Function

$$\log(g(x)) \text{ is defined only when } g(x) > 0$$

Also exclude any point where a denominator inside $g(x)$ vanishes

Range of Trigonometric Functions

$$-1 \leq \sin x \leq 1, \quad -1 \leq \cos x \leq 1$$

Used to bound the range of any function built from $\sin x$ or $\cos x$

Range of $a \sin x + b \cos x$

$$-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$$

Standard bound for a linear combination of sine and cosine

III. One-One, Onto and Bijective Functions (Ex 1.3)

One-One (Injective) Function

$$f(x_1) = f(x_2) \implies x_1 = x_2, \quad \forall x_1, x_2 \in \text{Domain}(f)$$

Equivalently: distinct inputs always give distinct outputs

Onto (Surjective) Function

$$\forall y \in \text{Codomain}, \exists x \in \text{Domain such that } f(x) = y$$

Equivalently: $\text{Range}(f) = \text{Codomain}(f)$

Bijective Function

$$f \text{ is bijective} \iff f \text{ is both one-one and onto}$$

A bijective function is precisely the kind of function that is invertible

Number of Functions from A to B

$$n(B)^{n(A)}$$

Every element of A can independently map to any of the $n(B)$ elements of B

Number of One-One Functions

$${}^{n(B)}P_{n(A)} = \frac{n(B)!}{(n(B) - n(A))!}, \quad \text{valid only when } n(A) \leq n(B)$$

Equals 0 when $n(A) > n(B)$, since no injective map can exist

Number of Bijections

$$n! \quad \text{when } n(A) = n(B) = n$$

Every bijection between two equal-sized finite sets is a permutation of one onto the other

IV. Composition of Functions (Ex 1.4)

Composition of Functions

$$(g \circ f)(x) = g(f(x))$$

Apply f first, then apply g to the result; $g \circ f$ is defined only where $\text{Range}(f) \subseteq \text{Domain}(g)$

Composition on Finite Sets

$$(g \circ f)(x) =$$

$g(f(x))$ computed pair-by-pair from the ordered-pair definitions of f, g

Trace each element of the domain through f first, then through g

V. Invertible Functions and Inverse of a Function (Ex 1.5)

Invertibility Criterion

$$f \text{ is invertible} \iff f \text{ is bijective (one-one and onto)}$$

This is the single test to apply before attempting to find f^{-1}

Finding the Inverse from Ordered Pairs

$$f = \{(x, y)\} \implies f^{-1} = \{(y, x) : (x, y) \in f\}$$

Reverse every ordered pair; valid only once bijectivity (or a roster-level one-one/onto check) is confirmed

Inverse via Equation Solving

$$y = f(x) \iff x = f^{-1}(y)$$

Solve the equation $y = f(x)$ for x in terms of y , then relabel $y \rightarrow x$ to state $f^{-1}(x)$

Composition Test for Inverse Functions

$$g \circ f = I_{\text{domain}} \text{ and } f \circ g = I_{\text{codomain}} \iff g = f^{-1}$$

The standard way to verify a claimed inverse is correct

Inverse of an Inverse

$$(f^{-1})^{-1} = f$$

Reversing the reverse of a bijection returns the original function

Self-Inverse Function

$$f^{-1} = f \iff (f \circ f)(x) = x \quad \forall x \in \text{Domain}(f)$$

Check by composing f with itself and confirming it returns the identity

Monotonicity Test for Injectivity

$$f'(x) \geq 0 \quad \forall x \text{ (or } f'(x) \leq 0 \quad \forall x) \implies f \text{ is monotonic} \implies f \text{ is one-one}$$

Useful for proving injectivity of continuous algebraic functions without testing $f(x_1) = f(x_2)$ directly

Chapter 2: Inverse Trigonometric Functions

I. Domain, Range and Basic Identities (Ex 2.1)

Domains and Principal Value Ranges

$$\begin{aligned}\sin^{-1} x &: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] & \cos^{-1} x &: [0, \pi] \\ \tan^{-1} x &: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) & \cot^{-1} x &: (0, \pi) \\ \sec^{-1} x &: [0, \pi] \setminus \left\{\frac{\pi}{2}\right\} & \operatorname{cosec}^{-1} x &: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}\end{aligned}$$

All domains: $\sin^{-1}, \cos^{-1} : [-1, 1]$; $\tan^{-1}, \cot^{-1} : \mathbb{R}$; $\sec^{-1}, \operatorname{cosec}^{-1} : |x| \geq 1$.

Property I – Inverse-of-Function Identities

$$\begin{aligned}\sin(\sin^{-1} x) &= x, \quad |x| \leq 1 & \cos(\cos^{-1} x) &= x, \quad |x| \leq 1 \\ \tan(\tan^{-1} x) &= x, \quad x \in \mathbb{R}\end{aligned}$$

Holds whenever the inverse function is defined at x .

Property II – Function-of-Inverse Identities

$$\begin{aligned}\sin^{-1}(\sin \theta) &= \theta, \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] & \cos^{-1}(\cos \theta) &= \theta, \quad \theta \in [0, \pi] \\ \tan^{-1}(\tan \theta) &= \theta, \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\end{aligned}$$

Valid *only* inside the stated principal-value interval; outside it, reduce θ into range first.

Property III – Reciprocal Identities

$$\begin{aligned}\sin^{-1}\left(\frac{1}{x}\right) &= \operatorname{cosec}^{-1} x, \quad |x| \geq 1 & \cos^{-1}\left(\frac{1}{x}\right) &= \sec^{-1} x, \quad |x| \geq 1 \\ \tan^{-1}\left(\frac{1}{x}\right) &= \cot^{-1} x, \quad x > 0\end{aligned}$$

For $x < 0$, $\tan^{-1}\left(\frac{1}{x}\right) = -\pi + \cot^{-1} x$.

Property IV – Negative-Argument (Odd/Even) Identities

$$\begin{aligned}\sin^{-1}(-x) &= -\sin^{-1} x & \tan^{-1}(-x) &= -\tan^{-1} x \\ \operatorname{cosec}^{-1}(-x) &= -\operatorname{cosec}^{-1} x\end{aligned}$$

$\sin^{-1}, \tan^{-1}, \operatorname{cosec}^{-1}$ are odd functions.

Property IV – Negative-Argument (Supplementary) Identities

$$\begin{aligned}\cos^{-1}(-x) &= \pi - \cos^{-1} x & \cot^{-1}(-x) &= \pi - \cot^{-1} x \\ \sec^{-1}(-x) &= \pi - \sec^{-1} x\end{aligned}$$

$\cos^{-1}, \cot^{-1}, \sec^{-1}$ satisfy this supplementary rule instead of being odd.

Property V – Complementary-Angle Identities

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad |x| \leq 1 \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, \quad x \in \mathbb{R}$$

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, \quad |x| \geq 1$$

The two inverse functions in each pair always sum to $\pi/2$ on their common domain.

II. Addition, Subtraction and Double-Angle Formulas (Ex 2.2)**Property VI(a) – Sum Formula for Tangent Inverse**

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right), \quad xy < 1$$

If $xy > 1$ with $x, y > 0$: add π ; if $xy > 1$ with $x, y < 0$: subtract π .

Property VI(b) – Difference Formula for Tangent Inverse

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x - y}{1 + xy} \right), \quad xy > -1$$

Used to combine or split two arctangent terms.

Property VII – Sum/Difference Formulas for Sine and Cosine Inverse

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1} \left(x\sqrt{1 - y^2} + y\sqrt{1 - x^2} \right)$$

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1} \left(xy - \sqrt{1 - x^2}\sqrt{1 - y^2} \right)$$

Valid for suitable x, y so that the resulting angle stays within the principal range; otherwise subtract from π or add π as needed.

Property VIII – Double-Angle (2x) Formulas

$$2 \tan^{-1} x = \sin^{-1} \left(\frac{2x}{1 + x^2} \right), \quad |x| \leq 1 \quad 2 \tan^{-1} x = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right), \quad x \geq 0$$

$$2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1 - x^2} \right), \quad -1 < x < 1$$

These come from substituting $x = \tan \theta$ into $\sin 2\theta, \cos 2\theta, \tan 2\theta$.

Triple-Tangent Identity

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

Used when three inverse-tangent terms are added and their sum is a known angle (e.g. π).

III. Conversion Techniques (Ex 2.1 & 2.2)

Right-Triangle Conversion Method

If $\theta = \sin^{-1}\left(\frac{p}{h}\right)$, construct a right triangle with

opposite = p , hypotenuse = h , adjacent = $\sqrt{h^2 - p^2}$

The standard technique for converting one inverse trig ratio into another (e.g. $\sin^{-1} \rightarrow \tan^{-1}$) by reading off sides of the constructed triangle.

Weierstrass (Half-Angle) Substitution

$$t = \tan \frac{\theta}{2} \Rightarrow \sin \theta = \frac{2t}{1+t^2}, \quad \cos \theta = \frac{1-t^2}{1+t^2}, \quad \tan \theta = \frac{2t}{1-t^2}$$

Useful for simplifying expressions of the form $\tan^{-1}(f(\sin \theta, \cos \theta))$.

Half-Angle Trigonometric Identities

$$1 - \cos x = 2 \sin^2 \frac{x}{2} \quad 1 + \cos x = 2 \cos^2 \frac{x}{2} \quad \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

Frequently used to simplify $\sqrt{\frac{1 - \cos x}{1 + \cos x}}$ -type expressions to $\tan \frac{x}{2}$ before applying $\tan^{-1}$.

Chapter 3: Matrices

I. Order, Elements and Construction of Matrices (Ex 3.1)

Order of a Matrix

$$A_{m \times n} \implies m \text{ rows, } n \text{ columns}$$

The order is always stated as rows $\times$ columns

General Element Notation

$$A = [a_{ij}]_{m \times n}$$

a_{ij} denotes the entry in the i -th row and j -th column

Number of Possible Matrices

Total matrices of order $m \times n$ from k available entries = $k^{m \times n}$

Each of the mn positions can independently take any of the k values

Equality of Matrices

$$A = [a_{ij}] = B = [b_{ij}] \iff \text{same order and } a_{ij} = b_{ij} \forall i, j$$

Both the order and every corresponding entry must match

II. Addition, Subtraction and Scalar Multiplication (Ex 3.2)

Matrix Addition and Subtraction

$$A \pm B = [a_{ij} \pm b_{ij}]$$

Defined only when A and B have the same order

Scalar Multiplication

$$kA = [k a_{ij}]$$

Every entry of A is multiplied by the scalar k

Linear Combination of Matrices

$$kA + mB = [k a_{ij} + m b_{ij}]$$

Combines scalar multiplication and addition entry-by-entry

Solving Simultaneous Matrix Equations

$$X + Y = A, X - Y = B \implies 2X = A + B, 2Y = A - B$$

Add and subtract the two equations, then divide by the scalar 2, exactly as with ordinary simultaneous equations

III. Multiplication of Matrices (Ex 3.3)

Compatibility for Multiplication

$$P_{m \times n} \cdot Q_{n \times p} = (PQ)_{m \times p}$$

The number of columns of P must equal the number of rows of Q ; the product takes the outer dimensions

Element of a Product Matrix

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}$$

Row i of A dotted with column j of B

Identity Matrix Property

$$AI = IA = A$$

Multiplying by the identity matrix leaves A unchanged

Distributive Property

$$A(B + C) = AB + AC$$

Matrix multiplication distributes over addition

Associative Property

$$(AB)C = A(BC)$$

Grouping does not affect the product

Non-Commutativity of Multiplication

$$AB \neq BA \quad (\text{in general})$$

Always verify the order of multiplication; do not assume $AB = BA$ unless proved

Matrix Polynomial

$$f(x) = a_n x^n + \cdots + a_1 x + a_0 \implies f(A) = a_n A^n + \cdots + a_1 A + a_0 I$$

The constant term becomes a multiple of the identity matrix, not a bare number

Reducing Higher Powers via a Matrix Relation

$$\text{If } A^2 =$$

$kA - mI$, use it repeatedly to express A^n in terms of A and I

Avoids computing high powers of A directly by hand

Principle of Mathematical Induction (on Matrices)

$$P(1) \text{ true, and } P(k) \implies P(k+1) \implies P(n) \text{ true } \forall n \in \mathbb{N}$$

Standard three-step structure: base case, inductive hypothesis, inductive step

IV. Transpose, Symmetric and Skew-Symmetric Matrices (Ex 3.4)**Transpose of a Matrix**

$$(A')_{ij} = A_{ji}$$

If A is $m \times n$, then A' is $n \times m$; rows become columns

Properties of Transpose

$$(A')' = A, \quad (kA)' = kA', \quad (A \pm B)' = A' \pm B', \quad (AB)' = B'A'$$

Note the reversal of order in $(AB)' = B'A'$

Symmetric Matrix

$$A' = A \iff a_{ij} = a_{ji} \quad \forall i, j$$

A symmetric matrix must be square

Skew-Symmetric Matrix

$$A' = -A \iff a_{ij} = -a_{ji}, \quad \text{so } a_{ii} = 0$$

Every diagonal entry of a skew-symmetric matrix is 0

Symmetric-Skew-Symmetric Decomposition

$$A = P + Q, \quad P = \frac{1}{2}(A + A') \text{ (symmetric)}, \quad Q = \frac{1}{2}(A - A') \text{ (skew-symmetric)}$$

Every square matrix can be written uniquely as this sum

Orthogonal Matrix

A is orthogonal $\iff AA' = I$

Equivalently, $A' = A^{-1}$

Chapter 4: Determinants

I. Evaluation of Determinants: Minors and Cofactors (Ex 4.1)

2 × 2 Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

The basic expansion for a second-order determinant

3 × 3 Determinant – Expansion Along First Row

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

Can be expanded along any row or column; signs alternate by the checkerboard pattern

Minor and Cofactor

$$M_{ij} = \text{determinant left after deleting row } i \text{ and column } j, \quad A_{ij} = (-1)^{i+j} M_{ij}$$

The cofactor is the minor with a sign attached based on position

Determinant via Cofactor Expansion (Any Row/Column)

$$\Delta = a_{i1}A_{i1} + a_{i2}A_{i2} + a_{i3}A_{i3}$$

Expanding along row i ; an analogous formula holds for any column

Effect of Scalar Multiplication on a Determinant

$$|kA| = k^n |A|, \quad \text{for an } n \times n \text{ matrix } A$$

Multiplying an $n \times n$ matrix by k scales its determinant by k^n , not k

II. Properties of Determinants and Proving Identities (Ex 4.2)

Row/Column Operations Preserve Value

$$R_i \rightarrow R_i +$$

$$kR_j \quad (\text{or the analogous column operation}) \text{ leaves } |A| \text{ unchanged}$$

The standard tool for simplifying a determinant before expanding it

Scalar Factored From a Single Row/Column

If every entry of one row (or column) has a common factor k , then k can be

Only one row/column at a time – not the whole determinant

Zero Determinant Conditions

$|A| = 0$ if any two rows (or columns) are identical or proportional

A quick way to prove a determinant vanishes without full expansion

Determinant of a Product

$$|AB| = |A| |B|$$

Also gives $|A^k| = |A|^k$ and $|A^{-1}| = \frac{1}{|A|}$

Determinant of a Transpose

$$|A^T| = |A|$$

The determinant is unaffected by transposing the matrix

III. Applications of Determinants: Area, Collinearity and Equation of a Line (Ex 4.3)

Area of a Triangle

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Take the absolute value, since area cannot be negative

Collinearity of Three Points

$$(x_1, y_1), (x_2, y_2), (x_3, y_3) \text{ are collinear} \iff \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Equivalent to the triangle formed having zero area

Equation of a Line Through Two Points

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Setting the determinant to zero forces (x, y) to be collinear with the two given points

IV. Adjoint and Inverse of a Matrix (Ex 4.4)

Singular and Non-Singular Matrices

A is singular $\iff |A| = 0$; A is non-singular $\iff |A| \neq 0$

A matrix has an inverse only when it is non-singular

Adjoint of a 2×2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Swap the leading diagonal entries and negate the off-diagonal entries

Adjoint via Cofactor Matrix

$$\text{adj } A = C', \quad \text{where } C = [C_{ij}] = [(-1)^{i+j} M_{ij}]$$

Adjoint is the transpose of the cofactor matrix

Inverse of a Matrix

$$A^{-1} = \frac{1}{|A|} \text{adj } A, \quad \text{defined only when } |A| \neq 0$$

Divide every entry of the adjoint by the determinant

Inverse of a 2×2 Matrix (Direct Form)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The fast shortcut for 2×2 matrices, skipping the general cofactor method

Fundamental Adjoint Identity

$$A(\text{adj } A) = (\text{adj } A)A = |A| I_n$$

The identity that most adjoint/inverse proofs are built on

Determinant of the Adjoint

$$|\text{adj } A| = |A|^{n-1}, \quad \text{for an } n \times n \text{ matrix } A$$

Also gives $\text{adj}(\text{adj } A) = |A|^{n-2} A$

Reversal Law for Inverse and Adjoint

$$(AB)^{-1} = B^{-1}A^{-1}, \quad \text{adj}(AB) = (\text{adj } B)(\text{adj } A)$$

The order reverses, just as with the transpose of a product

Inverse from a Matrix's Own Characteristic Relation

$$\text{If } A^2 - c_1 A - c_0 I = O \text{ with } c_0 \neq 0, \text{ then } A^{-1} = \frac{1}{c_0}(A - c_1 I)$$

Lets A^{-1} be found algebraically without computing the adjoint, using a relation the matrix itself satisfies

V. Solving Systems of Linear Equations Using Matrices (Ex 4.5)

Matrix Form of a Linear System

$$AX = B$$

A is the coefficient matrix, X the column of unknowns, B the column of constants

Solving a Consistent System

$$X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B, \quad \text{valid when } |A| \neq 0$$

Gives a unique solution; compute A^{-1} first, then multiply by B

Consistency of the System

$|A| \neq 0 \implies$ unique solution; $|A| = 0$ and $(\text{adj } A)B \neq O \implies$
no solution; $|A| = 0$ and $(\text{adj } A)B = O \implies$
infinitely many solutions (or none, needing further check)

Always check $|A|$ first before attempting to solve or classify the system

Homogeneous System

$$AX = O \implies x = y = z = 0 \text{ (trivial solution) when } |A| \neq 0$$

A homogeneous system has a non-trivial solution only when $|A| = 0$

Reducible (Non-Linear) Systems via Substitution

Substituting $u = \frac{1}{x}$, $v = \frac{1}{y}$, $w =$

$\frac{1}{z}$ converts a reciprocal system into a linear one $AU = B$

Solve for u, v, w first via the matrix method, then invert the substitution to recover x, y, z

Chapter 5: Continuity and Differentiability

I. Continuity (Ex 5.1 – 5.3)

Continuity at a Point

$$f \text{ is continuous at } x = a \iff \lim_{x \rightarrow a} f(x) = f(a)$$

Requires the limit to exist and equal the function value at that point

Continuity via Left- and Right-Hand Limits

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

The standard three-part test used for piecewise functions at a junction point

Continuity of Standard Functions

Polynomial, $\sin x$, $\cos x$, e^x are continuous everywhere; $\tan x$, $\sec x$, $\cot x$, $\csc x$
Discontinuities in trig functions occur only where the function itself is undefined

Algebra of Continuous Functions

$$f, g \text{ continuous at } a \implies f \pm g, fg, \frac{f}{g} (g(a) \neq 0) \text{ are all continuous at } a$$

Continuity is preserved under the usual arithmetic operations

Continuity of Composite Functions

$$f \text{ continuous at } a \text{ and } g \text{ continuous at } f(a) \implies g \circ f \text{ continuous at } a$$

Composing two continuous functions gives a continuous function

Removable Discontinuity

$$\lim_{x \rightarrow a} f(x) \text{ exists but } \neq f(a) \text{ (or } f(a) \text{ undefined)}$$

Can be "removed" by redefining $f(a)$ to equal the limit

II. Differentiability (Ex 5.4)

Differentiability at a Point

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided the limit exists}$$

The derivative at a point is defined as this two-sided limit

Left-Hand and Right-Hand Derivatives

$$Lf'(a) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}, \quad Rf'(a) =$$

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$$

f is differentiable at a only when $Lf'(a) = Rf'(a)$

Differentiability Implies Continuity

f differentiable at $a \implies f$ continuous at a

The converse is false – continuity does not guarantee differentiability (e.g. $|x|$ at $x = 0$)

III. Derivatives of Standard and Composite Functions (Ex 5.5 – 5.6)

Chain Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Differentiate the outer function first, then multiply by the derivative of the inner function

Derivative of $|x|$

$$\frac{d}{dx}|x| = \frac{x}{|x|} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}, \quad \text{not differentiable at } x = 0$$

Always split into cases based on the sign of the expression inside the modulus

Standard Derivatives

$$\begin{aligned} \frac{d}{dx} \sin x &= \cos x, & \frac{d}{dx} \cos x &= -\sin x, & \frac{d}{dx} \tan x &= \sec^2 x, \\ \frac{d}{dx} e^x &= e^x, & \frac{d}{dx} \log x &= \frac{1}{x} \end{aligned}$$

The core building-block derivatives every chain-rule question relies on

Derivatives of Inverse Trigonometric Functions

$$\begin{aligned} \frac{d}{dx} \sin^{-1} x &= \frac{1}{\sqrt{1-x^2}}, & \frac{d}{dx} \cos^{-1} x &= \frac{-1}{\sqrt{1-x^2}}, & \frac{d}{dx} \tan^{-1} x &= \frac{1}{1+x^2} \end{aligned}$$

Valid on the open interval $(-1, 1)$ for the first two; all reals for the third

Trigonometric Substitution Before Differentiation

Substitute $x =$

$\sin \theta, \cos \theta,$ or $\tan \theta$ to simplify an inverse-trig expression before differentiating

Converts a complicated nested inverse-trig expression into a simple linear one in θ

IV. Implicit Differentiation (Ex 5.7)

Implicit Differentiation

Differentiate both sides of $F(x, y) = c$ w.r.t. x , treating y as a function of x , then solve for $\frac{dy}{dx}$

Every term containing y picks up a factor of $\frac{dy}{dx}$ via the chain rule

Product Rule (needed inside implicit differentiation)

$$\frac{d}{dx}(uv) = u'v + uv'$$

Frequently needed when a term mixes x and y together, e.g. xy

V. Logarithmic Differentiation (Ex 5.8 – 5.9)

Logarithmic Differentiation

For $y = [f(x)]^{g(x)}$:

take log of both sides, then differentiate implicitly

The standard method whenever both the base and the exponent are functions of x

General Logarithmic Differentiation Formula

$$y = [f(x)]^{g(x)} \implies \frac{dy}{dx} = y \left[g'(x) \log f(x) + \frac{g(x)f'(x)}{f(x)} \right]$$

Derived from $\log y = g(x) \log f(x)$, differentiated implicitly

Logarithm Laws (needed before differentiating)

$$\log(ab) = \log a + \log b, \quad \log \frac{a}{b} = \log a - \log b, \quad \log a^n = n \log a$$

Always simplify using these laws before differentiating – it avoids a messy quotient/product rule

VI. Parametric Differentiation and Derivatives w.r.t. Another Function (Ex 5.10)

Parametric Differentiation

$$x = f(t), y = g(t) \implies \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Both derivatives are taken with respect to the parameter t , then divided

Derivative of One Function w.r.t. Another

$$\frac{du}{dv} = \frac{du/dx}{dv/dx}$$

Treat x as the underlying parameter for both u and v

VII. Second-Order Derivatives (Ex 5.11)

Second-Order Derivative

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

Differentiate the first derivative again with respect to x

Second-Order Derivative – Parametric Form

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}$$

Never differentiate dy/dx directly with respect to x in parametric form – always divide by dx/dt again

Second-Order Implicit Differentiation

Differentiate $\frac{dy}{dx}$ (already in terms of x, y) implicitly a second time, substituting $\frac{dy}{dx}$ for y' when necessary.

The first-derivative expression must be substituted back in before simplifying

Verifying a Second-Order Differential Equation

Compute y' , y'' from the given relation, then substitute into the given equation and simplify.

The standard structure for every "prove that" second-derivative identity question

Chapter 6: Applications of Derivatives

I. Rate of Change of Quantities (Ex 6.1)

Rate of Change

If $y = f(x)$, the rate of change of y w.r.t. x is $\frac{dy}{dx}$

For a quantity changing with time, differentiate w.r.t. t instead

Related Rates via Chain Rule

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

The core tool linking the rate of one quantity to the rate of another through a shared relation

Standard Mensuration Formulas Needed for Rates

$$A = \pi r^2, C = 2\pi r, V = \frac{4}{3}\pi r^3, S = 4\pi r^2 \text{ (sphere); } V = \pi r^2 h, S = 2\pi r h + 2\pi r^2 \text{ (cylinder); } V = \frac{1}{3}\pi r^2 h \text{ (cone)}$$

Write the known geometric formula first, then differentiate both sides w.r.t. t

Related Rates with a Geometric Setup

Use similar triangles or a trigonometric ratio to relate the two changing quantities

Common in shadow/kite/leaking-vessel problems, where the direct formula is not given outright and must be derived first

Marginal Cost and Marginal Revenue

$$MC = \frac{dC}{dx}, \quad MR = \frac{dR}{dx}$$

The derivative of total cost/revenue gives the approximate cost/revenue of producing one additional unit

II. Tangents and Normals (Ex 6.2)

Slope of Tangent

$$m_T = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$$

Evaluate the derivative at the specific point of contact

Slope of Normal

$$m_N = -\frac{1}{m_T}$$

The normal is perpendicular to the tangent at the point of contact

Equation of Tangent

$$y - y_1 = m_T(x - x_1)$$

Point-slope form using the tangent's slope

Equation of Normal

$$y - y_1 = m_N(x - x_1)$$

Point-slope form using the normal's slope

Horizontal and Vertical Tangents

$$\begin{aligned} \text{Tangent parallel to } x\text{-axis} &\iff \frac{dy}{dx} = 0; \\ \text{tangent parallel to } y\text{-axis} &\iff \frac{dx}{dy} = 0 \end{aligned}$$

Solve the appropriate derivative equation to locate these points

Parallel and Perpendicular Lines

$$\text{Parallel: } m_1 = m_2; \quad \text{Perpendicular: } m_1 m_2 = -1$$

Match the tangent/normal slope against the given line's slope

Tangent/Normal to a Parametric Curve

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Compute the slope in terms of the parameter, then substitute the specific parameter value

Angle of Intersection of Two Curves

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

m_1, m_2 are the tangent slopes of the two curves at their common point

Orthogonal Curves

$$m_1 m_2 = -1$$

Condition for two curves to intersect at right angles

III. Solving Inequalities via Sign-Scheme Method (Ex 6.3)**Sign-Scheme (Wavy Curve) Method**

$$f(x) = (x - a_1)(x - a_2) \cdots (x - a_n)$$

Sign alternates across each root on the number line; determine the sign in the rightmost interval first, then alternate

Repeated Factors

At a root of even multiplicity, the sign does NOT change; at odd multiplicity

Always check the multiplicity of each factor before alternating signs

Rational Inequalities

Bring all terms to one side, combine into a single fraction, then apply the sign-scheme method.
Denominator roots are always excluded from the solution set, even if the inequality is non-strict

IV. Increasing and Decreasing Functions (Ex 6.4)

Monotonicity Test

$f'(x) > 0 \implies f$ increasing; $f'(x) < 0 \implies f$ decreasing
Applied throughout an interval; equality at isolated points does not break strict monotonicity

Finding Intervals of Increase/Decrease

Find $f'(x)$, locate its roots, then use the sign-scheme method on $f'(x)$ across the domain.
The roots of $f'(x)$ partition the domain into intervals of constant sign

Parameter Range for Monotonicity

Require $f'(x) \geq 0$ (or ≤ 0) $\forall x$ in the given interval, then solve the resulting inequality for the parameter.
Often reduces to a condition on the discriminant or sign of a quadratic in the parameter

V. Maxima and Minima Without Derivative Tests (Ex 6.5)

Extrema by Inspection

For functions like $|x|$, $a \sin x + b \cos x$, or piecewise-defined functions, use the known range/behaviour of the function.
Faster than calculus when the function's shape is already familiar, e.g. $-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$

No Extrema on an Open Interval

If f is strictly monotonic throughout an interval, it has no local maximum or minimum in the interior.
A strictly increasing or decreasing function only attains extreme values at the endpoints, if at all

Absolute Extrema on a Closed Interval

Evaluate f at all critical points inside $[a, b]$ and at both endpoints a, b ; the largest and smallest values are the absolute extrema.
Critical points alone are not enough on a closed interval – the endpoints must always be checked too

VI. Maxima and Minima Using Derivative Tests (Ex 6.6)

First Derivative Test

$f'(x)$ changes sign from $+$ to $-$ at $c \implies$
local max at c ; $-$ to $+$ $\implies$ local min at c

Check the sign of $f'(x)$ just before and just after the critical point

Second Derivative Test

$f'(c) = 0$ and $f''(c) < 0 \implies$ local max; $f'(c) =$
 0 and $f''(c) > 0 \implies$ local min

Faster than the first derivative test when $f''(c) \neq 0$; if $f''(c) = 0$, fall back to the first derivative test

Stationary Points

$$f'(x) = 0$$

Every local extremum of a differentiable function occurs at a stationary point, but not every stationary point is an extremum

VII. Applications of Maxima and Minima (Ex 6.7)

Constrained Optimization Method

- (1) Express the quantity to be optimized as a function of one variable using the
0. (3) Confirm max/min using the first or second derivative test.

The recurring three-step structure behind every applied optimization problem, regardless of the specific geometric or physical context

Common Constraint Substitutions

Use the given fixed quantity (perimeter, surface area, volume, or a geometric
Reduces a two-variable objective function down to one variable so ordinary differentiation applies

ISC Class XII Mathematics

Chapter 1 Relations and Functions

138 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 1.1 – Types of Relations: Reflexive, Symmetric, Transitive & Equivalence Relations
- II. 1.2 – Functions: Domain and Range
- III. 1.3 – One-One, Onto and Bijective Functions
- IV. 1.4 – Composition of Functions
- V. 1.5 – Invertible Functions and Inverse of a Function

1.1 – Types of Relations: Reflexive, Symmetric, Transitive & Equivalence Relations

Question 1

If $A = \{-1, 1, 3\}$, then what is the number of relations on A ?

TYPE 1 Total Number of Relations

Given: $A = \{-1, 1, 3\}$

Concept / Formula Used: $n(A) = m$ $A \times A$ m^2 A 2^{m^2}

Solution:

The set A contains 3 elements:

$$n(A) = 3$$

The number of elements in the Cartesian product $A \times A$ is given by:

$$\begin{aligned} n(A \times A) &= n(A) \times n(A) \\ &= 3 \times 3 \\ &= 9 \end{aligned}$$

Since any relation on A is a subset of $A \times A$, the total number of relations on A is the total number of subsets of $A \times A$:

$$\begin{aligned} \text{Total number of relations} &= 2^{n(A \times A)} \\ &= 2^9 \\ &= 512 \end{aligned}$$

Final Answer

The total number of relations on A is 512.

Question 2

- (i) If $R = \{(x, y) : x + 2y = 8\}$ is a relation on $\mathbb{N}$, write the range of R .
(ii) Let $R = \{(a, a^3) : a \text{ is a prime number less than } 5\}$. Find the range of R .

TYPE 2 Domain and Range of a Relation

Given: $R_1 = \{(x, y) : x + 2y = 8, x, y \in \mathbb{N}\}$, $R_2 = \{(a, a^3) : a \text{ is prime}, a < 5\}$

Concept / Formula Used: R R $\text{Range}(R) = \{y : (x, y) \in R\}$

Solution:

Part (i): Given equation:

$$\begin{aligned}x + 2y &= 8 \\2y &= 8 - x \\y &= \frac{8 - x}{2}\end{aligned}$$

Since $x, y \in \mathbb{N} = \{1, 2, 3, \dots\}$, we substitute positive integer values for x :

$$\begin{aligned}\text{For } x = 1 : \quad y &= \frac{8 - 1}{2} = \frac{7}{2} \notin \mathbb{N} \\ \text{For } x = 2 : \quad y &= \frac{8 - 2}{2} = 3 \in \mathbb{N} \implies (2, 3) \in R \\ \text{For } x = 3 : \quad y &= \frac{8 - 3}{2} = \frac{5}{2} \notin \mathbb{N} \\ \text{For } x = 4 : \quad y &= \frac{8 - 4}{2} = 2 \in \mathbb{N} \implies (4, 2) \in R \\ \text{For } x = 5 : \quad y &= \frac{8 - 5}{2} = \frac{3}{2} \notin \mathbb{N} \\ \text{For } x = 6 : \quad y &= \frac{8 - 6}{2} = 1 \in \mathbb{N} \implies (6, 1) \in R \\ \text{For } x \geq 7 : \quad y &\leq 0 \notin \mathbb{N}\end{aligned}$$

Therefore, in roster form:

$$R = \{(2, 3), (4, 2), (6, 1)\}$$

The range of R consists of the second elements:

$$\text{Range}(R) = \{1, 2, 3\}$$

Part (ii): Prime numbers less than 5 are 2 and 3. Calculating a^3 for each prime a :

$$\begin{aligned}\text{For } a = 2 : \quad a^3 &= 2^3 = 8 \implies (2, 8) \in R \\ \text{For } a = 3 : \quad a^3 &= 3^3 = 27 \implies (3, 27) \in R\end{aligned}$$

In roster form:

$$R = \{(2, 8), (3, 27)\}$$

The range of R consists of the second elements:

$$\text{Range}(R) = \{8, 27\}$$

Final Answer

- (i) Range of $R = \{1, 2, 3\}$
- (ii) Range of $R = \{8, 27\}$

Question 3

Let A be any non-empty set. State true or false:

- (i) Identity relation on A is reflexive.
- (ii) Every reflexive relation on A is identity relation on A .

- (iii) Identity relation on A is symmetric.
- (iv) Identity relation on A is an equivalence relation.
- (v) Universal relation on A is an equivalence relation.

TYPE 3 Properties of Special Relations

Given: $A \neq \emptyset$

Concept / Formula Used: $I_A = \{(a, a) : a \in A\}$ $U = A \times A$

Solution:

- (i) **True.** The identity relation $I_A = \{(a, a) : a \in A\}$ contains (a, a) for every element $a \in A$. Hence, it satisfies the definition of reflexivity.
- (ii) **False.** A reflexive relation must contain all pairs (a, a) , but it can also contain other additional pairs (a, b) with $a \neq b$. An identity relation can contain *only* pairs of the form (a, a) .
- (iii) **True.** For every $(a, a) \in I_A$, reversing the order gives (a, a) , which is also in I_A . Hence, it is symmetric.
- (iv) **True.** I_A is reflexive, symmetric, and transitive, so it is an equivalence relation.
- (v) **True.** The universal relation $U = A \times A$ contains all possible ordered pairs of elements of A , so it is reflexive, symmetric, and transitive, making it an equivalence relation.

Final Answer

- (i) True (ii) False (iii) True (iv) True (v) True

Question 4

State the reason for the relation R in the set $\{1, 2, 3\}$ given by $R = \{(1, 2), (2, 1)\}$ not to be transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1)\}$

Concept / Formula Used: $R \quad (a, b) \in R \quad (b, c) \in R \implies (a, c) \in R$

Solution:

We examine the ordered pairs present in R :

$$(1, 2) \in R \quad \text{and} \quad (2, 1) \in R$$

For the relation R to be transitive, the pair formed by the first element of the first pair and the second element of the second pair, which is $(1, 1)$, must belong to R .

Checking the set R :

$$(1, 1) \notin R$$

Since $(1, 2) \in R$ and $(2, 1) \in R$ but $(1, 1) \notin R$, R is not transitive.

Final Answer

R is not transitive because $(1, 2) \in R$ and $(2, 1) \in R$, but $(1, 1) \notin R$.

Question 5

The relation R on $A = \{1, 2, 3\}$ is defined by $R = \{(1, 2), (2, 1)\}$. What kind of the relation R is?

TYPE 4 Verifying Relation Properties on Finite Sets

★ Likely Exam Question

Given: $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1)\}$

Concept / Formula Used: R is a relation on A if $(a, a) \in R$ for $a \in A$.
 $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$

Solution:

1. **Reflexivity:**

$$1 \in A \text{ but } (1, 1) \notin R$$

Since $(a, a) \notin R$ for $a = 1$, R is not reflexive.

2. **Symmetry:**

$$(1, 2) \in R \implies (2, 1) \in R$$

$$(2, 1) \in R \implies (1, 2) \in R$$

Since for every $(a, b) \in R$, $(b, a) \in R$, the relation R is symmetric.

3. **Transitivity:**

$$(1, 2) \in R \text{ and } (2, 1) \in R \text{ but } (1, 1) \notin R$$

Hence, R is not transitive.

Final Answer

R is a symmetric relation, but neither reflexive nor transitive.

Question 6

Write the smallest equivalence relation on A where $A = \{1, 2, 3\}$.

TYPE 3 Properties of Special Relations

★ Likely Exam Question

Given: $A = \{1, 2, 3\}$

Concept / Formula Used: $I_A = \{(a, a) : a \in A\}$

Solution:

For a relation R on $A = \{1, 2, 3\}$ to be an equivalence relation, it must be reflexive, symmetric, and transitive.

To satisfy reflexivity, R must contain:

$$(1, 1), (2, 2), \text{ and } (3, 3)$$

Consider the set:

$$R_0 = \{(1, 1), (2, 2), (3, 3)\}$$

Checking properties of R_0 : 1. Reflexive: $(a, a) \in R_0$ for all $a \in \{1, 2, 3\}$. 2. Symmetric: For each $(a, a) \in R_0$, reversing it gives $(a, a) \in R_0$. 3. Transitive: $(a, a) \in R_0$ and $(a, a) \in R_0 \implies (a, a) \in R_0$.

Thus, R_0 is an equivalence relation with the minimum possible number of elements (3 elements).

Final Answer

$$R = \{(1, 1), (2, 2), (3, 3)\}$$

Question 7

If $A = \{0, 1, 2, \dots, 9\}$ and the relation R on A is defined by $R = \{(x, y) : x, y \in A, y = 2x + 1\}$, then determine whether the relation R is (i) reflexive (ii) symmetric (iii) transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $y = 2x + 1$

Solution:

First, we list the elements of relation R by choosing $x \in A$ such that $y = 2x + 1 \in A$:

$$x = 0 \implies y = 2(0) + 1 = 1 \in A \implies (0, 1) \in R$$

$$x = 1 \implies y = 2(1) + 1 = 3 \in A \implies (1, 3) \in R$$

$$x = 2 \implies y = 2(2) + 1 = 5 \in A \implies (2, 5) \in R$$

$$x = 3 \implies y = 2(3) + 1 = 7 \in A \implies (3, 7) \in R$$

$$x = 4 \implies y = 2(4) + 1 = 9 \in A \implies (4, 9) \in R$$

$$x = 5 \implies y = 2(5) + 1 = 11 \notin A$$

Thus, in roster form:

$$R = \{(0, 1), (1, 3), (2, 5), (3, 7), (4, 9)\}$$

(i) **Reflexivity:**

$$0 \in A \quad \text{but} \quad (0, 0) \notin R$$

Hence, R is not reflexive.

(ii) **Symmetry:**

$$(0, 1) \in R \quad \text{but} \quad (1, 0) \notin R$$

Hence, R is not symmetric.

(iii) **Transitivity:**

$$(0, 1) \in R \quad \text{and} \quad (1, 3) \in R \quad \text{but} \quad (0, 3) \notin R$$

Hence, R is not transitive.

Final Answer

R is neither reflexive, nor symmetric, nor transitive.

Question 8

Is the relation R on the set $\mathbb{R}$ of real numbers defined by $R = \{(a, b) : a, b \in \mathbb{R}, 1 + ab \geq 0\}$ transitive? Justify your answer.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(a, b) \in \mathbb{R} \times \mathbb{R} : 1 + ab \geq 0\}$

Concept / Formula Used: $a, b, c \quad (a, b) \in R \quad (b, c) \in R \quad (a, c) \notin R$

Solution:

Let us test the relation with specific real numbers. Choose $a = 1$, $b = -\frac{1}{2}$, and $c = -3$.

1. Check if $(a, b) = \left(1, -\frac{1}{2}\right) \in R$:

$$\begin{aligned} 1 + ab &= 1 + (1) \left(-\frac{1}{2}\right) \\ &= 1 - \frac{1}{2} \\ &= \frac{1}{2} \geq 0 \end{aligned}$$

Thus, $\left(1, -\frac{1}{2}\right) \in R$.

2. Check if $(b, c) = \left(-\frac{1}{2}, -3\right) \in R$:

$$\begin{aligned} 1 + bc &= 1 + \left(-\frac{1}{2}\right)(-3) \\ &= 1 + \frac{3}{2} \\ &= \frac{5}{2} \geq 0 \end{aligned}$$

Thus, $\left(-\frac{1}{2}, -3\right) \in R$.

3. Check if $(a, c) = (1, -3) \in R$:

$$\begin{aligned} 1 + ac &= 1 + (1)(-3) \\ &= 1 - 3 \\ &= -2 < 0 \end{aligned}$$

Thus, $(1, -3) \notin R$.

Since $\left(1, -\frac{1}{2}\right) \in R$ and $\left(-\frac{1}{2}, -3\right) \in R$, but $(1, -3) \notin R$, the relation R fails transitivity.

Final Answer

No, R is not transitive because $\left(1, -\frac{1}{2}\right) \in R$ and $\left(-\frac{1}{2}, -3\right) \in R$, but $(1, -3) \notin R$.

Question 9

Is the relation R on the set $\mathbb{Q}$ of rational numbers defined by $R = \{(x, y) : x, y \in \mathbb{Q}, x < y^2\}$, symmetric? Justify your answer.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} : x < y^2\}$

Concept / Formula Used: $R \quad x, y \in \mathbb{Q} \quad (x, y) \in R \quad (y, x) \notin R$

Solution:

Let us choose a counterexample with rational numbers. Take $x = 1$ and $y = 5$.

1. Check if $(x, y) = (1, 5) \in R$:

$$y^2 = 5^2 = 25$$

Since $1 < 25$, the condition $x < y^2$ holds true. Thus, $(1, 5) \in R$.

2. Check if $(y, x) = (5, 1) \in R$:

$$x^2 = 1^2 = 1$$

Since $5 \not< 1$, the condition $y < x^2$ is false. Thus, $(5, 1) \notin R$.

Since $(1, 5) \in R$ but $(5, 1) \notin R$, the relation R is not symmetric.

Final Answer

No, R is not symmetric because $(1, 5) \in R$ as $1 < 5^2$, but $(5, 1) \notin R$ as $5 \not< 1^2$.

Question 10

If a relation R on the set $\{a, b, c\}$ is defined by $R = \{(b, b)\}$, then classify the relation.

TYPE 4 Verifying Relation Properties on Finite Sets

★ Likely Exam Question

Given: $A = \{a, b, c\}$, $R = \{(b, b)\}$

Concept / Formula Used: R

Solution:

1. Reflexivity:

$$a \in A \quad \text{and} \quad c \in A \quad \text{but} \quad (a, a) \notin R, (c, c) \notin R$$

Hence, R is not reflexive.

2. **Symmetry:** The only pair in R is (b, b) . Reversing it gives (b, b) , which is in R .

$$(b, b) \in R \implies (b, b) \in R$$

Hence, R is symmetric.

3. **Transitivity:** The only chain of pairs is $(b, b) \in R$ and $(b, b) \in R$, which gives $(b, b) \in R$. Hence, R is transitive.

Final Answer

R is symmetric and transitive, but not reflexive.

Question 11

If the relation R on the set $A = \{1, 2, 3\}$ is defined by $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (3, 2)\}$, then determine whether the relation R is (i) reflexive (ii) symmetric (iii) transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (3, 2)\}$

Concept / Formula Used: $A = \{1, 2, 3\}$

Solution:

(i) **Reflexivity:** For every $x \in A$:

$$(1, 1) \in R, \quad (2, 2) \in R, \quad (3, 3) \in R$$

Since $(x, x) \in R$ for all $x \in A$, R is reflexive.

(ii) **Symmetry:**

$$(2, 1) \in R \quad \text{but} \quad (1, 2) \notin R$$

Hence, R is not symmetric.

(iii) **Transitivity:** We observe:

$$(3, 2) \in R \quad \text{and} \quad (2, 1) \in R$$

For transitivity, we require $(3, 1) \in R$. Checking R :

$$(3, 1) \notin R$$

Hence, R is not transitive.

Final Answer

R is reflexive, but neither symmetric nor transitive.

Question 12

How many reflexive relations are possible on a set A of order 3?

TYPE 6 Counting Specific Types of Relations

Given: $n(A) = 3$

Concept / Formula Used: $n \quad 2^{n(n-1)} = 2^{n^2-n}$

Solution:

Given the number of elements in set A :

$$n = 3$$

The Cartesian product $A \times A$ contains $n^2 = 3^2 = 9$ ordered pairs.

For a relation R on A to be reflexive, it must contain all 3 diagonal pairs $(a_1, a_1), (a_2, a_2), (a_3, a_3)$.
The number of non-diagonal pairs is:

$$\begin{aligned} n^2 - n &= 9 - 3 \\ &= 6 \end{aligned}$$

Each non-diagonal pair can either be included or excluded from the relation independently.

$$\begin{aligned} \text{Total reflexive relations} &= 2^{n^2-n} \\ &= 2^6 \\ &= 64 \end{aligned}$$

Final Answer

64

Question 13

If R be the relation in the set $\{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}$, then determine whether

- R is reflexive and symmetric but not transitive.
- R is reflexive and transitive but not symmetric.
- R is symmetric and transitive but not reflexive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}$

Concept / Formula Used: R

Solution:

1. **Reflexivity:**

$$(1, 1) \in R, \quad (2, 2) \in R, \quad (3, 3) \in R, \quad (4, 4) \in R$$

Since $(a, a) \in R$ for every $a \in A$, R is reflexive.

2. **Symmetry:**

$$(1, 2) \in R \quad \text{but} \quad (2, 1) \notin R$$

Hence, R is not symmetric.

3. **Transitivity:** Checking non-trivial chains:

$$(1, 3) \in R \text{ and } (3, 2) \in R \implies (1, 2) \in R$$

$$(1, 1) \in R \text{ and } (1, 2) \in R \implies (1, 2) \in R$$

$$(1, 1) \in R \text{ and } (1, 3) \in R \implies (1, 3) \in R$$

$$(3, 3) \in R \text{ and } (3, 2) \in R \implies (3, 2) \in R$$

All pair combinations satisfy transitivity. Hence, R is transitive.

Thus, R is reflexive and transitive but not symmetric.

Final Answer

Option (ii) is correct: R is reflexive and transitive but not symmetric.

Question 14

Determine whether each of the following relations are reflexive, symmetric and transitive:

- (i) Relation R in the set $A = \{1, 2, 3, \dots, 10\}$ defined by $R = \{(x, y) : 2x - y = 0\}$
- (ii) Relation R in the set $\mathbb{Z}$ of all integers defined by $R = \{(x, y) : x - y \text{ is an integer}\}$
- (iii) Relation R on the set $A = \{1, 2, 3, 4, 5, 6\}$ defined by $R = \{(a, b) : b = a + 1\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: (i) $A = \{1, 2, \dots, 10\}, 2x - y = 0$, (ii) $x, y \in \mathbb{Z}, x - y \in \mathbb{Z}$, (iii) $A = \{1, \dots, 6\}, b = a + 1$

Solution:

Part (i): $A = \{1, 2, \dots, 10\}, 2x - y = 0 \implies y = 2x$.

$$R = \{(1, 2), (2, 4), (3, 6), (4, 8), (5, 10)\}$$

- Reflexive: $1 \in A$ but $(1, 1) \notin R$. Not reflexive. - Symmetric: $(1, 2) \in R$ but $(2, 1) \notin R$. Not symmetric. - Transitive: $(1, 2) \in R$ and $(2, 4) \in R$, but $(1, 4) \notin R$. Not transitive.

Part (ii): $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x - y \in \mathbb{Z}\}$. - Reflexive: For any $x \in \mathbb{Z}$:

$$x - x = 0 \in \mathbb{Z} \implies (x, x) \in R$$

Hence, R is reflexive. - Symmetric: Let $(x, y) \in R \implies x - y = k \in \mathbb{Z}$.

$$y - x = -(x - y) = -k \in \mathbb{Z} \implies (y, x) \in R$$

Hence, R is symmetric. - Transitive: Let $(x, y) \in R$ and $(y, z) \in R$. Then $x - y = k_1 \in \mathbb{Z}$ and $y - z = k_2 \in \mathbb{Z}$. Adding the equations:

$$\begin{aligned} (x - y) + (y - z) &= k_1 + k_2 \\ x - z &= k_1 + k_2 \in \mathbb{Z} \implies (x, z) \in R \end{aligned}$$

Hence, R is transitive.

Part (iii): $A = \{1, 2, 3, 4, 5, 6\}, b = a + 1$.

$$R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$$

- Reflexive: $(1, 1) \notin R$. Not reflexive. - Symmetric: $(1, 2) \in R$ but $(2, 1) \notin R$. Not symmetric. - Transitive: $(1, 2) \in R$ and $(2, 3) \in R$, but $(1, 3) \notin R$. Not transitive.

Final Answer

- (i) Neither reflexive, nor symmetric, nor transitive.
- (ii) Equivalence relation (reflexive, symmetric, and transitive).
- (iii) Neither reflexive, nor symmetric, nor transitive.

Question 15

If the relation R in the set A , where $A = \{1, 2, 3, 4, 5, 6\}$, is defined by $R = \{(x, y) : y \text{ is divisible by } x\}$, then express R in the roster form. Also determine whether the relation R is (i) reflexive (ii) symmetric (iii) transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3, 4, 5, 6\}$, $R = \{(x, y) : y \text{ is divisible by } x\}$

Concept / Formula Used: $y = kx \quad k \in \mathbb{Z}^+$

Solution:

1. **Roster Form:** Finding all pairs (x, y) such that y is divisible by x :

$$x = 1 \implies y \in \{1, 2, 3, 4, 5, 6\} \implies (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)$$

$$x = 2 \implies y \in \{2, 4, 6\} \implies (2, 2), (2, 4), (2, 6)$$

$$x = 3 \implies y \in \{3, 6\} \implies (3, 3), (3, 6)$$

$$x = 4 \implies y \in \{4\} \implies (4, 4)$$

$$x = 5 \implies y \in \{5\} \implies (5, 5)$$

$$x = 6 \implies y \in \{6\} \implies (6, 6)$$

Combining all pairs:

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

2. **Properties of R :** (i) **Reflexivity:** For every $x \in A$, x is divisible by x (since $x = 1 \cdot x$). Thus, $(x, x) \in R$ for all $x \in A$. R is reflexive.

(ii) **Symmetry:** $(1, 2) \in R$ since 2 is divisible by 1. But 1 is not divisible by 2, so $(2, 1) \notin R$. R is not symmetric.

(iii) **Transitivity:** Let $(x, y) \in R$ and $(y, z) \in R$. Then:

$$\begin{aligned} y &= k_1x \quad \text{for some } k_1 \in \mathbb{Z}^+ \\ z &= k_2y \quad \text{for some } k_2 \in \mathbb{Z}^+ \\ \implies z &= k_2(k_1x) = (k_1k_2)x \end{aligned}$$

Since $k_1k_2 \in \mathbb{Z}^+$, z is divisible by x , so $(x, z) \in R$. R is transitive.

Final Answer

Roster form: $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$
 R is reflexive and transitive, but not symmetric.

Question 16

If R is the relation defined on the set of natural numbers $\mathbb{N}$ as follows: $R = \{(x, y) : x, y \in \mathbb{N}, 2x + y = 41\}$, find the domain and the range of the relation R . Determine whether the relation is reflexive, symmetric and transitive.

TYPE 2 Domain and Range of a Relation

Given: $R = \{(x, y) \in \mathbb{N} \times \mathbb{N} : 2x + y = 41\}$

Solution:

Given equation:

$$\begin{aligned} 2x + y &= 41 \\ y &= 41 - 2x \end{aligned}$$

Since $x, y \in \mathbb{N} = \{1, 2, 3, \dots\}$, $y \geq 1$:

$$\begin{aligned} 41 - 2x &\geq 1 \\ 2x &\leq 40 \\ x &\leq 20 \end{aligned}$$

Calculating values of y for $x \in \{1, 2, \dots, 20\}$:

$$\begin{aligned} x = 1 &\implies y = 41 - 2(1) = 39 \\ x = 2 &\implies y = 41 - 2(2) = 37 \\ &\vdots \\ x = 20 &\implies y = 41 - 2(20) = 1 \end{aligned}$$

$$\text{Domain of } R = \{1, 2, 3, \dots, 20\}$$

$$\text{Range of } R = \{1, 3, 5, \dots, 39\}$$

Checking relation properties: 1. **Reflexivity:** Take $x = 1$.

$$2(1) + 1 = 3 \neq 41 \implies (1, 1) \notin R$$

Hence, R is not reflexive.

2. **Symmetry:** Take $(1, 39) \in R$ (since $2(1) + 39 = 41$).

$$2(39) + 1 = 78 + 1 = 79 \neq 41 \implies (39, 1) \notin R$$

Hence, R is not symmetric.

3. **Transitivity:** Take $(11, 19) \in R$ (since $2(11) + 19 = 41$) and $(19, 3) \in R$ (since $2(19) + 3 = 41$).

$$2(11) + 3 = 22 + 3 = 25 \neq 41 \implies (11, 3) \notin R$$

Hence, R is not transitive.

Final Answer

Domain = $\{1, 2, 3, \dots, 20\}$, Range = $\{1, 3, 5, \dots, 39\}$.

R is neither reflexive, nor symmetric, nor transitive.

Question 17

Determine whether each of the following relations in the set A of human beings in a city at a particular time are reflexive, symmetric and transitive:

- (i) $R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$
- (ii) $R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$
- (iii) $R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$
- (iv) $R = \{(x, y) : x \text{ and } y \text{ live within 2 kilometres}\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: A = set of human beings in a city

Solution:

Part (i): $R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$ - Reflexive: Every person x works at the same place as x . $(x, x) \in R$. - Symmetric: If x and y work at the same place, then y and x work at the same place. $(x, y) \in R \implies (y, x) \in R$. - Transitive: If x and y work at the same place, and y and z work at the same place, then x and z work at the same place. $(x, y), (y, z) \in R \implies (x, z) \in R$. Thus, R is reflexive, symmetric, and transitive (an equivalence relation).

Part (ii): $R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$ - Following the exact same logic as (i), R is reflexive, symmetric, and transitive (an equivalence relation).

Part (iii): $R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$ - Reflexive: A person x cannot be 7 cm taller than himself/herself. $(x, x) \notin R$. - Symmetric: If x is 7 cm taller than y , then y is 7 cm shorter than x . $(y, x) \notin R$. - Transitive: If x is 7 cm taller than y and y is 7 cm taller than z , then x is 14 cm taller than z . $(x, z) \notin R$. Thus, R is neither reflexive, nor symmetric, nor transitive.

Part (iv): $R = \{(x, y) : x \text{ and } y \text{ live within 2 kilometres}\}$ - Reflexive: The distance between x and x is $0 \text{ km} \leq 2 \text{ km}$. $(x, x) \in R$. - Symmetric: If distance between x and y is $\leq 2 \text{ km}$, distance between y and x is $\leq 2 \text{ km}$. $(x, y) \in R \implies (y, x) \in R$. - Transitive: Let x live at point 0 km, y at 1.5 km, and z at 3 km.

$$\text{dist}(x, y) = 1.5 \text{ km} \leq 2 \text{ km} \implies (x, y) \in R$$

$$\text{dist}(y, z) = 1.5 \text{ km} \leq 2 \text{ km} \implies (y, z) \in R$$

$$\text{dist}(x, z) = 3.0 \text{ km} > 2 \text{ km} \implies (x, z) \notin R$$

Thus, R is not transitive.

Final Answer

- (i) Equivalence relation (reflexive, symmetric, transitive).
- (ii) Equivalence relation (reflexive, symmetric, transitive).
- (iii) Neither reflexive, nor symmetric, nor transitive.
- (iv) Reflexive and symmetric, but not transitive.

Question 18

Determine whether each of the following relations in the set A of students at a particular time are symmetric and transitive:

- (i) $R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are honest}\}$
- (ii) $R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are obedient}\}$

(iii) $R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are hardworking}\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: A = set of students at a particular time

Concept / Formula Used: Let P denote the attribute in question (honest / obedient / hardworking). If $H \subseteq A$ is the subset of students possessing P , then $R = \{(x, y) : x, y \in H\} = H \times H$

Solution:

Let $H \subseteq A$ be the subset of students who possess the given attribute (honest, obedient, or hardworking, according to the part). Then

$$(x, y) \in R \iff x \in H \text{ and } y \in H$$

Symmetry: Let $(x, y) \in R$.

$$\begin{aligned} (x, y) \in R &\implies x \in H \text{ and } y \in H \\ &\implies y \in H \text{ and } x \in H \\ &\implies (y, x) \in R \end{aligned}$$

Hence, R is symmetric in all three cases (i), (ii), and (iii).

Transitivity: Let $(x, y) \in R$ and $(y, z) \in R$.

$$\begin{aligned} (x, y) \in R \text{ and } (y, z) \in R &\implies (x \in H \text{ and } y \in H) \text{ and } (y \in H \text{ and } z \in H) \\ &\implies x \in H \text{ and } z \in H \\ &\implies (x, z) \in R \end{aligned}$$

Hence, R is transitive in all three cases (i), (ii), and (iii).

Final Answer

The relations in (i), (ii) and (iii) are all symmetric and transitive.

Common Mistake: Testing symmetry/transitivity with a concrete numbered example (e.g. students labelled 1, 2, 3) is not valid here – A is the set of *all* students, and R has the special structure $H \times H$, which forces both symmetry and transitivity to hold in general. A single counter-example only disproves a property; it cannot be used to prove one holds only sometimes when the underlying structure guarantees it always.

Question 19

Show that the relation R in the set $A = \{1, 2, 3\}$ given by $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$ is reflexive but neither symmetric nor transitive. (NCERT)

TYPE 4 Verifying Relation Properties on Finite Sets

To Prove: R is reflexive but neither symmetric nor transitive

Solution:

Reflexivity: For R to be reflexive, $(a, a) \in R$ for every $a \in A = \{1, 2, 3\}$.

$$(1, 1) \in R, \quad (2, 2) \in R, \quad (3, 3) \in R$$

All three pairs are present, so R is reflexive.

Symmetry: Consider the pair $(1, 2) \in R$.

$$(1, 2) \in R \quad \text{but} \quad (2, 1) \notin R$$

Since $(1, 2) \in R$ does not imply $(2, 1) \in R$, R is not symmetric.

Transitivity: Consider the pairs $(1, 2) \in R$ and $(2, 3) \in R$.

$$(1, 2) \in R \text{ and } (2, 3) \in R \quad \text{but} \quad (1, 3) \notin R$$

Since $(1, 2) \in R$ and $(2, 3) \in R$ does not imply $(1, 3) \in R$, R is not transitive.

Hence Proved

R is reflexive but neither symmetric nor transitive.

Question 20

Explain why $x + 4y = 12$, $x, y \in N$ is NOT a symmetric relation.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $R = \{(x, y) : x, y \in N, x + 4y = 12\}$

Solution:

List the pairs (x, y) satisfying $x + 4y = 12$ with $x, y \in N$:

$$y = 1 \implies x = 8 \implies (8, 1) \in R$$

$$y = 2 \implies x = 4 \implies (4, 2) \in R$$

(For $y \geq 3$, $x = 12 - 4y \leq 0$, which is not a natural number, so no further pairs exist.)

Consider the pair $(4, 2) \in R$. For R to be symmetric, $(2, 4)$ must also belong to R . Check:

$$2 + 4(4) = 2 + 16 = 18 \neq 12 \implies (2, 4) \notin R$$

Final Answer

$(4, 2) \in R$ but $(2, 4) \notin R$, so R is not symmetric.

Question 21

Let R be a relation defined over N , where N is the set of natural numbers, defined as “ $m R n$ if and only if m is a multiple of n , where $m, n \in N$.” Find whether R is reflexive, symmetric and transitive or not.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $R = \{(m, n) : m, n \in N, m \text{ is a multiple of } n\}$

Solution:

Reflexivity: Every natural number m is a multiple of itself, since $m = 1 \cdot m$.

$$\implies (m, m) \in R \text{ for all } m \in N$$

Hence, R is reflexive.

Symmetry: Take $m = 4$, $n = 2$. Since $4 = 2 \times 2$, 4 is a multiple of 2, so $(4, 2) \in R$. Check whether $(2, 4) \in R$, i.e. whether 2 is a multiple of 4:

$$2 \neq 4k \text{ for any } k \in N \implies (2, 4) \notin R$$

Hence, R is not symmetric.

Transitivity: Let $(m, n) \in R$ and $(n, p) \in R$, i.e. m is a multiple of n and n is a multiple of p .

$$\begin{aligned} m &= kn \text{ for some } k \in N \quad \text{and} \quad n = lp \text{ for some } l \in N \\ \implies m &= k(lp) = (kl)p, \quad kl \in N \end{aligned}$$

So m is a multiple of p , i.e. $(m, p) \in R$. Hence, R is transitive.

Final Answer

R is reflexive and transitive but not symmetric.

Question 22

Let R be the relation on W (set of whole numbers) defined by

$$R = \{(x, y) : x, y \in W, 3x + 2y = 12\}$$

Check the relation R for reflexivity, symmetricity and transitivity.

TYPE 2 Domain and Range of a Relation

Given: $R = \{(x, y) : x, y \in W, 3x + 2y = 12\}$

Solution:

First, list all pairs (x, y) with $x, y \in W = \{0, 1, 2, 3, \dots\}$ satisfying $3x + 2y = 12$:

$$\begin{aligned} x = 0 &\implies 2y = 12 \implies y = 6 \implies (0, 6) \in R \\ x = 2 &\implies 6 + 2y = 12 \implies y = 3 \implies (2, 3) \in R \\ x = 4 &\implies 12 + 2y = 12 \implies y = 0 \implies (4, 0) \in R \end{aligned}$$

(For $x = 1$ or $x = 3$, y works out to a non-integer, so these are excluded.)

$$R = \{(0, 6), (2, 3), (4, 0)\}$$

Reflexivity: Check if $(0, 0) \in R$: $3(0) + 2(0) = 0 \neq 12$. So $(0, 0) \notin R$. Hence, R is not reflexive.

Symmetry: Consider $(0, 6) \in R$. Check whether $(6, 0) \in R$:

$$3(6) + 2(0) = 18 \neq 12 \implies (6, 0) \notin R$$

Hence, R is not symmetric.

Transitivity: Since no two pairs in R share a common middle element in the required chain form (e.g. $(4, 0)$ and $(0, 6)$ are both in R), check whether $(4, 6) \in R$:

$$3(4) + 2(6) = 12 + 12 = 24 \neq 12 \implies (4, 6) \notin R$$

Hence, R is not transitive.

Final Answer

R is neither reflexive, nor symmetric, nor transitive.

Common Mistake: The printed answer key for this question states “symmetric,” which does not hold for the actual relation $R = \{(0, 6), (2, 3), (4, 0)\}$ – verify by direct enumeration, as shown above, rather than accepting a stated answer without checking it against the listed ordered pairs.

Question 23

Let R be the relation defined on $\mathbb{Q}_0$ (set of non-zero rational numbers) defined by $R = \left\{ (a, b) : a, b \in \mathbb{Q}_0, b = \frac{1}{a} \right\}$. Check the relation R for reflexivity, symmetricity and transitivity.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $\mathbb{Q}_0 = \mathbb{Q} \setminus \{0\}$, $R = \{(a, b) \in \mathbb{Q}_0 \times \mathbb{Q}_0 : b = \frac{1}{a}\}$

Concept / Formula Used: $b = \frac{1}{a} \iff ab = 1$

Solution:

1. **Reflexivity:** For $(a, a) \in R$, we require:

$$\begin{aligned} a &= \frac{1}{a} \\ a^2 &= 1 \\ a &= \pm 1 \end{aligned}$$

For any $a \in \mathbb{Q}_0$ such that $a \neq \pm 1$ (for instance, $a = 2$):

$$2 \neq \frac{1}{2} \implies (2, 2) \notin R$$

Hence, R is not reflexive.

2. **Symmetry:** Let $(a, b) \in R$.

$$(a, b) \in R \implies b = \frac{1}{a}$$

Taking the reciprocal on both sides (since $a, b \neq 0$):

$$a = \frac{1}{b} \implies (b, a) \in R$$

Hence, R is symmetric.

3. **Transitivity:** Let $(a, b) \in R$ and $(b, c) \in R$.

$$b = \frac{1}{a} \quad \text{and} \quad c = \frac{1}{b}$$

Substituting $b = \frac{1}{a}$ into $c = \frac{1}{b}$:

$$c = \frac{1}{\frac{1}{a}} = a$$

For $(a, c) = (a, a) \in R$, we would require $a = \frac{1}{a} \implies a = \pm 1$. Counterexample: Take $a = 2$, $b = \frac{1}{2}$, $c = 2$:

$$\left(2, \frac{1}{2}\right) \in R \quad \text{and} \quad \left(\frac{1}{2}, 2\right) \in R \quad \text{but} \quad (2, 2) \notin R$$

Hence, R is not transitive.

Final Answer

R is symmetric, but neither reflexive nor transitive.

Question 24

A relation R is defined on a set of real numbers $\mathbb{R}$ as

$$R = \{(x, y) : xy \text{ is an irrational number}\}.$$

Check whether R is reflexive, symmetric and transitive.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : xy \in \mathbb{R} \setminus \mathbb{Q}\}$

Concept / Formula Used: $R \quad A \quad (x, x) \in R \quad x \in A \quad (x, y) \in R \implies (y, x) \in R$
 $(x, y) \in R \quad (y, z) \in R \implies (x, z) \in R$

Solution:

Reflexivity: Let $x = 0 \in \mathbb{R}$. We evaluate $x \cdot x$:

$$\begin{aligned} x \cdot x &= 0 \cdot 0 \\ &= 0 \end{aligned}$$

Since $0 \in \mathbb{Q}$ (a rational number), $x \cdot x$ is not an irrational number. Thus, $(0, 0) \notin R$. Therefore, R is not reflexive.

Symmetry: Let $(x, y) \in R$. By definition of relation R :

$$xy \in \mathbb{R} \setminus \mathbb{Q}$$

Using the commutative property of real number multiplication:

$$yx = xy$$

Therefore, yx is also an irrational number. This implies $(y, x) \in R$. Thus, R is symmetric.

Transitivity: Consider $x = \sqrt{2} \in \mathbb{R}$, $y = \sqrt{3} \in \mathbb{R}$, and $z = \sqrt{2} \in \mathbb{R}$. Evaluating xy and yz :

$$\begin{aligned} xy &= \sqrt{2} \cdot \sqrt{3} = \sqrt{6} \quad (\text{irrational}) \implies (x, y) \in R \\ yz &= \sqrt{3} \cdot \sqrt{2} = \sqrt{6} \quad (\text{irrational}) \implies (y, z) \in R \end{aligned}$$

Now, examine xz :

$$\begin{aligned}xz &= \sqrt{2} \cdot \sqrt{2} \\ &= 2 \quad (\text{rational})\end{aligned}$$

Since $2 \in \mathbb{Q}$, $(x, z) \notin R$. Thus, R is not transitive.

Final Answer

R is symmetric, but neither reflexive nor transitive.

Question 25

Check whether the relation R on the set $\mathbb{N}$ of natural numbers given by $R = \{(a, b) : a, b \in \mathbb{N}, a \text{ is a divisor of } b\}$ is reflexive, symmetric or transitive. Also determine whether R is an equivalence relation.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(a, b) : a, b \in \mathbb{N}, a \mid b\}$

Solution:

Reflexivity: Let $a \in \mathbb{N}$. Every natural number divides itself, since:

$$a = 1 \cdot a$$

Thus, $a \mid a$ for all $a \in \mathbb{N}$, which means $(a, a) \in R$. Therefore, R is reflexive.

Symmetry: Consider $a = 2 \in \mathbb{N}$ and $b = 4 \in \mathbb{N}$. Since 2 divides 4, we have $(2, 4) \in R$. However, 4 does not divide 2, so:

$$(4, 2) \notin R$$

Therefore, R is not symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$ for $a, b, c \in \mathbb{N}$. Then, there exist positive integers $k, m \in \mathbb{N}$ such that:

$$\begin{aligned}b &= ka \\ c &= mb\end{aligned}$$

Substituting $b = ka$ into $c = mb$:

$$\begin{aligned}c &= m(ka) \\ c &= (mk)a\end{aligned}$$

Since $mk \in \mathbb{N}$, a divides c , which means $(a, c) \in R$. Therefore, R is transitive.

Equivalence Relation Determination: Since R is not symmetric, it is not an equivalence relation.

Final Answer

R is reflexive and transitive, but not symmetric. Hence, R is not an equivalence relation.

Question 26

Let R be the relation on $\mathbb{R}$ (set of real numbers) defined by $R = \{(a, b) : a, b \in \mathbb{R}, a^2 + b^2 = 1\}$. Show that R is symmetric but neither reflexive nor transitive.

TYPE 5 Verifying Relation Properties on Infinite Sets

To Prove: R is symmetric, but R is neither reflexive nor transitive.

Concept / Formula Used: $R \quad (a, b) \in R \implies (b, a) \in R \quad R \quad (a, a) \in R \quad a \in \mathbb{R} \quad (a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$

Solution:

Symmetry: Let $(a, b) \in R$. Then,

$$a^2 + b^2 = 1$$

By commutativity of addition in $\mathbb{R}$:

$$b^2 + a^2 = 1$$

This implies $(b, a) \in R$. Hence, R is symmetric.

Reflexivity: Let $a = 2 \in \mathbb{R}$. We check if $(2, 2) \in R$:

$$\begin{aligned} 2^2 + 2^2 &= 4 + 4 \\ &= 8 \neq 1 \end{aligned}$$

Since $a^2 + a^2 \neq 1$ for $a = 2$, $(2, 2) \notin R$. Hence, R is not reflexive.

Transitivity: Consider $a = 1$, $b = 0$, and $c = 1$. Checking $(a, b) \in R$:

$$1^2 + 0^2 = 1 + 0 = 1 \implies (1, 0) \in R$$

Checking $(b, c) \in R$:

$$0^2 + 1^2 = 0 + 1 = 1 \implies (0, 1) \in R$$

Now checking $(a, c) \in R$:

$$1^2 + 1^2 = 1 + 1 = 2 \neq 1 \implies (1, 1) \notin R$$

Hence, R is not transitive.

Hence Proved

R is symmetric, but neither reflexive nor transitive.

Question 27

Determine whether each of the following relations are reflexive, symmetric and transitive:

- (i) $R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$
- (ii) $R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: Part (i): $R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$; Part (ii): $R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$

Concept / Formula Used: $(a, a) \in R \quad (a, b) \in R \implies (b, a) \in R \quad (a, b) \in R \quad (b, c) \in R \implies (a, c) \in R$

Solution:

Part (i): $R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$

Reflexivity: For any $a \in \mathbb{N}$, the statement $a < a$ is false. Thus, $(a, a) \notin R$. Hence, R is not reflexive.

Symmetry: Let $1, 2 \in \mathbb{N}$. Since $1 < 2$, we have $(1, 2) \in R$. However, $2 < 1$ is false, so $(2, 1) \notin R$. Hence, R is not symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$a < b \quad \text{and} \quad b < c$$

By the transitive property of order in natural numbers:

$$a < c$$

Thus, $(a, c) \in R$. Hence, R is transitive.

Part (ii): $R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$

Reflexivity: For any $a \in \mathbb{R}$, $a \leq a$ is true (since $a = a$). Thus, $(a, a) \in R$ for all $a \in \mathbb{R}$. Hence, R is reflexive.

Symmetry: Let $1, 2 \in \mathbb{R}$. Since $1 \leq 2$, we have $(1, 2) \in R$. However, $2 \leq 1$ is false, so $(2, 1) \notin R$. Hence, R is not symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$a \leq b \quad \text{and} \quad b \leq c$$

By the transitive property of inequalities:

$$a \leq c$$

Thus, $(a, c) \in R$. Hence, R is transitive.

Final Answer

- (i) R is transitive, but neither reflexive nor symmetric.
- (ii) R is reflexive and transitive, but not symmetric.

Question 28

Let T be the set of all triangles drawn in a plane with R as a relation in T given by $R = \{(T_1, T_2) : T_1 \cong T_2\}$. Show that R is an equivalence relation.

TYPE 7 Proving Equivalence Relations

To Prove: $R = \{(T_1, T_2) : T_1 \cong T_2\}$ is an equivalence relation.

Solution:

Reflexivity: Let $T_1 \in T$. Every triangle is congruent to itself.

$$T_1 \cong T_1$$

Thus, $(T_1, T_1) \in R$ for all $T_1 \in T$. Hence, R is reflexive.

Symmetry: Let $(T_1, T_2) \in R$. By definition of relation R :

$$T_1 \cong T_2$$

If triangle T_1 is congruent to triangle T_2 , then triangle T_2 is congruent to triangle T_1 :

$$T_2 \cong T_1$$

Thus, $(T_2, T_1) \in R$. Hence, R is symmetric.

Transitivity: Let $(T_1, T_2) \in R$ and $(T_2, T_3) \in R$. Then,

$$T_1 \cong T_2$$

$$T_2 \cong T_3$$

If T_1 is congruent to T_2 , and T_2 is congruent to T_3 , then T_1 is congruent to T_3 :

$$T_1 \cong T_3$$

Thus, $(T_1, T_3) \in R$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation on T .

Hence Proved

R is an equivalence relation on the set of triangles T .

Question 29

Show that the relation R in the set A of all books in a library of a school, given by $R = \{(x, y) : x \text{ and } y \text{ have same number of pages}\}$, is an equivalence relation.

TYPE 7 Proving Equivalence Relations

To Prove: R is an equivalence relation on the set A of all books.

Concept / Formula Used: R

Solution:

Reflexivity: Let $x \in A$. Book x has the same number of pages as itself.

$$(x, x) \in R \quad \forall x \in A$$

Hence, R is reflexive.

Symmetry: Let $(x, y) \in R$. Then, book x and book y have the same number of pages. This implies book y and book x have the same number of pages.

$$(y, x) \in R$$

Hence, R is symmetric.

Transitivity: Let $(x, y) \in R$ and $(y, z) \in R$. Then,

$$\text{Number of pages in } x = \text{Number of pages in } y$$

$$\text{Number of pages in } y = \text{Number of pages in } z$$

Equating the expressions:

$$\text{Number of pages in } x = \text{Number of pages in } z$$

Thus, $(x, z) \in R$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Hence Proved

R is an equivalence relation on set A .

Question 30

Show that the relation R on the set $\mathbb{Z}$ of all integers defined by $R = \{(a, b) : a - b \text{ is divisible by } 3, a, b \in \mathbb{Z}\}$ is an equivalence relation.

TYPE 7 Proving Equivalence Relations

To Prove: $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : 3 \mid (a - b)\}$ is an equivalence relation.

Concept / Formula Used: $3 \mid (a - b) \iff a - b = 3k \quad k \in \mathbb{Z}$

Solution:

Reflexivity: Let $a \in \mathbb{Z}$.

$$a - a = 0$$

Since $0 = 3 \times 0$, $a - a$ is divisible by 3. Thus, $(a, a) \in R$ for all $a \in \mathbb{Z}$. Hence, R is reflexive.

Symmetry: Let $(a, b) \in R$. Then,

$$a - b = 3k \quad \text{for some } k \in \mathbb{Z}$$

Multiplying both sides by -1 :

$$\begin{aligned} -(a - b) &= -3k \\ b - a &= 3(-k) \end{aligned}$$

Since $-k \in \mathbb{Z}$, $b - a$ is divisible by 3. Thus, $(b, a) \in R$. Hence, R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$\begin{aligned} a - b &= 3k \quad \text{for some } k \in \mathbb{Z} \\ b - c &= 3m \quad \text{for some } m \in \mathbb{Z} \end{aligned}$$

Adding the two equations:

$$\begin{aligned} (a - b) + (b - c) &= 3k + 3m \\ a - c &= 3(k + m) \end{aligned}$$

Since $k + m \in \mathbb{Z}$, $a - c$ is divisible by 3. Thus, $(a, c) \in R$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Hence Proved

R is an equivalence relation on $\mathbb{Z}$.

Question 31

Show that the relation R on the set $\mathbb{Z}$ of all integers defined by $R = \{(a, b) : a - b \text{ is divisible by } 2, a, b \in \mathbb{Z}\}$ is an equivalence relation. Also, write the equivalence class of 0.

TYPE 8 Equivalence Classes and Related Elements**★ Likely Exam Question**

Given: $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : 2 \mid (a - b)\}$

Concept / Formula Used: $x \quad [x] = \{y \in \mathbb{Z} : (y, x) \in R\}$

Solution:

Reflexivity: Let $a \in \mathbb{Z}$.

$$a - a = 0 = 2 \times 0$$

Since 0 is divisible by 2, $(a, a) \in R$ for all $a \in \mathbb{Z}$. Hence, R is reflexive.

Symmetry: Let $(a, b) \in R$. Then,

$$a - b = 2k \quad \text{for some } k \in \mathbb{Z}$$

Rearranging:

$$\begin{aligned} b - a &= -(a - b) \\ &= -2k \\ &= 2(-k) \end{aligned}$$

Since $-k \in \mathbb{Z}$, $b - a$ is divisible by 2. Thus, $(b, a) \in R$. Hence, R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$\begin{aligned} a - b &= 2k \quad \text{for some } k \in \mathbb{Z} \\ b - c &= 2m \quad \text{for some } m \in \mathbb{Z} \end{aligned}$$

Adding the two equations:

$$\begin{aligned} (a - b) + (b - c) &= 2k + 2m \\ a - c &= 2(k + m) \end{aligned}$$

Since $k + m \in \mathbb{Z}$, $a - c$ is divisible by 2. Thus, $(a, c) \in R$. Hence, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Finding the Equivalence Class of 0: By definition:

$$\begin{aligned} [0] &= \{a \in \mathbb{Z} : (a, 0) \in R\} \\ &= \{a \in \mathbb{Z} : a - 0 \text{ is divisible by } 2\} \\ &= \{a \in \mathbb{Z} : a = 2k, k \in \mathbb{Z}\} \\ &= \{\dots, -4, -2, 0, 2, 4, \dots\} \end{aligned}$$

Final Answer

R is an equivalence relation on $\mathbb{Z}$, and the equivalence class of 0 is $[0] = \{\dots, -4, -2, 0, 2, 4, \dots\}$ (the set of all even integers).

Question 32

Prove that the relation R in the set $A = \{5, 6, 7, 8, 9\}$, given by $R = \{(a, b) : |a - b| \text{ is divisible by } 2\}$, is an equivalence relation. Find all elements related to element 6.

★ Likely Exam Question

Given: $A = \{5, 6, 7, 8, 9\}$ and $R = \{(a, b) \in A \times A : |a - b| \text{ is even}\}$

Concept / Formula Used: $x \in A \quad \{y \in A : (y, x) \in R\}$

Solution:

Reflexivity: Let $a \in A$.

$$\begin{aligned} |a - a| &= |0| \\ &= 0 \end{aligned}$$

Since 0 is divisible by 2, $(a, a) \in R$ for all $a \in A$. Hence, R is reflexive.

Symmetry: Let $(a, b) \in R$. Then,

$$|a - b| = 2k \quad \text{for some } k \in \mathbb{W}$$

Since $|b - a| = |-(a - b)| = |a - b|$:

$$|b - a| = 2k$$

Thus, $(b, a) \in R$. Hence, R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then $a - b$ is even and $b - c$ is even. Let $a - b = 2k$ and $b - c = 2m$ where $k, m \in \mathbb{Z}$. Adding these equations:

$$\begin{aligned} (a - b) + (b - c) &= 2k + 2m \\ a - c &= 2(k + m) \end{aligned}$$

Thus, $|a - c| = |2(k + m)| = 2|k + m|$, which is divisible by 2. Hence, $(a, c) \in R$. Therefore, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Elements related to element 6: Let $x \in A$ be related to 6. Then $|x - 6|$ must be divisible by 2. Testing each element of $A = \{5, 6, 7, 8, 9\}$:

$$\begin{aligned} x = 5 &\implies |5 - 6| = 1 \quad (\text{not divisible by } 2) \\ x = 6 &\implies |6 - 6| = 0 \quad (\text{divisible by } 2) \\ x = 7 &\implies |7 - 6| = 1 \quad (\text{not divisible by } 2) \\ x = 8 &\implies |8 - 6| = 2 \quad (\text{divisible by } 2) \\ x = 9 &\implies |9 - 6| = 3 \quad (\text{not divisible by } 2) \end{aligned}$$

Thus, the elements related to 6 are 6 and 8.

Final Answer

R is an equivalence relation, and the set of elements related to 6 is $\{6, 8\}$.

Question 33

Give examples of relations which are

- symmetric but neither reflexive nor transitive.
- transitive but neither reflexive nor symmetric.

- (iii) symmetric and transitive but not reflexive.
- (iv) symmetric and reflexive but not transitive.
- (v) reflexive and transitive but not symmetric.

TYPE 9 Constructing Examples of Relations**Solution:**

Part (i): Symmetric but neither reflexive nor transitive. Let L be the set of all lines in a plane, and relation R on L defined by:

$$R = \{(l_1, l_2) : l_1 \text{ is perpendicular to } l_2\}$$

- Reflexive: A line l_1 cannot be perpendicular to itself, so $(l_1, l_1) \notin R$. - Symmetric: If $l_1 \perp l_2$, then $l_2 \perp l_1$, so $(l_1, l_2) \in R \implies (l_2, l_1) \in R$. - Transitive: If $l_1 \perp l_2$ and $l_2 \perp l_3$, then $l_1 \parallel l_3$, so $(l_1, l_3) \notin R$.

Part (ii): Transitive but neither reflexive nor symmetric. Let R be defined on $\mathbb{N}$ by:

$$R = \{(a, b) : a < b\}$$

- Reflexive: $a < a$ is false, so $(a, a) \notin R$. - Symmetric: $1 < 2 \implies (1, 2) \in R$, but $2 < 1$ is false, so $(2, 1) \notin R$. - Transitive: $a < b$ and $b < c \implies a < c$, so $(a, c) \in R$.

Part (iii): Symmetric and transitive but not reflexive. Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.

- Reflexive: $(3, 3) \notin R$, so R is not reflexive on A . - Symmetric: For every $(a, b) \in R$, $(b, a) \in R$. - Transitive: If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

Part (iv): Symmetric and reflexive but not transitive. Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$.

- Reflexive: $(1, 1), (2, 2), (3, 3) \in R$. - Symmetric: All pairs are symmetric. - Transitive: $(1, 2) \in R$ and $(2, 3) \in R$, but $(1, 3) \notin R$.

Part (v): Reflexive and transitive but not symmetric. Let R be defined on $\mathbb{R}$ by:

$$R = \{(a, b) : a \leq b\}$$

- Reflexive: $a \leq a$ is true for all $a \in \mathbb{R}$. - Symmetric: $1 \leq 2 \implies (1, 2) \in R$, but $2 \leq 1$ is false $\implies (2, 1) \notin R$. - Transitive: $a \leq b$ and $b \leq c \implies a \leq c$.

Final Answer

- (i) $R = \{(l_1, l_2) : l_1 \perp l_2\}$ on lines in a plane.
- (ii) $R = \{(a, b) : a < b\}$ on $\mathbb{N}$.
- (iii) $R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ on $A = \{1, 2, 3\}$.
- (iv) $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$ on $A = \{1, 2, 3\}$.
- (v) $R = \{(a, b) : a \leq b\}$ on $\mathbb{R}$.

Question 34

Give an example of a relation R on $A = \{a, b, c\}$ which is

- (i) neither reflexive nor symmetric but transitive.
- (ii) neither symmetric nor transitive but reflexive.
- (iii) neither transitive nor reflexive but symmetric.

TYPE 9 Constructing Examples of Relations

Given: $A = \{a, b, c\}$

Concept / Formula Used: $A \times A$

Solution:

Part (i): Neither reflexive nor symmetric but transitive. Let $R = \{(a, b)\}$. - Reflexivity: $(a, a) \notin R$, so not reflexive. - Symmetry: $(a, b) \in R$ but $(b, a) \notin R$, so not symmetric. - Transitivity: There are no pairs (x, y) and (y, z) to test non-vacuously; hence it is vacuously transitive.

Part (ii): Neither symmetric nor transitive but reflexive. Let $R = \{(a, a), (b, b), (c, c), (a, b), (b, c)\}$. - Reflexivity: Contains $(a, a), (b, b), (c, c)$, so reflexive. - Symmetry: $(a, b) \in R$ but $(b, a) \notin R$, so not symmetric. - Transitivity: $(a, b) \in R$ and $(b, c) \in R$, but $(a, c) \notin R$, so not transitive.

Part (iii): Neither transitive nor reflexive but symmetric. Let $R = \{(a, b), (b, a)\}$. - Reflexivity: $(a, a) \notin R$, so not reflexive. - Symmetry: $(a, b) \in R \implies (b, a) \in R$, so symmetric. - Transitivity: $(a, b) \in R$ and $(b, a) \in R$, but $(a, a) \notin R$, so not transitive.

Final Answer

- (i) $R = \{(a, b)\}$
- (ii) $R = \{(a, a), (b, b), (c, c), (a, b), (b, c)\}$
- (iii) $R = \{(a, b), (b, a)\}$

Question 35

Show that the relation S in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by

$$S = \{(a, b) : a, b \in A, |a - b| \text{ is divisible by } 4\}$$

is an equivalence relation. Find the set of all elements related to 1.

TYPE 8 Equivalence Classes and Related Elements

★ **Likely Exam Question**

Given: $A = \{0, 1, 2, \dots, 12\}$ and $S = \{(a, b) \in A \times A : 4 \mid |a - b|\}$

Concept / Formula Used: 1 $[1] = \{x \in A : (x, 1) \in S\}$

Solution:

Reflexivity: Let $a \in A$.

$$|a - a| = 0$$

Since $0 = 4 \times 0$, $|a - a|$ is divisible by 4. Thus, $(a, a) \in S$ for all $a \in A$. Hence, S is reflexive.

Symmetry: Let $(a, b) \in S$. Then,

$$|a - b| = 4k \quad \text{for some } k \in \mathbb{W}$$

Since $|b - a| = |-(a - b)| = |a - b|$:

$$|b - a| = 4k$$

Thus, $(b, a) \in S$. Hence, S is symmetric.

Transitivity: Let $(a, b) \in S$ and $(b, c) \in S$. Then,

$$a - b = 4k \quad \text{for some } k \in \mathbb{Z}$$

$$b - c = 4m \quad \text{for some } m \in \mathbb{Z}$$

Adding the two equations:

$$\begin{aligned}(a - b) + (b - c) &= 4k + 4m \\ a - c &= 4(k + m)\end{aligned}$$

Thus, $|a - c| = 4|k + m|$, which is divisible by 4. Hence, $(a, c) \in S$. Therefore, S is transitive. Since S is reflexive, symmetric, and transitive, S is an equivalence relation.

Finding elements related to 1:

$$[1] = \{x \in A : |x - 1| \text{ is a multiple of } 4\}$$

For $x \in \{0, 1, 2, \dots, 12\}$:

$$\begin{aligned}|x - 1| = 0 &\implies x = 1 \\ |x - 1| = 4 &\implies x - 1 = 4 \implies x = 5 \\ |x - 1| = 8 &\implies x - 1 = 8 \implies x = 9\end{aligned}$$

Thus, the elements related to 1 are $\{1, 5, 9\}$.

Final Answer

S is an equivalence relation, and the set of elements related to 1 is $\{1, 5, 9\}$.

Question 36

Show that the relation R in the set $A = \{x \in \mathbb{W}, 0 \leq x \leq 17\}$ given by

- (i) $R = \{(a, b) : |a - b| \text{ is a multiple of } 5\}$
- (ii) $R = \{(a, b) : a = b\}$

are equivalence relations. Find the set of all elements related to 2 in each case.

TYPE 8 Equivalence Classes and Related Elements

Given: $A = \{0, 1, 2, \dots, 17\}$

Concept / Formula Used: 2 $[2] = \{x \in A : (x, 2) \in R\}$

Solution:

Part (i): $R = \{(a, b) : |a - b| \text{ is a multiple of } 5\}$

Reflexivity: For any $a \in A$, $|a - a| = 0 = 5 \times 0$. Thus, $(a, a) \in R$. R is reflexive.

Symmetry: Let $(a, b) \in R$. Then $|a - b| = 5k$ for $k \in \mathbb{W}$. Since $|b - a| = |a - b| = 5k$, $(b, a) \in R$. R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then $a - b = 5k$ and $b - c = 5m$ for $k, m \in \mathbb{Z}$. Adding them:

$$a - c = (a - b) + (b - c) = 5(k + m)$$

Thus $|a - c|$ is a multiple of 5, so $(a, c) \in R$. R is transitive.

Hence, R is an equivalence relation.

Elements related to 2 in part (i):

$$[2] = \{x \in A : |x - 2| \text{ is a multiple of } 5\}$$

Testing values in $A = \{0, 1, \dots, 17\}$:

$$\begin{aligned}x = 2 &\implies |2 - 2| = 0 \\x = 7 &\implies |7 - 2| = 5 \\x = 12 &\implies |12 - 2| = 10 \\x = 17 &\implies |17 - 2| = 15\end{aligned}$$

Thus, the set of elements related to 2 is $\{2, 7, 12, 17\}$.

Part (ii): $R = \{(a, b) : a = b\}$

Reflexivity: For any $a \in A$, $a = a$ is true, so $(a, a) \in R$. R is reflexive.

Symmetry: If $(a, b) \in R$, then $a = b \implies b = a \implies (b, a) \in R$. R is symmetric.

Transitivity: If $(a, b) \in R$ and $(b, c) \in R$, then $a = b$ and $b = c \implies a = c \implies (a, c) \in R$. R is transitive.

Hence, R is an equivalence relation.

Elements related to 2 in part (ii):

$$[2] = \{x \in A : x = 2\} = \{2\}$$

Final Answer

- (i) R is an equivalence relation; set of elements related to 2 is $\{2, 7, 12, 17\}$.
(ii) R is an equivalence relation; set of elements related to 2 is $\{2\}$.

Question 37

Show that the relation R defined by $(a, b) R (c, d) \iff a + d = b + c$ on $A \times A$ where $A = \{1, 2, 3, \dots, 10\}$ is an equivalence relation. Hence, write the equivalence class $[(3, 4)]$.

TYPE 10 Equivalence Relations on Product Sets

★ Likely Exam Question

Given: $A = \{1, 2, 3, \dots, 10\}$ and relation R on $A \times A$ defined by $(a, b) R (c, d) \iff a + d = b + c$

Concept / Formula Used: $[(a, b)] = \{(x, y) \in A \times A : (x, y) R (a, b)\}$

Solution:

Reflexivity: Let $(a, b) \in A \times A$.

$$a + b = b + a$$

This satisfies the definition $(a, b) R (a, b)$. Thus, R is reflexive.

Symmetry: Let $(a, b) R (c, d)$. Then,

$$a + d = b + c$$

Rearranging the equality:

$$c + b = d + a$$

By definition of R , this means:

$$(c, d)R(a, b)$$

Thus, R is symmetric.

Transitivity: Let $(a, b)R(c, d)$ and $(c, d)R(e, f)$. Then,

$$a + d = b + c \quad \text{--- (1)}$$

$$c + f = d + e \quad \text{--- (2)}$$

Adding equation (1) and equation (2):

$$(a + d) + (c + f) = (b + c) + (d + e)$$

$$a + d + c + f = b + c + d + e$$

Subtracting $c + d$ from both sides:

$$a + f = b + e$$

This implies $(a, b)R(e, f)$. Thus, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation on $A \times A$.

Equivalence Class $[(3, 4)]$:

$$\begin{aligned} [(3, 4)] &= \{(x, y) \in A \times A : (x, y)R(3, 4)\} \\ &= \{(x, y) \in A \times A : x + 4 = y + 3\} \\ &= \{(x, y) \in A \times A : y = x + 1\} \end{aligned}$$

Since $x, y \in \{1, 2, 3, \dots, 10\}$, we evaluate $y = x + 1$ for $x \in \{1, 2, \dots, 9\}$:

$$x = 1 \implies y = 2 \implies (1, 2)$$

$$x = 2 \implies y = 3 \implies (2, 3)$$

$$x = 3 \implies y = 4 \implies (3, 4)$$

$$x = 4 \implies y = 5 \implies (4, 5)$$

$$x = 5 \implies y = 6 \implies (5, 6)$$

$$x = 6 \implies y = 7 \implies (6, 7)$$

$$x = 7 \implies y = 8 \implies (7, 8)$$

$$x = 8 \implies y = 9 \implies (8, 9)$$

$$x = 9 \implies y = 10 \implies (9, 10)$$

Final Answer

R is an equivalence relation, and $[(3, 4)] = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8), (8, 9), (9, 10)\}$.

Question 38

Show that the relation R in the set $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)R(c, d)$ iff $a^2 + d^2 = b^2 + c^2$ for all $a, b, c, d \in \mathbb{N}$ is an equivalence relation.

TYPE 10 Equivalence Relations on Product Sets

To Prove: R is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Concept / Formula Used: $(a, b)R(c, d) \iff a^2 + d^2 = b^2 + c^2$

Solution:

Reflexivity: Let $(a, b) \in \mathbb{N} \times \mathbb{N}$. We check if $(a, b)R(a, b)$:

$$a^2 + b^2 = b^2 + a^2$$

By commutativity of addition in $\mathbb{N}$, this statement holds. Thus, $(a, b)R(a, b)$ for all $(a, b) \in \mathbb{N} \times \mathbb{N}$. Hence, R is reflexive.

Symmetry: Let $(a, b)R(c, d)$. Then,

$$a^2 + d^2 = b^2 + c^2$$

Swapping left and right sides:

$$b^2 + c^2 = a^2 + d^2$$

By commutativity of addition:

$$c^2 + b^2 = d^2 + a^2$$

This implies $(c, d)R(a, b)$. Hence, R is symmetric.

Transitivity: Let $(a, b)R(c, d)$ and $(c, d)R(e, f)$. Then,

$$a^2 + d^2 = b^2 + c^2 \quad \text{--- (1)}$$

$$c^2 + f^2 = d^2 + e^2 \quad \text{--- (2)}$$

Adding equation (1) and equation (2):

$$\begin{aligned} (a^2 + d^2) + (c^2 + f^2) &= (b^2 + c^2) + (d^2 + e^2) \\ a^2 + d^2 + c^2 + f^2 &= b^2 + c^2 + d^2 + e^2 \end{aligned}$$

Subtracting $c^2 + d^2$ from both sides:

$$a^2 + f^2 = b^2 + e^2$$

This implies $(a, b)R(e, f)$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Hence Proved

R is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Question 39

The Earth has 24 time zones, defined by dividing the Earth into 24 equal longitudinal segments. These are the regions on Earth that have the same standard time. For example, USA and India fall in different time zones, but Sri Lanka and India are in the same time zone.

A relation R is defined on the set $U = \{\text{All people on the Earth}\}$ such that $R = \{(x, y) \mid \text{the time difference between the time zones } x \text{ and } y \text{ reside in is 6 hours}\}$.

(i) Check whether the relation R is reflexive, symmetric and transitive.

(ii) Is relation R an equivalence relation?

Show your work.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $U = \{\text{All people on the Earth}\}$, $R = \{(x, y) \in U \times U : \text{time difference between zones of } x \text{ and } y \text{ is 6 hours}\}$

Concept / Formula Used: $(x, x) \in R$ x $(x, y) \in R \implies (y, x) \in R$ $(x, y) \in R$ and $(y, z) \in R \implies (x, z) \in R$

Solution:

Part (i): Checking Reflexivity, Symmetry, and Transitivity

Reflexivity: Let $x \in U$ be any person on Earth. Person x resides in a specific time zone. The time difference between x 's time zone and x 's own time zone is:

$$|t_x - t_x| = 0 \text{ hours}$$

Since $0 \neq 6$ hours, $(x, x) \notin R$ for any $x \in U$. Thus, R is not reflexive.

Symmetry: Let $(x, y) \in R$. This means the time difference between the time zones of x and y is 6 hours:

$$|t_x - t_y| = 6 \text{ hours}$$

Since $|t_y - t_x| = |t_x - t_y|$:

$$|t_y - t_x| = 6 \text{ hours}$$

This implies $(y, x) \in R$. Thus, R is symmetric.

Transitivity: Suppose $(x, y) \in R$ and $(y, z) \in R$. This means:

$$|t_x - t_y| = 6 \text{ hours}$$

$$|t_y - t_z| = 6 \text{ hours}$$

Evaluating the possible values for $t_x - t_z$:

$$t_x - t_z = (t_x - t_y) + (t_y - t_z) = 6 + 6 = 12 \text{ hours}$$

$$\text{or } t_x - t_z = (t_x - t_y) + (t_y - t_z) = 6 + (-6) = 0 \text{ hours}$$

In either case (12 hours or 0 hours), the time difference $|t_x - t_z| \neq 6$ hours. Thus, $(x, z) \notin R$. Hence, R is not transitive.

Part (ii): Is R an equivalence relation?

Since R is neither reflexive nor transitive, R fails to satisfy the definition of an equivalence relation.

Final Answer

- (i) R is symmetric, but neither reflexive nor transitive.
- (ii) No, R is not an equivalence relation.

1.2 – Functions: Domain and Range

Question 1

A function is defined by $f(x) = \frac{3x^2 + 2x - 1}{x + 1}$, $x \in \mathbb{R}, x \neq -1$. Find the value of $f(-3) + 1$.

TYPE 11 Evaluating Function Values

Given: $f(x) = \frac{3x^2 + 2x - 1}{x + 1}, x \in \mathbb{R}, x \neq -1$

To Find: $f(-3) + 1$

Concept / Formula Used: $f(a) = f(x)|_{x=a}$

Solution:

Substitute $x = -3$ into the given function $f(x)$:

$$\begin{aligned} f(-3) &= \frac{3(-3)^2 + 2(-3) - 1}{(-3) + 1} \\ &= \frac{3(9) - 6 - 1}{-2} \\ &= \frac{27 - 7}{-2} \\ &= \frac{20}{-2} \\ &= -10 \end{aligned}$$

Now, add 1 to $f(-3)$:

$$\begin{aligned} f(-3) + 1 &= -10 + 1 \\ &= -9 \end{aligned}$$

Final Answer

$$f(-3) + 1 = -9$$

Question 2

(i) Is the relation f defined by $f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 2x, & 3 \leq x \leq 5 \end{cases}$ a function? Justify your answer.

(ii) A relation R in the set of real numbers $\mathbb{R}$ defined as $R = \{(a, b) : \sqrt{a} = b\}$ is a function or not. Justify.

TYPE 12 Testing Definition of a Function

Given: (i) $f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 2x, & 3 \leq x \leq 5 \end{cases}$, (ii) $R = \{(a, b) \in \mathbb{R} \times \mathbb{R} : b = \sqrt{a}\}$

To Find: Determine whether f and R are functions with justification

Concept / Formula Used: $f : A \rightarrow B \quad x \in A \quad B$

Solution:

(i) Check the value of $f(x)$ at the point of overlap $x = 3$:

$$\text{For } 0 \leq x \leq 3 : f(3) = 3^2 = 9$$

$$\text{For } 3 \leq x \leq 5 : f(3) = 2(3) = 6$$

Since $f(3)$ yields two distinct values (9 and 6), the domain element 3 does not have a unique image. Therefore, f is not a function.

(ii) For $R = \{(a, b) \in \mathbb{R} \times \mathbb{R} : b = \sqrt{a}\}$: Consider any negative real number $a \in \mathbb{R}$ such that $a < 0$, e.g., $a = -4$. The expression $\sqrt{-4}$ does not produce a real number. Hence, negative real numbers in the domain do not have any image in $\mathbb{R}$. Therefore, R is not a function on $\mathbb{R}$.

Final Answer

(i) No, f is not a function because $f(3)$ has two images (9 and 6).

(ii) No, R is not a function because $\sqrt{a}$ is undefined for negative real numbers $a < 0$.

Common Mistake: Assuming piecewise expressions are always functions without checking the boundary values for consistency is a common oversight.

Question 3

A real function f is defined by $f(x) = 3x - 2, x \in \mathbb{R}$, find the value of $f(-3) + f(0) \times f(7)$.

TYPE 11 Evaluating Function Values

Given: $f(x) = 3x - 2, x \in \mathbb{R}$

To Find: $f(-3) + f(0) \times f(7)$

Concept / Formula Used: $f(a) = 3a - 2$

Solution:

Evaluate each term separately:

$$f(-3) = 3(-3) - 2 = -9 - 2 = -11$$

$$f(0) = 3(0) - 2 = 0 - 2 = -2$$

$$f(7) = 3(7) - 2 = 21 - 2 = 19$$

Substitute these calculated values into the main expression:

$$\begin{aligned} f(-3) + f(0) \times f(7) &= -11 + (-2) \times 19 \\ &= -11 - 38 \\ &= -49 \end{aligned}$$

Final Answer

$$f(-3) + f(0) \times f(7) = -49$$

Question 4

If a real function f is defined by $f(x) = 2x^2 - 3$, find x such that $f(x) = 15$.

TYPE 13 Finding Pre-images of Function Values

Given: $f(x) = 2x^2 - 3$ and $f(x) = 15$

To Find: Values of x

Concept / Formula Used: $f(x) = y \implies 2x^2 - 3 = y$

Solution:

Set $f(x) = 15$:

$$\begin{aligned}2x^2 - 3 &= 15 \\2x^2 &= 15 + 3 \\2x^2 &= 18 \\x^2 &= \frac{18}{2} \\x^2 &= 9 \\x &= \pm\sqrt{9} \\x &= \pm 3\end{aligned}$$

Final Answer

$x = 3$ or $x = -3$

Question 5

Given $f(x) = x^3 - 1$, find x if $f(x)$ is 215.

TYPE 13 Finding Pre-images of Function Values

Given: $f(x) = x^3 - 1$ and $f(x) = 215$

To Find: Value of x

Concept / Formula Used: $x^3 - 1 = y \implies x = \sqrt[3]{y+1}$

Solution:

Set $f(x) = 215$:

$$\begin{aligned}x^3 - 1 &= 215 \\x^3 &= 215 + 1 \\x^3 &= 216 \\x &= \sqrt[3]{216} \\x &= 6\end{aligned}$$

Final Answer

$x = 6$

Question 6

A function is given by the formula $f(x) = 144 - 16x^2$. Calculate $f(2)$. Also find the values of x when $f(x) = 0$.

TYPE 13 Finding Pre-images of Function Values**Given:** $f(x) = 144 - 16x^2$ **To Find:** $f(2)$ and x such that $f(x) = 0$ **Concept / Formula Used:** $f(a) = 144 - 16a^2$ **Solution:**Calculate $f(2)$:

$$\begin{aligned}f(2) &= 144 - 16(2)^2 \\&= 144 - 16(4) \\&= 144 - 64 \\&= 80\end{aligned}$$

Now set $f(x) = 0$ to find the roots:

$$\begin{aligned}144 - 16x^2 &= 0 \\16x^2 &= 144 \\x^2 &= \frac{144}{16} \\x^2 &= 9 \\x &= \pm 3\end{aligned}$$

Final Answer $f(2) = 80$, and $x = \pm 3$ when $f(x) = 0$ **Question 7**

A function ' f ' is defined by $f(x) = 2x^2 + 3$, for all $x \in \mathbb{R}$. Find (i) image of -1 under f (ii) element (elements) of the domain which has image 35

TYPE 13 Finding Pre-images of Function Values**Given:** $f(x) = 2x^2 + 3, x \in \mathbb{R}$ **To Find:** (i) $f(-1)$, (ii) x such that $f(x) = 35$ **Concept / Formula Used:** $f(a) = 2a^2 + 3$ **Solution:**

(i) Image of -1 under f :

$$\begin{aligned} f(-1) &= 2(-1)^2 + 3 \\ &= 2(1) + 3 \\ &= 2 + 3 \\ &= 5 \end{aligned}$$

(ii) Elements of the domain with image 35:

$$\begin{aligned} f(x) &= 35 \\ 2x^2 + 3 &= 35 \\ 2x^2 &= 35 - 3 \\ 2x^2 &= 32 \\ x^2 &= 16 \\ x &= \pm 4 \end{aligned}$$

Final Answer

(i) Image of -1 is 5, (ii) Domain elements are 4 and -4

Question 8

If $f(x) = 2x^2 + 1$ and domain of $f = \{-2, -1, 0, 1, 3\}$, find (i) range of f (ii) $f(3) \times f(-2)$ (iii) x if $f(x) = 9$

TYPE 14 Domain and Range of Functions on Finite Sets

Given: $f(x) = 2x^2 + 1$, Domain = $\{-2, -1, 0, 1, 3\}$

To Find: (i) Range of f , (ii) $f(3) \times f(-2)$, (iii) x if $f(x) = 9$

Concept / Formula Used: $\text{Range}(f) = \{f(x) : x \in \text{Domain}\}$

Solution:

(i) Evaluate $f(x)$ for each element in the domain:

$$\begin{aligned} f(-2) &= 2(-2)^2 + 1 = 2(4) + 1 = 9 \\ f(-1) &= 2(-1)^2 + 1 = 2(1) + 1 = 3 \\ f(0) &= 2(0)^2 + 1 = 0 + 1 = 1 \\ f(1) &= 2(1)^2 + 1 = 2(1) + 1 = 3 \\ f(3) &= 2(3)^2 + 1 = 2(9) + 1 = 19 \end{aligned}$$

Gathering unique outputs gives the range:

$$\text{Range of } f = \{1, 3, 9, 19\}$$

(ii) Calculate $f(3) \times f(-2)$:

$$\begin{aligned} f(3) \times f(-2) &= 19 \times 9 \\ &= 171 \end{aligned}$$

(iii) Find $x \in \text{Domain}$ such that $f(x) = 9$:

$$\begin{aligned} 2x^2 + 1 &= 9 \\ 2x^2 &= 8 \\ x^2 &= 4 \\ x &= \pm 2 \end{aligned}$$

Cross-referencing with $\text{Domain} = \{-2, -1, 0, 1, 3\}$, the values belonging to the domain are $x = -2$. (Note: $x = 2$ is not in the given domain).

Final Answer

(i) $\text{Range} = \{1, 3, 9, 19\}$, (ii) 171, (iii) $x = -2$

Question 9

A function 'f' is defined by $f(x) = x^2 + 3, x \in \mathbb{N}$ and $x \leq 5$. (i) Find the range of f (ii) Find $f(2) \times f(5)$ (iii) Does $f(-3)$ exist? (iv) Find x when $f(x) = 7$

TYPE 14 Domain and Range of Functions on Finite Sets

Given: $f(x) = x^2 + 3, \text{Domain} = \{x \in \mathbb{N} : x \leq 5\} = \{1, 2, 3, 4, 5\}$

To Find: (i) Range of f, (ii) $f(2) \times f(5)$, (iii) Existence of $f(-3)$, (iv) x when $f(x) = 7$

Concept / Formula Used: $\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$

Solution:

(i) Calculate $f(x)$ for all elements in the domain $\{1, 2, 3, 4, 5\}$:

$$\begin{aligned} f(1) &= 1^2 + 3 = 4 \\ f(2) &= 2^2 + 3 = 7 \\ f(3) &= 3^2 + 3 = 12 \\ f(4) &= 4^2 + 3 = 19 \\ f(5) &= 5^2 + 3 = 28 \end{aligned}$$

Thus, $\text{Range}(f) = \{4, 7, 12, 19, 28\}$.

(ii) Calculate $f(2) \times f(5)$:

$$\begin{aligned} f(2) \times f(5) &= 7 \times 28 \\ &= 196 \end{aligned}$$

(iii) Check if $f(-3)$ exists: Since $-3 \notin \mathbb{N}$, -3 is not in the specified domain of f. Thus, $f(-3)$ does not exist.

(iv) Solve for x when $f(x) = 7$:

$$\begin{aligned} x^2 + 3 &= 7 \\ x^2 &= 4 \\ x &= 2 \quad (\text{since } x \in \mathbb{N}, x > 0) \end{aligned}$$

Final Answer

(i) $\text{Range} = \{4, 7, 12, 19, 28\}$, (ii) 196, (iii) No, $f(-3)$ does not exist, (iv) $x = 2$

Question 10

Find the domain of the following real functions: (i) $f(x) = |x + 2|$ (ii) $f(x) = -|x - 2|$ (iii) $f(x) = 2 - |x - 1|$

TYPE 15 Finding Domain and Range of Modulus Functions

Given: (i) $f(x) = |x + 2|$, (ii) $f(x) = -|x - 2|$, (iii) $f(x) = 2 - |x - 1|$

To Find: Domain of each function

Concept / Formula Used: $|g(x)| \in \mathbb{R} \quad x \in \mathbb{R}$

Solution:

(i) For $f(x) = |x + 2|$: The modulus expression $|x + 2|$ is defined and returns a real value for every real number x .

$$\text{Domain}(f) = \mathbb{R} = (-\infty, \infty)$$

(ii) For $f(x) = -|x - 2|$: The expression $-|x - 2|$ is defined for all real values of x .

$$\text{Domain}(f) = \mathbb{R} = (-\infty, \infty)$$

(iii) For $f(x) = 2 - |x - 1|$: The expression $2 - |x - 1|$ is defined for all real values of x .

$$\text{Domain}(f) = \mathbb{R} = (-\infty, \infty)$$

Final Answer

(i) Domain = $\mathbb{R}$, (ii) Domain = $\mathbb{R}$, (iii) Domain = $\mathbb{R}$

Question 11

Find the range of the following functions: (i) $f(x) = x^2 + 2, x \in \mathbb{R}$ (ii) $f(x) = 3 - 2x, x \in \mathbb{R}, x \geq 1$

TYPE 16 Finding Domain and Range of Polynomial Functions

Given: (i) $f(x) = x^2 + 2, x \in \mathbb{R}$, (ii) $f(x) = 3 - 2x, x \in \mathbb{R}, x \geq 1$

To Find: Range of each function

Concept / Formula Used: $x^2 \geq 0 \quad x \in \mathbb{R}$

Solution:

(i) For $f(x) = x^2 + 2, x \in \mathbb{R}$: For any real number x :

$$x^2 \geq 0$$

Add 2 to both sides:

$$x^2 + 2 \geq 2$$

$$f(x) \geq 2$$

Hence, $\text{Range}(f) = [2, \infty)$.

(ii) For $f(x) = 3 - 2x, x \geq 1$: Starting from the domain condition $x \geq 1$: Multiply by -2 (reversing the inequality direction):

$$-2x \leq -2$$

Add 3 to both sides:

$$\begin{aligned} 3 - 2x &\leq 3 - 2 \\ f(x) &\leq 1 \end{aligned}$$

Hence, $\text{Range}(f) = (-\infty, 1]$.

Final Answer

(i) $\text{Range} = [2, \infty)$, (ii) $\text{Range} = (-\infty, 1]$

Question 12

Find the domain and the range of the function $f(x) = 2x^2 + 1$. Also find $f(-2)$ and the numbers which are associated with the number 51 in its range.

TYPE 16 Finding Domain and Range of Polynomial Functions

Given: $f(x) = 2x^2 + 1$

To Find: Domain, Range, $f(-2)$, and numbers associated with 51

Concept / Formula Used: $x^2 \geq 0 \implies 2x^2 + 1 \geq 1$

Solution:

1. Domain: Since $2x^2 + 1$ is defined for every real number x :

$$\text{Domain}(f) = \mathbb{R}$$

2. Range: For all $x \in \mathbb{R}$:

$$\begin{aligned} x^2 &\geq 0 \\ 2x^2 &\geq 0 \\ 2x^2 + 1 &\geq 1 \\ f(x) &\geq 1 \end{aligned}$$

Thus, $\text{Range}(f) = [1, \infty)$.

3. Calculate $f(-2)$:

$$\begin{aligned} f(-2) &= 2(-2)^2 + 1 \\ &= 2(4) + 1 \\ &= 8 + 1 \\ &= 9 \end{aligned}$$

4. Find numbers associated with 51: Set $f(x) = 51$:

$$\begin{aligned} 2x^2 + 1 &= 51 \\ 2x^2 &= 50 \\ x^2 &= 25 \\ x &= \pm 5 \end{aligned}$$

Final Answer

Domain = $\mathbb{R}$, Range = $[1, \infty)$, $f(-2) = 9$, associated numbers = 5 and -5

Question 13

Find the domain and the range of the following functions: (i) $f(x) = \sqrt{x+2}$ (ii) $f(x) = \sqrt{3-2x}$ (iii) $f(x) = \frac{1}{\sqrt{x-5}}$

TYPE 17 Finding Domain and Range of Irrational Functions

★ Likely Exam Question

Given: (i) $f(x) = \sqrt{x+2}$, (ii) $f(x) = \sqrt{3-2x}$, (iii) $f(x) = \frac{1}{\sqrt{x-5}}$

To Find: Domain and Range of each function

Concept / Formula Used: $\sqrt{g(x)}$ $g(x) \geq 0$ $\frac{1}{\sqrt{g(x)}}$ $g(x) > 0$

Solution:

(i) For $f(x) = \sqrt{x+2}$: Domain condition:

$$\begin{aligned}x+2 &\geq 0 \\x &\geq -2\end{aligned}$$

Thus, $\text{Domain}(f) = [-2, \infty)$. Since square root outputs non-negative real numbers, as x increases from -2 to ∞ , $f(x)$ increases from 0 to ∞ :

$$\text{Range}(f) = [0, \infty)$$

(ii) For $f(x) = \sqrt{3-2x}$: Domain condition:

$$\begin{aligned}3-2x &\geq 0 \\-2x &\geq -3 \\x &\leq \frac{3}{2}\end{aligned}$$

Thus, $\text{Domain}(f) = (-\infty, \frac{3}{2}]$. Square root values are non-negative, so:

$$\text{Range}(f) = [0, \infty)$$

(iii) For $f(x) = \frac{1}{\sqrt{x-5}}$: Domain condition (strictly positive in denominator):

$$\begin{aligned}x-5 &> 0 \\x &> 5\end{aligned}$$

Thus, $\text{Domain}(f) = (5, \infty)$. As $x \rightarrow 5^+$, $\sqrt{x-5} \rightarrow 0^+$, so $f(x) \rightarrow \infty$. As $x \rightarrow \infty$, $f(x) \rightarrow 0^+$. Since $\sqrt{x-5} > 0$, $f(x) > 0$:

$$\text{Range}(f) = (0, \infty)$$

Final Answer

- (i) Domain = $[-2, \infty)$, Range = $[0, \infty)$
 (ii) Domain = $\left(-\infty, \frac{3}{2}\right]$, Range = $[0, \infty)$
 (iii) Domain = $(5, \infty)$, Range = $(0, \infty)$

Question 14

Find the domain and the range of the following functions: (i) $f(x) = \frac{4-x}{x-4}$ (ii) $f(x) = \frac{x^2-9}{x-3}$
 (iii) $f(x) = \frac{x+1}{2x+1}$ (iv) $f(x) = \frac{4+x}{4-x}$

TYPE 18 Finding Domain and Range of Rational Functions

Given: (i) $f(x) = \frac{4-x}{x-4}$, (ii) $f(x) = \frac{x^2-9}{x-3}$, (iii) $f(x) = \frac{x+1}{2x+1}$, (iv) $f(x) = \frac{4+x}{4-x}$

To Find: Domain and Range of each function

Concept / Formula Used: $\frac{P(x)}{Q(x)} \quad Q(x) \neq 0$

Solution:

(i) For $f(x) = \frac{4-x}{x-4}$: Denominator condition: $x-4 \neq 0 \implies x \neq 4$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{4\}$$

Simplifying $f(x)$ for $x \neq 4$:

$$f(x) = \frac{-(x-4)}{x-4} = -1$$

$$\text{Range}(f) = \{-1\}$$

(ii) For $f(x) = \frac{x^2-9}{x-3}$: Denominator condition: $x-3 \neq 0 \implies x \neq 3$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{3\}$$

Simplifying $f(x)$ for $x \neq 3$:

$$f(x) = \frac{(x-3)(x+3)}{x-3} = x+3$$

Since $x \neq 3$, $f(x) \neq 3+3 = 6$.

$$\text{Range}(f) = \mathbb{R} \setminus \{6\}$$

(iii) For $f(x) = \frac{x+1}{2x+1}$: Denominator condition: $2x+1 \neq 0 \implies x \neq -\frac{1}{2}$.

$$\text{Domain}(f) = \mathbb{R} \setminus \left\{-\frac{1}{2}\right\}$$

To find the range, set $y = \frac{x+1}{2x+1}$ and express x in terms of y :

$$\begin{aligned}y(2x+1) &= x+1 \\2xy+y &= x+1 \\2xy-x &= 1-y \\x(2y-1) &= 1-y \\x &= \frac{1-y}{2y-1}\end{aligned}$$

For x to be real, $2y-1 \neq 0 \implies y \neq \frac{1}{2}$.

$$\text{Range}(f) = \mathbb{R} \setminus \left\{ \frac{1}{2} \right\}$$

(iv) For $f(x) = \frac{4+x}{4-x}$: Denominator condition: $4-x \neq 0 \implies x \neq 4$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{4\}$$

Set $y = \frac{4+x}{4-x}$ and express x in terms of y :

$$\begin{aligned}y(4-x) &= 4+x \\4y-xy &= 4+x \\4y-4 &= x+xy \\x(1+y) &= 4y-4 \\x &= \frac{4y-4}{y+1}\end{aligned}$$

For x to be real, $y+1 \neq 0 \implies y \neq -1$.

$$\text{Range}(f) = \mathbb{R} \setminus \{-1\}$$

Final Answer

- (i) Domain = $\mathbb{R} \setminus \{4\}$, Range = $\{-1\}$
- (ii) Domain = $\mathbb{R} \setminus \{3\}$, Range = $\mathbb{R} \setminus \{6\}$
- (iii) Domain = $\mathbb{R} \setminus \left\{ -\frac{1}{2} \right\}$, Range = $\mathbb{R} \setminus \left\{ \frac{1}{2} \right\}$
- (iv) Domain = $\mathbb{R} \setminus \{4\}$, Range = $\mathbb{R} \setminus \{-1\}$

Question 15

Find the domain and the range of the following functions: (i) $f(x) = \sqrt{16-x^2}$ (ii) $f(x) = \sqrt{x^2-9}$ (iii) $f(x) = \frac{1}{\sqrt{4-x^2}}$

TYPE 17 Finding Domain and Range of Irrational Functions

★ Likely Exam Question

Given: (i) $f(x) = \sqrt{16-x^2}$, (ii) $f(x) = \sqrt{x^2-9}$, (iii) $f(x) = \frac{1}{\sqrt{4-x^2}}$

To Find: Domain and Range of each function

Concept / Formula Used: $a^2 - x^2 \geq 0 \implies -a \leq x \leq a$

Solution:

(i) For $f(x) = \sqrt{16 - x^2}$: Domain condition:

$$16 - x^2 \geq 0$$

$$x^2 \leq 16$$

$$-4 \leq x \leq 4$$

$$\text{Domain}(f) = [-4, 4]$$

Range: For $-4 \leq x \leq 4$, $0 \leq x^2 \leq 16$.

$$0 \leq 16 - x^2 \leq 16$$

$$0 \leq \sqrt{16 - x^2} \leq 4$$

$$\text{Range}(f) = [0, 4]$$

(ii) For $f(x) = \sqrt{x^2 - 9}$: Domain condition:

$$x^2 - 9 \geq 0$$

$$(x - 3)(x + 3) \geq 0$$

$$x \leq -3 \text{ or } x \geq 3$$

$$\text{Domain}(f) = (-\infty, -3] \cup [3, \infty)$$

Range: Since $x^2 \geq 9$, $x^2 - 9 \geq 0$, giving $\sqrt{x^2 - 9} \geq 0$. As $|x| \rightarrow \infty$, $f(x) \rightarrow \infty$.

$$\text{Range}(f) = [0, \infty)$$

(iii) For $f(x) = \frac{1}{\sqrt{4 - x^2}}$: Domain condition:

$$4 - x^2 > 0$$

$$x^2 < 4$$

$$-2 < x < 2$$

$$\text{Domain}(f) = (-2, 2)$$

Range: For $-2 < x < 2$, $0 < x^2 < 4$.

$$0 < 4 - x^2 \leq 4$$

$$0 < \sqrt{4 - x^2} \leq 2$$

$$\frac{1}{\sqrt{4 - x^2}} \geq \frac{1}{2}$$

As $x \rightarrow \pm 2$, $4 - x^2 \rightarrow 0^+$, so $f(x) \rightarrow \infty$.

$$\text{Range}(f) = \left[\frac{1}{2}, \infty \right)$$

Final Answer

(i) Domain = $[-4, 4]$, Range = $[0, 4]$

(ii) Domain = $(-\infty, -3] \cup [3, \infty)$, Range = $[0, \infty)$

(iii) Domain = $(-2, 2)$, Range = $\left[\frac{1}{2}, \infty \right)$

Question 16

Find the domain and the range of the following functions: (i) $f(x) = |x - 3|$ (ii) $f(x) = 3 - |x - 2|$

TYPE 15 Finding Domain and Range of Modulus Functions

Given: (i) $f(x) = |x - 3|$, (ii) $f(x) = 3 - |x - 2|$

To Find: Domain and Range of each function

Concept / Formula Used: $|g(x)| \geq 0 \quad x \in \mathbb{R}$

Solution:

(i) For $f(x) = |x - 3|$: The expression $|x - 3|$ is defined for all real numbers.

$$\text{Domain}(f) = \mathbb{R}$$

Since absolute values are non-negative:

$$|x - 3| \geq 0$$

$$\text{Range}(f) = [0, \infty)$$

(ii) For $f(x) = 3 - |x - 2|$: Defined for all real numbers:

$$\text{Domain}(f) = \mathbb{R}$$

Since $|x - 2| \geq 0$:

$$-|x - 2| \leq 0$$

$$3 - |x - 2| \leq 3$$

$$f(x) \leq 3$$

$$\text{Range}(f) = (-\infty, 3]$$

Final Answer

(i) Domain = $\mathbb{R}$, Range = $[0, \infty)$

(ii) Domain = $\mathbb{R}$, Range = $(-\infty, 3]$

Question 17

If a real function f is defined by $f(x) = \frac{|x| - x}{2x}$, then find its domain and range.

TYPE 15 Finding Domain and Range of Modulus Functions

Given: $f(x) = \frac{|x| - x}{2x}$

To Find: Domain and Range of f

Concept / Formula Used: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

Solution:

1. Domain: The denominator must not vanish:

$$2x \neq 0 \implies x \neq 0$$

Thus, $\text{Domain}(f) = \mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$.

2. Range: Consider the two cases for $x \in \text{Domain}(f)$: Case 1: When $x > 0$: Here, $|x| = x$.

$$\begin{aligned} f(x) &= \frac{x - x}{2x} \\ &= \frac{0}{2x} \\ &= 0 \end{aligned}$$

Case 2: When $x < 0$: Here, $|x| = -x$.

$$\begin{aligned} f(x) &= \frac{-x - x}{2x} \\ &= \frac{-2x}{2x} \\ &= -1 \end{aligned}$$

Thus, the function takes only two constant values: 0 and -1 .

$$\text{Range}(f) = \{-1, 0\}$$

Final Answer

Domain = $\mathbb{R} \setminus \{0\}$, Range = $\{-1, 0\}$

Question 18

Find the domain and the range of the function f defined by $f(x) = \frac{|x - 4|}{x - 4}$.

TYPE 15 Finding Domain and Range of Modulus Functions

★ Likely Exam Question

Given: $f(x) = \frac{|x - 4|}{x - 4}$

To Find: Domain and Range of f

Concept / Formula Used: $|x - 4| = \begin{cases} x - 4, & x > 4 \\ -(x - 4), & x < 4 \end{cases}$

Solution:

1. Domain: The denominator must not be zero:

$$x - 4 \neq 0 \implies x \neq 4$$

Thus, $\text{Domain}(f) = \mathbb{R} \setminus \{4\} = (-\infty, 4) \cup (4, \infty)$.

2. Range: Case 1: For $x > 4$: Since $x - 4 > 0$, we have $|x - 4| = x - 4$.

$$f(x) = \frac{x - 4}{x - 4} = 1$$

Case 2: For $x < 4$: Since $x - 4 < 0$, we have $|x - 4| = -(x - 4)$.

$$f(x) = \frac{-(x - 4)}{x - 4} = -1$$

The function yields only two values: 1 and -1 .

$$\text{Range}(f) = \{-1, 1\}$$

Final Answer

Domain = $\mathbb{R} \setminus \{4\}$, Range = $\{-1, 1\}$

Question 19

If $f(x) = \log \left(\frac{1+x}{1-x} \right)$, then show that $f(x) + f(y) = f \left(\frac{x+y}{1+xy} \right)$.

TYPE 19 Proving Functional Identities

★ Likely Exam Question

Given: $f(x) = \log \left(\frac{1+x}{1-x} \right)$

To Prove: $f(x) + f(y) = f \left(\frac{x+y}{1+xy} \right)$

Concept / Formula Used: $\log A + \log B = \log(A \cdot B)$

Solution:

Start with the Left-Hand Side (LHS):

$$\begin{aligned} \text{LHS} &= f(x) + f(y) \\ &= \log \left(\frac{1+x}{1-x} \right) + \log \left(\frac{1+y}{1-y} \right) \end{aligned}$$

Using the logarithmic property $\log A + \log B = \log(A \cdot B)$:

$$\begin{aligned} \text{LHS} &= \log \left[\left(\frac{1+x}{1-x} \right) \cdot \left(\frac{1+y}{1-y} \right) \right] \\ &= \log \left[\frac{(1+x)(1+y)}{(1-x)(1-y)} \right] \\ &= \log \left[\frac{1+x+y+xy}{1-x-y+xy} \right] \\ &= \log \left[\frac{(1+xy) + (x+y)}{(1+xy) - (x+y)} \right] \end{aligned}$$

Divide both numerator and denominator inside the log by $(1 + xy)$:

$$\begin{aligned}\text{LHS} &= \log \left[\frac{\frac{(1+xy)+(x+y)}{1+xy}}{\frac{(1+xy)-(x+y)}{1+xy}} \right] \\ &= \log \left[\frac{1 + \frac{x+y}{1+xy}}{1 - \frac{x+y}{1+xy}} \right]\end{aligned}$$

By the given definition of the function $f(t) = \log \left(\frac{1+t}{1-t} \right)$ where $t = \frac{x+y}{1+xy}$:

$$\text{LHS} = f \left(\frac{x+y}{1+xy} \right) = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 20

Find the domain of the following functions: (i) $\log(4x - 3)$ (ii) $\frac{1}{\log(9 - x^2)}$

TYPE 20 Finding Domain and Range of Logarithmic Functions

Given: (i) $f(x) = \log(4x - 3)$, (ii) $f(x) = \frac{1}{\log(9 - x^2)}$

To Find: Domain of each function

Concept / Formula Used: $\log(g(x))$ $g(x) > 0$ $\frac{1}{h(x)}$ $h(x) \neq 0$

Solution:

(i) For $f(x) = \log(4x - 3)$: Logarithm is defined only when its argument is positive:

$$4x - 3 > 0$$

$$4x > 3$$

$$x > \frac{3}{4}$$

$$\text{Domain}(f) = \left(\frac{3}{4}, \infty \right)$$

(ii) For $f(x) = \frac{1}{\log(9 - x^2)}$: We apply two separate constraints: 1. Positivity of log argument:

$$9 - x^2 > 0$$

$$x^2 < 9$$

$$-3 < x < 3$$

2. Non-zero denominator:

$$\begin{aligned}\log(9 - x^2) &\neq 0 \\ 9 - x^2 &\neq 1 \quad (\text{since } \log 1 = 0) \\ x^2 &\neq 8 \\ x &\neq \pm\sqrt{8} = \pm 2\sqrt{2}\end{aligned}$$

Combining both conditions:

$$\text{Domain}(f) = (-3, 3) \setminus \{-2\sqrt{2}, 2\sqrt{2}\}$$

Final Answer

$$\begin{aligned}\text{(i) Domain} &= \left(\frac{3}{4}, \infty\right) \\ \text{(ii) Domain} &= (-3, 3) \setminus \{-2\sqrt{2}, 2\sqrt{2}\}\end{aligned}$$

Question 21

Find the domain of the following functions: (i) $f(x) = \frac{1}{3 - 2\sin x}$ (ii) $f(x) = \frac{1}{\sqrt{1 - \cos x}}$

TYPE 21 Finding Domain and Range of Trigonometric Functions

Given: (i) $f(x) = \frac{1}{3 - 2\sin x}$, (ii) $f(x) = \frac{1}{\sqrt{1 - \cos x}}$

To Find: Domain of each function

Concept / Formula Used: $-1 \leq \sin x \leq 1$ $-1 \leq \cos x \leq 1$

Solution:

(i) For $f(x) = \frac{1}{3 - 2\sin x}$: Denominator must not be zero:

$$\begin{aligned}3 - 2\sin x &\neq 0 \\ 2\sin x &\neq 3 \\ \sin x &\neq \frac{3}{2}\end{aligned}$$

Since $-1 \leq \sin x \leq 1$ for all real numbers x , $\sin x$ can never equal $\frac{3}{2}$. Therefore, $3 - 2\sin x > 0$ for all $x \in \mathbb{R}$.

$$\text{Domain}(f) = \mathbb{R}$$

(ii) For $f(x) = \frac{1}{\sqrt{1 - \cos x}}$: The term inside the square root in the denominator must be strictly positive:

$$\begin{aligned}1 - \cos x &> 0 \\ \cos x &< 1\end{aligned}$$

Since $\cos x \leq 1$ for all real numbers x , $\cos x < 1$ holds everywhere except where $\cos x = 1$. Solve for $\cos x = 1$:

$$x = 2n\pi, \quad n \in \mathbb{Z}$$

Thus, $1 - \cos x > 0$ for all $x \in \mathbb{R} \setminus \{2n\pi : n \in \mathbb{Z}\}$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{2n\pi : n \in \mathbb{Z}\}$$

Final Answer

- (i) Domain = $\mathbb{R}$
 (ii) Domain = $\mathbb{R} \setminus \{2n\pi : n \in \mathbb{Z}\}$

Question 22

Find the range of the following functions: (i) $f(x) = 2 - 3 \cos x$ (ii) $f(x) = 2 + 5 \sin 3x$

TYPE 21 Finding Domain and Range of Trigonometric Functions

Given: (i) $f(x) = 2 - 3 \cos x$, (ii) $f(x) = 2 + 5 \sin 3x$

To Find: Range of each function

Concept / Formula Used: $-1 \leq \cos x \leq 1$ $-1 \leq \sin \theta \leq 1$

Solution:

(i) For $f(x) = 2 - 3 \cos x$: For all $x \in \mathbb{R}$:

$$-1 \leq \cos x \leq 1$$

Multiply by -3 (reversing the inequality signs):

$$\begin{aligned} 3 &\geq -3 \cos x \geq -3 \\ -3 &\leq -3 \cos x \leq 3 \end{aligned}$$

Add 2 throughout:

$$\begin{aligned} 2 - 3 &\leq 2 - 3 \cos x \leq 2 + 3 \\ -1 &\leq f(x) \leq 5 \\ \text{Range}(f) &= [-1, 5] \end{aligned}$$

(ii) For $f(x) = 2 + 5 \sin 3x$: For all $x \in \mathbb{R}$, $\theta = 3x \in \mathbb{R}$, so:

$$-1 \leq \sin 3x \leq 1$$

Multiply by 5:

$$-5 \leq 5 \sin 3x \leq 5$$

Add 2 throughout:

$$\begin{aligned} 2 - 5 &\leq 2 + 5 \sin 3x \leq 2 + 5 \\ -3 &\leq f(x) \leq 7 \\ \text{Range}(f) &= [-3, 7] \end{aligned}$$

Final Answer

- (i) Range = $[-1, 5]$, (ii) Range = $[-3, 7]$

1.3 – One-One, Onto and Bijective Functions

Question 1

Show that the function $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(x) = 3x - 2$ is one-one.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(x) = 3x - 2$ is one-one (injective).

Concept / Formula Used: $f : A \rightarrow B \quad f(x_1) = f(x_2) \implies x_1 = x_2 \quad x_1, x_2 \in A$

Solution:

Let $x_1, x_2 \in \mathbb{Q}$ such that $f(x_1) = f(x_2)$.

By definition of $f(x)$:

$$3x_1 - 2 = 3x_2 - 2$$

Adding 2 to both sides:

$$3x_1 = 3x_2$$

Dividing both sides by 3:

$$x_1 = x_2$$

Since $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in \mathbb{Q}$, the function f is one-one.

Hence Proved

The function $f : \mathbb{Q} \rightarrow \mathbb{Q}$ is one-one.

Question 2

Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 2x - 1$ is not onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 2x - 1$ is not onto (surjective).

Concept / Formula Used: $f : A \rightarrow B \quad y \in B \quad x \in A \quad f(x) = y$

Solution:

Let $y \in \mathbb{N}$ be an element in the codomain.

We set $f(x) = y$:

$$\begin{aligned} 2x - 1 &= y \\ 2x &= y + 1 \\ x &= \frac{y + 1}{2} \end{aligned}$$

Consider $y = 2 \in \mathbb{N}$ (codomain).

Substituting $y = 2$:

$$x = \frac{2+1}{2}$$

$$x = \frac{3}{2} \notin \mathbb{N}$$

Thus, there is no element $x \in \mathbb{N}$ (domain) such that $f(x) = 2$.

Hence, f is not onto.

Hence Proved

The function $f : \mathbb{N} \rightarrow \mathbb{N}$ is not onto.

Common Mistake: Assuming $\frac{y+1}{2}$ is always a natural number. For even values of y , $\frac{y+1}{2}$ is not an integer, so it does not belong to $\mathbb{N}$.

Question 3

Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n^2$ is injective.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n^2$ is injective.

Concept / Formula Used: $f : A \rightarrow B$ $f(n_1) = f(n_2) \implies n_1 = n_2$ $n_1, n_2 \in A$

Solution:

Let $n_1, n_2 \in \mathbb{N}$ such that $f(n_1) = f(n_2)$.

By definition of $f(n)$:

$$n_1^2 = n_2^2$$

Taking the square root on both sides:

$$n_1 = \pm n_2$$

Since $n_1, n_2 \in \mathbb{N}$, both $n_1 > 0$ and $n_2 > 0$.

Therefore, $n_1 = -n_2$ is impossible for natural numbers.

Hence:

$$n_1 = n_2$$

Thus, $f(n_1) = f(n_2) \implies n_1 = n_2$, so f is injective.

Hence Proved

The function $f : \mathbb{N} \rightarrow \mathbb{N}$ is injective.

Question 4

Is the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 2$ one-one? Justify your answer.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 2$.

Concept / Formula Used: $f : A \rightarrow B \quad f(m_1) = f(m_2) \implies m_1 = m_2 \quad m_1, m_2 \in A$

Solution:

Let $m_1, m_2 \in \mathbb{N}$ such that $f(m_1) = f(m_2)$.

Substitute into the function definition:

$$m_1^2 + m_1 + 2 = m_2^2 + m_2 + 2$$

Subtract 2 from both sides:

$$m_1^2 + m_1 = m_2^2 + m_2$$

Rearranging terms to one side:

$$\begin{aligned}(m_1^2 - m_2^2) + (m_1 - m_2) &= 0 \\(m_1 - m_2)(m_1 + m_2) + (m_1 - m_2) &= 0 \\(m_1 - m_2)(m_1 + m_2 + 1) &= 0\end{aligned}$$

Since $m_1, m_2 \in \mathbb{N}$, we have $m_1 \geq 1$ and $m_2 \geq 1$, which implies:

$$m_1 + m_2 + 1 \geq 3 \neq 0$$

Dividing both sides by $(m_1 + m_2 + 1)$:

$$\begin{aligned}m_1 - m_2 &= 0 \\m_1 &= m_2\end{aligned}$$

Therefore, $f(m_1) = f(m_2) \implies m_1 = m_2$ for all $m_1, m_2 \in \mathbb{N}$.

Hence, the function f is one-one.

Final Answer

Yes, the function f is one-one.

Question 5

What is the range of the function $f(x) = \frac{|x-1|}{x-1}$?

TYPE 15 Finding Domain and Range of Modulus Functions

To Find: The range of the function $f(x) = \frac{|x-1|}{x-1}$.

Concept / Formula Used: $|x-1| = \begin{cases} x-1, & \text{if } x > 1 \\ -(x-1), & \text{if } x < 1 \end{cases}$

Solution:

The function is defined for all $x \in \mathbb{R} \setminus \{1\}$.

Case 1: When $x > 1$:

$$|x - 1| = x - 1$$

$$f(x) = \frac{x - 1}{x - 1} = 1$$

Case 2: When $x < 1$:

$$|x - 1| = -(x - 1)$$

$$f(x) = \frac{-(x - 1)}{x - 1} = -1$$

Thus, the only values taken by $f(x)$ are -1 and 1 .

Hence, the range of $f(x)$ is $\{-1, 1\}$.

Final Answer

Range of $f = \{-1, 1\}$

Common Mistake: Including $x = 1$ in the domain. At $x = 1$, the expression is $\frac{0}{0}$ which is undefined.

Question 6

Find the domain of the function $f(x) = \frac{1}{5 - 3 \sin x}$.

TYPE 21 Finding Domain and Range of Trigonometric Functions

To Find: The domain of $f(x) = \frac{1}{5 - 3 \sin x}$.

Concept / Formula Used: $\frac{1}{g(x)}$ $g(x) \neq 0$ $-1 \leq \sin x \leq 1$

Solution:

The given function is:

$$f(x) = \frac{1}{5 - 3 \sin x}$$

For $f(x)$ to be defined, the denominator must not be zero:

$$5 - 3 \sin x \neq 0$$

$$3 \sin x \neq 5$$

$$\sin x \neq \frac{5}{3}$$

Using the standard result $-1 \leq \sin x \leq 1$ for all real x :

Multiplying by -3 :

$$-3 \leq -3 \sin x \leq 3$$

Adding 5 to all parts:

$$\begin{aligned} 5 - 3 &\leq 5 - 3 \sin x \leq 5 + 3 \\ 2 &\leq 5 - 3 \sin x \leq 8 \end{aligned}$$

Since $5 - 3 \sin x \geq 2 > 0$ for all real numbers x , the denominator is never zero for any $x \in \mathbb{R}$. Thus, $f(x)$ is defined for all $x \in \mathbb{R}$.

Final Answer

Domain of $f = \mathbb{R}$

Question 7

Find the range of the function $f(x) = \frac{1}{2 - \cos x}$.

TYPE 21 Finding Domain and Range of Trigonometric Functions

To Find: The range of $f(x) = \frac{1}{2 - \cos x}$.

Concept / Formula Used: $x \quad -1 \leq \cos x \leq 1$

Solution:

Using the standard result $-1 \leq \cos x \leq 1$ for all $x \in \mathbb{R}$:

Multiply the inequality by -1 (which reverses inequality signs):

$$-1 \leq -\cos x \leq 1$$

Add 2 to all parts of the inequality:

$$\begin{aligned} 2 - 1 &\leq 2 - \cos x \leq 2 + 1 \\ 1 &\leq 2 - \cos x \leq 3 \end{aligned}$$

Taking the reciprocal of all terms (since all terms are positive):

$$\begin{aligned} \frac{1}{3} &\leq \frac{1}{2 - \cos x} \leq \frac{1}{1} \\ \frac{1}{3} &\leq f(x) \leq 1 \end{aligned}$$

Therefore, the range of $f(x)$ is $\left[\frac{1}{3}, 1\right]$.

Final Answer

Range of $f = \left[\frac{1}{3}, 1\right]$

Question 8

Find the range of the following functions:

- (i) $f(x) = \sin x + \cos x$
- (ii) $f(x) = 3 \sin x + 4 \cos x$

TYPE 21 Finding Domain and Range of Trigonometric Functions

To Find: The range of (i) $f(x) = \sin x + \cos x$ and (ii) $f(x) = 3 \sin x + 4 \cos x$.

Concept / Formula Used: $a \sin x + b \cos x \quad -\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$

Solution:

Part (i): $f(x) = \sin x + \cos x$

Here, $a = 1$ and $b = 1$.

Calculate $\sqrt{a^2 + b^2}$:

$$\begin{aligned}\sqrt{a^2 + b^2} &= \sqrt{1^2 + 1^2} \\ &= \sqrt{1 + 1} \\ &= \sqrt{2}\end{aligned}$$

Using the standard result $-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$:

$$-\sqrt{2} \leq \sin x + \cos x \leq \sqrt{2}$$

Thus, the range of $f(x) = \sin x + \cos x$ is $[-\sqrt{2}, \sqrt{2}]$.

Part (ii): $f(x) = 3 \sin x + 4 \cos x$

Here, $a = 3$ and $b = 4$.

Calculate $\sqrt{a^2 + b^2}$:

$$\begin{aligned}\sqrt{a^2 + b^2} &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

Using the standard result $-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$:

$$-5 \leq 3 \sin x + 4 \cos x \leq 5$$

Thus, the range of $f(x) = 3 \sin x + 4 \cos x$ is $[-5, 5]$.

Final Answer

(i) Range of $f = [-\sqrt{2}, \sqrt{2}]$

(ii) Range of $f = [-5, 5]$

Question 9

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |2x - 3|$.

(i) Is f one-one?

(ii) Is f onto?

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = |2x - 3|$.

Concept / Formula Used: $f : A \rightarrow B \quad f(x_1) = f(x_2) \implies x_1 = x_2 \quad f \quad B$

Solution:

Part (i): Checking for One-One

Let us find two distinct elements in $\mathbb{R}$ with the same image.

$$\text{Set } 2x - 3 = 1 \implies 2x = 4 \implies x = 2.$$

$$\text{Set } 2x - 3 = -1 \implies 2x = 2 \implies x = 1.$$

Now evaluate $f(x)$ at these points:

$$f(2) = |2(2) - 3| = |4 - 3| = |1| = 1$$

$$f(1) = |2(1) - 3| = |2 - 3| = |-1| = 1$$

Since $f(2) = f(1) = 1$ but $2 \neq 1$, the function f is not one-one.

Part (ii): Checking for Onto

By definition of absolute value, $|2x - 3| \geq 0$ for all $x \in \mathbb{R}$.

Thus:

$$\text{Range}(f) = [0, \infty)$$

However, the codomain is $\mathbb{R}$ (which includes negative real numbers).

For example, for $y = -2 \in \mathbb{R}$ (codomain), there is no $x \in \mathbb{R}$ such that $f(x) = -2$.

Since $\text{Range}(f) = [0, \infty) \neq \mathbb{R}$ (codomain), the function f is not onto.

Final Answer

(i) f is not one-one.

(ii) f is not onto.

Question 10

If $A = \{1, 2, 3\}$ and $B = \{3, 5\}$, then write the number of functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{1, 2, 3\}$ and $B = \{3, 5\}$.

Concept / Formula Used: $|A| = m$ $|B| = n$ A B n^m

Solution:

Number of elements in set A :

$$m = |A| = 3$$

Number of elements in set B :

$$n = |B| = 2$$

Using the standard result for total number of functions $N = n^m$:

$$\begin{aligned} N &= 2^3 \\ &= 8 \end{aligned}$$

Final Answer

Number of functions from A to $B = 8$

Question 11

If $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$, then write the number of functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$.

Concept / Formula Used: $|A| = m$ $|B| = n$ A B n^m

Solution:

Number of elements in set A :

$$m = |A| = 3$$

Number of elements in set B :

$$n = |B| = 4$$

Using the formula $N = n^m$:

$$\begin{aligned} N &= 4^3 \\ &= 64 \end{aligned}$$

Final Answer

Number of functions from A to $B = 64$

Question 12

If $A = \{a, b\}$ and $B = \{1, 3, 5\}$, then write the number of one-one functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{a, b\}$ and $B = \{1, 3, 5\}$.

Concept / Formula Used: $|A| = m$ $|B| = n$ A B ${}^nP_m = \frac{n!}{(n-m)!}$ $m \leq n$ 0 $m > n$

Solution:

Number of elements in set A :

$$m = |A| = 2$$

Number of elements in set B :

$$n = |B| = 3$$

Since $m \leq n$, using the formula $N = {}^nP_m$:

$$\begin{aligned} N &= {}^3P_2 \\ &= \frac{3!}{(3-2)!} \\ &= \frac{6}{1!} \\ &= 6 \end{aligned}$$

Final Answer

Number of one-one functions from A to $B = 6$

Question 13

If $A = \{a, b, c\}$ and $B = \{p, q\}$, then write the number of onto functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{a, b, c\}$ and $B = \{p, q\}$.

Concept / Formula Used: $|A| = m \geq 2$ $|B| = 2$ A B $2^m - 2$

Solution:

Number of elements in A :

$$m = |A| = 3$$

Number of elements in B :

$$n = |B| = 2$$

Using the standard formula for $n = 2$:

$$\begin{aligned} \text{Number of onto functions} &= 2^m - 2 \\ &= 2^3 - 2 \\ &= 8 - 2 \\ &= 6 \end{aligned}$$

Final Answer

Number of onto functions from A to $B = 6$

Question 14

If $A = \{1, 2, 3, 4\}$ and $B = \{-1, 3\}$, then write the number of onto functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{1, 2, 3, 4\}$ and $B = \{-1, 3\}$.

Concept / Formula Used: $|A| = m \geq 2$ $|B| = 2$ A B $2^m - 2$

Solution:

Number of elements in A :

$$m = |A| = 4$$

Number of elements in B :

$$n = |B| = 2$$

Using the standard formula $N = 2^m - 2$:

$$\begin{aligned} N &= 2^4 - 2 \\ &= 16 - 2 \\ &= 14 \end{aligned}$$

Final Answer

Number of onto functions from A to $B = 14$

Question 15

If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, then write the number of

- (i) one-one onto functions from A to B
- (ii) bijections from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.

Concept / Formula Used: $|A| = |B| = n \quad A \quad B \quad n!$

Solution:

Number of elements in set A and set B :

$$\begin{aligned} m &= |A| = 3 \\ n &= |B| = 3 \end{aligned}$$

Since $m = n = 3$, using the formula $N = n!$:

$$\begin{aligned} N &= 3! \\ &= 3 \times 2 \times 1 \\ &= 6 \end{aligned}$$

Note that "one-one onto functions" and "bijections" refer to the exact same set of functions.

Part (i): Number of one-one onto functions = 6.

Part (ii): Number of bijections = 6.

Final Answer

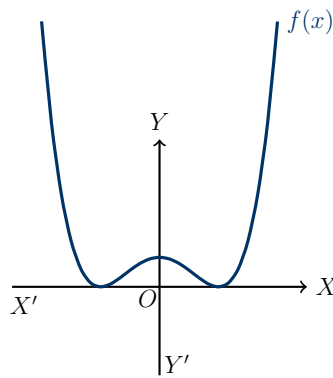
- (i) Number of one-one onto functions = 6
- (ii) Number of bijections = 6

Question 16

The given function $f : \mathbb{R} \rightarrow \mathbb{R}$ is not 'onto' function. Give reason. (ISC Specimen Paper 2025)

TYPE 22 Testing Injectivity and Surjectivity of Functions

★ Likely Exam Question

Graph of function $f : \mathbb{R} \rightarrow \mathbb{R}$

Given: A graph of a function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Concept / Formula Used: $f : \mathbb{R} \rightarrow \mathbb{R}$ $\mathbb{R}$ $y = c$ $c \in \mathbb{R}$

Solution:

From the given graph, we observe that:

1. As $x \rightarrow \infty$ and $x \rightarrow -\infty$, the curve extends upwards towards $+\infty$.
2. The graph has a global minimum value below which it does not exist (the curve does not extend downwards to $-\infty$).
3. Consequently, for negative values of y below the global minimum, there are no corresponding points (x, y) on the graph.

Therefore:

$$\text{Range}(f) \subset \mathbb{R} \quad \text{and} \quad \text{Range}(f) \neq \mathbb{R}$$

Since the range of the function is not equal to its codomain $\mathbb{R}$, f is not an onto function.

Hence Proved

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is not onto because its range is a proper subset of the codomain $\mathbb{R}$.

Question 17

Let $A = \{1, 2\}$, $B = \{3, 6\}$, function $f : A \rightarrow B$ be defined by $f(x) = 3x$ and function $g : A \rightarrow B$ be defined by $g(x) = x^2 + 2$. Show that $f = g$.

TYPE 24 Testing Equality of Functions

To Prove: $f = g$, where $f(x) = 3x$ and $g(x) = x^2 + 2$ on domain $A = \{1, 2\}$.

Concept / Formula Used: $f : A \rightarrow B$ $g : A \rightarrow B$ $f(x) = g(x)$ $x \in A$

Solution:

Domain of both functions is $A = \{1, 2\}$.

For $x = 1 \in A$:

$$\begin{aligned} f(1) &= 3(1) = 3 \\ g(1) &= (1)^2 + 2 = 1 + 2 = 3 \end{aligned}$$

Thus, $f(1) = g(1) = 3$.

For $x = 2 \in A$:

$$\begin{aligned} f(2) &= 3(2) = 6 \\ g(2) &= (2)^2 + 2 = 4 + 2 = 6 \end{aligned}$$

Thus, $f(2) = g(2) = 6$.

Since $f(x) = g(x)$ for all $x \in A$, the two functions are equal.

Hence Proved

$f = g$, as $f(x) = g(x)$ for all $x \in A$.

Question 18

Check the injectivity and surjectivity of the following functions:

- (i) $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2$
- (ii) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^2$
- (iii) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$
- (iv) $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^3$
- (v) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^3$
- (vi) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Six functions specified on different domains and codomains $(\mathbb{N}, \mathbb{Z}, \mathbb{R})$.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad f$

Solution:

Part (i): $f : \mathbb{N} \rightarrow \mathbb{N}, f(x) = x^2$

Injectivity: Let $f(x_1) = f(x_2) \implies x_1^2 = x_2^2 \implies x_1 = x_2$ (since $x_1, x_2 \in \mathbb{N}$). Thus, f is injective.

Surjectivity: $\text{Range}(f) = \{1, 4, 9, 16, \dots\} \neq \mathbb{N}$. For example, $2 \in \mathbb{N}$ has no pre-image in $\mathbb{N}$ since $x = \sqrt{2} \notin \mathbb{N}$. Thus, f is not surjective.

Part (ii): $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^2$

Injectivity: $f(-1) = (-1)^2 = 1$ and $f(1) = 1^2 = 1$. Since $f(-1) = f(1)$ but $-1 \neq 1$, f is not injective.

Surjectivity: $\text{Range}(f) = \{0, 1, 4, 9, \dots\} \neq \mathbb{Z}$. Negative integers like $-2 \in \mathbb{Z}$ have no pre-image. Thus, f is not surjective.

Part (iii): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$

Injectivity: $f(-2) = 4$ and $f(2) = 4$. Since $f(-2) = f(2)$ but $-2 \neq 2$, f is not injective.

Surjectivity: $\text{Range}(f) = [0, \infty) \neq \mathbb{R}$. Negative real numbers have no pre-image. Thus, f is not surjective.

Part (iv): $f : \mathbb{N} \rightarrow \mathbb{N}, f(x) = x^3$

Injectivity: $f(x_1) = f(x_2) \implies x_1^3 = x_2^3 \implies x_1 = x_2$. Thus, f is injective.

Surjectivity: $\text{Range}(f) = \{1, 8, 27, \dots\} \neq \mathbb{N}$. For example, $2 \in \mathbb{N}$ has no pre-image in $\mathbb{N}$. Thus, f is not surjective.

Part (v): $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^3$

Injectivity: $f(x_1) = f(x_2) \implies x_1^3 = x_2^3 \implies x_1 = x_2$ (over real/integer numbers). Thus, f is injective.

Surjectivity: $\text{Range}(f) = \{\dots, -8, -1, 0, 1, 8, \dots\} \neq \mathbb{Z}$. For example, $2 \in \mathbb{Z}$ has no pre-image in $\mathbb{Z}$ ($x = \sqrt[3]{2} \notin \mathbb{Z}$). Thus, f is not surjective.

Part (vi): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$

Injectivity: $f(x_1) = f(x_2) \implies x_1^3 = x_2^3 \implies x_1 = x_2$. Thus, f is injective.

Surjectivity: For any $y \in \mathbb{R}$, taking $x = y^{1/3} \in \mathbb{R}$, we get $f(x) = (y^{1/3})^3 = y$. Thus, $\text{Range}(f) = \mathbb{R}$, so f is surjective.

Final Answer

- (i) Injective, not surjective.
- (ii) Neither injective nor surjective.
- (iii) Neither injective nor surjective.
- (iv) Injective, not surjective.
- (v) Injective, not surjective.
- (vi) Both injective and surjective (bijective).

Question 19

Let $\mathbb{I}$ be the set of all integers. Show that the function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = |x|$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = |x|$ is neither one-one nor onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2$ f

Solution:

1. Injectivity (One-One) Test:

Consider $x_1 = 1 \in \mathbb{I}$ and $x_2 = -1 \in \mathbb{I}$.

Evaluating $f(x)$ at these points:

$$\begin{aligned} f(1) &= |1| = 1 \\ f(-1) &= |-1| = 1 \end{aligned}$$

Here $f(1) = f(-1) = 1$, but $1 \neq -1$.

Hence, f is not one-one.

2. Surjectivity (Onto) Test:

By definition, $|x| \geq 0$ for all $x \in \mathbb{I}$.

Thus:

$$\text{Range}(f) = \{0, 1, 2, 3, \dots\}$$

Consider $-3 \in \mathbb{I}$ (codomain).

There is no integer $x \in \mathbb{I}$ such that $f(x) = |x| = -3$.

Hence, $\text{Range}(f) \neq \mathbb{I}$, so f is not onto.

Hence Proved

The function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = |x|$ is neither one-one nor onto.

Question 20

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is many-one into.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is many-one into.

Concept / Formula Used: $\neq$

Solution:

1. Proving Many-One:

Consider $x = 1 \in \mathbb{R}$ and $x = -1 \in \mathbb{R}$:

$$\begin{aligned} f(1) &= 3(1)^2 - 2 = 3(1) - 2 = 1 \\ f(-1) &= 3(-1)^2 - 2 = 3(1) - 2 = 1 \end{aligned}$$

Since $f(1) = f(-1) = 1$ but $1 \neq -1$, f is not one-one, which means f is many-one.

2. Proving Into:

For any $x \in \mathbb{R}$, we know $x^2 \geq 0$.

Multiplying by 3:

$$3x^2 \geq 0$$

Subtracting 2:

$$\begin{aligned} 3x^2 - 2 &\geq -2 \\ f(x) &\geq -2 \end{aligned}$$

Thus, $\text{Range}(f) = [-2, \infty)$.

Since $\text{Range}(f) = [-2, \infty) \neq \mathbb{R}$ (codomain), negative numbers less than -2 (such as $-5 \in \mathbb{R}$) have no pre-image in $\mathbb{R}$.

Hence, f is an into function.

Combining both properties, f is many-one into.

Hence Proved

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is many-one into.

Question 21

Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and $f = \{(1, 4), (2, 5), (3, 6)\}$ be a function from A to B . Show that f is one-one but not onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f = \{(1, 4), (2, 5), (3, 6)\}$ from A to B is one-one but not onto.

Concept / Formula Used: $f: A \rightarrow B$

Solution:

Given function mapping as ordered pairs:

$$f(1) = 4$$

$$f(2) = 5$$

$$f(3) = 6$$

1. Injectivity (One-One):

Distinct elements in domain $A = \{1, 2, 3\}$ have distinct images in codomain B :

$$f(1) \neq f(2) \neq f(3)$$

Therefore, f is one-one.

2. Surjectivity (Onto):

The range of f is:

$$\text{Range}(f) = \{4, 5, 6\}$$

The codomain is $B = \{4, 5, 6, 7\}$.

Element $7 \in B$ has no pre-image in A .

Since $\text{Range}(f) = \{4, 5, 6\} \neq B$, f is not onto.

Hence Proved

The function f is one-one but not onto.

Question 22

In each of the following cases, state whether the function is bijective or not. Justify your answer.

- (i) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 1$
- (ii) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3 - 4x$
- (iii) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1 + x^2$

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Three functions defined on $\mathbb{R} \rightarrow \mathbb{R}$.

Solution:

Part (i): $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1$

Injectivity: Let $f(x_1) = f(x_2) \implies 2x_1 + 1 = 2x_2 + 1 \implies 2x_1 = 2x_2 \implies x_1 = x_2$. So f is one-one.

Surjectivity: Let $y \in \mathbb{R}$. Set $2x + 1 = y \implies x = \frac{y-1}{2} \in \mathbb{R}$. Then $f(x) = 2\left(\frac{y-1}{2}\right) + 1 = y$. So f is onto.

Since f is both one-one and onto, f is bijective.

Part (ii): $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3 - 4x$

Injectivity: Let $f(x_1) = f(x_2) \implies 3 - 4x_1 = 3 - 4x_2 \implies -4x_1 = -4x_2 \implies x_1 = x_2$. So f is one-one.

Surjectivity: Let $y \in \mathbb{R}$. Set $3 - 4x = y \implies 4x = 3 - y \implies x = \frac{3-y}{4} \in \mathbb{R}$. Then $f(x) = 3 - 4\left(\frac{3-y}{4}\right) = y$. So f is onto.

Since f is both one-one and onto, f is bijective.

Part (iii): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 1 + x^2$

Injectivity: $f(1) = 1 + 1^2 = 2$ and $f(-1) = 1 + (-1)^2 = 2$. Since $f(1) = f(-1) = 2$ but $1 \neq -1$, f is not one-one.

Since it is not one-one, f is not bijective.

Final Answer

- (i) Bijective.
- (ii) Bijective.
- (iii) Not bijective.

Question 23

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^4$. Show that f is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^4$ is neither one-one nor onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad f \quad \mathbb{R}$

Solution:

1. Injectivity (One-One) Test:

Consider $x_1 = 1 \in \mathbb{R}$ and $x_2 = -1 \in \mathbb{R}$.

$$\begin{aligned} f(1) &= (1)^4 = 1 \\ f(-1) &= (-1)^4 = 1 \end{aligned}$$

Since $f(1) = f(-1) = 1$ but $1 \neq -1$, f is not one-one.

2. Surjectivity (Onto) Test:

For any real number x , $x^4 \geq 0$.

$$\text{Range}(f) = [0, \infty)$$

Consider $-2 \in \mathbb{R}$ (codomain).

There exists no real number $x \in \mathbb{R}$ such that $x^4 = -2$.

Since $\text{Range}(f) = [0, \infty) \neq \mathbb{R}$, f is not onto.

Hence, f is neither one-one nor onto.

Hence Proved

The function $f(x) = x^4$ is neither one-one nor onto.

Question 24

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$ is neither one-one nor onto.

Concept / Formula Used: $\mathbb{R} \quad -1 \leq \cos x \leq 1$

Solution:

1. Injectivity (One-One) Test:

Consider $x_1 = 0 \in \mathbb{R}$ and $x_2 = 2\pi \in \mathbb{R}$:

$$\begin{aligned} f(0) &= \cos(0) = 1 \\ f(2\pi) &= \cos(2\pi) = 1 \end{aligned}$$

Since $f(0) = f(2\pi) = 1$ but $0 \neq 2\pi$, f is not one-one.

2. Surjectivity (Onto) Test:

Using the standard result $-1 \leq \cos x \leq 1$ for all $x \in \mathbb{R}$:

$$\text{Range}(f) = [-1, 1]$$

Consider $y = 2 \in \mathbb{R}$ (codomain).

There exists no $x \in \mathbb{R}$ such that $\cos x = 2$.

Since $\text{Range}(f) = [-1, 1] \neq \mathbb{R}$, f is not onto.

Hence, f is neither one-one nor onto.

Hence Proved

The function $f(x) = \cos x$ is neither one-one nor onto.

Question 25

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin^2 x + \cos^2 x$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin^2 x + \cos^2 x$ is neither one-one nor onto.

Concept / Formula Used: $\sin^2 x + \cos^2 x = 1 \quad x \in \mathbb{R}$

Solution:

Using the standard identity $\sin^2 x + \cos^2 x = 1$:

$$f(x) = 1 \quad \text{for all } x \in \mathbb{R}$$

Thus, $f(x)$ is a constant function.

1. Injectivity (One-One) Test:

Take $x_1 = 0$ and $x_2 = \frac{\pi}{2}$:

$$\begin{aligned} f(0) &= 1 \\ f\left(\frac{\pi}{2}\right) &= 1 \end{aligned}$$

Since $f(0) = f\left(\frac{\pi}{2}\right) = 1$ but $0 \neq \frac{\pi}{2}$, f is not one-one.

2. Surjectivity (Onto) Test:

The range of f contains only a single element:

$$\text{Range}(f) = \{1\}$$

Since $\text{Range}(f) = \{1\} \neq \mathbb{R}$ (codomain), any value $y \neq 1$ in $\mathbb{R}$ has no pre-image.

Hence, f is not onto.

Thus, f is neither one-one nor onto.

Hence Proved

The constant function $f(x) = \sin^2 x + \cos^2 x = 1$ is neither one-one nor onto.

Question 26

Show that the function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: Function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is neither one-one nor onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad f \quad \mathbb{I}$

Solution:**1. Injectivity (One-One) Test:**

Consider $x_1 = 0 \in \mathbb{I}$ and $x_2 = -1 \in \mathbb{I}$:

$$\begin{aligned} f(0) &= (0)^2 + 0 = 0 \\ f(-1) &= (-1)^2 + (-1) = 1 - 1 = 0 \end{aligned}$$

Since $f(0) = f(-1) = 0$ but $0 \neq -1$, f is not one-one.

2. Surjectivity (Onto) Test:

Rewrite $f(x)$:

$$f(x) = x(x+1)$$

For any integer x , one of x or $x+1$ must be even.

Thus, $x(x+1)$ is always an even integer.

Therefore, $f(x)$ takes only even integer values, so $\text{Range}(f) \subseteq \{\text{even integers}\}$.

Odd integers such as $1 \in \mathbb{I}$ have no pre-image in $\mathbb{I}$.

For example, setting $x^2 + x = 1$:

$$\begin{aligned} x^2 + x - 1 &= 0 \\ x &= \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2} \notin \mathbb{I} \end{aligned}$$

Since $\text{Range}(f) \neq \mathbb{I}$, f is not onto.

Hence, f is neither one-one nor onto.

Hence Proved

The function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is neither one-one nor onto.

Question 27

If the function $f : \mathbb{I} \rightarrow \mathbb{I}$ is defined by

$$f(x) = \begin{cases} 2x + 1, & \text{if } x \text{ is odd} \\ 2x - 1, & \text{if } x \text{ is even} \end{cases}$$

- (i) Find $f(3)$.
 - (ii) Find $f(4)$.
 - (iii) Find x , if $f(x) = -9$.
- Is this function one-one?

TYPE 25 Testing Injectivity and Surjectivity of Piecewise Functions

Given: Function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined piecewise based on parity of x .

Concept / Formula Used: $f(x)$ x

Solution:

Part (i): Finding $f(3)$

Since $x = 3$ is odd, we use the branch $f(x) = 2x + 1$:

$$\begin{aligned} f(3) &= 2(3) + 1 \\ &= 6 + 1 \\ &= 7 \end{aligned}$$

Part (ii): Finding $f(4)$

Since $x = 4$ is even, we use the branch $f(x) = 2x - 1$:

$$\begin{aligned} f(4) &= 2(4) - 1 \\ &= 8 - 1 \\ &= 7 \end{aligned}$$

Part (iii): Finding x when $f(x) = -9$

Case 1: If x is odd:

$$\begin{aligned} 2x + 1 &= -9 \\ 2x &= -10 \\ x &= -5 \quad (\text{which is odd, valid solution}) \end{aligned}$$

Case 2: If x is even:

$$\begin{aligned} 2x - 1 &= -9 \\ 2x &= -8 \\ x &= -4 \quad (\text{which is even, valid solution}) \end{aligned}$$

Thus, $x = -5$ or $x = -4$.

Checking if f is one-one:

From parts (i) and (ii), we have:

$$\begin{aligned} f(3) &= 7 \\ f(4) &= 7 \end{aligned}$$

Since $f(3) = f(4) = 7$ but $3 \neq 4$, the function f is not one-one.

Final Answer

(i) $f(3) = 7$

(ii) $f(4) = 7$

(iii) $x = -5$ or $x = -4$

Is f one-one? No, f is not one-one.

Question 28

(i) Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(n) = \begin{cases} n+1, & \text{if } n \text{ is odd} \\ n-1, & \text{if } n \text{ is even} \end{cases}$$

is both one-one and onto.

(ii) Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n - (-1)^n$ for all $n \in \mathbb{N}$ is bijective.

TYPE 25 Testing Injectivity and Surjectivity of Piecewise Functions

★ Likely Exam Question

To Prove: Both given representations define a bijective function $f : \mathbb{N} \rightarrow \mathbb{N}$.

Solution:

Part (i):

1. Injectivity (One-One):

Let $n_1, n_2 \in \mathbb{N}$ such that $f(n_1) = f(n_2)$.

- Case A: Both n_1, n_2 are odd. Then $n_1 + 1 = n_2 + 1 \implies n_1 = n_2$.

- Case B: Both n_1, n_2 are even. Then $n_1 - 1 = n_2 - 1 \implies n_1 = n_2$.

- Case C: n_1 is odd and n_2 is even. Then $f(n_1) = n_1 + 1$ (which is even) and $f(n_2) = n_2 - 1$ (which is odd). An even number cannot equal an odd number, so $f(n_1) = f(n_2)$ is impossible in this case.

Thus, $f(n_1) = f(n_2) \implies n_1 = n_2$ in all valid cases. So f is one-one.

2. Surjectivity (Onto):

Let $m \in \mathbb{N}$ (codomain).

- If m is odd, then $m + 1$ is even. Then $f(m + 1) = (m + 1) - 1 = m$.

- If m is even, then $m - 1$ is odd (and since $m \geq 2$, $m - 1 \in \mathbb{N}$). Then $f(m - 1) = (m - 1) + 1 = m$.

Thus, every $m \in \mathbb{N}$ has a pre-image in $\mathbb{N}$. So f is onto.

Hence, f is both one-one and onto.

Part (ii):

For $n \in \mathbb{N}$:

$$\text{If } n \text{ is odd, } (-1)^n = -1 \implies f(n) = n - (-1) = n + 1$$

$$\text{If } n \text{ is even, } (-1)^n = 1 \implies f(n) = n - 1$$

This is identical to the function defined in Part (i).

Since the function in Part (i) was shown to be both one-one and onto, $f(n) = n - (-1)^n$ is also bijective.

Hence Proved

Both functions are proved to be bijective on $\mathbb{N}$.

Question 29

Find the number of all one-one functions from the set $\{1, 2, 3, 4\}$ to itself.

TYPE 23 Counting Functions Between Finite Sets

To Find: The number of one-one functions from $A = \{1, 2, 3, 4\}$ to A .

Concept / Formula Used: $A \quad n \quad n!$

Solution:

Let $A = \{1, 2, 3, 4\}$.

The number of elements in set A is:

$$n = |A| = 4$$

A one-one function from a finite set to itself is a permutation of the set elements.

Using the formula $N = n!$:

$$\begin{aligned} N &= 4! \\ &= 4 \times 3 \times 2 \times 1 \\ &= 24 \end{aligned}$$

Final Answer

Number of one-one functions = 24

Question 30

Show that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{5x-3}{4}$ is both one-one and onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{5x-3}{4}$ is both one-one and onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad y \in \mathbb{R} \quad \exists x \in \mathbb{R} \quad f(x) = y$

Solution:

1. Injectivity (One-One) Proof:

Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$.

Substitute into the function definition:

$$\frac{5x_1 - 3}{4} = \frac{5x_2 - 3}{4}$$

Multiply both sides by 4:

$$5x_1 - 3 = 5x_2 - 3$$

Add 3 to both sides:

$$5x_1 = 5x_2$$

Divide both sides by 5:

$$x_1 = x_2$$

Therefore, f is one-one.

2. Surjectivity (Onto) Proof:

Let $y \in \mathbb{R}$ be an arbitrary element in the codomain.

Set $f(x) = y$:

$$\begin{aligned}\frac{5x-3}{4} &= y \\ 5x-3 &= 4y \\ 5x &= 4y+3 \\ x &= \frac{4y+3}{5}\end{aligned}$$

Since $y \in \mathbb{R}$, we have $x = \frac{4y+3}{5} \in \mathbb{R}$ (domain).

Now evaluate $f(x)$:

$$\begin{aligned}f(x) &= f\left(\frac{4y+3}{5}\right) \\ &= \frac{5\left(\frac{4y+3}{5}\right)-3}{4} \\ &= \frac{(4y+3)-3}{4} \\ &= \frac{4y}{4} \\ &= y\end{aligned}$$

Thus, for every $y \in \mathbb{R}$, there exists an $x \in \mathbb{R}$ such that $f(x) = y$.

Therefore, f is onto.

Hence, f is both one-one and onto.

Hence Proved

The function $f(x) = \frac{5x-3}{4}$ is both one-one and onto.

Question 31

Let $f: \mathbb{R} - \left\{-\frac{4}{3}\right\} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \frac{4x}{3x+4}$. Show that f is a one-one function. Also check whether f is an onto function or not.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: A function $f: \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{4x}{3x+4}$.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad x_1, x_2 \in \text{Domain} \quad y \in \text{Codomain} \quad x \in \text{Domain} \quad f(x) = y$

Solution:

Step 1: Proof for One-One (Injectivity)

Let $x_1, x_2 \in \mathbb{R} - \{-\frac{4}{3}\}$ such that:

$$f(x_1) = f(x_2)$$

Substituting the function definition:

$$\frac{4x_1}{3x_1+4} = \frac{4x_2}{3x_2+4}$$

Dividing both sides by 4:

$$\frac{x_1}{3x_1+4} = \frac{x_2}{3x_2+4}$$

Cross-multiplying:

$$\begin{aligned} x_1(3x_2+4) &= x_2(3x_1+4) \\ 3x_1x_2+4x_1 &= 3x_1x_2+4x_2 \end{aligned}$$

Subtracting $3x_1x_2$ from both sides:

$$\begin{aligned} 4x_1 &= 4x_2 \\ x_1 &= x_2 \end{aligned}$$

Since $f(x_1) = f(x_2) \implies x_1 = x_2$, the function f is a ****one-one (injective)**** function.

Step 2: Check for Onto (Surjectivity)

Let $y \in \mathbb{R}$ (codomain). Set $f(x) = y$:

$$\begin{aligned} y &= \frac{4x}{3x+4} \\ y(3x+4) &= 4x \\ 3xy+4y &= 4x \\ 4x-3xy &= 4y \\ x(4-3y) &= 4y \\ x &= \frac{4y}{4-3y} \end{aligned}$$

For x to be a well-defined real number in the domain $\mathbb{R} - \{-\frac{4}{3}\}$, the denominator cannot be zero:

$$\begin{aligned} 4-3y &\neq 0 \\ 3y &\neq 4 \\ y &\neq \frac{4}{3} \end{aligned}$$

If $y = \frac{4}{3} \in \mathbb{R}$ (codomain), let us check if there exists any pre-image x :

$$\begin{aligned}\frac{4x}{3x+4} &= \frac{4}{3} \\ 12x &= 4(3x+4) \\ 12x &= 12x+16 \\ 0 &= 16 \quad (\text{which is impossible})\end{aligned}$$

Thus, the element $y = \frac{4}{3}$ in the codomain has no pre-image in the domain $\mathbb{R} - \{-\frac{4}{3}\}$.
Therefore, f is **not** an onto function.

Final Answer

$f(x)$ is a **one-one** function, but it is **not** an onto function because $y = \frac{4}{3} \in \mathbb{R}$ has no pre-image in the domain.

Question 32

If $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$, then show that the function $f: A \rightarrow B$ defined by $f(x) = 2x$ is both one-one and onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Sets $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$, and function $f: A \rightarrow B$ defined by $f(x) = 2x$.

To Prove: The function f is both one-one and onto.

Concept / Formula Used: $f: A \rightarrow B$ A B $f: A \rightarrow B$ Range of $f =$
Codomain B

Solution:

Evaluating the image $f(x) = 2x$ for every element $x \in A$:

$$\begin{aligned}\text{For } x = 1, \quad f(1) &= 2(1) = 2 \\ \text{For } x = 2, \quad f(2) &= 2(2) = 4 \\ \text{For } x = 3, \quad f(3) &= 2(3) = 6 \\ \text{For } x = 4, \quad f(4) &= 2(4) = 8\end{aligned}$$

1. Proof of One-One (Injectivity): The images of the elements 1, 2, 3, 4 under f are 2, 4, 6, 8 respectively. Different elements in the domain A have different images in the codomain B :

$$1 \neq 2 \neq 3 \neq 4 \implies f(1) \neq f(2) \neq f(3) \neq f(4)$$

Alternatively, for any $x_1, x_2 \in A$:

$$\begin{aligned}f(x_1) &= f(x_2) \\ 2x_1 &= 2x_2 \\ x_1 &= x_2\end{aligned}$$

Hence, f is a one-one function.

2. Proof of Onto (Surjectivity): The range of f is the set of all image values:

$$\text{Range}(f) = \{f(1), f(2), f(3), f(4)\} = \{2, 4, 6, 8\}$$

The given codomain is:

$$\text{Codomain } B = \{2, 4, 6, 8\}$$

Since $\text{Range}(f) = B$, every element in B has a pre-image in A . Hence, f is an onto function.

Hence Proved

Since f is both one-one and onto, $f: A \rightarrow B$ is a bijective function. Hence proved.

Question 33

- (a) If a real-valued function is given by $f(x) = \sqrt{25 - x^2}$ is an onto function, then find the codomain for $f(x)$.
 (b) If the domain is given to be $[-5, 5]$, is $f(x)$ a one-one function?
 (c) Find all possible values of 'a' for which $f(a) = 4$.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Real-valued function $f(x) = \sqrt{25 - x^2}$ defined on domain $[-5, 5]$.

Concept / Formula Used: Codomain = Range $f(x)$ $x_1 \neq x_2$ $f(x_1) = f(x_2)$

Solution:

Part (a): Finding the codomain for $f(x)$ to be onto

Given domain is $[-5, 5]$, which means:

$$-5 \leq x \leq 5$$

Squaring all terms in the domain range:

$$\begin{aligned} 0 &\leq x^2 \leq 25 \\ -25 &\leq -x^2 \leq 0 \\ 25 - 25 &\leq 25 - x^2 \leq 25 - 0 \\ 0 &\leq 25 - x^2 \leq 25 \end{aligned}$$

Taking the non-negative principal square root:

$$\begin{aligned} \sqrt{0} &\leq \sqrt{25 - x^2} \leq \sqrt{25} \\ 0 &\leq f(x) \leq 5 \end{aligned}$$

Thus, the range of $f(x)$ is $[0, 5]$. For $f(x)$ to be an onto function, the codomain must equal its range:

$$\text{Codomain} = \text{Range} = [0, 5]$$

Part (b): Checking if $f(x)$ is a one-one function on $[-5, 5]$

Consider two distinct points $x_1 = 3$ and $x_2 = -3$ in the domain $[-5, 5]$. Evaluating $f(x)$ at these points:

$$\begin{aligned} f(3) &= \sqrt{25 - (3)^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \\ f(-3) &= \sqrt{25 - (-3)^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \end{aligned}$$

Here, $3 \neq -3$, but $f(3) = f(-3) = 4$. Since distinct elements in the domain have the same image, $f(x)$ is **not** a one-one function.

Part (c): Finding all possible values of 'a' for which $f(a) = 4$

Given equation:

$$f(a) = 4$$

Substituting the expression for $f(a)$:

$$\sqrt{25 - a^2} = 4$$

Squaring both sides:

$$\begin{aligned} 25 - a^2 &= 4^2 \\ 25 - a^2 &= 16 \\ a^2 &= 25 - 16 \\ a^2 &= 9 \\ a &= \pm\sqrt{9} \\ a &= \pm 3 \end{aligned}$$

Both values $a = 3$ and $a = -3$ belong to the domain $[-5, 5]$. Thus, the possible values of a are $\{-3, 3\}$.

Final Answer

- (a) Codomain = $[0, 5]$
- (b) No, $f(x)$ is not a one-one function.
- (c) $a \in \{-3, 3\}$

Common Mistake: In part (c), taking only the positive square root $a = 3$ and forgetting $a = -3$ is a common error; both values lie within the domain $[-5, 5]$.

1.4 – Composition of Functions

Question 1

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = (3 - x^3)^{1/3}$, then find $(f \circ f)(x)$.

TYPE 26 Finding Composition of Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (3 - x^3)^{1/3}$.

Concept / Formula Used: $f \quad (f \circ f)(x) = f(f(x))$

Solution:

By the definition of composite functions:

$$(f \circ f)(x) = f(f(x))$$

Substitute the expression for $f(x)$:

$$(f \circ f)(x) = f((3 - x^3)^{1/3})$$

Apply the rule $f(y) = (3 - y^3)^{1/3}$ with $y = (3 - x^3)^{1/3}$:

$$\begin{aligned}(f \circ f)(x) &= \left[3 - ((3 - x^3)^{1/3})^3\right]^{1/3} \\ &= [3 - (3 - x^3)]^{1/3} \\ &= [3 - 3 + x^3]^{1/3} \\ &= [x^3]^{1/3} \\ &= x\end{aligned}$$

Final Answer

$(f \circ f)(x) = x$ for all $x \in \mathbb{R}$.

Question 2

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = (5 - x^5)^{1/5}$, then find $f \circ f$.

TYPE 26 Finding Composition of Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = (5 - x^5)^{1/5}$.

Concept / Formula Used: $f \quad (f \circ f)(x) = f(f(x))$

Solution:

By definition of composite functions:

$$(f \circ f)(x) = f(f(x))$$

Substitute $f(x) = (5 - x^5)^{1/5}$ into the function:

$$(f \circ f)(x) = f((5 - x^5)^{1/5})$$

Applying the definition $f(y) = (5 - y^5)^{1/5}$ with $y = (5 - x^5)^{1/5}$:

$$\begin{aligned}(f \circ f)(x) &= \left[5 - ((5 - x^5)^{1/5})^5\right]^{1/5} \\ &= [5 - (5 - x^5)]^{1/5} \\ &= [5 - 5 + x^5]^{1/5} \\ &= [x^5]^{1/5} \\ &= x\end{aligned}$$

Final Answer

$(f \circ f)(x) = x$ for all $x \in \mathbb{R}$.

Question 3

If $f(x) = |x|$ and $g(x) = [x]$ for all $x \in \mathbb{R}$, find

- (i) $(g \circ f)\left(-\frac{5}{3}\right)$
 (ii) $(f \circ g)\left(-\frac{5}{3}\right)$

TYPE 27 Evaluating Composite Functions at a Point

Given: $f(x) = |x|$ and $g(x) = [x]$ (Greatest Integer Function) for all $x \in \mathbb{R}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$

Solution:

(i) **Evaluation of $(g \circ f)\left(-\frac{5}{3}\right)$:** By definition of composite functions:

$$(g \circ f)\left(-\frac{5}{3}\right) = g\left(f\left(-\frac{5}{3}\right)\right)$$

First, evaluate $f\left(-\frac{5}{3}\right)$:

$$f\left(-\frac{5}{3}\right) = \left|-\frac{5}{3}\right| = \frac{5}{3}$$

Now evaluate $g\left(\frac{5}{3}\right)$:

$$g\left(\frac{5}{3}\right) = \left[\frac{5}{3}\right] = [1.666\dots] = 1$$

Therefore, $(g \circ f)\left(-\frac{5}{3}\right) = 1$.

(ii) **Evaluation of $(f \circ g)\left(-\frac{5}{3}\right)$:** By definition of composite functions:

$$(f \circ g)\left(-\frac{5}{3}\right) = f\left(g\left(-\frac{5}{3}\right)\right)$$

First, evaluate $g\left(-\frac{5}{3}\right)$:

$$g\left(-\frac{5}{3}\right) = \left[-\frac{5}{3}\right] = [-1.666\dots] = -2$$

Now evaluate $f(-2)$:

$$f(-2) = |-2| = 2$$

Therefore, $(f \circ g)\left(-\frac{5}{3}\right) = 2$.

Final Answer

$$(i) (g \circ f) \left(-\frac{5}{3} \right) = 1$$

$$(ii) (f \circ g) \left(-\frac{5}{3} \right) = 2$$

Question 4

If $f(x) = x + 7$ and $g(x) = x - 7$, find $(f \circ g)(7)$.

TYPE 27 Evaluating Composite Functions at a Point

Given: $f(x) = x + 7$ and $g(x) = x - 7$.

Concept / Formula Used: $(f \circ g)(x) = f(g(x))$

Solution:

By definition of function composition:

$$(f \circ g)(7) = f(g(7))$$

First, calculate $g(7)$:

$$\begin{aligned} g(7) &= 7 - 7 \\ &= 0 \end{aligned}$$

Now substitute $g(7) = 0$ into f :

$$\begin{aligned} f(g(7)) &= f(0) \\ &= 0 + 7 \\ &= 7 \end{aligned}$$

Final Answer

$$(f \circ g)(7) = 7$$

Question 5

If $f : \{2, 3, 4, 5\} \rightarrow \{3, 4, 5, 9\}$ and $g : \{3, 4, 5, 9\} \rightarrow \{7, 11, 15\}$ are functions defined as $f(2) = 3$, $f(3) = 4$, $f(4) = f(5) = 5$ and $g(3) = g(4) = 7$, $g(5) = g(9) = 11$, find $g \circ f$.

TYPE 28 Composition of Functions on Finite Sets

Given: $f : \{2, 3, 4, 5\} \rightarrow \{3, 4, 5, 9\}$ with $f(2) = 3$, $f(3) = 4$, $f(4) = 5$, $f(5) = 5$.
 $g : \{3, 4, 5, 9\} \rightarrow \{7, 11, 15\}$ with $g(3) = 7$, $g(4) = 7$, $g(5) = 11$, $g(9) = 11$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x)) \quad x \in \text{Domain}(f) = \{2, 3, 4, 5\}$

Solution:

We evaluate $(g \circ f)(x)$ for each element in the domain $\{2, 3, 4, 5\}$:

$$(g \circ f)(2) = g(f(2)) = g(3) = 7$$

$$(g \circ f)(3) = g(f(3)) = g(4) = 7$$

$$(g \circ f)(4) = g(f(4)) = g(5) = 11$$

$$(g \circ f)(5) = g(f(5)) = g(5) = 11$$

Expressing $(g \circ f)$ as a set of ordered pairs:

$$(g \circ f) = \{(2, 7), (3, 7), (4, 11), (5, 11)\}$$

Final Answer

$$g \circ f = \{(2, 7), (3, 7), (4, 11), (5, 11)\}$$

Question 6

Let $A = \{3, 4, 5, 6\}$, $B = \{13, 14, 15, 16\}$ and $C = \{23, 24, 25\}$. If $f : A \rightarrow B$ and $g : B \rightarrow C$ are defined by

$$f = \{(3, 13), (4, 14), (5, 15), (6, 16)\}$$

and

$$g = \{(13, 23), (14, 23), (15, 24), (16, 25)\}$$

then find $g \circ f : A \rightarrow C$.

TYPE 28 Composition of Functions on Finite Sets

Given: $A = \{3, 4, 5, 6\}$, $B = \{13, 14, 15, 16\}$, $C = \{23, 24, 25\}$, $f : A \rightarrow B$, and $g : B \rightarrow C$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x)) \quad x \in A$

Solution:

We evaluate $(g \circ f)(x)$ for each element in the domain set $A = \{3, 4, 5, 6\}$:

$$(g \circ f)(3) = g(f(3)) = g(13) = 23$$

$$(g \circ f)(4) = g(f(4)) = g(14) = 23$$

$$(g \circ f)(5) = g(f(5)) = g(15) = 24$$

$$(g \circ f)(6) = g(f(6)) = g(16) = 25$$

Writing $g \circ f$ as a set of ordered pairs:

$$g \circ f = \{(3, 23), (4, 23), (5, 24), (6, 25)\}$$

Final Answer

$$g \circ f = \{(3, 23), (4, 23), (5, 24), (6, 25)\}$$

Question 7

Let $A = \{1, 2, 3, 4\}$. If the functions $f : A \rightarrow A$ and $g : A \rightarrow A$ are defined by

$$f = \{(1, 4), (2, 1), (3, 3), (4, 2)\}$$

and

$$g = \{(1, 3), (2, 1), (3, 2), (4, 4)\}$$

then find

- (i) $g \circ f$
- (ii) $f \circ g$
- (iii) $f \circ f$

TYPE 28 Composition of Functions on Finite Sets

Given: $A = \{1, 2, 3, 4\}$, $f = \{(1, 4), (2, 1), (3, 3), (4, 2)\}$, $g = \{(1, 3), (2, 1), (3, 2), (4, 4)\}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$ $(f \circ f)(x) = f(f(x))$

Solution:

(i) Finding $g \circ f$: Evaluating $(g \circ f)(x) = g(f(x))$ for all $x \in A$:

$$\begin{aligned}(g \circ f)(1) &= g(f(1)) = g(4) = 4 \\(g \circ f)(2) &= g(f(2)) = g(1) = 3 \\(g \circ f)(3) &= g(f(3)) = g(3) = 2 \\(g \circ f)(4) &= g(f(4)) = g(2) = 1\end{aligned}$$

Thus, $g \circ f = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$.

(ii) Finding $f \circ g$: Evaluating $(f \circ g)(x) = f(g(x))$ for all $x \in A$:

$$\begin{aligned}(f \circ g)(1) &= f(g(1)) = f(3) = 3 \\(f \circ g)(2) &= f(g(2)) = f(1) = 4 \\(f \circ g)(3) &= f(g(3)) = f(2) = 1 \\(f \circ g)(4) &= f(g(4)) = f(4) = 2\end{aligned}$$

Thus, $f \circ g = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$.

(iii) Finding $f \circ f$: Evaluating $(f \circ f)(x) = f(f(x))$ for all $x \in A$:

$$\begin{aligned}(f \circ f)(1) &= f(f(1)) = f(4) = 2 \\(f \circ f)(2) &= f(f(2)) = f(1) = 4 \\(f \circ f)(3) &= f(f(3)) = f(3) = 3 \\(f \circ f)(4) &= f(f(4)) = f(2) = 1\end{aligned}$$

Thus, $f \circ f = \{(1, 2), (2, 4), (3, 3), (4, 1)\}$.

Final Answer

- (i) $g \circ f = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$
- (ii) $f \circ g = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$
- (iii) $f \circ f = \{(1, 2), (2, 4), (3, 3), (4, 1)\}$

Question 8

If the functions f and g are given by $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(2, 3), (5, 1), (1, 3)\}$,

- (i) find the domain of f and g
- (ii) find the range of f and g
- (iii) write $f \circ g$ and $g \circ f$ as sets of ordered pairs

TYPE 28 Composition of Functions on Finite Sets

Given: $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(2, 3), (5, 1), (1, 3)\}$.

Concept / Formula Used: $R = \{(x, y)\}$ Domain = $\{x : (x, y) \in R\}$ Range = $\{y : (x, y) \in R\}$

Solution:

(i) **Domain of f and g :** The domain consists of the first elements of each ordered pair:

$$\text{Domain of } f = \{1, 3, 4\}$$

$$\text{Domain of } g = \{2, 5, 1\} = \{1, 2, 5\}$$

(ii) **Range of f and g :** The range consists of the second elements of each ordered pair:

$$\text{Range of } f = \{2, 5, 1\} = \{1, 2, 5\}$$

$$\text{Range of } g = \{3, 1, 3\} = \{1, 3\}$$

(iii) **Writing $f \circ g$ and $g \circ f$ as sets of ordered pairs:** For $f \circ g$, the domain is $\text{Domain}(g) = \{1, 2, 5\}$:

$$(f \circ g)(1) = f(g(1)) = f(3) = 5$$

$$(f \circ g)(2) = f(g(2)) = f(3) = 5$$

$$(f \circ g)(5) = f(g(5)) = f(1) = 2$$

Hence, $f \circ g = \{(1, 5), (2, 5), (5, 2)\}$.

For $g \circ f$, the domain is $\text{Domain}(f) = \{1, 3, 4\}$:

$$(g \circ f)(1) = g(f(1)) = g(2) = 3$$

$$(g \circ f)(3) = g(f(3)) = g(5) = 1$$

$$(g \circ f)(4) = g(f(4)) = g(1) = 3$$

Hence, $g \circ f = \{(1, 3), (3, 1), (4, 3)\}$.

Final Answer

(i) $\text{Domain}(f) = \{1, 3, 4\}$, $\text{Domain}(g) = \{1, 2, 5\}$

(ii) $\text{Range}(f) = \{1, 2, 5\}$, $\text{Range}(g) = \{1, 3\}$

(iii) $f \circ g = \{(1, 5), (2, 5), (5, 2)\}$, $g \circ f = \{(1, 3), (3, 1), (4, 3)\}$

Question 9

Let $\mathbb{R}$ be the set of all real numbers. Find $g \circ f$ and $f \circ g$ when $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by

(i) $f(x) = 2x - 1$ and $g(x) = x^2 + 3$ for all $x \in \mathbb{R}$

(ii) $f(x) = x + 1$ and $g(x) = |x|$ for all $x \in \mathbb{R}$

(iii) $f(x) = x^3$ and $g(x) = 2x^2 + 1$ for all $x \in \mathbb{R}$

(iv) $f(x) = \sin x$ and $g(x) = 5x^2$ for all $x \in \mathbb{R}$

TYPE 26 Finding Composition of Functions

★ Likely Exam Question

Given: Functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$

Solution:

(i) $f(x) = 2x - 1$ and $g(x) = x^2 + 3$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(2x - 1) \\ &= (2x - 1)^2 + 3 \\ &= 4x^2 - 4x + 1 + 3 \\ &= 4x^2 - 4x + 4\end{aligned}$$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(x^2 + 3) \\ &= 2(x^2 + 3) - 1 \\ &= 2x^2 + 6 - 1 \\ &= 2x^2 + 5\end{aligned}$$

(ii) $f(x) = x + 1$ and $g(x) = |x|$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(x + 1) = |x + 1| \\ (f \circ g)(x) &= f(g(x)) = f(|x|) = |x| + 1\end{aligned}$$

(iii) $f(x) = x^3$ and $g(x) = 2x^2 + 1$ (ISC 2019):

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(x^3) \\ &= 2(x^3)^2 + 1 \\ &= 2x^6 + 1 \\ (f \circ g)(x) &= f(g(x)) = f(2x^2 + 1) \\ &= (2x^2 + 1)^3\end{aligned}$$

(iv) $f(x) = \sin x$ and $g(x) = 5x^2$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(\sin x) \\ &= 5(\sin x)^2 \\ &= 5 \sin^2 x \\ (f \circ g)(x) &= f(g(x)) = f(5x^2) \\ &= \sin(5x^2)\end{aligned}$$

Final Answer

- (i) $(g \circ f)(x) = 4x^2 - 4x + 4$, $(f \circ g)(x) = 2x^2 + 5$
(ii) $(g \circ f)(x) = |x + 1|$, $(f \circ g)(x) = |x| + 1$
(iii) $(g \circ f)(x) = 2x^6 + 1$, $(f \circ g)(x) = (2x^2 + 1)^3$
(iv) $(g \circ f)(x) = 5 \sin^2 x$, $(f \circ g)(x) = \sin(5x^2)$

Question 10

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^2 - 3x + 2$ for all $x \in \mathbb{R}$, then find $f \circ f$.

TYPE 26 Finding Composition of Functions

★ Likely Exam Question

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2 - 3x + 2$.

Concept / Formula Used: $(f \circ f)(x) = f(f(x))$

Solution:

By definition of composite functions:

$$(f \circ f)(x) = f(f(x))$$

Substitute $f(x) = x^2 - 3x + 2$:

$$(f \circ f)(x) = (x^2 - 3x + 2)^2 - 3(x^2 - 3x + 2) + 2$$

Expanding $(x^2 - 3x + 2)^2$ using the expansion formula $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$:

$$\begin{aligned}(x^2 - 3x + 2)^2 &= (x^2)^2 + (-3x)^2 + (2)^2 + 2(x^2)(-3x) + 2(-3x)(2) + 2(2)(x^2) \\ &= x^4 + 9x^2 + 4 - 6x^3 - 12x + 4x^2 \\ &= x^4 - 6x^3 + 13x^2 - 12x + 4\end{aligned}$$

Expanding the second part:

$$-3(x^2 - 3x + 2) = -3x^2 + 9x - 6$$

Substitute back into the expression:

$$\begin{aligned}(f \circ f)(x) &= (x^4 - 6x^3 + 13x^2 - 12x + 4) + (-3x^2 + 9x - 6) + 2 \\ &= x^4 - 6x^3 + (13x^2 - 3x^2) + (-12x + 9x) + (4 - 6 + 2) \\ &= x^4 - 6x^3 + 10x^2 - 3x + 0 \\ &= x^4 - 6x^3 + 10x^2 - 3x\end{aligned}$$

Final Answer

$$(f \circ f)(x) = x^4 - 6x^3 + 10x^2 - 3x$$

Common Mistake: A common error occurs in expanding $(x^2 - 3x + 2)^2$; remember to keep track of signs when evaluating cross-terms like $2(x^2)(-3x) = -6x^3$.

Question 11

Let $\mathbb{R}$ be the set of all real numbers. If the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are given by $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$ for all $x \in \mathbb{R}$, then find

- (i) $f \circ g$
- (ii) $g \circ f$
- (iii) $f \circ f$

$$(iv) \ g \circ g$$

TYPE 26 Finding Composition of Functions

Given: $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $(f \circ g)(x) = f(g(x))$ $(g \circ f)(x) = g(f(x))$ $(f \circ f)(x) = f(f(x))$ $(g \circ g)(x) = g(g(x))$

Solution:

(i) Finding $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(2x - 3) \\ &= (2x - 3)^2 + 3(2x - 3) + 1 \\ &= (4x^2 - 12x + 9) + (6x - 9) + 1 \\ &= 4x^2 - 12x + 6x + 9 - 9 + 1 \\ &= 4x^2 - 6x + 1\end{aligned}$$

(ii) Finding $(g \circ f)(x)$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(x^2 + 3x + 1) \\ &= 2(x^2 + 3x + 1) - 3 \\ &= 2x^2 + 6x + 2 - 3 \\ &= 2x^2 + 6x - 1\end{aligned}$$

(iii) Finding $(f \circ f)(x)$:

$$\begin{aligned}(f \circ f)(x) &= f(f(x)) = f(x^2 + 3x + 1) \\ &= (x^2 + 3x + 1)^2 + 3(x^2 + 3x + 1) + 1 \\ &= [(x^2)^2 + (3x)^2 + (1)^2 + 2(x^2)(3x) + 2(3x)(1) + 2(1)(x^2)] + 3x^2 + 9x + 3 + 1 \\ &= [x^4 + 9x^2 + 1 + 6x^3 + 6x + 2x^2] + 3x^2 + 9x + 4 \\ &= x^4 + 6x^3 + (9 + 2 + 3)x^2 + (6 + 9)x + (1 + 4) \\ &= x^4 + 6x^3 + 14x^2 + 15x + 5\end{aligned}$$

(iv) Finding $(g \circ g)(x)$:

$$\begin{aligned}(g \circ g)(x) &= g(g(x)) = g(2x - 3) \\ &= 2(2x - 3) - 3 \\ &= 4x - 6 - 3 \\ &= 4x - 9\end{aligned}$$

Final Answer

$$(i) \ (f \circ g)(x) = 4x^2 - 6x + 1$$

$$(ii) \ (g \circ f)(x) = 2x^2 + 6x - 1$$

$$(iii) \ (f \circ f)(x) = x^4 + 6x^3 + 14x^2 + 15x + 5$$

$$(iv) \ (g \circ g)(x) = 4x - 9$$

Question 12

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^2$ and $g(x) = x + 1$ for all $x \in \mathbb{R}$, show that $g \circ f \neq f \circ g$.

TYPE 26 Finding Composition of Functions**To Prove:** $g \circ f \neq f \circ g$

Concept / Formula Used: $h_1 \quad h_2 \quad h_1(x) = h_2(x) \quad x \in \mathbb{R} \quad x \quad h_1(x) \neq h_2(x) \quad h_1 \neq h_2$

Solution:First, calculate $(g \circ f)(x)$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x^2) \\ &= x^2 + 1\end{aligned}$$

Next, calculate $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x + 1) \\ &= (x + 1)^2 \\ &= x^2 + 2x + 1\end{aligned}$$

Comparing $(g \circ f)(x)$ and $(f \circ g)(x)$:

$$\begin{aligned}(g \circ f)(x) &= x^2 + 1 \\ (f \circ g)(x) &= x^2 + 2x + 1\end{aligned}$$

For example, evaluate both at $x = 1$:

$$\begin{aligned}(g \circ f)(1) &= 1^2 + 1 = 2 \\ (f \circ g)(1) &= (1 + 1)^2 = 4\end{aligned}$$

Since $(g \circ f)(1) \neq (f \circ g)(1)$, the two composition functions are not equal.**Hence Proved** $(g \circ f)(x) \neq (f \circ g)(x)$, hence $g \circ f \neq f \circ g$ is proved.**Question 13**

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x + 1$ and $g(x) = x - 1$ for all $x \in \mathbb{R}$, show that $g \circ f = f \circ g = I_{\mathbb{R}}$.

TYPE 26 Finding Composition of Functions**To Prove:** $g \circ f = f \circ g = I_{\mathbb{R}}$, where $I_{\mathbb{R}}(x) = x$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $\mathbb{R} \quad I_{\mathbb{R}}(x) = x \quad x \in \mathbb{R}$

Solution:

Step 1: Evaluate $(g \circ f)(x)$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x + 1) \\ &= (x + 1) - 1 \\ &= x \\ &= I_{\mathbb{R}}(x)\end{aligned}$$

Thus, $g \circ f = I_{\mathbb{R}}$.

Step 2: Evaluate $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x - 1) \\ &= (x - 1) + 1 \\ &= x \\ &= I_{\mathbb{R}}(x)\end{aligned}$$

Thus, $f \circ g = I_{\mathbb{R}}$.

From Step 1 and Step 2, we conclude $g \circ f = f \circ g = I_{\mathbb{R}}$.

Hence Proved

$g \circ f = f \circ g = I_{\mathbb{R}}$, hence proved.

Question 14

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = 3x^2 - 2$ and $g(x) = \sin 2x$ respectively, find

- (i) $g \circ f$
- (ii) $f \circ g$
- (iii) $f \circ f$

TYPE 26 Finding Composition of Functions

Given: $f(x) = 3x^2 - 2$ and $g(x) = \sin 2x$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$ $(f \circ f)(x) = f(f(x))$

Solution:

(i) Finding $(g \circ f)(x)$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(3x^2 - 2) \\ &= \sin(2(3x^2 - 2)) \\ &= \sin(6x^2 - 4)\end{aligned}$$

(ii) Finding $(f \circ g)(x)$:

$$\begin{aligned}
 (f \circ g)(x) &= f(g(x)) \\
 &= f(\sin 2x) \\
 &= 3(\sin 2x)^2 - 2 \\
 &= 3 \sin^2 2x - 2
 \end{aligned}$$

(iii) Finding $(f \circ f)(x)$:

$$\begin{aligned}
 (f \circ f)(x) &= f(f(x)) \\
 &= f(3x^2 - 2) \\
 &= 3(3x^2 - 2)^2 - 2 \\
 &= 3(9x^4 - 12x^2 + 4) - 2 \\
 &= 27x^4 - 36x^2 + 12 - 2 \\
 &= 27x^4 - 36x^2 + 10
 \end{aligned}$$

Final Answer

(i) $(g \circ f)(x) = \sin(6x^2 - 4)$

(ii) $(f \circ g)(x) = 3 \sin^2 2x - 2$

(iii) $(f \circ f)(x) = 27x^4 - 36x^2 + 10$

Question 15

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2 + 1}$, for all $x \in \mathbb{R}$ is neither one-one nor onto. Also if $g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $g(x) = 2x - 1$, find $(f \circ g)(x)$.

TYPE 22 Testing Injectivity and Surjectivity of Functions**★ Likely Exam Question**

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = \frac{x}{x^2 + 1}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ where $g(x) = 2x - 1$.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2$ $\text{Range}(f) = \text{Codomain of } f = \mathbb{R}$

Solution:

Part 1: Show f is not one-one (injective): Consider two distinct domain values $x = 2$ and $x = \frac{1}{2}$ in $\mathbb{R}$:

$$\begin{aligned}
 f(2) &= \frac{2}{2^2 + 1} = \frac{2}{4 + 1} = \frac{2}{5} \\
 f\left(\frac{1}{2}\right) &= \frac{\frac{1}{2}}{\left(\frac{1}{2}\right)^2 + 1} = \frac{\frac{1}{2}}{\frac{1}{4} + 1} = \frac{\frac{1}{2}}{\frac{5}{4}} = \frac{2}{5}
 \end{aligned}$$

Since $f(2) = f\left(\frac{1}{2}\right) = \frac{2}{5}$ for $2 \neq \frac{1}{2}$, f is **not one-one**.

Part 2: Show f is not onto (surjective): Let $y = f(x) = \frac{x}{x^2 + 1}$. Cross-multiplying:

$$y(x^2 + 1) = x \implies yx^2 - x + y = 0$$

For x to be a real number, the discriminant D of this quadratic equation in x must be non-negative ($D \geq 0$):

$$\begin{aligned} D &= (-1)^2 - 4(y)(y) \\ &= 1 - 4y^2 \geq 0 \end{aligned}$$

Solving $1 - 4y^2 \geq 0$:

$$(1 - 2y)(1 + 2y) \geq 0 \implies -\frac{1}{2} \leq y \leq \frac{1}{2}$$

Thus, $\text{Range}(f) = \left[-\frac{1}{2}, \frac{1}{2}\right] \neq \mathbb{R}$ (Codomain). For instance, $y = 2 \in \mathbb{R}$ has no real pre-image because $1 - 4(2)^2 = -15 < 0$. Hence, f is **not onto**.

Part 3: Evaluate $(f \circ g)(x)$:

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(2x - 1) \\ &= \frac{2x - 1}{(2x - 1)^2 + 1} \\ &= \frac{2x - 1}{(4x^2 - 4x + 1) + 1} \\ &= \frac{2x - 1}{4x^2 - 4x + 2} \end{aligned}$$

Final Answer

Function f is neither one-one nor onto.

$$(f \circ g)(x) = \frac{2x - 1}{4x^2 - 4x + 2}$$

Common Mistake: When checking surjectivity for rational functions, always set up $y = f(x)$, solve for real x via the discriminant, and verify if the Range equals the Codomain.

1.5 – Invertible Functions and Inverse of a Function

Question 1

If $f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$, find f^{-1} .

TYPE 29 Finding Inverse Function on Finite Sets

Given: $f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$

To Find: f^{-1}

Concept / Formula Used: $f = \{(x, y)\}$ $f^{-1} = \{(y, x) \mid (x, y) \in f\}$

Solution:

Given the function defined as a set of ordered pairs:

$$f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$$

To find the inverse function f^{-1} , reverse the components of each ordered pair $(x, y) \mapsto (y, x)$:

$$(1, 5) \implies (5, 1)$$

$$(2, 3) \implies (3, 2)$$

$$(3, 0) \implies (0, 3)$$

$$(4, -2) \implies (-2, 4)$$

Combining these inverted pairs gives:

$$f^{-1} = \{(5, 1), (3, 2), (0, 3), (-2, 4)\}$$

Final Answer

$$f^{-1} = \{(5, 1), (3, 2), (0, 3), (-2, 4)\}$$

Question 2

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = 3x + 5$, then find f^{-1} .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x + 5$

To Find: f^{-1}

Concept / Formula Used: $f^{-1}(x)$ $y = f(x)$ x y

Solution:

Let $y = f(x)$.

$$y = 3x + 5$$

Subtract 5 from both sides:

$$y - 5 = 3x$$

Divide both sides by 3:

$$x = \frac{y - 5}{3}$$

Since $y = f(x) \iff x = f^{-1}(y)$, we have:

$$f^{-1}(y) = \frac{y - 5}{3}$$

Replacing y with x gives the inverse function:

$$f^{-1}(x) = \frac{x - 5}{3}$$

Final Answer

$$f^{-1}(x) = \frac{x-5}{3} \text{ for all } x \in \mathbb{R}$$

Question 3

If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{3x-2}{5}$ is invertible, find the inverse of f .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{3x-2}{5}$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \quad f^{-1}(y) = x$

Solution:

Let $y = f(x)$.

$$y = \frac{3x-2}{5}$$

Multiply both sides by 5:

$$5y = 3x - 2$$

Add 2 to both sides:

$$5y + 2 = 3x$$

Divide both sides by 3:

$$x = \frac{5y+2}{3}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{5y+2}{3}$$

Replacing the variable y with x :

$$f^{-1}(x) = \frac{5x+2}{3}$$

Final Answer

$$f^{-1}(x) = \frac{5x+2}{3}$$

Question 4

If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-7}{4}$ is invertible, find f^{-1} .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{2x-7}{4}$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \quad x \quad y$

Solution:

Let $y = f(x)$.

$$y = \frac{2x-7}{4}$$

Multiply both sides by 4:

$$4y = 2x - 7$$

Add 7 to both sides:

$$4y + 7 = 2x$$

Divide both sides by 2:

$$x = \frac{4y + 7}{2}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{4y + 7}{2}$$

Replacing y with x :

$$f^{-1}(x) = \frac{4x + 7}{2}$$

Final Answer

$$f^{-1}(x) = \frac{4x+7}{2}$$

Question 5

If f is an invertible function defined as $f(x) = \frac{3x-4}{5}$, write $f^{-1}(x)$.

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = \frac{3x-4}{5}$, f invertible

To Find: $f^{-1}(x)$

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = \frac{3x-4}{5}$$

Multiply both sides by 5:

$$5y = 3x - 4$$

Add 4 to both sides:

$$5y + 4 = 3x$$

Divide both sides by 3:

$$x = \frac{5y + 4}{3}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{5y + 4}{3}$$

Replacing y with x :

$$f^{-1}(x) = \frac{5x + 4}{3}$$

Final Answer

$$f^{-1}(x) = \frac{5x+4}{3}$$

Question 6

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = ax - b$, $a, b \in \mathbb{R}$, $a \neq 0$, then find f^{-1} .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = ax - b$, $a \neq 0$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = ax - b$$

Add b to both sides:

$$y + b = ax$$

Divide both sides by a (valid since $a \neq 0$):

$$x = \frac{y + b}{a}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{y + b}{a}$$

Final Answer

$f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f^{-1}(y) = \frac{y + b}{a}$

Question 7

If $f : \mathbb{R} - \left\{-\frac{3}{5}\right\} \rightarrow \mathbb{R}$ is a function defined by $f(x) = \frac{2x}{5x + 3}$, then find $f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{3}{5}\right\}$.

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = \frac{2x}{5x + 3}$, $x \neq -\frac{3}{5}$

To Find: $f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{3}{5}\right\}$

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = \frac{2x}{5x + 3}$$

Cross-multiply:

$$y(5x + 3) = 2x$$

Expand the left side:

$$5xy + 3y = 2x$$

Collect all x -terms on one side:

$$5xy - 2x = -3y$$

Factor out x :

$$x(5y - 2) = -3y$$

Divide both sides by $(5y - 2)$:

$$x = \frac{-3y}{5y - 2} = \frac{3y}{2 - 5y}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{3y}{2 - 5y}$$

Final Answer

$f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{3}{5}\right\}$ is given by $f^{-1}(y) = \frac{3y}{2 - 5y}$

Question 8

If $f : \mathbb{R} - \left\{-\frac{4}{3}\right\} \rightarrow \mathbb{R}$ is a function defined by $f(x) = \frac{4x}{3x + 4}$, then find f^{-1} . (NCERT)

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = \frac{4x}{3x + 4}$, $x \neq -\frac{4}{3}$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = \frac{4x}{3x + 4}$$

Cross-multiply:

$$y(3x + 4) = 4x$$

Expand the left side:

$$3xy + 4y = 4x$$

Collect all x -terms on one side:

$$3xy - 4x = -4y$$

Factor out x :

$$x(3y - 4) = -4y$$

Divide both sides by $(3y - 4)$:

$$x = \frac{-4y}{3y - 4} = \frac{4y}{4 - 3y}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{4y}{4-3y}$$

Final Answer

$f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{4}{3}\right\}$ is given by $f^{-1}(y) = \frac{4y}{4-3y}$

Question 9

If $f(x) = [4 - (x - 7)^3]^{1/5}$ is a real invertible function, then find $f^{-1}(x)$. (ISC 2023)

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = [4 - (x - 7)^3]^{1/5}$

To Find: $f^{-1}(x)$

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = [4 - (x - 7)^3]^{1/5}$$

Raise both sides to the power 5:

$$y^5 = 4 - (x - 7)^3$$

Rearrange to isolate the cube term:

$$(x - 7)^3 = 4 - y^5$$

Take the cube root of both sides:

$$x - 7 = (4 - y^5)^{1/3}$$

Add 7 to both sides:

$$x = 7 + (4 - y^5)^{1/3}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \sqrt[3]{4 - y^5} + 7$$

Replacing y with x :

$$f^{-1}(x) = \sqrt[3]{4 - x^5} + 7$$

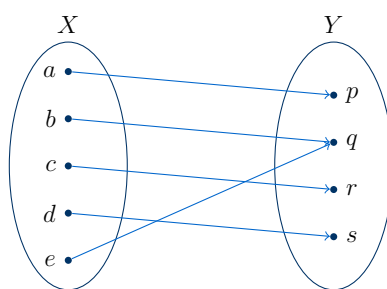
Final Answer

$$f^{-1}(x) = \sqrt[3]{4 - x^5} + 7$$

Question 10

The function $f : X \rightarrow Y$ is not invertible. State the reason. (ISC Specimen Paper 2024)

TYPE 31 Testing Invertibility of Functions



$f : X \rightarrow Y$ mapping five elements of X onto only four elements of Y , with two arrows landing on the same image

Given: $f : X \rightarrow Y$, $n(X) = 5$, $n(Y) = 4$ (as shown in the given arrow diagram)

Solution:

From the arrow diagram, X has more elements than Y . By the pigeonhole principle, at least two distinct elements of X must map to the same element of Y .

In the diagram, two arrows point to the same element of Y , i.e. there exist $x_1, x_2 \in X$ with $x_1 \neq x_2$ such that

$$f(x_1) = f(x_2)$$

This means distinct elements of X have the same image, so f is not one-one.

Since a function must be both one-one and onto (bijective) to be invertible, and f fails to be one-one, f cannot be invertible.

Final Answer

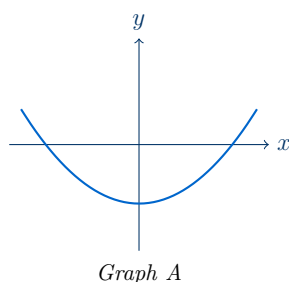
f is not invertible because f is not one-one.

Question 11

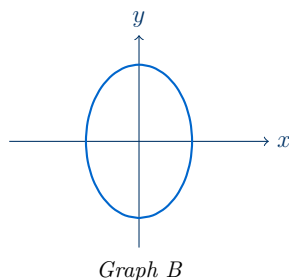
Which one of the following graphs is a function of x ?

(ISC 2024)

TYPE 32 Vertical Line Test for a Function



Graph A: a single continuous curve, dipping down and rising again – every vertical line meets it exactly once



Graph B: a sideways loop – a vertical line through the loop meets the curve more than once

Solution:

By the definition of a function, $f : X \rightarrow Y$ is a function iff every element $x \in X$ has *exactly one* image $y \in Y$. Graphically, this is tested using the **vertical line test**: a curve represents a function of x iff every vertical line $x = k$ intersects the curve in *at most one* point.

Graph A: Every vertical line drawn on Graph A crosses the curve exactly once. So for each x , there is a unique y .

$\implies$ Graph A represents a function of x

Graph B: Graph B is a closed loop. A vertical line drawn through the middle of the loop crosses the curve at *two* points, giving two different y -values for the same x .

$\implies$ Graph B does *not* represent a function of x

Final Answer

Graph A is the function of x .

Question 12

Let $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$ be a function defined by $f(1) = a$, $f(2) = b$ and $f(3) = c$. Show that f is invertible and find f^{-1} . Also show that $(f^{-1})^{-1} = f$. (NCERT)

TYPE 33a Proving Invertibility & Finding Inverse – Roster/Finite Functions

To Prove: f is invertible, and $(f^{-1})^{-1} = f$

Given: $f = \{(1, a), (2, b), (3, c)\}$

Solution:

One-one: The images a, b, c are all distinct (given), so distinct elements of the domain have distinct images.

$\implies f$ is one-one

Onto: The range of f is $\{a, b, c\}$, which equals the codomain.

$\implies f$ is onto

Since f is both one-one and onto, f is bijective, and hence invertible.

To find f^{-1} , reverse each ordered pair of f :

$$(1, a) \implies (a, 1)$$

$$(2, b) \implies (b, 2)$$

$$(3, c) \implies (c, 3)$$

$$f^{-1} = \{(a, 1), (b, 2), (c, 3)\}$$

Now, to show $(f^{-1})^{-1} = f$, reverse each ordered pair of f^{-1} :

$$(a, 1) \implies (1, a)$$

$$(b, 2) \implies (2, b)$$

$$(c, 3) \implies (3, c)$$

$$(f^{-1})^{-1} = \{(1, a), (2, b), (3, c)\} = f$$

Hence Proved

f is invertible, $f^{-1} = \{(a, 1), (b, 2), (c, 3)\}$, and $(f^{-1})^{-1} = f$.

Question 13

Let $S = \{1, 2, 3\}$. Determine whether the functions $f : S \rightarrow S$ defined as below have inverse. Find f^{-1} , if it exists.

(i) $f = \{(1, 1), (2, 2), (3, 3)\}$

(ii) $f = \{(1, 2), (2, 1), (3, 1)\}$

(iii) $f = \{(1, 3), (3, 2), (2, 1)\}$

(iv) $f = \{(1, 3), (2, 2), (3, 1)\}$

TYPE 33a Proving Invertibility & Finding Inverse – Roster/Finite Functions

Given: $S = \{1, 2, 3\}$

Solution:

(i) $f = \{(1, 1), (2, 2), (3, 3)\}$

Images 1, 2, 3 are all distinct $\implies f$ is one-one. Range = $\{1, 2, 3\} = S \implies f$ is onto. So f is bijective and invertible.

Reversing the pairs: $(1, 1) \implies (1, 1), (2, 2) \implies (2, 2), (3, 3) \implies (3, 3)$.

$$f^{-1} = \{(1, 1), (2, 2), (3, 3)\}$$

(ii) $f = \{(1, 2), (2, 1), (3, 1)\}$

Check one-one: $f(2) = 1$ and $f(3) = 1$, i.e. two distinct elements 2, 3 have the same image 1.

$\implies f$ is not one-one $\implies f$ is not invertible

(iii) $f = \{(1, 3), (3, 2), (2, 1)\}$

Images 3, 2, 1 (for $x = 1, 3, 2$ respectively) are all distinct $\implies f$ is one-one. Range = $\{3, 2, 1\} = S \implies f$ is onto. So f is bijective and invertible.

Reversing the pairs: $(1, 3) \implies (3, 1), (3, 2) \implies (2, 3), (2, 1) \implies (1, 2)$.

$$f^{-1} = \{(3, 1), (2, 3), (1, 2)\}$$

(iv) $f = \{(1, 3), (2, 2), (3, 1)\}$

Images 3, 2, 1 are all distinct $\implies f$ is one-one. Range = $\{3, 2, 1\} = S \implies f$ is onto. So f is bijective and invertible.

Reversing the pairs: $(1, 3) \implies (3, 1), (2, 2) \implies (2, 2), (3, 1) \implies (1, 3)$.

$$f^{-1} = \{(3, 1), (2, 2), (1, 3)\}$$

Final Answer

- (i) Invertible, $f^{-1} = \{(1, 1), (2, 2), (3, 3)\}$
- (ii) Not invertible, since f is not one-one
- (iii) Invertible, $f^{-1} = \{(3, 1), (2, 3), (1, 2)\}$
- (iv) Invertible, $f^{-1} = \{(3, 1), (2, 2), (1, 3)\}$

Question 14

Explain why for the following functions $f : X \rightarrow Y$, we cannot define f^{-1} :

- (i) $X = \mathbb{R} - \{0\}$, $Y = \mathbb{R}$ and $f(x) = \frac{1}{x}$ for all $x \in X$
- (ii) $X = \{-1, 0, 1, 2\}$, $Y = \{2, 3\}$ and $f(x) = 2$ for $x = -1, 0, 1$ and $f(2) = 3$
- (iii) $X = \mathbb{R} = Y$ and $f(x) = x^2 + 1$ for all $x \in \mathbb{R}$

TYPE 31 Testing Invertibility of Functions

To Prove: Explain why f^{-1} cannot be defined for each of the given functions.

Concept / Formula Used: $f : X \rightarrow Y$

Solution:

- (i) For $X = \mathbb{R} - \{0\}$, $Y = \mathbb{R}$, $f(x) = \frac{1}{x}$:

$$\text{Range of } f = \left\{ \frac{1}{x} \mid x \in \mathbb{R} - \{0\} \right\} = \mathbb{R} - \{0\}$$

Since $0 \in Y$ has no pre-image in X (as $\frac{1}{x} = 0$ has no solution), the range of f does not equal the codomain $Y = \mathbb{R}$. Thus, f is not onto. Therefore, f^{-1} cannot be defined.

- (ii) For $X = \{-1, 0, 1, 2\}$, $Y = \{2, 3\}$ with $f(-1) = 2$, $f(0) = 2$, $f(1) = 2$, $f(2) = 3$: Observe that:

$$f(-1) = f(0) = f(1) = 2$$

Since distinct domain elements $-1, 0, 1$ share the same image 2, f is not one-one. Therefore, f^{-1} cannot be defined.

- (iii) For $X = \mathbb{R}$, $Y = \mathbb{R}$, $f(x) = x^2 + 1$: Take $x_1 = 1$ and $x_2 = -1$:

$$\begin{aligned} f(1) &= 1^2 + 1 = 2 \\ f(-1) &= (-1)^2 + 1 = 2 \end{aligned}$$

Since $f(1) = f(-1) = 2$ while $1 \neq -1$, f is not one-one. Furthermore, $x^2 + 1 \geq 1$ for all $x \in \mathbb{R}$, so negative numbers in Y have no pre-images, making f not onto as well. Therefore, f^{-1} cannot be defined.

Hence Proved

In all three cases, f fails to be bijective (either not one-one or not onto), so f^{-1} cannot be defined.

Question 15

Let $S = \{a, b, c\}$ and $T = \{1, 2, 3\}$. Find f^{-1} of the following functions f from S to T , if it exists.

- (i) $f = \{(a, 3), (b, 2), (c, 1)\}$
- (ii) $f = \{(a, 2), (b, 1), (c, 1)\}$

TYPE 29 Finding Inverse Function on Finite Sets

Given: $S = \{a, b, c\}$, $T = \{1, 2, 3\}$, and functions $f : S \rightarrow T$

To Find: f^{-1} if it exists

Concept / Formula Used: f^{-1} f

Solution:

(i) For $f = \{(a, 3), (b, 2), (c, 1)\}$: - Injectivity: $f(a) = 3, f(b) = 2, f(c) = 1$. All images are distinct, so f is one-one. - Surjectivity: Range of $f = \{3, 2, 1\} = T$, so f is onto. Since f is bijective, f^{-1} exists. Reversing the ordered pairs $(x, y) \mapsto (y, x)$:

$$f^{-1} = \{(3, a), (2, b), (1, c)\} \quad \text{or} \quad \{(1, c), (2, b), (3, a)\}$$

(ii) For $f = \{(a, 2), (b, 1), (c, 1)\}$: Observe that $f(b) = 1$ and $f(c) = 1$. Since $f(b) = f(c)$ with $b \neq c$, f is not one-one. Since f is not one-one, f is not bijective. Therefore, f^{-1} does not exist.

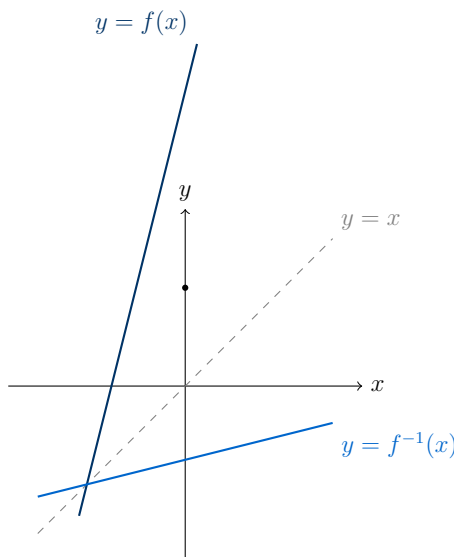
Final Answer

(i) $f^{-1} = \{(1, c), (2, b), (3, a)\}$

(ii) f^{-1} does not exist

Question 16

Let $\mathbb{R}$ be the set of all real numbers. Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 4x + 5$ is invertible. Also find inverse of f . Draw their sketch on one graph.

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

Graphs of $f(x) = 4x + 5$, $f^{-1}(x) = \frac{x-5}{4}$, and symmetry line $y = x$

To Prove: f is invertible, find $f^{-1}(x)$, and sketch both graphs

Concept / Formula Used: f $f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall y \in \text{Codomain}, \exists x \in \text{Domain} \quad f(x) = y$

Solution:

1. Proof of One-One (Injective): Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$.

$$4x_1 + 5 = 4x_2 + 5$$

$$4x_1 = 4x_2$$

$$x_1 = x_2$$

Hence, f is one-one.

2. Proof of Onto (Surjective): Let $y \in \mathbb{R}$ (codomain). Set $y = f(x)$:

$$y = 4x + 5$$

$$y - 5 = 4x$$

$$x = \frac{y - 5}{4}$$

For any $y \in \mathbb{R}$, $x = \frac{y-5}{4} \in \mathbb{R}$ (domain) and:

$$f(x) = f\left(\frac{y-5}{4}\right) = 4\left(\frac{y-5}{4}\right) + 5 = y - 5 + 5 = y$$

Hence, f is onto.

3. Finding f^{-1} : Since f is bijective, it is invertible. The inverse function $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is:

$$f^{-1}(y) = \frac{y-5}{4} \implies f^{-1}(x) = \frac{x-5}{4}$$

Hence Proved

f is invertible with inverse $f^{-1}(x) = \frac{x-5}{4}$.

Question 17

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-1}{3}$ is one-one and onto. Also find the inverse of the function f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is one-one and onto, and find f^{-1}

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall y \in \mathbb{R}, \exists x \in \mathbb{R} \quad f(x) = y$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$.

$$\frac{2x_1 - 1}{3} = \frac{2x_2 - 1}{3}$$

$$2x_1 - 1 = 2x_2 - 1$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

Therefore, f is one-one.

2. Proof of Onto: Let $y \in \mathbb{R}$ (codomain). Set $y = f(x)$:

$$y = \frac{2x - 1}{3}$$

$$3y = 2x - 1$$

$$2x = 3y + 1$$

$$x = \frac{3y + 1}{2}$$

For every $y \in \mathbb{R}$, $x = \frac{3y+1}{2} \in \mathbb{R}$ (domain) and:

$$f(x) = f\left(\frac{3y+1}{2}\right) = \frac{2\left(\frac{3y+1}{2}\right) - 1}{3} = \frac{3y + 1 - 1}{3} = \frac{3y}{3} = y$$

Therefore, f is onto.

3. Finding f^{-1} : Since f is bijective, $f^{-1}(y) = x = \frac{3y+1}{2}$. Replacing y with x :

$$f^{-1}(x) = \frac{3x + 1}{2}$$

Hence Proved

f is one-one and onto, with $f^{-1}(x) = \frac{3x+1}{2}$.

Question 18

Consider a bijective function $f : \mathbb{R}_0^+ \rightarrow (7, \infty)$ given by $f(x) = 16x^2 + 24x + 7$, where $\mathbb{R}_0^+$ is the set of positive real numbers. Find the inverse function of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

Given: $f : \mathbb{R}_0^+ \rightarrow (7, \infty)$, $f(x) = 16x^2 + 24x + 7$, where $\mathbb{R}_0^+ = (0, \infty)$

To Find: $f^{-1}(x)$

Concept / Formula Used: $ax^2 + bx + c$ x $y = f(x)$

Solution:

Let $y = f(x)$.

$$y = 16x^2 + 24x + 7$$

Complete the square on the right-hand side:

$$y = (4x)^2 + 2(4x)(3) + 3^2 - 9 + 7$$

$$y = (4x + 3)^2 - 2$$

Add 2 to both sides:

$$y + 2 = (4x + 3)^2$$

Since $x \in \mathbb{R}_0^+$ (i.e. $x > 0$), we have $4x + 3 > 3 > 0$. Taking the positive square root on both sides:

$$\sqrt{y + 2} = 4x + 3$$

Subtract 3 from both sides:

$$4x = \sqrt{y + 2} - 3$$

Divide by 4:

$$x = \frac{\sqrt{y+2} - 3}{4}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{\sqrt{y+2} - 3}{4}$$

Replacing dummy variable y with x :

$$f^{-1}(x) = \frac{\sqrt{x+2} - 3}{4}$$

Final Answer

$$f^{-1}(x) = \frac{\sqrt{x+2}-3}{4} \text{ for } x \in (7, \infty)$$

Question 19

If $A = \mathbb{R} - \{-3\}$ and $B = \mathbb{R} - \{2\}$ and a function $f : A \rightarrow B$ be given by $f(x) = \frac{2x+1}{x+3}$, then show that the function f is one-one and onto. Hence, find f^{-1} .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is one-one and onto, and find f^{-1}

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall y \in B, \exists x \in A \quad f(x) = y$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in A = \mathbb{R} - \{-3\}$ such that $f(x_1) = f(x_2)$.

$$\frac{2x_1 + 1}{x_1 + 3} = \frac{2x_2 + 1}{x_2 + 3}$$

Cross-multiplying:

$$(2x_1 + 1)(x_2 + 3) = (2x_2 + 1)(x_1 + 3)$$

Expanding both sides:

$$2x_1x_2 + 6x_1 + x_2 + 3 = 2x_1x_2 + 6x_2 + x_1 + 3$$

Subtract $2x_1x_2 + 3$ from both sides:

$$6x_1 + x_2 = 6x_2 + x_1$$

Rearranging terms:

$$6x_1 - x_1 = 6x_2 - x_2$$

$$5x_1 = 5x_2$$

$$x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Let $y \in B = \mathbb{R} - \{2\}$. Set $y = f(x)$:

$$\begin{aligned} y &= \frac{2x+1}{x+3} \\ y(x+3) &= 2x+1 \\ xy+3y &= 2x+1 \\ xy-2x &= 1-3y \\ x(y-2) &= 1-3y \\ x &= \frac{1-3y}{y-2} \end{aligned}$$

Since $y \neq 2$, $x = \frac{1-3y}{y-2}$ is a well-defined real number. Check if $x \in A$ (i.e., $x \neq -3$): If $x = -3$:

$$\frac{1-3y}{y-2} = -3 \implies 1-3y = -3y+6 \implies 1=6 \quad (\text{Impossible})$$

Thus, $x \neq -3$, so $x \in A$. Evaluating $f(x)$:

$$f(x) = f\left(\frac{1-3y}{y-2}\right) = \frac{2\left(\frac{1-3y}{y-2}\right) + 1}{\left(\frac{1-3y}{y-2}\right) + 3} = \frac{2-6y+y-2}{1-3y+3y-6} = \frac{-5y}{-5} = y$$

Thus, f is onto.

3. Finding f^{-1} : Since f is bijective, $f^{-1} : B \rightarrow A$ is:

$$f^{-1}(y) = \frac{1-3y}{y-2} \implies f^{-1}(x) = \frac{1-3x}{x-2}$$

Hence Proved

f is one-one and onto, with $f^{-1}(x) = \frac{1-3x}{x-2}$.

Question 20

If $f : A \rightarrow A$ and $A = \mathbb{R} - \left\{\frac{8}{5}\right\}$, show that the function $f(x) = \frac{8x+3}{5x-8}$ is one-one onto. Hence, find f^{-1} .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

★ Likely Exam Question

To Prove: f is one-one and onto, and find f^{-1}

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2$ $y = f(x)$ x

Solution:

1. Proof of One-One: Let $x_1, x_2 \in A$ such that $f(x_1) = f(x_2)$.

$$\frac{8x_1+3}{5x_1-8} = \frac{8x_2+3}{5x_2-8}$$

Cross-multiplying:

$$(8x_1+3)(5x_2-8) = (8x_2+3)(5x_1-8)$$

Expanding both sides:

$$40x_1x_2 - 64x_1 + 15x_2 - 24 = 40x_1x_2 - 64x_2 + 15x_1 - 24$$

Cancel $40x_1x_2 - 24$ from both sides:

$$-64x_1 + 15x_2 = -64x_2 + 15x_1$$

Rearranging terms:

$$15x_2 + 64x_2 = 15x_1 + 64x_1$$

$$79x_2 = 79x_1$$

$$x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Let $y \in A = \mathbb{R} - \{\frac{8}{5}\}$. Set $y = f(x)$:

$$y = \frac{8x+3}{5x-8}$$

$$y(5x-8) = 8x+3$$

$$5xy - 8y = 8x + 3$$

$$5xy - 8x = 8y + 3$$

$$x(5y-8) = 8y+3$$

$$x = \frac{8y+3}{5y-8}$$

Since $y \neq \frac{8}{5}$, $5y-8 \neq 0$, so x is well-defined. If $x = \frac{8}{5}$, then $\frac{8y+3}{5y-8} = \frac{8}{5} \implies 40y+15 = 40y-64 \implies 15 = -64$, which is impossible. Hence $x \in A$. Evaluating $f(x)$:

$$f(x) = f\left(\frac{8y+3}{5y-8}\right) = \frac{8\left(\frac{8y+3}{5y-8}\right) + 3}{5\left(\frac{8y+3}{5y-8}\right) - 8} = \frac{64y+24+15y-24}{40y+15-40y+64} = \frac{79y}{79} = y$$

Thus, f is onto.

3. Finding f^{-1} :

$$f^{-1}(y) = \frac{8y+3}{5y-8} \implies f^{-1}(x) = \frac{8x+3}{5x-8}$$

Notice that $f^{-1}(x) = f(x)$ for all $x \in A$.

Hence Proved

f is one-one and onto, and $f^{-1}(x) = \frac{8x+3}{5x-8}$.

Question 21

Consider the function $f : \mathbb{R}_+ \rightarrow [9, \infty)$ defined by $f(x) = x^2 + 9$, where $\mathbb{R}_+$ is the set of all non-negative real numbers. Show that f is invertible. Also find the inverse of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is invertible and find f^{-1}

Concept / Formula Used: f

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{R}_+ = [0, \infty)$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned}x_1^2 + 9 &= x_2^2 + 9 \\x_1^2 &= x_2^2\end{aligned}$$

Taking square roots on both sides, since $x_1, x_2 \geq 0$:

$$x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Let $y \in [9, \infty)$ (codomain). Set $y = f(x)$:

$$\begin{aligned}y &= x^2 + 9 \\y - 9 &= x^2 \\x &= \sqrt{y - 9} \quad (\text{since } x \geq 0 \text{ and } y \geq 9)\end{aligned}$$

Since $y \geq 9$, $y - 9 \geq 0$, so $x = \sqrt{y - 9} \in \mathbb{R}_+$. Evaluating $f(x)$:

$$f(\sqrt{y - 9}) = (\sqrt{y - 9})^2 + 9 = (y - 9) + 9 = y$$

Thus, f is onto.

3. Finding f^{-1} : Since f is one-one and onto, f is invertible:

$$f^{-1}(y) = \sqrt{y - 9} \implies f^{-1}(x) = \sqrt{x - 9}$$

Hence Proved

f is invertible with inverse $f^{-1}(x) = \sqrt{x - 9}$.

Question 22

If $\mathbb{R}_0$ is the set of all non-zero real numbers, show that the function $f : \mathbb{R}_0 \rightarrow \mathbb{R}_0$ defined by $f(x) = \frac{3}{x}$ is invertible and it is the inverse of itself.

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is invertible and $f^{-1}(x) = f(x)$

Concept / Formula Used: $f \quad f^{-1}(x) = f(x)$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{R}_0$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned}\frac{3}{x_1} &= \frac{3}{x_2} \\3x_2 &= 3x_1 \\x_1 &= x_2\end{aligned}$$

Hence, f is one-one.

2. Proof of Onto: Let $y \in \mathbb{R}_0$ (codomain). Set $y = f(x)$:

$$y = \frac{3}{x}$$

$$x = \frac{3}{y}$$

Since $y \neq 0$, $x = \frac{3}{y} \in \mathbb{R}_0$ (domain). Evaluating $f(x)$:

$$f\left(\frac{3}{y}\right) = \frac{3}{\frac{3}{y}} = y$$

Thus, f is onto.

3. Finding f^{-1} : Since f is bijective, f^{-1} exists. From $x = \frac{3}{y}$:

$$f^{-1}(y) = \frac{3}{y} \implies f^{-1}(x) = \frac{3}{x}$$

Therefore, $f^{-1}(x) = f(x) = \frac{3}{x}$ for all $x \in \mathbb{R}_0$. Thus, f is the inverse of itself.

Hence Proved

f is invertible and $f^{-1}(x) = f(x) = \frac{3}{x}$.

Question 23

Prove that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2 + x + 1$ is one-one but not onto. Find the inverse of $f : \mathbb{N} \rightarrow S$ where S is the range of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

★ Likely Exam Question

To Prove: $f : \mathbb{N} \rightarrow \mathbb{N}$ is one-one, not onto, and $f^{-1} : S \rightarrow \mathbb{N}$ is $f^{-1}(x) = \frac{\sqrt{4x-3}-1}{2}$

Concept / Formula Used: $f(x_1) - f(x_2) = 0$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{N}$ such that $f(x_1) = f(x_2)$.

$$x_1^2 + x_1 + 1 = x_2^2 + x_2 + 1$$

$$(x_1^2 - x_2^2) + (x_1 - x_2) = 0$$

$$(x_1 - x_2)(x_1 + x_2) + (x_1 - x_2) = 0$$

$$(x_1 - x_2)(x_1 + x_2 + 1) = 0$$

Since $x_1, x_2 \in \mathbb{N}$, $x_1 \geq 1$ and $x_2 \geq 1$, which implies $(x_1 + x_2 + 1) \geq 3 > 0$. Dividing by $(x_1 + x_2 + 1)$:

$$x_1 - x_2 = 0 \implies x_1 = x_2$$

Thus, f is one-one.

2. Proof that f is not onto on $\mathbb{N}$: For $x = 1 \in \mathbb{N}$, $f(1) = 1^2 + 1 + 1 = 3$. For $x = 2 \in \mathbb{N}$, $f(2) = 2^2 + 2 + 1 = 7$. Consider $2 \in \mathbb{N}$ (codomain). There is no $x \in \mathbb{N}$ such that $f(x) = 2$, since $f(x) \geq 3$ for all $x \in \mathbb{N}$. Thus, $f : \mathbb{N} \rightarrow \mathbb{N}$ is not onto.

3. Inverse of $f : \mathbb{N} \rightarrow S$ (where $S = \text{Range of } f$): By restricting the codomain to range S , $f : \mathbb{N} \rightarrow S$ becomes bijective and invertible. Let $y = f(x) = x^2 + x + 1$. Multiply both sides by 4:

$$\begin{aligned} 4y &= 4x^2 + 4x + 4 \\ 4y &= (2x + 1)^2 + 3 \\ (2x + 1)^2 &= 4y - 3 \end{aligned}$$

Since $x \in \mathbb{N}$, $2x + 1 > 0$, take the positive square root:

$$\begin{aligned} 2x + 1 &= \sqrt{4y - 3} \\ 2x &= \sqrt{4y - 3} - 1 \\ x &= \frac{\sqrt{4y - 3} - 1}{2} \end{aligned}$$

Therefore, $f^{-1} : S \rightarrow \mathbb{N}$ is:

$$f^{-1}(x) = \frac{\sqrt{4x - 3} - 1}{2}$$

Hence Proved

f is one-one but not onto on $\mathbb{N}$, and $f^{-1}(x) = \frac{\sqrt{4x-3}-1}{2}$ for domain S .

Question 24

Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 4x^2 + 12x + 15$, where S is the range of f . Show that f is invertible and find the inverse of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

★ Likely Exam Question

To Prove: f is invertible and find f^{-1}

Concept / Formula Used: S

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{N}$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} 4x_1^2 + 12x_1 + 15 &= 4x_2^2 + 12x_2 + 15 \\ 4(x_1^2 - x_2^2) + 12(x_1 - x_2) &= 0 \\ 4(x_1 - x_2)(x_1 + x_2) + 12(x_1 - x_2) &= 0 \\ 4(x_1 - x_2)(x_1 + x_2 + 3) &= 0 \end{aligned}$$

Since $x_1, x_2 \in \mathbb{N}$, $x_1 \geq 1$ and $x_2 \geq 1$, so $(x_1 + x_2 + 3) \geq 5 > 0$. Dividing by $4(x_1 + x_2 + 3)$:

$$x_1 - x_2 = 0 \implies x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Since S is defined explicitly as the range of f , every element $y \in S$ has at least one pre-image $x \in \mathbb{N}$ such that $f(x) = y$. Hence, $f : \mathbb{N} \rightarrow S$ is onto. Since f is both one-one and onto, f is invertible.

3. Finding f^{-1} : Let $y = f(x) = 4x^2 + 12x + 15$. Complete the square on the right side:

$$\begin{aligned}y &= (2x)^2 + 2(2x)(3) + 3^2 - 9 + 15 \\y &= (2x + 3)^2 + 6\end{aligned}$$

Subtract 6 from both sides:

$$(2x + 3)^2 = y - 6$$

Since $x \in \mathbb{N}$, $2x + 3 > 0$. Taking the positive square root:

$$\begin{aligned}2x + 3 &= \sqrt{y - 6} \\2x &= \sqrt{y - 6} - 3 \\x &= \frac{\sqrt{y - 6} - 3}{2}\end{aligned}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{\sqrt{y - 6} - 3}{2}$$

Replacing y with x :

$$f^{-1}(x) = \frac{\sqrt{x - 6} - 3}{2}$$

Hence Proved

f is invertible and $f^{-1}(x) = \frac{\sqrt{x-6}-3}{2}$.

Question 25

Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 9x^2 + 6x - 5$, where S is the range of f . Show that f is invertible. Find the inverse of f and hence find $f^{-1}(43)$ and $f^{-1}(163)$.

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

Given: $f : \mathbb{N} \rightarrow S$ defined by $f(x) = 9x^2 + 6x - 5$, where $S = \text{Range}(f)$.

To Find: Show that f is invertible, find $f^{-1}(y)$, and calculate $f^{-1}(43)$ and $f^{-1}(163)$.

Concept / Formula Used: $f : X \rightarrow Y$ $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$

Solution:

Step 1: Proof of Injectivity (One-One)

Let $x_1, x_2 \in \mathbb{N}$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned}9x_1^2 + 6x_1 - 5 &= 9x_2^2 + 6x_2 - 5 \\9x_1^2 - 9x_2^2 + 6x_1 - 6x_2 &= 0 \\9(x_1^2 - x_2^2) + 6(x_1 - x_2) &= 0 \\9(x_1 - x_2)(x_1 + x_2) + 6(x_1 - x_2) &= 0 \\(x_1 - x_2)[9(x_1 + x_2) + 6] &= 0\end{aligned}$$

Since $x_1, x_2 \in \mathbb{N}$, we have $x_1 \geq 1$ and $x_2 \geq 1$, which implies:

$$9(x_1 + x_2) + 6 \geq 9(1 + 1) + 6 = 24 > 0$$

Since $9(x_1 + x_2) + 6 \neq 0$, we must have:

$$x_1 - x_2 = 0 \implies x_1 = x_2$$

Therefore, f is a one-one (injective) function.

Step 2: Proof of Surjectivity (Onto)

The codomain of f is given to be S , where $S = \text{Range}(f)$. By definition of range, for every element $y \in S$, there exists an element $x \in \mathbf{N}$ such that $f(x) = y$. Therefore, f is an onto (surjective) function.

Since f is both one-one and onto, f is invertible.

Step 3: Finding the Inverse Function f^{-1}

Let $y \in S$ such that $y = f(x)$ for some $x \in \mathbf{N}$.

$$\begin{aligned} y &= 9x^2 + 6x - 5 \\ y &= (3x)^2 + 2(3x)(1) + 1^2 - 1 - 5 \\ y &= (3x + 1)^2 - 6 \\ y + 6 &= (3x + 1)^2 \\ \sqrt{y + 6} &= 3x + 1 \quad (\text{since } x \in \mathbf{N} \implies 3x + 1 > 0) \\ 3x &= \sqrt{y + 6} - 1 \\ x &= \frac{\sqrt{y + 6} - 1}{3} \end{aligned}$$

Since $f(x) = y \iff x = f^{-1}(y)$, we have:

$$f^{-1}(y) = \frac{\sqrt{y + 6} - 1}{3}$$

or expressed in terms of variable x :

$$f^{-1}(x) = \frac{\sqrt{x + 6} - 1}{3}$$

Step 4: Evaluating $f^{-1}(43)$ and $f^{-1}(163)$

For $y = 43$:

$$\begin{aligned} f^{-1}(43) &= \frac{\sqrt{43 + 6} - 1}{3} \\ &= \frac{\sqrt{49} - 1}{3} \\ &= \frac{7 - 1}{3} \\ &= \frac{6}{3} = 2 \end{aligned}$$

For $y = 163$:

$$\begin{aligned} f^{-1}(163) &= \frac{\sqrt{163 + 6} - 1}{3} \\ &= \frac{\sqrt{169} - 1}{3} \\ &= \frac{13 - 1}{3} \\ &= \frac{12}{3} = 4 \end{aligned}$$

Final Answer

The inverse function is $f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$.

The values are $f^{-1}(43) = 2$ and $f^{-1}(163) = 4$.

Common Mistake: Taking both positive and negative roots when solving $(3x+1)^2 = y+6$ is a common error. Since the domain is $\mathbf{N}$, $x > 0 \implies 3x+1 > 0$, so only the positive root is valid.

Question 26

If function $f : \mathbf{R} \rightarrow \mathbf{R}$ is defined by $f(x) = 5x - 3$, then find function $g : \mathbf{R} \rightarrow \mathbf{R}$ such that $g \circ f = I_{\mathbf{R}} = f \circ g$.

TYPE 34 Finding Function using Composite Identity

Given: $f : \mathbf{R} \rightarrow \mathbf{R}$ defined by $f(x) = 5x - 3$, and $g \circ f = I_{\mathbf{R}} = f \circ g$.

To Find: The function $g : \mathbf{R} \rightarrow \mathbf{R}$.

Concept / Formula Used: $g \circ f = I_{\mathbf{R}} \quad f \circ g = I_{\mathbf{R}} \quad g \quad f \quad g(x) = f^{-1}(x)$

Solution:

By definition of inverse function, since $g \circ f = I_{\mathbf{R}}$ and $f \circ g = I_{\mathbf{R}}$, g must be the inverse of f .

Let $y = f(x)$.

$$\begin{aligned} y &= 5x - 3 \\ y + 3 &= 5x \\ x &= \frac{y+3}{5} \end{aligned}$$

Since $f(x) = y \iff x = f^{-1}(y)$, we get:

$$f^{-1}(y) = \frac{y+3}{5}$$

Replacing the dummy variable y with x , the inverse function is:

$$g(x) = \frac{x+3}{5}$$

Verification of Composite Properties:

$$(g \circ f)(x) = g(f(x)) = g(5x - 3) = \frac{(5x - 3) + 3}{5} = \frac{5x}{5} = x = I_{\mathbf{R}}(x)$$

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{x+3}{5}\right) = 5\left(\frac{x+3}{5}\right) - 3 = x + 3 - 3 = x = I_{\mathbf{R}}(x)$$

Final Answer

The required function $g : \mathbf{R} \rightarrow \mathbf{R}$ is $g(x) = \frac{x+3}{5}$.

Question 27

A teacher wrote $f(x) = x^9$ on the board and asked her student to examine whether the following equation is true or false:

$$[f(x)]^{1/3} = [f^{-1}(x)]^3$$

Raghu said, "It is true". Is he correct? Justify your answer.

TYPE 35 Verifying Properties of Inverse Functions

Given: Function $f(x) = x^9$ on $\mathbf{R}$, and the equation $[f(x)]^{1/3} = [f^{-1}(x)]^3$.

To Find: Determine whether Raghu's claim that the equation is true is correct, with justification.

Concept / Formula Used: $f(x) = x^n$ $f^{-1}(x) = x^{1/n}$ $(x^a)^b = x^{a \cdot b}$

Solution:

First, let us find the inverse function $f^{-1}(x)$. Let $y = f(x) = x^9$. Taking the 9th root on both sides:

$$\begin{aligned} y &= x^9 \\ x &= y^{1/9} \end{aligned}$$

Thus, the inverse function is:

$$f^{-1}(x) = x^{1/9}$$

Now, let us evaluate the Left Hand Side (LHS) of the given equation:

$$\begin{aligned} \text{LHS} &= [f(x)]^{1/3} \\ &= (x^9)^{1/3} \\ &= x^{9 \cdot \frac{1}{3}} \\ &= x^3 \end{aligned}$$

Next, let us evaluate the Right Hand Side (RHS) of the given equation:

$$\begin{aligned} \text{RHS} &= [f^{-1}(x)]^3 \\ &= (x^{1/9})^3 \\ &= x^{\frac{1}{9} \cdot 3} \\ &= x^{1/3} \end{aligned}$$

Comparing LHS and RHS:

$$\text{LHS} = x^3 \quad \text{and} \quad \text{RHS} = x^{1/3}$$

For general values of $x \in \mathbf{R}$ (e.g., for $x = 8$, $\text{LHS} = 8^3 = 512$ while $\text{RHS} = 8^{1/3} = 2$), we have:

$$\text{LHS} \neq \text{RHS}$$

Therefore, the statement $[f(x)]^{1/3} = [f^{-1}(x)]^3$ is **FALSE**. Raghu is incorrect.

Final Answer

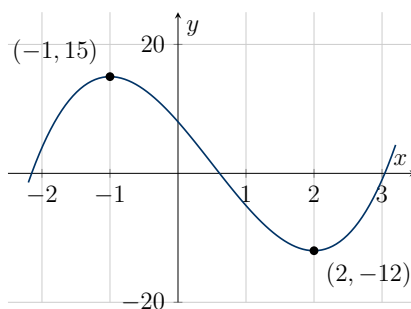
Raghu is **incorrect**. The given equation is **false** because $[f(x)]^{1/3} = x^3$, whereas $[f^{-1}(x)]^3 = x^{1/3}$, which are not equal for all $x \in \mathbf{R}$.

Common Mistake: Confusing exponent rules by assuming $(x^9)^{1/3}$ equals $(x^{1/9})^3$ is a common error; $(x^9)^{1/3} = x^3$ while $(x^{1/9})^3 = x^{1/3}$.

Question 28

A part of the graph of the function $f(x) = 2x^3 - 3x^2 - 12x + 8, x \in \mathbf{R}$ is shown alongside: Answer the following questions:

- Explain why ' f ' does not have an inverse.
- The domain of ' f ' is now restricted to $a \leq x \leq b$ where $a < 0$ and $b > 0$. a and b are chosen so that f has an inverse and the interval $[a, b]$ is as large as possible. Find the domain and range of f^{-1} .

TYPE 36 Invertibility and Inverse under Restricted Domain

Graph of $f(x) = 2x^3 - 3x^2 - 12x + 8$ showing local extrema at $x = -1$ and $x = 2$.

Given: Function $f(x) = 2x^3 - 3x^2 - 12x + 8, x \in \mathbf{R}$, with restricted domain $a \leq x \leq b$ where $a < 0 < b$.

To Find: (a) Explanation for why f lacks an inverse on $\mathbf{R}$.
(b) The domain and range of f^{-1} for the maximal interval $[a, b]$.

Concept / Formula Used: $f \quad f'(x) \geq 0 \quad f'(x) \leq 0$

Solution:

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (2x^3 - 3x^2 - 12x + 8) \\ &= 6x^2 - 6x - 12 \\ &= 6(x^2 - x - 2) \\ &= 6(x - 2)(x + 1) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points at $x = -1$ and $x = 2$.

Part (a): Why f does not have an inverse on $\mathbf{R}$

For $x < -1$, $f'(x) > 0$ (function is strictly increasing).

For $-1 < x < 2$, $f'(x) < 0$ (function is strictly decreasing).

For $x > 2$, $f'(x) > 0$ (function is strictly increasing).

Since $f'(x)$ changes sign, $f(x)$ has a local maximum at $x = -1$ with value $f(-1) = 2(-1)^3 - 3(-1)^2 - 12(-1) + 8 = 15$, and a local minimum at $x = 2$ with value $f(2) = 2(2)^3 - 3(2)^2 - 12(2) + 8 = -12$.

Because the function changes direction, horizontal lines $y = c$ (for $-12 < c < 15$) intersect the graph at three distinct points. Thus, multiple distinct x -values map to the same y -value (f is many-one). Since f is not one-one on $\mathbf{R}$, it is not bijective and does not have an inverse on $\mathbf{R}$.

Part (b): Finding domain and range of f^{-1}

We require $a < 0 < b$ such that f is strictly monotonic on $[a, b]$ and $[a, b]$ is maximal.

Since $a < 0 < b$, the interval $[a, b]$ contains $x = 0$.

The continuous interval containing $x = 0$ on which $f'(x) \leq 0$ (strictly decreasing) is $[-1, 2]$.

Thus, $a = -1 < 0$ and $b = 2 > 0$, giving the largest possible interval $[a, b] = [-1, 2]$.

On the restricted domain $[-1, 2]$:

$$f(-1) = 15 \quad (\text{maximum value})$$

$$f(2) = -12 \quad (\text{minimum value})$$

Since f is strictly decreasing on $[-1, 2]$, the range of f on this interval is $[-12, 15]$.

By the properties of inverse functions:

$$\text{Domain of } f^{-1} = \text{Range of } f = [-12, 15]$$

$$\text{Range of } f^{-1} = \text{Domain of } f = [-1, 2]$$

Final Answer

(a) f lacks an inverse on $\mathbf{R}$ because it has local extrema at $x = -1$ and $x = 2$, making it many-one.

(b) For the maximal interval $[-1, 2]$:

$$\text{Domain of } f^{-1} = [-12, 15]$$

$$\text{Range of } f^{-1} = [-1, 2].$$

Question 29

"A function f is called a self-inverse function if $f^{-1}(x) = f(x)$ for all values of x in the domain." Let $f(x) = \frac{\pi^2}{x}$, where $x \neq 0, x \in \mathbf{R}$. Show that $f(x)$ is a self-inverse function.

TYPE 37 Proving Self-Inverse Functions

To Prove: Show that $f(x) = \frac{\pi^2}{x}$ satisfies $f^{-1}(x) = f(x)$ for all $x \in \mathbf{R} \setminus \{0\}$.

Concept / Formula Used: $f \quad (f \circ f)(x) = x \quad f^{-1}(x) = f(x) \quad x \in \text{Domain}(f)$

Solution:

Method 1: By Composite Function $(f \circ f)(x)$

Let $x \in \mathbf{R} \setminus \{0\}$.

$$\begin{aligned}(f \circ f)(x) &= f(f(x)) \\ &= f\left(\frac{\pi^2}{x}\right) \\ &= \frac{\pi^2}{\left(\frac{\pi^2}{x}\right)} \\ &= \pi^2 \cdot \frac{x}{\pi^2} \\ &= x\end{aligned}$$

Since $(f \circ f)(x) = x$ for all $x \neq 0$, f is equal to its own inverse f^{-1} .

Method 2: By Finding $f^{-1}(x)$ Explicitly

Let $y = f(x)$.

$$\begin{aligned}y &= \frac{\pi^2}{x} \\ y \cdot x &= \pi^2 \\ x &= \frac{\pi^2}{y}\end{aligned}$$

Since $f(x) = y \iff x = f^{-1}(y)$, we have:

$$f^{-1}(y) = \frac{\pi^2}{y}$$

Replacing y with x :

$$f^{-1}(x) = \frac{\pi^2}{x}$$

Comparing $f^{-1}(x)$ with $f(x)$:

$$f^{-1}(x) = f(x)$$

Hence, $f(x)$ is a self-inverse function.

Hence Proved

Since $f^{-1}(x) = f(x) = \frac{\pi^2}{x}$ for all $x \in \mathbf{R} \setminus \{0\}$, $f(x)$ is proved to be a self-inverse function.

ISC Class XII Mathematics

Chapter 2 Inverse Trigonometric Functions

75 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 2.1 – Principal Values, Domain and Range of Inverse Trigonometric Functions
- II. 2.2 – Properties and Simplification of Inverse Trigonometric Functions

2.1 – Principal Values, Domain and Range of Inverse Trigonometric Functions

Question 1

Find the principal values of the following:

- (i) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$
- (ii) $\cos^{-1}\left(\frac{1}{2}\right)$
- (iii) $\tan^{-1}(\sqrt{3})$

TYPE 1 Principal Values of Inverse Trigonometric Functions

To Find: The principal values of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$, $\cos^{-1}\left(\frac{1}{2}\right)$, and $\tan^{-1}(\sqrt{3})$

Concept / Formula Used: $\sin^{-1} x$ $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\cos^{-1} x$ $[0, \pi]$ $\tan^{-1} x$ $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Solution:

(i) Let $y = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$.

$$\sin y = \frac{1}{\sqrt{2}}$$

Since $\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ and $\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$:

$$y = \frac{\pi}{4}$$

(ii) Let $y = \cos^{-1}\left(\frac{1}{2}\right)$.

$$\cos y = \frac{1}{2}$$

Since $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$ and $\frac{\pi}{3} \in [0, \pi]$:

$$y = \frac{\pi}{3}$$

(iii) Let $y = \tan^{-1}(\sqrt{3})$.

$$\tan y = \sqrt{3}$$

Since $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$ and $\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$y = \frac{\pi}{3}$$

Final Answer

$$(i) \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

$$(ii) \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$(iii) \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

Question 2

Find the principal values of the following:

$$(i) \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

$$(ii) \cot^{-1}(\sqrt{3})$$

$$(iii) \operatorname{cosec}^{-1}(\sqrt{2})$$

TYPE 1 Principal Values of Inverse Trigonometric Functions

To Find: The principal values of $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$, $\cot^{-1}(\sqrt{3})$, and $\operatorname{cosec}^{-1}(\sqrt{2})$

Concept / Formula Used: $\sec^{-1} x \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$ $\cot^{-1} x \in (0, \pi)$ $\operatorname{cosec}^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$

Solution:

$$(i) \text{ Let } y = \sec^{-1}\left(\frac{2}{\sqrt{3}}\right).$$

$$\sec y = \frac{2}{\sqrt{3}}$$

Since $\sec\left(\frac{\pi}{6}\right) = \frac{2}{\sqrt{3}}$ and $\frac{\pi}{6} \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$:

$$y = \frac{\pi}{6}$$

$$(ii) \text{ Let } y = \cot^{-1}(\sqrt{3}).$$

$$\cot y = \sqrt{3}$$

Since $\cot\left(\frac{\pi}{6}\right) = \sqrt{3}$ and $\frac{\pi}{6} \in (0, \pi)$:

$$y = \frac{\pi}{6}$$

(iii) Let $y = \operatorname{cosec}^{-1}(\sqrt{2})$.

$$\operatorname{cosec} y = \sqrt{2}$$

Since $\operatorname{cosec} \left(\frac{\pi}{4} \right) = \sqrt{2}$ and $\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \setminus \{0\}$:

$$y = \frac{\pi}{4}$$

Final Answer

$$(i) \sec^{-1} \left(\frac{2}{\sqrt{3}} \right) = \frac{\pi}{6}$$

$$(ii) \cot^{-1}(\sqrt{3}) = \frac{\pi}{6}$$

$$(iii) \operatorname{cosec}^{-1}(\sqrt{2}) = \frac{\pi}{4}$$

Question 3

Find the principal values of the following:

$$(i) \cos^{-1} \left(-\frac{1}{2} \right)$$

$$(ii) \tan^{-1}(-1)$$

$$(iii) \operatorname{cosec}^{-1}(-\sqrt{2})$$

TYPE 1 Principal Values of Inverse Trigonometric Functions

To Find: The principal values of $\cos^{-1} \left(-\frac{1}{2} \right)$, $\tan^{-1}(-1)$, and $\operatorname{cosec}^{-1}(-\sqrt{2})$

Concept / Formula Used: $\cos^{-1}(-x) = \pi - \cos^{-1} x$ $\tan^{-1}(-x) = -\tan^{-1} x$ $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$

Solution:

(i) Let $y = \cos^{-1} \left(-\frac{1}{2} \right)$. Using the identity $\cos^{-1}(-x) = \pi - \cos^{-1} x$:

$$\begin{aligned} \cos^{-1} \left(-\frac{1}{2} \right) &= \pi - \cos^{-1} \left(\frac{1}{2} \right) \\ &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3} \end{aligned}$$

Since $\frac{2\pi}{3} \in [0, \pi]$, this is the principal value.

(ii) Let $y = \tan^{-1}(-1)$. Using the identity $\tan^{-1}(-x) = -\tan^{-1} x$:

$$\begin{aligned} \tan^{-1}(-1) &= -\tan^{-1}(1) \\ &= -\frac{\pi}{4} \end{aligned}$$

Since $-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$, this is the principal value.

(iii) Let $y = \operatorname{cosec}^{-1}(-\sqrt{2})$. Using the identity $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$:

$$\begin{aligned}\operatorname{cosec}^{-1}(-\sqrt{2}) &= -\operatorname{cosec}^{-1}(\sqrt{2}) \\ &= -\frac{\pi}{4}\end{aligned}$$

Since $-\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$, this is the principal value.

Final Answer

$$(i) \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

$$(ii) \tan^{-1}(-1) = -\frac{\pi}{4}$$

$$(iii) \operatorname{cosec}^{-1}(-\sqrt{2}) = -\frac{\pi}{4}$$

Common Mistake: Writing $\cos^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{3}$ is a common error. For $\cos^{-1}(-x)$, the range is $[0, \pi]$, so it must evaluate to $\pi - \cos^{-1} x = \frac{2\pi}{3}$.

Question 4

Find the principal values of the following:

$$(i) \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$$

$$(ii) \sec^{-1}\left(-\frac{2}{\sqrt{3}}\right)$$

$$(iii) \sin^{-1}(-1)$$

TYPE 1 Principal Values of Inverse Trigonometric Functions

To Find: The principal values of $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$, $\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right)$, and $\sin^{-1}(-1)$

Concept / Formula Used: $\cot^{-1}(-x) = \pi - \cot^{-1} x$ $\sec^{-1}(-x) = \pi - \sec^{-1} x$ $\sin^{-1}(-x) = -\sin^{-1} x$

Solution:

(i) Using the identity $\cot^{-1}(-x) = \pi - \cot^{-1} x$:

$$\begin{aligned}\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) &= \pi - \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) \\ &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3}\end{aligned}$$

Since $\frac{2\pi}{3} \in (0, \pi)$, this is the principal value.

(ii) Using the identity $\sec^{-1}(-x) = \pi - \sec^{-1} x$:

$$\begin{aligned}\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) &= \pi - \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) \\ &= \pi - \frac{\pi}{6} \\ &= \frac{5\pi}{6}\end{aligned}$$

Since $\frac{5\pi}{6} \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$, this is the principal value.

(iii) Using the identity $\sin^{-1}(-x) = -\sin^{-1} x$:

$$\begin{aligned}\sin^{-1}(-1) &= -\sin^{-1}(1) \\ &= -\frac{\pi}{2}\end{aligned}$$

Since $-\frac{\pi}{2} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, this is the principal value.

Final Answer

$$(i) \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \frac{2\pi}{3}$$

$$(ii) \sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) = \frac{5\pi}{6}$$

$$(iii) \sin^{-1}(-1) = -\frac{\pi}{2}$$

Question 5

Evaluate the following:

$$(i) \cos^{-1}\left(\cos \frac{4\pi}{3}\right)$$

$$(ii) \sin^{-1}\left(\sin \frac{4\pi}{3}\right)$$

$$(iii) \tan^{-1}\left(\tan \frac{3\pi}{4}\right)$$

TYPE 2 Evaluating Composite Inverse Trigonometric Functions

To Find: The values of the given composite inverse trigonometric expressions

Concept / Formula Used: $f^{-1}(f(x)) = x \quad x \quad f^{-1}$

Solution:

(i) Since $\frac{4\pi}{3} \notin [0, \pi]$, we reduce the angle:

$$\begin{aligned}\cos\left(\frac{4\pi}{3}\right) &= \cos\left(2\pi - \frac{2\pi}{3}\right) \\ &= \cos\left(\frac{2\pi}{3}\right)\end{aligned}$$

Since $\frac{2\pi}{3} \in [0, \pi]$:

$$\begin{aligned}\cos^{-1}\left(\cos \frac{4\pi}{3}\right) &= \cos^{-1}\left(\cos \frac{2\pi}{3}\right) \\ &= \frac{2\pi}{3}\end{aligned}$$

(ii) Since $\frac{4\pi}{3} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we rewrite the angle:

$$\begin{aligned}\sin\left(\frac{4\pi}{3}\right) &= \sin\left(\pi + \frac{\pi}{3}\right) \\ &= -\sin\left(\frac{\pi}{3}\right) \\ &= \sin\left(-\frac{\pi}{3}\right)\end{aligned}$$

Since $-\frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$:

$$\begin{aligned}\sin^{-1}\left(\sin \frac{4\pi}{3}\right) &= \sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) \\ &= -\frac{\pi}{3}\end{aligned}$$

(iii) Since $\frac{3\pi}{4} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, we rewrite the angle:

$$\begin{aligned}\tan\left(\frac{3\pi}{4}\right) &= \tan\left(\pi - \frac{\pi}{4}\right) \\ &= -\tan\left(\frac{\pi}{4}\right) \\ &= \tan\left(-\frac{\pi}{4}\right)\end{aligned}$$

Since $-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\begin{aligned}\tan^{-1}\left(\tan \frac{3\pi}{4}\right) &= \tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right) \\ &= -\frac{\pi}{4}\end{aligned}$$

Final Answer

$$(i) \cos^{-1}\left(\cos \frac{4\pi}{3}\right) = \frac{2\pi}{3}$$

$$(ii) \sin^{-1}\left(\sin \frac{4\pi}{3}\right) = -\frac{\pi}{3}$$

$$(iii) \tan^{-1}\left(\tan \frac{3\pi}{4}\right) = -\frac{\pi}{4}$$

Common Mistake: Blindly writing $\cos^{-1}(\cos \theta) = \theta$ without checking if $\theta \in [0, \pi]$ is an extremely common error. Always reduce θ to the principal domain first.

Question 6

Evaluate the following:

- (i) $\cos^{-1}\left(\cos \frac{5\pi}{3}\right)$
 (ii) $\tan^{-1}\left(\tan \frac{9\pi}{8}\right)$
 (iii) $\sec^{-1}\left(\sec \frac{9\pi}{5}\right)$

TYPE 2 Evaluating Composite Inverse Trigonometric Functions**To Find:** The values of the given composite inverse trigonometric expressions**Concept / Formula Used:** $f^{-1}(f(x)) = x \quad x$ **Solution:**(i) Since $\frac{5\pi}{3} \notin [0, \pi]$:

$$\begin{aligned}\cos\left(\frac{5\pi}{3}\right) &= \cos\left(2\pi - \frac{\pi}{3}\right) \\ &= \cos\left(\frac{\pi}{3}\right)\end{aligned}$$

Since $\frac{\pi}{3} \in [0, \pi]$:

$$\begin{aligned}\cos^{-1}\left(\cos \frac{5\pi}{3}\right) &= \cos^{-1}\left(\cos \frac{\pi}{3}\right) \\ &= \frac{\pi}{3}\end{aligned}$$

(ii) Since $\frac{9\pi}{8} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\begin{aligned}\tan\left(\frac{9\pi}{8}\right) &= \tan\left(\pi + \frac{\pi}{8}\right) \\ &= \tan\left(\frac{\pi}{8}\right)\end{aligned}$$

Since $\frac{\pi}{8} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\begin{aligned}\tan^{-1}\left(\tan \frac{9\pi}{8}\right) &= \tan^{-1}\left(\tan \frac{\pi}{8}\right) \\ &= \frac{\pi}{8}\end{aligned}$$

(iii) Since $\frac{9\pi}{5} \notin [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$:

$$\begin{aligned}\sec\left(\frac{9\pi}{5}\right) &= \sec\left(2\pi - \frac{\pi}{5}\right) \\ &= \sec\left(\frac{\pi}{5}\right)\end{aligned}$$

Since $\frac{\pi}{5} \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$:

$$\begin{aligned}\sec^{-1}\left(\sec \frac{9\pi}{5}\right) &= \sec^{-1}\left(\sec \frac{\pi}{5}\right) \\ &= \frac{\pi}{5}\end{aligned}$$

Final Answer

(i) $\cos^{-1}\left(\cos \frac{5\pi}{3}\right) = \frac{\pi}{3}$

(ii) $\tan^{-1}\left(\tan \frac{9\pi}{8}\right) = \frac{\pi}{8}$

(iii) $\sec^{-1}\left(\sec \frac{9\pi}{5}\right) = \frac{\pi}{5}$

Question 7

Show that:

(i) $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) \neq \frac{5\pi}{6}$, what is its value?

(ii) $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$, what is its value?

(iii) $\sin^{-1}\left(\sin \frac{5\pi}{3}\right) \neq \frac{5\pi}{3}$, what is its value?

TYPE 2 Evaluating Composite Inverse Trigonometric Functions

To Prove: $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) \neq \frac{5\pi}{6}$, $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$, and $\sin^{-1}\left(\sin \frac{5\pi}{3}\right) \neq \frac{5\pi}{3}$, and evaluate each.

Concept / Formula Used: $f^{-1}(f(x)) = x \quad x \quad f^{-1}(x)$

Solution:

(i) The principal value branch of $\tan^{-1} x$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Since $\frac{5\pi}{6} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, it follows that $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) \neq \frac{5\pi}{6}$.

Now, rewriting the trigonometric argument:

$$\begin{aligned}\tan\left(\frac{5\pi}{6}\right) &= \tan\left(\pi - \frac{\pi}{6}\right) \\ &= -\tan\left(\frac{\pi}{6}\right) \\ &= \tan\left(-\frac{\pi}{6}\right)\end{aligned}$$

Since $-\frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\tan^{-1}\left(\tan \frac{5\pi}{6}\right) = \tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6}$$

(ii) The principal value branch of $\cos^{-1} x$ is $[0, \pi]$.

Since $-\frac{\pi}{6} \notin [0, \pi]$, it follows that $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$.

Now, using $\cos(-\theta) = \cos \theta$:

$$\cos\left(-\frac{\pi}{6}\right) = \cos\left(\frac{\pi}{6}\right)$$

Since $\frac{\pi}{6} \in [0, \pi]$:

$$\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) = \cos^{-1}\left(\cos\frac{\pi}{6}\right) = \frac{\pi}{6}$$

(iii) The principal value branch of $\sin^{-1} x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Since $\frac{5\pi}{3} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, it follows that $\sin^{-1}\left(\sin\frac{5\pi}{3}\right) \neq \frac{5\pi}{3}$.

Now, rewriting the trigonometric argument:

$$\begin{aligned}\sin\left(\frac{5\pi}{3}\right) &= \sin\left(2\pi - \frac{\pi}{3}\right) \\ &= -\sin\left(\frac{\pi}{3}\right) \\ &= \sin\left(-\frac{\pi}{3}\right)\end{aligned}$$

Since $-\frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$:

$$\sin^{-1}\left(\sin\frac{5\pi}{3}\right) = \sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3}$$

Final Answer

(i) Value $= -\frac{\pi}{6}$

(ii) Value $= \frac{\pi}{6}$

(iii) Value $= -\frac{\pi}{3}$

Question 8

Find the domain of the following functions:

(i) $\sin^{-1}(1-x)$

(ii) $\sec^{-1}(2x-3)$

(iii) $\cos^{-1} x + \sin x$

TYPE 3 Domain of Inverse Trigonometric Functions

★ Likely Exam Question

To Find: The domain of definition of each function

Concept / Formula Used: $\sin^{-1} u$ $\cos^{-1} u$ $-1 \leq u \leq 1$ $\sec^{-1} u$ $u \leq -1$ $u \geq 1$

Solution:

(i) For $\sin^{-1}(1-x)$ to be defined:

$$-1 \leq 1-x \leq 1$$

Subtracting 1 throughout:

$$\begin{aligned} -1-1 &\leq -x \leq 1-1 \\ -2 &\leq -x \leq 0 \end{aligned}$$

Multiplying by -1 (which reverses the inequality signs):

$$0 \leq x \leq 2$$

Thus, Domain = $[0, 2]$.

(ii) For $\sec^{-1}(2x-3)$ to be defined:

$$\begin{aligned} 2x-3 &\leq -1 \quad \text{or} \quad 2x-3 \geq 1 \\ 2x &\leq 2 \quad \text{or} \quad 2x \geq 4 \\ x &\leq 1 \quad \text{or} \quad x \geq 2 \end{aligned}$$

Thus, Domain = $(-\infty, 1] \cup [2, \infty)$.

(iii) For $f(x) = \cos^{-1} x + \sin x$:

- Domain of $\cos^{-1} x$ is $[-1, 1]$.
- Domain of $\sin x$ is $\mathbb{R} = (-\infty, \infty)$.

The domain of the sum is the intersection of the two domains:

$$\text{Domain} = [-1, 1] \cap (-\infty, \infty) = [-1, 1]$$

Final Answer

(i) Domain = $[0, 2]$

(ii) Domain = $(-\infty, 1] \cup [2, \infty)$

(iii) Domain = $[-1, 1]$

Question 9

- Write the range of one branch of $\sin^{-1} x$, other than the principal branch.
- Write the range of one branch of $\cos^{-1} x$, other than the principal branch.

TYPE 4 Ranges of Inverse Trigonometric Branches

To Find: One non-principal branch range for $\sin^{-1} x$ and $\cos^{-1} x$

Concept / Formula Used: $\sin^{-1} x$ $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\cos^{-1} x$ $[0, \pi]$ π

Solution:

(i) For $\sin^{-1} x$, the principal value branch is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Other branches on which $\sin x$ is bijective include:

$$\left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \quad \text{or} \quad \left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right]$$

Taking one such branch: $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$.

(ii) For $\cos^{-1} x$, the principal value branch is $[0, \pi]$.

Other branches on which $\cos x$ is bijective include:

$$[\pi, 2\pi] \quad \text{or} \quad [-\pi, 0]$$

Taking one such branch: $[\pi, 2\pi]$.

Final Answer

(i) A non-principal branch range of $\sin^{-1} x$ is $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

(ii) A non-principal branch range of $\cos^{-1} x$ is $[\pi, 2\pi]$

Question 10

Find the maximum value of $(\cos^{-1} x)^2$.

TYPE 5 Extrema of Inverse Trigonometric Functions

To Find: The maximum value of $(\cos^{-1} x)^2$

Concept / Formula Used: $\cos^{-1} x \quad [0, \pi]$

Solution:

For any $x \in [-1, 1]$, the range of the principal value branch of $\cos^{-1} x$ is:

$$0 \leq \cos^{-1} x \leq \pi$$

Since both bounds are non-negative, squaring all parts preserves the inequality:

$$0^2 \leq (\cos^{-1} x)^2 \leq \pi^2$$

$$0 \leq (\cos^{-1} x)^2 \leq \pi^2$$

The maximum value occurs at $x = -1$, where $\cos^{-1}(-1) = \pi$:

$$\text{Maximum value} = \pi^2$$

Final Answer

The maximum value of $(\cos^{-1} x)^2$ is π^2 .

Question 11

If $\cot^{-1}\left(\frac{3}{4}\right) = x$, find the value of $\cos x$.

TYPE 6 Evaluating Trigonometric Ratios from Inverse Functions

Given: $\cot^{-1}\left(\frac{3}{4}\right) = x$

To Find: The value of $\cos x$

Concept / Formula Used: $\cot x = \frac{\text{Base}}{\text{Perpendicular}}$ $\cos x = \frac{\text{Base}}{\text{Hypotenuse}}$

Solution:

Given:

$$\cot^{-1} \left(\frac{3}{4} \right) = x \implies \cot x = \frac{3}{4}$$

Since $\frac{3}{4} > 0$, x lies in the first quadrant, i.e., $x \in \left(0, \frac{\pi}{2} \right)$.

Using the trigonometric identity $\operatorname{cosec}^2 x = 1 + \cot^2 x$:

$$\begin{aligned} \operatorname{cosec}^2 x &= 1 + \left(\frac{3}{4} \right)^2 \\ &= 1 + \frac{9}{16} \\ &= \frac{25}{16} \\ \implies \sin^2 x &= \frac{16}{25} \implies \sin x = \frac{4}{5} \quad (\text{as } x \in (0, \pi/2)) \end{aligned}$$

Now, evaluating $\cos x$:

$$\begin{aligned} \cos x &= \cot x \cdot \sin x \\ &= \frac{3}{4} \cdot \frac{4}{5} \\ &= \frac{3}{5} \end{aligned}$$

Final Answer

$$\cos x = \frac{3}{5}$$

Question 12

- (i) If $\tan^{-1} \left(\frac{3}{4} \right) = x$, find the values of $\cos x$ and $\sin x$.
- (ii) If $\cot^{-1} \left(-\frac{1}{7} \right) = x$, find the values of $\sin x$ and $\cos x$.
- (iii) If $\tan^{-1} x = \sin^{-1} \left(\frac{1}{2} \right)$, find the value of x .

TYPE 6 Evaluating Trigonometric Ratios from Inverse Functions

To Find: The required trigonometric and algebraic values

Solution:

- (i) Given $\tan^{-1} \left(\frac{3}{4} \right) = x \implies \tan x = \frac{3}{4}$, with $x \in \left(0, \frac{\pi}{2} \right)$.

Using $\sec^2 x = 1 + \tan^2 x$:

$$\sec^2 x = 1 + \left(\frac{3}{4}\right)^2 = 1 + \frac{9}{16} = \frac{25}{16}$$

$$\sec x = \frac{5}{4} \implies \cos x = \frac{4}{5}$$

Then:

$$\sin x = \tan x \cdot \cos x = \frac{3}{4} \cdot \frac{4}{5} = \frac{3}{5}$$

(ii) Given $\cot^{-1}\left(-\frac{1}{7}\right) = x \implies \cot x = -\frac{1}{7}$.

Since the principal branch of $\cot^{-1}$ is $(0, \pi)$ and the argument is negative, $x \in \left(\frac{\pi}{2}, \pi\right)$ (Quadrant II).

Using $\operatorname{cosec}^2 x = 1 + \cot^2 x$:

$$\operatorname{cosec}^2 x = 1 + \left(-\frac{1}{7}\right)^2 = 1 + \frac{1}{49} = \frac{50}{49}$$

$$\operatorname{cosec} x = \frac{\sqrt{50}}{7} = \frac{5\sqrt{2}}{7} \quad (\text{since } \sin x > 0 \text{ in Quadrant II})$$

$$\implies \sin x = \frac{7}{5\sqrt{2}} = \frac{7\sqrt{2}}{10}$$

Now, $\cos x = \cot x \cdot \sin x$:

$$\cos x = \left(-\frac{1}{7}\right) \cdot \left(\frac{7}{5\sqrt{2}}\right) = -\frac{1}{5\sqrt{2}} = -\frac{\sqrt{2}}{10}$$

(iii) Given $\tan^{-1} x = \sin^{-1}\left(\frac{1}{2}\right)$.

First, evaluate the RHS:

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

Therefore:

$$\tan^{-1} x = \frac{\pi}{6}$$

$$x = \tan\left(\frac{\pi}{6}\right)$$

$$x = \frac{1}{\sqrt{3}}$$

Final Answer

(i) $\cos x = \frac{4}{5}, \sin x = \frac{3}{5}$

(ii) $\sin x = \frac{7}{5\sqrt{2}} = \frac{7\sqrt{2}}{10}, \cos x = -\frac{1}{5\sqrt{2}} = -\frac{\sqrt{2}}{10}$

(iii) $x = \frac{1}{\sqrt{3}}$

Question 13

Evaluate the following:

- (i) $\cot(\tan^{-1} \sqrt{3})$
- (ii) $\sin \left(\frac{\pi}{6} - \sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) \right)$
- (iii) $\cos \left(\cos^{-1} \left(-\frac{\sqrt{3}}{2} \right) + \frac{\pi}{6} \right)$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions**To Find:** The numerical value of each expression

Concept / Formula Used: $\cot(\tan^{-1} u) = \frac{1}{u}$ $\sin^{-1}(-u) = -\sin^{-1} u$ $\cos^{-1}(-u) = \pi - \cos^{-1} u$

Solution:(i) Since $\tan^{-1} \sqrt{3} = \frac{\pi}{3}$:

$$\begin{aligned} \cot(\tan^{-1} \sqrt{3}) &= \cot \left(\frac{\pi}{3} \right) \\ &= \frac{1}{\sqrt{3}} \end{aligned}$$

(ii) First, evaluate $\sin^{-1} \left(-\frac{\sqrt{3}}{2} \right)$:

$$\sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) = -\sin^{-1} \left(\frac{\sqrt{3}}{2} \right) = -\frac{\pi}{3}$$

Substituting this into the expression:

$$\begin{aligned} \sin \left(\frac{\pi}{6} - \sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) \right) &= \sin \left(\frac{\pi}{6} - \left(-\frac{\pi}{3} \right) \right) \\ &= \sin \left(\frac{\pi}{6} + \frac{\pi}{3} \right) \\ &= \sin \left(\frac{3\pi}{6} \right) \\ &= \sin \left(\frac{\pi}{2} \right) \\ &= 1 \end{aligned}$$

(iii) First, evaluate $\cos^{-1} \left(-\frac{\sqrt{3}}{2} \right)$:

$$\begin{aligned} \cos^{-1} \left(-\frac{\sqrt{3}}{2} \right) &= \pi - \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) \\ &= \pi - \frac{\pi}{6} \\ &= \frac{5\pi}{6} \end{aligned}$$

Substituting this into the expression:

$$\begin{aligned}\cos\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \frac{\pi}{6}\right) &= \cos\left(\frac{5\pi}{6} + \frac{\pi}{6}\right) \\ &= \cos\left(\frac{6\pi}{6}\right) \\ &= \cos(\pi) \\ &= -1\end{aligned}$$

Final Answer

- (i) $\cot(\tan^{-1} \sqrt{3}) = \frac{1}{\sqrt{3}}$
 (ii) $\sin\left(\frac{\pi}{6} - \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right) = 1$
 (iii) $\cos\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \frac{\pi}{6}\right) = -1$

Question 14

Prove the following:

- (i) $\sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{2\pi}{3}$
 (ii) $\tan^{-1}(-1) + \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{\pi}{2}$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

To Prove: The given inverse trigonometric equations hold true

Concept / Formula Used: $\sin^{-1}(-x) = -\sin^{-1} x$ $\cos^{-1}(-x) = \pi - \cos^{-1} x$
 $\tan^{-1}(-x) = -\tan^{-1} x$

Solution:

(i) Evaluating each term on the LHS:

$$\begin{aligned}\sin^{-1}\left(-\frac{1}{2}\right) &= -\sin^{-1}\left(\frac{1}{2}\right) = -\frac{\pi}{6} \\ \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) &= \pi - \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}\end{aligned}$$

Adding the two terms:

$$\begin{aligned}\text{LHS} &= -\frac{\pi}{6} + \frac{5\pi}{6} \\ &= \frac{4\pi}{6} \\ &= \frac{2\pi}{3} \\ &= \text{RHS}\end{aligned}$$

(ii) Evaluating each term on the LHS:

$$\begin{aligned}\tan^{-1}(-1) &= -\tan^{-1}(1) = -\frac{\pi}{4} \\ \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) &= \pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}\end{aligned}$$

Adding the two terms:

$$\begin{aligned}\text{LHS} &= -\frac{\pi}{4} + \frac{3\pi}{4} \\ &= \frac{2\pi}{4} \\ &= \frac{\pi}{2} \\ &= \text{RHS}\end{aligned}$$

Hence Proved

In both cases, LHS = RHS, hence proved.

Question 15

Using principal values, find the values of:

- (i) $\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(-\frac{1}{2}\right)$
- (ii) $\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3})$
- (iii) $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right)$
- (iv) $\operatorname{cosec}^{-1}(-1) + \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

To Find: The values of the given expressions

Solution:

(i) Evaluating individual terms:

$$\begin{aligned}\cos^{-1}\left(\frac{1}{2}\right) &= \frac{\pi}{3} \\ \sin^{-1}\left(-\frac{1}{2}\right) &= -\frac{\pi}{6}\end{aligned}$$

Substituting these into the expression:

$$\begin{aligned}\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(-\frac{1}{2}\right) &= \frac{\pi}{3} - 2\left(-\frac{\pi}{6}\right) \\ &= \frac{\pi}{3} + \frac{\pi}{3} \\ &= \frac{2\pi}{3}\end{aligned}$$

(ii) Evaluating individual terms:

$$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$\cot^{-1}(-\sqrt{3}) = \pi - \cot^{-1}(\sqrt{3}) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Substituting these into the expression:

$$\begin{aligned}\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3}) &= \frac{\pi}{3} - \frac{5\pi}{6} \\ &= \frac{2\pi - 5\pi}{6} \\ &= -\frac{3\pi}{6} \\ &= -\frac{\pi}{2}\end{aligned}$$

(iii) Evaluating individual terms:

$$\tan^{-1}(1) = \frac{\pi}{4}$$

$$\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

Substituting these into the expression:

$$\begin{aligned}\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) &= \frac{\pi}{4} + \frac{2\pi}{3} \\ &= \frac{3\pi + 8\pi}{12} \\ &= \frac{11\pi}{12}\end{aligned}$$

(iv) Evaluating individual terms:

$$\operatorname{cosec}^{-1}(-1) = -\operatorname{cosec}^{-1}(1) = -\frac{\pi}{2}$$

$$\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \pi - \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

Substituting these into the expression:

$$\begin{aligned}\operatorname{cosec}^{-1}(-1) + \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) &= -\frac{\pi}{2} + \frac{2\pi}{3} \\ &= \frac{-3\pi + 4\pi}{6} \\ &= \frac{\pi}{6}\end{aligned}$$

Final Answer

(i) $\frac{2\pi}{3}$

(ii) $-\frac{\pi}{2}$

(iii) $\frac{11\pi}{12}$

(iv) $\frac{\pi}{6}$

Question 16

Find the values of the following:

- (i) $\tan^{-1} \left(2 \cos \left(2 \sin^{-1} \frac{1}{2} \right) \right)$
 (ii) $\tan^{-1} \left(2 \sin \left(2 \cos^{-1} \frac{\sqrt{3}}{2} \right) \right)$
 (iii) $\sin^{-1} \left(\frac{1}{2} \sin (2 \cot^{-1}(-1)) \right)$
 (iv) $\cot(\sin^{-1}(\cos(\tan^{-1} 1)))$

TYPE 8 Evaluating Nested Inverse Trigonometric Functions**★ Likely Exam Question****To Find:** The values of the nested trigonometric expressions**Solution:**

(i) Innermost term: $\sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{6}$.

Working outwards:

$$\begin{aligned} 2 \sin^{-1} \left(\frac{1}{2} \right) &= 2 \cdot \frac{\pi}{6} = \frac{\pi}{3} \\ \cos \left(\frac{\pi}{3} \right) &= \frac{1}{2} \\ 2 \cos \left(\frac{\pi}{3} \right) &= 2 \cdot \frac{1}{2} = 1 \\ \tan^{-1}(1) &= \frac{\pi}{4} \end{aligned}$$

(ii) Innermost term: $\cos^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{6}$.

Working outwards:

$$\begin{aligned} 2 \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) &= 2 \cdot \frac{\pi}{6} = \frac{\pi}{3} \\ \sin \left(\frac{\pi}{3} \right) &= \frac{\sqrt{3}}{2} \\ 2 \sin \left(\frac{\pi}{3} \right) &= 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3} \\ \tan^{-1}(\sqrt{3}) &= \frac{\pi}{3} \end{aligned}$$

(iii) Innermost term: $\cot^{-1}(-1) = \pi - \cot^{-1}(1) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

Working outwards:

$$\begin{aligned} 2 \cot^{-1}(-1) &= 2 \cdot \frac{3\pi}{4} = \frac{3\pi}{2} \\ \sin\left(\frac{3\pi}{2}\right) &= -1 \\ \frac{1}{2} \sin\left(\frac{3\pi}{2}\right) &= \frac{1}{2}(-1) = -\frac{1}{2} \\ \sin^{-1}\left(-\frac{1}{2}\right) &= -\frac{\pi}{6} \end{aligned}$$

(iv) Innermost term: $\tan^{-1}(1) = \frac{\pi}{4}$.

Working outwards:

$$\begin{aligned} \cos(\tan^{-1} 1) &= \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \\ \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) &= \frac{\pi}{4} \\ \cot\left(\frac{\pi}{4}\right) &= 1 \end{aligned}$$

Final Answer

- (i) $\frac{\pi}{4}$
- (ii) $\frac{\pi}{3}$
- (iii) $-\frac{\pi}{6}$
- (iv) 1

Question 17

Find the value of:

- (i) $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) + \cos^{-1}\left(\cos \frac{13\pi}{6}\right)$
- (ii) $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right)$

TYPE 2 Evaluating Composite Inverse Trigonometric Functions

★ Likely Exam Question

To Find: The values of the given expressions

Solution:

(i) First evaluate each term individually:

$$\begin{aligned} \tan\left(\frac{5\pi}{6}\right) &= \tan\left(\pi - \frac{\pi}{6}\right) = -\tan\left(\frac{\pi}{6}\right) = \tan\left(-\frac{\pi}{6}\right) \\ \Rightarrow \tan^{-1}\left(\tan \frac{5\pi}{6}\right) &= \tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6} \quad \left(\text{since } -\frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right) \end{aligned}$$

For the second term:

$$\begin{aligned}\cos\left(\frac{13\pi}{6}\right) &= \cos\left(2\pi + \frac{\pi}{6}\right) = \cos\left(\frac{\pi}{6}\right) \\ \Rightarrow \cos^{-1}\left(\cos\frac{13\pi}{6}\right) &= \cos^{-1}\left(\cos\frac{\pi}{6}\right) = \frac{\pi}{6} \quad \left(\text{since } \frac{\pi}{6} \in [0, \pi]\right)\end{aligned}$$

Summing the terms:

$$\tan^{-1}\left(\tan\frac{5\pi}{6}\right) + \cos^{-1}\left(\cos\frac{13\pi}{6}\right) = -\frac{\pi}{6} + \frac{\pi}{6} = 0$$

(ii) Evaluate each term individually:

$$\begin{aligned}\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) &= -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6} \\ \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) &= \frac{\pi}{3} \\ \sin\left(-\frac{\pi}{2}\right) &= -1 \Rightarrow \tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right) = \tan^{-1}(-1) = -\frac{\pi}{4}\end{aligned}$$

Summing the three terms:

$$\begin{aligned}\text{Value} &= -\frac{\pi}{6} + \frac{\pi}{3} - \frac{\pi}{4} \\ &= \frac{-2\pi + 4\pi - 3\pi}{12} \\ &= \frac{-\pi}{12} \\ &= -\frac{\pi}{12}\end{aligned}$$

Final Answer

(i) 0

(ii) $-\frac{\pi}{12}$

Question 18 (i)

Evaluate:

$$\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

To Find: The value of $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$

Concept / Formula Used: $\sin^{-1}(\sin x) = x \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \cos^{-1}(\cos x) = x \quad x \in [0, \pi]$

Solution:

We evaluate each term separately using the principal value branches of inverse trigonometric functions.

First term:

$$\begin{aligned} \text{Since } \frac{3\pi}{4} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \text{ we rewrite } \sin \frac{3\pi}{4} : \\ \sin \frac{3\pi}{4} = \sin \left(\pi - \frac{\pi}{4}\right) = \sin \frac{\pi}{4} \\ \Rightarrow \sin^{-1} \left(\sin \frac{3\pi}{4}\right) = \sin^{-1} \left(\sin \frac{\pi}{4}\right) = \frac{\pi}{4} \quad \left(\text{since } \frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]\right) \end{aligned}$$

Second term:

$$\begin{aligned} \text{Since } \pi \in [0, \pi], \text{ we have:} \\ \cos^{-1}(\cos \pi) = \pi \end{aligned}$$

Third term:

$$\begin{aligned} \text{Since } \tan \frac{\pi}{4} = 1 \text{ and } \frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : \\ \tan^{-1}(1) = \frac{\pi}{4} \end{aligned}$$

Substituting these values back into the expression:

$$\begin{aligned} \sin^{-1} \left(\sin \frac{3\pi}{4}\right) + \cos^{-1}(\cos \pi) + \tan^{-1}(1) &= \frac{\pi}{4} + \pi + \frac{\pi}{4} \\ &= \frac{\pi}{2} + \pi \\ &= \frac{3\pi}{2} \end{aligned}$$

Final Answer

$$\sin^{-1} \left(\sin \frac{3\pi}{4}\right) + \cos^{-1}(\cos \pi) + \tan^{-1}(1) = \frac{3\pi}{2}$$

Common Mistake: A common error is writing $\sin^{-1} \left(\sin \frac{3\pi}{4}\right) = \frac{3\pi}{4}$ directly, ignoring the principal value range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ for $\sin^{-1}$.

Question 18 (ii)

Evaluate:

$$\sin^{-1} \left(\sin \frac{3\pi}{4}\right) + \cos^{-1} \left(\cos \frac{3\pi}{4}\right) + \tan^{-1}(1)$$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

To Find: The value of $\sin^{-1} \left(\sin \frac{3\pi}{4}\right) + \cos^{-1} \left(\cos \frac{3\pi}{4}\right) + \tan^{-1}(1)$

Concept / Formula Used: $\sin^{-1}(\sin x) = x \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \cos^{-1}(\cos x) = x \quad x \in [0, \pi]$

Solution:

We evaluate each term using its respective principal value branch.

First term:

$$\begin{aligned}\sin \frac{3\pi}{4} &= \sin \left(\pi - \frac{\pi}{4} \right) = \sin \frac{\pi}{4} \\ \Rightarrow \sin^{-1} \left(\sin \frac{3\pi}{4} \right) &= \sin^{-1} \left(\sin \frac{\pi}{4} \right) = \frac{\pi}{4} \quad \left(\text{since } \frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \right)\end{aligned}$$

Second term:

$$\begin{aligned}\text{Since } \frac{3\pi}{4} &\in [0, \pi], \text{ we have:} \\ \cos^{-1} \left(\cos \frac{3\pi}{4} \right) &= \frac{3\pi}{4}\end{aligned}$$

Third term:

$$\tan^{-1}(1) = \frac{\pi}{4} \quad \left(\text{since } \tan \frac{\pi}{4} = 1 \text{ and } \frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \right)$$

Adding all three terms together:

$$\begin{aligned}\sin^{-1} \left(\sin \frac{3\pi}{4} \right) + \cos^{-1} \left(\cos \frac{3\pi}{4} \right) + \tan^{-1}(1) &= \frac{\pi}{4} + \frac{3\pi}{4} + \frac{\pi}{4} \\ &= \pi + \frac{\pi}{4} \\ &= \frac{5\pi}{4}\end{aligned}$$

Final Answer

$$\sin^{-1} \left(\sin \frac{3\pi}{4} \right) + \cos^{-1} \left(\cos \frac{3\pi}{4} \right) + \tan^{-1}(1) = \frac{5\pi}{4}$$

Question 18 (iii)

Evaluate:

$$3 \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) + 2 \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) + \cos^{-1}(0)$$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

★ Likely Exam Question

To Find: The value of $3 \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) + 2 \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) + \cos^{-1}(0)$

Concept / Formula Used: $\sin^{-1}(y) = \theta \iff \sin \theta = y \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \quad \cos^{-1}(y) = \phi \iff \cos \phi = y \quad \phi \in [0, \pi]$

Solution:

Evaluating each term using principal values:

First term:

$$\begin{aligned}\text{Let } \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) &= \theta \\ \implies \sin \theta &= \frac{1}{\sqrt{2}} \implies \theta = \frac{\pi}{4} \\ \implies 3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) &= 3 \times \frac{\pi}{4} = \frac{3\pi}{4}\end{aligned}$$

Second term:

$$\begin{aligned}\text{Let } \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) &= \phi \\ \implies \cos \phi &= \frac{\sqrt{3}}{2} \implies \phi = \frac{\pi}{6} \\ \implies 2 \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) &= 2 \times \frac{\pi}{6} = \frac{\pi}{3}\end{aligned}$$

Third term:

$$\begin{aligned}\text{Let } \cos^{-1}(0) &= \alpha \\ \implies \cos \alpha &= 0 \implies \alpha = \frac{\pi}{2}\end{aligned}$$

Combining all three parts:

$$\begin{aligned}3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + 2 \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}(0) &= \frac{3\pi}{4} + \frac{\pi}{3} + \frac{\pi}{2} \\ &= \frac{9\pi + 4\pi + 6\pi}{12} \\ &= \frac{19\pi}{12}\end{aligned}$$

Final Answer

$$3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + 2 \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}(0) = \frac{19\pi}{12}$$

Question 19 (i)

Evaluate:

$$\cos\left(\sin^{-1}\left(-\frac{3}{5}\right)\right)$$

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\cos\left(\sin^{-1}\left(-\frac{3}{5}\right)\right)$

Concept / Formula Used: $\cos \theta = \sqrt{1 - \sin^2 \theta} \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Solution:

Let $\theta = \sin^{-1}\left(-\frac{3}{5}\right)$.

By the definition of inverse sine, we have:

$$\sin \theta = -\frac{3}{5} \quad \text{where } \theta \in \left[-\frac{\pi}{2}, 0\right)$$

Since θ lies in Quadrant IV, $\cos \theta > 0$. Using the fundamental trigonometric identity:

$$\begin{aligned} \cos \theta &= \sqrt{1 - \sin^2 \theta} \\ &= \sqrt{1 - \left(-\frac{3}{5}\right)^2} \\ &= \sqrt{1 - \frac{9}{25}} \\ &= \sqrt{\frac{25 - 9}{25}} \\ &= \sqrt{\frac{16}{25}} \\ &= \frac{4}{5} \end{aligned}$$

Back-substituting $\theta = \sin^{-1}\left(-\frac{3}{5}\right)$:

$$\cos \left(\sin^{-1} \left(-\frac{3}{5} \right) \right) = \frac{4}{5}$$

Final Answer

$$\cos \left(\sin^{-1} \left(-\frac{3}{5} \right) \right) = \frac{4}{5}$$

Question 19 (ii)

Evaluate:

$$\operatorname{cosec} \left(\cos^{-1} \left(-\frac{12}{13} \right) \right)$$

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\operatorname{cosec} \left(\cos^{-1} \left(-\frac{12}{13} \right) \right)$

Concept / Formula Used: $\sin \theta = \sqrt{1 - \cos^2 \theta} \quad \theta \in [0, \pi] \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta}$

Solution:

Let $\theta = \cos^{-1} \left(-\frac{12}{13} \right)$.

By definition of inverse cosine:

$$\cos \theta = -\frac{12}{13} \quad \text{where } \theta \in \left(\frac{\pi}{2}, \pi \right]$$

Since θ lies in Quadrant II, $\sin \theta$ is positive:

$$\begin{aligned}\sin \theta &= \sqrt{1 - \cos^2 \theta} \\ &= \sqrt{1 - \left(-\frac{12}{13}\right)^2} \\ &= \sqrt{1 - \frac{144}{169}} \\ &= \sqrt{\frac{169 - 144}{169}} \\ &= \sqrt{\frac{25}{169}} \\ &= \frac{5}{13}\end{aligned}$$

Now, using the reciprocal relation $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$:

$$\begin{aligned}\operatorname{cosec} \theta &= \frac{1}{\frac{5}{13}} \\ &= \frac{13}{5}\end{aligned}$$

Back-substituting $\theta = \cos^{-1}\left(-\frac{12}{13}\right)$:

$$\operatorname{cosec} \left(\cos^{-1} \left(-\frac{12}{13} \right) \right) = \frac{13}{5}$$

Final Answer

$$\operatorname{cosec} \left(\cos^{-1} \left(-\frac{12}{13} \right) \right) = \frac{13}{5}$$

Question 19 (iii)

Evaluate:

$$\sin \left(2 \cot^{-1} \frac{1}{5} \right)$$

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

★ Likely Exam Question

To Find: The value of $\sin \left(2 \cot^{-1} \frac{1}{5} \right)$

Concept / Formula Used: $\sin(2\theta) = \frac{2 \tan \theta}{1 + \tan^2 \theta}$ $\tan(\cot^{-1} x) = \frac{1}{x}$

Solution:

Let $\theta = \cot^{-1} \frac{1}{5}$.

Then:

$$\cot \theta = \frac{1}{5}$$

$$\implies \tan \theta = \frac{1}{\cot \theta} = 5$$

We need to compute $\sin(2\theta)$. Using the double-angle formula for sine in terms of tangent:

$$\sin(2\theta) = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

Substituting $\tan \theta = 5$:

$$\begin{aligned}\sin(2\theta) &= \frac{2(5)}{1 + (5)^2} \\ &= \frac{10}{1 + 25} \\ &= \frac{10}{26} \\ &= \frac{5}{13}\end{aligned}$$

Back-substituting $\theta = \cot^{-1} \frac{1}{5}$:

$$\sin \left(2 \cot^{-1} \frac{1}{5} \right) = \frac{5}{13}$$

Final Answer

$$\sin \left(2 \cot^{-1} \frac{1}{5} \right) = \frac{5}{13}$$

Question 20

Show that:

$$\tan \left(\frac{1}{2} \sin^{-1} \frac{3}{4} \right) = \frac{4 - \sqrt{7}}{3}$$

TYPE 10 Half-Angle Substitution in Inverse Trigonometric Functions

★ Likely Exam Question

To Prove: $\tan \left(\frac{1}{2} \sin^{-1} \frac{3}{4} \right) = \frac{4 - \sqrt{7}}{3}$

Concept / Formula Used: $\sin \theta = \frac{2t}{1+t^2}$ $t = \tan \left(\frac{\theta}{2} \right)$

Solution:

Let $\theta = \sin^{-1} \frac{3}{4}$.

By definition of inverse sine:

$$\sin \theta = \frac{3}{4} \quad \text{where } \theta \in \left(0, \frac{\pi}{2} \right)$$

Let $t = \tan\left(\frac{\theta}{2}\right)$. Since $0 < \theta < \frac{\pi}{2}$, we have $0 < \frac{\theta}{2} < \frac{\pi}{4}$, which implies $0 < t < 1$.
Expressing $\sin \theta$ in terms of $t = \tan\left(\frac{\theta}{2}\right)$:

$$\begin{aligned}\sin \theta &= \frac{2t}{1+t^2} \\ \Rightarrow \frac{3}{4} &= \frac{2t}{1+t^2}\end{aligned}$$

Cross-multiplying to form a quadratic equation in t :

$$\begin{aligned}3(1+t^2) &= 4(2t) \\ 3+3t^2 &= 8t \\ 3t^2-8t+3 &= 0\end{aligned}$$

Solving for t using the quadratic formula $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ with $a = 3$, $b = -8$, $c = 3$:

$$\begin{aligned}t &= \frac{-(-8) \pm \sqrt{(-8)^2 - 4(3)(3)}}{2(3)} \\ &= \frac{8 \pm \sqrt{64 - 36}}{6} \\ &= \frac{8 \pm \sqrt{28}}{6} \\ &= \frac{8 \pm 2\sqrt{7}}{6} \\ &= \frac{4 \pm \sqrt{7}}{3}\end{aligned}$$

Since $0 < t < 1$ and $\sqrt{7} \approx 2.646$:

$$\begin{aligned}\frac{4 + \sqrt{7}}{3} &\approx \frac{4 + 2.646}{3} = \frac{6.646}{3} \approx 2.215 > 1 \quad (\text{rejected}) \\ \frac{4 - \sqrt{7}}{3} &\approx \frac{4 - 2.646}{3} = \frac{1.354}{3} \approx 0.451 \in (0, 1) \quad (\text{accepted})\end{aligned}$$

Alternatively, using the half-angle formula $\tan\left(\frac{\theta}{2}\right) = \frac{1 - \cos \theta}{\sin \theta}$:

$$\begin{aligned}\cos \theta &= \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4} \\ \Rightarrow \tan\left(\frac{\theta}{2}\right) &= \frac{1 - \frac{\sqrt{7}}{4}}{\frac{3}{4}} = \frac{4 - \sqrt{7}}{3}\end{aligned}$$

Back-substituting $\theta = \sin^{-1} \frac{3}{4}$:

$$\tan\left(\frac{1}{2} \sin^{-1} \frac{3}{4}\right) = \frac{4 - \sqrt{7}}{3}$$

Hence Proved

LHS = RHS = $\frac{4 - \sqrt{7}}{3}$, hence proved.

Common Mistake: Taking the positive root $\frac{4 + \sqrt{7}}{3}$ without checking the domain condition $0 < \frac{\theta}{2} < \frac{\pi}{4} \Rightarrow 0 < t < 1$ is a common pitfall.

2.2 – Properties and Simplification of Inverse Trigonometric Functions

Question 1

Evaluate the following:

- (i) $\sin^{-1}(\sin 2)$
- (ii) $\sin^{-1}(\sin 5)$
- (iii) $\tan^{-1}(\tan 4)$
- (iv) $\cos^{-1}(\cos 10)$

TYPE 2 Evaluating Composite Inverse Trigonometric Functions

To Find: The principal value of each given inverse trigonometric expression.

Concept / Formula Used: $\sin^{-1}(\sin \theta) = \theta \iff \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\tan^{-1}(\tan \theta) = \theta \iff \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 $\cos^{-1}(\cos \theta) = \theta \iff \theta \in [0, \pi]$

Solution:

(i) For $\sin^{-1}(\sin 2)$:

Since $\frac{\pi}{2} \approx 1.57$ and $\pi \approx 3.14$, we have $2 \in \left(\frac{\pi}{2}, \pi\right)$.

Using the identity $\sin(\pi - \theta) = \sin \theta$:

$$\sin(2) = \sin(\pi - 2)$$

Since $-\frac{\pi}{2} \leq \pi - 2 \leq \frac{\pi}{2}$:

$$\begin{aligned}\sin^{-1}(\sin 2) &= \sin^{-1}(\sin(\pi - 2)) \\ &= \pi - 2\end{aligned}$$

(ii) For $\sin^{-1}(\sin 5)$:

Since $\frac{3\pi}{2} \approx 4.71$ and $2\pi \approx 6.28$, we have $5 \in \left(\frac{3\pi}{2}, 2\pi\right)$.

Using the identity $\sin(\theta - 2\pi) = \sin \theta$:

$$\sin(5) = \sin(5 - 2\pi)$$

Since $-\frac{\pi}{2} \leq 5 - 2\pi \leq \frac{\pi}{2}$:

$$\begin{aligned}\sin^{-1}(\sin 5) &= \sin^{-1}(\sin(5 - 2\pi)) \\ &= 5 - 2\pi\end{aligned}$$

(iii) For $\tan^{-1}(\tan 4)$:

Since $\pi \approx 3.14$ and $\frac{3\pi}{2} \approx 4.71$, we have $4 \in \left(\pi, \frac{3\pi}{2}\right)$.

Using the periodic identity $\tan(\theta - \pi) = \tan \theta$:

$$\tan(4) = \tan(4 - \pi)$$

Since $-\frac{\pi}{2} < 4 - \pi < \frac{\pi}{2}$:

$$\begin{aligned}\tan^{-1}(\tan 4) &= \tan^{-1}(\tan(4 - \pi)) \\ &= 4 - \pi\end{aligned}$$

(iv) For $\cos^{-1}(\cos 10)$:

Since $3\pi \approx 9.42$ and $4\pi \approx 12.57$, we have $10 \in (3\pi, 4\pi)$.

Using the identity $\cos(4\pi - \theta) = \cos \theta$:

$$\cos(10) = \cos(4\pi - 10)$$

Since $0 \leq 4\pi - 10 \leq \pi$:

$$\begin{aligned}\cos^{-1}(\cos 10) &= \cos^{-1}(\cos(4\pi - 10)) \\ &= 4\pi - 10\end{aligned}$$

Final Answer

(i) $\pi - 2$ (ii) $5 - 2\pi$ (iii) $4 - \pi$ (iv) $4\pi - 10$

Common Mistake: Directly writing $\sin^{-1}(\sin x) = x$ without checking whether x lies in the principal value interval $[-\pi/2, \pi/2]$ leads to incorrect answers.

Question 2

Find the values of:

- (i) $\sin(\sin^{-1} x + \cos^{-1} x), \quad |x| \leq 1$
- (ii) $\cot(\tan^{-1} x + \cot^{-1} x), \quad x \in \mathbb{R}$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

To Find: The value of each trigonometric expression.

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad \text{for } |x| \leq 1 \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, \quad \text{for } x \in \mathbb{R}$

Solution:

- (i) Using the standard result $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned}\sin(\sin^{-1} x + \cos^{-1} x) &= \sin\left(\frac{\pi}{2}\right) \\ &= 1\end{aligned}$$

(ii) Using the standard result $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned}\cot(\tan^{-1} x + \cot^{-1} x) &= \cot\left(\frac{\pi}{2}\right) \\ &= 0\end{aligned}$$

Final Answer

(i) 1 (ii) 0

Question 3

Find the values of:

(i) $\csc(\sin^{-1} x + \cos^{-1} x), \quad |x| \leq 1$

(ii) $\cos(\sec^{-1} x + \csc^{-1} x), \quad |x| \geq 1$

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

To Find: The value of each expression.

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad |x| \leq 1 \quad \sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}, \quad |x| \geq 1$

Solution:

(i) Using $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned}\csc(\sin^{-1} x + \cos^{-1} x) &= \csc\left(\frac{\pi}{2}\right) \\ &= 1\end{aligned}$$

(ii) Using $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned}\cos(\sec^{-1} x + \csc^{-1} x) &= \cos\left(\frac{\pi}{2}\right) \\ &= 0\end{aligned}$$

Final Answer

(i) 1 (ii) 0

Question 4

Find the values of:

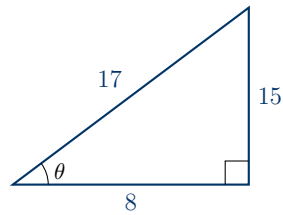
(i) $\tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right)$

(ii) $\cos\left(\tan^{-1}\frac{3}{4}\right)$

(iii) $\tan\left(\cos^{-1}\frac{8}{17}\right)$

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

★ Likely Exam Question



Right triangle construction for $\theta = \cos^{-1}\left(\frac{8}{17}\right)$

To Find: The exact value of each trigonometric composite function.

Concept / Formula Used: $\theta = \cos^{-1}\left(\frac{a}{c}\right)$ Base = a Hypotenuse = c
 Perpendicular = $\sqrt{c^2 - a^2}$ $\tan \theta = \frac{\sqrt{c^2 - a^2}}{a}$

Solution:

(i) Evaluating inner function first:

$$\begin{aligned}\sin\left(-\frac{\pi}{2}\right) &= -1 \\ \tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right) &= \tan^{-1}(-1) \\ &= -\frac{\pi}{4}\end{aligned}$$

(ii) Let $\theta = \tan^{-1} \frac{3}{4} \implies \tan \theta = \frac{3}{4}$ with $\theta \in (0, \frac{\pi}{2})$.

$$\text{Perpendicular} = 3, \quad \text{Base} = 4$$

$$\text{Hypotenuse} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

$$\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{4}{5}$$

$$\therefore \cos\left(\tan^{-1} \frac{3}{4}\right) = \frac{4}{5}$$

(iii) Let $\phi = \cos^{-1} \frac{8}{17} \implies \cos \phi = \frac{8}{17}$ with $\phi \in (0, \frac{\pi}{2})$.

$$\text{Base} = 8, \quad \text{Hypotenuse} = 17$$

$$\text{Perpendicular} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$$

$$\tan \phi = \frac{\text{Perpendicular}}{\text{Base}} = \frac{15}{8}$$

$$\therefore \tan\left(\cos^{-1} \frac{8}{17}\right) = \frac{15}{8}$$

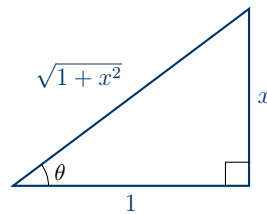
Final Answer

(i) $-\frac{\pi}{4}$ (ii) $\frac{4}{5}$ (iii) $\frac{15}{8}$

Question 5

Simplify the following:

- (i) $\tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right)$
 (ii) $\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right), \quad |x| < 1$
 (iii) $\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$
 (iv) $\csc^{-1} \left(\frac{\sqrt{1+x^2}}{x} \right)$
 (v) $\cot^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right), \quad |x| < 1$
 (vi) $\csc^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right), \quad |x| < 1$

TYPE 11 Simplifying Inverse Trigonometric Expressions

Right triangle for substitution $x = \tan \theta$ in $\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$

To Find: The simplified form of each expression.

Concept	/	Formula	Used:	$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$	$1 + \cos x = 2 \cos^2 \frac{x}{2}$
$\sqrt{1-x^2}$		$x = \sin \theta$	$x = \cos \theta$	$\sqrt{1+x^2}$	$x = \tan \theta$

Solution:

(i) Using double/half-angle formulas:

$$\begin{aligned}
 \tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right) &= \tan^{-1} \left(\frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \right) \\
 &= \tan^{-1} \left(\frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} \right) \\
 &= \tan^{-1} \left(\tan \frac{x}{2} \right) \\
 &= \frac{x}{2}
 \end{aligned}$$

(ii) Let $x = \sin \theta \implies \theta = \sin^{-1} x$.

$$\begin{aligned}\sqrt{1-x^2} &= \sqrt{1-\sin^2 \theta} = \cos \theta \\ \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) &= \tan^{-1} \left(\frac{\sin \theta}{\cos \theta} \right) \\ &= \tan^{-1}(\tan \theta) \\ &= \theta \\ &= \sin^{-1} x\end{aligned}$$

(iii) Let $x = \tan \theta \implies \theta = \tan^{-1} x$.

$$\begin{aligned}\sqrt{1+x^2} &= \sqrt{1+\tan^2 \theta} = \sec \theta \\ \sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) &= \sin^{-1} \left(\frac{\tan \theta}{\sec \theta} \right) \\ &= \sin^{-1} \left(\frac{\sin \theta / \cos \theta}{1/\cos \theta} \right) \\ &= \sin^{-1}(\sin \theta) \\ &= \theta \\ &= \tan^{-1} x\end{aligned}$$

(iv) Using the identity $\csc^{-1}(y) = \sin^{-1} \left(\frac{1}{y} \right)$:

$$\begin{aligned}\csc^{-1} \left(\frac{\sqrt{1+x^2}}{x} \right) &= \sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) \\ &= \tan^{-1} x \quad (\text{from part iii})\end{aligned}$$

(v) Let $x = \cos \theta \implies \theta = \cos^{-1} x$.

$$\begin{aligned}\sqrt{1-x^2} &= \sqrt{1-\cos^2 \theta} = \sin \theta \\ \cot^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) &= \cot^{-1} \left(\frac{\cos \theta}{\sin \theta} \right) \\ &= \cot^{-1}(\cot \theta) \\ &= \theta \\ &= \cos^{-1} x\end{aligned}$$

(vi) Using $\csc^{-1}(y) = \sin^{-1} \left(\frac{1}{y} \right)$:

$$\begin{aligned}\csc^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right) &= \sin^{-1} \left(\sqrt{1-x^2} \right) \\ \text{Let } x &= \sin \theta \implies \theta = \sin^{-1} x \\ \sin^{-1} \left(\sqrt{1-\sin^2 \theta} \right) &= \sin^{-1}(\cos \theta) \\ &= \sin^{-1} \left(\sin \left(\frac{\pi}{2} - \theta \right) \right) \\ &= \frac{\pi}{2} - \theta \\ &= \frac{\pi}{2} - \sin^{-1} x \\ &= \cos^{-1} x\end{aligned}$$

Final Answer

- (i) $\frac{x}{2}$ (ii) $\sin^{-1} x$ (iii) $\tan^{-1} x$ (iv) $\tan^{-1} x$ (v) $\cos^{-1} x$ (vi) $\cos^{-1} x$

Question 6

Solve the following equations for x :

- (i) $\tan^{-1} x = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right)$
 (ii) $\sin^{-1} x = \cos^{-1} \left(\frac{\sqrt{3}}{2} \right)$
 (iii) $\cos^{-1} x = \sin^{-1} \left(-\frac{1}{2} \right)$
 (iv) $\cos(\sin^{-1} x) = \frac{1}{2}$
 (v) $4 \sin^{-1} x + \cos^{-1} x = \pi$
 (vi) $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$
 (vii) $5 \tan^{-1} x + 3 \cot^{-1} x = 2\pi$
 (viii) $\tan^{-1} x^{-1} = \cot^{-1} \left(\frac{4}{x} \right), \quad x > 0$

TYPE 12 Solving Inverse Trigonometric Equations

★ **Likely Exam Question**

To Find: The value(s) of x satisfying each given equation.

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
 $\tan^{-1} \left(\frac{1}{x} \right) = \cot^{-1} x$ for $x > 0$

Solution:

- (i) Standard principal value:

$$\begin{aligned} \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) &= \frac{\pi}{4} \\ \tan^{-1} x &= \frac{\pi}{4} \\ x &= \tan \left(\frac{\pi}{4} \right) \\ x &= 1 \end{aligned}$$

(ii) Standard principal value:

$$\begin{aligned}\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) &= \frac{\pi}{6} \\ \sin^{-1} x &= \frac{\pi}{6} \\ x &= \sin\left(\frac{\pi}{6}\right) \\ x &= \frac{1}{2}\end{aligned}$$

(iii) Standard principal value:

$$\begin{aligned}\sin^{-1}\left(-\frac{1}{2}\right) &= -\frac{\pi}{6} \\ \cos^{-1} x &= -\frac{\pi}{6}\end{aligned}$$

Since the range of $\cos^{-1} x$ is $[0, \pi]$, $\cos^{-1} x$ cannot be negative. Therefore, there is no real solution for x .

(iv) Taking cosine inverse on both sides:

$$\begin{aligned}\cos(\sin^{-1} x) &= \frac{1}{2} \\ \sin^{-1} x &= \pm \frac{\pi}{3} \\ x &= \sin\left(\pm \frac{\pi}{3}\right) \\ x &= \pm \frac{\sqrt{3}}{2}\end{aligned}$$

(v) Using $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$:

$$\begin{aligned}4 \sin^{-1} x + \left(\frac{\pi}{2} - \sin^{-1} x\right) &= \pi \\ 3 \sin^{-1} x + \frac{\pi}{2} &= \pi \\ 3 \sin^{-1} x &= \pi - \frac{\pi}{2} \\ 3 \sin^{-1} x &= \frac{\pi}{2} \\ \sin^{-1} x &= \frac{\pi}{6} \\ x &= \sin\left(\frac{\pi}{6}\right) \\ x &= \frac{1}{2}\end{aligned}$$

(vi) Using $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$:

$$\begin{aligned}\sin^{-1} x - \left(\frac{\pi}{2} - \sin^{-1} x \right) &= \frac{\pi}{6} \\ 2 \sin^{-1} x - \frac{\pi}{2} &= \frac{\pi}{6} \\ 2 \sin^{-1} x &= \frac{\pi}{6} + \frac{\pi}{2} \\ 2 \sin^{-1} x &= \frac{2\pi}{3} \\ \sin^{-1} x &= \frac{\pi}{3} \\ x &= \sin \left(\frac{\pi}{3} \right) \\ x &= \frac{\sqrt{3}}{2}\end{aligned}$$

(vii) Using $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$:

$$\begin{aligned}5 \tan^{-1} x + 3 \left(\frac{\pi}{2} - \tan^{-1} x \right) &= 2\pi \\ 5 \tan^{-1} x + \frac{3\pi}{2} - 3 \tan^{-1} x &= 2\pi \\ 2 \tan^{-1} x &= 2\pi - \frac{3\pi}{2} \\ 2 \tan^{-1} x &= \frac{\pi}{2} \\ \tan^{-1} x &= \frac{\pi}{4} \\ x &= \tan \left(\frac{\pi}{4} \right) \\ x &= 1\end{aligned}$$

(viii) For $x > 0$, $\tan^{-1} \left(\frac{1}{x} \right) = \cot^{-1} x$:

$$\begin{aligned}\cot^{-1} x &= \cot^{-1} \left(\frac{4}{x} \right) \\ x &= \frac{4}{x} \\ x^2 &= 4 \\ x &= 2 \quad (\text{since } x > 0)\end{aligned}$$

Final Answer

$$\begin{array}{llll} \text{(i) } x = 1 & \text{(ii) } x = \frac{1}{2} & \text{(iii) No real solution} & \text{(iv) } x = \pm \frac{\sqrt{3}}{2} \\ \text{(v) } x = \frac{1}{2} & \text{(vi) } x = \frac{\sqrt{3}}{2} & \text{(vii) } x = 1 & \text{(viii) } x = 2 \end{array}$$

Common Mistake: In (iii), forgetting that $\cos^{-1} x \in [0, \pi]$ and incorrectly writing $x = \cos(-\pi/6) = \sqrt{3}/2$ is a standard trap.

Question 7

If $\sin \left(\sin^{-1} \frac{2}{3} + \cos^{-1} x \right) = 1$, then find the value of x .

TYPE 12 Solving Inverse Trigonometric Equations

Given: The equation $\sin \left(\sin^{-1} \frac{2}{3} + \cos^{-1} x \right) = 1$.

To Find: The value of x .

Concept / Formula Used: $\sin \theta = 1 \implies \theta = \frac{\pi}{2}$ $\sin^{-1} a + \cos^{-1} a = \frac{\pi}{2}$

Solution:

Taking sine inverse on both sides:

$$\begin{aligned}\sin^{-1} \frac{2}{3} + \cos^{-1} x &= \sin^{-1}(1) \\ \sin^{-1} \frac{2}{3} + \cos^{-1} x &= \frac{\pi}{2} \\ \cos^{-1} x &= \frac{\pi}{2} - \sin^{-1} \frac{2}{3}\end{aligned}$$

Using the standard identity $\cos^{-1} a = \frac{\pi}{2} - \sin^{-1} a$:

$$\begin{aligned}\cos^{-1} x &= \cos^{-1} \frac{2}{3} \\ x &= \frac{2}{3}\end{aligned}$$

Final Answer

$$x = \frac{2}{3}$$

Question 8

Find the values of:

- (i) $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$
- (ii) $\cot^{-1} 2 + \cot^{-1} 3$

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Find: The value of each sum.

Concept / Formula Used: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ for $xy < 1$

Solution:

(i) Here $x = \frac{1}{2}$ and $y = \frac{1}{3}$. Check product: $xy = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} < 1$. Using the addition identity:

$$\begin{aligned}\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} &= \tan^{-1} \left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} \right) \\ &= \tan^{-1} \left(\frac{\frac{5}{6}}{1 - \frac{1}{6}} \right) \\ &= \tan^{-1} \left(\frac{\frac{5}{6}}{\frac{5}{6}} \right) \\ &= \tan^{-1}(1) \\ &= \frac{\pi}{4}\end{aligned}$$

(ii) Using $\cot^{-1} a = \tan^{-1} \left(\frac{1}{a} \right)$ for $a > 0$:

$$\begin{aligned}\cot^{-1} 2 &= \tan^{-1} \frac{1}{2} \\ \cot^{-1} 3 &= \tan^{-1} \frac{1}{3} \\ \cot^{-1} 2 + \cot^{-1} 3 &= \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} \\ &= \frac{\pi}{4} \quad (\text{from part i})\end{aligned}$$

Final Answer

(i) $\frac{\pi}{4}$ (ii) $\frac{\pi}{4}$

Question 9

If $\tan^{-1} x - \tan^{-1} y = \frac{\pi}{4}$, $xy > -1$, then find the value of $x - y - xy$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

Given: $\tan^{-1} x - \tan^{-1} y = \frac{\pi}{4}$ with $xy > -1$.

To Find: The value of $x - y - xy$.

Concept / Formula Used: $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x - y}{1 + xy} \right)$ for $xy > -1$

Solution:

Applying the subtraction formula:

$$\begin{aligned}\tan^{-1} \left(\frac{x - y}{1 + xy} \right) &= \frac{\pi}{4} \\ \frac{x - y}{1 + xy} &= \tan \left(\frac{\pi}{4} \right) \\ \frac{x - y}{1 + xy} &= 1 \\ x - y &= 1 + xy \\ x - y - xy &= 1\end{aligned}$$

Final Answer

$$x - y - xy = 1$$

Question 10

- (i) If $\tan^{-1} x = \frac{\pi}{10}$ for some $x \in \mathbb{R}$, find the value of $\cot^{-1} x$.
- (ii) If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then find the value of $\cos^{-1} x + \cos^{-1} y$.
- (iii) If $\sec^{-1} x + \sec^{-1} y = \frac{2\pi}{3}$, then find the value of $\csc^{-1} x + \csc^{-1} y$.

TYPE 7 Evaluating Expressions with Inverse Trigonometric Functions

★ Likely Exam Question

To Find: The required values using complementary angle identities.

Concept / Formula Used: $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$

Solution:

- (i) Using $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$:

$$\begin{aligned} \frac{\pi}{10} + \cot^{-1} x &= \frac{\pi}{2} \\ \cot^{-1} x &= \frac{\pi}{2} - \frac{\pi}{10} \\ &= \frac{5\pi - \pi}{10} \\ &= \frac{4\pi}{10} \\ &= \frac{2\pi}{5} \end{aligned}$$

- (ii) Using $\sin^{-1} t = \frac{\pi}{2} - \cos^{-1} t$ for both x and y :

$$\begin{aligned} \left(\frac{\pi}{2} - \cos^{-1} x\right) + \left(\frac{\pi}{2} - \cos^{-1} y\right) &= \frac{\pi}{2} \\ \pi - (\cos^{-1} x + \cos^{-1} y) &= \frac{\pi}{2} \\ \cos^{-1} x + \cos^{-1} y &= \pi - \frac{\pi}{2} \\ &= \frac{\pi}{2} \end{aligned}$$

(iii) Using $\sec^{-1} t = \frac{\pi}{2} - \csc^{-1} t$ for both x and y :

$$\begin{aligned}\left(\frac{\pi}{2} - \csc^{-1} x\right) + \left(\frac{\pi}{2} - \csc^{-1} y\right) &= \frac{2\pi}{3} \\ \pi - (\csc^{-1} x + \csc^{-1} y) &= \frac{2\pi}{3} \\ \csc^{-1} x + \csc^{-1} y &= \pi - \frac{2\pi}{3} \\ &= \frac{\pi}{3}\end{aligned}$$

Final Answer

(i) $\frac{2\pi}{5}$ (ii) $\frac{\pi}{2}$ (iii) $\frac{\pi}{3}$

Question 11

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then find the value of

$$x^{100} + y^{100} + z^{100} - \frac{9}{x^{101} + y^{101} + z^{101}}$$

TYPE 5 Extrema of Inverse Trigonometric Functions

★ Likely Exam Question

Given: $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$.

To Find: The value of $x^{100} + y^{100} + z^{100} - \frac{9}{x^{101} + y^{101} + z^{101}}$.

Concept / Formula Used: $\cos^{-1} t$ $[0, \pi]$ $\cos^{-1} t$ π $t = -1$

Solution:

Since $0 \leq \cos^{-1} t \leq \pi$ for all $t \in [-1, 1]$:

$$\cos^{-1} x + \cos^{-1} y + \cos^{-1} z \leq \pi + \pi + \pi = 3\pi$$

The equality holds if and only if each term achieves its maximum value:

$$\cos^{-1} x = \pi \implies x = \cos \pi = -1$$

$$\cos^{-1} y = \pi \implies y = \cos \pi = -1$$

$$\cos^{-1} z = \pi \implies z = \cos \pi = -1$$

Now substituting $x = -1, y = -1, z = -1$ into the given expression:

$$\begin{aligned}
 x^{100} + y^{100} + z^{100} &= (-1)^{100} + (-1)^{100} + (-1)^{100} \\
 &= 1 + 1 + 1 = 3 \\
 x^{101} + y^{101} + z^{101} &= (-1)^{101} + (-1)^{101} + (-1)^{101} \\
 &= -1 - 1 - 1 = -3 \\
 \therefore \text{Expression} &= 3 - \frac{9}{-3} \\
 &= 3 - (-3) \\
 &= 3 + 3 \\
 &= 6
 \end{aligned}$$

Final Answer

6

Question 12

Prove the following:

- (i) $\tan^{-1} \left(\frac{1+x}{1-x} \right) = \frac{\pi}{4} + \tan^{-1} x, \quad x < 1$
- (ii) $\tan^{-1} x + \cot^{-1}(x+1) = \tan^{-1}(x^2 + x + 1)$
- (iii) $\cos [\tan^{-1} \{ \cot(\sin^{-1} x) \}] = x$

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

- To Prove:**
- (i) $\tan^{-1} \left(\frac{1+x}{1-x} \right) = \frac{\pi}{4} + \tan^{-1} x$
 - (ii) $\tan^{-1} x + \cot^{-1}(x+1) = \tan^{-1}(x^2 + x + 1)$
 - (iii) $\cos [\tan^{-1} \{ \cot(\sin^{-1} x) \}] = x$

Concept / Formula Used: $\tan^{-1} \left(\frac{a+b}{1-ab} \right) = \tan^{-1} a + \tan^{-1} b \quad \cot^{-1} y = \tan^{-1} \left(\frac{1}{y} \right) \quad \text{for } y > 0$

Solution:

- (i) Starting with the Left Hand Side (LHS):

$$\text{LHS} = \tan^{-1} \left(\frac{1+x}{1-1 \cdot x} \right)$$

Using the identity $\tan^{-1} \left(\frac{a+b}{1-ab} \right) = \tan^{-1} a + \tan^{-1} b$ with $a = 1$ and $b = x$:

$$\begin{aligned}
 \text{LHS} &= \tan^{-1}(1) + \tan^{-1} x \\
 &= \frac{\pi}{4} + \tan^{-1} x = \text{RHS}
 \end{aligned}$$

(ii) Convert $\cot^{-1}(x+1)$ to $\tan^{-1}\left(\frac{1}{x+1}\right)$:

$$\text{LHS} = \tan^{-1} x + \tan^{-1} \left(\frac{1}{x+1} \right)$$

Apply addition formula for $\tan^{-1}$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{x + \frac{1}{x+1}}{1 - x \cdot \frac{1}{x+1}} \right) \\ &= \tan^{-1} \left(\frac{\frac{x(x+1)+1}{x+1}}{\frac{(x+1)-x}{x+1}} \right) \\ &= \tan^{-1} \left(\frac{x^2 + x + 1}{1} \right) \\ &= \tan^{-1}(x^2 + x + 1) = \text{RHS} \end{aligned}$$

(iii) Work from the innermost term outwards:

$$\begin{aligned} \text{Let } \theta &= \sin^{-1} x \implies \sin \theta = x \\ \cos \theta &= \sqrt{1-x^2} \\ \cot(\sin^{-1} x) &= \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\sqrt{1-x^2}}{x} \end{aligned}$$

Now substitute this into the middle term:

$$\text{Let } \phi = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right) \implies \tan \phi = \frac{\sqrt{1-x^2}}{x}$$

Draw a right triangle for ϕ :

$$\begin{aligned} \text{Perpendicular} &= \sqrt{1-x^2}, \quad \text{Base} = x \\ \text{Hypotenuse} &= \sqrt{(\sqrt{1-x^2})^2 + x^2} = \sqrt{1-x^2+x^2} = 1 \\ \cos \phi &= \frac{\text{Base}}{\text{Hypotenuse}} = \frac{x}{1} = x \end{aligned}$$

Therefore:

$$\text{LHS} = \cos \phi = x = \text{RHS}$$

Hence Proved

LHS = RHS in all subparts. Hence proved.

Question 13

Prove the following (for suitable values of x and y):

$$(i) \tan^{-1} \sqrt{x} + \tan^{-1} \sqrt{y} = \tan^{-1} \frac{\sqrt{x} + \sqrt{y}}{1 + \sqrt{xy}}$$

$$(ii) \tan^{-1} \frac{x + \sqrt{x}}{1 - x^{3/2}} = \tan^{-1} x + \tan^{-1} \sqrt{x}$$

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: (i) $\tan^{-1} \sqrt{x} + \tan^{-1} \sqrt{y} = \tan^{-1} \frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{xy}}$

(ii) $\tan^{-1} \frac{x + \sqrt{x}}{1 - x^{3/2}} = \tan^{-1} x + \tan^{-1} \sqrt{x}$

Concept / Formula Used: $\tan^{-1} A + \tan^{-1} B = \tan^{-1} \left(\frac{A + B}{1 - AB} \right)$

Solution:

(i) Apply addition formula with $A = \sqrt{x}$ and $B = \sqrt{y}$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \sqrt{x} + \tan^{-1} \sqrt{y} \\ &= \tan^{-1} \left(\frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{x} \cdot \sqrt{y}} \right) \\ &= \tan^{-1} \left(\frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{xy}} \right) = \text{RHS} \end{aligned}$$

(ii) Rewrite the denominator of the LHS:

$$\begin{aligned} x^{3/2} &= x \cdot x^{1/2} = x\sqrt{x} \\ \text{LHS} &= \tan^{-1} \left(\frac{x + \sqrt{x}}{1 - x\sqrt{x}} \right) \end{aligned}$$

Using the identity $\tan^{-1} \left(\frac{A + B}{1 - AB} \right) = \tan^{-1} A + \tan^{-1} B$ with $A = x$ and $B = \sqrt{x}$:

$$\text{LHS} = \tan^{-1} x + \tan^{-1} \sqrt{x} = \text{RHS}$$

Hence Proved

LHS = RHS for both parts. Hence proved.

Question 14

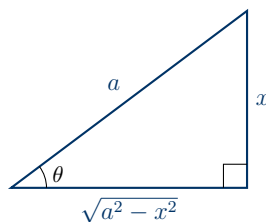
Write the following functions in simplest form:

- (i) $\tan^{-1} \left(\sqrt{\frac{1 - \cos x}{1 + \cos x}} \right)$
- (ii) $\tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right)$
- (iii) $\tan^{-1} \left(\frac{2\sqrt{x}}{1 - x} \right)$
- (iv) $\sin^{-1} \left(\sqrt{\frac{x}{1 + x}} \right)$
- (v) $\tan^{-1} \left(\sqrt{\frac{1 - x}{1 + x}} \right)$
- (vi) $\cos^{-1}(1 - 2x^2)$

$$\begin{aligned} \text{(vii)} \quad & \sec^{-1} \left(\frac{1}{2x^2 - 1} \right) \\ \text{(viii)} \quad & \tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right) \end{aligned}$$

TYPE 11 Simplifying Inverse Trigonometric Expressions

★ Likely Exam Question



Right triangle for substitution $x = a \sin \theta$ in $\tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right)$

To Find: The simplest simplified form of each expression.

Concept / Formula Used: $1 - \cos x = 2 \sin^2 \frac{x}{2}$, $1 + \cos x = 2 \cos^2 \frac{x}{2}$, $\tan \left(\frac{\pi}{4} - x \right) = \frac{1 - \tan x}{1 + \tan x}$, $2 \tan^{-1} u = \tan^{-1} \left(\frac{2u}{1 - u^2} \right)$, $\cos 2\theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$

Solution:

(i) Using standard half-angle forms:

$$\begin{aligned} \sqrt{\frac{1 - \cos x}{1 + \cos x}} &= \sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} = \sqrt{\tan^2 \frac{x}{2}} = \tan \frac{x}{2} \\ \tan^{-1} \left(\sqrt{\frac{1 - \cos x}{1 + \cos x}} \right) &= \tan^{-1} \left(\tan \frac{x}{2} \right) = \frac{x}{2} \end{aligned}$$

(ii) Divide numerator and denominator by $\cos x$:

$$\begin{aligned} \frac{\cos x - \sin x}{\cos x + \sin x} &= \frac{1 - \tan x}{1 + \tan x} = \tan \left(\frac{\pi}{4} - x \right) \\ \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) &= \tan^{-1} \left(\tan \left(\frac{\pi}{4} - x \right) \right) = \frac{\pi}{4} - x \end{aligned}$$

(iii) Let $\sqrt{x} = \tan \theta \implies \theta = \tan^{-1} \sqrt{x}$.

$$\begin{aligned} \frac{2\sqrt{x}}{1 - x} &= \frac{2 \tan \theta}{1 - \tan^2 \theta} = \tan 2\theta \\ \tan^{-1} \left(\frac{2\sqrt{x}}{1 - x} \right) &= \tan^{-1} (\tan 2\theta) = 2\theta = 2 \tan^{-1} \sqrt{x} \end{aligned}$$

(iv) Let $x = \tan^2 \theta \implies \tan \theta = \sqrt{x} \implies \theta = \tan^{-1} \sqrt{x}$.

$$\sqrt{\frac{x}{1+x}} = \sqrt{\frac{\tan^2 \theta}{1+\tan^2 \theta}} = \sqrt{\frac{\tan^2 \theta}{\sec^2 \theta}} = \sin \theta$$

$$\sin^{-1} \left(\sqrt{\frac{x}{1+x}} \right) = \sin^{-1}(\sin \theta) = \theta = \tan^{-1} \sqrt{x}$$

(v) Let $x = \cos 2\theta \implies 2\theta = \cos^{-1} x \implies \theta = \frac{1}{2} \cos^{-1} x$.

$$\sqrt{\frac{1-x}{1+x}} = \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} = \sqrt{\frac{2\sin^2 \theta}{2\cos^2 \theta}} = \tan \theta$$

$$\tan^{-1} \left(\sqrt{\frac{1-x}{1+x}} \right) = \tan^{-1}(\tan \theta) = \theta = \frac{1}{2} \cos^{-1} x$$

(vi) Let $x = \sin \theta \implies \theta = \sin^{-1} x$.

$$1 - 2x^2 = 1 - 2\sin^2 \theta = \cos 2\theta$$

$$\cos^{-1}(1 - 2x^2) = \cos^{-1}(\cos 2\theta) = 2\theta = 2\sin^{-1} x$$

(vii) Convert $\sec^{-1}$ to $\cos^{-1}$:

$$\sec^{-1} \left(\frac{1}{2x^2 - 1} \right) = \cos^{-1}(2x^2 - 1)$$

Let $x = \cos \theta \implies \theta = \cos^{-1} x$

$$2x^2 - 1 = 2\cos^2 \theta - 1 = \cos 2\theta$$

$$\cos^{-1}(2x^2 - 1) = \cos^{-1}(\cos 2\theta) = 2\theta = 2\cos^{-1} x$$

(viii) Let $x = a \sin \theta \implies \sin \theta = \frac{x}{a} \implies \theta = \sin^{-1} \left(\frac{x}{a} \right)$.

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = a \cos \theta$$

$$\frac{x}{\sqrt{a^2 - x^2}} = \frac{a \sin \theta}{a \cos \theta} = \tan \theta$$

$$\tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right) = \tan^{-1}(\tan \theta) = \theta = \sin^{-1} \left(\frac{x}{a} \right)$$

Final Answer

(i) $\frac{x}{2}$ (ii) $\frac{\pi}{4} - x$ (iii) $2 \tan^{-1} \sqrt{x}$ (iv) $\tan^{-1} \sqrt{x}$

(v) $\frac{1}{2} \cos^{-1} x$ (vi) $2 \sin^{-1} x$ (vii) $2 \cos^{-1} x$ (viii) $\sin^{-1} \left(\frac{x}{a} \right)$

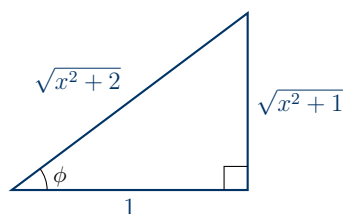
Question 15

Prove that:

$$\sin \left(\cot^{-1} \left(\cos \left(\tan^{-1} x \right) \right) \right) = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$$

TYPE 8 Evaluating Nested Inverse Trigonometric Functions

★ Likely Exam Question



Right triangle transformations for step-by-step nested inverse functions

To Prove: $\sin(\cot^{-1}(\cos(\tan^{-1} x))) = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$

Concept / Formula Used: $\tan \theta = x$ $\cos \theta = \frac{1}{\sqrt{1+x^2}}$ $\cot \phi = k$ $\sin \phi = \frac{1}{\sqrt{1+k^2}}$

Solution:

Work step-by-step from the innermost function to the outermost:

Step 1: Let $\theta = \tan^{-1} x \implies \tan \theta = x$.

$$\begin{aligned} \text{Perpendicular} &= x, \quad \text{Base} = 1, \quad \text{Hypotenuse} = \sqrt{1+x^2} \\ \cos(\tan^{-1} x) &= \cos \theta = \frac{1}{\sqrt{1+x^2}} \end{aligned}$$

Step 2: Substitute this back into the expression:

$$\text{Expression} = \sin \left(\cot^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) \right)$$

$$\text{Let } \phi = \cot^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) \implies \cot \phi = \frac{1}{\sqrt{1+x^2}}.$$

$$\begin{aligned} \text{Base} &= 1, \quad \text{Perpendicular} = \sqrt{1+x^2} \\ \text{Hypotenuse} &= \sqrt{\text{Base}^2 + \text{Perpendicular}^2} \\ &= \sqrt{1^2 + \left(\sqrt{1+x^2}\right)^2} \\ &= \sqrt{1+1+x^2} \\ &= \sqrt{x^2+2} \end{aligned}$$

Step 3: Evaluate $\sin \phi$:

$$\sin \phi = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{\sqrt{1+x^2}}{\sqrt{x^2+2}} = \sqrt{\frac{x^2+1}{x^2+2}}$$

Hence Proved

$$\text{LHS} = \text{RHS} = \sqrt{\frac{x^2+1}{x^2+2}}. \text{ Hence proved.}$$

Question 16

Find the value of

$$\tan \frac{1}{2} \left(\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right), \quad |x| < 1, y > 0, xy < 1$$

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions**★ Likely Exam Question****Given:** Expression $\tan \frac{1}{2} \left(\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right)$ with $|x| < 1, y > 0, xy < 1$.**To Find:** The value of the expression in terms of x and y .

Concept / Formula Used:

$$\begin{aligned} \sin^{-1} \left(\frac{2x}{1+x^2} \right) &= 2 \tan^{-1} x \quad \text{for } |x| < 1 \\ \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) &= 2 \tan^{-1} y \quad \text{for } y > 0 \\ \tan^{-1} x + \tan^{-1} y &= \tan^{-1} \left(\frac{x+y}{1-xy} \right) \quad \text{for } xy < 1 \end{aligned}$$

Solution:

Substitute the standard 2-angle identities into the expression:

$$\begin{aligned} \sin^{-1} \left(\frac{2x}{1+x^2} \right) &= 2 \tan^{-1} x \\ \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) &= 2 \tan^{-1} y \end{aligned}$$

Substituting these into the given expression:

$$\begin{aligned} \tan \left[\frac{1}{2} (2 \tan^{-1} x + 2 \tan^{-1} y) \right] &= \tan \left[\frac{1}{2} \cdot 2 (\tan^{-1} x + \tan^{-1} y) \right] \\ &= \tan (\tan^{-1} x + \tan^{-1} y) \end{aligned}$$

Now apply the addition formula $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$:

$$\begin{aligned} &= \tan \left(\tan^{-1} \left(\frac{x+y}{1-xy} \right) \right) \\ &= \frac{x+y}{1-xy} \end{aligned}$$

Final Answer

$$\frac{x+y}{1-xy}$$

Question 17

Prove the following:

- (i) $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{2}{11} = \tan^{-1} \frac{3}{4}$
(ii) $\tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} = \tan^{-1} \frac{2}{9}$
(iii) $\tan^{-1} 2 - \tan^{-1} 1 = \tan^{-1} \frac{1}{3}$
(iv) $\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{2}$

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions**★ Likely Exam Question**

- To Prove:** (i) $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{2}{11} = \tan^{-1} \frac{3}{4}$
(ii) $\tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} = \tan^{-1} \frac{2}{9}$
(iii) $\tan^{-1} 2 - \tan^{-1} 1 = \tan^{-1} \frac{1}{3}$
(iv) $\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{2}$

Concept / Formula Used: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$

Solution:

- (i) Apply addition formula with $x = \frac{1}{2}, y = \frac{2}{11}$:

$$\begin{aligned}
 \text{LHS} &= \tan^{-1} \left(\frac{\frac{1}{2} + \frac{2}{11}}{1 - \frac{1}{2} \cdot \frac{2}{11}} \right) \\
 &= \tan^{-1} \left(\frac{\frac{11+4}{22}}{1 - \frac{2}{22}} \right) \\
 &= \tan^{-1} \left(\frac{\frac{15}{22}}{\frac{20}{22}} \right) \\
 &= \tan^{-1} \left(\frac{15}{20} \right) \\
 &= \tan^{-1} \left(\frac{3}{4} \right) = \text{RHS}
 \end{aligned}$$

(ii) Apply addition formula with $x = \frac{1}{7}, y = \frac{1}{13}$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{1}{7} + \frac{1}{13}}{1 - \frac{1}{7} \cdot \frac{1}{13}} \right) \\ &= \tan^{-1} \left(\frac{\frac{13+7}{91}}{1 - \frac{1}{91}} \right) \\ &= \tan^{-1} \left(\frac{\frac{20}{91}}{\frac{90}{91}} \right) \\ &= \tan^{-1} \left(\frac{20}{90} \right) \\ &= \tan^{-1} \left(\frac{2}{9} \right) = \text{RHS} \end{aligned}$$

(iii) Apply subtraction formula with $x = 2, y = 1$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{2 - 1}{1 + 2 \cdot 1} \right) \\ &= \tan^{-1} \left(\frac{1}{1 + 2} \right) \\ &= \tan^{-1} \left(\frac{1}{3} \right) = \text{RHS} \end{aligned}$$

(iv) Combine terms stepwise:

$$\begin{aligned} \text{LHS} &= \tan^{-1}(1) + \left(\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} \right) \\ &= \frac{\pi}{4} + \tan^{-1} \left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} \right) \\ &= \frac{\pi}{4} + \tan^{-1} \left(\frac{\frac{5}{6}}{\frac{5}{6}} \right) \\ &= \frac{\pi}{4} + \tan^{-1}(1) \\ &= \frac{\pi}{4} + \frac{\pi}{4} \\ &= \frac{\pi}{2} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS for all four parts. Hence proved.

Question 17 (v)

Prove that $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$

Concept / Formula Used: $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

Solution:

Using the standard identity $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ for $x = \frac{1}{3}$:

$$\begin{aligned} 2 \tan^{-1} \frac{1}{3} &= \tan^{-1} \left(\frac{2 \times \frac{1}{3}}{1 - \left(\frac{1}{3}\right)^2} \right) \\ &= \tan^{-1} \left(\frac{\frac{2}{3}}{1 - \frac{1}{9}} \right) \\ &= \tan^{-1} \left(\frac{\frac{2}{3}}{\frac{8}{9}} \right) \\ &= \tan^{-1} \left(\frac{2}{3} \times \frac{9}{8} \right) \\ &= \tan^{-1} \frac{3}{4} \end{aligned}$$

Substitute this value into the left-hand side:

$$\text{LHS} = \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{1}{7}$$

Using the addition identity $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \times \frac{1}{7}} \right) \\ &= \tan^{-1} \left(\frac{\frac{21+4}{28}}{1 - \frac{3}{28}} \right) \\ &= \tan^{-1} \left(\frac{\frac{25}{28}}{\frac{25}{28}} \right) \\ &= \tan^{-1}(1) \\ &= \frac{\pi}{4} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (vi)

Prove that $\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{3}{5} - \tan^{-1} \frac{8}{19} = \frac{\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{3}{5} - \tan^{-1} \frac{8}{19} = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$

Solution:

First, combine the first two terms using $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$:

$$\begin{aligned} \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{3}{5} &= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{3}{5}}{1 - \frac{3}{4} \times \frac{3}{5}} \right) \\ &= \tan^{-1} \left(\frac{\frac{15+12}{20}}{1 - \frac{9}{20}} \right) \\ &= \tan^{-1} \left(\frac{\frac{27}{20}}{\frac{11}{20}} \right) \\ &= \tan^{-1} \frac{27}{11} \end{aligned}$$

Substitute this result back into the left-hand side:

$$\text{LHS} = \tan^{-1} \frac{27}{11} - \tan^{-1} \frac{8}{19}$$

Using the subtraction identity $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{27}{11} - \frac{8}{19}}{1 + \frac{27}{11} \times \frac{8}{19}} \right) \\ &= \tan^{-1} \left(\frac{\frac{27 \times 19 - 8 \times 11}{209}}{\frac{209 + 216}{209}} \right) \\ &= \tan^{-1} \left(\frac{513 - 88}{425} \right) \\ &= \tan^{-1} \left(\frac{425}{425} \right) \\ &= \tan^{-1}(1) \\ &= \frac{\pi}{4} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (vii)

Prove that $\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3 = \frac{\pi}{2}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3 = \frac{\pi}{2}$

Concept / Formula Used: $\cot^{-1} x = \tan^{-1} \left(\frac{1}{x} \right) \quad x > 0$

Solution:

Using the standard identity $\cot^{-1} x = \tan^{-1} \left(\frac{1}{x} \right)$ for $x > 0$:

$$\cot^{-1} 1 = \tan^{-1} 1$$

$$\cot^{-1} 2 = \tan^{-1} \frac{1}{2}$$

$$\cot^{-1} 3 = \tan^{-1} \frac{1}{3}$$

Rewrite the left-hand side:

$$\text{LHS} = \tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$$

Combine $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$ using $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$:

$$\begin{aligned} \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} &= \tan^{-1} \left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} \right) \\ &= \tan^{-1} \left(\frac{\frac{5}{6}}{1 - \frac{1}{6}} \right) \\ &= \tan^{-1} \left(\frac{\frac{5}{6}}{\frac{5}{6}} \right) \\ &= \tan^{-1}(1) \\ &= \frac{\pi}{4} \end{aligned}$$

Substitute this result back into the expression:

$$\begin{aligned} \text{LHS} &= \tan^{-1} 1 + \frac{\pi}{4} \\ &= \frac{\pi}{4} + \frac{\pi}{4} \\ &= \frac{\pi}{2} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (viii)

Prove that $\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ Likely Exam Question

To Prove: $\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$

Concept / Formula Used: $x > 0, y > 0 \quad xy > 1 \quad \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

Solution:

Here $x = 2 > 0$ and $y = 3 > 0$, so $xy = 2 \times 3 = 6 > 1$. Using the standard identity $\tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right)$:

$$\begin{aligned} \text{LHS} &= \pi + \tan^{-1} \left(\frac{2+3}{1-2 \times 3} \right) \\ &= \pi + \tan^{-1} \left(\frac{5}{1-6} \right) \\ &= \pi + \tan^{-1} \left(\frac{5}{-5} \right) \\ &= \pi + \tan^{-1}(-1) \\ &= \pi - \frac{\pi}{4} \\ &= \frac{3\pi}{4} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Applying $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ without adding π when $xy > 1$ yields $-\frac{\pi}{4}$, which is incorrect because both terms are positive angles.

Question 18

If $\tan^{-1} \left(\frac{x-1}{x+1} \right) + \tan^{-1} \left(\frac{2x-1}{2x+1} \right) = \tan^{-1} \left(\frac{23}{36} \right)$, then prove that $24x^2 - 23x - 12 = 0$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ Likely Exam Question

To Prove: $24x^2 - 23x - 12 = 0$ given $\tan^{-1} \left(\frac{x-1}{x+1} \right) + \tan^{-1} \left(\frac{2x-1}{2x+1} \right) = \tan^{-1} \left(\frac{23}{36} \right)$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Let $a = \frac{x-1}{x+1}$ and $b = \frac{2x-1}{2x+1}$. Applying the identity $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\tan^{-1} \left(\frac{\frac{x-1}{x+1} + \frac{2x-1}{2x+1}}{1 - \frac{x-1}{x+1} \cdot \frac{2x-1}{2x+1}} \right) = \tan^{-1} \left(\frac{23}{36} \right)$$

Taking tangent on both sides:

$$\frac{\frac{(x-1)(2x+1)+(2x-1)(x+1)}{(x+1)(2x+1)}}{\frac{(x+1)(2x+1)-(x-1)(2x-1)}{(x+1)(2x+1)}} = \frac{23}{36}$$

Cancelling the common denominator $(x+1)(2x+1)$:

$$\frac{(x-1)(2x+1) + (2x-1)(x+1)}{(x+1)(2x+1) - (x-1)(2x-1)} = \frac{23}{36}$$

Expand the numerator:

$$\begin{aligned}\text{Numerator} &= (2x^2 + x - 2x - 1) + (2x^2 + 2x - x - 1) \\ &= (2x^2 - x - 1) + (2x^2 + x - 1) \\ &= 4x^2 - 2\end{aligned}$$

Expand the denominator:

$$\begin{aligned}\text{Denominator} &= (2x^2 + 3x + 1) - (2x^2 - 3x + 1) \\ &= 2x^2 + 3x + 1 - 2x^2 + 3x - 1 \\ &= 6x\end{aligned}$$

Substitute these back into the fraction:

$$\frac{4x^2 - 2}{6x} = \frac{23}{36}$$

Divide numerator and denominator of the left side by 2:

$$\frac{2x^2 - 1}{3x} = \frac{23}{36}$$

Cross-multiply:

$$\begin{aligned}36(2x^2 - 1) &= 23(3x) \\ 72x^2 - 36 &= 69x \\ 72x^2 - 69x - 36 &= 0\end{aligned}$$

Divide the entire equation by 3:

$$24x^2 - 23x - 12 = 0$$

Hence Proved

Derived the equation $24x^2 - 23x - 12 = 0$, hence proved.

Question 19 (i)

Prove that $\sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{2}{\sqrt{5}} = \frac{\pi}{2}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{2}{\sqrt{5}} = \frac{\pi}{2}$

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ $\sin^{-1} y = \cos^{-1} \sqrt{1 - y^2}$ $y > 0$

Solution:

Let $\theta = \sin^{-1} \frac{2}{\sqrt{5}}$. Then $\sin \theta = \frac{2}{\sqrt{5}}$. Using $\cos \theta = \sqrt{1 - \sin^2 \theta}$:

$$\begin{aligned}\cos \theta &= \sqrt{1 - \left(\frac{2}{\sqrt{5}}\right)^2} \\ &= \sqrt{1 - \frac{4}{5}} \\ &= \sqrt{\frac{1}{5}} \\ &= \frac{1}{\sqrt{5}}\end{aligned}$$

Thus $\theta = \cos^{-1} \frac{1}{\sqrt{5}}$, which gives:

$$\sin^{-1} \frac{2}{\sqrt{5}} = \cos^{-1} \frac{1}{\sqrt{5}}$$

Substitute this into the left-hand side:

$$\text{LHS} = \sin^{-1} \frac{1}{\sqrt{5}} + \cos^{-1} \frac{1}{\sqrt{5}}$$

Using the standard result $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for $x = \frac{1}{\sqrt{5}}$:

$$\text{LHS} = \frac{\pi}{2} = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (ii)

Prove that $\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ Likely Exam Question

To Prove: $\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$

Concept / Formula Used: $\cos^{-1} x + \cos^{-1} y = \cos^{-1} (xy - \sqrt{1 - x^2} \sqrt{1 - y^2})$

Solution:

Let $x = \frac{4}{5}$ and $y = \frac{12}{13}$. Calculate the radical terms separately:

$$\sqrt{1-x^2} = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$$

$$\sqrt{1-y^2} = \sqrt{1 - \left(\frac{12}{13}\right)^2} = \sqrt{1 - \frac{144}{169}} = \sqrt{\frac{25}{169}} = \frac{5}{13}$$

Using the standard identity $\cos^{-1} x + \cos^{-1} y = \cos^{-1} (xy - \sqrt{1-x^2}\sqrt{1-y^2})$:

$$\begin{aligned} \text{LHS} &= \cos^{-1} \left(\frac{4}{5} \times \frac{12}{13} - \frac{3}{5} \times \frac{5}{13} \right) \\ &= \cos^{-1} \left(\frac{48}{65} - \frac{15}{65} \right) \\ &= \cos^{-1} \left(\frac{33}{65} \right) = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (iii)

Prove that $\cos^{-1} \frac{3}{5} + \sin^{-1} \frac{12}{13} = \sin^{-1} \frac{56}{65}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\cos^{-1} \frac{3}{5} + \sin^{-1} \frac{12}{13} = \sin^{-1} \frac{56}{65}$

Concept / Formula Used: $\sin^{-1} x + \sin^{-1} y = \sin^{-1} (x\sqrt{1-y^2} + y\sqrt{1-x^2})$

Solution:

Convert $\cos^{-1} \frac{3}{5}$ to sine form: Let $A = \cos^{-1} \frac{3}{5} \implies \cos A = \frac{3}{5}$. Then $\sin A = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$. Thus $A = \sin^{-1} \frac{4}{5}$. Substitute into the left-hand side:

$$\text{LHS} = \sin^{-1} \frac{4}{5} + \sin^{-1} \frac{12}{13}$$

Let $x = \frac{4}{5}$ and $y = \frac{12}{13}$. Compute the square root terms:

$$\sqrt{1-x^2} = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$$

$$\sqrt{1-y^2} = \sqrt{1 - \frac{144}{169}} = \frac{5}{13}$$

Using $\sin^{-1} x + \sin^{-1} y = \sin^{-1} (x\sqrt{1-y^2} + y\sqrt{1-x^2})$:

$$\begin{aligned}\text{LHS} &= \sin^{-1} \left(\frac{4}{5} \times \frac{5}{13} + \frac{12}{13} \times \frac{3}{5} \right) \\ &= \sin^{-1} \left(\frac{20}{65} + \frac{36}{65} \right) \\ &= \sin^{-1} \left(\frac{56}{65} \right) = \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (iv)

Prove that $\tan^{-1} \frac{1}{3} + \sec^{-1} \frac{\sqrt{5}}{2} = \frac{\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\tan^{-1} \frac{1}{3} + \sec^{-1} \frac{\sqrt{5}}{2} = \frac{\pi}{4}$

Concept / Formula Used: $\sec^{-1} x = \tan^{-1} \sqrt{x^2 - 1} \quad x > 1 \quad \tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Let $\theta = \sec^{-1} \frac{\sqrt{5}}{2} \implies \sec \theta = \frac{\sqrt{5}}{2}$. Using $\tan \theta = \sqrt{\sec^2 \theta - 1}$:

$$\begin{aligned}\tan \theta &= \sqrt{\left(\frac{\sqrt{5}}{2} \right)^2 - 1} \\ &= \sqrt{\frac{5}{4} - 1} \\ &= \sqrt{\frac{1}{4}} = \frac{1}{2}\end{aligned}$$

So $\sec^{-1} \frac{\sqrt{5}}{2} = \tan^{-1} \frac{1}{2}$. Substitute into the left-hand side:

$$\text{LHS} = \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{2}$$

Using $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{1}{3} + \frac{1}{2}}{1 - \frac{1}{3} \times \frac{1}{2}} \right) \\ &= \tan^{-1} \left(\frac{\frac{5}{6}}{1 - \frac{1}{6}} \right) \\ &= \tan^{-1} \left(\frac{\frac{5}{6}}{\frac{5}{6}} \right) \\ &= \tan^{-1}(1) = \frac{\pi}{4} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (v)

Prove that $\cos^{-1} \frac{4}{5} + \tan^{-1} \frac{3}{5} = \tan^{-1} \frac{27}{11}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\cos^{-1} \frac{4}{5} + \tan^{-1} \frac{3}{5} = \tan^{-1} \frac{27}{11}$

Concept / Formula Used: $\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right) \quad x > 0$

Solution:

Convert $\cos^{-1} \frac{4}{5}$ into tangent form: Let $\theta = \cos^{-1} \frac{4}{5} \implies \cos \theta = \frac{4}{5}$. Then $\sin \theta = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \frac{3}{5}$. Therefore $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{4/5} = \frac{3}{4}$. So $\cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4}$. Substitute into the left-hand side:

$$\text{LHS} = \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{3}{5}$$

Using $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{3}{5}}{1 - \frac{3}{4} \times \frac{3}{5}} \right) \\ &= \tan^{-1} \left(\frac{\frac{15+12}{20}}{1 - \frac{9}{20}} \right) \\ &= \tan^{-1} \left(\frac{\frac{27}{20}}{\frac{11}{20}} \right) \\ &= \tan^{-1} \frac{27}{11} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (vi)

Prove that $\cos^{-1} \frac{5}{\sqrt{41}} + \cot^{-1} \frac{4}{5} = \frac{\pi}{2}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\cos^{-1} \frac{5}{\sqrt{41}} + \cot^{-1} \frac{4}{5} = \frac{\pi}{2}$

Concept / Formula Used: $\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$ $\cot^{-1} y = \tan^{-1} \left(\frac{1}{y} \right)$

Solution:

Let $A = \cos^{-1} \frac{5}{\sqrt{41}} \Rightarrow \cos A = \frac{5}{\sqrt{41}}$. Then $\sin A = \sqrt{1 - \frac{25}{41}} = \sqrt{\frac{16}{41}} = \frac{4}{\sqrt{41}}$. So $\tan A = \frac{\sin A}{\cos A} = \frac{4/\sqrt{41}}{5/\sqrt{41}} = \frac{4}{5}$, giving $A = \tan^{-1} \frac{4}{5}$. Also, using $\cot^{-1} y = \tan^{-1} \left(\frac{1}{y} \right)$:

$$\cot^{-1} \frac{4}{5} = \tan^{-1} \frac{5}{4}$$

Substitute into the left-hand side:

$$\text{LHS} = \tan^{-1} \frac{4}{5} + \tan^{-1} \frac{5}{4}$$

Using the standard result $\tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right) = \frac{\pi}{2}$ for $x = \frac{4}{5}$:

$$\text{LHS} = \frac{\pi}{2} = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (vii)

Prove that $\sin^{-1} \frac{1}{\sqrt{17}} + \cos^{-1} \frac{9}{\sqrt{85}} = \tan^{-1} \frac{1}{2}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\sin^{-1} \frac{1}{\sqrt{17}} + \cos^{-1} \frac{9}{\sqrt{85}} = \tan^{-1} \frac{1}{2}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Convert both terms to tangent form: For $\sin^{-1} \frac{1}{\sqrt{17}}$: $\sin A = \frac{1}{\sqrt{17}} \Rightarrow \cos A = \frac{4}{\sqrt{17}} \Rightarrow \tan A = \frac{1}{4}$. So $\sin^{-1} \frac{1}{\sqrt{17}} = \tan^{-1} \frac{1}{4}$.

For $\cos^{-1} \frac{9}{\sqrt{85}}$: $\cos B = \frac{9}{\sqrt{85}} \implies \sin B = \frac{2}{\sqrt{85}} \implies \tan B = \frac{2}{9}$. So $\cos^{-1} \frac{9}{\sqrt{85}} = \tan^{-1} \frac{2}{9}$.

Substitute both into the left-hand side:

$$\text{LHS} = \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{2}{9}$$

Using $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{1}{4} + \frac{2}{9}}{1 - \frac{1}{4} \times \frac{2}{9}} \right) \\ &= \tan^{-1} \left(\frac{\frac{9+8}{36}}{1 - \frac{2}{36}} \right) \\ &= \tan^{-1} \left(\frac{\frac{17}{36}}{\frac{34}{36}} \right) \\ &= \tan^{-1} \left(\frac{17}{34} \right) \\ &= \tan^{-1} \frac{1}{2} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (viii)

Prove that $\sin^{-1} \frac{8}{17} + \cos^{-1} \frac{4}{5} = \cot^{-1} \frac{36}{77}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $\sin^{-1} \frac{8}{17} + \cos^{-1} \frac{4}{5} = \cot^{-1} \frac{36}{77}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$ $\cot^{-1} x = \tan^{-1} \left(\frac{1}{x} \right)$

Solution:

Convert terms to tangent form: For $\sin^{-1} \frac{8}{17}$: $\sin A = \frac{8}{17} \implies \cos A = \frac{15}{17} \implies \tan A = \frac{8}{15}$.

For $\cos^{-1} \frac{4}{5}$: $\cos B = \frac{4}{5} \implies \sin B = \frac{3}{5} \implies \tan B = \frac{3}{4}$.

Substitute into the left-hand side:

$$\text{LHS} = \tan^{-1} \frac{8}{15} + \tan^{-1} \frac{3}{4}$$

Using $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{8}{15} + \frac{3}{4}}{1 - \frac{8}{15} \times \frac{3}{4}} \right) \\ &= \tan^{-1} \left(\frac{\frac{32+45}{60}}{1 - \frac{24}{60}} \right) \\ &= \tan^{-1} \left(\frac{\frac{77}{60}}{\frac{36}{60}} \right) \\ &= \tan^{-1} \frac{77}{36} \end{aligned}$$

Using $\tan^{-1} \frac{77}{36} = \cot^{-1} \frac{36}{77}$:

$$\text{LHS} = \cot^{-1} \frac{36}{77} = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (ix)

Prove that $\sin^{-1} \frac{12}{13} + \cos^{-1} \frac{4}{5} + \tan^{-1} \frac{63}{16} = \pi$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ Likely Exam Question

To Prove: $\sin^{-1} \frac{12}{13} + \cos^{-1} \frac{4}{5} + \tan^{-1} \frac{63}{16} = \pi$

Concept / Formula Used: $x > 0, y > 0, xy > 1 \quad \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

Solution:

Convert the first two terms to tangent form: For $\sin^{-1} \frac{12}{13}$: $\sin A = \frac{12}{13} \Rightarrow \cos A = \frac{5}{13} \Rightarrow \tan A = \frac{12}{5}$. For $\cos^{-1} \frac{4}{5}$: $\cos B = \frac{4}{5} \Rightarrow \sin B = \frac{3}{5} \Rightarrow \tan B = \frac{3}{4}$.

Substitute into LHS:

$$\text{LHS} = \tan^{-1} \frac{12}{5} + \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{63}{16}$$

Consider $x = \frac{12}{5}$ and $y = \frac{3}{4}$. Here $xy = \frac{12}{5} \times \frac{3}{4} = \frac{9}{5} > 1$. Using $\tan^{-1} x + \tan^{-1} y =$

$$\pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right):$$

$$\begin{aligned} \tan^{-1} \frac{12}{5} + \tan^{-1} \frac{3}{4} &= \pi + \tan^{-1} \left(\frac{\frac{12}{5} + \frac{3}{4}}{1 - \frac{12}{5} \times \frac{3}{4}} \right) \\ &= \pi + \tan^{-1} \left(\frac{\frac{48+15}{20}}{1 - \frac{36}{20}} \right) \\ &= \pi + \tan^{-1} \left(\frac{\frac{63}{20}}{-\frac{16}{20}} \right) \\ &= \pi + \tan^{-1} \left(-\frac{63}{16} \right) \\ &= \pi - \tan^{-1} \frac{63}{16} \end{aligned}$$

Substitute this result back into the left-hand side:

$$\begin{aligned} \text{LHS} &= \left(\pi - \tan^{-1} \frac{63}{16} \right) + \tan^{-1} \frac{63}{16} \\ &= \pi = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (x)

Prove that $2 \tan^{-1} \frac{1}{5} + \sec^{-1} \frac{5\sqrt{2}}{7} + 2 \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

To Prove: $2 \tan^{-1} \frac{1}{5} + \sec^{-1} \frac{5\sqrt{2}}{7} + 2 \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$

Concept / Formula Used: $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ $\sec^{-1} y = \tan^{-1} \sqrt{y^2 - 1}$

Solution:

First group the terms with coefficient 2:

$$2 \left(\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{8} \right)$$

Using $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\begin{aligned} \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{8} &= \tan^{-1} \left(\frac{\frac{1}{5} + \frac{1}{8}}{1 - \frac{1}{5} \times \frac{1}{8}} \right) \\ &= \tan^{-1} \left(\frac{\frac{13}{40}}{\frac{39}{40}} \right) \\ &= \tan^{-1} \frac{1}{3} \end{aligned}$$

Now multiply by 2 using $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$:

$$2 \tan^{-1} \frac{1}{3} = \tan^{-1} \left(\frac{2 \times \frac{1}{3}}{1 - \frac{1}{9}} \right) = \tan^{-1} \left(\frac{\frac{2}{3}}{\frac{8}{9}} \right) = \tan^{-1} \frac{3}{4}$$

Next, convert $\sec^{-1} \frac{5\sqrt{2}}{7}$ to tangent form: Let $\theta = \sec^{-1} \frac{5\sqrt{2}}{7} \Rightarrow \sec \theta = \frac{5\sqrt{2}}{7}$.

$$\begin{aligned} \tan \theta &= \sqrt{\sec^2 \theta - 1} \\ &= \sqrt{\left(\frac{5\sqrt{2}}{7} \right)^2 - 1} \\ &= \sqrt{\frac{50}{49} - 1} \\ &= \sqrt{\frac{1}{49}} = \frac{1}{7} \end{aligned}$$

So $\sec^{-1} \frac{5\sqrt{2}}{7} = \tan^{-1} \frac{1}{7}$.

Now combine both parts into LHS:

$$\text{LHS} = \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{1}{7}$$

$$\begin{aligned} \text{LHS} &= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \times \frac{1}{7}} \right) \\ &= \tan^{-1} \left(\frac{\frac{25}{28}}{\frac{25}{28}} \right) \\ &= \tan^{-1}(1) = \frac{\pi}{4} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 20

If $\cos^{-1} \left(\frac{x}{a} \right) + \cos^{-1} \left(\frac{y}{b} \right) = \alpha$, prove that $\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \alpha + \frac{y^2}{b^2} = \sin^2 \alpha$.

TYPE 13 Addition and Subtraction Formulas for Inverse Trigonometric Functions

★ Likely Exam Question

To Prove: $\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \alpha + \frac{y^2}{b^2} = \sin^2 \alpha$ given $\cos^{-1} \left(\frac{x}{a} \right) + \cos^{-1} \left(\frac{y}{b} \right) = \alpha$

Concept / Formula Used: $\cos^{-1} u + \cos^{-1} v = \cos^{-1} (uv - \sqrt{1-u^2}\sqrt{1-v^2})$

Solution:

Given equation:

$$\cos^{-1}\left(\frac{x}{a}\right) + \cos^{-1}\left(\frac{y}{b}\right) = \alpha$$

Using the inverse cosine addition identity:

$$\cos^{-1}\left(\frac{x}{a} \cdot \frac{y}{b} - \sqrt{1 - \frac{x^2}{a^2}} \sqrt{1 - \frac{y^2}{b^2}}\right) = \alpha$$

Taking cosine on both sides:

$$\frac{xy}{ab} - \sqrt{1 - \frac{x^2}{a^2}} \sqrt{1 - \frac{y^2}{b^2}} = \cos \alpha$$

Isolating the square root term on the right-hand side:

$$\frac{xy}{ab} - \cos \alpha = \sqrt{1 - \frac{x^2}{a^2}} \sqrt{1 - \frac{y^2}{b^2}}$$

Squaring both sides:

$$\left(\frac{xy}{ab} - \cos \alpha\right)^2 = \left(1 - \frac{x^2}{a^2}\right) \left(1 - \frac{y^2}{b^2}\right)$$

Expanding both sides:

$$\frac{x^2 y^2}{a^2 b^2} - \frac{2xy}{ab} \cos \alpha + \cos^2 \alpha = 1 - \frac{y^2}{b^2} - \frac{x^2}{a^2} + \frac{x^2 y^2}{a^2 b^2}$$

Subtracting $\frac{x^2 y^2}{a^2 b^2}$ from both sides:

$$-\frac{2xy}{ab} \cos \alpha + \cos^2 \alpha = 1 - \frac{y^2}{b^2} - \frac{x^2}{a^2}$$

Rearranging terms:

$$\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \alpha + \frac{y^2}{b^2} = 1 - \cos^2 \alpha$$

Using $1 - \cos^2 \alpha = \sin^2 \alpha$:

$$\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \alpha + \frac{y^2}{b^2} = \sin^2 \alpha$$

Hence Proved

LHS = RHS, hence proved.

Question 21 (i)

Evaluate $\tan\left(2 \tan^{-1} \frac{1}{2} - \cot^{-1} 3\right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\tan\left(2 \tan^{-1} \frac{1}{2} - \cot^{-1} 3\right)$

Concept / Formula Used: $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2}\right)$ $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

Solution:First simplify $2 \tan^{-1} \frac{1}{2}$:

$$\begin{aligned}
 2 \tan^{-1} \frac{1}{2} &= \tan^{-1} \left(\frac{2 \times \frac{1}{2}}{1 - \left(\frac{1}{2}\right)^2} \right) \\
 &= \tan^{-1} \left(\frac{1}{1 - \frac{1}{4}} \right) \\
 &= \tan^{-1} \left(\frac{1}{\frac{3}{4}} \right) \\
 &= \tan^{-1} \frac{4}{3}
 \end{aligned}$$

Convert $\cot^{-1} 3$ to tangent form:

$$\cot^{-1} 3 = \tan^{-1} \frac{1}{3}$$

Let $A = 2 \tan^{-1} \frac{1}{2} = \tan^{-1} \frac{4}{3} \implies \tan A = \frac{4}{3}$. Let $B = \cot^{-1} 3 = \tan^{-1} \frac{1}{3} \implies \tan B = \frac{1}{3}$.Evaluating $\tan(A - B)$:

$$\begin{aligned}
 \tan(A - B) &= \frac{\tan A - \tan B}{1 + \tan A \tan B} \\
 &= \frac{\frac{4}{3} - \frac{1}{3}}{1 + \frac{4}{3} \times \frac{1}{3}} \\
 &= \frac{1}{1 + \frac{4}{9}} \\
 &= \frac{1}{\frac{13}{9}} \\
 &= \frac{9}{13}
 \end{aligned}$$

Final Answer

$$\tan \left(2 \tan^{-1} \frac{1}{2} - \cot^{-1} 3 \right) = \frac{9}{13}$$

Question 21 (ii)Evaluate $\tan \left(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4} \right)$.**TYPE 9** Evaluating Trigonometric Functions of Inverse Arguments**To Find:** The value of $\tan \left(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4} \right)$ **Concept / Formula Used:** $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1 - x^2} \right)$ $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$ **Solution:**

First simplify $2 \tan^{-1} \frac{1}{5}$:

$$\begin{aligned} 2 \tan^{-1} \frac{1}{5} &= \tan^{-1} \left(\frac{2 \times \frac{1}{5}}{1 - \left(\frac{1}{5}\right)^2} \right) \\ &= \tan^{-1} \left(\frac{\frac{2}{5}}{1 - \frac{1}{25}} \right) \\ &= \tan^{-1} \left(\frac{\frac{2}{5}}{\frac{24}{25}} \right) \\ &= \tan^{-1} \left(\frac{2}{5} \times \frac{25}{24} \right) \\ &= \tan^{-1} \frac{5}{12} \end{aligned}$$

Let $A = 2 \tan^{-1} \frac{1}{5} = \tan^{-1} \frac{5}{12} \implies \tan A = \frac{5}{12}$. Let $B = \frac{\pi}{4} \implies \tan B = 1$.

Evaluating $\tan \left(A - \frac{\pi}{4} \right)$:

$$\begin{aligned} \tan \left(A - \frac{\pi}{4} \right) &= \frac{\tan A - 1}{1 + \tan A} \\ &= \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} \\ &= \frac{-\frac{7}{12}}{\frac{17}{12}} \\ &= -\frac{7}{17} \end{aligned}$$

Final Answer

$$\tan \left(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4} \right) = -\frac{7}{17}$$

Question 22 (i)

Find the value of $\tan \left(\cos^{-1} \frac{3}{5} + \tan^{-1} \frac{1}{4} \right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\tan \left(\cos^{-1} \frac{3}{5} + \tan^{-1} \frac{1}{4} \right)$

Concept / Formula Used: $\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$ $x > 0$ $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

Solution:

Convert $\cos^{-1} \frac{3}{5}$ into tangent form: Let $A = \cos^{-1} \frac{3}{5} \implies \cos A = \frac{3}{5}$. Then $\sin A = \sqrt{1 - \frac{9}{25}} = \frac{4}{5} \implies \tan A = \frac{4}{3}$.

Let $B = \tan^{-1} \frac{1}{4} \Rightarrow \tan B = \frac{1}{4}$.

Evaluating $\tan(A + B)$:

$$\begin{aligned}\tan(A + B) &= \frac{\tan A + \tan B}{1 - \tan A \tan B} \\ &= \frac{\frac{4}{3} + \frac{1}{4}}{1 - \frac{4}{3} \times \frac{1}{4}} \\ &= \frac{\frac{16+3}{12}}{1 - \frac{4}{12}} \\ &= \frac{\frac{19}{12}}{\frac{8}{12}} \\ &= \frac{19}{8}\end{aligned}$$

Final Answer

$$\tan \left(\cos^{-1} \frac{3}{5} + \tan^{-1} \frac{1}{4} \right) = \frac{19}{8}$$

Question 22 (ii)

Find the value of $\cos \left(\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{5}{13} \right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\cos \left(\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{5}{13} \right)$

Concept / Formula Used: $\cos(A + B) = \cos A \cos B - \sin A \sin B$

Solution:

Let $A = \sin^{-1} \frac{3}{5} \Rightarrow \sin A = \frac{3}{5}$. Then $\cos A = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$.

Let $B = \sin^{-1} \frac{5}{13} \Rightarrow \sin B = \frac{5}{13}$. Then $\cos B = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$.

Applying the expansion formula for $\cos(A + B)$:

$$\begin{aligned}\cos(A + B) &= \cos A \cos B - \sin A \sin B \\ &= \frac{4}{5} \times \frac{12}{13} - \frac{3}{5} \times \frac{5}{13} \\ &= \frac{48}{65} - \frac{15}{65} \\ &= \frac{33}{65}\end{aligned}$$

Final Answer

$$\cos \left(\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{5}{13} \right) = \frac{33}{65}$$

Question 22 (iii)

Find the value of $\cot \left(\cos^{-1} \frac{4}{5} + \sin^{-1} \frac{2}{\sqrt{13}} \right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

To Find: The value of $\cot \left(\cos^{-1} \frac{4}{5} + \sin^{-1} \frac{2}{\sqrt{13}} \right)$

Concept / Formula Used: $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ $\cot \theta = \frac{1}{\tan \theta}$

Solution:

$$\text{Let } A = \cos^{-1} \frac{4}{5} \Rightarrow \cos A = \frac{4}{5}, \sin A = \frac{3}{5} \Rightarrow \tan A = \frac{3}{4}.$$

$$\text{Let } B = \sin^{-1} \frac{2}{\sqrt{13}} \Rightarrow \sin B = \frac{2}{\sqrt{13}}, \cos B = \sqrt{1 - \frac{4}{13}} = \frac{3}{\sqrt{13}} \Rightarrow \tan B = \frac{2}{3}.$$

Calculate $\tan(A + B)$:

$$\begin{aligned} \tan(A + B) &= \frac{\tan A + \tan B}{1 - \tan A \tan B} \\ &= \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \times \frac{2}{3}} \\ &= \frac{\frac{9+8}{12}}{1 - \frac{6}{12}} \\ &= \frac{\frac{17}{12}}{\frac{6}{12}} \\ &= \frac{17}{6} \end{aligned}$$

Using $\cot(A + B) = \frac{1}{\tan(A + B)}$:

$$\cot(A + B) = \frac{6}{17}$$

Final Answer

$$\cot \left(\cos^{-1} \frac{4}{5} + \sin^{-1} \frac{2}{\sqrt{13}} \right) = \frac{6}{17}$$

Question 22 (iv)

Find the value of $\sin \left(2 \tan^{-1} \frac{2}{3} \right) + \cos \left(\tan^{-1} \sqrt{3} \right)$.

TYPE 9 Evaluating Trigonometric Functions of Inverse Arguments

★ Likely Exam Question

To Find: The value of $\sin\left(2\tan^{-1}\frac{2}{3}\right) + \cos\left(\tan^{-1}\sqrt{3}\right)$

Concept / Formula Used: $\sin(2\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$ $\cos(\tan^{-1}x) = \frac{1}{\sqrt{1+x^2}}$

Solution:

Evaluate the first term $\sin\left(2\tan^{-1}\frac{2}{3}\right)$: Let $\theta = \tan^{-1}\frac{2}{3} \implies \tan\theta = \frac{2}{3}$. Using $\sin(2\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$:

$$\begin{aligned}\sin\left(2\tan^{-1}\frac{2}{3}\right) &= \frac{2 \times \frac{2}{3}}{1 + \left(\frac{2}{3}\right)^2} \\ &= \frac{\frac{4}{3}}{1 + \frac{4}{9}} \\ &= \frac{\frac{4}{3}}{\frac{13}{9}} \\ &= \frac{4}{3} \times \frac{9}{13} \\ &= \frac{12}{13}\end{aligned}$$

Evaluate the second term $\cos\left(\tan^{-1}\sqrt{3}\right)$: Since $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$:

$$\cos\left(\tan^{-1}\sqrt{3}\right) = \cos\frac{\pi}{3} = \frac{1}{2}$$

Summing both values:

$$\begin{aligned}\text{Total Value} &= \frac{12}{13} + \frac{1}{2} \\ &= \frac{24+13}{26} \\ &= \frac{37}{26}\end{aligned}$$

Final Answer

$$\sin\left(2\tan^{-1}\frac{2}{3}\right) + \cos\left(\tan^{-1}\sqrt{3}\right) = \frac{37}{26}$$

Question 23 (i)

Solve the equation for x : $\tan^{-1}\left(\frac{2-x}{2+x}\right) = \frac{1}{2}\tan^{-1}\frac{x}{2}, x > 0$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of $x > 0$ satisfying $\tan^{-1}\left(\frac{2-x}{2+x}\right) = \frac{1}{2}\tan^{-1}\frac{x}{2}$

Concept / Formula Used: $\tan^{-1} \left(\frac{a-b}{a+b} \right) = \tan^{-1} 1 - \tan^{-1} \left(\frac{b}{a} \right) = \frac{\pi}{4} - \tan^{-1} \left(\frac{b}{a} \right)$

Solution:

Rewrite the argument on the left-hand side:

$$\tan^{-1} \left(\frac{2-x}{2+x} \right) = \tan^{-1} \left(\frac{1 - \frac{x}{2}}{1 + \frac{x}{2}} \right)$$

Using $\tan^{-1} \left(\frac{1-y}{1+y} \right) = \tan^{-1} 1 - \tan^{-1} y = \frac{\pi}{4} - \tan^{-1} y$ for $y = \frac{x}{2}$:

$$\tan^{-1} \left(\frac{2-x}{2+x} \right) = \frac{\pi}{4} - \tan^{-1} \frac{x}{2}$$

Substitute into the given equation:

$$\frac{\pi}{4} - \tan^{-1} \frac{x}{2} = \frac{1}{2} \tan^{-1} \frac{x}{2}$$

Add $\tan^{-1} \frac{x}{2}$ to both sides:

$$\begin{aligned} \frac{\pi}{4} &= \frac{1}{2} \tan^{-1} \frac{x}{2} + \tan^{-1} \frac{x}{2} \\ \frac{\pi}{4} &= \frac{3}{2} \tan^{-1} \frac{x}{2} \end{aligned}$$

Multiply both sides by $\frac{2}{3}$:

$$\tan^{-1} \frac{x}{2} = \frac{\pi}{6}$$

Taking tangent on both sides:

$$\begin{aligned} \frac{x}{2} &= \tan \frac{\pi}{6} \\ \frac{x}{2} &= \frac{1}{\sqrt{3}} \end{aligned}$$

Multiply by 2:

$$x = \frac{2}{\sqrt{3}}$$

Since $x = \frac{2}{\sqrt{3}} > 0$, it satisfies the given domain condition.

Final Answer

$$x = \frac{2}{\sqrt{3}}$$

Question 23 (ii)

Solve the equation for x : $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{4}$.

TYPE 12 Solving Inverse Trigonometric Equations

★ **Likely Exam Question**

To Find: The value of x satisfying $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Applying the addition identity:

$$\tan^{-1} \left(\frac{4x + 6x}{1 - (4x)(6x)} \right) = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{10x}{1 - 24x^2} \right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{10x}{1 - 24x^2} = \tan \frac{\pi}{4} = 1$$

Cross-multiplying:

$$10x = 1 - 24x^2$$

Rearranging into standard quadratic form:

$$24x^2 + 10x - 1 = 0$$

Factorizing the quadratic expression:

$$\begin{aligned} 24x^2 + 12x - 2x - 1 &= 0 \\ 12x(2x + 1) - 1(2x + 1) &= 0 \\ (12x - 1)(2x + 1) &= 0 \end{aligned}$$

This gives two possible roots:

$$x = \frac{1}{12} \quad \text{or} \quad x = -\frac{1}{2}$$

Check validity: For $x = -\frac{1}{2}$, both $4x = -2 < 0$ and $6x = -3 < 0$, making $\tan^{-1} 4x + \tan^{-1} 6x < 0 \neq \frac{\pi}{4}$. Thus $x = -\frac{1}{2}$ is an extraneous root. For $x = \frac{1}{12}$, both $4x > 0$ and $6x > 0$ with product $(4x)(6x) = \frac{1}{6} < 1$, which satisfies the equation.

Final Answer

$$x = \frac{1}{12}$$

Common Mistake: Forgetting to verify roots in inverse trig equations: $x = -\frac{1}{2}$ makes both arguments negative, which produces a negative sum and must be rejected.

Question 23 (iii)

Solve the equation for x : $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1} \frac{6}{17}$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of x satisfying $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1} \frac{6}{17}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Apply the identity $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$ for $a = x+1$ and $b = x-1$:

$$\tan^{-1} \left(\frac{(x+1) + (x-1)}{1 - (x+1)(x-1)} \right) = \tan^{-1} \frac{6}{17}$$

Simplifying the argument on the left-hand side:

$$\tan^{-1} \left(\frac{2x}{1 - (x^2 - 1)} \right) = \tan^{-1} \frac{6}{17}$$

$$\tan^{-1} \left(\frac{2x}{2 - x^2} \right) = \tan^{-1} \frac{6}{17}$$

Equating the arguments:

$$\frac{2x}{2 - x^2} = \frac{6}{17}$$

Divide numerator of both sides by 2:

$$\frac{x}{2 - x^2} = \frac{3}{17}$$

Cross-multiplying:

$$\begin{aligned} 17x &= 3(2 - x^2) \\ 17x &= 6 - 3x^2 \\ 3x^2 + 17x - 6 &= 0 \end{aligned}$$

Factorizing:

$$\begin{aligned} 3x^2 + 18x - x - 6 &= 0 \\ 3x(x+6) - 1(x+6) &= 0 \\ (3x-1)(x+6) &= 0 \end{aligned}$$

This gives:

$$x = \frac{1}{3} \quad \text{or} \quad x = -6$$

Checking $x = -6$: LHS = $\tan^{-1}(-5) + \tan^{-1}(-7) < 0$, while RHS = $\tan^{-1} \frac{6}{17} > 0$, so $x = -6$ is extraneous. For $x = \frac{1}{3}$, LHS matches RHS.

Final Answer

$$x = \frac{1}{3}$$

Question 23 (iv)

Solve the equation for x : $\tan^{-1} \left(\frac{x-2}{x-1} \right) + \tan^{-1} \left(\frac{x+2}{x+1} \right) = \frac{\pi}{4}$.

TYPE 12 Solving Inverse Trigonometric Equations

★ Likely Exam Question

To Find: The value of x satisfying $\tan^{-1} \left(\frac{x-2}{x-1} \right) + \tan^{-1} \left(\frac{x+2}{x+1} \right) = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Applying $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\tan^{-1} \left(\frac{\frac{x-2}{x-1} + \frac{x+2}{x+1}}{1 - \frac{x-2}{x-1} \cdot \frac{x+2}{x+1}} \right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{\frac{(x-2)(x+1) + (x+2)(x-1)}{(x-1)(x+1)}}{\frac{(x-1)(x+1) - (x-2)(x+2)}{(x-1)(x+1)}} = \tan \frac{\pi}{4} = 1$$

Cancelling the common denominator $(x-1)(x+1)$:

$$\frac{(x-2)(x+1) + (x+2)(x-1)}{(x-1)(x+1) - (x-2)(x+2)} = 1$$

Expand numerator and denominator:

$$\text{Numerator} = (x^2 - x - 2) + (x^2 + x - 2) = 2x^2 - 4$$

$$\text{Denominator} = (x^2 - 1) - (x^2 - 4) = x^2 - 1 - x^2 + 4 = 3$$

Substitute these expressions back:

$$\frac{2x^2 - 4}{3} = 1$$

Multiply by 3:

$$2x^2 - 4 = 3$$

$$2x^2 = 7$$

$$x^2 = \frac{7}{2}$$

$$x = \pm \sqrt{\frac{7}{2}}$$

Final Answer

$$x = \pm \sqrt{\frac{7}{2}}$$

Question 23 (v)

Solve the equation for x : $\tan^{-1} \left(\frac{x-2}{x-3} \right) + \tan^{-1} \left(\frac{x+2}{x+3} \right) = \frac{\pi}{4}$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of x satisfying $\tan^{-1} \left(\frac{x-2}{x-3} \right) + \tan^{-1} \left(\frac{x+2}{x+3} \right) = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Applying the addition identity:

$$\tan^{-1} \left(\frac{\frac{x-2}{x-3} + \frac{x+2}{x+3}}{1 - \frac{x-2}{x-3} \cdot \frac{x+2}{x+3}} \right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{(x-2)(x+3) + (x+2)(x-3)}{(x-3)(x+3) - (x-2)(x+2)} = \tan \frac{\pi}{4} = 1$$

Expand numerator and denominator:

$$\begin{aligned} \text{Numerator} &= (x^2 + x - 6) + (x^2 - x - 6) = 2x^2 - 12 \\ \text{Denominator} &= (x^2 - 9) - (x^2 - 4) = x^2 - 9 - x^2 + 4 = -5 \end{aligned}$$

Substitute back:

$$\frac{2x^2 - 12}{-5} = 1$$

Multiply by -5 :

$$\begin{aligned} 2x^2 - 12 &= -5 \\ 2x^2 &= 7 \\ x^2 &= \frac{7}{2} \\ x &= \pm \sqrt{\frac{7}{2}} \end{aligned}$$

Final Answer

$$x = \pm \sqrt{\frac{7}{2}}$$

Question 23 (vi)

Solve the equation for x : $\tan^{-1} \left(\frac{x-3}{x-4} \right) + \tan^{-1} \left(\frac{x+3}{x+4} \right) = \frac{\pi}{4}$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of x satisfying $\tan^{-1} \left(\frac{x-3}{x-4} \right) + \tan^{-1} \left(\frac{x+3}{x+4} \right) = \frac{\pi}{4}$

Concept / Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Solution:

Applying the addition formula:

$$\tan^{-1} \left(\frac{\frac{x-3}{x-4} + \frac{x+3}{x+4}}{1 - \frac{x-3}{x-4} \cdot \frac{x+3}{x+4}} \right) = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\frac{(x-3)(x+4) + (x+3)(x-4)}{(x-4)(x+4) - (x-3)(x+3)} = \tan \frac{\pi}{4} = 1$$

Expand numerator and denominator:

$$\begin{aligned} \text{Numerator} &= (x^2 + x - 12) + (x^2 - x - 12) = 2x^2 - 24 \\ \text{Denominator} &= (x^2 - 16) - (x^2 - 9) = x^2 - 16 - x^2 + 9 = -7 \end{aligned}$$

Substitute back:

$$\frac{2x^2 - 24}{-7} = 1$$

Multiply by -7 :

$$\begin{aligned} 2x^2 - 24 &= -7 \\ 2x^2 &= 17 \\ x^2 &= \frac{17}{2} \\ x &= \pm \sqrt{\frac{17}{2}} \end{aligned}$$

Final Answer

$$x = \pm \sqrt{\frac{17}{2}}$$

Question 23 (vii)

Solve the equation for x : $\cos(\sin^{-1} x) = \frac{1}{9}$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of x satisfying $\cos(\sin^{-1} x) = \frac{1}{9}$

Concept / Formula Used: $\cos(\sin^{-1} x) = \sqrt{1-x^2}$

Solution:

Let $\theta = \sin^{-1} x \implies \sin \theta = x$. Using the standard identity $\cos \theta = \sqrt{1 - \sin^2 \theta}$:

$$\cos(\sin^{-1} x) = \sqrt{1 - x^2}$$

Given equation becomes:

$$\sqrt{1 - x^2} = \frac{1}{9}$$

Squaring both sides:

$$\begin{aligned} 1 - x^2 &= \frac{1}{81} \\ x^2 &= 1 - \frac{1}{81} \\ x^2 &= \frac{80}{81} \end{aligned}$$

Taking square root:

$$\begin{aligned} x &= \pm \frac{\sqrt{80}}{9} \\ &= \pm \frac{4\sqrt{5}}{9} \end{aligned}$$

Final Answer

$$x = \pm \frac{4\sqrt{5}}{9}$$

Question 23 (viii)

Solve the equation for x : $\cos(2 \sin^{-1} x) = \frac{1}{9}$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of x satisfying $\cos(2 \sin^{-1} x) = \frac{1}{9}$

Concept / Formula Used: $\cos(2\theta) = 1 - 2 \sin^2 \theta$

Solution:

Let $\theta = \sin^{-1} x \implies \sin \theta = x$. The equation becomes:

$$\cos(2\theta) = \frac{1}{9}$$

Using the double angle identity $\cos(2\theta) = 1 - 2 \sin^2 \theta$:

$$1 - 2 \sin^2 \theta = \frac{1}{9}$$

Substitute $\sin \theta = x$:

$$1 - 2x^2 = \frac{1}{9}$$

Rearranging terms:

$$2x^2 = 1 - \frac{1}{9}$$

$$2x^2 = \frac{8}{9}$$

$$x^2 = \frac{4}{9}$$

$$x = \pm \frac{2}{3}$$

Final Answer

$$x = \pm \frac{2}{3}$$

Question 23 (ix)

Solve the equation for x : $\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of x satisfying $\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$

Concept / Formula Used: $\sin(\cos^{-1} x) = \sqrt{1 - x^2}$

Solution:

Given equation:

$$\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$$

Taking cosine on both sides:

$$\sin(\cos^{-1} x) = \cos \frac{\pi}{6}$$

Since $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$:

$$\sin(\cos^{-1} x) = \frac{\sqrt{3}}{2}$$

Using the standard identity $\sin(\cos^{-1} x) = \sqrt{1 - x^2}$:

$$\sqrt{1 - x^2} = \frac{\sqrt{3}}{2}$$

Squaring both sides:

$$1 - x^2 = \frac{3}{4}$$

$$x^2 = 1 - \frac{3}{4}$$

$$x^2 = \frac{1}{4}$$

$$x = \pm \frac{1}{2}$$

Final Answer

$$x = \pm \frac{1}{2}$$

Question 23 (x)

Solve the equation for x : $\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$.

TYPE 12 Solving Inverse Trigonometric Equations

To Find: The value of x satisfying $\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$

Concept / Formula Used: $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$

Solution:

Given equation:

$$\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$$

Taking sine on both sides:

$$\cos(\sin^{-1} x) = \sin \frac{\pi}{3}$$

Since $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$:

$$\cos(\sin^{-1} x) = \frac{\sqrt{3}}{2}$$

Using the standard identity $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$:

$$\sqrt{1 - x^2} = \frac{\sqrt{3}}{2}$$

Squaring both sides:

$$\begin{aligned} 1 - x^2 &= \frac{3}{4} \\ x^2 &= 1 - \frac{3}{4} \\ x^2 &= \frac{1}{4} \\ x &= \pm \frac{1}{2} \end{aligned}$$

Final Answer

$$x = \pm \frac{1}{2}$$

Question 23 (xi)

Solve the equation for x : $\sin^{-1} \frac{8}{x} + \sin^{-1} \frac{15}{x} = \frac{\pi}{2}$.

TYPE 12 Solving Inverse Trigonometric Equations

★ Likely Exam Question

To Find: The value of x satisfying $\sin^{-1} \frac{8}{x} + \sin^{-1} \frac{15}{x} = \frac{\pi}{2}$

Concept / Formula Used: $\sin^{-1} A + \cos^{-1} A = \frac{\pi}{2}$

Solution:

Given equation:

$$\sin^{-1} \frac{8}{x} + \sin^{-1} \frac{15}{x} = \frac{\pi}{2}$$

Transpose $\sin^{-1} \frac{15}{x}$ to the right-hand side:

$$\sin^{-1} \frac{8}{x} = \frac{\pi}{2} - \sin^{-1} \frac{15}{x}$$

Using the co-function identity $\frac{\pi}{2} - \sin^{-1} y = \cos^{-1} y$:

$$\sin^{-1} \frac{8}{x} = \cos^{-1} \frac{15}{x}$$

Taking sine on both sides:

$$\frac{8}{x} = \sin \left(\cos^{-1} \frac{15}{x} \right)$$

Using $\sin(\cos^{-1} y) = \sqrt{1 - y^2}$:

$$\frac{8}{x} = \sqrt{1 - \left(\frac{15}{x} \right)^2}$$

Squaring both sides:

$$\begin{aligned} \frac{64}{x^2} &= 1 - \frac{225}{x^2} \\ \frac{64}{x^2} + \frac{225}{x^2} &= 1 \\ \frac{289}{x^2} &= 1 \\ x^2 &= 289 \\ x &= \pm 17 \end{aligned}$$

Since $\sin^{-1} \frac{8}{x}$ and $\sin^{-1} \frac{15}{x}$ must add up to positive $\frac{\pi}{2}$, x must be positive. Therefore $x = 17$.

Final Answer

$$x = 17$$

Common Mistake: Taking $x = -17$ leads to negative angles on the left side, summing to $-\frac{\pi}{2}$, which contradicts the right side $\frac{\pi}{2}$. Hence $x = -17$ must be discarded.

Question 23 (xii)

Solve the equation for x : $\sin^{-1} \frac{2a}{1+a^2} + \sin^{-1} \frac{2b}{1+b^2} = 2 \tan^{-1} x$.

TYPE 12 Solving Inverse Trigonometric Equations

★ Likely Exam Question

To Find: The value of x satisfying $\sin^{-1} \frac{2a}{1+a^2} + \sin^{-1} \frac{2b}{1+b^2} = 2 \tan^{-1} x$

Concept / Formula Used: $\sin^{-1} \left(\frac{2y}{1+y^2} \right) = 2 \tan^{-1} y \quad |y| \leq 1$

Solution:

Using the standard identity $\sin^{-1} \left(\frac{2y}{1+y^2} \right) = 2 \tan^{-1} y$:

$$\sin^{-1} \frac{2a}{1+a^2} = 2 \tan^{-1} a$$

$$\sin^{-1} \frac{2b}{1+b^2} = 2 \tan^{-1} b$$

Substitute these into the given equation:

$$2 \tan^{-1} a + 2 \tan^{-1} b = 2 \tan^{-1} x$$

Dividing the entire equation by 2:

$$\tan^{-1} a + \tan^{-1} b = \tan^{-1} x$$

Using the addition identity $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$:

$$\tan^{-1} \left(\frac{a+b}{1-ab} \right) = \tan^{-1} x$$

Taking tangent on both sides:

$$x = \frac{a+b}{1-ab}$$

Final Answer

$$x = \frac{a+b}{1-ab}$$

ISC Class XII Mathematics

Chapter 3 Matrices

101 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 3.1 – Order, Elements and Construction of Matrices
- II. 3.2 – Addition, Subtraction and Scalar Multiplication of Matrices
- III. 3.3 – Multiplication of Matrices
- IV. 3.4 – Transpose, Symmetric and Skew-Symmetric Matrices

3.1 – Order, Elements and Construction of Matrices

Question 1

If A is a matrix of type $p \times q$ and R is a row of A , then what is the type of R as a matrix?

TYPE 1 Matrix Order Analysis

Given: A is a matrix of order $p \times q$ and R is a row matrix of A .

Concept / Formula Used: $n \quad 1 \times n$

Solution:

Since A is a matrix of order $p \times q$, it has p rows and q columns.

Each row R of matrix A contains q elements corresponding to the q columns of A .

A single row of A , treated as a standalone matrix R , has 1 row and q columns.

Therefore, the type (order) of R as a matrix is $1 \times q$.

Final Answer

The type of R as a matrix is $1 \times q$.

Question 2

If A is a column matrix with 5 rows, then what type of matrix is a row of A ?

TYPE 1 Matrix Order Analysis

Given: A is a column matrix with 5 rows.

Concept / Formula Used: $m \quad m \times 1 \quad 1 \times 1$

Solution:

Since A is a column matrix with 5 rows, the order of matrix A is 5×1 .

Matrix A has only 1 column.

A row of matrix A contains elements across all columns of A . Since there is only 1 column in A , a row of A contains only 1 element.

A row of A , viewed as a matrix itself, has 1 row and 1 column.

Therefore, the order (type) of a row of A as a matrix is 1×1 .

Final Answer

A row of A is a 1×1 matrix.

Question 3

If a matrix has 18 elements, what the possible orders it can have? What, if it has 11 elements?

TYPE 2 Possible Matrix Orders from Total Elements

Given: Number of elements in a matrix is 18, or 11.

Concept / Formula Used: n (m, k) $m \times k = n$

Solution:

Case I: Matrix has 18 elements

We need to find all ordered pairs (m, k) of positive integers such that $m \times k = 18$.

The factor pairs of 18 are:

$$1 \times 18 = 18$$

$$2 \times 9 = 18$$

$$3 \times 6 = 18$$

$$6 \times 3 = 18$$

$$9 \times 2 = 18$$

$$18 \times 1 = 18$$

Hence, the possible orders are $1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1$.

Case II: Matrix has 11 elements

We need to find all ordered pairs (m, k) of positive integers such that $m \times k = 11$.

Since 11 is a prime number, its only positive integral factor pairs are:

$$1 \times 11 = 11$$

$$11 \times 1 = 11$$

Hence, the possible orders are $1 \times 11, 11 \times 1$.

Final Answer

For 18 elements: $1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1$.

For 11 elements: $1 \times 11, 11 \times 1$.

Question 4

(i) If A is a 3×3 matrix whose elements are given by $a_{ij} = \frac{1}{3}|-3i + j|$, write the value of a_{23} .

(ii) Write the element a_{12} of the matrix $A = [a_{ij}]_{2 \times 2}$, whose elements a_{ij} are given by $a_{ij} = 2^{2ix} \sin(jx)$.

TYPE 3 Evaluating Specific Matrix Elements

Given: (i) $a_{ij} = \frac{1}{3}|-3i + j|$ for $A_{3 \times 3}$.

(ii) $a_{ij} = 2^{2ix} \sin(jx)$ for $A_{2 \times 2}$.

Concept / Formula Used: a_{ij} i j a_{ij}

Solution:

Part (i):

We need to find a_{23} , so set $i = 2$ and $j = 3$.

Substitute $i = 2$ and $j = 3$ into $a_{ij} = \frac{1}{3}|-3i + j|$:

$$\begin{aligned} a_{23} &= \frac{1}{3}|-3(2) + 3| \\ &= \frac{1}{3}|-6 + 3| \\ &= \frac{1}{3}|-3| \\ &= \frac{1}{3} \times 3 \\ &= 1 \end{aligned}$$

Part (ii):

We need to find a_{12} , so set $i = 1$ and $j = 2$.

Substitute $i = 1$ and $j = 2$ into $a_{ij} = 2^{2ix} \sin(jx)$:

$$\begin{aligned} a_{12} &= 2^{2(1)x} \sin(2x) \\ &= 2^{2x} \sin(2x) \end{aligned}$$

Final Answer

- (i) $a_{23} = 1$
(ii) $a_{12} = 2^{2x} \sin(2x)$

Question 5

- (i) For a 2×2 matrix, $A = [a_{ij}]$, whose elements are given by $a_{ij} = \frac{i}{j}$, write the value of a_{12} .
(ii) Write the element a_{23} of a 3×3 matrix $A = [a_{ij}]$ whose elements a_{ij} are given by $a_{ij} = \frac{|i-j|}{2}$.

TYPE 3 Evaluating Specific Matrix Elements

Given: (i) $a_{ij} = \frac{i}{j}$ for $A_{2 \times 2}$.

(ii) $a_{ij} = \frac{|i-j|}{2}$ for $A_{3 \times 3}$.

Concept / Formula Used: $i \quad j \quad i \quad j \quad a_{ij}$

Solution:

Part (i):

To find a_{12} , substitute $i = 1$ and $j = 2$ into $a_{ij} = \frac{i}{j}$:

$$a_{12} = \frac{1}{2}$$

Part (ii):

To find a_{23} , substitute $i = 2$ and $j = 3$ into $a_{ij} = \frac{|i - j|}{2}$:

$$\begin{aligned} a_{23} &= \frac{|2 - 3|}{2} \\ &= \frac{|-1|}{2} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

(i) $a_{12} = \frac{1}{2}$

(ii) $a_{23} = \frac{1}{2}$

Question 6

Construct a 2×2 matrix $A = [a_{ij}]$ whose elements a_{ij} are given by:

(i) $a_{ij} = 2i - j$

(ii) $a_{ij} = |2i - 3j|$

TYPE 4 Matrix Construction from General Element

Given: $A = [a_{ij}]$ of order 2×2 , where (i) $a_{ij} = 2i - j$, (ii) $a_{ij} = |2i - 3j|$.

Concept / Formula Used: 2×2 $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

Solution:

Part (i): $a_{ij} = 2i - j$

Evaluating elements for $i = 1, 2$ and $j = 1, 2$:

$$a_{11} = 2(1) - 1 = 2 - 1 = 1$$

$$a_{12} = 2(1) - 2 = 2 - 2 = 0$$

$$a_{21} = 2(2) - 1 = 4 - 1 = 3$$

$$a_{22} = 2(2) - 2 = 4 - 2 = 2$$

Thus, $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$.

Part (ii): $a_{ij} = |2i - 3j|$

Evaluating elements for $i = 1, 2$ and $j = 1, 2$:

$$a_{11} = |2(1) - 3(1)| = |2 - 3| = |-1| = 1$$

$$a_{12} = |2(1) - 3(2)| = |2 - 6| = |-4| = 4$$

$$a_{21} = |2(2) - 3(1)| = |4 - 3| = |1| = 1$$

$$a_{22} = |2(2) - 3(2)| = |4 - 6| = |-2| = 2$$

Thus, $A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \end{bmatrix}$.

Final Answer

(i) $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$

(ii) $A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \end{bmatrix}$

Question 7

Construct a 2×3 matrix $B = [b_{ij}]$ whose elements b_{ij} are given by:

(i) $b_{ij} = i - 3j$

(ii) $b_{ij} = (i + 2j)^2$

TYPE 4 Matrix Construction from General Element

Given: $B = [b_{ij}]$ of order 2×3 , where (i) $b_{ij} = i - 3j$, (ii) $b_{ij} = (i + 2j)^2$.

Concept / Formula Used: $2 \times 3 \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$

Solution:

Part (i): $b_{ij} = i - 3j$

Evaluating elements for $i = 1, 2$ and $j = 1, 2, 3$:

$$b_{11} = 1 - 3(1) = 1 - 3 = -2$$

$$b_{12} = 1 - 3(2) = 1 - 6 = -5$$

$$b_{13} = 1 - 3(3) = 1 - 9 = -8$$

$$b_{21} = 2 - 3(1) = 2 - 3 = -1$$

$$b_{22} = 2 - 3(2) = 2 - 6 = -4$$

$$b_{23} = 2 - 3(3) = 2 - 9 = -7$$

Thus, $B = \begin{bmatrix} -2 & -5 & -8 \\ -1 & -4 & -7 \end{bmatrix}$.

Part (ii): $b_{ij} = (i + 2j)^2$

Evaluating elements for $i = 1, 2$ and $j = 1, 2, 3$:

$$b_{11} = (1 + 2(1))^2 = (1 + 2)^2 = 3^2 = 9$$

$$b_{12} = (1 + 2(2))^2 = (1 + 4)^2 = 5^2 = 25$$

$$b_{13} = (1 + 2(3))^2 = (1 + 6)^2 = 7^2 = 49$$

$$b_{21} = (2 + 2(1))^2 = (2 + 2)^2 = 4^2 = 16$$

$$b_{22} = (2 + 2(2))^2 = (2 + 4)^2 = 6^2 = 36$$

$$b_{23} = (2 + 2(3))^2 = (2 + 6)^2 = 8^2 = 64$$

Thus, $B = \begin{bmatrix} 9 & 25 & 49 \\ 16 & 36 & 64 \end{bmatrix}$.

Final Answer

$$(i) B = \begin{bmatrix} -2 & -5 & -8 \\ -1 & -4 & -7 \end{bmatrix}$$

$$(ii) B = \begin{bmatrix} 9 & 25 & 49 \\ 16 & 36 & 64 \end{bmatrix}$$

Question 8

$$(i) \text{ If } \begin{bmatrix} x+2y & -y \\ 3x & 4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 6 & 4 \end{bmatrix}, \text{ find the values of } x \text{ and } y.$$

$$(ii) \text{ If } \begin{bmatrix} x+3y & y \\ 7-x & 4 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 0 & 4 \end{bmatrix}, \text{ find the values of } x \text{ and } y.$$

$$(iii) \text{ Find the value of } y, \text{ if } \begin{bmatrix} x-y & 2 \\ x & 5 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}.$$

$$(iv) \text{ Find the value of } x, \text{ if } \begin{bmatrix} 3x+y & -y \\ 2y-x & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -5 & 3 \end{bmatrix}.$$

$$(v) \text{ If } \begin{bmatrix} x+2y & 3y \\ 4x & 2 \end{bmatrix} = \begin{bmatrix} 0 & -3 \\ 8 & 2 \end{bmatrix}, \text{ find the value of } (x-y).$$

$$(vi) \text{ If } \begin{bmatrix} x-y & z \\ 2x-y & w \end{bmatrix} = \begin{bmatrix} -1 & 4 \\ 0 & 5 \end{bmatrix}, \text{ find the value of } (x+y).$$

$$(vii) \text{ If } \begin{bmatrix} x+y & x+2 \\ 2x-y & 16 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ 1 & 3y+1 \end{bmatrix}, \text{ then write the value of } (y-x).$$

TYPE 5 Equality of Matrices**Given:** Seven pairs of equal 2×2 matrices with unknown variable entries.**Concept / Formula Used:** $A = [a_{ij}] \quad B = [b_{ij}] \quad a_{ij} = b_{ij} \quad i, j$ **Solution:****Part (i):**

Equating corresponding elements:

$$-y = 3 \implies y = -3$$

$$3x = 6 \implies x = 2$$

Part (ii):

Equating corresponding elements:

$$y = -1$$

$$7 - x = 0 \implies x = 7$$

Part (iii):

Equating corresponding elements:

$$x = 3$$

$$x - y = 2$$

Substitute $x = 3$:

$$3 - y = 2$$

$$y = 1$$

Part (iv):

Equating corresponding elements:

$$\begin{aligned} -y = 2 &\implies y = -2 \\ 3x + y = 1 \end{aligned}$$

Substitute $y = -2$:

$$\begin{aligned} 3x + (-2) &= 1 \\ 3x &= 3 \\ x &= 1 \end{aligned}$$

Part (v):

Equating corresponding elements:

$$\begin{aligned} 3y = -3 &\implies y = -1 \\ 4x = 8 &\implies x = 2 \end{aligned}$$

Now compute $x - y$:

$$\begin{aligned} x - y &= 2 - (-1) \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

Part (vi):

Equating corresponding elements:

$$\begin{aligned} x - y &= -1 & \text{--- (1)} \\ 2x - y &= 0 & \text{--- (2)} \end{aligned}$$

Subtracting equation (1) from equation (2):

$$\begin{aligned} (2x - y) - (x - y) &= 0 - (-1) \\ x &= 1 \end{aligned}$$

Substitute $x = 1$ into equation (1):

$$\begin{aligned} 1 - y &= -1 \\ y &= 2 \end{aligned}$$

Now compute $x + y$:

$$\begin{aligned} x + y &= 1 + 2 \\ &= 3 \end{aligned}$$

Part (vii):

Equating corresponding elements:

$$\begin{aligned} x + 2 = 5 &\implies x = 3 \\ 3y + 1 = 16 &\implies 3y = 15 \implies y = 5 \end{aligned}$$

Now compute $y - x$:

$$\begin{aligned}y - x &= 5 - 3 \\ &= 2\end{aligned}$$

Final Answer

- (i) $x = 2, y = -3$
- (ii) $x = 7, y = -1$
- (iii) $y = 1$
- (iv) $x = 1$
- (v) $x - y = 3$
- (vi) $x + y = 3$
- (vii) $y - x = 2$

Question 9

Write the value of $x - y + z$ from the following equation:

$$\begin{bmatrix} x + y + z \\ x + z \\ y + z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$$

TYPE 5 Equality of Matrices

Given: $\begin{bmatrix} x + y + z \\ x + z \\ y + z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$

Solution:

Equating corresponding elements:

$$x + y + z = 9 \quad \text{--- (1)}$$

$$x + z = 5 \quad \text{--- (2)}$$

$$y + z = 7 \quad \text{--- (3)}$$

Substitute equation (3) into equation (1):

$$x + (y + z) = 9$$

$$x + 7 = 9$$

$$x = 9 - 7$$

$$x = 2$$

Substitute $x = 2$ into equation (2):

$$2 + z = 5$$

$$z = 5 - 2$$

$$z = 3$$

Substitute $z = 3$ into equation (3):

$$y + 3 = 7$$

$$y = 7 - 3$$

$$y = 4$$

Now evaluate $x - y + z$:

$$\begin{aligned}x - y + z &= 2 - 4 + 3 \\&= 5 - 4 \\&= 1\end{aligned}$$

Final Answer

$$x - y + z = 1$$

Question 10

If $\begin{bmatrix} x & 3x - y \\ 2x + z & 3y - w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, find x, y, z and w .

TYPE 5 Equality of Matrices

Given: $\begin{bmatrix} x & 3x - y \\ 2x + z & 3y - w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$

Solution:

Equating corresponding elements:

$$\begin{aligned}x &= 3 & \text{--- (1)} \\3x - y &= 2 & \text{--- (2)} \\2x + z &= 4 & \text{--- (3)} \\3y - w &= 7 & \text{--- (4)}\end{aligned}$$

From equation (1):

$$x = 3$$

Substitute $x = 3$ into equation (2):

$$\begin{aligned}3(3) - y &= 2 \\9 - y &= 2 \\y &= 9 - 2 \\y &= 7\end{aligned}$$

Substitute $x = 3$ into equation (3):

$$\begin{aligned}2(3) + z &= 4 \\6 + z &= 4 \\z &= 4 - 6 \\z &= -2\end{aligned}$$

Substitute $y = 7$ into equation (4):

$$\begin{aligned}3(7) - w &= 7 \\21 - w &= 7 \\w &= 21 - 7 \\w &= 14\end{aligned}$$

Final Answer

$$x = 3, \quad y = 7, \quad z = -2, \quad w = 14$$

Question 11

Find the values of a, b, c and d from the following equations:

$$\begin{bmatrix} 2a + b & a - 2b \\ 5c - d & 4c + 3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$$

TYPE 5 Equality of Matrices

★ **Likely Exam Question**

Given: $\begin{bmatrix} 2a + b & a - 2b \\ 5c - d & 4c + 3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$

Solution:

Equating corresponding elements, we obtain two independent systems of equations:

System 1 (in a and b):

$$2a + b = 4 \quad \text{--- (1)}$$

$$a - 2b = -3 \quad \text{--- (2)}$$

Multiply equation (1) by 2:

$$4a + 2b = 8 \quad \text{--- (3)}$$

Add equation (2) and equation (3):

$$(a - 2b) + (4a + 2b) = -3 + 8$$

$$5a = 5$$

$$a = 1$$

Substitute $a = 1$ into equation (1):

$$2(1) + b = 4$$

$$2 + b = 4$$

$$b = 2$$

System 2 (in c and d):

$$5c - d = 11 \quad \text{--- (4)}$$

$$4c + 3d = 24 \quad \text{--- (5)}$$

From equation (4), express d in terms of c :

$$d = 5c - 11 \quad \text{--- (6)}$$

Substitute equation (6) into equation (5):

$$4c + 3(5c - 11) = 24$$

$$4c + 15c - 33 = 24$$

$$19c - 33 = 24$$

$$19c = 24 + 33$$

$$19c = 57$$

$$c = \frac{57}{19}$$

$$c = 3$$

Substitute $c = 3$ into equation (6):

$$d = 5(3) - 11$$

$$d = 15 - 11$$

$$d = 4$$

Final Answer

$$a = 1, \quad b = 2, \quad c = 3, \quad d = 4$$

Question 12

Find x, y, a and b if $\begin{bmatrix} 2x - 3y & a - b & 3 \\ 1 & x + 4y & 3a + 4b \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 6 & 29 \end{bmatrix}$

TYPE 5 Equality of Matrices

★ Likely Exam Question

Given: $\begin{bmatrix} 2x - 3y & a - b & 3 \\ 1 & x + 4y & 3a + 4b \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 6 & 29 \end{bmatrix}$

Concept / Formula Used: 2×3

Solution:

By equality of matrices, equate the corresponding entries:

System 1 (in x and y):

$$2x - 3y = 1 \quad \text{--- (1)}$$

$$x + 4y = 6 \quad \text{--- (2)}$$

From equation (2), express x in terms of y :

$$x = 6 - 4y \quad \text{--- (3)}$$

Substitute equation (3) into equation (1):

$$2(6 - 4y) - 3y = 1$$

$$12 - 8y - 3y = 1$$

$$12 - 11y = 1$$

$$11y = 12 - 1$$

$$11y = 11$$

$$y = 1$$

Substitute $y = 1$ into equation (3):

$$x = 6 - 4(1)$$

$$x = 6 - 4$$

$$x = 2$$

System 2 (in a and b):

$$a - b = -2 \quad \text{--- (4)}$$

$$3a + 4b = 29 \quad \text{--- (5)}$$

From equation (4), express a in terms of b :

$$a = b - 2 \quad \text{--- (6)}$$

Substitute equation (6) into equation (5):

$$3(b - 2) + 4b = 29$$

$$3b - 6 + 4b = 29$$

$$7b - 6 = 29$$

$$7b = 29 + 6$$

$$7b = 35$$

$$b = \frac{35}{7}$$

$$b = 5$$

Substitute $b = 5$ into equation (6):

$$a = 5 - 2$$

$$a = 3$$

Final Answer

$$x = 2, \quad y = 1, \quad a = 3, \quad b = 5$$

Question 13

- (i) Write the number of all possible matrices of order 2×3 with each entry 1 or 2.
- (ii) Write the number of all possible matrices of order 2×2 with each entry 1, 2 or 3.

TYPE 6 Number of Possible Matrices

To Find: The total number of possible matrices under the given conditions for parts (i) and (ii).

Concept / Formula Used: Total number of matrices = $k^{m \times n}$ $m \times n$ $m \times n$ k

Solution:

(i) For a matrix of order 2×3 with entries 1 or 2:

$$\text{Order of the matrix} = 2 \times 3$$

$$\text{Total number of elements in the matrix} = 2 \times 3 = 6$$

Each of the 6 positions can be independently filled by either 1 or 2.

$$\text{Number of choices for each entry} = 2$$

$$\text{Total number of possible matrices} = 2^6$$

$$= 64$$

(ii) For a matrix of order 2×2 with entries 1, 2 or 3:

$$\text{Order of the matrix} = 2 \times 2$$

$$\text{Total number of elements in the matrix} = 2 \times 2 = 4$$

Each of the 4 positions can be independently filled by 1, 2, or 3.

$$\text{Number of choices for each entry} = 3$$

$$\begin{aligned}\text{Total number of possible matrices} &= 3^4 \\ &= 81\end{aligned}$$

Final Answer

- (i) Number of possible matrices = 64
- (ii) Number of possible matrices = 81

Common Mistake: Confusing the base and the exponent by computing $(\text{number of elements})^{\text{number of choices}}$ instead of $(\text{number of choices})^{\text{number of elements}}$.

3.2 – Addition, Subtraction and Scalar Multiplication of Matrices

Question 1

If A and B are matrices of order $3 \times n$ and $m \times 5$ respectively, then find the order of the matrix $5A - 3B$, given that it is defined.

TYPE 1 Matrix Order Analysis

Given: Order of matrix $A = 3 \times n$, Order of matrix $B = m \times 5$, Matrix $5A - 3B$ is defined.

To Find: Order of matrix $5A - 3B$.

Concept / Formula Used: kA

Solution:

The order of scalar multiple matrix $5A$ is the same as the order of matrix A , which is $3 \times n$.
The order of scalar multiple matrix $3B$ is the same as the order of matrix B , which is $m \times 5$.
Since the matrix expression $5A - 3B$ is defined, matrices $5A$ and $3B$ must have identical orders.
Equating corresponding dimensions:

$$\text{Number of rows of } 5A = \text{Number of rows of } 3B \implies 3 = m$$

$$\text{Number of columns of } 5A = \text{Number of columns of } 3B \implies n = 5$$

Substituting $m = 3$ and $n = 5$, both matrices $5A$ and $3B$ have order 3×5 .

The resulting matrix $5A - 3B$ therefore retains the order 3×5 .

Final Answer

Order of matrix $5A - 3B = 3 \times 5$

Question 2

If $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix}$, find the following:

- (i) $A + B$
- (ii) $A - B$
- (iii) $3A - C$

TYPE 7 Matrix Addition Subtraction and Scalar Multiplication

★ **Likely Exam Question**

Given: $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$, $C = \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix}$.

To Find: Matrices $A + B$, $A - B$, and $3A - C$.

Concept / Formula Used: $A = [a_{ij}]$ $B = [b_{ij}]$ $A \pm B = [a_{ij} \pm b_{ij}]$ $kA = [ka_{ij}]$

Solution:

(i) **Evaluation of $A + B$:**

$$\begin{aligned} A + B &= \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 2+1 & 4+3 \\ 3+(-2) & 2+5 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 7 \\ 1 & 7 \end{bmatrix} \end{aligned}$$

(ii) **Evaluation of $A - B$:**

$$\begin{aligned} A - B &= \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 2-1 & 4-3 \\ 3-(-2) & 2-5 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 5 & -3 \end{bmatrix} \end{aligned}$$

(iii) **Evaluation of $3A - C$:** First, perform the scalar multiplication $3A$:

$$\begin{aligned} 3A &= 3 \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 \times 2 & 3 \times 4 \\ 3 \times 3 & 3 \times 2 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 12 \\ 9 & 6 \end{bmatrix} \end{aligned}$$

Now subtract C from $3A$:

$$\begin{aligned} 3A - C &= \begin{bmatrix} 6 & 12 \\ 9 & 6 \end{bmatrix} - \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 6 - (-2) & 12 - 5 \\ 9 - 3 & 6 - 4 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix} \end{aligned}$$

Final Answer

$$(i) A + B = \begin{bmatrix} 3 & 7 \\ 1 & 7 \end{bmatrix}$$

$$(ii) A - B = \begin{bmatrix} 1 & 1 \\ 5 & -3 \end{bmatrix}$$

$$(iii) 3A - C = \begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix}$$

Common Mistake: Watch out for double negatives during subtraction, such as $3 - (-2) = 5$ and $6 - (-2) = 8$.

Question 3

Compute the following:

$$(i) \begin{bmatrix} -1 & 4 & -6 \\ 8 & 5 & 16 \\ 2 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 12 & 7 & 6 \\ 8 & 0 & 5 \\ 3 & 2 & 4 \end{bmatrix}$$

$$(ii) \begin{bmatrix} \cos^2 x & \sin^2 x \\ \sin^2 x & \cos^2 x \end{bmatrix} + \begin{bmatrix} \sin^2 x & \cos^2 x \\ \cos^2 x & \sin^2 x \end{bmatrix}$$

TYPE 7 Matrix Addition Subtraction and Scalar Multiplication

★ Likely Exam Question

Concept / Formula Used: $\sin^2 x + \cos^2 x = 1$

Solution:

(i) Adding corresponding elements of the 3×3 matrices:

$$\begin{aligned} &\begin{bmatrix} -1 & 4 & -6 \\ 8 & 5 & 16 \\ 2 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 12 & 7 & 6 \\ 8 & 0 & 5 \\ 3 & 2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -1 + 12 & 4 + 7 & -6 + 6 \\ 8 + 8 & 5 + 0 & 16 + 5 \\ 2 + 3 & 8 + 2 & 5 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 11 & 11 & 0 \\ 16 & 5 & 21 \\ 5 & 10 & 9 \end{bmatrix} \end{aligned}$$

(ii) Adding corresponding elements of the 2×2 trigonometric matrices:

$$\begin{bmatrix} \cos^2 x & \sin^2 x \\ \sin^2 x & \cos^2 x \end{bmatrix} + \begin{bmatrix} \sin^2 x & \cos^2 x \\ \cos^2 x & \sin^2 x \end{bmatrix} \\ = \begin{bmatrix} \cos^2 x + \sin^2 x & \sin^2 x + \cos^2 x \\ \sin^2 x + \cos^2 x & \cos^2 x + \sin^2 x \end{bmatrix}$$

Applying the standard identity $\cos^2 x + \sin^2 x = 1$ to each entry:

$$= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Final Answer

(i) $\begin{bmatrix} 11 & 11 & 0 \\ 16 & 5 & 21 \\ 5 & 10 & 9 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Question 4

If $A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 4 & -7 \end{bmatrix}$, find $2A - 3B$.

TYPE 7 Matrix Addition Subtraction and Scalar Multiplication

Given: $A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 4 & -7 \end{bmatrix}$.

To Find: The matrix $2A - 3B$.

Concept / Formula Used: A, B $kA + mB = [ka_{ij} + mb_{ij}]$

Solution:

First, calculate $2A$ by multiplying each entry of A by 2:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 \times 1 & 2 \times 5 \\ 2 \times (-2) & 2 \times 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 10 \\ -4 & 6 \end{bmatrix} \end{aligned}$$

Next, calculate $3B$ by multiplying each entry of B by 3:

$$\begin{aligned} 3B &= 3 \begin{bmatrix} 0 & -1 \\ 4 & -7 \end{bmatrix} \\ &= \begin{bmatrix} 3 \times 0 & 3 \times (-1) \\ 3 \times 4 & 3 \times (-7) \end{bmatrix} \\ &= \begin{bmatrix} 0 & -3 \\ 12 & -21 \end{bmatrix} \end{aligned}$$

Subtract $3B$ from $2A$:

$$\begin{aligned} 2A - 3B &= \begin{bmatrix} 2 & 10 \\ -4 & 6 \end{bmatrix} - \begin{bmatrix} 0 & -3 \\ 12 & -21 \end{bmatrix} \\ &= \begin{bmatrix} 2-0 & 10-(-3) \\ -4-12 & 6-(-21) \end{bmatrix} \\ &= \begin{bmatrix} 2 & 13 \\ -16 & 27 \end{bmatrix} \end{aligned}$$

Final Answer

$$2A - 3B = \begin{bmatrix} 2 & 13 \\ -16 & 27 \end{bmatrix}$$

Question 5

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$, then find $2A - B$.

TYPE 7 Matrix Addition Subtraction and Scalar Multiplication

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$.

To Find: The matrix $2A - B$.

Concept / Formula Used: $(2A - B)_{ij} = 2a_{ij} - b_{ij}$

Solution:

Compute $2A$ by multiplying each entry of matrix A by 2:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} \end{aligned}$$

Subtract matrix B from $2A$:

$$\begin{aligned} 2A - B &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} - \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2-3 & 4-(-1) & 6-3 \\ 4-(-1) & 6-0 & 2-2 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 4+1 & 3 \\ 4+1 & 6 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix} \end{aligned}$$

Final Answer

$$2A - B = \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix}$$

Question 6

If $\begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} = A + \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$, then find the matrix A .

TYPE 8 Solving Linear Matrix Equations

Given: $\begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} = A + \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$.

To Find: The matrix A .

Concept / Formula Used: $P = A + Q$ $A = P - Q$

Solution:

Isolate A by subtracting $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$ from both sides of the matrix equation:

$$A = \begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$$

Subtract corresponding elements:

$$\begin{aligned} A &= \begin{bmatrix} 9-1 & -1-2 & 4-(-1) \\ -2-0 & 1-4 & 3-9 \end{bmatrix} \\ &= \begin{bmatrix} 8 & -3 & 4+1 \\ -2 & -3 & -6 \end{bmatrix} \\ &= \begin{bmatrix} 8 & -3 & 5 \\ -2 & -3 & -6 \end{bmatrix} \end{aligned}$$

Final Answer

$$A = \begin{bmatrix} 8 & -3 & 5 \\ -2 & -3 & -6 \end{bmatrix}$$

Question 7

If $A = \text{diagonal } [1, -2, 5]$, $B = \text{diagonal } [3, 0, -4]$ and $C = \text{diagonal } [-2, 7, 0]$, then find $A + 2B - 3C$.

TYPE 7 Matrix Addition Subtraction and Scalar Multiplication

Given: $A = \text{diag}[1, -2, 5]$, $B = \text{diag}[3, 0, -4]$, $C = \text{diag}[-2, 7, 0]$.

To Find: The diagonal matrix $A + 2B - 3C$.

Concept / Formula Used: $k_1 \text{diag}[d_1] + k_2 \text{diag}[d_2] = \text{diag}[k_1 d_1 + k_2 d_2]$

Solution:

Express scalar multiples of diagonal matrices component-wise:

$$2B = 2 \cdot \text{diag}[3, 0, -4] = \text{diag}[2(3), 2(0), 2(-4)] = \text{diag}[6, 0, -8]$$

$$3C = 3 \cdot \text{diag}[-2, 7, 0] = \text{diag}[3(-2), 3(7), 3(0)] = \text{diag}[-6, 21, 0]$$

Now substitute into the expression $A + 2B - 3C$:

$$\begin{aligned} A + 2B - 3C &= \text{diag}[1, -2, 5] + \text{diag}[6, 0, -8] - \text{diag}[-6, 21, 0] \\ &= \text{diag}[1 + 6 - (-6), -2 + 0 - 21, 5 + (-8) - 0] \\ &= \text{diag}[1 + 6 + 6, -2 - 21, 5 - 8] \\ &= \text{diag}[13, -23, -3] \end{aligned}$$

Final Answer

$$A + 2B - 3C = \text{diagonal } [13, -23, -3]$$

Question 8

If $A = \begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix}$ and $kA = \begin{bmatrix} 8 & 3a \\ -2b & c \end{bmatrix}$, find the values of k, a, b and c .

TYPE 5 Equality of Matrices

Given: $A = \begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix}$, $kA = \begin{bmatrix} 8 & 3a \\ -2b & c \end{bmatrix}$.

To Find: The scalar values k, a, b, c .

Solution:

Compute kA by multiplying matrix A by scalar k :

$$kA = k \begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 2k & -3k \\ 5k & 0 \end{bmatrix}$$

Equate kA to the given matrix:

$$\begin{bmatrix} 2k & -3k \\ 5k & 0 \end{bmatrix} = \begin{bmatrix} 8 & 3a \\ -2b & c \end{bmatrix}$$

Comparing corresponding entries step-by-step:

- 1) Entry (1, 1) : $2k = 8 \implies k = \frac{8}{2} \implies k = 4$
- 2) Entry (1, 2) : $-3k = 3a \implies -3(4) = 3a \implies -12 = 3a \implies a = -4$
- 3) Entry (2, 1) : $5k = -2b \implies 5(4) = -2b \implies 20 = -2b \implies b = -10$
- 4) Entry (2, 2) : $0 = c \implies c = 0$

Final Answer

$$k = 4$$

$$a = -4$$

$$b = -10$$

$$c = 0$$

Question 9

If $x \begin{bmatrix} 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$, find the values of x and y .

TYPE 5 Equality of Matrices

Given: $x \begin{bmatrix} 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$.

To Find: The values of x and y .

Concept / Formula Used: $xC_1 + yC_2 = B$ x y

Solution:

Perform scalar multiplication on the column vectors:

$$\begin{bmatrix} 2x \\ 3x \end{bmatrix} + \begin{bmatrix} -y \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

Combine LHS into a single matrix:

$$\begin{bmatrix} 2x - y \\ 3x + y \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

Equating corresponding entries yields two simultaneous linear equations:

$$2x - y = 10 \quad (1)$$

$$3x + y = 5 \quad (2)$$

Add equation (1) and equation (2) to eliminate y :

$$\begin{aligned} (2x - y) + (3x + y) &= 10 + 5 \\ 5x &= 15 \\ x &= \frac{15}{5} \\ x &= 3 \end{aligned}$$

Substitute $x = 3$ into equation (2):

$$\begin{aligned} 3(3) + y &= 5 \\ 9 + y &= 5 \\ y &= 5 - 9 \\ y &= -4 \end{aligned}$$

Final Answer

$$x = 3, \quad y = -4$$

Question 10

If $2 \begin{bmatrix} x & 5 \\ 7 & y - 3 \end{bmatrix} + \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$, find the value of $(x - y)$.

TYPE 5 Equality of Matrices

Given: $2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}.$

To Find: The value of $(x - y).$

Solution:

Multiply scalar 2 inside the matrix:

$$\begin{bmatrix} 2x & 10 \\ 14 & 2(y-3) \end{bmatrix} + \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

Expand $2(y - 3) = 2y - 6$ and perform matrix addition on LHS:

$$\begin{bmatrix} 2x-3 & 10-4 \\ 14+1 & 2y-6+2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 2x-3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

Equate corresponding elements:

$$2x - 3 = 7 \implies 2x = 10 \implies x = 5$$

$$2y - 4 = 14 \implies 2y = 18 \implies y = 9$$

Now compute $x - y$:

$$\begin{aligned} x - y &= 5 - 9 \\ &= -4 \end{aligned}$$

Final Answer

$$x - y = -4$$

Question 11

Given $2 \begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3 \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$, find x, y, z, t .

TYPE 5 Equality of Matrices

★ Likely Exam Question

Given: $2 \begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3 \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}.$

To Find: The values of x, y, z, t .

Solution:

Multiply scalars into each matrix:

$$\begin{bmatrix} 2x & 2z \\ 2y & 2t \end{bmatrix} + \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$$

Add the two matrices on the left-hand side:

$$\begin{bmatrix} 2x+3 & 2z-3 \\ 2y+0 & 2t+6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$$

Equate corresponding entries individually:

$$1) \text{ Entry } (1, 1) : 2x + 3 = 9 \implies 2x = 6 \implies x = 3$$

$$2) \text{ Entry } (1, 2) : 2z - 3 = 15 \implies 2z = 18 \implies z = 9$$

$$3) \text{ Entry } (2, 1) : 2y = 12 \implies y = 6$$

$$4) \text{ Entry } (2, 2) : 2t + 6 = 18 \implies 2t = 12 \implies t = 6$$

Final Answer

$$x = 3, \quad y = 6, \quad z = 9, \quad t = 6$$

Question 12

If $3 \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 6 \\ -1 & 2d \end{bmatrix} + \begin{bmatrix} 4 & a+b \\ c+d & 3 \end{bmatrix}$, find a, b, c and d .

TYPE 5 Equality of Matrices

★ Likely Exam Question

Given: $3 \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 6 \\ -1 & 2d \end{bmatrix} + \begin{bmatrix} 4 & a+b \\ c+d & 3 \end{bmatrix}$.

To Find: The values of a, b, c, d .

Solution:

Simplify LHS and RHS matrices:

$$\begin{bmatrix} 3a & 3b \\ 3c & 3d \end{bmatrix} = \begin{bmatrix} a+4 & 6+a+b \\ -1+c+d & 2d+3 \end{bmatrix}$$

Equating corresponding entries step-by-step:

1) From entry (1, 1):

$$3a = a + 4$$

$$3a - a = 4$$

$$2a = 4$$

$$a = 2$$

2) From entry (1, 2):

$$3b = 6 + a + b$$

$$3b - b = 6 + a$$

$$2b = 6 + 2 \quad (\text{substituting } a = 2)$$

$$2b = 8$$

$$b = 4$$

3) From entry (2, 2):

$$\begin{aligned}3d &= 2d + 3 \\3d - 2d &= 3 \\d &= 3\end{aligned}$$

4) From entry (2, 1):

$$\begin{aligned}3c &= -1 + c + d \\3c - c &= -1 + d \\2c &= -1 + 3 \quad (\text{substituting } d = 3) \\2c &= 2 \\c &= 1\end{aligned}$$

Final Answer

$$a = 2, \quad b = 4, \quad c = 1, \quad d = 3$$

Question 13

Find X if $Y = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$.

TYPE 8 Solving Linear Matrix Equations

★ Likely Exam Question

Given: $Y = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, $2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$.

To Find: The matrix X .

Concept / Formula Used: $2X + Y = B \quad 2X = B - Y \implies X = \frac{1}{2}(B - Y)$

Solution:

Substitute matrix Y into the equation $2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$:

$$2X + \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$$

Isolate $2X$ by subtracting $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ from both sides:

$$\begin{aligned}2X &= \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \\&= \begin{bmatrix} 1-3 & 0-2 \\ -3-1 & 2-4 \end{bmatrix} \\&= \begin{bmatrix} -2 & -2 \\ -4 & -2 \end{bmatrix}\end{aligned}$$

Multiply by $\frac{1}{2}$ to solve for X :

$$\begin{aligned} X &= \frac{1}{2} \begin{bmatrix} -2 & -2 \\ -4 & -2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{-2}{2} & \frac{-2}{2} \\ \frac{-4}{2} & \frac{-2}{2} \end{bmatrix} \\ &= \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix} \end{aligned}$$

Final Answer

$$X = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix}$$

Question 14

Find matrices X and Y , if $X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

TYPE 8 Solving Linear Matrix Equations

Given: $X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$, $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

To Find: The matrices X and Y .

Concept / Formula Used: $X + Y = A$ $X - Y = B$ $2X = A + B$ $2Y = A - B$

Solution:

Given simultaneous matrix equations:

$$X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \quad (1)$$

$$X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \quad (2)$$

Add equation (1) and equation (2) to eliminate Y :

$$(X + Y) + (X - Y) = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$2X = \begin{bmatrix} 7+3 & 0+0 \\ 2+0 & 5+3 \end{bmatrix}$$

$$2X = \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix}$$

$$X = \frac{1}{2} \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix}$$

$$X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$$

Subtract equation (2) from equation (1) to eliminate X :

$$\begin{aligned}(X + Y) - (X - Y) &= \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ 2Y &= \begin{bmatrix} 7-3 & 0-0 \\ 2-0 & 5-3 \end{bmatrix} \\ 2Y &= \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} \\ Y &= \frac{1}{2} \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} \\ Y &= \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}\end{aligned}$$

Final Answer

$$X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$$

$$Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

Question 15

Find matrices X and Y if $2X - Y = \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix}$ and $X + 2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix}$.

TYPE 8 Solving Linear Matrix Equations

Given: $2X - Y = \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix}$, $X + 2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix}$.

To Find: The matrices X and Y .

Solution:

Label the given matrix equations:

$$2X - Y = \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix} \quad (1)$$

$$X + 2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} \quad (2)$$

Multiply equation (1) by 2:

$$4X - 2Y = 2 \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 12 & -12 & 0 \\ -8 & 4 & 2 \end{bmatrix} \quad (3)$$

Add equation (2) and equation (3) to eliminate Y :

$$\begin{aligned}
 (X + 2Y) + (4X - 2Y) &= \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} + \begin{bmatrix} 12 & -12 & 0 \\ -8 & 4 & 2 \end{bmatrix} \\
 5X &= \begin{bmatrix} 3+12 & 2+(-12) & 5+0 \\ -2+(-8) & 1+4 & -7+2 \end{bmatrix} \\
 5X &= \begin{bmatrix} 15 & -10 & 5 \\ -10 & 5 & -5 \end{bmatrix} \\
 X &= \frac{1}{5} \begin{bmatrix} 15 & -10 & 5 \\ -10 & 5 & -5 \end{bmatrix} \\
 X &= \begin{bmatrix} 3 & -2 & 1 \\ -2 & 1 & -1 \end{bmatrix}
 \end{aligned}$$

Multiply equation (2) by 2:

$$2X + 4Y = 2 \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} = \begin{bmatrix} 6 & 4 & 10 \\ -4 & 2 & -14 \end{bmatrix} \quad (4)$$

Subtract equation (1) from equation (4) to eliminate X :

$$\begin{aligned}
 (2X + 4Y) - (2X - Y) &= \begin{bmatrix} 6 & 4 & 10 \\ -4 & 2 & -14 \end{bmatrix} - \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix} \\
 5Y &= \begin{bmatrix} 6-6 & 4-(-6) & 10-0 \\ -4-(-4) & 2-2 & -14-1 \end{bmatrix} \\
 5Y &= \begin{bmatrix} 0 & 10 & 10 \\ 0 & 0 & -15 \end{bmatrix} \\
 Y &= \frac{1}{5} \begin{bmatrix} 0 & 10 & 10 \\ 0 & 0 & -15 \end{bmatrix} \\
 Y &= \begin{bmatrix} 0 & 2 & 2 \\ 0 & 0 & -3 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$X = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 1 & -1 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & 2 & 2 \\ 0 & 0 & -3 \end{bmatrix}$$

Question 16

Find a matrix X such that $3A - 2B + X = O$, where $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$.

TYPE 8 Solving Linear Matrix Equations

Given: $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$, $3A - 2B + X = O$.

To Find: The matrix X .

Concept / Formula Used: $3A - 2B + X = O \quad X = 2B - 3A$

Solution:

Rearrange the matrix equation to isolate X :

$$\begin{aligned} 3A - 2B + X &= O \\ X &= 2B - 3A \end{aligned}$$

Calculate $2B$ by scalar multiplication:

$$\begin{aligned} 2B &= 2 \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 2 \\ 6 & 4 \end{bmatrix} \end{aligned}$$

Calculate $3A$ by scalar multiplication:

$$\begin{aligned} 3A &= 3 \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 12 & 6 \\ 3 & 9 \end{bmatrix} \end{aligned}$$

Subtract $3A$ from $2B$:

$$\begin{aligned} X &= \begin{bmatrix} -4 & 2 \\ 6 & 4 \end{bmatrix} - \begin{bmatrix} 12 & 6 \\ 3 & 9 \end{bmatrix} \\ &= \begin{bmatrix} -4 - 12 & 2 - 6 \\ 6 - 3 & 4 - 9 \end{bmatrix} \\ &= \begin{bmatrix} -16 & -4 \\ 3 & -5 \end{bmatrix} \end{aligned}$$

Final Answer

$$X = \begin{bmatrix} -16 & -4 \\ 3 & -5 \end{bmatrix}$$

Question 17

If $A = \begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix}$, find matrix C such that $5A + 3B + 2C$ is a null matrix.

TYPE 8 Solving Linear Matrix Equations

★ Likely Exam Question

Given: $A = \begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix}$, $5A + 3B + 2C = O$.

To Find: The matrix C .

Concept / Formula Used: $O \quad 5A + 3B + 2C = O \quad C = -\frac{1}{2}(5A + 3B)$

Solution:

Rearrange the equation to express $2C$:

$$\begin{aligned}5A + 3B + 2C &= O \\2C &= -(5A + 3B) \\C &= -\frac{1}{2}(5A + 3B)\end{aligned}$$

Compute $5A$:

$$\begin{aligned}5A &= 5 \begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix} \\&= \begin{bmatrix} 45 & 5 \\ 35 & 40 \end{bmatrix}\end{aligned}$$

Compute $3B$:

$$\begin{aligned}3B &= 3 \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix} \\&= \begin{bmatrix} 3 & 15 \\ 21 & 36 \end{bmatrix}\end{aligned}$$

Add $5A$ and $3B$:

$$\begin{aligned}5A + 3B &= \begin{bmatrix} 45 & 5 \\ 35 & 40 \end{bmatrix} + \begin{bmatrix} 3 & 15 \\ 21 & 36 \end{bmatrix} \\&= \begin{bmatrix} 45 + 3 & 5 + 15 \\ 35 + 21 & 40 + 36 \end{bmatrix} \\&= \begin{bmatrix} 48 & 20 \\ 56 & 76 \end{bmatrix}\end{aligned}$$

Multiply by $-\frac{1}{2}$ to get matrix C :

$$\begin{aligned}C &= -\frac{1}{2} \begin{bmatrix} 48 & 20 \\ 56 & 76 \end{bmatrix} \\&= \begin{bmatrix} -\frac{48}{2} & -\frac{20}{2} \\ -\frac{56}{2} & -\frac{76}{2} \end{bmatrix} \\&= \begin{bmatrix} -24 & -10 \\ -28 & -38 \end{bmatrix}\end{aligned}$$

Final Answer

$$C = \begin{bmatrix} -24 & -10 \\ -28 & -38 \end{bmatrix}$$

3.3 – Multiplication of Matrices

Question 1

If $A = \begin{bmatrix} -1 & 2 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ -1 \\ 7 \end{bmatrix}$, write the orders of AB and BA .

TYPE 9 Matrix Multiplication Compatibility and Order**Given:** Matrix A of order 1×3 and matrix B of order 3×1 **To Find:** The orders of the matrix products AB and BA **Concept / Formula Used:** $P \quad m \times n \quad Q \quad n \times p \quad PQ \quad m \times p$ **Solution:**The order of matrix A is 1×3 .The order of matrix B is 3×1 .For the product AB :

$$\text{Order of } A = 1 \times 3$$

$$\text{Order of } B = 3 \times 1$$

Using the matrix multiplication compatibility rule, since the number of columns in A (3) equals the number of rows in B (3), AB is defined and its order is:

$$\text{Order of } AB = 1 \times 1$$

For the product BA :

$$\text{Order of } B = 3 \times 1$$

$$\text{Order of } A = 1 \times 3$$

Since the number of columns in B (1) equals the number of rows in A (1), BA is defined and its order is:

$$\text{Order of } BA = 3 \times 3$$

Final AnswerThe order of AB is 1×1 and the order of BA is 3×3 .**Question 2**

If $A = \begin{bmatrix} 2 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$, then write the orders of AB and BA .

TYPE 9 Matrix Multiplication Compatibility and Order**Given:** Matrix $A = \begin{bmatrix} 2 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$ **To Find:** The orders of the products AB and BA

Concept / Formula Used: $m \times n$ $n \times p$ $m \times p$

Solution:

By counting rows and columns:

$$\text{Order of } A = 2 \times 3$$

$$\text{Order of } B = 3 \times 2$$

For AB :

$$\text{Number of columns of } A = 3$$

$$\text{Number of rows of } B = 3$$

Since these are equal, AB exists and its order is given by:

$$\text{Order of } AB = 2 \times 2$$

For BA :

$$\text{Number of columns of } B = 2$$

$$\text{Number of rows of } A = 2$$

Since these are equal, BA exists and its order is given by:

$$\text{Order of } BA = 3 \times 3$$

Final Answer

The order of AB is 2×2 and the order of BA is 3×3 .

Question 3

If A is a matrix of order 3×3 and B is a matrix of order 2×3 , can you find AB ?

TYPE 9 Matrix Multiplication Compatibility and Order

Given: Matrix A of order 3×3 and matrix B of order 2×3

To Find: Whether the matrix product AB can be found

Concept / Formula Used: PQ P Q

Solution:

Let us inspect the dimensions of A and B :

$$\text{Number of columns of } A = 3$$

$$\text{Number of rows of } B = 2$$

Comparing these two values:

$$3 \neq 2$$

Since the number of columns of A is not equal to the number of rows of B , the condition for matrix multiplication is not satisfied.

Final Answer

No, AB cannot be found because the number of columns in A (3) is not equal to the number of rows in B (2).

Question 4

If $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 4 & 2 \end{bmatrix}$ and $AB = [c_{ij}]$, find $c_{23} + c_{31}$.

TYPE 10 Evaluating Specific Product Matrix Elements

Given: $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 4 & 2 \end{bmatrix}$, $AB = [c_{ij}]$

To Find: The value of $c_{23} + c_{31}$

Concept / Formula Used: c_{ij} $C = AB$ i A j B

Solution:

The element c_{23} is the product of row 2 of A and column 3 of B :

$$\begin{aligned} \text{Row 2 of } A &= [-1 \quad 1] \\ \text{Column 3 of } B &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ c_{23} &= (-1)(1) + (1)(2) \\ c_{23} &= -1 + 2 \\ c_{23} &= 1 \end{aligned}$$

The element c_{31} is the product of row 3 of A and column 1 of B :

$$\begin{aligned} \text{Row 3 of } A &= [2 \quad 3] \\ \text{Column 1 of } B &= \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ c_{31} &= (2)(3) + (3)(1) \\ c_{31} &= 6 + 3 \\ c_{31} &= 9 \end{aligned}$$

Adding the two calculated entries:

$$\begin{aligned} c_{23} + c_{31} &= 1 + 9 \\ c_{23} + c_{31} &= 10 \end{aligned}$$

Final Answer

$$c_{23} + c_{31} = 10$$

Question 5

If $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \\ -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 4 \end{bmatrix}$ and $BA = [c_{ij}]$, find $c_{21} + c_{22}$.

TYPE 10 Evaluating Specific Product Matrix Elements

Given: $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \\ -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 4 \end{bmatrix}$, $BA = [c_{ij}]$

To Find: The value of $c_{21} + c_{22}$

Concept / Formula Used: c_{ij} BA i B j A

Solution:

To find c_{21} , multiply Row 2 of B with Column 1 of A :

$$\begin{aligned} \text{Row 2 of } B &= [2 \quad 1 \quad 4] \\ \text{Column 1 of } A &= \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \\ c_{21} &= (2)(2) + (1)(3) + (4)(-1) \\ c_{21} &= 4 + 3 - 4 \\ c_{21} &= 3 \end{aligned}$$

To find c_{22} , multiply Row 2 of B with Column 2 of A :

$$\begin{aligned} \text{Column 2 of } A &= \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} \\ c_{22} &= (2)(4) + (1)(1) + (4)(2) \\ c_{22} &= 8 + 1 + 8 \\ c_{22} &= 17 \end{aligned}$$

Summing the two components:

$$\begin{aligned} c_{21} + c_{22} &= 3 + 17 \\ c_{21} + c_{22} &= 20 \end{aligned}$$

Final Answer

$$c_{21} + c_{22} = 20$$

Question 6

If $\begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} = O$, find the value of x .

TYPE 11 Solving Matrix Equations using Multiplication

Given: $\begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$

To Find: The value of x

Solution:

Perform the matrix multiplication on the left hand side:

$$\begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} x(1) + 1(-2) & x(0) + 1(0) \end{bmatrix} \\ = \begin{bmatrix} x - 2 & 0 \end{bmatrix}$$

Equating the product to the zero matrix on the right hand side:

$$\begin{bmatrix} x - 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Equating corresponding entries:

$$x - 2 = 0 \\ x = 2$$

Final Answer

$$x = 2$$

Question 7

If $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$, write the value of x .

★ Likely Exam Question

TYPE 11 Solving Matrix Equations using Multiplication

Given: $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$

To Find: The value of x

Concept / Formula Used: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$

Solution:

Compute the left hand side matrix product:

$$\begin{aligned} \text{LHS} &= \begin{bmatrix} 2(1) + 3(-2) & 2(-3) + 3(4) \\ 5(1) + 7(-2) & 5(-3) + 7(4) \end{bmatrix} \\ &= \begin{bmatrix} 2 - 6 & -6 + 12 \\ 5 - 14 & -15 + 28 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 6 \\ -9 & 13 \end{bmatrix} \end{aligned}$$

Given that LHS = RHS:

$$\begin{bmatrix} -4 & 6 \\ -9 & 13 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$$

Comparing the entry at position (2, 2) on both sides:

$$x = 13$$

Final Answer

$$x = 13$$

Question 8

Write the value of $x + y + z$ if $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$.

TYPE 11 Solving Matrix Equations using Multiplication

Given: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

To Find: The value of $x + y + z$

Concept / Formula Used: $V \quad I \cdot V = V \quad I$

Solution:

Using the property $IX = X$ where $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 1(x) + 0(y) + 0(z) \\ 0(x) + 1(y) + 0(z) \\ 0(x) + 0(y) + 1(z) \end{bmatrix} \\ &= \begin{bmatrix} x \\ y \\ z \end{bmatrix} \end{aligned}$$

Equating to the given column vector:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Comparing elements:

$$\begin{aligned} x &= 1 \\ y &= -1 \\ z &= 0 \end{aligned}$$

Now compute $x + y + z$:

$$\begin{aligned} x + y + z &= 1 + (-1) + 0 \\ &= 0 \end{aligned}$$

Final Answer

$$x + y + z = 0$$

Question 9

Compute the indicated products:

$$\begin{aligned} \text{(i)} & \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \\ \text{(ii)} & \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix} \end{aligned}$$

TYPE 12 Matrix Multiplication

Given: Matrix expressions for parts (i) and (ii)

To Find: The product matrix for each part

Concept / Formula Used: $(i, j) = \sum_k a_{ik}b_{kj}$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$ of order 2×2 and $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ of order 2×3 .

The product AB will have order 2×3 :

$$\begin{aligned} AB &= \begin{bmatrix} 1(1) + (-2)(2) & 1(2) + (-2)(3) & 1(3) + (-2)(1) \\ 2(1) + 3(2) & 2(2) + 3(3) & 2(3) + 3(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 - 4 & 2 - 6 & 3 - 2 \\ 2 + 6 & 4 + 9 & 6 + 3 \end{bmatrix} \\ &= \begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix} \end{aligned}$$

Part (ii): Let $C = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$ of order 2×3 and $D = \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix}$ of order 3×2 .

The product CD will have order 2×2 :

$$\begin{aligned} CD &= \begin{bmatrix} 3(2) + (-1)(1) + 3(3) & 3(-3) + (-1)(0) + 3(1) \\ (-1)(2) + 0(1) + 2(3) & (-1)(-3) + 0(0) + 2(1) \end{bmatrix} \\ &= \begin{bmatrix} 6 - 1 + 9 & -9 + 0 + 3 \\ -2 + 0 + 6 & 3 + 0 + 2 \end{bmatrix} \\ &= \begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix} \end{aligned}$$

Final Answer

$$\begin{aligned} \text{(i)} & \begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix} \\ \text{(ii)} & \begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix} \end{aligned}$$

Question 10

If $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ and $A^2 = I$, find the value of $\alpha^2 + \beta\gamma$.

TYPE 11 Solving Matrix Equations using Multiplication

Given: $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ and $A^2 = I$

To Find: The value of $\alpha^2 + \beta\gamma$

Concept / Formula Used: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 2×2

Solution:

First calculate $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix} \\ &= \begin{bmatrix} \alpha(\alpha) + \beta(\gamma) & \alpha(\beta) + \beta(-\alpha) \\ \gamma(\alpha) + (-\alpha)(\gamma) & \gamma(\beta) + (-\alpha)(-\alpha) \end{bmatrix} \\ &= \begin{bmatrix} \alpha^2 + \beta\gamma & \alpha\beta - \alpha\beta \\ \alpha\gamma - \alpha\gamma & \beta\gamma + \alpha^2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \alpha^2 + \beta\gamma \end{bmatrix} \end{aligned}$$

Given $A^2 = I$:

$$\begin{bmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \alpha^2 + \beta\gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Comparing the entry at row 1, column 1:

$$\alpha^2 + \beta\gamma = 1$$

Final Answer

$$\alpha^2 + \beta\gamma = 1$$

Question 11

If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find AB and BA .

TYPE 12 Matrix Multiplication

Given: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

To Find: The matrices AB and BA

Concept / Formula Used: 2×2

Solution:

Calculating AB :

$$\begin{aligned}
 AB &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} 1(0) + 0(1) & 1(1) + 0(0) \\ 0(0) + (-1)(1) & 0(1) + (-1)(0) \end{bmatrix} \\
 &= \begin{bmatrix} 0 + 0 & 1 + 0 \\ 0 - 1 & 0 + 0 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}
 \end{aligned}$$

Calculating BA :

$$\begin{aligned}
 BA &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\
 &= \begin{bmatrix} 0(1) + 1(0) & 0(0) + 1(-1) \\ 1(1) + 0(0) & 1(0) + 0(-1) \end{bmatrix} \\
 &= \begin{bmatrix} 0 + 0 & 0 - 1 \\ 1 + 0 & 0 + 0 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Question 12

- (i) If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, find AB and BA .
- (ii) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ and $AB = BA$, then show that B is a scalar matrix.

TYPE 13 Matrix Multiplication Properties and Proofs

Given: (i) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

(ii) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, $AB = BA$

To Prove: (i) Find AB and BA

(ii) Show that B is a scalar matrix (i.e. $a = b$)

Concept / Formula Used: $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ $a = b$

Solution:

Part (i):

$$\begin{aligned}
 AB &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1(1) + 1(-1) & 1(-1) + 1(1) \\ 1(1) + 1(-1) & 1(-1) + 1(1) \end{bmatrix} \\
 &= \begin{bmatrix} 1 - 1 & -1 + 1 \\ 1 - 1 & -1 + 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O
 \end{aligned}$$

$$\begin{aligned}
 BA &= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1(1) + (-1)(1) & 1(1) + (-1)(1) \\ (-1)(1) + 1(1) & (-1)(1) + 1(1) \end{bmatrix} \\
 &= \begin{bmatrix} 1 - 1 & 1 - 1 \\ -1 + 1 & -1 + 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O
 \end{aligned}$$

Part (ii): Calculate product AB :

$$\begin{aligned}
 AB &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \\
 &= \begin{bmatrix} 1(a) + 2(0) & 1(0) + 2(b) \\ 3(a) + 4(0) & 3(0) + 4(b) \end{bmatrix} \\
 &= \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}
 \end{aligned}$$

Calculate product BA :

$$\begin{aligned}
 BA &= \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\
 &= \begin{bmatrix} a(1) + 0(3) & a(2) + 0(4) \\ 0(1) + b(3) & 0(2) + b(4) \end{bmatrix} \\
 &= \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}
 \end{aligned}$$

Since $AB = BA$:

$$\begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix} = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}$$

Equating corresponding entry at $(1, 2)$:

$$\begin{aligned}
 2b &= 2a \\
 a &= b
 \end{aligned}$$

Similarly, equating entry at $(2, 1)$:

$$\begin{aligned}
 3a &= 3b \\
 a &= b
 \end{aligned}$$

Since $a = b$, matrix B becomes $B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$, which is by definition a scalar matrix.

Hence Proved

Part (i) yields $AB = O$ and $BA = O$. Part (ii) shows $a = b$, hence B is a scalar matrix.

Question 13

If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ and $A^2 = B$, then find the value(s) of α .

TYPE 11 Solving Matrix Equations using Multiplication

Given: $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, and $A^2 = B$

To Find: The value(s) of α

Solution:

Compute $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \alpha(\alpha) + 0(1) & \alpha(0) + 0(1) \\ 1(\alpha) + 1(1) & 1(0) + 1(1) \end{bmatrix} \\ &= \begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix} \end{aligned}$$

Set $A^2 = B$:

$$\begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

Equating corresponding elements gives two conditions on α :

$$\alpha^2 = 1 \implies \alpha = 1 \text{ or } \alpha = -1 \quad (1)$$

$$\alpha + 1 = 2 \implies \alpha = 1 \quad (2)$$

For $A^2 = B$ to hold, α must satisfy both equations simultaneously. The common value satisfying both equations is:

$$\alpha = 1$$

Final Answer

$$\alpha = 1$$

Question 14

If A is a square matrix such that $A^2 = I$, then find the simplified value of $(A - I)^3 + (A + I)^3 - 7A$.

★ Likely Exam Question

TYPE 14 Simplifying Matrix Polynomial Expressions

Given: A is a square matrix satisfying $A^2 = I$

To Find: The simplified expression for $(A - I)^3 + (A + I)^3 - 7A$

Concept / Formula Used: $A \cdot I = AI = IA = A$ $(A - I)^3 = A^3 - 3A^2I + 3AI^2 - I^3$
 $(A + I)^3 = A^3 + 3A^2I + 3AI^2 + I^3$

Solution:

Given $A^2 = I$, we first compute higher powers of A :

$$\begin{aligned} A^3 &= A^2 \cdot A \\ &= I \cdot A \\ &= A \end{aligned}$$

Also note the properties of identity matrix powers:

$$\begin{aligned} I^2 &= I \\ I^3 &= I \\ A^2I &= I \cdot I = I \\ AI^2 &= A \cdot I = A \end{aligned}$$

Now expand $(A - I)^3$:

$$\begin{aligned} (A - I)^3 &= A^3 - 3A^2I + 3AI^2 - I^3 \\ &= A - 3I + 3A - I \\ &= 4A - 4I \end{aligned}$$

Next expand $(A + I)^3$:

$$\begin{aligned} (A + I)^3 &= A^3 + 3A^2I + 3AI^2 + I^3 \\ &= A + 3I + 3A + I \\ &= 4A + 4I \end{aligned}$$

Now substitute both expansions into the target expression:

$$\begin{aligned} (A - I)^3 + (A + I)^3 - 7A &= (4A - 4I) + (4A + 4I) - 7A \\ &= 4A + 4A - 7A - 4I + 4I \\ &= 8A - 7A + O \\ &= A \end{aligned}$$

Final Answer

$$(A - I)^3 + (A + I)^3 - 7A = A$$

Question 15

Write the following as a single matrix:

$$\begin{aligned} \text{(i)} \quad & \begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 5 \\ 0 & 2 & 4 \\ 7 & 5 & 0 \end{bmatrix} - \begin{bmatrix} 2 & -5 & 7 \end{bmatrix} \\ \text{(ii)} \quad & \begin{bmatrix} 3 & 2 & 5 \\ 7 & -4 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & -1 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 7 & -8 \\ 5 & 9 \end{bmatrix} \end{aligned}$$

TYPE 15 Matrix Operations and Simplification

Given: Matrix expressions combining multiplication and subtraction for parts (i) and (ii)

To Find: Single simplified matrix for each part

Solution:

Part (i): First perform the product of the row matrix 1×3 and 3×3 matrix:

$$\begin{aligned} \begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 5 \\ 0 & 2 & 4 \\ 7 & 5 & 0 \end{bmatrix} &= \begin{bmatrix} 1(2) + (-2)(0) + 3(7) & 1(-1) + (-2)(2) + 3(5) & 1(5) + (-2)(4) + 3(0) \end{bmatrix} \\ &= \begin{bmatrix} 2 + 0 + 21 & -1 - 4 + 15 & 5 - 8 + 0 \end{bmatrix} \\ &= \begin{bmatrix} 23 & 10 & -3 \end{bmatrix} \end{aligned}$$

Now perform matrix subtraction:

$$\begin{aligned} \begin{bmatrix} 23 & 10 & -3 \end{bmatrix} - \begin{bmatrix} 2 & -5 & 7 \end{bmatrix} &= \begin{bmatrix} 23 - 2 & 10 - (-5) & -3 - 7 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 15 & -10 \end{bmatrix} \end{aligned}$$

Part (ii): First evaluate the product of the 2×3 matrix and 3×2 matrix:

$$\begin{aligned} \begin{bmatrix} 3 & 2 & 5 \\ 7 & -4 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & -1 \\ 3 & 5 \end{bmatrix} &= \begin{bmatrix} 3(2) + 2(2) + 5(3) & 3(2) + 2(-1) + 5(5) \\ 7(2) + (-4)(2) + 0(3) & 7(2) + (-4)(-1) + 0(5) \end{bmatrix} \\ &= \begin{bmatrix} 6 + 4 + 15 & 6 - 2 + 25 \\ 14 - 8 + 0 & 14 + 4 + 0 \end{bmatrix} \\ &= \begin{bmatrix} 25 & 29 \\ 6 & 18 \end{bmatrix} \end{aligned}$$

Now perform matrix subtraction:

$$\begin{aligned} \begin{bmatrix} 25 & 29 \\ 6 & 18 \end{bmatrix} - \begin{bmatrix} 7 & -8 \\ 5 & 9 \end{bmatrix} &= \begin{bmatrix} 25 - 7 & 29 - (-8) \\ 6 - 5 & 18 - 9 \end{bmatrix} \\ &= \begin{bmatrix} 18 & 37 \\ 1 & 9 \end{bmatrix} \end{aligned}$$

Final Answer

(i) $\begin{bmatrix} 21 & 15 & -10 \end{bmatrix}$

(ii) $\begin{bmatrix} 18 & 37 \\ 1 & 9 \end{bmatrix}$

Question 16

(i) If $A = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$, find A^2 .

(ii) If $M(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, show that $M(x)M(y) = M(x + y)$.

TYPE 16 Matrix Multiplication with Trigonometric Identities

Given: (i) $A = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$
(ii) $M(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$

To Prove: (i) Find A^2
(ii) Show that $M(x)M(y) = M(x+y)$

Concept / Formula Used: *Trigonometric addition formulae* :
 $\cos A \cos B - \sin A \sin B = \cos(A+B)$
 $\sin A \cos B + \cos A \sin B = \sin(A+B)$

Solution:

Part (i):

$$\begin{aligned} A^2 &= \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix} \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 2\alpha - \sin^2 2\alpha & \cos 2\alpha \sin 2\alpha + \sin 2\alpha \cos 2\alpha \\ -\sin 2\alpha \cos 2\alpha - \cos 2\alpha \sin 2\alpha & -\sin^2 2\alpha + \cos^2 2\alpha \end{bmatrix} \end{aligned}$$

Using double-angle trigonometric identities:

$$\begin{aligned} \cos^2 2\alpha - \sin^2 2\alpha &= \cos(2\alpha + 2\alpha) = \cos 4\alpha \\ \sin 2\alpha \cos 2\alpha + \cos 2\alpha \sin 2\alpha &= \sin(2\alpha + 2\alpha) = \sin 4\alpha \end{aligned}$$

Substituting these identities back gives:

$$A^2 = \begin{bmatrix} \cos 4\alpha & \sin 4\alpha \\ -\sin 4\alpha & \cos 4\alpha \end{bmatrix}$$

Part (ii): Write down $M(x)$ and $M(y)$:

$$\begin{aligned} M(x) &= \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} \\ M(y) &= \begin{bmatrix} \cos y & \sin y \\ -\sin y & \cos y \end{bmatrix} \end{aligned}$$

Multiplying $M(x)$ and $M(y)$:

$$\begin{aligned} M(x)M(y) &= \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} \begin{bmatrix} \cos y & \sin y \\ -\sin y & \cos y \end{bmatrix} \\ &= \begin{bmatrix} \cos x \cos y - \sin x \sin y & \cos x \sin y + \sin x \cos y \\ -\sin x \cos y - \cos x \sin y & -\sin x \sin y + \cos x \cos y \end{bmatrix} \end{aligned}$$

Using standard trigonometric angle-addition identities:

$$\begin{aligned} \cos x \cos y - \sin x \sin y &= \cos(x+y) \\ \sin x \cos y + \cos x \sin y &= \sin(x+y) \\ -(\sin x \cos y + \cos x \sin y) &= -\sin(x+y) \\ \cos x \cos y - \sin x \sin y &= \cos(x+y) \end{aligned}$$

Substitute these into the product matrix:

$$\begin{aligned} M(x)M(y) &= \begin{bmatrix} \cos(x+y) & \sin(x+y) \\ -\sin(x+y) & \cos(x+y) \end{bmatrix} \\ &= M(x+y) \end{aligned}$$

Hence Proved

$$A^2 = \begin{bmatrix} \cos 4\alpha & \sin 4\alpha \\ -\sin 4\alpha & \cos 4\alpha \end{bmatrix} \text{ and } M(x)M(y) = M(x+y) \text{ is proved.}$$

Question 17

If $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$, find $3A^2 - 2B + I$.

TYPE 15 Matrix Operations and Simplification

Given: $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$, and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

To Find: The value of matrix expression $3A^2 - 2B + I$

Solution:

First evaluate $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2(2) + (-1)(3) & 2(-1) + (-1)(2) \\ 3(2) + 2(3) & 3(-1) + 2(2) \end{bmatrix} \\ &= \begin{bmatrix} 4 - 3 & -2 - 2 \\ 6 + 6 & -3 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} \end{aligned}$$

Calculate $3A^2$:

$$\begin{aligned} 3A^2 &= 3 \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & -12 \\ 36 & 3 \end{bmatrix} \end{aligned}$$

Calculate $2B$:

$$\begin{aligned} 2B &= 2 \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 8 \\ -2 & 14 \end{bmatrix} \end{aligned}$$

Now evaluate $3A^2 - 2B + I$:

$$\begin{aligned} 3A^2 - 2B + I &= \begin{bmatrix} 3 & -12 \\ 36 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 8 \\ -2 & 14 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 - 0 + 1 & -12 - 8 + 0 \\ 36 - (-2) + 0 & 3 - 14 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & -20 \\ 38 & -10 \end{bmatrix} \end{aligned}$$

Final Answer

$$3A^2 - 2B + I = \begin{bmatrix} 4 & -20 \\ 38 & -10 \end{bmatrix}$$

Question 18

If $A = \begin{bmatrix} 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$, verify that $A(B + C) = AB + AC$.

★ Likely Exam Question

TYPE 13 Matrix Multiplication Properties and Proofs

Given: $A = \begin{bmatrix} 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

To Prove: Verify the distributive property $A(B + C) = AB + AC$

Concept / Formula Used: $A(B + C) = AB + AC$

Solution:

First calculate LHS by evaluating $B + C$:

$$\begin{aligned} B + C &= \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix} + \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 5 + (-1) & 3 + 2 & 4 + 1 \\ 8 + 0 & 7 + 1 & 6 + 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 5 & 5 \\ 8 & 8 & 8 \end{bmatrix} \end{aligned}$$

Now compute LHS = $A(B + C)$:

$$\begin{aligned} A(B + C) &= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 5 & 5 \\ 8 & 8 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 2(4) + 1(8) & 2(5) + 1(8) & 2(5) + 1(8) \end{bmatrix} \\ &= \begin{bmatrix} 8 + 8 & 10 + 8 & 10 + 8 \end{bmatrix} \\ &= \begin{bmatrix} 16 & 18 & 18 \end{bmatrix} \end{aligned}$$

Next calculate RHS by evaluating AB and AC separately:

$$\begin{aligned} AB &= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 2(5) + 1(8) & 2(3) + 1(7) & 2(4) + 1(6) \end{bmatrix} \\ &= \begin{bmatrix} 10 + 8 & 6 + 7 & 8 + 6 \end{bmatrix} \\ &= \begin{bmatrix} 18 & 13 & 14 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} AC &= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2(-1) + 1(0) & 2(2) + 1(1) & 2(1) + 1(2) \end{bmatrix} \\ &= \begin{bmatrix} -2 + 0 & 4 + 1 & 2 + 2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 5 & 4 \end{bmatrix} \end{aligned}$$

Summing $AB + AC$:

$$\begin{aligned} AB + AC &= \begin{bmatrix} 18 & 13 & 14 \end{bmatrix} + \begin{bmatrix} -2 & 5 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 18 + (-2) & 13 + 5 & 14 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 16 & 18 & 18 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$\text{LHS} = \text{RHS} = \begin{bmatrix} 16 & 18 & 18 \end{bmatrix}$$

Hence Proved

LHS = RHS, hence $A(B + C) = AB + AC$ is verified.

Question 19

- (i) Solve for x and y , given that $\begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.
- (ii) Solve for x and y , given that $\begin{bmatrix} x & y \\ 3y & x \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.

TYPE 11 Solving Matrix Equations using Multiplication

Given: (i) $\begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

(ii) $\begin{bmatrix} x & y \\ 3y & x \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

To Find: The values of x and y for each part

Concept / Formula Used: $x \quad y$

Solution:

Part (i): Multiply the matrices on the LHS:

$$\begin{bmatrix} 2x - 3y \\ x + y \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Equating corresponding entries yields two linear equations:

$$2x - 3y = 1 \quad (3)$$

$$x + y = 3 \quad (4)$$

From equation (4), express x in terms of y :

$$x = 3 - y$$

Substitute $x = 3 - y$ into equation (3):

$$2(3 - y) - 3y = 1$$

$$6 - 2y - 3y = 1$$

$$6 - 5y = 1$$

$$-5y = 1 - 6$$

$$-5y = -5$$

$$y = 1$$

Substitute $y = 1$ back into $x = 3 - y$:

$$x = 3 - 1$$

$$x = 2$$

Part (ii): Multiply the matrices on the LHS:

$$\begin{bmatrix} x(1) + y(2) \\ 3y(1) + x(2) \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x + 2y \\ 2x + 3y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Equating corresponding elements:

$$x + 2y = 3 \quad (5)$$

$$2x + 3y = 5 \quad (6)$$

From equation (5):

$$x = 3 - 2y$$

Substitute $x = 3 - 2y$ into equation (6):

$$2(3 - 2y) + 3y = 5$$

$$6 - 4y + 3y = 5$$

$$6 - y = 5$$

$$-y = 5 - 6$$

$$-y = -1$$

$$y = 1$$

Substitute $y = 1$ into $x = 3 - 2y$:

$$x = 3 - 2(1)$$

$$x = 1$$

Final Answer

(i) $x = 2, y = 1$

(ii) $x = 1, y = 1$

Question 20

- (i) If $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$, show that $A^2 - 5A - 14I = O$.
- (ii) If $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$, find the value of $A^2 + 2A + 7I$.

TYPE 17 Evaluating Matrix Polynomials

Given: (i) $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$

(ii) $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$

To Prove: (i) Show that $A^2 - 5A - 14I = O$
 (ii) Compute the value of $A^2 + 2A + 7I$

Concept / Formula Used: 2×2

Solution:

Part (i): Calculate A^2 :

$$\begin{aligned}
 A^2 &= \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \\
 &= \begin{bmatrix} 3(3) + (-5)(-4) & 3(-5) + (-5)(2) \\ -4(3) + 2(-4) & -4(-5) + 2(2) \end{bmatrix} \\
 &= \begin{bmatrix} 9 + 20 & -15 - 10 \\ -12 - 8 & 20 + 4 \end{bmatrix} \\
 &= \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix}
 \end{aligned}$$

Calculate $5A$:

$$\begin{aligned}
 5A &= 5 \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \\
 &= \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix}
 \end{aligned}$$

Calculate $14I$:

$$\begin{aligned}
 14I &= 14 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}
 \end{aligned}$$

Now evaluate $A^2 - 5A - 14I$:

$$\begin{aligned}
 A^2 - 5A - 14I &= \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix} - \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix} - \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix} \\
 &= \begin{bmatrix} 29 - 15 - 14 & -25 - (-25) - 0 \\ -20 - (-20) - 0 & 24 - 10 - 14 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O
 \end{aligned}$$

Part (ii): Calculate A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(4) & 1(2) + 2(1) \\ 4(1) + 1(4) & 4(2) + 1(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 8 & 2 + 2 \\ 4 + 4 & 8 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 4 \\ 8 & 9 \end{bmatrix} \end{aligned}$$

Calculate $2A$ and $7I$:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 8 & 2 \end{bmatrix} \\ 7I &= 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \end{aligned}$$

Now evaluate $A^2 + 2A + 7I$:

$$\begin{aligned} A^2 + 2A + 7I &= \begin{bmatrix} 9 & 4 \\ 8 & 9 \end{bmatrix} + \begin{bmatrix} 2 & 4 \\ 8 & 2 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 9 + 2 + 7 & 4 + 4 + 0 \\ 8 + 8 + 0 & 9 + 2 + 7 \end{bmatrix} \\ &= \begin{bmatrix} 18 & 8 \\ 16 & 18 \end{bmatrix} \end{aligned}$$

Hence Proved

Part (i) shows $A^2 - 5A - 14I = O$. Part (ii) yields $A^2 + 2A + 7I = \begin{bmatrix} 18 & 8 \\ 16 & 18 \end{bmatrix}$.

Question 21

If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, show that $(aA + bB)(aA - bB) = (a^2 + b^2)A$.

TYPE 13 Matrix Multiplication Properties and Proofs

Given: $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

To Prove: $(aA + bB)(aA - bB) = (a^2 + b^2)A$

Concept / Formula Used: $A = I$ $AB = BA = B$ $A^2 = A = I$

Solution:

Notice that $A = I$, the 2×2 identity matrix.

First compute matrices $aA + bB$ and $aA - bB$:

$$\begin{aligned} aA + bB &= a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} + \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \end{aligned}$$

$$\begin{aligned} aA - bB &= a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - b \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} - \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} \\ &= \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \end{aligned}$$

Multiply LHS = $(aA + bB)(aA - bB)$:

$$\begin{aligned} \text{LHS} &= \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \\ &= \begin{bmatrix} a(a) + b(b) & a(-b) + b(a) \\ -b(a) + a(b) & -b(-b) + a(a) \end{bmatrix} \\ &= \begin{bmatrix} a^2 + b^2 & -ab + ab \\ -ab + ab & b^2 + a^2 \end{bmatrix} \\ &= \begin{bmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{bmatrix} \end{aligned}$$

Factor out $(a^2 + b^2)$:

$$\begin{aligned} \text{LHS} &= (a^2 + b^2) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= (a^2 + b^2)A = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence $(aA + bB)(aA - bB) = (a^2 + b^2)A$ is proved.

Question 22

If $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$, prove that $(A - 2I)(A - 3I) = O$.

TYPE 17 Evaluating Matrix Polynomials

Given: $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

To Prove: $(A - 2I)(A - 3I) = O$

Solution:

Compute $A - 2I$:

$$\begin{aligned} A - 2I &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4-2 & 2-0 \\ -1-0 & 1-2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \end{aligned}$$

Compute $A - 3I$:

$$\begin{aligned} A - 3I &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 4-3 & 2-0 \\ -1-0 & 1-3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \end{aligned}$$

Now calculate LHS = $(A - 2I)(A - 3I)$:

$$\begin{aligned} \text{LHS} &= \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2(1) + 2(-1) & 2(2) + 2(-2) \\ -1(1) + (-1)(-1) & -1(2) + (-1)(-2) \end{bmatrix} \\ &= \begin{bmatrix} 2-2 & 4-4 \\ -1+1 & -2+2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence $(A - 2I)(A - 3I) = O$ is proved.

Question 23

If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$, then find k so that $A^2 = kA - 2I$.

TYPE 11 Solving Matrix Equations using Multiplication

Given: $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$ and $A^2 = kA - 2I$

To Find: The value of scalar k

Concept / Formula Used: $A^2 = kA - 2I$

Solution:

Calculate $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 3(3) + (-2)(4) & 3(-2) + (-2)(-2) \\ 4(3) + (-2)(4) & 4(-2) + (-2)(-2) \end{bmatrix} \\ &= \begin{bmatrix} 9 - 8 & -6 + 4 \\ 12 - 8 & -8 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} \end{aligned}$$

Now compute the RHS = $kA - 2I$:

$$\begin{aligned} kA - 2I &= k \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3k & -2k \\ 4k & -2k \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3k - 2 & -2k \\ 4k & -2k - 2 \end{bmatrix} \end{aligned}$$

Equating A^2 to $kA - 2I$:

$$\begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} = \begin{bmatrix} 3k - 2 & -2k \\ 4k & -2k - 2 \end{bmatrix}$$

Comparing entry at position (1, 2):

$$\begin{aligned} -2k &= -2 \\ k &= 1 \end{aligned}$$

Check for consistency with other entries:

$$\begin{aligned} \text{Entry (1, 1): } & 3(1) - 2 = 1 \quad (\text{Matches}) \\ \text{Entry (2, 1): } & 4(1) = 4 \quad (\text{Matches}) \\ \text{Entry (2, 2): } & -2(1) - 2 = -4 \quad (\text{Matches}) \end{aligned}$$

The value $k = 1$ is consistent across all entries.

Final Answer

$$k = 1$$

Question 24 (i)

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, show that $f(A) = O$, where $f(x) = x^2 - 2x - 3$.

TYPE 17 Evaluating Matrix Polynomials

To Prove: $f(A) = A^2 - 2A - 3I = O$

Concept / Formula Used: $f(x) = a_n x^n + \cdots + a_1 x + a_0$ $f(A) = a_n A^n + \cdots + a_1 A + a_0 I$

Solution:

Given matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and identity matrix $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

First, evaluate A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (1)(1) + (2)(2) & (1)(2) + (2)(1) \\ (2)(1) + (1)(2) & (2)(2) + (1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 4 & 2 + 2 \\ 2 + 2 & 4 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \end{aligned}$$

Next, evaluate $2A$:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} \end{aligned}$$

Next, evaluate $3I$:

$$\begin{aligned} 3I &= 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \end{aligned}$$

Now, substitute A^2 , $2A$, and $3I$ into $f(A) = A^2 - 2A - 3I$:

$$\begin{aligned} f(A) &= \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 5 - 2 - 3 & 4 - 4 - 0 \\ 4 - 4 - 0 & 5 - 2 - 3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ &= O \end{aligned}$$

Hence Proved

LHS = RHS, hence $f(A) = O$ is proved.

Question 24 (ii)

If $A = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$, find $f(A)$, where $f(x) = x^2 - 2x + 3$.

TYPE 17 Evaluating Matrix Polynomials

Given: $A = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$, $f(x) = x^2 - 2x + 3$

To Find: $f(A) = A^2 - 2A + 3I$

Concept / Formula Used: $f(A) = A^2 - 2A + 3I$ $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Solution:

First, compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (-1)(-1) + (2)(3) & (-1)(2) + (2)(1) \\ (3)(-1) + (1)(3) & (3)(2) + (1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1+6 & -2+2 \\ -3+3 & 6+1 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \end{aligned}$$

Next, compute $2A$:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 4 \\ 6 & 2 \end{bmatrix} \end{aligned}$$

Next, compute $3I$:

$$\begin{aligned} 3I &= 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \end{aligned}$$

Substitute these into $f(A) = A^2 - 2A + 3I$:

$$\begin{aligned} f(A) &= \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} - \begin{bmatrix} -2 & 4 \\ 6 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 7 - (-2) + 3 & 0 - 4 + 0 \\ 0 - 6 + 0 & 7 - 2 + 3 \end{bmatrix} \\ &= \begin{bmatrix} 7 + 2 + 3 & -4 \\ -6 & 5 + 3 \end{bmatrix} \\ &= \begin{bmatrix} 12 & -4 \\ -6 & 8 \end{bmatrix} \end{aligned}$$

Final Answer

$$f(A) = \begin{bmatrix} 12 & -4 \\ -6 & 8 \end{bmatrix}$$

Question 25

If $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then:

- (i) find λ, μ so that $A^2 = \lambda A + \mu I$
 (ii) prove that $A^3 - 4A^2 + A = O$

TYPE 17 Evaluating Matrix Polynomials

Given: $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

To Find: (i) λ, μ such that $A^2 = \lambda A + \mu I$; (ii) Prove $A^3 - 4A^2 + A = O$

Solution:

Part (i): First, compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (2)(2) + (3)(1) & (2)(3) + (3)(2) \\ (1)(2) + (2)(1) & (1)(3) + (2)(2) \end{bmatrix} \\ &= \begin{bmatrix} 4 + 3 & 6 + 6 \\ 2 + 2 & 3 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} \end{aligned}$$

Now, express $\lambda A + \mu I$:

$$\begin{aligned} \lambda A + \mu I &= \lambda \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} + \mu \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2\lambda & 3\lambda \\ \lambda & 2\lambda \end{bmatrix} + \begin{bmatrix} \mu & 0 \\ 0 & \mu \end{bmatrix} \\ &= \begin{bmatrix} 2\lambda + \mu & 3\lambda \\ \lambda & 2\lambda + \mu \end{bmatrix} \end{aligned}$$

Equating $A^2 = \lambda A + \mu I$:

$$\begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 2\lambda + \mu & 3\lambda \\ \lambda & 2\lambda + \mu \end{bmatrix}$$

Equating corresponding elements:

$$\begin{aligned} \lambda &= 4 \\ 3\lambda &= 12 \implies \lambda = 4 \\ 2\lambda + \mu &= 7 \end{aligned}$$

Substitute $\lambda = 4$ into $2\lambda + \mu = 7$:

$$\begin{aligned} 2(4) + \mu &= 7 \\ 8 + \mu &= 7 \\ \mu &= 7 - 8 \\ \mu &= -1 \end{aligned}$$

Part (ii): From Part (i), we have $A^2 = 4A - I$. Rearranging this equation gives:

$$A^2 - 4A + I = O$$

Multiplying both sides of the equation by A :

$$A(A^2 - 4A + I) = A \cdot O$$

$$A^3 - 4A^2 + A \cdot I = O$$

$$A^3 - 4A^2 + A = O$$

Final Answer

(i) $\lambda = 4, \mu = -1$

(ii) $A^3 - 4A^2 + A = O$ is proved.

Question 26

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, verify that $A^2 - 4A - 5I = O$.

TYPE 17 Evaluating Matrix Polynomials

Given: $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

To Prove: $A^2 - 4A - 5I = O$

Concept / Formula Used: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$

Solution:

First, calculate A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} \end{aligned}$$

Next, calculate $4A$:

$$4A = 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix}$$

Next, calculate $5I$:

$$5I = 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Now compute $LHS = A^2 - 4A - 5I$:

$$\begin{aligned}
 LHS &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \\
 &= \begin{bmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-4-5 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 &= O = RHS
 \end{aligned}$$

Hence Proved

LHS = RHS, hence $A^2 - 4A - 5I = O$ is verified.

Question 27

- (i) Let A and B be square matrices of same order. Does $(A+B)^2 = A^2 + 2AB + B^2$ hold? If not, why?
- (ii) Let A and B be square matrices of order 3×3 . Is $(AB)^2 = A^2B^2$? Give reason.

TYPE 13 Matrix Multiplication Properties and Proofs

Given: A, B are square matrices of the same order.

To Find: Determine whether $(A+B)^2 = A^2 + 2AB + B^2$ and $(AB)^2 = A^2B^2$ hold, with reasons.

Concept / Formula Used: $AB \neq BA$

Solution:

Part (i): Expand the expression $(A+B)^2$:

$$\begin{aligned}
 (A+B)^2 &= (A+B)(A+B) \\
 &= A(A+B) + B(A+B) \\
 &= A^2 + AB + BA + B^2
 \end{aligned}$$

For $(A+B)^2$ to equal $A^2 + 2AB + B^2$, we must have:

$$AB + BA = 2AB \implies BA = AB$$

Since matrix multiplication is non-commutative in general ($AB \neq BA$), $(A+B)^2 = A^2 + 2AB + B^2$ does NOT hold in general.

Part (ii): Expand the expressions $(AB)^2$ and A^2B^2 :

$$\begin{aligned}
 (AB)^2 &= (AB)(AB) = A(BA)B \\
 A^2B^2 &= (AA)(BB) = A(AB)B
 \end{aligned}$$

For $(AB)^2$ to equal A^2B^2 , we must have:

$$A(BA)B = A(AB)B \iff BA = AB$$

Since matrix multiplication is not commutative in general, $BA \neq AB$, so $(AB)^2 = A^2B^2$ does NOT hold in general unless $AB = BA$.

Hence Proved

- (i) No, because matrix multiplication is not commutative in general ($AB \neq BA$).
 (ii) No, because $(AB)^2 = ABAB$ whereas $A^2B^2 = AABB$, which are equal only if $AB = BA$.

Question 28

If A and B are square matrices of the same order such that $AB = BA$, prove that:

- (i) $(A + B)(A - B) = A^2 - B^2$
 (ii) $(A + B)^2 = A^2 + 2AB + B^2$
 (iii) $(A - B)^2 = A^2 - 2AB + B^2$
 (iv) $(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$
 (v) $(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$

TYPE 13 Matrix Multiplication Properties and Proofs

Given: A and B are square matrices with $AB = BA$.

To Prove: Prove identity subparts (i) through (v).

Concept / Formula Used: $A(B + C) = AB + AC$ $AB = BA$

Solution:

Part (i):

$$\begin{aligned} LHS &= (A + B)(A - B) = A(A - B) + B(A - B) \\ &= A^2 - AB + BA - B^2 \end{aligned}$$

Since $AB = BA$, we have $-AB + BA = O$:

$$LHS = A^2 - AB + AB - B^2 = A^2 - B^2 = RHS$$

Part (ii):

$$\begin{aligned} LHS &= (A + B)^2 = (A + B)(A + B) = A(A + B) + B(A + B) \\ &= A^2 + AB + BA + B^2 \end{aligned}$$

Using $BA = AB$:

$$LHS = A^2 + AB + AB + B^2 = A^2 + 2AB + B^2 = RHS$$

Part (iii):

$$\begin{aligned} LHS &= (A - B)^2 = (A - B)(A - B) = A(A - B) - B(A - B) \\ &= A^2 - AB - BA + B^2 \end{aligned}$$

Using $BA = AB$:

$$LHS = A^2 - AB - AB + B^2 = A^2 - 2AB + B^2 = RHS$$

Part (iv): Using $(A + B)^2 = A^2 + 2AB + B^2$ from part (ii):

$$\begin{aligned} LHS &= (A + B)^3 = (A + B)^2(A + B) \\ &= (A^2 + 2AB + B^2)(A + B) \\ &= A^2(A + B) + 2AB(A + B) + B^2(A + B) \\ &= A^3 + A^2B + 2ABA + 2AB^2 + B^2A + B^3 \end{aligned}$$

Since $AB = BA$, we have $ABA = A(AB) = A(BA) = A^2B$ and $B^2A = B(BA) = B(AB) = (BA)B = (AB)B = AB^2$:

$$\begin{aligned} LHS &= A^3 + A^2B + 2A^2B + 2AB^2 + AB^2 + B^3 \\ &= A^3 + 3A^2B + 3AB^2 + B^3 = RHS \end{aligned}$$

Part (v): Using $(A - B)^2 = A^2 - 2AB + B^2$ from part (iii):

$$\begin{aligned} LHS &= (A - B)^3 = (A - B)^2(A - B) \\ &= (A^2 - 2AB + B^2)(A - B) \\ &= A^2(A - B) - 2AB(A - B) + B^2(A - B) \\ &= A^3 - A^2B - 2ABA + 2AB^2 + B^2A - B^3 \end{aligned}$$

Using $ABA = A^2B$ and $B^2A = AB^2$:

$$\begin{aligned} LHS &= A^3 - A^2B - 2A^2B + 2AB^2 + AB^2 - B^3 \\ &= A^3 - 3A^2B + 3AB^2 - B^3 = RHS \end{aligned}$$

Hence Proved

All five identities are proved using $AB = BA$.

Question 29

If $P = \begin{bmatrix} p & q \\ -q & p \end{bmatrix}$ and $Q = \begin{bmatrix} r & s \\ -s & r \end{bmatrix}$, prove that $PQ = QP$ for all values of p, q, r and s .

TYPE 13 Matrix Multiplication Properties and Proofs

Given: $P = \begin{bmatrix} p & q \\ -q & p \end{bmatrix}, Q = \begin{bmatrix} r & s \\ -s & r \end{bmatrix}$

To Prove: $PQ = QP$

Concept / Formula Used: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$

Solution:

First, compute $LHS = PQ$:

$$\begin{aligned} PQ &= \begin{bmatrix} p & q \\ -q & p \end{bmatrix} \begin{bmatrix} r & s \\ -s & r \end{bmatrix} \\ &= \begin{bmatrix} (p)(r) + (q)(-s) & (p)(s) + (q)(r) \\ (-q)(r) + (p)(-s) & (-q)(s) + (p)(r) \end{bmatrix} \\ &= \begin{bmatrix} pr - qs & ps + qr \\ -qr - ps & -qs + pr \end{bmatrix} \\ &= \begin{bmatrix} pr - qs & ps + qr \\ -(ps + qr) & pr - qs \end{bmatrix} \end{aligned}$$

Next, compute $RHS = QP$:

$$\begin{aligned}
 QP &= \begin{bmatrix} r & s \\ -s & r \end{bmatrix} \begin{bmatrix} p & q \\ -q & p \end{bmatrix} \\
 &= \begin{bmatrix} (r)(p) + (s)(-q) & (r)(q) + (s)(p) \\ (-s)(p) + (r)(-q) & (-s)(q) + (r)(p) \end{bmatrix} \\
 &= \begin{bmatrix} rp - sq & rq + sp \\ -sp - rq & -sq + rp \end{bmatrix} \\
 &= \begin{bmatrix} pr - qs & ps + qr \\ -(ps + qr) & pr - qs \end{bmatrix}
 \end{aligned}$$

Comparing LHS and RHS :

$$LHS = \begin{bmatrix} pr - qs & ps + qr \\ -(ps + qr) & pr - qs \end{bmatrix} = RHS$$

Hence Proved

$LHS = RHS$, hence $PQ = QP$ for all values of p, q, r and s .

Question 30

Find x , if:

$$\begin{aligned}
 \text{(i)} \quad [1 \quad x \quad 1] \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} &= O \\
 \text{(ii)} \quad [1 \quad 2 \quad 1] \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} &= O
 \end{aligned}$$

TYPE 11 Solving Matrix Equations using Multiplication

Given: Matrix equations involving x equal to zero matrix O .

To Find: The value(s) of x .

Concept / Formula Used: $(AB)C = A(BC)$

Solution:

Part (i): First, multiply the first two matrices (1×3 and 3×3):

$$\begin{aligned}
 [1 \quad x \quad 1] \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} &= [1(1) + x(2) + 1(15) \quad 1(3) + x(5) + 1(3) \quad 1(2) + x(1) + 1(2)] \\
 &= [1 + 2x + 15 \quad 3 + 5x + 3 \quad 2 + x + 2] \\
 &= [2x + 16 \quad 5x + 6 \quad x + 4]
 \end{aligned}$$

Now, multiply this result by the third matrix $\begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix}$:

$$\begin{bmatrix} 2x + 16 & 5x + 6 & x + 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = O$$

$$[(2x + 16)(1) + (5x + 6)(2) + (x + 4)(x)] = [0]$$

Expanding the equation:

$$2x + 16 + 10x + 12 + x^2 + 4x = 0$$

$$x^2 + (2x + 10x + 4x) + (16 + 12) = 0$$

$$x^2 + 16x + 28 = 0$$

Factoring the quadratic equation:

$$x^2 + 14x + 2x + 28 = 0$$

$$x(x + 14) + 2(x + 14) = 0$$

$$(x + 14)(x + 2) = 0$$

Thus, $x = -14$ or $x = -2$.

Part (ii): First, multiply the first two matrices:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1(1) + 2(2) + 1(1) & 1(2) + 2(0) + 1(0) & 1(0) + 2(1) + 1(2) \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 4 + 1 & 2 + 0 + 0 & 0 + 2 + 2 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 2 & 4 \end{bmatrix}$$

Now, multiply by the third matrix:

$$\begin{bmatrix} 6 & 2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = O$$

$$[6(0) + 2(2) + 4(x)] = [0]$$

$$0 + 4 + 4x = 0$$

$$4x = -4$$

$$x = -1$$

Final Answer

(i) $x = -2$ or $x = -14$

(ii) $x = -1$

Question 31

If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ and $f(x) = x^2 - 4x + 7$, show that $f(A) = O$. Use this to find A^3 and A^5 .

TYPE 18 Finding Higher Powers of Matrices

★ Likely Exam Question

Given: $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$, $f(x) = x^2 - 4x + 7$

To Find: Verify $f(A) = O$, and find A^3 and A^5 .

Concept / Formula Used: $A^2 - 4A + 7I = O$ $A^2 = 4A - 7I$

Solution:

First, calculate A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (2)(2) + (3)(-1) & (2)(3) + (3)(2) \\ (-1)(2) + (2)(-1) & (-1)(3) + (2)(2) \end{bmatrix} \\ &= \begin{bmatrix} 4 - 3 & 6 + 6 \\ -2 - 2 & -3 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} \end{aligned}$$

Now calculate $f(A) = A^2 - 4A + 7I$:

$$\begin{aligned} f(A) &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - 4 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 1 - 8 + 7 & 12 - 12 + 0 \\ -4 - (-4) + 0 & 1 - 8 + 7 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, $f(A) = O$ is shown.

To find A^3 , use $A^2 = 4A - 7I$:

$$\begin{aligned} A^3 &= A \cdot A^2 = A(4A - 7I) \\ &= 4A^2 - 7A \\ &= 4(4A - 7I) - 7A \\ &= 16A - 28I - 7A \\ &= 9A - 28I \end{aligned}$$

Substitute matrix A and I :

$$\begin{aligned} A^3 &= 9 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 28 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 18 & 27 \\ -9 & 18 \end{bmatrix} - \begin{bmatrix} 28 & 0 \\ 0 & 28 \end{bmatrix} \\ &= \begin{bmatrix} 18 - 28 & 27 - 0 \\ -9 - 0 & 18 - 28 \end{bmatrix} = \begin{bmatrix} -10 & 27 \\ -9 & -10 \end{bmatrix} \end{aligned}$$

To find A^5 , first find A^4 :

$$\begin{aligned} A^4 &= A \cdot A^3 = A(9A - 28I) = 9A^2 - 28A \\ &= 9(4A - 7I) - 28A = 36A - 63I - 28A \\ &= 8A - 63I \end{aligned}$$

Now, find A^5 :

$$\begin{aligned} A^5 &= A \cdot A^4 = A(8A - 63I) = 8A^2 - 63A \\ &= 8(4A - 7I) - 63A = 32A - 56I - 63A \\ &= -31A - 56I \end{aligned}$$

Substitute matrix A and I :

$$\begin{aligned} A^5 &= -31 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 56 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -62 & -93 \\ 31 & -62 \end{bmatrix} - \begin{bmatrix} 56 & 0 \\ 0 & 56 \end{bmatrix} \\ &= \begin{bmatrix} -62 - 56 & -93 - 0 \\ 31 - 0 & -62 - 56 \end{bmatrix} = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix} \end{aligned}$$

Final Answer

$$A^3 = \begin{bmatrix} -10 & 27 \\ -9 & -10 \end{bmatrix}$$

$$A^5 = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}$$

Question 32

If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, prove that $A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$, for all positive integers n .

TYPE 19 Mathematical Induction on Matrices

★ Likely Exam Question

To Prove: $P(n) : A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$ for all $n \in \mathbb{N}$

Concept / Formula Used: $P(1)$ $P(k)$ $P(k+1)$

Solution:

Let the given statement be $P(n) : A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$.

Step 1: Base Case ($n = 1$)

$$\begin{aligned} LHS &= A^1 = A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\ RHS &= \begin{bmatrix} 2^{1-1} & 2^{1-1} \\ 2^{1-1} & 2^{1-1} \end{bmatrix} = \begin{bmatrix} 2^0 & 2^0 \\ 2^0 & 2^0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \end{aligned}$$

Thus, $P(1)$ is true.

Step 2: Inductive Hypothesis Assume $P(k)$ is true for some positive integer k , i.e.,

$$A^k = \begin{bmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{bmatrix}$$

Step 3: Inductive Step We need to show that $P(k+1)$ is true, i.e., $A^{k+1} = \begin{bmatrix} 2^k & 2^k \\ 2^k & 2^k \end{bmatrix}$.

$$\begin{aligned} A^{k+1} &= A^k \cdot A \\ &= \begin{bmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2^{k-1}(1) + 2^{k-1}(1) & 2^{k-1}(1) + 2^{k-1}(1) \\ 2^{k-1}(1) + 2^{k-1}(1) & 2^{k-1}(1) + 2^{k-1}(1) \end{bmatrix} \\ &= \begin{bmatrix} 2 \cdot 2^{k-1} & 2 \cdot 2^{k-1} \\ 2 \cdot 2^{k-1} & 2 \cdot 2^{k-1} \end{bmatrix} \\ &= \begin{bmatrix} 2^k & 2^k \\ 2^k & 2^k \end{bmatrix} \end{aligned}$$

This shows that $P(k+1)$ is true whenever $P(k)$ is true. By the Principle of Mathematical Induction, $P(n)$ is true for all positive integers n .

Hence Proved

Result holds for all positive integers n .

Question 33

- (i) Find a 2×2 matrix B such that $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = 6I_2$.
- (ii) If $A = \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$, find matrix B such that $BA = I$.

TYPE 20 Solving for Unknown Matrix

Given: Matrix equations involving unknown matrix B .

To Find: Matrix B in both cases.

Solution:

Part (i): Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Given $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} &= \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} \\ \begin{bmatrix} a+b & -2a+4b \\ c+d & -2c+4d \end{bmatrix} &= \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} \end{aligned}$$

Equating elements for first row:

$$a + b = 6 \quad (7)$$

$$-2a + 4b = 0 \implies 2a = 4b \implies a = 2b \quad (8)$$

Substitute $a = 2b$ into equation (1):

$$\begin{aligned} 2b + b = 6 &\implies 3b = 6 \implies b = 2 \\ a = 2(2) &= 4 \end{aligned}$$

Equating elements for second row:

$$c + d = 0 \implies c = -d \quad (9)$$

$$-2c + 4d = 6 \quad (10)$$

Substitute $c = -d$ into equation (4):

$$\begin{aligned} -2(-d) + 4d = 6 &\implies 6d = 6 \implies d = 1 \\ c = -1 \end{aligned}$$

Thus, $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$.

Part (ii): Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Given $BA = I$:

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 2a + 3b & a + 2b \\ 2c + 3d & c + 2d \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Equating corresponding elements:

$$2a + 3b = 1 \quad (11)$$

$$a + 2b = 0 \implies a = -2b \quad (12)$$

Substitute $a = -2b$ into (5):

$$\begin{aligned} 2(-2b) + 3b = 1 &\implies -4b + 3b = 1 \implies -b = 1 \implies b = -1 \\ a = -2(-1) &= 2 \end{aligned}$$

Also,

$$2c + 3d = 0 \quad (13)$$

$$c + 2d = 1 \implies c = 1 - 2d \quad (14)$$

Substitute $c = 1 - 2d$ into (7):

$$\begin{aligned} 2(1 - 2d) + 3d &= 0 \\ 2 - 4d + 3d &= 0 \implies 2 - d = 0 \implies d = 2 \\ c = 1 - 2(2) &= 1 - 4 = -3 \end{aligned}$$

Thus, $B = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$.

Final Answer

(i) $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$, (ii) $B = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

Question 34

Find the matrix A such that $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$.

TYPE 20 Solving for Unknown Matrix

Given: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$

To Find: The matrix A .

Concept / Formula Used: Compatibility for the product: since the left factor is 2×2 and the product is 2×3 , A must be a 2×3 matrix.

Solution:

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, since the product $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} A$ must be 2×3 , so A is 2×3 .

Substituting into the given equation:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$$

Multiplying the matrices on the left-hand side:

$$\begin{bmatrix} a+d & b+e & c+f \\ d & e & f \end{bmatrix} = \begin{bmatrix} 3 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$$

Equating corresponding elements:

$$a + d = 3 \quad \text{--- (1)}$$

$$b + e = 3 \quad \text{--- (2)}$$

$$c + f = 5 \quad \text{--- (3)}$$

$$d = 1 \quad \text{--- (4)}$$

$$e = 0 \quad \text{--- (5)}$$

$$f = 1 \quad \text{--- (6)}$$

From equation (4), $d = 1$; substituting into equation (1):

$$a + 1 = 3$$

$$a = 2$$

From equation (5), $e = 0$; substituting into equation (2):

$$b + 0 = 3$$

$$b = 3$$

From equation (6), $f = 1$; substituting into equation (3):

$$c + 1 = 5$$

$$c = 4$$

Thus, $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix}$.

Final Answer

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix}$$

Common Mistake: Assuming A is 2×2 by default is a common slip – always determine the order of the unknown matrix from the compatibility of the given product first.

Question 35

Three students Ram, Mohan and Ankit go to a shop to buy stationary. Ram purchases 2 dozen notebooks, 1 dozen pens and 4 pencils, Mohan purchases 1 dozen notebook, 6 pens and 8 pencils and Ankit purchases 6 notebooks, 4 pens and 6 pencils. A notebook costs Rs. 45, a pen costs Rs. 13.50 and a pencil costs Rs. 4.50. Using matrix multiplication, find the total amount of bill by all the three students.

TYPE 21 Matrix Multiplication Word Problems

Given: Ram: 2 dozen notebooks, 1 dozen pens, 4 pencils. Mohan: 1 dozen notebooks, 6 pens, 8 pencils. Ankit: 6 notebooks, 4 pens, 6 pencils. Notebook = Rs. 45, pen = Rs. 13.50, pencil = Rs. 4.50.

To Find: Total amount of bill by all three students, using matrix multiplication.

Concept / Formula Used: Represent the quantities purchased as a matrix P (order 3×3) and the prices as a column matrix C (order 3×1); the product PC gives each student's total bill.

Solution:

Converting dozens to individual units: 1 dozen = 12 items.

Ram purchases: $2 \times 12 = 24$ notebooks, $1 \times 12 = 12$ pens, 4 pencils.

Mohan purchases: $1 \times 12 = 12$ notebooks, 6 pens, 8 pencils.

Ankit purchases: 6 notebooks, 4 pens, 6 pencils.

Let the quantity matrix (rows = students, columns = notebooks, pens, pencils) be

$$P = \begin{bmatrix} 24 & 12 & 4 \\ 12 & 6 & 8 \\ 6 & 4 & 6 \end{bmatrix}$$

and the price matrix (column vector, order 3×1) be

$$C = \begin{bmatrix} 45 \\ 13.50 \\ 4.50 \end{bmatrix}$$

Since P is 3×3 and C is 3×1 , the product PC is defined and gives a 3×1 matrix of total bills:

$$PC = \begin{bmatrix} 24 & 12 & 4 \\ 12 & 6 & 8 \\ 6 & 4 & 6 \end{bmatrix} \begin{bmatrix} 45 \\ 13.50 \\ 4.50 \end{bmatrix}$$

Computing each entry (row-by-column):

$$\begin{aligned}\text{Ram's bill} &= 24(45) + 12(13.50) + 4(4.50) \\ &= 1080 + 162 + 18 \\ &= 1260\end{aligned}$$

$$\begin{aligned}\text{Mohan's bill} &= 12(45) + 6(13.50) + 8(4.50) \\ &= 540 + 81 + 36 \\ &= 657\end{aligned}$$

$$\begin{aligned}\text{Ankit's bill} &= 6(45) + 4(13.50) + 6(4.50) \\ &= 270 + 54 + 27 \\ &= 351\end{aligned}$$

So,

$$PC = \begin{bmatrix} 1260 \\ 657 \\ 351 \end{bmatrix}$$

Total amount of bill by all three students:

$$\begin{aligned}\text{Total} &= 1260 + 657 + 351 \\ &= 2268\end{aligned}$$

Final Answer

Ram: Rs. 1260, Mohan: Rs. 657, Ankit: Rs. 351. Total bill by all three students = Rs. 2268

Common Mistake: Forgetting to convert "dozen" to 12 units before building the quantity matrix is the most common slip in this problem.

3.4 – Transpose, Symmetric and Skew-Symmetric Matrices

Question 1

If $A = \begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$, then find $(3A)'$.

TYPE 22 Matrix Transpose and Properties

Given: Matrix $A = \begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$.

Concept / Formula Used: $k \quad A \quad (kA)' = kA'$

Solution:

Given matrix A of order 2×3 :

$$A = \begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$$

First, multiply matrix A by the scalar 3:

$$\begin{aligned} 3A &= 3 \begin{bmatrix} 2 & -3 & 0 \\ -1 & 4 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 3(2) & 3(-3) & 3(0) \\ 3(-1) & 3(4) & 3(5) \end{bmatrix} \\ &= \begin{bmatrix} 6 & -9 & 0 \\ -3 & 12 & 15 \end{bmatrix} \end{aligned}$$

Now, taking the transpose $(3A)'$ by interchanging rows and columns:

$$\begin{aligned} (3A)' &= \begin{bmatrix} 6 & -9 & 0 \\ -3 & 12 & 15 \end{bmatrix}' \\ &= \begin{bmatrix} 6 & -3 \\ -9 & 12 \\ 0 & 15 \end{bmatrix} \end{aligned}$$

Final Answer

$$(3A)' = \begin{bmatrix} 6 & -3 \\ -9 & 12 \\ 0 & 15 \end{bmatrix}$$

Question 2

If $A = \begin{bmatrix} 2 & -1 & 5 \\ 4 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & -3 \end{bmatrix}$, find $A' + B'$.

TYPE 22 Matrix Transpose and Properties

Given: Matrices $A = \begin{bmatrix} 2 & -1 & 5 \\ 4 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & -3 \end{bmatrix}$.

Concept / Formula Used: $A' = A$

Solution:

First, find the transpose of matrix A :

$$\begin{aligned} A' &= \begin{bmatrix} 2 & -1 & 5 \\ 4 & 0 & 3 \end{bmatrix}' \\ &= \begin{bmatrix} 2 & 4 \\ -1 & 0 \\ 5 & 3 \end{bmatrix} \end{aligned}$$

Next, find the transpose of matrix B :

$$\begin{aligned} B' &= \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & -3 \end{bmatrix}' \\ &= \begin{bmatrix} -2 & -1 \\ 3 & 2 \\ 1 & -3 \end{bmatrix} \end{aligned}$$

Now, add the transposed matrices A' and B' element-wise:

$$\begin{aligned} A' + B' &= \begin{bmatrix} 2 & 4 \\ -1 & 0 \\ 5 & 3 \end{bmatrix} + \begin{bmatrix} -2 & -1 \\ 3 & 2 \\ 1 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 2 + (-2) & 4 + (-1) \\ -1 + 3 & 0 + 2 \\ 5 + 1 & 3 + (-3) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 3 \\ 2 & 2 \\ 6 & 0 \end{bmatrix} \end{aligned}$$

Final Answer

$$A' + B' = \begin{bmatrix} 0 & 3 \\ 2 & 2 \\ 6 & 0 \end{bmatrix}$$

Question 3

If $A = \begin{bmatrix} \sin x & -\cos x \\ \cos x & \sin x \end{bmatrix}$, $0 < x < \frac{\pi}{2}$ and $A + A' = I$ where I is unity matrix, find the value of x .

TYPE 22 Matrix Transpose and Properties

★ Likely Exam Question

Given: Matrix $A = \begin{bmatrix} \sin x & -\cos x \\ \cos x & \sin x \end{bmatrix}$ with $0 < x < \frac{\pi}{2}$ and $A + A' = I$.

Concept / Formula Used: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $A + A' = I$

Solution:

Find the transpose of A :

$$A' = \begin{bmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{bmatrix}$$

Compute $A + A'$:

$$\begin{aligned} A + A' &= \begin{bmatrix} \sin x & -\cos x \\ \cos x & \sin x \end{bmatrix} + \begin{bmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{bmatrix} \\ &= \begin{bmatrix} \sin x + \sin x & -\cos x + \cos x \\ \cos x - \cos x & \sin x + \sin x \end{bmatrix} \\ &= \begin{bmatrix} 2 \sin x & 0 \\ 0 & 2 \sin x \end{bmatrix} \end{aligned}$$

Given $A + A' = I_2$:

$$\begin{bmatrix} 2 \sin x & 0 \\ 0 & 2 \sin x \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Equating corresponding elements:

$$2 \sin x = 1$$

$$\sin x = \frac{1}{2}$$

Since $0 < x < \frac{\pi}{2}$, we solve for x :

$$x = \frac{\pi}{6}$$

Final Answer

$$x = \frac{\pi}{6}$$

Common Mistake: Watch out for element positions: $\sin x$ is on the principal diagonal, so adding $A + A'$ gives $2 \sin x$, not $2 \cos x$.

Question 4

If $A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, then find $A^T - B^T$.

TYPE 22 Matrix Transpose and Properties

Given: $A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$.

Concept / Formula Used: $B_{m \times n} \quad B_{n \times m}^T$

Solution:

The matrix A^T is already given as:

$$A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$$

Now, find the transpose of B :

$$\begin{aligned} B^T &= \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}^T \\ &= \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix} \end{aligned}$$

Subtract B^T from A^T element-wise:

$$\begin{aligned} A^T - B^T &= \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 3 - (-1) & 4 - 1 \\ -1 - 2 & 2 - 2 \\ 0 - 1 & 1 - 3 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix} \end{aligned}$$

Final Answer

$$A^T - B^T = \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix}$$

Question 5

For what value of k the matrix $\begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix}$ is a skew-symmetric?

TYPE 23 Symmetric and Skew-Symmetric Matrices

★ Likely Exam Question

Given: Matrix $M = \begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix}$ is skew-symmetric.

Concept / Formula Used: $M^T = -M$ $a_{ij} = -a_{ji}$

Solution:

Let $M = \begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix}$. Find the transpose M^T :

$$M^T = \begin{bmatrix} 0 & -6 \\ k & 0 \end{bmatrix}$$

Find $-M$:

$$\begin{aligned} -M &= -\begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -k \\ 6 & 0 \end{bmatrix} \end{aligned}$$

Since M is skew-symmetric, $M^T = -M$:

$$\begin{bmatrix} 0 & -6 \\ k & 0 \end{bmatrix} = \begin{bmatrix} 0 & -k \\ 6 & 0 \end{bmatrix}$$

Equating off-diagonal elements:

$$\begin{aligned} -k &= -6 \implies k = 6 \\ k &= 6 \end{aligned}$$

Final Answer

$$k = 6$$

Question 6 (i)

Show that $(A + A')$ is a symmetric matrix, if $A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$.

TYPE 23 Symmetric and Skew-Symmetric Matrices

★ Likely Exam Question

To Prove: $(A + A')$ is a symmetric matrix.

Concept / Formula Used: $X \quad X' = X$

Solution:

Given matrix A :

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$$

Find the transpose A' :

$$A' = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Compute $X = A + A'$:

$$\begin{aligned} X = A + A' &= \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 2+2 & 4+3 \\ 3+4 & 5+5 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 7 \\ 7 & 10 \end{bmatrix} \end{aligned}$$

Now find the transpose of X :

$$\begin{aligned} X' &= (A + A')' = \begin{bmatrix} 4 & 7 \\ 7 & 10 \end{bmatrix}' \\ &= \begin{bmatrix} 4 & 7 \\ 7 & 10 \end{bmatrix} \\ &= X \end{aligned}$$

Since $(A + A')' = A + A'$, $(A + A')$ is a symmetric matrix.

Hence Proved

$(A + A')$ is a symmetric matrix.

Question 6 (ii)

If $A = \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix}$, determine whether $A - A^T$ is symmetric or skew-symmetric.

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: Matrix $A = \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix}$.

Concept / Formula Used: $Y^T = Y$ $Y^T = -Y$

Solution:

Given matrix A :

$$A = \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix}$$

Find the transpose A^T :

$$A^T = \begin{bmatrix} 5 & -2 \\ -1 & 6 \end{bmatrix}$$

Compute $Y = A - A^T$:

$$\begin{aligned} Y = A - A^T &= \begin{bmatrix} 5 & -1 \\ -2 & 6 \end{bmatrix} - \begin{bmatrix} 5 & -2 \\ -1 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 5-5 & -1-(-2) \\ -2-(-1) & 6-6 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{aligned}$$

Take transpose of Y :

$$Y^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Notice that:

$$-Y = -\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Since $Y^T = -Y$, matrix $A - A^T$ is skew-symmetric.

Final Answer

$A - A^T$ is a skew-symmetric matrix.

Question 7 (i)

If $A = \begin{bmatrix} 5 & a \\ b & 0 \end{bmatrix}$ and A is symmetric matrix, show that $a = b$.

TYPE 23 Symmetric and Skew-Symmetric Matrices

★ Likely Exam Question

Given: Matrix $A = \begin{bmatrix} 5 & a \\ b & 0 \end{bmatrix}$ is symmetric.

To Prove: $a = b$

Concept / Formula Used: $A = A' = A$

Solution:

Find the transpose A' :

$$A' = \begin{bmatrix} 5 & b \\ a & 0 \end{bmatrix}$$

Since A is given to be a symmetric matrix, we have $A' = A$:

$$\begin{bmatrix} 5 & b \\ a & 0 \end{bmatrix} = \begin{bmatrix} 5 & a \\ b & 0 \end{bmatrix}$$

Equating the element at position $(1, 2)$ (or $(2, 1)$):

$$b = a$$

$$a = b$$

Hence Proved

$$a = b$$

Question 7 (ii)

If $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$ and A is symmetric, then write the relation between x and y .

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: Matrix $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$ is symmetric.

Concept / Formula Used: $A = A' = A$ $a_{12} = a_{21}$

Solution:

Given $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$. Find transpose A' :

$$A' = \begin{bmatrix} 5 & y \\ x & 0 \end{bmatrix}$$

Since A is symmetric, $A' = A$:

$$\begin{bmatrix} 5 & y \\ x & 0 \end{bmatrix} = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$$

Equating corresponding elements:

$$x = y$$

Final Answer

$$x = y$$

Question 7 (iii)

If the matrix $\begin{bmatrix} 6 & -x^2 \\ 2x - 15 & 10 \end{bmatrix}$ is symmetric, find the value(s) of x .

TYPE 23 Symmetric and Skew-Symmetric Matrices

★ Likely Exam Question

Given: Matrix $M = \begin{bmatrix} 6 & -x^2 \\ 2x - 15 & 10 \end{bmatrix}$ is symmetric.

Concept / Formula Used: $a_{12} = a_{21}$

Solution:

Equating elements at positions (1, 2) and (2, 1):

$$-x^2 = 2x - 15$$

Rearranging all terms to one side:

$$x^2 + 2x - 15 = 0$$

Factorize the quadratic expression:

$$x^2 + 5x - 3x - 15 = 0$$

$$x(x + 5) - 3(x + 5) = 0$$

$$(x + 5)(x - 3) = 0$$

Solving for x :

$$x + 5 = 0 \implies x = -5$$

$$x - 3 = 0 \implies x = 3$$

Final Answer

$$x = 3 \text{ or } x = -5$$

Question 8

If A is a square matrix, prove that $A'A$ is symmetric.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

To Prove: $A'A$ is a symmetric matrix for any square matrix A .

Concept / Formula Used: $(AB)' = B'A'$ $((A')') = A$

Solution:

Let $X = A'A$. To prove X is symmetric, we must show that $X' = X$. Take the transpose of X :

$$X' = (A'A)'$$

Using the reversal rule of transpose $(AB)' = B'A'$ where first matrix is A' and second is A :

$$X' = A'(A)'$$

Using the identity $(A')' = A$:

$$X' = A'A$$

Since $X' = X$, the matrix $A'A$ is symmetric.

Hence Proved

$A'A$ is a symmetric matrix.

Question 9

If $A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$ is written as $A = P + Q$, where P is a symmetric matrix and Q is a skew-symmetric matrix, then write the matrix P .

TYPE 25 Decomposing a Matrix into Symmetric and Skew-Symmetric Parts

Given: $A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$, with $A = P + Q$, P symmetric, Q skew-symmetric.

Concept / Formula Used: $A = P + Q$ $P = \frac{1}{2}(A + A')$ $Q = \frac{1}{2}(A - A')$

Solution:

Given matrix A :

$$A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$$

Find A' :

$$A' = \begin{bmatrix} 3 & 7 \\ 5 & 9 \end{bmatrix}$$

The symmetric matrix component P is given by:

$$\begin{aligned} P &= \frac{1}{2}(A + A') \\ &= \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix} + \begin{bmatrix} 3 & 7 \\ 5 & 9 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 3+3 & 5+7 \\ 7+5 & 9+9 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 6 & 12 \\ 12 & 18 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix} \end{aligned}$$

Final Answer

$$P = \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}$$

Question 10

If A, B are symmetric matrices and $AB = BA$, then show that AB is symmetric.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

Given: $A' = A$, $B' = B$, and $AB = BA$.

To Prove: AB is symmetric, i.e., $(AB)' = AB$.

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Given that A and B are symmetric matrices:

$$A' = A$$

$$B' = B$$

Consider the transpose of the matrix product AB :

$$(AB)' = B'A' \quad (\text{using reversal law of transpose})$$

Substitute $B' = B$ and $A' = A$:

$$(AB)' = BA$$

Given that $AB = BA$, substituting $BA = AB$:

$$(AB)' = AB$$

Since $(AB)' = AB$, the matrix AB is symmetric.

Hence Proved

AB is a symmetric matrix.

Question 11

If A, B and AB are all symmetric matrices, then show that $AB = BA$.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

Given: $A' = A$, $B' = B$, and $(AB)' = AB$.

To Prove: $AB = BA$

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Since A and B are symmetric matrices:

$$A' = A$$

$$B' = B$$

Since AB is also a symmetric matrix:

$$(AB)' = AB$$

By the reversal property of matrix transpose:

$$(AB)' = B'A'$$

Substitute $B' = B$ and $A' = A$:

$$(AB)' = BA$$

Equating both expressions for $(AB)'$:

$$AB = BA$$

Hence Proved

$$AB = BA$$

Question 12

If A, B are skew-symmetric matrices and $AB = BA$, then show that AB is symmetric.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

Given: $A' = -A$, $B' = -B$, and $AB = BA$.

To Prove: AB is symmetric, i.e., $(AB)' = AB$.

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Given A and B are skew-symmetric:

$$A' = -A$$

$$B' = -B$$

Take the transpose of AB :

$$(AB)' = B'A' \quad (\text{by reversal law of transpose})$$

Substitute $B' = -B$ and $A' = -A$:

$$\begin{aligned} (AB)' &= (-B)(-A) \\ &= BA \end{aligned}$$

Since $AB = BA$, we have:

$$(AB)' = AB$$

Since $(AB)' = AB$, AB is symmetric.

Hence Proved

AB is a symmetric matrix.

Question 13

If A, B are skew-symmetric matrices of same order and AB is symmetric, then show that $AB = BA$.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

Given: $A' = -A$, $B' = -B$, and $(AB)' = AB$.

To Prove: $AB = BA$

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Given that A and B are skew-symmetric matrices:

$$A' = -A$$

$$B' = -B$$

Since AB is symmetric, we have:

$$(AB)' = AB$$

Using the reversal property of transpose on LHS:

$$(AB)' = B'A'$$

Substitute $B' = -B$ and $A' = -A$:

$$\begin{aligned}(AB)' &= (-B)(-A) \\ &= BA\end{aligned}$$

Equating the two results for $(AB)'$:

$$AB = BA$$

Hence Proved

$$AB = BA$$

Question 14

Let $A = \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, verify that (i) $(A')' = A$ (ii) $(3A)' = 3A'$ (iii) $(A - B)' = A' - B'$ (iv) $(AB)' = B'A'$

TYPE 22 Matrix Transpose and Properties

Given: $A = \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$.

Concept / Formula Used: $(A')' = A$ $(kA)' = kA'$ $(A - B)' = A' - B'$ $(AB)' = B'A'$

Solution:

Find basic transposes first:

$$A' = \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix}, \quad B' = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

(i) **Verification of $(A')' = A$:**

$$\text{LHS} = (A')' = \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix}' = \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} = A = \text{RHS}$$

(ii) **Verification of $(3A)' = 3A'$:**

$$\begin{aligned} 3A &= 3 \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 15 & -3 \\ 18 & 21 \end{bmatrix} \\ \text{LHS} &= (3A)' = \begin{bmatrix} 15 & 18 \\ -3 & 21 \end{bmatrix} \\ \text{RHS} &= 3A' = 3 \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 18 \\ -3 & 21 \end{bmatrix} \\ \text{LHS} &= \text{RHS} \end{aligned}$$

(iii) **Verification of $(A - B)' = A' - B'$:**

$$\begin{aligned} A - B &= \begin{bmatrix} 5-2 & -1-1 \\ 6-3 & 7-4 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 3 & 3 \end{bmatrix} \\ \text{LHS} &= (A - B)' = \begin{bmatrix} 3 & 3 \\ -2 & 3 \end{bmatrix} \\ \text{RHS} &= A' - B' = \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 5-2 & 6-3 \\ -1-1 & 7-4 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ -2 & 3 \end{bmatrix} \\ \text{LHS} &= \text{RHS} \end{aligned}$$

(iv) **Verification of $(AB)' = B'A'$:**

$$\begin{aligned} AB &= \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5(2) + (-1)(3) & 5(1) + (-1)(4) \\ 6(2) + 7(3) & 6(1) + 7(4) \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 33 & 34 \end{bmatrix} \\ \text{LHS} &= (AB)' = \begin{bmatrix} 7 & 33 \\ 1 & 34 \end{bmatrix} \\ \text{RHS} &= B'A' = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 2(5) + 3(-1) & 2(6) + 3(7) \\ 1(5) + 4(-1) & 1(6) + 4(7) \end{bmatrix} = \begin{bmatrix} 7 & 33 \\ 1 & 34 \end{bmatrix} \\ \text{LHS} &= \text{RHS} \end{aligned}$$

Hence Proved

All four identities are verified.

Question 15

If $A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, then verify that (i) $(A')' = A$ (ii) $(A + B)' = A' + B'$ (iii) $(k\bar{B})' = k\bar{B}'$, where k is any real number.

Given: $A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$.

Concept / Formula Used: $2 \times 3 \quad 3 \times 2$

Solution:

Transposes of A and B :

$$A' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix}, \quad B' = \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix}$$

(i) **Verify** $(A')' = A$:

$$(A')' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix}' = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix} = A$$

(ii) **Verify** $(A + B)' = A' + B'$:

$$\begin{aligned} A + B &= \begin{bmatrix} 3+2 & \sqrt{3}-1 & 2+2 \\ 4+1 & 2+2 & 0+4 \end{bmatrix} = \begin{bmatrix} 5 & \sqrt{3}-1 & 4 \\ 5 & 4 & 4 \end{bmatrix} \\ \text{LHS} = (A + B)' &= \begin{bmatrix} 5 & 5 \\ \sqrt{3}-1 & 4 \\ 4 & 4 \end{bmatrix} \\ \text{RHS} = A' + B' &= \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 3+2 & 4+1 \\ \sqrt{3}-1 & 2+2 \\ 2+2 & 0+4 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ \sqrt{3}-1 & 4 \\ 4 & 4 \end{bmatrix} \\ \text{LHS} &= \text{RHS} \end{aligned}$$

(iii) **Verify** $(kB)' = kB'$:

$$\begin{aligned} kB &= \begin{bmatrix} 2k & -k & 2k \\ k & 2k & 4k \end{bmatrix} \\ \text{LHS} = (kB)' &= \begin{bmatrix} 2k & k \\ -k & 2k \\ 2k & 4k \end{bmatrix} \\ \text{RHS} = kB' &= k \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2k & k \\ -k & 2k \\ 2k & 4k \end{bmatrix} \\ \text{LHS} &= \text{RHS} \end{aligned}$$

Hence Proved

All three properties are verified.

Question 16

If $A' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$, then find $(A + 2B)'$.

TYPE 22 Matrix Transpose and Properties

Given: $A' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$.

Concept / Formula Used: $(A + 2B)' = A' + 2B'$

Solution:

Method 1: Find matrix A first.

$$A = (A')' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}' = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$$

Calculate $2B$:

$$2B = 2 \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 2 & 4 \end{bmatrix}$$

Compute $A + 2B$:

$$\begin{aligned} A + 2B &= \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} -2 & 0 \\ 2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -2 + (-2) & 1 + 0 \\ 3 + 2 & 2 + 4 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 1 \\ 5 & 6 \end{bmatrix} \end{aligned}$$

Take transpose of $A + 2B$:

$$\begin{aligned} (A + 2B)' &= \begin{bmatrix} -4 & 1 \\ 5 & 6 \end{bmatrix}' \\ &= \begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix} \end{aligned}$$

Final Answer

$$(A + 2B)' = \begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix}$$

Question 17

If $A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$, verify that $(AB)' = B'A'$.

TYPE 22 Matrix Transpose and Properties

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$ (3×1) and $B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$ (1×3).

Concept / Formula Used: $(AB)' = B'A'$

Solution:

First calculate the product matrix AB (3×3):

$$\begin{aligned} AB &= \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(-1) & 1(2) & 1(1) \\ -4(-1) & -4(2) & -4(1) \\ 3(-1) & 3(2) & 3(1) \end{bmatrix} \\ &= \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix} \end{aligned}$$

Now, find the LHS, $(AB)'$:

$$\begin{aligned} \text{LHS} = (AB)' &= \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix}' \\ &= \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix} \end{aligned}$$

Next, find B' and A' :

$$\begin{aligned} B' &= \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad (3 \times 1) \\ A' &= \begin{bmatrix} 1 & -4 & 3 \end{bmatrix} \quad (1 \times 3) \end{aligned}$$

Calculate the RHS, $B'A'$:

$$\begin{aligned} \text{RHS} = B'A' &= \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -4 & 3 \end{bmatrix} \\ &= \begin{bmatrix} -1(1) & -1(-4) & -1(3) \\ 2(1) & 2(-4) & 2(3) \\ 1(1) & 1(-4) & 1(3) \end{bmatrix} \\ &= \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$\text{LHS} = \text{RHS}$$

Hence Proved

$(AB)' = B'A'$ is verified.

Common Mistake: Keep track of matrix dimensions: A is 3×1 and B is 1×3 , making AB a 3×3 matrix.

Question 18

If $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 6 & 4 \\ 7 & 3 \end{bmatrix}$, then verify that (i) $(2A + B)' = 2A' + B'$ (ii) $(A - B)' = A' - B'$

TYPE 22 Matrix Transpose and Properties

Given: $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 6 & 4 \\ 7 & 3 \end{bmatrix}$.

Solution:

Find transposes A' and B' (2×3):

$$A' = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 1 & 6 \end{bmatrix}, \quad B' = \begin{bmatrix} 1 & 6 & 7 \\ 2 & 4 & 3 \end{bmatrix}$$

(i) **Verification of $(2A + B)' = 2A' + B'$:**

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 8 & 2 \\ 10 & 12 \end{bmatrix} \\ 2A + B &= \begin{bmatrix} 2+1 & 4+2 \\ 8+6 & 2+4 \\ 10+7 & 12+3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 14 & 6 \\ 17 & 15 \end{bmatrix} \\ \text{LHS} = (2A + B)' &= \begin{bmatrix} 3 & 14 & 17 \\ 6 & 6 & 15 \end{bmatrix} \\ 2A' &= 2 \begin{bmatrix} 1 & 4 & 5 \\ 2 & 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 8 & 10 \\ 4 & 2 & 12 \end{bmatrix} \\ \text{RHS} = 2A' + B' &= \begin{bmatrix} 2+1 & 8+6 & 10+7 \\ 4+2 & 2+4 & 12+3 \end{bmatrix} = \begin{bmatrix} 3 & 14 & 17 \\ 6 & 6 & 15 \end{bmatrix} \\ \text{LHS} &= \text{RHS} \end{aligned}$$

(ii) **Verification of $(A - B)' = A' - B'$:**

$$\begin{aligned} A - B &= \begin{bmatrix} 1-1 & 2-2 \\ 4-6 & 1-4 \\ 5-7 & 6-3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -2 & -3 \\ -2 & 3 \end{bmatrix} \\ \text{LHS} = (A - B)' &= \begin{bmatrix} 0 & -2 & -2 \\ 0 & -3 & 3 \end{bmatrix} \\ \text{RHS} = A' - B' &= \begin{bmatrix} 1-1 & 4-6 & 5-7 \\ 2-2 & 1-4 & 6-3 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -2 \\ 0 & -3 & 3 \end{bmatrix} \\ \text{LHS} &= \text{RHS} \end{aligned}$$

Hence Proved

Both identities are verified.

Question 19

If $A = \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix}$, verify that $(AB)' = B'A'$.

TYPE 22 Matrix Transpose and Properties

Given: $A = \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix}$ (2×3) and $B = \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix}$ (3×2).

Concept / Formula Used: $(AB)' = B'A'$

Solution:

Compute product AB (2×2):

$$\begin{aligned} AB &= \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2(1) + 4(2) + 0(1) & 2(4) + 4(8) + 0(3) \\ 3(1) + 9(2) + 6(1) & 3(4) + 9(8) + 6(3) \end{bmatrix} \\ &= \begin{bmatrix} 2 + 8 + 0 & 8 + 32 + 0 \\ 3 + 18 + 6 & 12 + 72 + 18 \end{bmatrix} \\ &= \begin{bmatrix} 10 & 40 \\ 27 & 102 \end{bmatrix} \end{aligned}$$

Find $\text{LHS} = (AB)'$:

$$\text{LHS} = (AB)' = \begin{bmatrix} 10 & 27 \\ 40 & 102 \end{bmatrix}$$

Find transposes B' (2×3) and A' (3×2):

$$B' = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 8 & 3 \end{bmatrix}, \quad A' = \begin{bmatrix} 2 & 3 \\ 4 & 9 \\ 0 & 6 \end{bmatrix}$$

Compute $\text{RHS} = B'A'$ (2×2):

$$\begin{aligned} \text{RHS} = B'A' &= \begin{bmatrix} 1 & 2 & 1 \\ 4 & 8 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 9 \\ 0 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 1(2) + 2(4) + 1(0) & 1(3) + 2(9) + 1(6) \\ 4(2) + 8(4) + 3(0) & 4(3) + 8(9) + 3(6) \end{bmatrix} \\ &= \begin{bmatrix} 2 + 8 + 0 & 3 + 18 + 6 \\ 8 + 32 + 0 & 12 + 72 + 18 \end{bmatrix} \\ &= \begin{bmatrix} 10 & 27 \\ 40 & 102 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$\text{LHS} = \text{RHS}$$

Hence Proved

$(AB)' = B'A'$ is verified.

Question 20

If $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 2 & 5 \\ 3 & 4 \end{bmatrix}$, find $(BA)'$.

TYPE 22 Matrix Transpose and Properties

Given: $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ (2×2) and $B = \begin{bmatrix} -1 & 0 \\ 2 & 5 \\ 3 & 4 \end{bmatrix}$ (3×2).

Concept / Formula Used: $(BA)' = BA' \quad (BA)' = A'B'$

Solution:

Matrix B has order 3×2 and matrix A has order 2×2 . The product BA is defined and will have order 3×2 :

$$\begin{aligned} BA &= \begin{bmatrix} -1 & 0 \\ 2 & 5 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -1(3) + 0(-1) & -1(2) + 0(1) \\ 2(3) + 5(-1) & 2(2) + 5(1) \\ 3(3) + 4(-1) & 3(2) + 4(1) \end{bmatrix} \\ &= \begin{bmatrix} -3 + 0 & -2 + 0 \\ 6 - 5 & 4 + 5 \\ 9 - 4 & 6 + 4 \end{bmatrix} \\ &= \begin{bmatrix} -3 & -2 \\ 1 & 9 \\ 5 & 10 \end{bmatrix} \end{aligned}$$

Now take transpose of BA :

$$\begin{aligned} (BA)' &= \begin{bmatrix} -3 & -2 \\ 1 & 9 \\ 5 & 10 \end{bmatrix}' \\ &= \begin{bmatrix} -3 & 1 & 5 \\ -2 & 9 & 10 \end{bmatrix} \end{aligned}$$

Final Answer

$$(BA)' = \begin{bmatrix} -3 & 1 & 5 \\ -2 & 9 & 10 \end{bmatrix}$$

Question 21

If $A = \begin{bmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{bmatrix}$, then verify that $A'A = I$.

TYPE 26 Orthogonal Matrix Properties

Given: $A = \begin{bmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{bmatrix}$

To Prove: $A'A = I$

Concept / Formula Used: $(A')_{ij} = A_{ji}$ $\sin^2 \alpha + \cos^2 \alpha = 1$

Solution:

By interchanging the rows and columns of matrix A , we obtain its transpose A' :

$$A' = \begin{bmatrix} \sin \alpha & -\cos \alpha \\ \cos \alpha & \sin \alpha \end{bmatrix}$$

Now, computing the product $A'A$:

$$\begin{aligned} A'A &= \begin{bmatrix} \sin \alpha & -\cos \alpha \\ \cos \alpha & \sin \alpha \end{bmatrix} \begin{bmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{bmatrix} \\ &= \begin{bmatrix} (\sin \alpha)(\sin \alpha) + (-\cos \alpha)(-\cos \alpha) & (\sin \alpha)(\cos \alpha) + (-\cos \alpha)(\sin \alpha) \\ (\cos \alpha)(\sin \alpha) + (\sin \alpha)(-\cos \alpha) & (\cos \alpha)(\cos \alpha) + (\sin \alpha)(\sin \alpha) \end{bmatrix} \\ &= \begin{bmatrix} \sin^2 \alpha + \cos^2 \alpha & \sin \alpha \cos \alpha - \cos \alpha \sin \alpha \\ \cos \alpha \sin \alpha - \sin \alpha \cos \alpha & \cos^2 \alpha + \sin^2 \alpha \end{bmatrix} \end{aligned}$$

Using the standard trigonometric identity $\sin^2 \alpha + \cos^2 \alpha = 1$:

$$\begin{aligned} A'A &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Hence Proved

$A'A = I$, hence verified.

Question 22

If $A = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$, show that $AA' = I$.

TYPE 26 Orthogonal Matrix Properties

Given: $A = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$

To Prove: $AA' = I$

Concept / Formula Used: $(kA)' = kA'$ $(AB)_{ij} = \sum_k a_{ik}b_{kj}$

Solution:

Taking the transpose of matrix A :

$$A' = \frac{1}{3} \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{bmatrix}$$

Now, computing the product AA' :

$$\begin{aligned} AA' &= \left(\frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix} \right) \left(\frac{1}{3} \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{bmatrix} \right) \\ &= \frac{1}{9} \begin{bmatrix} 1(1) + (-2)(-2) + 2(2) & 1(-2) + (-2)(1) + 2(2) & 1(-2) + (-2)(-2) + 2(-1) \\ -2(1) + 1(-2) + 2(2) & -2(-2) + 1(1) + 2(2) & -2(-2) + 1(-2) + 2(-1) \\ -2(1) + (-2)(-2) + (-1)(2) & -2(-2) + (-2)(1) + (-1)(2) & -2(-2) + (-2)(-2) + (-1)(-1) \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 1+4+4 & -2-2+4 & -2+4-2 \\ -2-2+4 & 4+1+4 & 4-2-2 \\ -2+4-2 & 4-2-2 & 4+4+1 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Hence Proved

$AA' = I$, hence proved.

Question 23

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & x \\ -2 & 2 & -1 \end{bmatrix}$ is a matrix satisfying $AA' = 9I_3$, find x .

TYPE 26 Orthogonal Matrix Properties

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & x \\ -2 & 2 & -1 \end{bmatrix}$ and $AA' = 9I_3$

To Find: Value of x

Concept / Formula Used: $AA' = 9I_3$ $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Solution:

The transpose of matrix A is given by:

$$A' = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & x & -1 \end{bmatrix}$$

Evaluating the matrix product AA' :

$$\begin{aligned} AA' &= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & x \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & x & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(2) + 2(2) & 1(2) + 2(1) + 2(x) & 1(-2) + 2(2) + 2(-1) \\ 2(1) + 1(2) + x(2) & 2(2) + 1(1) + x(x) & 2(-2) + 1(2) + x(-1) \\ -2(1) + 2(2) + (-1)(2) & -2(2) + 2(1) + (-1)(x) & -2(-2) + 2(2) + (-1)(-1) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 4 + 4 & 2 + 2 + 2x & -2 + 4 - 2 \\ 2 + 2 + 2x & 4 + 1 + x^2 & -4 + 2 - x \\ -2 + 4 - 2 & -4 + 2 - x & 4 + 4 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 4 + 2x & 0 \\ 4 + 2x & 5 + x^2 & -2 - x \\ 0 & -2 - x & 9 \end{bmatrix} \end{aligned}$$

We are given $AA' = 9I_3$:

$$\begin{bmatrix} 9 & 4 + 2x & 0 \\ 4 + 2x & 5 + x^2 & -2 - x \\ 0 & -2 - x & 9 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Equating corresponding elements: From element $(1, 2)$:

$$\begin{aligned} 4 + 2x &= 0 \\ 2x &= -4 \\ x &= -2 \end{aligned}$$

Verifying from element $(2, 3)$:

$$\begin{aligned} -2 - x &= 0 \\ x &= -2 \end{aligned}$$

Verifying from element $(2, 2)$:

$$\begin{aligned} 5 + x^2 &= 9 \\ 5 + (-2)^2 &= 5 + 4 = 9 \quad (\text{Satisfied}) \end{aligned}$$

Final Answer

$$x = -2$$

Question 24

Find the integral value of x if $\begin{bmatrix} x & 4 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x & 4 & -1 \end{bmatrix}' = O$.

TYPE 22 Matrix Transpose and Properties

Given: $[x \ 4 \ -1] \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 2 & 4 \end{bmatrix} [x \ 4 \ -1]' = O$

To Find: Integral value of x

Concept / Formula Used: $(XA)X' = O \quad ax^2 + bx + c = 0$

Solution:

The transpose of the row vector $[x \ 4 \ -1]$ is the column vector:

$$[x \ 4 \ -1]' = \begin{bmatrix} x \\ 4 \\ -1 \end{bmatrix}$$

First, compute the product of the first two matrices:

$$\begin{aligned} [x \ 4 \ -1] \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 2 & 4 \end{bmatrix} &= [2(x) + 1(4) + 2(-1) \quad 1(x) + 0(4) + 2(-1) \quad -1(x) + 0(4) + 4(-1)] \\ &= [2x + 4 - 2 \quad x + 0 - 2 \quad -x + 0 - 4] \\ &= [2x + 2 \quad x - 2 \quad -x - 4] \end{aligned}$$

Now, multiply this resulting row vector by the column vector:

$$\begin{aligned} [2x + 2 \quad x - 2 \quad -x - 4] \begin{bmatrix} x \\ 4 \\ -1 \end{bmatrix} &= [0] \\ (2x + 2)(x) + (x - 2)(4) + (-x - 4)(-1) &= 0 \\ 2x^2 + 2x + 4x - 8 + x + 4 &= 0 \\ 2x^2 + 7x - 4 &= 0 \end{aligned}$$

Factoring the quadratic equation:

$$\begin{aligned} 2x^2 + 8x - x - 4 &= 0 \\ 2x(x + 4) - 1(x + 4) &= 0 \\ (2x - 1)(x + 4) &= 0 \end{aligned}$$

Solving for x :

$$\begin{aligned} 2x - 1 = 0 &\implies x = \frac{1}{2} \\ x + 4 = 0 &\implies x = -4 \end{aligned}$$

Since x must be an integral value (an integer), we reject $x = \frac{1}{2}$.

Final Answer

$$x = -4$$

Common Mistake: Taking $x = \frac{1}{2}$ as part of the final answer without noticing the question explicitly asks for the integral value of x .

Question 25

Show that the matrix A , where $A = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, is a symmetric matrix.

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: $A = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$

To Prove: A is a symmetric matrix

Concept / Formula Used: $A = A' = A$

Solution:

Find the transpose A' by interchanging the rows and columns of matrix A :

$$A' = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$$

Comparing matrix A' with matrix A :

$$A' = A$$

Hence Proved

Since $A' = A$, the matrix A is a symmetric matrix.

Question 26

If the matrix $\begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$ is symmetric, find the values of a and b .

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: $A = \begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$ is symmetric

To Find: Values of a and b

Concept / Formula Used: $a_{ij} = a_{ji} \quad i \neq j$

Solution:

Let $A = \begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$. Since A is a symmetric matrix, we have $A' = A$, which implies $a_{ij} = a_{ji}$ for all i, j .

Equating a_{12} and a_{21} :

$$2b = 3$$

$$b = \frac{3}{2}$$

Equating a_{13} and a_{31} :

$$3a = -2$$

$$a = -\frac{2}{3}$$

Checking a_{23} and a_{32} :

$$3 = 3 \quad (\text{which holds naturally})$$

Final Answer

$$a = -\frac{2}{3}, \quad b = \frac{3}{2}$$

Question 27

If A and B are symmetric matrices of the same order, prove that $AB + BA$ is symmetric.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

Given: A and B are symmetric matrices of the same order, so $A' = A$ and $B' = B$

To Prove: $AB + BA$ is a symmetric matrix

Concept / Formula Used: $(X + Y)' = X' + Y'$ $(XY)' = Y'X'$

Solution:

To prove that $AB + BA$ is symmetric, we must show that $(AB + BA)' = AB + BA$.

Taking the transpose of $(AB + BA)$:

$$(AB + BA)' = (AB)' + (BA)' \quad [\text{using } (X + Y)' = X' + Y']$$

Applying the reversal law of transposes $(XY)' = Y'X'$:

$$= B'A' + A'B'$$

Since A and B are symmetric matrices, substitute $A' = A$ and $B' = B$:

$$= BA + AB$$

Using commutativity of matrix addition:

$$= AB + BA$$

Thus, $(AB + BA)' = AB + BA$.

Hence Proved

Hence, $AB + BA$ is a symmetric matrix.

Question 28

If A and B are symmetric matrices such that AB and BA are both defined, then prove that $AB - BA$ is a skew-symmetric matrix.

TYPE 24 Algebraic Proofs of Symmetric and Skew-Symmetric Matrices

★ Likely Exam Question

Given: A and B are symmetric matrices of the same order, so $A' = A$ and $B' = B$

To Prove: $AB - BA$ is a skew-symmetric matrix

Concept / Formula Used: $X \quad X' = -X \quad (X - Y)' = X' - Y' \quad (XY)' = Y'X'$

Solution:

To prove that $AB - BA$ is skew-symmetric, we must show that $(AB - BA)' = -(AB - BA)$.
Taking the transpose of $(AB - BA)$:

$$(AB - BA)' = (AB)' - (BA)' \quad [\text{using } (X - Y)' = X' - Y']$$

Applying the reversal law of transposes $(XY)' = Y'X'$:

$$= B'A' - A'B'$$

Since A and B are symmetric matrices, substitute $A' = A$ and $B' = B$:

$$= BA - AB$$

Factoring out a negative sign:

$$= -(AB - BA)$$

Thus, $(AB - BA)' = -(AB - BA)$.

Hence Proved

Hence, $AB - BA$ is a skew-symmetric matrix.

Question 29

Express each of the following matrices as the sum of a symmetric and a skew-symmetric matrices:

$$\begin{aligned} \text{(i)} & \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} \\ \text{(ii)} & \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \end{aligned}$$

TYPE 25 Decomposing a Matrix into Symmetric and Skew-Symmetric Parts

Given: Matrices $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ for part (i) and $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ for part (ii)

To Find: Decomposition $A = P + Q$ where $P' = P$ and $Q' = -Q$

Concept / Formula Used: $A = P + Q$ $P = \frac{1}{2}(A + A')$ $Q = \frac{1}{2}(A - A')$

Solution:

Part (i): Let $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$.

1. Find the transpose A' :

$$A' = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}$$

2. Compute the symmetric part $P = \frac{1}{2}(A + A')$:

$$\begin{aligned} A + A' &= \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix} \\ P &= \frac{1}{2} \begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} \end{aligned}$$

3. Compute the skew-symmetric part $Q = \frac{1}{2}(A - A')$:

$$\begin{aligned} A - A' &= \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} \\ Q &= \frac{1}{2} \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \end{aligned}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$.

1. Find the transpose A' :

$$A' = \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix}$$

2. Compute the symmetric part $P = \frac{1}{2}(A + A')$:

$$\begin{aligned} A + A' &= \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & -2 \end{bmatrix} \\ P &= \frac{1}{2} \begin{bmatrix} 6 & -3 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix} \end{aligned}$$

3. Compute the skew-symmetric part $Q = \frac{1}{2}(A - A')$:

$$\begin{aligned} A - A' &= \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} \\ Q &= \frac{1}{2} \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix} \end{aligned}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$$

Final Answer

$$(i) \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$$

Question 30

Express the following matrices as the sum of symmetric and skew-symmetric matrices:

$$(i) \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$$

TYPE 25 Decomposing a Matrix into Symmetric and Skew-Symmetric Parts

Given: Matrices $A = \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix}$ for part (i) and $A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$ for part (ii)

To Find: Decomposition $A = P + Q$ where $P' = P$ and $Q' = -Q$

Concept / Formula Used: $A = P + Q$ $P = \frac{1}{2}(A + A')$ $Q = \frac{1}{2}(A - A')$

Solution:

Part (i): Let $A = \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix}$.

1. Find the transpose A' :

$$A' = \begin{bmatrix} 3 & 4 & 0 \\ 2 & 1 & 6 \\ 5 & 3 & 7 \end{bmatrix}$$

2. Compute the symmetric matrix $P = \frac{1}{2}(A + A')$:

$$A + A' = \begin{bmatrix} 3+3 & 2+4 & 5+0 \\ 4+2 & 1+1 & 3+6 \\ 0+5 & 6+3 & 7+7 \end{bmatrix} = \begin{bmatrix} 6 & 6 & 5 \\ 6 & 2 & 9 \\ 5 & 9 & 14 \end{bmatrix}$$

$$P = \frac{1}{2} \begin{bmatrix} 6 & 6 & 5 \\ 6 & 2 & 9 \\ 5 & 9 & 14 \end{bmatrix} = \begin{bmatrix} 3 & 3 & \frac{5}{2} \\ 3 & 1 & \frac{9}{2} \\ \frac{5}{2} & \frac{9}{2} & 7 \end{bmatrix}$$

3. Compute the skew-symmetric matrix $Q = \frac{1}{2}(A - A')$:

$$A - A' = \begin{bmatrix} 3-3 & 2-4 & 5-0 \\ 4-2 & 1-1 & 3-6 \\ 0-5 & 6-3 & 7-7 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix}$$

$$Q = \frac{1}{2} \begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & \frac{5}{2} \\ 1 & 0 & -\frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 0 \end{bmatrix}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix} = \begin{bmatrix} 3 & 3 & \frac{5}{2} \\ 3 & 1 & \frac{9}{2} \\ \frac{5}{2} & \frac{9}{2} & 7 \end{bmatrix} + \begin{bmatrix} 0 & -1 & \frac{5}{2} \\ 1 & 0 & -\frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 0 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$.

1. Find the transpose A' :

$$A' = \begin{bmatrix} 2 & 7 & 1 \\ 4 & 3 & -2 \\ -6 & 5 & 4 \end{bmatrix}$$

2. Compute the symmetric matrix $P = \frac{1}{2}(A + A')$:

$$A + A' = \begin{bmatrix} 2+2 & 4+7 & -6+1 \\ 7+4 & 3+3 & 5-2 \\ 1-6 & -2+5 & 4+4 \end{bmatrix} = \begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix}$$

$$P = \frac{1}{2} \begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix} = \begin{bmatrix} 2 & \frac{11}{2} & -\frac{5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 4 \end{bmatrix}$$

3. Compute the skew-symmetric matrix $Q = \frac{1}{2}(A - A')$:

$$A - A' = \begin{bmatrix} 2-2 & 4-7 & -6-1 \\ 7-4 & 3-3 & 5-(-2) \\ 1-(-6) & -2-5 & 4-4 \end{bmatrix} = \begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix}$$

$$Q = \frac{1}{2} \begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{3}{2} & -\frac{7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & 0 \end{bmatrix}$$

4. Writing $A = P + Q$:

$$\begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & \frac{11}{2} & -\frac{5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 4 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{3}{2} & -\frac{7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & 0 \end{bmatrix}$$

Final Answer

$$(i) \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix} = \begin{bmatrix} 3 & 3 & \frac{5}{2} \\ 3 & 1 & \frac{9}{2} \\ \frac{5}{2} & \frac{9}{2} & 7 \end{bmatrix} + \begin{bmatrix} 0 & -1 & \frac{5}{2} \\ 1 & 0 & -\frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 0 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & \frac{11}{2} & -\frac{5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 4 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{3}{2} & -\frac{7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & 0 \end{bmatrix}$$

Question 31

For what value of x , is $A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ a skew symmetric matrix?

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: $A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ is a skew-symmetric matrix

To Find: Value of x

Concept / Formula Used: $A \quad A' = -A \implies a_{ij} = -a_{ji} \quad i, j$

Solution:

By definition, for any skew-symmetric matrix, off-diagonal elements satisfy $a_{ij} = -a_{ji}$.
Comparing element a_{31} with element a_{13} :

$$\begin{aligned} a_{31} &= -a_{13} \\ x &= -(-2) \\ x &= 2 \end{aligned}$$

Verifying with element a_{32} and element a_{23} :

$$\begin{aligned} a_{32} &= -a_{23} \\ -3 &= -(3) \quad (\text{which is consistent}) \end{aligned}$$

Final Answer

$$x = 2$$

Question 32

If the matrix $A = \begin{bmatrix} a & 6 & b \\ 2c & x+1 & 8 \\ -1 & -8 & y-3 \end{bmatrix}$ is a skew symmetric matrix, then find the value of $ab + bc - xy$.

TYPE 23 Symmetric and Skew-Symmetric Matrices

Given: $A = \begin{bmatrix} a & 6 & b \\ 2c & x+1 & 8 \\ -1 & -8 & y-3 \end{bmatrix}$ is a skew-symmetric matrix

To Find: Value of $ab + bc - xy$

Concept / Formula Used: $a_{ii} = 0 \quad a_{ij} = -a_{ji} \quad i \neq j$

Solution:

1. Setting diagonal elements to zero ($a_{ii} = 0$):

$$a_{11} = 0 \implies a = 0$$

$$a_{22} = 0 \implies x + 1 = 0 \implies x = -1$$

$$a_{33} = 0 \implies y - 3 = 0 \implies y = 3$$

2. Using $a_{ij} = -a_{ji}$ for off-diagonal elements:

$$a_{13} = -a_{31} \implies b = -(-1) \implies b = 1$$

$$a_{21} = -a_{12} \implies 2c = -(6) \implies 2c = -6 \implies c = -3$$

$$a_{23} = -a_{32} \implies 8 = -(-8) \implies 8 = 8 \quad (\text{Consistent})$$

3. Evaluating each term in the required expression $ab + bc - xy$:

$$ab = (0)(1) = 0$$

$$bc = (1)(-3) = -3$$

$$xy = (-1)(3) = -3$$

4. Combining the terms:

$$\begin{aligned} ab + bc - xy &= 0 + (-3) - (-3) \\ &= -3 + 3 \\ &= 0 \end{aligned}$$

Final Answer

$$ab + bc - xy = 0$$

ISC Class XII Mathematics

Chapter 4 Determinants

167 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 4.1 – Evaluation of Determinants: Minors and Cofactors
- II. 4.2 – Properties of Determinants and Proving Identities
- III. 4.3 – Applications of Determinants: Area, Collinearity and Equation of a Line
- IV. 4.4 – Adjoint and Inverse of a Matrix
- V. 4.5 – Solving Systems of Linear Equations Using Matrices

4.1 – Evaluation of Determinants: Minors and Cofactors

Question 1 (i)

Evaluate the determinant:

$$\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$$

TYPE 1 Evaluation of 2x2 Determinants

Given: $\Delta = \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Expanding the 2×2 determinant:

$$\Delta = (x^2 - x + 1)(x + 1) - (x - 1)(x + 1)$$

Applying algebraic expansion and identity $(x - 1)(x + 1) = x^2 - 1$:

$$\begin{aligned} (x^2 - x + 1)(x + 1) &= x^3 + 1 \\ (x - 1)(x + 1) &= x^2 - 1 \end{aligned}$$

Substituting these expressions back into the determinant expansion:

$$\begin{aligned} \Delta &= (x^3 + 1) - (x^2 - 1) \\ &= x^3 + 1 - x^2 + 1 \\ &= x^3 - x^2 + 2 \end{aligned}$$

Final Answer

$$\Delta = x^3 - x^2 + 2$$

Question 1 (ii)

Evaluate the determinant:

$$\begin{vmatrix} \cos 90^\circ & -\cos 45^\circ \\ \sin 90^\circ & \sin 45^\circ \end{vmatrix}$$

TYPE 1 Evaluation of 2x2 Determinants

Given: $\Delta = \begin{vmatrix} \cos 90^\circ & -\cos 45^\circ \\ \sin 90^\circ & \sin 45^\circ \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating standard trigonometric values:

$$\cos 90^\circ = 0$$

$$\sin 90^\circ = 1$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

Substituting these values into the determinant:

$$\begin{aligned}\Delta &= \begin{vmatrix} 0 & -\frac{1}{\sqrt{2}} \\ 1 & \frac{1}{\sqrt{2}} \end{vmatrix} \\ &= (0) \left(\frac{1}{\sqrt{2}} \right) - \left(-\frac{1}{\sqrt{2}} \right) \quad (1) \\ &= 0 + \frac{1}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}}\end{aligned}$$

Final Answer

$$\Delta = \frac{1}{\sqrt{2}}$$

Question 1 (iii)

Evaluate the determinant:

$$\begin{vmatrix} 2 \cos \theta & -2 \sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$$

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

Given: $\Delta = \begin{vmatrix} 2 \cos \theta & -2 \sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ $\sin^2 \theta + \cos^2 \theta = 1$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}\Delta &= (2 \cos \theta)(\cos \theta) - (-2 \sin \theta)(\sin \theta) \\ &= 2 \cos^2 \theta + 2 \sin^2 \theta \\ &= 2(\cos^2 \theta + \sin^2 \theta)\end{aligned}$$

Using the standard Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$:

$$\begin{aligned}\Delta &= 2(1) \\ &= 2\end{aligned}$$

Final Answer

$$\Delta = 2$$

Question 2 (i)

Evaluate the determinant:

$$\begin{vmatrix} \sin 30^\circ & \cos 30^\circ \\ -\sin 60^\circ & \cos 60^\circ \end{vmatrix}$$

TYPE 1 Evaluation of 2x2 Determinants

Given: $\Delta = \begin{vmatrix} \sin 30^\circ & \cos 30^\circ \\ -\sin 60^\circ & \cos 60^\circ \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Substituting standard angle trigonometric values:

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

Evaluating the 2×2 determinant:

$$\begin{aligned} \Delta &= \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) - \left(\frac{\sqrt{3}}{2}\right) \left(-\frac{\sqrt{3}}{2}\right) \\ &= \frac{1}{4} - \left(-\frac{3}{4}\right) \\ &= \frac{1}{4} + \frac{3}{4} \\ &= \frac{4}{4} \\ &= 1 \end{aligned}$$

Final Answer

$$\Delta = 1$$

Question 2 (ii)

Evaluate the determinant:

$$\begin{vmatrix} -1 & 31 & 40 \\ 0 & 5 & -432 \\ 0 & 0 & 20 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} -1 & 31 & 40 \\ 0 & 5 & -432 \\ 0 & 0 & 20 \end{vmatrix}$

Concept / Formula Used: $\Delta = a_{11}a_{22}a_{33}$

Solution:

Expanding along the first column C_1 :

$$\begin{aligned}\Delta &= -1 \cdot \begin{vmatrix} 5 & -432 \\ 0 & 20 \end{vmatrix} - 0 + 0 \\ &= -1 \cdot ((5)(20) - (-432)(0)) \\ &= -1 \cdot (100 - 0) \\ &= -1 \cdot 100 \\ &= -100\end{aligned}$$

Final Answer

$$\Delta = -100$$

Question 2 (iii)

Evaluate the determinant:

$$\begin{vmatrix} 2 & 0 & 0 \\ -25 & 3 & 0 \\ 71 & 64 & 7 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 2 & 0 & 0 \\ -25 & 3 & 0 \\ 71 & 64 & 7 \end{vmatrix}$

Concept / Formula Used: $\Delta = a_{11}a_{22}a_{33}$

Solution:

Expanding along the first row R_1 :

$$\begin{aligned}\Delta &= 2 \cdot \begin{vmatrix} 3 & 0 \\ 64 & 7 \end{vmatrix} - 0 + 0 \\ &= 2 \cdot ((3)(7) - (0)(64)) \\ &= 2 \cdot (21 - 0) \\ &= 2 \cdot 21 \\ &= 42\end{aligned}$$

Final Answer

$$\Delta = 42$$

Question 3 (i)

Find the value of $\begin{vmatrix} \sin A & -\sin B \\ \cos A & \cos B \end{vmatrix}$, where $A = 63^\circ$ and $B = 27^\circ$.

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

Given: $\Delta = \begin{vmatrix} \sin A & -\sin B \\ \cos A & \cos B \end{vmatrix}$, $A = 63^\circ$, $B = 27^\circ$

Concept / Formula Used: $\sin(A + B) = \sin A \cos B + \cos A \sin B$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}\Delta &= (\sin A)(\cos B) - (-\sin B)(\cos A) \\ &= \sin A \cos B + \cos A \sin B\end{aligned}$$

Applying the sine addition formula $\sin A \cos B + \cos A \sin B = \sin(A + B)$:

$$\Delta = \sin(A + B)$$

Substituting $A = 63^\circ$ and $B = 27^\circ$:

$$\begin{aligned}\Delta &= \sin(63^\circ + 27^\circ) \\ &= \sin(90^\circ) \\ &= 1\end{aligned}$$

Final Answer

$$\Delta = 1$$

Question 3 (ii)

Find the value of $\begin{vmatrix} \cos A & \sin A \\ -\sin B & \cos B \end{vmatrix}$, where $A = 75^\circ$ and $B = 45^\circ$.

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

Given: $\Delta = \begin{vmatrix} \cos A & \sin A \\ -\sin B & \cos B \end{vmatrix}$, $A = 75^\circ$, $B = 45^\circ$

Concept / Formula Used: $\cos(A - B) = \cos A \cos B + \sin A \sin B$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}\Delta &= (\cos A)(\cos B) - (\sin A)(-\sin B) \\ &= \cos A \cos B + \sin A \sin B\end{aligned}$$

Applying the cosine subtraction formula $\cos A \cos B + \sin A \sin B = \cos(A - B)$:

$$\Delta = \cos(A - B)$$

Substituting $A = 75^\circ$ and $B = 45^\circ$:

$$\begin{aligned}\Delta &= \cos(75^\circ - 45^\circ) \\ &= \cos(30^\circ) \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

Final Answer

$$\Delta = \frac{\sqrt{3}}{2}$$

Question 4 (i)

If $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$, then write the cofactor of the element a_{21} of its 2nd row.

TYPE 4 Minors and Cofactors of a Determinant

Given: Matrix $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$

Concept / Formula Used: $A_{ij} = (-1)^{i+j} M_{ij}$ M_{ij} a_{ij}

Solution:

The element at a_{21} is -4 . To find its minor M_{21} , delete the 2nd row and 1st column:

$$M_{21} = \begin{vmatrix} 6 & -3 \\ -7 & 3 \end{vmatrix}$$

Evaluating minor M_{21} :

$$\begin{aligned}M_{21} &= (6)(3) - (-3)(-7) \\ &= 18 - 21 \\ &= -3\end{aligned}$$

Calculating the cofactor A_{21} :

$$\begin{aligned}A_{21} &= (-1)^{2+1} M_{21} \\ &= (-1)^3 (-3) \\ &= (-1)(-3) \\ &= 3\end{aligned}$$

Final Answer

$$A_{21} = 3$$

Question 4 (ii)

If A_{ij} is the cofactor of the element a_{ij} of the determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$, then write the value of $a_{32}A_{32}$.

TYPE 4 Minors and Cofactors of a Determinant

Given: Determinant $\Delta = \begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$

Concept / Formula Used: $A_{32} = (-1)^{3+2}M_{32}$

Solution:

Identifying element a_{32} located at row 3, column 2:

$$a_{32} = 5$$

The minor M_{32} is formed by deleting row 3 and column 2:

$$M_{32} = \begin{vmatrix} 2 & 5 \\ 6 & 4 \end{vmatrix}$$

Evaluating M_{32} :

$$\begin{aligned} M_{32} &= (2)(4) - (5)(6) \\ &= 8 - 30 \\ &= -22 \end{aligned}$$

Calculating cofactor A_{32} :

$$\begin{aligned} A_{32} &= (-1)^{3+2}M_{32} \\ &= (-1)^5(-22) \\ &= (-1)(-22) \\ &= 22 \end{aligned}$$

Evaluating product $a_{32}A_{32}$:

$$\begin{aligned} a_{32}A_{32} &= 5 \times 22 \\ &= 110 \end{aligned}$$

Final Answer

$$a_{32}A_{32} = 110$$

Question 5 (i)

Find the value of x if:

$$\begin{vmatrix} 2x+5 & 3 \\ 5x+2 & 9 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 2x+5 & 3 \\ 5x+2 & 9 \end{vmatrix} = 0$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}(2x+5)(9) - (3)(5x+2) &= 0 \\ 18x + 45 - 15x - 6 &= 0\end{aligned}$$

Grouping like terms:

$$\begin{aligned}(18x - 15x) + (45 - 6) &= 0 \\ 3x + 39 &= 0 \\ 3x &= -39 \\ x &= \frac{-39}{3} \\ x &= -13\end{aligned}$$

Final Answer

$$x = -13$$

Question 5 (ii)

Find the value of x if:

$$\begin{vmatrix} x+2 & 3 \\ x+5 & 4 \end{vmatrix} = 3$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x+2 & 3 \\ x+5 & 4 \end{vmatrix} = 3$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}(x+2)(4) - (3)(x+5) &= 3 \\ 4x + 8 - 3x - 15 &= 3\end{aligned}$$

Combining like terms on the LHS:

$$\begin{aligned}(4x - 3x) + (8 - 15) &= 3 \\ x - 7 &= 3 \\ x &= 3 + 7 \\ x &= 10\end{aligned}$$

Final Answer

$$x = 10$$

Question 5 (iii)

Find the value of x if:

$$\begin{vmatrix} 3x & 7 \\ -2 & 4 \end{vmatrix} = \begin{vmatrix} 8 & 7 \\ 6 & 4 \end{vmatrix}$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 3x & 7 \\ -2 & 4 \end{vmatrix} = \begin{vmatrix} 8 & 7 \\ 6 & 4 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned} \text{LHS} &= (3x)(4) - (7)(-2) \\ &= 12x + 14 \end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned} \text{RHS} &= (8)(4) - (7)(6) \\ &= 32 - 42 \\ &= -10 \end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned} 12x + 14 &= -10 \\ 12x &= -10 - 14 \\ 12x &= -24 \\ x &= \frac{-24}{12} \\ x &= -2 \end{aligned}$$

Final Answer

$$x = -2$$

Question 6 (i)

If $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$, then write the value of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned} \text{LHS} &= (x+1)(x+2) - (x-1)(x-3) \\ &= (x^2 + 3x + 2) - (x^2 - 4x + 3) \\ &= x^2 + 3x + 2 - x^2 + 4x - 3 \\ &= 7x - 1 \end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned} \text{RHS} &= (4)(3) - (-1)(1) \\ &= 12 - (-1) \\ &= 12 + 1 \\ &= 13 \end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned} 7x - 1 &= 13 \\ 7x &= 13 + 1 \\ 7x &= 14 \\ x &= \frac{14}{7} \\ x &= 2 \end{aligned}$$

Final Answer

$$x = 2$$

Question 6 (ii)

If $\begin{vmatrix} \sqrt{6} & x \\ \sqrt{20} & \sqrt{24} \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 5 & 2 \end{vmatrix}$, then write the value of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} \sqrt{6} & x \\ \sqrt{20} & \sqrt{24} \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 5 & 2 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned} \text{LHS} &= (\sqrt{6})(\sqrt{24}) - (x)(\sqrt{20}) \\ &= \sqrt{144} - x\sqrt{20} \\ &= 12 - x(2\sqrt{5}) \\ &= 12 - 2\sqrt{5}x \end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned}\text{RHS} &= (6)(2) - (2)(5) \\ &= 12 - 10 \\ &= 2\end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned}12 - 2\sqrt{5}x &= 2 \\ -2\sqrt{5}x &= 2 - 12 \\ -2\sqrt{5}x &= -10 \\ x &= \frac{-10}{-2\sqrt{5}} \\ x &= \frac{5}{\sqrt{5}} \\ x &= \sqrt{5}\end{aligned}$$

Final Answer

$$x = \sqrt{5}$$

Question 7 (i)

If $x \in \mathbf{N}$ and $\begin{vmatrix} x & 3 \\ 4 & x \end{vmatrix} = \begin{vmatrix} 4 & -3 \\ 0 & 1 \end{vmatrix}$, find the value(s) of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x & 3 \\ 4 & x \end{vmatrix} = \begin{vmatrix} 4 & -3 \\ 0 & 1 \end{vmatrix}$, $x \in \mathbf{N}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned}\text{LHS} &= (x)(x) - (3)(4) \\ &= x^2 - 12\end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned}\text{RHS} &= (4)(1) - (-3)(0) \\ &= 4 - 0 \\ &= 4\end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned}x^2 - 12 &= 4 \\ x^2 &= 4 + 12 \\ x^2 &= 16 \\ x &= \pm 4\end{aligned}$$

Since $x \in \mathbf{N}$ (natural numbers), x must be positive:

$$x = 4$$

Final Answer

$$x = 4$$

Question 7 (ii)

If $x \in \mathbf{I}$ and $\begin{vmatrix} 2x & 3 \\ -1 & x \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ x & 3 \end{vmatrix}$, find the value(s) of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 2x & 3 \\ -1 & x \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ x & 3 \end{vmatrix}, x \in \mathbf{I}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned} \text{LHS} &= (2x)(x) - (3)(-1) \\ &= 2x^2 + 3 \end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned} \text{RHS} &= (3)(3) - (1)(x) \\ &= 9 - x \end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned} 2x^2 + 3 &= 9 - x \\ 2x^2 + x + 3 - 9 &= 0 \\ 2x^2 + x - 6 &= 0 \end{aligned}$$

Factorizing the quadratic equation $2x^2 + x - 6 = 0$:

$$\begin{aligned} 2x^2 + 4x - 3x - 6 &= 0 \\ 2x(x + 2) - 3(x + 2) &= 0 \\ (2x - 3)(x + 2) &= 0 \end{aligned}$$

Solving for roots:

$$\begin{aligned} 2x - 3 = 0 &\implies x = \frac{3}{2} \\ x + 2 = 0 &\implies x = -2 \end{aligned}$$

Since $x \in \mathbf{I}$ (set of integers), $x = \frac{3}{2} \notin \mathbf{I}$. Therefore, the only integer solution is:

$$x = -2$$

Final Answer

$$x = -2$$

Question 8

If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$.

TYPE 6 Determinant Properties of Scalar Multiplication

★ Likely Exam Question

To Prove: $|2A| = 4|A|$

Given: Matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$

Concept / Formula Used: $n \times n$ A $|kA| = k^n|A|$ 2×2 $|2A| = 2^2|A| = 4|A|$

Solution:

First, calculate the determinant of A :

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} \\ &= (1)(2) - (2)(4) \\ &= 2 - 8 \\ &= -6 \end{aligned}$$

Evaluating the RHS:

$$\begin{aligned} \text{RHS} &= 4|A| \\ &= 4 \times (-6) \\ &= -24 \end{aligned}$$

Now, determine the matrix $2A$:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 \\ 8 & 4 \end{bmatrix} \end{aligned}$$

Calculate the determinant of $2A$:

$$\begin{aligned} |2A| &= \begin{vmatrix} 2 & 4 \\ 8 & 4 \end{vmatrix} \\ &= (2)(4) - (4)(8) \\ &= 8 - 32 \\ &= -24 \end{aligned}$$

Comparing the two sides:

$$|2A| = -24 = 4 \times (-6) = 4|A|$$

Hence, $|2A| = 4|A|$ is verified.

Hence Proved

$|2A| = 4|A|$ – verified since $|2A| = -24$ and $4|A| = 4(-6) = -24$.

Question 9 (i)

Evaluate the determinant:

$$\begin{vmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{vmatrix}$

Concept / Formula Used: Expansion along the first row: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

Solution:

Expanding along the first row:

$$\Delta = 0 \begin{vmatrix} 0 & 3 \\ -3 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 1 & 3 \\ -2 & 0 \end{vmatrix} + 2 \begin{vmatrix} 1 & 0 \\ -2 & -3 \end{vmatrix}$$

Evaluating each 2×2 minor:

$$\begin{aligned} \begin{vmatrix} 0 & 3 \\ -3 & 0 \end{vmatrix} &= (0)(0) - (3)(-3) = 0 + 9 = 9 \\ \begin{vmatrix} 1 & 3 \\ -2 & 0 \end{vmatrix} &= (1)(0) - (3)(-2) = 0 + 6 = 6 \\ \begin{vmatrix} 1 & 0 \\ -2 & -3 \end{vmatrix} &= (1)(-3) - (0)(-2) = -3 - 0 = -3 \end{aligned}$$

Substituting back:

$$\begin{aligned} \Delta &= 0(9) + 1(6) + 2(-3) \\ &= 0 + 6 - 6 \\ &= 0 \end{aligned}$$

Final Answer

$$\Delta = 0$$

Question 9 (ii)

Evaluate the determinant:

$$\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$

Concept / Formula Used: Expansion along the first row: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

Solution:

Expanding along the first row:

$$\Delta = 3 \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} - (-4) \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} + 5 \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}$$

Evaluating each 2×2 minor:

$$\begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} = (1)(1) - (-2)(3) = 1 + 6 = 7$$

$$\begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = (1)(1) - (-2)(2) = 1 + 4 = 5$$

$$\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = (1)(3) - (1)(2) = 3 - 2 = 1$$

Substituting back:

$$\begin{aligned} \Delta &= 3(7) + 4(5) + 5(1) \\ &= 21 + 20 + 5 \\ &= 46 \end{aligned}$$

Final Answer

$$\Delta = 46$$

Question 9 (iii)

Evaluate the determinant:

$$\begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

Concept / Formula Used: Expansion along the first row: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

Solution:

Expanding along the first row:

$$\Delta = 2 \begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} + (-2) \begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix}$$

Evaluating each 2×2 minor:

$$\begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} = (2)(0) - (-1)(-5) = 0 - 5 = -5$$

$$\begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} = (0)(0) - (-1)(3) = 0 + 3 = 3$$

$$\begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix} = (0)(-5) - (2)(3) = 0 - 6 = -6$$

Substituting back:

$$\begin{aligned} \Delta &= 2(-5) + 1(3) - 2(-6) \\ &= -10 + 3 + 12 \\ &= 5 \end{aligned}$$

Final Answer

$$\Delta = 5$$

Question 10

Prove that:

$$\begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$$

TYPE 3 Evaluation of 3x3 Determinants

★ Likely Exam Question

To Prove: $\begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$

Given: $\Delta = \begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix}$

Concept / Formula Used: 3×3

Solution:

Expanding LHS determinant along the first row R_1 :

$$\text{LHS} = 1 \cdot \begin{vmatrix} 1 & c \\ -c & 1 \end{vmatrix} - a \cdot \begin{vmatrix} -a & c \\ -b & 1 \end{vmatrix} + b \cdot \begin{vmatrix} -a & 1 \\ -b & -c \end{vmatrix}$$

Evaluating each 2×2 minor separately:

$$\begin{vmatrix} 1 & c \\ -c & 1 \end{vmatrix} = (1)(1) - (c)(-c) = 1 + c^2$$

$$\begin{vmatrix} -a & c \\ -b & 1 \end{vmatrix} = (-a)(1) - (c)(-b) = -a + bc$$

$$\begin{vmatrix} -a & 1 \\ -b & -c \end{vmatrix} = (-a)(-c) - (1)(-b) = ac + b$$

Substituting these minors back into the expansion:

$$\begin{aligned} \text{LHS} &= 1(1 + c^2) - a(-a + bc) + b(ac + b) \\ &= 1 + c^2 + a^2 - abc + abc + b^2 \end{aligned}$$

Cancelling equal and opposite terms $-abc + abc$:

$$\begin{aligned} \text{LHS} &= 1 + c^2 + a^2 + b^2 \\ &= 1 + a^2 + b^2 + c^2 \\ &= \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS. Hence proved.

Question 11

Show that the value of the determinant $\begin{vmatrix} 0 & \tan x & 1 \\ 1 & -\sec x & 0 \\ \sec x & 0 & \tan x \end{vmatrix}$ is independent of x .

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

To Prove: $\Delta = \begin{vmatrix} 0 & \tan x & 1 \\ 1 & -\sec x & 0 \\ \sec x & 0 & \tan x \end{vmatrix}$ is independent of x .

Given: $\Delta = \begin{vmatrix} 0 & \tan x & 1 \\ 1 & -\sec x & 0 \\ \sec x & 0 & \tan x \end{vmatrix}$

Concept / Formula Used: $\sec^2 x - \tan^2 x = 1$

Solution:

Expanding the determinant along row 1 (R_1):

$$\Delta = 0 \cdot \begin{vmatrix} -\sec x & 0 \\ 0 & \tan x \end{vmatrix} - \tan x \cdot \begin{vmatrix} 1 & 0 \\ \sec x & \tan x \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & -\sec x \\ \sec x & 0 \end{vmatrix}$$

Evaluating the individual 2×2 minors:

$$\begin{aligned} \begin{vmatrix} 1 & 0 \\ \sec x & \tan x \end{vmatrix} &= (1)(\tan x) - (0)(\sec x) = \tan x \\ \begin{vmatrix} 1 & -\sec x \\ \sec x & 0 \end{vmatrix} &= (1)(0) - (-\sec x)(\sec x) = 0 + \sec^2 x = \sec^2 x \end{aligned}$$

Substituting these values back into the expanded expression:

$$\begin{aligned}\Delta &= 0 - \tan x(\tan x) + 1(\sec^2 x) \\ &= -\tan^2 x + \sec^2 x \\ &= \sec^2 x - \tan^2 x\end{aligned}$$

Using the standard identity $\sec^2 x - \tan^2 x = 1$:

$$\Delta = 1$$

Since 1 is a constant value containing no variable x , the determinant is independent of x .

Hence Proved

LHS = 1, which is independent of x . Hence proved.

Question 12

Using cofactors of elements of second row, evaluate $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$.

TYPE 7 Expansion of Determinant Using Cofactors

★ Likely Exam Question

Given: $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

Concept / Formula Used: $\Delta = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$ $A_{ij} = (-1)^{i+j}M_{ij}$

Solution:

Elements of the second row are:

$$a_{21} = 2, \quad a_{22} = 0, \quad a_{23} = 1$$

Calculating cofactor A_{21} :

$$\begin{aligned}A_{21} &= (-1)^{2+1} \begin{vmatrix} 3 & 8 \\ 2 & 3 \end{vmatrix} \\ &= (-1)^3 ((3)(3) - (8)(2)) \\ &= (-1)(9 - 16) \\ &= (-1)(-7) \\ &= 7\end{aligned}$$

Calculating cofactor A_{22} :

$$\begin{aligned}A_{22} &= (-1)^{2+2} \begin{vmatrix} 5 & 8 \\ 1 & 3 \end{vmatrix} \\ &= (-1)^4 ((5)(3) - (8)(1)) \\ &= (1)(15 - 8) \\ &= 7\end{aligned}$$

Calculating cofactor A_{23} :

$$\begin{aligned} A_{23} &= (-1)^{2+3} \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} \\ &= (-1)^5 ((5)(2) - (3)(1)) \\ &= (-1)(10 - 3) \\ &= (-1)(7) \\ &= -7 \end{aligned}$$

Expanding Δ along the second row using cofactors:

$$\begin{aligned} \Delta &= a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} \\ &= (2)(7) + (0)(7) + (1)(-7) \\ &= 14 + 0 - 7 \\ &= 7 \end{aligned}$$

Final Answer

$$\Delta = 7$$

4.2 – Properties of Determinants and Proving Identities

Question 1 (i)

Without expanding, find the value of:

$$\begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$$

Factoring out 6 from the first row R_1 :

$$\Delta = 6 \begin{vmatrix} 17 & 3 & 6 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$$

Observing row R_1 and row R_3 :

$$R_1 = [17 \quad 3 \quad 6]$$

$$R_3 = [17 \quad 3 \quad 6]$$

Since $R_1 = R_3$, by the property of determinants:

$$\Delta = 6 \times 0$$

$$\Delta = 0$$

Final Answer

0

Question 1 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$

Concept / Formula Used: $C_i \rightarrow C_i - kC_j$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$$

Applying the column operation $C_3 \rightarrow C_3 - 9C_2$:

$$\text{First row entry of } C_3 = 65 - 9(7) = 65 - 63 = 2$$

$$\text{Second row entry of } C_3 = 75 - 9(8) = 75 - 72 = 3$$

$$\text{Third row entry of } C_3 = 86 - 9(9) = 86 - 81 = 5$$

Substituting these transformed values back into C_3 :

$$\Delta = \begin{vmatrix} 2 & 7 & 2 \\ 3 & 8 & 3 \\ 5 & 9 & 5 \end{vmatrix}$$

Since column C_1 and column C_3 are identical ($C_1 = C_3$):

$$\Delta = 0$$

Final Answer

0

Question 1 (iii)

Without expanding, find the value of:

$$\begin{vmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{vmatrix}$$

Applying elementary row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = [7 - 1 \quad 9 - 3 \quad 11 - 5] = [6 \quad 6 \quad 6]$$

$$R_3 - R_1 = [13 - 1 \quad 15 - 3 \quad 17 - 5] = [12 \quad 12 \quad 12]$$

Substituting these into the determinant:

$$\Delta = \begin{vmatrix} 1 & 3 & 5 \\ 6 & 6 & 6 \\ 12 & 12 & 12 \end{vmatrix}$$

Since row R_2 and row R_3 are identical ($R_2 = R_3$):

$$\Delta = 0$$

Final Answer

0

Question 2 (i)

Without expanding, find the value of:

$$\begin{vmatrix} 5 & 5 & 5 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ **Likely Exam Question**

To Find: The value of $\begin{vmatrix} 5 & 5 & 5 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$

Concept / Formula Used: $|kA| = k|A|$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 5 & 5 & 5 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

Factoring out 5 from the first row R_1 :

$$\Delta = 5 \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

Applying the row operation $R_3 \rightarrow R_3 + R_2$:

$$\begin{aligned} R_3 &= [(b+c) + a \quad (c+a) + b \quad (a+b) + c] \\ &= [a+b+c \quad a+b+c \quad a+b+c] \end{aligned}$$

Substituting this back into the determinant:

$$\Delta = 5 \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a+b+c & a+b+c & a+b+c \end{vmatrix}$$

Factoring out $(a+b+c)$ from the third row R_3 :

$$\Delta = 5(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

Since row R_1 and row R_3 are identical:

$$\begin{aligned} \Delta &= 5(a+b+c) \times 0 \\ \Delta &= 0 \end{aligned}$$

Final Answer

0

Question 2 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

To Find: The value of $\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$$

Applying the row operation $R_2 \rightarrow R_2 - R_1 - 2R_3$:

$$\text{First entry of } R_2 = (a+2x) - a - 2(x) = 0$$

$$\text{Second entry of } R_2 = (b+2y) - b - 2(y) = 0$$

$$\text{Third entry of } R_2 = (c+2z) - c - 2(z) = 0$$

Substituting these entries into R_2 :

$$\Delta = \begin{vmatrix} a & b & c \\ 0 & 0 & 0 \\ x & y & z \end{vmatrix}$$

Since all elements of row R_2 are zero:

$$\Delta = 0$$

Final Answer

0

Question 3 (i)

Without expanding, find the value of:

$$\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

To Find: The value of $\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$

Concept / Formula Used: $(u-v) + (v-w) + (w-u) = 0$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$$

Applying the column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{First row entry of } C_1 = (a - b) + (b - c) + (c - a) = 0$$

$$\text{Second row entry of } C_1 = (x - y) + (y - z) + (z - x) = 0$$

$$\text{Third row entry of } C_1 = (p - q) + (q - r) + (r - p) = 0$$

Substituting these entries into C_1 :

$$\Delta = \begin{vmatrix} 0 & b - c & c - a \\ 0 & y - z & z - x \\ 0 & q - r & r - p \end{vmatrix}$$

Since all elements in the first column C_1 are zero:

$$\Delta = 0$$

Final Answer

0

Question 3 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} \beta\gamma & 1 & \alpha(\beta + \gamma) \\ \gamma\alpha & 1 & \beta(\gamma + \alpha) \\ \alpha\beta & 1 & \gamma(\alpha + \beta) \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

To Find: The value of $\begin{vmatrix} \beta\gamma & 1 & \alpha(\beta + \gamma) \\ \gamma\alpha & 1 & \beta(\gamma + \alpha) \\ \alpha\beta & 1 & \gamma(\alpha + \beta) \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} \beta\gamma & 1 & \alpha\beta + \alpha\gamma \\ \gamma\alpha & 1 & \beta\gamma + \beta\alpha \\ \alpha\beta & 1 & \gamma\alpha + \gamma\beta \end{vmatrix}$$

Applying the column operation $C_3 \rightarrow C_3 + C_1$:

$$\text{First row entry of } C_3 = (\alpha\beta + \alpha\gamma) + \beta\gamma = \alpha\beta + \beta\gamma + \gamma\alpha$$

$$\text{Second row entry of } C_3 = (\beta\gamma + \beta\alpha) + \gamma\alpha = \alpha\beta + \beta\gamma + \gamma\alpha$$

$$\text{Third row entry of } C_3 = (\gamma\alpha + \gamma\beta) + \alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha$$

Substituting back into column C_3 :

$$\Delta = \begin{vmatrix} \beta\gamma & 1 & \alpha\beta + \beta\gamma + \gamma\alpha \\ \gamma\alpha & 1 & \alpha\beta + \beta\gamma + \gamma\alpha \\ \alpha\beta & 1 & \alpha\beta + \beta\gamma + \gamma\alpha \end{vmatrix}$$

Factoring out $(\alpha\beta + \beta\gamma + \gamma\alpha)$ from column C_3 :

$$\Delta = (\alpha\beta + \beta\gamma + \gamma\alpha) \begin{vmatrix} \beta\gamma & 1 & 1 \\ \gamma\alpha & 1 & 1 \\ \alpha\beta & 1 & 1 \end{vmatrix}$$

Since column C_2 and column C_3 are identical:

$$\Delta = (\alpha\beta + \beta\gamma + \gamma\alpha) \times 0$$

$$\Delta = 0$$

Final Answer

0

Question 4 (i)

Without expanding, find the value of:

$$\begin{vmatrix} \frac{1}{a} & 1 & bc \\ \frac{1}{b} & 1 & ca \\ \frac{1}{c} & 1 & ab \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} \frac{1}{a} & 1 & bc \\ \frac{1}{b} & 1 & ca \\ \frac{1}{c} & 1 & ab \end{vmatrix}$

Concept / Formula Used: R_1, R_2, R_3 a, b, c abc

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} \frac{1}{a} & 1 & bc \\ \frac{1}{b} & 1 & ca \\ \frac{1}{c} & 1 & ab \end{vmatrix}$$

Multiplying rows R_1, R_2, R_3 by a, b, c respectively and dividing the determinant by abc :

$$\begin{aligned} \Delta &= \frac{1}{abc} \begin{vmatrix} a \cdot \frac{1}{a} & a \cdot 1 & a \cdot bc \\ b \cdot \frac{1}{b} & b \cdot 1 & b \cdot ca \\ c \cdot \frac{1}{c} & c \cdot 1 & c \cdot ab \end{vmatrix} \\ &= \frac{1}{abc} \begin{vmatrix} 1 & a & abc \\ 1 & b & abc \\ 1 & c & abc \end{vmatrix} \end{aligned}$$

Factoring out abc from column C_3 :

$$\begin{aligned} \Delta &= \frac{abc}{abc} \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} \\ &= 1 \cdot \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} \end{aligned}$$

Since column C_1 and column C_3 are identical:

$$\Delta = 0$$

Final Answer

0

Question 4 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} a+b & 2a+b & 3a+b \\ 2a+b & 3a+b & 4a+b \\ 4a+b & 5a+b & 6a+b \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} a+b & 2a+b & 3a+b \\ 2a+b & 3a+b & 4a+b \\ 4a+b & 5a+b & 6a+b \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a+b & 2a+b & 3a+b \\ 2a+b & 3a+b & 4a+b \\ 4a+b & 5a+b & 6a+b \end{vmatrix}$$

Applying the column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_2$:

$$\begin{aligned} C_2 - C_1 &= \begin{bmatrix} (2a+b) - (a+b) \\ (3a+b) - (2a+b) \\ (5a+b) - (4a+b) \end{bmatrix} = \begin{bmatrix} a \\ a \\ a \end{bmatrix} \\ C_3 - C_2 &= \begin{bmatrix} (3a+b) - (2a+b) \\ (4a+b) - (3a+b) \\ (6a+b) - (5a+b) \end{bmatrix} = \begin{bmatrix} a \\ a \\ a \end{bmatrix} \end{aligned}$$

Substituting these into the determinant:

$$\Delta = \begin{vmatrix} a+b & a & a \\ 2a+b & a & a \\ 4a+b & a & a \end{vmatrix}$$

Since column C_2 and column C_3 are identical ($C_2 = C_3$):

$$\Delta = 0$$

Final Answer

0

Question 5

If A is a square matrix of order 2 and $|A| = -5$, find the value of $|3A|$.

TYPE 6 Determinant Properties of Scalar Multiplication**Given:** Order of matrix $n = 2$, $|A| = -5$ **To Find:** The value of $|3A|$ **Concept / Formula Used:** A n $|kA| = k^n|A|$ **Solution:**

Given that A is a square matrix of order $n = 2$ and scalar $k = 3$: Using the standard result $|kA| = k^n|A|$ with $n = 2$ and $k = 3$:

$$\begin{aligned}|3A| &= 3^2|A| \\ &= 9|A|\end{aligned}$$

Substituting $|A| = -5$:

$$\begin{aligned}|3A| &= 9 \times (-5) \\ &= -45\end{aligned}$$

Final Answer

-45

Question 6If A is a square matrix of order 3 and $|A| = 4$, then write the value of $|2A|$.**TYPE 6 Determinant Properties of Scalar Multiplication****Given:** Order of matrix $n = 3$, $|A| = 4$ **To Find:** The value of $|2A|$ **Concept / Formula Used:** A n $|kA| = k^n|A|$ **Solution:**

Given $n = 3$, $k = 2$, and $|A| = 4$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|2A| &= 2^3|A| \\ &= 8|A|\end{aligned}$$

Substituting $|A| = 4$:

$$\begin{aligned}|2A| &= 8 \times 4 \\ &= 32\end{aligned}$$

Final Answer

32

Question 7If A is a square matrix of order 3 and $|A| = 5$, find the value of $|3A|$.

TYPE 6 Determinant Properties of Scalar Multiplication**Given:** Order of matrix $n = 3$, $|A| = 5$ **To Find:** The value of $|3A|$ **Concept / Formula Used:** A n $|kA| = k^n|A|$ **Solution:**Given $n = 3$, $k = 3$, and $|A| = 5$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|3A| &= 3^3|A| \\ &= 27|A|\end{aligned}$$

Substituting $|A| = 5$:

$$\begin{aligned}|3A| &= 27 \times 5 \\ &= 135\end{aligned}$$

Final Answer

135

Question 8If A is a square matrix of order 3 and $|A| = -2$, find the value of $|-5A|$.**TYPE 6 Determinant Properties of Scalar Multiplication****Given:** Order of matrix $n = 3$, $|A| = -2$ **To Find:** The value of $|-5A|$ **Concept / Formula Used:** A n $|kA| = k^n|A|$ **Solution:**Given $n = 3$, $k = -5$, and $|A| = -2$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|-5A| &= (-5)^3|A| \\ &= -125|A|\end{aligned}$$

Substituting $|A| = -2$:

$$\begin{aligned}|-5A| &= -125 \times (-2) \\ &= 250\end{aligned}$$

Final Answer

250

Question 9If A is a matrix of order 3×3 and its determinant is 4, then find $|3A|$.

TYPE 6 Determinant Properties of Scalar Multiplication**Given:** Order of matrix $n = 3$, $|A| = 4$ **To Find:** The value of $|3A|$ **Concept / Formula Used:** A n $|kA| = k^n|A|$ **Solution:**Given $n = 3$, $k = 3$, and $|A| = 4$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|3A| &= 3^3|A| \\ &= 27|A|\end{aligned}$$

Substituting $|A| = 4$:

$$\begin{aligned}|3A| &= 27 \times 4 \\ &= 108\end{aligned}$$

Final Answer

108

Question 10If A is a square matrix of order 3 and $|3A| = k|A|$, then find the value of k .**TYPE 6 Determinant Properties of Scalar Multiplication****Given:** Order of matrix $n = 3$, $|3A| = k|A|$ **To Find:** The value of k **Concept / Formula Used:** A n $|cA| = c^n|A|$ **Solution:**Using the property $|cA| = c^n|A|$ with $c = 3$ and $n = 3$:

$$\begin{aligned}|3A| &= 3^3|A| \\ |3A| &= 27|A|\end{aligned}$$

Equating this to the given relation $|3A| = k|A|$:

$$27|A| = k|A|$$

Comparing both sides gives:

$$k = 27$$

Final Answer

27

Question 11 (i)

If A is a square matrix such that $|A| = 5$, write the value of $|AA^T|$.

TYPE 9 Determinant Properties of Matrix Operations

Given: $|A| = 5$

To Find: The value of $|AA^T|$

Concept / Formula Used: $A \quad |A^T| = |A| \quad |AB| = |A||B|$

Solution:

Using the product rule for determinants $|AB| = |A||B|$:

$$|AA^T| = |A| \cdot |A^T|$$

Using the property that the determinant of a transpose equals the determinant of the matrix, $|A^T| = |A|$:

$$\begin{aligned} |AA^T| &= |A| \cdot |A| \\ &= |A|^2 \end{aligned}$$

Substituting $|A| = 5$:

$$\begin{aligned} |AA^T| &= 5^2 \\ &= 25 \end{aligned}$$

Final Answer

25

Question 11 (ii)

If A is a square matrix satisfying $AA' = I$, write the value of $|A|$.

TYPE 9 Determinant Properties of Matrix Operations

Given: $AA' = I$

To Find: The value of $|A|$

Concept / Formula Used: $|AB| = |A||B| \quad |A'| = |A| \quad |I| = 1$

Solution:

Taking the determinant on both sides of $AA' = I$:

$$|AA'| = |I|$$

Applying product property $|AB| = |A||B|$:

$$|A| \cdot |A'| = 1$$

Since $|A'| = |A|$:

$$\begin{aligned}|A| \cdot |A| &= 1 \\ |A|^2 &= 1 \\ |A| &= \pm 1\end{aligned}$$

Final Answer

± 1

Question 12

If A is a square matrix of order 3 such that $A^2 = 2A$, then find the value of $|A|$.

TYPE 9 Determinant Properties of Matrix Operations

Given: Order of matrix $n = 3$, $A^2 = 2A$

To Find: The value of $|A|$

Concept / Formula Used: $|A^2| = |A|^2$ $|kA| = k^n |A|$ n

Solution:

Given equation:

$$A^2 = 2A$$

Taking determinant on both sides:

$$|A^2| = |2A|$$

Using properties $|A^2| = |A|^2$ and $|2A| = 2^3 |A|$ for $n = 3$:

$$\begin{aligned}|A|^2 &= 8|A| \\ |A|^2 - 8|A| &= 0 \\ |A|(|A| - 8) &= 0\end{aligned}$$

Solving for $|A|$ gives two possible solutions:

$$|A| = 0 \quad \text{or} \quad |A| = 8$$

Final Answer

0 or 8

Question 13

If $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -4 \\ 3 & -2 \end{bmatrix}$, find $|AB|$.

TYPE 9 Determinant Properties of Matrix Operations

Given: $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -4 \\ 3 & -2 \end{bmatrix}$

To Find: The value of $|AB|$

Concept / Formula Used: $2 \times 2 \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad ad - bc \quad |AB| = |A||B|$

Solution:

Calculating $|A|$:

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} \\ &= (1)(-1) - (2)(3) \\ &= -1 - 6 \\ &= -7 \end{aligned}$$

Calculating $|B|$:

$$\begin{aligned} |B| &= \begin{vmatrix} 1 & -4 \\ 3 & -2 \end{vmatrix} \\ &= (1)(-2) - (-4)(3) \\ &= -2 + 12 \\ &= 10 \end{aligned}$$

Using the determinant product property $|AB| = |A||B|$:

$$\begin{aligned} |AB| &= (-7) \times 10 \\ &= -70 \end{aligned}$$

Final Answer

-70

Question 14

If $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$ and $|A^3| = 125$, find the value(s) of x .

TYPE 9 Determinant Properties of Matrix Operations

Given: $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$, $|A^3| = 125$

To Find: The value(s) of x

Concept / Formula Used: $|A^k| = |A|^k \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Using the property $|A^3| = |A|^3$:

$$\begin{aligned} |A|^3 &= 125 \\ |A| &= \sqrt[3]{125} \\ |A| &= 5 \end{aligned}$$

Now, evaluating $|A|$ directly from A :

$$\begin{aligned}|A| &= \begin{vmatrix} x & 2 \\ 2 & x \end{vmatrix} \\ &= x \cdot x - 2 \cdot 2 \\ &= x^2 - 4\end{aligned}$$

Equating the two expressions for $|A|$:

$$\begin{aligned}x^2 - 4 &= 5 \\ x^2 &= 5 + 4 \\ x^2 &= 9 \\ x &= \pm 3\end{aligned}$$

Final Answer

$$x = \pm 3$$

Question 15

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$

To Find: The value of x

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$

Solution:

Let the given determinant equation be $\Delta = 0$:

$$\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\begin{vmatrix} (x+a) + x + x & x & x \\ x + (x+a) + x & x+a & x \\ x + x + (x+a) & x & x+a \end{vmatrix} = 0$$

$$\begin{vmatrix} 3x+a & x & x \\ 3x+a & x+a & x \\ 3x+a & x & x+a \end{vmatrix} = 0$$

Factoring out $(3x + a)$ from column C_1 :

$$(3x + a) \begin{vmatrix} 1 & x & x \\ 1 & x + a & x \\ 1 & x & x + a \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(3x + a) \begin{vmatrix} 1 & x & x \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix} = 0$$

Expanding along column C_1 :

$$\begin{aligned} (3x + a) \cdot 1 \cdot (a \cdot a - 0 \cdot 0) &= 0 \\ a^2(3x + a) &= 0 \end{aligned}$$

Assuming $a \neq 0$:

$$\begin{aligned} 3x + a &= 0 \\ 3x &= -a \\ x &= -\frac{a}{3} \end{aligned}$$

Final Answer

$$x = -\frac{a}{3}$$

Question 16

Show that one root of the equation

$$\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$$

is $-(a + b + c)$.

TYPE 5 Solving Determinant Equations

To Prove: One root of $\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$ is $x = -(a + b + c)$

Concept / Formula Used: $(x - r) \quad \Delta(x) = 0 \quad r$

Solution:

Consider the given equation:

$$\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{First row entry of } C_1 = (x + a) + b + c = x + a + b + c$$

$$\text{Second row entry of } C_1 = b + (x + c) + a = x + a + b + c$$

$$\text{Third row entry of } C_1 = c + a + (x + b) = x + a + b + c$$

Substituting these into column C_1 :

$$\begin{vmatrix} x + a + b + c & b & c \\ x + a + b + c & x + c & a \\ x + a + b + c & a & x + b \end{vmatrix} = 0$$

Factoring out $(x + a + b + c)$ from column C_1 :

$$(x + a + b + c) \begin{vmatrix} 1 & b & c \\ 1 & x + c & a \\ 1 & a & x + b \end{vmatrix} = 0$$

Setting the linear factor to zero:

$$x + a + b + c = 0$$

$$x = -(a + b + c)$$

Thus, $x = -(a + b + c)$ is a root of the equation.

Hence Proved

One root of the given equation is $-(a + b + c)$.

Question 17

Without expanding, prove that

$$\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

Concept / Formula Used: (-1)

Solution:

$$\text{Let } \Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}.$$

Step 1: Transform Δ_1 to Δ_2 : Interchanging $R_1 \leftrightarrow R_2$:

$$\Delta_1 = - \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix}$$

Interchanging $R_2 \leftrightarrow R_3$:

$$\begin{aligned}\Delta_1 &= (-1)(-1) \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} \\ &= \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} = \Delta_2\end{aligned}$$

Step 2: Transform Δ_1 to Δ_3 : Taking transpose of Δ_1 (since $|A^T| = |A|$):

$$\Delta_1 = \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix}$$

Interchanging $R_1 \leftrightarrow R_2$:

$$\Delta_1 = - \begin{vmatrix} b & y & q \\ a & x & p \\ c & z & r \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\begin{aligned}\Delta_1 &= (-1)(-1) \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} \\ &= \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} = \Delta_3\end{aligned}$$

Thus, $\Delta_1 = \Delta_2 = \Delta_3$.

Hence Proved

$\Delta_1 = \Delta_2 = \Delta_3$, hence proved.

Question 18

Without expanding, prove that

$$\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

Concept / Formula Used: a, b, c abc

Solution:

Let $LHS = \Delta$:

$$\Delta = \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix}$$

Multiplying rows R_1, R_2, R_3 by a, b, c respectively and dividing by abc :

$$\begin{aligned} \Delta &= \frac{1}{abc} \begin{vmatrix} a \cdot a & a \cdot a^2 & a \cdot bc \\ b \cdot b & b \cdot b^2 & b \cdot ca \\ c \cdot c & c \cdot c^2 & c \cdot ab \end{vmatrix} \\ &= \frac{1}{abc} \begin{vmatrix} a^2 & a^3 & abc \\ b^2 & b^3 & abc \\ c^2 & c^3 & abc \end{vmatrix} \end{aligned}$$

Factoring out abc from column C_3 :

$$\begin{aligned} \Delta &= \frac{abc}{abc} \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix} \\ &= \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix} \end{aligned}$$

Interchanging $C_2 \leftrightarrow C_3$:

$$\Delta = - \begin{vmatrix} a^2 & 1 & a^3 \\ b^2 & 1 & b^3 \\ c^2 & 1 & c^3 \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\begin{aligned} \Delta &= (-1)(-1) \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} \\ &= \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (i)

Use properties of determinants to solve for x :

$$\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

★ **Likely Exam Question**

Given: $\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$

To Find: The values of x

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \quad (x + a + b + c)$

Solution:

Consider the given equation:

$$\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\begin{vmatrix} x+a+b+c & b & c \\ x+a+b+c & x+b & a \\ x+a+b+c & b & x+c \end{vmatrix} = 0$$

Factoring out $(x + a + b + c)$ from column C_1 :

$$(x + a + b + c) \begin{vmatrix} 1 & b & c \\ 1 & x+b & a \\ 1 & b & x+c \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(x + a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & x & a-c \\ 0 & 0 & x \end{vmatrix} = 0$$

Expanding along column C_1 :

$$\begin{aligned} (x + a + b + c) \cdot 1 \cdot (x \cdot x - (a - c) \cdot 0) &= 0 \\ x^2(x + a + b + c) &= 0 \end{aligned}$$

Equating each factor to zero:

$$\begin{aligned} x^2 = 0 &\implies x = 0 \\ x + a + b + c = 0 &\implies x = -(a + b + c) \end{aligned}$$

Final Answer

$$x = 0 \text{ or } x = -(a + b + c)$$

Question 19 (ii)

Use properties of determinants to solve for x :

$$\begin{vmatrix} 3-x & -1 & 1 \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 3-x & -1 & 1 \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$

To Find: The values of x

Solution:

Consider the equation:

$$\begin{vmatrix} 3-x & -1 & 1 \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$$

Applying $R_1 \rightarrow R_1 + R_3$:

$$\begin{aligned} R_1 &= [(3-x) + 1 \quad -1 + (-1) \quad 1 + (3-x)] \\ &= [4-x \quad -2 \quad 4-x] \end{aligned}$$

Substituting into R_1 :

$$\begin{vmatrix} 4-x & -2 & 4-x \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$$

Factoring out $(4-x)$ from row R_1 :

$$(4-x) \begin{vmatrix} 1 & -2 & 1 \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$$

Applying $C_3 \rightarrow C_3 - C_1$:

$$(4-x) \begin{vmatrix} 1 & -2 & 0 \\ -1 & 5-x & 0 \\ 1 & -1 & 2-x \end{vmatrix} = 0$$

Expanding along column C_3 :

$$\begin{aligned} (4-x)(2-x) \begin{vmatrix} 1 & -2 \\ -1 & 5-x \end{vmatrix} &= 0 \\ (4-x)(2-x)[(1)(5-x) - (-2)(-1)] &= 0 \\ (4-x)(2-x)(5-x-2) &= 0 \\ (4-x)(2-x)(3-x) &= 0 \end{aligned}$$

Solving for roots:

$$x = 2, 3, 4$$

Final Answer

$$x = 2, 3, 4$$

Question 20

If $s = a + b + c$, then prove that

$$\begin{vmatrix} s+c & a & b \\ c & s+a & b \\ c & a & s+b \end{vmatrix} = 2s^3$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} s+c & a & b \\ c & s+a & b \\ c & a & s+b \end{vmatrix} = 2s^3$ given $s = a + b + c$

Solution:

Let $LHS = \Delta$:

$$\Delta = \begin{vmatrix} s+c & a & b \\ c & s+a & b \\ c & a & s+b \end{vmatrix}$$

Applying row operations $R_1 \rightarrow R_1 - R_3$ and $R_2 \rightarrow R_2 - R_3$:

$$R_1 - R_3 = [(s+c) - c \quad a - a \quad b - (s+b)] = [s \quad 0 \quad -s]$$

$$R_2 - R_3 = [c - c \quad (s+a) - a \quad b - (s+b)] = [0 \quad s \quad -s]$$

Substituting these into the determinant:

$$\Delta = \begin{vmatrix} s & 0 & -s \\ 0 & s & -s \\ c & a & s+b \end{vmatrix}$$

Factoring out s from R_1 and s from R_2 :

$$\Delta = s^2 \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ c & a & s+b \end{vmatrix}$$

Applying column operation $C_3 \rightarrow C_3 + C_1 + C_2$:

$$\text{Third entry of } C_3 = (s+b) + c + a = s + (a+b+c)$$

Given $a + b + c = s$, this entry becomes $s + s = 2s$:

$$\Delta = s^2 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & a & 2s \end{vmatrix}$$

Expanding along row R_1 :

$$\begin{aligned} \Delta &= s^2 \cdot 1 \cdot (1 \cdot 2s - 0 \cdot a) \\ &= s^2 \cdot 2s \\ &= 2s^3 = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 21

If a, b and c are in A.P., then find the value of

$$\begin{vmatrix} 2y+4 & 5y+7 & 8y+a \\ 3y+5 & 6y+8 & 9y+b \\ 4y+6 & 7y+9 & 10y+c \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

Given: a, b, c are in A.P.

To Find: The value of $\begin{vmatrix} 2y+4 & 5y+7 & 8y+a \\ 3y+5 & 6y+8 & 9y+b \\ 4y+6 & 7y+9 & 10y+c \end{vmatrix}$

Concept / Formula Used: a, b, c $2b = a + c \implies a - 2b + c = 0$

Solution:

Since a, b, c are in Arithmetic Progression:

$$\begin{aligned} 2b &= a + c \\ a - 2b + c &= 0 \end{aligned}$$

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 2y+4 & 5y+7 & 8y+a \\ 3y+5 & 6y+8 & 9y+b \\ 4y+6 & 7y+9 & 10y+c \end{vmatrix}$$

Applying the row operation $R_1 \rightarrow R_1 - 2R_2 + R_3$:

$$\begin{aligned} \text{First entry of } R_1 &= (2y+4) - 2(3y+5) + (4y+6) \\ &= 2y - 6y + 4y + 4 - 10 + 6 = 0 \end{aligned}$$

$$\begin{aligned} \text{Second entry of } R_1 &= (5y+7) - 2(6y+8) + (7y+9) \\ &= 5y - 12y + 7y + 7 - 16 + 9 = 0 \end{aligned}$$

$$\begin{aligned} \text{Third entry of } R_1 &= (8y+a) - 2(9y+b) + (10y+c) \\ &= 8y - 18y + 10y + a - 2b + c \\ &= a - 2b + c = 0 \end{aligned}$$

Substituting these entries into R_1 :

$$\Delta = \begin{vmatrix} 0 & 0 & 0 \\ 3y+5 & 6y+8 & 9y+b \\ 4y+6 & 7y+9 & 10y+c \end{vmatrix}$$

Since row R_1 is entirely composed of zeros:

$$\Delta = 0$$

Final Answer

0

Question 22

Using the properties of determinant, prove that

$$\begin{vmatrix} x-3 & x-4 & x-\alpha \\ x-2 & x-3 & x-\beta \\ x-1 & x-2 & x-\gamma \end{vmatrix} = 0$$

where α, β and γ are in A.P.

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x-3 & x-4 & x-\alpha \\ x-2 & x-3 & x-\beta \\ x-1 & x-2 & x-\gamma \end{vmatrix} = 0$ given α, β, γ in A.P.

Concept / Formula Used: $\alpha, \beta, \gamma \implies 2\beta = \alpha + \gamma \implies \alpha - 2\beta + \gamma = 0$

Solution:

Given α, β, γ are in A.P.:

$$\begin{aligned} 2\beta &= \alpha + \gamma \\ \alpha - 2\beta + \gamma &= 0 \end{aligned}$$

Let LHS = Δ :

$$\Delta = \begin{vmatrix} x-3 & x-4 & x-\alpha \\ x-2 & x-3 & x-\beta \\ x-1 & x-2 & x-\gamma \end{vmatrix}$$

Applying the row operation $R_1 \rightarrow R_1 - 2R_2 + R_3$:

$$\begin{aligned} \text{First entry of } R_1 &= (x-3) - 2(x-2) + (x-1) \\ &= x - 2x + x - 3 + 4 - 1 = 0 \end{aligned}$$

$$\begin{aligned} \text{Second entry of } R_1 &= (x-4) - 2(x-3) + (x-2) \\ &= x - 2x + x - 4 + 6 - 2 = 0 \end{aligned}$$

$$\begin{aligned} \text{Third entry of } R_1 &= (x-\alpha) - 2(x-\beta) + (x-\gamma) \\ &= x - 2x + x - \alpha + 2\beta - \gamma \\ &= -(\alpha - 2\beta + \gamma) = 0 \end{aligned}$$

Substituting these entries back into R_1 :

$$\Delta = \begin{vmatrix} 0 & 0 & 0 \\ x-2 & x-3 & x-\beta \\ x-1 & x-2 & x-\gamma \end{vmatrix}$$

Since all elements of row R_1 are zero:

$$\Delta = 0 = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 23

Using properties of determinants, prove that:

$$\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} 2(a+b+c) & b+c & c+a \\ 2(a+b+c) & c+a & a+b \\ 2(a+b+c) & a+b & b+c \end{vmatrix}$$

Factoring out 2 from column C_1 :

$$\Delta = 2 \begin{vmatrix} a+b+c & b+c & c+a \\ a+b+c & c+a & a+b \\ a+b+c & a+b & b+c \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$C_2 - C_1 = \begin{bmatrix} (b+c) - (a+b+c) \\ (c+a) - (a+b+c) \\ (a+b) - (a+b+c) \end{bmatrix} = \begin{bmatrix} -a \\ -b \\ -c \end{bmatrix}$$

$$C_3 - C_1 = \begin{bmatrix} (c+a) - (a+b+c) \\ (a+b) - (a+b+c) \\ (b+c) - (a+b+c) \end{bmatrix} = \begin{bmatrix} -b \\ -c \\ -a \end{bmatrix}$$

Substituting these into column C_2 and C_3 :

$$\Delta = 2 \begin{vmatrix} a+b+c & -a & -b \\ a+b+c & -b & -c \\ a+b+c & -c & -a \end{vmatrix}$$

Factoring out (-1) from C_2 and (-1) from C_3 :

$$\Delta = 2(-1)(-1) \begin{vmatrix} a+b+c & a & b \\ a+b+c & b & c \\ a+b+c & c & a \end{vmatrix}$$

$$= 2 \begin{vmatrix} a+b+c & a & b \\ a+b+c & b & c \\ a+b+c & c & a \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_2 - C_3$:

$$C_1 = \begin{bmatrix} (a+b+c) - a - b \\ (a+b+c) - b - c \\ (a+b+c) - c - a \end{bmatrix} = \begin{bmatrix} c \\ a \\ b \end{bmatrix}$$

So we have:

$$\Delta = 2 \begin{vmatrix} c & a & b \\ a & b & c \\ b & c & a \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_3$:

$$\Delta = -2 \begin{vmatrix} b & a & c \\ c & b & a \\ a & c & b \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\begin{aligned} \Delta &= (-2)(-1) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \\ &= 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (i)

Prove that

$$\begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} = 1+a+b+c$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} = 1+a+b+c$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} 1+a+b+c & b & c \\ 1+a+b+c & 1+b & c \\ 1+a+b+c & b & 1+c \end{vmatrix}$$

Factoring out $(1+a+b+c)$ from column C_1 :

$$\Delta = (1+a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & 1+b & c \\ 1 & b & 1+c \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= [1-1 \quad (1+b)-b \quad c-c] = [0 \quad 1 \quad 0] \\ R_3 - R_1 &= [1-1 \quad b-b \quad (1+c)-c] = [0 \quad 0 \quad 1] \end{aligned}$$

Substituting these into the determinant:

$$\Delta = (1+a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (1+a+b+c) \cdot 1 \cdot (1 \cdot 1 - 0 \cdot 0) \\ &= 1+a+b+c = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (ii)

Prove that

$$\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$ $(a+x+y+z)$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} a+x+y+z & y & z \\ a+x+y+z & a+y & z \\ a+x+y+z & y & a+z \end{vmatrix}$$

Factoring out $(a+x+y+z)$ from column C_1 :

$$\Delta = (a+x+y+z) \begin{vmatrix} 1 & y & z \\ 1 & a+y & z \\ 1 & y & a+z \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = [1-1 \quad (a+y)-y \quad z-z] = [0 \quad a \quad 0]$$

$$R_3 - R_1 = [1-1 \quad y-y \quad (a+z)-z] = [0 \quad 0 \quad a]$$

Substituting these into the determinant:

$$\Delta = (a+x+y+z) \begin{vmatrix} 1 & y & z \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (a+x+y+z) \cdot 1 \cdot (a \cdot a - 0 \cdot 0) \\ &= a^2(a+x+y+z) = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (iii)

Prove that

$$\begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3y+k)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3y+k)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \quad (3y+k)$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} 3y+k & y & y \\ 3y+k & y+k & y \\ 3y+k & y & y+k \end{vmatrix}$$

Factoring out $(3y+k)$ from column C_1 :

$$\Delta = (3y+k) \begin{vmatrix} 1 & y & y \\ 1 & y+k & y \\ 1 & y & y+k \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = [1-1 \quad (y+k)-y \quad y-y] = [0 \quad k \quad 0]$$

$$R_3 - R_1 = [1-1 \quad y-y \quad (y+k)-y] = [0 \quad 0 \quad k]$$

Substituting these into the determinant:

$$\Delta = (3y+k) \begin{vmatrix} 1 & y & y \\ 0 & k & 0 \\ 0 & 0 & k \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (3y+k) \cdot 1 \cdot (k \cdot k - 0 \cdot 0) \\ &= k^2(3y+k) = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (iv)

Prove that

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$ $(5x+4)$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{First entry of } C_1 = (x + 4) + 2x + 2x = 5x + 4$$

$$\text{Second entry of } C_1 = 2x + (x + 4) + 2x = 5x + 4$$

$$\text{Third entry of } C_1 = 2x + 2x + (x + 4) = 5x + 4$$

Substituting into C_1 :

$$\Delta = \begin{vmatrix} 5x+4 & 2x & 2x \\ 5x+4 & x+4 & 2x \\ 5x+4 & 2x & x+4 \end{vmatrix}$$

Factoring out $(5x + 4)$ from column C_1 :

$$\Delta = (5x + 4) \begin{vmatrix} 1 & 2x & 2x \\ 1 & x+4 & 2x \\ 1 & 2x & x+4 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = [1 - 1 \quad (x + 4) - 2x \quad 2x - 2x] = [0 \quad 4 - x \quad 0]$$

$$R_3 - R_1 = [1 - 1 \quad 2x - 2x \quad (x + 4) - 2x] = [0 \quad 0 \quad 4 - x]$$

Substituting these into the determinant:

$$\Delta = (5x + 4) \begin{vmatrix} 1 & 2x & 2x \\ 0 & 4 - x & 0 \\ 0 & 0 & 4 - x \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (5x + 4) \cdot 1 \cdot [(4 - x)(4 - x) - 0 \cdot 0] \\ &= (5x + 4)(4 - x)^2 = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 25

Show that:

$$\begin{vmatrix} a+b+nc & na-a & nb-b \\ nc-c & b+c+na & nb-b \\ nc-c & na-a & c+a+nb \end{vmatrix} = n(a+b+c)^3$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a+b+nc & na-a & nb-b \\ nc-c & b+c+na & nb-b \\ nc-c & na-a & c+a+nb \end{vmatrix} = n(a+b+c)^3$

Concept / Formula Used: $\det(A) = \det(A^T)$

$$\begin{vmatrix} ka_1 & b_1 & c_1 \\ ka_2 & b_2 & c_2 \\ ka_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a+b+nc & (n-1)a & (n-1)b \\ (n-1)c & b+c+na & (n-1)b \\ (n-1)c & (n-1)a & c+a+nb \end{vmatrix}$$

Applying the column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

Row 1 element of $C_1 = (a+b+nc) + (na-a) + (nb-b) = n(a+b+c)$

Row 2 element of $C_1 = (nc-c) + (b+c+na) + (nb-b) = n(a+b+c)$

Row 3 element of $C_1 = (nc-c) + (na-a) + (c+a+nb) = n(a+b+c)$

Substituting these into C_1 :

$$\Delta = \begin{vmatrix} n(a+b+c) & (n-1)a & (n-1)b \\ n(a+b+c) & b+c+na & (n-1)b \\ n(a+b+c) & (n-1)a & c+a+nb \end{vmatrix}$$

Factoring out $n(a+b+c)$ from C_1 :

$$\Delta = n(a+b+c) \begin{vmatrix} 1 & (n-1)a & (n-1)b \\ 1 & b+c+na & (n-1)b \\ 1 & (n-1)a & c+a+nb \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= \begin{pmatrix} 1-1 & (b+c+na)-(na-a) & (nb-b)-(nb-b) \end{pmatrix} \\ &= \begin{pmatrix} 0 & a+b+c & 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} R_3 - R_1 &= \begin{pmatrix} 1-1 & (na-a)-(na-a) & (c+a+nb)-(nb-b) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & a+b+c \end{pmatrix} \end{aligned}$$

The determinant becomes:

$$\Delta = n(a+b+c) \begin{vmatrix} 1 & (n-1)a & (n-1)b \\ 0 & a+b+c & 0 \\ 0 & 0 & a+b+c \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta &= n(a+b+c) \cdot 1 \cdot [(a+b+c)(a+b+c) - 0] \\ &= n(a+b+c) \cdot (a+b+c)^2 \\ &= n(a+b+c)^3 \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 26 (i)

Prove that:

$$\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

TYPE 10 Proving Determinant Identities Using Properties**★ Likely Exam Question**

To Prove: $\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$

Concept / Formula Used: $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

Solution:

Let $\Delta_1 = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$.

Multiplying R_1 by x , R_2 by y , and R_3 by z :

$$\Delta_1 = \frac{1}{xyz} \begin{vmatrix} x & x^2 & xyz \\ y & y^2 & xyz \\ z & z^2 & xyz \end{vmatrix}$$

Factoring out xyz from C_3 :

$$\Delta_1 = \frac{xyz}{xyz} \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix}$$

Interchanging $C_3 \leftrightarrow C_2$ (introducing a negative sign):

$$\Delta_1 = - \begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$ (introducing a second negative sign):

$$\Delta_1 = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = \Delta_2$$

Now evaluating $\Delta_2 = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$:

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta_2 = \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix}$$

Factoring out $(y - x)$ from R_2 and $(z - x)$ from R_3 :

$$\Delta_2 = (y - x)(z - x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y + x \\ 0 & 1 & z + x \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta_2 &= (y - x)(z - x) [1 \cdot ((z + x) - (y + x))] \\ &= (y - x)(z - x)(z - y) \\ &= [-(x - y)][-(z - x)][-(y - z)] \\ &= (x - y)(y - z)(z - x) \end{aligned}$$

Hence Proved

LHS = MiddleTerm = RHS, hence proved.

Question 26 (ii)

Prove that:

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz(x - y)(y - z)(z - x)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz(x - y)(y - z)(z - x)$

Concept / Formula Used: $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$

Solution:

Let $\Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix}$.

Factoring out x from C_1 , y from C_2 , and z from C_3 :

$$\Delta = xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Applying column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$\Delta = xyz \begin{vmatrix} 1 & 0 & 0 \\ x & y - x & z - x \\ x^2 & y^2 - x^2 & z^2 - x^2 \end{vmatrix}$$

Factoring out $(y - x)$ from C_2 and $(z - x)$ from C_3 :

$$\Delta = xyz(y - x)(z - x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ x^2 & y + x & z + x \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned}\Delta &= xyz(y-x)(z-x)[1 \cdot ((z+x) - (y+x))] \\ &= xyz(y-x)(z-x)(z-y) \\ &= xyz[-(x-y)][-(z-x)][-(y-z)] \\ &= xyz(x-y)(y-z)(z-x)\end{aligned}$$

Hence Proved

$LHS = RHS$, hence proved.

Question 27 (i)

Prove that:

$$\begin{vmatrix} x & y & 1 \\ \alpha & x & 1 \\ \alpha & \beta & 1 \end{vmatrix} = (x - \alpha)(x - \beta)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x & y & 1 \\ \alpha & x & 1 \\ \alpha & \beta & 1 \end{vmatrix} = (x - \alpha)(x - \beta)$

Concept / Formula Used: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Solution:

Let $\Delta = \begin{vmatrix} x & y & 1 \\ \alpha & x & 1 \\ \alpha & \beta & 1 \end{vmatrix}$.

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta = \begin{vmatrix} x & y & 1 \\ \alpha - x & x - y & 0 \\ \alpha - x & \beta - y & 0 \end{vmatrix}$$

Expanding along C_3 :

$$\begin{aligned}\Delta &= 1 \cdot [(\alpha - x)(\beta - y) - (x - y)(\alpha - x)] \\ &= (\alpha - x)[(\beta - y) - (x - y)] \\ &= (\alpha - x)[\beta - y - x + y] \\ &= (\alpha - x)(\beta - x) \\ &= [-(x - \alpha)][-(x - \beta)] \\ &= (x - \alpha)(x - \beta)\end{aligned}$$

Hence Proved

$LHS = RHS$, hence proved.

Question 27 (ii)

Prove that:

$$\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix} = xy$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix} = xy$

Concept / Formula Used: $\det(\text{Upper Triangular Matrix}) =$
Product of diagonal elements

Solution:

Let $\Delta = \begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$.

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= (1-1 \quad (x+y)-x \quad y-y) = (0 \quad y \quad 0) \\ R_3 - R_1 &= (1-1 \quad x-x \quad (x+y)-y) = (0 \quad 0 \quad x) \end{aligned}$$

Substituting these rows back into the determinant:

$$\Delta = \begin{vmatrix} 1 & x & y \\ 0 & y & 0 \\ 0 & 0 & x \end{vmatrix}$$

Since this is an upper triangular matrix, the determinant is the product of its diagonal elements:

$$\begin{aligned} \Delta &= 1 \cdot y \cdot x \\ &= xy \end{aligned}$$

Hence Proved*LHS = RHS, hence proved.***Question 27 (iii)**

Prove that:

$$\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = (x+y+z)(x-z)^2$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = (x+y+z)(x-z)^2$

Concept / Formula Used: $R_1 \rightarrow R_1 + R_2 + R_3 \implies$ factor out $(x + y + z)$

Solution:

Let $\Delta = \begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$.

Applying $R_1 \rightarrow R_1 + R_2 + R_3$:

$$\text{Column 1 element} = (y+z) + (z+x) + (x+y) = 2(x+y+z)$$

$$\text{Column 2 element} = x + z + y = x + y + z$$

$$\text{Column 3 element} = y + x + z = x + y + z$$

Substituting into R_1 :

$$\Delta = \begin{vmatrix} 2(x+y+z) & x+y+z & x+y+z \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$$

Factoring out $(x + y + z)$ from R_1 :

$$\Delta = (x+y+z) \begin{vmatrix} 2 & 1 & 1 \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$$

Applying column operations $C_1 \rightarrow C_1 - 2C_2$ and $C_3 \rightarrow C_3 - C_2$:

$$C_1 - 2C_2 = \begin{pmatrix} 2 - 2(1) \\ (z+x) - 2z \\ (x+y) - 2y \end{pmatrix} = \begin{pmatrix} 0 \\ x-z \\ x-y \end{pmatrix}$$

$$C_3 - C_2 = \begin{pmatrix} 1 - 1 \\ x - z \\ z - y \end{pmatrix} = \begin{pmatrix} 0 \\ x - z \\ z - y \end{pmatrix}$$

Substituting these back into the determinant:

$$\Delta = (x+y+z) \begin{vmatrix} 0 & 1 & 0 \\ x-z & z & x-z \\ x-y & y & z-y \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned} \Delta &= (x+y+z) \cdot (-1) \cdot [(x-z)(z-y) - (x-z)(x-y)] \\ &= (x+y+z) \cdot (-1)(x-z) \cdot [(z-y) - (x-y)] \\ &= -(x+y+z)(x-z)(z-x) \\ &= -(x+y+z)(x-z)[- (x-z)] \\ &= (x+y+z)(x-z)^2 \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 28

Prove that:

$$\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ca & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ca & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$

Concept / Formula Used: $\det(kA) = k^n \det(A)$ $n \times n$ A **Solution:**

Let $\Delta = \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ca & bc & -c^2 \end{vmatrix}$.

Factoring out a from R_1 , b from R_2 , and c from R_3 :

$$\Delta = abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$$

Now factoring out a from C_1 , b from C_2 , and c from C_3 :

$$\Delta = a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 + R_1$ and $R_3 \rightarrow R_3 + R_1$:

$$R_2 + R_1 = (1 + (-1) \quad -1 + 1 \quad 1 + 1) = (0 \quad 0 \quad 2)$$

$$R_3 + R_1 = (1 + (-1) \quad 1 + 1 \quad -1 + 1) = (0 \quad 2 \quad 0)$$

Substituting into the determinant:

$$\Delta = a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta &= a^2b^2c^2 \cdot (-1) \cdot [(0)(0) - (2)(2)] \\ &= a^2b^2c^2 \cdot (-1) \cdot (-4) \\ &= 4a^2b^2c^2 \end{aligned}$$

Hence Proved*LHS = RHS, hence proved.*

Question 29

Prove that:

$$\begin{vmatrix} 1 & \alpha & \alpha^2 + \beta\gamma \\ 1 & \beta & \beta^2 + \gamma\alpha \\ 1 & \gamma & \gamma^2 + \alpha\beta \end{vmatrix} = 2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1 & \alpha & \alpha^2 + \beta\gamma \\ 1 & \beta & \beta^2 + \gamma\alpha \\ 1 & \gamma & \gamma^2 + \alpha\beta \end{vmatrix} = 2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$

Concept / Formula Used: $\begin{vmatrix} a & b & c+d \\ e & f & g+h \\ i & j & k+l \end{vmatrix} = \begin{vmatrix} a & b & c \\ e & f & g \\ i & j & k \end{vmatrix} + \begin{vmatrix} a & b & d \\ e & f & h \\ i & j & l \end{vmatrix}$

Solution:

Let $\Delta = \begin{vmatrix} 1 & \alpha & \alpha^2 + \beta\gamma \\ 1 & \beta & \beta^2 + \gamma\alpha \\ 1 & \gamma & \gamma^2 + \alpha\beta \end{vmatrix}$.

Splitting C_3 into two determinants:

$$\Delta = \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix} + \begin{vmatrix} 1 & \alpha & \beta\gamma \\ 1 & \beta & \gamma\alpha \\ 1 & \gamma & \alpha\beta \end{vmatrix} = \Delta_1 + \Delta_2$$

Using the standard result for Δ_1 :

$$\Delta_1 = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

For $\Delta_2 = \begin{vmatrix} 1 & \alpha & \beta\gamma \\ 1 & \beta & \gamma\alpha \\ 1 & \gamma & \alpha\beta \end{vmatrix}$, multiply R_1 by α , R_2 by β , and R_3 by γ :

$$\Delta_2 = \frac{1}{\alpha\beta\gamma} \begin{vmatrix} \alpha & \alpha^2 & \alpha\beta\gamma \\ \beta & \beta^2 & \alpha\beta\gamma \\ \gamma & \gamma^2 & \alpha\beta\gamma \end{vmatrix}$$

Factoring out $\alpha\beta\gamma$ from C_3 :

$$\Delta_2 = \frac{\alpha\beta\gamma}{\alpha\beta\gamma} \begin{vmatrix} \alpha & \alpha^2 & 1 \\ \beta & \beta^2 & 1 \\ \gamma & \gamma^2 & 1 \end{vmatrix} = \begin{vmatrix} \alpha & \alpha^2 & 1 \\ \beta & \beta^2 & 1 \\ \gamma & \gamma^2 & 1 \end{vmatrix}$$

Interchanging $C_3 \leftrightarrow C_2$, then $C_2 \leftrightarrow C_1$:

$$\Delta_2 = - \begin{vmatrix} \alpha & 1 & \alpha^2 \\ \beta & 1 & \beta^2 \\ \gamma & 1 & \gamma^2 \end{vmatrix} = \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix} = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

Adding Δ_1 and Δ_2 :

$$\begin{aligned} \Delta &= \Delta_1 + \Delta_2 \\ &= (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) + (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) \\ &= 2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) \end{aligned}$$

Hence Proved*LHS = RHS, hence proved.***Question 30**

Prove that:

$$\begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix} = \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$$

TYPE 10 Proving Determinant Identities Using Properties**★ Likely Exam Question**

To Prove: $\begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix} = \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(1 + a^2 + b^2 + c^2)$ **Solution:**

Let $\Delta_1 = \begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix}$.

Multiplying R_1 by a , R_2 by b , and R_3 by c :

$$\Delta_1 = \frac{1}{abc} \begin{vmatrix} a(a^2 + 1) & a^2b & a^2c \\ ab^2 & b(b^2 + 1) & b^2c \\ ac^2 & bc^2 & c(c^2 + 1) \end{vmatrix}$$

Factoring out a from C_1 , b from C_2 , and c from C_3 :

$$\Delta_1 = \frac{abc}{abc} \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = \Delta_2$$

Now evaluating Δ_2 :Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta_2 = \begin{vmatrix} 1 + a^2 + b^2 + c^2 & b^2 & c^2 \\ 1 + a^2 + b^2 + c^2 & b^2 + 1 & c^2 \\ 1 + a^2 + b^2 + c^2 & b^2 & c^2 + 1 \end{vmatrix}$$

Factoring out $(1 + a^2 + b^2 + c^2)$ from C_1 :

$$\Delta_2 = (1 + a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 & c^2 \\ 1 & b^2 + 1 & c^2 \\ 1 & b^2 & c^2 + 1 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta_2 = (1 + a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 & c^2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned}\Delta_2 &= (1 + a^2 + b^2 + c^2) \cdot 1 \cdot (1 - 0) \\ &= 1 + a^2 + b^2 + c^2\end{aligned}$$

Hence Proved

$LHS = MiddleTerm = RHS$, hence proved.

Question 31

Prove that:

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

Concept / Formula Used: $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

Solution:

Let $\Delta_1 = \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$.

Multiplying C_1 by a , C_2 by b , and C_3 by c :

$$\Delta_1 = \frac{1}{abc} \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ abc & abc & abc \end{vmatrix}$$

Factoring out abc from R_3 :

$$\Delta_1 = \frac{abc}{abc} \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix}$$

Interchanging $R_3 \leftrightarrow R_2$, then $R_2 \leftrightarrow R_1$:

$$\Delta_1 = - \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a^3 & b^3 & c^3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

Taking transpose ($\det(A) = \det(A^T)$):

$$\Delta_1 = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \Delta_2$$

Now evaluating Δ_2 :

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta_2 = \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2 - a^2 & b^3 - a^3 \\ 0 & c^2 - a^2 & c^3 - a^3 \end{vmatrix}$$

Factoring out $(b - a)$ from R_2 and $(c - a)$ from R_3 :

$$\Delta_2 = (b - a)(c - a) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b + a & b^2 + ab + a^2 \\ 0 & c + a & c^2 + ac + a^2 \end{vmatrix}$$

Expanding along C_1 :

$$\Delta_2 = (b - a)(c - a) [(b + a)(c^2 + ac + a^2) - (c + a)(b^2 + ab + a^2)]$$

Simplifying the terms inside the square bracket:

$$\text{First product} = (b + a)(c^2 + ac + a^2) = bc^2 + abc + ba^2 + ac^2 + a^2c + a^3$$

$$\text{Second product} = (c + a)(b^2 + ab + a^2) = cb^2 + abc + ca^2 + ab^2 + a^2b + a^3$$

$$\text{Difference} = (bc^2 - cb^2) + (ac^2 - ab^2)$$

$$= bc(c - b) + a(c^2 - b^2)$$

$$= (c - b)[bc + a(c + b)]$$

$$= (c - b)(ab + bc + ca)$$

Substituting back into Δ_2 :

$$\begin{aligned} \Delta_2 &= (b - a)(c - a)(c - b)(ab + bc + ca) \\ &= [-(a - b)][-(c - a)][-(b - c)](ab + bc + ca) \\ &= (a - b)(b - c)(c - a)(ab + bc + ca) \end{aligned}$$

Hence Proved

LHS = MiddleTerm = RHS, hence proved.

Question 32

Prove that:

$$\begin{vmatrix} a^2 & a^2 - (b - c)^2 & bc \\ b^2 & b^2 - (c - a)^2 & ca \\ c^2 & c^2 - (a - b)^2 & ab \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)(a^2 + b^2 + c^2)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a^2 & a^2 - (b - c)^2 & bc \\ b^2 & b^2 - (c - a)^2 & ca \\ c^2 & c^2 - (a - b)^2 & ab \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)(a^2 + b^2 + c^2)$

Concept / Formula Used: $(b - c)^2 + 2bc = b^2 + c^2$

$$\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$$

Solution:

$$\text{Let } \Delta = \begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}.$$

Applying $C_2 \rightarrow C_1 - C_2 + 2C_3$:

$$\text{Row 1} = a^2 - [a^2 - (b-c)^2] + 2bc = (b-c)^2 + 2bc = b^2 + c^2$$

$$\text{Row 2} = b^2 - [b^2 - (c-a)^2] + 2ca = (c-a)^2 + 2ca = c^2 + a^2$$

$$\text{Row 3} = c^2 - [c^2 - (a-b)^2] + 2ab = (a-b)^2 + 2ab = a^2 + b^2$$

The determinant becomes:

$$\Delta = \begin{vmatrix} a^2 & b^2 + c^2 & bc \\ b^2 & c^2 + a^2 & ca \\ c^2 & a^2 + b^2 & ab \end{vmatrix}$$

Now applying $C_2 \rightarrow C_2 + C_1$:

$$\Delta = \begin{vmatrix} a^2 & a^2 + b^2 + c^2 & bc \\ b^2 & a^2 + b^2 + c^2 & ca \\ c^2 & a^2 + b^2 + c^2 & ab \end{vmatrix}$$

Factoring out $(a^2 + b^2 + c^2)$ from C_2 :

$$\Delta = (a^2 + b^2 + c^2) \begin{vmatrix} a^2 & 1 & bc \\ b^2 & 1 & ca \\ c^2 & 1 & ab \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\Delta = -(a^2 + b^2 + c^2) \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$$

$$\text{Evaluating } \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}: \text{Applying } R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1:$$

$$\begin{vmatrix} 1 & a^2 & bc \\ 0 & b^2 - a^2 & ca - bc \\ 0 & c^2 - a^2 & ab - bc \end{vmatrix} = \begin{vmatrix} 1 & a^2 & bc \\ 0 & (b-a)(b+a) & -c(b-a) \\ 0 & (c-a)(c+a) & -b(c-a) \end{vmatrix}$$

Factoring out $(b-a)$ from R_2 and $(c-a)$ from R_3 :

$$= (b-a)(c-a) \begin{vmatrix} 1 & a^2 & bc \\ 0 & b+a & -c \\ 0 & c+a & -b \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} &= (b-a)(c-a) [-b(b+a) - (-c)(c+a)] \\ &= (b-a)(c-a) [-b^2 - ab + c^2 + ac] \\ &= (b-a)(c-a) [(c^2 - b^2) + a(c-b)] \\ &= (b-a)(c-a)(c-b)(a+b+c) \\ &= -(a-b)(b-c)(c-a)(a+b+c) \end{aligned}$$

Substituting this back into Δ :

$$\begin{aligned}\Delta &= -(a^2 + b^2 + c^2) [-(a-b)(b-c)(c-a)(a+b+c)] \\ &= (a-b)(b-c)(c-a)(a+b+c)(a^2 + b^2 + c^2)\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 33

Prove that:

$$\begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)(a^2 + b^2 + c^2)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)(a^2 + b^2 + c^2)$

Concept / Formula Used: $(b+c)^2 - 2bc = b^2 + c^2$

Solution:

Let $\Delta = \begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix}$.

Applying $C_1 \rightarrow C_1 - C_2 - 2C_3$:

$$\text{Row 1} = (b+c)^2 - a^2 - 2bc = b^2 + c^2 + 2bc - a^2 - 2bc = b^2 + c^2 - a^2$$

$$\text{Row 2} = (c+a)^2 - b^2 - 2ca = c^2 + a^2 + 2ca - b^2 - 2ca = c^2 + a^2 - b^2$$

$$\text{Row 3} = (a+b)^2 - c^2 - 2ab = a^2 + b^2 + 2ab - c^2 - 2ab = a^2 + b^2 - c^2$$

Substituting back into C_1 :

$$\Delta = \begin{vmatrix} b^2 + c^2 - a^2 & a^2 & bc \\ c^2 + a^2 - b^2 & b^2 & ca \\ a^2 + b^2 - c^2 & c^2 & ab \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_2$:

$$\Delta = \begin{vmatrix} b^2 + c^2 & a^2 & bc \\ c^2 + a^2 & b^2 & ca \\ a^2 + b^2 & c^2 & ab \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_2$:

$$\Delta = \begin{vmatrix} a^2 + b^2 + c^2 & a^2 & bc \\ a^2 + b^2 + c^2 & b^2 & ca \\ a^2 + b^2 + c^2 & c^2 & ab \end{vmatrix}$$

Factoring out $(a^2 + b^2 + c^2)$ from C_1 :

$$\Delta = (a^2 + b^2 + c^2) \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$$

Using the standard result derived in Question 32 for $\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$:

$$\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

Thus:

$$\Delta = (a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$$

Hence Proved

LHS = RHS, hence proved.

Question 34 (i)

Prove that:

$$\begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix} = 2(a+b)(b+c)(c+a)$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix} = 2(a+b)(b+c)(c+a)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ first column elements become a, b, c

Solution:

Let $\Delta = \begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix}$.

Applying $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{Row 1} = (a+b+c) - c - b = a$$

$$\text{Row 2} = -c + (a+b+c) - a = b$$

$$\text{Row 3} = -b - a + (a+b+c) = c$$

The determinant becomes:

$$\Delta = \begin{vmatrix} a & -c & -b \\ b & a+b+c & -a \\ c & -a & a+b+c \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2$:

$$\Delta = \begin{vmatrix} a+b & a+b & -(a+b) \\ b & a+b+c & -a \\ c & -a & a+b+c \end{vmatrix}$$

Factoring out $(a+b)$ from R_1 :

$$\Delta = (a+b) \begin{vmatrix} 1 & 1 & -1 \\ b & a+b+c & -a \\ c & -a & a+b+c \end{vmatrix}$$

Applying column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 + C_1$:

$$C_2 - C_1 = \begin{pmatrix} 1-1 \\ (a+b+c)-b \\ -a-c \end{pmatrix} = \begin{pmatrix} 0 \\ a+c \\ -(a+c) \end{pmatrix}$$

$$C_3 + C_1 = \begin{pmatrix} -1+1 \\ -a+b \\ (a+b+c)+c \end{pmatrix} = \begin{pmatrix} 0 \\ b-a \\ a+b+2c \end{pmatrix}$$

Substituting these into the determinant:

$$\Delta = (a+b) \begin{vmatrix} 1 & 0 & 0 \\ b & a+c & b-a \\ c & -(a+c) & a+b+2c \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned} \Delta &= (a+b) [(a+c)(a+b+2c) - (b-a)(-(a+c))] \\ &= (a+b) [(a+c)(a+b+2c) + (b-a)(a+c)] \end{aligned}$$

Factoring out $(a+c)$ from the square bracket:

$$\begin{aligned} \Delta &= (a+b)(a+c) [(a+b+2c) + (b-a)] \\ &= (a+b)(a+c)[2b+2c] \\ &= 2(a+b)(b+c)(c+a) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 34 (ii)

Prove that:

$$\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(a+x+y+z)$

Solution:

Let $\Delta = \begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix}$.

Applying $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} a+x+y+z & y & z \\ a+x+y+z & a+y & z \\ a+x+y+z & y & a+z \end{vmatrix}$$

Factoring out $(a + x + y + z)$ from C_1 :

$$\Delta = (a + x + y + z) \begin{vmatrix} 1 & y & z \\ 1 & a + y & z \\ 1 & y & a + z \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= (1 - 1 \quad (a + y) - y \quad z - z) = (0 \quad a \quad 0) \\ R_3 - R_1 &= (1 - 1 \quad y - y \quad (a + z) - z) = (0 \quad 0 \quad a) \end{aligned}$$

Substituting these rows into the determinant:

$$\Delta = (a + x + y + z) \begin{vmatrix} 1 & y & z \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta &= (a + x + y + z) \cdot 1 \cdot (a \cdot a - 0) \\ &= a^2(a + x + y + z) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 35

Prove that:

$$\begin{vmatrix} y + z & x + y & x \\ z + x & y + z & y \\ x + y & z + x & z \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} y + z & x + y & x \\ z + x & y + z & y \\ x + y & z + x & z \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz$

Concept / Formula Used: $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

Solution:

Let $\Delta = \begin{vmatrix} y + z & x + y & x \\ z + x & y + z & y \\ x + y & z + x & z \end{vmatrix}$.

Applying $C_2 \rightarrow C_2 - C_3$:

$$\Delta = \begin{vmatrix} y + z & y & x \\ z + x & z & y \\ x + y & x & z \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_2$:

$$\Delta = \begin{vmatrix} z & y & x \\ x & z & y \\ y & x & z \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$:

$$\Delta = \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ x & z & y \\ y & x & z \end{vmatrix}$$

Factoring out $(x+y+z)$ from R_1 :

$$\Delta = (x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ x & z & y \\ y & x & z \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$\Delta = (x+y+z) \begin{vmatrix} 1 & 0 & 0 \\ x & z-x & y-x \\ y & x-y & z-y \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned} \Delta &= (x+y+z) [(z-x)(z-y) - (y-x)(x-y)] \\ &= (x+y+z) [(z^2 - zy - zx + xy) + (x-y)^2] \\ &= (x+y+z) [z^2 - yz - zx + xy + x^2 - 2xy + y^2] \\ &= (x+y+z) [x^2 + y^2 + z^2 - xy - yz - zx] \\ &= x^3 + y^3 + z^3 - 3xyz \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 36

Prove that:

$$\begin{vmatrix} a & b-c & c-b \\ a-c & b & c-a \\ a-b & b-a & c \end{vmatrix} = (a+b-c)(b+c-a)(c+a-b)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a & b-c & c-b \\ a-c & b & c-a \\ a-b & b-a & c \end{vmatrix} = (a+b-c)(b+c-a)(c+a-b)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 \implies$ factor out $(a+b-c)$

Solution:

Let $\Delta = \begin{vmatrix} a & b-c & c-b \\ a-c & b & c-a \\ a-b & b-a & c \end{vmatrix}$.

Applying $C_1 \rightarrow C_1 + C_2$:

$$\text{Row 1} = a + (b-c) = a+b-c$$

$$\text{Row 2} = (a-c) + b = a+b-c$$

$$\text{Row 3} = (a-b) + (b-a) = 0$$

Substituting into C_1 :

$$\Delta = \begin{vmatrix} a+b-c & b-c & c-b \\ a+b-c & b & c-a \\ 0 & b-a & c \end{vmatrix}$$

Factoring out $(a+b-c)$ from C_1 :

$$\Delta = (a+b-c) \begin{vmatrix} 1 & b-c & c-b \\ 1 & b & c-a \\ 0 & b-a & c \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$:

$$\begin{aligned} R_2 - R_1 &= (1-1 \quad b-(b-c) \quad (c-a)-(c-b)) \\ &= (0 \quad c \quad b-a) \end{aligned}$$

Substituting back into R_2 :

$$\Delta = (a+b-c) \begin{vmatrix} 1 & b-c & c-b \\ 0 & c & b-a \\ 0 & b-a & c \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta &= (a+b-c) \cdot 1 \cdot [c^2 - (b-a)^2] \\ &= (a+b-c) [c - (b-a)] [c + (b-a)] \\ &= (a+b-c)(c-b+a)(c+b-a) \\ &= (a+b-c)(b+c-a)(c+a-b) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 37

Prove that:

$$\begin{vmatrix} 1 & bc+ad & b^2c^2+a^2d^2 \\ 1 & ca+bd & c^2a^2+b^2d^2 \\ 1 & ab+cd & a^2b^2+c^2d^2 \end{vmatrix} = (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1 & bc+ad & b^2c^2+a^2d^2 \\ 1 & ca+bd & c^2a^2+b^2d^2 \\ 1 & ab+cd & a^2b^2+c^2d^2 \end{vmatrix} = (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$

Concept / Formula Used: $(bc+ad)^2 - 2abcd = b^2c^2 + a^2d^2$

Solution:

Let $x = bc + ad$, $y = ca + bd$, and $z = ab + cd$.

Note that:

$$\begin{aligned} x^2 &= (bc+ad)^2 = b^2c^2 + a^2d^2 + 2abcd \implies b^2c^2 + a^2d^2 = x^2 - 2abcd \\ y^2 &= (ca+bd)^2 = c^2a^2 + b^2d^2 + 2abcd \implies c^2a^2 + b^2d^2 = y^2 - 2abcd \\ z^2 &= (ab+cd)^2 = a^2b^2 + c^2d^2 + 2abcd \implies a^2b^2 + c^2d^2 = z^2 - 2abcd \end{aligned}$$

Substituting these into the determinant Δ :

$$\Delta = \begin{vmatrix} 1 & x & x^2 - 2abcd \\ 1 & y & y^2 - 2abcd \\ 1 & z & z^2 - 2abcd \end{vmatrix}$$

Applying $C_3 \rightarrow C_3 + 2abcdC_1$:

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x - y)(y - z)(z - x)$$

Now computing the factors $x - y$, $y - z$, and $z - x$:

$$x - y = (bc + ad) - (ca + bd) = c(b - a) - d(b - a) = (b - a)(c - d) = -(a - b)(c - d)$$

$$y - z = (ca + bd) - (ab + cd) = a(c - b) - d(c - b) = (c - b)(a - d) = -(b - c)(a - d)$$

$$z - x = (ab + cd) - (bc + ad) = b(a - c) - d(a - c) = (a - c)(b - d)$$

Multiplying these factors together:

$$\begin{aligned} \Delta &= [-(a - b)(c - d)] \cdot [-(b - c)(a - d)] \cdot [(a - c)(b - d)] \\ &= (a - b)(b - c)(a - d)(c - d)(a - c)(b - d) \\ &= (a - b)(a - c)(a - d)(b - c)(b - d)(c - d) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 38

If a, b, c are all non-zero and:

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$$

then prove that:

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

Given: $a, b, c \neq 0$ and $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$

To Prove: $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$

Concept / Formula Used: $R_1 \rightarrow R_1 + R_2 + R_3 \implies$ factor out $(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c})$

Solution:

Let $\Delta = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$.

Factoring out a from R_1 , b from R_2 , and c from R_3 :

$$\Delta = abc \begin{vmatrix} \frac{1}{a} + 1 & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & \frac{1}{b} + 1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c} + 1 \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$:

$$\Delta = abc \begin{vmatrix} 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \\ \frac{1}{b} & \frac{1}{b} + 1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c} + 1 \end{vmatrix}$$

Factoring out $(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c})$ from R_1 :

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & \frac{1}{b} + 1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c} + 1 \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$$

Expanding along R_1 :

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \cdot (1 - 0) = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

We are given that $\Delta = 0$. Since a, b, c are non-zero, $abc \neq 0$:

$$abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = 0 \implies 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$$

Hence Proved

Hence proved.

Question 39

Without expanding any of the determinants given below, prove that:

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$

Concept / Formula Used: $(a + 1)^2 - a^2 = 2a + 1$ and $(a - 1)^2 - a^2 = -2a + 1$

Solution:

$$\text{Let } \Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}.$$

Expanding the algebraic squares in R_2 and R_3 :

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^2 + 2a + 1 & b^2 + 2b + 1 & c^2 + 2c + 1 \\ a^2 - 2a + 1 & b^2 - 2b + 1 & c^2 - 2c + 1 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ 2a + 1 & 2b + 1 & 2c + 1 \\ -2a + 1 & -2b + 1 & -2c + 1 \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 + R_3$:

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ 2 & 2 & 2 \\ -2a + 1 & -2b + 1 & -2c + 1 \end{vmatrix}$$

Factoring out 2 from R_2 :

$$\Delta = 2 \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ -2a + 1 & -2b + 1 & -2c + 1 \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - R_2$:

$$\Delta = 2 \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ -2a & -2b & -2c \end{vmatrix}$$

Factoring out -2 from R_3 :

$$\Delta = 2 \cdot (-2) \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix} = -4 \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}$$

Interchanging $R_2 \leftrightarrow R_3$ (multiplying by -1):

$$\Delta = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

Hence Proved

LHS = RHS, hence proved.

Question 40 (i)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} 3x - 8 & 3 & 3 \\ 3 & 3x - 8 & 3 \\ 3 & 3 & 3x - 8 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The values of x satisfying the determinant equation.

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(3x - 2)$

Solution:

Given equation:

$$\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{Row 1} = (3x-8) + 3 + 3 = 3x-2$$

$$\text{Row 2} = 3 + (3x-8) + 3 = 3x-2$$

$$\text{Row 3} = 3 + 3 + (3x-8) = 3x-2$$

Substituting into C_1 :

$$\begin{vmatrix} 3x-2 & 3 & 3 \\ 3x-2 & 3x-8 & 3 \\ 3x-2 & 3 & 3x-8 \end{vmatrix} = 0$$

Factoring out $(3x-2)$ from C_1 :

$$(3x-2) \begin{vmatrix} 1 & 3 & 3 \\ 1 & 3x-8 & 3 \\ 1 & 3 & 3x-8 \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(3x-2) \begin{vmatrix} 1 & 3 & 3 \\ 0 & 3x-11 & 0 \\ 0 & 0 & 3x-11 \end{vmatrix} = 0$$

Expanding along C_1 :

$$(3x-2) \cdot 1 \cdot [(3x-11)(3x-11) - 0] = 0$$

$$(3x-2)(3x-11)^2 = 0$$

Equating each factor to zero:

$$3x-2=0 \implies x = \frac{2}{3}$$

$$(3x-11)^2=0 \implies x = \frac{11}{3}$$

Final Answer

$$x = \frac{2}{3} \quad \text{or} \quad x = \frac{11}{3}$$

Question 40 (ii)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The values of x satisfying the determinant equation.

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(x+9)$

Solution:

Given equation:

$$\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{Row 1} = (x+1) + 3 + 5 = x+9$$

$$\text{Row 2} = 2 + (x+2) + 5 = x+9$$

$$\text{Row 3} = 2 + 3 + (x+4) = x+9$$

Substituting into C_1 :

$$\begin{vmatrix} x+9 & 3 & 5 \\ x+9 & x+2 & 5 \\ x+9 & 3 & x+4 \end{vmatrix} = 0$$

Factoring out $(x+9)$ from C_1 :

$$(x+9) \begin{vmatrix} 1 & 3 & 5 \\ 1 & x+2 & 5 \\ 1 & 3 & x+4 \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(x+9) \begin{vmatrix} 1 & 3 & 5 \\ 0 & x-1 & 0 \\ 0 & 0 & x-1 \end{vmatrix} = 0$$

Expanding along C_1 :

$$(x+9) \cdot 1 \cdot [(x-1)(x-1) - 0] = 0$$

$$(x+9)(x-1)^2 = 0$$

Equating each factor to zero:

$$x+9=0 \implies x=-9$$

$$(x-1)^2=0 \implies x=1$$

Final Answer

$$x = -9 \text{ or } x = 1$$

Question 40 (iii)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The values of x satisfying the determinant equation.

Concept / Formula Used: $\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$

Solution:

Given equation:

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$$

Expanding along R_1 :

$$x \begin{vmatrix} -3x & x-3 \\ 2x & x+2 \end{vmatrix} - (-6) \begin{vmatrix} 2 & x-3 \\ -3 & x+2 \end{vmatrix} + (-1) \begin{vmatrix} 2 & -3x \\ -3 & 2x \end{vmatrix} = 0$$

Evaluating each 2×2 determinant separately:

$$\begin{aligned} \text{First term} &= x [(-3x)(x+2) - (2x)(x-3)] \\ &= x [-3x^2 - 6x - 2x^2 + 6x] \\ &= x [-5x^2] = -5x^3 \end{aligned}$$

$$\begin{aligned} \text{Second term} &= +6 [2(x+2) - (-3)(x-3)] \\ &= 6 [2x + 4 + 3x - 9] \\ &= 6 [5x - 5] = 30x - 30 \end{aligned}$$

$$\begin{aligned} \text{Third term} &= -1 [2(2x) - (-3)(-3x)] \\ &= -1 [4x - 9x] \\ &= -1 [-5x] = 5x \end{aligned}$$

Summing all terms together:

$$\begin{aligned} -5x^3 + (30x - 30) + 5x &= 0 \\ -5x^3 + 35x - 30 &= 0 \end{aligned}$$

Dividing both sides by -5 :

$$x^3 - 7x + 6 = 0$$

Testing $x = 1$: $1^3 - 7(1) + 6 = 0$, so $(x - 1)$ is a factor. Factorizing $x^3 - 7x + 6$ using synthetic division / grouping:

$$\begin{aligned} x^3 - x^2 + x^2 - x - 6x + 6 &= 0 \\ x^2(x-1) + x(x-1) - 6(x-1) &= 0 \\ (x-1)(x^2 + x - 6) &= 0 \\ (x-1)(x+3)(x-2) &= 0 \end{aligned}$$

Setting each factor to zero:

$$x = 1, \quad x = 2, \quad x = -3$$

Final Answer

$$x = 1, \quad x = 2, \quad \text{or} \quad x = -3$$

Question 40 (iv)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} 15 - 2x & 11 & 10 \\ 11 - 3x & 17 & 16 \\ 7 - x & 14 & 13 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The value of x satisfying the determinant equation.

Concept / Formula Used: $C_2 \rightarrow C_2 - C_3 \implies$ column 2 becomes $[1, 1, 1]^T$

Solution:

Given equation:

$$\begin{vmatrix} 15 - 2x & 11 & 10 \\ 11 - 3x & 17 & 16 \\ 7 - x & 14 & 13 \end{vmatrix} = 0$$

Applying column operation $C_2 \rightarrow C_2 - C_3$:

$$\begin{vmatrix} 15 - 2x & 1 & 10 \\ 11 - 3x & 1 & 16 \\ 7 - x & 1 & 13 \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= ((11 - 3x) - (15 - 2x) \quad 1 - 1 \quad 16 - 10) = (-4 - x \quad 0 \quad 6) \\ R_3 - R_1 &= ((7 - x) - (15 - 2x) \quad 1 - 1 \quad 13 - 10) = (x - 8 \quad 0 \quad 3) \end{aligned}$$

Substituting these into the determinant:

$$\begin{vmatrix} 15 - 2x & 1 & 10 \\ -4 - x & 0 & 6 \\ x - 8 & 0 & 3 \end{vmatrix} = 0$$

Expanding along C_2 :

$$\begin{aligned} -1 \cdot [(-4 - x)(3) - 6(x - 8)] &= 0 \\ -3(-4 - x) + 6(x - 8) &= 0 \\ 12 + 3x + 6x - 48 &= 0 \\ 9x - 36 &= 0 \\ 9x &= 36 \\ x &= 4 \end{aligned}$$

Final Answer

$$x = 4$$

Question 41

If a, b, c are all different and:

$$\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$$

prove that $abc(ab + bc + ca) = a + b + c$.

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

Given: a, b, c are distinct and $\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$

To Prove: $abc(ab + bc + ca) = a + b + c$

Concept / Formula Used: $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$

Solution:

Let $\Delta = \begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$.

Splitting C_3 into two determinants:

$$\Delta = \begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} - \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix} = 0$$

For the first determinant, factoring out a from R_1 , b from R_2 , and c from R_3 :

$$\begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} = abc \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

For the second determinant, performing column interchanges $C_3 \leftrightarrow C_2$ then $C_2 \leftrightarrow C_1$:

$$\begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix} = - \begin{vmatrix} a & 1 & a^3 \\ b & 1 & b^3 \\ c & 1 & c^3 \end{vmatrix} = \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$$

Substituting both back into the equation:

$$abc \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} - \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = 0$$

Using standard cyclic determinant identities:

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

Substituting these into the equation:

$$abc(a-b)(b-c)(c-a)(ab+bc+ca) - (a-b)(b-c)(c-a)(a+b+c) = 0$$

Factoring out $(a-b)(b-c)(c-a)$:

$$(a-b)(b-c)(c-a)[abc(ab+bc+ca) - (a+b+c)] = 0$$

Since a, b, c are distinct, $(a-b) \neq 0$, $(b-c) \neq 0$, and $(c-a) \neq 0$:

$$abc(ab+bc+ca) - (a+b+c) = 0$$

$$abc(ab+bc+ca) = a+b+c$$

Hence Proved

Hence proved.

Question 42

In a triangle ABC , if:

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos A + \cos^2 A & \cos B + \cos^2 B & \cos C + \cos^2 C \end{vmatrix} = 0$$

then prove that ABC is an isosceles triangle.

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

Given: In $\triangle ABC$, $\begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos A + \cos^2 A & \cos B + \cos^2 B & \cos C + \cos^2 C \end{vmatrix} = 0$

To Prove: $\triangle ABC$ is an isosceles triangle.

Concept / Formula Used: $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$

Solution:

Let $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos A + \cos^2 A & \cos B + \cos^2 B & \cos C + \cos^2 C \end{vmatrix} = 0$.

Applying row operation $R_2 \rightarrow R_2 - R_1$:

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \cos A & \cos B & \cos C \\ \cos A(1 + \cos A) & \cos B(1 + \cos B) & \cos C(1 + \cos C) \end{vmatrix} = 0$$

Applying row operation $R_3 \rightarrow R_3 - R_2$:

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \cos A & \cos B & \cos C \\ \cos^2 A & \cos^2 B & \cos^2 C \end{vmatrix} = 0$$

This is the standard Vandermonde determinant in terms of $\cos A, \cos B, \cos C$:

$$(\cos A - \cos B)(\cos B - \cos C)(\cos C - \cos A) = 0$$

Equating each factor to zero:

$$\begin{aligned} \cos A - \cos B = 0 &\implies \cos A = \cos B \implies A = B \\ \text{or } \cos B - \cos C = 0 &\implies \cos B = \cos C \implies B = C \\ \text{or } \cos C - \cos A = 0 &\implies \cos C = \cos A \implies C = A \end{aligned}$$

Since at least two angles of $\triangle ABC$ are equal ($A = B$, $B = C$, or $C = A$), triangle ABC is isosceles.

Hence Proved

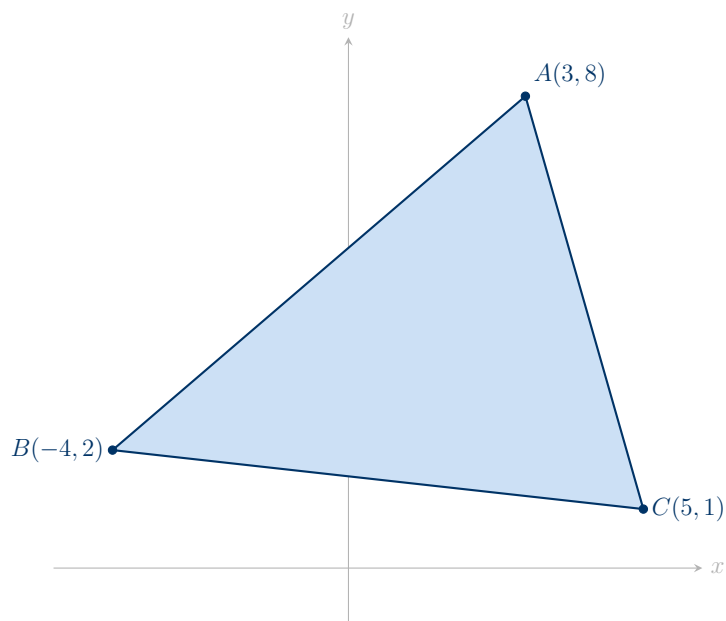
Hence proved.

4.3 – Applications of Determinants: Area, Collinearity and Equation of a Line

Question 1 (i)

Using determinants, find the area of the triangle whose vertices are $(3, 8)$, $(-4, 2)$, and $(5, 1)$.

TYPE 11 Area of Triangle Using Determinants



Triangle with vertices $A(3, 8)$, $B(-4, 2)$, $C(5, 1)$

Given: Vertices of the triangle are $A(3, 8)$, $B(-4, 2)$, and $C(5, 1)$.

Concept / Formula Used: Area of triangle $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Solution:

Let $(x_1, y_1) = (3, 8)$, $(x_2, y_2) = (-4, 2)$, and $(x_3, y_3) = (5, 1)$.

The area of triangle ABC is given by:

$$\Delta = \frac{1}{2} \begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix}$$

Evaluating the determinant by expanding along the first row:

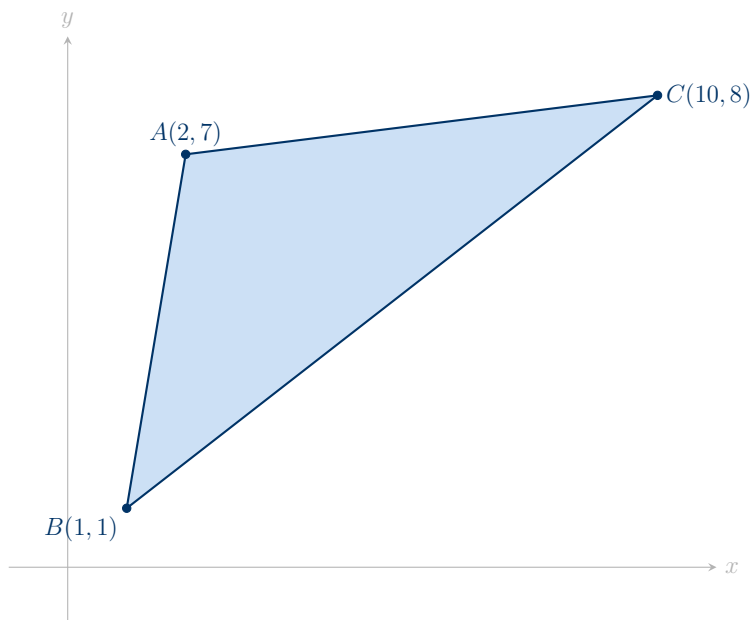
$$\begin{aligned} \begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix} &= 3 \cdot \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} - 8 \cdot \begin{vmatrix} -4 & 1 \\ 5 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} -4 & 2 \\ 5 & 1 \end{vmatrix} \\ &= 3(2 \cdot 1 - 1 \cdot 1) - 8((-4) \cdot 1 - 1 \cdot 5) + 1((-4) \cdot 1 - 2 \cdot 5) \\ &= 3(2 - 1) - 8(-4 - 5) + 1(-4 - 10) \\ &= 3(1) - 8(-9) + 1(-14) \\ &= 3 + 72 - 14 \\ &= 61 \end{aligned}$$

Substituting this determinant value back into the area expression:

$$\begin{aligned} \Delta &= \frac{1}{2} \times 61 \\ &= \frac{61}{2} \\ &= 30.5 \text{ sq. units} \end{aligned}$$

Final Answer

Area = 30.5 sq. units

Question 1 (ii)Using determinants, find the area of the triangle whose vertices are $(2, 7)$, $(1, 1)$, and $(10, 8)$.**TYPE 11 Area of Triangle Using Determinants***Triangle with vertices $A(2, 7)$, $B(1, 1)$, $C(10, 8)$* **Given:** Vertices of the triangle are $A(2, 7)$, $B(1, 1)$, and $C(10, 8)$.**Concept / Formula Used:** Area of triangle $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ **Solution:**Let $(x_1, y_1) = (2, 7)$, $(x_2, y_2) = (1, 1)$, and $(x_3, y_3) = (10, 8)$.The area of triangle ABC is given by:

$$\Delta = \frac{1}{2} \begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix}$$

Evaluating the determinant by expanding along the first row:

$$\begin{aligned}
 \begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix} &= 2 \cdot \begin{vmatrix} 1 & 1 \\ 8 & 1 \end{vmatrix} - 7 \cdot \begin{vmatrix} 1 & 1 \\ 10 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 1 \\ 10 & 8 \end{vmatrix} \\
 &= 2(1 \cdot 1 - 8 \cdot 1) - 7(1 \cdot 1 - 10 \cdot 1) + 1(1 \cdot 8 - 10 \cdot 1) \\
 &= 2(1 - 8) - 7(1 - 10) + 1(8 - 10) \\
 &= 2(-7) - 7(-9) + 1(-2) \\
 &= -14 + 63 - 2 \\
 &= 47
 \end{aligned}$$

Substituting this value back into the area expression:

$$\begin{aligned}
 \Delta &= \frac{1}{2} \times 47 \\
 &= \frac{47}{2} \\
 &= 23.5 \text{ sq. units}
 \end{aligned}$$

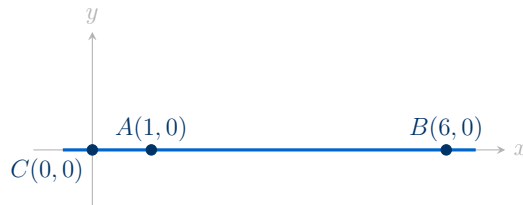
Final Answer

Area = 23.5 sq. units

Question 2

Show that the points $(1, 0)$, $(6, 0)$, $(0, 0)$ are collinear.

TYPE 12 Collinearity of Points Using Determinants



Collinear points $(1, 0)$, $(6, 0)$, $(0, 0)$ on the x -axis

To Prove: The points $A(1, 0)$, $B(6, 0)$, and $C(0, 0)$ are collinear.

Concept / Formula Used: $(x_1, y_1) \quad (x_2, y_2) \quad (x_3, y_3) \quad \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Solution:

Let $(x_1, y_1) = (1, 0)$, $(x_2, y_2) = (6, 0)$, and $(x_3, y_3) = (0, 0)$.

We set up the determinant D for the area of the triangle formed by these three points:

$$D = \begin{vmatrix} 1 & 0 & 1 \\ 6 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along the second column C_2 :

$$\begin{aligned} D &= -0 \cdot \begin{vmatrix} 6 & 1 \\ 0 & 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} - 0 \cdot \begin{vmatrix} 1 & 1 \\ 6 & 1 \end{vmatrix} \\ &= 0 + 0 - 0 \\ &= 0 \end{aligned}$$

Since the value of the determinant is equal to 0, the area of the triangle formed by these points is zero.

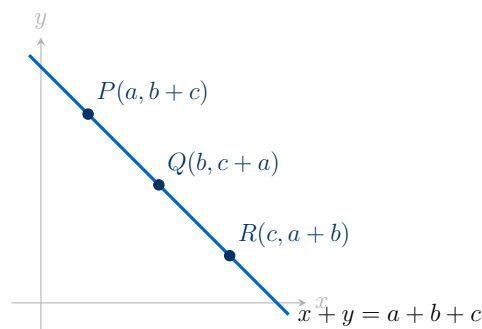
Hence Proved

Since the determinant vanishes, the points $(1, 0)$, $(6, 0)$, and $(0, 0)$ are collinear.

Question 3

Using determinants, determine whether the points $P(a, b + c)$, $Q(b, c + a)$ and $R(c, a + b)$ form a triangle or not.

TYPE 12 Collinearity of Points Using Determinants



Points $P(a, b + c)$, $Q(b, c + a)$, $R(c, a + b)$ on the line $x + y = a + b + c$

Given: Points $P(a, b + c)$, $Q(b, c + a)$, and $R(c, a + b)$.

Concept / Formula Used: $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \neq 0 \quad \Delta = 0$

Solution:

Consider the determinant expression for the area Δ :

$$\Delta = \frac{1}{2} \begin{vmatrix} a & b + c & 1 \\ b & c + a & 1 \\ c & a + b & 1 \end{vmatrix}$$

Applying the column operation $C_2 \rightarrow C_2 + C_1$:

$$\Delta = \frac{1}{2} \begin{vmatrix} a & a + b + c & 1 \\ b & b + c + a & 1 \\ c & c + a + b & 1 \end{vmatrix}$$

Taking out the common factor $(a + b + c)$ from column C_2 :

$$\Delta = \frac{1}{2}(a + b + c) \begin{vmatrix} a & 1 & 1 \\ b & 1 & 1 \\ c & 1 & 1 \end{vmatrix}$$

Using the property that if any two columns of a determinant are identical, the value of the determinant is zero (here $C_2 = C_3$):

$$\begin{aligned} \Delta &= \frac{1}{2}(a + b + c) \times 0 \\ &= 0 \end{aligned}$$

Since the area of the triangle is 0, the points P, Q, R are collinear.

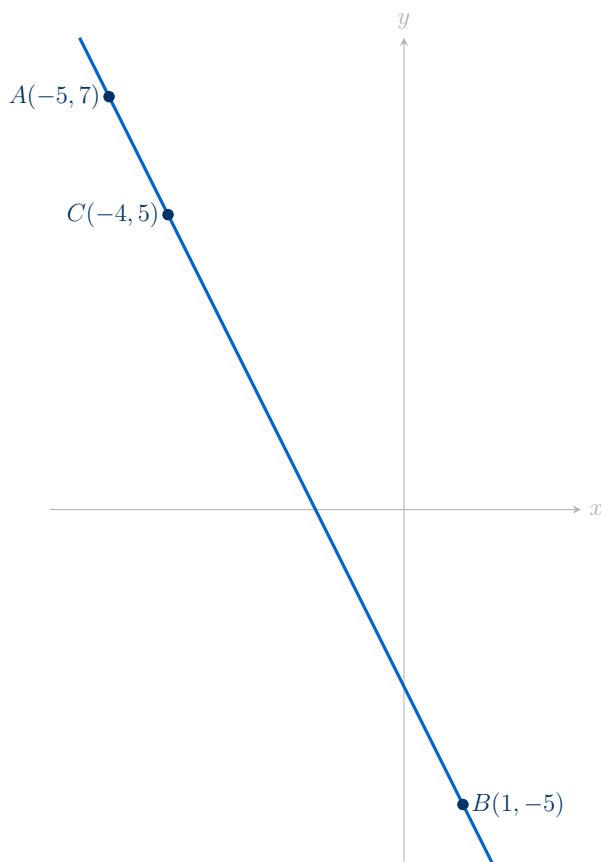
Final Answer

The given points P, Q, R are collinear and do not form a triangle.

Question 4

Using determinants, find the value of λ so that the points $(\lambda, 7)$, $(1, -5)$ and $(-4, 5)$ are collinear.

TYPE 12 Collinearity of Points Using Determinants



Collinear points $A(-5, 7)$, $B(1, -5)$, $C(-4, 5)$

Given: Points $A(\lambda, 7)$, $B(1, -5)$, and $C(-4, 5)$ are collinear.

Concept / Formula Used: $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Solution:

Since $A(\lambda, 7)$, $B(1, -5)$, and $C(-4, 5)$ are collinear, the determinant of their coordinates must equal zero:

$$\begin{vmatrix} \lambda & 7 & 1 \\ 1 & -5 & 1 \\ -4 & 5 & 1 \end{vmatrix} = 0$$

Expanding the determinant along the first row:

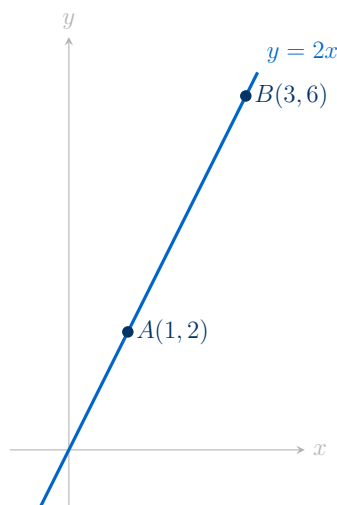
$$\begin{aligned} \lambda \cdot \begin{vmatrix} -5 & 1 \\ 5 & 1 \end{vmatrix} - 7 \cdot \begin{vmatrix} 1 & 1 \\ -4 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & -5 \\ -4 & 5 \end{vmatrix} &= 0 \\ \lambda((-5) \cdot 1 - 5 \cdot 1) - 7(1 \cdot 1 - 1 \cdot (-4)) + 1(1 \cdot 5 - (-5) \cdot (-4)) &= 0 \\ \lambda(-5 - 5) - 7(1 + 4) + 1(5 - 20) &= 0 \\ \lambda(-10) - 7(5) + 1(-15) &= 0 \\ -10\lambda - 35 - 15 &= 0 \\ -10\lambda - 50 &= 0 \\ -10\lambda &= 50 \\ \lambda &= \frac{50}{-10} \\ \lambda &= -5 \end{aligned}$$

Final Answer

$$\lambda = -5$$

Question 5

Using determinants, find the equation of the line passing through the points $(1, 2)$ and $(3, 6)$.

TYPE 13 Equation of a Line Using Determinants

Line $y = 2x$ passing through $(1, 2)$ and $(3, 6)$

Given: Points $A(1, 2)$ and $B(3, 6)$.

Concept / Formula Used: $(x_1, y_1) \quad (x_2, y_2) \quad \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

Solution:

Let $P(x, y)$ be any arbitrary point on the line joining $A(1, 2)$ and $B(3, 6)$.

Since $P(x, y)$, $A(1, 2)$, and $B(3, 6)$ are collinear, the determinant of their coordinates is zero:

$$\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 0$$

Expanding along the first row:

$$\begin{aligned} x \cdot \begin{vmatrix} 2 & 1 \\ 6 & 1 \end{vmatrix} - y \cdot \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} &= 0 \\ x(2 \cdot 1 - 6 \cdot 1) - y(1 \cdot 1 - 3 \cdot 1) + 1(1 \cdot 6 - 3 \cdot 2) &= 0 \\ x(2 - 6) - y(1 - 3) + 1(6 - 6) &= 0 \\ x(-4) - y(-2) + 1(0) &= 0 \\ -4x + 2y &= 0 \end{aligned}$$

Dividing both sides by -2 :

$$\begin{aligned} 2x - y &= 0 \\ y &= 2x \end{aligned}$$

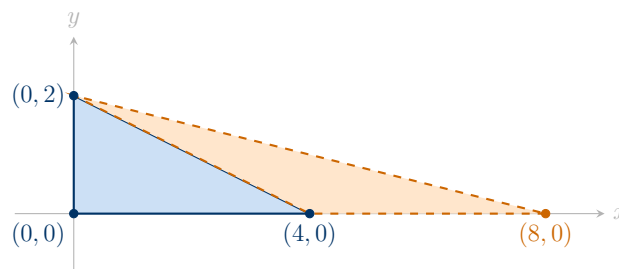
Final Answer

$$y = 2x$$

Question 6

Find the value(s) of k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$ and $(0, 2)$ is 4 sq. units.

TYPE 11 Area of Triangle Using Determinants



Triangles with vertices $(k, 0)$, $(4, 0)$, $(0, 2)$ for $k = 0$ and $k = 8$

Given: Vertices $A(k, 0)$, $B(4, 0)$, $C(0, 2)$, and Area = 4 sq. units.

Concept / Formula Used: $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \pm \text{given area}$

Solution:

Setting up the determinant equation with ± 4 since area is absolute:

$$\frac{1}{2} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} = \pm 4$$

Expanding the determinant along the first row:

$$\begin{aligned} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} &= k \cdot \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} - 0 + 1 \cdot \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} \\ &= k(0 \cdot 1 - 2 \cdot 1) + 1(4 \cdot 2 - 0 \cdot 0) \\ &= k(-2) + 8 \\ &= -2k + 8 \end{aligned}$$

Substituting back into the area equation:

$$\begin{aligned} \frac{1}{2}(-2k + 8) &= \pm 4 \\ -k + 4 &= \pm 4 \end{aligned}$$

Solving the two cases separately:

$$\begin{aligned} \text{Case 1: } -k + 4 &= 4 \\ -k &= 0 \\ k &= 0 \end{aligned}$$

$$\begin{aligned} \text{Case 2: } -k + 4 &= -4 \\ -k &= -8 \\ k &= 8 \end{aligned}$$

Final Answer

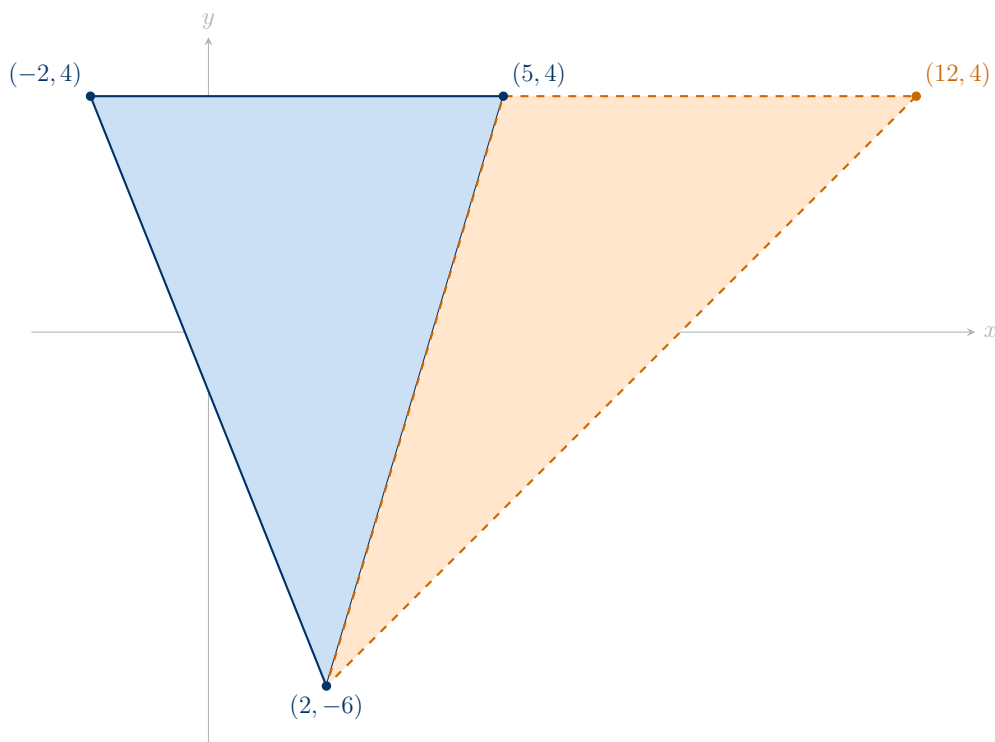
$$k = 0, 8$$

Common Mistake: Taking only +4 and neglecting -4 when setting up the determinant equation will result in missing the second value of k .

Question 7

Find the value(s) of x if the area of the triangle with vertices $(x, 4)$, $(2, -6)$ and $(5, 4)$ is 35 sq. units.

TYPE 11 Area of Triangle Using Determinants



Triangles with vertices $(x, 4)$, $(2, -6)$, $(5, 4)$ for $x = -2$ and $x = 12$

Given: Vertices $A(x, 4)$, $B(2, -6)$, $C(5, 4)$, and Area = 35 sq. units.

Concept / Formula Used: $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \pm \text{given area}$

Solution:

Setting up the determinant equation with ± 35 :

$$\frac{1}{2} \begin{vmatrix} x & 4 & 1 \\ 2 & -6 & 1 \\ 5 & 4 & 1 \end{vmatrix} = \pm 35$$

Multiplying both sides by 2:

$$\begin{vmatrix} x & 4 & 1 \\ 2 & -6 & 1 \\ 5 & 4 & 1 \end{vmatrix} = \pm 70$$

Expanding the determinant along the first row:

$$\begin{aligned} x \cdot \begin{vmatrix} -6 & 1 \\ 4 & 1 \end{vmatrix} - 4 \cdot \begin{vmatrix} 2 & 1 \\ 5 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & -6 \\ 5 & 4 \end{vmatrix} &= \pm 70 \\ x((-6) \cdot 1 - 4 \cdot 1) - 4(2 \cdot 1 - 5 \cdot 1) + 1(2 \cdot 4 - (-6) \cdot 5) &= \pm 70 \\ x(-6 - 4) - 4(2 - 5) + 1(8 + 30) &= \pm 70 \\ x(-10) - 4(-3) + 1(38) &= \pm 70 \\ -10x + 12 + 38 &= \pm 70 \\ -10x + 50 &= \pm 70 \end{aligned}$$

Solving the two cases separately:

$$\text{Case 1: } -10x + 50 = 70$$

$$-10x = 20$$

$$x = -2$$

$$\text{Case 2: } -10x + 50 = -70$$

$$-10x = -120$$

$$x = 12$$

Final Answer

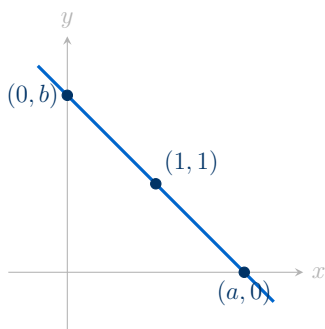
$$x = -2, 12$$

Common Mistake: Forgetting that the value of the determinant can be negative leads to missing $x = 12$.

Question 8

If the points $(a, 0)$, $(0, b)$ and $(1, 1)$ are collinear, prove that $a + b = ab$.

TYPE 12 Collinearity of Points Using Determinants



Collinear points $(a, 0)$, $(0, b)$, $(1, 1)$

To Prove: $a + b = ab$

Concept / Formula Used: $(x_1, y_1) \quad (x_2, y_2) \quad (x_3, y_3) \quad \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Solution:

Given that the points $A(a, 0)$, $B(0, b)$, and $C(1, 1)$ are collinear, the determinant formed by their coordinates must equal zero:

$$\begin{vmatrix} a & 0 & 1 \\ 0 & b & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Expanding along the first row:

$$\begin{aligned} a \cdot \begin{vmatrix} b & 1 \\ 1 & 1 \end{vmatrix} - 0 \cdot \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 0 & b \\ 1 & 1 \end{vmatrix} &= 0 \\ a(b \cdot 1 - 1 \cdot 1) - 0 + 1(0 \cdot 1 - b \cdot 1) &= 0 \\ a(b - 1) + 1(0 - b) &= 0 \\ ab - a - b &= 0 \end{aligned}$$

Adding $a + b$ to both sides of the equation:

$$\begin{aligned} ab &= a + b \\ a + b &= ab \end{aligned}$$

Hence Proved

LHS = RHS, hence proved that $a + b = ab$.

4.4 – Adjoint and Inverse of a Matrix

Question 1 (i)

For what value of x , is the following matrix singular?

$$\begin{bmatrix} 3 - 2x & x + 1 \\ 2 & 4 \end{bmatrix}$$

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} 3 - 2x & x + 1 \\ 2 & 4 \end{bmatrix}$ is singular.

Concept / Formula Used: $A \quad |A| = 0$

Solution:

Let the given matrix be denoted by A :

$$A = \begin{bmatrix} 3 - 2x & x + 1 \\ 2 & 4 \end{bmatrix}$$

Since the matrix A is singular, its determinant must be equal to zero:

$$|A| = 0$$

We calculate the determinant of the 2×2 matrix:

$$|A| = (3 - 2x)(4) - (x + 1)(2)$$

Setting the determinant to zero:

$$\begin{aligned}(3 - 2x)(4) - 2(x + 1) &= 0 \\ 12 - 8x - 2x - 2 &= 0 \\ 10 - 10x &= 0 \\ -10x &= -10 \\ x &= 1\end{aligned}$$

Final Answer

$$x = 1$$

Question 1 (ii)

If $0 < x < \pi$ and the matrix $\begin{bmatrix} \cos x & 1 \\ 3 & 4 \cos x \end{bmatrix}$ is singular, find the values of x .

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} \cos x & 1 \\ 3 & 4 \cos x \end{bmatrix}$ is singular, where $0 < x < \pi$.

Concept / Formula Used: A $|A| = 0$

Solution:

Let the given matrix be denoted by A :

$$A = \begin{bmatrix} \cos x & 1 \\ 3 & 4 \cos x \end{bmatrix}$$

Since the matrix A is singular, its determinant must be equal to zero:

$$|A| = 0$$

We calculate the determinant of the 2×2 matrix:

$$|A| = (\cos x)(4 \cos x) - (1)(3)$$

Setting the determinant to zero:

$$\begin{aligned}4 \cos^2 x - 3 &= 0 \\ 4 \cos^2 x &= 3 \\ \cos^2 x &= \frac{3}{4} \\ \cos x &= \pm \frac{\sqrt{3}}{2}\end{aligned}$$

We are given the interval $0 < x < \pi$. Let us solve for x in this interval:

$$\text{For } \cos x = \frac{\sqrt{3}}{2} \implies x = \frac{\pi}{6}$$

$$\text{For } \cos x = -\frac{\sqrt{3}}{2} \implies x = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Both values lie within the interval $(0, \pi)$.

Final Answer

$$x = \frac{\pi}{6} \text{ or } x = \frac{5\pi}{6}$$

Question 2 (i)

Determine if the following matrix is non-singular:

$$\begin{bmatrix} 6 & -3 & 2 \\ 2 & -1 & 2 \\ -10 & 5 & 2 \end{bmatrix}$$

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} 6 & -3 & 2 \\ 2 & -1 & 2 \\ -10 & 5 & 2 \end{bmatrix}$.

Concept / Formula Used: $A \quad |A| \neq 0$

Solution:

Let us evaluate the determinant of the matrix A :

$$|A| = \begin{vmatrix} 6 & -3 & 2 \\ 2 & -1 & 2 \\ -10 & 5 & 2 \end{vmatrix}$$

Expanding along the first row:

$$|A| = 6 \begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} - (-3) \begin{vmatrix} 2 & 2 \\ -10 & 2 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ -10 & 5 \end{vmatrix}$$

We evaluate each of the 2×2 determinants:

$$\begin{aligned} \begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} &= (-1)(2) - (2)(5) = -2 - 10 = -12 \\ \begin{vmatrix} 2 & 2 \\ -10 & 2 \end{vmatrix} &= (2)(2) - (2)(-10) = 4 + 20 = 24 \\ \begin{vmatrix} 2 & -1 \\ -10 & 5 \end{vmatrix} &= (2)(5) - (-1)(-10) = 10 - 10 = 0 \end{aligned}$$

Substituting these values back into the determinant expression:

$$\begin{aligned} |A| &= 6(-12) + 3(24) + 2(0) \\ |A| &= -72 + 72 + 0 \\ |A| &= 0 \end{aligned}$$

Since $|A| = 0$, the matrix is singular.

Final Answer

The matrix is singular.

Question 2 (ii)

Determine if the following matrix is non-singular:

$$\begin{bmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{bmatrix}$$

TYPE 14 Singular Matrix Condition

Given: The matrix $B = \begin{bmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{bmatrix}$.

Concept / Formula Used: $B \quad |B| \neq 0$

Solution:

Let us evaluate the determinant of the matrix B :

$$|B| = \begin{vmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{vmatrix}$$

Expanding along the first row:

$$|B| = 1 \begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} - (-3) \begin{vmatrix} 4 & 2 \\ 3 & 2 \end{vmatrix} + 2 \begin{vmatrix} 4 & -1 \\ 3 & 5 \end{vmatrix}$$

We evaluate each of the 2×2 determinants:

$$\begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} = (-1)(2) - (2)(5) = -2 - 10 = -12$$

$$\begin{vmatrix} 4 & 2 \\ 3 & 2 \end{vmatrix} = (4)(2) - (2)(3) = 8 - 6 = 2$$

$$\begin{vmatrix} 4 & -1 \\ 3 & 5 \end{vmatrix} = (4)(5) - (-1)(3) = 20 + 3 = 23$$

Substituting these values back into the determinant expression:

$$|B| = 1(-12) + 3(2) + 2(23)$$

$$|B| = -12 + 6 + 46$$

$$|B| = 40$$

Since $|B| = 40 \neq 0$, the matrix is non-singular.

Final Answer

The matrix is non-singular.

Question 3 (i)

Write the adjoint of the matrix:

$$\begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$$

TYPE 15 Adjoint of a Matrix

Given: The matrix $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$.

Concept / Formula Used: 2×2 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

Let the given matrix be:

$$A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$$

Let C_{ij} denote the cofactor of the element a_{ij} of matrix A :

$$C_{11} = (-1)^{1+1}(3) = 3$$

$$C_{12} = (-1)^{1+2}(4) = -4$$

$$C_{21} = (-1)^{2+1}(-1) = 1$$

$$C_{22} = (-1)^{2+2}(2) = 2$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\begin{aligned} \text{adj } A &= C^T \\ &= \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix} \end{aligned}$$

Final Answer

$$\text{adj } A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$$

Question 3 (ii)

Write the adjoint of the matrix:

$$\begin{bmatrix} 3 & 0 \\ -2 & -5 \end{bmatrix}$$

TYPE 15 Adjoint of a Matrix

Given: The matrix $B = \begin{bmatrix} 3 & 0 \\ -2 & -5 \end{bmatrix}$.

Concept / Formula Used: 2×2 $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\text{adj } B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

Let the given matrix be:

$$B = \begin{bmatrix} 3 & 0 \\ -2 & -5 \end{bmatrix}$$

Let C_{ij} denote the cofactor of the element b_{ij} of matrix B :

$$C_{11} = (-1)^{1+1}(-5) = -5$$

$$C_{12} = (-1)^{1+2}(-2) = 2$$

$$C_{21} = (-1)^{2+1}(0) = 0$$

$$C_{22} = (-1)^{2+2}(3) = 3$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -5 & 2 \\ 0 & 3 \end{bmatrix}$$

The adjoint of B is the transpose of the cofactor matrix C :

$$\begin{aligned} \text{adj } B &= C^T \\ &= \begin{bmatrix} -5 & 0 \\ 2 & 3 \end{bmatrix} \end{aligned}$$

Final Answer

$$\text{adj } B = \begin{bmatrix} -5 & 0 \\ 2 & 3 \end{bmatrix}$$

Question 4 (i)

For what value of k , the matrix $\begin{bmatrix} 2 & k \\ 3 & 5 \end{bmatrix}$ has no inverse?

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} 2 & k \\ 3 & 5 \end{bmatrix}$ has no inverse.

Concept / Formula Used: $A \quad |A| = 0$

Solution:

Let the given matrix be:

$$A = \begin{bmatrix} 2 & k \\ 3 & 5 \end{bmatrix}$$

For A to have no inverse, its determinant must be equal to zero:

$$|A| = 0$$

We calculate the determinant of the 2×2 matrix:

$$|A| = (2)(5) - (k)(3)$$

Setting the determinant to zero:

$$10 - 3k = 0$$

$$-3k = -10$$

$$k = \frac{10}{3}$$

Final Answer

$$k = \frac{10}{3}$$

Question 4 (ii)

If $A = \begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix}$, then write A^{-1} .

TYPE 16 Inverse of a Matrix Using Adjoint

Given: The matrix $A = \begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix}$.

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

First, we find the determinant of A :

$$\begin{aligned}|A| &= (3)(7) - (10)(2) \\ &= 21 - 20 \\ &= 1\end{aligned}$$

Since $|A| = 1 \neq 0$, the inverse A^{-1} exists. Next, we find the cofactors of A :

$$\begin{aligned}C_{11} &= 7 \\ C_{12} &= -2 \\ C_{21} &= -10 \\ C_{22} &= 3\end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 7 & -2 \\ -10 & 3 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix}$$

Using the formula for the inverse:

$$\begin{aligned}A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \frac{1}{1} \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix}\end{aligned}$$

Final Answer

$$A^{-1} = \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix}$$

Question 4 (iii)

Write the inverse of the matrix $\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$.

TYPE 16 Inverse of a Matrix Using Adjoint

Given: The matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$.

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

Let the given matrix be:

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$$

First, we find the determinant of A :

$$\begin{aligned} |A| &= (3)(2) - (-1)(1) \\ &= 6 + 1 \\ &= 7 \end{aligned}$$

Since $|A| = 7 \neq 0$, the inverse A^{-1} exists. Next, we find the cofactors of A :

$$\begin{aligned} C_{11} &= 2 \\ C_{12} &= -1 \\ C_{21} &= 1 \\ C_{22} &= 3 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Using the formula for the inverse:

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{3}{7} \end{bmatrix} \end{aligned}$$

Final Answer

$$A^{-1} = \begin{bmatrix} \frac{2}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{3}{7} \end{bmatrix}$$

Question 5 (i)

If $A = \begin{bmatrix} 8 & 2 \\ 3 & 2 \end{bmatrix}$, then find $|\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 8 & 2 \\ 3 & 2 \end{bmatrix}$.

Concept / Formula Used: A n $|\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 2$. First, we calculate the determinant of A :

$$\begin{aligned} |A| &= (8)(2) - (2)(3) \\ &= 16 - 6 \\ &= 10 \end{aligned}$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 2$:

$$\begin{aligned} |\text{adj } A| &= |A|^{2-1} \\ &= |A|^1 \\ &= 10 \end{aligned}$$

Final Answer

$$|\text{adj } A| = 10$$

Question 5 (ii)

If A is a square matrix of order 3 and $|A| = 5$, then find $|\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|A| = 5$.

Concept / Formula Used: A n $|\text{adj } A| = |A|^{n-1}$

Solution:

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$|\text{adj } A| = |A|^{3-1} = |A|^2$$

Substituting the given value $|A| = 5$:

$$\begin{aligned} |\text{adj } A| &= 5^2 \\ &= 25 \end{aligned}$$

Final Answer

$$|\text{adj } A| = 25$$

Question 5 (iii)

If A is a non-singular square matrix of order 3 such that $|\text{adj } A| = 64$, find $|A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|\text{adj } A| = 64$.

Concept / Formula Used: A n $|\text{adj } A| = |A|^{n-1}$

Solution:

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$|\text{adj } A| = |A|^{3-1} = |A|^2$$

We are given that $|\text{adj } A| = 64$. Therefore:

$$|A|^2 = 64$$

$$|A| = \pm\sqrt{64}$$

$$|A| = \pm 8$$

Final Answer

$$|A| = \pm 8$$

Question 5 (iv)

If A is a square matrix of order 3 such that $|A| = -2$, then find $|3\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|A| = -2$.

Concept / Formula Used: M n k $|kM| = k^n|M|$ $|\text{adj } A| = |A|^{n-1}$

Solution:

We need to find $|3\text{adj } A|$. Using the property $|kM| = k^n|M|$ with $k = 3$, $n = 3$, and $M = \text{adj } A$:

$$|3\text{adj } A| = 3^3|\text{adj } A| = 27|\text{adj } A|$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$|\text{adj } A| = |A|^{3-1} = |A|^2$$

Substituting $|A| = -2$:

$$|\text{adj } A| = (-2)^2 = 4$$

Now, substituting this back into the expression:

$$\begin{aligned} |3\text{adj } A| &= 27 \times 4 \\ &= 108 \end{aligned}$$

Final Answer

$$|3\text{adj } A| = 108$$

Question 5 (v)

If $|A^{-1}| = 5$, then find the value of $|A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: An invertible matrix A with $|A^{-1}| = 5$.

Concept / Formula Used: $A \quad |A^{-1}| = \frac{1}{|A|}$

Solution:

Using the standard result $|A^{-1}| = \frac{1}{|A|}$:

$$5 = \frac{1}{|A|}$$

$$|A| = \frac{1}{5}$$

Final Answer

$$|A| = \frac{1}{5}$$

Question 5 (vi)

If $|A| = -4$ and $A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{5}{4} & \frac{1}{2} \end{bmatrix}$, then write $\text{adj } A$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: $|A| = -4$ and $A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{5}{4} & \frac{1}{2} \end{bmatrix}$.

Concept / Formula Used: $A \quad A^{-1} = \frac{1}{|A|} \text{adj } A \quad \text{adj } A = |A|A^{-1}$

Solution:

Using the standard result $\text{adj } A = |A|A^{-1}$:

$$\text{adj } A = -4 \begin{bmatrix} -2 & 1 \\ \frac{5}{4} & \frac{1}{2} \end{bmatrix}$$

Multiplying each element of the matrix by -4 :

$$\begin{aligned} \text{adj } A &= \begin{bmatrix} (-4)(-2) & (-4)(1) \\ (-4)\left(\frac{5}{4}\right) & (-4)\left(\frac{1}{2}\right) \end{bmatrix} \\ &= \begin{bmatrix} 8 & -4 \\ -5 & -2 \end{bmatrix} \end{aligned}$$

Final Answer

$$\text{adj } A = \begin{bmatrix} 8 & -4 \\ -5 & -2 \end{bmatrix}$$

Question 5 (vii)

If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $A^{-1} = kA$, then find the value of k .

TYPE 16 Inverse of a Matrix Using Adjoint

Given: $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ and $A^{-1} = kA$.

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

First, we find the determinant of A :

$$\begin{aligned} |A| &= (2)(-2) - (3)(5) \\ &= -4 - 15 \\ &= -19 \end{aligned}$$

Next, we find the cofactors of A :

$$\begin{aligned} C_{11} &= -2 \\ C_{12} &= -5 \\ C_{21} &= -3 \\ C_{22} &= 2 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -2 & -5 \\ -3 & 2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix}$$

Using the formula for the inverse:

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \frac{1}{-19} \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} \end{aligned}$$

Since $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, we can write:

$$A^{-1} = \frac{1}{19} A$$

Comparing this with the given equation $A^{-1} = kA$:

$$k = \frac{1}{19}$$

Final Answer

$$k = \frac{1}{19}$$

Question 5 (viii)

If $A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then find the value of $|\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: $A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad A(\text{adj } A) = |A|I_n \quad |\text{adj } A| = |A|^{n-1}$

Solution:

We are given:

$$A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 3I_3$$

Using the standard result $A(\text{adj } A) = |A|I_3$:

$$|A|I_3 = 3I_3$$

Comparing the coefficients:

$$|A| = 3$$

Since the order of the matrix is $n = 3$, we use the standard result $|\text{adj } A| = |A|^{n-1}$:

$$\begin{aligned} |\text{adj } A| &= |A|^{3-1} \\ &= |A|^2 \\ &= 3^2 \\ &= 9 \end{aligned}$$

Final Answer

$$|\text{adj } A| = 9$$

Question 5 (ix)

If the value of a determinant of third order is 12, then what is the value of the determinant formed by replacing each element by its co-factor?

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|A| = 12$.

Concept / Formula Used: $A \quad n \quad C \quad |C| = |A|^{n-1}$

Solution:Let C be the cofactor matrix of A . Using the standard result $|C| = |A|^{n-1}$ with $n = 3$:

$$|C| = |A|^{3-1} = |A|^2$$

Substituting the given value $|A| = 12$:

$$\begin{aligned}|C| &= 12^2 \\ &= 144\end{aligned}$$

Final Answer

144

Question 6 (i)

Without computing $\text{adj } A$, find $|\text{adj } A|$ if:

$$A = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}$$

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}$.

Concept / Formula Used: A n $|\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$\begin{aligned}|A| &= (1)(4) - (3)(-1) \\ &= 4 + 3 \\ &= 7\end{aligned}$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 2$:

$$\begin{aligned}|\text{adj } A| &= |A|^{2-1} \\ &= |A|^1 \\ &= 7\end{aligned}$$

Final Answer

$|\text{adj } A| = 7$

Question 6 (ii)

Without computing $\text{adj } A$, find $|\text{adj } A|$ if:

$$A = \begin{bmatrix} -2 & 0 & 0 \\ 3 & 4 & 0 \\ 10 & -7 & 3 \end{bmatrix}$$

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} -2 & 0 & 0 \\ 3 & 4 & 0 \\ 10 & -7 & 3 \end{bmatrix}$.

Concept / Formula Used: A n $|\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 3$. Since A is a lower triangular matrix, its determinant is the product of its diagonal elements:

$$\begin{aligned}|A| &= (-2)(4)(3) \\ &= -24\end{aligned}$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$\begin{aligned}|\text{adj } A| &= |A|^{3-1} \\ &= |A|^2 \\ &= (-24)^2 \\ &= 576\end{aligned}$$

Final Answer

$$|\text{adj } A| = 576$$

Question 6 (iii)

Without computing $\text{adj } A$, find $|\text{adj } A|$ if:

$$A = \begin{bmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{bmatrix}$$

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{bmatrix}$.

Concept / Formula Used: A n $|\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 3$. Let us find the determinant of A by expanding along the first row:

$$|A| = 2 \begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} + (-2) \begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix}$$

We evaluate each of the 2×2 determinants:

$$\begin{aligned}\begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} &= (2)(0) - (-1)(-5) = 0 - 5 = -5 \\ \begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} &= (0)(0) - (-1)(3) = 0 + 3 = 3 \\ \begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix} &= (0)(-5) - (2)(3) = 0 - 6 = -6\end{aligned}$$

Substituting these values back into the determinant expression:

$$|A| = 2(-5) + 1(3) - 2(-6)$$

$$|A| = -10 + 3 + 12$$

$$|A| = 5$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$|\text{adj } A| = |A|^{3-1}$$

$$= |A|^2$$

$$= 5^2$$

$$= 25$$

Final Answer

$$|\text{adj } A| = 25$$

Question 7 (i)

If $A = \begin{bmatrix} 2 & 1 \\ 7 & 5 \end{bmatrix}$, find $|A \text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 2 & 1 \\ 7 & 5 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad A \text{adj } A = |A|I_n \quad |kI_n| = k^n$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$|A| = (2)(5) - (1)(7)$$

$$= 10 - 7$$

$$= 3$$

Using the standard result $A \text{adj } A = |A|I_2$:

$$A \text{adj } A = 3I_2$$

Taking the determinant on both sides:

$$|A \text{adj } A| = |3I_2|$$

Using the property $|kI_n| = k^n$ with $k = 3$ and $n = 2$:

$$|3I_2| = 3^2$$

$$= 9$$

Final Answer

$$|A \text{adj } A| = 9$$

Question 7 (ii)

If $A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$, $a \neq 0$, find $|A \text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$ with $a \neq 0$.

Concept / Formula Used: A n $A \text{adj } A = |A|I_n$ $|kI_n| = k^n$

Solution:

The given matrix A is of order $n = 3$. Since A is a diagonal matrix, its determinant is the product of its diagonal elements:

$$\begin{aligned} |A| &= a \cdot a \cdot a \\ &= a^3 \end{aligned}$$

Using the standard result $A \text{adj } A = |A|I_3$:

$$A \text{adj } A = a^3 I_3$$

Taking the determinant on both sides:

$$|A \text{adj } A| = |a^3 I_3|$$

Using the property $|kI_n| = k^n$ with $k = a^3$ and $n = 3$:

$$\begin{aligned} |a^3 I_3| &= (a^3)^3 \\ &= a^9 \end{aligned}$$

Final Answer

$$|A \text{adj } A| = a^9$$

Question 7 (iii)

If $A = \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}$, find $\text{adj}(\text{adj } A)$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}$.

Concept / Formula Used: A n $\text{adj}(\text{adj } A) = |A|^{n-2}A$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$\begin{aligned}|A| &= (3)(5) - (-1)(4) \\ &= 15 + 4 \\ &= 19\end{aligned}$$

Using the standard result $\text{adj}(\text{adj } A) = |A|^{n-2}A$ with $n = 2$:

$$\begin{aligned}\text{adj}(\text{adj } A) &= |A|^{2-2}A \\ &= |A|^0A \\ &= 1 \cdot A \\ &= \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}\end{aligned}$$

Final Answer

$$\text{adj}(\text{adj } A) = \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}$$

Question 8 (i)

If A is a non-singular matrix, then show that $|A^{-1}| = \frac{1}{|A|}$.

TYPE 18 Matrix Inverse Properties and Proofs

To Prove: $|A^{-1}| = \frac{1}{|A|}$

Concept / Formula Used: $A \quad AA^{-1} = I \quad |AB| = |A||B|$

Solution:

Since A is non-singular, its inverse A^{-1} exists and satisfies the relation:

$$AA^{-1} = I$$

Taking the determinant on both sides:

$$|AA^{-1}| = |I|$$

Using the multiplicative property of determinants $|AB| = |A||B|$:

$$|A||A^{-1}| = |I|$$

Since the determinant of the identity matrix I is 1:

$$|A||A^{-1}| = 1$$

Since A is non-singular, $|A| \neq 0$. Dividing both sides by $|A|$:

$$|A^{-1}| = \frac{1}{|A|}$$

LHS = RHS, hence proved.

Hence Proved

The relation $|A^{-1}| = \frac{1}{|A|}$ is proved.

Question 8 (ii)

If A is a matrix such that $A^3 = I$, then show that A is invertible.

TYPE 18 Matrix Inverse Properties and Proofs

To Prove: A is invertible.

Concept / Formula Used: $A \quad B \quad AB = BA = I$

Solution:

We are given the matrix equation:

$$A^3 = I$$

We can rewrite A^3 as:

$$A(A^2) = I \quad \text{and} \quad (A^2)A = I$$

Let us define a matrix B such that:

$$B = A^2$$

Substituting B into the equations:

$$AB = I \quad \text{and} \quad BA = I$$

By the definition of the inverse of a matrix, since there exists a matrix $B = A^2$ such that $AB = BA = I$, the matrix A is invertible, and its inverse is:

$$A^{-1} = A^2$$

LHS = RHS, hence proved.

Hence Proved

The matrix A is invertible.

Question 8 (iii)

If A is a matrix such that $A^4 = I$, then show that $A^{-1} = A^3$.

TYPE 18 Matrix Inverse Properties and Proofs

To Prove: $A^{-1} = A^3$

Concept / Formula Used: $B \quad A \quad AB = BA = I$

Solution:

We are given the matrix equation:

$$A^4 = I$$

We can rewrite A^4 as:

$$A(A^3) = I \quad \text{and} \quad (A^3)A = I$$

By the definition of the inverse of a matrix, if there exists a matrix B such that $AB = BA = I$, then $B = A^{-1}$. Here, the matrix $B = A^3$ satisfies this condition. Therefore:

$$A^{-1} = A^3$$

LHS = RHS, hence proved.

Hence Proved

The relation $A^{-1} = A^3$ is proved.

Question 9

If $A = \begin{bmatrix} 3 & -5 \\ 4 & 2 \end{bmatrix}$, find $A(\text{adj } A)$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 3 & -5 \\ 4 & 2 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad A(\text{adj } A) = |A|I_n$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$\begin{aligned} |A| &= (3)(2) - (-5)(4) \\ &= 6 + 20 \\ &= 26 \end{aligned}$$

Using the standard result $A(\text{adj } A) = |A|I_2$:

$$\begin{aligned} A(\text{adj } A) &= 26I_2 \\ &= 26 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 26 & 0 \\ 0 & 26 \end{bmatrix} \end{aligned}$$

Final Answer

$$A(\text{adj } A) = \begin{bmatrix} 26 & 0 \\ 0 & 26 \end{bmatrix}$$

Question 10

If the matrix $\begin{bmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{bmatrix}$ is singular, find x .

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{bmatrix}$ is singular.

Concept / Formula Used: $A \quad |A| = 0$

Solution:

Let the given matrix be denoted by A :

$$A = \begin{bmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{bmatrix}$$

Since A is singular, its determinant must be equal to zero:

$$|A| = \begin{vmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix} = 0$$

Let us apply row operations to simplify the determinant. Perform $R_1 \rightarrow R_1 + R_2 + R_3$:

$$|A| = \begin{vmatrix} (x+4) + x + x & x + (x+4) + x & x + x + (x+4) \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

$$|A| = \begin{vmatrix} 3x+4 & 3x+4 & 3x+4 \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

Taking out the common factor $(3x+4)$ from the first row:

$$|A| = (3x+4) \begin{vmatrix} 1 & 1 & 1 \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

Now, perform column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$|A| = (3x+4) \begin{vmatrix} 1 & 1-1 & 1-1 \\ x & (x+4)-x & x-x \\ x & x-x & (x+4)-x \end{vmatrix}$$

$$|A| = (3x+4) \begin{vmatrix} 1 & 0 & 0 \\ x & 4 & 0 \\ x & 0 & 4 \end{vmatrix}$$

Expanding along the first row:

$$|A| = (3x+4) \cdot 1 \cdot \begin{vmatrix} 4 & 0 \\ 0 & 4 \end{vmatrix}$$

$$= (3x+4) \cdot (16-0)$$

$$= 16(3x+4)$$

Setting the determinant to zero:

$$16(3x+4) = 0$$

$$3x+4 = 0$$

$$3x = -4$$

$$x = -\frac{4}{3}$$

Final Answer

$$x = -\frac{4}{3}$$

Question 11 (i)

Find the adjoint of the matrix:

$$A = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix}$$

Also verify that $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix}$.

Concept / Formula Used: $A(\text{adj } A) = |A|I_n = (\text{adj } A)A$

Solution:

First, we find the determinant of A :

$$\begin{aligned} |A| &= (-2)(4) - (1)(3) \\ &= -8 - 3 \\ &= -11 \end{aligned}$$

Next, we find the cofactors of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1}(4) = 4 \\ C_{12} &= (-1)^{1+2}(3) = -3 \\ C_{21} &= (-1)^{2+1}(1) = -1 \\ C_{22} &= (-1)^{2+2}(-2) = -2 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 4 & -3 \\ -1 & -2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix}$$

Now, let us verify the relation $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$:

****Step 1: Compute $A(\text{adj } A)$ ****

$$\begin{aligned} A(\text{adj } A) &= \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix} \\ &= \begin{bmatrix} (-2)(4) + (1)(-3) & (-2)(-1) + (1)(-2) \\ (3)(4) + (4)(-3) & (3)(-1) + (4)(-2) \end{bmatrix} \\ &= \begin{bmatrix} -8 - 3 & 2 - 2 \\ 12 - 12 & -3 - 8 \end{bmatrix} \\ &= \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix} \end{aligned}$$

****Step 2: Compute $|A|I_2$ ****

$$\begin{aligned}|A|I_2 &= -11 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}\end{aligned}$$

****Step 3: Compute $(\text{adj } A)A$ ****

$$\begin{aligned}(\text{adj } A)A &= \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} \\ &= \begin{bmatrix} (4)(-2) + (-1)(3) & (4)(1) + (-1)(4) \\ (-3)(-2) + (-2)(3) & (-3)(1) + (-2)(4) \end{bmatrix} \\ &= \begin{bmatrix} -8 - 3 & 4 - 4 \\ 6 - 6 & -3 - 8 \end{bmatrix} \\ &= \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}\end{aligned}$$

Since all three expressions yield the same matrix:

$$A(\text{adj } A) = |A|I_2 = (\text{adj } A)A = \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}$$

The relation is verified.

Final Answer

$$\text{adj } A = \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix}$$

Question 11 (ii)

Find the adjoint of the matrix:

$$A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$$

Also verify that $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$.

Concept / Formula Used: $A(\text{adj } A) = |A|I_n = (\text{adj } A)A$

Solution:

First, we find the determinant of A :

$$\begin{aligned}|A| &= (2)(-6) - (3)(-4) \\ &= -12 + 12 \\ &= 0\end{aligned}$$

Next, we find the cofactors of A :

$$C_{11} = (-1)^{1+1}(-6) = -6$$

$$C_{12} = (-1)^{1+2}(-4) = 4$$

$$C_{21} = (-1)^{2+1}(3) = -3$$

$$C_{22} = (-1)^{2+2}(2) = 2$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -6 & 4 \\ -3 & 2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

Now, let us verify the relation $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$:

****Step 1: Compute $A(\text{adj } A)$ ****

$$\begin{aligned} A(\text{adj } A) &= \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (2)(-6) + (3)(4) & (2)(-3) + (3)(2) \\ (-4)(-6) + (-6)(4) & (-4)(-3) + (-6)(2) \end{bmatrix} \\ &= \begin{bmatrix} -12 + 12 & -6 + 6 \\ 24 - 24 & 12 - 12 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

****Step 2: Compute $|A|I_2$ ****

$$\begin{aligned} |A|I_2 &= 0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

****Step 3: Compute $(\text{adj } A)A$ ****

$$\begin{aligned} (\text{adj } A)A &= \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \\ &= \begin{bmatrix} (-6)(2) + (-3)(-4) & (-6)(3) + (-3)(-6) \\ (4)(2) + (2)(-4) & (4)(3) + (2)(-6) \end{bmatrix} \\ &= \begin{bmatrix} -12 + 12 & -18 + 18 \\ 8 - 8 & 12 - 12 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Since all three expressions yield the same matrix:

$$A(\text{adj } A) = |A|I_2 = (\text{adj } A)A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The relation is verified.

Final Answer

$$\text{adj } A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

Question 12

If $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$, find $A(\text{adj } A)$ without calculating $\text{adj } A$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$

To Find: $A(\text{adj } A)$ without calculating $\text{adj } A$

Concept / Formula Used: $A \quad n \quad A(\text{adj } A) = |A|I_n$

Solution:

First, we calculate the determinant of A , denoted by $|A|$:

$$|A| = \begin{vmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{vmatrix}$$

Expanding along the first row:

$$|A| = 3 \begin{vmatrix} -3 & -1 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix}$$

Evaluating the 2×2 determinants:

$$\begin{aligned} \begin{vmatrix} -3 & -1 \\ 2 & 1 \end{vmatrix} &= (-3)(1) - (-1)(2) = -3 + 2 = -1 \\ \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} &= (2)(1) - (-1)(1) = 2 + 1 = 3 \\ \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} &= (2)(2) - (-3)(1) = 4 + 3 = 7 \end{aligned}$$

Substituting these values back into the determinant expression:

$$\begin{aligned} |A| &= 3(-1) - 1(3) + 2(7) \\ &= -3 - 3 + 14 \\ &= 8 \end{aligned}$$

Using the standard result $A(\text{adj } A) = |A|I_3$:

$$\begin{aligned} A(\text{adj } A) &= 8 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix} \end{aligned}$$

Final Answer

$$A(\text{adj } A) = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Question 13

Find the adjoint of the following matrices:

$$(i) \ A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$$

$$(ii) \ A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$$

Also verify that $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$.**TYPE 19 Verification of Matrix Adjoint Properties**

Given: The matrices (i) $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$

To Find: The adjoint of each matrix and verification of the identity $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$

Concept / Formula Used: $A \quad \text{adj } A = C' = [C_{ji}] \quad C_{ij} = (-1)^{i+j}M_{ij}$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$. We first find the cofactors C_{ij} of each element a_{ij} :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 5 \\ 0 & 1 \end{vmatrix} = 1(3 - 0) = 3$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 5 \\ -2 & 1 \end{vmatrix} = -1(2 - (-10)) = -12$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 2 & 3 \\ -2 & 0 \end{vmatrix} = 1(0 - (-6)) = 6$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -1(-1 - 0) = 1$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = 1(1 - (-4)) = 5$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & -1 \\ -2 & 0 \end{vmatrix} = -1(0 - 2) = 2$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -1 & 2 \\ 3 & 5 \end{vmatrix} = 1(-5 - 6) = -11$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = -1(5 - 4) = -1$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = 1(3 - (-2)) = 5$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 3 & -12 & 6 \\ 1 & 5 & 2 \\ -11 & -1 & 5 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix}$$

Now, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= 1(3) + (-1)(-12) + 2(6) \\ &= 3 + 12 + 12 \\ &= 27 \end{aligned}$$

Next, we verify $A(\text{adj } A)$:

$$\begin{aligned} A(\text{adj } A) &= \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 1(3) + (-1)(-12) + 2(6) & 1(1) + (-1)(5) + 2(2) & 1(-11) + (-1)(-1) + 2(5) \\ 2(3) + 3(-12) + 5(6) & 2(1) + 3(5) + 5(2) & 2(-11) + 3(-1) + 5(5) \\ -2(3) + 0(-12) + 1(6) & -2(1) + 0(5) + 1(2) & -2(-11) + 0(-1) + 1(5) \end{bmatrix} \\ &= \begin{bmatrix} 3 + 12 + 12 & 1 - 5 + 4 & -11 + 1 + 10 \\ 6 - 36 + 30 & 2 + 15 + 10 & -22 - 3 + 25 \\ -6 + 0 + 6 & -2 + 0 + 2 & 22 + 0 + 5 \end{bmatrix} \\ &= \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix} = 27I_3 \end{aligned}$$

Now, we verify $(\text{adj } A)A$:

$$\begin{aligned} (\text{adj } A)A &= \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3(1) + 1(2) + (-11)(-2) & 3(-1) + 1(3) + (-11)(0) & 3(2) + 1(5) + (-11)(1) \\ -12(1) + 5(2) + (-1)(-2) & -12(-1) + 5(3) + (-1)(0) & -12(2) + 5(5) + (-1)(1) \\ 6(1) + 2(2) + 5(-2) & 6(-1) + 2(3) + 5(0) & 6(2) + 2(5) + 5(1) \end{bmatrix} \\ &= \begin{bmatrix} 3 + 2 + 22 & -3 + 3 + 0 & 6 + 5 - 11 \\ -12 + 10 + 2 & 12 + 15 + 0 & -24 + 25 - 1 \\ 6 + 4 - 10 & -6 + 6 + 0 & 12 + 10 + 5 \end{bmatrix} \\ &= \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix} = 27I_3 \end{aligned}$$

Thus, $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$ is verified for part (i).

Part (ii): Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$. We find the cofactors C_{ij} of each element a_{ij} :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 0 & -2 \\ 0 & 3 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 3 & -2 \\ 1 & 3 \end{vmatrix} = -1(9 - (-2)) = -11$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 0 \\ 1 & 0 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -1 & 2 \\ 0 & 3 \end{vmatrix} = -1(-3 - 0) = 3$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1(3 - 2) = 1$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = -1(0 - (-1)) = -1$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -1 & 2 \\ 0 & -2 \end{vmatrix} = 1(2 - 0) = 2$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix} = -1(-2 - 6) = 8$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & -1 \\ 3 & 0 \end{vmatrix} = 1(0 - (-3)) = 3$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 0 & -11 & 0 \\ 3 & 1 & -1 \\ 2 & 8 & 3 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$$

Now, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= 1(0) + (-1)(-11) + 2(0) \\ &= 11 \end{aligned}$$

Next, we verify $A(\text{adj } A)$:

$$\begin{aligned}
 A(\text{adj } A) &= \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 1(0) + (-1)(-11) + 2(0) & 1(3) + (-1)(1) + 2(-1) & 1(2) + (-1)(8) + 2(3) \\ 3(0) + 0(-11) + (-2)(0) & 3(3) + 0(1) + (-2)(-1) & 3(2) + 0(8) + (-2)(3) \\ 1(0) + 0(-11) + 3(0) & 1(3) + 0(1) + 3(-1) & 1(2) + 0(8) + 3(3) \end{bmatrix} \\
 &= \begin{bmatrix} 11 & 3 - 1 - 2 & 2 - 8 + 6 \\ 0 & 9 + 2 & 6 - 6 \\ 0 & 3 - 3 & 2 + 9 \end{bmatrix} \\
 &= \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = 11I_3
 \end{aligned}$$

Now, we verify $(\text{adj } A)A$:

$$\begin{aligned}
 (\text{adj } A)A &= \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 0(1) + 3(3) + 2(1) & 0(-1) + 3(0) + 2(0) & 0(2) + 3(-2) + 2(3) \\ -11(1) + 1(3) + 8(1) & -11(-1) + 1(0) + 8(0) & -11(2) + 1(-2) + 8(3) \\ 0(1) + (-1)(3) + 3(1) & 0(-1) + (-1)(0) + 3(0) & 0(2) + (-1)(-2) + 3(3) \end{bmatrix} \\
 &= \begin{bmatrix} 9 + 2 & 0 & -6 + 6 \\ -11 + 3 + 8 & 11 & -22 - 2 + 24 \\ -3 + 3 & 0 & 2 + 9 \end{bmatrix} \\
 &= \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = 11I_3
 \end{aligned}$$

Thus, $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$ is verified for part (ii).

Final Answer

$$(i) \text{ adj } A = \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix}$$

$$(ii) \text{ adj } A = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$$

Question 14

If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, find $\text{adj } A$ and verify that $A(\text{adj } A) = (\text{adj } A)A = |A|I_3$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

To Find: $\text{adj } A$ and verification of the identity $A(\text{adj } A) = (\text{adj } A)A = |A|I_3$

Concept / Formula Used: $A \quad \text{adj } A = C'$

Solution:

We first find the cofactors C_{ij} of each element a_{ij} :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} \cos \alpha & 0 \\ 0 & 1 \end{vmatrix} = 1(\cos \alpha - 0) = \cos \alpha$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} \sin \alpha & 0 \\ 0 & 1 \end{vmatrix} = -1(\sin \alpha - 0) = -\sin \alpha$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} \sin \alpha & \cos \alpha \\ 0 & 0 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -\sin \alpha & 0 \\ 0 & 1 \end{vmatrix} = -1(-\sin \alpha - 0) = \sin \alpha$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} \cos \alpha & 0 \\ 0 & 1 \end{vmatrix} = 1(\cos \alpha - 0) = \cos \alpha$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} \cos \alpha & -\sin \alpha \\ 0 & 0 \end{vmatrix} = -1(0 - 0) = 0$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -\sin \alpha & 0 \\ \cos \alpha & 0 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} \cos \alpha & 0 \\ \sin \alpha & 0 \end{vmatrix} = -1(0 - 0) = 0$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{vmatrix} = 1(\cos^2 \alpha - (-\sin^2 \alpha)) = \cos^2 \alpha + \sin^2 \alpha = 1$$

The cofactor matrix C is:

$$C = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, we calculate the determinant $|A|$:

$$\begin{aligned}
 |A| &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\
 &= \cos \alpha (\cos \alpha) + (-\sin \alpha)(-\sin \alpha) + 0 \\
 &= \cos^2 \alpha + \sin^2 \alpha \\
 &= 1
 \end{aligned}$$

Next, we verify $A(\text{adj } A)$:

$$\begin{aligned}
 A(\text{adj } A) &= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & \cos \alpha \sin \alpha - \sin \alpha \cos \alpha & 0 \\ \sin \alpha \cos \alpha - \cos \alpha \sin \alpha & \sin^2 \alpha + \cos^2 \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3
 \end{aligned}$$

Now, we verify $(\text{adj } A)A$:

$$\begin{aligned}
 (\text{adj } A)A &= \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & -\cos \alpha \sin \alpha + \sin \alpha \cos \alpha & 0 \\ -\sin \alpha \cos \alpha + \cos \alpha \sin \alpha & \sin^2 \alpha + \cos^2 \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3
 \end{aligned}$$

Since $|A| = 1$, we have $|A|I_3 = I_3$. Thus, $A(\text{adj } A) = (\text{adj } A)A = |A|I_3$ is verified.

Final Answer

$$\text{adj } A = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 15

For the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 5 \\ 0 & 4 & -1 \end{bmatrix}$, verify that $A(\text{adj } A) = O$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 5 \\ 0 & 4 & -1 \end{bmatrix}$

To Prove: $A(\text{adj } A) = O$

Concept / Formula Used: $A(\text{adj } A) = |A|I_n$ $|A| = 0$ $A(\text{adj } A) = O$

Solution:

First, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= 2 \begin{vmatrix} 2 & 5 \\ 4 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 4 & 5 \\ 0 & -1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 2 \\ 0 & 4 \end{vmatrix} \\ &= 2(2(-1) - 5(4)) + 1(4(-1) - 5(0)) + 3(4(4) - 2(0)) \\ &= 2(-2 - 20) + 1(-4 - 0) + 3(16 - 0) \\ &= 2(-22) - 4 + 48 \\ &= -44 - 4 + 48 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, the matrix is singular. By the standard result $A(\text{adj } A) = |A|I_3$, we must have $A(\text{adj } A) = O$.

Let us verify this explicitly by calculating $\text{adj } A$: Cofactors of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1} \begin{vmatrix} 2 & 5 \\ 4 & -1 \end{vmatrix} = 1(-2 - 20) = -22 \\ C_{12} &= (-1)^{1+2} \begin{vmatrix} 4 & 5 \\ 0 & -1 \end{vmatrix} = -1(-4 - 0) = 4 \\ C_{13} &= (-1)^{1+3} \begin{vmatrix} 4 & 2 \\ 0 & 4 \end{vmatrix} = 1(16 - 0) = 16 \\ C_{21} &= (-1)^{2+1} \begin{vmatrix} -1 & 3 \\ 4 & -1 \end{vmatrix} = -1(1 - 12) = 11 \\ C_{22} &= (-1)^{2+2} \begin{vmatrix} 2 & 3 \\ 0 & -1 \end{vmatrix} = 1(-2 - 0) = -2 \\ C_{23} &= (-1)^{2+3} \begin{vmatrix} 2 & -1 \\ 0 & 4 \end{vmatrix} = -1(8 - 0) = -8 \\ C_{31} &= (-1)^{3+1} \begin{vmatrix} -1 & 3 \\ 2 & 5 \end{vmatrix} = 1(-5 - 6) = -11 \\ C_{32} &= (-1)^{3+2} \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -1(10 - 12) = 2 \\ C_{33} &= (-1)^{3+3} \begin{vmatrix} 2 & -1 \\ 4 & 2 \end{vmatrix} = 1(4 - (-4)) = 8 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -22 & 4 & 16 \\ 11 & -2 & -8 \\ -11 & 2 & 8 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -22 & 11 & -11 \\ 4 & -2 & 2 \\ 16 & -8 & 8 \end{bmatrix}$$

Now, we compute $A(\text{adj } A)$:

$$\begin{aligned} A(\text{adj } A) &= \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 5 \\ 0 & 4 & -1 \end{bmatrix} \begin{bmatrix} -22 & 11 & -11 \\ 4 & -2 & 2 \\ 16 & -8 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 2(-22) + (-1)(4) + 3(16) & 2(11) + (-1)(-2) + 3(-8) & 2(-11) + (-1)(2) + 3(8) \\ 4(-22) + 2(4) + 5(16) & 4(11) + 2(-2) + 5(-8) & 4(-11) + 2(2) + 5(8) \\ 0(-22) + 4(4) + (-1)(16) & 0(11) + 4(-2) + (-1)(-8) & 0(-11) + 4(2) + (-1)(8) \end{bmatrix} \\ &= \begin{bmatrix} -44 - 4 + 48 & 22 + 2 - 24 & -22 - 2 + 24 \\ -88 + 8 + 80 & 44 - 4 - 40 & -44 + 4 + 40 \\ 0 + 16 - 16 & 0 - 8 + 8 & 0 + 8 - 8 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

LHS = RHS, hence proved.

Hence Proved

LHS = RHS, hence proved.

Question 16

For the matrix $A = \begin{bmatrix} 5 & 2 \\ -3 & -1 \end{bmatrix}$, verify that $\text{adj } A' = (\text{adj } A)'$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 5 & 2 \\ -3 & -1 \end{bmatrix}$

To Prove: $\text{adj } A' = (\text{adj } A)'$

Concept / Formula Used: 2×2 $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\text{adj } M = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

First, we find the transpose of A , denoted by A' :

$$A' = \begin{bmatrix} 5 & -3 \\ 2 & -1 \end{bmatrix}$$

Using the standard formula for the adjoint of a 2×2 matrix:

$$\text{adj } A' = \begin{bmatrix} -1 & -(-3) \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

Now, we find the adjoint of A :

$$\text{adj } A = \begin{bmatrix} -1 & -2 \\ -(-3) & 5 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 3 & 5 \end{bmatrix}$$

Taking the transpose of $\text{adj } A$:

$$(\text{adj } A)' = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

Comparing the two results:

$$\text{adj } A' = (\text{adj } A)' = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

LHS = RHS, hence proved.

Question 17

- (i) Given $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$, compute A^{-1} and show that $2A^{-1} = 9I - A$.
- (ii) If $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$, show that $A - 3I = 2(I + 3A^{-1})$.

TYPE 18 Matrix Inverse Properties and Proofs

Given: The matrices (i) $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$

To Prove: (i) $2A^{-1} = 9I - A$
(ii) $A - 3I = 2(I + 3A^{-1})$

Concept / Formula Used: 2×2 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

Part (i): Let $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$. First, we calculate the determinant $|A|$:

$$|A| = (2)(7) - (-3)(-4) = 14 - 12 = 2$$

Since $|A| \neq 0$, A^{-1} exists. The adjoint of A is:

$$\text{adj } A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{2} \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

Now, we compute the LHS: $2A^{-1}$:

$$2A^{-1} = 2 \left(\frac{1}{2} \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \right) = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

Next, we compute the RHS: $9I - A$:

$$\begin{aligned} 9I - A &= 9 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 9-2 & 0-(-3) \\ 0-(-4) & 9-7 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$2A^{-1} = 9I - A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

LHS = RHS, hence proved.

Part (ii): Let $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$. First, we calculate the determinant $|A|$:

$$|A| = (4)(1) - (5)(2) = 4 - 10 = -6$$

The adjoint of A is:

$$\text{adj } A = \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = -\frac{1}{6} \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix}$$

Now, we compute the RHS: $2(I + 3A^{-1})$:

$$\begin{aligned} 3A^{-1} &= 3 \left(-\frac{1}{6} \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix} \right) = -\frac{1}{2} \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -1/2 & 5/2 \\ 1 & -2 \end{bmatrix} \\ I + 3A^{-1} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -1/2 & 5/2 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1/2 & 5/2 \\ 1 & -1 \end{bmatrix} \\ 2(I + 3A^{-1}) &= 2 \begin{bmatrix} 1/2 & 5/2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

Next, we compute the LHS: $A - 3I$:

$$\begin{aligned} A - 3I &= \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4-3 & 5-0 \\ 2-0 & 1-3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 5 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$A - 3I = 2(I + 3A^{-1}) = \begin{bmatrix} 1 & 5 \\ 2 & -2 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

Both parts (i) and (ii) are verified and proved.

Question 18

Find the inverse of the matrix $A = \begin{bmatrix} a & b \\ c & \frac{1+bc}{a} \end{bmatrix}$ and show that $aA^{-1} = (a^2 + bc + 1)I - aA$.

TYPE 18 Matrix Inverse Properties and Proofs

Given: The matrix $A = \begin{bmatrix} a & b \\ c & \frac{1+bc}{a} \end{bmatrix}$

To Prove: $aA^{-1} = (a^2 + bc + 1)I - aA$

Concept / Formula Used: 2×2 $A = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$ $A^{-1} = \frac{1}{wz - xy} \begin{bmatrix} z & -x \\ -y & w \end{bmatrix}$

Solution:

First, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= a \left(\frac{1+bc}{a} \right) - b(c) \\ &= 1 + bc - bc \\ &= 1 \end{aligned}$$

Since $|A| = 1$, the inverse A^{-1} is equal to the adjoint of A :

$$A^{-1} = \begin{bmatrix} \frac{1+bc}{a} & -b \\ -c & a \end{bmatrix}$$

Now, we compute the LHS: aA^{-1} :

$$aA^{-1} = a \begin{bmatrix} \frac{1+bc}{a} & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 1+bc & -ab \\ -ac & a^2 \end{bmatrix}$$

Next, we compute the RHS: $(a^2 + bc + 1)I - aA$:

$$\begin{aligned} (a^2 + bc + 1)I &= \begin{bmatrix} a^2 + bc + 1 & 0 \\ 0 & a^2 + bc + 1 \end{bmatrix} \\ aA &= a \begin{bmatrix} a & b \\ c & \frac{1+bc}{a} \end{bmatrix} = \begin{bmatrix} a^2 & ab \\ ac & 1+bc \end{bmatrix} \\ (a^2 + bc + 1)I - aA &= \begin{bmatrix} (a^2 + bc + 1) - a^2 & 0 - ab \\ 0 - ac & (a^2 + bc + 1) - (1 + bc) \end{bmatrix} \\ &= \begin{bmatrix} 1 + bc & -ab \\ -ac & a^2 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$aA^{-1} = (a^2 + bc + 1)I - aA = \begin{bmatrix} 1 + bc & -ab \\ -ac & a^2 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

LHS = RHS, hence proved.

Question 19

If $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix}$, find the value(s) of x and y .

TYPE 18 Matrix Inverse Properties and Proofs

Given: The matrix $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$ and its inverse $A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix}$

To Find: The values of x and y

Concept / Formula Used: $AA^{-1} = I_3$

Solution:

Using the definition $AA^{-1} = I_3$:

$$\begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix} \left(\frac{1}{2} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiplying both sides by 2:

$$\begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Let us compute the product of the two matrices on the LHS and equate it to the RHS matrix:

$$\text{Let } P = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix}.$$

Equating the element at position (1,2):

$$\begin{aligned} P_{12} &= 0(-8) + 1(6) + 3(2y) = 0 \\ 6 + 6y &= 0 \\ 6y &= -6 \\ y &= -1 \end{aligned}$$

Equating the element at position (2,1):

$$\begin{aligned} P_{21} &= 1(1) + 2(-1) + x(1) = 0 \\ 1 - 2 + x &= 0 \\ x - 1 &= 0 \\ x &= 1 \end{aligned}$$

Let us verify these values using other elements of the product matrix P :

$$\begin{aligned}P_{22} &= 1(-8) + 2(6) + x(2y) \\&= -8 + 12 + 1(2(-1)) \\&= 4 - 2 = 2 \quad (\text{matches the RHS element at } (2, 2))\end{aligned}$$

$$\begin{aligned}P_{23} &= 1(5) + 2(-3) + x(1) \\&= 5 - 6 + 1(1) \\&= 0 \quad (\text{matches the RHS element at } (2, 3))\end{aligned}$$

$$\begin{aligned}P_{32} &= 2(-8) + 3(6) + 1(2y) \\&= -16 + 18 + 2(-1) \\&= 2 - 2 = 0 \quad (\text{matches the RHS element at } (3, 2))\end{aligned}$$

All elements match perfectly.

Final Answer

$$x = 1, \quad y = -1$$

Question 20

If $A = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 \\ 4 & 0 \end{bmatrix}$, verify that $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$.

TYPE 21 Reversal Law for Matrix Inverse and Adjoint

Given: The matrices $A = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 \\ 4 & 0 \end{bmatrix}$

To Prove: $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

Concept / Formula Used: $A \quad B \quad \text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

Solution:

First, we compute the product AB :

$$\begin{aligned} AB &= \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2(-1) + (-3)(4) & 2(3) + (-3)(0) \\ 5(-1) + 1(4) & 5(3) + 1(0) \end{bmatrix} \\ &= \begin{bmatrix} -2 - 12 & 6 + 0 \\ -5 + 4 & 15 + 0 \end{bmatrix} \\ &= \begin{bmatrix} -14 & 6 \\ -1 & 15 \end{bmatrix} \end{aligned}$$

Now, we find $\text{adj}(AB)$ using the 2×2 adjoint formula:

$$\text{adj}(AB) = \begin{bmatrix} 15 & -6 \\ -(-1) & -14 \end{bmatrix} = \begin{bmatrix} 15 & -6 \\ 1 & -14 \end{bmatrix}$$

Next, we find $\text{adj } A$ and $\text{adj } B$:

$$\text{adj } A = \begin{bmatrix} 1 & -(-3) \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ -5 & 2 \end{bmatrix}$$

$$\text{adj } B = \begin{bmatrix} 0 & -3 \\ -4 & -1 \end{bmatrix}$$

Now, we compute the product $(\text{adj } B)(\text{adj } A)$:

$$\begin{aligned} (\text{adj } B)(\text{adj } A) &= \begin{bmatrix} 0 & -3 \\ -4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ -5 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0(1) + (-3)(-5) & 0(3) + (-3)(2) \\ -4(1) + (-1)(-5) & -4(3) + (-1)(2) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 15 & 0 - 6 \\ -4 + 5 & -12 - 2 \end{bmatrix} \\ &= \begin{bmatrix} 15 & -6 \\ 1 & -14 \end{bmatrix} \end{aligned}$$

Comparing the two results:

$$\text{adj}(AB) = (\text{adj } B)(\text{adj } A) = \begin{bmatrix} 15 & -6 \\ 1 & -14 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

LHS = RHS, hence proved.

Question 21

Find the inverse of each of the following matrices:

$$(i) A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$$

$$(ii) A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$$

Also verify that $A^{-1}A = I_3$.

TYPE 16 Inverse of a Matrix Using Adjoint

Given: The matrices (i) $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$

To Find: The inverse of each matrix and verification of $A^{-1}A = I_3$

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$. Since A is a lower triangular matrix, its determinant is the product of its diagonal elements:

$$|A| = 1 \times 3 \times (-1) = -3$$

Cofactors C_{ij} of A :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 0 \\ 2 & -1 \end{vmatrix} = 1(-3 - 0) = -3$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 3 & 0 \\ 5 & -1 \end{vmatrix} = -1(-3 - 0) = 3$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 3 \\ 5 & 2 \end{vmatrix} = 1(6 - 15) = -9$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 0 \\ 2 & -1 \end{vmatrix} = -1(0 - 0) = 0$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 0 \\ 5 & -1 \end{vmatrix} = 1(-1 - 0) = -1$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 0 \\ 5 & 2 \end{vmatrix} = -1(2 - 0) = -2$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 0 \\ 3 & 0 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 0 \\ 3 & 0 \end{vmatrix} = -1(0 - 0) = 0$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 0 \\ 3 & 3 \end{vmatrix} = 1(3 - 0) = 3$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -3 & 3 & -9 \\ 0 & -1 & -2 \\ 0 & 0 & 3 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = -\frac{1}{3} \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix}$$

Verification of $A^{-1}A = I_3$:

$$\begin{aligned} A^{-1}A &= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 0 + 0 & 1(0) + 0 + 0 & 1(0) + 0 + 0 \\ -1(1) + (1/3)(3) + 0 & -1(0) + (1/3)(3) + 0 & -1(0) + 0 + 0 \\ 3(1) + (2/3)(3) - 1(5) & 3(0) + (2/3)(2) - 1(2) & 3(0) + 0 - 1(-1) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ -1 + 1 & 1 & 0 \\ 3 + 2 - 5 & 4/3 - 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Hence verified.

Part (ii): Let $A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$. First, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= -1 \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} - 1 \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} + 2 \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} \\ &= -1(-4 - 3) - 1(12 - (-1)) + 2(9 - 1) \\ &= -1(-7) - 1(13) + 2(8) \\ &= 7 - 13 + 16 \\ &= 10 \end{aligned}$$

Cofactors C_{ij} of A :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} = 1(-4 - 3) = -7$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} = -1(12 - (-1)) = -13$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} = 1(9 - 1) = 8$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -1(4 - 6) = 2$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} -1 & 2 \\ -1 & 4 \end{vmatrix} = 1(-4 - (-2)) = -2$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} -1 & 1 \\ -1 & 3 \end{vmatrix} = -1(-3 - (-1)) = 2$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} = 1(1 - (-2)) = 3$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} -1 & 2 \\ 3 & 1 \end{vmatrix} = -1(-1 - 6) = 7$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} -1 & 1 \\ 3 & -1 \end{vmatrix} = 1(1 - 3) = -2$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -7 & -13 & 8 \\ 2 & -2 & 2 \\ 3 & 7 & -2 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{10} \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

Verification of $A^{-1}A = I_3$:

$$\begin{aligned} A^{-1}A &= \frac{1}{10} \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} -7(-1) + 2(3) + 3(-1) & -7(1) + 2(-1) + 3(3) & -7(2) + 2(1) + 3(4) \\ -13(-1) + (-2)(3) + 7(-1) & -13(1) + (-2)(-1) + 7(3) & -13(2) + (-2)(1) + 7(4) \\ 8(-1) + 2(3) + (-2)(-1) & 8(1) + 2(-1) + (-2)(3) & 8(2) + 2(1) + (-2)(4) \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 7 + 6 - 3 & -7 - 2 + 9 & -14 + 2 + 12 \\ 13 - 6 - 7 & -13 + 2 + 21 & -26 - 2 + 28 \\ -8 + 6 + 2 & 8 - 2 - 6 & 16 + 2 - 8 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{bmatrix} = I_3 \end{aligned}$$

Hence verified.

Final Answer

$$(i) A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix}$$

$$(ii) A^{-1} = \frac{1}{10} \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

Question 22

Given $A = \begin{bmatrix} 5 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. Compute $(AB)^{-1}$.

TYPE 21 Reversal Law for Matrix Inverse and Adjoint

Given: The matrix $A = \begin{bmatrix} 5 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

To Find: The matrix $(AB)^{-1}$

Concept / Formula Used: $(AB)^{-1} = B^{-1}A^{-1}$

Solution:

First, we need to find A^{-1} . We calculate the determinant $|A|$:

$$\begin{aligned} |A| &= 5 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} - 0 + 4 \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} \\ &= 5(3 - 4) + 4(4 - 3) \\ &= 5(-1) + 4(1) \\ &= -1 \end{aligned}$$

Cofactors C_{ij} of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1} \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = 1(3 - 4) = -1 \\ C_{12} &= (-1)^{1+2} \begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix} = -1(2 - 2) = 0 \\ C_{13} &= (-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1(4 - 3) = 1 \\ C_{21} &= (-1)^{2+1} \begin{vmatrix} 0 & 4 \\ 2 & 1 \end{vmatrix} = -1(0 - 8) = 8 \\ C_{22} &= (-1)^{2+2} \begin{vmatrix} 5 & 4 \\ 1 & 1 \end{vmatrix} = 1(5 - 4) = 1 \\ C_{23} &= (-1)^{2+3} \begin{vmatrix} 5 & 0 \\ 1 & 2 \end{vmatrix} = -1(10 - 0) = -10 \\ C_{31} &= (-1)^{3+1} \begin{vmatrix} 0 & 4 \\ 3 & 2 \end{vmatrix} = 1(0 - 12) = -12 \\ C_{32} &= (-1)^{3+2} \begin{vmatrix} 5 & 4 \\ 2 & 2 \end{vmatrix} = -1(10 - 8) = -2 \\ C_{33} &= (-1)^{3+3} \begin{vmatrix} 5 & 0 \\ 2 & 3 \end{vmatrix} = 1(15 - 0) = 15 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -1 & 0 & 1 \\ 8 & 1 & -10 \\ -12 & -2 & 15 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -1 & 8 & -12 \\ 0 & 1 & -2 \\ 1 & -10 & 15 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{-1} \begin{bmatrix} -1 & 8 & -12 \\ 0 & 1 & -2 \\ 1 & -10 & 15 \end{bmatrix} = \begin{bmatrix} 1 & -8 & 12 \\ 0 & -1 & 2 \\ -1 & 10 & -15 \end{bmatrix}$$

Now, we compute $(AB)^{-1} = B^{-1}A^{-1}$:

$$\begin{aligned} (AB)^{-1} &= \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -8 & 12 \\ 0 & -1 & 2 \\ -1 & 10 & -15 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 3(0) + 3(-1) & 1(-8) + 3(-1) + 3(10) & 1(12) + 3(2) + 3(-15) \\ 1(1) + 4(0) + 3(-1) & 1(-8) + 4(-1) + 3(10) & 1(12) + 4(2) + 3(-15) \\ 1(1) + 3(0) + 4(-1) & 1(-8) + 3(-1) + 4(10) & 1(12) + 3(2) + 4(-15) \end{bmatrix} \\ &= \begin{bmatrix} 1-3 & -8-3+30 & 12+6-45 \\ 1-3 & -8-4+30 & 12+8-45 \\ 1-4 & -8-3+40 & 12+6-60 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 19 & -27 \\ -2 & 18 & -25 \\ -3 & 29 & -42 \end{bmatrix} \end{aligned}$$

Final Answer

$$(AB)^{-1} = \begin{bmatrix} -2 & 19 & -27 \\ -2 & 18 & -25 \\ -3 & 29 & -42 \end{bmatrix}$$

Question 23

- (i) Show that $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$ satisfies the equation $A^2 - 3A - 7I = O$ and hence, find A^{-1} .
- (ii) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and I is the identity matrix of order 2, then show that $A^2 = 4A - 3I$. Hence, find A^{-1} .
- (iii) If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$, then find the value of λ so that $A^2 = \lambda A - 2I$. Hence, find A^{-1} .
- (iv) If $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$, then show that $A^3 - 4A^2 - 3A + 11I = O$. Hence, find A^{-1} .

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrices defined in parts (i), (ii), (iii), and (iv)

To Prove: The respective matrix equations and finding A^{-1} in each case

Concept / Formula Used: $c_k A^k + \dots + c_1 A + c_0 I = O \quad c_0 \neq 0 \quad A^{-1} = -\frac{1}{c_0}(c_k A^{k-1} + \dots + c_1 I)$

Solution:

Part (i): Let $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 5(5) + 3(-1) & 5(3) + 3(-2) \\ -1(5) + (-2)(-1) & -1(3) + (-2)(-2) \end{bmatrix} = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix}$$

Now, we evaluate $A^2 - 3A - 7I$:

$$\begin{aligned} A^2 - 3A - 7I &= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} - 3 \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} - \begin{bmatrix} 15 & 9 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 22 - 15 - 7 & 9 - 9 - 0 \\ -3 - (-3) - 0 & 1 - (-6) - 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, A satisfies the equation. To find A^{-1} , we multiply the equation $A^2 - 3A - 7I = O$ by A^{-1} :

$$\begin{aligned} A^{-1}(A^2 - 3A - 7I) &= O \\ A - 3I - 7A^{-1} &= O \\ 7A^{-1} &= A - 3I \\ A^{-1} &= \frac{1}{7}(A - 3I) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{7} \left(\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2(2) + (-1)(-1) & 2(-1) + (-1)(2) \\ -1(2) + 2(-1) & -1(-1) + 2(2) \end{bmatrix} = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$$

Now, we compute $4A - 3I$:

$$4A - 3I = 4 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$$

Clearly, $A^2 = 4A - 3I$. To find A^{-1} , we multiply the equation $A^2 = 4A - 3I$ by A^{-1} :

$$\begin{aligned} A &= 4I - 3A^{-1} \\ 3A^{-1} &= 4I - A \\ A^{-1} &= \frac{1}{3}(4I - A) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{3} \left(\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \right) = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Part (iii): Let $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 3(3) + (-2)(4) & 3(-2) + (-2)(-2) \\ 4(3) + (-2)(4) & 4(-2) + (-2)(-2) \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}$$

We are given $A^2 = \lambda A - 2I$:

$$\begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} = \lambda \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3\lambda - 2 & -2\lambda \\ 4\lambda & -2\lambda - 2 \end{bmatrix}$$

Comparing the corresponding elements at position (1, 2):

$$-2 = -2\lambda \implies \lambda = 1$$

Let us verify with other elements: - (1, 1): $3(1) - 2 = 1$ (verified) - (2, 1): $4(1) = 4$ (verified) - (2, 2): $-2(1) - 2 = -4$ (verified) Thus, $\lambda = 1$. The equation is $A^2 = A - 2I$. To find A^{-1} , we multiply by A^{-1} :

$$\begin{aligned} A &= I - 2A^{-1} \\ 2A^{-1} &= I - A \\ A^{-1} &= \frac{1}{2}(I - A) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{2} \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} -2 & 2 \\ -4 & 3 \end{bmatrix}$$

Part (iv): Let $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$. First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 3(2) + 2(1) & 1(3) + 3(0) + 2(2) & 1(2) + 3(-1) + 2(3) \\ 2(1) + 0(2) - 1(1) & 2(3) + 0(0) - 1(2) & 2(2) + 0(-1) - 1(3) \\ 1(1) + 2(2) + 3(1) & 1(3) + 2(0) + 3(2) & 1(2) + 2(-1) + 3(3) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \end{aligned}$$

Now, we compute $A^3 = A^2A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 9(1) + 7(2) + 5(1) & 9(3) + 7(0) + 5(2) & 9(2) + 7(-1) + 5(3) \\ 1(1) + 4(2) + 1(1) & 1(3) + 4(0) + 1(2) & 1(2) + 4(-1) + 1(3) \\ 8(1) + 9(2) + 9(1) & 8(3) + 9(0) + 9(2) & 8(2) + 9(-1) + 9(3) \end{bmatrix} \\ &= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} \end{aligned}$$

Now, we evaluate $A^3 - 4A^2 - 3A + 11I$:

$$\begin{aligned} A^3 - 4A^2 - 3A + 11I &= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - 4 \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} - 3 \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - \begin{bmatrix} 36 & 28 & 20 \\ 4 & 16 & 4 \\ 32 & 36 & 36 \end{bmatrix} - \begin{bmatrix} 3 & 9 & 6 \\ 6 & 0 & -3 \\ 3 & 6 & 9 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} \\ &= \begin{bmatrix} 28 - 36 - 3 + 11 & 37 - 28 - 9 + 0 & 26 - 20 - 6 + 0 \\ 10 - 4 - 6 + 0 & 5 - 16 - 0 + 11 & 1 - 4 - (-3) + 0 \\ 35 - 32 - 3 + 0 & 42 - 36 - 6 + 0 & 34 - 36 - 9 + 11 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, the equation is satisfied. To find A^{-1} , we multiply the equation by A^{-1} :

$$\begin{aligned} A^2 - 4A - 3I + 11A^{-1} &= O \\ 11A^{-1} &= 4A + 3I - A^2 \\ A^{-1} &= \frac{1}{11}(4A + 3I - A^2) \end{aligned}$$

First, we compute $4A + 3I$:

$$4A + 3I = 4 \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 12 & 8 \\ 8 & 3 & -4 \\ 4 & 8 & 15 \end{bmatrix}$$

Now, we subtract A^2 :

$$A^{-1} = \frac{1}{11} \left(\begin{bmatrix} 7 & 12 & 8 \\ 8 & 3 & -4 \\ 4 & 8 & 15 \end{bmatrix} - \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \right) = \frac{1}{11} \begin{bmatrix} -2 & 5 & 3 \\ 7 & -1 & -5 \\ -4 & -1 & 6 \end{bmatrix}$$

Final Answer

(i) $A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$

(ii) $A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(iii) $\lambda = 1, \quad A^{-1} = \frac{1}{2} \begin{bmatrix} -2 & 2 \\ -4 & 3 \end{bmatrix}$

(iv) $A^{-1} = \frac{1}{11} \begin{bmatrix} -2 & 5 & 3 \\ 7 & -1 & -5 \\ -4 & -1 & 6 \end{bmatrix}$

Question 24

(i) If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, prove that $A^2 - 4A - 5I = O$. Hence, find A^{-1} .

(ii) Find A^{-1} if $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Also show that $A^{-1} = \frac{A^2 - 3I}{2}$.

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrices (i) $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

To Prove: (i) $A^2 - 4A - 5I = O$ and finding A^{-1}

(ii) Finding A^{-1} and showing $A^{-1} = \frac{A^2 - 3I}{2}$

Concept / Formula Used: $A^2 - c_1A - c_0I = O$ $A^{-1} = \frac{1}{c_0}(A - c_1I)$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$. First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(2) + 2(2) & 1(2) + 2(1) + 2(2) & 1(2) + 2(2) + 2(1) \\ 2(1) + 1(2) + 2(2) & 2(2) + 1(1) + 2(2) & 2(2) + 1(2) + 2(1) \\ 2(1) + 2(2) + 1(2) & 2(2) + 2(1) + 1(2) & 2(2) + 2(2) + 1(1) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} \end{aligned}$$

Now, we evaluate $A^2 - 4A - 5I$:

$$\begin{aligned} A^2 - 4A - 5I &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-4-5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, $A^2 - 4A - 5I = O$ is proved. To find A^{-1} , we multiply the equation by A^{-1} :

$$\begin{aligned} A - 4I - 5A^{-1} &= O \\ 5A^{-1} &= A - 4I \\ A^{-1} &= \frac{1}{5}(A - 4I) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{5} \left(\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \right) = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. First, we find A^{-1} using the standard cofactor method. We calculate the determinant $|A|$:

$$|A| = 0(0 - 1) - 1(0 - 1) + 1(1 - 0) = 1 + 1 = 2$$

Cofactors C_{ij} of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1(0 - 1) = -1 \\ C_{12} &= (-1)^{1+2} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{13} &= (-1)^{1+3} \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1(1 - 0) = 1 \\ C_{21} &= (-1)^{2+1} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{22} &= (-1)^{2+2} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1(0 - 1) = -1 \\ C_{23} &= (-1)^{2+3} \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{31} &= (-1)^{3+1} \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1(1 - 0) = 1 \\ C_{32} &= (-1)^{3+2} \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{33} &= (-1)^{3+3} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1(0 - 1) = -1 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

Since C is symmetric, $\text{adj } A = C' = C$. Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

Now, let us show that $A^{-1} = \frac{A^2 - 3I}{2}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Now, we compute $\frac{A^2 - 3I}{2}$:

$$A^2 - 3I = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$
$$\frac{A^2 - 3I}{2} = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

This is exactly equal to A^{-1} . Hence proved.

Final Answer

$$(i) A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

$$(ii) A^{-1} = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

Question 25

- (i) If $A = \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$, prove that $A^{-1} = A'$.
- (ii) If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, prove that $A^{-1} = A^2$.
- (iii) If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, prove that $(F(x))^{-1} = F(-x)$.

TYPE 18 Matrix Inverse Properties and Proofs

Given: The matrices defined in parts (i), (ii), and (iii)

To Prove: The respective inverse relations

Concept / Formula Used: $M \quad N \quad MN = NM = I$

Solution:

Part (i): To prove $A^{-1} = A'$, we show that $AA' = I_3$. First, we write the transpose A' :

$$A' = \frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}$$

Now, we compute AA' :

$$\begin{aligned} AA' &= \left(\frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix} \right) \left(\frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix} \right) \\ &= \frac{1}{81} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix} \\ &= \frac{1}{81} \begin{bmatrix} (-8)(-8) + 1(1) + 4(4) & (-8)(4) + 1(4) + 4(7) & (-8)(1) + 1(-8) + 4(4) \\ 4(-8) + 4(1) + 7(4) & 4(4) + 4(4) + 7(7) & 4(1) + 4(-8) + 7(4) \\ 1(-8) + (-8)(1) + 4(4) & 1(4) + (-8)(4) + 4(7) & 1(1) + (-8)(-8) + 4(4) \end{bmatrix} \\ &= \frac{1}{81} \begin{bmatrix} 64 + 1 + 16 & -32 + 4 + 28 & -8 - 8 + 16 \\ -32 + 4 + 28 & 16 + 16 + 49 & 4 - 32 + 28 \\ -8 - 8 + 16 & 4 - 32 + 28 & 1 + 64 + 16 \end{bmatrix} \\ &= \frac{1}{81} \begin{bmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Since $AA' = I_3$, it follows that $A^{-1} = A'$. Hence proved.

Part (ii): To prove $A^{-1} = A^2$, we show that $A^3 = I_3$. First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + (-1)(2) + 1(1) & 1(-1) + (-1)(-1) + 1(0) & 1(1) + (-1)(0) + 1(0) \\ 2(1) + (-1)(2) + 0(1) & 2(-1) + (-1)(-1) + 0(0) & 2(1) + (-1)(0) + 0(0) \\ 1(1) + 0(2) + 0(1) & 1(-1) + 0(-1) + 0(0) & 1(1) + 0(0) + 0(0) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \end{aligned}$$

Now, we compute $A^3 = A^2A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0(1) + 0(2) + 1(1) & 0(-1) + 0(-1) + 1(0) & 0(1) + 0(0) + 1(0) \\ 0(1) + (-1)(2) + 2(1) & 0(-1) + (-1)(-1) + 2(0) & 0(1) + (-1)(0) + 2(0) \\ 1(1) + (-1)(2) + 1(1) & 1(-1) + (-1)(-1) + 1(0) & 1(1) + (-1)(0) + 1(0) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Since $A^3 = I_3$, multiplying both sides by A^{-1} gives $A^2 = A^{-1}$. Hence proved.

Part (iii): To prove $(F(x))^{-1} = F(-x)$, we show that $F(x)F(-x) = I_3$. First, we write $F(-x)$:

$$F(-x) = \begin{bmatrix} \cos(-x) & -\sin(-x) & 0 \\ \sin(-x) & \cos(-x) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, we compute $F(x)F(-x)$:

$$\begin{aligned} F(x)F(-x) &= \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 x + \sin^2 x & \cos x \sin x - \sin x \cos x & 0 \\ \sin x \cos x - \cos x \sin x & \sin^2 x + \cos^2 x & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Since $F(x)F(-x) = I_3$, it follows that $(F(x))^{-1} = F(-x)$. Hence proved.

Hence Proved

All parts (i), (ii), and (iii) are proved.

Question 26

Show that for the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$, $A^3 - 6A^2 + 5A + 11I = O$. Hence, find A^{-1} .

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$

To Prove: $A^3 - 6A^2 + 5A + 11I = O$ and finding A^{-1}

Concept / Formula Used: $A^3 - c_2A^2 - c_1A - c_0I = O$ $A^{-1} = \frac{1}{c_0}(A^2 - c_2A - c_1I)$

Solution:

First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 1(1) + 1(2) & 1(1) + 1(2) + 1(-1) & 1(1) + 1(-3) + 1(3) \\ 1(1) + 2(1) - 3(2) & 1(1) + 2(2) - 3(-1) & 1(1) + 2(-3) - 3(3) \\ 2(1) - 1(1) + 3(2) & 2(1) - 1(2) + 3(-1) & 2(1) - 1(-3) + 3(3) \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \end{aligned}$$

Now, we compute $A^3 = A^2A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 4(1) + 2(1) + 1(2) & 4(1) + 2(2) + 1(-1) & 4(1) + 2(-3) + 1(3) \\ -3(1) + 8(1) - 14(2) & -3(1) + 8(2) - 14(-1) & -3(1) + 8(-3) - 14(3) \\ 7(1) - 3(1) + 14(2) & 7(1) - 3(2) + 14(-1) & 7(1) - 3(-3) + 14(3) \end{bmatrix} \\ &= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} \end{aligned}$$

Now, we evaluate $A^3 - 6A^2 + 5A + 11I$:

$$\begin{aligned}
 A^3 - 6A^2 + 5A + 11I &= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - 6 \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} + 5 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} + \begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & -15 \\ 10 & -5 & 15 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} \\
 &= \begin{bmatrix} 8 - 24 + 5 + 11 & 7 - 12 + 5 + 0 & 1 - 6 + 5 + 0 \\ -23 - (-18) + 5 + 0 & 27 - 48 + 10 + 11 & -69 - (-84) - 15 + 0 \\ 32 - 42 + 10 + 0 & -13 - (-18) - 5 + 0 & 58 - 84 + 15 + 11 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O
 \end{aligned}$$

Thus, the equation is satisfied. To find A^{-1} , we multiply the equation by A^{-1} :

$$\begin{aligned}
 A^2 - 6A + 5I + 11A^{-1} &= O \\
 11A^{-1} &= 6A - 5I - A^2 \\
 A^{-1} &= \frac{1}{11}(6A - 5I - A^2)
 \end{aligned}$$

First, we compute $6A - 5I$:

$$6A - 5I = 6 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 6 \\ 6 & 7 & -18 \\ 12 & -6 & 13 \end{bmatrix}$$

Now, we subtract A^2 :

$$A^{-1} = \frac{1}{11} \left(\begin{bmatrix} 1 & 6 & 6 \\ 6 & 7 & -18 \\ 12 & -6 & 13 \end{bmatrix} - \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \right) = \frac{1}{11} \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

Final Answer

$$A^{-1} = \frac{1}{11} \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

Question 27

- (i) Find a 2×2 matrix B such that $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$.
- (ii) Solve the matrix equation $\begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$.

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrix equations (i) $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$ and (ii) $\begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$

To Find: The matrices B and A

Concept / Formula Used: $MX = N \quad M = NX^{-1} \quad X^{-1} = \frac{1}{|X|} \text{adj } X \quad YA = Z \quad A = Y^{-1}Z$

Solution:

Part (i): Let $X = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$. The equation is $BX = 6I_2$. Thus, $B = 6X^{-1}$. First, we find X^{-1} :

$$|X| = (1)(4) - (-2)(1) = 4 + 2 = 6$$

The adjoint of X is:

$$\text{adj } X = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

Thus:

$$X^{-1} = \frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

Therefore:

$$B = 6 \left(\frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} \right) = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

Part (ii): Let $Y = \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix}$ and $Z = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$. The equation is $YA = Z$. Thus, $A = Y^{-1}Z$.

First, we find Y^{-1} :

$$|Y| = (5)(1) - (4)(1) = 5 - 4 = 1$$

The adjoint of Y is:

$$\text{adj } Y = \begin{bmatrix} 1 & -4 \\ -1 & 5 \end{bmatrix}$$

Since $|Y| = 1$, we have:

$$Y^{-1} = \begin{bmatrix} 1 & -4 \\ -1 & 5 \end{bmatrix}$$

Now, we compute $A = Y^{-1}Z$:

$$\begin{aligned} A &= \begin{bmatrix} 1 & -4 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + (-4)(1) & 1(-2) + (-4)(3) \\ -1(1) + 5(1) & -1(-2) + 5(3) \end{bmatrix} \\ &= \begin{bmatrix} 1 - 4 & -2 - 12 \\ -1 + 5 & 2 + 15 \end{bmatrix} \\ &= \begin{bmatrix} -3 & -14 \\ 4 & 17 \end{bmatrix} \end{aligned}$$

Final Answer

(i) $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$

(ii) $A = \begin{bmatrix} -3 & -14 \\ 4 & 17 \end{bmatrix}$

Question 28

- (i) Find the matrix A satisfying the matrix equation $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} A \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- (ii) Find the matrix P satisfying the matrix equation $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} P \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$.

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrix equations (i) $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} A \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix} = I_2$ and (ii) $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} P \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$

To Find: The matrices A and P

Concept / Formula Used: $XY = Z \Rightarrow X = ZY^{-1}$ $XY = Z \Rightarrow Y = X^{-1}Z$ $XY = Z \Rightarrow Y = X^{-1}Z$

Solution:

Part (i): Let $X = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, $Y = \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix}$, and $Z = I_2$. The equation is $XAY = I_2$. Thus, $A = X^{-1}Y^{-1}$. First, we find X^{-1} :

$$|X| = (1)(3) - (2)(2) = 3 - 4 = -1$$

$$\text{adj } X = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}$$

$$X^{-1} = \frac{1}{-1} \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$$

Next, we find Y^{-1} :

$$|Y| = (4)(5) - (7)(3) = 20 - 21 = -1$$

$$\text{adj } Y = \begin{bmatrix} 5 & -7 \\ -3 & 4 \end{bmatrix}$$

$$Y^{-1} = \frac{1}{-1} \begin{bmatrix} 5 & -7 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix}$$

Now, we compute $A = X^{-1}Y^{-1}$:

$$\begin{aligned} A &= \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix} \\ &= \begin{bmatrix} -3(-5) + 2(3) & -3(7) + 2(-4) \\ 2(-5) + (-1)(3) & 2(7) + (-1)(-4) \end{bmatrix} \\ &= \begin{bmatrix} 15 + 6 & -21 - 8 \\ -10 - 3 & 14 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 21 & -29 \\ -13 & 18 \end{bmatrix} \end{aligned}$$

Part (ii): Let $X = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, $Y = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$, and $Z = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. The equation is $XPY = Z$. Thus, $P = X^{-1}ZY^{-1}$. First, we find X^{-1} :

$$|X| = (2)(2) - (1)(3) = 4 - 3 = 1$$

$$\text{adj } X = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

$$X^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

Next, we find Y^{-1} :

$$|Y| = (-3)(-3) - (2)(5) = 9 - 10 = -1$$

$$\text{adj } Y = \begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix}$$

$$Y^{-1} = \frac{1}{-1} \begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

Now, we compute $X^{-1}Z$:

$$\begin{aligned} X^{-1}Z &= \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2(1) + (-1)(2) & 2(2) + (-1)(-1) \\ -3(1) + 2(2) & -3(2) + 2(-1) \end{bmatrix} \\ &= \begin{bmatrix} 2 - 2 & 4 + 1 \\ -3 + 4 & -6 - 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 5 \\ 1 & -8 \end{bmatrix} \end{aligned}$$

Finally, we compute $P = (X^{-1}Z)Y^{-1}$:

$$\begin{aligned} P &= \begin{bmatrix} 0 & 5 \\ 1 & -8 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 0(3) + 5(5) & 0(2) + 5(3) \\ 1(3) + (-8)(5) & 1(2) + (-8)(3) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 25 & 0 + 15 \\ 3 - 40 & 2 - 24 \end{bmatrix} \\ &= \begin{bmatrix} 25 & 15 \\ -37 & -22 \end{bmatrix} \end{aligned}$$

Final Answer

(i) $A = \begin{bmatrix} 21 & -29 \\ -13 & 18 \end{bmatrix}$

(ii) $P = \begin{bmatrix} 25 & 15 \\ -37 & -22 \end{bmatrix}$

4.5 – Solving Systems of Linear Equations Using Matrices

Use matrix method to solve the following system of equations:

$$5x + 2y = 4$$

$$7x + 3y = 5$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $5x + 2y = 4$ and $7x + 3y = 5$

Concept / Formula Used: $AX = B$ $X = A^{-1}B$ $A^{-1} = \frac{1}{|A|}\text{adj}(A)$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (5)(3) - (2)(7) \\ &= 15 - 14 \\ &= 1 \neq 0 \end{aligned}$$

Since $|A| \neq 0$, A^{-1} exists and the system has a unique solution.

Find the cofactors of matrix A :

$$\begin{aligned} C_{11} &= 3 \\ C_{12} &= -7 \\ C_{21} &= -2 \\ C_{22} &= 5 \end{aligned}$$

The adjugate matrix $\text{adj}(A)$ is:

$$\text{adj}(A) = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

The inverse matrix A^{-1} is:

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

Now, calculate $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} (3)(4) + (-2)(5) \\ (-7)(4) + (5)(5) \end{bmatrix} \\ &= \begin{bmatrix} 12 - 10 \\ -28 + 25 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3 \end{bmatrix} \end{aligned}$$

Equating corresponding entries:

$$x = 2, \quad y = -3$$

Final Answer

$$x = 2, \quad y = -3$$

Question 1 (ii)

Use matrix method to solve the following system of equations:

$$4x - 3y = 3$$

$$3x - 5y = 7$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $4x - 3y = 3$ and $3x - 5y = 7$

Concept / Formula Used: $AX = B$ $X = A^{-1}B$ $A^{-1} = \frac{1}{|A|}\text{adj}(A)$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 4 & -3 \\ 3 & -5 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (4)(-5) - (-3)(3) \\ &= -20 + 9 \\ &= -11 \neq 0 \end{aligned}$$

Find the cofactors of matrix A :

$$\begin{aligned} C_{11} &= -5 \\ C_{12} &= -3 \\ C_{21} &= 3 \\ C_{22} &= 4 \end{aligned}$$

The adjugate matrix $\text{adj}(A)$ is:

$$\text{adj}(A) = \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix}$$

The inverse matrix A^{-1} is:

$$A^{-1} = -\frac{1}{11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix}$$

Now, calculate $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \end{bmatrix} &= -\frac{1}{11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} (-5)(3) + (3)(7) \\ (-3)(3) + (4)(7) \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} -15 + 21 \\ -9 + 28 \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} 6 \\ 19 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{6}{11} \\ -\frac{19}{11} \end{bmatrix}\end{aligned}$$

Equating corresponding entries:

$$x = -\frac{6}{11}, \quad y = -\frac{19}{11}$$

Final Answer

$$x = -\frac{6}{11}, \quad y = -\frac{19}{11}$$

Question 2 (i)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$x + 2y = 2$$

$$2x + 3y = 3$$

TYPE 23 Consistency of System of Linear Equations

★ Likely Exam Question

Given: $x + 2y = 2$ and $2x + 3y = 3$

Concept / Formula Used: $AX = B$ $|A| \neq 0$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned}|A| &= (1)(3) - (2)(2) \\ &= 3 - 4 \\ &= -1\end{aligned}$$

Since $|A| = -1 \neq 0$, matrix A is non-singular and A^{-1} exists. Therefore, the given system of linear equations is consistent and has a unique solution.

Final Answer

The system of equations is consistent.

Question 2 (ii)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$x + 2y = 9$$

$$2x + 4y = 7$$

TYPE 23 Consistency of System of Linear Equations

Given: $x + 2y = 9$ and $2x + 4y = 7$

Concept / Formula Used: $|A| = 0$ $(\text{adj } A)B \neq O$ $AX = B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (1)(4) - (2)(2) \\ &= 4 - 4 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, calculate $(\text{adj } A)B$.

The cofactors of A are:

$$C_{11} = 4, \quad C_{12} = -2, \quad C_{21} = -2, \quad C_{22} = 1$$

$$\text{adj}(A) = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$$

Calculate the product $(\text{adj } A)B$:

$$\begin{aligned} (\text{adj } A)B &= \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 4(9) + (-2)(7) \\ -2(9) + 1(7) \end{bmatrix} \\ &= \begin{bmatrix} 36 - 14 \\ -18 + 7 \end{bmatrix} \\ &= \begin{bmatrix} 22 \\ -11 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

Since $|A| = 0$ and $(\text{adj } A)B \neq O$, the system has no solution.

Final Answer

The system of equations is inconsistent.

Question 2 (iii)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$3x - y + 2z = 3$$

$$2x + y + 3z = 5$$

$$x - 2y - z = 1$$

TYPE 23 Consistency of System of Linear Equations

Given: $3x - y + 2z = 3$, $2x + y + 3z = 5$, and $x - 2y - z = 1$

Concept / Formula Used: $|A| = 0$ $(\text{adj } A)B \neq O$ $AX = B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 1 & 3 \\ 1 & -2 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$$

Calculate the determinant $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 3 \begin{vmatrix} 1 & 3 \\ -2 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 1 \\ 1 & -2 \end{vmatrix} \\ &= 3(-1 - (-6)) + 1(-2 - 3) + 2(-4 - 1) \\ &= 3(5) + 1(-5) + 2(-5) \\ &= 15 - 5 - 10 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, calculate $(\text{adj } A)B$.

Find the cofactors of A :

$$\begin{aligned} C_{11} &= +(-1 - (-6)) = 5, & C_{12} &= -(-2 - 3) = 5, & C_{13} &= +(-4 - 1) = -5 \\ C_{21} &= -(1 - (-4)) = -5, & C_{22} &= +(-3 - 2) = -5, & C_{23} &= -(-6 - (-1)) = 5 \\ C_{31} &= +(-3 - 2) = -5, & C_{32} &= -(9 - 4) = -5, & C_{33} &= +(3 - (-2)) = 5 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 5 & -5 & -5 \\ 5 & -5 & -5 \\ -5 & 5 & 5 \end{bmatrix}$$

Calculate $(\text{adj } A)B$:

$$\begin{aligned}
 (\text{adj } A)B &= \begin{bmatrix} 5 & -5 & -5 \\ 5 & -5 & -5 \\ -5 & 5 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 5(3) - 5(5) - 5(1) \\ 5(3) - 5(5) - 5(1) \\ -5(3) + 5(5) + 5(1) \end{bmatrix} \\
 &= \begin{bmatrix} 15 - 25 - 5 \\ 15 - 25 - 5 \\ -15 + 25 + 5 \end{bmatrix} \\
 &= \begin{bmatrix} -15 \\ -15 \\ 15 \end{bmatrix} \neq O
 \end{aligned}$$

Since $|A| = 0$ and $(\text{adj } A)B \neq O$, the system has no solution.

Final Answer

The system of equations is inconsistent.

Question 2 (iv)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$\begin{aligned}
 5x - y + 4z &= 5 \\
 2x + 3y + 5z &= 2 \\
 5x - 2y + 6z &= -1
 \end{aligned}$$

TYPE 23 Consistency of System of Linear Equations

★ Likely Exam Question

Given: $5x - y + 4z = 5$, $2x + 3y + 5z = 2$, and $5x - 2y + 6z = -1$

Concept / Formula Used: $AX = B$ $|A| \neq 0$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned}
 |A| &= 5(18 - (-10)) - (-1)(12 - 25) + 4(-4 - 15) \\
 &= 5(28) + 1(-13) + 4(-19) \\
 &= 140 - 13 - 76 \\
 &= 51 \neq 0
 \end{aligned}$$

Since $|A| = 51 \neq 0$, A^{-1} exists. Therefore, the given system of linear equations is consistent and has a unique solution.

Final Answer

The system of equations is consistent.

Question 3

Solve the following system of equations by using matrix method:

$$x + y = 1$$

$$7x + 7y = 7$$

TYPE 24 Solving Dependent System of Linear Equations

Given: $x + y = 1$ and $7x + 7y = 7$

Concept / Formula Used: $|A| = 0$ $(\text{adj } A)B = O$ $AX = B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (1)(7) - (1)(7) \\ &= 7 - 7 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, compute $(\text{adj } A)B$.

The cofactors of matrix A are:

$$C_{11} = 7, \quad C_{12} = -7, \quad C_{21} = -1, \quad C_{22} = 1$$

$$\text{adj}(A) = \begin{bmatrix} 7 & -1 \\ -7 & 1 \end{bmatrix}$$

Calculate $(\text{adj } A)B$:

$$\begin{aligned} (\text{adj } A)B &= \begin{bmatrix} 7 & -1 \\ -7 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 7(1) + (-1)(7) \\ -7(1) + 1(7) \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} = O \end{aligned}$$

Since $|A| = 0$ and $(\text{adj } A)B = O$, the system has infinitely many solutions.

Let $y = k$, where $k \in \mathbb{R}$. Substitute $y = k$ into the first equation $x + y = 1$:

$$x + k = 1 \implies x = 1 - k$$

Final Answer

$x = 1 - k$, $y = k$, where k is any real number.

Question 4 (i)

Using matrix method, solve the following system of equations:

$$x - 2y + 3z = 6$$

$$x + 4y + z = 12$$

$$x - 3y + 2z = 1$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x - 2y + 3z = 6$, $x + 4y + z = 12$, and $x - 3y + 2z = 1$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 1(8 - (-3)) - (-2)(2 - 1) + 3(-3 - 4) \\ &= 1(11) + 2(1) + 3(-7) \\ &= 11 + 2 - 21 \\ &= -8 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(8 - (-3)) = 11, & C_{12} &= -(2 - 1) = -1, & C_{13} &= +(-3 - 4) = -7 \\ C_{21} &= -(-4 - (-9)) = -5, & C_{22} &= +(2 - 3) = -1, & C_{23} &= -(-3 - (-2)) = 1 \\ C_{31} &= +(-2 - 12) = -14, & C_{32} &= -(1 - 3) = 2, & C_{33} &= +(4 - (-2)) = 6 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = -\frac{1}{8} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}
 \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= -\frac{1}{8} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix} \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix} \\
 &= -\frac{1}{8} \begin{bmatrix} 11(6) - 5(12) - 14(1) \\ -1(6) - 1(12) + 2(1) \\ -7(6) + 1(12) + 6(1) \end{bmatrix} \\
 &= -\frac{1}{8} \begin{bmatrix} 66 - 60 - 14 \\ -6 - 12 + 2 \\ -42 + 12 + 6 \end{bmatrix} \\
 &= -\frac{1}{8} \begin{bmatrix} -8 \\ -16 \\ -24 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$x = 1, \quad y = 2, \quad z = 3$$

Question 4 (ii)

Using matrix method, solve the following system of equations:

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x + y - z = 1$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x + y + z = 6$, $x - y + z = 2$, and $2x + y - z = 1$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned}
 |A| &= 1(1 - 1) - 1(-1 - 2) + 1(1 - (-2)) \\
 &= 1(0) - 1(-3) + 1(3) \\
 &= 0 + 3 + 3 \\
 &= 6 \neq 0
 \end{aligned}$$

Find cofactors of matrix A :

$$C_{11} = +(1 - 1) = 0,$$

$$C_{12} = -(-1 - 2) = 3,$$

$$C_{13} = +(1 - (-2)) = 3$$

$$C_{21} = -(-1 - 1) = 2,$$

$$C_{22} = +(-1 - 2) = -3,$$

$$C_{23} = -(1 - 2) = 1$$

$$C_{31} = +(1 - (-1)) = 2,$$

$$C_{32} = -(1 - 1) = 0,$$

$$C_{33} = +(-1 - 1) = -2$$

$$\text{adj}(A) = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \\ &= \frac{1}{6} \begin{bmatrix} 0(6) + 2(2) + 2(1) \\ 3(6) - 3(2) + 0(1) \\ 3(6) + 1(2) - 2(1) \end{bmatrix} \\ &= \frac{1}{6} \begin{bmatrix} 0 + 4 + 2 \\ 18 - 6 + 0 \\ 18 + 2 - 2 \end{bmatrix} \\ &= \frac{1}{6} \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \end{aligned}$$

Final Answer

$$x = 1, \quad y = 2, \quad z = 3$$

Question 4 (iii)

Using matrix method, solve the following system of equations:

$$x + y + z = 9$$

$$2x + 5y + 7z = 52$$

$$2x + y - z = 0$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x + y + z = 9$, $2x + 5y + 7z = 52$, and $2x + y - z = 0$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 1(-5 - 7) - 1(-2 - 14) + 1(2 - 10) \\ &= 1(-12) - 1(-16) + 1(-8) \\ &= -12 + 16 - 8 \\ &= -4 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{array}{lll} C_{11} = +(-5 - 7) = -12, & C_{12} = -(-2 - 14) = 16, & C_{13} = +(2 - 10) = -8 \\ C_{21} = -(-1 - 1) = 2, & C_{22} = +(-1 - 2) = -3, & C_{23} = -(1 - 2) = 1 \\ C_{31} = +(7 - 5) = 2, & C_{32} = -(7 - 2) = -5, & C_{33} = +(5 - 2) = 3 \end{array}$$

$$\text{adj}(A) = \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -12(9) + 2(52) + 2(0) \\ 16(9) - 3(52) - 5(0) \\ -8(9) + 1(52) + 3(0) \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -108 + 104 + 0 \\ 144 - 156 - 0 \\ -72 + 52 + 0 \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -4 \\ -12 \\ -20 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \end{aligned}$$

Final Answer

$$x = 1, \quad y = 3, \quad z = 5$$

Question 4 (iv)

Using matrix method, solve the following system of equations:

$$x - 2y = 10$$

$$2x - y - z = 8$$

$$-2y + z = 7$$

TYPE 22 Solving Linear Equations Using Matrix Method

★ Likely Exam Question

Given: $x - 2y + 0z = 10$, $2x - y - z = 8$, and $0x - 2y + z = 7$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 1(-1 - 2) - (-2)(2 - 0) + 0(-4 - 0) \\ &= 1(-3) + 2(2) + 0 \\ &= -3 + 4 \\ &= 1 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{array}{lll} C_{11} = +(-1 - 2) = -3, & C_{12} = -(2 - 0) = -2, & C_{13} = +(-4 - 0) = -4 \\ C_{21} = -(-2 - 0) = 2, & C_{22} = +(1 - 0) = 1, & C_{23} = -(-2 - 0) = 2 \\ C_{31} = +(2 - 0) = 2, & C_{32} = -(-1 - 0) = 1, & C_{33} = +(-1 - (-4)) = 3 \end{array}$$

$$\text{adj}(A) = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}$$

Since $|A| = 1$:

$$A^{-1} = \text{adj}(A) = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} -3(10) + 2(8) + 2(7) \\ -2(10) + 1(8) + 1(7) \\ -4(10) + 2(8) + 3(7) \end{bmatrix} \\ &= \begin{bmatrix} -30 + 16 + 14 \\ -20 + 8 + 7 \\ -40 + 16 + 21 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ -5 \\ -3 \end{bmatrix}\end{aligned}$$

Final Answer

$$x = 0, \quad y = -5, \quad z = -3$$

Question 4 (v)

Using matrix method, solve the following system of equations:

$$3x + y + z = 1$$

$$2x + 2z = 0$$

$$5x + y + 2z = 2$$

TYPE 22 Solving Linear Equations Using Matrix Method

★ Likely Exam Question

Given: $3x + y + z = 1$, $2x + 0y + 2z = 0$, and $5x + y + 2z = 2$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 0 & 2 \\ 5 & 1 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned}|A| &= 3(0 - 2) - 1(4 - 10) + 1(2 - 0) \\ &= 3(-2) - 1(-6) + 1(2) \\ &= -6 + 6 + 2 \\ &= 2 \neq 0\end{aligned}$$

Find cofactors of matrix A :

$$C_{11} = +(0 - 2) = -2,$$

$$C_{12} = -(4 - 10) = 6,$$

$$C_{13} = +(2 - 0) = 2$$

$$C_{21} = -(2 - 1) = -1,$$

$$C_{22} = +(6 - 5) = 1,$$

$$C_{23} = -(3 - 5) = 2$$

$$C_{31} = +(2 - 0) = 2,$$

$$C_{32} = -(6 - 2) = -4,$$

$$C_{33} = +(0 - 2) = -2$$

$$\text{adj}(A) = \begin{bmatrix} -2 & -1 & 2 \\ 6 & 1 & -4 \\ 2 & 2 & -2 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{2} \begin{bmatrix} -2 & -1 & 2 \\ 6 & 1 & -4 \\ 2 & 2 & -2 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} -2 & -1 & 2 \\ 6 & 1 & -4 \\ 2 & 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} -2(1) - 1(0) + 2(2) \\ 6(1) + 1(0) - 4(2) \\ 2(1) + 2(0) - 2(2) \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} -2 + 0 + 4 \\ 6 + 0 - 8 \\ 2 + 0 - 4 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} \end{aligned}$$

Final Answer

$$x = 1, \quad y = -1, \quad z = -1$$

Question 4 (vi)

Using matrix method, solve the following system of equations:

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = 3$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $2x - 3y + 5z = 11$, $3x + 2y - 4z = -5$, and $x + y - 2z = 3$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 11 \\ -5 \\ 3 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 2(-4 - (-4)) - (-3)(-6 - (-4)) + 5(3 - 2) \\ &= 2(0) + 3(-2) + 5(1) \\ &= 0 - 6 + 5 \\ &= -1 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(-4 - (-4)) = 0, & C_{12} &= -(-6 - (-4)) = 2, & C_{13} &= +(3 - 2) = 1 \\ C_{21} &= -(6 - 5) = -1, & C_{22} &= +(-4 - 5) = -9, & C_{23} &= -(2 - (-3)) = -5 \\ C_{31} &= +(12 - 10) = 2, & C_{32} &= -(-8 - 15) = 23, & C_{33} &= +(4 - (-9)) = 13 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \begin{bmatrix} 11 \\ -5 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 0(11) + 1(-5) - 2(3) \\ -2(11) + 9(-5) - 23(3) \\ -1(11) + 5(-5) - 13(3) \end{bmatrix} \\ &= \begin{bmatrix} 0 - 5 - 6 \\ -22 - 45 - 69 \\ -11 - 25 - 39 \end{bmatrix} \\ &= \begin{bmatrix} -11 \\ -136 \\ -75 \end{bmatrix} \end{aligned}$$

Final Answer

$$x = -11, \quad y = -136, \quad z = -75$$

Question 5

Solve the following system of linear equations using matrix method:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 9$$

$$\frac{2}{x} + \frac{5}{y} + \frac{7}{z} = 52$$

$$\frac{2}{x} + \frac{1}{y} - \frac{1}{z} = 0$$

TYPE 25 Solving Reducible System of Linear Equations

★ Likely Exam Question

Given: $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 9$, $\frac{2}{x} + \frac{5}{y} + \frac{7}{z} = 52$, and $\frac{2}{x} + \frac{1}{y} - \frac{1}{z} = 0$

Concept / Formula Used: $u = \frac{1}{x}, v = \frac{1}{y}, w = \frac{1}{z}$ $AU = B \implies U = A^{-1}B$

Solution:

Let:

$$u = \frac{1}{x}, \quad v = \frac{1}{y}, \quad w = \frac{1}{z}$$

Substitute u, v, w into the system of equations:

$$\begin{aligned} u + v + w &= 9 \\ 2u + 5v + 7w &= 52 \\ 2u + v - w &= 0 \end{aligned}$$

Write in matrix form $AU = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix}, \quad U = \begin{bmatrix} u \\ v \\ w \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

Calculate $|A|$:

$$\begin{aligned} |A| &= 1(-5 - 7) - 1(-2 - 14) + 1(2 - 10) \\ &= -12 + 16 - 8 \\ &= -4 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{array}{lll} C_{11} = -12, & C_{12} = 16, & C_{13} = -8 \\ C_{21} = 2, & C_{22} = -3, & C_{23} = 1 \\ C_{31} = 2, & C_{32} = -5, & C_{33} = 3 \end{array}$$

$$\text{adj}(A) = \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Compute $U = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} u \\ v \\ w \end{bmatrix} &= -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -12(9) + 2(52) + 2(0) \\ 16(9) - 3(52) - 5(0) \\ -8(9) + 1(52) + 3(0) \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -108 + 104 + 0 \\ 144 - 156 - 0 \\ -72 + 52 + 0 \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -4 \\ -12 \\ -20 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \end{aligned}$$

So $u = 1$, $v = 3$, $w = 5$.

Perform back-substitution:

$$\begin{aligned} x &= \frac{1}{u} = \frac{1}{1} = 1 \\ y &= \frac{1}{v} = \frac{1}{3} \\ z &= \frac{1}{w} = \frac{1}{5} \end{aligned}$$

Final Answer

$$x = 1, \quad y = \frac{1}{3}, \quad z = \frac{1}{5}$$

Common Mistake: Forgetting to back-substitute $x = \frac{1}{u}$, $y = \frac{1}{v}$, $z = \frac{1}{w}$ and leaving u, v, w as the final answer.

Question 6

State the condition under which the following equations have a unique solution:

$$x + 2y - 2z + 5 = 0$$

$$-x + 3y + 4 = 0$$

$$-2y + z - 4 = 0$$

Using Martin's rule, find the unique solution of the above system of linear equations.

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x + 2y - 2z = -5$, $-x + 3y + 0z = -4$, and $0x - 2y + z = 4$

Concept / Formula Used: $AX = B$ $|A| \neq 0$ $X = A^{-1}B$

Solution:

Rearrange the given equations into standard form $AX = B$:

$$\begin{aligned}x + 2y - 2z &= -5 \\ -x + 3y + 0z &= -4 \\ 0x - 2y + z &= 4\end{aligned}$$

Matrix representation:

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} -5 \\ -4 \\ 4 \end{bmatrix}$$

Condition for Unique Solution: The system of equations has a unique solution if and only if $|A| \neq 0$.

Calculate determinant $|A|$:

$$\begin{aligned}|A| &= 1(3 - 0) - 2(-1 - 0) + (-2)(2 - 0) \\ &= 1(3) - 2(-1) - 2(2) \\ &= 3 + 2 - 4 \\ &= 1 \neq 0\end{aligned}$$

Since $|A| = 1 \neq 0$, the system possesses a unique solution.

Find cofactors of matrix A :

$$\begin{aligned}C_{11} &= +(3 - 0) = 3, & C_{12} &= -(-1 - 0) = 1, & C_{13} &= +(2 - 0) = 2 \\ C_{21} &= -(2 - 4) = 2, & C_{22} &= +(1 - 0) = 1, & C_{23} &= -(-2 - 0) = 2 \\ C_{31} &= +(0 - (-6)) = 6, & C_{32} &= -(0 - 2) = 2, & C_{33} &= +(3 - (-2)) = 5\end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

Since $|A| = 1$, $A^{-1} = \text{adj}(A)$.

Using Martin's Rule $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix} \begin{bmatrix} -5 \\ -4 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 3(-5) + 2(-4) + 6(4) \\ 1(-5) + 1(-4) + 2(4) \\ 2(-5) + 2(-4) + 5(4) \end{bmatrix} \\ &= \begin{bmatrix} -15 - 8 + 24 \\ -5 - 4 + 8 \\ -10 - 8 + 20 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}\end{aligned}$$

Final Answer**Condition:** $|A| \neq 0$.**Solution:** $x = 1, \quad y = -1, \quad z = 2$ **Question 7**

The sum of three numbers is 6. If we multiply third number by 3 and add second to it we get 11. By adding first and third numbers, we get double of the second number. Represent it algebraically and find the numbers using matrix method.

TYPE 26 Word Problems on System of Linear Equations**★ Likely Exam Question****Given:** Let x, y, z be the first, second, and third numbers respectively**Concept / Formula Used:** $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$ **Solution:**

Formulate equations from the given conditions: 1. "The sum of three numbers is 6":

$$x + y + z = 6$$

2. "Multiply third number by 3 and add second to it we get 11":

$$y + 3z = 11 \implies 0x + y + 3z = 11$$

3. "Adding first and third numbers, we get double of the second number":

$$x + z = 2y \implies x - 2y + z = 0$$

Write system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 1(1 - (-6)) - 1(0 - 3) + 1(0 - 1) \\ &= 1(7) - 1(-3) + 1(-1) \\ &= 7 + 3 - 1 \\ &= 9 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{array}{lll} C_{11} = +(1 - (-6)) = 7, & C_{12} = -(0 - 3) = 3, & C_{13} = +(0 - 1) = -1 \\ C_{21} = -(1 - (-2)) = -3, & C_{22} = +(1 - 1) = 0, & C_{23} = -(-2 - 1) = 3 \\ C_{31} = +(3 - 1) = 2, & C_{32} = -(3 - 0) = -3, & C_{33} = +(1 - 0) = 1 \end{array}$$

$$\text{adj}(A) = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 7(6) - 3(11) + 2(0) \\ 3(6) + 0(11) - 3(0) \\ -1(6) + 3(11) + 1(0) \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 42 - 33 + 0 \\ 18 + 0 - 0 \\ -6 + 33 + 0 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \end{aligned}$$

Final Answer

Algebraic representation:

$$x + y + z = 6$$

$$y + 3z = 11$$

$$x - 2y + z = 0$$

Numbers: First number $x = 1$, Second number $y = 2$, Third number $z = 3$

Question 8

Garima owns a cafe that serves burgers, samosas, and juice bottles. On a particular day, 100 items in total were sold. The number of burgers sold was thrice the number of samosas sold. Also, the number of samosas sold was 10 more than the number of juice bottles sold.

- Form a set of simultaneous equations for the above information.
- Solve the set of equations formed in (i) by matrix method.

TYPE 26 Word Problems on System of Linear Equations

★ Likely Exam Question

Given: Let x, y, z be the number of burgers, samosas, and juice bottles sold respectively

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

(i) **Formulating simultaneous equations:** 1. Total items sold:

$$x + y + z = 100$$

2. Burgers sold was thrice samosas sold:

$$x = 3y \implies x - 3y + 0z = 0$$

3. Samosas sold was 10 more than juice bottles sold:

$$y = z + 10 \implies 0x + y - z = 10$$

System of equations:

$$\begin{aligned} x + y + z &= 100 \\ x - 3y + 0z &= 0 \\ 0x + y - z &= 10 \end{aligned}$$

(ii) **Solving by matrix method:** Write in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -3 & 0 \\ 0 & 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 100 \\ 0 \\ 10 \end{bmatrix}$$

Calculate $|A|$:

$$\begin{aligned} |A| &= 1(3 - 0) - 1(-1 - 0) + 1(1 - 0) \\ &= 1(3) - 1(-1) + 1(1) \\ &= 3 + 1 + 1 \\ &= 5 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(3 - 0) = 3, & C_{12} &= -(-1 - 0) = 1, & C_{13} &= +(1 - 0) = 1 \\ C_{21} &= -(-1 - 1) = 2, & C_{22} &= +(-1 - 0) = -1, & C_{23} &= -(1 - 0) = -1 \\ C_{31} &= +(0 - (-3)) = 3, & C_{32} &= -(0 - 1) = 1, & C_{33} &= +(-3 - 1) = -4 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 1 \\ 1 & -1 & -4 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{5} \begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 1 \\ 1 & -1 & -4 \end{bmatrix} \begin{bmatrix} 100 \\ 0 \\ 10 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 3(100) + 2(0) + 3(10) \\ 1(100) - 1(0) + 1(10) \\ 1(100) - 1(0) - 4(10) \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 300 + 0 + 30 \\ 100 - 0 + 10 \\ 100 - 0 - 40 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 330 \\ 110 \\ 60 \end{bmatrix} \\ &= \begin{bmatrix} 66 \\ 22 \\ 12 \end{bmatrix} \end{aligned}$$

Final Answer**(i) Equations:**

$$x + y + z = 100$$

$$x - 3y = 0$$

$$y - z = 10$$

(ii) Solution: Burgers $x = 66$, Samosas $y = 22$, Juice bottles $z = 12$ **Question 9**

If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$, find A^{-1} . Using A^{-1} , solve the system of linear equations:

$$x + 2y - 3z = -4$$

$$2x + 3y + 2z = 2$$

$$3x - 3y - 4z = 11$$

TYPE 27 Solving Linear Equations Using Matrix Inverse**★ Likely Exam Question**

Given: $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$ and system equations $x + 2y - 3z = -4, 2x + 3y + 2z = 2, 3x - 3y - 4z = 11$

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj}(A)$ $X = A^{-1}B$

Solution:

First, calculate determinant $|A|$:

$$\begin{aligned} |A| &= 1(-12 - (-6)) - 2(-8 - 6) + (-3)(-6 - 9) \\ &= 1(-6) - 2(-14) - 3(-15) \\ &= -6 + 28 + 45 \\ &= 67 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(-12 - (-6)) = -6, & C_{12} &= -(-8 - 6) = 14, & C_{13} &= +(-6 - 9) = -15 \\ C_{21} &= -(-8 - 9) = 17, & C_{22} &= +(-4 - (-9)) = 5, & C_{23} &= -(-3 - 6) = 9 \\ C_{31} &= +(4 - (-9)) = 13, & C_{32} &= -(2 - (-6)) = -8, & C_{33} &= +(3 - 4) = -1 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

The given system of linear equations in matrix form is $AX = B$:

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix} \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix} \\ &= \frac{1}{67} \begin{bmatrix} -6(-4) + 17(2) + 13(11) \\ 14(-4) + 5(2) - 8(11) \\ -15(-4) + 9(2) - 1(11) \end{bmatrix} \\ &= \frac{1}{67} \begin{bmatrix} 24 + 34 + 143 \\ -56 + 10 - 88 \\ 60 + 18 - 11 \end{bmatrix} \\ &= \frac{1}{67} \begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \end{aligned}$$

Final Answer

$$A^{-1} = \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

$$x = 3, \quad y = -2, \quad z = 1$$

Question 10

If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} . Using A^{-1} , solve the following system of equations:

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = -3$$

TYPE 27 Solving Linear Equations Using Matrix Inverse

★ Likely Exam Question

Given: $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$ and system equations $2x - 3y + 5z = 11, 3x + 2y - 4z = -5, x + y - 2z = -3$

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj}(A)$ $X = A^{-1}B$

Solution:

Calculate determinant $|A|$:

$$\begin{aligned} |A| &= 2(-4 - (-4)) - (-3)(-6 - (-4)) + 5(3 - 2) \\ &= 2(0) + 3(-2) + 5(1) \\ &= 0 - 6 + 5 \\ &= -1 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(-4 - (-4)) = 0, & C_{12} &= -(-6 - (-4)) = 2, & C_{13} &= +(3 - 2) = 1 \\ C_{21} &= -(6 - 5) = -1, & C_{22} &= +(-4 - 5) = -9, & C_{23} &= -(2 - (-3)) = -5 \\ C_{31} &= +(12 - 10) = 2, & C_{32} &= -(-8 - 15) = 23, & C_{33} &= +(4 - (-9)) = 13 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

Write the system of equations in matrix form $AX = B$:

$$\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix} \\ &= \begin{bmatrix} 0(11) + 1(-5) - 2(-3) \\ -2(11) + 9(-5) - 23(-3) \\ -1(11) + 5(-5) - 13(-3) \end{bmatrix} \\ &= \begin{bmatrix} 0 - 5 + 6 \\ -22 - 45 + 69 \\ -11 - 25 + 39 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \end{aligned}$$

Final Answer

$$A^{-1} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

$$x = 1, \quad y = 2, \quad z = 3$$

Question 11

If $A = \begin{bmatrix} 3 & 5 \\ -2 & 3 \end{bmatrix}$, find A^{-1} and use it to solve the system of equations:

$$3x - 2y = 7$$

$$5x + 3y = 1$$

TYPE 28 Solving Linear Equations Using Transpose Matrix Inverse

Given: $A = \begin{bmatrix} 3 & 5 \\ -2 & 3 \end{bmatrix}$ and system equations $3x - 2y = 7, 5x + 3y = 1$

Concept / Formula Used: $C = A^T \quad C^{-1} = (A^T)^{-1} = (A^{-1})^T$

Solution:

Calculate determinant $|A|$:

$$\begin{aligned} |A| &= (3)(3) - (5)(-2) \\ &= 9 + 10 \\ &= 19 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$C_{11} = 3, \quad C_{12} = 2, \quad C_{21} = -5, \quad C_{22} = 3$$

$$\text{adj}(A) = \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{19} \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix}$$

Notice that the coefficient matrix C of the system $3x - 2y = 7, 5x + 3y = 1$ is:

$$C = \begin{bmatrix} 3 & -2 \\ 5 & 3 \end{bmatrix} = A^T$$

Using the property $(A^T)^{-1} = (A^{-1})^T$:

$$C^{-1} = (A^{-1})^T = \frac{1}{19} \begin{bmatrix} 3 & 2 \\ -5 & 3 \end{bmatrix}$$

Solve $CX = B$ where $B = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} x \\ y \end{bmatrix} &= C^{-1}B \\ &= \frac{1}{19} \begin{bmatrix} 3 & 2 \\ -5 & 3 \end{bmatrix} \begin{bmatrix} 7 \\ 1 \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 3(7) + 2(1) \\ -5(7) + 3(1) \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 21 + 2 \\ -35 + 3 \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 23 \\ -32 \end{bmatrix} \\ &= \begin{bmatrix} \frac{23}{19} \\ -\frac{32}{19} \end{bmatrix} \end{aligned}$$

Final Answer

$$\begin{aligned} A^{-1} &= \frac{1}{19} \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix} \\ x &= \frac{23}{19}, \quad y = -\frac{32}{19} \end{aligned}$$

Question 12

If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$, find A^{-1} and use it to solve the system of equations:

$$x + 2y + z = 4$$

$$-x + y + z = 0$$

$$x - 3y + z = 2$$

TYPE 28 Solving Linear Equations Using Transpose Matrix Inverse

Given: $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and system equations $x + 2y + z = 4, -x + y + z = 0, x - 3y + z = 2$

Concept / Formula Used: $C = A^T \quad C^{-1} = (A^T)^{-1} = (A^{-1})^T$

Solution:

Calculate determinant $|A|$:

$$\begin{aligned} |A| &= 1(1 - (-3)) - (-1)(2 - (-3)) + 1(2 - 1) \\ &= 1(4) + 1(5) + 1(1) \\ &= 4 + 5 + 1 \\ &= 10 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(1 - (-3)) = 4, & C_{12} &= -(2 - (-3)) = -5, & C_{13} &= +(2 - 1) = 1 \\ C_{21} &= -(-1 - 1) = 2, & C_{22} &= +(1 - 1) = 0, & C_{23} &= -(1 - (-1)) = -2 \\ C_{31} &= +(3 - 1) = 2, & C_{32} &= -(-3 - 2) = 5, & C_{33} &= +(1 - (-2)) = 3 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

Observe the coefficient matrix C of the given system:

$$C = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix} = A^T$$

Therefore:

$$C^{-1} = (A^T)^{-1} = (A^{-1})^T = \frac{1}{10} \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$$

Now solve $CX = B$ where $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{10} \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 4(4) - 5(0) + 1(2) \\ 2(4) + 0(0) - 2(2) \\ 2(4) + 5(0) + 3(2) \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 16 + 0 + 2 \\ 8 + 0 - 4 \\ 8 + 0 + 6 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 18 \\ 4 \\ 14 \end{bmatrix} \\ &= \begin{bmatrix} \frac{9}{5} \\ \frac{2}{5} \\ \frac{7}{5} \end{bmatrix} \end{aligned}$$

Final Answer

$$\begin{aligned} A^{-1} &= \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix} \\ x &= \frac{9}{5}, \quad y = \frac{2}{5}, \quad z = \frac{7}{5} \end{aligned}$$

Question 13

Given two matrices A and B where $A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$, find AB and use this result to solve the following system of equations:

$$x - 2y + 3z = 6$$

$$x + 4y + z = 12$$

$$x - 3y + 2z = 1$$

TYPE 29 Solving Linear Equations Using Matrix Product

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$

Concept / Formula Used: $AB = kI$ $A^{-1} = \frac{1}{k}B$

Solution:

First, compute the product AB :

$$\begin{aligned} AB &= \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 1(11) + (-2)(-1) + 3(-7) & 1(-5) + (-2)(-1) + 3(1) & 1(-14) + (-2)(2) + 3(6) \\ 1(11) + 4(-1) + 1(-7) & 1(-5) + 4(-1) + 1(1) & 1(-14) + 4(2) + 1(6) \\ 1(11) + (-3)(-1) + 2(-7) & 1(-5) + (-3)(-1) + 2(1) & 1(-14) + (-3)(2) + 2(6) \end{bmatrix} \\ &= \begin{bmatrix} 11 + 2 - 21 & -5 + 2 + 3 & -14 - 4 + 18 \\ 11 - 4 - 7 & -5 - 4 + 1 & -14 + 8 + 6 \\ 11 + 3 - 14 & -5 + 3 + 2 & -14 - 6 + 12 \end{bmatrix} \\ &= \begin{bmatrix} -8 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & -8 \end{bmatrix} = -8I \end{aligned}$$

Since $AB = -8I$, we have:

$$A \left(-\frac{1}{8}B \right) = I \implies A^{-1} = -\frac{1}{8}B$$

Write the given system of linear equations in matrix form $AX = C$:

$$\begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix}$$

Now, compute $X = A^{-1}C = -\frac{1}{8}BC$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= -\frac{1}{8} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix} \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix} \\ &= -\frac{1}{8} \begin{bmatrix} 11(6) - 5(12) - 14(1) \\ -1(6) - 1(12) + 2(1) \\ -7(6) + 1(12) + 6(1) \end{bmatrix} \\ &= -\frac{1}{8} \begin{bmatrix} 66 - 60 - 14 \\ -6 - 12 + 2 \\ -42 + 12 + 6 \end{bmatrix} \\ &= -\frac{1}{8} \begin{bmatrix} -8 \\ -16 \\ -24 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\end{aligned}$$

Final Answer

$$AB = -8I$$

$$x = 1, \quad y = 2, \quad z = 3$$

Question 14

If $A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, then find AB and use it to solve the following system of equations:

$$x - 2y = 3$$

$$2x - y - z = 2$$

$$-2y + z = 3$$

TYPE 29 Solving Linear Equations Using Matrix Product

★ Likely Exam Question

Given: $A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, and system $x - 2y = 3$, $2x - y - z = 2$, $-2y + z = 3$.

Concept / Formula Used: $AB = I$ $B^{-1} = A$ $(B^T)^{-1} = A^T$ $CX = D$ $X = C^{-1}D$

Solution:

First, compute the matrix product AB :

$$\begin{aligned}
 AB &= \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} (-3)(1) + (-2)(-2) + (-4)(0) & (-3)(2) + (-2)(-1) + (-4)(-1) & (-3)(0) + (-2)(-2) + (-4)(1) \\ (2)(1) + (1)(-2) + (2)(0) & (2)(2) + (1)(-1) + (2)(-1) & (2)(0) + (1)(-2) + (2)(1) \\ (2)(1) + (1)(-2) + (3)(0) & (2)(2) + (1)(-1) + (3)(-1) & (2)(0) + (1)(-2) + (3)(1) \end{bmatrix} \\
 &= \begin{bmatrix} -3 + 4 + 0 & -6 + 2 + 4 & 0 + 4 - 4 \\ 2 - 2 + 0 & 4 - 1 - 2 & 0 - 2 + 2 \\ 2 - 2 + 0 & 4 - 1 - 3 & 0 - 2 + 3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I
 \end{aligned}$$

Since $AB = I$, by definition of matrix inverse:

$$B^{-1} = A$$

Now, write the given system of linear equations in matrix form $CX = D$:

$$\begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$$

Observe the coefficient matrix C :

$$C = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix}$$

Comparing matrix C with matrix B :

$$B^T = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix} = C$$

Thus, $C = B^T$.

Using the identity $(B^T)^{-1} = (B^{-1})^T$:

$$C^{-1} = (B^T)^{-1} = (B^{-1})^T = A^T$$

Taking the transpose of matrix A :

$$A^T = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}$$

The solution vector X is given by $X = C^{-1}D = A^T D$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} (-3)(3) + 2(2) + 2(3) \\ (-2)(3) + 1(2) + 1(3) \\ (-4)(3) + 2(2) + 3(3) \end{bmatrix} \\ &= \begin{bmatrix} -9 + 4 + 6 \\ -6 + 2 + 3 \\ -12 + 4 + 9 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \end{aligned}$$

Equating corresponding elements:

$$x = 1, \quad y = -1, \quad z = 1$$

Final Answer

$$AB = I; \quad x = 1, \quad y = -1, \quad z = 1$$

Common Mistake: Confusing the coefficient matrix C with B directly instead of recognizing $C = B^T$, which requires using $C^{-1} = A^T$ rather than A .

Question 15 (i)

Using matrices, solve the following system of homogeneous equations:

$$5x + 5y + 2z = 0$$

$$2x + 5y + 4z = 0$$

$$4x + 5y + 2z = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

Given: System of homogeneous equations: $5x + 5y + 2z = 0$, $2x + 5y + 4z = 0$, and $4x + 5y + 2z = 0$.

Concept / Formula Used: $AX = O \quad x = 0, y = 0, z = 0 \quad |A| \neq 0$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 5 & 5 & 2 \\ 2 & 5 & 4 \\ 4 & 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 5 & 5 & 2 \\ 2 & 5 & 4 \\ 4 & 5 & 2 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned}
 |A| &= 5 \begin{vmatrix} 5 & 4 \\ 5 & 2 \end{vmatrix} - 5 \begin{vmatrix} 2 & 4 \\ 4 & 2 \end{vmatrix} + 2 \begin{vmatrix} 2 & 5 \\ 4 & 5 \end{vmatrix} \\
 &= 5[(5)(2) - (4)(5)] - 5[(2)(2) - (4)(4)] + 2[(2)(5) - (5)(4)] \\
 &= 5[10 - 20] - 5[4 - 16] + 2[10 - 20] \\
 &= 5(-10) - 5(-12) + 2(-10) \\
 &= -50 + 60 - 20 \\
 &= -10
 \end{aligned}$$

Since $|A| = -10 \neq 0$, the coefficient matrix A is non-singular.

Using the standard result that if $|A| \neq 0$, the system $AX = O$ has only the trivial solution:

$$x = 0, \quad y = 0, \quad z = 0$$

Final Answer

$$x = 0, \quad y = 0, \quad z = 0$$

Question 15 (ii)

Using matrices, solve the following system of homogeneous equations:

$$3x + y - 2z = 0$$

$$x + y + z = 0$$

$$x - 2y + z = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

Given: System of homogeneous equations: $3x + y - 2z = 0$, $x + y + z = 0$, and $x - 2y + z = 0$.

Concept / Formula Used: $AX = O \quad x = 0, y = 0, z = 0 \quad |A| \neq 0$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 3 & 1 & -2 \\ 1 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned}
 |A| &= 3 \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + (-2) \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} \\
 &= 3[(1)(1) - (1)(-2)] - 1[(1)(1) - (1)(1)] - 2[(1)(-2) - (1)(1)] \\
 &= 3[1 + 2] - 1[1 - 1] - 2[-2 - 1] \\
 &= 3(3) - 1(0) - 2(-3) \\
 &= 9 - 0 + 6 \\
 &= 15
 \end{aligned}$$

Since $|A| = 15 \neq 0$, the coefficient matrix A is non-singular.

Using the standard result that if $|A| \neq 0$, the system $AX = O$ has only the trivial solution:

$$x = 0, \quad y = 0, \quad z = 0$$

Final Answer

$$x = 0, \quad y = 0, \quad z = 0$$

Question 16 (i)

Using matrices, solve the following system of homogeneous equations:

$$2x - 3y - z = 0$$

$$x + 3y - 2z = 0$$

$$x - 3y = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

★ Likely Exam Question

Given: System of homogeneous equations: $2x - 3y - z = 0$, $x + 3y - 2z = 0$, and $x - 3y + 0z = 0$.

Concept / Formula Used: $|A| = 0 \quad AX = O$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 2 & -3 & -1 \\ 1 & 3 & -2 \\ 1 & -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 2 & -3 & -1 \\ 1 & 3 & -2 \\ 1 & -3 & 0 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned}
 |A| &= 2 \begin{vmatrix} 3 & -2 \\ -3 & 0 \end{vmatrix} - (-3) \begin{vmatrix} 1 & -2 \\ 1 & 0 \end{vmatrix} + (-1) \begin{vmatrix} 1 & 3 \\ 1 & -3 \end{vmatrix} \\
 &= 2[(3)(0) - (-2)(-3)] + 3[(1)(0) - (-2)(1)] - 1[(1)(-3) - (3)(1)] \\
 &= 2[0 - 6] + 3[0 + 2] - 1[-3 - 3] \\
 &= 2(-6) + 3(2) - 1(-6) \\
 &= -12 + 6 + 6 \\
 &= 0
 \end{aligned}$$

Since $|A| = 0$, the system has infinitely many non-trivial solutions.

Let $z = k$, where k is any real parameter ($k \in \mathbb{R}$).

Substituting $z = k$ into the third equation $x - 3y = 0$:

$$x = 3y \quad \text{--- (Equation 1)}$$

Substituting $z = k$ into the second equation $x + 3y - 2z = 0$:

$$x + 3y = 2k \quad \text{--- (Equation 2)}$$

Substituting $x = 3y$ from Equation 1 into Equation 2:

$$3y + 3y = 2k$$

$$6y = 2k$$

$$y = \frac{2k}{6}$$

$$y = \frac{1}{3}k$$

Now substitute $y = \frac{1}{3}k$ back into Equation 1:

$$x = 3 \left(\frac{1}{3}k \right)$$

$$x = k$$

Check these values in the first equation $2x - 3y - z = 0$:

$$2(k) - 3 \left(\frac{1}{3}k \right) - k = 2k - k - k = 0$$

This satisfies all equations.

Final Answer

$x = k$, $y = \frac{1}{3}k$, $z = k$, where k is any real number.

Common Mistake: Stopping after finding $|A| = 0$ and stating 'no solution' is incorrect; homogeneous systems with $|A| = 0$ always have infinitely many non-trivial solutions.

Question 16 (ii)

Using matrices, solve the following system of homogeneous equations:

$$x + y - z = 0$$

$$x - 2y + z = 0$$

$$3x + 6y - 5z = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

Given: System of homogeneous equations: $x + y - z = 0$, $x - 2y + z = 0$, and $3x + 6y - 5z = 0$.

Concept / Formula Used: $|A| = 0$ $AX = O$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 1 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 1 \\ 3 & 6 & -5 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned} |A| &= 1 \begin{vmatrix} -2 & 1 \\ 6 & -5 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 3 & -5 \end{vmatrix} + (-1) \begin{vmatrix} 1 & -2 \\ 3 & 6 \end{vmatrix} \\ &= 1[(-2)(-5) - (1)(6)] - 1[(1)(-5) - (1)(3)] - 1[(1)(6) - (-2)(3)] \\ &= 1[10 - 6] - 1[-5 - 3] - 1[6 + 6] \\ &= 1(4) - 1(-8) - 1(12) \\ &= 4 + 8 - 12 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, the system has infinitely many non-trivial solutions.

Let $z = k$, where k is any real parameter ($k \in \mathbb{R}$).

Substitute $z = k$ into the first two equations:

$$x + y = k \quad \text{--- (Equation 1)}$$

$$x - 2y = -k \quad \text{--- (Equation 2)}$$

Subtracting Equation 2 from Equation 1:

$$(x + y) - (x - 2y) = k - (-k)$$

$$x + y - x + 2y = k + k$$

$$3y = 2k$$

$$y = \frac{2}{3}k$$

Substituting $y = \frac{2}{3}k$ into Equation 1:

$$x + \frac{2}{3}k = k$$

$$x = k - \frac{2}{3}k$$

$$x = \frac{1}{3}k$$

Check these values in the third equation $3x + 6y - 5z = 0$:

$$3\left(\frac{1}{3}k\right) + 6\left(\frac{2}{3}k\right) - 5k = k + 4k - 5k = 0$$

This satisfies all equations.

Final Answer

$x = \frac{1}{3}k$, $y = \frac{2}{3}k$, $z = k$, where k is any real number.

ISC Class XII Mathematics

Chapter 5 Differentiation

350 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- | | |
|--|--|
| I. 5.1 – Continuity: Checking Continuity at a Point | II. 5.2 – Continuity: Piecewise Functions and Finding Constants |
| III. 5.3 – Continuity: Algebra and Composition of Continuous Functions | IV. 5.4 – Differentiability: Definition and Relation to Continuity |
| V. 5.5 – Derivatives of Standard and Absolute Value Functions | VI. 5.6 – Derivatives of Inverse Trigonometric Functions |
| VII. 5.7 – Implicit Differentiation | VIII. 5.8 – Differentiation of Exponential and Logarithmic Functions |
| IX. 5.9 – Logarithmic Differentiation | X. 5.10 – Parametric Differentiation |
| XI. 5.11 – Second-Order Derivatives (Miscellaneous) | |

5.1 – Continuity: Checking Continuity at a Point

Question 1 (i)

Examine the following function for continuity at the indicated point:

$$f(x) = \begin{cases} x^3 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases} \quad \text{at } x = 0$$

TYPE 1 Checking Continuity at a Point

Given: $f(x) = \begin{cases} x^3 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases} \text{ at } x = 0$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a) \quad x = a$

Solution:

First, evaluate the limit of $f(x)$ as $x \rightarrow 0$:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} (x^3 + 3) \\ &= 0^3 + 3 \\ &= 3 \end{aligned}$$

Next, find the function value at $x = 0$:

$$f(0) = 1$$

Comparing the limit value with the point value:

$$\lim_{x \rightarrow 0} f(x) = 3 \neq f(0) = 1$$

Since the limit as $x \rightarrow 0$ does not equal $f(0)$, the function is discontinuous at $x = 0$.

Final Answer

$f(x)$ is discontinuous at $x = 0$ because $\lim_{x \rightarrow 0} f(x) = 3 \neq f(0) = 1$.

Question 1 (ii)

Examine the following function for continuity at the indicated point:

$$f(x) = x^3 + 2x^2 - 1 \quad \text{at } x = 1$$

TYPE 1 Checking Continuity at a Point

Given: $f(x) = x^3 + 2x^2 - 1$ at $x = 1$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a) \quad x = a$

Solution:

Evaluate the limit of $f(x)$ as $x \rightarrow 1$:

$$\begin{aligned}\lim_{x \rightarrow 1} f(x) &= \lim_{x \rightarrow 1} (x^3 + 2x^2 - 1) \\ &= 1^3 + 2(1)^2 - 1 \\ &= 1 + 2 - 1 \\ &= 2\end{aligned}$$

Evaluate the function value at $x = 1$:

$$\begin{aligned}f(1) &= 1^3 + 2(1)^2 - 1 \\ &= 1 + 2 - 1 \\ &= 2\end{aligned}$$

Since $\lim_{x \rightarrow 1} f(x) = f(1) = 2$, the function $f(x)$ is continuous at $x = 1$.

Final Answer

$f(x)$ is continuous at $x = 1$.

Question 2

If $f(x) = \frac{x^2 - 1}{x - 1}$ for $x \neq 1$ and $f(x) = 2$ when $x = 1$, show that the function is continuous at $x = 1$.

TYPE 1 Checking Continuity at a Point

To Prove: $f(x)$ is continuous at $x = 1$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a) \quad x = a$

Solution:

Calculate the limit of $f(x)$ as $x \rightarrow 1$:

$$\begin{aligned}\lim_{x \rightarrow 1} f(x) &= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \\ &= \lim_{x \rightarrow 1} (x + 1) \\ &= 1 + 1 \\ &= 2\end{aligned}$$

The given value of the function at $x = 1$ is:

$$f(1) = 2$$

Comparing both values:

$$\lim_{x \rightarrow 1} f(x) = 2 = f(1)$$

LHS equals RHS, hence $f(x)$ is continuous at $x = 1$.

Hence Proved

Since $\lim_{x \rightarrow 1} f(x) = f(1) = 2$, $f(x)$ is continuous at $x = 1$.

Question 3

A function f is defined as $f(x) = \begin{cases} \frac{x^2 - x - 6}{x - 3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$. Show that f is continuous at $x = 3$.

TYPE 1 Checking Continuity at a Point

To Prove: $f(x)$ is continuous at $x = 3$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a) \quad x = a$

Solution:

Evaluate the limit of $f(x)$ as $x \rightarrow 3$:

$$\begin{aligned} \lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 2)}{x - 3} \\ &= \lim_{x \rightarrow 3} (x + 2) \\ &= 3 + 2 \\ &= 5 \end{aligned}$$

Given value at $x = 3$:

$$f(3) = 5$$

Thus:

$$\lim_{x \rightarrow 3} f(x) = f(3) = 5$$

Hence, f is continuous at $x = 3$.

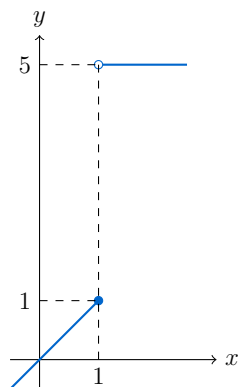
Hence Proved

Since $\lim_{x \rightarrow 3} f(x) = f(3) = 5$, the function f is continuous at $x = 3$.

Question 4

Is the function f defined by $f(x) = \begin{cases} x, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$ continuous at (i) $x = 0$ (ii) $x = 1$ (iii) $x = 2$?

TYPE 2 Continuity of Piecewise Functions



Graph of piecewise function $f(x)$ showing a step discontinuity at $x = 1$

Given: $f(x) = \begin{cases} x, & x \leq 1 \\ 5, & x > 1 \end{cases}$

Concept / Formula Used: $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a) \quad x = a$

Solution:

(i) At $x = 0$: In a small neighborhood of $x = 0$ ($x < 1$), $f(x) = x$.

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} (x) = 0 \\ f(0) &= 0 \end{aligned}$$

Since $\lim_{x \rightarrow 0} f(x) = f(0) = 0$, f is continuous at $x = 0$.

(ii) At $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x) = 1 \\ \text{RHL} &= \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (5) = 5 \end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$, $\lim_{x \rightarrow 1} f(x)$ does not exist. Hence, f is discontinuous at $x = 1$.

(iii) At $x = 2$: In a small neighborhood of $x = 2$ ($x > 1$), $f(x) = 5$.

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} (5) = 5 \\ f(2) &= 5 \end{aligned}$$

Since $\lim_{x \rightarrow 2} f(x) = f(2) = 5$, f is continuous at $x = 2$.

Final Answer

- (i) Continuous at $x = 0$
- (ii) Discontinuous at $x = 1$
- (iii) Continuous at $x = 2$

Question 5

Is the function f defined by $f(x) = \begin{cases} 3x + 5, & \text{if } x \geq 2 \\ x^2, & \text{if } x < 2 \end{cases}$ continuous at $x = 2$?

TYPE 1 Checking Continuity at a Point

Given: $f(x) = \begin{cases} 3x + 5, & x \geq 2 \\ x^2, & x < 2 \end{cases}$ at $x = 2$

Concept / Formula Used: $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a) \quad x = a$

Solution:

Find the Left-Hand Limit (LHL) at $x = 2$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 2^-} f(x) \\ &= \lim_{x \rightarrow 2^-} (x^2) \\ &= 2^2 \\ &= 4 \end{aligned}$$

Find the Right-Hand Limit (RHL) at $x = 2$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 2^+} f(x) \\ &= \lim_{x \rightarrow 2^+} (3x + 5) \\ &= 3(2) + 5 \\ &= 6 + 5 \\ &= 11 \end{aligned}$$

Since $\text{LHL} = 4 \neq 11 = \text{RHL}$, the limit does not exist. Hence, $f(x)$ is not continuous at $x = 2$.

Final Answer

No, $f(x)$ is not continuous at $x = 2$ because $\lim_{x \rightarrow 2^-} f(x) = 4 \neq 11 = \lim_{x \rightarrow 2^+} f(x)$.

Question 6

Is the function f defined by $f(x) = \begin{cases} \frac{2x^2 - 3x - 2}{x - 2}, & \text{if } x \neq 2 \\ 5, & \text{if } x = 2 \end{cases}$ continuous at $x = 2$?

TYPE 1 Checking Continuity at a Point

Given: $f(x) = \begin{cases} \frac{2x^2 - 3x - 2}{x - 2}, & x \neq 2 \\ 5, & x = 2 \end{cases}$ at $x = 2$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a) \quad x = a$

Solution:

Evaluate the limit of $f(x)$ as $x \rightarrow 2$:

$$\begin{aligned}
 \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{2x^2 - 3x - 2}{x - 2} \\
 &= \lim_{x \rightarrow 2} \frac{2x^2 - 4x + x - 2}{x - 2} \\
 &= \lim_{x \rightarrow 2} \frac{2x(x - 2) + 1(x - 2)}{x - 2} \\
 &= \lim_{x \rightarrow 2} \frac{(2x + 1)(x - 2)}{x - 2} \\
 &= \lim_{x \rightarrow 2} (2x + 1) \\
 &= 2(2) + 1 \\
 &= 5
 \end{aligned}$$

Given value of function at $x = 2$:

$$f(2) = 5$$

Since $\lim_{x \rightarrow 2} f(x) = f(2) = 5$, the function $f(x)$ is continuous at $x = 2$.

Final Answer

Yes, $f(x)$ is continuous at $x = 2$.

Question 7

If $f(x) = \begin{cases} kx^2 + 5, & x \leq 1 \\ 2, & x > 1 \end{cases}$, find k so that f may be continuous at $x = 1$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} kx^2 + 5, & x \leq 1 \\ 2, & x > 1 \end{cases}$ continuous at $x = 1$

Concept / Formula Used: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$

Solution:

Calculate Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned}
 \text{LHL} &= \lim_{x \rightarrow 1^-} (kx^2 + 5) \\
 &= k(1)^2 + 5 \\
 &= k + 5
 \end{aligned}$$

Calculate Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned}
 \text{RHL} &= \lim_{x \rightarrow 1^+} (2) \\
 &= 2
 \end{aligned}$$

Also, $f(1) = k(1)^2 + 5 = k + 5$. For f to be continuous at $x = 1$, we must have:

$$\text{LHL} = \text{RHL}$$

$$k + 5 = 2$$

$$k = 2 - 5$$

$$k = -3$$

Final Answer

$$k = -3$$

Question 8

If $f(x) = \begin{cases} 3x - 8, & x \leq 5 \\ 2k, & x > 5 \end{cases}$, find k so that f may be continuous at $x = 5$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} 3x - 8, & x \leq 5 \\ 2k, & x > 5 \end{cases}$ continuous at $x = 5$

Concept / Formula Used: $\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^+} f(x) = f(5)$

Solution:

Calculate Left-Hand Limit (LHL) at $x = 5$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 5^-} (3x - 8) \\ &= 3(5) - 8 \\ &= 15 - 8 \\ &= 7 \end{aligned}$$

Calculate Right-Hand Limit (RHL) at $x = 5$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 5^+} (2k) \\ &= 2k \end{aligned}$$

For continuity at $x = 5$, equate LHL and RHL:

$$\begin{aligned} 2k &= 7 \\ k &= \frac{7}{2} \end{aligned}$$

Final Answer

$$k = \frac{7}{2}$$

Question 9

If $f(x) = \begin{cases} 3x - 4, & 0 \leq x \leq 2 \\ 2x + \lambda, & 2 < x \leq 5 \end{cases}$, find λ so that f may be continuous at $x = 2$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} 3x - 4, & 0 \leq x \leq 2 \\ 2x + \lambda, & 2 < x \leq 5 \end{cases}$ continuous at $x = 2$

Concept / Formula Used: $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$

Solution:

Calculate Left-Hand Limit (LHL) at $x = 2$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 2^-} (3x - 4) \\ &= 3(2) - 4 \\ &= 6 - 4 \\ &= 2 \end{aligned}$$

Calculate Right-Hand Limit (RHL) at $x = 2$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 2^+} (2x + \lambda) \\ &= 2(2) + \lambda \\ &= 4 + \lambda \end{aligned}$$

Also, $f(2) = 3(2) - 4 = 2$. For $f(x)$ to be continuous at $x = 2$:

$$\begin{aligned} \text{RHL} &= \text{LHL} \\ 4 + \lambda &= 2 \\ \lambda &= 2 - 4 \\ \lambda &= -2 \end{aligned}$$

Final Answer

$$\lambda = -2$$

Question 10 (i)

$f(x) = \frac{x^2 - 9}{x - 3}$ is not defined at $x = 3$. What type of removable discontinuity is there? What value should be assigned to $f(3)$ for continuity of $f(x)$ at $x = 3$?

TYPE 4 Removable Discontinuity Analysis

Given: $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a)$

Solution:

Calculate the limit of $f(x)$ as $x \rightarrow 3$:

$$\begin{aligned}\lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} \\ &= \lim_{x \rightarrow 3} (x + 3) \\ &= 3 + 3 \\ &= 6\end{aligned}$$

Since $\lim_{x \rightarrow 3} f(x)$ exists finitely ($= 6$) but $f(3)$ is undefined, the discontinuity is a **missing point removable discontinuity**. To redefine $f(x)$ so that it is continuous at $x = 3$, we must assign $f(3) = \lim_{x \rightarrow 3} f(x) = 6$.

Final Answer

Type of discontinuity: Removable (Missing Point) Discontinuity.
Value assigned to $f(3) = 6$.

Question 10 (ii)

If $f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ k, & x = -1 \end{cases}$, find k so that the function f may be continuous at $x = -1$.

TYPE 3 Finding Constants for Continuity**★ Likely Exam Question**

Given: $f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ k, & x = -1 \end{cases}$ continuous at $x = -1$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a)$

Solution:

Find the limit of $f(x)$ as $x \rightarrow -1$:

$$\begin{aligned}\lim_{x \rightarrow -1} f(x) &= \lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{x + 1} \\ &= \lim_{x \rightarrow -1} \frac{(x + 1)(x - 3)}{x + 1} \\ &= \lim_{x \rightarrow -1} (x - 3) \\ &= -1 - 3 \\ &= -4\end{aligned}$$

Given that $f(-1) = k$. For continuity at $x = -1$:

$$\begin{aligned} f(-1) &= \lim_{x \rightarrow -1} f(x) \\ k &= -4 \end{aligned}$$

Final Answer

$$k = -4$$

Question 10 (iii)

Determine the value of ' k ' for which the following function is continuous at $x = 3$:

$$f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$ continuous at $x = 3$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a)$

Solution:

Evaluate the limit of $f(x)$ as $x \rightarrow 3$:

$$\begin{aligned} \lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} \frac{(x+3)^2 - 6^2}{x-3} \\ &= \lim_{x \rightarrow 3} \frac{[(x+3) - 6][(x+3) + 6]}{x-3} \\ &= \lim_{x \rightarrow 3} \frac{(x-3)(x+9)}{x-3} \\ &= \lim_{x \rightarrow 3} (x+9) \\ &= 3 + 9 \\ &= 12 \end{aligned}$$

Since $f(3) = k$, for $f(x)$ to be continuous at $x = 3$:

$$\begin{aligned} f(3) &= \lim_{x \rightarrow 3} f(x) \\ k &= 12 \end{aligned}$$

Final Answer

$$k = 12$$

Question 10 (iv)

If $f(x) = \begin{cases} \frac{x^2 - x - 6}{x^2 - 2x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$, find k so that the function f may be continuous at $x = 3$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} \frac{x^2 - x - 6}{x^2 - 2x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$ continuous at $x = 3$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a)$

Solution:

Factorize both numerator and denominator:

$$\begin{aligned} x^2 - x - 6 &= (x - 3)(x + 2) \\ x^2 - 2x - 3 &= (x - 3)(x + 1) \end{aligned}$$

Evaluate the limit of $f(x)$ as $x \rightarrow 3$:

$$\begin{aligned} \lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 2)}{(x - 3)(x + 1)} \\ &= \lim_{x \rightarrow 3} \frac{x + 2}{x + 1} \\ &= \frac{3 + 2}{3 + 1} \\ &= \frac{5}{4} \end{aligned}$$

For continuity at $x = 3$:

$$\begin{aligned} f(3) &= \lim_{x \rightarrow 3} f(x) \\ k &= \frac{5}{4} \end{aligned}$$

Final Answer

$$k = \frac{5}{4}$$

Question 11

If $f(x) = \begin{cases} \frac{\tan 3x}{kx}, & x \neq 0 \\ 1, & x = 0 \end{cases}$, find k so that the function f may be continuous at $x = 0$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} \frac{\tan 3x}{kx}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ continuous at $x = 0$

Concept / Formula Used: $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$

Solution:

Evaluate the limit as $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{\tan 3x}{kx} \\ &= \lim_{x \rightarrow 0} \left(\frac{\tan 3x}{3x} \cdot \frac{3}{k} \right) \\ &= \frac{3}{k} \cdot \lim_{3x \rightarrow 0} \frac{\tan 3x}{3x}\end{aligned}$$

Applying the standard result $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \frac{3}{k} \cdot 1 \\ &= \frac{3}{k}\end{aligned}$$

For $f(x)$ to be continuous at $x = 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= f(0) \\ \frac{3}{k} &= 1 \\ k &= 3\end{aligned}$$

Final Answer

$$k = 3$$

Question 12

Is the function f defined by $f(x) = \tan x$ continuous at $x = \frac{\pi}{2}$?

TYPE 1 Checking Continuity at a Point

Given: $f(x) = \tan x$ at $x = \frac{\pi}{2}$

Concept / Formula Used: $\lim_{x \rightarrow a} f(x) = f(a) \quad x = a$

Solution:

Evaluate the function at $x = \frac{\pi}{2}$:

$$f\left(\frac{\pi}{2}\right) = \tan\left(\frac{\pi}{2}\right) \quad (\text{undefined})$$

Also, evaluating left-hand and right-hand limits:

$$\begin{aligned}\lim_{x \rightarrow (\pi/2)^-} \tan x &= +\infty \\ \lim_{x \rightarrow (\pi/2)^+} \tan x &= -\infty\end{aligned}$$

Since $f\left(\frac{\pi}{2}\right)$ is not defined and the limit does not exist, $f(x)$ is not continuous at $x = \frac{\pi}{2}$.

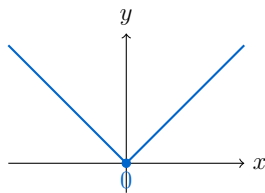
Final Answer

No, $f(x) = \tan x$ is not continuous at $x = \frac{\pi}{2}$ because $f\left(\frac{\pi}{2}\right)$ is undefined.

Question 13

Is the function f defined by $f(x) = |x|$ continuous at $x = 0$?

TYPE 5 Continuity of Absolute Value Functions



Graph of $f(x) = |x|$ continuous at the origin $x = 0$

Given: $f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$ at $x = 0$

Concept / Formula Used: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$

Solution:

Evaluate Left-Hand Limit (LHL) at $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} |x| \\ &= \lim_{x \rightarrow 0^-} (-x) \\ &= -0 \\ &= 0 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 0$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} |x| \\ &= \lim_{x \rightarrow 0^+} (x) \\ &= 0 \end{aligned}$$

Evaluate function value at $x = 0$:

$$f(0) = |0| = 0$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x) = |x|$ is continuous at $x = 0$.

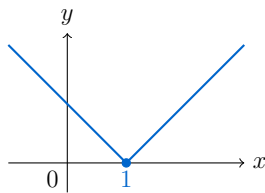
Final Answer

Yes, $f(x) = |x|$ is continuous at $x = 0$.

Question 14

Is the function f defined by $f(x) = |x - 1|$ continuous at $x = 1$?

TYPE 5 Continuity of Absolute Value Functions



Graph of $f(x) = |x - 1|$ continuous at $x = 1$

Given: $f(x) = |x - 1| = \begin{cases} x - 1, & x \geq 1 \\ -(x - 1), & x < 1 \end{cases}$ at $x = 1$

Concept / Formula Used: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$

Solution:

Evaluate Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 1^-} |x - 1| \\ &= \lim_{x \rightarrow 1^-} [-(x - 1)] \\ &= \lim_{x \rightarrow 1^-} (1 - x) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 1^+} |x - 1| \\ &= \lim_{x \rightarrow 1^+} (x - 1) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Evaluate function value at $x = 1$:

$$f(1) = |1 - 1| = 0$$

Since $\text{LHL} = \text{RHL} = f(1) = 0$, $f(x) = |x - 1|$ is continuous at $x = 1$.

Final Answer

Yes, $f(x) = |x - 1|$ is continuous at $x = 1$.

Question 15

Is the function f defined by $f(x) = x - |x|$ continuous at $x = 0$?

TYPE 5 Continuity of Absolute Value Functions

Given: $f(x) = x - |x|$ at $x = 0$

Concept / Formula Used: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$

Solution:

Redefining $f(x)$ using the definition of $|x|$:

$$\text{For } x \geq 0, \quad |x| = x \implies f(x) = x - x = 0$$

$$\text{For } x < 0, \quad |x| = -x \implies f(x) = x - (-x) = 2x$$

$$\text{Thus, } f(x) = \begin{cases} 0, & x \geq 0 \\ 2x, & x < 0 \end{cases}.$$

Evaluate Left-Hand Limit (LHL) at $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{x \rightarrow 0^-} (2x) \\ &= 2(0) \\ &= 0 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 0$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{x \rightarrow 0^+} (0) \\ &= 0 \end{aligned}$$

Evaluate function value at $x = 0$:

$$f(0) = 0 - |0| = 0$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Final Answer

Yes, $f(x) = x - |x|$ is continuous at $x = 0$.

Question 16 (i)

Examine the following function for continuity at the indicated point:

$$f(x) = \begin{cases} x^2, & x \geq 0 \\ -x, & x < 0 \end{cases} \quad \text{at } x = 0$$

TYPE 2 Continuity of Piecewise Functions

Given: $f(x) = \begin{cases} x^2, & x \geq 0 \\ -x, & x < 0 \end{cases}$ at $x = 0$

Concept / Formula Used: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$

Solution:

Evaluate Left-Hand Limit (LHL) at $x = 0$:

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} (-x) \\ &= -0 \\ &= 0\end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 0$:

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} (x^2) \\ &= 0^2 \\ &= 0\end{aligned}$$

Evaluate function value at $x = 0$:

$$f(0) = 0^2 = 0$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Final Answer

$f(x)$ is continuous at $x = 0$.

Question 16 (ii)

Examine the following function for continuity at the indicated point:

$$f(x) = \begin{cases} 5x - 4, & \text{if } x < 1 \\ 4x^2 - 3x, & \text{if } x \geq 1 \end{cases} \quad \text{at } x = 1$$

TYPE 2 Continuity of Piecewise Functions

Given: $f(x) = \begin{cases} 5x - 4, & x < 1 \\ 4x^2 - 3x, & x \geq 1 \end{cases} \quad \text{at } x = 1$

Concept / Formula Used: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$

Solution:

Evaluate Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 1^-} (5x - 4) \\ &= 5(1) - 4 \\ &= 5 - 4 \\ &= 1\end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 1^+} (4x^2 - 3x) \\ &= 4(1)^2 - 3(1) \\ &= 4 - 3 \\ &= 1\end{aligned}$$

Evaluate function value at $x = 1$:

$$\begin{aligned} f(1) &= 4(1)^2 - 3(1) \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(1) = 1$, $f(x)$ is continuous at $x = 1$.

Final Answer

$f(x)$ is continuous at $x = 1$.

Question 17

Is the function f defined by

$$f(x) = \begin{cases} 2x^2 - 3, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$$

continuous at $x = 0$? What about its continuity at $x = 1$ and at $x = 2$?

TYPE 2 Continuity of Piecewise Functions

Given: $f(x) = \begin{cases} 2x^2 - 3, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$

Concept / Formula Used: $f(x)$ is continuous at $x = c$ if $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$

Solution:

1. Continuity at $x = 0$: Since $0 < 1$, in a neighbourhood of $x = 0$, the function is defined as $f(x) = 2x^2 - 3$.

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} (2x^2 - 3) \\ &= 2(0)^2 - 3 \\ &= -3 \end{aligned}$$

The value of the function at $x = 0$ is:

$$\begin{aligned} f(0) &= 2(0)^2 - 3 \\ &= -3 \end{aligned}$$

Since $\lim_{x \rightarrow 0} f(x) = f(0) = -3$, the function f is continuous at $x = 0$.

2. Continuity at $x = 1$: Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 1^-} f(x) \\ &= \lim_{x \rightarrow 1^-} (2x^2 - 3) \\ &= 2(1)^2 - 3 \\ &= 2 - 3 \\ &= -1 \end{aligned}$$

Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 1^+} f(x) \\ &= \lim_{x \rightarrow 1^+} (5) \\ &= 5\end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$ (i.e., $-1 \neq 5$), $\lim_{x \rightarrow 1} f(x)$ does not exist. Hence, f is discontinuous at $x = 1$.

3. Continuity at $x = 2$: Since $2 > 1$, in a neighbourhood of $x = 2$, the function is defined as $f(x) = 5$.

$$\begin{aligned}\lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} (5) \\ &= 5\end{aligned}$$

The value of the function at $x = 2$ is:

$$f(2) = 5$$

Since $\lim_{x \rightarrow 2} f(x) = f(2) = 5$, the function f is continuous at $x = 2$.

Final Answer

The function f is continuous at $x = 0$ and $x = 2$, but discontinuous at $x = 1$.

Question 18 (i)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

TYPE 5 Continuity of Absolute Value Functions

Given: $f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ at $c = 0$

Concept / Formula Used: $|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$

Solution:

Left-Hand Limit (LHL) at $x = 0$: For $x < 0$, $|x| = -x$.

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} \frac{x}{|x|} \\ &= \lim_{x \rightarrow 0^-} \frac{x}{-x} \\ &= \lim_{x \rightarrow 0^-} (-1) \\ &= -1\end{aligned}$$

Right-Hand Limit (RHL) at $x = 0$: For $x > 0$, $|x| = x$.

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} \frac{x}{|x|} \\ &= \lim_{x \rightarrow 0^+} \frac{x}{x} \\ &= \lim_{x \rightarrow 0^+} (1) \\ &= 1\end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$ ($-1 \neq 1$), the limit $\lim_{x \rightarrow 0} f(x)$ does not exist.

Final Answer

$f(x)$ is discontinuous at $x = 0$.

Common Mistake: Evaluating $\frac{0}{|0|}$ directly leads to an indeterminate form. Always evaluate left-hand and right-hand limits separately when absolute value functions are involved.

Question 18 (ii)

Examine the following function for continuity at $x = 4$:

$$f(x) = \begin{cases} \frac{x-4}{2|x-4|}, & x \neq 4 \\ 0, & x = 4 \end{cases}$$

TYPE 5 Continuity of Absolute Value Functions

Given: $f(x) = \begin{cases} \frac{x-4}{2|x-4|}, & x \neq 4 \\ 0, & x = 4 \end{cases}$ at $c = 4$

Concept / Formula Used: $|x - a| = \begin{cases} x - a, & x > a \\ -(x - a), & x < a \end{cases}$

Solution:

Left-Hand Limit (LHL) at $x = 4$: For $x < 4$, $(x - 4) < 0$, so $|x - 4| = -(x - 4)$.

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 4^-} \frac{x-4}{2|x-4|} \\ &= \lim_{x \rightarrow 4^-} \frac{x-4}{2(-(x-4))} \\ &= \lim_{x \rightarrow 4^-} \left(-\frac{1}{2}\right) \\ &= -\frac{1}{2}\end{aligned}$$

Right-Hand Limit (RHL) at $x = 4$: For $x > 4$, $(x - 4) > 0$, so $|x - 4| = x - 4$.

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 4^+} \frac{x - 4}{2|x - 4|} \\ &= \lim_{x \rightarrow 4^+} \frac{x - 4}{2(x - 4)} \\ &= \lim_{x \rightarrow 4^+} \left(\frac{1}{2} \right) \\ &= \frac{1}{2}\end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$ (i.e., $-\frac{1}{2} \neq \frac{1}{2}$), $\lim_{x \rightarrow 4} f(x)$ does not exist.

Final Answer

$f(x)$ is discontinuous at $x = 4$.

Question 19 (i)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \frac{x}{\sin 3x}, & \text{when } x \neq 0 \\ 3, & \text{when } x = 0 \end{cases}$$

TYPE 6 Continuity Using Trigonometric Limits

Given: $f(x) = \begin{cases} \frac{x}{\sin 3x}, & x \neq 0 \\ 3, & x = 0 \end{cases}$ at $c = 0$

Concept / Formula Used: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

We evaluate the limit of $f(x)$ as $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{x}{\sin 3x} \\ &= \lim_{x \rightarrow 0} \frac{1}{3} \cdot \frac{3x}{\sin 3x} \\ &= \frac{1}{3} \cdot \frac{1}{\lim_{x \rightarrow 0} \frac{\sin 3x}{3x}}\end{aligned}$$

Using the standard result $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ with $\theta = 3x$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \frac{1}{3} \cdot \frac{1}{1} \\ &= \frac{1}{3}\end{aligned}$$

The given value of the function at $x = 0$ is:

$$f(0) = 3$$

Since $\lim_{x \rightarrow 0} f(x) = \frac{1}{3} \neq 3 = f(0)$, the function is not continuous at $x = 0$.

Final Answer

$f(x)$ is discontinuous at $x = 0$.

Question 19 (ii)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \frac{\sin 2x}{\sin 3x}, & \text{when } x \neq 0 \\ 2, & \text{when } x = 0 \end{cases}$$

TYPE 6 Continuity Using Trigonometric Limits

Given: $f(x) = \begin{cases} \frac{\sin 2x}{\sin 3x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$ at $c = 0$

Concept / Formula Used: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

Evaluating the limit of $f(x)$ as $x \rightarrow 0$:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 3x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{\sin 2x}{2x} \cdot 2x}{\frac{\sin 3x}{3x} \cdot 3x} \\ &= \lim_{x \rightarrow 0} \left(\frac{2}{3} \cdot \frac{\frac{\sin 2x}{2x}}{\frac{\sin 3x}{3x}} \right) \\ &= \frac{2}{3} \cdot \frac{\lim_{x \rightarrow 0} \frac{\sin 2x}{2x}}{\lim_{x \rightarrow 0} \frac{\sin 3x}{3x}} \end{aligned}$$

Applying the standard limit $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \frac{2}{3} \cdot \frac{1}{1} \\ &= \frac{2}{3} \end{aligned}$$

Given value of $f(0)$:

$$f(0) = 2$$

Since $\lim_{x \rightarrow 0} f(x) = \frac{2}{3} \neq 2 = f(0)$, the function is not continuous at $x = 0$.

Final Answer

$f(x)$ is discontinuous at $x = 0$.

Question 19 (iii)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \frac{\cos 3x - \cos x}{x^2}, & \text{if } x \neq 0 \\ -4, & \text{if } x = 0 \end{cases}$$

TYPE 6 Continuity Using Trigonometric Limits

★ Likely Exam Question

Given: $f(x) = \begin{cases} \frac{\cos 3x - \cos x}{x^2}, & x \neq 0 \\ -4, & x = 0 \end{cases}$ at $c = 0$

Concept / Formula Used: $\cos C - \cos D = -2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$ $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

Applying the identity $\cos 3x - \cos x = -2 \sin \left(\frac{3x+x}{2} \right) \sin \left(\frac{3x-x}{2} \right)$:

$$\cos 3x - \cos x = -2 \sin(2x) \sin(x)$$

Now evaluating the limit as $x \rightarrow 0$:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{-2 \sin 2x \sin x}{x^2} \\ &= -2 \cdot \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{x} \cdot \frac{\sin x}{x} \right) \\ &= -2 \cdot \lim_{x \rightarrow 0} \left(2 \cdot \frac{\sin 2x}{2x} \cdot \frac{\sin x}{x} \right) \\ &= -4 \cdot \left(\lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \right) \cdot \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \end{aligned}$$

Using the standard result $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= -4 \cdot (1) \cdot (1) \\ &= -4 \end{aligned}$$

The given functional value is $f(0) = -4$. Since $\lim_{x \rightarrow 0} f(x) = f(0) = -4$, the function is continuous at $x = 0$.

Final Answer

$f(x)$ is continuous at $x = 0$.

Question 20 (i)

If $f(x) = \begin{cases} \frac{x^2 - 25}{x - 5}, & x \neq 5 \\ k, & x = 5 \end{cases}$ is continuous at $x = 5$, find the value of k .

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} \frac{x^2 - 25}{x - 5}, & x \neq 5 \\ k, & x = 5 \end{cases}$ is continuous at $x = 5$

Concept / Formula Used: $f(x)$ $x = c$ $\lim_{x \rightarrow c} f(x) = f(c)$

Solution:

Evaluating the limit of $f(x)$ as $x \rightarrow 5$:

$$\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$$

Factoring the numerator: $x^2 - 25 = (x - 5)(x + 5)$.

$$\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} \frac{(x - 5)(x + 5)}{x - 5}$$

Since $x \neq 5$, we divide numerator and denominator by $(x - 5)$:

$$\begin{aligned} \lim_{x \rightarrow 5} f(x) &= \lim_{x \rightarrow 5} (x + 5) \\ &= 5 + 5 \\ &= 10 \end{aligned}$$

Given $f(5) = k$. For continuity at $x = 5$:

$$\begin{aligned} f(5) &= \lim_{x \rightarrow 5} f(x) \\ k &= 10 \end{aligned}$$

Final Answer

$$k = 10$$

Question 20 (ii)

Find the value of k so that the function $f(x) = \begin{cases} kx + 1, & \text{if } x \leq 5 \\ 3x - 5, & \text{if } x > 5 \end{cases}$ is continuous at $x = 5$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} kx + 1, & x \leq 5 \\ 3x - 5, & x > 5 \end{cases}$ is continuous at $x = 5$

Concept / Formula Used: LHL = RHL = $f(5)$

Solution:

Left-Hand Limit (LHL) and functional value at $x = 5$:

$$\begin{aligned}\text{LHL} = f(5) &= \lim_{x \rightarrow 5^-} (kx + 1) \\ &= k(5) + 1 \\ &= 5k + 1\end{aligned}$$

Right-Hand Limit (RHL) at $x = 5$:

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 5^+} (3x - 5) \\ &= 3(5) - 5 \\ &= 15 - 5 \\ &= 10\end{aligned}$$

Since f is continuous at $x = 5$, we equate LHL and RHL:

$$\begin{aligned}5k + 1 &= 10 \\ 5k &= 10 - 1 \\ 5k &= 9 \\ k &= \frac{9}{5}\end{aligned}$$

Final Answer

$$k = \frac{9}{5}$$

Question 20 (iii)

For what value of k is the following function continuous at $x = 2$?

$$f(x) = \begin{cases} 2x + 1, & x < 2 \\ k, & x = 2 \\ 3x - 1, & x > 2 \end{cases}$$

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} 2x + 1, & x < 2 \\ k, & x = 2 \\ 3x - 1, & x > 2 \end{cases}$ is continuous at $x = 2$

Concept / Formula Used: $\text{LHL} = \text{RHL} = f(2)$

Solution:

Evaluating Left-Hand Limit (LHL) at $x = 2$:

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 2^-} (2x + 1) \\ &= 2(2) + 1 \\ &= 4 + 1 \\ &= 5\end{aligned}$$

Evaluating Right-Hand Limit (RHL) at $x = 2$:

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 2^+} (3x - 1) \\ &= 3(2) - 1 \\ &= 6 - 1 \\ &= 5\end{aligned}$$

Given $f(2) = k$. For continuity at $x = 2$:

$$\begin{aligned}f(2) &= \text{LHL} = \text{RHL} \\ k &= 5\end{aligned}$$

Final Answer

$$k = 5$$

Question 20 (iv)

Find the value of k so that the function f defined by

$$f(x) = \begin{cases} kx + 1, & \text{if } x \leq \pi \\ \cos x, & \text{if } x > \pi \end{cases}$$

is continuous at $x = \pi$.

TYPE 3 Finding Constants for Continuity

★ Likely Exam Question

Given: $f(x) = \begin{cases} kx + 1, & x \leq \pi \\ \cos x, & x > \pi \end{cases}$ is continuous at $x = \pi$

Concept / Formula Used: $\text{LHL} = \text{RHL} = f(\pi)$

Solution:

Left-Hand Limit (LHL) and value of function at $x = \pi$:

$$\begin{aligned}\text{LHL} &= f(\pi) = \lim_{x \rightarrow \pi^-} (kx + 1) \\ &= k\pi + 1\end{aligned}$$

Right-Hand Limit (RHL) at $x = \pi$:

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow \pi^+} \cos x \\ &= \cos \pi\end{aligned}$$

Using the trigonometric value $\cos \pi = -1$:

$$\text{RHL} = -1$$

Equating LHL and RHL for continuity:

$$\begin{aligned}k\pi + 1 &= -1 \\k\pi &= -1 - 1 \\k\pi &= -2 \\k &= -\frac{2}{\pi}\end{aligned}$$

Final Answer

$$k = -\frac{2}{\pi}$$

Question 20 (v)

For what value of k is the function

$$f(x) = \begin{cases} \frac{\tan 5x}{\sin 2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$$

continuous at $x = 0$?

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} \frac{\tan 5x}{\sin 2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$

Concept / Formula Used: $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1 \quad \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

Evaluating the limit of $f(x)$ as $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{\tan 5x}{\sin 2x} \\&= \lim_{x \rightarrow 0} \frac{\frac{\tan 5x}{5x} \cdot 5x}{\frac{\sin 2x}{2x} \cdot 2x} \\&= \lim_{x \rightarrow 0} \left(\frac{5}{2} \cdot \frac{\frac{\tan 5x}{5x}}{\frac{\sin 2x}{2x}} \right) \\&= \frac{5}{2} \cdot \frac{\lim_{x \rightarrow 0} \frac{\tan 5x}{5x}}{\lim_{x \rightarrow 0} \frac{\sin 2x}{2x}}\end{aligned}$$

Using $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$ and $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \frac{5}{2} \cdot \frac{1}{1} \\ &= \frac{5}{2}\end{aligned}$$

For continuity at $x = 0$, $f(0) = \lim_{x \rightarrow 0} f(x)$:

$$k = \frac{5}{2}$$

Final Answer

$$k = \frac{5}{2}$$

Question 20 (vi)

Find the value of k for which the function f given as

$$f(x) = \begin{cases} \frac{1 - \cos x}{2x^2}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$$

is continuous at $x = 0$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} \frac{1 - \cos x}{2x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$

Concept / Formula Used: $1 - \cos x = 2 \sin^2 \left(\frac{x}{2} \right)$ $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

Substituting $1 - \cos x = 2 \sin^2 \left(\frac{x}{2} \right)$ into the limit expression:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{2 \sin^2 \left(\frac{x}{2} \right)}{2x^2} \\ &= \lim_{x \rightarrow 0} \frac{\sin^2 \left(\frac{x}{2} \right)}{x^2} \\ &= \lim_{x \rightarrow 0} \left[\frac{\sin \left(\frac{x}{2} \right)}{x} \right]^2 \\ &= \lim_{x \rightarrow 0} \left[\frac{1}{2} \cdot \frac{\sin \left(\frac{x}{2} \right)}{\frac{x}{2}} \right]^2 \\ &= \frac{1}{4} \cdot \left[\lim_{x \rightarrow 0} \frac{\sin \left(\frac{x}{2} \right)}{\frac{x}{2}} \right]^2\end{aligned}$$

Using the standard result $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \frac{1}{4} \cdot (1)^2 \\ &= \frac{1}{4}\end{aligned}$$

For continuity, $f(0) = \lim_{x \rightarrow 0} f(x)$:

$$k = \frac{1}{4}$$

Final Answer

$$k = \frac{1}{4}$$

Question 20 (vii)

Find the value(s) of ' λ ' for which the function f given as

$$f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

is continuous at $x = 0$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ is continuous at $x = 0$

Concept / Formula Used: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

Evaluating the limit of $f(x)$ as $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{\sin^2 \lambda x}{x^2} \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin \lambda x}{x} \right)^2 \\ &= \lim_{x \rightarrow 0} \left(\lambda \cdot \frac{\sin \lambda x}{\lambda x} \right)^2 \\ &= \lambda^2 \cdot \left(\lim_{x \rightarrow 0} \frac{\sin \lambda x}{\lambda x} \right)^2\end{aligned}$$

Applying the standard limit $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ with $\theta = \lambda x$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lambda^2 \cdot (1)^2 \\ &= \lambda^2\end{aligned}$$

For $f(x)$ to be continuous at $x = 0$:

$$\lim_{x \rightarrow 0} f(x) = f(0)$$

$$\lambda^2 = 1$$

$$\lambda = \pm 1$$

Final Answer

$$\lambda = \pm 1$$

Question 21

Find the relationship between a and b so that the function f defined by

$$f(x) = \begin{cases} ax + 1, & \text{if } x \leq 3 \\ bx + 3, & \text{if } x > 3 \end{cases}$$

is continuous at $x = 3$.

TYPE 3 Finding Constants for Continuity

Given: $f(x) = \begin{cases} ax + 1, & x \leq 3 \\ bx + 3, & x > 3 \end{cases}$ is continuous at $x = 3$

Concept / Formula Used: $x = 3$ LHL = RHL = $f(3)$

Solution:

Evaluating Left-Hand Limit (LHL) and $f(3)$:

$$\begin{aligned} \text{LHL} = f(3) &= \lim_{x \rightarrow 3^-} (ax + 1) \\ &= a(3) + 1 \\ &= 3a + 1 \end{aligned}$$

Evaluating Right-Hand Limit (RHL) at $x = 3$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 3^+} (bx + 3) \\ &= b(3) + 3 \\ &= 3b + 3 \end{aligned}$$

Since the function f is continuous at $x = 3$, we set LHL = RHL:

$$\begin{aligned} 3a + 1 &= 3b + 3 \\ 3a - 3b &= 3 - 1 \\ 3a - 3b &= 2 \\ 3a &= 3b + 2 \quad \text{or} \quad a &= b + \frac{2}{3} \end{aligned}$$

Final Answer

$$3a = 3b + 2 \quad \left(\text{or } a - b = \frac{2}{3} \right)$$

Question 22

If the function f defined by

$$f(x) = \begin{cases} 3ax + b, & \text{if } x > 1 \\ 11, & \text{if } x = 1 \\ 5ax - 2b, & \text{if } x < 1 \end{cases}$$

is continuous at $x = 1$, find the values of a and b .

TYPE 3 Finding Constants for Continuity

★ Likely Exam Question

Given: $f(x) = \begin{cases} 3ax + b, & x > 1 \\ 11, & x = 1 \\ 5ax - 2b, & x < 1 \end{cases}$ is continuous at $x = 1$

Concept / Formula Used: $x = 1$ LHL = RHL = $f(1)$

Solution:

Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 1^-} (5ax - 2b) \\ &= 5a(1) - 2b \\ &= 5a - 2b \end{aligned}$$

Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 1^+} (3ax + b) \\ &= 3a(1) + b \\ &= 3a + b \end{aligned}$$

Given functional value $f(1) = 11$. Since f is continuous at $x = 1$:

$$\text{RHL} = f(1) \implies 3a + b = 11 \quad \text{--- (Equation 1)}$$

$$\text{LHL} = f(1) \implies 5a - 2b = 11 \quad \text{--- (Equation 2)}$$

Solving simultaneous equations step-by-step: From Equation 1, express b in terms of a :

$$b = 11 - 3a$$

Substitute $b = 11 - 3a$ into Equation 2:

$$\begin{aligned} 5a - 2(11 - 3a) &= 11 \\ 5a - 22 + 6a &= 11 \\ 11a - 22 &= 11 \\ 11a &= 11 + 22 \\ 11a &= 33 \\ a &= 3 \end{aligned}$$

Substitute $a = 3$ back into $b = 11 - 3a$:

$$b = 11 - 3(3)$$

$$b = 11 - 9$$

$$b = 2$$

Final Answer

$$a = 3, \quad b = 2$$

Question 23

If $f(x) = \begin{cases} \frac{\sin(a+1)x + 2\sin x}{x}, & x < 0 \\ 2, & x = 0 \\ \frac{\sqrt{1+bx} - 1}{x}, & x > 0 \end{cases}$ is continuous at $x = 0$, then find the values of a and b .

TYPE 3 Finding Constants for Continuity

★ Likely Exam Question

Given: $f(x) = \begin{cases} \frac{\sin(a+1)x + 2\sin x}{x}, & x < 0 \\ 2, & x = 0 \\ \frac{\sqrt{1+bx} - 1}{x}, & x > 0 \end{cases}$ is continuous at $x = 0$

Concept / Formula Used: $x = 0$ LHL = $f(0)$ = RHL

Solution:

1. Evaluating Left-Hand Limit (LHL) at $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} \frac{\sin(a+1)x + 2\sin x}{x} \\ &= \lim_{x \rightarrow 0^-} \left(\frac{\sin(a+1)x}{x} + \frac{2\sin x}{x} \right) \\ &= \lim_{x \rightarrow 0^-} \left((a+1) \cdot \frac{\sin(a+1)x}{(a+1)x} + 2 \cdot \frac{\sin x}{x} \right) \\ &= (a+1) \cdot \lim_{x \rightarrow 0^-} \frac{\sin(a+1)x}{(a+1)x} + 2 \cdot \lim_{x \rightarrow 0^-} \frac{\sin x}{x} \end{aligned}$$

Using the standard result $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$:

$$\begin{aligned} \text{LHL} &= (a+1)(1) + 2(1) \\ &= a + 1 + 2 \\ &= a + 3 \end{aligned}$$

Since f is continuous at $x = 0$, $\text{LHL} = f(0)$:

$$a + 3 = 2$$

$$a = 2 - 3$$

$$a = -1$$

2. Evaluating Right-Hand Limit (RHL) at $x = 0$:

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{\sqrt{1+bx} - 1}{x}$$

Rationalizing the numerator by multiplying and dividing by $(\sqrt{1+bx} + 1)$:

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} \frac{(\sqrt{1+bx} - 1)(\sqrt{1+bx} + 1)}{x(\sqrt{1+bx} + 1)} \\ &= \lim_{x \rightarrow 0^+} \frac{(1+bx) - 1}{x(\sqrt{1+bx} + 1)} \\ &= \lim_{x \rightarrow 0^+} \frac{bx}{x(\sqrt{1+bx} + 1)} \\ &= \lim_{x \rightarrow 0^+} \frac{b}{\sqrt{1+bx} + 1} \\ &= \frac{b}{\sqrt{1+b(0)} + 1} \\ &= \frac{b}{1+1} \\ &= \frac{b}{2} \end{aligned}$$

Since f is continuous at $x = 0$, $\text{RHL} = f(0)$:

$$\frac{b}{2} = 2$$

$$b = 4$$

Final Answer

$$a = -1, \quad b = 4$$

Question 24 (i)

Consider the function defined as follows, for $x \neq 0$:

$$f(x) = \frac{\sqrt[3]{1+x} - 1}{x}$$

What choice (if any) of $f(0)$ will make the function continuous at $x = 0$?

TYPE 4 Removable Discontinuity Analysis

Given: $f(x) = \frac{\sqrt[3]{1+x} - 1}{x}$ for $x \neq 0$

Concept / Formula Used: $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$ $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

Solution:

For $f(x)$ to be continuous at $x = 0$, $f(0)$ must be equal to $\lim_{x \rightarrow 0} f(x)$.

Let $1 + x = y$. As $x \rightarrow 0$, $y \rightarrow 1$. Substituting $x = y - 1$:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - 1}{x} = \lim_{y \rightarrow 1} \frac{y^{1/3} - 1^{1/3}}{y - 1}$$

Using the standard result $\lim_{y \rightarrow a} \frac{y^n - a^n}{y - a} = na^{n-1}$ with $a = 1$ and $n = \frac{1}{3}$:

$$\begin{aligned} \lim_{y \rightarrow 1} \frac{y^{1/3} - 1^{1/3}}{y - 1} &= \frac{1}{3}(1)^{\frac{1}{3}-1} \\ &= \frac{1}{3}(1)^{-2/3} \\ &= \frac{1}{3} \end{aligned}$$

Thus, setting $f(0) = \frac{1}{3}$ will make $f(x)$ continuous at $x = 0$.

Final Answer

$$f(0) = \frac{1}{3}$$

Question 24 (ii)

Consider the function defined as follows, for $x \neq 0$:

$$f(x) = \frac{5|x| - 3 \tan x + \sin 2x}{x}$$

What choice (if any) of $f(0)$ will make the function continuous at $x = 0$?

TYPE 4 Removable Discontinuity Analysis

Given: $f(x) = \frac{5|x| - 3 \tan x + \sin 2x}{x}$ for $x \neq 0$

Concept / Formula Used: $|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$ $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1, \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} = 1$

Solution:

1. Left-Hand Limit (LHL) at $x = 0$: For $x < 0$, $|x| = -x$.

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} \frac{5(-x) - 3 \tan x + \sin 2x}{x} \\ &= \lim_{x \rightarrow 0^-} \left(\frac{-5x}{x} - \frac{3 \tan x}{x} + \frac{\sin 2x}{x} \right) \\ &= \lim_{x \rightarrow 0^-} \left(-5 - 3 \cdot \frac{\tan x}{x} + 2 \cdot \frac{\sin 2x}{2x} \right) \\ &= -5 - 3(1) + 2(1) \\ &= -5 - 3 + 2 \\ &= -6\end{aligned}$$

2. Right-Hand Limit (RHL) at $x = 0$: For $x > 0$, $|x| = x$.

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} \frac{5x - 3 \tan x + \sin 2x}{x} \\ &= \lim_{x \rightarrow 0^+} \left(\frac{5x}{x} - \frac{3 \tan x}{x} + \frac{\sin 2x}{x} \right) \\ &= \lim_{x \rightarrow 0^+} \left(5 - 3 \cdot \frac{\tan x}{x} + 2 \cdot \frac{\sin 2x}{2x} \right) \\ &= 5 - 3(1) + 2(1) \\ &= 5 - 3 + 2 \\ &= 4\end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$ (i.e., $-6 \neq 4$), the limit $\lim_{x \rightarrow 0} f(x)$ does not exist. Hence, no choice of $f(0)$ can make $f(x)$ continuous at $x = 0$.

Final Answer

No choice of $f(0)$ will make the function continuous at $x = 0$.

Question 25

Examine the function

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

for continuity at $x = 0$.

TYPE 7 Squeeze Theorem for Continuity

★ Likely Exam Question

Given: $f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ at $c = 0$

Concept / Formula Used: $g(x) \leq f(x) \leq h(x) \quad \lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L \quad \lim_{x \rightarrow c} f(x) = L$

Solution:

For any $x \neq 0$, the sine function is bounded:

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

Multiplying the inequality by $|x| > 0$:

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$$

Taking limits as $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0}(-|x|) &= 0 \\ \lim_{x \rightarrow 0}|x| &= 0\end{aligned}$$

By the Squeeze Theorem:

$$\lim_{x \rightarrow 0}\left(x \sin \frac{1}{x}\right) = 0$$

Thus:

$$\lim_{x \rightarrow 0} f(x) = 0$$

The defined value of the function at $x = 0$ is $f(0) = 0$. Since $\lim_{x \rightarrow 0} f(x) = f(0) = 0$, the function $f(x)$ is continuous at $x = 0$.

Final Answer

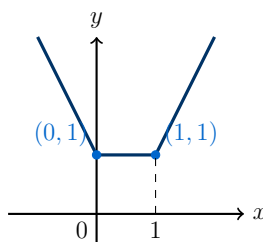
$f(x)$ is continuous at $x = 0$.

Question 26

Show that the function $f(x) = |x| + |x - 1|$, $x \in \mathbb{R}$, is continuous both at $x = 0$ and $x = 1$.

TYPE 5 Continuity of Absolute Value Functions**★ Likely Exam Question**

To Prove: $f(x) = |x| + |x - 1|$ is continuous at $x = 0$ and $x = 1$.



Graph of $f(x) = |x| + |x - 1|$ showing continuity at corner points $x = 0$ and $x = 1$

Concept / Formula Used: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$ $|x - 1| = \begin{cases} x - 1, & x \geq 1 \\ -(x - 1), & x < 1 \end{cases}$

Solution:

Redefining $f(x) = |x| + |x - 1|$ piecewise:

- Case 1: For $x < 0$, $|x| = -x$ and $|x - 1| = -(x - 1) = 1 - x$.

$$f(x) = -x + 1 - x = 1 - 2x$$

- Case 2: For $0 \leq x < 1$, $|x| = x$ and $|x - 1| = -(x - 1) = 1 - x$.

$$f(x) = x + 1 - x = 1$$

- Case 3: For $x \geq 1$, $|x| = x$ and $|x - 1| = x - 1$.

$$f(x) = x + x - 1 = 2x - 1$$

Thus:

$$f(x) = \begin{cases} 1 - 2x, & x < 0 \\ 1, & 0 \leq x < 1 \\ 2x - 1, & x \geq 1 \end{cases}$$

1. Continuity at $x = 0$:

$$\text{LHL} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (1 - 2x) = 1 - 2(0) = 1$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (1) = 1$$

$$f(0) = |0| + |0 - 1| = 0 + 1 = 1$$

Since $\text{LHL} = \text{RHL} = f(0) = 1$, $f(x)$ is continuous at $x = 0$.

2. Continuity at $x = 1$:

$$\text{LHL} = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (1) = 1$$

$$\text{RHL} = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2x - 1) = 2(1) - 1 = 1$$

$$f(1) = |1| + |1 - 1| = 1 + 0 = 1$$

Since $\text{LHL} = \text{RHL} = f(1) = 1$, $f(x)$ is continuous at $x = 1$.

Hence Proved

LHS = RHS in limit evaluations at $x = 0$ and $x = 1$. Thus, $f(x)$ is continuous at both $x = 0$ and $x = 1$.

Question 27

Examine the function

$$f(x) = \begin{cases} e^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

for continuity at $x = 0$.

TYPE 8 Continuity of Exponential Functions

Given: $f(x) = \begin{cases} e^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $c = 0$

Concept / Formula Used: $\lim_{y \rightarrow -\infty} e^y = 0$ $\lim_{y \rightarrow +\infty} e^y = +\infty$

Solution:

Evaluating Left-Hand Limit (LHL) at $x = 0$: Let $x = 0 - h$, where $h > 0$ and $h \rightarrow 0$.

$$\begin{aligned}\text{LHL} &= \lim_{h \rightarrow 0} e^{1/(0-h)} \\ &= \lim_{h \rightarrow 0} e^{-1/h} \\ &= \lim_{h \rightarrow 0} \frac{1}{e^{1/h}}\end{aligned}$$

As $h \rightarrow 0^+$, $\frac{1}{h} \rightarrow +\infty \implies e^{1/h} \rightarrow +\infty$.

$$\begin{aligned}\text{LHL} &= \frac{1}{\infty} \\ &= 0\end{aligned}$$

Evaluating Right-Hand Limit (RHL) at $x = 0$: Let $x = 0 + h$, where $h > 0$ and $h \rightarrow 0$.

$$\begin{aligned}\text{RHL} &= \lim_{h \rightarrow 0} e^{1/(0+h)} \\ &= \lim_{h \rightarrow 0} e^{1/h}\end{aligned}$$

As $h \rightarrow 0^+$, $\frac{1}{h} \rightarrow +\infty \implies e^{1/h} \rightarrow +\infty$.

$$\text{RHL} = +\infty$$

Since $\text{LHL} = 0 \neq +\infty = \text{RHL}$, $\lim_{x \rightarrow 0} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 0$.

Final Answer

$f(x)$ is discontinuous at $x = 0$.

5.2 – Continuity: Piecewise Functions and Finding Constants

Question 1

Is the function f defined by $f(x) = [x]$ continuous at:

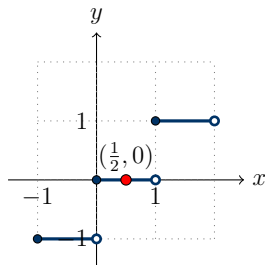
- (i) $x = 0$
- (ii) $x = 1$
- (iii) $x = \frac{1}{2}$?

TYPE 1 Checking Continuity at a Point

Given: The function $f(x) = [x]$, where $[x]$ denotes the greatest integer less than or equal to x .

To Find: Check continuity at $x = 0$, $x = 1$, and $x = \frac{1}{2}$.

Concept / Formula Used: $f(x)$ $x = c$ $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$



Graph of Greatest Integer Function $f(x) = [x]$ showing step discontinuities at integers

Solution:

(i) Continuity at $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} [x] \\ &= -1\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} [x] \\ &= 0\end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = [0] = 0$$

Since $\text{LHL} \neq \text{RHL}$, $\lim_{x \rightarrow 0} f(x)$ does not exist. Hence, $f(x)$ is not continuous at $x = 0$.

(ii) Continuity at $x = 1$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 1^-} [x] \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 1^+} [x] \\ &= 1\end{aligned}$$

Value of the function at $x = 1$:

$$f(1) = [1] = 1$$

Since $\text{LHL} \neq \text{RHL}$, $\lim_{x \rightarrow 1} f(x)$ does not exist. Hence, $f(x)$ is not continuous at $x = 1$.

(iii) **Continuity at $x = \frac{1}{2}$:**

For any $x \in (0, 1)$, the value of $[x] = 0$.

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow (1/2)^-} [x] \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow (1/2)^+} [x] \\ &= 0\end{aligned}$$

Value of the function at $x = \frac{1}{2}$:

$$f\left(\frac{1}{2}\right) = \left[\frac{1}{2}\right] = 0$$

Since $\text{LHL} = \text{RHL} = f\left(\frac{1}{2}\right) = 0$, $f(x)$ is continuous at $x = \frac{1}{2}$.

Final Answer

- (i) No, f is not continuous at $x = 0$.
- (ii) No, f is not continuous at $x = 1$.
- (iii) Yes, f is continuous at $x = \frac{1}{2}$.

Common Mistake: Assuming that greatest integer function is discontinuous everywhere. It is discontinuous only at integer values, but continuous at all non-integer values.

Question 2 (i)

Examine the following function for continuity:

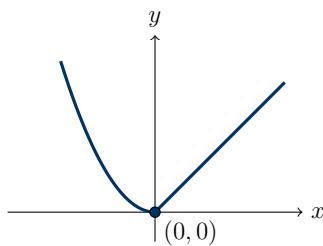
$$f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ x^2, & \text{if } x < 0 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ x^2, & \text{if } x < 0 \end{cases}$

To Find: Examine the continuity of $f(x)$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $x = c \quad \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$



Graph of piecewise function $f(x) = x$ for $x \geq 0$ and $f(x) = x^2$ for $x < 0$

Solution:

Case 1: For $x < 0$, $f(x) = x^2$, which is a polynomial function and hence continuous for all $x < 0$.

Case 2: For $x > 0$, $f(x) = x$, which is a polynomial function and hence continuous for all $x > 0$.

Case 3: At the boundary point $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{x \rightarrow 0^-} (x^2) \\ &= 0^2 \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{x \rightarrow 0^+} (x) \\ &= 0 \end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = 0$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Combining all three cases, $f(x)$ is continuous at all points $x \in \mathbb{R}$.

Final Answer

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

Question 2 (ii)

Examine the following function for continuity:

$$f(x) = \begin{cases} x + 5, & \text{if } x \leq 1 \\ x - 5, & \text{if } x > 1 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} x + 5, & \text{if } x \leq 1 \\ x - 5, & \text{if } x > 1 \end{cases}$

To Find: Examine the continuity of $f(x)$ for all real numbers.

Concept / Formula Used: $x = 1$ $\lim_{x \rightarrow 1^-} f(x)$ $\lim_{x \rightarrow 1^+} f(x)$

Solution:

Case 1: For $x < 1$, $f(x) = x + 5$, which is a polynomial function and hence continuous.

Case 2: For $x > 1$, $f(x) = x - 5$, which is a polynomial function and hence continuous.

Case 3: At $x = 1$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 1^-} (x + 5) \\ &= 1 + 5 \\ &= 6\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 1^+} (x - 5) \\ &= 1 - 5 \\ &= -4\end{aligned}$$

Value of the function at $x = 1$:

$$\begin{aligned}f(1) &= 1 + 5 \\ &= 6\end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$, $\lim_{x \rightarrow 1} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 1$.

Final Answer

The function $f(x)$ is continuous for all real numbers except at $x = 1$.

Question 3 (i)

Examine the following function for continuity:

$$f(x) = \begin{cases} 2x - 1, & \text{if } x < 2 \\ \frac{3}{2}x, & \text{if } x \geq 2 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} 2x - 1, & \text{if } x < 2 \\ \frac{3}{2}x, & \text{if } x \geq 2 \end{cases}$

To Find: Examine the continuity of $f(x)$.

Concept / Formula Used: $f(x)$ $x = 2$ $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$

Solution:

For $x < 2$, $f(x) = 2x - 1$ is continuous as it is linear polynomial. For $x > 2$, $f(x) = \frac{3}{2}x$ is continuous as it is linear polynomial.

At the boundary point $x = 2$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 2^-} (2x - 1) \\ &= 2(2) - 1 \\ &= 4 - 1 \\ &= 3\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 2^+} \left(\frac{3}{2}x \right) \\ &= \frac{3}{2}(2) \\ &= 3\end{aligned}$$

Value of the function at $x = 2$:

$$\begin{aligned}f(2) &= \frac{3}{2}(2) \\ &= 3\end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(2) = 3$, $f(x)$ is continuous at $x = 2$. Therefore, $f(x)$ is continuous for all $x \in \mathbb{R}$.

Final Answer

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

Question 3 (ii)

Examine the following function for continuity:

$$f(x) = \begin{cases} \sin x - \cos x, & x \neq 0 \\ -1, & x = 0 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} \sin x - \cos x, & x \neq 0 \\ -1, & x = 0 \end{cases}$

To Find: Examine the continuity of $f(x)$.

Concept / Formula Used: $f(x)$ is continuous at $x = 0$ if $\lim_{x \rightarrow 0} f(x) = f(0)$

Solution:

For $x \neq 0$, $f(x) = \sin x - \cos x$, which is the difference of two continuous trigonometric functions, hence continuous for all $x \neq 0$.

At $x = 0$:

Limit as $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} (\sin x - \cos x) \\ &= \sin 0 - \cos 0 \\ &= 0 - 1 \\ &= -1\end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = -1$$

Since $\lim_{x \rightarrow 0} f(x) = f(0) = -1$, the function $f(x)$ is continuous at $x = 0$.

Hence, $f(x)$ is continuous for all $x \in \mathbb{R}$.

Final Answer

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

Question 4 (i)

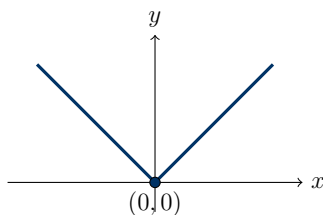
Examine the continuity of $f(x) = |x|$.

TYPE 5 Continuity of Absolute Value Functions

Given: The modulus function $f(x) = |x|$.

To Find: Examine the continuity of $f(x)$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$



Graph of modulus function $f(x) = |x|$

Solution:

By definition, $f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$

Case 1: For $x < 0$, $f(x) = -x$, a polynomial function, continuous for all $x < 0$.

Case 2: For $x > 0$, $f(x) = x$, a polynomial function, continuous for all $x > 0$.

Case 3: At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} (-x) \\ &= -0 \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} (x) \\ &= 0\end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = |0| = 0$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Therefore, $f(x) = |x|$ is continuous for all real numbers $x \in \mathbb{R}$.

Final Answer

The function $f(x) = |x|$ is continuous for all $x \in \mathbb{R}$.

Question 4 (ii)

Examine the continuity of $f(x) = |x - 5|$.

TYPE 5 Continuity of Absolute Value Functions

Given: The modulus function $f(x) = |x - 5|$.

To Find: Examine the continuity of $f(x)$.

Concept / Formula Used: $|x - 5| = \begin{cases} x - 5, & x \geq 5 \\ -(x - 5), & x < 5 \end{cases}$

Solution:

Rewrite $f(x)$ as a piecewise function:

$$f(x) = \begin{cases} -(x - 5), & \text{if } x < 5 \\ x - 5, & \text{if } x \geq 5 \end{cases}$$

Case 1: For $x < 5$, $f(x) = -(x - 5) = 5 - x$, which is polynomial and hence continuous.

Case 2: For $x > 5$, $f(x) = x - 5$, which is polynomial and hence continuous.

Case 3: At $x = 5$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 5^-} (5 - x) \\ &= 5 - 5 \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 5^+} (x - 5) \\ &= 5 - 5 \\ &= 0\end{aligned}$$

Value of the function at $x = 5$:

$$f(5) = |5 - 5| = 0$$

Since $\text{LHL} = \text{RHL} = f(5) = 0$, $f(x)$ is continuous at $x = 5$.

Hence, $f(x)$ is continuous for all real numbers $x \in \mathbb{R}$.

Final Answer

The function $f(x) = |x - 5|$ is continuous for all $x \in \mathbb{R}$.

Question 5 (i)

Examine the continuity of $f(x) = \frac{x^2 - 25}{x + 5}$.

TYPE 1 Checking Continuity at a Point

Given: The rational function $f(x) = \frac{x^2 - 25}{x + 5}$.

To Find: Determine the domain and examine the continuity of $f(x)$.

Concept / Formula Used: $\frac{P(x)}{Q(x)} \quad Q(x) \neq 0$

Solution:

The denominator $x + 5 \neq 0 \implies x \neq -5$. Thus, the domain of $f(x)$ is $\mathbb{R} \setminus \{-5\}$.

Let c be any arbitrary real number in the domain $(\mathbb{R} \setminus \{-5\})$, so $c \neq -5$.

Value of function at $x = c$:

$$f(c) = \frac{c^2 - 25}{c + 5}$$

Limit at $x = c$:

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \frac{x^2 - 25}{x + 5} \\ &= \frac{c^2 - 25}{c + 5} \\ &= f(c)\end{aligned}$$

Since $\lim_{x \rightarrow c} f(x) = f(c)$ for every c in the domain of f , $f(x)$ is continuous at all points in its domain.

Final Answer

The function $f(x)$ is continuous for all $x \in \mathbb{R} \setminus \{-5\}$.

Question 5 (ii)

Examine the continuity of $f(x) = \frac{1}{x-5}$.

TYPE 1 Checking Continuity at a Point

Given: The function $f(x) = \frac{1}{x-5}$.

To Find: Examine continuity in its domain.

Solution:

The denominator $x - 5 = 0 \implies x = 5$. So $f(x)$ is defined for all $x \in \mathbb{R} \setminus \{5\}$.

For any point $c \in \mathbb{R} \setminus \{5\}$:

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \frac{1}{x-5} \\ &= \frac{1}{c-5} \\ &= f(c)\end{aligned}$$

Since $\lim_{x \rightarrow c} f(x) = f(c)$ for all $c \in \mathbb{R} \setminus \{5\}$, the function is continuous on its domain.

Final Answer

The function $f(x)$ is continuous for all $x \in \mathbb{R} \setminus \{5\}$.

Question 6 (i)

Examine the following function for continuity:

$$f(x) = \begin{cases} \frac{\sin 2x}{x}, & \text{if } x > 0 \\ 2, & \text{if } x \leq 0 \end{cases}$$

TYPE 6 Continuity Using Trigonometric Limits

Given: The piecewise function $f(x) = \begin{cases} \frac{\sin 2x}{x}, & \text{if } x > 0 \\ 2, & \text{if } x \leq 0 \end{cases}$

To Find: Examine the continuity of $f(x)$ at $x = 0$ and over $\mathbb{R}$.

Concept / Formula Used: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

For $x < 0$, $f(x) = 2$ is a constant function, hence continuous. For $x > 0$, $f(x) = \frac{\sin 2x}{x}$ is a quotient of continuous trigonometric and polynomial functions with $x \neq 0$, hence continuous.

At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{x \rightarrow 0^-} (2) \\ &= 2\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{x \rightarrow 0^+} \frac{\sin 2x}{x} \\ &= \lim_{x \rightarrow 0^+} \left(2 \cdot \frac{\sin 2x}{2x} \right)\end{aligned}$$

Using standard result $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ with $\theta = 2x$:

$$\begin{aligned}&= 2 \times 1 \\ &= 2\end{aligned}$$

Value of function at $x = 0$:

$$f(0) = 2$$

Since $\text{LHL} = \text{RHL} = f(0) = 2$, $f(x)$ is continuous at $x = 0$.

Hence, $f(x)$ is continuous for all $x \in \mathbb{R}$.

Final Answer

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

Question 6 (ii)

Examine the following function for continuity:

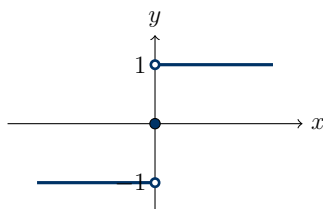
$$f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

TYPE 5 Continuity of Absolute Value Functions

Given: The function $f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

To Find: Examine continuity for all $x \in \mathbb{R}$.

Concept / Formula Used: $|x| = x \quad x > 0 \quad |x| = -x \quad x < 0$



Graph of signum-type function $f(x) = \frac{x}{|x|}$ showing jump discontinuity at $x = 0$

Solution:

Rewrite $f(x)$ for $x < 0$ and $x > 0$:

For $x < 0$, $|x| = -x$, so:

$$f(x) = \frac{x}{-x} = -1$$

For $x > 0$, $|x| = x$, so:

$$f(x) = \frac{x}{x} = 1$$

At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} (-1) \\ &= -1 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} (1) \\ &= 1 \end{aligned}$$

Value of function at $x = 0$:

$$f(0) = 0$$

Since $\text{LHL} \neq \text{RHL}$, $\lim_{x \rightarrow 0} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 0$.

Final Answer

The function $f(x)$ is continuous for all real numbers except at $x = 0$.

Question 7 (i)

Examine the following function for continuity:

$$f(x) = \begin{cases} x^{10} - 1, & \text{if } x \leq 1 \\ x^2, & \text{if } x > 1 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The piecewise function $f(x) = \begin{cases} x^{10} - 1, & \text{if } x \leq 1 \\ x^2, & \text{if } x > 1 \end{cases}$

To Find: Examine the continuity of $f(x)$.

Concept / Formula Used: $f(x)$ $x = 1$ $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$

Solution:

For $x < 1$, $f(x) = x^{10} - 1$ is a polynomial, continuous everywhere on $(-\infty, 1)$. For $x > 1$, $f(x) = x^2$ is a polynomial, continuous everywhere on $(1, \infty)$.

At $x = 1$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 1^-} (x^{10} - 1) \\ &= 1^{10} - 1 \\ &= 1 - 1 \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 1^+} (x^2) \\ &= 1^2 \\ &= 1\end{aligned}$$

Value of function at $x = 1$:

$$\begin{aligned}f(1) &= 1^{10} - 1 \\ &= 0\end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$, $\lim_{x \rightarrow 1} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 1$.

Final Answer

The function $f(x)$ is continuous for all $x \in \mathbb{R} \setminus \{1\}$.

Question 7 (ii)

Examine the continuity of $f(x) = \tan x$ for $0 \leq x \leq \frac{\pi}{4}$.

TYPE 1 Checking Continuity at a Point

Given: The function $f(x) = \tan x$ defined on the closed interval $\left[0, \frac{\pi}{4}\right]$.

To Find: Examine the continuity of $f(x)$ on $\left[0, \frac{\pi}{4}\right]$.

Concept / Formula Used: $\tan x = \frac{\sin x}{\cos x}$ $\cos x \neq 0$

Solution:

We know $\tan x = \frac{\sin x}{\cos x}$. The function $\tan x$ is undefined only when $\cos x = 0$, which occurs at $x = \frac{\pi}{2} + n\pi, n \in \mathbb{Z}$.

On the given closed interval $\left[0, \frac{\pi}{4}\right]$: $\cos x \neq 0$ for all $x \in \left[0, \frac{\pi}{4}\right]$ since $\cos 0 = 1$ and $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$.

For any point $c \in \left[0, \frac{\pi}{4}\right]$:

$$\lim_{x \rightarrow c} \tan x = \tan c = f(c)$$

Therefore, $f(x) = \tan x$ is continuous at every point in the interval $\left[0, \frac{\pi}{4}\right]$.

Final Answer

The function $f(x) = \tan x$ is continuous on the interval $\left[0, \frac{\pi}{4}\right]$.

Question 8 (i)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} 2x + 3, & \text{if } x \leq 2 \\ 2x - 3, & \text{if } x > 2 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} 2x + 3, & \text{if } x \leq 2 \\ 2x - 3, & \text{if } x > 2 \end{cases}$

To Find: Locate all points of discontinuity.

Concept / Formula Used: $x = c$ $\lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x)$ $\lim_{x \rightarrow c} f(x) \neq f(c)$

Solution:

For $x < 2$, $f(x) = 2x + 3$ is continuous as it is a polynomial function. For $x > 2$, $f(x) = 2x - 3$ is continuous as it is a polynomial function.

At $x = 2$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 2^-} (2x + 3) \\ &= 2(2) + 3 \\ &= 4 + 3 \\ &= 7 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 2^+} (2x - 3) \\ &= 2(2) - 3 \\ &= 4 - 3 \\ &= 1\end{aligned}$$

Value of function at $x = 2$:

$$f(2) = 2(2) + 3 = 7$$

Since $\text{LHL} \neq \text{RHL}$, $f(x)$ is not continuous at $x = 2$.

Final Answer

The point of discontinuity is $x = 2$.

Question 8 (ii)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} x^3 - 3, & \text{if } x \leq 2 \\ x^2 + 1, & \text{if } x > 2 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} x^3 - 3, & \text{if } x \leq 2 \\ x^2 + 1, & \text{if } x > 2 \end{cases}$

To Find: Find all points of discontinuity.

Concept / Formula Used: $x = 2$ LHL RHL $f(2)$

Solution:

For $x < 2$, $f(x) = x^3 - 3$ is continuous (polynomial). For $x > 2$, $f(x) = x^2 + 1$ is continuous (polynomial).

At $x = 2$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 2^-} (x^3 - 3) \\ &= 2^3 - 3 \\ &= 8 - 3 \\ &= 5\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 2^+} (x^2 + 1) \\ &= 2^2 + 1 \\ &= 4 + 1 \\ &= 5\end{aligned}$$

Value of function at $x = 2$:

$$\begin{aligned} f(2) &= 2^3 - 3 \\ &= 5 \end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(2) = 5$, $f(x)$ is continuous at $x = 2$.

Thus, $f(x)$ is continuous for all real numbers.

Final Answer

There are no points of discontinuity.

Question 9 (i)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} \frac{x^4 - 16}{x - 2}, & x \neq 2 \\ 16, & x = 2 \end{cases}$$

TYPE 1 Checking Continuity at a Point

Given: The function $f(x) = \begin{cases} \frac{x^4 - 16}{x - 2}, & x \neq 2 \\ 16, & x = 2 \end{cases}$

To Find: Find all points of discontinuity.

Concept / Formula Used: $x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$

Solution:

For $x \neq 2$, $f(x) = \frac{x^4 - 16}{x - 2}$ is continuous as a quotient of continuous polynomials.

At $x = 2$:

Evaluating the limit as $x \rightarrow 2$:

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x^2 - 4)(x^2 + 4)}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)(x^2 + 4)}{x - 2} \end{aligned}$$

Cancelling $(x - 2)$ as $x \neq 2$:

$$\begin{aligned} &= \lim_{x \rightarrow 2} (x + 2)(x^2 + 4) \\ &= (2 + 2)(2^2 + 4) \\ &= (4)(4 + 4) \\ &= 4 \times 8 \\ &= 32 \end{aligned}$$

Value of function at $x = 2$:

$$f(2) = 16$$

Since $\lim_{x \rightarrow 2} f(x) = 32 \neq f(2) = 16$, the function is not continuous at $x = 2$.

Final Answer

The point of discontinuity is $x = 2$.

Question 9 (ii)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} \frac{\sin 3x}{x} + \cos x, & x > 0 \\ 4 - 3x, & x \leq 0 \end{cases}$$

TYPE 6 Continuity Using Trigonometric Limits

Given: The function $f(x) = \begin{cases} \frac{\sin 3x}{x} + \cos x, & x > 0 \\ 4 - 3x, & x \leq 0 \end{cases}$

To Find: Locate points of discontinuity.

Concept / Formula Used: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Solution:

For $x < 0$, $f(x) = 4 - 3x$ is a polynomial function, continuous everywhere. For $x > 0$, $f(x) = \frac{\sin 3x}{x} + \cos x$ is a sum of continuous functions, continuous everywhere.

At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} (4 - 3x) \\ &= 4 - 3(0) \\ &= 4 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} \left(\frac{\sin 3x}{x} + \cos x \right) \\ &= \lim_{x \rightarrow 0^+} \left(3 \cdot \frac{\sin 3x}{3x} + \cos x \right) \end{aligned}$$

Using $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ with $\theta = 3x$:

$$\begin{aligned} &= 3(1) + \cos 0 \\ &= 3 + 1 \\ &= 4 \end{aligned}$$

Value of function at $x = 0$:

$$\begin{aligned} f(0) &= 4 - 3(0) \\ &= 4 \end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(0) = 4$, $f(x)$ is continuous at $x = 0$.

Final Answer

There are no points of discontinuity.

Question 10 (i)

Locate the points of discontinuity (if any) of the function:

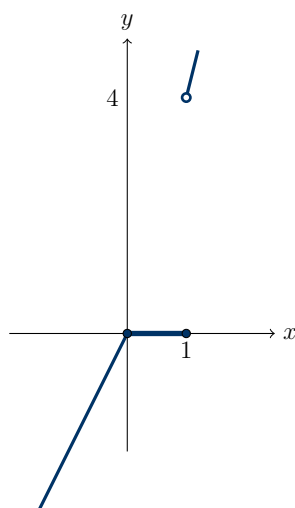
$$f(x) = \begin{cases} 2x, & \text{if } x < 0 \\ 0, & \text{if } 0 \leq x \leq 1 \\ 4x, & \text{if } x > 1 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} 2x, & \text{if } x < 0 \\ 0, & \text{if } 0 \leq x \leq 1 \\ 4x, & \text{if } x > 1 \end{cases}$

To Find: Find all points of discontinuity.

Concept / Formula Used: $x = 0$ $x = 1$



Graph of three-piece function showing continuity at $x = 0$ and jump discontinuity at $x = 1$

Solution:

Check continuity at boundary point $x = 0$:

Left-Hand Limit (LHL at $x = 0$):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} (2x) \\ &= 2(0) \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL at $x = 0$):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} (0) \\ &= 0\end{aligned}$$

Value of function $f(0) = 0$. Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Check continuity at boundary point $x = 1$:

Left-Hand Limit (LHL at $x = 1$):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 1^-} (0) \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL at $x = 1$):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 1^+} (4x) \\ &= 4(1) \\ &= 4\end{aligned}$$

Value of function $f(1) = 0$. Since $\text{LHL} \neq \text{RHL}$ at $x = 1$, $f(x)$ is discontinuous at $x = 1$.

Final Answer

The point of discontinuity is $x = 1$.

Question 10 (ii)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} x + 2, & x \leq 1 \\ x - 2, & 1 < x < 2 \\ 0, & x \geq 2 \end{cases}$$

TYPE 2 Continuity of Piecewise Functions

Given: The function $f(x) = \begin{cases} x + 2, & x \leq 1 \\ x - 2, & 1 < x < 2 \\ 0, & x \geq 2 \end{cases}$

To Find: Find all points of discontinuity.

Concept / Formula Used: $x = 1$ $x = 2$

Solution:

Check continuity at $x = 1$:

Left-Hand Limit (LHL at $x = 1$):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 1^-} (x + 2) \\ &= 1 + 2 \\ &= 3\end{aligned}$$

Right-Hand Limit (RHL at $x = 1$):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 1^+} (x - 2) \\ &= 1 - 2 \\ &= -1\end{aligned}$$

Since $\text{LHL} \neq \text{RHL}$ at $x = 1$, $f(x)$ is discontinuous at $x = 1$.

Check continuity at $x = 2$:

Left-Hand Limit (LHL at $x = 2$):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 2^-} (x - 2) \\ &= 2 - 2 \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL at $x = 2$):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 2^+} (0) \\ &= 0\end{aligned}$$

Value of function $f(2) = 0$. Since $\text{LHL} = \text{RHL} = f(2) = 0$, $f(x)$ is continuous at $x = 2$.

Final Answer

The point of discontinuity is $x = 1$.

Question 11

Find the value of the constant k so that the function f defined by

$$f(x) = \begin{cases} kx^2, & x \leq 2 \\ x - 3, & x > 2 \end{cases}$$

may be continuous.

TYPE 3 Finding Constants for Continuity

Given: The function $f(x) = \begin{cases} kx^2, & x \leq 2 \\ x - 3, & x > 2 \end{cases}$ is continuous everywhere.

To Find: The value of constant k .

Concept / Formula Used: $x = 2$ $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$

Solution:

Since $f(x)$ is given to be continuous, it must be continuous at $x = 2$.

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 2^-} f(x) \\ &= \lim_{x \rightarrow 2^-} (kx^2) \\ &= k(2)^2 \\ &= 4k\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 2^+} f(x) \\ &= \lim_{x \rightarrow 2^+} (x - 3) \\ &= 2 - 3 \\ &= -1\end{aligned}$$

Value of function at $x = 2$:

$$f(2) = k(2)^2 = 4k$$

For $f(x)$ to be continuous at $x = 2$:

$$\begin{aligned}\text{LHL} &= \text{RHL} \\ 4k &= -1 \\ k &= -\frac{1}{4}\end{aligned}$$

Final Answer

$$k = -\frac{1}{4}$$

Question 12

Determine the value of k for which the following function is continuous at $x = 3$:

$$f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

★ Likely Exam Question

TYPE 3 Finding Constants for Continuity

Given: The function $f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$ is continuous at $x = 3$.

To Find: The value of k .

Concept / Formula Used: $f(x)$ $x = 3$ $\lim_{x \rightarrow 3} f(x) = f(3) = k$

Solution:

We evaluate $\lim_{x \rightarrow 3} f(x)$:

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{(x+3)^2 - 36}{x-3}$$

Expanding the numerator $(x+3)^2 - 36$:

$$\begin{aligned}(x+3)^2 - 36 &= x^2 + 6x + 9 - 36 \\ &= x^2 + 6x - 27\end{aligned}$$

Factorizing $x^2 + 6x - 27$:

$$\begin{aligned}x^2 + 6x - 27 &= x^2 + 9x - 3x - 27 \\ &= x(x+9) - 3(x+9) \\ &= (x-3)(x+9)\end{aligned}$$

Substituting back into the limit:

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{(x-3)(x+9)}{x-3}$$

Cancelling $(x-3)$ as $x \neq 3$:

$$\begin{aligned}&= \lim_{x \rightarrow 3} (x+9) \\ &= 3+9 \\ &= 12\end{aligned}$$

Since $f(x)$ is continuous at $x = 3$:

$$\begin{aligned}f(3) &= \lim_{x \rightarrow 3} f(x) \\ k &= 12\end{aligned}$$

Final Answer

$$k = 12$$

Question 13 (i)

For what choice of a and b is the following function continuous:

$$f(x) = \begin{cases} -5, & \text{if } x \leq -1 \\ ax - b, & \text{if } -1 < x < 3 \\ 7, & \text{if } x \geq 3 \end{cases}$$

★ Likely Exam Question

TYPE 3 Finding Constants for Continuity

Given: The function $f(x) = \begin{cases} -5, & \text{if } x \leq -1 \\ ax - b, & \text{if } -1 < x < 3 \\ 7, & \text{if } x \geq 3 \end{cases}$ is continuous everywhere.

To Find: The values of constants a and b .

Concept / Formula Used: $x = -1$ $x = 3$

Solution:

Since $f(x)$ is continuous everywhere, it must be continuous at $x = -1$ and $x = 3$.

Continuity at $x = -1$:

Left-Hand Limit (LHL):

$$\text{LHL} = \lim_{x \rightarrow -1^-} (-5) = -5$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow -1^+} (ax - b) \\ &= a(-1) - b \\ &= -a - b \end{aligned}$$

Value of function $f(-1) = -5$.

Equating LHL and RHL for continuity at $x = -1$:

$$\begin{aligned} -a - b &= -5 \\ a + b &= 5 \quad \text{--- (Equation 1)} \end{aligned}$$

Continuity at $x = 3$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 3^-} (ax - b) \\ &= a(3) - b \\ &= 3a - b \end{aligned}$$

Right-Hand Limit (RHL):

$$\text{RHL} = \lim_{x \rightarrow 3^+} (7) = 7$$

Value of function $f(3) = 7$.

Equating LHL and RHL for continuity at $x = 3$:

$$3a - b = 7 \quad \text{--- (Equation 2)}$$

Solving Equation 1 and Equation 2 simultaneously:

Add Equation 1 and Equation 2:

$$\begin{aligned}(a + b) + (3a - b) &= 5 + 7 \\ 4a &= 12 \\ a &= \frac{12}{4} \\ a &= 3\end{aligned}$$

Substitute $a = 3$ into Equation 1:

$$\begin{aligned}3 + b &= 5 \\ b &= 5 - 3 \\ b &= 2\end{aligned}$$

Final Answer

$$a = 3, \quad b = 2$$

Question 13 (ii)

For what choice of a and b is the following function continuous:

$$f(x) = \begin{cases} x^2, & x \leq 0 \\ ax + b, & x > 0 \end{cases}$$

TYPE 3 Finding Constants for Continuity

Given: The function $f(x) = \begin{cases} x^2, & x \leq 0 \\ ax + b, & x > 0 \end{cases}$ is continuous everywhere.

To Find: The values of a and b .

Concept / Formula Used: $x = 0 \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$

Solution:

Check continuity at $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} (x^2) \\ &= 0^2 \\ &= 0\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow 0^+} (ax + b) \\ &= a(0) + b \\ &= b\end{aligned}$$

Value of function at $x = 0$:

$$f(0) = 0^2 = 0$$

Equating LHL and RHL for continuity at $x = 0$:

$$b = 0$$

Since $ax + b$ becomes ax when $b = 0$, $\lim_{x \rightarrow 0^+} ax = 0$ regardless of the value of a . Hence, a can take any real value ($a \in \mathbb{R}$).

Final Answer

$b = 0$, and a may be any real number ($a \in \mathbb{R}$).

Question 14

Consider $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$

- Determine if there is a removable discontinuity at $x = 3$.
- Classify it as missing point or isolated point.
- Redefine the function to make it continuous.

TYPE 4 Removable Discontinuity Analysis

Given: The function $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$

To Find:

- Check for removable discontinuity at $x = 3$.
- Classify the discontinuity.
- Redefine $f(x)$ for continuity.

Concept / Formula Used: $x = c \quad \lim_{x \rightarrow c} f(x) \quad f(c) \quad f(c)$

Solution:

(i) Check for Removable Discontinuity at $x = 3$:

Evaluate $\lim_{x \rightarrow 3} f(x)$:

$$\begin{aligned} \lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} \end{aligned}$$

Cancelling $(x - 3)$ since $x \neq 3$:

$$\begin{aligned} &= \lim_{x \rightarrow 3} (x + 3) \\ &= 3 + 3 \\ &= 6 \end{aligned}$$

Value of function at $x = 3$:

$$f(3) = 5$$

Since $\lim_{x \rightarrow 3} f(x) = 6$ exists but $\lim_{x \rightarrow 3} f(x) \neq f(3)$, there is a removable discontinuity at $x = 3$.

(ii) Classification:

Since $f(3)$ is defined ($f(3) = 5$) but does not match the limit ($\lim_{x \rightarrow 3} f(x) = 6$), the discontinuity is an **isolated point discontinuity**.

(iii) Redefining the function for continuity:

To make $f(x)$ continuous at $x = 3$, we set $f(3) = \lim_{x \rightarrow 3} f(x) = 6$.

The redefined continuous function is:

$$f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & x \neq 3 \\ 6, & x = 3 \end{cases}$$

Final Answer

(i) Yes, there is a removable discontinuity at $x = 3$.

(ii) Isolated point discontinuity.

(iii) Redefined function: $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & x \neq 3 \\ 6, & x = 3 \end{cases}$

5.3 – Continuity: Algebra and Composition of Continuous Functions

Question 1

Is $\tan x$ a continuous function?

TYPE 9 Algebra of Continuous Functions

Given: $f(x) = \tan x$

Concept / Formula Used: $\frac{g(x)}{h(x)}$ $g(x)$ $h(x)$ $h(x) \neq 0$

Solution:

The given function is:

$$f(x) = \tan x = \frac{\sin x}{\cos x}$$

Let $g(x) = \sin x$ and $h(x) = \cos x$. Both $g(x)$ and $h(x)$ are standard trigonometric functions that are continuous for all real numbers $x \in \mathbb{R}$.

The domain of $f(x) = \tan x$ consists of all real numbers x for which $\cos x \neq 0$:

$$\text{Domain}(f) = \mathbb{R} \setminus \left\{ (2k+1)\frac{\pi}{2} \mid k \in \mathbb{Z} \right\}$$

By the quotient rule of continuous functions, if $g(x)$ and $h(x)$ are continuous at $x = c$, then $\frac{g(x)}{h(x)}$ is continuous at $x = c$ provided $h(c) \neq 0$.

Therefore, $f(x) = \tan x$ is continuous at every point in its domain.

Final Answer

Yes, $\tan x$ is a continuous function in its domain $\mathbb{R} \setminus \{(2k+1)\frac{\pi}{2} \mid k \in \mathbb{Z}\}$.

Question 2

State which of the following functions are continuous?

- (i) $\sin x + \cos x$
- (ii) $\sin x - \cos x$
- (iii) $\sin x \cdot \cos x$

TYPE 9 Algebra of Continuous Functions

Given: The functions (i) $f_1(x) = \sin x + \cos x$, (ii) $f_2(x) = \sin x - \cos x$, and (iii) $f_3(x) = \sin x \cdot \cos x$.

Concept / Formula Used: $g(x) \quad h(x) \quad \mathbb{R} \quad (g+h)(x) = g(x)+h(x) \quad \mathbb{R} \quad (g-h)(x) = g(x) - h(x) \quad \mathbb{R} \quad (g \cdot h)(x) = g(x) \cdot h(x) \quad \mathbb{R}$

Solution:

Let $g(x) = \sin x$ and $h(x) = \cos x$. Both $g(x)$ and $h(x)$ are continuous functions for all $x \in \mathbb{R}$.

Part (i):

$$\begin{aligned} f_1(x) &= \sin x + \cos x \\ &= g(x) + h(x) \end{aligned}$$

Since the sum of two continuous functions is continuous, $f_1(x)$ is continuous for all $x \in \mathbb{R}$.

Part (ii):

$$\begin{aligned} f_2(x) &= \sin x - \cos x \\ &= g(x) - h(x) \end{aligned}$$

Since the difference of two continuous functions is continuous, $f_2(x)$ is continuous for all $x \in \mathbb{R}$.

Part (iii):

$$\begin{aligned} f_3(x) &= \sin x \cdot \cos x \\ &= g(x) \cdot h(x) \end{aligned}$$

Since the product of two continuous functions is continuous, $f_3(x)$ is continuous for all $x \in \mathbb{R}$.

Final Answer

All three functions (i), (ii), and (iii) are continuous for all $x \in \mathbb{R}$.

Question 3

Is the function f defined by $f(x) = 2 - 3x + |x|$ a continuous function?

TYPE 5 Continuity of Absolute Value Functions

Given: $f(x) = 2 - 3x + |x|$

Concept / Formula Used: $\mathbb{R}$ $g(x) = |x|$ $\mathbb{R}$ $\mathbb{R}$

Solution:

Method 1: Using Algebra of Continuous Functions

Let $p(x) = 2 - 3x$ and $q(x) = |x|$.

$$f(x) = p(x) + q(x)$$

- $p(x) = 2 - 3x$ is a polynomial function, so it is continuous for all $x \in \mathbb{R}$. - $q(x) = |x|$ is the absolute value function, which is continuous for all $x \in \mathbb{R}$.

Since $f(x)$ is the sum of two continuous functions, $f(x)$ is continuous for all $x \in \mathbb{R}$.

Method 2: By Definition at $x = 0$

Redefining $f(x)$:

$$f(x) = \begin{cases} 2 - 3x + x = 2 - 2x, & \text{if } x \geq 0 \\ 2 - 3x - x = 2 - 4x, & \text{if } x < 0 \end{cases}$$

At $x = 0$:

$$\text{LHL} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (2 - 4x) = 2 - 4(0) = 2$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (2 - 2x) = 2 - 2(0) = 2$$

$$f(0) = 2 - 2(0) = 2$$

Since $\text{LHL} = \text{RHL} = f(0)$, $f(x)$ is continuous at $x = 0$. For $x > 0$ and $x < 0$, $f(x)$ is a linear polynomial and hence continuous.

Final Answer

Yes, $f(x) = 2 - 3x + |x|$ is a continuous function on $\mathbb{R}$.

Question 4

Is the function f defined by $f(x) = |\sin x|$ a continuous function?

TYPE 10 Continuity of Composite Functions

★ Likely Exam Question

Given: $f(x) = |\sin x|$

Concept / Formula Used: $h(x)$ c $g(x)$ $h(c)$ $(g \circ h)(x) = g(h(x))$ c

Solution:

Let $h(x) = \sin x$ and $g(x) = |x|$.

The composition $(g \circ h)(x)$ is given by:

$$\begin{aligned} (g \circ h)(x) &= g(h(x)) \\ &= g(\sin x) \\ &= |\sin x| \\ &= f(x) \end{aligned}$$

We know that: 1. $h(x) = \sin x$ is continuous for all $x \in \mathbb{R}$. 2. $g(x) = |x|$ is continuous for all $x \in \mathbb{R}$.

By the composite function theorem, since both g and h are continuous on $\mathbb{R}$, their composition $(g \circ h)(x) = |\sin x|$ is also continuous for all $x \in \mathbb{R}$.

Final Answer

Yes, $f(x) = |\sin x|$ is a continuous function on $\mathbb{R}$.

Question 5

Is the function f defined by $f(x) = \cos |x|$ a continuous function?

TYPE 10 Continuity of Composite Functions

★ Likely Exam Question

Given: $f(x) = \cos |x|$

Concept / Formula Used: $h : \mathbb{R} \rightarrow \mathbb{R}$ $g : \mathbb{R} \rightarrow \mathbb{R}$ $(g \circ h)(x) = g(h(x))$ $\mathbb{R}$

Solution:

Let $h(x) = |x|$ and $g(x) = \cos x$.

The composite function $(g \circ h)(x)$ is defined as:

$$\begin{aligned}(g \circ h)(x) &= g(h(x)) \\ &= g(|x|) \\ &= \cos |x| \\ &= f(x)\end{aligned}$$

We check the continuity of individual functions: 1. $h(x) = |x|$ is continuous for all $x \in \mathbb{R}$. 2. $g(x) = \cos x$ is continuous for all $x \in \mathbb{R}$.

Since the composition of two continuous functions is continuous, $f(x) = (g \circ h)(x) = \cos |x|$ is continuous for all $x \in \mathbb{R}$.

Final Answer

Yes, $f(x) = \cos |x|$ is a continuous function on $\mathbb{R}$.

Question 6

Prove that the following functions are continuous:

- (i) $\frac{2x^3 - 7x^2 + 3}{(x-1)(x+3)}$
- (ii) $|\sec x + \tan x|$

TYPE 9 Algebra of Continuous Functions

To Prove: That (i) $f(x) = \frac{2x^3 - 7x^2 + 3}{(x-1)(x+3)}$ and (ii) $g(x) = |\sec x + \tan x|$ are continuous in their respective domains.

Concept / Formula Used: $\frac{P(x)}{Q(x)}$ $Q(x) \neq 0$

Solution:

Part (i): Let $f(x) = \frac{2x^3 - 7x^2 + 3}{(x-1)(x+3)}$.

Let $P(x) = 2x^3 - 7x^2 + 3$ and $Q(x) = (x-1)(x+3) = x^2 + 2x - 3$.

- $P(x)$ is a polynomial function, so it is continuous for all $x \in \mathbb{R}$. - $Q(x)$ is a polynomial function, so it is continuous for all $x \in \mathbb{R}$.

The domain of $f(x)$ is where the denominator is non-zero:

$$\begin{aligned} Q(x) &\neq 0 \\ (x-1)(x+3) &\neq 0 \\ x &\neq 1 \quad \text{and} \quad x \neq -3 \end{aligned}$$

$$\text{Domain}(f) = \mathbb{R} \setminus \{1, -3\}$$

By the quotient property of continuous functions, $f(x) = \frac{P(x)}{Q(x)}$ is continuous at every point in its domain $\mathbb{R} \setminus \{1, -3\}$.

Part (ii): Let $g(x) = |\sec x + \tan x|$.

Let $u(x) = \sec x + \tan x$ and $v(y) = |y|$.

- $u(x) = \sec x + \tan x$ is continuous for all $x \in \mathbb{R} \setminus \{(2k+1)\frac{\pi}{2} \mid k \in \mathbb{Z}\}$. - $v(y) = |y|$ is continuous for all $y \in \mathbb{R}$.

The composite function is:

$$\begin{aligned} (v \circ u)(x) &= v(u(x)) \\ &= v(\sec x + \tan x) \\ &= |\sec x + \tan x| \\ &= g(x) \end{aligned}$$

By the composite function theorem, $g(x)$ is continuous at all points in its domain $\mathbb{R} \setminus \{(2k+1)\frac{\pi}{2} \mid k \in \mathbb{Z}\}$.

Hence Proved

Both functions are continuous in their respective domains, as proved above.

Question 7

Examine the following functions for continuity:

- (i) $\cos(x^2)$
- (ii) $\sin(3x^2 - 5)$
- (iii) $\tan^{-1}(2x^2 + 3)$

TYPE 10 Continuity of Composite Functions

★ Likely Exam Question

Given: The functions (i) $f_1(x) = \cos(x^2)$, (ii) $f_2(x) = \sin(3x^2 - 5)$, and (iii) $f_3(x) = \tan^{-1}(2x^2 + 3)$.

Concept / Formula Used: $h : \mathbb{R} \rightarrow \mathbb{R} \quad \mathbb{R} \quad g : \mathbb{R} \rightarrow \mathbb{R} \quad \mathbb{R} \quad (g \circ h)(x) = g(h(x)) \quad \mathbb{R}$

Solution:

Part (i): $f_1(x) = \cos(x^2)$ Let $h_1(x) = x^2$ and $g_1(x) = \cos x$.

$$(g_1 \circ h_1)(x) = g_1(h_1(x)) = \cos(x^2) = f_1(x)$$

- $h_1(x) = x^2$ is a polynomial function, hence continuous on $\mathbb{R}$. - $g_1(x) = \cos x$ is a trigonometric function, hence continuous on $\mathbb{R}$.

By the composite function theorem, $f_1(x) = \cos(x^2)$ is continuous for all $x \in \mathbb{R}$.

Part (ii): $f_2(x) = \sin(3x^2 - 5)$ Let $h_2(x) = 3x^2 - 5$ and $g_2(x) = \sin x$.

$$(g_2 \circ h_2)(x) = g_2(h_2(x)) = \sin(3x^2 - 5) = f_2(x)$$

- $h_2(x) = 3x^2 - 5$ is a polynomial function, hence continuous on $\mathbb{R}$. - $g_2(x) = \sin x$ is a trigonometric function, hence continuous on $\mathbb{R}$.

By the composite function theorem, $f_2(x) = \sin(3x^2 - 5)$ is continuous for all $x \in \mathbb{R}$.

Part (iii): $f_3(x) = \tan^{-1}(2x^2 + 3)$ Let $h_3(x) = 2x^2 + 3$ and $g_3(x) = \tan^{-1} x$.

$$(g_3 \circ h_3)(x) = g_3(h_3(x)) = \tan^{-1}(2x^2 + 3) = f_3(x)$$

- $h_3(x) = 2x^2 + 3$ is a polynomial function, hence continuous on $\mathbb{R}$. - $g_3(x) = \tan^{-1} x$ is an inverse trigonometric function, continuous for all $x \in \mathbb{R}$.

By the composite function theorem, $f_3(x) = \tan^{-1}(2x^2 + 3)$ is continuous for all $x \in \mathbb{R}$.

Final Answer

- (i) $f(x) = \cos(x^2)$ is continuous for all $x \in \mathbb{R}$.
- (ii) $f(x) = \sin(3x^2 - 5)$ is continuous for all $x \in \mathbb{R}$.
- (iii) $f(x) = \tan^{-1}(2x^2 + 3)$ is continuous for all $x \in \mathbb{R}$.

5.4 – Differentiability: Definition and Relation to Continuity

Question 1

If a function is derivable at a point, is it necessary that it must be continuous at that point?

TYPE 11 Differentiability and Continuity Relationship

Given: A function $f(x)$ is derivable at a point $x = c$.

To Find: Whether $f(x)$ must be continuous at $x = c$.

Concept / Formula Used: $f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$

Solution:

Let $f(x)$ be derivable at $x = c$. By definition, the limit

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists finitely. Now, consider the difference $f(x) - f(c)$ for $x \neq c$:

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c)$$

Taking the limit as $x \rightarrow c$ on both sides:

$$\begin{aligned} \lim_{x \rightarrow c} [f(x) - f(c)] &= \lim_{x \rightarrow c} \left[\frac{f(x) - f(c)}{x - c} \cdot (x - c) \right] \\ &= \left(\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right) \cdot \left(\lim_{x \rightarrow c} (x - c) \right) \\ &= f'(c) \cdot 0 \\ &= 0 \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow c} f(x) - f(c) = 0 \implies \lim_{x \rightarrow c} f(x) = f(c)$$

This is the precise condition for $f(x)$ to be continuous at $x = c$.

Final Answer

Yes, if a function is derivable at a point, it is necessary that it must be continuous at that point.

Question 2

If a function is continuous at a point, is it necessary that it must be derivable at that point?

TYPE 11 Differentiability and Continuity Relationship

Given: A function $f(x)$ is continuous at a point $x = c$.

To Find: Whether $f(x)$ must be derivable at $x = c$.

Solution:

Consider the absolute value function $f(x) = |x|$ at the point $x = 0$.

1. Checking continuity at $x = 0$:

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} (-x) = 0 \\ \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} x = 0 \\ f(0) &= |0| = 0 \end{aligned}$$

Since $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

2. Checking derivability at $x = 0$:

$$\begin{aligned} \text{LHD} = Lf'(0) &= \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{|h| - 0}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1 \\ \text{RHD} = Rf'(0) &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{|h| - 0}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \end{aligned}$$

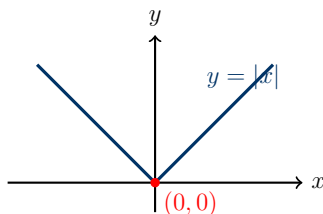
Since $Lf'(0) \neq Rf'(0)$, $f(x)$ is not derivable at $x = 0$.

Final Answer

No, it is not necessary. A function may be continuous at a point without being derivable at that point (e.g., $f(x) = |x|$ at $x = 0$).

Question 3

Is the function $f(x) = |x|$ derivable at $x = 0$?

TYPE 12 Differentiability of Absolute Value Functions

Graph of $f(x) = |x|$ showing a sharp corner at $x = 0$

Given: Function $f(x) = |x|$ at $x = 0$.

To Find: Whether $f(x)$ is derivable at $x = 0$.

Concept / Formula Used: $Lf'(c) = \lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h}$ $Rf'(c) = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h}$

Solution:

To test derivability at $x = 0$, we calculate the Left-Hand Derivative (LHD) and Right-Hand Derivative (RHD):

1. Left-Hand Derivative at $x = 0$:

$$\begin{aligned} Lf'(0) &= \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{|0+h| - |0|}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-h}{h} \quad (\text{since } h < 0 \implies |h| = -h) \\ &= -1 \end{aligned}$$

2. Right-Hand Derivative at $x = 0$:

$$\begin{aligned} Rf'(0) &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{|0+h| - |0|}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{h}{h} \quad (\text{since } h > 0 \implies |h| = h) \\ &= 1 \end{aligned}$$

Since $Lf'(0) = -1 \neq 1 = Rf'(0)$, the limit does not exist.

Final Answer

No, the function $f(x) = |x|$ is not derivable at $x = 0$.

Question 4

Is the function $f(x) = \cot x$ derivable at $x = 0$?

TYPE 13 Differentiability of Trigonometric Functions

Given: Function $f(x) = \cot x$ at $x = 0$.

To Find: Whether $f(x)$ is derivable at $x = 0$.

Solution:

We examine the definition and continuity of $f(x) = \cot x$ at $x = 0$:

$$\cot x = \frac{\cos x}{\sin x}$$

At $x = 0$:

$$\sin 0 = 0 \implies \cot 0 \text{ is undefined.}$$

Since $f(x) = \cot x$ is not defined at $x = 0$, it cannot be continuous at $x = 0$. Consequently, it cannot be derivable at $x = 0$.

Final Answer

No, $f(x) = \cot x$ is not derivable at $x = 0$ because it is undefined at $x = 0$.

Question 5

Is the function $f(x) = \sec x$ derivable at $x = \frac{\pi}{2}$?

TYPE 13 Differentiability of Trigonometric Functions

Given: Function $f(x) = \sec x$ at $x = \frac{\pi}{2}$.

To Find: Whether $f(x)$ is derivable at $x = \frac{\pi}{2}$.

Solution:

We evaluate $f(x) = \sec x$ at $x = \frac{\pi}{2}$:

$$\sec x = \frac{1}{\cos x}$$

At $x = \frac{\pi}{2}$:

$$\cos\left(\frac{\pi}{2}\right) = 0 \implies \sec\left(\frac{\pi}{2}\right) \text{ is undefined.}$$

Since $f(x) = \sec x$ is not defined at $x = \frac{\pi}{2}$, it is not continuous, and therefore not derivable at $x = \frac{\pi}{2}$.

Final Answer

No, $f(x) = \sec x$ is not derivable at $x = \frac{\pi}{2}$ because it is undefined at $x = \frac{\pi}{2}$.

Question 6

When a function is called derivable?

TYPE 14 Definition of Differentiability

Given: A general real-valued function $f(x)$.

To Find: The definition of when a function is derivable at a point and on an interval.

Concept / Formula Used: $f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$

Solution:

1. **Derivability at a point:** A function $f(x)$ is called derivable (or differentiable) at a point $x = c$ in its domain if the limit

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} \quad \text{or equivalently} \quad \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists finitely. This requires that both the Left-Hand Derivative ($Lf'(c)$) and Right-Hand Derivative ($Rf'(c)$) exist and are equal:

$$Lf'(c) = Rf'(c) < \infty$$

2. **Derivability on an interval:** A function is called derivable on an open interval (a, b) if it is derivable at every point in (a, b) .

Final Answer

A function $f(x)$ is called derivable at $x = c$ if $\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$ exists finitely (i.e., LHD = RHD). It is derivable on an interval if it is derivable at every point of that interval.

Question 7

If a function is derivable at $x = c$, then write the value of $\lim_{x \rightarrow c} f(x)$.

TYPE 11 Differentiability and Continuity Relationship

Given: Function $f(x)$ is derivable at $x = c$.

To Find: The value of $\lim_{x \rightarrow c} f(x)$.

Solution:

Since $f(x)$ is derivable at $x = c$, it is necessarily continuous at $x = c$.

By definition of continuity at $x = c$:

$$\lim_{x \rightarrow c} f(x) = f(c)$$

Final Answer

$$\lim_{x \rightarrow c} f(x) = f(c)$$

Question 8

Which of the following functions are derivable?

- (i) $\sin x$
- (ii) $\cos x$
- (iii) $\tan x$
- (iv) $\cot x$
- (v) $\sec x$
- (vi) $\csc x$

TYPE 13 Differentiability of Trigonometric Functions

Given: Trigonometric functions: $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$.

To Find: The derivability domain for each function.

Solution:

We evaluate each function over its domain:

- $\sin x$ **and** $\cos x$: Defined and continuous for all $x \in \mathbb{R}$. Their derivatives $\cos x$ and $-\sin x$ exist everywhere on $\mathbb{R}$.
- $\tan x$ **and** $\sec x$: Defined for all $x \in \mathbb{R} \setminus \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$. They are derivable at all points in their domain.
- $\cot x$ **and** $\csc x$: Defined for all $x \in \mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$. They are derivable at all points in their domain.

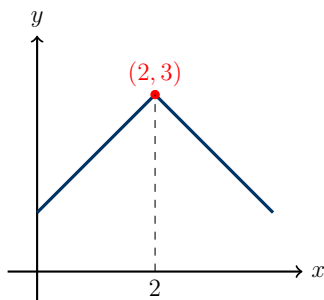
Final Answer

- (i) $\sin x$: Derivable for all $x \in \mathbb{R}$.
- (ii) $\cos x$: Derivable for all $x \in \mathbb{R}$.
- (iii) $\tan x$: Derivable for all $x \in \mathbb{R} \setminus \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$.
- (iv) $\cot x$: Derivable for all $x \in \mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$.
- (v) $\sec x$: Derivable for all $x \in \mathbb{R} \setminus \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$.
- (vi) $\csc x$: Derivable for all $x \in \mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$.

Question 9

Examine the function $f(x) = \begin{cases} 1+x, & \text{if } x \leq 2 \\ 5-x, & \text{if } x > 2 \end{cases}$ for differentiability at $x = 2$.

TYPE 15 Differentiability of Piecewise Functions



Graph of $f(x)$ showing a non-differentiable corner at $x = 2$

Given: Function $f(x) = \begin{cases} 1 + x, & x \leq 2 \\ 5 - x, & x > 2 \end{cases}$ at $x = 2$.

To Find: Differentiability of $f(x)$ at $x = 2$.

Concept / Formula Used: $Lf'(c) = \lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h}$ $Rf'(c) = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h}$

Solution:

First, compute $f(2)$:

$$f(2) = 1 + 2 = 3$$

1. Left-Hand Derivative at $x = 2$:

$$Lf'(2) = \lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h}$$

For $h < 0$, $2 + h < 2$, so $f(2 + h) = 1 + (2 + h) = 3 + h$:

$$\begin{aligned} Lf'(2) &= \lim_{h \rightarrow 0^-} \frac{(3 + h) - 3}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h}{h} \\ &= 1 \end{aligned}$$

2. Right-Hand Derivative at $x = 2$:

$$Rf'(2) = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$$

For $h > 0$, $2 + h > 2$, so $f(2 + h) = 5 - (2 + h) = 3 - h$:

$$\begin{aligned} Rf'(2) &= \lim_{h \rightarrow 0^+} \frac{(3 - h) - 3}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{-h}{h} \\ &= -1 \end{aligned}$$

Since $Lf'(2) = 1 \neq -1 = Rf'(2)$, $f(x)$ is not differentiable at $x = 2$.

Final Answer

The function $f(x)$ is not differentiable at $x = 2$.

Question 10

If $f(2) = 4$ and $f'(2) = 4$, then evaluate $\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2}$.

TYPE 16 Differentiability Using Limits

★ Likely Exam Question

Given: $f(2) = 4$ and $f'(2) = 4$.

To Find: $\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2}$

Concept / Formula Used: $f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$

Solution:

$$\text{Let } L = \lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2}.$$

Add and subtract $2f(2)$ in the numerator:

$$\begin{aligned} L &= \lim_{x \rightarrow 2} \frac{xf(2) - 2f(2) + 2f(2) - 2f(x)}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x - 2)f(2) - 2[f(x) - f(2)]}{x - 2} \end{aligned}$$

Split the fraction into two separate terms:

$$\begin{aligned} L &= \lim_{x \rightarrow 2} \left[\frac{(x - 2)f(2)}{x - 2} - 2 \cdot \frac{f(x) - f(2)}{x - 2} \right] \\ &= \lim_{x \rightarrow 2} f(2) - 2 \cdot \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} \end{aligned}$$

Using the given definition $f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$:

$$L = f(2) - 2f'(2)$$

Substitute the given values $f(2) = 4$ and $f'(2) = 4$:

$$\begin{aligned} L &= 4 - 2(4) \\ &= 4 - 8 \\ &= -4 \end{aligned}$$

Final Answer

$$\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2} = -4$$

Question 11

Examine the following function for continuity at $x = 1$ and differentiability at $x = 2$:

$$f(x) = \begin{cases} 5x - 4, & 0 < x < 1 \\ 4x^2 - 3x, & 1 \leq x < 2 \\ 3x + 4, & x \geq 2 \end{cases}$$

TYPE 15 Differentiability of Piecewise Functions

Given: Function $f(x) = \begin{cases} 5x - 4, & 0 < x < 1 \\ 4x^2 - 3x, & 1 \leq x < 2 \\ 3x + 4, & x \geq 2 \end{cases}$

To Find: Continuity at $x = 1$ and differentiability at $x = 2$.

Concept / Formula Used: $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$ $Lf'(c) = Rf'(c)$

Solution:

Part 1: Continuity at $x = 1$

1. Left-Hand Limit (LHL) at $x = 1$:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (5x - 4) = 5(1) - 4 = 1$$

2. Right-Hand Limit (RHL) at $x = 1$:

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (4x^2 - 3x) = 4(1)^2 - 3(1) = 1$$

3. Value of function at $x = 1$:

$$f(1) = 4(1)^2 - 3(1) = 1$$

Since $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1) = 1$, the function $f(x)$ is continuous at $x = 1$.

Part 2: Differentiability at $x = 2$

Value of function at $x = 2$:

$$f(2) = 3(2) + 4 = 10$$

1. Left-Hand Derivative at $x = 2$:

$$Lf'(2) = \lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h}$$

For $h < 0$, $2 + h < 2$, so $f(2 + h) = 4(2 + h)^2 - 3(2 + h)$:

$$\begin{aligned} Lf'(2) &= \lim_{h \rightarrow 0^-} \frac{4(4 + 4h + h^2) - 3(2 + h) - 10}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{16 + 16h + 4h^2 - 6 - 3h - 10}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{13h + 4h^2}{h} \\ &= \lim_{h \rightarrow 0^-} (13 + 4h) \\ &= 13 \end{aligned}$$

2. Right-Hand Derivative at $x = 2$:

$$Rf'(2) = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$$

For $h > 0$, $2 + h > 2$, so $f(2 + h) = 3(2 + h) + 4$:

$$\begin{aligned} Rf'(2) &= \lim_{h \rightarrow 0^+} \frac{[3(2+h) + 4] - 10}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{6 + 3h + 4 - 10}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{3h}{h} \\ &= 3 \end{aligned}$$

Since $Lf'(2) = 13 \neq 3 = Rf'(2)$, $f(x)$ is not differentiable at $x = 2$.

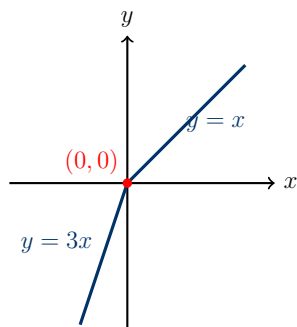
Final Answer

$f(x)$ is continuous at $x = 1$, but not differentiable at $x = 2$.

Question 12

Show that the function $f(x) = 2x - |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$.

TYPE 12 Differentiability of Absolute Value Functions



Graph of $f(x) = 2x - |x|$ with a sharp peak/corner at $x = 0$

To Prove: $f(x) = 2x - |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$.

Concept / Formula Used: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

Solution:

Redefining $f(x)$ explicitly without absolute value:

$$f(x) = \begin{cases} 2x - x = x, & x \geq 0 \\ 2x - (-x) = 3x, & x < 0 \end{cases}$$

Step 1: Proof of Continuity at $x = 0$

$$\begin{aligned}\text{LHL} &= \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (3x) = 3(0) = 0 \\ \text{RHL} &= \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x) = 0 \\ f(0) &= 2(0) - |0| = 0\end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Step 2: Proof of Non-Differentiability at $x = 0$

$$\begin{aligned}Lf'(0) &= \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{3h - 0}{h} = 3 \\ Rf'(0) &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{h - 0}{h} = 1\end{aligned}$$

Since $Lf'(0) = 3 \neq 1 = Rf'(0)$, $f(x)$ is not differentiable at $x = 0$.

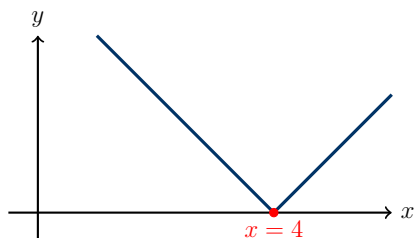
Hence Proved

LHL = RHL = $f(0)$ proves continuity at $x = 0$; LHD $\neq$ RHD proves non-differentiability at $x = 0$.

Question 13

Show that $f(x) = |x - 4|$ is continuous but not differentiable at $x = 4$.

TYPE 12 Differentiability of Absolute Value Functions

★ Likely Exam Question

Graph of $f(x) = |x - 4|$ with a corner at $x = 4$

To Prove: $f(x) = |x - 4|$ is continuous but not differentiable at $x = 4$.

Concept / Formula Used: $|x - 4| = \begin{cases} x - 4, & x \geq 4 \\ -(x - 4) = 4 - x, & x < 4 \end{cases}$

Solution:

Redefining $f(x)$:

$$f(x) = \begin{cases} 4 - x, & x < 4 \\ x - 4, & x \geq 4 \end{cases}$$

Step 1: Proof of Continuity at $x = 4$

$$\text{LHL} = \lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} (4 - x) = 4 - 4 = 0$$

$$\text{RHL} = \lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} (x - 4) = 4 - 4 = 0$$

$$f(4) = |4 - 4| = 0$$

Since $\text{LHL} = \text{RHL} = f(4) = 0$, $f(x)$ is continuous at $x = 4$.

Step 2: Proof of Non-Differentiability at $x = 4$

$$Lf'(4) = \lim_{h \rightarrow 0^-} \frac{f(4+h) - f(4)}{h} = \lim_{h \rightarrow 0^-} \frac{|4+h-4| - 0}{h} = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

$$Rf'(4) = \lim_{h \rightarrow 0^+} \frac{f(4+h) - f(4)}{h} = \lim_{h \rightarrow 0^+} \frac{|4+h-4| - 0}{h} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

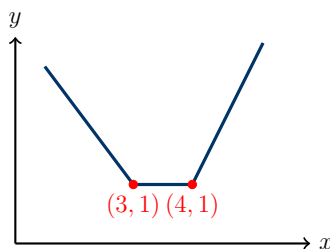
Since $Lf'(4) = -1 \neq 1 = Rf'(4)$, $f(x)$ is not differentiable at $x = 4$.

Hence Proved

$f(x) = |x - 4|$ is continuous at $x = 4$ but not differentiable at $x = 4$.

Question 14

If the function $f(x) = |x - 3| + |x - 4|$, then show that f is not differentiable at $x = 3$ and $x = 4$.

TYPE 12 Differentiability of Absolute Value Functions

Graph of $f(x) = |x - 3| + |x - 4|$ showing corners at $x = 3$ and $x = 4$

To Prove: $f(x) = |x - 3| + |x - 4|$ is not differentiable at $x = 3$ and $x = 4$.

Concept / Formula Used: $x = 3$ $x = 4$

Solution:

Expressing $f(x)$ explicitly in intervals:

$$f(x) = \begin{cases} -(x - 3) - (x - 4) = 7 - 2x, & x < 3 \\ (x - 3) - (x - 4) = 1, & 3 \leq x < 4 \\ (x - 3) + (x - 4) = 2x - 7, & x \geq 4 \end{cases}$$

Part 1: Non-Differentiability at $x = 3$

Calculate $f(3) = |3 - 3| + |3 - 4| = 0 + 1 = 1$.

$$Lf'(3) = \lim_{h \rightarrow 0^-} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0^-} \frac{[7 - 2(3+h)] - 1}{h} = \lim_{h \rightarrow 0^-} \frac{1 - 2h - 1}{h} = -2$$

$$Rf'(3) = \lim_{h \rightarrow 0^+} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0^+} \frac{1 - 1}{h} = 0$$

Since $Lf'(3) = -2 \neq 0 = Rf'(3)$, $f(x)$ is not differentiable at $x = 3$.

Part 2: Non-Differentiability at $x = 4$

Calculate $f(4) = |4 - 3| + |4 - 4| = 1 + 0 = 1$.

$$Lf'(4) = \lim_{h \rightarrow 0^-} \frac{f(4+h) - f(4)}{h} = \lim_{h \rightarrow 0^-} \frac{1-1}{h} = 0$$

$$Rf'(4) = \lim_{h \rightarrow 0^+} \frac{f(4+h) - f(4)}{h} = \lim_{h \rightarrow 0^+} \frac{[2(4+h) - 7] - 1}{h} = \lim_{h \rightarrow 0^+} \frac{1+2h-1}{h} = 2$$

Since $Lf'(4) = 0 \neq 2 = Rf'(4)$, $f(x)$ is not differentiable at $x = 4$.

Hence Proved

$f(x)$ is not differentiable at $x = 3$ and $x = 4$.

Question 15

Show that the function f is continuous at $x = 1$ for all values of a where

$$f(x) = \begin{cases} ax^2 + 1, & x \geq 1 \\ x + a, & x < 1 \end{cases}$$

Find its right and left hand derivatives at $x = 1$. Hence, find the condition for the existence of the derivative at $x = 1$.

TYPE 17 Finding Constants for Differentiability

Given: Function $f(x) = \begin{cases} ax^2 + 1, & x \geq 1 \\ x + a, & x < 1 \end{cases}$

To Find: (i) Proof of continuity at $x = 1$ for all a , (ii) LHD and RHD at $x = 1$, (iii) Condition for derivative to exist at $x = 1$.

Concept / Formula Used: $Lf'(1) = Rf'(1)$

Solution:

Step 1: Proof of Continuity at $x = 1$ for all a

$$\text{LHL} = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x + a) = 1 + a$$

$$\text{RHL} = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (ax^2 + 1) = a(1)^2 + 1 = a + 1$$

$$f(1) = a(1)^2 + 1 = a + 1$$

Since $\text{LHL} = \text{RHL} = f(1) = a + 1$ for every real number a , $f(x)$ is continuous at $x = 1$ for all values of a .

Step 2: Finding Left and Right Hand Derivatives at $x = 1$

1. Left-Hand Derivative ($Lf'(1)$):

$$\begin{aligned} Lf'(1) &= \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{[(1+h) + a] - (a+1)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h}{h} \\ &= 1 \end{aligned}$$

2. Right-Hand Derivative ($Rf'(1)$):

$$\begin{aligned} Rf'(1) &= \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{[a(1+h)^2 + 1] - (a+1)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{a(1+2h+h^2) + 1 - a - 1}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{2ah + ah^2}{h} \\ &= \lim_{h \rightarrow 0^+} (2a + ah) \\ &= 2a \end{aligned}$$

Step 3: Condition for existence of $f'(1)$

The derivative $f'(1)$ exists if and only if $Lf'(1) = Rf'(1)$:

$$1 = 2a \implies a = \frac{1}{2}$$

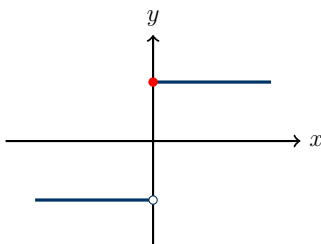
Final Answer

$Lf'(1) = 1$, $Rf'(1) = 2a$. The derivative $f'(1)$ exists if and only if $a = \frac{1}{2}$.

Question 16

Prove that the function $f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ is not differentiable at $x = 0$.

TYPE 11 Differentiability and Continuity Relationship



Graph of the signum function $f(x)$ showing jump discontinuity at $x = 0$

To Prove: $f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ is not differentiable at $x = 0$.

Solution:

Let us test continuity of $f(x)$ at $x = 0$.

Simplifying $f(x)$ for non-zero x :

$$f(x) = \begin{cases} \frac{x}{x} = 1, & x > 0 \\ \frac{x}{-x} = -1, & x < 0 \\ 1, & x = 0 \end{cases}$$

1. Left-Hand Limit at $x = 0$:

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-1) = -1$$

2. Right-Hand Limit at $x = 0$:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (1) = 1$$

Since $\lim_{x \rightarrow 0^-} f(x) = -1 \neq 1 = \lim_{x \rightarrow 0^+} f(x)$, the limit $\lim_{x \rightarrow 0} f(x)$ does not exist.

Hence, $f(x)$ is discontinuous at $x = 0$.

Since continuity is a necessary condition for differentiability, a discontinuous function cannot be differentiable.

Hence Proved

Since $f(x)$ is discontinuous at $x = 0$, it is not differentiable at $x = 0$.

Question 17

If $f(x) = \begin{cases} x^2, & \text{if } x \geq 1 \\ x, & \text{if } x < 1 \end{cases}$, then show that f is not differentiable at $x = 1$.

TYPE 15 Differentiability of Piecewise Functions

To Prove: $f(x) = \begin{cases} x^2, & x \geq 1 \\ x, & x < 1 \end{cases}$ is not differentiable at $x = 1$.

Concept / Formula Used: $Lf'(c) = Rf'(c)$

Solution:

Compute $f(1) = 1^2 = 1$.

1. Left-Hand Derivative at $x = 1$:

$$\begin{aligned} Lf'(1) &= \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{(1+h) - 1}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h}{h} \\ &= 1 \end{aligned}$$

2. Right-Hand Derivative at $x = 1$:

$$\begin{aligned}
 Rf'(1) &= \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{(1+h)^2 - 1}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{1 + 2h + h^2 - 1}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{2h + h^2}{h} \\
 &= \lim_{h \rightarrow 0^+} (2 + h) \\
 &= 2
 \end{aligned}$$

Since $Lf'(1) = 1 \neq 2 = Rf'(1)$, $f(x)$ is not differentiable at $x = 1$.

Hence Proved

$Lf'(1) = 1 \neq 2 = Rf'(1)$, hence f is not differentiable at $x = 1$.

Question 18

If $f(x) = \begin{cases} ax + b, & 0 < x \leq 1 \\ 2x^2 - x, & 1 < x < 2 \end{cases}$ is a differentiable function in $(0, 2)$, then find the values of a and b .

TYPE 17 Finding Constants for Differentiability

★ Likely Exam Question

Given: $f(x) = \begin{cases} ax + b, & 0 < x \leq 1 \\ 2x^2 - x, & 1 < x < 2 \end{cases}$ is differentiable in $(0, 2)$.

To Find: The values of a and b .

Concept / Formula Used: $x = c$ LHL = RHL = $f(c)$ LHD = RHD $x = c$

Solution:

Since $f(x)$ is differentiable in $(0, 2)$, it must be continuous and differentiable at $x = 1$.

Step 1: Applying Continuity at $x = 1$

$$\begin{aligned}
 \text{LHL} &= \lim_{x \rightarrow 1^-} (ax + b) = a(1) + b = a + b \\
 \text{RHL} &= \lim_{x \rightarrow 1^+} (2x^2 - x) = 2(1)^2 - 1 = 1 \\
 f(1) &= a(1) + b = a + b
 \end{aligned}$$

Equating LHL and RHL:

$$a + b = 1 \quad (15)$$

Step 2: Applying Differentiability at $x = 1$

$$\begin{aligned}
 Lf'(1) &= \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^-} \frac{[a(1+h) + b] - (a+b)}{h} = \lim_{h \rightarrow 0^-} \frac{ah}{h} = a \\
 Rf'(1) &= \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{[2(1+h)^2 - (1+h)] - 1}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{2(1+2h+h^2) - 1 - h - 1}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{2 + 4h + 2h^2 - 2 - h}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{3h + 2h^2}{h} = 3
 \end{aligned}$$

Equating $Lf'(1)$ and $Rf'(1)$:

$$a = 3 \quad (16)$$

Step 3: Solving for b

Substitute $a = 3$ from Equation (16) into Equation (15):

$$\begin{aligned}
 3 + b &= 1 \\
 b &= 1 - 3 \\
 b &= -2
 \end{aligned}$$

Final Answer

$$a = 3, \quad b = -2$$

Question 19

If the function $f(x) = \begin{cases} -\ln(4-x), & x \leq 3 \\ p - qx, & x > 3 \end{cases}$ is differentiable at $x = 3$, find the values of p and q .

TYPE 17 Finding Constants for Differentiability

Given: Function $f(x) = \begin{cases} -\ln(4-x), & x \leq 3 \\ p - qx, & x > 3 \end{cases}$ is differentiable at $x = 3$.

To Find: The values of p and q .

Concept / Formula Used: $x = c \implies \quad \text{LHL} = \text{RHL} \quad \text{LHD} = \text{RHD}$

Solution:

Since $f(x)$ is differentiable at $x = 3$, it must be continuous and differentiable at $x = 3$.

Step 1: Applying Continuity at $x = 3$

$$\begin{aligned}
 \text{LHL} &= \lim_{x \rightarrow 3^-} [-\ln(4-x)] = -\ln(4-3) = -\ln(1) = 0 \\
 \text{RHL} &= \lim_{x \rightarrow 3^+} (p - qx) = p - 3q \\
 f(3) &= -\ln(4-3) = 0
 \end{aligned}$$

Equating LHL and RHL:

$$p - 3q = 0 \implies p = 3q \quad (17)$$

Step 2: Applying Differentiability at $x = 3$

$$Lf'(3) = \lim_{x \rightarrow 3^-} \frac{d}{dx} [-\ln(4-x)] = -\frac{1}{4-x} \cdot (-1) \Big|_{x=3} = \frac{1}{4-3} = 1$$

$$Rf'(3) = \lim_{x \rightarrow 3^+} \frac{d}{dx} (p - qx) = -q$$

Equating $Lf'(3)$ and $Rf'(3)$:

$$-q = 1 \implies q = -1 \quad (18)$$

Step 3: Solving for p

Substitute $q = -1$ into Equation (17):

$$p = 3(-1)$$

$$p = -3$$

Final Answer

$$p = -3, \quad q = -1$$

Question 20 (i)

Examine for continuity and differentiability:

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x < 0 \\ 0, & x \geq 0 \end{cases} \quad \text{at } x = 0$$

TYPE 18 Squeeze Theorem for Differentiability

Given: Function $f(x) = \begin{cases} x \sin \frac{1}{x}, & x < 0 \\ 0, & x \geq 0 \end{cases}$ at $x = 0$.

To Find: Continuity and differentiability at $x = 0$.

Concept / Formula Used: $-|x| \leq x \sin \frac{1}{x} \leq |x|$

Solution:

Part 1: Continuity at $x = 0$

$$\text{LHL} = \lim_{x \rightarrow 0^-} x \sin \frac{1}{x}$$

Since $-1 \leq \sin \frac{1}{x} \leq 1$ for all $x \neq 0$:

$$\lim_{x \rightarrow 0^-} \left| x \sin \frac{1}{x} \right| \leq \lim_{x \rightarrow 0^-} |x| \cdot 1 = 0 \implies \text{LHL} = 0$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} 0 = 0$$

$$f(0) = 0$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Part 2: Differentiability at $x = 0$

$$\begin{aligned} Lf'(0) &= \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h \sin \frac{1}{h} - 0}{h} \\ &= \lim_{h \rightarrow 0^-} \sin \frac{1}{h} \end{aligned}$$

As $h \rightarrow 0^-$, $\sin \frac{1}{h}$ oscillates endlessly between -1 and 1 , so this limit does not exist.

Final Answer

$f(x)$ is continuous at $x = 0$, but not differentiable at $x = 0$.

Question 20 (ii)

Examine for continuity and differentiability:

$$f(x) = \begin{cases} |x| \sin \frac{1}{x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad \text{at } x = 0$$

TYPE 18 Squeeze Theorem for Differentiability

Given: Function $f(x) = \begin{cases} |x| \sin \frac{1}{x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$ at $x = 0$.

To Find: Continuity and differentiability at $x = 0$.

Concept / Formula Used: $x > 0 \quad |x| = x$

Solution:

For $x > 0$, $f(x) = x \sin \frac{1}{x}$.

Part 1: Continuity at $x = 0$

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} x \sin \frac{1}{x} = 0 \quad (\text{by Squeeze Theorem, since } -x \leq x \sin \frac{1}{x} \leq x) \\ \text{LHL} &= \lim_{x \rightarrow 0^-} 0 = 0 \\ f(0) &= 0 \end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Part 2: Differentiability at $x = 0$

$$\begin{aligned} Rf'(0) &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{h \sin \frac{1}{h} - 0}{h} \\ &= \lim_{h \rightarrow 0^+} \sin \frac{1}{h} \end{aligned}$$

As $h \rightarrow 0^+$, $\sin \frac{1}{h}$ oscillates infinitely between -1 and 1 , so $Rf'(0)$ does not exist.

Final Answer

$f(x)$ is continuous at $x = 0$, but not differentiable at $x = 0$.

Question 20 (iii)

Examine for continuity and differentiability:

$$f(x) = \begin{cases} (x - c)^2 \cos \frac{1}{x - c}, & x \neq c \\ 0, & x = c \end{cases} \quad \text{at } x = c$$

TYPE 18 Squeeze Theorem for Differentiability

★ Likely Exam Question

Given: Function $f(x) = \begin{cases} (x - c)^2 \cos \frac{1}{x - c}, & x \neq c \\ 0, & x = c \end{cases}$ at $x = c$.

To Find: Continuity and differentiability at $x = c$.

Concept / Formula Used: $(x - c)^2 \cos \frac{1}{x - c}$

Solution:

Part 1: Continuity at $x = c$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x - c)^2 \cos \frac{1}{x - c}$$

Since $-1 \leq \cos \frac{1}{x - c} \leq 1$:

$$-(x - c)^2 \leq (x - c)^2 \cos \frac{1}{x - c} \leq (x - c)^2$$

Taking limits as $x \rightarrow c$:

$$\lim_{x \rightarrow c} [-(x - c)^2] = 0 \quad \text{and} \quad \lim_{x \rightarrow c} (x - c)^2 = 0$$

By the Squeeze Theorem:

$$\lim_{x \rightarrow c} f(x) = 0 = f(c)$$

Hence, $f(x)$ is continuous at $x = c$.

Part 2: Differentiability at $x = c$

$$\begin{aligned} f'(c) &= \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(c + h - c)^2 \cos \frac{1}{c + h - c} - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 \cos \frac{1}{h}}{h} \\ &= \lim_{h \rightarrow 0} h \cos \frac{1}{h} \end{aligned}$$

Using Squeeze Theorem again:

$$-|h| \leq h \cos \frac{1}{h} \leq |h| \implies \lim_{h \rightarrow 0} h \cos \frac{1}{h} = 0$$

Since $f'(c) = 0$ exists finitely, $f(x)$ is differentiable at $x = c$.

Final Answer

$f(x)$ is both continuous and differentiable at $x = c$, with $f'(c) = 0$.

Question 21

Find the values of a and b so that the following function is differentiable for all values of x :

$$f(x) = \begin{cases} ax + b, & x > -1 \\ bx^2 - 3, & x \leq -1 \end{cases}$$

TYPE 17 Finding Constants for Differentiability

★ Likely Exam Question

Given: Function $f(x) = \begin{cases} ax + b, & x > -1 \\ bx^2 - 3, & x \leq -1 \end{cases}$ is differentiable for all $x \in \mathbb{R}$.

To Find: The values of a and b .

Concept / Formula Used: $x = -1 \implies \quad x = -1 \quad Lf'(-1) = Rf'(-1)$

Solution:

The function is differentiable for all x , so it must be continuous and differentiable at the transition point $x = -1$.

Step 1: Applying Continuity at $x = -1$

$$\text{LHL} = \lim_{x \rightarrow -1^-} (bx^2 - 3) = b(-1)^2 - 3 = b - 3$$

$$\text{RHL} = \lim_{x \rightarrow -1^+} (ax + b) = a(-1) + b = -a + b$$

$$f(-1) = b(-1)^2 - 3 = b - 3$$

Equating LHL and RHL:

$$b - 3 = -a + b$$

$$-3 = -a$$

$$a = 3$$

Step 2: Applying Differentiability at $x = -1$

$$Lf'(-1) = \left. \frac{d}{dx} (bx^2 - 3) \right|_{x=-1} = 2bx|_{x=-1} = -2b$$

$$Rf'(-1) = \left. \frac{d}{dx} (ax + b) \right|_{x=-1} = a$$

Equating $Lf'(-1)$ and $Rf'(-1)$:

$$-2b = a$$

Substitute $a = 3$:

$$\begin{aligned} -2b &= 3 \\ b &= -\frac{3}{2} \end{aligned}$$

Final Answer

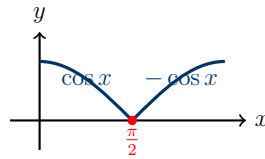
$$a = 3, \quad b = -\frac{3}{2}$$

Question 22

Check the differentiability of $f(x) = |\cos x|$ at $x = \frac{\pi}{2}$.

TYPE 12 Differentiability of Absolute Value Functions

★ Likely Exam Question



Graph of $f(x) = |\cos x|$ showing sharp corner at $x = \frac{\pi}{2}$

Given: Function $f(x) = |\cos x|$ at $x = \frac{\pi}{2}$.

To Find: Differentiability of $f(x)$ at $x = \frac{\pi}{2}$.

Concept / Formula Used: $\cos x > 0$ for $x < \frac{\pi}{2}$ and $\cos x < 0$ for $x > \frac{\pi}{2}$.

Solution:

Redefine $f(x)$ in the neighborhood of $x = \frac{\pi}{2}$:

$$f(x) = \begin{cases} \cos x, & \text{for } x \leq \frac{\pi}{2} \\ -\cos x, & \text{for } x > \frac{\pi}{2} \end{cases}$$

Value at point $x = \frac{\pi}{2}$: $f\left(\frac{\pi}{2}\right) = \left|\cos \frac{\pi}{2}\right| = 0$.

1. Left-Hand Derivative at $x = \frac{\pi}{2}$:

$$\begin{aligned} Lf' \left(\frac{\pi}{2} \right) &= \lim_{h \rightarrow 0^-} \frac{f \left(\frac{\pi}{2} + h \right) - f \left(\frac{\pi}{2} \right)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{\cos \left(\frac{\pi}{2} + h \right) - 0}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-\sin h}{h} \\ &= -1 \end{aligned}$$

2. Right-Hand Derivative at $x = \frac{\pi}{2}$:

$$\begin{aligned} Rf' \left(\frac{\pi}{2} \right) &= \lim_{h \rightarrow 0^+} \frac{f \left(\frac{\pi}{2} + h \right) - f \left(\frac{\pi}{2} \right)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{-\cos \left(\frac{\pi}{2} + h \right) - 0}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{-(-\sin h)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{\sin h}{h} \\ &= 1 \end{aligned}$$

Since $Lf' \left(\frac{\pi}{2} \right) = -1 \neq 1 = Rf' \left(\frac{\pi}{2} \right)$, $f(x)$ is not differentiable at $x = \frac{\pi}{2}$.

Final Answer

$f(x) = |\cos x|$ is not differentiable at $x = \frac{\pi}{2}$.

5.5 – Derivatives of Standard and Absolute Value Functions

Question 1 (i)

If $f(x) = \sqrt{4 - x^2}$, $x \in (-2, 2)$, find $f'(x)$.

TYPE 19 Chain Rule for Differentiation

Given: $f(x) = \sqrt{4 - x^2}$ for $x \in (-2, 2)$

To Find: $f'(x)$

Concept / Formula Used: $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}} \frac{du}{dx}$

Solution:

Expressing the square root as a fractional power:

$$f(x) = (4 - x^2)^{1/2}$$

Differentiating with respect to x using the chain rule:

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} [(4 - x^2)^{1/2}] \\
 &= \frac{1}{2}(4 - x^2)^{1/2-1} \cdot \frac{d}{dx}(4 - x^2) \\
 &= \frac{1}{2}(4 - x^2)^{-1/2} \cdot (0 - 2x) \\
 &= \frac{-2x}{2\sqrt{4 - x^2}} \\
 &= -\frac{x}{\sqrt{4 - x^2}}
 \end{aligned}$$

Final Answer

$$f'(x) = -\frac{x}{\sqrt{4-x^2}}$$

Question 1 (ii)

If $f(x) = \sqrt{1 - x^2}$, $x \in (0, 1)$, find $f'(x)$.

TYPE 19 Chain Rule for Differentiation

Given: $f(x) = \sqrt{1 - x^2}$ for $x \in (0, 1)$

To Find: $f'(x)$

Concept / Formula Used: $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}} \frac{du}{dx}$

Solution:

Expressing $f(x)$ as a power function:

$$f(x) = (1 - x^2)^{1/2}$$

Differentiating with respect to x using the chain rule:

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} [(1 - x^2)^{1/2}] \\
 &= \frac{1}{2}(1 - x^2)^{-1/2} \cdot \frac{d}{dx}(1 - x^2) \\
 &= \frac{1}{2\sqrt{1 - x^2}} \cdot (-2x) \\
 &= -\frac{x}{\sqrt{1 - x^2}}
 \end{aligned}$$

Final Answer

$$f'(x) = -\frac{x}{\sqrt{1-x^2}}$$

Question 2 (i)

Differentiate $\sqrt{2x - 1}$ w.r.t. x .

TYPE 19 Chain Rule for Differentiation

Given: $y = \sqrt{2x - 1}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}} \frac{du}{dx}$

Solution:

Let $y = (2x - 1)^{1/2}$. Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2}(2x - 1)^{-1/2} \cdot \frac{d}{dx}(2x - 1) \\ &= \frac{1}{2\sqrt{2x - 1}} \cdot (2) \\ &= \frac{1}{\sqrt{2x - 1}}\end{aligned}$$

Final Answer

$$\frac{1}{\sqrt{2x-1}}$$

Question 2 (ii)

Differentiate $(3x^2 - 9x + 5)^9$ w.r.t. x .

TYPE 19 Chain Rule for Differentiation

Given: $y = (3x^2 - 9x + 5)^9$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$

Solution:

Differentiating with respect to x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= 9(3x^2 - 9x + 5)^8 \cdot \frac{d}{dx}(3x^2 - 9x + 5) \\ &= 9(3x^2 - 9x + 5)^8 \cdot (6x - 9) \\ &= 9 \cdot 3(2x - 3)(3x^2 - 9x + 5)^8 \\ &= 27(2x - 3)(3x^2 - 9x + 5)^8\end{aligned}$$

Final Answer

$$27(2x - 3)(3x^2 - 9x + 5)^8$$

Question 3 (i)

Differentiate $\sqrt{3x + 2} + \frac{1}{\sqrt{2x^2 + 4}}$ w.r.t. x .

TYPE 20 Sum and Product Rules

Given: $y = \sqrt{3x+2} + \frac{1}{\sqrt{2x^2+4}}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx}(u+v) = \frac{du}{dx} + \frac{dv}{dx}$

Solution:

Rewriting y using fractional exponents:

$$y = (3x+2)^{1/2} + (2x^2+4)^{-1/2}$$

Differentiating term by term with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} [(3x+2)^{1/2}] + \frac{d}{dx} [(2x^2+4)^{-1/2}] \\ &= \frac{1}{2}(3x+2)^{-1/2} \cdot \frac{d}{dx}(3x+2) + \left(-\frac{1}{2}\right)(2x^2+4)^{-3/2} \cdot \frac{d}{dx}(2x^2+4) \\ &= \frac{1}{2\sqrt{3x+2}} \cdot (3) - \frac{1}{2(2x^2+4)^{3/2}} \cdot (4x) \\ &= \frac{3}{2\sqrt{3x+2}} - \frac{2x}{(2x^2+4)^{3/2}}\end{aligned}$$

Final Answer

$$\frac{3}{2\sqrt{3x+2}} - \frac{2x}{(2x^2+4)^{3/2}}$$

Question 3 (ii)

Differentiate $\sin^3 x + \cos^6 x$ w.r.t. x .

TYPE 20 Sum and Product Rules

Given: $y = \sin^3 x + \cos^6 x$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1}f'(x)$

Solution:

Differentiating term by term using the power rule for trigonometric functions:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(\sin^3 x) + \frac{d}{dx}(\cos^6 x) \\ &= 3\sin^2 x \cdot \frac{d}{dx}(\sin x) + 6\cos^5 x \cdot \frac{d}{dx}(\cos x) \\ &= 3\sin^2 x \cos x + 6\cos^5 x(-\sin x) \\ &= 3\sin^2 x \cos x - 6\sin x \cos^5 x \\ &= 3\sin x \cos x(\sin x - 2\cos^4 x)\end{aligned}$$

Final Answer

$$3\sin x \cos x(\sin x - 2\cos^4 x)$$

Question 4 (i)Differentiate $\sin(x^2)$ w.r.t. x .**TYPE 21** Chain Rule for Trigonometric Functions**Given:** $y = \sin(x^2)$ **To Find:** $\frac{dy}{dx}$ **Concept / Formula Used:** $\frac{d}{dx} \sin(u) = \cos(u) \frac{du}{dx}$ **Solution:**Differentiating with respect to x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \cos(x^2) \cdot \frac{d}{dx}(x^2) \\ &= \cos(x^2) \cdot (2x) \\ &= 2x \cos(x^2)\end{aligned}$$

Final Answer

$$2x \cos(x^2)$$

Question 4 (ii)Differentiate $\cos \sqrt{x}$ w.r.t. x .**TYPE 21** Chain Rule for Trigonometric Functions**Given:** $y = \cos \sqrt{x}$ **To Find:** $\frac{dy}{dx}$ **Concept / Formula Used:** $\frac{d}{dx} \cos(u) = -\sin(u) \frac{du}{dx}$ **Solution:**Differentiating with respect to x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= -\sin \sqrt{x} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= -\sin \sqrt{x} \cdot \left(\frac{1}{2\sqrt{x}}\right) \\ &= -\frac{\sin \sqrt{x}}{2\sqrt{x}}\end{aligned}$$

Final Answer

$$-\frac{\sin \sqrt{x}}{2\sqrt{x}}$$

Question 5 (i)Differentiate $\sin(ax + b)$ w.r.t. x .**TYPE 19** Chain Rule for Differentiation**Given:** $y = \sin(ax + b)$ **To Find:** $\frac{dy}{dx}$ **Concept / Formula Used:** $\frac{d}{dx} \sin(u) = \cos(u) \frac{du}{dx}$ **Solution:**Differentiating with respect to x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \cos(ax + b) \cdot \frac{d}{dx}(ax + b) \\ &= \cos(ax + b) \cdot a \\ &= a \cos(ax + b)\end{aligned}$$

Final Answer

$$a \cos(ax + b)$$

Question 5 (ii)Differentiate $\tan(2x + 3)$ w.r.t. x .**TYPE 19** Chain Rule for Differentiation**Given:** $y = \tan(2x + 3)$ **To Find:** $\frac{dy}{dx}$ **Concept / Formula Used:** $\frac{d}{dx} \tan(u) = \sec^2(u) \frac{du}{dx}$ **Solution:**Differentiating with respect to x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \sec^2(2x + 3) \cdot \frac{d}{dx}(2x + 3) \\ &= \sec^2(2x + 3) \cdot 2 \\ &= 2 \sec^2(2x + 3)\end{aligned}$$

Final Answer

$$2 \sec^2(2x + 3)$$

Question 6 (i)Differentiate $\cos(\sin x)$ w.r.t. x .**TYPE 21** Chain Rule for Trigonometric Functions

Given: $y = \cos(\sin x)$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx} \cos(u) = -\sin(u) \frac{du}{dx}$

Solution:

Differentiating with respect to x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= -\sin(\sin x) \cdot \frac{d}{dx}(\sin x) \\ &= -\sin(\sin x) \cdot \cos x \\ &= -\cos x \sin(\sin x)\end{aligned}$$

Final Answer

$$-\cos x \sin(\sin x)$$

Question 6 (ii)

Differentiate $\sin(\cos(x^2))$ w.r.t. x .

TYPE 22 Repeated Chain Rule

Given: $y = \sin(\cos(x^2))$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx} \sin(u) = \cos(u) \frac{du}{dx}$

Solution:

Applying chain rule repeatedly:

$$\begin{aligned}\frac{dy}{dx} &= \cos(\cos(x^2)) \cdot \frac{d}{dx}(\cos(x^2)) \\ &= \cos(\cos(x^2)) \cdot \left[-\sin(x^2) \cdot \frac{d}{dx}(x^2) \right] \\ &= \cos(\cos(x^2)) \cdot [-\sin(x^2) \cdot (2x)] \\ &= -2x \sin(x^2) \cos(\cos(x^2))\end{aligned}$$

Final Answer

$$-2x \sin(x^2) \cos(\cos(x^2))$$

Question 7 (i)

Differentiate $\cos(\tan \sqrt{x+1})$ w.r.t. x .

TYPE 22 Repeated Chain Rule

Given: $y = \cos(\tan \sqrt{x+1})$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx} \cos(u) = -\sin(u) \frac{du}{dx}$, $\frac{d}{dx} \tan(v) = \sec^2(v) \frac{dv}{dx}$

Solution:

Applying chain rule step-by-step:

$$\begin{aligned}\frac{dy}{dx} &= -\sin(\tan \sqrt{x+1}) \cdot \frac{d}{dx}(\tan \sqrt{x+1}) \\ &= -\sin(\tan \sqrt{x+1}) \cdot \sec^2(\sqrt{x+1}) \cdot \frac{d}{dx}(\sqrt{x+1}) \\ &= -\sin(\tan \sqrt{x+1}) \cdot \sec^2(\sqrt{x+1}) \cdot \frac{1}{2\sqrt{x+1}} \\ &= -\frac{\sec^2(\sqrt{x+1}) \sin(\tan \sqrt{x+1})}{2\sqrt{x+1}}\end{aligned}$$

Final Answer

$$-\frac{\sec^2(\sqrt{x+1}) \sin(\tan \sqrt{x+1})}{2\sqrt{x+1}}$$

Question 7 (ii)

Differentiate $\sqrt{\tan \sqrt{x}}$ w.r.t. x .

TYPE 22 Repeated Chain Rule

Given: $y = \sqrt{\tan \sqrt{x}}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx} \sqrt{u} = \frac{1}{2\sqrt{u}} \frac{du}{dx}$, $\frac{d}{dx} \tan(v) = \sec^2(v) \frac{dv}{dx}$

Solution:

Applying chain rule step-by-step:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{\tan \sqrt{x}}} \cdot \frac{d}{dx}(\tan \sqrt{x}) \\ &= \frac{1}{2\sqrt{\tan \sqrt{x}}} \cdot \sec^2 \sqrt{x} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= \frac{1}{2\sqrt{\tan \sqrt{x}}} \cdot \sec^2 \sqrt{x} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{\sec^2 \sqrt{x}}{4\sqrt{x} \sqrt{\tan \sqrt{x}}}\end{aligned}$$

Final Answer

$$\frac{\sec^2 \sqrt{x}}{4\sqrt{x} \sqrt{\tan \sqrt{x}}}$$

Question 8 (i)

If $f(x) = x - 1$, find $\frac{d}{dx}((f \circ g)(x))$.

TYPE 23 Derivative of Composite Functions

Given: $f(x) = x - 1$

To Find: $\frac{d}{dx}((f \circ g)(x))$

Concept / Formula Used: $(f \circ g)(x) = f(g(x))$

Solution:

Evaluating the composite function $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= g(x) - 1\end{aligned}$$

Differentiating with respect to x :

$$\begin{aligned}\frac{d}{dx}((f \circ g)(x)) &= \frac{d}{dx}(g(x) - 1) \\ &= g'(x) - 0 \\ &= g'(x)\end{aligned}$$

Final Answer

$$g'(x)$$

Question 8 (ii)

If $f(x) = x + 7$ and $g(x) = x - 7$, find $\frac{d}{dx}((f \circ f)(x))$.

TYPE 23 Derivative of Composite Functions

Given: $f(x) = x + 7$, $g(x) = x - 7$

To Find: $\frac{d}{dx}((f \circ f)(x))$

Concept / Formula Used: $(f \circ f)(x) = f(f(x))$

Solution:

Evaluating the composite function $(f \circ f)(x)$:

$$\begin{aligned}(f \circ f)(x) &= f(f(x)) \\ &= f(x + 7) \\ &= (x + 7) + 7 \\ &= x + 14\end{aligned}$$

Differentiating with respect to x :

$$\begin{aligned}\frac{d}{dx}((f \circ f)(x)) &= \frac{d}{dx}(x + 14) \\ &= 1 + 0 \\ &= 1\end{aligned}$$

Final Answer

1

Question 9

If $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \frac{x^{98}}{98} + \cdots + x + 1$, show that $f'(1) = 100 f'(0)$.

★ Likely Exam Question

TYPE 24 Term-by-Term Polynomial Differentiation

Given: $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \frac{x^{98}}{98} + \cdots + x + 1$

To Prove: $f'(1) = 100 f'(0)$

Concept / Formula Used: $\frac{d}{dx}(x^n) = nx^{n-1}$

Solution:

Differentiating $f(x)$ term by term with respect to x :

$$\begin{aligned}f'(x) &= \frac{d}{dx} \left(\frac{x^{100}}{100} + \frac{x^{99}}{99} + \cdots + x + 1 \right) \\ &= \frac{100x^{99}}{100} + \frac{99x^{98}}{99} + \frac{98x^{97}}{98} + \cdots + 1 + 0 \\ &= x^{99} + x^{98} + x^{97} + \cdots + x + 1\end{aligned}$$

Evaluating $f'(1)$ by substituting $x = 1$:

$$\begin{aligned}f'(1) &= 1^{99} + 1^{98} + 1^{97} + \cdots + 1 + 1 \\ &= \underbrace{1 + 1 + 1 + \cdots + 1}_{100 \text{ terms}} \\ &= 100\end{aligned}$$

Evaluating $f'(0)$ by substituting $x = 0$:

$$\begin{aligned}f'(0) &= 0^{99} + 0^{98} + \cdots + 0 + 1 \\ &= 1\end{aligned}$$

Evaluating $100 f'(0)$:

$$\begin{aligned}100 f'(0) &= 100 \times 1 \\ &= 100\end{aligned}$$

Since $f'(1) = 100$ and $100 f'(0) = 100$, we have $f'(1) = 100 f'(0)$.

Hence Proved

$f'(1) = 100 f'(0)$, hence proved.

Question 10

Find the derivative of $|2x^2 - 3|$ with respect to x .

TYPE 25 Derivative of Absolute Value Functions

Given: $y = |2x^2 - 3|$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx}|u| = \frac{u}{|u|} \frac{du}{dx} \quad u \neq 0$

Solution:

Using the property $|u| = \sqrt{u^2}$, we express y as:

$$y = \sqrt{(2x^2 - 3)^2}$$

Differentiating with respect to x using chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2\sqrt{(2x^2 - 3)^2}} \cdot \frac{d}{dx} [(2x^2 - 3)^2] \\ &= \frac{1}{2|2x^2 - 3|} \cdot 2(2x^2 - 3) \cdot \frac{d}{dx}(2x^2 - 3) \\ &= \frac{2x^2 - 3}{|2x^2 - 3|} \cdot (4x) \\ &= \frac{4x(2x^2 - 3)}{|2x^2 - 3|}, \quad \text{for } x \neq \pm\sqrt{\frac{3}{2}} \end{aligned}$$

Final Answer

$$\frac{4x(2x^2 - 3)}{|2x^2 - 3|}, \quad x \neq \pm\sqrt{\frac{3}{2}}$$

Question 11

If $f(x) = |\cos x|$, find $f'\left(\frac{3\pi}{4}\right)$.

★ Likely Exam Question

TYPE 26 Derivative of Absolute Value Functions at a Point

Given: $f(x) = |\cos x|$, point $x = \frac{3\pi}{4}$

To Find: $f'\left(\frac{3\pi}{4}\right)$

Concept / Formula Used: $\cos x < 0 \quad |\cos x| = -\cos x$

Solution:

Evaluating $\cos x$ at $x = \frac{3\pi}{4}$:

$$\cos\left(\frac{3\pi}{4}\right) = -\frac{1}{\sqrt{2}} < 0$$

Since $\cos x < 0$ in the second quadrant, in a neighborhood around $x = \frac{3\pi}{4}$:

$$f(x) = -\cos x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(-\cos x) \\ &= -(-\sin x) \\ &= \sin x \end{aligned}$$

Substituting $x = \frac{3\pi}{4}$:

$$\begin{aligned} f'\left(\frac{3\pi}{4}\right) &= \sin\left(\frac{3\pi}{4}\right) \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

Final Answer

$$\frac{1}{\sqrt{2}}$$

Question 12

The table below shows some of the values of differentiable functions u and v and their derivatives.

x	$u(x)$	$u'(x)$	$v(x)$	$v'(x)$
1	0	2	-1	5
2	4	5	3	2
3	2	-1	3	0

If $r(x) = u(v(x))$, find $r'(2)$.

★ Likely Exam Question

TYPE 27 Chain Rule with Tabular Values

Given: Values of $u(x)$, $u'(x)$, $v(x)$, $v'(x)$ in table; $r(x) = u(v(x))$

To Find: $r'(2)$

Concept / Formula Used: $r'(x) = u'(v(x)) \cdot v'(x)$

Solution:

Applying the chain rule to $r(x) = u(v(x))$:

$$r'(x) = u'(v(x)) \cdot v'(x)$$

Substituting $x = 2$:

$$r'(2) = u'(v(2)) \cdot v'(2)$$

Reading values from the table at $x = 2$:

$$\begin{aligned} v(2) &= 3 \\ v'(2) &= 2 \end{aligned}$$

Substituting $v(2) = 3$ into the expression:

$$r'(2) = u'(3) \cdot 2$$

Reading $u'(3)$ from the table at $x = 3$:

$$u'(3) = -1$$

Substituting $u'(3) = -1$:

$$\begin{aligned} r'(2) &= (-1) \cdot 2 \\ &= -2 \end{aligned}$$

Final Answer

-2

Question 13 (i)

Differentiate $\sqrt{\frac{a^2-x^2}{a^2+x^2}}$ w.r.t. x .

TYPE 28 Quotient Rule for Differentiation

Given: $y = \sqrt{\frac{a^2-x^2}{a^2+x^2}}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$

Solution:

Expressing y with fractional exponent:

$$y = \left(\frac{a^2 - x^2}{a^2 + x^2} \right)^{1/2}$$

Differentiating using chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} \left(\frac{a^2 - x^2}{a^2 + x^2} \right)^{-1/2} \cdot \frac{d}{dx} \left(\frac{a^2 - x^2}{a^2 + x^2} \right) \\ &= \frac{1}{2} \sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{(a^2 + x^2) \frac{d}{dx}(a^2 - x^2) - (a^2 - x^2) \frac{d}{dx}(a^2 + x^2)}{(a^2 + x^2)^2} \\ &= \frac{1}{2} \sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{(a^2 + x^2)(-2x) - (a^2 - x^2)(2x)}{(a^2 + x^2)^2} \\ &= \frac{1}{2} \sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{-2a^2x - 2x^3 - 2a^2x + 2x^3}{(a^2 + x^2)^2} \\ &= \frac{1}{2} \sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{-4a^2x}{(a^2 + x^2)^2} \\ &= \frac{-2a^2x}{\sqrt{a^2 - x^2}(a^2 + x^2)^{3/2}} \end{aligned}$$

Final Answer

$$-\frac{2a^2x}{\sqrt{a^2-x^2}(a^2+x^2)^{3/2}}$$

Question 13 (ii)Differentiate $\sin \sqrt{x} + \cos^2 \sqrt{x}$ w.r.t. x .**TYPE 20** Sum and Product Rules**Given:** $y = \sin \sqrt{x} + \cos^2 \sqrt{x}$ **To Find:** $\frac{dy}{dx}$ **Concept / Formula Used:** $\frac{d}{dx} \sin(u) = \cos(u) \frac{du}{dx}$, $\frac{d}{dx}(u^2) = 2u \frac{du}{dx}$ **Solution:**Differentiating term by term with respect to x :

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx}(\sin \sqrt{x}) + \frac{d}{dx}(\cos^2 \sqrt{x}) \\
 &= \cos \sqrt{x} \cdot \frac{d}{dx}(\sqrt{x}) + 2 \cos \sqrt{x} \cdot \frac{d}{dx}(\cos \sqrt{x}) \\
 &= \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}} + 2 \cos \sqrt{x} \cdot \left(-\sin \sqrt{x} \cdot \frac{1}{2\sqrt{x}}\right) \\
 &= \frac{\cos \sqrt{x}}{2\sqrt{x}} - \frac{2 \sin \sqrt{x} \cos \sqrt{x}}{2\sqrt{x}} \\
 &= \frac{\cos \sqrt{x}(1 - 2 \sin \sqrt{x})}{2\sqrt{x}}
 \end{aligned}$$

Final Answer

$$\frac{\cos \sqrt{x}(1 - 2 \sin \sqrt{x})}{2\sqrt{x}}$$

Question 14 (i)Differentiate $\frac{\sin(ax+b)}{\cos(cx+d)}$ w.r.t. x .**TYPE 28** Quotient Rule for Differentiation**Given:** $y = \frac{\sin(ax+b)}{\cos(cx+d)}$ **To Find:** $\frac{dy}{dx}$ **Concept / Formula Used:** $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$ **Solution:**Let $u = \sin(ax + b)$ and $v = \cos(cx + d)$. Differentiating u and v :

$$\begin{aligned}
 u' &= a \cos(ax + b) \\
 v' &= -c \sin(cx + d)
 \end{aligned}$$

Applying the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\cos(cx+d) \cdot [a \cos(ax+b)] - \sin(ax+b) \cdot [-c \sin(cx+d)]}{\cos^2(cx+d)} \\ &= \frac{a \cos(ax+b) \cos(cx+d) + c \sin(ax+b) \sin(cx+d)}{\cos^2(cx+d)} \\ &= \frac{a \cos(ax+b) \cos(cx+d)}{\cos^2(cx+d)} + \frac{c \sin(ax+b) \sin(cx+d)}{\cos^2(cx+d)} \\ &= a \cos(ax+b) \sec(cx+d) + c \sin(ax+b) \tan(cx+d) \sec(cx+d)\end{aligned}$$

Final Answer

$$a \cos(ax+b) \sec(cx+d) + c \sin(ax+b) \tan(cx+d) \sec(cx+d)$$

Question 14 (ii)

Differentiate $\frac{\sin x + x^2}{\cot 2x}$ w.r.t. x .

TYPE 28 Quotient Rule for Differentiation

Given: $y = \frac{\sin x + x^2}{\cot 2x}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{1}{\cot 2x} = \tan 2x$, $\frac{d}{dx}(uv) = u'v + uv'$

Solution:

Rewriting y using trigonometric reciprocal identity $\frac{1}{\cot 2x} = \tan 2x$:

$$y = (\sin x + x^2) \tan 2x$$

Applying the product rule with $u = \sin x + x^2$ and $v = \tan 2x$:

$$\begin{aligned}u' &= \cos x + 2x \\ v' &= 2 \sec^2 2x\end{aligned}$$

Differentiating:

$$\begin{aligned}\frac{dy}{dx} &= u'v + uv' \\ &= (\cos x + 2x) \tan 2x + (\sin x + x^2)(2 \sec^2 2x) \\ &= (\cos x + 2x) \tan 2x + 2(\sin x + x^2) \sec^2 2x\end{aligned}$$

Final Answer

$$(\cos x + 2x) \tan 2x + 2(\sin x + x^2) \sec^2 2x$$

Question 15 (i)

Differentiate $\sin^m x \cos^n x$ w.r.t. x .

TYPE 20 Sum and Product Rules

Given: $y = \sin^m x \cos^n x$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx}(uv) = u'v + uv'$

Solution:

Let $u = \sin^m x$ and $v = \cos^n x$. Differentiating u and v using chain rule:

$$u' = m \sin^{m-1} x \cos x$$

$$v' = n \cos^{n-1} x (-\sin x) = -n \cos^{n-1} x \sin x$$

Applying the product rule:

$$\begin{aligned} \frac{dy}{dx} &= (m \sin^{m-1} x \cos x)(\cos^n x) + (\sin^m x)(-n \cos^{n-1} x \sin x) \\ &= m \sin^{m-1} x \cos^{n+1} x - n \sin^{m+1} x \cos^{n-1} x \\ &= \sin^{m-1} x \cos^{n-1} x (m \cos^2 x - n \sin^2 x) \end{aligned}$$

Final Answer

$$\sin^{m-1} x \cos^{n-1} x (m \cos^2 x - n \sin^2 x)$$

Question 15 (ii)

Differentiate $\sin^n(ax^2 + bx + c)$ w.r.t. x .

TYPE 22 Repeated Chain Rule

Given: $y = \sin^n(ax^2 + bx + c)$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{d}{dx}[f(u)]^n = n[f(u)]^{n-1} f'(u) \frac{du}{dx}$

Solution:

Differentiating using repeated chain rule applications:

$$\begin{aligned} \frac{dy}{dx} &= n \sin^{n-1}(ax^2 + bx + c) \cdot \frac{d}{dx}[\sin(ax^2 + bx + c)] \\ &= n \sin^{n-1}(ax^2 + bx + c) \cdot \cos(ax^2 + bx + c) \cdot \frac{d}{dx}(ax^2 + bx + c) \\ &= n \sin^{n-1}(ax^2 + bx + c) \cdot \cos(ax^2 + bx + c) \cdot (2ax + b) \\ &= n(2ax + b) \sin^{n-1}(ax^2 + bx + c) \cos(ax^2 + bx + c) \end{aligned}$$

Final Answer

$$n(2ax + b) \sin^{n-1}(ax^2 + bx + c) \cos(ax^2 + bx + c)$$

Question 16 (i)

Differentiate $\frac{\sqrt{a+x}-\sqrt{a-x}}{\sqrt{a+x}+\sqrt{a-x}}$ w.r.t. x .

TYPE 29 Rationalization before Differentiation

Given: $y = \frac{\sqrt{a+x}-\sqrt{a-x}}{\sqrt{a+x}+\sqrt{a-x}}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $(\sqrt{a+x} - \sqrt{a-x})$

Solution:

Rationalizing the denominator:

$$\begin{aligned} y &= \frac{(\sqrt{a+x} - \sqrt{a-x})^2}{(\sqrt{a+x} + \sqrt{a-x})(\sqrt{a+x} - \sqrt{a-x})} \\ &= \frac{(a+x) + (a-x) - 2\sqrt{(a+x)(a-x)}}{(a+x) - (a-x)} \\ &= \frac{2a - 2\sqrt{a^2 - x^2}}{2x} \\ &= \frac{a - \sqrt{a^2 - x^2}}{x} \end{aligned}$$

Differentiating $y = \frac{a - \sqrt{a^2 - x^2}}{x}$ using quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{x \cdot \frac{d}{dx}(a - \sqrt{a^2 - x^2}) - (a - \sqrt{a^2 - x^2}) \cdot \frac{d}{dx}(x)}{x^2} \\ &= \frac{x \cdot \left(-\frac{-2x}{2\sqrt{a^2 - x^2}}\right) - (a - \sqrt{a^2 - x^2}) \cdot 1}{x^2} \\ &= \frac{\frac{x^2}{\sqrt{a^2 - x^2}} - a + \sqrt{a^2 - x^2}}{x^2} \\ &= \frac{x^2 - a\sqrt{a^2 - x^2} + (a^2 - x^2)}{x^2\sqrt{a^2 - x^2}} \\ &= \frac{a^2 - a\sqrt{a^2 - x^2}}{x^2\sqrt{a^2 - x^2}} \\ &= \frac{a(a - \sqrt{a^2 - x^2})}{x^2\sqrt{a^2 - x^2}} \end{aligned}$$

Final Answer

$$\frac{a(a - \sqrt{a^2 - x^2})}{x^2\sqrt{a^2 - x^2}}$$

Question 16 (ii)

Differentiate $\frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{\sqrt{x^2+1}-\sqrt{x^2-1}}$ w.r.t. x .

TYPE 29 Rationalization before Differentiation

Given: $y = \frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{\sqrt{x^2+1}-\sqrt{x^2-1}}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $(\sqrt{x^2 + 1} + \sqrt{x^2 - 1})$

Solution:

Rationalizing the denominator:

$$\begin{aligned} y &= \frac{(\sqrt{x^2 + 1} + \sqrt{x^2 - 1})^2}{(\sqrt{x^2 + 1} - \sqrt{x^2 - 1})(\sqrt{x^2 + 1} + \sqrt{x^2 - 1})} \\ &= \frac{(x^2 + 1) + (x^2 - 1) + 2\sqrt{(x^2 + 1)(x^2 - 1)}}{(x^2 + 1) - (x^2 - 1)} \\ &= \frac{2x^2 + 2\sqrt{x^4 - 1}}{2} \\ &= x^2 + \sqrt{x^4 - 1} \end{aligned}$$

Differentiating with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^2) + \frac{d}{dx}(\sqrt{x^4 - 1}) \\ &= 2x + \frac{1}{2\sqrt{x^4 - 1}} \cdot \frac{d}{dx}(x^4 - 1) \\ &= 2x + \frac{4x^3}{2\sqrt{x^4 - 1}} \\ &= 2x + \frac{2x^3}{\sqrt{x^4 - 1}} \end{aligned}$$

Final Answer

$$2x + \frac{2x^3}{\sqrt{x^4 - 1}}$$

Question 17

If $f(x) = \sqrt{\frac{\sec x - 1}{\sec x + 1}}$, find $f'(x)$. Also find $f'(\frac{\pi}{3})$.

TYPE 30 Trigonometric Simplification before Differentiation

Given: $f(x) = \sqrt{\frac{\sec x - 1}{\sec x + 1}}$

To Find: $f'(x)$ and $f'(\frac{\pi}{3})$

Concept / Formula Used: $\sec x = \frac{1}{\cos x}$, $\frac{1 - \cos x}{1 + \cos x} = \tan^2\left(\frac{x}{2}\right)$

Solution:

Simplifying $f(x)$ using trigonometric identities:

$$\begin{aligned} f(x) &= \sqrt{\frac{\frac{1}{\cos x} - 1}{\frac{1}{\cos x} + 1}} \\ &= \sqrt{\frac{1 - \cos x}{1 + \cos x}} \\ &= \sqrt{\tan^2 \left(\frac{x}{2}\right)} \\ &= \tan \left(\frac{x}{2}\right) \quad \text{for } x \in \left(0, \frac{\pi}{2}\right) \end{aligned}$$

Differentiating $f(x) = \tan \left(\frac{x}{2}\right)$ with respect to x :

$$\begin{aligned} f'(x) &= \sec^2 \left(\frac{x}{2}\right) \cdot \frac{d}{dx} \left(\frac{x}{2}\right) \\ &= \frac{1}{2} \sec^2 \left(\frac{x}{2}\right) \end{aligned}$$

Evaluating at $x = \frac{\pi}{3}$:

$$\begin{aligned} f' \left(\frac{\pi}{3}\right) &= \frac{1}{2} \sec^2 \left(\frac{\pi}{6}\right) \\ &= \frac{1}{2} \cdot \left(\frac{2}{\sqrt{3}}\right)^2 \\ &= \frac{1}{2} \cdot \frac{4}{3} \\ &= \frac{2}{3} \end{aligned}$$

Final Answer

$$\begin{aligned} f'(x) &= \frac{1}{2} \sec^2 \left(\frac{x}{2}\right), \\ f' \left(\frac{\pi}{3}\right) &= \frac{2}{3} \end{aligned}$$

Question 18 (i)

Find the derivative of $\cos \left(2x + \frac{\pi}{3}\right)$ at $x = \frac{\pi}{3}$.

TYPE 31 Chain Rule and Evaluation

Given: $f(x) = \cos \left(2x + \frac{\pi}{3}\right)$, point $x = \frac{\pi}{3}$

To Find: $f' \left(\frac{\pi}{3}\right)$

Concept / Formula Used: $\frac{d}{dx} \cos(u) = -\sin(u) \frac{du}{dx}$

Solution:

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= -\sin \left(2x + \frac{\pi}{3}\right) \cdot \frac{d}{dx} \left(2x + \frac{\pi}{3}\right) \\ &= -2 \sin \left(2x + \frac{\pi}{3}\right) \end{aligned}$$

Substituting $x = \frac{\pi}{3}$:

$$\begin{aligned} f'\left(\frac{\pi}{3}\right) &= -2 \sin\left(2\left(\frac{\pi}{3}\right) + \frac{\pi}{3}\right) \\ &= -2 \sin\left(\frac{2\pi}{3} + \frac{\pi}{3}\right) \\ &= -2 \sin(\pi) \\ &= -2(0) \\ &= 0 \end{aligned}$$

Final Answer

0

Question 18 (ii)

Find the derivative of $\frac{1-\sin x}{1+\cos x}$ at $x = \frac{\pi}{2}$.

TYPE 28 Quotient Rule for Differentiation

Given: $f(x) = \frac{1-\sin x}{1+\cos x}$, point $x = \frac{\pi}{2}$

To Find: $f'\left(\frac{\pi}{2}\right)$

Concept / Formula Used: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

Solution:

Let $u = 1 - \sin x$ and $v = 1 + \cos x$. Differentiating u and v :

$$u' = -\cos x$$

$$v' = -\sin x$$

Applying quotient rule:

$$\begin{aligned} f'(x) &= \frac{(1 + \cos x)(-\cos x) - (1 - \sin x)(-\sin x)}{(1 + \cos x)^2} \\ &= \frac{-\cos x - \cos^2 x + \sin x - \sin^2 x}{(1 + \cos x)^2} \\ &= \frac{\sin x - \cos x - (\sin^2 x + \cos^2 x)}{(1 + \cos x)^2} \\ &= \frac{\sin x - \cos x - 1}{(1 + \cos x)^2} \end{aligned}$$

Evaluating at $x = \frac{\pi}{2}$ using $\sin\left(\frac{\pi}{2}\right) = 1$ and $\cos\left(\frac{\pi}{2}\right) = 0$:

$$\begin{aligned} f'\left(\frac{\pi}{2}\right) &= \frac{1 - 0 - 1}{(1 + 0)^2} \\ &= \frac{0}{1} \\ &= 0 \end{aligned}$$

Final Answer

0

Question 19

If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, then show that $2x \frac{dy}{dx} + y = 2\sqrt{x}$.

TYPE 32 Verifying Differential Relations

Given: $y = x^{1/2} + x^{-1/2}$

To Prove: $2x \frac{dy}{dx} + y = 2\sqrt{x}$

Concept / Formula Used: $\frac{d}{dx}(x^n) = nx^{n-1}$

Solution:

Differentiating $y = x^{1/2} + x^{-1/2}$ with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2} \\ &= \frac{1}{2\sqrt{x}} - \frac{1}{2x\sqrt{x}}\end{aligned}$$

Multiplying $\frac{dy}{dx}$ by $2x$:

$$\begin{aligned}2x \frac{dy}{dx} &= 2x \left(\frac{1}{2\sqrt{x}} - \frac{1}{2x\sqrt{x}} \right) \\ &= \sqrt{x} - \frac{1}{\sqrt{x}}\end{aligned}$$

Substituting into the left-hand side LHS = $2x \frac{dy}{dx} + y$:

$$\begin{aligned}\text{LHS} &= \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right) + \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) \\ &= \sqrt{x} + \sqrt{x} \\ &= 2\sqrt{x} \\ &= \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 20

If $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, prove that $\frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0$.

TYPE 30 Trigonometric Simplification before Differentiation

Given: $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$

To Prove: $\frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0$

Concept / Formula Used: $1 - \sin 2x = (\cos x - \sin x)^2$, $1 + \sin 2x = (\cos x + \sin x)^2$

Solution:

Expressing $1 \pm \sin 2x$ as perfect squares:

$$1 - \sin 2x = \cos^2 x + \sin^2 x - 2 \sin x \cos x = (\cos x - \sin x)^2$$

$$1 + \sin 2x = \cos^2 x + \sin^2 x + 2 \sin x \cos x = (\cos x + \sin x)^2$$

Substituting these into y :

$$\begin{aligned} y &= \sqrt{\frac{(\cos x - \sin x)^2}{(\cos x + \sin x)^2}} \\ &= \frac{\cos x - \sin x}{\cos x + \sin x} \end{aligned}$$

Dividing numerator and denominator by $\cos x$:

$$\begin{aligned} y &= \frac{1 - \tan x}{1 + \tan x} \\ &= \tan \left(\frac{\pi}{4} - x \right) \end{aligned}$$

Differentiating y with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \sec^2 \left(\frac{\pi}{4} - x \right) \cdot \frac{d}{dx} \left(\frac{\pi}{4} - x \right) \\ &= \sec^2 \left(\frac{\pi}{4} - x \right) \cdot (-1) \\ &= -\sec^2 \left(\frac{\pi}{4} - x \right) \end{aligned}$$

Transposing terms:

$$\frac{dy}{dx} + \sec^2 \left(\frac{\pi}{4} - x \right) = 0$$

Hence Proved

$\frac{dy}{dx} + \sec^2 \left(\frac{\pi}{4} - x \right) = 0$, hence proved.

Question 21

Differentiate $|\cos x|$ w.r.t. x . Is this function differentiable? What can you say about the differentiability of $\cos |x|$?

TYPE 25 Derivative of Absolute Value Functions

Given: Functions $f(x) = |\cos x|$ and $g(x) = \cos |x|$

To Find: Derivative of $|\cos x|$ and differentiability analysis of both functions

Concept / Formula Used: $\frac{d}{dx}|u| = \frac{u}{|u|}u'$ $u \neq 0$

Solution:

First, differentiating $f(x) = |\cos x|$:

$$\begin{aligned}\frac{d}{dx}|\cos x| &= \frac{\cos x}{|\cos x|} \cdot \frac{d}{dx}(\cos x) \\ &= \frac{\cos x}{|\cos x|} \cdot (-\sin x) \\ &= -\frac{\sin x \cos x}{|\cos x|}, \quad \text{for } \cos x \neq 0\end{aligned}$$

Second, analyzing differentiability of $f(x) = |\cos x|$: The function $|\cos x|$ is not differentiable at points where $\cos x = 0$, i.e., at $x = (2n+1)\frac{\pi}{2}$ for $n \in \mathbb{Z}$, because the absolute value creates sharp corners at the roots of $\cos x$.

Third, analyzing differentiability of $g(x) = \cos |x|$: Since $\cos(-\theta) = \cos \theta$ for all $\theta \in \mathbb{R}$, we have:

$$g(x) = \cos |x| = \cos x \quad \text{for all } x \in \mathbb{R}$$

Since $\cos x$ is differentiable everywhere on $\mathbb{R}$, $\cos |x|$ is differentiable everywhere on $\mathbb{R}$.

Final Answer

$$\frac{d}{dx}|\cos x| = -\frac{\sin x \cos x}{|\cos x|} \text{ for } x \neq (2n+1)\frac{\pi}{2};$$

$|\cos x|$ is not differentiable at $x = (2n+1)\frac{\pi}{2}$;

$\cos |x|$ is differentiable everywhere on $\mathbb{R}$.

Question 22

Prove that the derivative of an odd function is always an even function.

TYPE 33 Proofs Using Chain Rule

Given: An odd function $f(x)$ satisfying $f(-x) = -f(x)$ for all x

To Prove: $f'(x)$ is an even function, i.e., $f'(-x) = f'(x)$

Concept / Formula Used: $f(-x) = -f(x)$

Solution:

Since $f(x)$ is an odd function, by definition:

$$f(-x) = -f(x)$$

Differentiating both sides with respect to x :

$$\frac{d}{dx}[f(-x)] = \frac{d}{dx}[-f(x)]$$

Applying the chain rule to the left-hand side:

$$\begin{aligned}f'(-x) \cdot \frac{d}{dx}(-x) &= -f'(x) \\ f'(-x) \cdot (-1) &= -f'(x) \\ -f'(-x) &= -f'(x)\end{aligned}$$

Multiplying both sides by -1 :

$$f'(-x) = f'(x)$$

By definition, $f'(x)$ is an even function.

Hence Proved

$f'(-x) = f'(x)$, hence the derivative of an odd function is an even function.

Question 23

If $2f(x) + 3f(-x) = x^2 + x + 1$, find $f'(1)$.

TYPE 34 Solving Functional Equations for Derivatives

Given: $2f(x) + 3f(-x) = x^2 + x + 1$

To Find: $f'(1)$

Concept / Formula Used: $f(x) \quad -x \quad x$

Solution:

Given functional equation:

$$2f(x) + 3f(-x) = x^2 + x + 1 \quad (1)$$

Replacing x with $-x$ in equation (1):

$$\begin{aligned} 2f(-x) + 3f(x) &= (-x)^2 + (-x) + 1 \\ 3f(x) + 2f(-x) &= x^2 - x + 1 \end{aligned} \quad (2)$$

Multiplying equation (1) by 2:

$$4f(x) + 6f(-x) = 2x^2 + 2x + 2 \quad (3)$$

Multiplying equation (2) by 3:

$$9f(x) + 6f(-x) = 3x^2 - 3x + 3 \quad (4)$$

Subtracting equation (3) from equation (4):

$$\begin{aligned} [9f(x) + 6f(-x)] - [4f(x) + 6f(-x)] &= (3x^2 - 3x + 3) - (2x^2 + 2x + 2) \\ 5f(x) &= x^2 - 5x + 1 \\ f(x) &= \frac{1}{5}x^2 - x + \frac{1}{5} \end{aligned}$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\frac{1}{5}x^2 - x + \frac{1}{5} \right) \\ &= \frac{2}{5}x - 1 \end{aligned}$$

Substituting $x = 1$:

$$\begin{aligned} f'(1) &= \frac{2}{5}(1) - 1 \\ &= \frac{2}{5} - 1 \\ &= -\frac{3}{5} \end{aligned}$$

Final Answer

$$f'(1) = -\frac{3}{5}$$

Question 24

If $f(x) = |\tan 2x|$, then find the value of $f'(x)$ at $x = \frac{\pi}{3}$.

TYPE 26 Derivative of Absolute Value Functions at a Point

Given: $f(x) = |\tan 2x|$, point $x = \frac{\pi}{3}$

To Find: $f'(\frac{\pi}{3})$

Concept / Formula Used: $\tan(\frac{2\pi}{3}) < 0 \implies |\tan 2x| = -\tan 2x$

Solution:

Evaluating $\tan 2x$ at $x = \frac{\pi}{3}$:

$$\begin{aligned} 2x &= \frac{2\pi}{3} \\ \tan\left(\frac{2\pi}{3}\right) &= -\sqrt{3} < 0 \end{aligned}$$

Since $\tan 2x < 0$ in a neighborhood of $x = \frac{\pi}{3}$:

$$f(x) = -\tan 2x$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(-\tan 2x) \\ &= -\sec^2 2x \cdot \frac{d}{dx}(2x) \\ &= -2\sec^2 2x \end{aligned}$$

Substituting $x = \frac{\pi}{3}$:

$$\begin{aligned} f'\left(\frac{\pi}{3}\right) &= -2\sec^2\left(\frac{2\pi}{3}\right) \\ &= -2 \cdot (-2)^2 \\ &= -2 \cdot 4 \\ &= -8 \end{aligned}$$

Final Answer

$$-8$$

Question 25

If $y = \cos^3(\sec^2 2t)$, find $\frac{dy}{dt}$.

TYPE 22 Repeated Chain Rule

Given: $y = \cos^3(\sec^2 2t)$

To Find: $\frac{dy}{dt}$

Concept / Formula Used: $\frac{d}{dt}[u^n] = nu^{n-1}u'$, $\frac{d}{du} \cos u = -\sin u$, $\frac{d}{dv} \sec^2 v = 2 \sec^2 v \tan v$

Solution:

Differentiating $y = [\cos(\sec^2 2t)]^3$ with respect to t using the chain rule:

$$\frac{dy}{dt} = 3 \cos^2(\sec^2 2t) \cdot \frac{d}{dt} [\cos(\sec^2 2t)]$$

Differentiating the inner function $\cos(\sec^2 2t)$:

$$\frac{d}{dt} [\cos(\sec^2 2t)] = -\sin(\sec^2 2t) \cdot \frac{d}{dt}(\sec^2 2t)$$

Differentiating $\sec^2 2t = (\sec 2t)^2$:

$$\begin{aligned} \frac{d}{dt}(\sec^2 2t) &= 2 \sec 2t \cdot \frac{d}{dt}(\sec 2t) \\ &= 2 \sec 2t \cdot (\sec 2t \tan 2t \cdot 2) \\ &= 4 \sec^2 2t \tan 2t \end{aligned}$$

Combining all the results:

$$\begin{aligned} \frac{dy}{dt} &= 3 \cos^2(\sec^2 2t) \cdot [-\sin(\sec^2 2t)] \cdot [4 \sec^2 2t \tan 2t] \\ &= -12 \sec^2 2t \tan 2t \cos^2(\sec^2 2t) \sin(\sec^2 2t) \end{aligned}$$

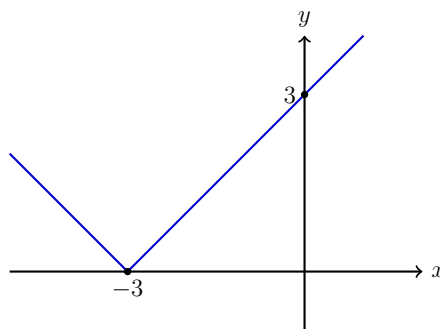
Final Answer

$$-12 \sec^2 2t \tan 2t \cos^2(\sec^2 2t) \sin(\sec^2 2t)$$

Question 26

The figure given below shows the graph of a function $y = f(x)$. What is the derivative of the function?

TYPE 25 Derivative of Absolute Value Functions



V-shaped graph of $y = f(x)$ with corner at $(-3, 0)$

Given: V-shaped graph passing through $(-3, 0)$, $(0, 3)$, and $(-6, 3)$

To Find: The derivative $f'(x)$

Concept / Formula Used: $f'(x)$ $y = f(x)$

Solution:

From the given graph, the function is represented by two linear rays meeting at the vertex $(-3, 0)$:

For $x > -3$, the line passes through $(-3, 0)$ and $(0, 3)$. Its slope is:

$$\begin{aligned} m_1 &= \frac{3 - 0}{0 - (-3)} \\ &= \frac{3}{3} \\ &= 1 \end{aligned}$$

For $x < -3$, the line passes through $(-3, 0)$ and $(-6, 3)$. Its slope is:

$$\begin{aligned} m_2 &= \frac{3 - 0}{-6 - (-3)} \\ &= \frac{3}{-3} \\ &= -1 \end{aligned}$$

Thus, $f(x) = |x + 3|$. Differentiating $f(x)$ for $x \neq -3$:

$$f'(x) = \begin{cases} 1, & \text{if } x > -3 \\ -1, & \text{if } x < -3 \end{cases}$$

Equivalently, in absolute value notation:

$$f'(x) = \frac{x + 3}{|x + 3|}, \quad \text{for } x \neq -3$$

At $x = -3$, $f'(x)$ does not exist due to a sharp corner.

Final Answer

$$f'(x) = \frac{x+3}{|x+3|}, \quad x \neq -3$$

5.6 – Derivatives of Inverse Trigonometric Functions

Question 1

Find the domain of differentiability of the following functions:

- (i) $\sin^{-1} x$
- (ii) $\cos^{-1} x$
- (iii) $\sec^{-1} x$
- (iv) $\csc^{-1} x$
- (v) $\tan^{-1} x$
- (vi) $\cot^{-1} x$

TYPE 35 Domain of Differentiability

To Find: The domain of differentiability for each given inverse trigonometric function.

Concept / Formula Used: $f(x)$ $f'(x)$

Solution:

- (i) Let $f(x) = \sin^{-1} x$. Using the standard derivative formula:

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

For $f'(x)$ to be defined and real, we require $1-x^2 > 0$, which gives $x^2 < 1$, i.e., $x \in (-1, 1)$.

- (ii) Let $f(x) = \cos^{-1} x$. Using the standard derivative formula:

$$f'(x) = -\frac{1}{\sqrt{1-x^2}}$$

For $f'(x)$ to be defined and real, we require $1-x^2 > 0$, which gives $x \in (-1, 1)$.

- (iii) Let $f(x) = \sec^{-1} x$. Using the standard derivative formula:

$$f'(x) = \frac{1}{|x|\sqrt{x^2-1}}$$

For $f'(x)$ to be defined and real, we require $x \neq 0$ and $x^2 - 1 > 0$, which gives $|x| > 1$, i.e., $x \in (-\infty, -1) \cup (1, \infty)$.

- (iv) Let $f(x) = \csc^{-1} x$. Using the standard derivative formula:

$$f'(x) = -\frac{1}{|x|\sqrt{x^2-1}}$$

For $f'(x)$ to be defined and real, we require $x \neq 0$ and $x^2 - 1 > 0$, which gives $x \in (-\infty, -1) \cup (1, \infty)$.

- (v) Let $f(x) = \tan^{-1} x$. Using the standard derivative formula:

$$f'(x) = \frac{1}{1+x^2}$$

Since $1+x^2 \geq 1 > 0$ for all real numbers x , $f'(x)$ exists for all $x \in \mathbb{R}$.

(vi) Let $f(x) = \cot^{-1} x$. Using the standard derivative formula:

$$f'(x) = -\frac{1}{1+x^2}$$

Since $1+x^2 \geq 1 > 0$ for all real numbers x , $f'(x)$ exists for all $x \in \mathbb{R}$.

Final Answer

- (i) $(-1, 1)$
- (ii) $(-1, 1)$
- (iii) $(-\infty, -1) \cup (1, \infty)$
- (iv) $(-\infty, -1) \cup (1, \infty)$
- (v) $\mathbb{R}$ or $(-\infty, \infty)$
- (vi) $\mathbb{R}$ or $(-\infty, \infty)$

Question 2

Find the derivatives of the following functions:

- (i) $\tan^{-1}(x\sqrt{x})$
- (ii) $\cos^{-1}(3x)$
- (iii) $\sqrt{\sin^{-1}(2x)}$
- (iv) $\sin(\tan^{-1} x)$
- (v) $\cot^{-1} \sqrt{x}$
- (vi) $(\tan^{-1} x)^2$

TYPE 36 Chain Rule for Inverse Trigonometric Functions

To Find: The derivative $\frac{dy}{dx}$ for each function.

Concept / Formula Used: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Solution:

(i) Let $y = \tan^{-1}(x\sqrt{x}) = \tan^{-1}(x^{3/2})$.

Let $u = x^{3/2}$.

Differentiating u w.r.t. x :

$$\frac{du}{dx} = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x}$$

Using the standard result $\frac{d}{du} \tan^{-1} u = \frac{1}{1+u^2}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1+u^2} \cdot \frac{du}{dx} \\ &= \frac{1}{1+(x^{3/2})^2} \cdot \left(\frac{3}{2}\sqrt{x}\right) \\ &= \frac{3\sqrt{x}}{2(1+x^3)} \end{aligned}$$

(ii) Let $y = \cos^{-1}(3x)$.

Let $u = 3x$.

Differentiating u w.r.t. x :

$$\frac{du}{dx} = 3$$

Using the standard result $\frac{d}{du} \cos^{-1} u = -\frac{1}{\sqrt{1-u^2}}$:

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx} \\ &= -\frac{1}{\sqrt{1-(3x)^2}} \cdot 3 \\ &= -\frac{3}{\sqrt{1-9x^2}} \end{aligned}$$

(iii) Let $y = \sqrt{\sin^{-1}(2x)}$.

Let $u = \sin^{-1}(2x)$.

Let $v = 2x \implies \frac{dv}{dx} = 2$.

Using the standard result $\frac{d}{dv} \sin^{-1} v = \frac{1}{\sqrt{1-v^2}}$:

$$\frac{du}{dx} = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2 = \frac{2}{\sqrt{1-4x^2}}$$

Now for $y = \sqrt{u}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2\sqrt{u}} \cdot \frac{du}{dx} \\ &= \frac{1}{2\sqrt{\sin^{-1}(2x)}} \cdot \frac{2}{\sqrt{1-4x^2}} \\ &= \frac{1}{\sqrt{1-4x^2}\sqrt{\sin^{-1}(2x)}} \end{aligned}$$

(iv) Let $y = \sin(\tan^{-1} x)$.

Let $\tan^{-1} x = \theta \implies \tan \theta = x$.

Using the trigonometric identity $\sin \theta = \frac{\tan \theta}{\sqrt{1+\tan^2 \theta}}$:

$$y = \frac{x}{\sqrt{1+x^2}}$$

Differentiating y w.r.t. x using the Quotient Rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sqrt{1+x^2} \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(\sqrt{1+x^2})}{(\sqrt{1+x^2})^2} \\ &= \frac{\sqrt{1+x^2}(1) - x \cdot \left(\frac{2x}{2\sqrt{1+x^2}}\right)}{1+x^2} \\ &= \frac{\sqrt{1+x^2} - \frac{x^2}{\sqrt{1+x^2}}}{1+x^2} \\ &= \frac{1+x^2 - x^2}{(1+x^2)^{3/2}} \\ &= \frac{1}{(1+x^2)^{3/2}} \end{aligned}$$

(v) Let $y = \cot^{-1} \sqrt{x}$.

Let $u = \sqrt{x}$.

Differentiating u w.r.t. x :

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

Using the standard result $\frac{d}{du} \cot^{-1} u = -\frac{1}{1+u^2}$:

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{1+u^2} \cdot \frac{du}{dx} \\ &= -\frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}} \\ &= -\frac{1}{2\sqrt{x}(1+x)} \end{aligned}$$

(vi) Let $y = (\tan^{-1} x)^2$.

Using the Power Rule combined with Chain Rule:

$$\begin{aligned} \frac{dy}{dx} &= 2(\tan^{-1} x) \cdot \frac{d}{dx}(\tan^{-1} x) \\ &= 2(\tan^{-1} x) \cdot \frac{1}{1+x^2} \\ &= \frac{2 \tan^{-1} x}{1+x^2} \end{aligned}$$

Final Answer

- (i) $\frac{3\sqrt{x}}{2(1+x^3)}$
- (ii) $-\frac{3}{\sqrt{1-9x^2}}$
- (iii) $\frac{1}{\sqrt{1-4x^2}\sqrt{\sin^{-1}(2x)}}$
- (iv) $\frac{1}{(1+x^2)^{3/2}}$
- (v) $-\frac{1}{2\sqrt{x}(1+x)}$
- (vi) $\frac{2 \tan^{-1} x}{1+x^2}$

Question 3

Find the derivatives of the following functions:

- (i) $\sec^{-1}(\csc x)$
- (ii) $\tan^{-1}(\cot x)$

TYPE 37 Inverse Trigonometric Simplification before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each given expression.

Concept / Formula Used: $\csc x = \sec\left(\frac{\pi}{2} - x\right)$ $\cot x = \tan\left(\frac{\pi}{2} - x\right)$

Solution:

(i) Let $y = \sec^{-1}(\csc x)$.

Using the identity $\csc x = \sec\left(\frac{\pi}{2} - x\right)$:

$$\begin{aligned} y &= \sec^{-1}\left(\sec\left(\frac{\pi}{2} - x\right)\right) \\ &= \frac{\pi}{2} - x \end{aligned}$$

Differentiating w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\frac{\pi}{2}\right) - \frac{d}{dx}(x) \\ &= 0 - 1 \\ &= -1 \end{aligned}$$

(ii) Let $y = \tan^{-1}(\cot x)$.

Using the identity $\cot x = \tan\left(\frac{\pi}{2} - x\right)$:

$$\begin{aligned} y &= \tan^{-1}\left(\tan\left(\frac{\pi}{2} - x\right)\right) \\ &= \frac{\pi}{2} - x \end{aligned}$$

Differentiating w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\frac{\pi}{2}\right) - \frac{d}{dx}(x) \\ &= 0 - 1 \\ &= -1 \end{aligned}$$

Final Answer

(i) -1

(ii) -1

Question 4

If $f(x) = \cot^{-1}(\tan x)$, show that $f'(x) = -1$.

TYPE 37 Inverse Trigonometric Simplification before Differentiation

To Prove: $f'(x) = -1$ where $f(x) = \cot^{-1}(\tan x)$.

Concept / Formula Used: $\tan x = \cot\left(\frac{\pi}{2} - x\right)$

Solution:

Given function:

$$f(x) = \cot^{-1}(\tan x)$$

Using the complementary angle identity $\tan x = \cot\left(\frac{\pi}{2} - x\right)$:

$$\begin{aligned} f(x) &= \cot^{-1}\left(\cot\left(\frac{\pi}{2} - x\right)\right) \\ &= \frac{\pi}{2} - x \end{aligned}$$

Differentiating $f(x)$ w.r.t. x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\frac{\pi}{2} - x \right) \\ &= 0 - 1 \\ &= -1 \end{aligned}$$

Hence Proved

LHS = $f'(x) = -1$, hence proved.

Question 5

If $y = \sin^{-1} x + \cos^{-1} x$, find $\frac{dy}{dx}$.

TYPE 37 Inverse Trigonometric Simplification before Differentiation

Given: $y = \sin^{-1} x + \cos^{-1} x$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad x \in [-1, 1]$

Solution:

Given function:

$$y = \sin^{-1} x + \cos^{-1} x$$

Using the standard inverse trigonometric identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$y = \frac{\pi}{2}$$

Differentiating w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{\pi}{2} \right) \\ &= 0 \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = 0$$

Question 6

If $f(x) = \sin^{-1} x + \sec^{-1} \frac{1}{x}$, find $f'(x)$.

TYPE 37 Inverse Trigonometric Simplification before Differentiation

Given: $f(x) = \sin^{-1} x + \sec^{-1} \frac{1}{x}$

To Find: $f'(x)$

Concept / Formula Used: $\sec^{-1} \left(\frac{1}{x} \right) = \cos^{-1} x \quad \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

Solution:

Given function:

$$f(x) = \sin^{-1} x + \sec^{-1} \left(\frac{1}{x} \right)$$

Using the reciprocal property $\sec^{-1} \left(\frac{1}{x} \right) = \cos^{-1} x$:

$$f(x) = \sin^{-1} x + \cos^{-1} x$$

Using the identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$f(x) = \frac{\pi}{2}$$

Differentiating w.r.t. x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\frac{\pi}{2} \right) \\ &= 0 \end{aligned}$$

Final Answer

$$f'(x) = 0$$

Question 7

If $y = \tan^{-1} x + \cot^{-1} \frac{1}{x}$, $x > 0$, find $\frac{dy}{dx}$.

TYPE 37 Inverse Trigonometric Simplification before Differentiation

Given: $y = \tan^{-1} x + \cot^{-1} \frac{1}{x}$ for $x > 0$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $x > 0$ $\cot^{-1} \left(\frac{1}{x} \right) = \tan^{-1} x$

Solution:

Given function for $x > 0$:

$$y = \tan^{-1} x + \cot^{-1} \left(\frac{1}{x} \right)$$

Using the identity $\cot^{-1} \left(\frac{1}{x} \right) = \tan^{-1} x$ for $x > 0$:

$$\begin{aligned} y &= \tan^{-1} x + \tan^{-1} x \\ &= 2 \tan^{-1} x \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 2 \cdot \frac{d}{dx} (\tan^{-1} x)$$

Using the standard result $\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$:

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

Final Answer

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

Question 8

If $y = \sin^{-1}(\cos x) + 3 \csc^{-1}(\sec x)$, find $\frac{dy}{dx}$.

TYPE 37 Inverse Trigonometric Simplification before Differentiation

Given: $y = \sin^{-1}(\cos x) + 3 \csc^{-1}(\sec x)$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\cos x = \sin\left(\frac{\pi}{2} - x\right)$ $\sec x = \csc\left(\frac{\pi}{2} - x\right)$

Solution:

Given function:

$$y = \sin^{-1}(\cos x) + 3 \csc^{-1}(\sec x)$$

Using complementary angle transformations:

$$\begin{aligned}\cos x &= \sin\left(\frac{\pi}{2} - x\right) \implies \sin^{-1}(\cos x) = \frac{\pi}{2} - x \\ \sec x &= \csc\left(\frac{\pi}{2} - x\right) \implies \csc^{-1}(\sec x) = \frac{\pi}{2} - x\end{aligned}$$

Substituting these into y :

$$\begin{aligned}y &= \left(\frac{\pi}{2} - x\right) + 3\left(\frac{\pi}{2} - x\right) \\ &= 4\left(\frac{\pi}{2} - x\right) \\ &= 2\pi - 4x\end{aligned}$$

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(2\pi - 4x) \\ &= 0 - 4 \\ &= -4\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -4$$

Question 9

If $y = 2 \cos^{-1}(\sin x) + 3 \cot^{-1}(\tan x)$, find $\frac{dy}{dx}$.

TYPE 37 Inverse Trigonometric Simplification before Differentiation

Given: $y = 2 \cos^{-1}(\sin x) + 3 \cot^{-1}(\tan x)$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\sin x = \cos\left(\frac{\pi}{2} - x\right)$ $\tan x = \cot\left(\frac{\pi}{2} - x\right)$

Solution:

Given function:

$$y = 2 \cos^{-1}(\sin x) + 3 \cot^{-1}(\tan x)$$

Using complementary angle transformations:

$$\begin{aligned}\sin x = \cos\left(\frac{\pi}{2} - x\right) &\implies \cos^{-1}(\sin x) = \frac{\pi}{2} - x \\ \tan x = \cot\left(\frac{\pi}{2} - x\right) &\implies \cot^{-1}(\tan x) = \frac{\pi}{2} - x\end{aligned}$$

Substituting these into y :

$$\begin{aligned}y &= 2\left(\frac{\pi}{2} - x\right) + 3\left(\frac{\pi}{2} - x\right) \\ &= 5\left(\frac{\pi}{2} - x\right) \\ &= \frac{5\pi}{2} - 5x\end{aligned}$$

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left(\frac{5\pi}{2} - 5x\right) \\ &= 0 - 5 \\ &= -5\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -5$$

Question 10

Differentiate the following functions w.r.t. x :

- (i) $x \tan^{-1} x$
- (ii) $x \cos^{-1} \sqrt{x}$
- (iii) $\frac{\sin^{-1} x}{x}$

TYPE 38 Product and Quotient Rules with Inverse Trigonometric Functions

To Find: The derivative $\frac{dy}{dx}$ for each given expression.

Concept / Formula Used: $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$ $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Solution:

- (i) Let $y = x \tan^{-1} x$.

Using Product Rule with $u = x$ and $v = \tan^{-1} x$:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \frac{d}{dx}(\tan^{-1} x) + \tan^{-1} x \cdot \frac{d}{dx}(x) \\ &= x \cdot \frac{1}{1+x^2} + (\tan^{-1} x) \cdot 1 \\ &= \frac{x}{1+x^2} + \tan^{-1} x\end{aligned}$$

(ii) Let $y = x \cos^{-1} \sqrt{x}$.

Using Product Rule with $u = x$ and $v = \cos^{-1} \sqrt{x}$:

$$\frac{dy}{dx} = x \cdot \frac{d}{dx}(\cos^{-1} \sqrt{x}) + \cos^{-1} \sqrt{x} \cdot \frac{d}{dx}(x)$$

To find $\frac{d}{dx}(\cos^{-1} \sqrt{x})$, let $w = \sqrt{x} \implies \frac{dw}{dx} = \frac{1}{2\sqrt{x}}$:

$$\begin{aligned}\frac{d}{dx}(\cos^{-1} \sqrt{x}) &= -\frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}} \\ &= -\frac{1}{2\sqrt{x}\sqrt{1-x}}\end{aligned}$$

Substituting back:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \left(-\frac{1}{2\sqrt{x}\sqrt{1-x}}\right) + \cos^{-1} \sqrt{x} \\ &= -\frac{\sqrt{x}}{2\sqrt{1-x}} + \cos^{-1} \sqrt{x}\end{aligned}$$

(iii) Let $y = \frac{\sin^{-1} x}{x}$.

Using Quotient Rule with $u = \sin^{-1} x$ and $v = x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{x \cdot \frac{d}{dx}(\sin^{-1} x) - \sin^{-1} x \cdot \frac{d}{dx}(x)}{x^2} \\ &= \frac{x \cdot \frac{1}{\sqrt{1-x^2}} - \sin^{-1} x \cdot 1}{x^2} \\ &= \frac{\frac{x}{\sqrt{1-x^2}} - \sin^{-1} x}{x^2} \\ &= \frac{x - \sqrt{1-x^2} \sin^{-1} x}{x^2 \sqrt{1-x^2}}\end{aligned}$$

Final Answer

- (i) $\frac{x}{1+x^2} + \tan^{-1} x$
 (ii) $\cos^{-1} \sqrt{x} - \frac{\sqrt{x}}{2\sqrt{1-x}}$
 (iii) $\frac{x - \sqrt{1-x^2} \sin^{-1} x}{x^2 \sqrt{1-x^2}}$

Question 11

Differentiate the following functions w.r.t. x :

- (i) $\cos^{-1} \left(\frac{1-x}{1+x} \right)$

- (ii) $\sec^{-1} \left(\frac{1}{x-1} \right)$
 (iii) $\sin^{-1} \left(\frac{1}{\sqrt{x+1}} \right)$

★ Likely Exam Question

TYPE 39 Trigonometric Substitution before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each function.

Concept / Formula Used: $\cos 2\theta = \frac{1-\tan^2 \theta}{1+\tan^2 \theta}$ $\sec^{-1} \left(\frac{1}{u} \right) = \cos^{-1} u$

Solution:

- (i) Let $y = \cos^{-1} \left(\frac{1-x}{1+x} \right)$.
 Let $x = \tan^2 \theta \implies \sqrt{x} = \tan \theta$.
 Differentiating $\tan \theta = \sqrt{x}$:

$$\theta = \tan^{-1} \sqrt{x}$$

Using double-angle identity:

$$\frac{1-x}{1+x} = \frac{1-\tan^2 \theta}{1+\tan^2 \theta} = \cos 2\theta$$

Substituting this into y :

$$y = \cos^{-1}(\cos 2\theta) = 2\theta = 2 \tan^{-1} \sqrt{x}$$

Differentiating y w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot \frac{1}{1+(\sqrt{x})^2} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= \frac{2}{1+x} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{1}{\sqrt{x}(1+x)} \end{aligned}$$

- (ii) Let $y = \sec^{-1} \left(\frac{1}{x-1} \right)$.
 Using identity $\sec^{-1} \left(\frac{1}{u} \right) = \cos^{-1} u$:

$$y = \cos^{-1}(x-1)$$

Differentiating w.r.t. x : Using standard result $\frac{d}{du} \cos^{-1} u = -\frac{1}{\sqrt{1-u^2}}$ with $u = x-1$:

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{\sqrt{1-(x-1)^2}} \cdot \frac{d}{dx}(x-1) \\ &= -\frac{1}{\sqrt{1-(x^2-2x+1)}} \cdot 1 \\ &= -\frac{1}{\sqrt{2x-x^2}} \end{aligned}$$

(iii) Let $y = \sin^{-1} \left(\frac{1}{\sqrt{x+1}} \right)$.

Let $x = \cot^2 \theta \implies \sqrt{x} = \cot \theta \implies \theta = \cot^{-1} \sqrt{x}$.

Then:

$$\begin{aligned} x + 1 &= \cot^2 \theta + 1 = \csc^2 \theta \\ \sqrt{x + 1} &= \csc \theta \\ \frac{1}{\sqrt{x + 1}} &= \frac{1}{\csc \theta} = \sin \theta \end{aligned}$$

Substituting this into y :

$$y = \sin^{-1}(\sin \theta) = \theta = \cot^{-1} \sqrt{x}$$

Differentiating y w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{1 + (\sqrt{x})^2} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= -\frac{1}{1 + x} \cdot \frac{1}{2\sqrt{x}} \\ &= -\frac{1}{2\sqrt{x}(1 + x)} \end{aligned}$$

Final Answer

- (i) $\frac{1}{\sqrt{x(1+x)}}$
 (ii) $-\frac{1}{\sqrt{2x-x^2}}$
 (iii) $-\frac{1}{2\sqrt{x}(1+x)}$

Question 12

Differentiate the following functions w.r.t. x :

- (i) $\sin^{-1} x + \sin^{-1} \sqrt{1 - x^2}$
 (ii) $x(\sin^{-1} x)^2 + 2\sqrt{1 - x^2} \sin^{-1} x - 2x$

TYPE 37 Inverse Trigonometric Simplification before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each given expression.

Concept / Formula Used: $\sin^{-1} \sqrt{1 - x^2} = \cos^{-1} x \quad x \in [0, 1]$

Solution:

(i) Let $y = \sin^{-1} x + \sin^{-1} \sqrt{1 - x^2}$.

For $x \in [0, 1]$, let $x = \sin \theta \implies \theta = \sin^{-1} x$.

Then $\sqrt{1 - x^2} = \cos \theta = \sin \left(\frac{\pi}{2} - \theta \right)$.

Thus $\sin^{-1} \sqrt{1 - x^2} = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \sin^{-1} x$.

Substituting into y :

$$\begin{aligned} y &= \sin^{-1} x + \left(\frac{\pi}{2} - \sin^{-1} x \right) \\ &= \frac{\pi}{2} \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 0$$

(ii) Let $y = x(\sin^{-1} x)^2 + 2\sqrt{1-x^2} \sin^{-1} x - 2x$.

Differentiating term by term w.r.t. x :

$$\begin{aligned}\frac{d}{dx} [x(\sin^{-1} x)^2] &= 1 \cdot (\sin^{-1} x)^2 + x \cdot 2(\sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}} \\ &= (\sin^{-1} x)^2 + \frac{2x \sin^{-1} x}{\sqrt{1-x^2}} \\ \frac{d}{dx} [2\sqrt{1-x^2} \sin^{-1} x] &= 2 \left[\frac{-2x}{2\sqrt{1-x^2}} \sin^{-1} x + \sqrt{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}} \right] \\ &= -\frac{2x \sin^{-1} x}{\sqrt{1-x^2}} + 2 \\ \frac{d}{dx} [-2x] &= -2\end{aligned}$$

Summing all three derivatives:

$$\begin{aligned}\frac{dy}{dx} &= (\sin^{-1} x)^2 + \frac{2x \sin^{-1} x}{\sqrt{1-x^2}} - \frac{2x \sin^{-1} x}{\sqrt{1-x^2}} + 2 - 2 \\ &= (\sin^{-1} x)^2\end{aligned}$$

Final Answer

(i) 0

(ii) $(\sin^{-1} x)^2$

Question 13

Differentiate the following functions w.r.t. x :

(i) $\tan^{-1}(\sin x + \cos x)$

(ii) $\tan^{-1} \left(\frac{\sin x}{1+\cos x} \right)$

(iii) $\tan^{-1} \left(\frac{1+\cos x}{\sin x} \right)$

(iv) $\tan^{-1} \left(\frac{\cos x}{1+\sin x} \right)$

TYPE 37 Inverse Trigonometric Simplification before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each given expression.

Concept / Formula Used: $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$ $1 + \cos x = 2 \cos^2 \frac{x}{2}$

Solution:

(i) Let $y = \tan^{-1}(\sin x + \cos x)$.

Let $u = \sin x + \cos x \implies \frac{du}{dx} = \cos x - \sin x$.

Using standard result $\frac{d}{du} \tan^{-1} u = \frac{1}{1+u^2}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1 + (\sin x + \cos x)^2} \cdot (\cos x - \sin x) \\ &= \frac{\cos x - \sin x}{1 + \sin^2 x + \cos^2 x + 2 \sin x \cos x} \\ &= \frac{\cos x - \sin x}{1 + 1 + \sin 2x} \\ &= \frac{\cos x - \sin x}{2 + \sin 2x} \end{aligned}$$

(ii) Let $y = \tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right)$.

Using half-angle identities $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$ and $1 + \cos x = 2 \cos^2 \frac{x}{2}$:

$$\begin{aligned} \frac{\sin x}{1 + \cos x} &= \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} = \tan \frac{x}{2} \\ y &= \tan^{-1} \left(\tan \frac{x}{2} \right) = \frac{x}{2} \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{2}$$

(iii) Let $y = \tan^{-1} \left(\frac{1 + \cos x}{\sin x} \right)$.

Using half-angle identities:

$$\begin{aligned} \frac{1 + \cos x}{\sin x} &= \frac{2 \cos^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \cot \frac{x}{2} = \tan \left(\frac{\pi}{2} - \frac{x}{2} \right) \\ y &= \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \frac{x}{2} \right) \right) = \frac{\pi}{2} - \frac{x}{2} \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 0 - \frac{1}{2} = -\frac{1}{2}$$

(iv) Let $y = \tan^{-1} \left(\frac{\cos x}{1 + \sin x} \right)$.

Converting sine and cosine to complementary angles:

$$\begin{aligned} \cos x &= \sin \left(\frac{\pi}{2} - x \right) = 2 \sin \left(\frac{\pi}{4} - \frac{x}{2} \right) \cos \left(\frac{\pi}{4} - \frac{x}{2} \right) \\ 1 + \sin x &= 1 + \cos \left(\frac{\pi}{2} - x \right) = 2 \cos^2 \left(\frac{\pi}{4} - \frac{x}{2} \right) \\ \frac{\cos x}{1 + \sin x} &= \frac{2 \sin \left(\frac{\pi}{4} - \frac{x}{2} \right) \cos \left(\frac{\pi}{4} - \frac{x}{2} \right)}{2 \cos^2 \left(\frac{\pi}{4} - \frac{x}{2} \right)} = \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \\ y &= \tan^{-1} \left(\tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right) = \frac{\pi}{4} - \frac{x}{2} \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{1}{2}$$

Final Answer

- (i) $\frac{\cos x - \sin x}{2 + \sin 2x}$
 (ii) $\frac{1}{2}$
 (iii) $-\frac{1}{2}$
 (iv) $-\frac{1}{2}$

Question 14

Differentiate the following functions w.r.t. x :

- (i) $\cot^{-1}(\csc x + \cot x)$
 (ii) $\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$
 (iii) $\cot^{-1}\left(\sqrt{\frac{1-\sin x}{1+\sin x}}\right)$
 (iv) $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$

TYPE 37 Inverse Trigonometric Simplification before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each given expression.

Concept / Formula Used: $1 - \cos x = 2 \sin^2 \frac{x}{2}$ $1 + \cos x = 2 \cos^2 \frac{x}{2}$

Solution:

- (i) Let $y = \cot^{-1}(\csc x + \cot x)$.

Expressing inside terms in sine and cosine:

$$\begin{aligned}\csc x + \cot x &= \frac{1}{\sin x} + \frac{\cos x}{\sin x} = \frac{1 + \cos x}{\sin x} \\ &= \frac{2 \cos^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \cot \frac{x}{2} \\ y &= \cot^{-1}\left(\cot \frac{x}{2}\right) = \frac{x}{2}\end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{2}$$

- (ii) Let $y = \tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$.

Using standard identities $1 - \cos x = 2 \sin^2 \frac{x}{2}$ and $1 + \cos x = 2 \cos^2 \frac{x}{2}$:

$$\begin{aligned}\sqrt{\frac{1-\cos x}{1+\cos x}} &= \sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} = \sqrt{\tan^2 \frac{x}{2}} = \tan \frac{x}{2} \\ y &= \tan^{-1}\left(\tan \frac{x}{2}\right) = \frac{x}{2}\end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{2}$$

(iii) Let $y = \cot^{-1} \left(\sqrt{\frac{1-\sin x}{1+\sin x}} \right)$.

Using $1 - \sin x = \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)^2$ and $1 + \sin x = \left(\cos \frac{x}{2} + \sin \frac{x}{2} \right)^2$:

$$\begin{aligned} \sqrt{\frac{1-\sin x}{1+\sin x}} &= \frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} = \frac{1 - \tan \frac{x}{2}}{1 + \tan \frac{x}{2}} = \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \\ y &= \cot^{-1} \left(\tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right) \\ &= \cot^{-1} \left(\cot \left(\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{x}{2} \right) \right) \right) = \frac{\pi}{4} + \frac{x}{2} \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{2}$$

(iv) Let $y = \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right)$.

Dividing numerator and denominator by $\cos x$:

$$\begin{aligned} \frac{\cos x - \sin x}{\cos x + \sin x} &= \frac{1 - \tan x}{1 + \tan x} = \tan \left(\frac{\pi}{4} - x \right) \\ y &= \tan^{-1} \left(\tan \left(\frac{\pi}{4} - x \right) \right) = \frac{\pi}{4} - x \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -1$$

Final Answer

- (i) $\frac{1}{2}$
- (ii) $\frac{1}{2}$
- (iii) $\frac{1}{2}$
- (iv) -1

Question 15

- (i) If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, prove that $(1-x^2) \frac{dy}{dx} - xy = 1$.
- (ii) If $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$, prove that $(1+x^2) \frac{dy}{dx} = 2$.

★ Likely Exam Question

TYPE 32 Verifying Differential Relations

- To Prove:** (i) $(1-x^2) \frac{dy}{dx} - xy = 1$
 (ii) $(1+x^2) \frac{dy}{dx} = 2$

Concept / Formula Used: $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$

Solution:

- (i) Given $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$.
Cross-multiplying:

$$y\sqrt{1-x^2} = \sin^{-1} x$$

Differentiating both sides w.r.t. x using Product Rule on LHS:

$$\begin{aligned}\frac{d}{dx} [y\sqrt{1-x^2}] &= \frac{d}{dx} (\sin^{-1} x) \\ y \cdot \frac{d}{dx} (\sqrt{1-x^2}) + \sqrt{1-x^2} \cdot \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} \\ y \cdot \left(\frac{-2x}{2\sqrt{1-x^2}} \right) + \sqrt{1-x^2} \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} \\ -\frac{xy}{\sqrt{1-x^2}} + \sqrt{1-x^2} \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}}\end{aligned}$$

Multiplying the entire equation by $\sqrt{1-x^2}$:

$$\begin{aligned}-xy + (1-x^2) \frac{dy}{dx} &= 1 \\ (1-x^2) \frac{dy}{dx} - xy &= 1\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

- (ii) Given $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$.
Let $x = \tan \theta \implies \theta = \tan^{-1} x$.
Using identity $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$:

$$\begin{aligned}y &= \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \\ &= \sin^{-1} (\sin 2\theta) \\ &= 2\theta \\ &= 2 \tan^{-1} x\end{aligned}$$

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= 2 \cdot \frac{d}{dx} (\tan^{-1} x) \\ \frac{dy}{dx} &= 2 \cdot \frac{1}{1+x^2}\end{aligned}$$

Multiplying both sides by $(1+x^2)$:

$$(1+x^2) \frac{dy}{dx} = 2$$

Hence Proved

LHS = RHS, hence proved.

Question 16Differentiate the following functions w.r.t. x :

- (i) $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$
- (ii) $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$
- (iii) $\cos^{-1}(1-2x^2)$
- (iv) $\sin^{-1}(2x\sqrt{1-x^2})$

★ Likely Exam Question**TYPE 39** Trigonometric Substitution before Differentiation**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

Concept / Formula Used: $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = 1 - 2 \sin^2 \theta$ $\sin 2\theta = 2 \sin \theta \cos \theta$

Solution:

- (i) Let $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$.
 Let $x = \tan \theta \implies \theta = \tan^{-1} x$.
 Using identity $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$:

$$y = \tan^{-1}(\tan 2\theta) = 2\theta = 2 \tan^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 2 \cdot \frac{1}{1+x^2} = \frac{2}{1+x^2}$$

- (ii) Let $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$.
 Let $x = \tan \theta \implies \theta = \tan^{-1} x$.
 Using identity $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$:

$$y = \cos^{-1}(\cos 2\theta) = 2\theta = 2 \tan^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

- (iii) Let $y = \cos^{-1}(1-2x^2)$.
 Let $x = \sin \theta \implies \theta = \sin^{-1} x$.
 Using identity $\cos 2\theta = 1 - 2 \sin^2 \theta$:

$$y = \cos^{-1}(\cos 2\theta) = 2\theta = 2 \sin^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}$$

(iv) Let $y = \sin^{-1}(2x\sqrt{1-x^2})$.

Let $x = \sin \theta \implies \theta = \sin^{-1} x$.

Then $\sqrt{1-x^2} = \cos \theta$.

Using identity $\sin 2\theta = 2 \sin \theta \cos \theta$:

$$y = \sin^{-1}(\sin 2\theta) = 2\theta = 2 \sin^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$$

Final Answer

(i) $\frac{2}{1+x^2}$

(ii) $\frac{2}{1+x^2}$

(iii) $\frac{2}{\sqrt{1-x^2}}$

(iv) $\frac{2}{\sqrt{1-x^2}}$

Question 17

Differentiate the following functions w.r.t. x :

(i) $\sin^{-1}(3x - 4x^3)$

(ii) $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$

(iii) $\cos^{-1}\left(\frac{2x}{1+x^2}\right)$

(iv) $\sec^{-1}\left(\frac{1}{4x^3-3x}\right)$

★ Likely Exam Question

TYPE 39 Trigonometric Substitution before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each function.

Concept / Formula Used: $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$ $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$

Solution:

(i) Let $y = \sin^{-1}(3x - 4x^3)$.

Let $x = \sin \theta \implies \theta = \sin^{-1} x$.

Using triple-angle identity $3 \sin \theta - 4 \sin^3 \theta = \sin 3\theta$:

$$y = \sin^{-1}(\sin 3\theta) = 3\theta = 3 \sin^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 3 \cdot \frac{1}{\sqrt{1-x^2}} = \frac{3}{\sqrt{1-x^2}}$$

(ii) Let $y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$.

Let $x = \tan \theta \implies \theta = \tan^{-1} x$.

Using triple-angle identity $\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = \tan 3\theta$:

$$y = \tan^{-1}(\tan 3\theta) = 3\theta = 3 \tan^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 3 \cdot \frac{1}{1+x^2} = \frac{3}{1+x^2}$$

(iii) Let $y = \cos^{-1} \left(\frac{2x}{1+x^2} \right)$.

Let $x = \tan \theta \implies \theta = \tan^{-1} x$.

Since $\frac{2 \tan \theta}{1+\tan^2 \theta} = \sin 2\theta = \cos \left(\frac{\pi}{2} - 2\theta \right)$:

$$y = \cos^{-1} \left(\cos \left(\frac{\pi}{2} - 2\theta \right) \right) = \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2 \tan^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{2}{1+x^2}$$

(iv) Let $y = \sec^{-1} \left(\frac{1}{4x^3-3x} \right)$.

Using identity $\sec^{-1} \left(\frac{1}{u} \right) = \cos^{-1} u$:

$$y = \cos^{-1}(4x^3 - 3x)$$

Let $x = \cos \theta \implies \theta = \cos^{-1} x$.

Using triple-angle identity $4 \cos^3 \theta - 3 \cos \theta = \cos 3\theta$:

$$y = \cos^{-1}(\cos 3\theta) = 3\theta = 3 \cos^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 3 \cdot \left(-\frac{1}{\sqrt{1-x^2}} \right) = -\frac{3}{\sqrt{1-x^2}}$$

Final Answer

- (i) $\frac{3}{\sqrt{1-x^2}}$
- (ii) $\frac{3}{1+x^2}$
- (iii) $-\frac{2}{1+x^2}$
- (iv) $-\frac{3}{\sqrt{1-x^2}}$

Question 18

Differentiate the following functions w.r.t. x :

- (i) $\sec^{-1} \left(\frac{x^2+1}{x^2-1} \right)$
- (ii) $\sin^{-1} \left(\frac{1-x^2}{1+x^2} \right)$
- (iii) $\tan^{-1}(\sqrt{1+x^2} - x)$
- (iv) $\cot^{-1}(\sqrt{1+x^2} + x)$

TYPE 39 Trigonometric Substitution before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each given function.

Concept / Formula Used: $\csc \theta - \cot \theta = \tan \frac{\theta}{2}$ $\csc \theta + \cot \theta = \cot \frac{\theta}{2}$

Solution:

(i) Let $y = \sec^{-1} \left(\frac{x^2+1}{x^2-1} \right)$.

Using reciprocal identity $\sec^{-1} u = \cos^{-1} \left(\frac{1}{u} \right)$:

$$\begin{aligned} y &= \cos^{-1} \left(\frac{x^2-1}{x^2+1} \right) = \cos^{-1} \left(-\frac{1-x^2}{1+x^2} \right) \\ &= \pi - \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \end{aligned}$$

Let $x = \tan \theta \implies \theta = \tan^{-1} x$.

Using $\frac{1-\tan^2 \theta}{1+\tan^2 \theta} = \cos 2\theta$:

$$y = \pi - \cos^{-1}(\cos 2\theta) = \pi - 2\theta = \pi - 2 \tan^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{2}{1+x^2}$$

(ii) Let $y = \sin^{-1} \left(\frac{1-x^2}{1+x^2} \right)$.

Let $x = \tan \theta \implies \theta = \tan^{-1} x$.

Since $\frac{1-\tan^2 \theta}{1+\tan^2 \theta} = \cos 2\theta = \sin \left(\frac{\pi}{2} - 2\theta \right)$:

$$y = \sin^{-1} \left(\sin \left(\frac{\pi}{2} - 2\theta \right) \right) = \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2 \tan^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{2}{1+x^2}$$

(iii) Let $y = \tan^{-1}(\sqrt{1+x^2} - x)$.

Let $x = \cot \theta \implies \theta = \cot^{-1} x$.

Then $\sqrt{1+x^2} = \sqrt{1+\cot^2 \theta} = \csc \theta$.

Expression inside inverse tan:

$$\begin{aligned} \csc \theta - \cot \theta &= \frac{1 - \cos \theta}{\sin \theta} = \frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \tan \frac{\theta}{2} \\ y &= \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2} = \frac{1}{2} \cot^{-1} x \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{2} \left(-\frac{1}{1+x^2} \right) = -\frac{1}{2(1+x^2)}$$

(iv) Let $y = \cot^{-1}(\sqrt{1+x^2} + x)$.

Let $x = \cot \theta \implies \theta = \cot^{-1} x$.

Then $\sqrt{1+x^2} = \csc \theta$.

Expression inside inverse cot:

$$\begin{aligned} \csc \theta + \cot \theta &= \frac{1 + \cos \theta}{\sin \theta} = \frac{2 \cos^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \cot \frac{\theta}{2} \\ y &= \cot^{-1} \left(\cot \frac{\theta}{2} \right) = \frac{\theta}{2} = \frac{1}{2} \cot^{-1} x \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{1}{2(1+x^2)}$$

Final Answer

- (i) $-\frac{2}{1+x^2}$
- (ii) $-\frac{2}{1+x^2}$
- (iii) $-\frac{1}{2(1+x^2)}$
- (iv) $-\frac{1}{2(1+x^2)}$

Question 19

Differentiate the following functions w.r.t. x :

- (i) $\tan^{-1} \left(\frac{x}{\sqrt{a^2-x^2}} \right)$
- (ii) $\sin^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$
- (iii) $\cos^{-1} \left(\sqrt{\frac{1+x}{2}} \right)$
- (iv) $\sin^{-1} \left(\frac{x}{\sqrt{x^2+a^2}} \right)$

TYPE 39 Trigonometric Substitution before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each expression.

Concept / Formula Used: $\frac{d}{dx} \sin^{-1} \left(\frac{x}{a} \right) = \frac{1}{\sqrt{a^2-x^2}}$ $\frac{d}{dx} \tan^{-1} \left(\frac{x}{a} \right) = \frac{a}{a^2+x^2}$

Solution:

(i) Let $y = \tan^{-1} \left(\frac{x}{\sqrt{a^2-x^2}} \right)$.

Let $x = a \sin \theta \implies \sin \theta = \frac{x}{a} \implies \theta = \sin^{-1} \left(\frac{x}{a} \right)$.

Then $\sqrt{a^2-x^2} = \sqrt{a^2-a^2 \sin^2 \theta} = a \cos \theta$.

Substituting into y :

$$y = \tan^{-1} \left(\frac{a \sin \theta}{a \cos \theta} \right) = \tan^{-1}(\tan \theta) = \theta = \sin^{-1} \left(\frac{x}{a} \right)$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x/a)^2}} \cdot \frac{1}{a} = \frac{1}{\sqrt{a^2-x^2}}$$

(ii) Let $y = \sin^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$.

Let $x = \cot \theta \implies \theta = \cot^{-1} x$.

Then $\sqrt{1+x^2} = \csc \theta \implies \frac{1}{\sqrt{1+x^2}} = \sin \theta$.

Substituting into y :

$$y = \sin^{-1}(\sin \theta) = \theta = \cot^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{1}{1+x^2}$$

(iii) Let $y = \cos^{-1} \left(\sqrt{\frac{1+x}{2}} \right)$.

Let $x = \cos \theta \implies \theta = \cos^{-1} x$.

Using identity $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$:

$$\begin{aligned} \sqrt{\frac{1+\cos \theta}{2}} &= \sqrt{\cos^2 \frac{\theta}{2}} = \cos \frac{\theta}{2} \\ y &= \cos^{-1} \left(\cos \frac{\theta}{2} \right) = \frac{\theta}{2} = \frac{1}{2} \cos^{-1} x \end{aligned}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{2} \left(-\frac{1}{\sqrt{1-x^2}} \right) = -\frac{1}{2\sqrt{1-x^2}}$$

(iv) Let $y = \sin^{-1} \left(\frac{x}{\sqrt{x^2+a^2}} \right)$.

Let $x = a \tan \theta \implies \tan \theta = \frac{x}{a} \implies \theta = \tan^{-1} \left(\frac{x}{a} \right)$.

Then $\sqrt{x^2+a^2} = a \sec \theta \implies \frac{x}{\sqrt{x^2+a^2}} = \frac{a \tan \theta}{a \sec \theta} = \sin \theta$.

Substituting into y :

$$y = \sin^{-1}(\sin \theta) = \theta = \tan^{-1} \left(\frac{x}{a} \right)$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{1+(x/a)^2} \cdot \frac{1}{a} = \frac{a}{a^2+x^2}$$

Final Answer

- (i) $\frac{1}{\sqrt{a^2-x^2}}$
- (ii) $-\frac{1}{1+x^2}$
- (iii) $-\frac{1}{2\sqrt{1-x^2}}$
- (iv) $\frac{a}{a^2+x^2}$

Question 20

Differentiate the following functions w.r.t. x :

- (i) $\sin^{-1} \left(\sqrt{\frac{1+x^2}{2}} \right)$
- (ii) $\tan^{-1} \left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}} \right)$
- (iii) $\sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right)$
- (iv) $\sin^{-1} \left(\frac{2 \sec x}{1+\sec^2 x} \right)$

To Find: The derivative $\frac{dy}{dx}$ for each given expression.

Concept / Formula Used: $1 + \cos 2\theta = 2 \cos^2 \theta$ $1 - \cos 2\theta = 2 \sin^2 \theta$

Solution:

(i) Let $y = \sin^{-1} \left(\sqrt{\frac{1+x^2}{2}} \right)$.

Let $x^2 = \cos 2\theta \implies 2\theta = \cos^{-1}(x^2) \implies \theta = \frac{1}{2} \cos^{-1}(x^2)$.

Then $\frac{1+x^2}{2} = \frac{1+\cos 2\theta}{2} = \cos^2 \theta \implies \sqrt{\frac{1+x^2}{2}} = \cos \theta = \sin \left(\frac{\pi}{2} - \theta \right)$.

$$y = \sin^{-1} \left(\sin \left(\frac{\pi}{2} - \theta \right) \right) = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \frac{1}{2} \cos^{-1}(x^2)$$

Differentiating w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{2} \cdot \left(-\frac{1}{\sqrt{1-(x^2)^2}} \right) \cdot \frac{d}{dx}(x^2) \\ &= \frac{1}{2\sqrt{1-x^4}} \cdot 2x \\ &= \frac{x}{\sqrt{1-x^4}} \end{aligned}$$

(ii) Let $y = \tan^{-1} \left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}} \right)$.

Let $x = \cos 2\theta \implies \theta = \frac{1}{2} \cos^{-1} x$.

Then $\sqrt{1+x} = \sqrt{2} \cos \theta$ and $\sqrt{1-x} = \sqrt{2} \sin \theta$.

$$\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}} = \frac{\sqrt{2} \cos \theta - \sqrt{2} \sin \theta}{\sqrt{2} \cos \theta + \sqrt{2} \sin \theta} = \frac{1 - \tan \theta}{1 + \tan \theta} = \tan \left(\frac{\pi}{4} - \theta \right)$$

$$y = \tan^{-1} \left(\tan \left(\frac{\pi}{4} - \theta \right) \right) = \frac{\pi}{4} - \theta = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{1}{2} \cdot \left(-\frac{1}{\sqrt{1-x^2}} \right) = \frac{1}{2\sqrt{1-x^2}}$$

(iii) Let $y = \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right)$.

Let $x = \cos 2\theta \implies 2\theta = \cos^{-1} x$.

Then $\sqrt{\frac{1+x}{1-x}} = \sqrt{\frac{2 \cos^2 \theta}{2 \sin^2 \theta}} = \cot \theta \implies \cot^{-1}(\cot \theta) = \theta$.

$$y = \sin^2 \theta = \frac{1 - \cos 2\theta}{2} = \frac{1-x}{2} = \frac{1}{2} - \frac{x}{2}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{1}{2}$$

(iv) Let $y = \sin^{-1} \left(\frac{2 \sec x}{1 + \sec^2 x} \right)$.

Let $\sec x = \tan \theta \implies \theta = \tan^{-1}(\sec x)$.

Using identity $\frac{2 \tan \theta}{1 + \tan^2 \theta} = \sin 2\theta$:

$$y = \sin^{-1}(\sin 2\theta) = 2\theta = 2 \tan^{-1}(\sec x)$$

Differentiating w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot \frac{1}{1 + \sec^2 x} \cdot \frac{d}{dx}(\sec x) \\ &= \frac{2 \sec x \tan x}{1 + \sec^2 x} \end{aligned}$$

Final Answer

(i) $\frac{x}{\sqrt{1-x^4}}$

(ii) $\frac{1}{2\sqrt{1-x^2}}$

(iii) $-\frac{1}{2}$

(iv) $\frac{2 \sec x \tan x}{1 + \sec^2 x}$

Question 21

Differentiate the following functions w.r.t. x :

(i) $\cos^{-1} \left(\frac{\sin x + \cos x}{\sqrt{2}} \right), -\frac{\pi}{4} < x < \frac{\pi}{4}$

(ii) $\cos^{-1} \left(\frac{3 \cos x - 4 \sin x}{5} \right)$

(iii) $\sin^{-1} \left(\frac{x + \sqrt{1-x^2}}{\sqrt{2}} \right)$

(iv) $\cos^{-1} \left(\frac{3x + 4\sqrt{1-x^2}}{5} \right)$

★ Likely Exam Question

TYPE 39 Trigonometric Substitution before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each expression.

Concept / Formula Used: $A \cos \theta + B \sin \theta = R \cos(\theta - \alpha)$ $R = \sqrt{A^2 + B^2}$

Solution:

(i) Let $y = \cos^{-1} \left(\frac{\sin x + \cos x}{\sqrt{2}} \right)$.

Expanding inside terms:

$$\frac{\cos x + \sin x}{\sqrt{2}} = \cos x \cos \frac{\pi}{4} + \sin x \sin \frac{\pi}{4} = \cos \left(x - \frac{\pi}{4} \right)$$

For $-\frac{\pi}{4} < x < \frac{\pi}{4}$, we have $-\frac{\pi}{2} < x - \frac{\pi}{4} < 0$.

Since $\cos(-\theta) = \cos \theta$, we write $\cos \left(x - \frac{\pi}{4} \right) = \cos \left(\frac{\pi}{4} - x \right)$ where $0 < \frac{\pi}{4} - x < \frac{\pi}{2}$.

$$y = \cos^{-1} \left(\cos \left(\frac{\pi}{4} - x \right) \right) = \frac{\pi}{4} - x$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -1$$

- (ii) Let $y = \cos^{-1} \left(\frac{3 \cos x - 4 \sin x}{5} \right)$.
 Let $\cos \alpha = \frac{3}{5}$ and $\sin \alpha = \frac{4}{5}$, so $\alpha = \tan^{-1} \frac{4}{3}$ is a constant.

$$\frac{3 \cos x - 4 \sin x}{5} = \cos x \cos \alpha - \sin x \sin \alpha = \cos(x + \alpha)$$

$$y = \cos^{-1}(\cos(x + \alpha)) = x + \alpha$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = 1 + 0 = 1$$

- (iii) Let $y = \sin^{-1} \left(\frac{x + \sqrt{1-x^2}}{\sqrt{2}} \right)$.

$$\text{Let } x = \sin \theta \implies \theta = \sin^{-1} x \implies \sqrt{1-x^2} = \cos \theta.$$

$$\frac{x + \sqrt{1-x^2}}{\sqrt{2}} = \frac{\sin \theta + \cos \theta}{\sqrt{2}} = \sin \theta \cos \frac{\pi}{4} + \cos \theta \sin \frac{\pi}{4} = \sin \left(\theta + \frac{\pi}{4} \right)$$

$$y = \sin^{-1} \left(\sin \left(\theta + \frac{\pi}{4} \right) \right) = \theta + \frac{\pi}{4} = \sin^{-1} x + \frac{\pi}{4}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

- (iv) Let $y = \cos^{-1} \left(\frac{3x + 4\sqrt{1-x^2}}{5} \right)$.

$$\text{Let } x = \cos \theta \implies \theta = \cos^{-1} x \implies \sqrt{1-x^2} = \sin \theta.$$

$$\text{Let } \cos \alpha = \frac{3}{5} \text{ and } \sin \alpha = \frac{4}{5} \implies \alpha = \cos^{-1} \frac{3}{5} \text{ (constant).}$$

$$\frac{3x + 4\sqrt{1-x^2}}{5} = \cos \alpha \cos \theta + \sin \alpha \sin \theta = \cos(\theta - \alpha)$$

$$y = \cos^{-1}(\cos(\theta - \alpha)) = \theta - \alpha = \cos^{-1} x - \alpha$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

Final Answer

- (i) -1
 (ii) 1
 (iii) $\frac{1}{\sqrt{1-x^2}}$
 (iv) $-\frac{1}{\sqrt{1-x^2}}$

Question 22

Differentiate the following functions w.r.t. x :

- (i) $\tan^{-1} \left(\frac{4\sqrt{x}}{1-4x} \right)$
 (ii) $\tan^{-1} \left(\frac{x^{1/3} + a^{1/3}}{1 - x^{1/3}a^{1/3}} \right)$
 (iii) $\sin(2 \sin^{-1} x)$

$$(iv) \cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right)$$

TYPE 39 Trigonometric Substitution before Differentiation

To Find: The derivative $\frac{dy}{dx}$ for each expression.

Concept / Formula Used: $\tan^{-1} \left(\frac{u+v}{1-uv} \right) = \tan^{-1} u + \tan^{-1} v$ $\sin(2\theta) = 2 \sin \theta \cos \theta$

Solution:

(i) Let $y = \tan^{-1} \left(\frac{4\sqrt{x}}{1-4x} \right)$.

Rewriting: $4\sqrt{x} = 2(2\sqrt{x})$ and $4x = (2\sqrt{x})^2$.

Let $u = 2\sqrt{x}$.

$$y = \tan^{-1} \left(\frac{2u}{1-u^2} \right) = 2 \tan^{-1} u = 2 \tan^{-1}(2\sqrt{x})$$

Differentiating w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot \frac{1}{1+(2\sqrt{x})^2} \cdot \frac{d}{dx}(2\sqrt{x}) \\ &= \frac{2}{1+4x} \cdot \left(2 \cdot \frac{1}{2\sqrt{x}} \right) \\ &= \frac{2}{\sqrt{x}(1+4x)} \end{aligned}$$

(ii) Let $y = \tan^{-1} \left(\frac{x^{1/3} + a^{1/3}}{1 - x^{1/3}a^{1/3}} \right)$.

Using identity $\tan^{-1} \left(\frac{u+v}{1-uv} \right) = \tan^{-1} u + \tan^{-1} v$:

$$y = \tan^{-1}(x^{1/3}) + \tan^{-1}(a^{1/3})$$

Differentiating w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1+(x^{1/3})^2} \cdot \frac{d}{dx}(x^{1/3}) + 0 \\ &= \frac{1}{1+x^{2/3}} \cdot \frac{1}{3}x^{-2/3} \\ &= \frac{1}{3x^{2/3}(1+x^{2/3})} \end{aligned}$$

(iii) Let $y = \sin(2 \sin^{-1} x)$.

Let $\sin^{-1} x = \theta \implies \sin \theta = x$ and $\cos \theta = \sqrt{1-x^2}$.

$$y = \sin(2\theta) = 2 \sin \theta \cos \theta = 2x\sqrt{1-x^2}$$

Differentiating w.r.t. x using Product Rule:

$$\begin{aligned}\frac{dy}{dx} &= 2 \cdot \left[\frac{d}{dx}(x) \cdot \sqrt{1-x^2} + x \cdot \frac{d}{dx}(\sqrt{1-x^2}) \right] \\ &= 2 \left[1 \cdot \sqrt{1-x^2} + x \cdot \frac{-2x}{2\sqrt{1-x^2}} \right] \\ &= 2 \left[\sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}} \right] \\ &= 2 \left[\frac{1-x^2-x^2}{\sqrt{1-x^2}} \right] \\ &= \frac{2(1-2x^2)}{\sqrt{1-x^2}}\end{aligned}$$

(iv) Let $y = \cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right)$.

Using $1 + \sin x = \left(\cos \frac{x}{2} + \sin \frac{x}{2} \right)^2$ and $1 - \sin x = \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)^2$:

$$\sqrt{1 + \sin x} = \cos \frac{x}{2} + \sin \frac{x}{2}$$

$$\sqrt{1 - \sin x} = \cos \frac{x}{2} - \sin \frac{x}{2}$$

$$\text{Numerator} = 2 \cos \frac{x}{2}$$

$$\text{Denominator} = 2 \sin \frac{x}{2}$$

$$\frac{\text{Numerator}}{\text{Denominator}} = \frac{2 \cos \frac{x}{2}}{2 \sin \frac{x}{2}} = \cot \frac{x}{2}$$

$$y = \cot^{-1} \left(\cot \frac{x}{2} \right) = \frac{x}{2}$$

Differentiating w.r.t. x :

$$\frac{dy}{dx} = \frac{1}{2}$$

Final Answer

(i) $\frac{2}{\sqrt{x}(1+4x)}$

(ii) $\frac{1}{3x^{2/3}(1+x^{2/3})}$

(iii) $\frac{2(1-2x^2)}{\sqrt{1-x^2}}$

(iv) $\frac{1}{2}$

Question 23

If $y = \sin(\tan^{-1} e^x)$, then find $\frac{dy}{dx}$ at $x = 0$.

TYPE 31 Chain Rule and Evaluation

Given: $y = \sin(\tan^{-1} e^x)$

To Find: $\frac{dy}{dx} \Big|_{x=0}$

Concept / Formula Used: $\sin(\tan^{-1} u) = \frac{u}{\sqrt{1+u^2}}$ $\frac{d}{dx}(e^x) = e^x$

Solution:

Let $u = e^x$.

Using trigonometric simplification $\sin(\tan^{-1} u) = \frac{u}{\sqrt{1+u^2}}$:

$$y = \frac{e^x}{\sqrt{1+(e^x)^2}} = \frac{e^x}{\sqrt{1+e^{2x}}}$$

Differentiating y w.r.t. x using Quotient Rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sqrt{1+e^{2x}} \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}(\sqrt{1+e^{2x}})}{1+e^{2x}} \\ &= \frac{\sqrt{1+e^{2x}} \cdot e^x - e^x \cdot \left(\frac{2e^{2x}}{2\sqrt{1+e^{2x}}}\right)}{1+e^{2x}} \\ &= \frac{\frac{e^x(1+e^{2x}) - e^{3x}}{\sqrt{1+e^{2x}}}}{1+e^{2x}} \\ &= \frac{e^x + e^{3x} - e^{3x}}{(1+e^{2x})^{3/2}} \\ &= \frac{e^x}{(1+e^{2x})^{3/2}} \end{aligned}$$

Now evaluating at $x = 0$:

$$\begin{aligned} e^0 &= 1 \\ \left. \frac{dy}{dx} \right|_{x=0} &= \frac{1}{(1+1^2)^{3/2}} \\ &= \frac{1}{2^{3/2}} \\ &= \frac{1}{2\sqrt{2}} \end{aligned}$$

Final Answer

$$\left. \frac{dy}{dx} \right|_{x=0} = \frac{1}{2\sqrt{2}}$$

Question 24

If $y = \csc(\cot^{-1} x)$, then prove that $\sqrt{1+x^2} \frac{dy}{dx} - x = 0$.

TYPE 32 Verifying Differential Relations

Given: $y = \csc(\cot^{-1} x)$

To Prove: $\sqrt{1+x^2} \frac{dy}{dx} - x = 0$

Concept / Formula Used: $\csc(\cot^{-1} x) = \sqrt{1+x^2}$

Solution:

Given function:

$$y = \csc(\cot^{-1} x)$$

Let $\theta = \cot^{-1} x \implies \cot \theta = x$.

Using the identity $\csc \theta = \sqrt{1 + \cot^2 \theta}$:

$$y = \sqrt{1 + x^2}$$

Differentiating both sides w.r.t. x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} (\sqrt{1 + x^2}) \\ &= \frac{1}{2\sqrt{1 + x^2}} \cdot \frac{d}{dx} (1 + x^2) \\ &= \frac{2x}{2\sqrt{1 + x^2}} \\ &= \frac{x}{\sqrt{1 + x^2}} \end{aligned}$$

Multiplying both sides by $\sqrt{1 + x^2}$:

$$\begin{aligned} \sqrt{1 + x^2} \frac{dy}{dx} &= x \\ \sqrt{1 + x^2} \frac{dy}{dx} - x &= 0 \end{aligned}$$

Hence Proved

LHS = $\sqrt{1 + x^2} \frac{dy}{dx} - x = 0$, hence proved.

5.7 – Implicit Differentiation

Question 1 (i)

Find $\frac{dy}{dx}$ in $x - y = \pi$.

TYPE 40 Implicit Differentiation

Given: The implicit equation $x - y = \pi$.

Concept / Formula Used: $\frac{d}{dx}(x) = 1$ $\frac{d}{dx}(y) = \frac{dy}{dx}$ $\frac{d}{dx}(c) = 0$ c

Solution:

Given equation:

$$x - y = \pi$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(x - y) &= \frac{d}{dx}(\pi) \\ \frac{d}{dx}(x) - \frac{d}{dx}(y) &= 0 \\ 1 - \frac{dy}{dx} &= 0\end{aligned}$$

Rearranging to solve for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = 1$$

Final Answer

$$\frac{dy}{dx} = 1$$

Question 1 (ii)

Find $\frac{dy}{dx}$ in $x^2 + y^2 = 25$.

TYPE 40 Implicit Differentiation

Given: The equation of circle $x^2 + y^2 = 25$.

Concept / Formula Used: $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$

Solution:

Given equation:

$$x^2 + y^2 = 25$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(25) \\ 2x + 2y\frac{dy}{dx} &= 0\end{aligned}$$

Subtracting $2x$ from both sides:

$$2y\frac{dy}{dx} = -2x$$

Dividing both sides by $2y$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{2x}{2y} \\ \frac{dy}{dx} &= -\frac{x}{y}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\frac{x}{y}$$

Question 2 (i)

Find $\frac{dy}{dx}$ in $xy = c^2$.

TYPE 40 Implicit Differentiation

Given: The hyperbola equation $xy = c^2$.

Concept / Formula Used: $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

Solution:

Given equation:

$$xy = c^2$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(xy) &= \frac{d}{dx}(c^2) \\ x \frac{d}{dx}(y) + y \frac{d}{dx}(x) &= 0 \\ x \frac{dy}{dx} + y(1) &= 0 \\ x \frac{dy}{dx} + y &= 0\end{aligned}$$

Subtracting y from both sides:

$$x \frac{dy}{dx} = -y$$

Dividing both sides by x :

$$\frac{dy}{dx} = -\frac{y}{x}$$

Final Answer

$$\frac{dy}{dx} = -\frac{y}{x}$$

Question 2 (ii)

Find $\frac{dy}{dx}$ in $\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$.

TYPE 40 Implicit Differentiation

Given: The equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$.

Concept / Formula Used: $\frac{d}{dx}(x^2) = 2x$ $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$

Solution:

Given equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{1}{a^2} \frac{d}{dx}(x^2) + \frac{1}{b^2} \frac{d}{dx}(y^2) &= \frac{d}{dx}(c^2) \\ \frac{1}{a^2}(2x) + \frac{1}{b^2} \left(2y \frac{dy}{dx} \right) &= 0 \\ \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} &= 0\end{aligned}$$

Subtracting $\frac{2x}{a^2}$ from both sides:

$$\frac{2y}{b^2} \frac{dy}{dx} = -\frac{2x}{a^2}$$

Multiplying both sides by $\frac{b^2}{2y}$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{2x}{a^2} \cdot \frac{b^2}{2y} \\ \frac{dy}{dx} &= -\frac{b^2 x}{a^2 y}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

Question 3 (i)

Find $\frac{dy}{dx}$ in $\sqrt{x} + \sqrt{y} = \sqrt{c}$.

TYPE 40 Implicit Differentiation

Given: The equation $\sqrt{x} + \sqrt{y} = \sqrt{c}$.

Concept / Formula Used: $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$ $\frac{d}{dx}(\sqrt{y}) = \frac{1}{2\sqrt{y}} \frac{dy}{dx}$

Solution:

Given equation:

$$\sqrt{x} + \sqrt{y} = \sqrt{c}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(\sqrt{x}) + \frac{d}{dx}(\sqrt{y}) &= \frac{d}{dx}(\sqrt{c}) \\ \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} &= 0\end{aligned}$$

Subtracting $\frac{1}{2\sqrt{x}}$ from both sides:

$$\frac{1}{2\sqrt{y}} \frac{dy}{dx} = -\frac{1}{2\sqrt{x}}$$

Multiplying both sides by $2\sqrt{y}$:

$$\frac{dy}{dx} = -\frac{2\sqrt{y}}{2\sqrt{x}}$$

$$\frac{dy}{dx} = -\sqrt{\frac{y}{x}}$$

Final Answer

$$\frac{dy}{dx} = -\sqrt{\frac{y}{x}}$$

Question 3 (ii)

Find $\frac{dy}{dx}$ in $x^3 + y^3 = 3axy$.

TYPE 40 Implicit Differentiation

★ Likely Exam Question

Given: The folium equation $x^3 + y^3 = 3axy$.

Concept / Formula Used: $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$ $\frac{d}{dx}(y^3) = 3y^2\frac{dy}{dx}$

Solution:

Given equation:

$$x^3 + y^3 = 3axy$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) &= 3a\frac{d}{dx}(xy) \\ 3x^2 + 3y^2\frac{dy}{dx} &= 3a\left(x\frac{dy}{dx} + y(1)\right) \\ 3x^2 + 3y^2\frac{dy}{dx} &= 3ax\frac{dy}{dx} + 3ay\end{aligned}$$

Dividing both sides by 3:

$$x^2 + y^2\frac{dy}{dx} = ax\frac{dy}{dx} + ay$$

Rearranging terms with $\frac{dy}{dx}$ to the left side:

$$y^2\frac{dy}{dx} - ax\frac{dy}{dx} = ay - x^2$$

Factoring out $\frac{dy}{dx}$:

$$(y^2 - ax)\frac{dy}{dx} = ay - x^2$$

Dividing by $(y^2 - ax)$:

$$\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

Final Answer

$$\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

Question 4 (i)

Find $\frac{dy}{dx}$ in $2x + 3y = \sin x$.

TYPE 40 Implicit Differentiation

Given: The equation $2x + 3y = \sin x$.

Concept / Formula Used: $\frac{d}{dx}(\sin x) = \cos x$

Solution:

Given equation:

$$2x + 3y = \sin x$$

Differentiating both sides with respect to x :

$$\frac{d}{dx}(2x + 3y) = \frac{d}{dx}(\sin x)$$

$$2(1) + 3\frac{dy}{dx} = \cos x$$

$$2 + 3\frac{dy}{dx} = \cos x$$

Subtracting 2 from both sides:

$$3\frac{dy}{dx} = \cos x - 2$$

Dividing by 3:

$$\frac{dy}{dx} = \frac{\cos x - 2}{3}$$

Final Answer

$$\frac{dy}{dx} = \frac{\cos x - 2}{3}$$

Question 4 (ii)

Find $\frac{dy}{dx}$ in $2x + 3y = \sin y$.

TYPE 40 Implicit Differentiation

Given: The equation $2x + 3y = \sin y$.

Concept / Formula Used: $\frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx}$

Solution:

Given equation:

$$2x + 3y = \sin y$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(2x + 3y) &= \frac{d}{dx}(\sin y) \\ 2 + 3\frac{dy}{dx} &= \cos y \frac{dy}{dx}\end{aligned}$$

Rearranging terms containing $\frac{dy}{dx}$ to the right side:

$$\begin{aligned}2 &= \cos y \frac{dy}{dx} - 3\frac{dy}{dx} \\ 2 &= (\cos y - 3)\frac{dy}{dx}\end{aligned}$$

Dividing both sides by $(\cos y - 3)$:

$$\frac{dy}{dx} = \frac{2}{\cos y - 3}$$

Final Answer

$$\frac{dy}{dx} = \frac{2}{\cos y - 3}$$

Question 5 (i)

Find $\frac{dy}{dx}$ when $(x^2 + y^2)^2 = xy$.

TYPE 40 Implicit Differentiation

Given: The equation $(x^2 + y^2)^2 = xy$.

Concept / Formula Used: $\frac{d}{dx}(u^2) = 2u\frac{du}{dx}$ $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$

Solution:

Given equation:

$$(x^2 + y^2)^2 = xy$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}((x^2 + y^2)^2) &= \frac{d}{dx}(xy) \\ 2(x^2 + y^2)\frac{d}{dx}(x^2 + y^2) &= x\frac{dy}{dx} + y(1) \\ 2(x^2 + y^2)\left(2x + 2y\frac{dy}{dx}\right) &= x\frac{dy}{dx} + y\end{aligned}$$

Expanding the left-hand side:

$$\begin{aligned}4x(x^2 + y^2) + 4y(x^2 + y^2)\frac{dy}{dx} &= x\frac{dy}{dx} + y \\ (4x^3 + 4xy^2) + (4x^2y + 4y^3)\frac{dy}{dx} &= x\frac{dy}{dx} + y\end{aligned}$$

Grouping all terms containing $\frac{dy}{dx}$ on the left-hand side:

$$(4x^2y + 4y^3) \frac{dy}{dx} - x \frac{dy}{dx} = y - (4x^3 + 4xy^2)$$

$$(4x^2y + 4y^3 - x) \frac{dy}{dx} = y - 4x^3 - 4xy^2$$

Dividing both sides by $(4x^2y + 4y^3 - x)$:

$$\frac{dy}{dx} = \frac{y - 4x^3 - 4xy^2}{4x^2y + 4y^3 - x}$$

Final Answer

$$\frac{dy}{dx} = \frac{y - 4x^3 - 4xy^2}{4x^2y + 4y^3 - x}$$

Question 5 (ii)

Find $\frac{dy}{dx}$ when $\sin(x + y) = \frac{2}{3}$.

TYPE 40 Implicit Differentiation

Given: The equation $\sin(x + y) = \frac{2}{3}$.

Concept / Formula Used: $\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$

Solution:

Given equation:

$$\sin(x + y) = \frac{2}{3}$$

Differentiating both sides with respect to x :

$$\frac{d}{dx}(\sin(x + y)) = \frac{d}{dx} \left(\frac{2}{3} \right)$$

$$\cos(x + y) \cdot \frac{d}{dx}(x + y) = 0$$

$$\cos(x + y) \left(1 + \frac{dy}{dx} \right) = 0$$

Since $\sin(x + y) = \frac{2}{3}$, $\cos(x + y) = \pm \sqrt{1 - \left(\frac{2}{3} \right)^2} = \pm \frac{\sqrt{5}}{3} \neq 0$. Dividing both sides by $\cos(x + y)$:

$$1 + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -1$$

Final Answer

$$\frac{dy}{dx} = -1$$

Question 6

If $x^{2/3} + y^{2/3} = 2$, find $\frac{dy}{dx}$ at $(1, 1)$.

TYPE 41 Implicit Differentiation at a Point

★ Likely Exam Question

Given: The astroid equation $x^{2/3} + y^{2/3} = 2$ and the point $(x, y) = (1, 1)$.

Concept / Formula Used: $\frac{d}{dx}(x^n) = nx^{n-1}$

Solution:

Given equation:

$$x^{2/3} + y^{2/3} = 2$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^{2/3}) + \frac{d}{dx}(y^{2/3}) &= \frac{d}{dx}(2) \\ \frac{2}{3}x^{2/3-1} + \frac{2}{3}y^{2/3-1}\frac{dy}{dx} &= 0 \\ \frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} &= 0\end{aligned}$$

Multiplying through by $\frac{3}{2}$:

$$x^{-1/3} + y^{-1/3}\frac{dy}{dx} = 0$$

Isolating $\frac{dy}{dx}$:

$$\begin{aligned}y^{-1/3}\frac{dy}{dx} &= -x^{-1/3} \\ \frac{dy}{dx} &= -\frac{x^{-1/3}}{y^{-1/3}} \\ \frac{dy}{dx} &= -\left(\frac{y}{x}\right)^{1/3}\end{aligned}$$

Substituting $(x, y) = (1, 1)$:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{(1,1)} &= -\left(\frac{1}{1}\right)^{1/3} \\ &= -1\end{aligned}$$

Final Answer

$$\left.\frac{dy}{dx}\right|_{(1,1)} = -1$$

Question 7 (i)

Use implicit differentiation to verify that $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$, when $y^2 = 4ax$.

TYPE 42 Verifying Reciprocal Derivative Relation

To Prove: $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ for the parabola $y^2 = 4ax$.

Concept / Formula Used: x y

Solution:

Given equation:

$$y^2 = 4ax$$

Differentiating with respect to x :

$$\begin{aligned}\frac{d}{dx}(y^2) &= \frac{d}{dx}(4ax) \\ 2y \frac{dy}{dx} &= 4a \\ \frac{dy}{dx} &= \frac{4a}{2y} = \frac{2a}{y} \quad \text{--- (Equation 1)}\end{aligned}$$

Differentiating $y^2 = 4ax$ with respect to y :

$$\begin{aligned}\frac{d}{dy}(y^2) &= \frac{d}{dy}(4ax) \\ 2y &= 4a \frac{dx}{dy} \\ \frac{dx}{dy} &= \frac{2y}{4a} = \frac{y}{2a} \quad \text{--- (Equation 2)}\end{aligned}$$

Multiplying Equation 1 and Equation 2:

$$\begin{aligned}\frac{dy}{dx} \cdot \frac{dx}{dy} &= \left(\frac{2a}{y}\right) \cdot \left(\frac{y}{2a}\right) \\ &= 1\end{aligned}$$

Hence Proved

$\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ is verified.

Question 7 (ii)

Use implicit differentiation to verify that $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$, when $x^4 + y^4 = 3x^2y^2$.

TYPE 42 Verifying Reciprocal Derivative Relation

To Prove: $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ for $x^4 + y^4 = 3x^2y^2$.

Concept / Formula Used: $x \quad y$

Solution:

Given equation:

$$x^4 + y^4 = 3x^2y^2$$

First, differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^4 + y^4) &= \frac{d}{dx}(3x^2y^2) \\ 4x^3 + 4y^3 \frac{dy}{dx} &= 3 \left(x^2 \cdot 2y \frac{dy}{dx} + y^2 \cdot 2x \right) \\ 4x^3 + 4y^3 \frac{dy}{dx} &= 6x^2y \frac{dy}{dx} + 6xy^2\end{aligned}$$

Grouping terms with $\frac{dy}{dx}$:

$$\begin{aligned}(4y^3 - 6x^2y) \frac{dy}{dx} &= 6xy^2 - 4x^3 \\ 2y(2y^2 - 3x^2) \frac{dy}{dx} &= 2x(3y^2 - 2x^2) \\ \frac{dy}{dx} &= \frac{x(3y^2 - 2x^2)}{y(2y^2 - 3x^2)} \quad \text{--- (Equation 1)}\end{aligned}$$

Next, differentiating $x^4 + y^4 = 3x^2y^2$ with respect to y :

$$\begin{aligned}\frac{d}{dy}(x^4 + y^4) &= \frac{d}{dy}(3x^2y^2) \\ 4x^3 \frac{dx}{dy} + 4y^3 &= 3 \left(x^2 \cdot 2y + y^2 \cdot 2x \frac{dx}{dy} \right) \\ 4x^3 \frac{dx}{dy} + 4y^3 &= 6x^2y + 6xy^2 \frac{dx}{dy}\end{aligned}$$

Grouping terms with $\frac{dx}{dy}$:

$$\begin{aligned}(4x^3 - 6xy^2) \frac{dx}{dy} &= 6x^2y - 4y^3 \\ 2x(2x^2 - 3y^2) \frac{dx}{dy} &= 2y(3x^2 - 2y^2) \\ \frac{dx}{dy} &= \frac{y(3x^2 - 2y^2)}{x(2x^2 - 3y^2)} \quad \text{--- (Equation 2)}\end{aligned}$$

Note that $(3x^2 - 2y^2) = -(2y^2 - 3x^2)$ and $(2x^2 - 3y^2) = -(3y^2 - 2x^2)$. Thus:

$$\frac{dx}{dy} = \frac{y(2y^2 - 3x^2)}{x(3y^2 - 2x^2)}$$

Multiplying Equation 1 and Equation 2:

$$\begin{aligned}\frac{dy}{dx} \cdot \frac{dx}{dy} &= \left[\frac{x(3y^2 - 2x^2)}{y(2y^2 - 3x^2)} \right] \cdot \left[\frac{y(2y^2 - 3x^2)}{x(3y^2 - 2x^2)} \right] \\ &= 1\end{aligned}$$

Hence Proved

$$\frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \text{ is verified.}$$

Question 8 (i)

Find $\frac{dy}{dx}$, when $y + \sin y = \cos x$.

TYPE 40 Implicit Differentiation

Given: The equation $y + \sin y = \cos x$.

Concept / Formula Used: $\frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx}$ $\frac{d}{dx}(\cos x) = -\sin x$

Solution:

Given equation:

$$y + \sin y = \cos x$$

Differentiating both sides with respect to x :

$$\begin{aligned} \frac{d}{dx}(y) + \frac{d}{dx}(\sin y) &= \frac{d}{dx}(\cos x) \\ \frac{dy}{dx} + \cos y \frac{dy}{dx} &= -\sin x \end{aligned}$$

Factoring out $\frac{dy}{dx}$:

$$(1 + \cos y) \frac{dy}{dx} = -\sin x$$

Dividing both sides by $(1 + \cos y)$:

$$\frac{dy}{dx} = -\frac{\sin x}{1 + \cos y}$$

Final Answer

$$\frac{dy}{dx} = -\frac{\sin x}{1 + \cos y}$$

Question 8 (ii)

Find $\frac{dy}{dx}$, when $ax + by^2 = \cos y$.

TYPE 40 Implicit Differentiation

Given: The equation $ax + by^2 = \cos y$.

Concept / Formula Used: $\frac{d}{dx}(by^2) = 2by \frac{dy}{dx}$ $\frac{d}{dx}(\cos y) = -\sin y \frac{dy}{dx}$

Solution:

Given equation:

$$ax + by^2 = \cos y$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(ax + by^2) &= \frac{d}{dx}(\cos y) \\ a(1) + 2by \frac{dy}{dx} &= -\sin y \frac{dy}{dx} \\ a + 2by \frac{dy}{dx} &= -\sin y \frac{dy}{dx}\end{aligned}$$

Rearranging terms with $\frac{dy}{dx}$ to one side:

$$\begin{aligned}2by \frac{dy}{dx} + \sin y \frac{dy}{dx} &= -a \\ (2by + \sin y) \frac{dy}{dx} &= -a\end{aligned}$$

Dividing both sides by $(2by + \sin y)$:

$$\frac{dy}{dx} = -\frac{a}{2by + \sin y} = \frac{a}{-\sin y - 2by}$$

Final Answer

$$\frac{dy}{dx} = \frac{a}{-\sin y - 2by}$$

Question 8 (iii)

Find $\frac{dy}{dx}$, when $y \sec x + \tan x + x^2 y = 0$.

TYPE 40 Implicit Differentiation

Given: The equation $y \sec x + \tan x + x^2 y = 0$.

Concept / Formula Used: $\frac{d}{dx}(\sec x) = \sec x \tan x$ $\frac{d}{dx}(\tan x) = \sec^2 x$

Solution:

Given equation:

$$y \sec x + \tan x + x^2 y = 0$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(y \sec x) + \frac{d}{dx}(\tan x) + \frac{d}{dx}(x^2 y) &= 0 \\ \left(y \cdot \sec x \tan x + \sec x \frac{dy}{dx}\right) + \sec^2 x + \left(x^2 \frac{dy}{dx} + y \cdot 2x\right) &= 0\end{aligned}$$

Grouping terms containing $\frac{dy}{dx}$:

$$(\sec x + x^2) \frac{dy}{dx} + y \sec x \tan x + \sec^2 x + 2xy = 0$$

Transferring non- $\frac{dy}{dx}$ terms to the right side:

$$(\sec x + x^2) \frac{dy}{dx} = -(2xy + \sec^2 x + y \sec x \tan x)$$

Dividing both sides by $(\sec x + x^2)$:

$$\frac{dy}{dx} = -\frac{2xy + \sec^2 x + y \sec x \tan x}{x^2 + \sec x}$$

Final Answer

$$\frac{dy}{dx} = -\frac{2xy + \sec^2 x + y \sec x \tan x}{x^2 + \sec x}$$

Question 8 (iv)

Find $\frac{dy}{dx}$, when $\sin^2 x + \cos^2 y = 1$.

TYPE 40 Implicit Differentiation

Given: The equation $\sin^2 x + \cos^2 y = 1$.

Concept / Formula Used: $\sin 2\theta = 2 \sin \theta \cos \theta$

Solution:

Given equation:

$$\sin^2 x + \cos^2 y = 1$$

Differentiating both sides with respect to x :

$$\begin{aligned} \frac{d}{dx}(\sin^2 x) + \frac{d}{dx}(\cos^2 y) &= \frac{d}{dx}(1) \\ 2 \sin x \frac{d}{dx}(\sin x) + 2 \cos y \frac{d}{dx}(\cos y) &= 0 \\ 2 \sin x \cos x + 2 \cos y \left(-\sin y \frac{dy}{dx} \right) &= 0 \\ 2 \sin x \cos x - 2 \sin y \cos y \frac{dy}{dx} &= 0 \end{aligned}$$

Applying $\sin 2\theta = 2 \sin \theta \cos \theta$:

$$\sin 2x - \sin 2y \frac{dy}{dx} = 0$$

Isolating $\frac{dy}{dx}$:

$$\begin{aligned} \sin 2y \frac{dy}{dx} &= \sin 2x \\ \frac{dy}{dx} &= \frac{\sin 2x}{\sin 2y} \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \frac{\sin 2x}{\sin 2y}$$

Question 8 (v)

Find $\frac{dy}{dx}$, when $xy + y^2 = \tan x + y$.

TYPE 40 Implicit Differentiation

Given: The equation $xy + y^2 = \tan x + y$.

Concept / Formula Used: $\frac{d}{dx}(xy) = x \frac{dy}{dx} + y$ $\frac{d}{dx}(\tan x) = \sec^2 x$

Solution:

Given equation:

$$xy + y^2 = \tan x + y$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(xy) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(\tan x) + \frac{d}{dx}(y) \\ \left(x \frac{dy}{dx} + y(1)\right) + 2y \frac{dy}{dx} &= \sec^2 x + \frac{dy}{dx} \\ x \frac{dy}{dx} + y + 2y \frac{dy}{dx} &= \sec^2 x + \frac{dy}{dx}\end{aligned}$$

Grouping all terms containing $\frac{dy}{dx}$ on the left side:

$$\begin{aligned}x \frac{dy}{dx} + 2y \frac{dy}{dx} - \frac{dy}{dx} &= \sec^2 x - y \\ (x + 2y - 1) \frac{dy}{dx} &= \sec^2 x - y\end{aligned}$$

Dividing both sides by $(x + 2y - 1)$:

$$\frac{dy}{dx} = \frac{\sec^2 x - y}{x + 2y - 1}$$

Final Answer

$$\frac{dy}{dx} = \frac{\sec^2 x - y}{x + 2y - 1}$$

Question 8 (vi)

Find $\frac{dy}{dx}$, when $y = \tan(x + y)$.

TYPE 40 Implicit Differentiation

Given: The equation $y = \tan(x + y)$.

Concept / Formula Used: $\frac{d}{dx}(\tan u) = \sec^2 u \frac{du}{dx}$ $1 - \sec^2 u = -\tan^2 u$

Solution:

Given equation:

$$y = \tan(x + y)$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(\tan(x + y)) \\ \frac{dy}{dx} &= \sec^2(x + y) \cdot \frac{d}{dx}(x + y) \\ \frac{dy}{dx} &= \sec^2(x + y) \left(1 + \frac{dy}{dx}\right) \\ \frac{dy}{dx} &= \sec^2(x + y) + \sec^2(x + y) \frac{dy}{dx}\end{aligned}$$

Grouping terms containing $\frac{dy}{dx}$ on the left side:

$$\begin{aligned}\frac{dy}{dx} - \sec^2(x + y) \frac{dy}{dx} &= \sec^2(x + y) \\ (1 - \sec^2(x + y)) \frac{dy}{dx} &= \sec^2(x + y)\end{aligned}$$

Using $1 - \sec^2(x + y) = -\tan^2(x + y)$:

$$-\tan^2(x + y) \frac{dy}{dx} = \sec^2(x + y)$$

Dividing both sides by $-\tan^2(x + y)$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{\sec^2(x + y)}{\tan^2(x + y)} \\ &= -\frac{\frac{1}{\cos^2(x + y)}}{\frac{\sin^2(x + y)}{\cos^2(x + y)}} \\ &= -\frac{1}{\sin^2(x + y)} \\ &= -\csc^2(x + y)\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\csc^2(x + y)$$

Question 8 (vii)

Find $\frac{dy}{dx}$, when $\sin(xy) + \frac{x}{y} = x^2 - y$.

TYPE 40 Implicit Differentiation

Given: The equation $\sin(xy) + \frac{x}{y} = x^2 - y$.

Concept / Formula Used: $\frac{d}{dx} \left(\frac{x}{y} \right) = \frac{y(1) - x \frac{dy}{dx}}{y^2}$

Solution:

Given equation:

$$\sin(xy) + \frac{x}{y} = x^2 - y$$

Differentiating both sides with respect to x :

$$\begin{aligned} \frac{d}{dx}(\sin(xy)) + \frac{d}{dx} \left(\frac{x}{y} \right) &= \frac{d}{dx}(x^2) - \frac{d}{dx}(y) \\ \cos(xy) \left(x \frac{dy}{dx} + y \right) + \frac{y(1) - x \frac{dy}{dx}}{y^2} &= 2x - \frac{dy}{dx} \\ x \cos(xy) \frac{dy}{dx} + y \cos(xy) + \frac{1}{y} - \frac{x}{y^2} \frac{dy}{dx} &= 2x - \frac{dy}{dx} \end{aligned}$$

Multiplying the entire equation by y^2 :

$$xy^2 \cos(xy) \frac{dy}{dx} + y^3 \cos(xy) + y - x \frac{dy}{dx} = 2xy^2 - y^2 \frac{dy}{dx}$$

Grouping terms containing $\frac{dy}{dx}$ on the left-hand side:

$$(xy^2 \cos(xy) - x + y^2) \frac{dy}{dx} = 2xy^2 - y - y^3 \cos(xy)$$

Dividing by $(xy^2 \cos(xy) - x + y^2)$:

$$\frac{dy}{dx} = \frac{2xy^2 - y - y^3 \cos(xy)}{xy^2 \cos(xy) - x + y^2}$$

Final Answer

$$\frac{dy}{dx} = \frac{2xy^2 - y - y^3 \cos(xy)}{xy^2 \cos(xy) - x + y^2}$$

Question 8 (viii)

Find $\frac{dy}{dx}$, when $\sec(x + y) = xy$.

TYPE 40 Implicit Differentiation

Given: The equation $\sec(x + y) = xy$.

Concept / Formula Used: $\frac{d}{dx}(\sec u) = \sec u \tan u \frac{du}{dx}$

Solution:

Given equation:

$$\sec(x + y) = xy$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(\sec(x+y)) &= \frac{d}{dx}(xy) \\ \sec(x+y) \tan(x+y) \cdot \left(1 + \frac{dy}{dx}\right) &= x \frac{dy}{dx} + y(1) \\ \sec(x+y) \tan(x+y) + \sec(x+y) \tan(x+y) \frac{dy}{dx} &= x \frac{dy}{dx} + y\end{aligned}$$

Grouping terms containing $\frac{dy}{dx}$ on the left-hand side:

$$(\sec(x+y) \tan(x+y) - x) \frac{dy}{dx} = y - \sec(x+y) \tan(x+y)$$

Dividing both sides by $(\sec(x+y) \tan(x+y) - x)$:

$$\frac{dy}{dx} = \frac{y - \sec(x+y) \tan(x+y)}{\sec(x+y) \tan(x+y) - x}$$

Final Answer

$$\frac{dy}{dx} = \frac{y - \sec(x+y) \tan(x+y)}{\sec(x+y) \tan(x+y) - x}$$

Question 9

If $\sec\left(\frac{x+y}{x-y}\right) = a$, prove that $\frac{dy}{dx} = \frac{y}{x}$.

TYPE 43 Simplification before Implicit Differentiation

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{y}{x}$ given $\sec\left(\frac{x+y}{x-y}\right) = a$.

Concept / Formula Used: $f(x, y) = \text{constant} \quad \frac{x+y}{x-y} = \text{constant}$

Solution:

Given equation:

$$\sec\left(\frac{x+y}{x-y}\right) = a$$

Taking inverse secant on both sides:

$$\frac{x+y}{x-y} = \sec^{-1}(a)$$

Since a is a constant, $\sec^{-1}(a)$ is also a constant. Let $\sec^{-1}(a) = k$.

$$\frac{x+y}{x-y} = k$$

Cross-multiplying:

$$x + y = k(x - y)$$

$$x + y = kx - ky$$

Rearranging to isolate y terms on one side:

$$y + ky = kx - x$$

$$y(1 + k) = x(k - 1)$$

$$\frac{y}{x} = \frac{k - 1}{k + 1}$$

Let the constant $\frac{k - 1}{k + 1} = m$. Thus:

$$y = mx$$

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = m$$

Since $m = \frac{y}{x}$:

$$\frac{dy}{dx} = \frac{y}{x}$$

Hence Proved

Hence proved.

Question 10 (i)

If $y = x \sin(a + y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin(a + y) - y \cos(a + y)}$.

TYPE 44 Differentiation by Isolating Variables

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin(a + y) - y \cos(a + y)}$ given $y = x \sin(a + y)$.

Concept / Formula Used: $\frac{d}{dy} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$ $\frac{dy}{dx} = \frac{1}{dx/dy}$

Solution:

Given relation:

$$y = x \sin(a + y)$$

Isolating x in terms of y :

$$x = \frac{y}{\sin(a + y)}$$

Differentiating x with respect to y using the Quotient Rule:

$$\begin{aligned}\frac{dx}{dy} &= \frac{\sin(a+y) \cdot \frac{d}{dy}(y) - y \cdot \frac{d}{dy}(\sin(a+y))}{\sin^2(a+y)} \\ \frac{dx}{dy} &= \frac{\sin(a+y) \cdot 1 - y \cdot \cos(a+y)}{\sin^2(a+y)} \\ \frac{dx}{dy} &= \frac{\sin(a+y) - y \cos(a+y)}{\sin^2(a+y)}\end{aligned}$$

Taking the reciprocal to find $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{\sin^2(a+y)}{\sin(a+y) - y \cos(a+y)}$$

Hence Proved

Hence proved.

Question 10 (ii)

If $\sin x = y \sin(x+a)$, prove that $\frac{dy}{dx} = \frac{\sin a}{\sin^2(x+a)}$.

TYPE 44 Differentiation by Isolating Variables

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{\sin a}{\sin^2(x+a)}$ given $\sin x = y \sin(x+a)$.

Concept / Formula Used: $\sin(A-B) = \sin A \cos B - \cos A \sin B$

Solution:

Given equation:

$$\sin x = y \sin(x+a)$$

Isolating y in terms of x :

$$y = \frac{\sin x}{\sin(x+a)}$$

Differentiating y with respect to x using the Quotient Rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\sin(x+a) \cdot \frac{d}{dx}(\sin x) - \sin x \cdot \frac{d}{dx}(\sin(x+a))}{\sin^2(x+a)} \\ \frac{dy}{dx} &= \frac{\sin(x+a) \cos x - \sin x \cos(x+a)}{\sin^2(x+a)}\end{aligned}$$

Applying $\sin(A-B) = \sin A \cos B - \cos A \sin B$ with $A = x+a$ and $B = x$:

$$\begin{aligned}\sin(x+a) \cos x - \sin x \cos(x+a) &= \sin((x+a) - x) \\ &= \sin a\end{aligned}$$

Substituting this into the numerator:

$$\frac{dy}{dx} = \frac{\sin a}{\sin^2(x+a)}$$

Hence Proved

Hence proved.

Question 10 (iii)

If $\sin y = x \sin(a+y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$.

TYPE 44 Differentiation by Isolating Variables

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$ given $\sin y = x \sin(a+y)$.

Concept / Formula Used: $\sin(A-B) = \sin A \cos B - \cos A \sin B$

Solution:

Given equation:

$$\sin y = x \sin(a+y)$$

Isolating x in terms of y :

$$x = \frac{\sin y}{\sin(a+y)}$$

Differentiating x with respect to y using the Quotient Rule:

$$\begin{aligned} \frac{dx}{dy} &= \frac{\sin(a+y) \cdot \frac{d}{dy}(\sin y) - \sin y \cdot \frac{d}{dy}(\sin(a+y))}{\sin^2(a+y)} \\ \frac{dx}{dy} &= \frac{\sin(a+y) \cos y - \sin y \cos(a+y)}{\sin^2(a+y)} \end{aligned}$$

Applying $\sin(A-B) = \sin A \cos B - \cos A \sin B$ with $A = a+y$ and $B = y$:

$$\begin{aligned} \sin(a+y) \cos y - \cos(a+y) \sin y &= \sin((a+y) - y) \\ &= \sin a \end{aligned}$$

Thus:

$$\frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)}$$

Taking the reciprocal to get $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{\sin^2(a+y)}{\sin a}$$

Hence Proved

Hence proved.

Question 11

If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$, prove that $\frac{dy}{dx} = \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$.

TYPE 39 Trigonometric Substitution before Differentiation

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$ given $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$.

Concept / Formula Used: $x^3 = \sin \theta$ $y^3 = \sin \phi$ $\cos \theta + \cos \phi = 2 \cos \left(\frac{\theta+\phi}{2}\right) \cos \left(\frac{\theta-\phi}{2}\right)$

Solution:

Let $x^3 = \sin \theta \implies \theta = \sin^{-1}(x^3)$

and $y^3 = \sin \phi \implies \phi = \sin^{-1}(y^3)$.

Substituting these into the given equation:

$$\begin{aligned}\sqrt{1-\sin^2 \theta} + \sqrt{1-\sin^2 \phi} &= a^3(\sin \theta - \sin \phi) \\ \cos \theta + \cos \phi &= a^3(\sin \theta - \sin \phi)\end{aligned}$$

Applying the sum-to-product formulas:

$$2 \cos \left(\frac{\theta+\phi}{2}\right) \cos \left(\frac{\theta-\phi}{2}\right) = a^3 \cdot 2 \cos \left(\frac{\theta+\phi}{2}\right) \sin \left(\frac{\theta-\phi}{2}\right)$$

Cancelling $2 \cos \left(\frac{\theta+\phi}{2}\right)$ on both sides:

$$\begin{aligned}\cos \left(\frac{\theta-\phi}{2}\right) &= a^3 \sin \left(\frac{\theta-\phi}{2}\right) \\ \cot \left(\frac{\theta-\phi}{2}\right) &= a^3 \\ \frac{\theta-\phi}{2} &= \cot^{-1}(a^3) \\ \theta - \phi &= 2 \cot^{-1}(a^3)\end{aligned}$$

Substituting back $\theta = \sin^{-1}(x^3)$ and $\phi = \sin^{-1}(y^3)$:

$$\sin^{-1}(x^3) - \sin^{-1}(y^3) = 2 \cot^{-1}(a^3)$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(\sin^{-1}(x^3)) - \frac{d}{dx}(\sin^{-1}(y^3)) &= \frac{d}{dx}(2 \cot^{-1}(a^3)) \\ \frac{1}{\sqrt{1-(x^3)^2}} \cdot 3x^2 - \frac{1}{\sqrt{1-(y^3)^2}} \cdot 3y^2 \frac{dy}{dx} &= 0 \\ \frac{3x^2}{\sqrt{1-x^6}} - \frac{3y^2}{\sqrt{1-y^6}} \frac{dy}{dx} &= 0\end{aligned}$$

Isolating $\frac{dy}{dx}$:

$$\begin{aligned}\frac{3y^2}{\sqrt{1-y^6}} \frac{dy}{dx} &= \frac{3x^2}{\sqrt{1-x^6}} \\ \frac{dy}{dx} &= \frac{3x^2}{\sqrt{1-x^6}} \cdot \frac{\sqrt{1-y^6}}{3y^2} \\ \frac{dy}{dx} &= \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}\end{aligned}$$

Hence Proved

Hence proved.

Question 12 (i)

If $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$, prove that $(2y - 1)\frac{dy}{dx} = 1$.

TYPE 45 Differentiation of Infinite Recursive Functions

★ Likely Exam Question

To Prove: $(2y - 1)\frac{dy}{dx} = 1$ given $y = \sqrt{x + \sqrt{x + \dots \infty}}$.

Concept / Formula Used: $y = \sqrt{x + y}$

Solution:

Given infinite radical series:

$$y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$$

Using self-similarity, rewrite as:

$$y = \sqrt{x + y}$$

Squaring both sides:

$$y^2 = x + y$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(y^2) &= \frac{d}{dx}(x + y) \\ 2y \frac{dy}{dx} &= 1 + \frac{dy}{dx}\end{aligned}$$

Subtracting $\frac{dy}{dx}$ from both sides:

$$\begin{aligned}2y \frac{dy}{dx} - \frac{dy}{dx} &= 1 \\ (2y - 1) \frac{dy}{dx} &= 1\end{aligned}$$

Hence Proved

Hence proved.

Question 12 (ii)

If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$, prove that $(2y - 1)\frac{dy}{dx} = \cos x$.

TYPE 45 Differentiation of Infinite Recursive Functions

★ Likely Exam Question

To Prove: $(2y - 1)\frac{dy}{dx} = \cos x$ given $y = \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}$.

Concept / Formula Used: $y = \sqrt{\sin x + y}$

Solution:

Given infinite radical series:

$$y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$$

Using self-similarity, rewrite as:

$$y = \sqrt{\sin x + y}$$

Squaring both sides:

$$y^2 = \sin x + y$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(y^2) &= \frac{d}{dx}(\sin x + y) \\ 2y\frac{dy}{dx} &= \cos x + \frac{dy}{dx}\end{aligned}$$

Subtracting $\frac{dy}{dx}$ from both sides:

$$\begin{aligned}2y\frac{dy}{dx} - \frac{dy}{dx} &= \cos x \\ (2y - 1)\frac{dy}{dx} &= \cos x\end{aligned}$$

Hence Proved

Hence proved.

Question 12 (iii)

If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots \infty}}}$, prove that $(2y - 1)\frac{dy}{dx} = \sec^2 x$.

TYPE 45 Differentiation of Infinite Recursive Functions

★ Likely Exam Question

To Prove: $(2y - 1)\frac{dy}{dx} = \sec^2 x$ given $y = \sqrt{\tan x + \sqrt{\tan x + \dots \infty}}$.

Concept / Formula Used: $y = \sqrt{\tan x + y}$

Solution:

Given infinite radical series:

$$y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots \infty}}}$$

Using self-similarity, rewrite as:

$$y = \sqrt{\tan x + y}$$

Squaring both sides:

$$y^2 = \tan x + y$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}(y^2) &= \frac{d}{dx}(\tan x + y) \\ 2y \frac{dy}{dx} &= \sec^2 x + \frac{dy}{dx}\end{aligned}$$

Subtracting $\frac{dy}{dx}$ from both sides:

$$\begin{aligned}2y \frac{dy}{dx} - \frac{dy}{dx} &= \sec^2 x \\ (2y - 1) \frac{dy}{dx} &= \sec^2 x\end{aligned}$$

Hence Proved

Hence proved.

Question 13

If $y = x + \frac{1}{x + \frac{1}{x + \dots \text{ to } \infty}}$, prove that $\frac{dy}{dx} = \frac{y^2}{1 + y^2}$.

TYPE 45 Differentiation of Infinite Recursive Functions

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{y^2}{1 + y^2}$ given $y = x + \frac{1}{x + \frac{1}{x + \dots \text{ to } \infty}}$.

Concept / Formula Used: $y = x + \frac{1}{y}$

Solution:

Given infinite continued fraction:

$$y = x + \frac{1}{x + \frac{1}{x + \dots \text{ to } \infty}}$$

Using self-similarity, rewrite as:

$$y = x + \frac{1}{y}$$

Isolating x in terms of y :

$$x = y - \frac{1}{y}$$

Differentiating x with respect to y :

$$\frac{dx}{dy} = \frac{d}{dy} (y - y^{-1})$$

$$\frac{dx}{dy} = 1 - (-1)y^{-2}$$

$$\frac{dx}{dy} = 1 + \frac{1}{y^2}$$

$$\frac{dx}{dy} = \frac{y^2 + 1}{y^2}$$

Taking the reciprocal to evaluate $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{y^2}{1 + y^2}$$

Hence Proved

Hence proved.

5.8 – Differentiation of Exponential and Logarithmic Functions

Question 1

Differentiate the following functions w.r.t. x :

(i) $\log x + \frac{1}{x}$

(ii) $10^x + \frac{1}{3}e^x - 2\log x$

TYPE 20 Sum and Product Rules

To Find: The derivative of (i) $y = \log x + \frac{1}{x}$ and (ii) $y = 10^x + \frac{1}{3}e^x - 2\log x$ with respect to x .

Concept / Formula Used: $\frac{d}{dx}(x^n) = nx^{n-1}$ $\frac{d}{dx}(\log x) = \frac{1}{x}$ $\frac{d}{dx}(a^x) = a^x \log a$ $\frac{d}{dx}(e^x) = e^x$

Solution:

Part (i): Let $y = \log x + \frac{1}{x} = \log x + x^{-1}$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(\log x) + \frac{d}{dx}(x^{-1}) \\ &= \frac{1}{x} + (-1)x^{-2} \\ &= \frac{1}{x} - \frac{1}{x^2}\end{aligned}$$

Part (ii): Let $y = 10^x + \frac{1}{3}e^x - 2\log x$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(10^x) + \frac{1}{3}\frac{d}{dx}(e^x) - 2\frac{d}{dx}(\log x) \\ &= 10^x \log 10 + \frac{1}{3}e^x - 2 \cdot \frac{1}{x} \\ &= 10^x \log 10 + \frac{1}{3}e^x - \frac{2}{x}\end{aligned}$$

Final Answer

(i) $\frac{dy}{dx} = \frac{1}{x} - \frac{1}{x^2}$

(ii) $\frac{dy}{dx} = 10^x \log 10 + \frac{1}{3}e^x - \frac{2}{x}$

Question 2

Differentiate the following functions w.r.t. x :

(i) e^{-x}

(ii) $\sin(\log x), \quad x > 0$

TYPE 19 Chain Rule for Differentiation

To Find: The derivative of (i) $y = e^{-x}$ and (ii) $y = \sin(\log x)$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$

Solution:

Part (i): Let $y = e^{-x}$.

Differentiating w.r.t. x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= e^{-x} \cdot \frac{d}{dx}(-x) \\ &= e^{-x} \cdot (-1) \\ &= -e^{-x}\end{aligned}$$

Part (ii): Let $y = \sin(\log x)$.

Differentiating w.r.t. x using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \cos(\log x) \cdot \frac{d}{dx}(\log x) \\ &= \cos(\log x) \cdot \frac{1}{x} \\ &= \frac{\cos(\log x)}{x}\end{aligned}$$

Final Answer

$$\begin{aligned}\text{(i)} \quad \frac{dy}{dx} &= -e^{-x} \\ \text{(ii)} \quad \frac{dy}{dx} &= \frac{\cos(\log x)}{x}\end{aligned}$$

Question 3

Differentiate the following functions w.r.t. x :

- (i) $e^{\cos x}$
- (ii) $e^{\sin^{-1} x}$

TYPE 19 Chain Rule for Differentiation

To Find: The derivative of (i) $y = e^{\cos x}$ and (ii) $y = e^{\sin^{-1} x}$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$ $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

Solution:

Part (i): Let $y = e^{\cos x}$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= e^{\cos x} \cdot \frac{d}{dx}(\cos x) \\ &= e^{\cos x} \cdot (-\sin x) \\ &= -\sin x \cdot e^{\cos x}\end{aligned}$$

Part (ii): Let $y = e^{\sin^{-1} x}$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= e^{\sin^{-1} x} \cdot \frac{d}{dx}(\sin^{-1} x) \\ &= e^{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}\end{aligned}$$

Final Answer

$$\begin{aligned}\text{(i)} \quad \frac{dy}{dx} &= -\sin x \cdot e^{\cos x} \\ \text{(ii)} \quad \frac{dy}{dx} &= \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}\end{aligned}$$

Question 4

Differentiate the following functions w.r.t. x :

- (i) e^{x^3}
 (ii) $\sqrt{e^{\sqrt{x}}}$, $x > 0$

TYPE 22 Repeated Chain Rule

To Find: The derivative of (i) $y = e^{x^3}$ and (ii) $y = \sqrt{e^{\sqrt{x}}}$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}} \frac{du}{dx}$

Solution:

Part (i): Let $y = e^{x^3}$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= e^{x^3} \cdot \frac{d}{dx}(x^3) \\ &= e^{x^3} \cdot (3x^2) \\ &= 3x^2 e^{x^3}\end{aligned}$$

Part (ii): Let $y = \sqrt{e^{\sqrt{x}}}$.

Differentiating w.r.t. x using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \cdot \frac{d}{dx}(e^{\sqrt{x}}) \\ &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \cdot e^{\sqrt{x}} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= \frac{e^{\sqrt{x}}}{2\sqrt{e^{\sqrt{x}}}} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{\sqrt{e^{\sqrt{x}}}}{4\sqrt{x}}\end{aligned}$$

Final Answer

- (i) $\frac{dy}{dx} = 3x^2 e^{x^3}$
 (ii) $\frac{dy}{dx} = \frac{\sqrt{e^{\sqrt{x}}}}{4\sqrt{x}}$

Question 5

Differentiate the following functions w.r.t. x :

- (i) $\log(\log x)$, $x > 1$
 (ii) $\log_7(\log x)$, $x > 1$

TYPE 19 Chain Rule for Differentiation

To Find: The derivative of (i) $y = \log(\log x)$ and (ii) $y = \log_7(\log x)$ w.r.t. x .

Concept / Formula Used: $\log_a b = \frac{\log b}{\log a}$

Solution:

Part (i): Let $y = \log(\log x)$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\log x} \cdot \frac{d}{dx}(\log x) \\ &= \frac{1}{\log x} \cdot \frac{1}{x} \\ &= \frac{1}{x \log x}\end{aligned}$$

Part (ii): Let $y = \log_7(\log x)$.

Applying change of base formula to natural logarithm:

$$y = \frac{\log(\log x)}{\log 7}$$

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\log 7} \cdot \frac{d}{dx}(\log(\log x)) \\ &= \frac{1}{\log 7} \cdot \frac{1}{\log x} \cdot \frac{1}{x} \\ &= \frac{1}{x \log 7 \log x}\end{aligned}$$

Final Answer

$$\begin{aligned}\text{(i)} \quad \frac{dy}{dx} &= \frac{1}{x \log x} \\ \text{(ii)} \quad \frac{dy}{dx} &= \frac{1}{x \log 7 \log x}\end{aligned}$$

Question 6

Differentiate the following functions w.r.t. x :

- (i) $\log(\cos e^x)$
- (ii) $\tan^{-1}(e^{\sin x})$

TYPE 22 Repeated Chain Rule

To Find: The derivative of (i) $y = \log(\cos e^x)$ and (ii) $y = \tan^{-1}(e^{\sin x})$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}(\tan^{-1} u) = \frac{1}{1+u^2} \frac{du}{dx}$

Solution:

Part (i): Let $y = \log(\cos e^x)$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\cos e^x} \cdot \frac{d}{dx}(\cos e^x) \\ &= \frac{1}{\cos e^x} \cdot (-\sin e^x) \cdot \frac{d}{dx}(e^x) \\ &= -\frac{\sin e^x}{\cos e^x} \cdot e^x \\ &= -e^x \tan(e^x)\end{aligned}$$

Part (ii): Let $y = \tan^{-1}(e^{\sin x})$.

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{1 + (e^{\sin x})^2} \cdot \frac{d}{dx}(e^{\sin x}) \\ &= \frac{1}{1 + e^{2 \sin x}} \cdot e^{\sin x} \cdot \frac{d}{dx}(\sin x) \\ &= \frac{e^{\sin x} \cos x}{1 + e^{2 \sin x}}\end{aligned}$$

Final Answer

(i) $\frac{dy}{dx} = -e^x \tan(e^x)$

(ii) $\frac{dy}{dx} = \frac{e^{\sin x} \cos x}{1 + e^{2 \sin x}}$

Question 7

Is the function $f(x) = \log(2x - 1)$ derivable at $x = 0$?

TYPE 35 Domain of Differentiability

To Find: Whether $f(x) = \log(2x - 1)$ is derivable at $x = 0$.

Concept / Formula Used: $\log u \quad u > 0$

Solution:

For $f(x) = \log(2x - 1)$, the argument of the logarithm must be strictly positive:

$$\begin{aligned}2x - 1 &> 0 \\ 2x &> 1 \\ x &> \frac{1}{2}\end{aligned}$$

Thus, the domain of $f(x)$ is $(\frac{1}{2}, \infty)$.

Evaluating the argument at $x = 0$:

$$2(0) - 1 = -1 \leq 0$$

Since $f(0) = \log(-1)$ is undefined in real numbers, $f(x)$ is not defined in any neighbourhood of $x = 0$.

Therefore, $f(x)$ is not derivable at $x = 0$.

Final Answer

No, the function $f(x) = \log(2x - 1)$ is not derivable at $x = 0$ because it is not defined at $x = 0$.

Question 8

If $f(x) = \log(\log x)$, find $f'(e)$.

TYPE 31 Chain Rule and Evaluation

Given: $f(x) = \log(\log x)$

To Find: $f'(e)$

Concept / Formula Used: $\frac{d}{dx}(\log u) = \frac{1}{u} \frac{du}{dx}$ $\log e = 1$

Solution:

Given function:

$$f(x) = \log(\log x)$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{1}{\log x} \cdot \frac{d}{dx}(\log x) \\ &= \frac{1}{\log x} \cdot \frac{1}{x} \\ &= \frac{1}{x \log x} \end{aligned}$$

Evaluating at $x = e$:

$$f'(e) = \frac{1}{e \log e}$$

Since $\log e = 1$:

$$\begin{aligned} f'(e) &= \frac{1}{e \cdot 1} \\ &= \frac{1}{e} \end{aligned}$$

Final Answer

$$f'(e) = \frac{1}{e}$$

Question 9

If $y = \log(\tan x)$, show that $\frac{dy}{dx} = 2 \csc 2x$.

TYPE 33 Proofs Using Chain Rule

To Prove: $\frac{dy}{dx} = 2 \csc 2x$

Concept / Formula Used: $\frac{d}{dx}(\tan x) = \sec^2 x$ $\sin 2x = 2 \sin x \cos x$ $\csc \theta = \frac{1}{\sin \theta}$

Solution:

Given:

$$y = \log(\tan x)$$

Differentiating y w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\tan x} \cdot \frac{d}{dx}(\tan x) \\ &= \frac{1}{\tan x} \cdot \sec^2 x\end{aligned}$$

Expressing in terms of $\sin x$ and $\cos x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x} \\ &= \frac{1}{\sin x \cos x}\end{aligned}$$

Multiplying numerator and denominator by 2:

$$\frac{dy}{dx} = \frac{2}{2 \sin x \cos x}$$

Using the identity $\sin 2x = 2 \sin x \cos x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{2}{\sin 2x} \\ &= 2 \csc 2x\end{aligned}$$

Hence Proved

$\frac{dy}{dx} = 2 \csc 2x$, hence proved.

Question 10

If $f(1) = 4$ and $f'(1) = 2$, find the value of the derivative of $\log f(e^x)$ w.r.t. x at $x = 0$.

TYPE 31 Chain Rule and Evaluation

Given: $f(1) = 4$ and $f'(1) = 2$

To Find: $\left. \frac{d}{dx}[\log f(e^x)] \right|_{x=0}$

Concept / Formula Used: $\frac{d}{dx}[g(h(x))] = g'(h(x)) \cdot h'(x)$

Solution:

Let $y = \log f(e^x)$.

Differentiating y w.r.t. x using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{f(e^x)} \cdot \frac{d}{dx} [f(e^x)] \\ &= \frac{1}{f(e^x)} \cdot f'(e^x) \cdot \frac{d}{dx} (e^x) \\ &= \frac{f'(e^x) \cdot e^x}{f(e^x)}\end{aligned}$$

Evaluating at $x = 0$:

$$\left. \frac{dy}{dx} \right|_{x=0} = \frac{f'(e^0) \cdot e^0}{f(e^0)}$$

Since $e^0 = 1$:

$$\begin{aligned}\left. \frac{dy}{dx} \right|_{x=0} &= \frac{f'(1) \cdot 1}{f(1)} \\ &= \frac{f'(1)}{f(1)}\end{aligned}$$

Substituting the given values $f(1) = 4$ and $f'(1) = 2$:

$$\begin{aligned}\left. \frac{dy}{dx} \right|_{x=0} &= \frac{2}{4} \\ &= \frac{1}{2}\end{aligned}$$

Final Answer

$$\left. \frac{d}{dx} [\log f(e^x)] \right|_{x=0} = \frac{1}{2}$$

Question 11

If $f(x) = e^{xg(x)}$, $g(0) = 2$ and $g'(0) = 1$, then find $f'(0)$.

TYPE 31 Chain Rule and Evaluation

Given: $f(x) = e^{xg(x)}$, $g(0) = 2$, $g'(0) = 1$

To Find: $f'(0)$

Concept / Formula Used: $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

Solution:

Given function:

$$f(x) = e^{xg(x)}$$

Differentiating $f(x)$ w.r.t. x using chain rule:

$$f'(x) = e^{xg(x)} \cdot \frac{d}{dx} [xg(x)]$$

Applying product rule to $\frac{d}{dx}[xg(x)]$:

$$\begin{aligned}\frac{d}{dx}[xg(x)] &= x \cdot g'(x) + g(x) \cdot \frac{d}{dx}(x) \\ &= xg'(x) + g(x) \cdot 1 \\ &= xg'(x) + g(x)\end{aligned}$$

So,

$$f'(x) = e^{xg(x)} [xg'(x) + g(x)]$$

Evaluating at $x = 0$:

$$\begin{aligned}f'(0) &= e^{0 \cdot g(0)} [0 \cdot g'(0) + g(0)] \\ &= e^0 [0 + g(0)] \\ &= 1 \cdot g(0)\end{aligned}$$

Substituting $g(0) = 2$:

$$f'(0) = 2$$

Final Answer

$$f'(0) = 2$$

Question 12

Differentiate the following functions w.r.t. x :

- (i) $\frac{e^x}{\sin x}$
 (ii) $\frac{e^x}{1 + \sin x}$

TYPE 28 Quotient Rule for Differentiation

To Find: The derivative of (i) $y = \frac{e^x}{\sin x}$ and (ii) $y = \frac{e^x}{1 + \sin x}$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Solution:

Part (i): Let $y = \frac{e^x}{\sin x}$.

Using quotient rule with $u = e^x$ and $v = \sin x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\sin x \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}(\sin x)}{(\sin x)^2} \\ &= \frac{\sin x \cdot e^x - e^x \cdot \cos x}{\sin^2 x} \\ &= \frac{e^x(\sin x - \cos x)}{\sin^2 x}\end{aligned}$$

Part (ii): Let $y = \frac{e^x}{1 + \sin x}$.

Using quotient rule with $u = e^x$ and $v = 1 + \sin x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1 + \sin x) \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}(1 + \sin x)}{(1 + \sin x)^2} \\ &= \frac{(1 + \sin x) \cdot e^x - e^x \cdot (0 + \cos x)}{(1 + \sin x)^2} \\ &= \frac{e^x(1 + \sin x - \cos x)}{(1 + \sin x)^2}\end{aligned}$$

Final Answer

$$\begin{aligned}\text{(i)} \quad \frac{dy}{dx} &= \frac{e^x(\sin x - \cos x)}{\sin^2 x} \\ \text{(ii)} \quad \frac{dy}{dx} &= \frac{e^x(1 + \sin x - \cos x)}{(1 + \sin x)^2}\end{aligned}$$

Question 13

Differentiate the following functions w.r.t. x :

- (i) $2x \tan^{-1} x - \log(1 + x^2)$
 (ii) $\frac{\cos x}{\log x}, \quad x > 0$

TYPE 38 Product and Quotient Rules with Inverse Trigonometric Functions

To Find: The derivative of (i) $y = 2x \tan^{-1} x - \log(1 + x^2)$ and (ii) $y = \frac{\cos x}{\log x}$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2}$

Solution:

Part (i): Let $y = 2x \tan^{-1} x - \log(1 + x^2)$.

Differentiating $2x \tan^{-1} x$ using product rule:

$$\begin{aligned}\frac{d}{dx}(2x \tan^{-1} x) &= 2x \cdot \frac{d}{dx}(\tan^{-1} x) + \tan^{-1} x \cdot \frac{d}{dx}(2x) \\ &= 2x \cdot \frac{1}{1 + x^2} + \tan^{-1} x \cdot 2 \\ &= \frac{2x}{1 + x^2} + 2 \tan^{-1} x\end{aligned}$$

Differentiating $\log(1 + x^2)$ using chain rule:

$$\begin{aligned}\frac{d}{dx}(\log(1 + x^2)) &= \frac{1}{1 + x^2} \cdot \frac{d}{dx}(1 + x^2) \\ &= \frac{1}{1 + x^2} \cdot 2x \\ &= \frac{2x}{1 + x^2}\end{aligned}$$

Combining both terms:

$$\begin{aligned}\frac{dy}{dx} &= \left(\frac{2x}{1+x^2} + 2 \tan^{-1} x \right) - \frac{2x}{1+x^2} \\ &= 2 \tan^{-1} x\end{aligned}$$

Part (ii): Let $y = \frac{\cos x}{\log x}$.

Using quotient rule with $u = \cos x$ and $v = \log x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\log x \cdot \frac{d}{dx}(\cos x) - \cos x \cdot \frac{d}{dx}(\log x)}{(\log x)^2} \\ &= \frac{\log x \cdot (-\sin x) - \cos x \cdot \frac{1}{x}}{(\log x)^2} \\ &= \frac{-\sin x \log x - \frac{\cos x}{x}}{(\log x)^2} \\ &= \frac{-(x \sin x \log x + \cos x)}{x(\log x)^2} \\ &= -\frac{x \sin x \log x + \cos x}{x(\log x)^2}\end{aligned}$$

Final Answer

(i) $\frac{dy}{dx} = 2 \tan^{-1} x$

(ii) $\frac{dy}{dx} = -\frac{x \sin x \log x + \cos x}{x(\log x)^2}$

Question 14

Differentiate the following functions w.r.t. x :

(i) $e^{\sec^2 x} + 3 \cos^{-1} x$

(ii) $e^x + e^{x^2} + \dots + e^{x^5}$

TYPE 20 Sum and Product Rules

To Find: The derivative of (i) $y = e^{\sec^2 x} + 3 \cos^{-1} x$ and (ii) $y = e^x + e^{x^2} + \dots + e^{x^5}$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$ $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$

Solution:

Part (i): Let $y = e^{\sec^2 x} + 3 \cos^{-1} x$.

Differentiating first term:

$$\begin{aligned}\frac{d}{dx}(e^{\sec^2 x}) &= e^{\sec^2 x} \cdot \frac{d}{dx}(\sec^2 x) \\ &= e^{\sec^2 x} \cdot 2 \sec x \cdot \frac{d}{dx}(\sec x) \\ &= e^{\sec^2 x} \cdot 2 \sec x \cdot (\sec x \tan x) \\ &= 2 \sec^2 x \tan x e^{\sec^2 x}\end{aligned}$$

Differentiating second term:

$$\frac{d}{dx}(3 \cos^{-1} x) = 3 \cdot \left(-\frac{1}{\sqrt{1-x^2}} \right) = -\frac{3}{\sqrt{1-x^2}}$$

Thus,

$$\frac{dy}{dx} = 2 \sec^2 x \tan x e^{\sec^2 x} - \frac{3}{\sqrt{1-x^2}}$$

Part (ii): Let $y = e^x + e^{x^2} + e^{x^3} + e^{x^4} + e^{x^5}$.

Differentiating term by term using chain rule:

$$\begin{aligned}\frac{d}{dx}(e^x) &= e^x \\ \frac{d}{dx}(e^{x^2}) &= e^{x^2} \cdot (2x) = 2xe^{x^2} \\ \frac{d}{dx}(e^{x^3}) &= e^{x^3} \cdot (3x^2) = 3x^2e^{x^3} \\ \frac{d}{dx}(e^{x^4}) &= e^{x^4} \cdot (4x^3) = 4x^3e^{x^4} \\ \frac{d}{dx}(e^{x^5}) &= e^{x^5} \cdot (5x^4) = 5x^4e^{x^5}\end{aligned}$$

Summing all terms:

$$\frac{dy}{dx} = e^x + 2xe^{x^2} + 3x^2e^{x^3} + 4x^3e^{x^4} + 5x^4e^{x^5}$$

Final Answer

$$\begin{aligned}\text{(i)} \quad \frac{dy}{dx} &= 2 \sec^2 x \tan x e^{\sec^2 x} - \frac{3}{\sqrt{1-x^2}} \\ \text{(ii)} \quad \frac{dy}{dx} &= e^x + 2xe^{x^2} + 3x^2e^{x^3} + 4x^3e^{x^4} + 5x^4e^{x^5}\end{aligned}$$

Question 15

Differentiate the following functions w.r.t. x :

- (i) $\frac{5^x \log x}{x^2 + 1}$
- (ii) $\cos(\log x + e^x)$

TYPE 28 Quotient Rule for Differentiation

To Find: The derivative of (i) $y = \frac{5^x \log x}{x^2 + 1}$ and (ii) $y = \cos(\log x + e^x)$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}(5^x) = 5^x \log 5$

Solution:

Part (i): Let $y = \frac{5^x \log x}{x^2 + 1}$.

Let numerator $u = 5^x \log x$ and denominator $v = x^2 + 1$.

Differentiating u using product rule:

$$\begin{aligned}\frac{du}{dx} &= 5^x \cdot \frac{d}{dx}(\log x) + \log x \cdot \frac{d}{dx}(5^x) \\ &= 5^x \cdot \frac{1}{x} + \log x \cdot (5^x \log 5) \\ &= 5^x \left(\frac{1}{x} + \log 5 \log x \right) \\ &= 5^x \left(\frac{1 + x \log 5 \log x}{x} \right)\end{aligned}$$

Applying quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \\ &= \frac{(x^2 + 1) \cdot 5^x \left(\frac{1 + x \log 5 \log x}{x} \right) - (5^x \log x) \cdot (2x)}{(x^2 + 1)^2} \\ &= \frac{5^x [(x^2 + 1)(1 + x \log 5 \log x) - 2x^2 \log x]}{x(x^2 + 1)^2}\end{aligned}$$

Part (ii): Let $y = \cos(\log x + e^x)$.

Differentiating w.r.t. x using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= -\sin(\log x + e^x) \cdot \frac{d}{dx}(\log x + e^x) \\ &= -\sin(\log x + e^x) \cdot \left(\frac{1}{x} + e^x \right) \\ &= -\left(\frac{1}{x} + e^x \right) \sin(\log x + e^x)\end{aligned}$$

Final Answer

$$(i) \frac{dy}{dx} = \frac{5^x [(x^2 + 1)(1 + x \log 5 \log x) - 2x^2 \log x]}{x(x^2 + 1)^2}$$

$$(ii) \frac{dy}{dx} = -\left(\frac{1}{x} + e^x \right) \sin(\log x + e^x)$$

Question 16

Differentiate the following functions w.r.t. x :

$$(i) \log \left(\frac{x + \sqrt{x^2 - a^2}}{x - \sqrt{x^2 - a^2}} \right)$$

$$(ii) \log \sqrt{\frac{1 - \cos x}{1 + \cos x}}$$

TYPE 30 Trigonometric Simplification before Differentiation

To Find: The derivative of (i) $y = \log \left(\frac{x + \sqrt{x^2 - a^2}}{x - \sqrt{x^2 - a^2}} \right)$ and (ii) $y = \log \sqrt{\frac{1 - \cos x}{1 + \cos x}}$ w.r.t. x .

Concept / Formula Used: $\tan^2\left(\frac{x}{2}\right) = \frac{1 - \cos x}{1 + \cos x}$ $\log(u^2) = 2 \log u$

Solution:

Part (i): Let $y = \log\left(\frac{x + \sqrt{x^2 - a^2}}{x - \sqrt{x^2 - a^2}}\right)$.

Rationalizing the denominator inside the logarithm:

$$\begin{aligned}\frac{x + \sqrt{x^2 - a^2}}{x - \sqrt{x^2 - a^2}} &= \frac{(x + \sqrt{x^2 - a^2})^2}{(x)^2 - (\sqrt{x^2 - a^2})^2} \\ &= \frac{(x + \sqrt{x^2 - a^2})^2}{x^2 - (x^2 - a^2)} \\ &= \frac{(x + \sqrt{x^2 - a^2})^2}{a^2}\end{aligned}$$

Substitute back into y :

$$\begin{aligned}y &= \log\left(\frac{(x + \sqrt{x^2 - a^2})^2}{a^2}\right) \\ &= 2 \log(x + \sqrt{x^2 - a^2}) - \log(a^2)\end{aligned}$$

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= 2 \cdot \frac{1}{x + \sqrt{x^2 - a^2}} \cdot \frac{d}{dx}(x + \sqrt{x^2 - a^2}) - 0 \\ &= \frac{2}{x + \sqrt{x^2 - a^2}} \cdot \left(1 + \frac{1}{2\sqrt{x^2 - a^2}} \cdot 2x\right) \\ &= \frac{2}{x + \sqrt{x^2 - a^2}} \cdot \left(1 + \frac{x}{\sqrt{x^2 - a^2}}\right) \\ &= \frac{2}{x + \sqrt{x^2 - a^2}} \cdot \left(\frac{\sqrt{x^2 - a^2} + x}{\sqrt{x^2 - a^2}}\right) \\ &= \frac{2}{\sqrt{x^2 - a^2}}\end{aligned}$$

Part (ii): Let $y = \log \sqrt{\frac{1 - \cos x}{1 + \cos x}}$.

Using trigonometric identity $\frac{1 - \cos x}{1 + \cos x} = \tan^2\left(\frac{x}{2}\right)$:

$$\begin{aligned}y &= \log \sqrt{\tan^2\left(\frac{x}{2}\right)} \\ &= \log \tan\left(\frac{x}{2}\right)\end{aligned}$$

Differentiating w.r.t. x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\tan(x/2)} \cdot \frac{d}{dx} \left[\tan \left(\frac{x}{2} \right) \right] \\ &= \frac{1}{\tan(x/2)} \cdot \sec^2 \left(\frac{x}{2} \right) \cdot \frac{1}{2} \\ &= \frac{\cos(x/2)}{\sin(x/2)} \cdot \frac{1}{\cos^2(x/2)} \cdot \frac{1}{2} \\ &= \frac{1}{2 \sin(x/2) \cos(x/2)} \\ &= \frac{1}{\sin x} \\ &= \csc x\end{aligned}$$

Final Answer

$$(i) \frac{dy}{dx} = \frac{2}{\sqrt{x^2 - a^2}}$$

$$(ii) \frac{dy}{dx} = \csc x$$

Question 17

Differentiate the following functions w.r.t. x :

$$(i) e^{\cot^{-1} x^2}$$

$$(ii) x\sqrt{1+x^2} + \log(x + \sqrt{x^2+1})$$

TYPE 36 Chain Rule for Inverse Trigonometric Functions

To Find: The derivative of (i) $y = e^{\cot^{-1} x^2}$ and (ii) $y = x\sqrt{1+x^2} + \log(x + \sqrt{x^2+1})$ w.r.t. x .

Concept / Formula Used: $\frac{d}{dx}(\cot^{-1} u) = -\frac{1}{1+u^2} \frac{du}{dx}$ $\frac{d}{dx} \left[\log(x + \sqrt{x^2 + a^2}) \right] = \frac{1}{\sqrt{x^2 + a^2}}$

Solution:

Part (i): Let $y = e^{\cot^{-1} x^2}$.

Differentiating w.r.t. x using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= e^{\cot^{-1} x^2} \cdot \frac{d}{dx} (\cot^{-1} x^2) \\ &= e^{\cot^{-1} x^2} \cdot \left(-\frac{1}{1+(x^2)^2} \right) \cdot \frac{d}{dx} (x^2) \\ &= e^{\cot^{-1} x^2} \cdot \left(-\frac{1}{1+x^4} \right) \cdot (2x) \\ &= -\frac{2xe^{\cot^{-1} x^2}}{1+x^4}\end{aligned}$$

Part (ii): Let $y = x\sqrt{1+x^2} + \log(x + \sqrt{x^2+1})$.

Differentiating the first term using product rule:

$$\begin{aligned}
 \frac{d}{dx} \left(x\sqrt{1+x^2} \right) &= x \cdot \frac{d}{dx} \left(\sqrt{1+x^2} \right) + \sqrt{1+x^2} \cdot \frac{d}{dx}(x) \\
 &= x \cdot \frac{1}{2\sqrt{1+x^2}} \cdot (2x) + \sqrt{1+x^2} \cdot 1 \\
 &= \frac{x^2}{\sqrt{1+x^2}} + \sqrt{1+x^2} \\
 &= \frac{x^2 + (1+x^2)}{\sqrt{1+x^2}} \\
 &= \frac{2x^2 + 1}{\sqrt{1+x^2}}
 \end{aligned}$$

Differentiating the second term:

$$\begin{aligned}
 \frac{d}{dx} \left[\log(x + \sqrt{x^2 + 1}) \right] &= \frac{1}{x + \sqrt{x^2 + 1}} \cdot \left(1 + \frac{2x}{2\sqrt{x^2 + 1}} \right) \\
 &= \frac{1}{x + \sqrt{x^2 + 1}} \cdot \left(\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}} \right) \\
 &= \frac{1}{\sqrt{x^2 + 1}}
 \end{aligned}$$

Adding both derivatives:

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{2x^2 + 1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \\
 &= \frac{2x^2 + 2}{\sqrt{1+x^2}} \\
 &= \frac{2(x^2 + 1)}{\sqrt{x^2 + 1}} \\
 &= 2\sqrt{x^2 + 1}
 \end{aligned}$$

Final Answer

- (i) $\frac{dy}{dx} = -\frac{2xe^{\cot^{-1}x^2}}{1+x^4}$
- (ii) $\frac{dy}{dx} = 2\sqrt{x^2 + 1}$

Question 18

- (i) If $y = \frac{\log(x + \sqrt{x^2 + 1})}{\sqrt{x^2 + 1}}$, prove that $(x^2 + 1)\frac{dy}{dx} + xy = 1$.
- (ii) If $y = e^{2\log x + 3x}$, prove that $\frac{dy}{dx} = x(2 + 3x)e^{3x}$.

TYPE 32 Verifying Differential Relations

★ Likely Exam Question

To Prove: (i) $(x^2 + 1)\frac{dy}{dx} + xy = 1$

(ii) $\frac{dy}{dx} = x(2 + 3x)e^{3x}$

Concept / Formula Used: $e^{a \log b} = b^a$

Solution:

Part (i): Given:

$$y = \frac{\log(x + \sqrt{x^2 + 1})}{\sqrt{x^2 + 1}}$$

Cross-multiplying:

$$y\sqrt{x^2 + 1} = \log(x + \sqrt{x^2 + 1})$$

Differentiating both sides w.r.t. x :

$$\frac{d}{dx} (y\sqrt{x^2 + 1}) = \frac{d}{dx} [\log(x + \sqrt{x^2 + 1})]$$

Using product rule on LHS:

$$\begin{aligned} \sqrt{x^2 + 1} \frac{dy}{dx} + y \cdot \frac{d}{dx} (\sqrt{x^2 + 1}) &= \frac{1}{x + \sqrt{x^2 + 1}} \cdot \left(1 + \frac{2x}{2\sqrt{x^2 + 1}}\right) \\ \sqrt{x^2 + 1} \frac{dy}{dx} + y \cdot \frac{2x}{2\sqrt{x^2 + 1}} &= \frac{1}{x + \sqrt{x^2 + 1}} \cdot \left(\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}}\right) \\ \sqrt{x^2 + 1} \frac{dy}{dx} + \frac{xy}{\sqrt{x^2 + 1}} &= \frac{1}{\sqrt{x^2 + 1}} \end{aligned}$$

Multiplying the entire equation by $\sqrt{x^2 + 1}$:

$$(x^2 + 1)\frac{dy}{dx} + xy = 1$$

Part (ii): Given:

$$y = e^{2 \log x + 3x}$$

Simplifying using logarithm properties:

$$\begin{aligned} y &= e^{\log(x^2) + 3x} \\ &= e^{\log(x^2)} \cdot e^{3x} \\ &= x^2 e^{3x} \end{aligned}$$

Differentiating y w.r.t. x using product rule:

$$\begin{aligned} \frac{dy}{dx} &= x^2 \cdot \frac{d}{dx}(e^{3x}) + e^{3x} \cdot \frac{d}{dx}(x^2) \\ &= x^2 \cdot (3e^{3x}) + e^{3x} \cdot (2x) \\ &= 3x^2 e^{3x} + 2x e^{3x} \\ &= x(2 + 3x)e^{3x} \end{aligned}$$

Hence Proved

Both identities (i) and (ii) are proved.

Question 19

Differentiate $\tan^{-1}\left(\frac{2^{x+1}}{1-4^x}\right)$ w.r.t. x .

TYPE 39 Trigonometric Substitution before Differentiation

To Find: The derivative of $y = \tan^{-1}\left(\frac{2^{x+1}}{1-4^x}\right)$ w.r.t. x .

Concept / Formula Used: $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ $\frac{d}{dx}(a^x) = a^x \log a$

Solution:

Let $y = \tan^{-1}\left(\frac{2^{x+1}}{1-4^x}\right)$.

Rewrite the terms inside the argument:

$$2^{x+1} = 2 \cdot 2^x$$

$$4^x = (2^2)^x = (2^x)^2$$

Substituting these into y :

$$y = \tan^{-1}\left(\frac{2 \cdot 2^x}{1 - (2^x)^2}\right)$$

Let $2^x = \tan \theta$, which gives $\theta = \tan^{-1}(2^x)$.

Substituting $2^x = \tan \theta$:

$$y = \tan^{-1}\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right)$$

Using standard identity $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$:

$$y = \tan^{-1}(\tan 2\theta)$$

$$= 2\theta$$

Back-substituting $\theta = \tan^{-1}(2^x)$:

$$y = 2 \tan^{-1}(2^x)$$

Differentiating y w.r.t. x using chain rule:

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot \frac{1}{1 + (2^x)^2} \cdot \frac{d}{dx}(2^x) \\ &= \frac{2}{1 + 4^x} \cdot (2^x \log 2) \\ &= \frac{2 \cdot 2^x \log 2}{1 + 4^x} \\ &= \frac{2^{x+1} \log 2}{1 + 4^x} \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \frac{2^{x+1} \log 2}{1 + 4^x}$$

Question 20

Find $\frac{dy}{dx}$ when

- (i) $xy + xe^{-y} + ye^x = x^2$
- (ii) $e^{x-y} = \log\left(\frac{x}{y}\right)$
- (iii) $5^x + 5^y = 5^{x+y}$

TYPE 40 Implicit Differentiation

To Find: $\frac{dy}{dx}$ for each given implicit equation.

Concept / Formula Used: $x \quad \frac{dy}{dx}$

Solution:

Part (i): Given: $xy + xe^{-y} + ye^x = x^2$.

Differentiating term-by-term w.r.t. x :

$$\begin{aligned} \frac{d}{dx}(xy) + \frac{d}{dx}(xe^{-y}) + \frac{d}{dx}(ye^x) &= \frac{d}{dx}(x^2) \\ \left(x \frac{dy}{dx} + y \cdot 1\right) + \left(x \left(-e^{-y} \frac{dy}{dx}\right) + e^{-y} \cdot 1\right) + \left(ye^x + e^x \frac{dy}{dx}\right) &= 2x \\ x \frac{dy}{dx} + y - xe^{-y} \frac{dy}{dx} + e^{-y} + ye^x + e^x \frac{dy}{dx} &= 2x \end{aligned}$$

Grouping terms containing $\frac{dy}{dx}$ on LHS:

$$(x - xe^{-y} + e^x) \frac{dy}{dx} = 2x - y - e^{-y} - ye^x$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{2x - y - e^{-y} - ye^x}{x - xe^{-y} + e^x}$$

Part (ii): Given: $e^{x-y} = \log\left(\frac{x}{y}\right) = \log x - \log y$.

Differentiating both sides w.r.t. x :

$$\begin{aligned} \frac{d}{dx}(e^{x-y}) &= \frac{d}{dx}(\log x - \log y) \\ e^{x-y} \left(1 - \frac{dy}{dx}\right) &= \frac{1}{x} - \frac{1}{y} \frac{dy}{dx} \\ e^{x-y} - e^{x-y} \frac{dy}{dx} &= \frac{1}{x} - \frac{1}{y} \frac{dy}{dx} \end{aligned}$$

Rearranging terms with $\frac{dy}{dx}$:

$$\begin{aligned} \left(\frac{1}{y} - e^{x-y}\right) \frac{dy}{dx} &= \frac{1}{x} - e^{x-y} \\ \left(\frac{1 - ye^{x-y}}{y}\right) \frac{dy}{dx} &= \frac{1 - xe^{x-y}}{x} \end{aligned}$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{y(1 - xe^{x-y})}{x(1 - ye^{x-y})}$$

Part (iii): Given: $5^x + 5^y = 5^{x+y}$.

Differentiating both sides w.r.t. x :

$$5^x \log 5 + 5^y \log 5 \frac{dy}{dx} = 5^{x+y} \log 5 \left(1 + \frac{dy}{dx}\right)$$

Dividing both sides by $\log 5$:

$$5^x + 5^y \frac{dy}{dx} = 5^{x+y} + 5^{x+y} \frac{dy}{dx}$$

Rearranging terms with $\frac{dy}{dx}$:

$$\begin{aligned} (5^y - 5^{x+y}) \frac{dy}{dx} &= 5^{x+y} - 5^x \\ \frac{dy}{dx} &= \frac{5^{x+y} - 5^x}{5^y - 5^{x+y}} \end{aligned}$$

Substituting $5^{x+y} = 5^x + 5^y$ into the expression:

$$\begin{aligned} \frac{dy}{dx} &= \frac{(5^x + 5^y) - 5^x}{5^y - (5^x + 5^y)} \\ &= \frac{5^y}{-5^x} \\ &= -5^{y-x} \end{aligned}$$

Final Answer

$$(i) \frac{dy}{dx} = \frac{2x - y - e^{-y} - ye^x}{x - xe^{-y} + e^x}$$

$$(ii) \frac{dy}{dx} = \frac{y(1 - xe^{x-y})}{x(1 - ye^{x-y})}$$

$$(iii) \frac{dy}{dx} = -5^{y-x}$$

Question 21

If $\log(x^2 + y^2) = 2 \tan^{-1} \left(\frac{y}{x} \right)$, show that $\frac{dy}{dx} = \frac{x+y}{x-y}$.

TYPE 40 Implicit Differentiation

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{x+y}{x-y}$

Concept / Formula Used: $\frac{d}{dx}(\tan^{-1} u) = \frac{1}{1+u^2} \frac{du}{dx}$

Solution:

Given equation:

$$\log(x^2 + y^2) = 2 \tan^{-1} \left(\frac{y}{x} \right)$$

Differentiating both sides w.r.t. x :

$$\frac{d}{dx} [\log(x^2 + y^2)] = \frac{d}{dx} \left[2 \tan^{-1} \left(\frac{y}{x} \right) \right]$$

Differentiating LHS:

$$\frac{1}{x^2 + y^2} \cdot \left(2x + 2y \frac{dy}{dx} \right) = \frac{2(x + y \frac{dy}{dx})}{x^2 + y^2}$$

Differentiating RHS using chain rule and quotient rule:

$$\begin{aligned} 2 \cdot \frac{1}{1 + \left(\frac{y}{x} \right)^2} \cdot \frac{d}{dx} \left(\frac{y}{x} \right) &= 2 \cdot \frac{1}{\frac{x^2 + y^2}{x^2}} \cdot \left(\frac{x \frac{dy}{dx} - y}{x^2} \right) \\ &= \frac{2x^2}{x^2 + y^2} \cdot \left(\frac{x \frac{dy}{dx} - y}{x^2} \right) \\ &= \frac{2(x \frac{dy}{dx} - y)}{x^2 + y^2} \end{aligned}$$

Equating LHS and RHS:

$$\frac{2(x + y \frac{dy}{dx})}{x^2 + y^2} = \frac{2(x \frac{dy}{dx} - y)}{x^2 + y^2}$$

Multiplying both sides by $\frac{x^2 + y^2}{2}$:

$$x + y \frac{dy}{dx} = x \frac{dy}{dx} - y$$

Rearranging terms:

$$\begin{aligned} x + y &= x \frac{dy}{dx} - y \frac{dy}{dx} \\ x + y &= (x - y) \frac{dy}{dx} \end{aligned}$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{x + y}{x - y}$$

Hence Proved

$\frac{dy}{dx} = \frac{x + y}{x - y}$, hence proved.

Question 22

If $y \log x = x - y$, prove that $\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$.

TYPE 44 Differentiation by Isolating Variables

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$

Concept / Formula Used: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$

Solution:

Given equation:

$$y \log x = x - y$$

Grouping terms involving y on the LHS:

$$\begin{aligned} y \log x + y &= x \\ y(1 + \log x) &= x \\ y &= \frac{x}{1 + \log x} \end{aligned}$$

Differentiating y w.r.t. x using the quotient rule with $u = x$ and $v = 1 + \log x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{(1 + \log x) \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(1 + \log x)}{(1 + \log x)^2} \\ &= \frac{(1 + \log x) \cdot 1 - x \cdot (0 + \frac{1}{x})}{(1 + \log x)^2} \\ &= \frac{1 + \log x - 1}{(1 + \log x)^2} \\ &= \frac{\log x}{(1 + \log x)^2} \end{aligned}$$

Hence Proved

$$\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}, \text{ hence proved.}$$

Question 23

Differentiate the following functions w.r.t. x :

- (i) $\log_x(2x - 3)$
- (ii) $\log_{\cos x} \sin x$

TYPE 28 Quotient Rule for Differentiation

To Find: The derivative of (i) $y = \log_x(2x - 3)$ and (ii) $y = \log_{\cos x} \sin x$ w.r.t. x .

Concept / Formula Used: $\log_a b = \frac{\log b}{\log a}$

Solution:

Part (i): Let $y = \log_x(2x - 3)$.

Applying change of base formula to express y using natural logarithms:

$$y = \frac{\log(2x - 3)}{\log x}$$

Differentiating y w.r.t. x using quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\log x \cdot \frac{d}{dx} [\log(2x - 3)] - \log(2x - 3) \cdot \frac{d}{dx} (\log x)}{(\log x)^2} \\ &= \frac{\log x \cdot \left(\frac{2}{2x-3}\right) - \log(2x - 3) \cdot \left(\frac{1}{x}\right)}{(\log x)^2} \\ &= \frac{\frac{2 \log x}{2x-3} - \frac{\log(2x-3)}{x}}{(\log x)^2} \\ &= \frac{2x \log x - (2x - 3) \log(2x - 3)}{x(2x - 3)(\log x)^2} \end{aligned}$$

Part (ii): Let $y = \log_{\cos x} \sin x$.

Applying change of base formula to express y using natural logarithms:

$$y = \frac{\log \sin x}{\log \cos x}$$

Differentiating y w.r.t. x using quotient rule:

$$\frac{dy}{dx} = \frac{\log \cos x \cdot \frac{d}{dx} (\log \sin x) - \log \sin x \cdot \frac{d}{dx} (\log \cos x)}{(\log \cos x)^2}$$

Evaluating the individual derivatives:

$$\begin{aligned} \frac{d}{dx} (\log \sin x) &= \frac{1}{\sin x} \cdot \cos x = \cot x \\ \frac{d}{dx} (\log \cos x) &= \frac{1}{\cos x} \cdot (-\sin x) = -\tan x \end{aligned}$$

Substituting these back into $\frac{dy}{dx}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\log \cos x \cdot (\cot x) - \log \sin x \cdot (-\tan x)}{(\log \cos x)^2} \\ &= \frac{\cot x \log \cos x + \tan x \log \sin x}{(\log \cos x)^2} \end{aligned}$$

Final Answer

$$(i) \quad \frac{dy}{dx} = \frac{2x \log x - (2x - 3) \log(2x - 3)}{x(2x - 3)(\log x)^2}$$

$$(ii) \quad \frac{dy}{dx} = \frac{\cot x \log \cos x + \tan x \log \sin x}{(\log \cos x)^2}$$

5.9 – Logarithmic Differentiation

Question 1 (i)

Differentiate $(x+3)^2(x+4)^3(x+5)^4$ w.r.t. x .

TYPE 46 Logarithmic Differentiation of Products

Given: Let $y = (x+3)^2(x+4)^3(x+5)^4$

Concept / Formula Used: $\log(abc) = \log a + \log b + \log c$ $\log(u^k) = k \log u$

Solution:

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log((x+3)^2(x+4)^3(x+5)^4) \\ &= \log((x+3)^2) + \log((x+4)^3) + \log((x+5)^4) \\ &= 2 \log(x+3) + 3 \log(x+4) + 4 \log(x+5)\end{aligned}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= 2 \cdot \frac{1}{x+3} + 3 \cdot \frac{1}{x+4} + 4 \cdot \frac{1}{x+5} \\ \frac{dy}{dx} &= y \left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5} \right]\end{aligned}$$

Substituting $y = (x+3)^2(x+4)^3(x+5)^4$:

$$\frac{dy}{dx} = (x+3)^2(x+4)^3(x+5)^4 \left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5} \right]$$

Final Answer

$$\frac{dy}{dx} = (x+3)^2(x+4)^3(x+5)^4 \left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5} \right]$$

Question 1 (ii)

Differentiate $\cos x \cos 2x \cos 3x$ w.r.t. x .

TYPE 46 Logarithmic Differentiation of Products

★ **Likely Exam Question**

Given: Let $y = \cos x \cos 2x \cos 3x$

Concept / Formula Used: $\log(uvw) = \log u + \log v + \log w$

Solution:

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log(\cos x \cos 2x \cos 3x) \\ &= \log(\cos x) + \log(\cos 2x) + \log(\cos 3x)\end{aligned}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= \frac{1}{\cos x} \cdot (-\sin x) + \frac{1}{\cos 2x} \cdot (-\sin 2x) \cdot 2 + \frac{1}{\cos 3x} \cdot (-\sin 3x) \cdot 3 \\ \frac{1}{y} \frac{dy}{dx} &= -\tan x - 2 \tan 2x - 3 \tan 3x \\ \frac{dy}{dx} &= -y(\tan x + 2 \tan 2x + 3 \tan 3x)\end{aligned}$$

Substituting $y = \cos x \cos 2x \cos 3x$:

$$\frac{dy}{dx} = -\cos x \cos 2x \cos 3x (\tan x + 2 \tan 2x + 3 \tan 3x)$$

Final Answer

$$\frac{dy}{dx} = -\cos x \cos 2x \cos 3x (\tan x + 2 \tan 2x + 3 \tan 3x)$$

Question 2 (i)

Differentiate $e^x \cos^3 x \sin^2 x$ w.r.t. x .

TYPE 46 Logarithmic Differentiation of Products

Given: Let $y = e^x \cos^3 x \sin^2 x$

Concept / Formula Used: $\log(abc) = \log a + \log b + \log c$

Solution:

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log(e^x \cos^3 x \sin^2 x) \\ &= \log(e^x) + \log(\cos^3 x) + \log(\sin^2 x) \\ &= x + 3 \log(\cos x) + 2 \log(\sin x)\end{aligned}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= 1 + 3 \cdot \frac{1}{\cos x} \cdot (-\sin x) + 2 \cdot \frac{1}{\sin x} \cdot (\cos x) \\ \frac{1}{y} \frac{dy}{dx} &= 1 - 3 \tan x + 2 \cot x \\ \frac{dy}{dx} &= y(1 - 3 \tan x + 2 \cot x)\end{aligned}$$

Substituting $y = e^x \cos^3 x \sin^2 x$:

$$\frac{dy}{dx} = e^x \cos^3 x \sin^2 x (1 - 3 \tan x + 2 \cot x)$$

Final Answer

$$\frac{dy}{dx} = e^x \cos^3 x \sin^2 x (1 - 3 \tan x + 2 \cot x)$$

Question 2 (ii)

Differentiate $\frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$ w.r.t. x .

TYPE 46 Logarithmic Differentiation of Products

Given: Let $y = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$

Concept / Formula Used: $\log\left(\frac{uv}{w}\right) = \log u + \log v - \log w$

Solution:

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log \left(\frac{e^{x^2} \tan^{-1} x}{(1+x^2)^{1/2}} \right) \\ &= \log(e^{x^2}) + \log(\tan^{-1} x) - \log((1+x^2)^{1/2}) \\ &= x^2 + \log(\tan^{-1} x) - \frac{1}{2} \log(1+x^2)\end{aligned}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= 2x + \frac{1}{\tan^{-1} x} \cdot \frac{1}{1+x^2} - \frac{1}{2} \cdot \frac{1}{1+x^2} \cdot 2x \\ \frac{1}{y} \frac{dy}{dx} &= 2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2} \\ \frac{dy}{dx} &= y \left[2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2} \right]\end{aligned}$$

Substituting $y = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$:

$$\frac{dy}{dx} = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}} \left[2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2} \right]$$

Final Answer

$$\frac{dy}{dx} = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}} \left[2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2} \right]$$

Question 3 (i)

Differentiate $(\sin x)^{\cos x}$, $0 < x < \pi$ w.r.t. x .

TYPE 47 Logarithmic Differentiation of Variable Exponents

Given: Let $y = (\sin x)^{\cos x}$, where $0 < x < \pi$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log y = \cos x \cdot \log(\sin x)$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= \cos x \cdot \frac{d}{dx}[\log(\sin x)] + \log(\sin x) \cdot \frac{d}{dx}[\cos x] \\ &= \cos x \cdot \left(\frac{1}{\sin x} \cdot \cos x \right) + \log(\sin x) \cdot (-\sin x) \\ &= \cos x \cot x - \sin x \log(\sin x) \\ \frac{dy}{dx} &= y [\cos x \cot x - \sin x \log(\sin x)]\end{aligned}$$

Substituting $y = (\sin x)^{\cos x}$:

$$\frac{dy}{dx} = (\sin x)^{\cos x} [\cos x \cot x - \sin x \log(\sin x)]$$

Final Answer

$$\frac{dy}{dx} = (\sin x)^{\cos x} [\cos x \cot x - \sin x \log(\sin x)]$$

Question 3 (ii)

Differentiate $(\sin x)^{\sin x}$, $0 < x < \pi$ w.r.t. x .

TYPE 47 Logarithmic Differentiation of Variable Exponents

★ Likely Exam Question

Given: Let $y = (\sin x)^{\sin x}$, where $0 < x < \pi$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log y = \sin x \cdot \log(\sin x)$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= \sin x \cdot \frac{d}{dx}[\log(\sin x)] + \log(\sin x) \cdot \frac{d}{dx}[\sin x] \\ &= \sin x \cdot \left(\frac{1}{\sin x} \cdot \cos x \right) + \log(\sin x) \cdot \cos x \\ &= \cos x + \cos x \log(\sin x) \\ &= \cos x (1 + \log(\sin x)) \\ \frac{dy}{dx} &= y \cdot \cos x (1 + \log(\sin x))\end{aligned}$$

Substituting $y = (\sin x)^{\sin x}$:

$$\frac{dy}{dx} = (\sin x)^{\sin x} \cos x (1 + \log(\sin x))$$

Final Answer

$$\frac{dy}{dx} = (\sin x)^{\sin x} (1 + \log(\sin x)) \cos x$$

Question 3 (iii)

Differentiate $(\tan x)^{\tan x}$, $0 < x < \frac{\pi}{2}$ w.r.t. x .

TYPE 47 Logarithmic Differentiation of Variable Exponents

Given: Let $y = (\tan x)^{\tan x}$, where $0 < x < \frac{\pi}{2}$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log y = \tan x \cdot \log(\tan x)$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= \tan x \cdot \frac{d}{dx} [\log(\tan x)] + \log(\tan x) \cdot \frac{d}{dx} [\tan x] \\ &= \tan x \cdot \left(\frac{1}{\tan x} \cdot \sec^2 x \right) + \log(\tan x) \cdot \sec^2 x \\ &= \sec^2 x + \sec^2 x \log(\tan x) \\ &= \sec^2 x (1 + \log(\tan x)) \\ \frac{dy}{dx} &= y \cdot \sec^2 x (1 + \log(\tan x)) \end{aligned}$$

Substituting $y = (\tan x)^{\tan x}$:

$$\frac{dy}{dx} = (\tan x)^{\tan x} (1 + \log(\tan x)) \sec^2 x$$

Final Answer

$$\frac{dy}{dx} = (\tan x)^{\tan x} (1 + \log(\tan x)) \sec^2 x$$

Question 4 (i)

Differentiate $x^{\sin x}$, $x > 0$ w.r.t. x .

TYPE 47 Logarithmic Differentiation of Variable Exponents

★ Likely Exam Question

Given: Let $y = x^{\sin x}$, where $x > 0$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log y = \sin x \cdot \log x$$

Differentiating both sides with respect to x :

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= \sin x \cdot \frac{d}{dx} [\log x] + \log x \cdot \frac{d}{dx} [\sin x] \\ &= \sin x \cdot \frac{1}{x} + \log x \cdot \cos x \\ &= \frac{\sin x}{x} + \cos x \log x \\ \frac{dy}{dx} &= y \left[\frac{\sin x}{x} + \cos x \log x \right] \end{aligned}$$

Substituting $y = x^{\sin x}$:

$$\frac{dy}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right]$$

Final Answer

$$\frac{dy}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right]$$

Question 4 (ii)

Differentiate $(2x + 3)^{x-5}$, $x > -\frac{3}{2}$ w.r.t. x .

TYPE 47 Logarithmic Differentiation of Variable Exponents

Given: Let $y = (2x + 3)^{x-5}$, where $x > -\frac{3}{2}$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log y = (x - 5) \log(2x + 3)$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= (x - 5) \cdot \frac{d}{dx} [\log(2x + 3)] + \log(2x + 3) \cdot \frac{d}{dx} [x - 5] \\ &= (x - 5) \cdot \left(\frac{1}{2x + 3} \cdot 2 \right) + \log(2x + 3) \cdot 1 \\ &= \frac{2(x - 5)}{2x + 3} + \log(2x + 3) \\ \frac{dy}{dx} &= y \left[\frac{2(x - 5)}{2x + 3} + \log(2x + 3) \right] \end{aligned}$$

Substituting $y = (2x + 3)^{x-5}$:

$$\frac{dy}{dx} = (2x + 3)^{x-5} \left[\frac{2(x-5)}{2x+3} + \log(2x+3) \right]$$

Final Answer

$$\frac{dy}{dx} = (2x + 3)^{x-5} \left[\frac{2(x-5)}{2x+3} + \log(2x+3) \right]$$

Question 5 (i)

Differentiate $(\log x)^{\cos x}$, $x > 1$ w.r.t. x .

TYPE 47 Logarithmic Differentiation of Variable Exponents

★ Likely Exam Question

Given: Let $y = (\log x)^{\cos x}$, where $x > 1$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log y = \cos x \cdot \log(\log x)$$

Differentiating both sides with respect to x :

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= \cos x \cdot \frac{d}{dx} [\log(\log x)] + \log(\log x) \cdot \frac{d}{dx} [\cos x] \\ &= \cos x \cdot \left(\frac{1}{\log x} \cdot \frac{1}{x} \right) + \log(\log x) \cdot (-\sin x) \\ &= \frac{\cos x}{x \log x} - \sin x \log(\log x) \\ \frac{dy}{dx} &= y \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right] \end{aligned}$$

Substituting $y = (\log x)^{\cos x}$:

$$\frac{dy}{dx} = (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right]$$

Final Answer

$$\frac{dy}{dx} = (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right]$$

Question 5 (ii)

Differentiate $(\log x)^{\log x}$, $x > 1$ w.r.t. x .

TYPE 47 Logarithmic Differentiation of Variable Exponents

★ Likely Exam Question

Given: Let $y = (\log x)^{\log x}$, where $x > 1$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log y = \log x \cdot \log(\log x)$$

Differentiating both sides with respect to x :

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= \log x \cdot \frac{d}{dx} [\log(\log x)] + \log(\log x) \cdot \frac{d}{dx} [\log x] \\ &= \log x \cdot \left(\frac{1}{\log x} \cdot \frac{1}{x} \right) + \log(\log x) \cdot \frac{1}{x} \\ &= \frac{1}{x} + \frac{\log(\log x)}{x} \\ &= \frac{1}{x} (1 + \log(\log x)) \\ \frac{dy}{dx} &= y \cdot \frac{1}{x} (1 + \log(\log x)) \end{aligned}$$

Substituting $y = (\log x)^{\log x}$:

$$\frac{dy}{dx} = (\log x)^{\log x} \cdot \frac{1}{x} (1 + \log(\log x))$$

Final Answer

$$\frac{dy}{dx} = (\log x)^{\log x} \cdot \frac{1}{x} (1 + \log(\log x))$$

Question 6 (i)

If $y = x^y$, prove that $x \frac{dy}{dx} = \frac{y^2}{1 - y \log x}$.

TYPE 48 Implicit Logarithmic Differentiation

★ Likely Exam Question

To Prove: $x \frac{dy}{dx} = \frac{y^2}{1 - y \log x}$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Given:

$$y = x^y$$

Taking natural logarithm on both sides:

$$\log y = \log(x^y)$$

$$\log y = y \log x$$

Differentiating both sides with respect to x :

$$\frac{1}{y} \frac{dy}{dx} = y \cdot \frac{d}{dx}[\log x] + \log x \cdot \frac{dy}{dx}$$

$$\frac{1}{y} \frac{dy}{dx} = y \cdot \frac{1}{x} + \log x \frac{dy}{dx}$$

Grouping terms containing $\frac{dy}{dx}$ on the left-hand side:

$$\frac{1}{y} \frac{dy}{dx} - \log x \frac{dy}{dx} = \frac{y}{x}$$

$$\left(\frac{1}{y} - \log x\right) \frac{dy}{dx} = \frac{y}{x}$$

$$\left(\frac{1 - y \log x}{y}\right) \frac{dy}{dx} = \frac{y}{x}$$

Multiplying both sides by x :

$$x \left(\frac{1 - y \log x}{y}\right) \frac{dy}{dx} = y$$

$$x \frac{dy}{dx} = \frac{y^2}{1 - y \log x}$$

Hence Proved

$$x \frac{dy}{dx} = \frac{y^2}{1 - y \log x}, \text{ hence proved.}$$

Question 6 (ii)

If $x = e^{x/y}$, prove that $\frac{dy}{dx} = \frac{x - y}{x \log x}$.

TYPE 48 Implicit Logarithmic Differentiation

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{x - y}{x \log x}$

Concept / Formula Used: $\log(e^z) = z$

Solution:

Given:

$$x = e^{x/y}$$

Taking natural logarithm on both sides:

$$\begin{aligned}\log x &= \log(e^{x/y}) \\ \log x &= \frac{x}{y}\end{aligned}$$

From this, we can write:

$$\begin{aligned}y \log x &= x \\ y &= \frac{x}{\log x}\end{aligned}$$

Differentiating y with respect to x using the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\log x \cdot \frac{d}{dx}[x] - x \cdot \frac{d}{dx}[\log x]}{(\log x)^2} \\ &= \frac{\log x \cdot 1 - x \cdot \frac{1}{x}}{(\log x)^2} \\ &= \frac{\log x - 1}{(\log x)^2}\end{aligned}$$

Since $\log x = \frac{x}{y}$, we substitute $\log x = \frac{x}{y}$ into the numerator:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{x}{y} - 1}{\frac{x}{y} \cdot \log x} \\ &= \frac{\frac{x-y}{y}}{\frac{x \log x}{y}} \\ &= \frac{x-y}{x \log x}\end{aligned}$$

Hence Proved

$$\frac{dy}{dx} = \frac{x-y}{x \log x}, \text{ hence proved.}$$

Question 6 (iii)

If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\log x}{(\log xe)^2}$.

TYPE 48 Implicit Logarithmic Differentiation

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{\log x}{(\log xe)^2}$

Concept / Formula Used: $\log(ab) = \log a + \log b$ $\log e = 1$

Solution:

Given equation:

$$x^y = e^{x-y}$$

Taking natural logarithm on both sides:

$$\begin{aligned}\log(x^y) &= \log(e^{x-y}) \\ y \log x &= (x - y) \log e \\ y \log x &= x - y \quad (\text{since } \log e = 1)\end{aligned}$$

Rearranging to express y explicitly in terms of x :

$$\begin{aligned}y \log x + y &= x \\ y(1 + \log x) &= x \\ y &= \frac{x}{1 + \log x}\end{aligned}$$

Differentiating y with respect to x using the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1 + \log x) \cdot \frac{d}{dx}[x] - x \cdot \frac{d}{dx}[1 + \log x]}{(1 + \log x)^2} \\ &= \frac{(1 + \log x) \cdot 1 - x \cdot \frac{1}{x}}{(1 + \log x)^2} \\ &= \frac{1 + \log x - 1}{(1 + \log x)^2} \\ &= \frac{\log x}{(1 + \log x)^2}\end{aligned}$$

Since $1 + \log x = \log e + \log x = \log(xe)$:

$$\frac{dy}{dx} = \frac{\log x}{(\log xe)^2}$$

Hence Proved

$$\frac{dy}{dx} = \frac{\log x}{(\log xe)^2}, \text{ hence proved.}$$

Question 6 (iv)

If $x^{16}y^9 = (x^2 + y)^{17}$, prove that $\frac{dy}{dx} = \frac{2y}{x}$.

TYPE 48 Implicit Logarithmic Differentiation

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{2y}{x}$

Concept / Formula Used: $\log(u \cdot v) = \log u + \log v$ $\log(u^k) = k \log u$

Solution:

Given equation:

$$x^{16}y^9 = (x^2 + y)^{17}$$

Taking natural logarithm on both sides:

$$\begin{aligned}\log(x^{16}y^9) &= \log((x^2 + y)^{17}) \\ \log(x^{16}) + \log(y^9) &= 17 \log(x^2 + y) \\ 16 \log x + 9 \log y &= 17 \log(x^2 + y)\end{aligned}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{16}{x} + \frac{9}{y} \frac{dy}{dx} &= \frac{17}{x^2 + y} \cdot \left(2x + \frac{dy}{dx} \right) \\ \frac{16}{x} + \frac{9}{y} \frac{dy}{dx} &= \frac{34x}{x^2 + y} + \frac{17}{x^2 + y} \frac{dy}{dx}\end{aligned}$$

Grouping terms involving $\frac{dy}{dx}$ on one side:

$$\begin{aligned}\left(\frac{9}{y} - \frac{17}{x^2 + y} \right) \frac{dy}{dx} &= \frac{34x}{x^2 + y} - \frac{16}{x} \\ \left(\frac{9(x^2 + y) - 17y}{y(x^2 + y)} \right) \frac{dy}{dx} &= \frac{34x^2 - 16(x^2 + y)}{x(x^2 + y)} \\ \left(\frac{9x^2 + 9y - 17y}{y(x^2 + y)} \right) \frac{dy}{dx} &= \frac{34x^2 - 16x^2 - 16y}{x(x^2 + y)} \\ \left(\frac{9x^2 - 8y}{y(x^2 + y)} \right) \frac{dy}{dx} &= \frac{18x^2 - 16y}{x(x^2 + y)} \\ \left(\frac{9x^2 - 8y}{y(x^2 + y)} \right) \frac{dy}{dx} &= \frac{2(9x^2 - 8y)}{x(x^2 + y)}\end{aligned}$$

Cancelling $(x^2 + y)$ from both denominators and $(9x^2 - 8y)$ from both numerators:

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= \frac{2}{x} \\ \frac{dy}{dx} &= \frac{2y}{x}\end{aligned}$$

Hence Proved

$$\frac{dy}{dx} = \frac{2y}{x}, \text{ hence proved.}$$

Question 7

Find the derivative of $x^x + a^x + x^a + a^a$ for some fixed $a > 0, x > 0$.

TYPE 49 Logarithmic Differentiation of Sums

★ Likely Exam Question

Given: Let $y = x^x + a^x + x^a + a^a$, where $a > 0, x > 0$

Concept / Formula Used: $\frac{d}{dx}(a^x) = a^x \log a$ $\frac{d}{dx}(x^a) = ax^{a-1}$ $\frac{d}{dx}(a^a) = 0$ $\frac{d}{dx}(x^x) = x^x(1 + \log x)$

Solution:

Let $y = u + v + w + z$, where:

$$u = x^x, \quad v = a^x, \quad w = x^a, \quad z = a^a$$

To find $\frac{du}{dx}$, take logarithm on both sides of $u = x^x$:

$$\begin{aligned} \log u &= x \log x \\ \frac{1}{u} \frac{du}{dx} &= x \cdot \frac{1}{x} + \log x \cdot 1 \\ \frac{du}{dx} &= u(1 + \log x) = x^x(1 + \log x) \end{aligned}$$

For standard terms:

$$\begin{aligned} \frac{dv}{dx} &= \frac{d}{dx}(a^x) = a^x \log a \\ \frac{dw}{dx} &= \frac{d}{dx}(x^a) = ax^{a-1} \\ \frac{dz}{dx} &= \frac{d}{dx}(a^a) = 0 \quad (\text{since } a^a \text{ is constant}) \end{aligned}$$

Therefore, differentiating y with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx} + \frac{dz}{dx} \\ &= x^x(1 + \log x) + a^x \log a + ax^{a-1} + 0 \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = x^x(1 + \log x) + a^x \log a + ax^{a-1}$$

Question 8 (i)

Differentiate $x^{\log x} + (\log x)^x$ w.r.t. x .

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = x^{\log x}$ and $v = (\log x)^x$

Concept / Formula Used: $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

Solution:

For $u = x^{\log x}$:

$$\begin{aligned} \log u &= \log x \cdot \log x = (\log x)^2 \\ \frac{1}{u} \frac{du}{dx} &= 2 \log x \cdot \frac{1}{x} \\ \frac{du}{dx} &= u \cdot \frac{2 \log x}{x} = x^{\log x} \cdot \frac{2 \log x}{x} \end{aligned}$$

For $v = (\log x)^x$:

$$\begin{aligned}\log v &= x \log(\log x) \\ \frac{1}{v} \frac{dv}{dx} &= x \cdot \left(\frac{1}{\log x} \cdot \frac{1}{x} \right) + \log(\log x) \cdot 1 \\ \frac{1}{v} \frac{dv}{dx} &= \frac{1}{\log x} + \log(\log x) \\ \frac{dv}{dx} &= v \left[\frac{1}{\log x} + \log(\log x) \right] = (\log x)^x \left[\frac{1}{\log x} + \log(\log x) \right]\end{aligned}$$

Combining both derivatives:

$$\begin{aligned}\frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \\ &= x^{\log x} \cdot \frac{2 \log x}{x} + (\log x)^x \left[\frac{1}{\log x} + \log(\log x) \right]\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = x^{\log x} \cdot \frac{2 \log x}{x} + (\log x)^x \left[\frac{1}{\log x} + \log(\log x) \right]$$

Question 8 (ii)

Differentiate $(\sin x)^{\cos x} + x^{\sin x}$ w.r.t. x .

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = (\sin x)^{\cos x}$ and $v = x^{\sin x}$

Concept / Formula Used: $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

Solution:

For $u = (\sin x)^{\cos x}$:

$$\begin{aligned}\log u &= \cos x \log(\sin x) \\ \frac{1}{u} \frac{du}{dx} &= \cos x \cdot \frac{\cos x}{\sin x} + \log(\sin x) \cdot (-\sin x) \\ &= \cos x \cot x - \sin x \log(\sin x) \\ \frac{du}{dx} &= (\sin x)^{\cos x} [\cos x \cot x - \sin x \log(\sin x)]\end{aligned}$$

For $v = x^{\sin x}$:

$$\begin{aligned}\log v &= \sin x \log x \\ \frac{1}{v} \frac{dv}{dx} &= \sin x \cdot \frac{1}{x} + \log x \cdot \cos x \\ \frac{dv}{dx} &= x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right]\end{aligned}$$

Combining the results:

$$\begin{aligned}\frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \\ &= (\sin x)^{\cos x} [\cos x \cot x - \sin x \log(\sin x)] + x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right]\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = (\sin x)^{\cos x} [\cos x \cot x - \sin x \log(\sin x)] + x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right]$$

Question 8 (iii)

Differentiate $x^{\cos x} + (\cos x)^x$ w.r.t. x .

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = x^{\cos x}$ and $v = (\cos x)^x$

Concept / Formula Used: $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

Solution:

For $u = x^{\cos x}$:

$$\begin{aligned} \log u &= \cos x \log x \\ \frac{1}{u} \frac{du}{dx} &= \cos x \cdot \frac{1}{x} + \log x \cdot (-\sin x) \\ &= \frac{\cos x}{x} - \sin x \log x \\ \frac{du}{dx} &= x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right] \end{aligned}$$

For $v = (\cos x)^x$:

$$\begin{aligned} \log v &= x \log(\cos x) \\ \frac{1}{v} \frac{dv}{dx} &= x \cdot \frac{1}{\cos x} (-\sin x) + \log(\cos x) \cdot 1 \\ &= -x \tan x + \log(\cos x) \\ \frac{dv}{dx} &= (\cos x)^x [\log(\cos x) - x \tan x] \end{aligned}$$

Combining the two derivatives:

$$\frac{dy}{dx} = x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right] + (\cos x)^x [\log(\cos x) - x \tan x]$$

Final Answer

$$\frac{dy}{dx} = x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right] + (\cos x)^x [\log(\cos x) - x \tan x]$$

Question 8 (iv)

Differentiate $x^{\cos x} + (\sin x)^{\tan x}$ w.r.t. x .

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = x^{\cos x}$ and $v = (\sin x)^{\tan x}$

Concept / Formula Used: $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

Solution:For $u = x^{\cos x}$:

$$\begin{aligned}\log u &= \cos x \log x \\ \frac{1}{u} \frac{du}{dx} &= \cos x \cdot \frac{1}{x} - \sin x \log x \\ \frac{du}{dx} &= x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right]\end{aligned}$$

For $v = (\sin x)^{\tan x}$:

$$\begin{aligned}\log v &= \tan x \log(\sin x) \\ \frac{1}{v} \frac{dv}{dx} &= \tan x \cdot \frac{\cos x}{\sin x} + \log(\sin x) \cdot \sec^2 x \\ &= \frac{\sin x}{\cos x} \cdot \frac{\cos x}{\sin x} + \sec^2 x \log(\sin x) \\ &= 1 + \sec^2 x \log(\sin x) \\ \frac{dv}{dx} &= (\sin x)^{\tan x} [1 + \sec^2 x \log(\sin x)]\end{aligned}$$

Combining both:

$$\frac{dy}{dx} = x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right] + (\sin x)^{\tan x} [1 + \sec^2 x \log(\sin x)]$$

Final Answer

$$\frac{dy}{dx} = x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right] + (\sin x)^{\tan x} [1 + \sec^2 x \log(\sin x)]$$

Question 8 (v)Differentiate $(\sin 2x)^x + \sin^{-1} \sqrt{3x}$ w.r.t. x .**TYPE 49** Logarithmic Differentiation of Sums**Given:** Let $y = u + v$, where $u = (\sin 2x)^x$ and $v = \sin^{-1} \sqrt{3x}$ **Concept / Formula Used:** $\frac{d}{dx}(\sin^{-1} z) = \frac{1}{\sqrt{1-z^2}} \cdot \frac{dz}{dx}$ **Solution:**For $u = (\sin 2x)^x$:

$$\begin{aligned}\log u &= x \log(\sin 2x) \\ \frac{1}{u} \frac{du}{dx} &= x \cdot \frac{1}{\sin 2x} \cdot (2 \cos 2x) + \log(\sin 2x) \cdot 1 \\ &= 2x \cot 2x + \log(\sin 2x) \\ \frac{du}{dx} &= (\sin 2x)^x [2x \cot 2x + \log(\sin 2x)]\end{aligned}$$

For $v = \sin^{-1} \sqrt{3x}$:

$$\begin{aligned}\frac{dv}{dx} &= \frac{1}{\sqrt{1 - (\sqrt{3x})^2}} \cdot \frac{d}{dx}(\sqrt{3x}) \\ &= \frac{1}{\sqrt{1 - 3x}} \cdot \frac{\sqrt{3}}{2\sqrt{x}} \\ &= \frac{\sqrt{3}}{2\sqrt{x(1 - 3x)}} \\ &= \frac{3}{2\sqrt{3x - 9x^2}}\end{aligned}$$

Combining both parts:

$$\frac{dy}{dx} = (\sin 2x)^x [2x \cot 2x + \log(\sin 2x)] + \frac{3}{2\sqrt{3x - 9x^2}}$$

Final Answer

$$\frac{dy}{dx} = (\sin 2x)^x [2x \cot 2x + \log(\sin 2x)] + \frac{3}{2\sqrt{3x - 9x^2}}$$

Question 8 (vi)

Differentiate $e^{\sin x} + (\tan x)^x$ w.r.t. x .

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = e^{\sin x}$ and $v = (\tan x)^x$

Concept / Formula Used: $\frac{d}{dx}(e^{g(x)}) = e^{g(x)}g'(x)$

Solution:

For $u = e^{\sin x}$:

$$\frac{du}{dx} = e^{\sin x} \cdot \frac{d}{dx}(\sin x) = e^{\sin x} \cos x$$

For $v = (\tan x)^x$:

$$\begin{aligned}\log v &= x \log(\tan x) \\ \frac{1}{v} \frac{dv}{dx} &= x \cdot \frac{1}{\tan x} \cdot \sec^2 x + \log(\tan x) \cdot 1 \\ &= x \cdot \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x} + \log(\tan x) \\ &= \frac{x}{\sin x \cos x} + \log(\tan x) \\ &= \frac{2x}{2 \sin x \cos x} + \log(\tan x) \\ &= 2x \csc 2x + \log(\tan x) \\ \frac{dv}{dx} &= (\tan x)^x [2x \csc 2x + \log(\tan x)]\end{aligned}$$

Combining both:

$$\frac{dy}{dx} = e^{\sin x} \cos x + (\tan x)^x [2x \csc 2x + \log(\tan x)]$$

Final Answer

$$\frac{dy}{dx} = e^{\sin x} \cos x + (\tan x)^x [2x \csc 2x + \log(\tan x)]$$

Question 8 (vii)

Differentiate $x^x - 2^{\sin x}$ w.r.t. x .

TYPE 49 Logarithmic Differentiation of Sums

★ Likely Exam Question

Given: Let $y = u - v$, where $u = x^x$ and $v = 2^{\sin x}$

Concept / Formula Used: $\frac{d}{dx}(a^{g(x)}) = a^{g(x)} \log a \cdot g'(x)$

Solution:

For $u = x^x$:

$$\begin{aligned} \log u &= x \log x \\ \frac{1}{u} \frac{du}{dx} &= x \cdot \frac{1}{x} + \log x \cdot 1 = 1 + \log x \\ \frac{du}{dx} &= x^x (1 + \log x) \end{aligned}$$

For $v = 2^{\sin x}$:

$$\begin{aligned} \frac{dv}{dx} &= 2^{\sin x} \cdot \log 2 \cdot \frac{d}{dx}(\sin x) \\ &= 2^{\sin x} \cdot \log 2 \cdot \cos x \end{aligned}$$

Combining the results:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dx} - \frac{dv}{dx} \\ &= x^x (1 + \log x) - 2^{\sin x} \cos x \log 2 \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = x^x (1 + \log x) - 2^{\sin x} \cos x \log 2$$

Question 8 (viii)

Differentiate $x^{x \cos x} + \frac{x^2 + 1}{x^2 - 1}$ w.r.t. x .

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = x^{x \cos x}$ and $v = \frac{x^2 + 1}{x^2 - 1}$

Concept / Formula Used: $\frac{d}{dx} \left(\frac{f}{g} \right) = \frac{gf' - fg'}{g^2}$

Solution:

For $u = x^{x \cos x}$:

$$\log u = x \cos x \log x$$

Differentiating w.r.t. x using the product rule for three terms:

$$\begin{aligned} \frac{1}{u} \frac{du}{dx} &= \frac{d}{dx}(x) \cdot \cos x \log x + x \cdot \frac{d}{dx}(\cos x) \cdot \log x + x \cos x \cdot \frac{d}{dx}(\log x) \\ &= 1 \cdot \cos x \log x + x(-\sin x) \log x + x \cos x \cdot \frac{1}{x} \\ &= \cos x \log x - x \sin x \log x + \cos x \\ &= (1 + \log x) \cos x - x \sin x \log x \\ \frac{du}{dx} &= x^{x \cos x} [(1 + \log x) \cos x - x \sin x \log x] \end{aligned}$$

For $v = \frac{x^2 + 1}{x^2 - 1}$:

$$\begin{aligned} \frac{dv}{dx} &= \frac{(x^2 - 1) \frac{d}{dx}(x^2 + 1) - (x^2 + 1) \frac{d}{dx}(x^2 - 1)}{(x^2 - 1)^2} \\ &= \frac{(x^2 - 1)(2x) - (x^2 + 1)(2x)}{(x^2 - 1)^2} \\ &= \frac{2x^3 - 2x - 2x^3 - 2x}{(x^2 - 1)^2} \\ &= \frac{-4x}{(x^2 - 1)^2} \end{aligned}$$

Combining both derivatives:

$$\frac{dy}{dx} = x^{x \cos x} [(1 + \log x) \cos x - x \sin x \log x] - \frac{4x}{(x^2 - 1)^2}$$

Final Answer

$$\frac{dy}{dx} = x^{x \cos x} [(1 + \log x) \cos x - x \sin x \log x] - \frac{4x}{(x^2 - 1)^2}$$

Question 9 (i)

If $y = (\log x)^{\cos x} + \frac{x^2 + 1}{x^2 - 1}$, find $\frac{dy}{dx}$.

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = (\log x)^{\cos x}$ and $v = \frac{x^2 + 1}{x^2 - 1}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

Solution:

For $u = (\log x)^{\cos x}$:

$$\begin{aligned}\log u &= \cos x \log(\log x) \\ \frac{1}{u} \frac{du}{dx} &= \cos x \cdot \frac{1}{\log x} \cdot \frac{1}{x} + \log(\log x) \cdot (-\sin x) \\ &= \frac{\cos x}{x \log x} - \sin x \log(\log x) \\ \frac{du}{dx} &= (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right]\end{aligned}$$

For $v = \frac{x^2 + 1}{x^2 - 1}$:

$$\begin{aligned}\frac{dv}{dx} &= \frac{(x^2 - 1)(2x) - (x^2 + 1)(2x)}{(x^2 - 1)^2} \\ &= \frac{-4x}{(x^2 - 1)^2}\end{aligned}$$

Combining the two derivatives:

$$\begin{aligned}\frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \\ &= (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right] - \frac{4x}{(x^2 - 1)^2}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right] - \frac{4x}{(x^2 - 1)^2}$$

Question 9 (ii)

If $y = e^{x^2 \cos x} + (\cos x)^x$, find $\frac{dy}{dx}$.

TYPE 49 Logarithmic Differentiation of Sums

Given: Let $y = u + v$, where $u = e^{x^2 \cos x}$ and $v = (\cos x)^x$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

Solution:

For $u = e^{x^2 \cos x}$:

$$\begin{aligned}\frac{du}{dx} &= e^{x^2 \cos x} \cdot \frac{d}{dx}(x^2 \cos x) \\ &= e^{x^2 \cos x} [x^2(-\sin x) + \cos x(2x)] \\ &= e^{x^2 \cos x} (2x \cos x - x^2 \sin x)\end{aligned}$$

For $v = (\cos x)^x$:

$$\begin{aligned}\log v &= x \log(\cos x) \\ \frac{1}{v} \frac{dv}{dx} &= x \cdot \frac{-\sin x}{\cos x} + \log(\cos x) \cdot 1 \\ &= -x \tan x + \log(\cos x) \\ \frac{dv}{dx} &= (\cos x)^x [\log(\cos x) - x \tan x]\end{aligned}$$

Combining both parts:

$$\frac{dy}{dx} = e^{x^2 \cos x} (2x \cos x - x^2 \sin x) + (\cos x)^x [\log(\cos x) - x \tan x]$$

Final Answer

$$\frac{dy}{dx} = e^{x^2 \cos x} (2x \cos x - x^2 \sin x) + (\cos x)^x [\log(\cos x) - x \tan x]$$

Question 10 (i)

Find $\frac{dy}{dx}$ when $x^y y^x = a^b$.

TYPE 48 Implicit Logarithmic Differentiation

Given: $x^y y^x = a^b$, where a, b are constants

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\log(uv) = \log u + \log v$

Solution:

Taking natural logarithm on both sides:

$$\begin{aligned}\log(x^y y^x) &= \log(a^b) \\ \log(x^y) + \log(y^x) &= \log(a^b) \\ y \log x + x \log y &= b \log a\end{aligned}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx}[y \log x] + \frac{d}{dx}[x \log y] &= \frac{d}{dx}[b \log a] \\ \left(y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx}\right) + \left(x \cdot \frac{1}{y} \frac{dy}{dx} + \log y \cdot 1\right) &= 0 \\ \frac{y}{x} + \log x \frac{dy}{dx} + \frac{x}{y} \frac{dy}{dx} + \log y &= 0\end{aligned}$$

Grouping terms containing $\frac{dy}{dx}$:

$$\begin{aligned}\left(\log x + \frac{x}{y}\right) \frac{dy}{dx} &= -\left(\log y + \frac{y}{x}\right) \\ \left(\frac{y \log x + x}{y}\right) \frac{dy}{dx} &= -\left(\frac{x \log y + y}{x}\right) \\ \frac{dy}{dx} &= -\frac{y}{x} \cdot \frac{x \log y + y}{y \log x + x}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\frac{y(y + x \log y)}{x(x + y \log x)}$$

Question 10 (ii)

Find $\frac{dy}{dx}$ when $x^y + y^x = a^b$.

TYPE 49 Logarithmic Differentiation of Sums

★ Likely Exam Question

Given: $x^y + y^x = a^b$, where a, b are constants

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $u = x^y$ $v = y^x$ $\frac{du}{dx} + \frac{dv}{dx} = 0$

Solution:

Let $u = x^y$ and $v = y^x$. Then $u + v = a^b$. Differentiating with respect to x :

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

For $u = x^y$:

$$\begin{aligned}\log u &= y \log x \\ \frac{1}{u} \frac{du}{dx} &= y \cdot \frac{1}{x} + \log x \frac{dy}{dx} \\ \frac{du}{dx} &= x^y \left(\frac{y}{x} + \log x \frac{dy}{dx} \right) = yx^{y-1} + x^y \log x \frac{dy}{dx}\end{aligned}$$

For $v = y^x$:

$$\begin{aligned}\log v &= x \log y \\ \frac{1}{v} \frac{dv}{dx} &= x \cdot \frac{1}{y} \frac{dy}{dx} + \log y \cdot 1 \\ \frac{dv}{dx} &= y^x \left(\frac{x}{y} \frac{dy}{dx} + \log y \right) = xy^{x-1} \frac{dy}{dx} + y^x \log y\end{aligned}$$

Substituting $\frac{du}{dx}$ and $\frac{dv}{dx}$ into $\frac{du}{dx} + \frac{dv}{dx} = 0$:

$$\begin{aligned}
 yx^{y-1} + x^y \log x \frac{dy}{dx} + xy^{x-1} \frac{dy}{dx} + y^x \log y &= 0 \\
 (x^y \log x + xy^{x-1}) \frac{dy}{dx} &= -(yx^{y-1} + y^x \log y) \\
 \frac{dy}{dx} &= -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}
 \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$$

Question 10 (iii)

Find $\frac{dy}{dx}$ when $x^x + y^x = 1$.

TYPE 49 Logarithmic Differentiation of Sums

Given: $x^x + y^x = 1$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $u = x^x$ $v = y^x$ $\frac{du}{dx} + \frac{dv}{dx} = 0$

Solution:

Let $u = x^x$ and $v = y^x$. Then $u + v = 1$. Differentiating with respect to x :

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

For $u = x^x$:

$$\begin{aligned}
 \log u &= x \log x \\
 \frac{1}{u} \frac{du}{dx} &= x \cdot \frac{1}{x} + \log x \cdot 1 = 1 + \log x \\
 \frac{du}{dx} &= x^x (1 + \log x)
 \end{aligned}$$

For $v = y^x$:

$$\begin{aligned}
 \log v &= x \log y \\
 \frac{1}{v} \frac{dv}{dx} &= x \cdot \frac{1}{y} \frac{dy}{dx} + \log y \cdot 1 \\
 \frac{dv}{dx} &= y^x \left(\frac{x}{y} \frac{dy}{dx} + \log y \right) = xy^{x-1} \frac{dy}{dx} + y^x \log y
 \end{aligned}$$

Substituting into $\frac{du}{dx} + \frac{dv}{dx} = 0$:

$$x^x(1 + \log x) + xy^{x-1}\frac{dy}{dx} + y^x \log y = 0$$

$$xy^{x-1}\frac{dy}{dx} = -[x^x(1 + \log x) + y^x \log y]$$

$$\frac{dy}{dx} = -\frac{x^x(1 + \log x) + y^x \log y}{xy^{x-1}}$$

Final Answer

$$\frac{dy}{dx} = -\frac{x^x(1 + \log x) + y^x \log y}{xy^{x-1}}$$

Question 10 (iv)

Find $\frac{dy}{dx}$ when $(\sin x)^y = x + y$.

TYPE 48 Implicit Logarithmic Differentiation

Given: $(\sin x)^y = x + y$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Taking natural logarithm on both sides:

$$\log((\sin x)^y) = \log(x + y)$$

$$y \log(\sin x) = \log(x + y)$$

Differentiating both sides with respect to x :

$$y \cdot \frac{d}{dx}[\log(\sin x)] + \log(\sin x) \cdot \frac{dy}{dx} = \frac{d}{dx}[\log(x + y)]$$

$$y \cdot \left(\frac{1}{\sin x} \cdot \cos x \right) + \log(\sin x) \frac{dy}{dx} = \frac{1}{x + y} \left(1 + \frac{dy}{dx} \right)$$

$$y \cot x + \log(\sin x) \frac{dy}{dx} = \frac{1}{x + y} + \frac{1}{x + y} \frac{dy}{dx}$$

Grouping terms containing $\frac{dy}{dx}$ on the left side:

$$\left(\log(\sin x) - \frac{1}{x + y} \right) \frac{dy}{dx} = \frac{1}{x + y} - y \cot x$$

$$\left(\frac{(x + y) \log(\sin x) - 1}{x + y} \right) \frac{dy}{dx} = \frac{1 - y(x + y) \cot x}{x + y}$$

$$((x + y) \log(\sin x) - 1) \frac{dy}{dx} = 1 - y(x + y) \cot x$$

$$\frac{dy}{dx} = \frac{1 - y(x + y) \cot x}{(x + y) \log(\sin x) - 1}$$

Final Answer

$$\frac{dy}{dx} = \frac{1 - y(x+y) \cot x}{(x+y) \log(\sin x) - 1}$$

Question 10 (v)

Find $\frac{dy}{dx}$ when $xy = e^{x-y}$.

TYPE 48 Implicit Logarithmic Differentiation

★ Likely Exam Question

Given: $xy = e^{x-y}$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\log(xy) = \log x + \log y$ $\log(e^{x-y}) = x - y$

Solution:

Taking natural logarithm on both sides:

$$\begin{aligned}\log(xy) &= \log(e^{x-y}) \\ \log x + \log y &= x - y\end{aligned}$$

Differentiating both sides with respect to x :

$$\frac{1}{x} + \frac{1}{y} \frac{dy}{dx} = 1 - \frac{dy}{dx}$$

Grouping terms involving $\frac{dy}{dx}$ on one side:

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} + \frac{dy}{dx} &= 1 - \frac{1}{x} \\ \left(\frac{1}{y} + 1\right) \frac{dy}{dx} &= \frac{x-1}{x} \\ \left(\frac{1+y}{y}\right) \frac{dy}{dx} &= \frac{x-1}{x} \\ \frac{dy}{dx} &= \frac{y(x-1)}{x(1+y)}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \frac{y(x-1)}{x(y+1)}$$

Question 10 (vi)

Find $\frac{dy}{dx}$ when $(\cos x)^y = (\cos y)^x$.

Given: $(\cos x)^y = (\cos y)^x$

To Find: $\frac{dy}{dx}$

Concept / Formula Used: $\log(u^v) = v \log u$

Taking natural logarithm on both sides:

$$\begin{aligned}\log((\cos x)^y) &= \log((\cos y)^x) \\ y \log(\cos x) &= x \log(\cos y)\end{aligned}$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned} y \cdot \frac{d}{dx}[\log(\cos x)] + \log(\cos x) \cdot \frac{dy}{dx} &= x \cdot \frac{d}{dx}[\log(\cos y)] + \log(\cos y) \cdot \frac{d}{dx}[x] \\ y \cdot \left(\frac{-\sin x}{\cos x} \right) + \log(\cos x) \frac{dy}{dx} &= x \cdot \left(\frac{-\sin y}{\cos y} \frac{dy}{dx} \right) + \log(\cos y) \cdot 1 \\ -y \tan x + \log(\cos x) \frac{dy}{dx} &= -x \tan y \frac{dy}{dx} + \log(\cos y) \end{aligned}$$

Grouping $\frac{dy}{dx}$ terms:

$$\begin{aligned} (\log(\cos x) + x \tan y) \frac{dy}{dx} &= \log(\cos y) + y \tan x \\ \frac{dy}{dx} &= \frac{\log(\cos y) + y \tan x}{\log(\cos x) + x \tan y} \end{aligned}$$

$$\frac{dy}{dx} = \frac{\log(\cos y) + y \tan x}{\log(\cos x) + x \tan y}$$

If $y = x^{x^{\cdot^{\cdot^{\cdot^{\infty}}}}}$, prove that $\frac{dy}{dx} = \frac{y^2}{x(1 - y \log x)}$.

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = \frac{y^2}{x(1 - y \log x)}$

Concept / Formula Used: $y = x^y$

783

Given the infinite power tower:

$$y = x^{x^{x^{\cdot^{\cdot^{\cdot^{\infty}}}}}}$$

Since the exponent tower is infinite, we can rewrite the relation as:

$$y = x^y$$

Taking natural logarithm on both sides:

$$\log y = \log(x^y)$$

$$\log y = y \log x$$

Differentiating both sides with respect to x :

$$\frac{1}{y} \frac{dy}{dx} = y \cdot \frac{d}{dx}[\log x] + \log x \cdot \frac{dy}{dx}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{y}{x} + \log x \frac{dy}{dx}$$

Collecting terms involving $\frac{dy}{dx}$ on the left-hand side:

$$\begin{aligned} \left(\frac{1}{y} - \log x\right) \frac{dy}{dx} &= \frac{y}{x} \\ \left(\frac{1 - y \log x}{y}\right) \frac{dy}{dx} &= \frac{y}{x} \\ \frac{dy}{dx} &= \frac{y}{x} \cdot \frac{y}{1 - y \log x} \\ \frac{dy}{dx} &= \frac{y^2}{x(1 - y \log x)} \end{aligned}$$

Hence Proved

$$\frac{dy}{dx} = \frac{y^2}{x(1 - y \log x)}, \text{ hence proved.}$$

Question 12

If $y = x^x$, then find $\frac{dy}{dx}$ at $x = 1$.

TYPE 50 Logarithmic Differentiation with Evaluation

Given: $y = x^x$

To Find: The value of $\frac{dy}{dx}$ at $x = 1$

Concept / Formula Used: $\log(u^v) = v \log u$

Solution:

Given function:

$$y = x^x$$

Taking natural logarithm on both sides:

$$\log y = x \log x$$

Differentiating both sides with respect to x :

$$\frac{1}{y} \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x \cdot 1$$

$$\frac{1}{y} \frac{dy}{dx} = 1 + \log x$$

$$\frac{dy}{dx} = y(1 + \log x)$$

Substituting back $y = x^x$:

$$\frac{dy}{dx} = x^x(1 + \log x)$$

Now, evaluating $\frac{dy}{dx}$ at $x = 1$:

$$\left. \frac{dy}{dx} \right|_{x=1} = (1)^1 \cdot (1 + \log 1)$$

Since $\log 1 = 0$:

$$\left. \frac{dy}{dx} \right|_{x=1} = 1 \cdot (1 + 0) = 1$$

Final Answer

$$\left. \frac{dy}{dx} \right|_{x=1} = 1$$

Question 9(i)

Find $\frac{dy}{dx}$ if $y = (\log x)^{\cos x}$.

TYPE 61 Logarithmic Differentiation

Given: $y = (\log x)^{\cos x}$

Concept / Formula Used: $\frac{d}{dx}[u^v] = u^v \left(v' \ln u + v \frac{u'}{u} \right)$ $\log$

Solution:

Taking natural logarithm on both sides of the given equation:

$$\log y = \log ((\log x)^{\cos x})$$

Using the logarithmic property $\log(a^b) = b \log a$:

$$\log y = \cos x \cdot \log(\log x)$$

Differentiating both sides with respect to x using the chain rule on the left and the product rule on the right:

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx}[\cos x] \cdot \log(\log x) + \cos x \cdot \frac{d}{dx}[\log(\log x)]$$

Evaluating the derivatives on the right-hand side:

$$\frac{1}{y} \frac{dy}{dx} = (-\sin x) \cdot \log(\log x) + \cos x \cdot \frac{1}{\log x} \cdot \frac{d}{dx}[\log x]$$

$$\frac{1}{y} \frac{dy}{dx} = -\sin x \log(\log x) + \cos x \cdot \frac{1}{\log x} \cdot \frac{1}{x}$$

Multiplying both sides by y to isolate $\frac{dy}{dx}$:

$$\frac{dy}{dx} = y \left(-\sin x \log(\log x) + \frac{\cos x}{x \log x} \right)$$

Back-substituting $y = (\log x)^{\cos x}$:

$$\frac{dy}{dx} = (\log x)^{\cos x} \left(\frac{\cos x}{x \log x} - \sin x \log(\log x) \right)$$

Final Answer

$$\frac{dy}{dx} = (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right]$$

Common Mistake: Forgetting to apply the chain rule when differentiating $\log(\log x)$ with respect to x .

Question 9(ii)

Find $\frac{dy}{dx}$ if $y = x^{e^{x^2} \cos x}$.

TYPE 61 Logarithmic Differentiation

Given: $y = x^{e^{x^2} \cos x}$

Concept / Formula Used: $\frac{d}{dx}[u^v] = u^v \left(v' \ln u + v \frac{u'}{u} \right)$

Solution:

Taking natural logarithm on both sides:

$$\log y = \log \left(x^{e^{x^2} \cos x} \right)$$

Using the property $\log(a^b) = b \log a$:

$$\log y = e^{x^2} \cos x \cdot \log x$$

Differentiating both sides with respect to x using the product rule for three functions on the right-hand side:

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx}[e^{x^2}] \cdot \cos x \cdot \log x + e^{x^2} \cdot \frac{d}{dx}[\cos x] \cdot \log x + e^{x^2} \cdot \cos x \cdot \frac{d}{dx}[\log x]$$

Evaluating each derivative term:

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= (e^{x^2} \cdot 2x) \cdot \cos x \cdot \log x + e^{x^2} \cdot (-\sin x) \cdot \log x + e^{x^2} \cdot \cos x \cdot \frac{1}{x} \\ &= e^{x^2} \left[2x \cos x \log x - \sin x \log x + \frac{\cos x}{x} \right]\end{aligned}$$

Multiplying both sides by $y = x^{e^{x^2} \cos x}$:

$$\frac{dy}{dx} = x^{e^{x^2} \cos x} \cdot e^{x^2} \left(2x \cos x \log x - \sin x \log x + \frac{\cos x}{x} \right)$$

Final Answer

$$\frac{dy}{dx} = x^{e^{x^2} \cos x} e^{x^2} \left(2x \cos x \log x - \sin x \log x + \frac{\cos x}{x} \right)$$

Common Mistake: Treating the exponent as a simple function and failing to apply the product rule correctly for the product of three terms.

Question 10(i)

Find $\frac{dy}{dx}$ if $x^y + y^x = a^b$. (Implicit differentiation / Logarithmic differentiation)

TYPE 62 Implicit Differentiation with Exponential/Logarithmic Terms

Given: $x^y + y^x = a^b$

Concept / Formula Used: $u = x^y$ $v = y^x$ $u + v = a^b$ $\frac{du}{dx} + \frac{dv}{dx} = 0$

Solution:

Let $u = x^y$ and $v = y^x$. Then the given equation becomes:

$$u + v = a^b$$

Differentiating implicitly with respect to x :

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

Now, considering $u = x^y$:

$$\log u = y \log x$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{1}{u} \frac{du}{dx} &= \frac{dy}{dx} \log x + y \cdot \frac{1}{x} \\ \frac{du}{dx} &= x^y \left(\log x \frac{dy}{dx} + \frac{y}{x} \right)\end{aligned}$$

Next, considering $v = y^x$:

$$\log v = x \log y$$

Differentiating both sides with respect to x :

$$\frac{1}{v} \frac{dv}{dx} = 1 \cdot \log y + x \cdot \frac{1}{y} \frac{dy}{dx}$$

$$\frac{dv}{dx} = y^x \left(\log y + \frac{x}{y} \frac{dy}{dx} \right)$$

Substituting $\frac{du}{dx}$ and $\frac{dv}{dx}$ into $\frac{du}{dx} + \frac{dv}{dx} = 0$:

$$x^y \left(\log x \frac{dy}{dx} + \frac{y}{x} \right) + y^x \left(\log y + \frac{x}{y} \frac{dy}{dx} \right) = 0$$

Grouping the terms containing $\frac{dy}{dx}$:

$$\frac{dy}{dx} \left(x^y \log x + y^x \frac{x}{y} \right) = - \left(x^y \frac{y}{x} + y^x \log y \right)$$

Isolating $\frac{dy}{dx}$:

$$\frac{dy}{dx} = - \frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$$

Final Answer

$$\frac{dy}{dx} = - \frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$$

Common Mistake: Incorrectly differentiating x^y directly as yx^{y-1} without accounting for the variable exponent using logarithmic differentiation.

Question 10(ii)

Find $\frac{dy}{dx}$ if $x^{y-1} + y^{x-1} = 1$.

TYPE 62 Implicit Differentiation with Exponential/Logarithmic Terms

Given: $x^{y-1} + y^{x-1} = 1$

Concept / Formula Used: $u(x, y) + v(x, y) = C \quad \frac{d}{dx}[u] + \frac{d}{dx}[v] = 0$

Solution:

Let $u = x^{y-1}$ and $v = y^{x-1}$. The equation is $u + v = 1$, so:

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

For $u = x^{y-1}$:

$$\log u = (y-1) \log x$$

Differentiating with respect to x :

$$\frac{1}{u} \frac{du}{dx} = \frac{dy}{dx} \log x + (y-1) \frac{1}{x}$$

$$\frac{du}{dx} = x^{y-1} \left(\log x \frac{dy}{dx} + \frac{y-1}{x} \right)$$

For $v = y^{x-1}$:

$$\log v = (x-1) \log y$$

Differentiating with respect to x :

$$\frac{1}{v} \frac{dv}{dx} = 1 \cdot \log y + (x-1) \frac{1}{y} \frac{dy}{dx}$$

$$\frac{dv}{dx} = y^{x-1} \left(\log y + \frac{x-1}{y} \frac{dy}{dx} \right)$$

Substituting $\frac{du}{dx}$ and $\frac{dv}{dx}$ into the sum derivative equation:

$$x^{y-1} \left(\log x \frac{dy}{dx} + \frac{y-1}{x} \right) + y^{x-1} \left(\log y + \frac{x-1}{y} \frac{dy}{dx} \right) = 0$$

Rearranging terms to collect coefficients of $\frac{dy}{dx}$:

$$\frac{dy}{dx} \left[x^{y-1} \log x + y^{x-1} \frac{x-1}{y} \right] = - \left[x^{y-1} \frac{y-1}{x} + y^{x-1} \log y \right]$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = - \frac{y^{x-1} \log y + x^{y-2}(y-1)}{x^{y-1} \log x + y^{x-2}(x-1)}$$

Final Answer

$$\frac{dy}{dx} = - \frac{y^{x-1} \log y + x^{y-1} \left(\frac{y-1}{x} \right)}{x^{y-1} \log x + y^{x-1} \left(\frac{x-1}{y} \right)}$$

Common Mistake: Algebraic errors when combining fractional powers of x and y after differentiation.

Question 10(iii)

Find $\frac{dy}{dx}$ if $x^x + y^x = xy$.

TYPE 62 Implicit Differentiation with Exponential/Logarithmic Terms

Given: $x^x + y^x = xy$

Concept / Formula Used: x^x $x^x(1 + \ln x)$ y^x

Solution:

Let $u = x^x$ and $v = y^x$. The equation can be written as $u + v = xy$. Differentiating both sides with respect to x :

$$\frac{du}{dx} + \frac{dv}{dx} = x \frac{dy}{dx} + y \cdot 1$$

For $u = x^x$:

$$\log u = x \log x \implies \frac{1}{u} \frac{du}{dx} = 1 \cdot \log x + x \cdot \frac{1}{x} = \log x + 1$$

$$\frac{du}{dx} = x^x(1 + \log x)$$

For $v = y^x$:

$$\log v = x \log y \implies \frac{1}{v} \frac{dv}{dx} = 1 \cdot \log y + x \cdot \frac{1}{y} \frac{dy}{dx}$$

$$\frac{dv}{dx} = y^x \left(\log y + \frac{x}{y} \frac{dy}{dx} \right)$$

Substituting $\frac{du}{dx}$ and $\frac{dv}{dx}$ into the derivative relation:

$$x^x(1 + \log x) + y^x \left(\log y + \frac{x}{y} \frac{dy}{dx} \right) = x \frac{dy}{dx} + y$$

Collecting all terms involving $\frac{dy}{dx}$ on one side:

$$y^x \frac{x}{y} \frac{dy}{dx} - x \frac{dy}{dx} = y - x^x(1 + \log x) - y^x \log y$$

$$\frac{dy}{dx} (xy^{x-1} - x) = y - x^x(1 + \log x) - y^x \log y$$

Isolating $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{y - x^x(1 + \log x) - y^x \log y}{x(y^{x-1} - 1)}$$

Final Answer

$$\frac{dy}{dx} = \frac{y - x^x(1 + \log x) - y^x \log y}{xy^{x-1} - x}$$

Common Mistake: Dropping the product rule terms on the right-hand side when differentiating xy .

Question 10(iv)

Find $\frac{dy}{dx}$ if $(x + y)^y = \log(\sin x)$.

TYPE 61 Logarithmic Differentiation

Given: $(x + y)^y = \log(\sin x)$

Concept / Formula Used: $\frac{d}{dx}[f(x)^{g(x)}]$

Solution:

Taking natural logarithm on both sides:

$$\log((x + y)^y) = \log(\log(\sin x))$$

Using the power property of logarithms:

$$y \log(x + y) = \log(\log(\sin x))$$

Differentiating both sides with respect to x using the product rule on the left-hand side and the chain rule on the right-hand side:

$$\frac{d}{dx}[y] \cdot \log(x + y) + y \cdot \frac{d}{dx}[\log(x + y)] = \frac{1}{\log(\sin x)} \cdot \frac{d}{dx}[\log(\sin x)]$$

Evaluating the derivatives:

$$\frac{dy}{dx} \log(x + y) + y \cdot \frac{1}{x + y} \left(1 + \frac{dy}{dx}\right) = \frac{1}{\log(\sin x)} \cdot \frac{1}{\sin x} \cdot \cos x$$

Expanding the left-hand side:

$$\frac{dy}{dx} \log(x + y) + \frac{y}{x + y} + \frac{y}{x + y} \frac{dy}{dx} = \frac{\cot x}{\log(\sin x)}$$

Grouping terms with $\frac{dy}{dx}$:

$$\frac{dy}{dx} \left[\log(x + y) + \frac{y}{x + y} \right] = \frac{\cot x}{\log(\sin x)} - \frac{y}{x + y}$$

Writing over common denominators and solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\frac{\cot x}{\log(\sin x)} - \frac{y}{x+y}}{\log(x + y) + \frac{y}{x+y}}$$

Multiplying numerator and denominator by $(x + y) \log(\sin x)$:

$$\frac{dy}{dx} = \frac{(x + y) \cot x - y \log(\sin x)}{(x + y) \log(\sin x) \left[\log(x + y) + \frac{y}{x+y} \right]}$$

$$\frac{dy}{dx} = \frac{(x + y) \cot x - y \log(\sin x)}{(x + y) \log(\sin x) \log(x + y) + y \log(\sin x)}$$

Final Answer

$$\frac{dy}{dx} = \frac{(x + y) \cot x - y \log(\sin x)}{\log(\sin x) [(x + y) \log(x + y) + y]}$$

Common Mistake: Incorrectly applying the chain rule to the composite logarithmic term $\log(\sin x)$.

Question 10(v)

Find $\frac{dy}{dx}$ if $y^x = x^y$.

TYPE 61 Logarithmic Differentiation

Given: $y^x = x^y$

Concept / Formula Used: $\ln(a^b) = b \ln a$

Solution:

Taking natural logarithm on both sides of the equation:

$$\log(y^x) = \log(x^y)$$

Using the power property of logarithms:

$$x \log y = y \log x$$

Differentiating both sides with respect to x using the product rule on both sides:

$$\frac{d}{dx}[x] \cdot \log y + x \cdot \frac{d}{dx}[\log y] = \frac{d}{dx}[y] \cdot \log x + y \cdot \frac{d}{dx}[\log x]$$

Evaluating the derivatives:

$$1 \cdot \log y + x \cdot \frac{1}{y} \frac{dy}{dx} = \frac{dy}{dx} \log x + y \cdot \frac{1}{x}$$

$$\log y + \frac{x}{y} \frac{dy}{dx} = \log x \frac{dy}{dx} + \frac{y}{x}$$

Rearranging terms to collect $\frac{dy}{dx}$ on one side:

$$\frac{x}{y} \frac{dy}{dx} - \log x \frac{dy}{dx} = \frac{y}{x} - \log y$$

$$\frac{dy}{dx} \left(\frac{x}{y} - \log x \right) = \frac{y}{x} - \log y$$

$$\frac{dy}{dx} \left(\frac{x - y \log x}{y} \right) = \frac{y - x \log y}{x}$$

Isolating $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{y}{x} \left(\frac{y - x \log y}{x - y \log x} \right)$$

Final Answer

$$\frac{dy}{dx} = \frac{y(y - x \log y)}{x(x - y \log x)}$$

Common Mistake: Forgetting to apply the product rule to both sides of the logarithmically simplified equation.

Question 10(vi)

Find $\frac{dy}{dx}$ if $y = \frac{y \tan x + \log \cos y}{x \tan y + \log \cos x}$. (Implicit / Logarithmic structure)

TYPE 62 Implicit Differentiation with Exponential/Logarithmic Terms

Given: $y = \frac{y \tan x + \log \cos y}{x \tan y + \log \cos x}$

Solution:

Clearing the denominator by cross-multiplication:

$$y(x \tan y + \log \cos x) = y \tan x + \log \cos y$$

$$xy \tan y + y \log \cos x = y \tan x + \log \cos y$$

Differentiating both sides with respect to x : For the first term $xy \tan y$, use product rule on xy and chain rule on $\tan y$:

$$\frac{d}{dx}[xy \tan y] = \left(x \frac{dy}{dx} + y\right) \tan y + xy \sec^2 y \frac{dy}{dx}$$

For the second term $y \log \cos x$:

$$\frac{d}{dx}[y \log \cos x] = \frac{dy}{dx} \log \cos x + y \cdot \frac{1}{\cos x}(-\sin x) = \frac{dy}{dx} \log \cos x - y \tan x$$

For the right-hand side $y \tan x + \log \cos y$:

$$\frac{d}{dx}[y \tan x] = \frac{dy}{dx} \tan x + y \sec^2 x$$

$$\frac{d}{dx}[\log \cos y] = \frac{1}{\cos y}(-\sin y) \frac{dy}{dx} = -\tan y \frac{dy}{dx}$$

Combining all derivatives:

$$\left(x \tan y \frac{dy}{dx} + y \tan y + xy \sec^2 y \frac{dy}{dx}\right) + \left(\log \cos x \frac{dy}{dx} - y \tan x\right) = \left(\tan x \frac{dy}{dx} + y \sec^2 x\right) - \tan y \frac{dy}{dx}$$

Grouping all terms containing $\frac{dy}{dx}$ on the left-hand side:

$$\frac{dy}{dx} (x \tan y + xy \sec^2 y + \log \cos x - \tan x + \tan y) = y \sec^2 x + y \tan x - y \tan y$$

Isolating $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{y \sec^2 x + y \tan x - y \tan y}{x \tan y + xy \sec^2 y + \log \cos x - \tan x + \tan y}$$

Final Answer

$$\frac{dy}{dx} = \frac{y(\sec^2 x + \tan x - \tan y)}{x \tan y + xy \sec^2 y + \log \cos x - \tan x + \tan y}$$

Common Mistake: Algebraic omission of terms when differentiating products involving trigonometric and logarithmic functions simultaneously.

5.10 – Parametric Differentiation

Question 1 (i)

Find $\frac{dy}{dx}$ when $x = at^2$ and $y = 2at$.

TYPE 51 Parametric Differentiation

Given: $x = at^2$ and $y = 2at$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= at^2 \\ y &= 2at\end{aligned}$$

Differentiating x with respect to t :

$$\begin{aligned}\frac{dx}{dt} &= \frac{d}{dt}(at^2) \\ \frac{dx}{dt} &= a \cdot (2t) \\ \frac{dx}{dt} &= 2at\end{aligned}$$

Differentiating y with respect to t :

$$\begin{aligned}\frac{dy}{dt} &= \frac{d}{dt}(2at) \\ \frac{dy}{dt} &= 2a \cdot (1) \\ \frac{dy}{dt} &= 2a\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{2a}{2at} \\ \frac{dy}{dx} &= \frac{1}{t}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \frac{1}{t}$$

Question 1 (ii)

Find $\frac{dy}{dx}$ when $x = 2at^2$ and $y = at^4$.

TYPE 51 Parametric Differentiation

Given: $x = 2at^2$ and $y = at^4$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= 2at^2 \\ y &= at^4\end{aligned}$$

Differentiating x with respect to t :

$$\begin{aligned}\frac{dx}{dt} &= \frac{d}{dt}(2at^2) \\ \frac{dx}{dt} &= 2a \cdot (2t) \\ \frac{dx}{dt} &= 4at\end{aligned}$$

Differentiating y with respect to t :

$$\begin{aligned}\frac{dy}{dt} &= \frac{d}{dt}(at^4) \\ \frac{dy}{dt} &= a \cdot (4t^3) \\ \frac{dy}{dt} &= 4at^3\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{4at^3}{4at} \\ \frac{dy}{dx} &= t^2\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = t^2$$

Question 2 (i)

Find $\frac{dy}{dx}$ when $x = a \cos \theta$ and $y = b \cos \theta$.

TYPE 51 Parametric Differentiation

Given: $x = a \cos \theta$ and $y = b \cos \theta$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$x = a \cos \theta$$

$$y = b \cos \theta$$

Differentiating x with respect to θ :

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(a \cos \theta)$$

$$\frac{dx}{d\theta} = -a \sin \theta$$

Differentiating y with respect to θ :

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(b \cos \theta)$$

$$\frac{dy}{d\theta} = -b \sin \theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$\frac{dy}{dx} = \frac{-b \sin \theta}{-a \sin \theta}$$

$$\frac{dy}{dx} = \frac{b}{a}$$

Final Answer

$$\frac{dy}{dx} = \frac{b}{a}$$

Question 2 (ii)

Find $\frac{dy}{dx}$ when $x = a \cos \theta$ and $y = a \sin \theta$.

TYPE 51 Parametric Differentiation

Given: $x = a \cos \theta$ and $y = a \sin \theta$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= a \cos \theta \\y &= a \sin \theta\end{aligned}$$

Differentiating x with respect to θ :

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{d}{d\theta}(a \cos \theta) \\ \frac{dx}{d\theta} &= -a \sin \theta\end{aligned}$$

Differentiating y with respect to θ :

$$\begin{aligned}\frac{dy}{d\theta} &= \frac{d}{d\theta}(a \sin \theta) \\ \frac{dy}{d\theta} &= a \cos \theta\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ \frac{dy}{dx} &= \frac{a \cos \theta}{-a \sin \theta} \\ \frac{dy}{dx} &= -\cot \theta\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\cot \theta$$

Question 3 (i)

Find $\frac{dy}{dx}$ when $x = a \sec \theta$ and $y = b \tan \theta$.

TYPE 51 Parametric Differentiation

Given: $x = a \sec \theta$ and $y = b \tan \theta$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= a \sec \theta \\y &= b \tan \theta\end{aligned}$$

Differentiating x with respect to θ :

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{d}{d\theta}(a \sec \theta) \\ \frac{dx}{d\theta} &= a \sec \theta \tan \theta\end{aligned}$$

Differentiating y with respect to θ :

$$\begin{aligned}\frac{dy}{d\theta} &= \frac{d}{d\theta}(b \tan \theta) \\ \frac{dy}{d\theta} &= b \sec^2 \theta\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ \frac{dy}{dx} &= \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} \\ \frac{dy}{dx} &= \frac{b \sec \theta}{a \tan \theta}\end{aligned}$$

Expressing in terms of sine and cosine:

$$\begin{aligned}\frac{dy}{dx} &= \frac{b \left(\frac{1}{\cos \theta} \right)}{a \left(\frac{\sin \theta}{\cos \theta} \right)} \\ \frac{dy}{dx} &= \frac{b}{a \sin \theta} \\ \frac{dy}{dx} &= \frac{b}{a} \csc \theta\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \frac{b}{a} \csc \theta$$

Question 3 (ii)

Find $\frac{dy}{dx}$ when $x = \sin t$ and $y = \cos 2t$.

TYPE 51 Parametric Differentiation

Given: $x = \sin t$ and $y = \cos 2t$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= \sin t \\ y &= \cos 2t\end{aligned}$$

Differentiating x with respect to t :

$$\begin{aligned}\frac{dx}{dt} &= \frac{d}{dt}(\sin t) \\ \frac{dx}{dt} &= \cos t\end{aligned}$$

Differentiating y with respect to t :

$$\begin{aligned}\frac{dy}{dt} &= \frac{d}{dt}(\cos 2t) \\ \frac{dy}{dt} &= -\sin 2t \cdot \frac{d}{dt}(2t) \\ \frac{dy}{dt} &= -2 \sin 2t\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{-2 \sin 2t}{\cos t}\end{aligned}$$

Using the standard double-angle identity $\sin 2t = 2 \sin t \cos t$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{-2(2 \sin t \cos t)}{\cos t} \\ \frac{dy}{dx} &= -4 \sin t\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -4 \sin t$$

Question 4 (i)

Find $\frac{dy}{dx}$ when $x = 4t$ and $y = \frac{4}{t}$.

TYPE 51 Parametric Differentiation

Given: $x = 4t$ and $y = \frac{4}{t}$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= 4t \\ y &= 4t^{-1}\end{aligned}$$

Differentiating x with respect to t :

$$\begin{aligned}\frac{dx}{dt} &= \frac{d}{dt}(4t) \\ \frac{dx}{dt} &= 4\end{aligned}$$

Differentiating y with respect to t :

$$\begin{aligned}\frac{dy}{dt} &= \frac{d}{dt}(4t^{-1}) \\ \frac{dy}{dt} &= -4t^{-2} \\ \frac{dy}{dt} &= -\frac{4}{t^2}\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{-\frac{4}{t^2}}{4} \\ \frac{dy}{dx} &= -\frac{1}{t^2}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\frac{1}{t^2}$$

Question 4 (ii)

Find $\frac{dy}{dx}$ when $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$.

TYPE 51 Parametric Differentiation

Given: $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= t + t^{-1} \\ y &= t - t^{-1}\end{aligned}$$

Differentiating x with respect to t :

$$\frac{dx}{dt} = \frac{d}{dt}(t + t^{-1})$$

$$\frac{dx}{dt} = 1 - t^{-2}$$

$$\frac{dx}{dt} = 1 - \frac{1}{t^2}$$

$$\frac{dx}{dt} = \frac{t^2 - 1}{t^2}$$

Differentiating y with respect to t :

$$\frac{dy}{dt} = \frac{d}{dt}(t - t^{-1})$$

$$\frac{dy}{dt} = 1 - (-t^{-2})$$

$$\frac{dy}{dt} = 1 + \frac{1}{t^2}$$

$$\frac{dy}{dt} = \frac{t^2 + 1}{t^2}$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{dy}{dx} = \frac{\frac{t^2+1}{t^2}}{\frac{t^2-1}{t^2}}$$

$$\frac{dy}{dx} = \frac{t^2 + 1}{t^2 - 1}$$

Final Answer

$$\frac{dy}{dx} = \frac{t^2 + 1}{t^2 - 1}$$

Question 4 (iii)

Find $\frac{dy}{dx}$ when $x = a(t - \sin t)$ and $y = a(1 + \cos t)$.

TYPE 51 Parametric Differentiation

Given: $x = a(t - \sin t)$ and $y = a(1 + \cos t)$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$x = a(t - \sin t)$$

$$y = a(1 + \cos t)$$

Differentiating x with respect to t :

$$\frac{dx}{dt} = a(1 - \cos t)$$

Differentiating y with respect to t :

$$\begin{aligned}\frac{dy}{dt} &= a(0 - \sin t) \\ \frac{dy}{dt} &= -a \sin t\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{-a \sin t}{a(1 - \cos t)} \\ \frac{dy}{dx} &= -\frac{\sin t}{1 - \cos t}\end{aligned}$$

Using the standard half-angle identities $\sin t = 2 \sin \frac{t}{2} \cos \frac{t}{2}$ and $1 - \cos t = 2 \sin^2 \frac{t}{2}$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \sin^2 \frac{t}{2}} \\ \frac{dy}{dx} &= -\frac{\cos \frac{t}{2}}{\sin \frac{t}{2}} \\ \frac{dy}{dx} &= -\cot \frac{t}{2}\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\cot \frac{t}{2}$$

Common Mistake: The textbook answer key lists $-\tan \frac{t}{2}$ for this question. This is a typographical error in the textbook's key; the correct mathematical derivative is indeed $-\cot \frac{t}{2}$.

Question 5 (i)

Find $\frac{dy}{dx}$ when $x = a \cos^3 t$ and $y = b \sin^3 t$.

TYPE 51 Parametric Differentiation

Given: $x = a \cos^3 t$ and $y = b \sin^3 t$

Concept / Formula Used: $x = f(t) \quad y = g(t) \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$x = a \cos^3 t$$

$$y = b \sin^3 t$$

Differentiating x with respect to t using the chain rule:

$$\frac{dx}{dt} = a \cdot 3 \cos^2 t \cdot \frac{d}{dt}(\cos t)$$

$$\frac{dx}{dt} = 3a \cos^2 t \cdot (-\sin t)$$

$$\frac{dx}{dt} = -3a \cos^2 t \sin t$$

Differentiating y with respect to t using the chain rule:

$$\frac{dy}{dt} = b \cdot 3 \sin^2 t \cdot \frac{d}{dt}(\sin t)$$

$$\frac{dy}{dt} = 3b \sin^2 t \cdot (\cos t)$$

$$\frac{dy}{dt} = 3b \sin^2 t \cos t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{dy}{dx} = \frac{3b \sin^2 t \cos t}{-3a \cos^2 t \sin t}$$

$$\frac{dy}{dx} = -\frac{b \sin t}{a \cos t}$$

$$\frac{dy}{dx} = -\frac{b}{a} \tan t$$

Final Answer

$$\frac{dy}{dx} = -\frac{b}{a} \tan t$$

Question 5 (ii)

Find $\frac{dy}{dx}$ when $x = a(1 - \sin t)$ and $y = a(1 + \cos t)$.

TYPE 51 Parametric Differentiation

Given: $x = a(1 - \sin t)$ and $y = a(1 + \cos t)$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$x = a(1 - \sin t)$$

$$y = a(1 + \cos t)$$

Differentiating x with respect to t :

$$\frac{dx}{dt} = a(0 - \cos t)$$

$$\frac{dx}{dt} = -a \cos t$$

Differentiating y with respect to t :

$$\frac{dy}{dt} = a(0 - \sin t)$$

$$\frac{dy}{dt} = -a \sin t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{dy}{dx} = \frac{-a \sin t}{-a \cos t}$$

$$\frac{dy}{dx} = \tan t$$

Final Answer

$$\frac{dy}{dx} = \tan t$$

Common Mistake: The textbook answer key lists $-\cot \frac{t}{2}$ for this question. This indicates that the question in the textbook was likely intended to be $x = a(t - \sin t)$ instead of $x = a(1 - \sin t)$. If $x = a(t - \sin t)$, then $\frac{dx}{dt} = a(1 - \cos t)$, which yields $\frac{dy}{dx} = -\cot \frac{t}{2}$.

Question 6 (i)

Find $\frac{dy}{dx}$ when $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$.

TYPE 51 Parametric Differentiation

Given: $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$x = a(\theta + \sin \theta)$$

$$y = a(1 - \cos \theta)$$

Differentiating x with respect to θ :

$$\frac{dx}{d\theta} = a(1 + \cos \theta)$$

Differentiating y with respect to θ :

$$\frac{dy}{d\theta} = a(0 - (-\sin \theta))$$

$$\frac{dy}{d\theta} = a \sin \theta$$

Now, using the parametric differentiation formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ \frac{dy}{dx} &= \frac{a \sin \theta}{a(1 + \cos \theta)} \\ \frac{dy}{dx} &= \frac{\sin \theta}{1 + \cos \theta} \end{aligned}$$

Using the standard half-angle identities $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ and $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} \\ \frac{dy}{dx} &= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \\ \frac{dy}{dx} &= \tan \frac{\theta}{2} \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \tan \frac{\theta}{2}$$

Question 6 (ii)

Find $\frac{dy}{dx}$ when $x = a(1 - \cos \theta)$ and $y = a(\theta + \sin \theta)$.

TYPE 51 Parametric Differentiation

Given: $x = a(1 - \cos \theta)$ and $y = a(\theta + \sin \theta)$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$x = a(1 - \cos \theta)$$

$$y = a(\theta + \sin \theta)$$

Differentiating x with respect to θ :

$$\frac{dx}{d\theta} = a(0 - (-\sin \theta))$$

$$\frac{dx}{d\theta} = a \sin \theta$$

Differentiating y with respect to θ :

$$\frac{dy}{d\theta} = a(1 + \cos \theta)$$

Now, using the parametric differentiation formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ \frac{dy}{dx} &= \frac{a(1 + \cos \theta)}{a \sin \theta} \\ \frac{dy}{dx} &= \frac{1 + \cos \theta}{\sin \theta} \end{aligned}$$

Using the standard half-angle identities $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$ and $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{2 \cos^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \\ \frac{dy}{dx} &= \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \\ \frac{dy}{dx} &= \cot \frac{\theta}{2} \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \cot \frac{\theta}{2}$$

Question 7 (i)

Find $\frac{dy}{dx}$ when $x = e^t(\sin t + \cos t)$ and $y = e^t(\sin t - \cos t)$.

TYPE 51 Parametric Differentiation

Given: $x = e^t(\sin t + \cos t)$ and $y = e^t(\sin t - \cos t)$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$x = e^t(\sin t + \cos t)$$

$$y = e^t(\sin t - \cos t)$$

Differentiating x with respect to t using the product rule:

$$\frac{dx}{dt} = e^t \frac{d}{dt}(\sin t + \cos t) + (\sin t + \cos t) \frac{d}{dt}(e^t)$$

$$\frac{dx}{dt} = e^t(\cos t - \sin t) + e^t(\sin t + \cos t)$$

$$\frac{dx}{dt} = e^t(\cos t - \sin t + \sin t + \cos t)$$

$$\frac{dx}{dt} = e^t(2 \cos t)$$

$$\frac{dx}{dt} = 2e^t \cos t$$

Differentiating y with respect to t using the product rule:

$$\frac{dy}{dt} = e^t \frac{d}{dt}(\sin t - \cos t) + (\sin t - \cos t) \frac{d}{dt}(e^t)$$

$$\frac{dy}{dt} = e^t(\cos t - (-\sin t)) + e^t(\sin t - \cos t)$$

$$\frac{dy}{dt} = e^t(\cos t + \sin t) + e^t(\sin t - \cos t)$$

$$\frac{dy}{dt} = e^t(\cos t + \sin t + \sin t - \cos t)$$

$$\frac{dy}{dt} = e^t(2 \sin t)$$

$$\frac{dy}{dt} = 2e^t \sin t$$

Now, using the parametric differentiation formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{2e^t \sin t}{2e^t \cos t} \\ \frac{dy}{dx} &= \tan t \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \tan t$$

Question 7 (ii)

Find $\frac{dy}{dx}$ when $x = \cos \theta - \cos 2\theta$ and $y = \sin \theta - \sin 2\theta$.

TYPE 51 Parametric Differentiation

Given: $x = \cos \theta - \cos 2\theta$ and $y = \sin \theta - \sin 2\theta$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$x = \cos \theta - \cos 2\theta$$

$$y = \sin \theta - \sin 2\theta$$

Differentiating x with respect to θ :

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(\cos \theta - \cos 2\theta)$$

$$\frac{dx}{d\theta} = -\sin \theta - (-2 \sin 2\theta)$$

$$\frac{dx}{d\theta} = 2 \sin 2\theta - \sin \theta$$

Differentiating y with respect to θ :

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(\sin \theta - \sin 2\theta)$$

$$\frac{dy}{d\theta} = \cos \theta - 2 \cos 2\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$\frac{dy}{dx} = \frac{\cos \theta - 2 \cos 2\theta}{2 \sin 2\theta - \sin \theta}$$

Final Answer

$$\frac{dy}{dx} = \frac{\cos \theta - 2 \cos 2\theta}{2 \sin 2\theta - \sin \theta}$$

Question 8 (i)

Find $\frac{dy}{dx}$ when $x = a(\cos \theta + \cos 2\theta)$ and $y = b(\sin \theta + \sin 2\theta)$.

TYPE 51 Parametric Differentiation

Given: $x = a(\cos \theta + \cos 2\theta)$ and $y = b(\sin \theta + \sin 2\theta)$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$x = a(\cos \theta + \cos 2\theta)$$

$$y = b(\sin \theta + \sin 2\theta)$$

Differentiating x with respect to θ :

$$\frac{dx}{d\theta} = a(-\sin \theta - 2 \sin 2\theta)$$

$$\frac{dx}{d\theta} = -a(\sin \theta + 2 \sin 2\theta)$$

Differentiating y with respect to θ :

$$\frac{dy}{d\theta} = b(\cos \theta + 2 \cos 2\theta)$$

Now, using the parametric differentiation formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ \frac{dy}{dx} &= \frac{b(\cos \theta + 2 \cos 2\theta)}{-a(\sin \theta + 2 \sin 2\theta)} \\ \frac{dy}{dx} &= -\frac{b}{a} \left(\frac{\cos \theta + 2 \cos 2\theta}{\sin \theta + 2 \sin 2\theta} \right) \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = -\frac{b}{a} \left(\frac{\cos \theta + 2 \cos 2\theta}{\sin \theta + 2 \sin 2\theta} \right)$$

Question 8 (ii)

Find $\frac{dy}{dx}$ when $x = \frac{1 + \log t}{t^2}$ and $y = \frac{3 + 2 \log t}{t}$.

TYPE 51 Parametric Differentiation

Given: $x = \frac{1 + \log t}{t^2}$ and $y = \frac{3 + 2 \log t}{t}$

Concept / Formula Used: $x = f(t) \quad y = g(t) \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$x = \frac{1 + \log t}{t^2}$$

$$y = \frac{3 + 2 \log t}{t}$$

Differentiating x with respect to t using the quotient rule:

$$\begin{aligned} \frac{dx}{dt} &= \frac{t^2 \frac{d}{dt}(1 + \log t) - (1 + \log t) \frac{d}{dt}(t^2)}{(t^2)^2} \\ \frac{dx}{dt} &= \frac{t^2 \left(\frac{1}{t}\right) - (1 + \log t)(2t)}{t^4} \\ \frac{dx}{dt} &= \frac{t - 2t(1 + \log t)}{t^4} \\ \frac{dx}{dt} &= \frac{t(1 - 2 - 2 \log t)}{t^4} \\ \frac{dx}{dt} &= \frac{-1 - 2 \log t}{t^3} \\ \frac{dx}{dt} &= -\frac{1 + 2 \log t}{t^3} \end{aligned}$$

Differentiating y with respect to t using the quotient rule:

$$\begin{aligned} \frac{dy}{dt} &= \frac{t \frac{d}{dt}(3 + 2 \log t) - (3 + 2 \log t) \frac{d}{dt}(t)}{t^2} \\ \frac{dy}{dt} &= \frac{t \left(\frac{2}{t}\right) - (3 + 2 \log t)(1)}{t^2} \\ \frac{dy}{dt} &= \frac{2 - 3 - 2 \log t}{t^2} \\ \frac{dy}{dt} &= \frac{-1 - 2 \log t}{t^2} \\ \frac{dy}{dt} &= -\frac{1 + 2 \log t}{t^2} \end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{-\frac{1+2 \log t}{t^2}}{-\frac{1+2 \log t}{t^3}} \\ \frac{dy}{dx} &= \frac{t^3}{t^2} \\ \frac{dy}{dx} &= t \end{aligned}$$

Final Answer

$$\frac{dy}{dx} = t$$

Question 9

Find $\frac{dy}{dx}$ when $x = 3 \cos \theta - 2 \cos^3 \theta$ and $y = 3 \sin \theta - 2 \sin^3 \theta$.

TYPE 51 Parametric Differentiation

Given: $x = 3 \cos \theta - 2 \cos^3 \theta$ and $y = 3 \sin \theta - 2 \sin^3 \theta$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$\begin{aligned} x &= 3 \cos \theta - 2 \cos^3 \theta \\ y &= 3 \sin \theta - 2 \sin^3 \theta \end{aligned}$$

Differentiating x with respect to θ :

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{d}{d\theta}(3 \cos \theta - 2 \cos^3 \theta) \\ \frac{dx}{d\theta} &= -3 \sin \theta - 2 \cdot 3 \cos^2 \theta \cdot (-\sin \theta) \\ \frac{dx}{d\theta} &= -3 \sin \theta + 6 \cos^2 \theta \sin \theta \\ \frac{dx}{d\theta} &= 3 \sin \theta (2 \cos^2 \theta - 1) \end{aligned}$$

Using the standard double-angle identity $\cos 2\theta = 2 \cos^2 \theta - 1$:

$$\frac{dx}{d\theta} = 3 \sin \theta \cos 2\theta$$

Differentiating y with respect to θ :

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{d}{d\theta}(3 \sin \theta - 2 \sin^3 \theta) \\ \frac{dy}{d\theta} &= 3 \cos \theta - 2 \cdot 3 \sin^2 \theta \cdot (\cos \theta) \\ \frac{dy}{d\theta} &= 3 \cos \theta - 6 \sin^2 \theta \cos \theta \\ \frac{dy}{d\theta} &= 3 \cos \theta (1 - 2 \sin^2 \theta) \end{aligned}$$

Using the standard double-angle identity $\cos 2\theta = 1 - 2 \sin^2 \theta$:

$$\frac{dy}{d\theta} = 3 \cos \theta \cos 2\theta$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ \frac{dy}{dx} &= \frac{3 \cos \theta \cos 2\theta}{3 \sin \theta \cos 2\theta} \\ \frac{dy}{dx} &= \frac{\cos \theta}{\sin \theta} \\ \frac{dy}{dx} &= \cot \theta\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \cot \theta$$

Question 10 (i)

If $x = 2 \cos t - \cos 2t$ and $y = 2 \sin t - \sin 2t$, find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$.

TYPE 52 Parametric Differentiation with Evaluation

Given: $x = 2 \cos t - \cos 2t$, $y = 2 \sin t - \sin 2t$, and $t = \frac{\pi}{4}$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= 2 \cos t - \cos 2t \\ y &= 2 \sin t - \sin 2t\end{aligned}$$

Differentiating x with respect to t :

$$\begin{aligned}\frac{dx}{dt} &= -2 \sin t - (-2 \sin 2t) \\ \frac{dx}{dt} &= 2 \sin 2t - 2 \sin t\end{aligned}$$

Differentiating y with respect to t :

$$\frac{dy}{dt} = 2 \cos t - 2 \cos 2t$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{2 \cos t - 2 \cos 2t}{2 \sin 2t - 2 \sin t} \\ \frac{dy}{dx} &= \frac{\cos t - \cos 2t}{\sin 2t - \sin t}\end{aligned}$$

Evaluating the derivative at $t = \frac{\pi}{4}$:

$$\begin{aligned}\cos\left(\frac{\pi}{4}\right) &= \frac{1}{\sqrt{2}} \\ \cos\left(2 \cdot \frac{\pi}{4}\right) &= \cos\left(\frac{\pi}{2}\right) = 0 \\ \sin\left(\frac{\pi}{4}\right) &= \frac{1}{\sqrt{2}} \\ \sin\left(2 \cdot \frac{\pi}{4}\right) &= \sin\left(\frac{\pi}{2}\right) = 1\end{aligned}$$

Substituting these values into the expression for $\frac{dy}{dx}$:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} &= \frac{\frac{1}{\sqrt{2}} - 0}{1 - \frac{1}{\sqrt{2}}} \\ \left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} &= \frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{2}-1}{\sqrt{2}}} \\ \left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} &= \frac{1}{\sqrt{2}-1}\end{aligned}$$

Rationalizing the denominator:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} &= \frac{1}{\sqrt{2}-1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1} \\ \left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} &= \frac{\sqrt{2}+1}{2-1} \\ \left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} &= \sqrt{2}+1\end{aligned}$$

Final Answer

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \sqrt{2} + 1$$

Question 10 (ii)

If $x = 3 \sin t - \sin 3t$ and $y = 3 \cos t - \cos 3t$, find $\frac{dy}{dx}$ at $t = \frac{\pi}{3}$.

TYPE 52 Parametric Differentiation with Evaluation

Given: $x = 3 \sin t - \sin 3t$, $y = 3 \cos t - \cos 3t$, and $t = \frac{\pi}{3}$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$x = 3 \sin t - \sin 3t$$

$$y = 3 \cos t - \cos 3t$$

Differentiating x with respect to t :

$$\frac{dx}{dt} = 3 \cos t - 3 \cos 3t$$

Differentiating y with respect to t :

$$\frac{dy}{dt} = -3 \sin t - (-3 \sin 3t)$$

$$\frac{dy}{dt} = 3 \sin 3t - 3 \sin t$$

Now, using the parametric differentiation formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{3 \sin 3t - 3 \sin t}{3 \cos t - 3 \cos 3t} \\ \frac{dy}{dx} &= \frac{\sin 3t - \sin t}{\cos t - \cos 3t} \end{aligned}$$

Evaluating the derivative at $t = \frac{\pi}{3}$:

$$\sin \left(3 \cdot \frac{\pi}{3} \right) = \sin \pi = 0$$

$$\sin \left(\frac{\pi}{3} \right) = \frac{\sqrt{3}}{2}$$

$$\cos \left(\frac{\pi}{3} \right) = \frac{1}{2}$$

$$\cos \left(3 \cdot \frac{\pi}{3} \right) = \cos \pi = -1$$

Substituting these values into the expression for $\frac{dy}{dx}$:

$$\left. \frac{dy}{dx} \right|_{t=\frac{\pi}{3}} = \frac{0 - \frac{\sqrt{3}}{2}}{\frac{1}{2} - (-1)}$$

$$\left. \frac{dy}{dx} \right|_{t=\frac{\pi}{3}} = \frac{-\frac{\sqrt{3}}{2}}{\frac{3}{2}}$$

$$\left. \frac{dy}{dx} \right|_{t=\frac{\pi}{3}} = -\frac{\sqrt{3}}{3}$$

$$\left. \frac{dy}{dx} \right|_{t=\frac{\pi}{3}} = -\frac{1}{\sqrt{3}}$$

Final Answer

$$\left. \frac{dy}{dx} \right|_{t=\frac{\pi}{3}} = -\frac{1}{\sqrt{3}}$$

Question 10 (iii)

If $x = a \sin 2t$ and $y = a(\cos 2t + \log \tan t)$, find $\frac{dy}{dx}$.

TYPE 51 Parametric Differentiation

Given: $x = a \sin 2t$ and $y = a(\cos 2t + \log \tan t)$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$x = a \sin 2t$$

$$y = a(\cos 2t + \log \tan t)$$

Differentiating x with respect to t :

$$\frac{dx}{dt} = 2a \cos 2t$$

Differentiating y with respect to t :

$$\frac{dy}{dt} = a \left(-2 \sin 2t + \frac{1}{\tan t} \cdot \sec^2 t \right)$$

Simplifying the term $\frac{\sec^2 t}{\tan t}$:

$$\begin{aligned} \frac{\sec^2 t}{\tan t} &= \frac{\frac{1}{\cos^2 t}}{\frac{\sin t}{\cos t}} \\ &= \frac{1}{\sin t \cos t} \\ &= \frac{2}{2 \sin t \cos t} \\ &= \frac{2}{\sin 2t} \end{aligned}$$

Substituting this back into $\frac{dy}{dt}$:

$$\frac{dy}{dt} = a \left(-2 \sin 2t + \frac{2}{\sin 2t} \right)$$

$$\frac{dy}{dt} = 2a \left(\frac{1 - \sin^2 2t}{\sin 2t} \right)$$

$$\frac{dy}{dt} = 2a \frac{\cos^2 2t}{\sin 2t}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{2a \frac{\cos^2 2t}{\sin 2t}}{2a \cos 2t} \\ \frac{dy}{dx} &= \frac{\cos 2t}{\sin 2t} \\ \frac{dy}{dx} &= \cot 2t\end{aligned}$$

Final Answer

$$\frac{dy}{dx} = \cot 2t$$

Question 11 (i)

If $x = \frac{2bt}{1+t^2}$ and $y = \frac{a(1-t^2)}{1+t^2}$, find $\frac{dy}{dx}$ at $t = 2$.

TYPE 52 Parametric Differentiation with Evaluation

Given: $x = \frac{2bt}{1+t^2}$, $y = \frac{a(1-t^2)}{1+t^2}$, and $t = 2$

Concept / Formula Used: $x = f(t)$ $y = g(t)$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Solution:

We are given the parametric equations:

$$\begin{aligned}x &= \frac{2bt}{1+t^2} \\ y &= \frac{a(1-t^2)}{1+t^2}\end{aligned}$$

Differentiating x with respect to t using the quotient rule:

$$\begin{aligned}\frac{dx}{dt} &= 2b \cdot \frac{(1+t^2)(1) - t(2t)}{(1+t^2)^2} \\ \frac{dx}{dt} &= 2b \cdot \frac{1+t^2 - 2t^2}{(1+t^2)^2} \\ \frac{dx}{dt} &= 2b \frac{1-t^2}{(1+t^2)^2}\end{aligned}$$

Differentiating y with respect to t using the quotient rule:

$$\begin{aligned}\frac{dy}{dt} &= a \cdot \frac{(1+t^2)(-2t) - (1-t^2)(2t)}{(1+t^2)^2} \\ \frac{dy}{dt} &= a \cdot \frac{-2t - 2t^3 - 2t + 2t^3}{(1+t^2)^2} \\ \frac{dy}{dt} &= -\frac{4at}{(1+t^2)^2}\end{aligned}$$

Now, using the parametric differentiation formula:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ \frac{dy}{dx} &= \frac{-\frac{4at}{(1+t^2)^2}}{2b\frac{1-t^2}{(1+t^2)^2}} \\ \frac{dy}{dx} &= -\frac{4at}{2b(1-t^2)} \\ \frac{dy}{dx} &= -\frac{2at}{b(1-t^2)}\end{aligned}$$

Evaluating the derivative at $t = 2$:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{t=2} &= -\frac{2a(2)}{b(1-2^2)} \\ \left.\frac{dy}{dx}\right|_{t=2} &= -\frac{4a}{b(1-4)} \\ \left.\frac{dy}{dx}\right|_{t=2} &= -\frac{4a}{-3b} \\ \left.\frac{dy}{dx}\right|_{t=2} &= \frac{4a}{3b}\end{aligned}$$

Final Answer

$$\left.\frac{dy}{dx}\right|_{t=2} = \frac{4a}{3b}$$

Question 11 (ii)

If $x = ae^\theta(\sin \theta - \cos \theta)$ and $y = ae^\theta(\sin \theta + \cos \theta)$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$.

TYPE 52 Parametric Differentiation with Evaluation

Given: $x = ae^\theta(\sin \theta - \cos \theta)$, $y = ae^\theta(\sin \theta + \cos \theta)$, and $\theta = \frac{\pi}{4}$

Concept / Formula Used: $x = f(\theta)$ $y = g(\theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Solution:

We are given the parametric equations:

$$x = ae^\theta(\sin \theta - \cos \theta)$$

$$y = ae^\theta(\sin \theta + \cos \theta)$$

Differentiating x with respect to θ using the product rule:

$$\frac{dx}{d\theta} = ae^\theta \frac{d}{d\theta}(\sin \theta - \cos \theta) + a(\sin \theta - \cos \theta) \frac{d}{d\theta}(e^\theta)$$

$$\frac{dx}{d\theta} = ae^\theta(\cos \theta - (-\sin \theta)) + ae^\theta(\sin \theta - \cos \theta)$$

$$\frac{dx}{d\theta} = ae^\theta(\cos \theta + \sin \theta + \sin \theta - \cos \theta)$$

$$\frac{dx}{d\theta} = 2ae^\theta \sin \theta$$

Differentiating y with respect to θ using the product rule:

$$\frac{dy}{d\theta} = ae^\theta \frac{d}{d\theta}(\sin \theta + \cos \theta) + a(\sin \theta + \cos \theta) \frac{d}{d\theta}(e^\theta)$$

$$\frac{dy}{d\theta} = ae^\theta(\cos \theta - \sin \theta) + ae^\theta(\sin \theta + \cos \theta)$$

$$\frac{dy}{d\theta} = ae^\theta(\cos \theta - \sin \theta + \sin \theta + \cos \theta)$$

$$\frac{dy}{d\theta} = 2ae^\theta \cos \theta$$

Now, using the parametric differentiation formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ \frac{dy}{dx} &= \frac{2ae^\theta \cos \theta}{2ae^\theta \sin \theta} \\ \frac{dy}{dx} &= \cot \theta \end{aligned}$$

Evaluating the derivative at $\theta = \frac{\pi}{4}$:

$$\left. \frac{dy}{dx} \right|_{\theta=\frac{\pi}{4}} = \cot \left(\frac{\pi}{4} \right) = 1$$

Final Answer

$$\left. \frac{dy}{dx} \right|_{\theta=\frac{\pi}{4}} = 1$$

Question 12 (i)

Differentiate $\frac{x^2}{1-x^2}$ with respect to x^2 .

Given: Let $u = \frac{x^2}{1-x^2}$ and $v = x^2$

Concept / Formula Used: $u \quad v \quad \frac{du}{dv}$

Solution:

Let:

$$u = \frac{x^2}{1-x^2}$$

$$v = x^2$$

We can express u directly in terms of v :

$$u = \frac{v}{1-v}$$

Differentiating u with respect to v using the quotient rule:

$$\frac{du}{dv} = \frac{(1-v)\frac{d}{dv}(v) - v\frac{d}{dv}(1-v)}{(1-v)^2}$$

$$\frac{du}{dv} = \frac{(1-v)(1) - v(-1)}{(1-v)^2}$$

$$\frac{du}{dv} = \frac{1-v+v}{(1-v)^2}$$

$$\frac{du}{dv} = \frac{1}{(1-v)^2}$$

Back-substituting $v = x^2$:

$$\frac{du}{dv} = \frac{1}{(1-x^2)^2}$$

Final Answer

$$\frac{d}{d(x^2)} \left(\frac{x^2}{1-x^2} \right) = \frac{1}{(1-x^2)^2}$$

Question 12 (ii)

Differentiate $\sin x^2$ with respect to x^3 .

TYPE 53 Differentiation of One Function with Respect to Another

Given: Let $u = \sin x^2$ and $v = x^3$

Concept / Formula Used: $u \quad v \quad \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$

Solution:

Let:

$$\begin{aligned}u &= \sin x^2 \\v &= x^3\end{aligned}$$

Differentiating u with respect to x using the chain rule:

$$\begin{aligned}\frac{du}{dx} &= \cos x^2 \cdot \frac{d}{dx}(x^2) \\ \frac{du}{dx} &= 2x \cos x^2\end{aligned}$$

Differentiating v with respect to x :

$$\frac{dv}{dx} = 3x^2$$

Now, using the derivative formula:

$$\begin{aligned}\frac{du}{dv} &= \frac{\frac{du}{dx}}{\frac{dv}{dx}} \\ \frac{du}{dv} &= \frac{2x \cos x^2}{3x^2} \\ \frac{du}{dv} &= \frac{2 \cos x^2}{3x}\end{aligned}$$

Final Answer

$$\frac{d(\sin x^2)}{d(x^3)} = \frac{2 \cos x^2}{3x}$$

Question 12 (iii)

Differentiate $\cot^3(2x + 1)$ with respect to $x^2 + 1$.

TYPE 53 Differentiation of One Function with Respect to Another

Given: Let $u = \cot^3(2x + 1)$ and $v = x^2 + 1$

Concept / Formula Used: $u \quad v \quad \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$

Solution:

Let:

$$\begin{aligned}u &= \cot^3(2x + 1) \\v &= x^2 + 1\end{aligned}$$

Differentiating u with respect to x using the chain rule:

$$\begin{aligned}\frac{du}{dx} &= 3 \cot^2(2x+1) \cdot \frac{d}{dx}(\cot(2x+1)) \\ \frac{du}{dx} &= 3 \cot^2(2x+1) \cdot \left(-\csc^2(2x+1) \cdot \frac{d}{dx}(2x+1) \right) \\ \frac{du}{dx} &= 3 \cot^2(2x+1) \cdot (-\csc^2(2x+1) \cdot 2) \\ \frac{du}{dx} &= -6 \cot^2(2x+1) \csc^2(2x+1)\end{aligned}$$

Differentiating v with respect to x :

$$\frac{dv}{dx} = 2x$$

Now, using the derivative formula:

$$\begin{aligned}\frac{du}{dv} &= \frac{\frac{du}{dx}}{\frac{dv}{dx}} \\ \frac{du}{dv} &= \frac{-6 \cot^2(2x+1) \csc^2(2x+1)}{2x} \\ \frac{du}{dv} &= -\frac{3 \cot^2(2x+1) \csc^2(2x+1)}{x}\end{aligned}$$

Final Answer

$$\frac{du}{dv} = -\frac{3 \cot^2(2x+1) \csc^2(2x+1)}{x}$$

Question 12 (iv)

Differentiate $\sin^2 x$ with respect to $e^{\cos x}$.

TYPE 53 Differentiation of One Function with Respect to Another

Given: Let $u = \sin^2 x$ and $v = e^{\cos x}$

Concept / Formula Used: $u \quad v \quad \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$

Solution:

Let:

$$\begin{aligned}u &= \sin^2 x \\ v &= e^{\cos x}\end{aligned}$$

Differentiating u with respect to x :

$$\frac{du}{dx} = 2 \sin x \cos x$$

Differentiating v with respect to x using the chain rule:

$$\begin{aligned}\frac{dv}{dx} &= e^{\cos x} \cdot \frac{d}{dx}(\cos x) \\ \frac{dv}{dx} &= e^{\cos x} \cdot (-\sin x) \\ \frac{dv}{dx} &= -\sin x e^{\cos x}\end{aligned}$$

Now, using the derivative formula:

$$\begin{aligned}\frac{du}{dv} &= \frac{\frac{du}{dx}}{\frac{dv}{dx}} \\ \frac{du}{dv} &= \frac{2 \sin x \cos x}{-\sin x e^{\cos x}} \\ \frac{du}{dv} &= -\frac{2 \cos x}{e^{\cos x}}\end{aligned}$$

Final Answer

$$\frac{du}{dv} = -\frac{2 \cos x}{e^{\cos x}}$$

Question 13 (i)

Differentiate $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\tan^{-1}x$.

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

Given: Let $u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $v = \tan^{-1}x$

Concept / Formula Used: $x = \tan \theta$ $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$

Solution:

Let:

$$\begin{aligned}u &= \sin^{-1}\left(\frac{2x}{1+x^2}\right) \\ v &= \tan^{-1}x\end{aligned}$$

Let $x = \tan \theta$, which implies $\theta = \tan^{-1}x$. Substituting $x = \tan \theta$ into the expression for u :

$$u = \sin^{-1}\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right)$$

Using the standard identity $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$:

$$\begin{aligned}u &= \sin^{-1}(\sin 2\theta) \\ u &= 2\theta\end{aligned}$$

Back-substituting $\theta = \tan^{-1} x$:

$$u = 2 \tan^{-1} x$$

Since $v = \tan^{-1} x$, we can write:

$$u = 2v$$

Differentiating u with respect to v :

$$\begin{aligned}\frac{du}{dv} &= \frac{d}{dv}(2v) \\ \frac{du}{dv} &= 2\end{aligned}$$

Final Answer

$$\frac{du}{dv} = 2$$

Question 13 (ii)

Differentiate $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$ with respect to $\tan^{-1} x$.

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

Given: Let $u = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ and $v = \tan^{-1} x$

Concept / Formula Used: $x = \tan \theta$ $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

Solution:

Let:

$$\begin{aligned}u &= \tan^{-1} \left(\frac{2x}{1-x^2} \right) \\ v &= \tan^{-1} x\end{aligned}$$

Let $x = \tan \theta$, which implies $\theta = \tan^{-1} x$. Substituting $x = \tan \theta$ into the expression for u :

$$u = \tan^{-1} \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)$$

Using the standard identity $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$:

$$\begin{aligned}u &= \tan^{-1}(\tan 2\theta) \\ u &= 2\theta\end{aligned}$$

Back-substituting $\theta = \tan^{-1} x$:

$$u = 2 \tan^{-1} x$$

Since $v = \tan^{-1} x$, we can write:

$$u = 2v$$

Differentiating u with respect to v :

$$\begin{aligned}\frac{du}{dv} &= \frac{d}{dv}(2v) \\ \frac{du}{dv} &= 2\end{aligned}$$

Final Answer

$$\frac{du}{dv} = 2$$

Question 14

Differentiate $\cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$ with respect to $\sin^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$.

TYPE 53 Differentiation of One Function with Respect to Another

Given: Let $u = \cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$ and $v = \sin^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$

Concept / Formula Used: $\sin^{-1} z + \cos^{-1} z = \frac{\pi}{2} \quad z \in [-1, 1]$

Solution:

Let:

$$\begin{aligned}u &= \cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right) \\ v &= \sin^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)\end{aligned}$$

Let $z = \frac{1}{\sqrt{1+t^2}}$. Since $t^2 \geq 0$, we have $1+t^2 \geq 1$, which implies $0 < \frac{1}{\sqrt{1+t^2}} \leq 1$. Thus, $z \in (0, 1]$.

Using the standard inverse trigonometric identity:

$$\sin^{-1} z + \cos^{-1} z = \frac{\pi}{2}$$

Substituting $z = \frac{1}{\sqrt{1+t^2}}$:

$$\begin{aligned}v + u &= \frac{\pi}{2} \\ u &= \frac{\pi}{2} - v\end{aligned}$$

Differentiating u with respect to v :

$$\begin{aligned}\frac{du}{dv} &= \frac{d}{dv}\left(\frac{\pi}{2} - v\right) \\ \frac{du}{dv} &= -1\end{aligned}$$

Final Answer

$$\frac{du}{dv} = -1$$

Common Mistake: The textbook answer key lists 1 for this question. This is a common sign error in the textbook key; since $u + v = \frac{\pi}{2}$, their derivatives must be equal in magnitude but opposite in sign, yielding $\frac{du}{dv} = -1$.

Question 15 (i)

Differentiate $\sin^{-1} \left(\frac{2x}{1+x^2} \right)$ with respect to $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$.

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

Given: Let $u = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$ and $v = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ for $x \geq 0$

Concept / Formula Used: $x = \tan \theta$ $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$ $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

Solution:

Let:

$$u = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$v = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

Let $x = \tan \theta$, which implies $\theta = \tan^{-1} x$. For $x \geq 0$, we have $\theta \in [0, \pi/2)$. Substituting $x = \tan \theta$ into the expression for u :

$$u = \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right)$$

$$u = \sin^{-1}(\sin 2\theta)$$

$$u = 2\theta$$

Substituting $x = \tan \theta$ into the expression for v :

$$v = \cos^{-1} \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right)$$

$$v = \cos^{-1}(\cos 2\theta)$$

$$v = 2\theta$$

Thus, we have:

$$u = v$$

Differentiating u with respect to v :

$$\frac{du}{dv} = 1$$

Final Answer

$$\frac{du}{dv} = 1$$

Question 15 (ii)

Differentiate $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ with respect to $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$.

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

Given: Let $u = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ and $v = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

Concept / Formula Used: $x = \tan \theta$ $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$ $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

Solution:

Let:

$$u = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$$

$$v = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

Let $x = \tan \theta$, which implies $\theta = \tan^{-1} x$. Substituting $x = \tan \theta$ into the expression for u :

$$u = \tan^{-1}\left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}\right)$$

$$u = \tan^{-1}(\tan 3\theta)$$

$$u = 3\theta$$

$$u = 3 \tan^{-1} x$$

Substituting $x = \tan \theta$ into the expression for v :

$$v = \tan^{-1}\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right)$$

$$v = \tan^{-1}(\tan 2\theta)$$

$$v = 2\theta$$

$$v = 2 \tan^{-1} x$$

We can express u in terms of v :

$$u = \frac{3}{2}v$$

Differentiating u with respect to v :

$$\frac{du}{dv} = \frac{3}{2}$$

Final Answer

$$\frac{du}{dv} = \frac{3}{2}$$

Question 16 (i)

Differentiate $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ with respect to $\tan^{-1} x$.

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

Given: Let $u = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ and $v = \tan^{-1} x$

Concept / Formula Used: $x = \tan \theta$ $1 + \tan^2 \theta = \sec^2 \theta$

Solution:

Let:

$$u = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

$$v = \tan^{-1} x$$

Let $x = \tan \theta$, which implies $\theta = \tan^{-1} x$. Substituting $x = \tan \theta$ into the expression for u :

$$u = \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta} \right)$$

$$u = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right)$$

$$u = \tan^{-1} \left(\frac{\frac{1}{\cos \theta} - 1}{\frac{\sin \theta}{\cos \theta}} \right)$$

$$u = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right)$$

Using the standard half-angle identities $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$ and $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$:

$$u = \tan^{-1} \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right)$$

$$u = \tan^{-1} \left(\tan \frac{\theta}{2} \right)$$

$$u = \frac{\theta}{2}$$

Back-substituting $\theta = \tan^{-1} x$:

$$u = \frac{1}{2} \tan^{-1} x$$

Since $v = \tan^{-1} x$, we can write:

$$u = \frac{1}{2}v$$

Differentiating u with respect to v :

$$\frac{du}{dv} = \frac{1}{2}$$

Final Answer

$$\frac{du}{dv} = \frac{1}{2}$$

Question 16 (ii)

Differentiate $\tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$ with respect to $\cos^{-1}(2x\sqrt{1-x^2})$.

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

Given: Let $u = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$ and $v = \cos^{-1}(2x\sqrt{1-x^2})$

Concept / Formula Used: $x = \cos \theta$ $\sin 2\theta = 2 \sin \theta \cos \theta$

Solution:

Let:

$$u = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$$

$$v = \cos^{-1}(2x\sqrt{1-x^2})$$

Let $x = \cos \theta$, which implies $\theta = \cos^{-1} x$. Substituting $x = \cos \theta$ into the expression for u :

$$u = \tan^{-1} \left(\frac{\sqrt{1-\cos^2 \theta}}{\cos \theta} \right)$$

$$u = \tan^{-1} \left(\frac{\sin \theta}{\cos \theta} \right)$$

$$u = \tan^{-1}(\tan \theta)$$

$$u = \theta$$

Substituting $x = \cos \theta$ into the expression for v :

$$v = \cos^{-1}(2 \cos \theta \sqrt{1-\cos^2 \theta})$$

$$v = \cos^{-1}(2 \cos \theta \sin \theta)$$

$$v = \cos^{-1}(\sin 2\theta)$$

Using the standard identity $\sin 2\theta = \cos \left(\frac{\pi}{2} - 2\theta \right)$:

$$v = \cos^{-1} \left(\cos \left(\frac{\pi}{2} - 2\theta \right) \right)$$

$$v = \frac{\pi}{2} - 2\theta$$

Since $u = \theta$, we can write:

$$\begin{aligned}v &= \frac{\pi}{2} - 2u \\2u &= \frac{\pi}{2} - v \\u &= \frac{\pi}{4} - \frac{v}{2}\end{aligned}$$

Differentiating u with respect to v :

$$\begin{aligned}\frac{du}{dv} &= \frac{d}{dv} \left(\frac{\pi}{4} - \frac{v}{2} \right) \\ \frac{du}{dv} &= -\frac{1}{2}\end{aligned}$$

Final Answer

$$\frac{du}{dv} = -\frac{1}{2}$$

Question 16 (iii)

Differentiate $\sec^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right)$ with respect to $\sin^{-1}(2x\sqrt{1-x^2})$.

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

Given: Let $u = \sec^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right)$ and $v = \sin^{-1}(2x\sqrt{1-x^2})$

Concept / Formula Used: $x = \sin \theta$ $\sin 2\theta = 2 \sin \theta \cos \theta$

Solution:

Let:

$$\begin{aligned}u &= \sec^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right) \\ v &= \sin^{-1}(2x\sqrt{1-x^2})\end{aligned}$$

Let $x = \sin \theta$, which implies $\theta = \sin^{-1} x$. Substituting $x = \sin \theta$ into the expression for u :

$$\begin{aligned}u &= \sec^{-1} \left(\frac{1}{\sqrt{1-\sin^2 \theta}} \right) \\ u &= \sec^{-1} \left(\frac{1}{\cos \theta} \right) \\ u &= \sec^{-1}(\sec \theta) \\ u &= \theta\end{aligned}$$

Substituting $x = \sin \theta$ into the expression for v :

$$\begin{aligned}v &= \sin^{-1}(2 \sin \theta \sqrt{1-\sin^2 \theta}) \\ v &= \sin^{-1}(2 \sin \theta \cos \theta) \\ v &= \sin^{-1}(\sin 2\theta) \\ v &= 2\theta\end{aligned}$$

Since $u = \theta$ and $v = 2\theta$, we can write:

$$u = \frac{1}{2}v$$

Differentiating u with respect to v :

$$\frac{du}{dv} = \frac{1}{2}$$

Final Answer

$$\frac{du}{dv} = \frac{1}{2}$$

Question 17

Prove that the derivative of $\tan^{-1} \left(\frac{x}{1 + \sqrt{1 - x^2}} \right)$ with respect to $\sin^{-1} x$ is independent of x .

TYPE 54 Differentiation of One Function with Respect to Another with Substitution

To Prove: The derivative of $u = \tan^{-1} \left(\frac{x}{1 + \sqrt{1 - x^2}} \right)$ with respect to $v = \sin^{-1} x$ is a constant.

Concept / Formula Used: $x = \sin \theta$ $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$ $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$

Solution:

Let:

$$u = \tan^{-1} \left(\frac{x}{1 + \sqrt{1 - x^2}} \right)$$

$$v = \sin^{-1} x$$

Let $x = \sin \theta$, which implies $\theta = \sin^{-1} x$. Substituting $x = \sin \theta$ into the expression for u :

$$u = \tan^{-1} \left(\frac{\sin \theta}{1 + \sqrt{1 - \sin^2 \theta}} \right)$$

$$u = \tan^{-1} \left(\frac{\sin \theta}{1 + \cos \theta} \right)$$

Using the standard half-angle identities $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ and $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$:

$$u = \tan^{-1} \left(\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} \right)$$

$$u = \tan^{-1} \left(\tan \frac{\theta}{2} \right)$$

$$u = \frac{\theta}{2}$$

Since $v = \sin^{-1} x = \theta$, we can write:

$$u = \frac{1}{2}v$$

Differentiating u with respect to v :

$$\begin{aligned}\frac{du}{dv} &= \frac{d}{dv} \left(\frac{1}{2}v \right) \\ \frac{du}{dv} &= \frac{1}{2}\end{aligned}$$

Since the derivative is $\frac{1}{2}$, which is a constant, it is independent of x .

Hence Proved

The derivative of $\tan^{-1} \left(\frac{x}{1 + \sqrt{1 - x^2}} \right)$ with respect to $\sin^{-1} x$ is $\frac{1}{2}$, which is independent of x .

Question 18

Differentiate x^x with respect to $x \log x$.

TYPE 53 Differentiation of One Function with Respect to Another

Given: Let $u = x^x$ and $v = x \log x$

Concept / Formula Used: $u \quad v \quad \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$

Solution:

Let:

$$\begin{aligned}u &= x^x \\ v &= x \log x\end{aligned}$$

To differentiate $u = x^x$ with respect to x , we write u as:

$$u = e^{x \log x}$$

Differentiating u with respect to x using the chain rule and product rule:

$$\begin{aligned}\frac{du}{dx} &= e^{x \log x} \cdot \frac{d}{dx}(x \log x) \\ \frac{du}{dx} &= x^x \cdot \left(x \cdot \frac{1}{x} + \log x \cdot 1 \right) \\ \frac{du}{dx} &= x^x(1 + \log x)\end{aligned}$$

Differentiating $v = x \log x$ with respect to x using the product rule:

$$\begin{aligned}\frac{dv}{dx} &= \frac{d}{dx}(x \log x) \\ \frac{dv}{dx} &= x \cdot \frac{1}{x} + \log x \cdot 1 \\ \frac{dv}{dx} &= 1 + \log x\end{aligned}$$

Now, using the derivative formula:

$$\begin{aligned}\frac{du}{dv} &= \frac{\frac{du}{dx}}{\frac{dv}{dx}} \\ \frac{du}{dv} &= \frac{x^x(1 + \log x)}{1 + \log x} \\ \frac{du}{dv} &= x^x\end{aligned}$$

Final Answer

$$\frac{d(x^x)}{d(x \log x)} = x^x$$

Question 19

If $x = \sqrt{a^{\tan^{-1} t}}$ and $y = \sqrt{a^{\cot^{-1} t}}$, show that $x \frac{dy}{dx} + y = 0$.

TYPE 43 Simplification before Implicit Differentiation

To Prove: $x \frac{dy}{dx} + y = 0$

Concept / Formula Used: $\tan^{-1} t + \cot^{-1} t = \frac{\pi}{2}$

Solution:

We are given:

$$\begin{aligned}x &= \sqrt{a^{\tan^{-1} t}} \\ y &= \sqrt{a^{\cot^{-1} t}}\end{aligned}$$

Multiplying x and y :

$$\begin{aligned}x \cdot y &= \sqrt{a^{\tan^{-1} t}} \cdot \sqrt{a^{\cot^{-1} t}} \\ x \cdot y &= \sqrt{a^{\tan^{-1} t} \cdot a^{\cot^{-1} t}} \\ x \cdot y &= \sqrt{a^{\tan^{-1} t + \cot^{-1} t}}\end{aligned}$$

Using the standard inverse trigonometric identity $\tan^{-1} t + \cot^{-1} t = \frac{\pi}{2}$:

$$x \cdot y = \sqrt{a^{\pi/2}}$$

Since a is a constant, $\sqrt{a^{\pi/2}}$ is also a constant. Let $C = \sqrt{a^{\pi/2}}$:

$$xy = C$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned}\frac{d}{dx}(xy) &= \frac{d}{dx}(C) \\ x \cdot \frac{dy}{dx} + y \cdot \frac{d}{dx}(x) &= 0 \\ x \frac{dy}{dx} + y(1) &= 0 \\ x \frac{dy}{dx} + y &= 0\end{aligned}$$

Hence Proved

LHS = $x \frac{dy}{dx} + y = 0 = \text{RHS}$. Hence Proved.

5.11 – Second-Order Derivatives (Miscellaneous)

Question 1(i)

Find the second order derivative of $x^3 + \tan x$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = x^3 + \tan x$

Concept / Formula Used: $\frac{d}{dx}(x^n) = nx^{n-1}$ $\frac{d}{dx}(\tan x) = \sec^2 x$

Solution:

Differentiating both sides with respect to x to find the first derivative:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^3) + \frac{d}{dx}(\tan x) \\ \frac{dy}{dx} &= 3x^2 + \sec^2 x\end{aligned}$$

Differentiating again with respect to x to find the second order derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(3x^2) + \frac{d}{dx}(\sec^2 x)$$

Using the chain rule for $\sec^2 x = (\sec x)^2$:

$$\frac{d}{dx}(\sec^2 x) = 2 \sec x \cdot \frac{d}{dx}(\sec x) = 2 \sec x \cdot (\sec x \tan x) = 2 \sec^2 x \tan x$$

Thus:

$$\frac{d^2y}{dx^2} = 6x + 2 \sec^2 x \tan x$$

Final Answer

$$\frac{d^2y}{dx^2} = 6x + 2 \sec^2 x \tan x$$

Common Mistake: Forgetting to apply the chain rule when differentiating $\sec^2 x$ with respect to x .

Question 1(ii)

Find the second order derivative of $\log x$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = \log x$

Concept / Formula Used: $\frac{d}{dx}(\log x) = \frac{1}{x}$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}(\log x) = \frac{1}{x} = x^{-1}$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(x^{-1})$$

Using the power rule $\frac{d}{dx}(x^n) = nx^{n-1}$:

$$\frac{d^2y}{dx^2} = -1 \cdot x^{-2} = -\frac{1}{x^2}$$

Final Answer

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}$$

Common Mistake: Applying the power rule incorrectly for negative exponents by writing x^0 instead of x^{-2} .

Question 1(iii)

Find the second order derivative of $\tan^{-1} x$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = \tan^{-1} x$

Concept / Formula Used: $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2} = (1+x^2)^{-1}$$

Differentiating again with respect to x using the chain rule and power rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}((1+x^2)^{-1})$$

$$\frac{d^2y}{dx^2} = -1(1+x^2)^{-2} \cdot \frac{d}{dx}(1+x^2)$$

$$\frac{d^2y}{dx^2} = -1(1+x^2)^{-2} \cdot (0+2x)$$

$$\frac{d^2y}{dx^2} = -\frac{2x}{(1+x^2)^2}$$

Final Answer

$$\frac{d^2y}{dx^2} = -\frac{2x}{(1+x^2)^2}$$

Common Mistake: Omitting the derivative of the inner function $(1+x^2)$, which is $2x$, due to missing chain rule application.

Question 2(i)

If $y = \log(x-2)$, $x > 2$, find $\frac{d^2y}{dx^2}$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: $y = \log(x-2)$ for $x > 2$

Concept / Formula Used: $\frac{d}{dx}(\log u) = \frac{1}{u} \cdot \frac{du}{dx}$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}[\log(x-2)]$$

$$\frac{dy}{dx} = \frac{1}{x-2} \cdot \frac{d}{dx}(x-2)$$

$$\frac{dy}{dx} = \frac{1}{x-2} \cdot (1-0) = (x-2)^{-1}$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}((x-2)^{-1})$$

$$\frac{d^2y}{dx^2} = -1(x-2)^{-2} \cdot \frac{d}{dx}(x-2)$$

$$\frac{d^2y}{dx^2} = -1(x-2)^{-2} \cdot (1) = -\frac{1}{(x-2)^2}$$

Final Answer

$$\frac{d^2y}{dx^2} = -\frac{1}{(x-2)^2}$$

Common Mistake: Forgetting to multiply by the derivative of the inner function $(x-2)$, though it evaluates to 1 here.

Question 2(ii)

If $y = \cot x$, find $\frac{d^2y}{dx^2}$ at $x = \frac{\pi}{4}$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: $y = \cot x$

Concept / Formula Used: $\frac{d}{dx}(\cot x) = -\csc^2 x$ $\frac{d}{dx}(\csc x) = -\csc x \cot x$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}(\cot x) = -\csc^2 x$$

Differentiating again with respect to x using the chain rule for $(\csc x)^2$:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} (-(\csc x)^2) \\ \frac{d^2y}{dx^2} &= -2(\csc x) \cdot \frac{d}{dx}(\csc x) \\ \frac{d^2y}{dx^2} &= -2(\csc x) \cdot (-\csc x \cot x) \\ \frac{d^2y}{dx^2} &= 2 \csc^2 x \cot x\end{aligned}$$

Now, substituting $x = \frac{\pi}{4}$:

$$\begin{aligned}\csc\left(\frac{\pi}{4}\right) &= \frac{1}{\sin(\pi/4)} = \sqrt{2} \\ \cot\left(\frac{\pi}{4}\right) &= \frac{\cos(\pi/4)}{\sin(\pi/4)} = 1\end{aligned}$$

Substituting these values into the second derivative expression:

$$\left. \frac{d^2y}{dx^2} \right|_{x=\pi/4} = 2(\sqrt{2})^2(1) = 2(2)(1) = 4$$

Final Answer

$$\left. \frac{d^2y}{dx^2} \right|_{x=\pi/4} = 4$$

Common Mistake: Incorrectly evaluating trigonometric values at $\pi/4$ or missing the negative sign when differentiating cosecant.

Question 2(iii)

If $x = at^2$ and $y = 2at$, then find $\frac{d^2y}{dx^2}$.

TYPE 64a Parametric Second-Order Derivatives – General Form

Given: $x = at^2$, $y = 2at$

Concept / Formula Used: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ $\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) / \frac{dx}{dt}$

Solution:

First, find the derivatives with respect to t :

$$\frac{dx}{dt} = \frac{d}{dt}(at^2) = 2at$$

$$\frac{dy}{dt} = \frac{d}{dt}(2at) = 2a$$

Now, find the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t} = t^{-1}$$

Now, find the second order derivative $\frac{d^2y}{dx^2}$ with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

$$\frac{d}{dt}(t^{-1}) = -1 \cdot t^{-2} = -\frac{1}{t^2}$$

Dividing by $\frac{dx}{dt} = 2at$:

$$\frac{d^2y}{dx^2} = \frac{-\frac{1}{t^2}}{2at} = -\frac{1}{2at^3}$$

Final Answer

$$\frac{d^2y}{dx^2} = -\frac{1}{2at^3}$$

Common Mistake: Differentiating $\frac{dy}{dx}$ directly with respect to x without dividing by $\frac{dx}{dt}$ (forgetting the chain rule in parametric form).

Question 3(i)

Find the second order derivatives of $\sin^{-1} x$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = \sin^{-1} x$

Concept / Formula Used: $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2}$$

Differentiating again with respect to x using the chain rule:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} ((1-x^2)^{-1/2}) \\ \frac{d^2y}{dx^2} &= -\frac{1}{2}(1-x^2)^{-3/2} \cdot \frac{d}{dx}(1-x^2) \\ \frac{d^2y}{dx^2} &= -\frac{1}{2}(1-x^2)^{-3/2} \cdot (0-2x) \\ \frac{d^2y}{dx^2} &= -\frac{1}{2}(1-x^2)^{-3/2} \cdot (-2x) \\ \frac{d^2y}{dx^2} &= \frac{x}{(1-x^2)^{3/2}} = \frac{x}{(1-x^2)\sqrt{1-x^2}}\end{aligned}$$

Final Answer

$$\frac{d^2y}{dx^2} = \frac{x}{(1-x^2)^{3/2}}$$

Common Mistake: Forgetting the negative sign from the power rule when differentiating $(1-x^2)^{-1/2}$.

Question 3(ii)

Find the second order derivatives of $x \cos x$.

TYPE 65 Product Rule Differentiation

Given: Let $y = x \cos x$

Concept / Formula Used: $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$

Solution:

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \frac{d}{dx}(\cos x) + \cos x \cdot \frac{d}{dx}(x) \\ \frac{dy}{dx} &= x(-\sin x) + \cos x(1)\end{aligned}$$

$$\frac{dy}{dx} = -x \sin x + \cos x$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(-x \sin x) + \frac{d}{dx}(\cos x)$$

For the first term, apply the product rule again:

$$\begin{aligned} \frac{d}{dx}(-x \sin x) &= - \left[x \cdot \frac{d}{dx}(\sin x) + \sin x \cdot \frac{d}{dx}(x) \right] \\ &= - [x \cos x + \sin x(1)] = -x \cos x - \sin x \end{aligned}$$

For the second term:

$$\frac{d}{dx}(\cos x) = -\sin x$$

Combining both terms:

$$\frac{d^2y}{dx^2} = -x \cos x - \sin x - \sin x = -x \cos x - 2 \sin x$$

Final Answer

$$\frac{d^2y}{dx^2} = -x \cos x - 2 \sin x$$

Common Mistake: Distributing the negative sign incorrectly while applying the product rule to the second derivative.

Question 3(iii)

Find the second order derivatives of $x \sin 2x$.

TYPE 65 Product Rule Differentiation

Given: Let $y = x \sin 2x$

Concept / Formula Used: $\frac{d}{dx}(\sin 2x) = 2 \cos 2x$

Solution:

Differentiating both sides with respect to x using the product rule:

$$\frac{dy}{dx} = x \cdot \frac{d}{dx}(\sin 2x) + \sin 2x \cdot \frac{d}{dx}(x)$$

$$\frac{dy}{dx} = x(2 \cos 2x) + \sin 2x(1)$$

$$\frac{dy}{dx} = 2x \cos 2x + \sin 2x$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(2x \cos 2x) + \frac{d}{dx}(\sin 2x)$$

Applying the product rule to the first term:

$$\begin{aligned}\frac{d}{dx}(2x \cos 2x) &= 2x \cdot \frac{d}{dx}(\cos 2x) + \cos 2x \cdot \frac{d}{dx}(2x) \\ &= 2x(-2 \sin 2x) + \cos 2x(2) \\ &= -4x \sin 2x + 2 \cos 2x\end{aligned}$$

Adding the derivative of the second term, $\frac{d}{dx}(\sin 2x) = 2 \cos 2x$:

$$\begin{aligned}\frac{d^2y}{dx^2} &= (-4x \sin 2x + 2 \cos 2x) + 2 \cos 2x \\ \frac{d^2y}{dx^2} &= -4x \sin 2x + 4 \cos 2x\end{aligned}$$

Final Answer

$$\frac{d^2y}{dx^2} = -4x \sin 2x + 4 \cos 2x$$

Common Mistake: Forgetting the factor of 2 from the chain rule when differentiating $\sin 2x$.

Question 3(iv)

Find the second order derivatives of $e^x \sin 5x$.

TYPE 65 Product Rule Differentiation

Given: Let $y = e^x \sin 5x$

Concept / Formula Used: $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$

Solution:

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned}\frac{dy}{dx} &= e^x \cdot \frac{d}{dx}(\sin 5x) + \sin 5x \cdot \frac{d}{dx}(e^x) \\ \frac{dy}{dx} &= e^x(5 \cos 5x) + \sin 5x(e^x) \\ \frac{dy}{dx} &= e^x(5 \cos 5x + \sin 5x)\end{aligned}$$

Differentiating again with respect to x using the product rule:

$$\frac{d^2y}{dx^2} = e^x \cdot \frac{d}{dx}(5 \cos 5x + \sin 5x) + (5 \cos 5x + \sin 5x) \cdot \frac{d}{dx}(e^x)$$

Evaluating the derivative of the bracket:

$$\frac{d}{dx}(5 \cos 5x + \sin 5x) = 5(-5 \sin 5x) + 5 \cos 5x = -25 \sin 5x + 5 \cos 5x$$

Substituting back into the product rule expression:

$$\frac{d^2y}{dx^2} = e^x(-25 \sin 5x + 5 \cos 5x) + (5 \cos 5x + \sin 5x)e^x$$

Factorizing e^x :

$$\frac{d^2y}{dx^2} = e^x [-25 \sin 5x + 5 \cos 5x + 5 \cos 5x + \sin 5x]$$

$$\frac{d^2y}{dx^2} = e^x [-24 \sin 5x + 10 \cos 5x]$$

Final Answer

$$\frac{d^2y}{dx^2} = e^x(10 \cos 5x - 24 \sin 5x)$$

Common Mistake: Arithmetic errors when collecting and combining like terms of $\sin 5x$ and $\cos 5x$.

Question 3(v)

Find the second order derivatives of $e^{2x} \sin 3x$.

TYPE 65 Product Rule Differentiation

Given: Let $y = e^{2x} \sin 3x$

Concept / Formula Used: $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$

Solution:

Differentiating both sides with respect to x using the product rule:

$$\frac{dy}{dx} = e^{2x} \cdot \frac{d}{dx}(\sin 3x) + \sin 3x \cdot \frac{d}{dx}(e^{2x})$$

$$\frac{dy}{dx} = e^{2x}(3 \cos 3x) + \sin 3x(2e^{2x})$$

$$\frac{dy}{dx} = e^{2x}(3 \cos 3x + 2 \sin 3x)$$

Differentiating again with respect to x using the product rule:

$$\frac{d^2y}{dx^2} = e^{2x} \cdot \frac{d}{dx}(3 \cos 3x + 2 \sin 3x) + (3 \cos 3x + 2 \sin 3x) \cdot \frac{d}{dx}(e^{2x})$$

Evaluating the derivative of the bracket:

$$\frac{d}{dx}(3 \cos 3x + 2 \sin 3x) = 3(-3 \sin 3x) + 2(3 \cos 3x) = -9 \sin 3x + 6 \cos 3x$$

And $\frac{d}{dx}(e^{2x}) = 2e^{2x}$. Substituting these:

$$\frac{d^2y}{dx^2} = e^{2x}(-9 \sin 3x + 6 \cos 3x) + (3 \cos 3x + 2 \sin 3x)(2e^{2x})$$

Factorizing e^{2x} :

$$\frac{d^2y}{dx^2} = e^{2x} [-9 \sin 3x + 6 \cos 3x + 6 \cos 3x + 4 \sin 3x]$$

$$\frac{d^2y}{dx^2} = e^{2x} [-5 \sin 3x + 12 \cos 3x]$$

Final Answer

$$\frac{d^2y}{dx^2} = e^{2x} (12 \cos 3x - 5 \sin 3x)$$

Common Mistake: Missing the coefficient multipliers from the chain rule when differentiating exponential and trigonometric functions.

Question 3(vi)

Find the second order derivatives of $\log(\log x)$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = \log(\log x)$

Concept / Formula Used: $\frac{d}{dx}[\log(\log x)] = \frac{1}{\log x} \cdot \frac{1}{x}$

Solution:

Differentiating both sides with respect to x using the chain rule:

$$\frac{dy}{dx} = \frac{1}{\log x} \cdot \frac{d}{dx}(\log x)$$

$$\frac{dy}{dx} = \frac{1}{x \log x} = (x \log x)^{-1}$$

Differentiating again with respect to x using the product rule and chain rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x \log x} \right)$$

Using the quotient rule $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ with $u = 1$ and $v = x \log x$:

$$\frac{d^2y}{dx^2} = \frac{(x \log x)(0) - 1 \cdot \frac{d}{dx}(x \log x)}{(x \log x)^2}$$

$$\frac{d}{dx}(x \log x) = x \cdot \frac{1}{x} + \log x \cdot 1 = 1 + \log x$$

Thus:

$$\frac{d^2y}{dx^2} = \frac{-(1 + \log x)}{(x \log x)^2}$$

Final Answer

$$\frac{d^2y}{dx^2} = -\frac{1 + \log x}{(x \log x)^2}$$

Common Mistake: Applying the chain rule incorrectly on nested logarithmic functions.

Question 3(vii)

Find the second order derivatives of $\frac{\log x}{x}$.

TYPE 66 Quotient Rule Differentiation

Given: Let $y = \frac{\log x}{x}$

Concept / Formula Used: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Solution:

Differentiating both sides with respect to x using the quotient rule:

$$\frac{dy}{dx} = \frac{x \cdot \frac{d}{dx}(\log x) - \log x \cdot \frac{d}{dx}(x)}{x^2}$$

$$\frac{dy}{dx} = \frac{x \left(\frac{1}{x} \right) - \log x(1)}{x^2}$$

$$\frac{dy}{dx} = \frac{1 - \log x}{x^2}$$

Differentiating again with respect to x using the quotient rule:

$$\frac{d^2y}{dx^2} = \frac{x^2 \cdot \frac{d}{dx}(1 - \log x) - (1 - \log x) \cdot \frac{d}{dx}(x^2)}{(x^2)^2}$$

$$\frac{d}{dx}(1 - \log x) = 0 - \frac{1}{x} = -\frac{1}{x}$$

$$\frac{d}{dx}(x^2) = 2x$$

Substituting these into the quotient rule formula:

$$\frac{d^2y}{dx^2} = \frac{x^2 \left(-\frac{1}{x} \right) - (1 - \log x)(2x)}{x^4}$$

$$\frac{d^2y}{dx^2} = \frac{-x - 2x(1 - \log x)}{x^4}$$

Factorizing x from the numerator:

$$\frac{d^2y}{dx^2} = \frac{-x[1 + 2(1 - \log x)]}{x^4} = \frac{-[1 + 2 - 2 \log x]}{x^3} = \frac{2 \log x - 3}{x^3}$$

Final Answer

$$\frac{d^2y}{dx^2} = \frac{2 \log x - 3}{x^3}$$

Common Mistake: Algebraic simplification errors when expanding and canceling terms in the numerator.

Question 3(viii)

Find the second order derivatives of $x^2 \log |\cos x|$.

TYPE 65 Product Rule Differentiation

Given: Let $y = x^2 \log |\cos x|$

Concept / Formula Used: $\frac{d}{dx}[\log |\cos x|] = \frac{-\sin x}{\cos x} = -\tan x$

Solution:

Differentiating both sides with respect to x using the product rule:

$$\frac{dy}{dx} = x^2 \cdot \frac{d}{dx}(\log |\cos x|) + \log |\cos x| \cdot \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = x^2 \left(\frac{-\sin x}{\cos x} \right) + \log |\cos x| (2x)$$

$$\frac{dy}{dx} = -x^2 \tan x + 2x \log |\cos x|$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(-x^2 \tan x) + \frac{d}{dx}(2x \log |\cos x|)$$

Applying the product rule to the first term:

$$\begin{aligned} \frac{d}{dx}(-x^2 \tan x) &= - \left[x^2 \cdot \frac{d}{dx}(\tan x) + \tan x \cdot \frac{d}{dx}(x^2) \right] \\ &= - [x^2 \sec^2 x + \tan x (2x)] = -x^2 \sec^2 x - 2x \tan x \end{aligned}$$

Applying the product rule to the second term:

$$\begin{aligned} \frac{d}{dx}(2x \log |\cos x|) &= 2x \cdot \frac{d}{dx}(\log |\cos x|) + \log |\cos x| \cdot \frac{d}{dx}(2x) \\ &= 2x(-\tan x) + \log |\cos x|(2) = -2x \tan x + 2 \log |\cos x| \end{aligned}$$

Combining all terms:

$$\frac{d^2y}{dx^2} = -x^2 \sec^2 x - 2x \tan x - 2x \tan x + 2 \log |\cos x|$$

$$\frac{d^2y}{dx^2} = -x^2 \sec^2 x - 4x \tan x + 2 \log |\cos x|$$

Final Answer

$$\frac{d^2y}{dx^2} = 2 \log |\cos x| - 4x \tan x - x^2 \sec^2 x$$

Common Mistake: Forgetting to apply the product rule separately to both terms of the first derivative.

Question 3(ix)

Find the second order derivatives of $\frac{2x+1}{2x+3}$.

TYPE 66 Quotient Rule Differentiation

Given: Let $y = \frac{2x+1}{2x+3}$

Concept / Formula Used: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Solution:

Differentiating both sides with respect to x using the quotient rule:

$$\frac{dy}{dx} = \frac{(2x+3) \cdot \frac{d}{dx}(2x+1) - (2x+1) \cdot \frac{d}{dx}(2x+3)}{(2x+3)^2}$$

$$\frac{dy}{dx} = \frac{(2x+3)(2) - (2x+1)(2)}{(2x+3)^2}$$

$$\frac{dy}{dx} = \frac{4x+6-4x-2}{(2x+3)^2} = \frac{4}{(2x+3)^2} = 4(2x+3)^{-2}$$

Differentiating again with respect to x using the chain rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} (4(2x+3)^{-2})$$

$$\frac{d^2y}{dx^2} = 4 \cdot (-2)(2x+3)^{-3} \cdot \frac{d}{dx}(2x+3)$$

$$\frac{d^2y}{dx^2} = -8(2x+3)^{-3} \cdot (2) = -\frac{16}{(2x+3)^3}$$

Final Answer

$$\frac{d^2y}{dx^2} = -\frac{16}{(2x+3)^3}$$

Common Mistake: Failing to multiply by the inner derivative 2 when differentiating $(2x+3)^{-2}$.

Question 4(i)

Find the second order derivative of $\sec ax$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = \sec ax$

Concept / Formula Used: $\frac{d}{dx}(\sec x) = \sec x \tan x$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}(\sec ax) = a \sec ax \tan ax$$

Differentiating again with respect to x using the product rule and chain rule:

$$\begin{aligned}\frac{d^2y}{dx^2} &= a \cdot \frac{d}{dx}(\sec ax \tan ax) \\ \frac{d}{dx}(\sec ax \tan ax) &= \sec ax \cdot \frac{d}{dx}(\tan ax) + \tan ax \cdot \frac{d}{dx}(\sec ax) \\ &= \sec ax(a \sec^2 ax) + \tan ax(a \sec ax \tan ax) \\ &= a \sec^3 ax + a \sec ax \tan^2 ax\end{aligned}$$

Multiplying by the outer constant a :

$$\frac{d^2y}{dx^2} = a^2 \sec^3 ax + a^2 \sec ax \tan^2 ax = a^2 \sec ax(\sec^2 ax + \tan^2 ax)$$

Using the trigonometric identity $\tan^2 ax = \sec^2 ax - 1$:

$$\frac{d^2y}{dx^2} = a^2 \sec ax(\sec^2 ax + \sec^2 ax - 1) = a^2 \sec ax(2 \sec^2 ax - 1)$$

Final Answer

$$\frac{d^2y}{dx^2} = a^2 \sec ax(2 \sec^2 ax - 1)$$

Common Mistake: Forgetting the factor of a^2 produced by applying the chain rule twice.

Question 4(ii)

Find the second order derivative of $\cot(1 - 2x)$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = \cot(1 - 2x)$

Concept / Formula Used: $\frac{d}{dx}[\cot(1 - 2x)] = -\csc^2(1 - 2x) \cdot (-2) = 2 \csc^2(1 - 2x)$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = -\csc^2(1 - 2x) \cdot \frac{d}{dx}(1 - 2x) = -\csc^2(1 - 2x) \cdot (-2) = 2 \csc^2(1 - 2x)$$

Differentiating again with respect to x using the chain rule for $(\csc(1 - 2x))^2$:

$$\begin{aligned}\frac{d^2y}{dx^2} &= 2 \cdot 2 \csc(1 - 2x) \cdot \frac{d}{dx}(\csc(1 - 2x)) \\ \frac{d}{dx}(\csc(1 - 2x)) &= -\csc(1 - 2x) \cot(1 - 2x) \cdot \frac{d}{dx}(1 - 2x)\end{aligned}$$

$$= -\csc(1-2x) \cot(1-2x) \cdot (-2) = 2 \csc(1-2x) \cot(1-2x)$$

Substituting back into the second derivative expression:

$$\frac{d^2y}{dx^2} = 4 \csc(1-2x) \cdot [2 \csc(1-2x) \cot(1-2x)]$$

$$\frac{d^2y}{dx^2} = 8 \csc^2(1-2x) \cot(1-2x)$$

Final Answer

$$\frac{d^2y}{dx^2} = 8 \csc^2(1-2x) \cot(1-2x)$$

Common Mistake: Dropping negative signs during successive applications of the chain rule.

Question 4(iii)

Find the second order derivative of $\sin 3x \cos 5x$.

TYPE 65 Product Rule Differentiation

Given: Let $y = \sin 3x \cos 5x$

Concept / Formula Used: $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$

Solution:

Method 1: Using the product rule directly.

$$\frac{dy}{dx} = \sin 3x \cdot \frac{d}{dx}(\cos 5x) + \cos 5x \cdot \frac{d}{dx}(\sin 3x)$$

$$\frac{dy}{dx} = \sin 3x(-5 \sin 5x) + \cos 5x(3 \cos 3x)$$

$$\frac{dy}{dx} = 3 \cos 3x \cos 5x - 5 \sin 3x \sin 5x$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = 3 \cdot \frac{d}{dx}(\cos 3x \cos 5x) - 5 \cdot \frac{d}{dx}(\sin 3x \sin 5x)$$

Applying the product rule to each term:

$$\frac{d}{dx}(\cos 3x \cos 5x) = \cos 3x(-5 \sin 5x) + \cos 5x(-3 \sin 3x) = -5 \cos 3x \sin 5x - 3 \sin 3x \cos 5x$$

$$\frac{d}{dx}(\sin 3x \sin 5x) = \sin 3x(5 \cos 5x) + \sin 5x(3 \cos 3x) = 5 \sin 3x \cos 5x + 3 \cos 3x \sin 5x$$

Combining both parts:

$$\frac{d^2y}{dx^2} = 3(-5 \cos 3x \sin 5x - 3 \sin 3x \cos 5x) - 5(5 \sin 3x \cos 5x + 3 \cos 3x \sin 5x)$$

Expanding and collecting like terms:

$$= -15 \cos 3x \sin 5x - 9 \sin 3x \cos 5x - 25 \sin 3x \cos 5x - 15 \cos 3x \sin 5x$$

$$= -30 \cos 3x \sin 5x - 34 \sin 3x \cos 5x$$

Final Answer

$$\frac{d^2y}{dx^2} = -34 \sin 3x \cos 5x - 30 \cos 3x \sin 5x$$

Common Mistake: Sign errors while differentiating products of sine and cosine functions.

Question 4(iv)

Find the second order derivative of $\sin^3 x$.

TYPE 63 Second Order Derivative – Direct Differentiation

Given: Let $y = \sin^3 x = (\sin x)^3$

Concept / Formula Used: $\frac{d}{dx}[(\sin x)^3] = 3 \sin^2 x \cos x$

Solution:

Differentiating both sides with respect to x using the chain rule:

$$\frac{dy}{dx} = 3 \sin^2 x \cdot \frac{d}{dx}(\sin x) = 3 \sin^2 x \cos x$$

Differentiating again with respect to x using the product rule:

$$\begin{aligned} \frac{d^2y}{dx^2} &= 3 \left[\sin^2 x \cdot \frac{d}{dx}(\cos x) + \cos x \cdot \frac{d}{dx}(\sin^2 x) \right] \\ \frac{d}{dx}(\sin^2 x) &= 2 \sin x \cos x \end{aligned}$$

Substituting this back:

$$\begin{aligned} \frac{d^2y}{dx^2} &= 3 [\sin^2 x (-\sin x) + \cos x (2 \sin x \cos x)] \\ \frac{d^2y}{dx^2} &= 3 [-\sin^3 x + 2 \sin x \cos^2 x] \end{aligned}$$

Using $\cos^2 x = 1 - \sin^2 x$:

$$\begin{aligned} \frac{d^2y}{dx^2} &= 3 [-\sin^3 x + 2 \sin x (1 - \sin^2 x)] \\ \frac{d^2y}{dx^2} &= 3 [-\sin^3 x + 2 \sin x - 2 \sin^3 x] = 3(2 \sin x - 3 \sin^3 x) = 6 \sin x - 9 \sin^3 x \end{aligned}$$

Final Answer

$$\frac{d^2y}{dx^2} = 6 \sin x - 9 \sin^3 x$$

Common Mistake: Forgetting to apply the product rule for the second derivative of $\sin^2 x \cos x$.

Question 4(v)Find the second order derivative of $\cos(2x^2 - 1)$.**TYPE 63** Second Order Derivative – Direct Differentiation**Given:** Let $y = \cos(2x^2 - 1)$ **Concept / Formula Used:** $\frac{d}{dx}[\cos(f(x))] = -\sin(f(x)) \cdot f'(x)$ **Solution:**Differentiating both sides with respect to x :

$$\frac{dy}{dx} = -\sin(2x^2 - 1) \cdot \frac{d}{dx}(2x^2 - 1)$$

$$\frac{dy}{dx} = -\sin(2x^2 - 1) \cdot (4x) = -4x \sin(2x^2 - 1)$$

Differentiating again with respect to x using the product rule:

$$\frac{d^2y}{dx^2} = -4 \left[x \cdot \frac{d}{dx}(\sin(2x^2 - 1)) + \sin(2x^2 - 1) \cdot \frac{d}{dx}(x) \right]$$

Evaluating the derivative of the sine term using the chain rule:

$$\frac{d}{dx}(\sin(2x^2 - 1)) = \cos(2x^2 - 1) \cdot \frac{d}{dx}(2x^2 - 1) = \cos(2x^2 - 1) \cdot (4x) = 4x \cos(2x^2 - 1)$$

Substituting this back:

$$\frac{d^2y}{dx^2} = -4 [x(4x \cos(2x^2 - 1)) + \sin(2x^2 - 1)(1)]$$

$$\frac{d^2y}{dx^2} = -4 [4x^2 \cos(2x^2 - 1) + \sin(2x^2 - 1)]$$

$$\frac{d^2y}{dx^2} = -16x^2 \cos(2x^2 - 1) - 4 \sin(2x^2 - 1)$$

Final Answer

$$\frac{d^2y}{dx^2} = -16x^2 \cos(2x^2 - 1) - 4 \sin(2x^2 - 1)$$

Common Mistake: Omitting the product rule when differentiating the product of $-4x$ and $\sin(2x^2 - 1)$.**Question 4(vi)**Find the second order derivative of $\sqrt{1 - x^2}$.**TYPE 63** Second Order Derivative – Direct Differentiation**Given:** Let $y = \sqrt{1 - x^2} = (1 - x^2)^{1/2}$

Concept / Formula Used: $\frac{d}{dx}[u^{1/2}] = \frac{1}{2\sqrt{u}} \cdot \frac{du}{dx}$

Solution:

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}} \cdot \frac{d}{dx}(1-x^2) = \frac{-2x}{2\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}}$$

Differentiating again with respect to x using the quotient rule:

$$\begin{aligned}\frac{d^2y}{dx^2} &= -\frac{d}{dx} \left(\frac{x}{\sqrt{1-x^2}} \right) \\ \frac{d^2y}{dx^2} &= - \left[\frac{\sqrt{1-x^2} \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(\sqrt{1-x^2})}{(\sqrt{1-x^2})^2} \right] \\ \frac{d}{dx}(x) &= 1, \quad \frac{d}{dx}(\sqrt{1-x^2}) = \frac{-x}{\sqrt{1-x^2}}\end{aligned}$$

Substituting these into the quotient rule:

$$\frac{d^2y}{dx^2} = - \left[\frac{\sqrt{1-x^2}(1) - x \left(\frac{-x}{\sqrt{1-x^2}} \right)}{1-x^2} \right]$$

Simplifying the numerator by finding a common denominator $\sqrt{1-x^2}$:

$$\frac{d^2y}{dx^2} = - \left[\frac{(1-x^2) + x^2}{(1-x^2)\sqrt{1-x^2}} \right] = - \left[\frac{1}{(1-x^2)^{3/2}} \right] = -\frac{1}{(1-x^2)^{3/2}}$$

Final Answer

$$\frac{d^2y}{dx^2} = -\frac{1}{(1-x^2)^{3/2}}$$

Common Mistake: Errors in fraction simplification within the numerator during quotient rule execution.

Question 5

If $y = \cos^{-1} x$, find $\frac{d^2y}{dx^2}$ in terms of y alone.

TYPE 67 Implicit / Inverse Trig Second Derivative in terms of y

Given: $y = \cos^{-1} x$

Concept / Formula Used: $\sin y = \sqrt{1 - \cos^2 y}$ $\cos y = x$

Solution:

From $y = \cos^{-1} x$, we have $x = \cos y$. Differentiating both sides with respect to x :

$$1 = -\sin y \cdot \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\sin y} = -\csc y$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(-\csc y) = -(-\csc y \cot y) \cdot \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \csc y \cot y \cdot (-\csc y) = -\csc^2 y \cot y$$

Final Answer

$$\frac{d^2y}{dx^2} = -\csc^2 y \cot y$$

Common Mistake: Failing to substitute $\frac{dy}{dx}$ in terms of y when differentiating with respect to x .

Question 6(i)

If $y = \cot x$, prove that $\frac{d^2y}{dx^2} + 2y \frac{dy}{dx} = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\frac{d^2y}{dx^2} + 2y \frac{dy}{dx} = 0$ given $y = \cot x$

Concept / Formula Used: $\frac{dy}{dx} = -\csc^2 x$ $\frac{d^2y}{dx^2} = 2 \csc^2 x \cot x$

Solution:

Given $y = \cot x$. First derivative:

$$\frac{dy}{dx} = -\csc^2 x$$

Second derivative:

$$\frac{d^2y}{dx^2} = 2 \csc^2 x \cot x$$

Substitute $y = \cot x$ and $\frac{dy}{dx} = -\csc^2 x$ into the expression $\frac{d^2y}{dx^2} + 2y \frac{dy}{dx}$:

$$\text{LHS} = 2 \csc^2 x \cot x + 2(\cot x)(-\csc^2 x)$$

$$= 2 \csc^2 x \cot x - 2 \csc^2 x \cot x = 0 = \text{RHS}$$

Hence Proved

$LHS = RHS$, hence proved.

Common Mistake: Sign errors when substituting the first derivative back into the expression.

Question 6(ii)

If $y = 5 \cos x - 3 \sin x$, prove that $\frac{d^2y}{dx^2} + y = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\frac{d^2y}{dx^2} + y = 0$ given $y = 5 \cos x - 3 \sin x$

Concept / Formula Used: $\frac{d}{dx}(\cos x) = -\sin x$ $\frac{d}{dx}(\sin x) = \cos x$

Solution:

Given $y = 5 \cos x - 3 \sin x$. Differentiating with respect to x :

$$\frac{dy}{dx} = 5(-\sin x) - 3(\cos x) = -5 \sin x - 3 \cos x$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = -5(\cos x) - 3(-\sin x) = -5 \cos x + 3 \sin x$$

Now substitute $\frac{d^2y}{dx^2}$ and y into the LHS of the expression:

$$\text{LHS} = \frac{d^2y}{dx^2} + y = (-5 \cos x + 3 \sin x) + (5 \cos x - 3 \sin x)$$

Combining terms:

$$\text{LHS} = -5 \cos x + 5 \cos x + 3 \sin x - 3 \sin x = 0 = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Incorrectly distributing negative signs during the second differentiation.

Question 7(i)

If $y = x + \tan x$, prove that $\cos^2 x \frac{d^2y}{dx^2} - 2y + 2x = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\cos^2 x \frac{d^2y}{dx^2} - 2y + 2x = 0$ given $y = x + \tan x$

Concept / Formula Used: $\frac{dy}{dx} = 1 + \sec^2 x$ $\frac{d^2y}{dx^2} = 2 \sec^2 x \tan x$

Solution:

Given $y = x + \tan x$. First derivative:

$$\frac{dy}{dx} = 1 + \sec^2 x$$

Second derivative:

$$\frac{d^2y}{dx^2} = 0 + 2 \sec x \cdot (\sec x \tan x) = 2 \sec^2 x \tan x$$

Substitute $\frac{d^2y}{dx^2}$ and y into the LHS:

$$\text{LHS} = \cos^2 x (2 \sec^2 x \tan x) - 2(x + \tan x) + 2x$$

Since $\cos^2 x \cdot \sec^2 x = 1$:

$$\text{LHS} = 2(1) \tan x - 2x - 2 \tan x + 2x$$

$$\text{LHS} = 2 \tan x - 2 \tan x - 2x + 2x = 0 = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Forgetting to substitute $y = x + \tan x$ fully into the expression.

Question 7(ii)

If $y = \tan x + \sec x$, prove that $\frac{d^2y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\frac{d^2y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$ given $y = \tan x + \sec x$

Concept / Formula Used: $\tan x = \frac{\sin x}{\cos x}$ $\sec x = \frac{1}{\cos x}$

Solution:

Given $y = \tan x + \sec x = \frac{\sin x + 1}{\cos x}$. First derivative:

$$\frac{dy}{dx} = \sec^2 x + \sec x \tan x = \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} = \frac{1 + \sin x}{\cos^2 x}$$

Second derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{1 + \sin x}{\cos^2 x} \right)$$

Using the quotient rule:

$$\frac{d^2y}{dx^2} = \frac{\cos^2 x \cdot \frac{d}{dx}(1 + \sin x) - (1 + \sin x) \cdot \frac{d}{dx}(\cos^2 x)}{(\cos^2 x)^2}$$

$$\frac{d}{dx}(1 + \sin x) = \cos x$$

$$\frac{d}{dx}(\cos^2 x) = 2 \cos x (-\sin x) = -2 \sin x \cos x$$

Substituting these:

$$\frac{d^2y}{dx^2} = \frac{\cos^2 x (\cos x) - (1 + \sin x)(-2 \sin x \cos x)}{\cos^4 x}$$

Factorizing $\cos x$ from the numerator:

$$\frac{d^2y}{dx^2} = \frac{\cos x [\cos^2 x + 2 \sin x (1 + \sin x)]}{\cos^4 x} = \frac{\cos^2 x + 2 \sin x + 2 \sin^2 x}{\cos^3 x}$$

Using $\cos^2 x = 1 - \sin^2 x$:

$$\frac{d^2y}{dx^2} = \frac{1 - \sin^2 x + 2 \sin x + 2 \sin^2 x}{\cos^3 x} = \frac{1 + 2 \sin x + \sin^2 x}{\cos^3 x} = \frac{(1 + \sin x)^2}{\cos^3 x}$$

Expressing $\cos^3 x$ as $\cos x \cdot \cos^2 x = \cos x(1 - \sin^2 x) = \cos x(1 - \sin x)(1 + \sin x)$:

$$\frac{d^2y}{dx^2} = \frac{(1 + \sin x)^2}{\cos x(1 - \sin x)(1 + \sin x)} = \frac{1 + \sin x}{\cos x(1 - \sin x)}$$

Multiplying numerator and denominator by $(1 - \sin x)$:

$$\frac{d^2y}{dx^2} = \frac{(1 + \sin x)(1 - \sin x)}{\cos x(1 - \sin x)^2} = \frac{1 - \sin^2 x}{\cos x(1 - \sin x)^2} = \frac{\cos^2 x}{\cos x(1 - \sin x)^2} = \frac{\cos x}{(1 - \sin x)^2}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Algebraic simplification errors in trigonometric identities.

Question 8(i)

If $y = \sec x - \tan x$, prove that $\cos x \frac{d^2y}{dx^2} = y^2$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\cos x \frac{d^2y}{dx^2} = y^2$ given $y = \sec x - \tan x$

Concept / Formula Used: $\frac{dy}{dx} = \sec x \tan x - \sec^2 x$

Solution:

Given $y = \sec x - \tan x$. First derivative:

$$\frac{dy}{dx} = \sec x \tan x - \sec^2 x = \sec x(\tan x - \sec x) = -\sec x(\sec x - \tan x) = -y \sec x$$

Second derivative:

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx}(-y \sec x) = - \left[y \cdot \frac{d}{dx}(\sec x) + \sec x \cdot \frac{dy}{dx} \right] \\ &= - [y \sec x \tan x + \sec x(-y \sec x)] = -y \sec x \tan x + y \sec^2 x \\ &= y \sec x(\sec x - \tan x) = y \sec x(y) = y^2 \sec x \end{aligned}$$

Multiplying both sides by $\cos x$:

$$\cos x \frac{d^2y}{dx^2} = \cos x(y^2 \sec x) = y^2(\cos x \sec x) = y^2(1) = y^2$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Not substituting y back into the derivative expression to simplify.

Question 8(ii)

If $y = x + \cot x$, prove that $\sin^2 x \frac{d^2 y}{dx^2} - 2y + 2x = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\sin^2 x \frac{d^2 y}{dx^2} - 2y + 2x = 0$ given $y = x + \cot x$

Concept / Formula Used: $\frac{dy}{dx} = 1 - \csc^2 x$ $\frac{d^2 y}{dx^2} = 2 \csc^2 x \cot x$

Solution:

Given $y = x + \cot x$. First derivative:

$$\frac{dy}{dx} = 1 - \csc^2 x$$

Second derivative:

$$\frac{d^2 y}{dx^2} = 0 - 2 \csc x (-\csc x \cot x) = 2 \csc^2 x \cot x$$

Substitute $\frac{d^2 y}{dx^2}$ and y into the LHS:

$$\text{LHS} = \sin^2 x (2 \csc^2 x \cot x) - 2(x + \cot x) + 2x$$

Since $\sin^2 x \cdot \csc^2 x = 1$:

$$\text{LHS} = 2(1) \cot x - 2x - 2 \cot x + 2x = 0 = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Sign errors while evaluating the derivative of $\cot x$.

Question 9(i)

If $y = ae^{mx} + be^{-mx}$, prove that $y_2 - m^2 y = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $y_2 - m^2 y = 0$ given $y = ae^{mx} + be^{-mx}$

Concept / Formula Used: $\frac{d}{dx}(e^{mx}) = me^{mx}$ $\frac{d}{dx}(e^{-mx}) = -me^{-mx}$

Solution:

Given $y = ae^{mx} + be^{-mx}$. First derivative (y_1):

$$y_1 = a(me^{mx}) + b(-me^{-mx}) = m(ae^{mx} - be^{-mx})$$

Second derivative (y_2):

$$y_2 = m(a(me^{mx}) - b(-me^{-mx})) = m^2(ae^{mx} + be^{-mx})$$

Since $y = ae^{mx} + be^{-mx}$, we have:

$$y_2 = m^2 y \implies y_2 - m^2 y = 0$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Forgetting the negative sign from the exponent in e^{-mx} during differentiation.

Question 9(ii)

If $y = ae^{2x} + be^{-x}$, prove that $\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 2y = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 2y = 0$ given $y = ae^{2x} + be^{-x}$

Concept / Formula Used: $\frac{d}{dx}(e^{2x}) = 2e^{2x}$ $\frac{d}{dx}(e^{-x}) = -e^{-x}$

Solution:

Given $y = ae^{2x} + be^{-x}$. First derivative:

$$\frac{dy}{dx} = 2ae^{2x} - be^{-x}$$

Second derivative:

$$\frac{d^2 y}{dx^2} = 4ae^{2x} + be^{-x}$$

Substitute y , $\frac{dy}{dx}$, and $\frac{d^2 y}{dx^2}$ into the LHS:

$$\text{LHS} = (4ae^{2x} + be^{-x}) - (2ae^{2x} - be^{-x}) - 2(ae^{2x} + be^{-x})$$

Expanding and grouping terms:

$$\begin{aligned} \text{LHS} &= 4ae^{2x} + be^{-x} - 2ae^{2x} + be^{-x} - 2ae^{2x} - 2be^{-x} \\ &= (4a - 2a - 2a)e^{2x} + (1 + 1 - 2)be^{-x} = 0 \cdot ae^{2x} + 0 \cdot be^{-x} = 0 = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Algebraic errors when combining coefficients of e^{2x} and e^{-x} .

Question 10(i)

If $y = A \cos nx + B \sin nx$, show that $\frac{d^2 y}{dx^2} + n^2 y = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $\frac{d^2y}{dx^2} + n^2y = 0$ given $y = A \cos nx + B \sin nx$

Concept / Formula Used: $\frac{d}{dx}(\cos nx) = -n \sin nx$ $\frac{d}{dx}(\sin nx) = n \cos nx$

Solution:

Given $y = A \cos nx + B \sin nx$. First derivative:

$$\frac{dy}{dx} = -nA \sin nx + nB \cos nx$$

Second derivative:

$$\frac{d^2y}{dx^2} = -n^2A \cos nx - n^2B \sin nx = -n^2(A \cos nx + B \sin nx)$$

Since $y = A \cos nx + B \sin nx$, we have:

$$\frac{d^2y}{dx^2} = -n^2y \implies \frac{d^2y}{dx^2} + n^2y = 0$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Missing the factor of n^2 produced by the chain rule upon second differentiation.

Question 10(ii)

If $y = A \sin 2x + B \cos 2x$ and $\frac{d^2y}{dx^2} - ky = 0$, find the value of k .

TYPE 69 Parameter Finding via Second Derivatives

Given: $y = A \sin 2x + B \cos 2x$ and $\frac{d^2y}{dx^2} - ky = 0$

Concept / Formula Used: $\frac{d}{dx}(\sin 2x) = 2 \cos 2x$ $\frac{d}{dx}(\cos 2x) = -2 \sin 2x$

Solution:

Given $y = A \sin 2x + B \cos 2x$. First derivative:

$$\frac{dy}{dx} = 2A \cos 2x - 2B \sin 2x$$

Second derivative:

$$\frac{d^2y}{dx^2} = -4A \sin 2x - 4B \cos 2x = -4(A \sin 2x + B \cos 2x) = -4y$$

Substitute $\frac{d^2y}{dx^2} = -4y$ into the given equation $\frac{d^2y}{dx^2} - ky = 0$:

$$-4y - ky = 0 \implies (-4 - k)y = 0$$

Since this holds for all x , the coefficient must vanish:

$$-4 - k = 0 \implies k = -4$$

Final Answer

$$k = -4$$

Common Mistake: Incorrectly equating coefficients or missing the negative sign in the second derivative.

Question 11(i)

If $y = \tan^{-1} x$, prove that $(1 + x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $(1 + x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 0$ given $y = \tan^{-1} x$

Concept / Formula Used: $\frac{dy}{dx} = \frac{1}{1 + x^2}$

Solution:

Given $y = \tan^{-1} x$. First derivative:

$$\frac{dy}{dx} = \frac{1}{1 + x^2} \implies (1 + x^2) \frac{dy}{dx} = 1$$

Differentiating both sides with respect to x using the product rule:

$$(1 + x^2) \cdot \frac{d}{dx} \left(\frac{dy}{dx} \right) + \frac{dy}{dx} \cdot \frac{d}{dx} (1 + x^2) = \frac{d}{dx} (1)$$

$$(1 + x^2) \frac{d^2 y}{dx^2} + \frac{dy}{dx} (2x) = 0$$

$$(1 + x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 0$$

Hence Proved

$LHS = RHS$, hence proved.

Common Mistake: Attempting to differentiate the second order derivative directly without cross-multiplying $1 + x^2$.

Question 11(ii)

If $y = \sin^{-1} x$, prove that $(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$ given $y = \sin^{-1} x$

Concept / Formula Used: $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

Solution:

Given $y = \sin^{-1} x$. First derivative:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} \implies \sqrt{1-x^2} \frac{dy}{dx} = 1$$

Squaring both sides:

$$(1-x^2) \left(\frac{dy}{dx} \right)^2 = 1$$

Differentiating both sides with respect to x using the product rule and chain rule:

$$(1-x^2) \cdot 2 \left(\frac{dy}{dx} \right) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \cdot (-2x) = 0$$

Dividing both sides by $2 \frac{dy}{dx}$ (assuming $\frac{dy}{dx} \neq 0$):

$$(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 0$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Forgetting to divide by $2 \frac{dy}{dx}$ correctly after differentiation.

Question 12(i)

If $y = \sin(\log x)$, prove that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$ given $y = \sin(\log x)$

Concept / Formula Used: $\frac{dy}{dx} = \frac{\cos(\log x)}{x}$

Solution:

Given $y = \sin(\log x)$. First derivative:

$$\frac{dy}{dx} = \cos(\log x) \cdot \frac{1}{x} \implies x \frac{dy}{dx} = \cos(\log x)$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned} x \cdot \frac{d}{dx} \left(\frac{dy}{dx} \right) + \frac{dy}{dx} \cdot \frac{d}{dx}(x) &= \frac{d}{dx}(\cos(\log x)) \\ x \frac{d^2y}{dx^2} + \frac{dy}{dx}(1) &= -\sin(\log x) \cdot \frac{1}{x} \end{aligned}$$

Multiply the entire equation by x :

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -\sin(\log x)$$

Since $y = \sin(\log x)$, substitute y on the RHS:

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -y \implies x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Omitting the multiplication by x after the second differentiation step.

Question 12(ii)

If $y = 3 \cos(\log x) + 4 \sin(\log x)$, prove that $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$ given $y = 3 \cos(\log x) + 4 \sin(\log x)$

Concept / Formula Used: $\frac{dy}{dx} = \frac{-3 \sin(\log x) + 4 \cos(\log x)}{x}$

Solution:

Given $y = 3 \cos(\log x) + 4 \sin(\log x)$. First derivative:

$$\frac{dy}{dx} = \frac{-3 \sin(\log x) + 4 \cos(\log x)}{x} \implies x \frac{dy}{dx} = -3 \sin(\log x) + 4 \cos(\log x)$$

Differentiating both sides with respect to x using the product rule:

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx}(1) = -3 \cos(\log x) \cdot \frac{1}{x} - 4 \sin(\log x) \cdot \frac{1}{x}$$

Multiply the entire equation by x :

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -3 \cos(\log x) - 4 \sin(\log x) = -(3 \cos(\log x) + 4 \sin(\log x))$$

Since $y = 3 \cos(\log x) + 4 \sin(\log x)$, we get:

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -y \implies x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Sign errors while differentiating logarithmic trigonometric terms.

Question 13

If $y = x \cos x$, prove that $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + (2 + x^2)y = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + (2 + x^2)y = 0$ given $y = x \cos x$

Concept / Formula Used: $\frac{dy}{dx} = \cos x - x \sin x$ $\frac{d^2y}{dx^2} = -2 \sin x - x \cos x$

Solution:

Given $y = x \cos x$. First derivative:

$$\frac{dy}{dx} = x(-\sin x) + \cos x(1) = \cos x - x \sin x$$

Second derivative:

$$\frac{d^2y}{dx^2} = -\sin x - (x \cos x + \sin x) = -2 \sin x - x \cos x$$

Substitute y , $\frac{dy}{dx}$, and $\frac{d^2y}{dx^2}$ into the LHS:

$$\text{LHS} = x^2(-2 \sin x - x \cos x) - 2x(\cos x - x \sin x) + (2 + x^2)(x \cos x)$$

Expanding all terms:

$$\text{LHS} = -2x^2 \sin x - x^3 \cos x - 2x \cos x + 2x^2 \sin x + 2x \cos x + x^3 \cos x$$

Grouping and canceling opposite terms:

$$\text{LHS} = (-2x^2 \sin x + 2x^2 \sin x) + (-x^3 \cos x + x^3 \cos x) + (-2x \cos x + 2x \cos x) = 0 = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Algebraic expansion errors when multiplying out terms with x^2 and x .

Question 14(i)

If $y = (\cos^{-1} x)^2$, prove that $(1 - x^2)y_2 - xy_1 = 2$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $(1 - x^2)y_2 - xy_1 = 2$ given $y = (\cos^{-1} x)^2$

Concept / Formula Used: $\frac{dy}{dx} = 2(\cos^{-1} x) \left(-\frac{1}{\sqrt{1-x^2}} \right)$

Solution:

Given $y = (\cos^{-1} x)^2$. First derivative (y_1):

$$y_1 = 2(\cos^{-1} x) \cdot \left(-\frac{1}{\sqrt{1-x^2}} \right)$$

$$\sqrt{1-x^2} y_1 = -2 \cos^{-1} x$$

Squaring both sides:

$$(1-x^2)y_1^2 = 4(\cos^{-1} x)^2 = 4y$$

Differentiating both sides with respect to x using the product rule:

$$(1-x^2) \cdot 2y_1 y_2 + y_1^2 \cdot (-2x) = 4y_1$$

Dividing both sides by $2y_1$ (assuming $y_1 \neq 0$):

$$(1-x^2)y_2 - xy_1 = 2$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Forgetting to square both sides to eliminate the square root before differentiating.

Question 14(ii)

If $y = (\tan^{-1} x)^2$, prove that $(x^2 + 1)^2 \frac{d^2 y}{dx^2} + 2x(x^2 + 1) \frac{dy}{dx} = 2$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $(x^2 + 1)^2 \frac{d^2 y}{dx^2} + 2x(x^2 + 1) \frac{dy}{dx} = 2$ given $y = (\tan^{-1} x)^2$

Concept / Formula Used: $\frac{dy}{dx} = \frac{2 \tan^{-1} x}{1+x^2}$

Solution:

Given $y = (\tan^{-1} x)^2$. First derivative:

$$\frac{dy}{dx} = \frac{2 \tan^{-1} x}{1+x^2} \implies (1+x^2) \frac{dy}{dx} = 2 \tan^{-1} x$$

Differentiating both sides with respect to x using the product rule:

$$(1+x^2) \cdot \frac{d^2 y}{dx^2} + \frac{dy}{dx} \cdot (2x) = 2 \cdot \frac{1}{1+x^2}$$

Multiply the entire equation by $(1+x^2)$:

$$(1+x^2)^2 \frac{d^2 y}{dx^2} + 2x(1+x^2) \frac{dy}{dx} = 2$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Failing to multiply through by $(1+x^2)$ at the final step to match the target equation.

Question 14(iii)

If $y = (\sec^{-1} x)^2$, $x > 1$, show that $x^2(x^2 - 1)\frac{d^2y}{dx^2} + (2x^3 - x)\frac{dy}{dx} - 2 = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $x^2(x^2 - 1)\frac{d^2y}{dx^2} + (2x^3 - x)\frac{dy}{dx} - 2 = 0$ given $y = (\sec^{-1} x)^2$

Concept / Formula Used: $\frac{dy}{dx} = \frac{2 \sec^{-1} x}{x\sqrt{x^2 - 1}}$

Solution:

Given $y = (\sec^{-1} x)^2$. First derivative:

$$\frac{dy}{dx} = \frac{2 \sec^{-1} x}{x\sqrt{x^2 - 1}} \implies x\sqrt{x^2 - 1}\frac{dy}{dx} = 2 \sec^{-1} x$$

Squaring both sides:

$$x^2(x^2 - 1) \left(\frac{dy}{dx} \right)^2 = 4(\sec^{-1} x)^2 = 4y$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned} \frac{d}{dx} [x^2(x^2 - 1)] \left(\frac{dy}{dx} \right)^2 + x^2(x^2 - 1) \cdot 2 \left(\frac{dy}{dx} \right) \frac{d^2y}{dx^2} &= 4 \frac{dy}{dx} \\ \frac{d}{dx} (x^4 - x^2) &= 4x^3 - 2x = 2x(2x^2 - 1) \end{aligned}$$

Substituting this back and dividing by $2\frac{dy}{dx}$:

$$2x(2x^2 - 1)\frac{dy}{dx} + 2x^2(x^2 - 1)\frac{d^2y}{dx^2} = 4$$

Dividing the entire equation by 2:

$$x^2(x^2 - 1)\frac{d^2y}{dx^2} + (2x^3 - x)\frac{dy}{dx} - 2 = 0$$

Hence Proved

$LHS = RHS$, hence proved.

Common Mistake: Algebraic errors when expanding and differentiating $x^2(x^2 - 1)$.

Question 15(i)

If $y = \log(x + \sqrt{x^2 + 1})$, prove that $(x^2 + 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $(x^2 + 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$ given $y = \log(x + \sqrt{x^2 + 1})$

Concept / Formula Used: $\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2 + 1}} \cdot \left(1 + \frac{x}{\sqrt{x^2 + 1}}\right)$

Solution:

Given $y = \log(x + \sqrt{x^2 + 1})$. First derivative:

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2 + 1}} \cdot \left(1 + \frac{2x}{2\sqrt{x^2 + 1}}\right)$$

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2 + 1}} \cdot \left(\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}}\right) = \frac{1}{\sqrt{x^2 + 1}}$$

Cross-multiplying:

$$\sqrt{x^2 + 1} \frac{dy}{dx} = 1$$

Differentiating both sides with respect to x (product rule on the left):

$$\sqrt{x^2 + 1} \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \left(\frac{x}{\sqrt{x^2 + 1}}\right) = 0$$

Multiplying through by $\sqrt{x^2 + 1}$:

$$(x^2 + 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = 0$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Failing to simplify $\frac{dy}{dx}$ to $\frac{1}{\sqrt{x^2 + 1}}$ before differentiating the second time.

Question 15(ii)

If $y = \log(x + \sqrt{x^2 + a^2})$, prove that $(x^2 + a^2)y_2 + xy_1 = 0$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $(x^2 + a^2)y_2 + xy_1 = 0$ given $y = \log(x + \sqrt{x^2 + a^2})$

Concept / Formula Used: $y_1 = \frac{1}{\sqrt{x^2 + a^2}}$

Solution:

Given $y = \log(x + \sqrt{x^2 + a^2})$. First derivative (y_1):

$$y_1 = \frac{1}{x + \sqrt{x^2 + a^2}} \cdot \left(1 + \frac{x}{\sqrt{x^2 + a^2}}\right) = \frac{1}{\sqrt{x^2 + a^2}}$$

Cross-multiplying:

$$\sqrt{x^2 + a^2} y_1 = 1$$

Differentiating both sides with respect to x using the product rule:

$$\sqrt{x^2 + a^2} \cdot y_2 + y_1 \cdot \left(\frac{x}{\sqrt{x^2 + a^2}} \right) = 0$$

Multiply the entire equation by $\sqrt{x^2 + a^2}$:

$$(x^2 + a^2)y_2 + xy_1 = 0$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Omitting the multiplication by $\sqrt{x^2 + a^2}$ to clear the denominator.

Question 15(iii)

If $y = (\log x)^2$, prove that $x^2y'' + xy' = 2$.

TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function

To Prove: $x^2y'' + xy' = 2$ given $y = (\log x)^2$

Concept / Formula Used: $y' = \frac{2 \log x}{x}$

Solution:

Given $y = (\log x)^2$. First derivative (y'):

$$y' = 2(\log x) \cdot \frac{1}{x} = \frac{2 \log x}{x}$$

Cross-multiplying by x :

$$xy' = 2 \log x$$

Differentiating both sides with respect to x using the product rule:

$$x \cdot y'' + y' \cdot (1) = 2 \cdot \frac{1}{x}$$

Multiply the entire equation by x :

$$x^2y'' + xy' = 2$$

Hence Proved

LHS = RHS, hence proved.

Common Mistake: Forgetting to multiply the final equation by x to clear the denominator.

Question 16

If $y = \cos(\sin x)$, prove that $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y \cos^2 x = 0$.

TYPE 68b Proving Second-Order Differential Equation Identities – Composite Trigonometric Function

★ **Likely Exam Question**

To Prove: (1) $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y \cos^2 x = 0$

Concept / Formula Used: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Solution:

Given equation:

$$y = \cos(\sin x)$$

Differentiating both sides with respect to x using the chain rule:

$$\frac{dy}{dx} = -\sin(\sin x) \cdot \frac{d}{dx}(\sin x)$$

$$\frac{dy}{dx} = -\sin(\sin x) \cdot \cos x$$

Differentiating again with respect to x using the product rule:

$$\begin{aligned} \frac{d^2y}{dx^2} &= - \left[\sin(\sin x) \cdot \frac{d}{dx}(\cos x) + \cos x \cdot \frac{d}{dx}(\sin(\sin x)) \right] \\ &= - [\sin(\sin x) \cdot (-\sin x) + \cos x \cdot \cos(\sin x) \cdot \cos x] \\ &= \sin x \sin(\sin x) - \cos^2 x \cos(\sin x) \end{aligned}$$

Since $y = \cos(\sin x)$, the second term contains y :

$$\frac{d^2y}{dx^2} = \sin x \sin(\sin x) - y \cos^2 x$$

From the first derivative expression, we have $\sin(\sin x) = -\frac{1}{\cos x} \frac{dy}{dx}$. Substituting this:

$$\begin{aligned} \frac{d^2y}{dx^2} &= \sin x \left(-\frac{1}{\cos x} \frac{dy}{dx} \right) - y \cos^2 x \\ &= -\frac{\sin x}{\cos x} \frac{dy}{dx} - y \cos^2 x \\ &= -\tan x \frac{dy}{dx} - y \cos^2 x \end{aligned}$$

Rearranging the terms:

$$\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y \cos^2 x = 0$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (i)

If $y = e^{\tan^{-1} x}$, prove that $(1 + x^2) \frac{d^2y}{dx^2} + (2x - 1) \frac{dy}{dx} = 0$.

TYPE 68c Proving Second-Order Differential Equation Identities – Exponential of Inverse Trig Function

★ Likely Exam Question

To Prove: $(1 + x^2) \frac{d^2y}{dx^2} + (2x - 1) \frac{dy}{dx} = 0$

Concept / Formula Used: $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2}$

Solution:

Given equation:

$$y = e^{\tan^{-1} x}$$

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = e^{\tan^{-1} x} \cdot \frac{d}{dx}(\tan^{-1} x)$$

$$\frac{dy}{dx} = e^{\tan^{-1} x} \cdot \frac{1}{1 + x^2}$$

Since $y = e^{\tan^{-1} x}$, we can write:

$$\frac{dy}{dx} = \frac{y}{1 + x^2}$$

Cross-multiplying:

$$(1 + x^2) \frac{dy}{dx} = y$$

Differentiating both sides again with respect to x using the product rule on the left side:

$$(1 + x^2) \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \frac{d}{dx}(1 + x^2) = \frac{dy}{dx}$$

$$(1 + x^2) \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot (2x) = \frac{dy}{dx}$$

$$(1 + x^2) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - \frac{dy}{dx} = 0$$

$$(1 + x^2) \frac{d^2y}{dx^2} + (2x - 1) \frac{dy}{dx} = 0$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (ii)

If $x = \tan\left(\frac{1}{a} \log y\right)$, then prove that $(1 + x^2) \frac{d^2y}{dx^2} + (2x - a) \frac{dy}{dx} = 0$.

TYPE 68d Proving Second-Order Differential Equation Identities – Implicit Logarithmic Substitution

★ Likely Exam Question

To Prove: $(1 + x^2) \frac{d^2y}{dx^2} + (2x - a) \frac{dy}{dx} = 0$

Solution:

Given equation:

$$x = \tan \left(\frac{1}{a} \log y \right)$$

Taking $\tan^{-1}$ on both sides:

$$\tan^{-1} x = \frac{1}{a} \log y$$

$$\log y = a \tan^{-1} x$$

Differentiating both sides with respect to x :

$$\frac{1}{y} \frac{dy}{dx} = a \cdot \frac{1}{1+x^2}$$

$$(1+x^2) \frac{1}{y} \frac{dy}{dx} = a$$

$$(1+x^2) \frac{dy}{dx} = ay$$

Differentiating both sides again with respect to x using the product rule:

$$(1+x^2) \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \frac{d}{dx}(1+x^2) = a \frac{dy}{dx}$$

$$(1+x^2) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = a \frac{dy}{dx}$$

$$(1+x^2) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - a \frac{dy}{dx} = 0$$

$$(1+x^2) \frac{d^2y}{dx^2} + (2x-a) \frac{dy}{dx} = 0$$

Hence Proved

LHS = RHS, hence proved.

Question 17 (iii)

If $y = e^{a \cos^{-1} x}$, $-1 < x < 1$, prove that $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$.

TYPE 68e Proving Second-Order Differential Equation Identities – Inverse Trig Exponential Function

★ Likely Exam Question

To Prove: $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$

Concept / Formula Used: $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$

Solution:

Given equation:

$$y = e^{a \cos^{-1} x}$$

Differentiating both sides with respect to x :

$$\frac{dy}{dx} = e^{a \cos^{-1} x} \cdot \frac{d}{dx}(a \cos^{-1} x)$$

$$\frac{dy}{dx} = e^{a \cos^{-1} x} \cdot \left(-\frac{a}{\sqrt{1-x^2}} \right)$$

$$\frac{dy}{dx} = -\frac{ay}{\sqrt{1-x^2}}$$

Squaring and cross-multiplying:

$$(1-x^2) \left(\frac{dy}{dx} \right)^2 = a^2 y^2$$

Differentiating both sides with respect to x :

$$(1-x^2) \cdot 2 \left(\frac{dy}{dx} \right) \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \cdot (-2x) = a^2 \cdot 2y \frac{dy}{dx}$$

Dividing the entire equation by $2 \frac{dy}{dx}$ (assuming $\frac{dy}{dx} \neq 0$):

$$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = a^2 y$$

$$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$$

Hence Proved

LHS = RHS, hence proved.

Question 18

If $y = \log \left(\frac{x}{a+bx} \right)^x$, prove that $x^3 \frac{d^2 y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2$.

TYPE 68f Proving Second-Order Differential Equation Identities – Logarithmic Power-Rule Identity

★ Likely Exam Question

To Prove: $x^3 \frac{d^2 y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2$

Concept / Formula Used: $\log(A^B) = B \log A$ $\log \left(\frac{A}{B} \right) = \log A - \log B$

Solution:

Given equation:

$$y = \log \left(\frac{x}{a+bx} \right)^x$$

Using logarithm properties to simplify the function:

$$y = x [\log x - \log(a+bx)]$$

Differentiating both sides with respect to x using the product rule:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \frac{d}{dx} [\log x - \log(a + bx)] + [\log x - \log(a + bx)] \cdot \frac{d}{dx}(x) \\ &= x \left(\frac{1}{x} - \frac{b}{a + bx} \right) + \log x - \log(a + bx) \\ &= 1 - \frac{bx}{a + bx} + \log \left(\frac{x}{a + bx} \right)\end{aligned}$$

Since $y = x \log \left(\frac{x}{a + bx} \right)$, the last term can be replaced by $\frac{y}{x}$:

$$\frac{dy}{dx} = \frac{a + bx - bx}{a + bx} + \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{a}{a + bx} + \frac{y}{x}$$

Rearranging to isolate the term without y :

$$x \frac{dy}{dx} - y = \frac{ax}{a + bx}$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{d}{dx} \left(x \frac{dy}{dx} - y \right) &= \frac{d}{dx} \left(\frac{ax}{a + bx} \right) \\ x \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 1 - \frac{dy}{dx} &= \frac{(a + bx)(a) - ax(b)}{(a + bx)^2} \\ x \frac{d^2y}{dx^2} &= \frac{a^2 + abx - abx}{(a + bx)^2} \\ x \frac{d^2y}{dx^2} &= \frac{a^2}{(a + bx)^2}\end{aligned}$$

From earlier, we had $\frac{ax}{a + bx} = x \frac{dy}{dx} - y$, which gives $\frac{a}{a + bx} = \frac{1}{x} (x \frac{dy}{dx} - y)$. Substituting this into our expression for $x \frac{d^2y}{dx^2}$:

$$\begin{aligned}x \frac{d^2y}{dx^2} &= \left[\frac{1}{x} \left(x \frac{dy}{dx} - y \right) \right]^2 \\ x \frac{d^2y}{dx^2} &= \frac{1}{x^2} \left(x \frac{dy}{dx} - y \right)^2\end{aligned}$$

Multiply both sides by x^2 :

$$x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2$$

Hence Proved

LHS = RHS, hence proved.

Question 19

If $\sqrt{x} + \sqrt{y} = \sqrt{a}$, find $\frac{d^2y}{dx^2}$ at $x = a$.

TYPE 75 Implicit differentiation with radical substitution evaluation

To Find: $\frac{d^2y}{dx^2}$ at $x = a$

Concept / Formula Used: $\frac{d}{dx}(x^n) = nx^{n-1}$

Solution:

Given equation:

$$\sqrt{x} + \sqrt{y} = \sqrt{a}$$

Differentiating implicitly with respect to x :

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$$

$$\frac{1}{2\sqrt{y}} \frac{dy}{dx} = -\frac{1}{2\sqrt{x}}$$

$$\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$$

To find the value of y when $x = a$:

$$\sqrt{a} + \sqrt{y} = \sqrt{a} \implies \sqrt{y} = 0 \implies y = 0$$

Differentiating $\frac{dy}{dx} = -\frac{y^{1/2}}{x^{1/2}}$ a second time would require substituting $y = 0$ into a term containing $\frac{1}{\sqrt{y}}$, which is undefined. Since the implicit form breaks down at this point, we instead solve the original equation explicitly for y near $x = a$.

$$\sqrt{y} = \sqrt{a} - \sqrt{x} \implies y = (\sqrt{a} - \sqrt{x})^2$$

Differentiating with respect to x :

$$\frac{dy}{dx} = 2(\sqrt{a} - \sqrt{x}) \cdot \left(-\frac{1}{2\sqrt{x}}\right) = -\frac{\sqrt{a} - \sqrt{x}}{\sqrt{x}} = 1 - \frac{\sqrt{a}}{\sqrt{x}}$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = 0 - a \cdot \left(-\frac{1}{2}x^{-3/2}\right) = \frac{a}{2x^{3/2}}$$

Evaluating at $x = a$:

$$\left. \frac{d^2y}{dx^2} \right|_{x=a} = \frac{a}{2a^{3/2}} = \frac{1}{2\sqrt{a}}$$

Final Answer

$$\frac{1}{2\sqrt{a}}$$

Question 20

If $x^m y^n = (x + y)^{m+n}$, then prove that: (i) $\frac{dy}{dx} = \frac{y}{x}$ (ii) $\frac{d^2y}{dx^2} = 0$

TYPE 68g Proving Second-Order Differential Equation Identities – Homogeneous Logarithmic Power Functions

★ **Likely Exam Question**

To Prove: (i) $\frac{dy}{dx} = \frac{y}{x}$ (ii) $\frac{d^2y}{dx^2} = 0$

Concept / Formula Used: $\log(AB) = \log A + \log B$

Solution:

Given equation:

$$x^m y^n = (x + y)^{m+n}$$

Taking natural logarithm on both sides:

$$\log(x^m y^n) = \log((x + y)^{m+n})$$

$$m \log x + n \log y = (m + n) \log(x + y)$$

Differentiating both sides implicitly with respect to x :

$$\frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{m + n}{x + y} \left(1 + \frac{dy}{dx} \right)$$

$$\frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{m + n}{x + y} + \frac{m + n}{x + y} \frac{dy}{dx}$$

Grouping the terms containing $\frac{dy}{dx}$ on one side:

$$\left(\frac{n}{y} - \frac{m + n}{x + y} \right) \frac{dy}{dx} = \frac{m + n}{x + y} - \frac{m}{x}$$

Simplifying both sides by taking common denominators:

$$\left(\frac{n(x + y) - y(m + n)}{y(x + y)} \right) \frac{dy}{dx} = \frac{x(m + n) - m(x + y)}{x(x + y)}$$

$$\left(\frac{nx + ny - my - ny}{y(x + y)} \right) \frac{dy}{dx} = \frac{mx + nx - mx - my}{x(x + y)}$$

$$\left(\frac{nx - my}{y(x + y)} \right) \frac{dy}{dx} = \frac{nx - my}{x(x + y)}$$

Multiplying both sides by $\frac{y(x+y)}{nx-my}$ (assuming $nx - my \neq 0$):

$$\frac{dy}{dx} = \frac{y}{x}$$

This proves part (i).

Now, differentiating $\frac{dy}{dx} = \frac{y}{x}$ with respect to x to find $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{x \cdot \frac{dy}{dx} - y \cdot 1}{x^2}$$

Substitute $\frac{dy}{dx} = \frac{y}{x}$:

$$\frac{d^2y}{dx^2} = \frac{x \left(\frac{y}{x} \right) - y}{x^2}$$

$$\frac{d^2y}{dx^2} = \frac{y - y}{x^2} = 0$$

This proves part (ii).

Hence Proved

(i) $\frac{dy}{dx} = \frac{y}{x}$ and (ii) $\frac{d^2y}{dx^2} = 0$, hence proved.

Question 21

If $x = a(\theta - \sin \theta)$, $y = a(1 + \cos \theta)$, find $\frac{d^2y}{dx^2}$.

TYPE 64b Parametric Second-Order Derivatives – Cycloid-Type Parameters

★ Likely Exam Question

To Find: $\frac{d^2y}{dx^2}$

Concept / Formula Used: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{d\theta} \left(\frac{dy}{dx} \right)}{dx/d\theta}$

Solution:

Given parametric equations:

$$x = a(\theta - \sin \theta)$$

$$y = a(1 + \cos \theta)$$

Differentiating x and y with respect to θ :

$$\frac{dx}{d\theta} = a(1 - \cos \theta)$$

$$\frac{dy}{d\theta} = a(0 - \sin \theta) = -a \sin \theta$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{-a \sin \theta}{a(1 - \cos \theta)} = -\frac{\sin \theta}{1 - \cos \theta}$$

Using half-angle trigonometric identities ($\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ and $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$):

$$\frac{dy}{dx} = -\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} = -\cot \frac{\theta}{2}$$

Now, finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{d\theta} \left(-\cot \frac{\theta}{2} \right)}{dx/d\theta}$$

Using the derivative of $\cot u$: $\frac{d}{d\theta} \left(-\cot \frac{\theta}{2} \right) = - \left(-\csc^2 \frac{\theta}{2} \cdot \frac{1}{2} \right) = \frac{1}{2} \csc^2 \frac{\theta}{2}$. Thus:

$$\frac{d^2y}{dx^2} = \frac{\frac{1}{2} \csc^2 \frac{\theta}{2}}{a(1 - \cos \theta)}$$

Since $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2} = \frac{2}{\csc^2(\theta/2)}$:

$$\frac{d^2y}{dx^2} = \frac{\frac{1}{2} \csc^2 \frac{\theta}{2}}{a \cdot 2 \sin^2 \frac{\theta}{2}} = \frac{1}{4a} \csc^2 \frac{\theta}{2} \cdot \csc^2 \frac{\theta}{2} = \frac{1}{4a} \csc^4 \frac{\theta}{2}$$

Final Answer

$$\frac{1}{4a} \csc^4 \frac{\theta}{2}$$

Question 22

If $x = \log t$ and $y = \frac{1}{t}$, prove that $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$.

TYPE 64c Parametric Second-Order Derivatives – Logarithmic/Reciprocal Parameters

★ Likely Exam Question

To Prove: $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Solution:

Given parametric equations:

$$x = \log t \implies \frac{dx}{dt} = \frac{1}{t}$$

$$y = \frac{1}{t} \implies \frac{dy}{dt} = -\frac{1}{t^2}$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-1/t^2}{1/t} = -\frac{1}{t}$$

Since $y = \frac{1}{t}$, we have $\frac{dy}{dx} = -y$. Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(-\frac{1}{t} \right) = \frac{d}{dt} \left(-\frac{1}{t} \right) \cdot \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \left(\frac{1}{t^2} \right) \cdot t = \frac{1}{t}$$

Substituting $\frac{d^2y}{dx^2}$ and $\frac{dy}{dx}$:

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} = \frac{1}{t} + \left(-\frac{1}{t} \right) = 0$$

Hence Proved

LHS = RHS, hence proved.

Question 23

If $x = a \left(\frac{1-t^2}{1+t^2} \right)$ and $y = \frac{2at}{1+t^2}$, prove that $\frac{dy}{dx} = -\frac{x}{y}$. Hence, prove that $\frac{d^2y}{dx^2} = -\left(\frac{x^2+y^2}{y^3} \right)$.

TYPE 64d Parametric Second-Order Derivatives – Trigonometric Substitution Proof

★ Likely Exam Question

To Prove: $\frac{dy}{dx} = -\frac{x}{y}$ and $\frac{d^2y}{dx^2} = -\left(\frac{x^2+y^2}{y^3} \right)$

Solution:

Given parametric equations:

$$x = a \left(\frac{1-t^2}{1+t^2} \right), \quad y = \frac{2at}{1+t^2}$$

Differentiating both x and y with respect to t :

$$\begin{aligned} \frac{dx}{dt} &= a \cdot \frac{(1+t^2)(-2t) - (1-t^2)(2t)}{(1+t^2)^2} \\ &= a \cdot \frac{-2t - 2t^3 - 2t + 2t^3}{(1+t^2)^2} \\ &= -\frac{4at}{(1+t^2)^2} \end{aligned}$$

$$\begin{aligned} \frac{dy}{dt} &= a \cdot \frac{(1+t^2)(2) - (2t)(2t)}{(1+t^2)^2} \\ &= a \cdot \frac{2+2t^2-4t^2}{(1+t^2)^2} \\ &= \frac{2a(1-t^2)}{(1+t^2)^2} \end{aligned}$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\frac{2a(1-t^2)}{(1+t^2)^2}}{-\frac{4at}{(1+t^2)^2}} = -\frac{2a(1-t^2)}{4at} = -\frac{1-t^2}{2t}$$

Now, consider $-\frac{x}{y}$:

$$-\frac{x}{y} = -\frac{a \left(\frac{1-t^2}{1+t^2} \right)}{\frac{2at}{1+t^2}} = -\frac{1-t^2}{2t}$$

Thus, $\frac{dy}{dx} = -\frac{x}{y}$. This proves the first part.

Now, finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(-\frac{x}{y} \right) = -\frac{y \cdot 1 - x \cdot \frac{dy}{dx}}{y^2}$$

Substitute $\frac{dy}{dx} = -\frac{x}{y}$:

$$\frac{d^2y}{dx^2} = -\frac{y - x\left(-\frac{x}{y}\right)}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{\frac{y^2+x^2}{y}}{y^2} = -\frac{x^2+y^2}{y^3}$$

Hence Proved

$\frac{dy}{dx} = -\frac{x}{y}$ and $\frac{d^2y}{dx^2} = -\left(\frac{x^2+y^2}{y^3}\right)$, hence proved.

Question 24

If $x = a \sin pt$ and $y = b \cos pt$, find the value of $\frac{d^2y}{dx^2}$ at $t = 0$.

TYPE 64e Parametric Second-Order Derivatives – Evaluation at a Point

★ Likely Exam Question

To Find: Value of $\frac{d^2y}{dx^2}$ at $t = 0$

Solution:

Given parametric equations:

$$x = a \sin pt \implies \frac{dx}{dt} = ap \cos pt$$

$$y = b \cos pt \implies \frac{dy}{dt} = -bp \sin pt$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-bp \sin pt}{ap \cos pt} = -\frac{b}{a} \tan pt$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(-\frac{b}{a} \tan pt \right) = \frac{\frac{d}{dt} \left(-\frac{b}{a} \tan pt \right)}{dx/dt}$$

$$\frac{d}{dt} \left(-\frac{b}{a} \tan pt \right) = -\frac{b}{a} \cdot p \sec^2 pt = -\frac{bp}{a} \sec^2 pt$$

Thus:

$$\frac{d^2y}{dx^2} = \frac{-\frac{bp}{a} \sec^2 pt}{ap \cos pt} = -\frac{b}{a^2 p} \cdot \frac{1}{\cos^3 pt} = -\frac{b}{a^2} \sec^3 pt$$

Evaluating at $t = 0$:

$$\left. \frac{d^2y}{dx^2} \right|_{t=0} = -\frac{b}{a^2} \sec^3(0) = -\frac{b}{a^2} (1)^3 = -\frac{b}{a^2}$$

Final Answer

$$-\frac{b}{a^2}$$

Question 25

If $x = a \sec^3 \theta$ and $y = a \tan^3 \theta$, find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{3}$.

TYPE 64f Parametric Second-Order Derivatives – Trigonometric Evaluation

★ Likely Exam Question

To Find: $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{3}$

Solution:

Given parametric equations:

$$x = a \sec^3 \theta \implies \frac{dx}{d\theta} = 3a \sec^2 \theta \cdot (\sec \theta \tan \theta) = 3a \sec^3 \theta \tan \theta$$

$$y = a \tan^3 \theta \implies \frac{dy}{d\theta} = 3a \tan^2 \theta \cdot \sec^2 \theta$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3a \tan^2 \theta \sec^2 \theta}{3a \sec^3 \theta \tan \theta} = \frac{\tan \theta}{\sec \theta} = \sin \theta$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(\sin \theta) = \frac{\frac{d}{d\theta}(\sin \theta)}{dx/d\theta} = \frac{\cos \theta}{3a \sec^3 \theta \tan \theta} = \frac{\cos \theta \cdot \cos^3 \theta}{3a \tan \theta} = \frac{\cos^4 \theta}{3a \left(\frac{\sin \theta}{\cos \theta}\right)} = \frac{\cos^5 \theta}{3a \sin \theta}$$

Evaluating at $\theta = \frac{\pi}{3}$:

$$\begin{aligned} \cos\left(\frac{\pi}{3}\right) &= \frac{1}{2}, \quad \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} \\ \frac{d^2y}{dx^2}\bigg|_{\theta=\pi/3} &= \frac{(1/2)^5}{3a \left(\frac{\sqrt{3}}{2}\right)} = \frac{1/32}{3\sqrt{3}a/2} = \frac{1}{32} \cdot \frac{2}{3\sqrt{3}a} = \frac{1}{48\sqrt{3}a} \end{aligned}$$

Final Answer

$$\frac{1}{48\sqrt{3}a}$$

Question 26

If $x = a(1 + \cos t)$, $y = a(t + \sin t)$, find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{2}$.

TYPE 64g Parametric Second-Order Derivatives – Trigonometric Sum Evaluation

★ Likely Exam Question

To Find: $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{2}$

Solution:

Given parametric equations:

$$x = a(1 + \cos t) \implies \frac{dx}{dt} = -a \sin t$$

$$y = a(t + \sin t) \implies \frac{dy}{dt} = a(1 + \cos t)$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a(1 + \cos t)}{-a \sin t} = -\frac{1 + \cos t}{\sin t}$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(-\frac{1 + \cos t}{\sin t} \right)}{dx/dt}$$

Using the quotient rule for $\frac{d}{dt} \left(-\frac{1 + \cos t}{\sin t} \right)$:

$$\begin{aligned} \frac{d}{dt} \left(-\frac{1 + \cos t}{\sin t} \right) &= -\frac{\sin t(-\sin t) - (1 + \cos t)(\cos t)}{\sin^2 t} \\ &= -\frac{-\sin^2 t - \cos t - \cos^2 t}{\sin^2 t} \\ &= -\frac{-(\sin^2 t + \cos^2 t) - \cos t}{\sin^2 t} \\ &= -\frac{-1 - \cos t}{\sin^2 t} = \frac{1 + \cos t}{\sin^2 t} \end{aligned}$$

Thus:

$$\frac{d^2y}{dx^2} = \frac{\frac{1 + \cos t}{\sin^2 t}}{-a \sin t} = -\frac{1 + \cos t}{a \sin^3 t}$$

Evaluating at $t = \frac{\pi}{2}$:

$$\cos \left(\frac{\pi}{2} \right) = 0, \quad \sin \left(\frac{\pi}{2} \right) = 1$$

$$\left. \frac{d^2y}{dx^2} \right|_{t=\pi/2} = -\frac{1 + 0}{a(1)^3} = -\frac{1}{a}$$

Final Answer

$$-\frac{1}{a}$$

Question 27

If $x = \cos t + \log \left(\tan \frac{t}{2} \right)$, $y = \sin t$, then find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

TYPE 64h Parametric Second-Order Derivatives – Complex Logarithmic-Trigonometric Evaluation

★ Likely Exam Question

To Find: $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$

Solution:

Given parametric equations:

$$x = \cos t + \log \left(\tan \frac{t}{2} \right)$$

Differentiating x with respect to t :

$$\begin{aligned} \frac{dx}{dt} &= -\sin t + \frac{1}{\tan \frac{t}{2}} \cdot \sec^2 \frac{t}{2} \cdot \frac{1}{2} \\ &= -\sin t + \frac{\cos \frac{t}{2}}{\sin \frac{t}{2}} \cdot \frac{1}{\cos^2 \frac{t}{2}} \cdot \frac{1}{2} \\ &= -\sin t + \frac{1}{2 \sin \frac{t}{2} \cos \frac{t}{2}} \\ &= -\sin t + \frac{1}{\sin t} = \frac{1 - \sin^2 t}{\sin t} = \frac{\cos^2 t}{\sin t} \end{aligned}$$

Differentiating $y = \sin t$ with respect to t :

$$\frac{dy}{dt} = \cos t$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{\frac{\cos^2 t}{\sin t}} = \frac{\sin t}{\cos t} = \tan t$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(\tan t)}{dx/dt} = \frac{\sec^2 t}{\frac{\cos^2 t}{\sin t}} = \frac{\frac{1}{\cos^2 t}}{\frac{\cos^2 t}{\sin t}} = \frac{\sin t}{\cos^4 t} = \tan t \sec^3 t$$

Evaluating at $t = \frac{\pi}{4}$:

$$\tan \left(\frac{\pi}{4} \right) = 1, \quad \sec \left(\frac{\pi}{4} \right) = \sqrt{2}$$

$$\left. \frac{d^2y}{dx^2} \right|_{t=\pi/4} = (1)(\sqrt{2})^3 = 2\sqrt{2}$$

Final Answer

General result: $\frac{d^2y}{dx^2} = \tan t \sec^3 t$

At $t = \frac{\pi}{4}$: $\frac{d^2y}{dx^2} = 2\sqrt{2}$

ISC Class XII Mathematics

Chapter 6 Applications of Derivatives

168 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

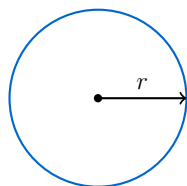
- I. 6.1 – Rate of Change of Quantities
- II. 6.2 – Tangents and Normals
- III. 6.3 – Solving Inequalities via Sign-Scheme Method
- IV. 6.4 – Increasing and Decreasing Functions
- V. 6.5 – Maxima and Minima Without Derivative Tests
- VI. 6.6 – Maxima and Minima Using Derivative Tests
- VII. 6.7 – Applications of Maxima and Minima

6.1 – Rate of Change of Quantities

Question 1

Find the rate of change of the area of a circle with respect to its radius when the radius is 6 cm.

TYPE 1a Rate of Change of Standard Geometric Quantities – Rate of Change of Circle Area with respect to Radius



Circle of radius r

Given: $r = 6$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let A be the area and r be the radius of the circle. The formula for the area of a circle is:

$$A = \pi r^2$$

Differentiating A with respect to radius r :

$$\begin{aligned}\frac{dA}{dr} &= \frac{d}{dr}(\pi r^2) \\ &= \pi \frac{d}{dr}(r^2) \\ &= 2\pi r\end{aligned}$$

Using the standard derivative formula $\frac{d}{dr}(r^n) = nr^{n-1}$ with $n = 2$.

Now, substitute the given value $r = 6$ cm:

$$\begin{aligned}\left. \frac{dA}{dr} \right|_{r=6} &= 2\pi(6) \\ &= 12\pi \text{ cm}^2/\text{cm}\end{aligned}$$

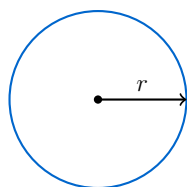
Final Answer

$12\pi \text{ cm}^2/\text{cm}$

Question 2

If the radius of a circle is increasing at the rate of 0.5 cm/sec, at what rate is its circumference increasing?

TYPE 1b Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Circumference from Radius R



Circle with radius r changing with time

Given: $\frac{dr}{dt} = 0.5$ cm/sec

Concept / Formula Used: $C = 2\pi r$

Solution:

Let C be the circumference and r be the radius of the circle at any time t . The circumference of a circle is given by:

$$C = 2\pi r$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dC}{dt} &= \frac{d}{dt}(2\pi r) \\ &= 2\pi \frac{dr}{dt}\end{aligned}$$

Given that $\frac{dr}{dt} = 0.5$ cm/sec, substituting this value:

$$\begin{aligned}\frac{dC}{dt} &= 2\pi \times 0.5 \\ &= \pi \text{ cm/sec}\end{aligned}$$

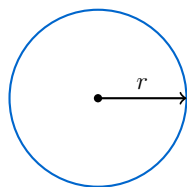
Final Answer

π cm/sec

Question 3

If the radius of a circle is increasing at the rate of 3 cm/sec, at what rate is its area increasing when its radius is 2 cm?

TYPE 1c Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Area from Radius Rate



Expanding circle with radius r

Given: $\frac{dr}{dt} = 3$ cm/sec, $r = 2$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let A be the area and r be the radius of the circle at time t . The area of a circle is given by:

$$A = \pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dA}{dt} &= \frac{d}{dt}(\pi r^2) \\ &= \pi \cdot 2r \frac{dr}{dt} \\ &= 2\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 2$ cm and $\frac{dr}{dt} = 3$ cm/sec:

$$\begin{aligned}\frac{dA}{dt} &= 2\pi(2)(3) \\ &= 12\pi \text{ cm}^2/\text{sec}\end{aligned}$$

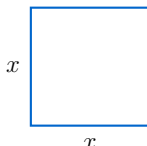
Final Answer

$12\pi \text{ cm}^2/\text{sec}$

Question 4

If the sides of a square are decreasing at the rate of 1.5 cm/sec, at what rate is its perimeter decreasing?

TYPE 1d Rate of Change of Standard Geometric Quantities – Time Rate of Change of Square Perimeter from Side Rate



Square of side x

Given: $\frac{dx}{dt} = -1.5$ cm/sec

Concept / Formula Used: $P = 4x$

Solution:

Let x be the length of a side of the square and P be its perimeter at time t . The perimeter of a square is given by:

$$P = 4x$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dP}{dt} &= \frac{d}{dt}(4x) \\ &= 4 \frac{dx}{dt}\end{aligned}$$

Given that the side is decreasing at 1.5 cm/sec, we have $\frac{dx}{dt} = -1.5$ cm/sec. Substituting this value:

$$\begin{aligned}\frac{dP}{dt} &= 4(-1.5) \\ &= -6 \text{ cm/sec}\end{aligned}$$

The negative sign indicates a decrease in perimeter. Thus, the rate at which the perimeter is decreasing is 6 cm/sec.

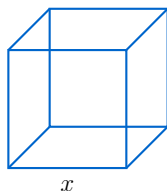
Final Answer

6 cm/sec

Question 5

If an edge of a variable cube is increasing at the rate of 10 cm/sec, at what rate is its volume increasing when its edge is 5 cm?

TYPE 1e Rate of Change of Standard Geometric Quantities – Time Rate of Change of Cube Volume from Edge Rate



Cube with edge x

Given: $\frac{dx}{dt} = 10$ cm/sec, $x = 5$ cm

Concept / Formula Used: $V = x^3$

Solution:

Let x be the side edge length and V be the volume of the cube at time t . The volume of a cube is given by:

$$V = x^3$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt}(x^3) \\ &= 3x^2 \frac{dx}{dt}\end{aligned}$$

Substitute $x = 5$ cm and $\frac{dx}{dt} = 10$ cm/sec:

$$\begin{aligned}\frac{dV}{dt} &= 3(5)^2(10) \\ &= 3(25)(10) \\ &= 750 \text{ cm}^3/\text{sec}\end{aligned}$$

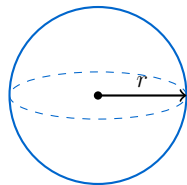
Final Answer

750 cm³/sec

Question 6

A balloon which always remains spherical has a variable radius. Find the rate at which its volume is increasing with respect to radius when the radius is 10 cm.

TYPE 1f Rate of Change of Standard Geometric Quantities – Rate of Change of Sphere Volume with respect to Radius



Spherical balloon of radius r

Given: $r = 10$ cm

Concept / Formula Used: $V = \frac{4}{3}\pi r^3$

Solution:

Let V be the volume and r be the radius of the spherical balloon. The volume of a sphere is given by:

$$V = \frac{4}{3}\pi r^3$$

Differentiating V with respect to radius r :

$$\begin{aligned}\frac{dV}{dr} &= \frac{d}{dr} \left(\frac{4}{3}\pi r^3 \right) \\ &= \frac{4}{3}\pi \cdot \frac{d}{dr}(r^3) \\ &= \frac{4}{3}\pi \cdot (3r^2) \\ &= 4\pi r^2\end{aligned}$$

Substitute $r = 10$ cm:

$$\begin{aligned}\left. \frac{dV}{dr} \right|_{r=10} &= 4\pi(10)^2 \\ &= 4\pi(100) \\ &= 400\pi \text{ cm}^3/\text{cm}\end{aligned}$$

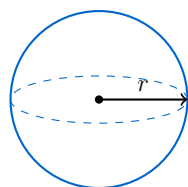
Final Answer

$400\pi \text{ cm}^3/\text{cm}$

Question 7

If the radius of a balloon which always remains spherical is increasing at the rate of 1.5 cm/sec, at what rate is its surface area increasing when its radius is 12 cm?

TYPE 1g Rate of Change of Standard Geometric Quantities – Time Rate of Change of Sphere Surface Area from Radius Ra

Sphere of radius r

Given: $\frac{dr}{dt} = 1.5$ cm/sec, $r = 12$ cm

Concept / Formula Used: $S = 4\pi r^2$

Solution:

Let S be the surface area and r be the radius of the spherical balloon at time t . The surface area of a sphere is given by:

$$S = 4\pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dS}{dt} &= \frac{d}{dt}(4\pi r^2) \\ &= 4\pi \cdot 2r \frac{dr}{dt} \\ &= 8\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 12$ cm and $\frac{dr}{dt} = 1.5$ cm/sec:

$$\begin{aligned}\frac{dS}{dt} &= 8\pi(12)(1.5) \\ &= 96\pi \times 1.5 \\ &= 144\pi \text{ cm}^2/\text{sec}\end{aligned}$$

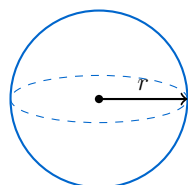
Final Answer

$144\pi \text{ cm}^2/\text{sec}$

Question 8

If the radius of a soap bubble is increasing at the rate of $\frac{1}{2}$ cm/sec, at what rate is its volume increasing when the radius is 1 cm?

TYPE 1h Rate of Change of Standard Geometric Quantities – Time Rate of Change of Sphere Volume from Radius Rate

Spherical soap bubble of radius r

Given: $\frac{dr}{dt} = \frac{1}{2}$ cm/sec, $r = 1$ cm

Concept / Formula Used: $V = \frac{4}{3}\pi r^3$

Solution:

Let V be the volume and r be the radius of the soap bubble at time t . The volume of a sphere is given by:

$$V = \frac{4}{3}\pi r^3$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} \left(\frac{4}{3}\pi r^3 \right) \\ &= \frac{4}{3}\pi \cdot 3r^2 \frac{dr}{dt} \\ &= 4\pi r^2 \frac{dr}{dt}\end{aligned}$$

Substitute $r = 1$ cm and $\frac{dr}{dt} = \frac{1}{2}$ cm/sec:

$$\begin{aligned}\frac{dV}{dt} &= 4\pi(1)^2 \left(\frac{1}{2} \right) \\ &= 4\pi \cdot \frac{1}{2} \\ &= 2\pi \text{ cm}^3/\text{sec}\end{aligned}$$

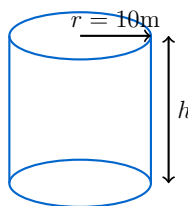
Final Answer

$2\pi \text{ cm}^3/\text{sec}$

Question 9

A cylindrical tank of radius 10 m is being filled with wheat at the rate of 314 cubic metre per hour. Find the rate at which the depth of the wheat is increasing, take $\pi = 3.14$.

TYPE 1i Rate of Change of Standard Geometric Quantities – Time Rate of Change of Cylinder Height with Constant Radius



Cylindrical tank filled with wheat

Given: $r = 10$ m, $\frac{dV}{dt} = 314 \text{ m}^3/\text{hr}$, $\pi = 3.14$

Concept / Formula Used: $V = \pi r^2 h$

Solution:

Let r be the radius, h be the depth of wheat, and V be the volume of wheat in the cylindrical tank at time t . The volume of the cylindrical tank is given by:

$$V = \pi r^2 h$$

Since the radius of the tank remains constant at $r = 10$ m:

$$\begin{aligned} V &= \pi(10)^2 h \\ &= 100\pi h \end{aligned}$$

Differentiating both sides with respect to time t :

$$\begin{aligned} \frac{dV}{dt} &= \frac{d}{dt}(100\pi h) \\ &= 100\pi \frac{dh}{dt} \end{aligned}$$

Given that $\frac{dV}{dt} = 314 \text{ m}^3/\text{hr}$ and $\pi = 3.14$:

$$\begin{aligned} 314 &= 100(3.14) \frac{dh}{dt} \\ 314 &= 314 \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{314}{314} \\ \frac{dh}{dt} &= 1 \text{ m/hr} \end{aligned}$$

Final Answer

1 m/hr

Question 10

- (i) The total revenue in rupees received from the sale of x units of a product is given by $R(x) = 3x^2 + 36x + 5$. Find the marginal revenue when $x = 5$.
 (ii) The total cost $C(x)$ associated with the production of x units of an item is given by $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$. Find the marginal cost when 3 units are produced.

TYPE 10 Marginal Cost and Marginal Revenue Calculations

- Given:** (i) $R(x) = 3x^2 + 36x + 5$, $x = 5$
 (ii) $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$, $x = 3$

Concept / Formula Used: $MR = \frac{dR}{dx}$ $MC = \frac{dC}{dx}$

Solution:

Part (i): Calculation of Marginal Revenue Marginal Revenue (MR) is defined as the rate of change of total revenue with respect to the number of units sold:

$$MR = \frac{d}{dx} R(x)$$

Given $R(x) = 3x^2 + 36x + 5$, differentiate with respect to x :

$$\begin{aligned} \frac{d}{dx} R(x) &= \frac{d}{dx} (3x^2 + 36x + 5) \\ &= 3(2x) + 36(1) + 0 \\ &= 6x + 36 \end{aligned}$$

Substitute $x = 5$:

$$\begin{aligned}MR &= 6(5) + 36 \\&= 30 + 36 \\&= 66\end{aligned}$$

Part (ii): Calculation of Marginal Cost Marginal Cost (MC) is defined as the rate of change of total cost with respect to the output produced:

$$MC = \frac{d}{dx}C(x)$$

Given $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$, differentiate with respect to x :

$$\begin{aligned}\frac{d}{dx}C(x) &= \frac{d}{dx}(0.005x^3 - 0.02x^2 + 30x + 5000) \\&= 0.005(3x^2) - 0.02(2x) + 30(1) + 0 \\&= 0.015x^2 - 0.04x + 30\end{aligned}$$

Substitute $x = 3$:

$$\begin{aligned}MC &= 0.015(3)^2 - 0.04(3) + 30 \\&= 0.015(9) - 0.12 + 30 \\&= 0.135 - 0.12 + 30 \\&= 30.015\end{aligned}$$

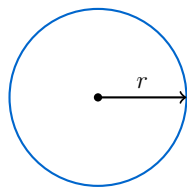
Final Answer

- (i) Marginal Revenue = Rs. 66
(ii) Marginal Cost = Rs. 30.015

Question 11

Find the rate of change of area of a circle of radius r with regard to its radius. How fast is the area changing with respect to the radius when the radius is 5 cm?

TYPE 1a Rate of Change of Standard Geometric Quantities – Rate of Change of Circle Area with respect to Radius



Circle with radius r

Given: $r = 5$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let A be the area of a circle of radius r . The area formula is:

$$A = \pi r^2$$

Differentiating A with respect to radius r :

$$\begin{aligned}\frac{dA}{dr} &= \frac{d}{dr}(\pi r^2) \\ &= \pi \cdot 2r \\ &= 2\pi r\end{aligned}$$

This gives the general rate of change of area with respect to the radius as $\frac{dA}{dr} = 2\pi r$.

When the radius $r = 5$ cm:

$$\begin{aligned}\left. \frac{dA}{dr} \right|_{r=5} &= 2\pi(5) \\ &= 10\pi \text{ cm}^2/\text{cm}\end{aligned}$$

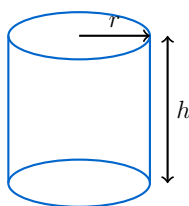
Final Answer

$$\frac{dA}{dr} = 2\pi r; \text{ when } r = 5 \text{ cm, the rate of change is } 10\pi \text{ cm}^2/\text{cm}$$

Question 12

Find the rate of change of the whole surface of a closed circular cylinder of radius r and height h with respect to change in radius.

TYPE 1j Rate of Change of Standard Geometric Quantities – Rate of Change of Cylinder Surface Area with respect to Radius



Closed circular cylinder

Given: Closed circular cylinder with radius r and height h

Concept / Formula Used: $S = 2\pi rh + 2\pi r^2$

Solution:

Let S be the total (whole) surface area of a closed circular cylinder. The formula for total surface area is:

$$S = 2\pi rh + 2\pi r^2$$

Differentiating S with respect to radius r , treating height h as constant:

$$\begin{aligned}\frac{dS}{dr} &= \frac{d}{dr}(2\pi rh + 2\pi r^2) \\ &= 2\pi h \frac{d}{dr}(r) + 2\pi \frac{d}{dr}(r^2) \\ &= 2\pi h(1) + 2\pi(2r) \\ &= 2\pi h + 4\pi r \\ &= 2\pi(h + 2r)\end{aligned}$$

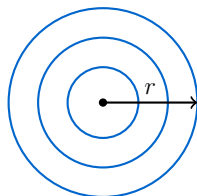
Final Answer

$$2\pi(h + 2r)$$

Question 13

A stone is dropped into a quiet lake and waves move in circles at the speed of 5 cm/sec. At the instant when the radius of the circular wave is 8 cm, how fast is the enclosed area increasing?

TYPE 1c Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Area from Radius Rate



Concentric circular ripples in water

Given: $\frac{dr}{dt} = 5$ cm/sec, $r = 8$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let r be the radius of the circular wave and A be its enclosed area at time t . The area of a circle is given by:

$$A = \pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dA}{dt} &= \frac{d}{dt}(\pi r^2) \\ &= 2\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 8$ cm and $\frac{dr}{dt} = 5$ cm/sec:

$$\begin{aligned}\frac{dA}{dt} &= 2\pi(8)(5) \\ &= 80\pi \text{ cm}^2/\text{sec}\end{aligned}$$

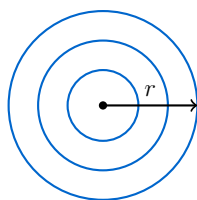
Final Answer

$$80\pi \text{ cm}^2/\text{sec}$$

Question 14

A stone is dropped into a quiet lake and waves move in circles at a speed of 4 cm/sec. At the instant when the radius of the circular wave is 10 cm, how fast is the enclosed area increasing?

TYPE 1c Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Area from Radius Rate

*Concentric circular ripples in water*

Given: $\frac{dr}{dt} = 4$ cm/sec, $r = 10$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let r be the radius of the circular wave and A be its enclosed area at time t . The area of a circle is given by:

$$A = \pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dA}{dt} &= \frac{d}{dt}(\pi r^2) \\ &= 2\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 10$ cm and $\frac{dr}{dt} = 4$ cm/sec:

$$\begin{aligned}\frac{dA}{dt} &= 2\pi(10)(4) \\ &= 80\pi \text{ cm}^2/\text{sec}\end{aligned}$$

Final Answer

$80\pi \text{ cm}^2/\text{sec}$

Question 15

If the circumference of circle is increasing at the constant rate, prove that rate of change of area of circle is directly proportional to its radius.

TYPE 12a Relations Between Coordinate Rates of Change – Proving Proportionality of Rate of Change of Circle Area

To Prove: $\frac{dA}{dt} \propto r$

Concept / Formula Used: $C = 2\pi r$ $A = \pi r^2$

Solution:

Let r be the radius, C be the circumference, and A be the area of the circle at time t . Since the circumference is increasing at a constant rate, let:

$$\frac{dC}{dt} = k \quad (\text{where } k \text{ is a constant})$$

We know that the formula for circumference is:

$$C = 2\pi r$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dC}{dt} &= 2\pi \frac{dr}{dt} \\ k &= 2\pi \frac{dr}{dt} \\ \frac{dr}{dt} &= \frac{k}{2\pi}\end{aligned}$$

Now, the formula for the area of the circle is:

$$A = \pi r^2$$

Differentiating both sides with respect to time t :

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Substituting $\frac{dr}{dt} = \frac{k}{2\pi}$ into the expression for $\frac{dA}{dt}$:

$$\begin{aligned}\frac{dA}{dt} &= 2\pi r \left(\frac{k}{2\pi} \right) \\ \frac{dA}{dt} &= kr\end{aligned}$$

Since k is a constant, we conclude:

$$\frac{dA}{dt} \propto r$$

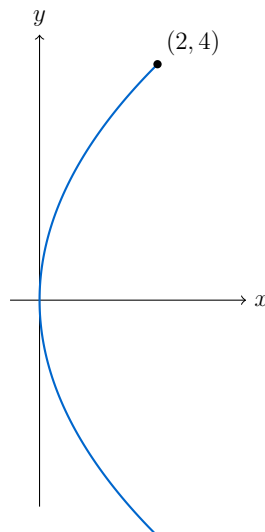
Hence Proved

LHS = RHS, hence proved that the rate of change of area is directly proportional to its radius.

Question 16

Find the point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same rate.

TYPE 13 Finding Curve Points with Equal Coordinate Rates of Change



Parabola $y^2 = 8x$ with required point

Given: Curve: $y^2 = 8x$, rate condition: $\frac{dx}{dt} = \frac{dy}{dt} \neq 0$

Concept / Formula Used: t

Solution:

The equation of the curve is given as:

$$y^2 = 8x$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{d}{dt}(y^2) &= \frac{d}{dt}(8x) \\ 2y \frac{dy}{dt} &= 8 \frac{dx}{dt}\end{aligned}$$

Given that the abscissa (x) and ordinate (y) change at the same rate:

$$\frac{dx}{dt} = \frac{dy}{dt}$$

Substitute $\frac{dy}{dt} = \frac{dx}{dt}$ into the differentiated equation:

$$2y \frac{dx}{dt} = 8 \frac{dx}{dt}$$

Dividing both sides by $\frac{dx}{dt}$ (since $\frac{dx}{dt} \neq 0$):

$$\begin{aligned}2y &= 8 \\ y &= \frac{8}{2} \\ y &= 4\end{aligned}$$

Now, substitute $y = 4$ back into the original curve equation $y^2 = 8x$:

$$\begin{aligned}(4)^2 &= 8x \\ 16 &= 8x \\ x &= \frac{16}{8} \\ x &= 2\end{aligned}$$

Therefore, the required point on the curve is $(2, 4)$.

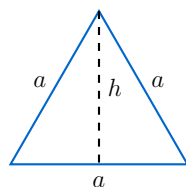
Final Answer

$(2, 4)$

Question 17

The median of an equilateral triangle is increasing at the rate of $2\sqrt{3}$ cm/sec. Find the rate at which the side is increasing.

TYPE 1k Rate of Change of Standard Geometric Quantities – Time Rate of Change of Equilateral Triangle Side from Media



Equilateral triangle showing median h and side a

Given: $\frac{dh}{dt} = 2\sqrt{3}$ cm/sec

Concept / Formula Used: $h = \frac{\sqrt{3}}{2}a$

Solution:

Let a be the side length and h be the median (altitude) of the equilateral triangle at time t . The height/median of an equilateral triangle in terms of its side a is:

$$h = \frac{\sqrt{3}}{2}a$$

Differentiating both sides with respect to time t :

$$\frac{dh}{dt} = \frac{\sqrt{3}}{2} \frac{da}{dt}$$

Substitute the given value $\frac{dh}{dt} = 2\sqrt{3}$ cm/sec:

$$2\sqrt{3} = \frac{\sqrt{3}}{2} \frac{da}{dt}$$

Dividing both sides by $\sqrt{3}$:

$$\begin{aligned} 2 &= \frac{1}{2} \frac{da}{dt} \\ \frac{da}{dt} &= 2 \times 2 \\ \frac{da}{dt} &= 4 \text{ cm/sec} \end{aligned}$$

Final Answer

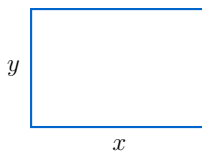
4 cm/sec

Question 18

The length x of a rectangle is decreasing at the rate of 5 cm/minute and the width y is increasing at the rate of 4 cm/minute. When $x = 8$ cm and $y = 6$ cm, find the rate of change of

- the perimeter
- the area of the rectangle.

TYPE 11 Rate of Change of Standard Geometric Quantities – Time Rate of Change of Rectangle Perimeter and Area

Rectangle of length x and width y

Given: $\frac{dx}{dt} = -5$ cm/min, $\frac{dy}{dt} = 4$ cm/min, $x = 8$ cm, $y = 6$ cm

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Part (i): Rate of change of perimeter Let P be the perimeter of the rectangle.

$$P = 2(x + y)$$

Differentiating with respect to time t :

$$\frac{dP}{dt} = 2 \left(\frac{dx}{dt} + \frac{dy}{dt} \right)$$

Substitute $\frac{dx}{dt} = -5$ cm/min and $\frac{dy}{dt} = 4$ cm/min:

$$\begin{aligned} \frac{dP}{dt} &= 2(-5 + 4) \\ &= 2(-1) \\ &= -2 \text{ cm/min} \end{aligned}$$

Thus, the perimeter is decreasing at the rate of 2 cm/min.

Part (ii): Rate of change of area Let A be the area of the rectangle.

$$A = xy$$

Differentiating with respect to time t using the product rule:

$$\frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$$

Substitute $x = 8$ cm, $y = 6$ cm, $\frac{dx}{dt} = -5$ cm/min, and $\frac{dy}{dt} = 4$ cm/min:

$$\begin{aligned} \frac{dA}{dt} &= (8)(4) + (6)(-5) \\ &= 32 - 30 \\ &= 2 \text{ cm}^2/\text{min} \end{aligned}$$

Thus, the area is increasing at the rate of 2 cm²/min.

Final Answer

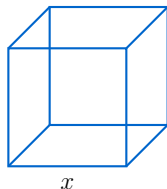
(i) Rate of change of perimeter = -2 cm/min (decreasing at 2 cm/min)

(ii) Rate of change of area = 2 cm²/min

Question 19

The volume of a cube is increasing at the rate of 9 cubic centimetres per second. How fast is the surface area increasing when the length of the edge is 10 centimetres?

TYPE 1m Rate of Change of Standard Geometric Quantities – Time Rate of Change of Cube Surface Area from Volume Rate



Cube of edge x

Given: $\frac{dV}{dt} = 9 \text{ cm}^3/\text{sec}$, $x = 10 \text{ cm}$

Concept / Formula Used: $V = x^3$ $S = 6x^2$

Solution:

Let x be the edge length, V be the volume, and S be the total surface area of the cube at time t . The volume of a cube is:

$$V = x^3$$

Differentiating with respect to time t :

$$\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$$

Given that $\frac{dV}{dt} = 9 \text{ cm}^3/\text{sec}$:

$$\begin{aligned} 9 &= 3x^2 \frac{dx}{dt} \\ \frac{dx}{dt} &= \frac{9}{3x^2} \\ \frac{dx}{dt} &= \frac{3}{x^2} \end{aligned}$$

Now, the surface area of the cube is given by:

$$S = 6x^2$$

Differentiating with respect to time t :

$$\frac{dS}{dt} = 12x \frac{dx}{dt}$$

Substitute $\frac{dx}{dt} = \frac{3}{x^2}$ into the derivative of surface area:

$$\begin{aligned} \frac{dS}{dt} &= 12x \left(\frac{3}{x^2} \right) \\ &= \frac{36}{x} \end{aligned}$$

Substitute $x = 10$ cm:

$$\begin{aligned}\frac{dS}{dt} &= \frac{36}{10} \\ &= 3.6 \text{ cm}^2/\text{sec}\end{aligned}$$

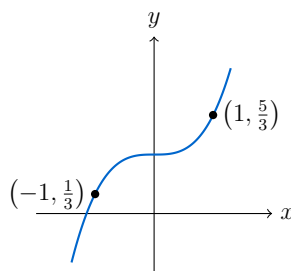
Final Answer

$3.6 \text{ cm}^2/\text{sec}$

Question 20

A particle moves along the curve $y = \frac{2}{3}x^3 + 1$. Find the points on the curve at which the y -coordinate is changing twice as fast as the x -coordinate.

TYPE 12c Relations Between Coordinate Rates of Change – Finding Curve Points with Proportional Coordinate Rates of C



Graph of curve $y = \frac{2}{3}x^3 + 1$ showing the two points

Given: Curve $y = \frac{2}{3}x^3 + 1$, rate condition $\frac{dy}{dt} = 2\frac{dx}{dt}$

Concept / Formula Used: $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$

Solution:

The given curve equation is:

$$y = \frac{2}{3}x^3 + 1$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dy}{dt} &= \frac{d}{dt} \left(\frac{2}{3}x^3 + 1 \right) \\ &= \frac{2}{3} \cdot 3x^2 \frac{dx}{dt} + 0 \\ &= 2x^2 \frac{dx}{dt}\end{aligned}$$

We are given that the y -coordinate changes twice as fast as the x -coordinate:

$$\frac{dy}{dt} = 2\frac{dx}{dt}$$

Substituting $\frac{dy}{dt} = 2\frac{dx}{dt}$ into the equation:

$$2\frac{dx}{dt} = 2x^2 \frac{dx}{dt}$$

Assuming $\frac{dx}{dt} \neq 0$, dividing both sides by $2\frac{dx}{dt}$:

$$\begin{aligned} 1 &= x^2 \\ x^2 &= 1 \\ x &= \pm 1 \end{aligned}$$

Case 1: When $x = 1$ Substitute $x = 1$ into the curve equation $y = \frac{2}{3}x^3 + 1$:

$$\begin{aligned} y &= \frac{2}{3}(1)^3 + 1 \\ &= \frac{2}{3} + 1 \\ &= \frac{5}{3} \end{aligned}$$

This gives the first point $(1, \frac{5}{3})$.

Case 2: When $x = -1$ Substitute $x = -1$ into the curve equation $y = \frac{2}{3}x^3 + 1$:

$$\begin{aligned} y &= \frac{2}{3}(-1)^3 + 1 \\ &= -\frac{2}{3} + 1 \\ &= \frac{1}{3} \end{aligned}$$

This gives the second point $(-1, \frac{1}{3})$.

Therefore, the required points on the curve are $(1, \frac{5}{3})$ and $(-1, \frac{1}{3})$.

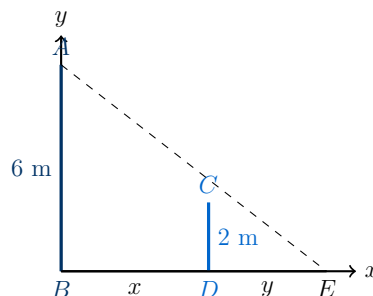
Final Answer

$$\left(1, \frac{5}{3}\right) \text{ and } \left(-1, \frac{1}{3}\right)$$

Question 21

A man of height 2 metres walks at a uniform speed of 5 km/h away from a lamp post which is 6 metres high. Find the rate at which the length of shadow increases.

TYPE 18a Related Rates via Geometric/Trigonometric Setup – Rate of Shadow Length Increase for a Walking Person



Man walking away from a lamp post

Given: Height of lamp post $AB = 6$ m, height of man $CD = 2$ m, rate of walking $\frac{dx}{dt} = 5$ km/h.

To Find: The rate at which the length of the shadow increases $\left(\frac{dy}{dt}\right)$.

Concept / Formula Used: $\frac{CD}{AB} = \frac{DE}{BE}$

Solution:

Let AB be the lamp post of height 6 m, and CD be the man of height 2 m.

Let x be the distance of the man from the lamp post ($BD = x$), and let y be the length of his shadow ($DE = y$).

The total distance from the base of the lamp post to the tip of the shadow is:

$$BE = BD + DE = x + y$$

From the geometric arrangement, $\triangle ECD$ and $\triangle EAB$ are similar triangles. Setting up the ratio of corresponding sides:

$$\begin{aligned}\frac{CD}{AB} &= \frac{DE}{BE} \\ \frac{2}{6} &= \frac{y}{x+y} \\ \frac{1}{3} &= \frac{y}{x+y}\end{aligned}$$

Cross-multiplying gives:

$$\begin{aligned}x + y &= 3y \\ x &= 3y - y \\ x &= 2y \\ y &= \frac{1}{2}x\end{aligned}$$

Differentiating both sides with respect to time t :

$$\frac{dy}{dt} = \frac{1}{2} \frac{dx}{dt}$$

Substitute the given walking speed $\frac{dx}{dt} = 5$ km/h:

$$\begin{aligned}\frac{dy}{dt} &= \frac{1}{2} \times 5 \\ \frac{dy}{dt} &= \frac{5}{2} \text{ km/h} = 2.5 \text{ km/h}\end{aligned}$$

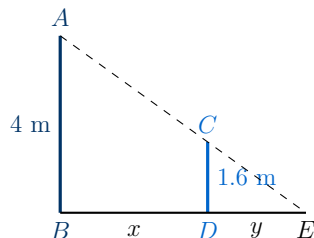
Final Answer

The length of the shadow increases at a rate of $\frac{5}{2}$ km/h (or 2.5 km/h).

Question 22

A girl of height 1.6 m walks at the rate of 50 metres per minute away from a lamp which is 4 m above the ground. How fast is the girl's shadow lengthening?

TYPE 18a Related Rates via Geometric/Trigonometric Setup – Rate of Shadow Length Increase for a Walking Person



Girl walking away from a lamp

Given: Height of lamp $AB = 4$ m, height of girl $CD = 1.6$ m, speed of girl $\frac{dx}{dt} = 50$ m/min.

To Find: The rate at which the girl's shadow is lengthening $\left(\frac{dy}{dt}\right)$.

Concept / Formula Used: $\triangle ECD \sim \triangle EAB$ $\frac{CD}{AB} = \frac{DE}{BE}$

Solution:

Let AB be the height of the lamp (4 m) and CD be the height of the girl (1.6 m).

Let x be the distance of the girl from the lamp post ($BD = x$), and let y be the length of her shadow ($DE = y$).

The total length BE is $x + y$.

Using the similar triangles $\triangle ECD$ and $\triangle EAB$:

$$\begin{aligned}\frac{CD}{AB} &= \frac{DE}{BE} \\ \frac{1.6}{4} &= \frac{y}{x+y} \\ \frac{2}{5} &= \frac{y}{x+y}\end{aligned}$$

Cross-multiplying:

$$\begin{aligned}2(x+y) &= 5y \\ 2x + 2y &= 5y \\ 2x &= 5y - 2y \\ 2x &= 3y \\ 3y &= 2x \\ y &= \frac{2}{3}x\end{aligned}$$

Differentiating both sides with respect to time t :

$$\frac{dy}{dt} = \frac{2}{3} \frac{dx}{dt}$$

Substitute the given rate $\frac{dx}{dt} = 50$ m/min:

$$\frac{dy}{dt} = \frac{2}{3} \times 50$$

$$\frac{dy}{dt} = \frac{100}{3} \text{ m/minute} = 33\frac{1}{3} \text{ m/minute}$$

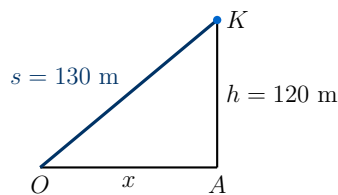
Final Answer

The girl's shadow is lengthening at the rate of $33\frac{1}{3}$ m/minute (or $\frac{100}{3}$ m/min).

Question 23

A kite is 120 m high and 130 m of string is out. If the kite is moving horizontally at the rate of 5.2 m/sec, find the rate at which the string is being paid out at that instant.

TYPE 18b Related Rates via Geometric/Trigonometric Setup – Rate of Change of String Length for a Flying Kite



Kite flying horizontally

Given: Height of kite $h = 120$ m (constant), length of string $s = 130$ m, rate of horizontal movement $\frac{dx}{dt} = 5.2$ m/sec.

To Find: The rate at which the string is being paid out $\left(\frac{ds}{dt}\right)$.

Concept / Formula Used: $x^2 + h^2 = s^2$ h

Solution:

Let x be the horizontal distance of the kite from the observer, $h = 120$ m be the constant height, and s be the length of the string.

Applying Pythagoras Theorem to the right-angled triangle formed:

$$x^2 + h^2 = s^2$$

$$x^2 + 120^2 = s^2$$

When the length of the string $s = 130$ m, we determine the horizontal distance x :

$$\begin{aligned}x^2 + 120^2 &= 130^2 \\x^2 + 14400 &= 16900 \\x^2 &= 16900 - 14400 \\x^2 &= 2500 \\x &= 50 \text{ m}\end{aligned}$$

Now, differentiating the equation $x^2 + 120^2 = s^2$ with respect to time t :

$$\begin{aligned}\frac{d}{dt}(x^2 + 120^2) &= \frac{d}{dt}(s^2) \\2x \frac{dx}{dt} + 0 &= 2s \frac{ds}{dt} \\x \frac{dx}{dt} &= s \frac{ds}{dt}\end{aligned}$$

Substitute $x = 50$ m, $\frac{dx}{dt} = 5.2$ m/sec, and $s = 130$ m:

$$\begin{aligned}50 \times 5.2 &= 130 \times \frac{ds}{dt} \\260 &= 130 \frac{ds}{dt} \\\frac{ds}{dt} &= \frac{260}{130} \\\frac{ds}{dt} &= 2 \text{ m/sec}\end{aligned}$$

Final Answer

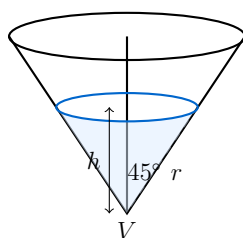
The rate at which the string is being paid out is 2 m/sec.

Question 24

A circular cone, with semi-vertical angle 45° , is fixed with its axis vertical and its vertex downwards. Water is poured into the cone at the rate of 2 cm^3 per second. Find the rate at which the depth of the water is increasing when the depth is 4 cm.

TYPE 18c Related Rates via Geometric/Trigonometric Setup – Rate of Water Level Rise in a Cone with Given Semi-Vertical

★ Likely Exam Question



Inverted conical vessel containing water

Given: Semi-vertical angle $\alpha = 45^\circ$, rate of volume increase $\frac{dV}{dt} = 2 \text{ cm}^3/\text{sec}$, depth $h = 4 \text{ cm}$.

To Find: The rate of increase of depth of water $\left(\frac{dh}{dt}\right)$ when $h = 4 \text{ cm}$.

Concept / Formula Used: $V = \frac{1}{3}\pi r^2 h$ $\tan \alpha = \frac{r}{h}$

Solution:

Let r be the radius and h be the depth of water in the conical vessel at any time t .

Given semi-vertical angle $\alpha = 45^\circ$, we relate r and h :

$$\begin{aligned}\tan 45^\circ &= \frac{r}{h} \\ 1 &= \frac{r}{h} \\ r &= h\end{aligned}$$

Substituting $r = h$ into the volume formula of the cone:

$$\begin{aligned}V &= \frac{1}{3}\pi r^2 h \\ V &= \frac{1}{3}\pi (h)^2 h \\ V &= \frac{1}{3}\pi h^3\end{aligned}$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} \left(\frac{1}{3}\pi h^3 \right) \\ \frac{dV}{dt} &= \frac{1}{3}\pi \cdot 3h^2 \frac{dh}{dt} \\ \frac{dV}{dt} &= \pi h^2 \frac{dh}{dt}\end{aligned}$$

Substitute $\frac{dV}{dt} = 2 \text{ cm}^3/\text{sec}$ and $h = 4 \text{ cm}$:

$$\begin{aligned}2 &= \pi (4)^2 \frac{dh}{dt} \\ 2 &= 16\pi \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{2}{16\pi} \\ \frac{dh}{dt} &= \frac{1}{8\pi} \text{ cm/sec}\end{aligned}$$

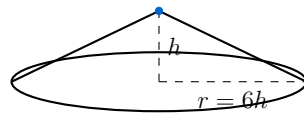
Final Answer

The depth of the water is increasing at the rate of $\frac{1}{8\pi}$ cm/sec.

Question 25

Sand is pouring from a pipe at the rate of $12 \text{ cm}^3/\text{sec}$. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm?

TYPE 18d Related Rates via Geometric/Trigonometric Setup – Rate of Height Increase of a Sand Cone with Proportional D

★ Likely Exam Question

Cone of falling sand

Given: Rate of volume increase $\frac{dV}{dt} = 12 \text{ cm}^3/\text{sec}$, relation $h = \frac{1}{6}r$, height $h = 4 \text{ cm}$.

To Find: The rate at which the height of the sand cone is increasing $\left(\frac{dh}{dt}\right)$.

Concept / Formula Used: $V = \frac{1}{3}\pi r^2 h$

Solution:

Let r be the base radius and h be the height of the sand cone at any time t .

Given relation between height and radius:

$$h = \frac{1}{6}r$$

$$r = 6h$$

Substitute $r = 6h$ into the volume formula of the cone:

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi(6h)^2 h$$

$$V = \frac{1}{3}\pi(36h^2)h$$

$$V = 12\pi h^3$$

Differentiating with respect to time t :

$$\frac{dV}{dt} = \frac{d}{dt}(12\pi h^3)$$

$$\frac{dV}{dt} = 12\pi \cdot 3h^2 \frac{dh}{dt}$$

$$\frac{dV}{dt} = 36\pi h^2 \frac{dh}{dt}$$

Substitute the given values $\frac{dV}{dt} = 12 \text{ cm}^3/\text{sec}$ and $h = 4 \text{ cm}$:

$$12 = 36\pi(4)^2 \frac{dh}{dt}$$

$$12 = 36\pi(16) \frac{dh}{dt}$$

$$12 = 576\pi \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{12}{576\pi}$$

$$\frac{dh}{dt} = \frac{1}{48\pi} \text{ cm/sec}$$

Final Answer

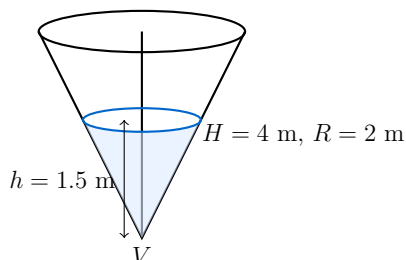
The height of the sand cone is increasing at the rate of $\frac{1}{48\pi} \text{ cm/sec}$.

Common Mistake: Failing to substitute $r = 6h$ before differentiating introduces $\frac{dr}{dt}$ needlessly, complicating the derivative.

Question 26

A conical vessel whose height is 4 metres and of base radius 2 metres is being filled with water at the rate of 0.75 cubic metres per minute. Find the rate at which the level of the water is rising when the depth of water is 1.5 metres.

TYPE 18e Related Rates via Geometric/Trigonometric Setup – Rate of Water Level Rise in a Conical Vessel of Given Dimensions



Water level in a conical vessel

Given: Height of conical vessel $H = 4 \text{ m}$, base radius $R = 2 \text{ m}$, rate of filling $\frac{dV}{dt} = 0.75 \text{ m}^3/\text{min}$, water depth $h = 1.5 \text{ m}$.

To Find: The rate of rise of water level $\left(\frac{dh}{dt}\right)$.

Concept / Formula Used: $\frac{r}{h} = \frac{R}{H}$ $V = \frac{1}{3}\pi r^2 h$

Solution:

Let r and h be the radius of the top surface of water and the depth of water at time t , respectively.
By similar vertical cross-sectional triangles:

$$\begin{aligned}\frac{r}{h} &= \frac{R}{H} \\ \frac{r}{h} &= \frac{2}{4} = \frac{1}{2} \\ r &= \frac{1}{2}h\end{aligned}$$

Substitute $r = \frac{1}{2}h$ into the volume equation:

$$\begin{aligned}V &= \frac{1}{3}\pi r^2 h \\ V &= \frac{1}{3}\pi \left(\frac{1}{2}h\right)^2 h \\ V &= \frac{1}{3}\pi \left(\frac{1}{4}h^2\right) h \\ V &= \frac{1}{12}\pi h^3\end{aligned}$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} \left(\frac{1}{12}\pi h^3 \right) \\ \frac{dV}{dt} &= \frac{1}{12}\pi \cdot 3h^2 \frac{dh}{dt} \\ \frac{dV}{dt} &= \frac{1}{4}\pi h^2 \frac{dh}{dt}\end{aligned}$$

Convert the given values to fractions: $\frac{dV}{dt} = 0.75 = \frac{3}{4}$ m³/min, and $h = 1.5 = \frac{3}{2}$ m.

Substitute these into the differentiated equation:

$$\begin{aligned}\frac{3}{4} &= \frac{1}{4}\pi \left(\frac{3}{2}\right)^2 \frac{dh}{dt} \\ \frac{3}{4} &= \frac{1}{4}\pi \cdot \frac{9}{4} \frac{dh}{dt} \\ \frac{3}{4} &= \frac{9\pi}{16} \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{3}{4} \times \frac{16}{9\pi} \\ \frac{dh}{dt} &= \frac{4}{3\pi} \text{ m/min}\end{aligned}$$

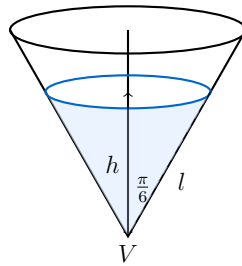
Final Answer

The level of the water is rising at a rate of $\frac{4}{3\pi}$ m/min.

Question 27

Water is dripping out at a steady rate of 1 cu cm/sec through a tiny hole at the vertex of the conical vessel, whose axis is vertical. When the slant height of water in the vessel is 4 cm, find the rate of decrease of slant height, where the semi-vertical angle of the cone is $\frac{\pi}{6}$.

TYPE 18f Related Rates via Geometric/Trigonometric Setup – Rate of Decrease of Slant Height in a Leaking Conical Vessel

★ Likely Exam Question

Water leaking from vertex of conical vessel

Given: Rate of volume change $\frac{dV}{dt} = -1$ cm³/sec, semi-vertical angle $\alpha = \frac{\pi}{6}$, slant height $l = 4$ cm.

To Find: The rate of decrease of slant height $\left(-\frac{dl}{dt}\right)$.

Concept / Formula Used: α l $r = l \sin \alpha$ $h = l \cos \alpha$ $V = \frac{1}{3}\pi r^2 h$

Solution:

Let l be the slant height, r be the radius, and h be the height of the cone of water at time t . Expressing radius r and height h in terms of slant height l and semi-vertical angle $\alpha = \frac{\pi}{6}$:

$$r = l \sin \left(\frac{\pi}{6}\right) = l \cdot \frac{1}{2} = \frac{l}{2}$$

$$h = l \cos \left(\frac{\pi}{6}\right) = l \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}l}{2}$$

Substitute r and h into the volume formula $V = \frac{1}{3}\pi r^2 h$:

$$V = \frac{1}{3}\pi \left(\frac{l}{2}\right)^2 \left(\frac{\sqrt{3}l}{2}\right)$$

$$V = \frac{1}{3}\pi \cdot \frac{l^2}{4} \cdot \frac{\sqrt{3}l}{2}$$

$$V = \frac{\sqrt{3}\pi}{24}l^3$$

Differentiating with respect to time t :

$$\frac{dV}{dt} = \frac{d}{dt} \left(\frac{\sqrt{3}\pi}{24} l^3 \right)$$

$$\frac{dV}{dt} = \frac{\sqrt{3}\pi}{24} \cdot 3l^2 \frac{dl}{dt}$$

$$\frac{dV}{dt} = \frac{\sqrt{3}\pi}{8} l^2 \frac{dl}{dt}$$

Since water is dripping out at $1 \text{ cm}^3/\text{sec}$, $\frac{dV}{dt} = -1 \text{ cm}^3/\text{sec}$. Substitute $\frac{dV}{dt} = -1$ and $l = 4 \text{ cm}$:

$$-1 = \frac{\sqrt{3}\pi}{8} (4)^2 \frac{dl}{dt}$$

$$-1 = \frac{\sqrt{3}\pi}{8} (16) \frac{dl}{dt}$$

$$-1 = 2\sqrt{3}\pi \frac{dl}{dt}$$

$$\frac{dl}{dt} = -\frac{1}{2\sqrt{3}\pi} \text{ cm/sec}$$

The negative sign signifies a decrease in slant height.

Final Answer

The rate of decrease of slant height is $\frac{1}{2\sqrt{3}\pi} \text{ cm/sec}$.

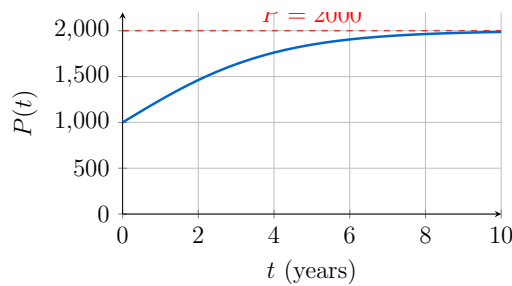
Question 28

The population of rabbits in a forest is modelled by the function below:

$$P(t) = \frac{2000}{1 + e^{-0.5t}}$$

where P represents the population of rabbits in t years. Determine whether the rabbit population is increasing or not, and justify your answer.

TYPE 24 Determining Growth Behavior of a Logistic Population Model



$$\text{Rabbit population } P(t) = \frac{2000}{1+e^{-0.5t}} \text{ over time}$$

Given: Population model $P(t) = \frac{2000}{1+e^{-0.5t}}$ for time $t \geq 0$.

To Find: Determine if $P(t)$ is increasing by analyzing the sign of the first derivative $\frac{dP}{dt}$.

Concept / Formula Used: $f(t) \quad f'(t) > 0 \quad t$

Solution:

To determine whether the rabbit population is increasing, we evaluate the derivative of $P(t)$ with respect to t .

Rewrite $P(t)$ in index form:

$$P(t) = 2000 (1 + e^{-0.5t})^{-1}$$

Differentiating with respect to t using the chain rule:

$$\frac{dP}{dt} = 2000 \cdot (-1) (1 + e^{-0.5t})^{-2} \cdot \frac{d}{dt} (1 + e^{-0.5t})$$

Evaluating the inner derivative $\frac{d}{dt} (1 + e^{-0.5t})$:

$$\begin{aligned} \frac{d}{dt} (1 + e^{-0.5t}) &= 0 + e^{-0.5t} \cdot (-0.5) \\ &= -0.5e^{-0.5t} \end{aligned}$$

Substituting this back into the expression for $\frac{dP}{dt}$:

$$\begin{aligned} \frac{dP}{dt} &= -2000 (1 + e^{-0.5t})^{-2} \cdot (-0.5e^{-0.5t}) \\ &= \frac{2000 \times 0.5 e^{-0.5t}}{(1 + e^{-0.5t})^2} \\ &= \frac{1000e^{-0.5t}}{(1 + e^{-0.5t})^2} \end{aligned}$$

Now, analyze the components of $\frac{dP}{dt}$ for all $t \geq 0$:

$$e^{-0.5t} > 0 \quad \text{for all real } t$$

$$(1 + e^{-0.5t})^2 > 0 \quad \text{for all real } t$$

$$1000 > 0$$

Therefore:

$$\frac{dP}{dt} > 0 \quad \text{for all } t \geq 0$$

Since the derivative $\frac{dP}{dt}$ is strictly positive for all $t \geq 0$, the function $P(t)$ is strictly increasing.

Hence Proved

Since $\frac{dP}{dt} > 0$ for all $t \geq 0$, the rabbit population is increasing.

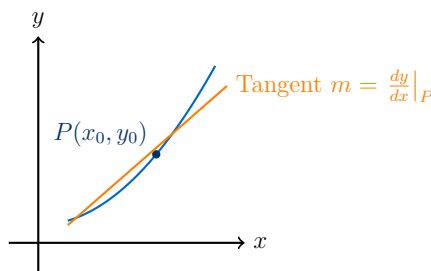
6.2 – Tangents and Normals

Question 1

What is the slope of the tangent to the following curves:

- $y = x^3 - x$ at $x = 2$?
- $y = 3x^4 - 4x$ at $x = 4$?
- $x^2 + 3y + y^2 = 5$ at the point $(1, 1)$?
- $y = x^3 - 3x + 2$ at the point whose x -coordinate is 3?
- $y = 3x^2 + 4x$ at the point whose x -coordinate is -2 ?
- $y = 2 \sin^2(3x)$ at $x = \frac{\pi}{6}$?

TYPE 25a Finding Slope of Tangent or Normal at a Point – Finding Slope of Tangent to Curve at Given Point



Slope of tangent to a curve $y = f(x)$ at a given point $P(x_0, y_0)$

Given: Curves $y = f(x)$ and specific points or x -coordinates as given in parts (i) through (vi).

Concept / Formula Used: m $y = f(x)$ (x_0, y_0) $m = \left. \frac{dy}{dx} \right|_{(x_0, y_0)}$

Solution:

- (i) Given curve: $y = x^3 - x$

Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^3 - x) \\ &= 3x^2 - 1\end{aligned}$$

Evaluating at $x = 2$:

$$\begin{aligned}m &= 3(2)^2 - 1 \\ &= 3(4) - 1 \\ &= 12 - 1 = 11\end{aligned}$$

(ii) Given curve: $y = 3x^4 - 4x$

Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(3x^4 - 4x) \\ &= 12x^3 - 4\end{aligned}$$

Evaluating at $x = 4$:

$$\begin{aligned}m &= 12(4)^3 - 4 \\ &= 12(64) - 4 \\ &= 768 - 4 = 764\end{aligned}$$

(iii) Given curve: $x^2 + 3y + y^2 = 5$

Differentiating implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^2) + \frac{d}{dx}(3y) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(5) \\ 2x + 3\frac{dy}{dx} + 2y\frac{dy}{dx} &= 0 \\ (3 + 2y)\frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{2x}{3 + 2y}\end{aligned}$$

Evaluating at $(x, y) = (1, 1)$:

$$\begin{aligned}m &= -\frac{2(1)}{3 + 2(1)} \\ &= -\frac{2}{3 + 2} = -\frac{2}{5}\end{aligned}$$

(iv) Given curve: $y = x^3 - 3x + 2$

Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^3 - 3x + 2) \\ &= 3x^2 - 3\end{aligned}$$

Evaluating at $x = 3$:

$$\begin{aligned}m &= 3(3)^2 - 3 \\ &= 3(9) - 3 \\ &= 27 - 3 = 24\end{aligned}$$

(v) Given curve: $y = 3x^2 + 4x$

Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(3x^2 + 4x) \\ &= 6x + 4\end{aligned}$$

Evaluating at $x = -2$:

$$\begin{aligned}m &= 6(-2) + 4 \\ &= -12 + 4 = -8\end{aligned}$$

(vi) Given curve: $y = 2 \sin^2(3x)$

Using chain rule to differentiate with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= 2 \cdot 2 \sin(3x) \cdot \frac{d}{dx}(\sin 3x) \\ &= 4 \sin(3x) \cdot \cos(3x) \cdot 3 \\ &= 6(2 \sin 3x \cos 3x) \\ &= 6 \sin(6x)\end{aligned}$$

Evaluating at $x = \frac{\pi}{6}$:

$$\begin{aligned}m &= 6 \sin\left(6 \cdot \frac{\pi}{6}\right) \\ &= 6 \sin(\pi) \\ &= 6(0) = 0\end{aligned}$$

Final Answer

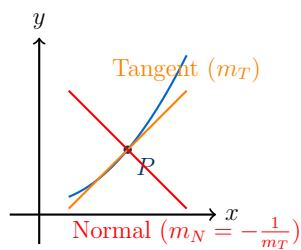
- (i) Slope $m = 11$
- (ii) Slope $m = 764$
- (iii) Slope $m = -\frac{2}{5}$
- (iv) Slope $m = 24$
- (v) Slope $m = -8$
- (vi) Slope $m = 0$

Question 2

What is the slope of the normal to the following curves:

- (i) $y = x^3 - 5x^2 - x + 1$ at the point $(1, -4)$?
- (ii) $y = 2x^2 + 3 \sin x$ at $x = 0$?

TYPE 25b Finding Slope of Tangent or Normal at a Point – Finding Slope of Normal to Curve at Given Point



Relationship between tangent and normal at point P

Given: Curves and points for parts (i) and (ii).

Concept / Formula Used: m_N m_T $m_N = -\frac{1}{m_T} = -\frac{1}{\left.\frac{dy}{dx}\right|_{(x_0, y_0)}}$

Solution:

(i) Given curve: $y = x^3 - 5x^2 - x + 1$

Differentiating with respect to x :

$$\frac{dy}{dx} = 3x^2 - 10x - 1$$

At the point $(1, -4)$, the slope of the tangent is:

$$\begin{aligned} m_T &= 3(1)^2 - 10(1) - 1 \\ &= 3 - 10 - 1 = -8 \end{aligned}$$

Therefore, the slope of the normal is:

$$\begin{aligned} m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{-8} = \frac{1}{8} \end{aligned}$$

(ii) Given curve: $y = 2x^2 + 3\sin x$

Differentiating with respect to x :

$$\frac{dy}{dx} = 4x + 3\cos x$$

At $x = 0$, the slope of the tangent is:

$$\begin{aligned} m_T &= 4(0) + 3\cos(0) \\ &= 0 + 3(1) = 3 \end{aligned}$$

Therefore, the slope of the normal is:

$$\begin{aligned} m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{3} \end{aligned}$$

Final Answer

- (i) Slope of normal $m_N = \frac{1}{8}$
 (ii) Slope of normal $m_N = -\frac{1}{3}$

Question 3

Find the point on the curve $y = 3x^2 - 2x + 1$ at which the slope of the tangent is 4.

TYPE 27a Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Given Tangent Slope

Given: Curve equation $y = 3x^2 - 2x + 1$ and slope of tangent $m = 4$.

Concept / Formula Used: $m = \frac{dy}{dx}$ $\frac{dy}{dx} = m$ x

Solution:

The given curve equation is:

$$y = 3x^2 - 2x + 1$$

Differentiating with respect to x :

$$\frac{dy}{dx} = 6x - 2$$

Given that the slope of the tangent is 4:

$$\begin{aligned} \frac{dy}{dx} = 4 &\implies 6x - 2 = 4 \\ 6x &= 6 \\ x &= 1 \end{aligned}$$

Substituting $x = 1$ back into the curve equation to find y :

$$\begin{aligned} y &= 3(1)^2 - 2(1) + 1 \\ &= 3 - 2 + 1 \\ &= 2 \end{aligned}$$

Final Answer

The required point on the curve is $(1, 2)$.

Question 4

Determine the point on the curve $y = 3x^2 - 1$ at which the slope of the tangent is 3.

TYPE 27a Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Given Tangent Slope

Given: Curve $y = 3x^2 - 1$ and target tangent slope $m = 3$.

Concept / Formula Used: $\frac{dy}{dx} = m$

Solution:

Given curve:

$$y = 3x^2 - 1$$

Differentiating with respect to x :

$$\frac{dy}{dx} = 6x$$

We are given that the slope of the tangent is 3:

$$\begin{aligned} 6x &= 3 \\ x &= \frac{3}{6} = \frac{1}{2} \end{aligned}$$

Substituting $x = \frac{1}{2}$ into the original equation:

$$\begin{aligned} y &= 3\left(\frac{1}{2}\right)^2 - 1 \\ &= 3\left(\frac{1}{4}\right) - 1 \\ &= \frac{3}{4} - 1 = -\frac{1}{4} \end{aligned}$$

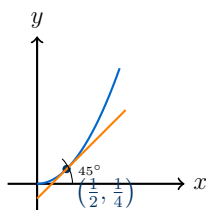
Final Answer

The required point on the curve is $\left(\frac{1}{2}, -\frac{1}{4}\right)$.

Question 5

At what point on the curve $y = x^2$ does the tangent make an angle of 45° with the x -axis?

TYPE 27b Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Given Tangent Inclination



Tangent line making angle $\theta = 45^\circ$ with positive x -axis

Given: Curve equation $y = x^2$ and angle $\theta = 45^\circ$ made by tangent with the positive x -axis.

Concept / Formula Used: $m = \tan \theta = \frac{dy}{dx}$

Solution:

Given angle $\theta = 45^\circ$, the slope of the tangent is:

$$m = \tan(45^\circ) = 1$$

Differentiating the curve $y = x^2$ with respect to x :

$$\frac{dy}{dx} = 2x$$

Equating the derivative to the slope:

$$\begin{aligned} 2x &= 1 \\ x &= \frac{1}{2} \end{aligned}$$

Substituting $x = \frac{1}{2}$ into $y = x^2$:

$$y = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Final Answer

The point on the curve is $\left(\frac{1}{2}, \frac{1}{4}\right)$.

Question 6

- (i) Find the point on the curve $y = x^2 - 2x + 3$ at which the tangent is parallel to x -axis.
- (ii) Find the point on the curve $y = x^2 + 2x - 1$, where the tangent is parallel to x -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to C

Given: Curves given in (i) and (ii), with tangent condition being parallel to x -axis.

Concept / Formula Used: $x \quad m = 0 \quad \frac{dy}{dx} = 0$

Solution:

- (i) Given curve: $y = x^2 - 2x + 3$

Differentiating with respect to x :

$$\frac{dy}{dx} = 2x - 2$$

Since the tangent is parallel to the x -axis:

$$\begin{aligned} \frac{dy}{dx} &= 0 \\ 2x - 2 &= 0 \\ 2x &= 2 \implies x = 1 \end{aligned}$$

Substituting $x = 1$ into $y = x^2 - 2x + 3$:

$$\begin{aligned} y &= (1)^2 - 2(1) + 3 \\ &= 1 - 2 + 3 = 2 \end{aligned}$$

Thus, the required point is $(1, 2)$.

(ii) Given curve: $y = x^2 + 2x - 1$

Differentiating with respect to x :

$$\frac{dy}{dx} = 2x + 2$$

Since the tangent is parallel to the x -axis:

$$\begin{aligned}\frac{dy}{dx} &= 0 \\ 2x + 2 &= 0 \\ 2x &= -2 \implies x = -1\end{aligned}$$

Substituting $x = -1$ into $y = x^2 + 2x - 1$:

$$\begin{aligned}y &= (-1)^2 + 2(-1) - 1 \\ &= 1 - 2 - 1 = -2\end{aligned}$$

Thus, the required point is $(-1, -2)$.

Final Answer

- (i) Point is $(1, 2)$
(ii) Point is $(-1, -2)$

Question 7

Find the slope of the normal to the curve $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ at $\theta = \frac{\pi}{4}$.

TYPE 25c Finding Slope of Tangent or Normal at a Point – Finding Slope of Normal to Parametric Curve

Given: Parametric equations $x = a \cos^3 \theta$, $y = a \sin^3 \theta$, and parameter value $\theta = \frac{\pi}{4}$.

Concept / Formula Used: $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$ $m_N = -\frac{1}{m_T}$

Solution:

Differentiating x with respect to θ :

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{d}{d\theta}(a \cos^3 \theta) \\ &= a \cdot 3 \cos^2 \theta \cdot (-\sin \theta) \\ &= -3a \cos^2 \theta \sin \theta\end{aligned}$$

Differentiating y with respect to θ :

$$\begin{aligned}\frac{dy}{d\theta} &= \frac{d}{d\theta}(a \sin^3 \theta) \\ &= a \cdot 3 \sin^2 \theta \cdot (\cos \theta) \\ &= 3a \sin^2 \theta \cos \theta\end{aligned}$$

Now calculating $\frac{dy}{dx}$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} \\ &= \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} \\ &= -\frac{\sin \theta}{\cos \theta} = -\tan \theta\end{aligned}$$

Slope of tangent m_T at $\theta = \frac{\pi}{4}$:

$$m_T = -\tan\left(\frac{\pi}{4}\right) = -1$$

Therefore, the slope of the normal m_N is:

$$\begin{aligned}m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{-1} = 1\end{aligned}$$

Final Answer

The slope of the normal at $\theta = \frac{\pi}{4}$ is 1.

Question 8

Find the slope of the normal to the curve $x = 1 - a \sin \theta$, $y = b \cos^2 \theta$ at $\theta = \frac{\pi}{2}$.

TYPE 25c Finding Slope of Tangent or Normal at a Point – Finding Slope of Normal to Parametric Curve

Given: Parametric equations $x = 1 - a \sin \theta$, $y = b \cos^2 \theta$ at $\theta = \frac{\pi}{2}$.

Concept / Formula Used: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ $m_N = -\frac{1}{m_T}$

Solution:

Differentiating x with respect to θ :

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{d}{d\theta}(1 - a \sin \theta) \\ &= -a \cos \theta\end{aligned}$$

Differentiating y with respect to θ :

$$\begin{aligned}\frac{dy}{d\theta} &= \frac{d}{d\theta}(b \cos^2 \theta) \\ &= b \cdot 2 \cos \theta \cdot (-\sin \theta) \\ &= -2b \sin \theta \cos \theta\end{aligned}$$

Now finding $\frac{dy}{dx}$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{-2b \sin \theta \cos \theta}{-a \cos \theta} \\ &= \frac{2b \sin \theta}{a}\end{aligned}$$

Evaluating the slope of tangent m_T at $\theta = \frac{\pi}{2}$:

$$\begin{aligned}m_T &= \frac{2b \sin(\pi/2)}{a} \\ &= \frac{2b(1)}{a} = \frac{2b}{a}\end{aligned}$$

Therefore, the slope of the normal m_N is:

$$\begin{aligned}m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{\frac{2b}{a}} = -\frac{a}{2b}\end{aligned}$$

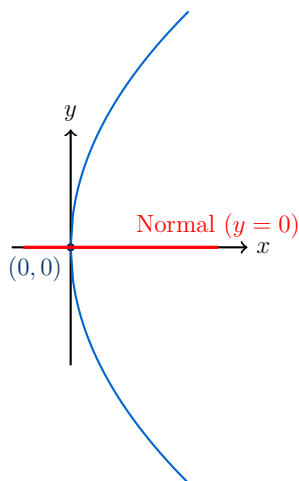
Final Answer

The slope of the normal at $\theta = \frac{\pi}{2}$ is $-\frac{a}{2b}$.

Question 9

Write the equation of the normal to the curve $y^2 = 8x$ at the origin.

TYPE 37a Finding Equation of Tangent or Normal – Finding Equation of Normal at Origin with Vertical Tangent



Parabola $y^2 = 8x$ with tangent (vertical line $x = 0$) and normal (horizontal line $y = 0$) at origin

Given: Curve equation $y^2 = 8x$ and point of contact $(0, 0)$.

Concept / Formula Used: (x_1, y_1) m $y - y_1 = m(x - x_1)$

Solution:

Differentiating $y^2 = 8x$ implicitly with respect to x :

$$2y \frac{dy}{dx} = 8$$

$$\frac{dy}{dx} = \frac{4}{y}$$

At the origin $(0, 0)$, as $y \rightarrow 0$, $\frac{dy}{dx} \rightarrow \infty$. This means the tangent line at $(0, 0)$ is vertical ($x = 0$, the y -axis).

Since the normal is perpendicular to the tangent line at $(0, 0)$, the normal line must be horizontal. A horizontal line passing through the origin $(0, 0)$ has slope $m_N = 0$.

Equation of normal line passing through $(0, 0)$ with slope $m_N = 0$:

$$y - 0 = 0(x - 0)$$

$$y = 0$$

Final Answer

The equation of the normal at the origin is $y = 0$ (the x -axis).

Question 10

Find the slopes of the tangents at the three points where the curve $y = x(x^2 - 3x - 4)$ cuts the x -axis.

TYPE 25d Finding Slope of Tangent or Normal at a Point – Finding Tangent Slopes at Curve x-Intercepts

Given: Curve equation $y = x(x^2 - 3x - 4)$.

Concept / Formula Used: x $y = 0$ $m = \frac{dy}{dx}$

Solution:

First, find the points where the curve cuts the x -axis by setting $y = 0$:

$$x(x^2 - 3x - 4) = 0$$

$$x(x - 4)(x + 1) = 0$$

This gives three roots:

$$x = 0, \quad x = 4, \quad x = -1$$

So the curve cuts the x -axis at $(0, 0)$, $(4, 0)$, and $(-1, 0)$.

Next, expand the curve equation for differentiation:

$$y = x^3 - 3x^2 - 4x$$

Differentiating with respect to x :

$$\frac{dy}{dx} = 3x^2 - 6x - 4$$

Now, evaluate the slope of tangent at each of the three points:

(i) At $(0, 0)$:

$$m_1 = 3(0)^2 - 6(0) - 4 = -4$$

(ii) At $(4, 0)$:

$$\begin{aligned} m_2 &= 3(4)^2 - 6(4) - 4 \\ &= 3(16) - 24 - 4 \\ &= 48 - 28 = 20 \end{aligned}$$

(iii) At $(-1, 0)$:

$$\begin{aligned} m_3 &= 3(-1)^2 - 6(-1) - 4 \\ &= 3(1) + 6 - 4 \\ &= 9 - 4 = 5 \end{aligned}$$

Final Answer

The slopes of the tangents at the three points are -4 , 20 , and 5 .

Question 11

Show that the tangents to the curve $y = 7x^3 + 11$ at the points where $x = 2$ and $x = -2$ are parallel.

TYPE 27d Finding Curve Points Satisfying a Tangent/Normal Condition – Proving Tangents at Two Points are Parallel

To Prove: The slopes of the tangents to $y = 7x^3 + 11$ at $x = 2$ and $x = -2$ are equal.

Concept / Formula Used: $m_1 = m_2$

Solution:

The equation of the given curve is:

$$y = 7x^3 + 11$$

Differentiating with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(7x^3 + 11) \\ &= 21x^2 \end{aligned}$$

Let m_1 be the slope of the tangent at $x = 2$:

$$\begin{aligned} m_1 &= \left. \frac{dy}{dx} \right|_{x=2} \\ &= 21(2)^2 \\ &= 21(4) = 84 \end{aligned}$$

Let m_2 be the slope of the tangent at $x = -2$:

$$\begin{aligned} m_2 &= \left. \frac{dy}{dx} \right|_{x=-2} \\ &= 21(-2)^2 \\ &= 21(4) = 84 \end{aligned}$$

Since $m_1 = m_2 = 84$, the slopes of the tangents at $x = 2$ and $x = -2$ are equal.

Hence Proved

Since $m_1 = m_2 = 84$, the tangents to the curve at $x = 2$ and $x = -2$ are parallel.

Question 12

If the tangent to the curve $y = x^3 + ax + b$ at $(1, -6)$ is parallel to the line $2x - 2y + 7 = 0$, find a and b .

TYPE 27e Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Parameters from Tangent Parallel

★ Likely Exam Question

Given: Curve $y = x^3 + ax + b$, point of contact $(1, -6)$, and line $2x - 2y + 7 = 0$.

Solution:

First, since $(1, -6)$ lies on the curve $y = x^3 + ax + b$:

$$\begin{aligned} -6 &= (1)^3 + a(1) + b \\ -6 &= 1 + a + b \\ a + b &= -7 \quad \text{--- (Equation 1)} \end{aligned}$$

Next, find the slope of the given line $2x - 2y + 7 = 0$:

$$\begin{aligned} 2y &= 2x + 7 \\ y &= x + \frac{7}{2} \end{aligned}$$

Comparing with $y = mx + c$, slope of the line $m_L = 1$.

Now differentiate the curve $y = x^3 + ax + b$ with respect to x :

$$\frac{dy}{dx} = 3x^2 + a$$

The slope of the tangent at $(1, -6)$ is:

$$m_T = 3(1)^2 + a = 3 + a$$

Since the tangent is parallel to the line $2x - 2y + 7 = 0$:

$$\begin{aligned} m_T &= m_L \\ 3 + a &= 1 \\ a &= 1 - 3 = -2 \end{aligned}$$

Substitute $a = -2$ into Equation 1:

$$\begin{aligned} -2 + b &= -7 \\ b &= -7 + 2 = -5 \end{aligned}$$

Final Answer

$$a = -2, \quad b = -5$$

Question 13

Determine the point of the curve $y = 3x^2 - 5$ at which the tangent is perpendicular to a line whose slope is $-\frac{1}{3}$.

TYPE 27f Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Perpendicular

Given: Curve $y = 3x^2 - 5$ and slope of line $m_L = -\frac{1}{3}$.

Concept / Formula Used: $-1 \quad m_T \cdot m_L = -1 \Rightarrow m_T = -\frac{1}{m_L}$

Solution:

Given slope of the line $m_L = -\frac{1}{3}$. Since the tangent is perpendicular to this line, the slope of the tangent m_T is:

$$m_T = -\frac{1}{-\frac{1}{3}} = 3$$

Differentiating the curve equation $y = 3x^2 - 5$ with respect to x :

$$\frac{dy}{dx} = 6x$$

Set $\frac{dy}{dx}$ equal to m_T :

$$\begin{aligned} 6x &= 3 \\ x &= \frac{3}{6} = \frac{1}{2} \end{aligned}$$

Substitute $x = \frac{1}{2}$ into $y = 3x^2 - 5$:

$$\begin{aligned} y &= 3 \left(\frac{1}{2} \right)^2 - 5 \\ &= 3 \left(\frac{1}{4} \right) - 5 \\ &= \frac{3}{4} - 5 = -\frac{17}{4} \end{aligned}$$

Final Answer

The required point on the curve is $\left(\frac{1}{2}, -\frac{17}{4} \right)$.

Question 14

Determine the points on the curve $y = x^3 - 3x^2 - 9x + 7$ at which the tangents are parallel to x -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to C

Given: Curve $y = x^3 - 3x^2 - 9x + 7$ and tangent condition $\frac{dy}{dx} = 0$.

Concept / Formula Used: $\Rightarrow \frac{dy}{dx} = 0$

Solution:

Differentiating $y = x^3 - 3x^2 - 9x + 7$ with respect to x :

$$\frac{dy}{dx} = 3x^2 - 6x - 9$$

Setting $\frac{dy}{dx} = 0$:

$$\begin{aligned} 3x^2 - 6x - 9 &= 0 \\ 3(x^2 - 2x - 3) &= 0 \\ x^2 - 2x - 3 &= 0 \\ (x - 3)(x + 1) &= 0 \end{aligned}$$

This gives two values for x :

$$x = 3 \quad \text{or} \quad x = -1$$

Case 1: When $x = 3$,

$$\begin{aligned} y &= (3)^3 - 3(3)^2 - 9(3) + 7 \\ &= 27 - 27 - 27 + 7 \\ &= -20 \end{aligned}$$

Case 2: When $x = -1$,

$$\begin{aligned} y &= (-1)^3 - 3(-1)^2 - 9(-1) + 7 \\ &= -1 - 3(1) + 9 + 7 \\ &= -1 - 3 + 9 + 7 = 12 \end{aligned}$$

Final Answer

The required points are $(3, -20)$ and $(-1, 12)$.

Question 15

Find the points on the curve $x^2 + y^2 - 2x - 3 = 0$ at which the tangents are parallel to x -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to C

Given: Circle $x^2 + y^2 - 2x - 3 = 0$ and horizontal tangent condition $\frac{dy}{dx} = 0$.

Concept / Formula Used: $\frac{d}{dx}(x^2 + y^2 - 2x - 3) = 0$

Solution:

Differentiating $x^2 + y^2 - 2x - 3 = 0$ implicitly with respect to x :

$$\begin{aligned} 2x + 2y \frac{dy}{dx} - 2 &= 0 \\ 2y \frac{dy}{dx} &= 2 - 2x \\ \frac{dy}{dx} &= \frac{1-x}{y} \quad (\text{for } y \neq 0) \end{aligned}$$

For tangents parallel to the x -axis, $\frac{dy}{dx} = 0$:

$$\frac{1-x}{y} = 0 \implies 1-x = 0 \implies x = 1$$

Substitute $x = 1$ into the original circle equation:

$$\begin{aligned} (1)^2 + y^2 - 2(1) - 3 &= 0 \\ 1 + y^2 - 2 - 3 &= 0 \\ y^2 - 4 &= 0 \\ y^2 = 4 &\implies y = \pm 2 \end{aligned}$$

Both points $(1, 2)$ and $(1, -2)$ have non-zero y -coordinates, satisfying $y \neq 0$.

Final Answer

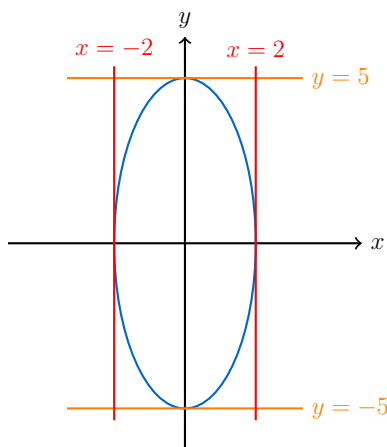
The points on the curve are $(1, 2)$ and $(1, -2)$.

Question 16

Find the points on the curve $\frac{x^2}{4} + \frac{y^2}{25} = 1$ at which the tangents are

- (i) parallel to x -axis
- (ii) parallel to y -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to C



Ellipse $\frac{x^2}{4} + \frac{y^2}{25} = 1$ showing vertical and horizontal tangents

Given: Ellipse equation $\frac{x^2}{4} + \frac{y^2}{25} = 1$.

Concept / Formula Used: $x \implies \frac{dy}{dx} = 0 \quad y \implies \frac{dx}{dy} = 0$

Solution:

Differentiating $\frac{x^2}{4} + \frac{y^2}{25} = 1$ implicitly with respect to x :

$$\begin{aligned}\frac{2x}{4} + \frac{2y}{25} \frac{dy}{dx} &= 0 \\ \frac{x}{2} + \frac{2y}{25} \frac{dy}{dx} &= 0 \\ \frac{2y}{25} \frac{dy}{dx} &= -\frac{x}{2} \\ \frac{dy}{dx} &= -\frac{25x}{4y}\end{aligned}$$

(i) Tangents parallel to x -axis ($\frac{dy}{dx} = 0$):

$$-\frac{25x}{4y} = 0 \implies x = 0$$

Substituting $x = 0$ into the ellipse equation:

$$\begin{aligned}\frac{0^2}{4} + \frac{y^2}{25} &= 1 \\ \frac{y^2}{25} &= 1 \implies y^2 = 25 \implies y = \pm 5\end{aligned}$$

The points are $(0, 5)$ and $(0, -5)$.

(ii) Tangents parallel to y -axis ($\frac{dx}{dy} = 0$, i.e., denominator of $\frac{dy}{dx}$ is zero):

$$4y = 0 \implies y = 0$$

Substituting $y = 0$ into the ellipse equation:

$$\begin{aligned}\frac{x^2}{4} + \frac{0^2}{25} &= 1 \\ \frac{x^2}{4} &= 1 \implies x^2 = 4 \implies x = \pm 2\end{aligned}$$

The points are $(2, 0)$ and $(-2, 0)$.

Final Answer

- (i) Points parallel to x -axis: $(0, 5)$ and $(0, -5)$
 (ii) Points parallel to y -axis: $(2, 0)$ and $(-2, 0)$

Question 17

Find the points on the curve $x^2 + y^2 = 13$, the tangent at each one of which is parallel to the line $2x + 3y = 7$.

Given: Circle $x^2 + y^2 = 13$ and line $2x + 3y = 7$.

Concept / Formula Used: $m_T = \left. \frac{dy}{dx} \right|_P \quad m_L$

Solution:

First, find slope of the line $2x + 3y = 7$:

$$\begin{aligned} 3y &= -2x + 7 \\ y &= -\frac{2}{3}x + \frac{7}{3} \implies m_L = -\frac{2}{3} \end{aligned}$$

Now differentiate $x^2 + y^2 = 13$ implicitly with respect to x :

$$\begin{aligned} 2x + 2y \frac{dy}{dx} &= 0 \\ 2y \frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{x}{y} \end{aligned}$$

Equating the slope of the tangent to the slope of the given line:

$$\begin{aligned} -\frac{x}{y} &= -\frac{2}{3} \\ \frac{x}{y} &= \frac{2}{3} \implies y = \frac{3}{2}x \end{aligned}$$

Substitute $y = \frac{3}{2}x$ into the curve $x^2 + y^2 = 13$:

$$\begin{aligned} x^2 + \left(\frac{3}{2}x\right)^2 &= 13 \\ x^2 + \frac{9}{4}x^2 &= 13 \\ \frac{13}{4}x^2 &= 13 \\ x^2 &= 4 \implies x = \pm 2 \end{aligned}$$

Finding corresponding y -values:

$$\begin{aligned} \text{If } x &= 2, \quad y = \frac{3}{2}(2) = 3 \\ \text{If } x &= -2, \quad y = \frac{3}{2}(-2) = -3 \end{aligned}$$

Final Answer

The required points are $(2, 3)$ and $(-2, -3)$.

Question 18

Find the points on the curve $y = x^3 - 3x^2 - 4x$ at which the tangent lines are parallel to the line $4x + y - 3 = 0$.

TYPE 27g Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to G

Given: Curve $y = x^3 - 3x^2 - 4x$ and parallel line $4x + y - 3 = 0$.

Concept / Formula Used: $\frac{dy}{dx} = m_L$

Solution:

First, find the slope of the line $4x + y - 3 = 0$:

$$y = -4x + 3 \implies m_L = -4$$

Differentiating $y = x^3 - 3x^2 - 4x$ with respect to x :

$$\frac{dy}{dx} = 3x^2 - 6x - 4$$

Equating $\frac{dy}{dx}$ to -4 :

$$3x^2 - 6x - 4 = -4$$

$$3x^2 - 6x = 0$$

$$3x(x - 2) = 0$$

This gives two roots:

$$x = 0 \quad \text{or} \quad x = 2$$

Evaluating corresponding y -values:

$$\text{When } x = 0 : \quad y = 0^3 - 3(0)^2 - 4(0) = 0$$

$$\begin{aligned} \text{When } x = 2 : \quad y &= (2)^3 - 3(2)^2 - 4(2) \\ &= 8 - 12 - 8 = -12 \end{aligned}$$

Final Answer

The required points are $(0, 0)$ and $(2, -12)$.

Question 19

Find the equations of the tangent and the normal to each of the following curves at the given point:

- (i) $y = x^3$ at $(1, 1)$
- (ii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(1, 3)$
- (iii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$
- (iv) $y = x^2$ at $(0, 0)$
- (v) $y = \sin^2 x$ at $x = \frac{\pi}{2}$

TYPE 37b Finding Equation of Tangent or Normal – Finding Equations of Tangent and Normal at Given Point**Given:** Curves and points specified in parts (i) through (v).

Concept / Formula Used: $y - y_1 = m_T(x - x_1)$ $m_T = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$ $y - y_1 = m_N(x - x_1)$
 $m_N = -\frac{1}{m_T}$

Solution:(i) $y = x^3$ at $(1, 1)$:

$$\frac{dy}{dx} = 3x^2 \implies m_T = 3(1)^2 = 3.$$

$$\text{Slope of normal } m_N = -\frac{1}{3}.$$

$$\text{Tangent: } y - 1 = 3(x - 1)$$

$$y - 1 = 3x - 3 \implies 3x - y - 2 = 0$$

$$\text{Normal: } y - 1 = -\frac{1}{3}(x - 1)$$

$$3y - 3 = -x + 1 \implies x + 3y - 4 = 0$$

(ii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(1, 3)$:

$$\frac{dy}{dx} = 4x^3 - 18x^2 + 26x - 10.$$

$$\text{At } x = 1: m_T = 4(1)^3 - 18(1)^2 + 26(1) - 10 = 4 - 18 + 26 - 10 = 2.$$

$$\text{Slope of normal } m_N = -\frac{1}{2}.$$

$$\text{Tangent: } y - 3 = 2(x - 1)$$

$$y - 3 = 2x - 2 \implies 2x - y + 1 = 0$$

$$\text{Normal: } y - 3 = -\frac{1}{2}(x - 1)$$

$$2y - 6 = -x + 1 \implies x + 2y - 7 = 0$$

(iii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$:

$$\text{At } x = 0: m_T = 4(0)^3 - 18(0)^2 + 26(0) - 10 = -10.$$

$$\text{Slope of normal } m_N = -\frac{1}{-10} = \frac{1}{10}.$$

$$\text{Tangent: } y - 5 = -10(x - 0)$$

$$y - 5 = -10x \implies 10x + y - 5 = 0$$

$$\text{Normal: } y - 5 = \frac{1}{10}(x - 0)$$

$$10y - 50 = x \implies x - 10y + 50 = 0$$

(iv) $y = x^2$ at $(0, 0)$:

$$\frac{dy}{dx} = 2x \implies m_T = 2(0) = 0.$$

Since tangent has slope 0, it is horizontal; normal is vertical.

$$\text{Tangent: } y - 0 = 0(x - 0) \implies y = 0$$

$$\text{Normal: } x = 0$$

(v) $y = \sin^2 x$ at $x = \frac{\pi}{2}$:

First find y : $y = \sin^2\left(\frac{\pi}{2}\right) = (1)^2 = 1$. Point is $\left(\frac{\pi}{2}, 1\right)$.

$$\frac{dy}{dx} = 2 \sin x \cos x = \sin 2x.$$

$$\text{At } x = \frac{\pi}{2}: m_T = \sin\left(2 \cdot \frac{\pi}{2}\right) = \sin \pi = 0.$$

Since $m_T = 0$, tangent is horizontal and normal is vertical.

$$\text{Tangent: } y = 1 \implies y - 1 = 0$$

$$\text{Normal: } x = \frac{\pi}{2} \implies 2x - \pi = 0$$

Final Answer

(i) Tangent: $3x - y - 2 = 0$; Normal: $x + 3y - 4 = 0$

(ii) Tangent: $2x - y + 1 = 0$; Normal: $x + 2y - 7 = 0$

(iii) Tangent: $10x + y - 5 = 0$; Normal: $x - 10y + 50 = 0$

(iv) Tangent: $y = 0$; Normal: $x = 0$

(v) Tangent: $y = 1$; Normal: $x = \frac{\pi}{2}$

Question 20

Find the equation of the normal to the curve $ay^2 = x^3$ at the point (am^2, am^3) .

TYPE 37c Finding Equation of Tangent or Normal – Finding Equation of Normal to Curve at Parametric Point

Given: Curve equation $ay^2 = x^3$ and point (am^2, am^3) .

Concept / Formula Used: $m_N = -\frac{1}{\frac{dy}{dx}\big|_P}$ $y - y_1 = m_N(x - x_1)$

Solution:

Differentiating $ay^2 = x^3$ implicitly with respect to x :

$$\frac{d}{dx}(ay^2) = \frac{d}{dx}(x^3)$$

$$2ay \frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dx} = \frac{3x^2}{2ay}$$

Evaluating the slope of tangent m_T at (am^2, am^3) :

$$\begin{aligned} m_T &= \frac{3(am^2)^2}{2a(am^3)} \\ &= \frac{3a^2m^4}{2a^2m^3} = \frac{3m}{2} \end{aligned}$$

Therefore, the slope of the normal m_N is:

$$m_N = -\frac{1}{m_T} = -\frac{2}{3m}$$

Equation of the normal line at (am^2, am^3) :

$$\begin{aligned} y - am^3 &= -\frac{2}{3m}(x - am^2) \\ 3m(y - am^3) &= -2(x - am^2) \\ 3my - 3am^4 &= -2x + 2am^2 \\ 2x + 3my - 3am^4 - 2am^2 &= 0 \\ 2x + 3my - am^2(2 + 3m^2) &= 0 \end{aligned}$$

Final Answer

The equation of the normal is $2x + 3my - am^2(2 + 3m^2) = 0$.

Question 21

- (i) Find the equation of the tangent to the curve $\sqrt{x} + \sqrt{y} = a$ at the point $\left(\frac{a^2}{4}, \frac{a^2}{4}\right)$.
 (ii) Find the equations of the tangent and the normal to the curve $x^{2/3} + y^{2/3} = 2$ at $(1, 1)$.

TYPE 37d Finding Equation of Tangent or Normal – Finding Equations of Tangent and Normal for Implicit Curves

Given: Curves and points for parts (i) and (ii).

Concept / Formula Used: $\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}$

Solution:

- (i) Given curve: $\sqrt{x} + \sqrt{y} = a$

Differentiating implicitly with respect to x :

$$\begin{aligned} \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} &= 0 \\ \frac{1}{2\sqrt{y}} \frac{dy}{dx} &= -\frac{1}{2\sqrt{x}} \\ \frac{dy}{dx} &= -\frac{\sqrt{y}}{\sqrt{x}} \end{aligned}$$

Evaluating m_T at $\left(\frac{a^2}{4}, \frac{a^2}{4}\right)$:

$$m_T = -\frac{\sqrt{a^2/4}}{\sqrt{a^2/4}} = -1$$

Equation of tangent:

$$y - \frac{a^2}{4} = -1 \left(x - \frac{a^2}{4} \right)$$

$$y - \frac{a^2}{4} = -x + \frac{a^2}{4}$$

$$x + y = \frac{a^2}{4} + \frac{a^2}{4} = \frac{a^2}{2}$$

(ii) Given curve: $x^{2/3} + y^{2/3} = 2$

Differentiating implicitly with respect to x :

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} = 0$$

$$\frac{2}{3}y^{-1/3}\frac{dy}{dx} = -\frac{2}{3}x^{-1/3}$$

$$\frac{dy}{dx} = -\frac{x^{-1/3}}{y^{-1/3}} = -\left(\frac{y}{x}\right)^{1/3}$$

Evaluating at $(1, 1)$:

$$m_T = -\left(\frac{1}{1}\right)^{1/3} = -1$$

Slope of normal:

$$m_N = -\frac{1}{-1} = 1$$

Equation of tangent:

$$y - 1 = -1(x - 1)$$

$$y - 1 = -x + 1 \implies x + y - 2 = 0$$

Equation of normal:

$$y - 1 = 1(x - 1)$$

$$y - 1 = x - 1 \implies x - y = 0$$

Final Answer

(i) Tangent: $x + y = \frac{a^2}{2}$

(ii) Tangent: $x + y - 2 = 0$; Normal: $x - y = 0$

Question 22

(i) Find the equation of tangent to the curve given by $x = a \sin^3 t$, $y = b \cos^3 t$ at a point where $t = \frac{\pi}{2}$.

(ii) Find the equation of the tangent to the curve $x = \sin 3t$, $y = \cos 2t$ at $t = \frac{\pi}{4}$.

Given: Parametric curves and parameter values for parts (i) and (ii).

Concept / Formula Used: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

Solution:

(i) Given $x = a \sin^3 t$ and $y = b \cos^3 t$ at $t = \frac{\pi}{2}$:

First find coordinates of the point at $t = \frac{\pi}{2}$:

$$x = a \sin^3 \left(\frac{\pi}{2} \right) = a(1)^3 = a$$

$$y = b \cos^3 \left(\frac{\pi}{2} \right) = b(0)^3 = 0$$

The point is $(a, 0)$. Now calculate derivatives with respect to t :

$$\frac{dx}{dt} = 3a \sin^2 t \cos t$$

$$\frac{dy}{dt} = 3b \cos^2 t (-\sin t) = -3b \cos^2 t \sin t$$

$$\frac{dy}{dx} = \frac{-3b \cos^2 t \sin t}{3a \sin^2 t \cos t} = -\frac{b \cos t}{a \sin t} = -\frac{b}{a} \cot t$$

At $t = \frac{\pi}{2}$:

$$m_T = -\frac{b}{a} \cot \left(\frac{\pi}{2} \right) = -\frac{b}{a}(0) = 0$$

Equation of tangent at $(a, 0)$ with slope 0:

$$y - 0 = 0(x - a)$$

$$y = 0$$

(ii) Given $x = \sin 3t$ and $y = \cos 2t$ at $t = \frac{\pi}{4}$:

First find coordinates of the point at $t = \frac{\pi}{4}$:

$$x = \sin \left(\frac{3\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$y = \cos \left(2 \cdot \frac{\pi}{4} \right) = \cos \left(\frac{\pi}{2} \right) = 0$$

The point is $\left(\frac{1}{\sqrt{2}}, 0 \right)$. Now calculate derivatives:

$$\frac{dx}{dt} = 3 \cos 3t$$

$$\frac{dy}{dt} = -2 \sin 2t$$

$$\frac{dy}{dx} = \frac{-2 \sin 2t}{3 \cos 3t}$$

At $t = \frac{\pi}{4}$:

$$\sin 2t = \sin \left(\frac{\pi}{2} \right) = 1$$

$$\cos 3t = \cos \left(\frac{3\pi}{4} \right) = -\frac{1}{\sqrt{2}}$$

$$m_T = \frac{-2(1)}{3(-1/\sqrt{2})} = \frac{2\sqrt{2}}{3}$$

Equation of tangent at $\left(\frac{1}{\sqrt{2}}, 0 \right)$:

$$y - 0 = \frac{2\sqrt{2}}{3} \left(x - \frac{1}{\sqrt{2}} \right)$$

$$y = \frac{2\sqrt{2}}{3}x - \frac{2}{3}$$

$$3y = 2\sqrt{2}x - 2$$

$$2\sqrt{2}x - 3y - 2 = 0$$

Final Answer

- (i) Tangent equation: $y = 0$
 (ii) Tangent equation: $2\sqrt{2}x - 3y - 2 = 0$

Question 23

- (i) Find the equations of all lines having slope 2 and that are tangents to the curve

$$y = \frac{1}{x-3}.$$

- (ii) Find the equation of the tangent to the curve $y = x + \frac{4}{x^2}$ which is parallel to x -axis.

TYPE 37f Finding Equation of Tangent or Normal – Finding Equations of Tangents with Given Slope or Parallelism

Given: Curves and slope conditions in parts (i) and (ii).

Concept / Formula Used: $\frac{dy}{dx} = m$

Solution:

- (i) Curve $y = \frac{1}{x-3} = (x-3)^{-1}$:

Differentiating with respect to x :

$$\frac{dy}{dx} = -(x-3)^{-2} = -\frac{1}{(x-3)^2}$$

Set slope equal to 2:

$$-\frac{1}{(x-3)^2} = 2$$

$$(x-3)^2 = -\frac{1}{2}$$

Since $(x - 3)^2 \geq 0$ for all real numbers x , there is no real solution for x . Thus, no such tangent lines exist.

- (ii) Curve $y = x + \frac{4}{x^2} = x + 4x^{-2}$:

Differentiating with respect to x :

$$\frac{dy}{dx} = 1 - 8x^{-3} = 1 - \frac{8}{x^3}$$

For tangent parallel to x -axis, $\frac{dy}{dx} = 0$:

$$1 - \frac{8}{x^3} = 0$$

$$\frac{8}{x^3} = 1 \implies x^3 = 8 \implies x = 2$$

Substituting $x = 2$ into curve equation:

$$\begin{aligned} y &= 2 + \frac{4}{2^2} \\ &= 2 + \frac{4}{4} = 2 + 1 = 3 \end{aligned}$$

Since the tangent line has slope 0 and passes through $(2, 3)$, its equation is:

$$y - 3 = 0(x - 2) \implies y = 3$$

Final Answer

- (i) No such line exists having slope 2.
(ii) Equation of tangent: $y = 3$

Question 24

Find the equation of the tangent to the curve $y = x^2 - 2x + 7$ which is

- (i) parallel to the line $2x - y + 9 = 0$
(ii) perpendicular to the line $5y - 15x = 13$.

TYPE 42a Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Tangents Parallel or

Given: Curve $y = x^2 - 2x + 7$ and lines in parts (i) and (ii).

Concept / Formula Used: $m_T = \frac{dy}{dx}$ $m_T = m_L$ $m_T = -\frac{1}{m_L}$

Solution:

Differentiating $y = x^2 - 2x + 7$ with respect to x :

$$\frac{dy}{dx} = 2x - 2$$

- (i) Line equation: $2x - y + 9 = 0 \implies y = 2x + 9$.

Slope of line $m_L = 2$.

Since tangent is parallel, $m_T = 2$:

$$2x - 2 = 2$$

$$2x = 4 \implies x = 2$$

When $x = 2$:

$$y = (2)^2 - 2(2) + 7 = 4 - 4 + 7 = 7$$

Equation of tangent at $(2, 7)$ with slope 2:

$$y - 7 = 2(x - 2)$$

$$y - 7 = 2x - 4 \implies 2x - y + 3 = 0$$

(ii) Line equation: $5y - 15x = 13 \implies 5y = 15x + 13 \implies y = 3x + \frac{13}{5}$.

Slope of line $m_L = 3$.

Since tangent is perpendicular, $m_T = -\frac{1}{3}$:

$$2x - 2 = -\frac{1}{3}$$

$$2x = 2 - \frac{1}{3} = \frac{5}{3}$$

$$x = \frac{5}{6}$$

When $x = \frac{5}{6}$:

$$\begin{aligned} y &= \left(\frac{5}{6}\right)^2 - 2\left(\frac{5}{6}\right) + 7 \\ &= \frac{25}{36} - \frac{10}{6} + 7 \\ &= \frac{25 - 60 + 252}{36} = \frac{217}{36} \end{aligned}$$

Equation of tangent at $\left(\frac{5}{6}, \frac{217}{36}\right)$ with slope $-\frac{1}{3}$:

$$y - \frac{217}{36} = -\frac{1}{3}\left(x - \frac{5}{6}\right)$$

$$36y - 217 = -12\left(x - \frac{5}{6}\right)$$

$$36y - 217 = -12x + 10$$

$$12x + 36y - 227 = 0$$

Final Answer

(i) $2x - y + 3 = 0$

(ii) $12x + 36y - 227 = 0$

Question 25

(i) Find the equations of the tangents to the curve $y = 4x^3 - 3x + 5$ which are perpendicular to the line $9y + x + 3 = 0$.

(ii) Find the equations of the tangents to the curve $3x^2 - y^2 = 2$ which are perpendicular to the line $x + 3y = 2$.

Given: Curves and perpendicular line equations given in parts (i) and (ii).

Concept / Formula Used: $m_T = -\frac{1}{m_L}$

Solution:

(i) Line equation: $9y + x + 3 = 0 \implies 9y = -x - 3 \implies y = -\frac{1}{9}x - \frac{1}{3}$.

Slope of line $m_L = -\frac{1}{9}$.

Slope of perpendicular tangent $m_T = -\frac{1}{-1/9} = 9$.

Differentiating $y = 4x^3 - 3x + 5$:

$$\frac{dy}{dx} = 12x^2 - 3$$

Setting $\frac{dy}{dx} = 9$:

$$12x^2 - 3 = 9$$

$$12x^2 = 12 \implies x^2 = 1 \implies x = \pm 1$$

Case 1: $x = 1 \implies y = 4(1)^3 - 3(1) + 5 = 6$. Point is $(1, 6)$.

$$y - 6 = 9(x - 1)$$

$$y - 6 = 9x - 9 \implies 9x - y - 3 = 0$$

Case 2: $x = -1 \implies y = 4(-1)^3 - 3(-1) + 5 = -4 + 3 + 5 = 4$. Point is $(-1, 4)$.

$$y - 4 = 9(x - (-1))$$

$$y - 4 = 9x + 9 \implies 9x - y + 13 = 0$$

(ii) Line equation: $x + 3y = 2 \implies 3y = -x + 2 \implies y = -\frac{1}{3}x + \frac{2}{3}$.

Slope of line $m_L = -\frac{1}{3}$.

Slope of perpendicular tangent $m_T = -\frac{1}{-1/3} = 3$.

Differentiating $3x^2 - y^2 = 2$ implicitly with respect to x :

$$6x - 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = 6x \implies \frac{dy}{dx} = \frac{3x}{y}$$

Setting $\frac{dy}{dx} = 3$:

$$\frac{3x}{y} = 3 \implies y = x$$

Substituting $y = x$ into hyperbola $3x^2 - y^2 = 2$:

$$3x^2 - x^2 = 2$$

$$2x^2 = 2 \implies x^2 = 1 \implies x = \pm 1$$

Case 1: $x = 1 \Rightarrow y = 1$. Point is $(1, 1)$.

$$y - 1 = 3(x - 1)$$

$$y - 1 = 3x - 3 \Rightarrow 3x - y - 2 = 0$$

Case 2: $x = -1 \Rightarrow y = -1$. Point is $(-1, -1)$.

$$y - (-1) = 3(x - (-1))$$

$$y + 1 = 3x + 3 \Rightarrow 3x - y + 2 = 0$$

Final Answer

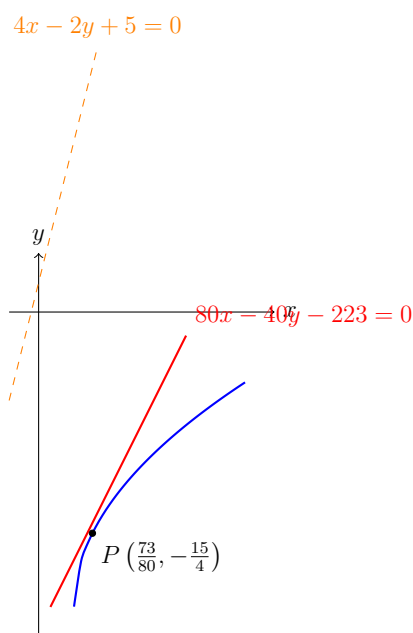
(i) $9x - y - 3 = 0$ and $9x - y + 13 = 0$

(ii) $3x - y - 2 = 0$ and $3x - y + 2 = 0$

Question 26

Find the equation of the tangent line to the curve $y = \sqrt{5x - 3} - 5$, which is parallel to the line $4x - 2y + 5 = 0$.

TYPE 42a Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Tangents Parallel to



Tangent to $y = \sqrt{5x - 3} - 5$ parallel to the line $4x - 2y + 5 = 0$

Given: Curve $y = \sqrt{5x - 3} - 5$ and parallel line $4x - 2y + 5 = 0$.

Concept / Formula Used: $m = \frac{dy}{dx}$ $m_1 = m_2$ (x_1, y_1) $y - y_1 = m(x - x_1)$

Solution:

Given the equation of the line:

$$4x - 2y + 5 = 0$$

$$2y = 4x + 5$$

$$y = 2x + \frac{5}{2}$$

Comparing with $y = mx + c$, the slope of the given line is:

$$m_L = 2$$

Now, given the equation of the curve:

$$y = \sqrt{5x - 3} - 5$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} [(5x - 3)^{1/2} - 5] \\ &= \frac{1}{2}(5x - 3)^{-1/2} \cdot \frac{d}{dx}(5x - 3) - 0 \\ &= \frac{1}{2\sqrt{5x - 3}} \cdot 5 \\ &= \frac{5}{2\sqrt{5x - 3}}\end{aligned}$$

Since the tangent line is parallel to the given line, their slopes must be equal:

$$\begin{aligned}\frac{dy}{dx} &= 2 \\ \frac{5}{2\sqrt{5x - 3}} &= 2 \\ 5 &= 4\sqrt{5x - 3} \\ \sqrt{5x - 3} &= \frac{5}{4}\end{aligned}$$

Squaring both sides:

$$\begin{aligned}5x - 3 &= \left(\frac{5}{4}\right)^2 \\ 5x - 3 &= \frac{25}{16} \\ 5x &= \frac{25}{16} + 3 \\ 5x &= \frac{25 + 48}{16} \\ 5x &= \frac{73}{16} \\ x &= \frac{73}{80}\end{aligned}$$

Substitute $x = \frac{73}{80}$ into the curve equation to find y :

$$\begin{aligned}y &= \sqrt{5\left(\frac{73}{80}\right) - 3} - 5 \\ &= \sqrt{\frac{73}{16}} - 3 - 5 \\ &= \sqrt{\frac{25}{16}} - 5 \\ &= \frac{5}{4} - 5 \\ &= -\frac{15}{4}\end{aligned}$$

Thus, the point of contact is $P\left(\frac{73}{80}, -\frac{15}{4}\right)$ and the slope of the tangent is $m = 2$.

Using the point-slope form $y - y_1 = m(x - x_1)$:

$$\begin{aligned} y - \left(-\frac{15}{4}\right) &= 2\left(x - \frac{73}{80}\right) \\ y + \frac{15}{4} &= 2x - \frac{73}{40} \end{aligned}$$

Multiplying the entire equation by 40:

$$\begin{aligned} 40y + 150 &= 80x - 73 \\ 80x - 40y - 73 - 150 &= 0 \\ 80x - 40y - 223 &= 0 \end{aligned}$$

Final Answer

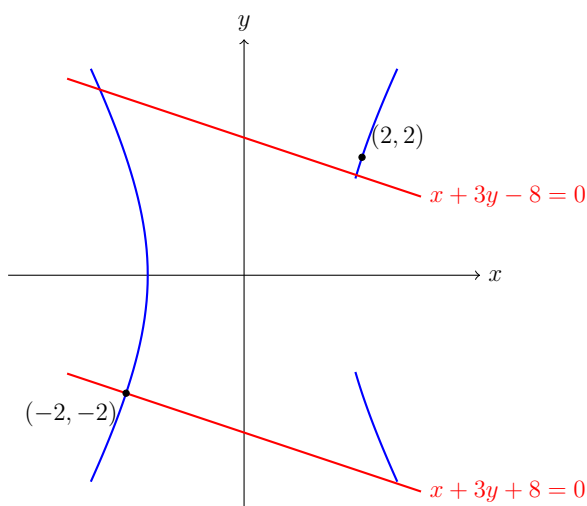
$$80x - 40y - 223 = 0$$

Question 27

Find the equations of the normals to the curve $3x^2 - y^2 = 8$ which are parallel to the line $x + 3y = 4$.

TYPE 42b Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Normals Parallel o

★ Likely Exam Question



Normals to hyperbola $3x^2 - y^2 = 8$ parallel to $x + 3y = 4$

Given: Curve $3x^2 - y^2 = 8$ and parallel line $x + 3y = 4$.

Concept / Formula Used: $m_N = -\frac{1}{\frac{dy}{dx}}$

Solution:

Given the line equation:

$$\begin{aligned}x + 3y &= 4 \\3y &= -x + 4 \\y &= -\frac{1}{3}x + \frac{4}{3}\end{aligned}$$

The slope of the given line is:

$$m_L = -\frac{1}{3}$$

Given the curve equation:

$$3x^2 - y^2 = 8$$

Differentiating both sides implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(3x^2) - \frac{d}{dx}(y^2) &= \frac{d}{dx}(8) \\6x - 2y \frac{dy}{dx} &= 0 \\2y \frac{dy}{dx} &= 6x \\\frac{dy}{dx} &= \frac{3x}{y}\end{aligned}$$

The slope of the normal at any point (x, y) is:

$$m_N = -\frac{1}{\frac{dy}{dx}} = -\frac{y}{3x}$$

Since the normal is parallel to the line $x + 3y = 4$:

$$\begin{aligned}m_N &= m_L \\-\frac{y}{3x} &= -\frac{1}{3} \\\frac{y}{x} &= 1 \\y &= x\end{aligned}$$

Substitute $y = x$ into the curve equation $3x^2 - y^2 = 8$:

$$\begin{aligned}3x^2 - x^2 &= 8 \\2x^2 &= 8 \\x^2 &= 4 \\x &= \pm 2\end{aligned}$$

When $x = 2$, $y = 2$, giving the point $(2, 2)$. When $x = -2$, $y = -2$, giving the point $(-2, -2)$.

Case 1: Normal at $(2, 2)$ with slope $m = -\frac{1}{3}$:

$$\begin{aligned}y - 2 &= -\frac{1}{3}(x - 2) \\3(y - 2) &= -(x - 2) \\3y - 6 &= -x + 2 \\x + 3y - 8 &= 0\end{aligned}$$

Case 2: Normal at $(-2, -2)$ with slope $m = -\frac{1}{3}$:

$$\begin{aligned}y - (-2) &= -\frac{1}{3}(x - (-2)) \\y + 2 &= -\frac{1}{3}(x + 2) \\3(y + 2) &= -(x + 2) \\3y + 6 &= -x - 2 \\x + 3y + 8 &= 0\end{aligned}$$

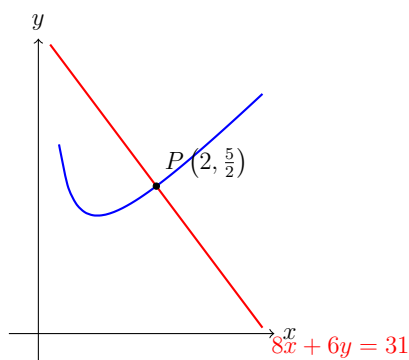
Final Answer

$$x + 3y - 8 = 0 \quad \text{and} \quad x + 3y + 8 = 0$$

Question 28

Find the equation of the normal to the curve $y = x + \frac{1}{x}$, $x > 0$, perpendicular to the line $3x - 4y = 7$.

TYPE 42b Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Normals Parallel o



Normal to $y = x + \frac{1}{x}$ perpendicular to line $3x - 4y = 7$

Given: Curve $y = x + \frac{1}{x}$ for $x > 0$, and perpendicular line $3x - 4y = 7$.

Concept / Formula Used: $m_1 \cdot m_2 = -1$ $m_N = -\frac{1}{\frac{dy}{dx}}$

Solution:

Given the equation of the line:

$$\begin{aligned}3x - 4y &= 7 \\4y &= 3x - 7 \\y &= \frac{3}{4}x - \frac{7}{4}\end{aligned}$$

The slope of the given line is:

$$m_L = \frac{3}{4}$$

Given the curve equation for $x > 0$:

$$y = x + \frac{1}{x}$$

Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x) + \frac{d}{dx}(x^{-1}) \\ &= 1 - \frac{1}{x^2} \\ &= \frac{x^2 - 1}{x^2}\end{aligned}$$

The slope of the normal m_N to the curve is:

$$m_N = -\frac{1}{\frac{dy}{dx}} = -\frac{x^2}{x^2 - 1}$$

Since the normal is perpendicular to the given line:

$$\begin{aligned}m_N \cdot m_L &= -1 \\ \left(-\frac{x^2}{x^2 - 1}\right) \cdot \left(\frac{3}{4}\right) &= -1 \\ \frac{3x^2}{4(x^2 - 1)} &= 1 \\ 3x^2 &= 4x^2 - 4 \\ 4x^2 - 3x^2 &= 4 \\ x^2 &= 4\end{aligned}$$

Since $x > 0$, we choose:

$$x = 2$$

Substitute $x = 2$ into the curve equation:

$$\begin{aligned}y &= 2 + \frac{1}{2} \\ &= \frac{5}{2}\end{aligned}$$

The point of contact is $\left(2, \frac{5}{2}\right)$. The slope of the normal is:

$$m_N = -\frac{2^2}{2^2 - 1} = -\frac{4}{3}$$

Now, equation of the normal line at $\left(2, \frac{5}{2}\right)$:

$$\begin{aligned}y - \frac{5}{2} &= -\frac{4}{3}(x - 2) \\ 3\left(y - \frac{5}{2}\right) &= -4(x - 2) \\ 3y - \frac{15}{2} &= -4x + 8 \\ 4x + 3y - \frac{15}{2} - 8 &= 0 \\ 4x + 3y - \frac{31}{2} &= 0 \\ 8x + 6y - 31 &= 0\end{aligned}$$

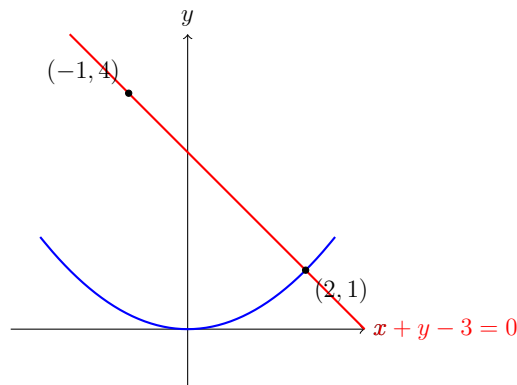
Final Answer

$$8x + 6y = 31$$

Question 29

Find the equation of the normal to the curve $x^2 = 4y$ which passes through the point $(-1, 4)$.

TYPE 37g Finding Equation of Tangent or Normal – Finding Equation of Normal Passing Through External Point



$$\text{Normal to } x^2 = 4y \text{ passing through } (-1, 4)$$

Given: Parabola $x^2 = 4y$ and external point $(-1, 4)$ through which the normal passes.

Concept / Formula Used: $m_T = \frac{dy}{dx}$ $m_N = -\frac{1}{m_T}$ (x_1, y_1) $y - y_1 = m_N(x - x_1)$

Solution:

Let (x_1, y_1) be the point of contact on the curve $x^2 = 4y$. Thus, (x_1, y_1) satisfies the curve:

$$x_1^2 = 4y_1 \implies y_1 = \frac{x_1^2}{4}$$

Differentiating the curve equation $x^2 = 4y$ with respect to x :

$$\begin{aligned} 2x &= 4 \frac{dy}{dx} \\ \frac{dy}{dx} &= \frac{x}{2} \end{aligned}$$

The slope of the tangent at (x_1, y_1) is $\left. \frac{dy}{dx} \right|_{(x_1, y_1)} = \frac{x_1}{2}$. Therefore, the slope of the normal at (x_1, y_1) is:

$$m_N = -\frac{2}{x_1}$$

The equation of the normal line at (x_1, y_1) is:

$$\begin{aligned} y - y_1 &= m_N(x - x_1) \\ y - \frac{x_1^2}{4} &= -\frac{2}{x_1}(x - x_1) \end{aligned}$$

Since this normal passes through the point $(-1, 4)$, we substitute $x = -1$ and $y = 4$:

$$4 - \frac{x_1^2}{4} = -\frac{2}{x_1}(-1 - x_1)$$

$$4 - \frac{x_1^2}{4} = \frac{2(1 + x_1)}{x_1}$$

Multiplying both sides by $4x_1$:

$$4x_1 \left(4 - \frac{x_1^2}{4} \right) = 4x_1 \cdot \frac{2(1 + x_1)}{x_1}$$

$$16x_1 - x_1^3 = 8(1 + x_1)$$

$$16x_1 - x_1^3 = 8 + 8x_1$$

$$x_1^3 - 8x_1 + 8 = 0$$

To solve $x_1^3 - 8x_1 + 8 = 0$, we test integer roots: For $x_1 = 2$:

$$2^3 - 8(2) + 8 = 8 - 16 + 8 = 0$$

Thus, $x_1 = 2$ is a root. Factoring out $(x_1 - 2)$:

$$(x_1 - 2)(x_1^2 + 2x_1 - 4) = 0$$

Using $x_1 = 2$:

$$y_1 = \frac{2^2}{4} = 1$$

The point of contact is $(2, 1)$, and the slope of the normal is:

$$m_N = -\frac{2}{2} = -1$$

Substitute $(x_1, y_1) = (2, 1)$ and $m_N = -1$ into the normal line equation:

$$y - 1 = -1(x - 2)$$

$$y - 1 = -x + 2$$

$$x + y - 3 = 0$$

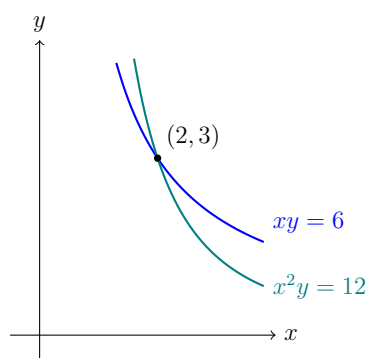
Final Answer

$$x + y - 3 = 0$$

Question 30 (i)

Find the angle of intersection between the curves $xy = 6$ and $x^2y = 12$.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves



Intersection of curves $xy = 6$ and $x^2y = 12$

Given: Curves $C_1 : xy = 6$ and $C_2 : x^2y = 12$.

Concept / Formula Used: θ m_1, m_2 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, we find the point(s) of intersection:

$$C_1 : xy = 6 \implies y = \frac{6}{x}$$

$$C_2 : x^2y = 12$$

Substitute $y = \frac{6}{x}$ into C_2 :

$$\begin{aligned} x^2 \left(\frac{6}{x} \right) &= 12 \\ 6x &= 12 \\ x &= 2 \end{aligned}$$

Substituting $x = 2$ back into $xy = 6$:

$$\begin{aligned} 2y &= 6 \\ y &= 3 \end{aligned}$$

The point of intersection is $P(2, 3)$.

Now, find the slopes of the tangents to both curves at $P(2, 3)$:

For curve $C_1 : xy = 6$: Differentiating implicitly with respect to x :

$$\begin{aligned} y + x \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{y}{x} \end{aligned}$$

At $P(2, 3)$, slope m_1 :

$$m_1 = -\frac{3}{2}$$

For curve $C_2 : x^2y = 12$: Differentiating implicitly with respect to x :

$$\begin{aligned} 2xy + x^2 \frac{dy}{dx} &= 0 \\ x^2 \frac{dy}{dx} &= -2xy \\ \frac{dy}{dx} &= -\frac{2y}{x} \end{aligned}$$

At $P(2, 3)$, slope m_2 :

$$m_2 = -\frac{2(3)}{2} = -3$$

Now compute the angle of intersection θ :

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{-\frac{3}{2} - (-3)}{1 + \left(-\frac{3}{2}\right)(-3)} \right| \\ &= \left| \frac{-\frac{3}{2} + 3}{1 + \frac{9}{2}} \right| \\ &= \left| \frac{\frac{3}{2}}{\frac{11}{2}} \right| \\ &= \frac{3}{11} \end{aligned}$$

Taking inverse tangent:

$$\theta = \tan^{-1} \left(\frac{3}{11} \right)$$

Final Answer

$$\theta = \tan^{-1} \left(\frac{3}{11} \right)$$

Question 30 (ii)

Find the angle of intersection between the curves $y = 4 - x^2$ and $y = x^2$.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves

★ Likely Exam Question

Given: Curves $C_1 : y = 4 - x^2$ and $C_2 : y = x^2$.

Concept / Formula Used: θ m_1, m_2 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, find the points of intersection by equating the two curves:

$$\begin{aligned}4 - x^2 &= x^2 \\2x^2 &= 4 \\x^2 &= 2 \\x &= \pm\sqrt{2}\end{aligned}$$

When $x = \sqrt{2}$, $y = (\sqrt{2})^2 = 2$, giving point $P_1(\sqrt{2}, 2)$. When $x = -\sqrt{2}$, $y = (-\sqrt{2})^2 = 2$, giving point $P_2(-\sqrt{2}, 2)$.

Differentiating $C_1 : y = 4 - x^2$:

$$\frac{dy}{dx} = -2x$$

Differentiating $C_2 : y = x^2$:

$$\frac{dy}{dx} = 2x$$

At $P_1(\sqrt{2}, 2)$:

$$\begin{aligned}m_1 &= -2\sqrt{2} \\m_2 &= 2\sqrt{2}\end{aligned}$$

Calculate $\tan \theta$:

$$\begin{aligned}\tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\&= \left| \frac{-2\sqrt{2} - 2\sqrt{2}}{1 + (-2\sqrt{2})(2\sqrt{2})} \right| \\&= \left| \frac{-4\sqrt{2}}{1 - 8} \right| \\&= \left| \frac{-4\sqrt{2}}{-7} \right| \\&= \frac{4\sqrt{2}}{7}\end{aligned}$$

Thus, $\theta = \tan^{-1} \left(\frac{4\sqrt{2}}{7} \right)$. By symmetry, the angle at $P_2(-\sqrt{2}, 2)$ is identical.

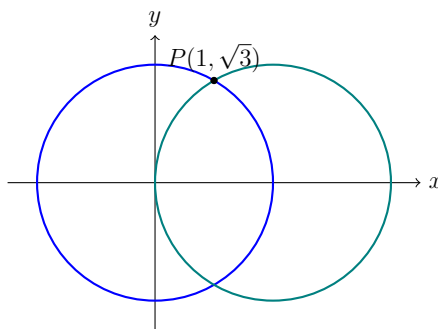
Final Answer

$$\theta = \tan^{-1} \left(\frac{4\sqrt{2}}{7} \right)$$

Question 31

Find the angle of intersection of the curves $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$ at the point in the first quadrant.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves



Intersection of circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$

Given: Circles $C_1 : x^2 + y^2 = 4$ and $C_2 : (x - 2)^2 + y^2 = 4$.

Concept / Formula Used: θ m_1, m_2 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, solve for the point of intersection in the first quadrant:

$$x^2 + y^2 = 4 \quad \text{--- (1)}$$

$$(x - 2)^2 + y^2 = 4 \quad \text{--- (2)}$$

Subtracting equation (2) from equation (1):

$$x^2 - (x - 2)^2 = 0$$

$$x^2 - (x^2 - 4x + 4) = 0$$

$$4x - 4 = 0$$

$$4x = 4$$

$$x = 1$$

Substitute $x = 1$ into equation (1):

$$1^2 + y^2 = 4$$

$$y^2 = 3$$

$$y = \sqrt{3} \quad (\text{since } y > 0 \text{ in the first quadrant})$$

The point of intersection in the first quadrant is $P(1, \sqrt{3})$.

Differentiating $C_1 : x^2 + y^2 = 4$ implicitly with respect to x :

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

At $P(1, \sqrt{3})$, slope m_1 :

$$m_1 = -\frac{1}{\sqrt{3}}$$

Differentiating $C_2 : (x - 2)^2 + y^2 = 4$ implicitly with respect to x :

$$2(x - 2) + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x - 2}{y} = \frac{2 - x}{y}$$

At $P(1, \sqrt{3})$, slope m_2 :

$$m_2 = \frac{2-1}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

Now compute $\tan \theta$:

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{-\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}}}{1 + \left(-\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right)} \right| \\ &= \left| \frac{-\frac{2}{\sqrt{3}}}{1 - \frac{1}{3}} \right| \\ &= \left| \frac{-\frac{2}{\sqrt{3}}}{\frac{2}{3}} \right| \\ &= \frac{2}{\sqrt{3}} \cdot \frac{3}{2} \\ &= \sqrt{3} \end{aligned}$$

Since $\tan \theta = \sqrt{3}$:

$$\theta = 60^\circ$$

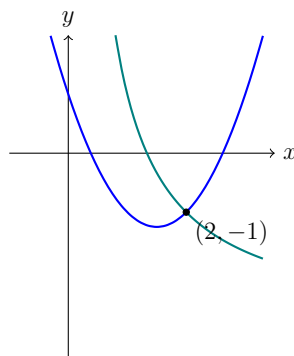
Final Answer

$\theta = 60^\circ$ (or $\frac{\pi}{3}$ radians)

Question 32

Prove that the curves $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ intersect at right angles at the point $(2, -1)$.

TYPE 45b Angle of Intersection and Orthogonality of Curves – Proving or Finding Condition for Orthogonal Intersection of



Orthogonal intersection of $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ at $(2, -1)$

To Prove: The curves $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ intersect orthogonally at $(2, -1)$, i.e., $m_1 \cdot m_2 = -1$.

Concept / Formula Used: $m_1 \cdot m_2 = -1$

Solution:

First, check that $(2, -1)$ lies on both curves: For $C_1 : y = x^2 - 3x + 1$:

$$\text{RHS} = 2^2 - 3(2) + 1 = 4 - 6 + 1 = -1 = \text{LHS}$$

For $C_2 : x(y + 3) = 4$:

$$\text{LHS} = 2(-1 + 3) = 2(2) = 4 = \text{RHS}$$

Thus, $(2, -1)$ is a point of intersection of both curves.

Next, calculate slope m_1 for curve $C_1 : y = x^2 - 3x + 1$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^2 - 3x + 1) \\ &= 2x - 3\end{aligned}$$

At point $(2, -1)$:

$$\begin{aligned}m_1 &= 2(2) - 3 \\ &= 4 - 3 \\ &= 1\end{aligned}$$

Now, calculate slope m_2 for curve $C_2 : x(y + 3) = 4$: Rewriting C_2 :

$$\begin{aligned}y + 3 &= \frac{4}{x} \\ y &= \frac{4}{x} - 3\end{aligned}$$

Differentiating with respect to x :

$$\frac{dy}{dx} = -\frac{4}{x^2}$$

At point $(2, -1)$:

$$\begin{aligned}m_2 &= -\frac{4}{2^2} \\ &= -\frac{4}{4} \\ &= -1\end{aligned}$$

Now, find the product of slopes m_1 and m_2 :

$$\begin{aligned}m_1 \cdot m_2 &= (1) \cdot (-1) \\ &= -1\end{aligned}$$

Since the product of the slopes of the tangents at $(2, -1)$ is -1 , the curves intersect at right angles.

Hence Proved

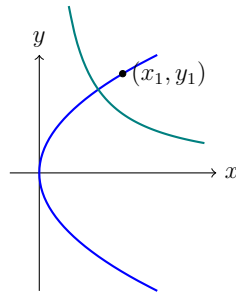
LHS = RHS. The given curves intersect orthogonally at $(2, -1)$. Hence proved.

Question 33

Find the condition that the curves $2x = y^2$ and $2xy = k$ intersect orthogonally.

TYPE 45b Angle of Intersection and Orthogonality of Curves – Proving or Finding Condition for Orthogonal Intersection of

★ Likely Exam Question



Orthogonal intersection of $y^2 = 2x$ and $2xy = k$

Given: Curves $C_1 : y^2 = 2x$ and $C_2 : 2xy = k$.

Concept / Formula Used: $m_1 \cdot m_2 = -1$ (x_1, y_1)

Solution:

Let (x_1, y_1) be a point of intersection of curves C_1 and C_2 . Then (x_1, y_1) satisfies both equations:

$$y_1^2 = 2x_1 \quad \text{--- (1)}$$

$$2x_1y_1 = k \quad \text{--- (2)}$$

Substitute $2x_1 = y_1^2$ from equation (1) into equation (2):

$$(y_1^2)y_1 = k$$

$$y_1^3 = k$$

$$y_1 = k^{1/3}$$

From equation (1):

$$2x_1 = (k^{1/3})^2$$

$$2x_1 = k^{2/3}$$

$$x_1 = \frac{k^{2/3}}{2}$$

Differentiating $C_1 : y^2 = 2x$ with respect to x :

$$2y \frac{dy}{dx} = 2$$

$$\frac{dy}{dx} = \frac{1}{y}$$

Slope of tangent m_1 at (x_1, y_1) :

$$m_1 = \frac{1}{y_1}$$

Differentiating $C_2 : 2xy = k \implies xy = \frac{k}{2}$ with respect to x :

$$y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

Slope of tangent m_2 at (x_1, y_1) :

$$m_2 = -\frac{y_1}{x_1}$$

For the curves to intersect orthogonally:

$$\begin{aligned} m_1 \cdot m_2 &= -1 \\ \left(\frac{1}{y_1}\right) \cdot \left(-\frac{y_1}{x_1}\right) &= -1 \\ -\frac{1}{x_1} &= -1 \\ x_1 &= 1 \end{aligned}$$

Substitute $x_1 = 1$ into $x_1 = \frac{k^{2/3}}{2}$:

$$\begin{aligned} 1 &= \frac{k^{2/3}}{2} \\ k^{2/3} &= 2 \end{aligned}$$

Cubing both sides:

$$\begin{aligned} (k^{2/3})^3 &= 2^3 \\ k^2 &= 8 \end{aligned}$$

Final Answer

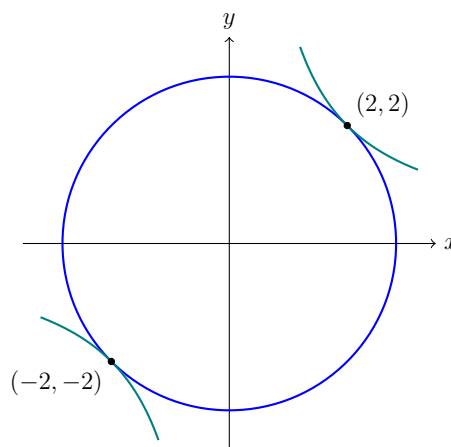
$$k^2 = 8$$

Question 34

Prove that the curves $xy = 4$ and $x^2 + y^2 = 8$ touch each other.

TYPE 45c Angle of Intersection and Orthogonality of Curves – Proving Two Curves Touch Each Other

★ Likely Exam Question



Curves $xy = 4$ and $x^2 + y^2 = 8$ touching at $(2, 2)$ and $(-2, -2)$

To Prove: The curves $xy = 4$ and $x^2 + y^2 = 8$ touch each other, i.e., they have equal tangent slopes ($m_1 = m_2$) at their common points of intersection.

Concept / Formula Used: $m_1 = m_2$

Solution:

First, find the points of intersection of $C_1 : xy = 4$ and $C_2 : x^2 + y^2 = 8$: From C_1 , $y = \frac{4}{x}$. Substitute into C_2 :

$$\begin{aligned}x^2 + \left(\frac{4}{x}\right)^2 &= 8 \\x^2 + \frac{16}{x^2} &= 8 \\x^4 + 16 &= 8x^2 \\x^4 - 8x^2 + 16 &= 0 \\(x^2 - 4)^2 &= 0 \\x^2 - 4 &= 0 \\x^2 &= 4 \\x &= \pm 2\end{aligned}$$

When $x = 2$, $y = \frac{4}{2} = 2$, giving point $P_1(2, 2)$. When $x = -2$, $y = \frac{4}{-2} = -2$, giving point $P_2(-2, -2)$.

Differentiating $C_1 : xy = 4$ with respect to x :

$$\begin{aligned}y + x \frac{dy}{dx} &= 0 \\\frac{dy}{dx} &= -\frac{y}{x}\end{aligned}$$

Differentiating $C_2 : x^2 + y^2 = 8$ with respect to x :

$$\begin{aligned}2x + 2y \frac{dy}{dx} &= 0 \\\frac{dy}{dx} &= -\frac{x}{y}\end{aligned}$$

At point $P_1(2, 2)$:

$$\text{Slope for } C_1 : m_1 = -\frac{2}{2} = -1$$

$$\text{Slope for } C_2 : m_2 = -\frac{2}{2} = -1$$

Since $m_1 = m_2 = -1$, both curves share a common tangent at $P_1(2, 2)$.

At point $P_2(-2, -2)$:

$$\text{Slope for } C_1 : m_1 = -\frac{-2}{-2} = -1$$

$$\text{Slope for } C_2 : m_2 = -\frac{-2}{-2} = -1$$

Since $m_1 = m_2 = -1$, both curves share a common tangent at $P_2(-2, -2)$.

Since the slopes of the tangents to both curves are equal at each point of intersection, the two curves touch each other.

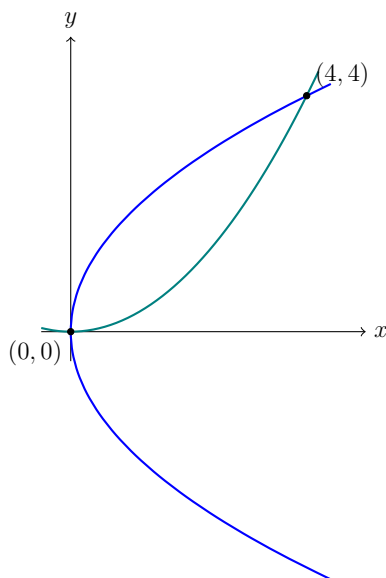
Hence Proved

Since $m_1 = m_2 = -1$ at $(2, 2)$ and $(-2, -2)$, the curves touch each other. Hence proved.

Question 35

Find the angle of intersection between the curves $y^2 = 4x$ and $x^2 = 4y$.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves



Intersection of parabolas $y^2 = 4x$ and $x^2 = 4y$

Given: Curves $C_1 : y^2 = 4x$ and $C_2 : x^2 = 4y$.

Concept / Formula Used: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, find the points of intersection: From $C_2 : y = \frac{x^2}{4}$. Substitute into $C_1 : y^2 = 4x$:

$$\begin{aligned} \left(\frac{x^2}{4} \right)^2 &= 4x \\ \frac{x^4}{16} &= 4x \\ x^4 &= 64x \\ x^4 - 64x &= 0 \\ x(x^3 - 64) &= 0 \end{aligned}$$

So $x = 0$ or $x^3 = 64 \implies x = 4$. When $x = 0$, $y = 0$, giving $O(0, 0)$. When $x = 4$, $y = \frac{4^2}{4} = 4$, giving $P(4, 4)$.

Differentiating $C_1 : y^2 = 4x$:

$$\begin{aligned} 2y \frac{dy}{dx} &= 4 \\ \frac{dy}{dx} &= \frac{2}{y} \end{aligned}$$

Differentiating $C_2 : x^2 = 4y$:

$$2x = 4 \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{2}$$

Case 1: At $O(0, 0)$: For C_1 , $\frac{dy}{dx} \rightarrow \infty$, so the tangent is the y-axis ($x = 0$). For C_2 , $\frac{dy}{dx} = \frac{0}{2} = 0$, so the tangent is the x-axis ($y = 0$). Since the x-axis and y-axis are perpendicular, the angle of intersection at $(0, 0)$ is 90° (they cut orthogonally).

Case 2: At $P(4, 4)$:

$$m_1 = \frac{2}{4} = \frac{1}{2}$$

$$m_2 = \frac{4}{2} = 2$$

Calculating $\tan \theta$:

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{\frac{1}{2} - 2}{1 + \left(\frac{1}{2}\right)(2)} \right| \\ &= \left| \frac{-\frac{3}{2}}{1 + 1} \right| \\ &= \left| \frac{-\frac{3}{2}}{2} \right| \\ &= \frac{3}{4} \end{aligned}$$

Thus, $\theta = \tan^{-1} \left(\frac{3}{4} \right)$.

Final Answer

At $(0, 0)$, the curves cut orthogonally ($\theta = 90^\circ$); at $(4, 4)$, the angle of intersection is $\theta = \tan^{-1} \left(\frac{3}{4} \right)$.

6.3 – Solving Inequalities via Sign-Scheme Method

Question 1

Solve the following inequalities for $x \in \mathbb{R}$:

- (i) $x(x - 2)(x - 5)(x + 3) > 0$
- (ii) $x^4 - 5x^2 + 4 \geq 0$



Sign schemes on the real number line for parts (i) and (ii)

To Find: The set of real values of x satisfying the given polynomial inequalities.

Concept / Formula Used: $P(x) = (x - a_1)(x - a_2) \dots (x - a_n)$ $a_1 < a_2 < \dots < a_n$
 $P(x)$

Solution:

Part (i): Solve $x(x - 2)(x - 5)(x + 3) > 0$.

Step 1: Find the critical points by setting each factor to zero.

$$\begin{aligned}x &= 0 \\x - 2 &= 0 \implies x = 2 \\x - 5 &= 0 \implies x = 5 \\x + 3 &= 0 \implies x = -3\end{aligned}$$

Step 2: Arrange the critical points in ascending order:

$$-3 < 0 < 2 < 5$$

Step 3: Test the sign of the product $f(x) = x(x - 2)(x - 5)(x + 3)$ in each interval:

- For $x \in (5, \infty)$: Choose $x = 6 \implies (6)(4)(1)(9) = 216 > 0$ (+).
- For $x \in (2, 5)$: Choose $x = 3 \implies (3)(1)(-2)(6) = -36 < 0$ (-).
- For $x \in (0, 2)$: Choose $x = 1 \implies (1)(-1)(-4)(4) = 16 > 0$ (+).
- For $x \in (-3, 0)$: Choose $x = -1 \implies (-1)(-3)(-6)(2) = -36 < 0$ (-).
- For $x \in (-\infty, -3)$: Choose $x = -4 \implies (-4)(-6)(-9)(-1) = 216 > 0$ (+).

Step 4: Since the inequality is strictly positive (> 0), we select the positive intervals and exclude the critical points.

$$x \in (-\infty, -3) \cup (0, 2) \cup (5, \infty)$$

Part (ii): Solve $x^4 - 5x^2 + 4 \geq 0$.

Step 1: Let $y = x^2$. Substitute y into the inequality:

$$y^2 - 5y + 4 \geq 0$$

Step 2: Factorize the quadratic expression:

$$\begin{aligned}y^2 - 4y - y + 4 &\geq 0 \\y(y - 4) - 1(y - 4) &\geq 0 \\(y - 1)(y - 4) &\geq 0\end{aligned}$$

Step 3: Back-substitute $y = x^2$:

$$(x^2 - 1)(x^2 - 4) \geq 0$$

Step 4: Express as linear factors using $a^2 - b^2 = (a - b)(a + b)$:

$$(x - 1)(x + 1)(x - 2)(x + 2) \geq 0$$

Step 5: Identify critical points in ascending order:

$$-2 < -1 < 1 < 2$$

Step 6: Test the sign in each interval:

- For $x \in (2, \infty)$: $(+)(+)(+)(+) > 0$ (+).
- For $x \in (1, 2)$: $(+)(+)(-)(+) < 0$ (-).
- For $x \in (-1, 1)$: $(-)(+)(-)(+) > 0$ (+).
- For $x \in (-2, -1)$: $(-)(+)(-)(-) < 0$ (-).
- For $x \in (-\infty, -2)$: $(-)(-)(-)(-) > 0$ (+).

Step 7: Since the inequality is non-negative (≥ 0), include the critical points where the expression equals zero:

$$x \in (-\infty, -2] \cup [-1, 1] \cup [2, \infty)$$

Final Answer

- (i) $x \in (-\infty, -3) \cup (0, 2) \cup (5, \infty)$
 (ii) $x \in (-\infty, -2] \cup [-1, 1] \cup [2, \infty)$

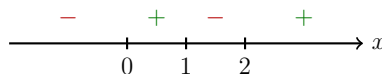
Common Mistake: In part (ii), forgetting to include the endpoints for non-strict inequalities ($\geq$) or failing to back-substitute $y = x^2$ into linear factors of x leads to incorrect solution sets.

Question 2

Find all real values of x which satisfy:

- (i) $x^3(x-1)(x-2) > 0$
 (ii) $x^2(x-1)(x-2) \leq 0$

TYPE 48b Solving Inequalities via Sign-Scheme Method – Solving Polynomial Inequalities with Repeated Linear Factors



Sign behavior around critical points with odd and even multiplicities

To Find: The set of all real values of x satisfying each given inequality.

Concept / Formula Used: $(x-a)^m$ m $x=a$ m $x=a$

Solution:

Part (i): Solve $x^3(x-1)(x-2) > 0$.

Step 1: Identify critical points:

$$\begin{aligned} x^3 &= 0 \implies x = 0 \quad (\text{multiplicity } 3, \text{ odd}) \\ x-1 &= 0 \implies x = 1 \quad (\text{multiplicity } 1, \text{ odd}) \\ x-2 &= 0 \implies x = 2 \quad (\text{multiplicity } 1, \text{ odd}) \end{aligned}$$

Step 2: Arrange critical points in ascending order:

$$0 < 1 < 2$$

Step 3: Test the sign of $f(x) = x^3(x-1)(x-2)$ across the intervals:

- For $x \in (2, \infty)$: Choose $x = 3 \implies (27)(2)(1) = 54 > 0$ (+).
- For $x \in (1, 2)$: Choose $x = 1.5 \implies (1.5)^3(0.5)(-0.5) = -0.84375 < 0$ (-).
- For $x \in (0, 1)$: Choose $x = 0.5 \implies (0.5)^3(-0.5)(-1.5) = 0.09375 > 0$ (+).
- For $x \in (-\infty, 0)$: Choose $x = -1 \implies (-1)^3(-2)(-3) = -6 < 0$ (-).

Step 4: Collect intervals where $f(x) > 0$:

$$x \in (0, 1) \cup (2, \infty)$$

Part (ii): Solve $x^2(x-1)(x-2) \leq 0$.

Step 1: Analyze the factor x^2 : Since $x^2 \geq 0$ for all real x , x^2 does not change the sign of the expression for $x \neq 0$.

Step 2: Consider two cases based on the value of x^2 :

Case 1: $x = 0$. Substituting $x = 0$ into the inequality:

$$0^2(0-1)(0-2) = 0 \leq 0$$

This is a true statement, so $x = 0$ is a valid solution.

Case 2: $x \neq 0$. Since $x^2 > 0$, we can divide both sides of the inequality by x^2 without changing the inequality sign:

$$(x-1)(x-2) \leq 0$$

Step 3: Solve $(x-1)(x-2) \leq 0$: The critical points are $x = 1$ and $x = 2$. The quadratic $(x-1)(x-2)$ is non-positive between its roots:

$$1 \leq x \leq 2 \implies x \in [1, 2]$$

Step 4: Combine the solutions from Case 1 and Case 2:

$$x \in \{0\} \cup [1, 2]$$

Final Answer

- (i) $x \in (0, 1) \cup (2, \infty)$
- (ii) $x \in \{0\} \cup [1, 2]$

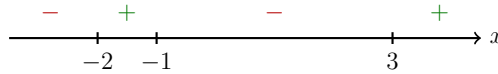
Common Mistake: In part (ii), students often divide by x^2 immediately and forget to check $x = 0$, thereby missing the isolated solution point $\{0\}$.

Question 3

Solve the following rational inequalities for $x \in \mathbb{R}$:

- (i) $\frac{1}{x-2} \leq 1$
- (ii) $\frac{(x+1)(x-3)}{x+2} \geq 0$

TYPE 48c Solving Inequalities via Sign-Scheme Method – Solving Rational Inequalities via Sign Scheme



Sign scheme on the real number line for part (ii)

To Find: The set of real values of x satisfying each rational inequality.

Concept / Formula Used: $\frac{N(x)}{D(x)} \geq 0$ $N(x) = 0$ $D(x) = 0$

Solution:

Part (i): Solve $\frac{1}{x-2} \leq 1$.

Step 1: Identify domain restrictions. Denominator cannot be zero:

$$x - 2 \neq 0 \implies x \neq 2$$

Step 2: Transpose 1 to the left-hand side:

$$\frac{1}{x-2} - 1 \leq 0$$

Step 3: Combine into a single fraction over the common denominator:

$$\begin{aligned} \frac{1 - 1(x-2)}{x-2} &\leq 0 \\ \frac{1 - x + 2}{x-2} &\leq 0 \\ \frac{3 - x}{x-2} &\leq 0 \end{aligned}$$

Step 4: Multiply the numerator by -1 to make the coefficient of x positive (reversing the inequality sign):

$$\frac{x-3}{x-2} \geq 0$$

Step 5: Determine critical points:

$$\text{Numerator zero: } x - 3 = 0 \implies x = 3$$

$$\text{Denominator zero: } x - 2 = 0 \implies x = 2 \quad (\text{excluded})$$

Step 6: Test signs across the intervals $(-\infty, 2)$, $(2, 3)$, and $(3, \infty)$:

- For $x \in (3, \infty)$: Choose $x = 4 \implies \frac{4-3}{4-2} = \frac{1}{2} > 0$ (+).
- For $x \in (2, 3)$: Choose $x = 2.5 \implies \frac{2.5-3}{2.5-2} = \frac{-0.5}{0.5} = -1 < 0$ (-).
- For $x \in (-\infty, 2)$: Choose $x = 0 \implies \frac{0-3}{0-2} = \frac{3}{2} > 0$ (+).

Step 7: Select intervals where the expression is ≥ 0 , including $x = 3$ and excluding $x = 2$:

$$x \in (-\infty, 2) \cup [3, \infty)$$

Part (ii): Solve $\frac{(x+1)(x-3)}{x+2} \geq 0$.

Step 1: Identify domain restrictions:

$$x + 2 \neq 0 \implies x \neq -2$$

Step 2: Determine all critical points from numerator and denominator:

$$x + 1 = 0 \implies x = -1$$

$$x - 3 = 0 \implies x = 3$$

$$x + 2 = 0 \implies x = -2 \quad (\text{excluded})$$

Step 3: Arrange critical points in ascending order:

$$-2 < -1 < 3$$

Step 4: Test the sign of $f(x) = \frac{(x+1)(x-3)}{x+2}$ in each interval:

- For $x \in (3, \infty)$: Choose $x = 4 \implies \frac{(5)(1)}{6} = \frac{5}{6} > 0 \quad (+)$.
- For $x \in (-1, 3)$: Choose $x = 0 \implies \frac{(1)(-3)}{2} = -\frac{3}{2} < 0 \quad (-)$.
- For $x \in (-2, -1)$: Choose $x = -1.5 \implies \frac{(-0.5)(-4.5)}{0.5} = \frac{2.25}{0.5} = 4.5 > 0 \quad (+)$.
- For $x \in (-\infty, -2)$: Choose $x = -3 \implies \frac{(-2)(-6)}{-1} = -12 < 0 \quad (-)$.

Step 5: Select non-negative regions (≥ 0). Include numerator roots $x = -1$ and $x = 3$; exclude denominator root $x = -2$:

$$x \in (-2, -1] \cup [3, \infty)$$

Final Answer

- (i) $x \in (-\infty, 2) \cup [3, \infty)$
- (ii) $x \in (-2, -1] \cup [3, \infty)$

Common Mistake: Cross-multiplying by $(x-2)$ in part (i) without considering its sign is a standard conceptual error; rational inequalities must always be rearranged to have 0 on one side.

6.4 – Increasing and Decreasing Functions

Question 1

Show that the following functions are strictly increasing on $\mathbb{R}$:

- (i) $f(x) = 3x + 17$
- (ii) $f(x) = 7x - 3$

TYPE 51a Proving Monotonicity on an Interval – Proving Monotonicity of Linear Functions

To Prove: Show that $f'(x) > 0$ for all $x \in \mathbb{R}$ for both given functions.

Concept / Formula Used: $f(x) \quad I \quad f'(x) > 0 \quad x \in I$

Solution:

Part (i): Let the given function be:

$$f(x) = 3x + 17$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(3x + 17) \\ &= 3 \cdot 1 + 0 \\ &= 3 \end{aligned}$$

Since $3 > 0$ for all $x \in \mathbb{R}$, we have:

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 3x + 17$ is strictly increasing on $\mathbb{R}$.

Part (ii): Let the given function be:

$$f(x) = 7x - 3$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(7x - 3) \\ &= 7 \cdot 1 - 0 \\ &= 7 \end{aligned}$$

Since $7 > 0$ for all $x \in \mathbb{R}$, we have:

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 7x - 3$ is strictly increasing on $\mathbb{R}$.

Hence Proved

Both functions $f(x) = 3x + 17$ and $f(x) = 7x - 3$ satisfy $f'(x) > 0$ for all $x \in \mathbb{R}$, so they are strictly increasing on $\mathbb{R}$.

Question 2

- (i) Show that the function f given by $f(x) = 3 - 7x$ is strictly decreasing.
- (ii) Show that the function $f(x) = 1 - \frac{1}{x}$ is strictly increasing.

TYPE 51b Proving Monotonicity on an Interval – Proving Monotonicity of Simple Linear and Rational Functions

To Prove:

- (i) Show that $f'(x) < 0$ for all $x \in \mathbb{R}$.
- (ii) Show that $f'(x) > 0$ for all $x \in \mathbb{R} \setminus \{0\}$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$

Solution:

Part (i): Given function:

$$f(x) = 3 - 7x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(3 - 7x) \\ &= 0 - 7 \\ &= -7 \end{aligned}$$

Since $-7 < 0$ for all real numbers x :

$$f'(x) < 0 \quad \forall x \in \mathbb{R}$$

Therefore, $f(x) = 3 - 7x$ is strictly decreasing on $\mathbb{R}$.

Part (ii): Given function:

$$f(x) = 1 - \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(1 - x^{-1}) \\ &= 0 - (-1)x^{-2} \\ &= \frac{1}{x^2} \end{aligned}$$

For any non-zero real number x :

$$x^2 > 0 \implies \frac{1}{x^2} > 0$$

Thus, $f'(x) > 0$ for all $x \in \mathbb{R} \setminus \{0\}$. Therefore, $f(x) = 1 - \frac{1}{x}$ is strictly increasing on its domain.

Hence Proved

Part (i) shows $f'(x) = -7 < 0$, so $f(x)$ is strictly decreasing. Part (ii) shows $f'(x) = \frac{1}{x^2} > 0$, so $f(x)$ is strictly increasing.

Question 3

- (i) Prove that the function $f(x) = 3 + \frac{1}{x}$ is strictly decreasing.
- (ii) Show that the function $f(x) = x - \frac{1}{x}$ is strictly increasing.

TYPE 51c Proving Monotonicity on an Interval – Proving Monotonicity of Simple Rational Functions

To Prove:

- (i) Show that $f'(x) < 0$ for all $x \neq 0$.
- (ii) Show that $f'(x) > 0$ for all $x \neq 0$.

Concept / Formula Used: $x \in \mathbb{R} \setminus \{0\}$ $x^2 > 0$

Solution:

Part (i): Given function:

$$f(x) = 3 + \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (3 + x^{-1}) \\ &= 0 + (-1)x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

Since $x^2 > 0$ for all $x \neq 0$:

$$\frac{1}{x^2} > 0 \implies -\frac{1}{x^2} < 0$$

Thus, $f'(x) < 0$ for all $x \neq 0$. Hence, $f(x) = 3 + \frac{1}{x}$ is strictly decreasing on its domain.

Part (ii): Given function:

$$f(x) = x - \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (x - x^{-1}) \\ &= 1 - (-1)x^{-2} \\ &= 1 + \frac{1}{x^2} \end{aligned}$$

Since $x^2 > 0$ for all $x \neq 0$, we have $\frac{1}{x^2} > 0$. Adding 1 to both sides:

$$1 + \frac{1}{x^2} > 1 > 0$$

Thus, $f'(x) > 0$ for all $x \neq 0$. Hence, $f(x) = x - \frac{1}{x}$ is strictly increasing on its domain.

Hence Proved

LHS derivations show $f'(x) < 0$ for part (i) and $f'(x) > 0$ for part (ii), establishing the required monotonicity.

Question 4

- (i) Show that the function $f(x) = e^{-2x}$ is strictly decreasing on $\mathbb{R}$.
- (ii) Show that the function $f(x) = 5^x$ is strictly increasing.
- (iii) Show that the function $f(x) = \log x$ is strictly increasing.
- (iv) Show that the function $f(x) = -x + \cos x$ is decreasing.

TYPE 51d Proving Monotonicity on an Interval – Proving Monotonicity of Basic Exponential, Logarithmic, and Trigonometric Functions

To Prove: Verify the sign of $f'(x)$ on the respective domains for all four functions.

Concept / Formula Used: $\frac{d}{dx}(e^{kx}) = ke^{kx}$, $\frac{d}{dx}(a^x) = a^x \ln a$, $\frac{d}{dx}(\log x) = \frac{1}{x}$

Solution:

Part (i): Given $f(x) = e^{-2x}$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (e^{-2x}) \\ &= -2e^{-2x} \end{aligned}$$

Since exponential function $e^{-2x} > 0$ for all $x \in \mathbb{R}$:

$$-2e^{-2x} < 0 \implies f'(x) < 0 \quad \forall x \in \mathbb{R}$$

Thus, $f(x) = e^{-2x}$ is strictly decreasing on $\mathbb{R}$.

Part (ii): Given $f(x) = 5^x$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (5^x) \\ &= 5^x \log 5 \end{aligned}$$

Since $5^x > 0$ for all $x \in \mathbb{R}$ and $\log 5 > 0$:

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

Thus, $f(x) = 5^x$ is strictly increasing on $\mathbb{R}$.

Part (iii): Given $f(x) = \log x$, defined for $x > 0$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (\log x) \\ &= \frac{1}{x} \end{aligned}$$

For all $x \in (0, \infty)$, $x > 0 \implies \frac{1}{x} > 0$. Thus, $f'(x) > 0$ for all $x > 0$. Hence, $f(x) = \log x$ is strictly increasing on its domain.

Part (iv): Given $f(x) = -x + \cos x$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (-x + \cos x) \\ &= -1 - \sin x \\ &= -(1 + \sin x) \end{aligned}$$

Using the standard trigonometric property $-1 \leq \sin x \leq 1$:

$$1 + \sin x \geq 0 \implies -(1 + \sin x) \leq 0$$

Thus, $f'(x) \leq 0$ for all $x \in \mathbb{R}$. Hence, $f(x) = -x + \cos x$ is a decreasing function on $\mathbb{R}$.

Hence Proved

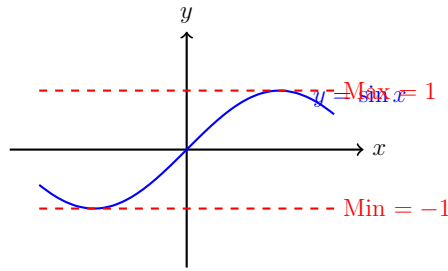
All four sub-parts are verified using derivative tests on their respective domains.

Question 5

Find the values of 'a' for which the function $f(x) = b - ax + \sin x$ is increasing on $\mathbb{R}$.

TYPE 55a Finding Parameter Range for Monotonicity – Finding Parameter Range for Monotonicity of Trigonometric-Linear

★ **Likely Exam Question**



Sign condition for parameter a

Given: Function $f(x) = b - ax + \sin x$ is increasing on $\mathbb{R}$, where b is a real constant.

To Find: The set of real values of a .

Concept / Formula Used: $f(x) \quad \mathbb{R} \quad f'(x) \geq 0 \quad x \in \mathbb{R}$

Solution:

Given function:

$$f(x) = b - ax + \sin x$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(b - ax + \sin x) \\ &= 0 - a + \cos x \\ &= \cos x - a \end{aligned}$$

For $f(x)$ to be an increasing function on $\mathbb{R}$, we must have:

$$f'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

Substituting $f'(x) = \cos x - a$:

$$\begin{aligned} \cos x - a &\geq 0 \quad \forall x \in \mathbb{R} \\ \cos x &\geq a \quad \forall x \in \mathbb{R} \end{aligned}$$

This inequality must hold true for all $x \in \mathbb{R}$. Therefore, a must be less than or equal to the minimum value of $\cos x$ on $\mathbb{R}$.

We know that:

$$-1 \leq \cos x \leq 1 \quad \forall x \in \mathbb{R}$$

Thus, the minimum value of $\cos x$ is -1 .

Setting $a \leq \min(\cos x)$:

$$a \leq -1$$

Final Answer

The function $f(x)$ is increasing on $\mathbb{R}$ for all $a \leq -1$, i.e., $a \in (-\infty, -1]$.

Question 6

- Find the interval in which the function f given by $f(x) = 7 - 4x - x^2$ is strictly increasing.
- If $f(x) = x^3 - 48x$, find the interval where $f : \mathbb{R} \rightarrow \mathbb{R}$ is decreasing.

(iii) Find the interval in which the function $f(x) = 2x^3 - 3x$ is strictly increasing.

TYPE 56a Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quadratic and Cubic Polynomials

To Find: The intervals of monotonicity for each polynomial function.

Concept / Formula Used: $f'(x) > 0$ $f'(x) \leq 0$ $f'(x) < 0$

Solution:

Part (i): Given function:

$$f(x) = 7 - 4x - x^2$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(7 - 4x - x^2) \\ &= -4 - 2x \\ &= -2(x + 2) \end{aligned}$$

For $f(x)$ to be strictly increasing:

$$\begin{aligned} f'(x) &> 0 \\ -2(x + 2) &> 0 \end{aligned}$$

Dividing by -2 (and reversing the inequality sign):

$$\begin{aligned} x + 2 &< 0 \\ x &< -2 \end{aligned}$$

Thus, $x \in (-\infty, -2)$.

Part (ii): Given function:

$$f(x) = x^3 - 48x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 - 48x) \\ &= 3x^2 - 48 \\ &= 3(x^2 - 16) \\ &= 3(x - 4)(x + 4) \end{aligned}$$

For $f(x)$ to be decreasing:

$$\begin{aligned} f'(x) &\leq 0 \\ 3(x - 4)(x + 4) &\leq 0 \\ (x - 4)(x + 4) &\leq 0 \end{aligned}$$

Solving the quadratic inequality gives:

$$-4 \leq x \leq 4$$

Thus, $x \in [-4, 4]$.

Part (iii): Given function:

$$f(x) = 2x^3 - 3x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(2x^3 - 3x) \\ &= 6x^2 - 3 \\ &= 3(2x^2 - 1) \end{aligned}$$

For $f(x)$ to be strictly increasing:

$$\begin{aligned} f'(x) &> 0 \\ 3(2x^2 - 1) &> 0 \\ 2x^2 - 1 &> 0 \\ x^2 &> \frac{1}{2} \end{aligned}$$

Taking square root on both sides:

$$|x| > \frac{1}{\sqrt{2}} \implies x < -\frac{1}{\sqrt{2}} \quad \text{or} \quad x > \frac{1}{\sqrt{2}}$$

$$\text{Thus, } x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right).$$

Final Answer

- (i) Strictly increasing in $(-\infty, -2)$
- (ii) Decreasing in $[-4, 4]$
- (iii) Strictly increasing in $\left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right)$

Question 7

Show that the function f defined by $f(x) = (x - 1)e^x + 1$ is an increasing function for all $x > 0$.

TYPE 51e Proving Monotonicity on an Interval – Proving Monotonicity of Product of Polynomial and Exponential Function

To Prove: Show that $f'(x) > 0$ for all $x > 0$.

Concept / Formula Used: $\frac{d}{dx}(uv) = u'v + uv'$

Solution:

Given function:

$$f(x) = (x - 1)e^x + 1$$

Differentiating with respect to x using the product rule:

$$\begin{aligned} f'(x) &= \frac{d}{dx} [(x-1)e^x + 1] \\ &= \frac{d}{dx} (x-1) \cdot e^x + (x-1) \frac{d}{dx} (e^x) + 0 \\ &= 1 \cdot e^x + (x-1)e^x \\ &= e^x(1+x-1) \\ &= xe^x \end{aligned}$$

Now, examine the signs of the terms for $x > 0$:

$$\begin{aligned} x &> 0 \\ e^x &> 0 \quad \forall x \in \mathbb{R} \end{aligned}$$

Multiplying two positive quantities:

$$f'(x) = xe^x > 0 \quad \forall x > 0$$

Since $f'(x) > 0$ for all $x > 0$, the function $f(x)$ is strictly increasing for all $x > 0$.

Hence Proved

Since $f'(x) = xe^x > 0$ for all $x > 0$, $f(x) = (x-1)e^x + 1$ is an increasing function for all $x > 0$.

Question 8

For what value of k , the function $f(x) = kx^3 + 5$ is decreasing?

TYPE 55b Finding Parameter Range for Monotonicity – Finding Parameter Range for Monotonicity of Cubic Polynomial

Given: Function $f(x) = kx^3 + 5$ is decreasing on $\mathbb{R}$.

To Find: The condition on parameter k .

Concept / Formula Used: $f(x) \quad f'(x) \leq 0 \quad x \in \mathbb{R}$

Solution:

Given function:

$$f(x) = kx^3 + 5$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (kx^3 + 5) \\ &= 3kx^2 \end{aligned}$$

For $f(x)$ to be decreasing on $\mathbb{R}$:

$$f'(x) \leq 0 \quad \forall x \in \mathbb{R}$$

Substituting $f'(x)$:

$$3kx^2 \leq 0 \quad \forall x \in \mathbb{R}$$

Since $x^2 \geq 0$ for all real numbers x :

- If $k < 0$, then $3k < 0$, so $3kx^2 \leq 0$ for all $x \in \mathbb{R}$.
- If $k = 0$, then $f(x) = 5$, a constant function, which is non-increasing.
- If $k > 0$, then $3kx^2 \geq 0$, which makes $f(x)$ increasing.

For non-trivial strict decreasing behavior (or general decreasing behavior):

$$k < 0 \quad (\text{or } k \leq 0)$$

Final Answer

$k < 0$ (or $k \leq 0$ for non-strictly decreasing).

Question 9

- Prove that the function $f(x) = ax + b$ is strictly decreasing iff $a < 0$.
- Prove that the function $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.

TYPE 51f Proving Monotonicity on an Interval – Proving Monotonicity of Parametric Linear and Greatest Integer Function

To Prove:

- $f(x) = ax + b$ is strictly decreasing if and only if $a < 0$.
- $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.

Concept / Formula Used: $x \in [0, 1)$ $[x] = 0$

Solution:

Part (i): Let $f(x) = ax + b$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(ax + b) \\ &= a \end{aligned}$$

By definition of monotonicity:

- $f(x)$ is strictly decreasing $\iff f'(x) < 0 \quad \forall x \in \mathbb{R}$.
- Since $f'(x) = a$, this condition directly gives $a < 0$.

Thus, $f(x) = ax + b$ is strictly decreasing if and only if $a < 0$.

Part (ii): For $x \in [0, 1)$, by definition of the greatest integer function:

$$[x] = 0$$

Substitute $[x] = 0$ into $f(x)$:

$$\begin{aligned} f(x) &= 0 - x \\ &= -x \end{aligned}$$

Differentiating $f(x)$ with respect to x for $x \in (0, 1)$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}(-x) \\ &= -1 \end{aligned}$$

Since $-1 < 0$ for all $x \in (0, 1)$:

$$f'(x) < 0 \quad \forall x \in (0, 1)$$

Alternatively, pick any $x_1, x_2 \in [0, 1)$ such that $x_1 < x_2$:

$$\begin{aligned} f(x_1) &= -x_1 \\ f(x_2) &= -x_2 \\ x_1 < x_2 &\implies -x_1 > -x_2 \implies f(x_1) > f(x_2) \end{aligned}$$

Hence, $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.

Hence Proved

Both assertions are proven: (i) $f'(x) = a < 0 \iff a < 0$, and (ii) $f(x) = -x$ on $[0, 1)$ has derivative $-1 < 0$.

Question 10

Prove that the function $f(x) = x^3 - 6x^2 + 12x + 5$ is increasing on $\mathbb{R}$.

TYPE 51g Proving Monotonicity on an Interval – Proving Monotonicity of Cubic Polynomial with Perfect Square Derivative

★ Likely Exam Question

To Prove: Show that $f'(x) \geq 0$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $(x - k)^2 \geq 0 \quad x$

Solution:

Given function:

$$f(x) = x^3 - 6x^2 + 12x + 5$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (x^3 - 6x^2 + 12x + 5) \\ &= 3x^2 - 12x + 12 \end{aligned}$$

Factoring out 3:

$$\begin{aligned} f'(x) &= 3(x^2 - 4x + 4) \\ &= 3(x - 2)^2 \end{aligned}$$

Since the square of any real number is non-negative:

$$(x - 2)^2 \geq 0 \quad \forall x \in \mathbb{R}$$

Multiplying by $3 > 0$:

$$3(x - 2)^2 \geq 0 \quad \forall x \in \mathbb{R}$$

Therefore:

$$f'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

Since $f'(x) \geq 0$ for all $x \in \mathbb{R}$ and vanishes only at the point $x = 2$ (which does not contain any sub-interval), $f(x)$ is increasing on $\mathbb{R}$.

Hence Proved

$f'(x) = 3(x - 2)^2 \geq 0$ for all $x \in \mathbb{R}$, so $f(x)$ is an increasing function on $\mathbb{R}$.

Question 11

Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ is strictly increasing on $\mathbb{R}$.

TYPE 51g Proving Monotonicity on an Interval – Proving Monotonicity of Cubic Polynomial with Perfect Square Derivative

To Prove: Show that $f'(x) > 0$ for almost all $x \in \mathbb{R}$.

Concept / Formula Used: $a(bx - c)^2 \geq 0 \quad \mathbb{R}$

Solution:

Given function:

$$f(x) = 4x^3 - 18x^2 + 27x - 7$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (4x^3 - 18x^2 + 27x - 7) \\ &= 12x^2 - 36x + 27 \end{aligned}$$

Factoring out 3:

$$f'(x) = 3(4x^2 - 12x + 9)$$

Recognizing the perfect square inside the bracket:

$$4x^2 - 12x + 9 = (2x - 3)^2$$

Thus:

$$f'(x) = 3(2x - 3)^2$$

For all $x \in \mathbb{R}$:

$$(2x - 3)^2 \geq 0 \implies f'(x) \geq 0$$

Furthermore, $f'(x) = 0$ holds only at the isolated point:

$$2x - 3 = 0 \implies x = \frac{3}{2}$$

Since $f'(x) > 0$ for all $x \in \mathbb{R} \setminus \{\frac{3}{2}\}$, $f(x)$ does not remain constant on any interval. Hence, $f(x)$ is strictly increasing on $\mathbb{R}$.

Hence Proved

Since $f'(x) = 3(2x - 3)^2 \geq 0$ with equality at a single point, $f(x)$ is strictly increasing on $\mathbb{R}$.

Question 12

Prove that the function $f(x) = 5 - 3x + 3x^2 - x^3$ is decreasing on $\mathbb{R}$.

TYPE 51g Proving Monotonicity on an Interval – Proving Monotonicity of Cubic Polynomial with Perfect Square Derivative

To Prove: Show that $f'(x) \leq 0$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $f'(x) \leq 0 \implies f(x)$

Solution:

Given function:

$$f(x) = 5 - 3x + 3x^2 - x^3$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (5 - 3x + 3x^2 - x^3) \\ &= 0 - 3 + 6x - 3x^2 \\ &= -3(x^2 - 2x + 1) \end{aligned}$$

Recognizing the perfect square:

$$x^2 - 2x + 1 = (x - 1)^2$$

Thus:

$$f'(x) = -3(x - 1)^2$$

For all $x \in \mathbb{R}$:

$$(x - 1)^2 \geq 0$$

Multiplying by $-3 < 0$:

$$-3(x - 1)^2 \leq 0$$

Therefore:

$$f'(x) \leq 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 5 - 3x + 3x^2 - x^3$ is a decreasing function on $\mathbb{R}$.

Hence Proved

Since $f'(x) = -3(x - 1)^2 \leq 0$ for all $x \in \mathbb{R}$, $f(x)$ is decreasing on $\mathbb{R}$.

Question 13

- Show that $f(x) = 2x - \sin 2x$ is an increasing function.
- Show that the function $f(x) = \frac{16 \sin x}{4 + \cos x} - x$ is strictly decreasing in $\left(\frac{\pi}{2}, \pi\right)$.

TYPE 51h Proving Monotonicity on an Interval – Proving Monotonicity of Trigonometric and Rational Trigonometric Functions

To Prove:

- $f'(x) \geq 0$ for all $x \in \mathbb{R}$.
- $f'(x) < 0$ for all $x \in \left(\frac{\pi}{2}, \pi\right)$.

Concept / Formula Used: $\frac{d}{dx} \left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$ $\cos^2 x + \sin^2 x = 1$

Solution:

Part (i): Given function:

$$f(x) = 2x - \sin 2x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (2x - \sin 2x) \\ &= 2 - 2 \cos 2x \\ &= 2(1 - \cos 2x) \end{aligned}$$

Using the standard identity $1 - \cos 2x = 2 \sin^2 x$:

$$\begin{aligned} f'(x) &= 2(2 \sin^2 x) \\ &= 4 \sin^2 x \end{aligned}$$

Since $\sin^2 x \geq 0$ for all $x \in \mathbb{R}$:

$$f'(x) = 4 \sin^2 x \geq 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 2x - \sin 2x$ is an increasing function.

Part (ii): Given function:

$$f(x) = \frac{16 \sin x}{4 + \cos x} - x$$

Differentiating using the quotient rule:

$$\begin{aligned} f'(x) &= \frac{(4 + \cos x) \frac{d}{dx}(16 \sin x) - 16 \sin x \frac{d}{dx}(4 + \cos x)}{(4 + \cos x)^2} - 1 \\ &= \frac{(4 + \cos x)(16 \cos x) - 16 \sin x(-\sin x)}{(4 + \cos x)^2} - 1 \\ &= \frac{64 \cos x + 16 \cos^2 x + 16 \sin^2 x}{(4 + \cos x)^2} - 1 \end{aligned}$$

Using $\cos^2 x + \sin^2 x = 1$:

$$\begin{aligned} f'(x) &= \frac{64 \cos x + 16}{(4 + \cos x)^2} - 1 \\ &= \frac{64 \cos x + 16 - (4 + \cos x)^2}{(4 + \cos x)^2} \\ &= \frac{64 \cos x + 16 - (16 + 8 \cos x + \cos^2 x)}{(4 + \cos x)^2} \\ &= \frac{56 \cos x - \cos^2 x}{(4 + \cos x)^2} \\ &= \frac{\cos x(56 - \cos x)}{(4 + \cos x)^2} \end{aligned}$$

Now analyze the signs of the factors in the interval $x \in (\frac{\pi}{2}, \pi)$:

- In the second quadrant $(\frac{\pi}{2}, \pi)$, $\cos x < 0$.
- Since $-1 < \cos x < 0$, we have $56 - \cos x > 56 - 0 > 0$.
- The denominator $(4 + \cos x)^2 > 0$.

Combining these signs:

$$f'(x) = \frac{(\text{negative}) \cdot (\text{positive})}{\text{positive}} < 0$$

Thus, $f'(x) < 0$ for all $x \in (\frac{\pi}{2}, \pi)$. Hence, $f(x)$ is strictly decreasing in $(\frac{\pi}{2}, \pi)$.

Hence Proved

Both parts are proved using explicit derivative evaluation and interval sign analysis.

Question 14

- (i) Prove that the function f given by $f(x) = \log \cos x$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$.
 (ii) If $f(x) = a(\tan x - \cot x)$, where $a > 0$, then find whether $f(x)$ is increasing or decreasing function in its domain.

TYPE 62a Determining/Classifying Monotonicity of a Function – Determining Monotonicity of Logarithmic-Trigonometric a**To Prove:**

- (i) Show $f'(x) < 0$ for $x \in \left(0, \frac{\pi}{2}\right)$.
 (ii) Determine the sign of $f'(x)$ for $f(x) = a(\tan x - \cot x)$ with $a > 0$.

Concept / Formula Used: $\frac{d}{dx}(\log(\cos x)) = \frac{1}{\cos x}(-\sin x) = -\tan x$ $\frac{d}{dx}(\tan x) = \sec^2 x$, $\frac{d}{dx}(\cot x) = -\csc^2 x$

Solution:**Part (i):** Given function:

$$f(x) = \log \cos x$$

Differentiating with respect to x using chain rule:

$$\begin{aligned} f'(x) &= \frac{1}{\cos x} \cdot \frac{d}{dx}(\cos x) \\ &= \frac{1}{\cos x} \cdot (-\sin x) \\ &= -\tan x \end{aligned}$$

For $x \in \left(0, \frac{\pi}{2}\right)$:

$$\tan x > 0 \implies -\tan x < 0$$

Thus:

$$f'(x) < 0 \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

Hence, $f(x) = \log \cos x$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$.**Part (ii):** Given function:

$$f(x) = a(\tan x - \cot x), \quad a > 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= a \cdot \frac{d}{dx}(\tan x - \cot x) \\ &= a(\sec^2 x - (-\csc^2 x)) \\ &= a(\sec^2 x + \csc^2 x) \end{aligned}$$

Since $\sec^2 x > 0$ and $\csc^2 x > 0$ for all x in the domain of definition, and given $a > 0$:

$$f'(x) = a(\sec^2 x + \csc^2 x) > 0$$

Therefore, $f'(x) > 0$ for all x in its domain. Hence, $f(x)$ is an increasing function in its domain.**Final Answer**

- (i) Proved that $f(x)$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$.
 (ii) $f(x)$ is an ****increasing**** function in its domain.

Question 15

Determine whether the following functions are increasing or decreasing for the stated values of x :

(i) $f(x) = \frac{1}{x}, \quad x < 0$

(ii) $f(x) = \frac{1}{1+x^2}, \quad x \geq 0$

(iii) $f(x) = \cos^2 x$ in $\left[0, \frac{\pi}{2}\right]$

(iv) $f(x) = -\frac{x}{2} + \sin x, \quad -\frac{\pi}{3} < x < \frac{\pi}{3}$

TYPE 62b Determining/Classifying Monotonicity of a Function – Determining Monotonicity of Rational and Trigonometric

To Find: Monotonicity behavior (increasing/decreasing) for each given function on its specified interval.

Concept / Formula Used: $f'(x) > 0$ < 0

Solution:

Part (i): Given $f(x) = \frac{1}{x} = x^{-1}$ for $x < 0$. Differentiating with respect to x :

$$f'(x) = -\frac{1}{x^2}$$

For any $x < 0$, $x^2 > 0 \implies \frac{1}{x^2} > 0 \implies -\frac{1}{x^2} < 0$. Thus, $f'(x) < 0$ for $x < 0$. So, $f(x)$ is ****decreasing**** for $x < 0$.

Part (ii): Given $f(x) = \frac{1}{1+x^2} = (1+x^2)^{-1}$ for $x \geq 0$. Differentiating using chain rule:

$$\begin{aligned} f'(x) &= -(1+x^2)^{-2} \cdot (2x) \\ &= -\frac{2x}{(1+x^2)^2} \end{aligned}$$

For $x > 0$: $2x > 0$ and $(1+x^2)^2 > 0$, so $f'(x) < 0$. At $x = 0$: $f'(0) = 0$. Thus, $f'(x) \leq 0$ for all $x \geq 0$. So, $f(x)$ is ****decreasing**** for $x \geq 0$.

Part (iii): Given $f(x) = \cos^2 x$ for $x \in \left[0, \frac{\pi}{2}\right]$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 2 \cos x (-\sin x) \\ &= -\sin 2x \end{aligned}$$

For $x \in \left(0, \frac{\pi}{2}\right) \implies 2x \in (0, \pi) \implies \sin 2x > 0$. Therefore, $f'(x) = -\sin 2x < 0$ for $x \in \left(0, \frac{\pi}{2}\right)$. So, $f(x)$ is ****decreasing**** in $\left[0, \frac{\pi}{2}\right]$.

Part (iv): Given $f(x) = -\frac{x}{2} + \sin x$ for $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$. Differentiating with respect to x :

$$f'(x) = -\frac{1}{2} + \cos x$$

For $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$:

$$\cos x > \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Thus:

$$f'(x) = \cos x - \frac{1}{2} > 0$$

So, $f(x)$ is **increasing** in $(-\frac{\pi}{3}, \frac{\pi}{3})$.

Final Answer

- (i) Decreasing
- (ii) Decreasing
- (iii) Decreasing
- (iv) Increasing

Question 16

Which of the following functions are strictly decreasing on $(0, \frac{\pi}{2})$?

- (i) $\cos x$
- (ii) $\cos 2x$
- (iii) $\sin^2 x$
- (iv) $\tan x$

TYPE 62c Determining/Classifying Monotonicity of a Function – Identifying Strictly Decreasing Trigonometric Functions on

To Find: Identify which of the four given trigonometric functions are strictly decreasing on $(0, \frac{\pi}{2})$.

Concept / Formula Used: $f(x)$ $f'(x) < 0$

Solution:

Let us evaluate $f'(x)$ for each function on the open interval $x \in (0, \frac{\pi}{2})$:

Case (i): Let $f(x) = \cos x$.

$$f'(x) = -\sin x$$

In $(0, \frac{\pi}{2})$, $\sin x > 0$, so $f'(x) = -\sin x < 0$. Hence, $\cos x$ is **strictly decreasing** on $(0, \frac{\pi}{2})$.

Case (ii): Let $f(x) = \cos 2x$.

$$f'(x) = -2 \sin 2x$$

For $x \in (0, \frac{\pi}{2})$, we have $2x \in (0, \pi)$. Since $\sin 2x > 0$ for $2x \in (0, \pi)$, we get $f'(x) = -2 \sin 2x < 0$. Hence, $\cos 2x$ is **strictly decreasing** on $(0, \frac{\pi}{2})$.

Case (iii): Let $f(x) = \sin^2 x$.

$$f'(x) = 2 \sin x \cos x = \sin 2x$$

In $(0, \frac{\pi}{2})$, $\sin 2x > 0$, so $f'(x) > 0$. Hence, $\sin^2 x$ is strictly increasing (not decreasing).

Case (iv): Let $f(x) = \tan x$.

$$f'(x) = \sec^2 x$$

Since $\sec^2 x > 0$ everywhere in $(0, \frac{\pi}{2})$, $f'(x) > 0$. Hence, $\tan x$ is strictly increasing (not decreasing).

Final Answer

The functions that are strictly decreasing on $(0, \frac{\pi}{2})$ are **(i) $\cos x$** and **(ii) $\cos 2x$** .

Question 17

Consider the statement "There exists at least one value of $b \in \mathbb{R}$ for which $f(x) = \frac{b}{x}, b \neq 0$ is strictly increasing in $\mathbb{R} - \{0\}$." State True or False. Justify.

TYPE 55c Finding Parameter Range for Monotonicity – Analyzing Monotonicity of Rational Function with Parameter

To Find: State whether the statement is True or False and provide mathematical justification.

Concept / Formula Used: $f(x) = \frac{b}{x}$ $f'(x) = -\frac{b}{x^2}$

Solution:

Given function:

$$f(x) = \frac{b}{x}, \quad b \neq 0, \quad x \in \mathbb{R} - \{0\}$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (bx^{-1}) \\ &= -bx^{-2} \\ &= -\frac{b}{x^2} \end{aligned}$$

For $f(x)$ to be strictly increasing on $\mathbb{R} - \{0\}$, we require:

$$f'(x) > 0 \quad \forall x \in \mathbb{R} - \{0\}$$

Substituting $f'(x) = -\frac{b}{x^2}$:

$$-\frac{b}{x^2} > 0$$

Since $x^2 > 0$ for all $x \neq 0$, dividing/multiplying by x^2 preserves the inequality direction:

$$-b > 0 \implies b < 0$$

Thus, if we choose any negative real number for b (e.g., $b = -1$), then:

$$f(x) = -\frac{1}{x} \implies f'(x) = \frac{1}{x^2} > 0 \quad \forall x \neq 0$$

This means there indeed exists at least one value of $b \in \mathbb{R}$ (in fact, any $b < 0$) for which $f(x)$ is strictly increasing on $\mathbb{R} - \{0\}$.

Therefore, the given statement is ****True****.

Final Answer

True. Justification: For any $b < 0$, $f'(x) = -\frac{b}{x^2} > 0$ for all $x \in \mathbb{R} - \{0\}$, making $f(x)$ strictly increasing.

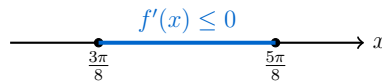
Question 18

Prove that the function $f(x) = \sin\left(2x + \frac{\pi}{4}\right)$ is decreasing for $\frac{3\pi}{8} \leq x \leq \frac{5\pi}{8}$.

TYPE 51i Proving Monotonicity on an Interval – Proving Monotonicity of Composite Sine Function on Closed Interval

To Prove: Function $f(x) = \sin\left(2x + \frac{\pi}{4}\right)$ is decreasing for $x \in \left[\frac{3\pi}{8}, \frac{5\pi}{8}\right]$.

Concept / Formula Used: $f(x)$ $[a, b]$ $f'(x) \leq 0$ $x \in (a, b)$



Interval evaluation for $f'(x)$

Solution:

Given function:

$$f(x) = \sin\left(2x + \frac{\pi}{4}\right)$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left[\sin\left(2x + \frac{\pi}{4}\right) \right] \\ &= \cos\left(2x + \frac{\pi}{4}\right) \cdot \frac{d}{dx} \left(2x + \frac{\pi}{4}\right) \\ &= 2 \cos\left(2x + \frac{\pi}{4}\right) \end{aligned}$$

Given interval for x :

$$\frac{3\pi}{8} \leq x \leq \frac{5\pi}{8}$$

Multiplying by 2:

$$\frac{3\pi}{4} \leq 2x \leq \frac{5\pi}{4}$$

Adding $\frac{\pi}{4}$ throughout:

$$\begin{aligned} \frac{3\pi}{4} + \frac{\pi}{4} &\leq 2x + \frac{\pi}{4} \leq \frac{5\pi}{4} + \frac{\pi}{4} \\ \pi &\leq 2x + \frac{\pi}{4} \leq \frac{3\pi}{2} \end{aligned}$$

For angle $\theta = 2x + \frac{\pi}{4}$ lying in the third quadrant $\left[\pi, \frac{3\pi}{2}\right]$:

$$\cos\left(2x + \frac{\pi}{4}\right) \leq 0$$

Multiplying by 2:

$$f'(x) = 2 \cos\left(2x + \frac{\pi}{4}\right) \leq 0$$

Since $f'(x) \leq 0$ for all $x \in \left[\frac{3\pi}{8}, \frac{5\pi}{8}\right]$, the function $f(x)$ is decreasing in this interval.

Hence Proved

Since $f'(x) \leq 0$ for all $x \in \left[\frac{3\pi}{8}, \frac{5\pi}{8}\right]$, $f(x) = \sin\left(2x + \frac{\pi}{4}\right)$ is decreasing in the given interval.

Question 19

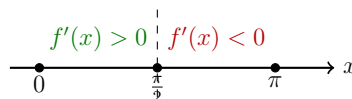
Prove that the function $f(x) = \sin x$ is:

- (i) strictly increasing in $(0, \frac{\pi}{2})$
- (ii) strictly decreasing in $(\frac{\pi}{2}, \pi)$
- (iii) neither increasing nor decreasing in $(0, \pi)$

TYPE 51j Proving Monotonicity on an Interval – Proving Monotonicity of Sine Function on Different Quadrants

To Prove: Monotonicity of $f(x) = \sin x$ on $(0, \frac{\pi}{2})$, $(\frac{\pi}{2}, \pi)$, and $(0, \pi)$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$



Sign of $f'(x)$ = $\cos x$ in different intervals

Solution:

The given function is:

$$f(x) = \sin x$$

Differentiating with respect to x :

$$f'(x) = \cos x$$

- (i) For $x \in (0, \frac{\pi}{2})$: This represents the first quadrant where cosine is positive.

$$\cos x > 0 \implies f'(x) > 0$$

Hence, $f(x) = \sin x$ is strictly increasing in $(0, \frac{\pi}{2})$.

- (ii) For $x \in (\frac{\pi}{2}, \pi)$: This represents the second quadrant where cosine is negative.

$$\cos x < 0 \implies f'(x) < 0$$

Hence, $f(x) = \sin x$ is strictly decreasing in $(\frac{\pi}{2}, \pi)$.

- (iii) For $x \in (0, \pi)$: In $(0, \frac{\pi}{2})$, $f'(x) > 0$, while in $(\frac{\pi}{2}, \pi)$, $f'(x) < 0$. Since $f'(x)$ changes sign over the interval $(0, \pi)$, $f(x)$ is neither strictly increasing nor strictly decreasing in $(0, \pi)$.

Hence Proved

The function $f(x) = \sin x$ is strictly increasing in $(0, \frac{\pi}{2})$, strictly decreasing in $(\frac{\pi}{2}, \pi)$, and neither increasing nor decreasing in $(0, \pi)$.

Question 20

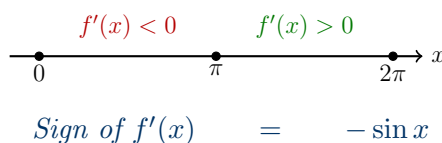
Prove that the function f defined by $f(x) = \cos x$ is:

- (i) strictly increasing in $(\pi, 2\pi)$
- (ii) strictly decreasing in $(0, \pi)$
- (iii) neither increasing nor decreasing in $(0, 2\pi)$

TYPE 51k Proving Monotonicity on an Interval – Proving Monotonicity of Cosine Function on Different Quadrants

To Prove: Monotonicity properties of $f(x) = \cos x$ on $(\pi, 2\pi)$, $(0, \pi)$, and $(0, 2\pi)$.

Concept / Formula Used: $f(x) = \cos x$ $f'(x) = -\sin x$ $f'(x) > 0$ $f'(x) < 0$



Solution:

Given function:

$$f(x) = \cos x$$

Differentiating with respect to x :

$$f'(x) = -\sin x$$

(i) For $x \in (\pi, 2\pi)$: In the third and fourth quadrants, $\sin x < 0$.

$$f'(x) = -\sin x > 0$$

Hence, $f(x) = \cos x$ is strictly increasing in $(\pi, 2\pi)$.

(ii) For $x \in (0, \pi)$: In the first and second quadrants, $\sin x > 0$.

$$f'(x) = -\sin x < 0$$

Hence, $f(x) = \cos x$ is strictly decreasing in $(0, \pi)$.

(iii) For $x \in (0, 2\pi)$: $f'(x) < 0$ in $(0, \pi)$ and $f'(x) > 0$ in $(\pi, 2\pi)$. Since $f'(x)$ changes sign over $(0, 2\pi)$, $f(x)$ is neither increasing nor decreasing in $(0, 2\pi)$.

Hence Proved

The function $f(x) = \cos x$ is strictly increasing in $(\pi, 2\pi)$, strictly decreasing in $(0, \pi)$, and neither increasing nor decreasing in $(0, 2\pi)$.

Question 21

Prove that the following functions are neither increasing nor decreasing:

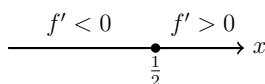
(i) $f(x) = x^2 - x + 1$ on $(-1, 1)$

(ii) $f(x) = \frac{1}{x^2+1}$ on $\mathbb{R}$

TYPE 69 Proving Non-Monotonicity of Quadratic and Rational Functions

To Prove: Show that $f'(x)$ changes sign within the given domains for both functions.

Concept / Formula Used: $f'(x)$ $f(x)$



Sign change of $f'(x)$ across critical point

Solution:

- (i) Given $f(x) = x^2 - x + 1$ on $(-1, 1)$. Differentiating with respect to x :

$$f'(x) = 2x - 1$$

Setting $f'(x) = 0$:

$$2x - 1 = 0$$

$$x = \frac{1}{2}$$

Note that $x = \frac{1}{2} \in (-1, 1)$. For $x \in (-1, \frac{1}{2})$:

$$f'(x) = 2x - 1 < 0$$

For $x \in (\frac{1}{2}, 1)$:

$$f'(x) = 2x - 1 > 0$$

Since $f'(x)$ changes sign within $(-1, 1)$, $f(x)$ is neither increasing nor decreasing on $(-1, 1)$.

- (ii) Given $f(x) = \frac{1}{x^2+1}$ on $\mathbb{R}$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} [(x^2 + 1)^{-1}] \\ &= -(x^2 + 1)^{-2} \cdot 2x \\ &= -\frac{2x}{(x^2 + 1)^2} \end{aligned}$$

Since $(x^2 + 1)^2 > 0$ for all real x : For $x \in (-\infty, 0)$, $x < 0 \implies f'(x) > 0$. For $x \in (0, \infty)$, $x > 0 \implies f'(x) < 0$. Since $f'(x)$ changes sign at $x = 0$, $f(x)$ is neither increasing nor decreasing on $\mathbb{R}$.

Hence Proved

Both functions exhibit sign changes in $f'(x)$ within their domains, proving they are neither increasing nor decreasing.

Question 22

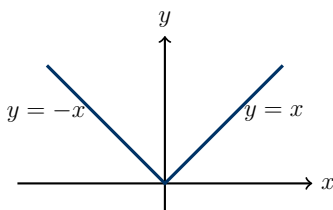
Prove that the function $f(x) = |x|$ is:

- (i) strictly increasing in $(0, \infty)$
- (ii) strictly decreasing in $(-\infty, 0)$

TYPE 511 Proving Monotonicity on an Interval – Proving Monotonicity of Absolute Value Function

To Prove: Monotonicity of $f(x) = |x|$ on $(0, \infty)$ and $(-\infty, 0)$.

Concept / Formula Used: $|x| = x$ $x > 0$ $|x| = -x$ $x < 0$



$$\text{Graph of } f(x) = |x|$$

Solution:

The absolute value function is defined piecewise:

$$f(x) = |x| = \begin{cases} x, & \text{if } x > 0 \\ -x, & \text{if } x < 0 \end{cases}$$

(i) For $x \in (0, \infty)$: Here $f(x) = x$.

$$f'(x) = \frac{d}{dx}(x) = 1 > 0$$

Since $f'(x) > 0$ for all $x \in (0, \infty)$, $f(x) = |x|$ is strictly increasing in $(0, \infty)$.

(ii) For $x \in (-\infty, 0)$: Here $f(x) = -x$.

$$f'(x) = \frac{d}{dx}(-x) = -1 < 0$$

Since $f'(x) < 0$ for all $x \in (-\infty, 0)$, $f(x) = |x|$ is strictly decreasing in $(-\infty, 0)$.

Hence Proved

The function $f(x) = |x|$ is strictly increasing in $(0, \infty)$ and strictly decreasing in $(-\infty, 0)$.

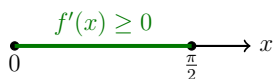
Question 23

Prove that the function f given by $f(x) = \frac{4 \sin x}{2 + \cos x} - x$ is increasing in $[0, \frac{\pi}{2}]$.

TYPE 51m Proving Monotonicity on an Interval – Proving Monotonicity of Rational Trigonometric Function with Linear Term

To Prove: $f(x) = \frac{4 \sin x}{2 + \cos x} - x$ is increasing in $[0, \frac{\pi}{2}]$.

Concept / Formula Used: $\frac{d}{dx} \left[\frac{u}{v} \right] = \frac{vu' - uv'}{v^2}$ $f'(x) \geq 0$



Domain interval for $f'(x)$

Solution:

Given function:

$$f(x) = \frac{4 \sin x}{2 + \cos x} - x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\frac{4 \sin x}{2 + \cos x} \right) - \frac{d}{dx}(x) \\ &= \frac{(2 + \cos x) \frac{d}{dx}(4 \sin x) - (4 \sin x) \frac{d}{dx}(2 + \cos x)}{(2 + \cos x)^2} - 1 \\ &= \frac{(2 + \cos x)(4 \cos x) - (4 \sin x)(-\sin x)}{(2 + \cos x)^2} - 1 \\ &= \frac{8 \cos x + 4 \cos^2 x + 4 \sin^2 x}{(2 + \cos x)^2} - 1 \end{aligned}$$

Using $\cos^2 x + \sin^2 x = 1$:

$$\begin{aligned} f'(x) &= \frac{8 \cos x + 4}{(2 + \cos x)^2} - 1 \\ &= \frac{8 \cos x + 4 - (2 + \cos x)^2}{(2 + \cos x)^2} \\ &= \frac{8 \cos x + 4 - (4 + 4 \cos x + \cos^2 x)}{(2 + \cos x)^2} \\ &= \frac{4 \cos x - \cos^2 x}{(2 + \cos x)^2} \\ &= \frac{\cos x(4 - \cos x)}{(2 + \cos x)^2} \end{aligned}$$

Analyzing the sign of each factor on $\left[0, \frac{\pi}{2}\right]$:

$$\begin{aligned} \cos x &\geq 0 \quad \text{for all } x \in \left[0, \frac{\pi}{2}\right] \\ 4 - \cos x &\geq 3 > 0 \quad \text{since } -1 \leq \cos x \leq 1 \\ (2 + \cos x)^2 &> 0 \quad \text{for all real } x \end{aligned}$$

Therefore:

$$f'(x) = \frac{\cos x(4 - \cos x)}{(2 + \cos x)^2} \geq 0 \quad \text{for all } x \in \left[0, \frac{\pi}{2}\right]$$

Hence Proved

Since $f'(x) \geq 0$ for all $x \in \left[0, \frac{\pi}{2}\right]$, the function $f(x)$ is increasing in $\left[0, \frac{\pi}{2}\right]$.

Question 24 (i)

Find the intervals in which the function $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = 2x^3 - 3x^2 - 36x + 7$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$

$$\begin{array}{ccccccc} & + & & - & & + & \\ & \bullet & & & & \bullet & \\ & -2 & & & & 3 & \\ \text{Sign scheme of } f'(x) & = & 6(x - 3)(x + 2) & & & & \end{array}$$

Solution:

Given function:

$$f(x) = 2x^3 - 3x^2 - 36x + 7$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 6x^2 - 6x - 36 \\ &= 6(x^2 - x - 6) \\ &= 6(x - 3)(x + 2) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points:

$$x = -2, \quad x = 3$$

These points divide the real line into three intervals: $(-\infty, -2)$, $(-2, 3)$, and $(3, \infty)$.

$$\text{For } x \in (-\infty, -2) : (x - 3) < 0, (x + 2) < 0 \implies f'(x) > 0$$

$$\text{For } x \in (-2, 3) : (x - 3) < 0, (x + 2) > 0 \implies f'(x) < 0$$

$$\text{For } x \in (3, \infty) : (x - 3) > 0, (x + 2) > 0 \implies f'(x) > 0$$

Final Answer

Strictly increasing in $(-\infty, -2] \cup [3, \infty)$

Strictly decreasing in $[-2, 3]$

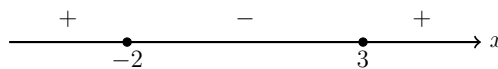
Question 24 (ii)

Find the intervals in which the function $f(x) = 4x^3 - 6x^2 - 72x + 30$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = 4x^3 - 6x^2 - 72x + 30$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$



$$\text{Sign scheme for } f'(x) = 12(x - 3)(x + 2)$$

Solution:

Given function:

$$f(x) = 4x^3 - 6x^2 - 72x + 30$$

Differentiating with respect to x :

$$f'(x) = 12x^2 - 12x - 72$$

$$= 12(x^2 - x - 6)$$

$$= 12(x - 3)(x + 2)$$

Setting $f'(x) = 0$ gives critical points:

$$x = -2, \quad x = 3$$

Evaluating signs of $f'(x)$:

$$\text{On } (-\infty, -2) : f'(x) > 0$$

$$\text{On } (-2, 3) : f'(x) < 0$$

$$\text{On } (3, \infty) : f'(x) > 0$$

Final Answer

Strictly increasing in $(-\infty, -2] \cup [3, \infty)$

Strictly decreasing in $[-2, 3]$

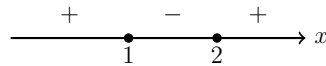
Question 24 (iii)

Find the intervals in which the function $f(x) = 2x^3 - 9x^2 + 12x + 15$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = 2x^3 - 9x^2 + 12x + 15$.

Concept / Formula Used: $f'(x) = 0$ $f'(x)$



$$\text{Sign scheme for } f'(x) = 6(x - 1)(x - 2)$$

Solution:

Given function:

$$f(x) = 2x^3 - 9x^2 + 12x + 15$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 6x^2 - 18x + 12 \\ &= 6(x^2 - 3x + 2) \\ &= 6(x - 1)(x - 2) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points:

$$x = 1, \quad x = 2$$

Evaluating signs of $f'(x)$:

$$\text{On } (-\infty, 1) : f'(x) > 0$$

$$\text{On } (1, 2) : f'(x) < 0$$

$$\text{On } (2, \infty) : f'(x) > 0$$

Final Answer

Strictly increasing in $(-\infty, 1] \cup [2, \infty)$

Strictly decreasing in $[1, 2]$

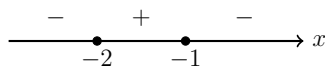
Question 24 (iv)

Find the intervals in which the function $f(x) = -2x^3 - 9x^2 - 12x + 1$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = -2x^3 - 9x^2 - 12x + 1$.

Concept / Formula Used: $f'(x) = 0$



$$\text{Sign scheme for } f'(x) = -6(x+1)(x+2)$$

Solution:

Given function:

$$f(x) = -2x^3 - 9x^2 - 12x + 1$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= -6x^2 - 18x - 12 \\ &= -6(x^2 + 3x + 2) \\ &= -6(x+1)(x+2) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points:

$$x = -2, \quad x = -1$$

Testing interval signs:

$$\text{For } x < -2 : (x+1) < 0, (x+2) < 0 \implies f'(x) < 0$$

$$\text{For } -2 < x < -1 : (x+1) < 0, (x+2) > 0 \implies f'(x) > 0$$

$$\text{For } x > -1 : (x+1) > 0, (x+2) > 0 \implies f'(x) < 0$$

Final Answer

Strictly increasing in $[-2, -1]$

Strictly decreasing in $(-\infty, -2] \cup [-1, \infty)$

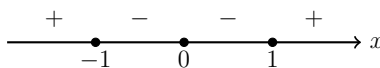
Question 24 (v)

Find the intervals in which the function $f(x) = x + \frac{1}{x}, x \neq 0$ is strictly increasing or strictly decreasing.

TYPE 56c Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Rational Functions

To Find: Intervals of strict monotonicity for $f(x) = x + \frac{1}{x}$.

Concept / Formula Used: $f'(x) = \frac{d}{dx} \left(x + \frac{1}{x} \right) = 1 - \frac{1}{x^2}$



$$\text{Sign scheme for } f'(x) = \frac{x^2-1}{x^2}$$

Solution:

Given function:

$$f(x) = x + \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 1 - \frac{1}{x^2} \\ &= \frac{x^2 - 1}{x^2} \\ &= \frac{(x-1)(x+1)}{x^2} \end{aligned}$$

Since $x^2 > 0$ for all $x \neq 0$, the sign of $f'(x)$ is governed by $(x-1)(x+1)$. Setting $f'(x) = 0 \implies x = -1, 1$. Domain excludes $x = 0$.

$$\text{On } (-\infty, -1) : f'(x) > 0$$

$$\text{On } (-1, 0) : f'(x) < 0$$

$$\text{On } (0, 1) : f'(x) < 0$$

$$\text{On } (1, \infty) : f'(x) > 0$$

Final Answer

Strictly increasing in $(-\infty, -1] \cup [1, \infty)$

Strictly decreasing in $[-1, 0) \cup (0, 1]$

Question 24 (vi)

Find the intervals in which the function $f(x) = \frac{4x^2+1}{x}$, $x \neq 0$ is strictly increasing or strictly decreasing.

TYPE 56c Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Rational Functions

To Find: Intervals of strict monotonicity for $f(x) = \frac{4x^2+1}{x}$.

Concept / Formula Used: $f(x) = 4x + \frac{1}{x}$

$$\begin{array}{ccccccc} & + & & - & & - & & + \\ & \bullet & & \bullet & & \bullet & & \\ & -\frac{1}{2} & & 0 & & \frac{1}{2} & & \end{array} \rightarrow x$$

Sign scheme for $f'(x)$ = $\frac{4x^2-1}{x^2}$

Solution:

Rewrite the function:

$$f(x) = 4x + \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 4 - \frac{1}{x^2} \\ &= \frac{4x^2 - 1}{x^2} \\ &= \frac{(2x-1)(2x+1)}{x^2} \end{aligned}$$

Critical points: $x = -\frac{1}{2}, \frac{1}{2}$. Point of discontinuity: $x = 0$. Since $x^2 > 0$ for $x \neq 0$:

$$\text{On } \left(-\infty, -\frac{1}{2}\right) : f'(x) > 0$$

$$\text{On } \left(-\frac{1}{2}, 0\right) : f'(x) < 0$$

$$\text{On } \left(0, \frac{1}{2}\right) : f'(x) < 0$$

$$\text{On } \left(\frac{1}{2}, \infty\right) : f'(x) > 0$$

Final Answer

Strictly increasing in $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$
 Strictly decreasing in $\left[-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right]$

Question 24 (vii)

Find the intervals in which the function $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$ is strictly increasing or strictly decreasing.

TYPE 56d Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quartic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$.

Concept / Formula Used: $f'(x) = 4x^3 - 24x^2 + 44x - 24$

$$\begin{array}{ccccccc} & - & & + & & - & & + \\ & \bullet & & \bullet & & \bullet & & \bullet \\ & 1 & & 2 & & 3 & & \end{array} \rightarrow x$$

$$\text{Sign scheme for } f'(x) = 4(x-1)(x-2)(x-3)$$

Solution:

Given function:

$$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 4x^3 - 24x^2 + 44x - 24 \\ &= 4(x^3 - 6x^2 + 11x - 6) \end{aligned}$$

By inspection, $x = 1$ is a root of $x^3 - 6x^2 + 11x - 6 = 0$. Factorizing:

$$\begin{aligned} f'(x) &= 4(x-1)(x^2 - 5x + 6) \\ &= 4(x-1)(x-2)(x-3) \end{aligned}$$

Critical points: $x = 1, 2, 3$. Testing interval signs:

$$\text{For } x \in (-\infty, 1) : f'(x) < 0$$

$$\text{For } x \in (1, 2) : f'(x) > 0$$

$$\text{For } x \in (2, 3) : f'(x) < 0$$

$$\text{For } x \in (3, \infty) : f'(x) > 0$$

Final AnswerStrictly increasing in $[1, 2] \cup [3, \infty)$ Strictly decreasing in $(-\infty, 1] \cup [2, 3]$ **Question 24 (viii)**

Find the intervals in which the function $f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$ is strictly increasing or strictly decreasing.

TYPE 56d Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quartic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$.

Concept / Formula Used: $f'(x) = x^3 - 3x^2 - 10x + 24$

$$\begin{array}{ccccccc} & - & & + & & - & & + \\ & \bullet & & & & \bullet & & \bullet \\ & -3 & & & & 2 & & 4 \end{array} \xrightarrow{x}$$

Sign scheme for $f'(x) = (x + 3)(x - 2)(x - 4)$

Solution:

Given function:

$$f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$$

Differentiating with respect to x :

$$f'(x) = x^3 - 3x^2 - 10x + 24$$

Testing $x = 2$: $2^3 - 3(2)^2 - 10(2) + 24 = 8 - 12 - 20 + 24 = 0$. So $(x - 2)$ is a factor:

$$\begin{aligned} f'(x) &= (x - 2)(x^2 - x - 12) \\ &= (x - 2)(x - 4)(x + 3) \end{aligned}$$

Critical points: $x = -3, 2, 4$. Testing interval signs:

$$\text{For } x \in (-\infty, -3) : f'(x) < 0$$

$$\text{For } x \in (-3, 2) : f'(x) > 0$$

$$\text{For } x \in (2, 4) : f'(x) < 0$$

$$\text{For } x \in (4, \infty) : f'(x) > 0$$

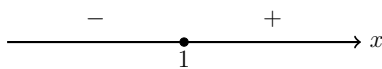
Final AnswerStrictly increasing in $[-3, 2] \cup [4, \infty)$ Strictly decreasing in $(-\infty, -3] \cup [2, 4]$ **Question 24 (ix)**

Find the intervals in which the function $f(x) = x^4 - 4x$ is strictly increasing or strictly decreasing.

TYPE 56d Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quartic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = x^4 - 4x$.

Concept / Formula Used: $f'(x) = 4(x - 1)(x^2 + x + 1)$



$$\text{Sign scheme for } f'(x) = 4(x - 1)(x^2 + x + 1)$$

Solution:

Given function:

$$f(x) = x^4 - 4x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 4x^3 - 4 \\ &= 4(x^3 - 1) \\ &= 4(x - 1)(x^2 + x + 1) \end{aligned}$$

For the quadratic factor $x^2 + x + 1$:

$$\text{Discriminant } D = 1^2 - 4(1)(1) = -3 < 0$$

Thus $x^2 + x + 1 > 0$ for all real x . Hence, the sign of $f'(x)$ depends solely on $(x - 1)$.

$$\text{For } x < 1 : f'(x) < 0$$

$$\text{For } x > 1 : f'(x) > 0$$

Final Answer

Strictly increasing in $[1, \infty)$

Strictly decreasing in $(-\infty, 1]$

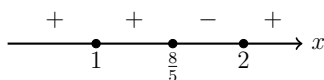
Question 24 (x)

Find the intervals in which the function $f(x) = (x - 1)^3(x - 2)^2$ is strictly increasing or strictly decreasing.

TYPE 56e Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Polynomials with Repeated Factors

To Find: Intervals of strict monotonicity for $f(x) = (x - 1)^3(x - 2)^2$.

Concept / Formula Used: $\frac{d}{dx}[uv] = u'v + uv'$



$$\text{Sign scheme for } f'(x) = (x - 1)^2(x - 2)(5x - 8)$$

Solution:

Given function:

$$f(x) = (x - 1)^3(x - 2)^2$$

Differentiating with respect to x using product rule:

$$\begin{aligned} f'(x) &= 3(x-1)^2(x-2)^2 + (x-1)^3 \cdot 2(x-2) \\ &= (x-1)^2(x-2)[3(x-2) + 2(x-1)] \\ &= (x-1)^2(x-2)[3x-6+2x-2] \\ &= (x-1)^2(x-2)(5x-8) \end{aligned}$$

Critical points: $x = 1, \frac{8}{5}, 2$. Since $(x-1)^2 \geq 0$ for all real x , it does not change the sign of $f'(x)$. Evaluating the sign of $(x-2)(5x-8)$:

$$\text{For } x \in \left(-\infty, \frac{8}{5}\right) \setminus \{1\}: (x-2) < 0, (5x-8) < 0 \implies f'(x) > 0$$

$$\text{For } x \in \left(\frac{8}{5}, 2\right): (x-2) < 0, (5x-8) > 0 \implies f'(x) < 0$$

$$\text{For } x \in (2, \infty): (x-2) > 0, (5x-8) > 0 \implies f'(x) > 0$$

Final Answer

Strictly increasing in $\left(-\infty, \frac{8}{5}\right] \cup [2, \infty)$

Strictly decreasing in $\left[\frac{8}{5}, 2\right]$

Question 25 (i)

Find the interval in which the function $f(x) = x^2e^{-x}$ is increasing.

TYPE 56f Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Product of Quadratic and Exponential Functions

To Find: Interval where $f(x) = x^2e^{-x}$ is increasing.

Concept / Formula Used: $\frac{d}{dx}[uv] = u'v + uv' \quad f(x) \quad f'(x) \geq 0$

$$\begin{array}{c} - \qquad \qquad + \qquad \qquad - \\ \hline \qquad \qquad 0 \qquad \qquad 2 \qquad \qquad \rightarrow x \\ \text{Sign scheme for } f'(x) \quad = \quad x(2-x)e^{-x} \end{array}$$

Solution:

Given function:

$$f(x) = x^2e^{-x}$$

Differentiating with respect to x using product rule:

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^2) \cdot e^{-x} + x^2 \cdot \frac{d}{dx}(e^{-x}) \\ &= 2xe^{-x} + x^2(-e^{-x}) \\ &= e^{-x}(2x - x^2) \\ &= x(2-x)e^{-x} \end{aligned}$$

Since $e^{-x} > 0$ for all real x , the sign of $f'(x)$ depends on $x(2-x)$. For $f(x)$ to be increasing:

$$\begin{aligned}f'(x) &\geq 0 \\x(2-x)e^{-x} &\geq 0 \\x(2-x) &\geq 0 \\x(x-2) &\leq 0 \\0 &\leq x \leq 2\end{aligned}$$

Final Answer

The function $f(x) = x^2e^{-x}$ is increasing in $[0, 2]$.

Question 25 (ii)

Find the intervals in which the function given by $f(x) = x^2e^x$ is strictly increasing or strictly decreasing.

TYPE 56f Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Product of Quadratic and Exponential Functions

To Find: Intervals of strict monotonicity for $f(x) = x^2e^x$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$

$$\begin{array}{ccccccc} & + & & - & & + & \\ & \bullet & & \bullet & & & \\ & -2 & & 0 & & & \\ \text{Sign scheme for } f'(x) & = & x(x+2)e^x\end{array}$$

Solution:

Given function:

$$f(x) = x^2e^x$$

Differentiating with respect to x using product rule:

$$\begin{aligned}f'(x) &= 2xe^x + x^2e^x \\&= e^x(x^2 + 2x) \\&= x(x+2)e^x\end{aligned}$$

Since $e^x > 0$ for all real x , $f'(x) = 0 \implies x = 0, -2$. Testing interval signs:

$$\text{For } x \in (-\infty, -2) : x < 0, (x+2) < 0 \implies f'(x) > 0$$

$$\text{For } x \in (-2, 0) : x < 0, (x+2) > 0 \implies f'(x) < 0$$

$$\text{For } x \in (0, \infty) : x > 0, (x+2) > 0 \implies f'(x) > 0$$

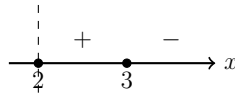
Final Answer

Strictly increasing in $(-\infty, -2] \cup [0, \infty)$

Strictly decreasing in $[-2, 0]$

Question 26

Find the intervals in which the function $f(x) = 2 \log(x-2) - x^2 + 4x - 5$ is strictly increasing or strictly decreasing.

TYPE 56g Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Logarithmic-Polynomial Combination**To Find:** Intervals of strict monotonicity for $f(x) = 2 \log(x - 2) - x^2 + 4x - 5$.**Concept / Formula Used:** $x - 2 > 0 \implies x > 2$ *Sign scheme of $f'(x)$ on domain $(2, \infty)$* **Solution:**The domain of $f(x)$ requires:

$$x - 2 > 0 \implies x > 2$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}[2 \log(x - 2) - x^2 + 4x - 5] \\ &= \frac{2}{x - 2} - 2x + 4 \\ &= \frac{2}{x - 2} - 2(x - 2) \\ &= \frac{2 - 2(x - 2)^2}{x - 2} \\ &= \frac{2[1 - (x - 2)^2]}{x - 2} \\ &= \frac{2(1 - (x - 2))(1 + (x - 2))}{x - 2} \\ &= \frac{2(3 - x)(x - 1)}{x - 2} \end{aligned}$$

Since $x > 2$, both $(x - 2) > 0$ and $(x - 1) > 0$. Thus, the sign of $f'(x)$ depends solely on $(3 - x)$:

$$\text{For } x \in (2, 3) : 3 - x > 0 \implies f'(x) > 0$$

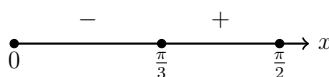
$$\text{For } x \in (3, \infty) : 3 - x < 0 \implies f'(x) < 0$$

Final AnswerStrictly increasing in $(2, 3]$ Strictly decreasing in $[3, \infty)$ **Question 27**Find the intervals in which the function f given by $f(x) = \tan x - 4x$, $x \in (0, \frac{\pi}{2})$ is:

- (i) strictly increasing
- (ii) strictly decreasing

TYPE 56h Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Trigonometric-Linear Function**To Find:** Intervals of strict monotonicity for $f(x) = \tan x - 4x$ on $(0, \frac{\pi}{2})$.

Concept / Formula Used: $\frac{d}{dx}(\tan x) = \sec^2 x$



Sign scheme for $f'(x) = \sec^2 x - 4$ on $(0, \pi/2)$

Solution:

Given function:

$$f(x) = \tan x - 4x, \quad x \in \left(0, \frac{\pi}{2}\right)$$

Differentiating with respect to x :

$$f'(x) = \sec^2 x - 4$$

Setting $f'(x) = 0$:

$$\sec^2 x - 4 = 0$$

$$\sec^2 x = 4$$

$$\sec x = 2 \quad \left(\text{since } \sec x > 0 \text{ in } \left(0, \frac{\pi}{2}\right)\right)$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}$$

This critical point $x = \frac{\pi}{3}$ divides $\left(0, \frac{\pi}{2}\right)$ into two sub-intervals:

(i) For $x \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$:

$$\cos x < \frac{1}{2} \implies \sec x > 2 \implies \sec^2 x > 4 \implies f'(x) > 0$$

Hence, $f(x)$ is strictly increasing in $\left[\frac{\pi}{3}, \frac{\pi}{2}\right)$.

(ii) For $x \in \left(0, \frac{\pi}{3}\right)$:

$$\cos x > \frac{1}{2} \implies \sec x < 2 \implies \sec^2 x < 4 \implies f'(x) < 0$$

Hence, $f(x)$ is strictly decreasing in $\left(0, \frac{\pi}{3}\right]$.

Final Answer

(i) Strictly increasing in $\left[\frac{\pi}{3}, \frac{\pi}{2}\right)$

(ii) Strictly decreasing in $\left(0, \frac{\pi}{3}\right]$

6.5 – Maxima and Minima Without Derivative Tests

Question 1

Find the maximum and the minimum values (if any) of the following functions:

(i) $f(x) = x^2$

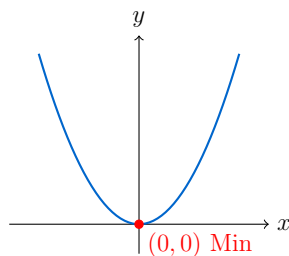
(ii) $f(x) = (2x - 1)^2 + 3$

TYPE 79a Finding Local Extrema by Inspection – Finding Extrema of Quadratic Functions via Inspection

Given: Functions $f(x) = x^2$ and $f(x) = (2x - 1)^2 + 3$ defined for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of $f(x)$.

Concept / Formula Used: $A \quad A^2 \geq 0 \quad 0 \quad A = 0 \quad A \rightarrow \pm\infty$



Graph of $y = x^2$ illustrating global minimum at origin $(0, 0)$

Solution:

Part (i): $f(x) = x^2$

For all real numbers $x \in \mathbb{R}$, we have:

$$x^2 \geq 0$$

Thus, $f(x) \geq 0$ for all $x \in \mathbb{R}$.

The minimum value occurs when $x = 0$:

$$f(0) = 0^2 = 0$$

As $x \rightarrow \pm\infty$, $f(x) \rightarrow \infty$. Therefore, $f(x)$ has no maximum value.

Part (ii): $f(x) = (2x - 1)^2 + 3$

For all $x \in \mathbb{R}$, the squared term satisfies:

$$(2x - 1)^2 \geq 0$$

Adding 3 to both sides of the inequality:

$$(2x - 1)^2 + 3 \geq 0 + 3$$

$$f(x) \geq 3$$

The minimum value occurs when $2x - 1 = 0 \implies x = \frac{1}{2}$:

Evaluating $f(x)$ at $x = \frac{1}{2}$:

$$\begin{aligned} f\left(\frac{1}{2}\right) &= \left(2\left(\frac{1}{2}\right) - 1\right)^2 + 3 \\ &= (1 - 1)^2 + 3 \\ &= 0 + 3 = 3 \end{aligned}$$

Since $(2x - 1)^2 \rightarrow \infty$ as $x \rightarrow \pm\infty$, there is no maximum value.

Final Answer

(i) Minimum value = 0 at $x = 0$; No maximum value.

(ii) Minimum value = 3 at $x = \frac{1}{2}$; No maximum value.

Question 2

Find the maximum and the minimum values (if any) of the following functions:

(i) $f(x) = -(x - 1)^2 + 10$

(ii) $f(x) = 9x^2 + 12x + 2$

TYPE 79a Finding Local Extrema by Inspection – Finding Extrema of Quadratic Functions via Inspection

Given: Functions $f(x) = -(x - 1)^2 + 10$ and $f(x) = 9x^2 + 12x + 2$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $A \quad A^2 \geq 0 \implies -A^2 \leq 0 \quad ax^2 + bx + c$

Solution:

Part (i): $f(x) = -(x - 1)^2 + 10$

For all $x \in \mathbb{R}$, we have:

$$(x - 1)^2 \geq 0$$

Multiplying by -1 reverses the inequality:

$$-(x - 1)^2 \leq 0$$

Adding 10 to both sides:

$$\begin{aligned} -(x - 1)^2 + 10 &\leq 10 \\ f(x) &\leq 10 \end{aligned}$$

The maximum value is attained when $x - 1 = 0 \implies x = 1$:

$$f(1) = -(1 - 1)^2 + 10 = 10$$

Since $-(x - 1)^2 \rightarrow -\infty$ as $x \rightarrow \pm\infty$, there is no minimum value.

Part (ii): $f(x) = 9x^2 + 12x + 2$

Completing the square for the quadratic expression:

$$\begin{aligned} f(x) &= (3x)^2 + 2(3x)(2) + 2^2 - 2^2 + 2 \\ &= (3x + 2)^2 - 4 + 2 \\ &= (3x + 2)^2 - 2 \end{aligned}$$

Since $(3x + 2)^2 \geq 0$ for all $x \in \mathbb{R}$:

$$\begin{aligned} (3x + 2)^2 - 2 &\geq -2 \\ f(x) &\geq -2 \end{aligned}$$

The minimum value occurs when $3x + 2 = 0 \implies x = -\frac{2}{3}$:

$$f\left(-\frac{2}{3}\right) = \left(3\left(-\frac{2}{3}\right) + 2\right)^2 - 2 = 0 - 2 = -2$$

As $x \rightarrow \pm\infty$, $f(x) \rightarrow \infty$, so there is no maximum value.

Final Answer

- (i) Maximum value = 10 at $x = 1$; No minimum value.
- (ii) Minimum value = -2 at $x = -\frac{2}{3}$; No maximum value.

Question 3

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = |x|$
- (ii) $f(x) = x^3 + 1$

TYPE 79b Finding Local Extrema by Inspection – Finding Extrema of Power and Absolute Value Functions

Given: Functions $f(x) = |x|$ and $f(x) = x^3 + 1$ defined for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $|x| \geq 0 \quad x \in \mathbb{R}$

Solution:

Part (i): $f(x) = |x|$

By definition of the absolute value function, for all $x \in \mathbb{R}$:

$$|x| \geq 0$$

Thus, $f(x) \geq 0$ for all $x \in \mathbb{R}$.

The minimum value occurs at $x = 0$:

$$f(0) = |0| = 0$$

As $x \rightarrow \pm\infty$, $|x| \rightarrow \infty$, so $f(x)$ has no maximum value.

Part (ii): $f(x) = x^3 + 1$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 + 1) \\ &= 3x^2 \end{aligned}$$

Since $3x^2 \geq 0$ for all $x \in \mathbb{R}$ and $f'(x) = 0$ only at $x = 0$, $f(x)$ is strictly increasing on $\mathbb{R}$.

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

Therefore, $f(x)$ has neither a maximum value nor a minimum value.

Final Answer

- (i) Minimum value = 0 at $x = 0$; No maximum value.
- (ii) Neither maximum value nor minimum value exists.

Question 4

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = |x + 2| - 1$
- (ii) $f(x) = 3 + |x|$

TYPE 79b Finding Local Extrema by Inspection – Finding Extrema of Power and Absolute Value Functions

Given: Functions $f(x) = |x + 2| - 1$ and $f(x) = 3 + |x|$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $|A| \geq 0 \quad A \in \mathbb{R} \quad 0 \quad A = 0$

Solution:

Part (i): $f(x) = |x + 2| - 1$

For all $x \in \mathbb{R}$, we have:

$$|x + 2| \geq 0$$

Subtracting 1 from both sides:

$$\begin{aligned} |x + 2| - 1 &\geq 0 - 1 \\ f(x) &\geq -1 \end{aligned}$$

The minimum value occurs when $x + 2 = 0 \implies x = -2$:

$$f(-2) = |-2 + 2| - 1 = 0 - 1 = -1$$

Since $|x + 2| \rightarrow \infty$ as $x \rightarrow \pm\infty$, $f(x)$ has no maximum value.

Part (ii): $f(x) = 3 + |x|$

For all $x \in \mathbb{R}$, we have:

$$|x| \geq 0$$

Adding 3 to both sides:

$$\begin{aligned} 3 + |x| &\geq 3 + 0 \\ f(x) &\geq 3 \end{aligned}$$

The minimum value occurs when $x = 0$:

$$f(0) = 3 + |0| = 3$$

Since $|x| \rightarrow \infty$ as $x \rightarrow \pm\infty$, there is no maximum value.

Final Answer

- (i) Minimum value = -1 at $x = -2$; No maximum value.
 (ii) Minimum value = 3 at $x = 0$; No maximum value.

Question 5

Find the maximum and the minimum values (if any) of the following functions:

(i) $f(x) = -|x + 1| + 3$

(ii) $f(x) = \sin 2x$

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions

Given: Functions $f(x) = -|x + 1| + 3$ and $f(x) = \sin 2x$ defined on $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $|A| \geq 0 \implies -|A| \leq 0 \quad \theta \quad -1 \leq \sin \theta \leq 1$

Solution:

Part (i): $f(x) = -|x + 1| + 3$

For all $x \in \mathbb{R}$, we have:

$$|x + 1| \geq 0$$

Multiplying by -1 reverses the inequality:

$$-|x + 1| \leq 0$$

Adding 3 to both sides:

$$\begin{aligned} -|x + 1| + 3 &\leq 3 \\ f(x) &\leq 3 \end{aligned}$$

The maximum value occurs when $x + 1 = 0 \implies x = -1$:

$$f(-1) = -|-1 + 1| + 3 = 0 + 3 = 3$$

As $x \rightarrow \pm\infty$, $-|x + 1| \rightarrow -\infty$, so there is no minimum value.

Part (ii): $f(x) = \sin 2x$

For all $x \in \mathbb{R}$, the standard range of the sine function is $[-1, 1]$:

$$-1 \leq \sin 2x \leq 1$$

Therefore:

$$\text{Maximum value of } f(x) = 1$$

$$\text{Minimum value of } f(x) = -1$$

Maximum value 1 occurs at $2x = \frac{\pi}{2} + 2k\pi \implies x = \frac{\pi}{4} + k\pi, k \in \mathbb{Z}$.

Minimum value -1 occurs at $2x = \frac{3\pi}{2} + 2k\pi \implies x = \frac{3\pi}{4} + k\pi, k \in \mathbb{Z}$.

Final Answer

- (i) Maximum value = 3 at $x = -1$; No minimum value.
 (ii) Maximum value = 1, Minimum value = -1 .

Question 6

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = \sin 2x + 5$
 (ii) $f(x) = |\sin 4x + 3|$

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions

Given: Functions $f(x) = \sin 2x + 5$ and $f(x) = |\sin 4x + 3|$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $\theta \quad -1 \leq \sin \theta \leq 1$

Solution:

Part (i): $f(x) = \sin 2x + 5$

For all $x \in \mathbb{R}$, we know:

$$-1 \leq \sin 2x \leq 1$$

Adding 5 across the inequality:

$$\begin{aligned} -1 + 5 &\leq \sin 2x + 5 \leq 1 + 5 \\ 4 &\leq f(x) \leq 6 \end{aligned}$$

Therefore:

Maximum value = 6

Minimum value = 4

Part (ii): $f(x) = |\sin 4x + 3|$

For all $x \in \mathbb{R}$, we know:

$$-1 \leq \sin 4x \leq 1$$

Adding 3 across the inequality:

$$\begin{aligned} -1 + 3 &\leq \sin 4x + 3 \leq 1 + 3 \\ 2 &\leq \sin 4x + 3 \leq 4 \end{aligned}$$

Since the range $[2, 4]$ consists entirely of positive numbers, taking the absolute value preserves the bounds:

$$2 \leq |\sin 4x + 3| \leq 4$$

Therefore:

$$\text{Maximum value} = 4$$

$$\text{Minimum value} = 2$$

Final Answer

(i) Maximum value = 6, Minimum value = 4.

(ii) Maximum value = 4, Minimum value = 2.

Question 7

Find the maximum and the minimum values (if any) of the following functions:

(i) $f(x) = |\sin 3x| - 3$

(ii) $f(x) = 4 - |\cos 3x|$

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions

Given: Functions $f(x) = |\sin 3x| - 3$ and $f(x) = 4 - |\cos 3x|$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $-1 \leq \sin \theta \leq 1$ $-1 \leq \cos \theta \leq 1$ $0 \leq |\sin \theta| \leq 1$ $0 \leq |\cos \theta| \leq 1$

Solution:

Part (i): $f(x) = |\sin 3x| - 3$

For all $x \in \mathbb{R}$, the absolute value of sine is bounded as:

$$0 \leq |\sin 3x| \leq 1$$

Subtracting 3 across the inequality:

$$\begin{aligned} 0 - 3 &\leq |\sin 3x| - 3 \leq 1 - 3 \\ -3 &\leq f(x) \leq -2 \end{aligned}$$

Therefore:

$$\text{Maximum value} = -2$$

$$\text{Minimum value} = -3$$

Part (ii): $f(x) = 4 - |\cos 3x|$

For all $x \in \mathbb{R}$, the absolute value of cosine is bounded as:

$$0 \leq |\cos 3x| \leq 1$$

Multiplying by -1 reverses the inequality:

$$-1 \leq -|\cos 3x| \leq 0$$

Adding 4 across the inequality:

$$\begin{aligned} 4 - 1 &\leq 4 - |\cos 3x| \leq 4 + 0 \\ 3 &\leq f(x) \leq 4 \end{aligned}$$

Therefore:

$$\text{Maximum value} = 4$$

$$\text{Minimum value} = 3$$

Final Answer

- (i) Maximum value = -2 , Minimum value = -3 .
 (ii) Maximum value = 4 , Minimum value = 3 .

Question 8

- (i) Is the minimum value of $\cos x$ zero?
 (ii) What are the minimum and maximum values of $2 - 3 \cos x$?

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions

Given: Expressions $\cos x$ and $g(x) = 2 - 3 \cos x$ for real $x \in \mathbb{R}$.

To Find: (i) Verification of whether the minimum value of $\cos x$ is zero. (ii) Extreme values of $2 - 3 \cos x$.

Concept / Formula Used: $x \quad -1 \leq \cos x \leq 1$

Solution:

Part (i):

For all real numbers $x \in \mathbb{R}$, the domain-wide bounds of $\cos x$ are:

$$-1 \leq \cos x \leq 1$$

The minimum value attained by $\cos x$ is -1 , which occurs at $x = \pi + 2k\pi, k \in \mathbb{Z}$.

Hence, the minimum value of $\cos x$ is NOT zero; it is -1 .

Part (ii):

Let $g(x) = 2 - 3 \cos x$. We start from the standard inequality:

$$-1 \leq \cos x \leq 1$$

Multiplying all parts by -3 (reversing inequality direction):

$$\begin{aligned} (-3)(1) &\leq -3 \cos x \leq (-3)(-1) \\ -3 &\leq -3 \cos x \leq 3 \end{aligned}$$

Adding 2 to all parts:

$$\begin{aligned} 2 - 3 &\leq 2 - 3 \cos x \leq 2 + 3 \\ -1 &\leq 2 - 3 \cos x \leq 5 \end{aligned}$$

Therefore, the minimum value is -1 and the maximum value is 5 .

Final Answer

- (i) No, the minimum value of $\cos x$ is -1 .
 (ii) Minimum value = -1 , Maximum value = 5 .

Question 9

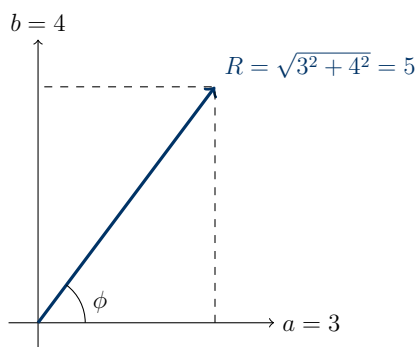
What are the maximum and minimum values of $3 \sin x + 4 \cos x$?

TYPE 79d Finding Local Extrema by Inspection – Finding Extrema of Linear Combination of Sine and Cosine

Given: Function $f(x) = 3 \sin x + 4 \cos x$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values of $f(x)$.

Concept / **Formula Used:** $a \sin x + b \cos x = R \sin(x + \phi)$ $R = \sqrt{a^2 + b^2}$
 $\mathbb{R} \quad [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$



Phasor representation of $3 \sin x + 4 \cos x$ showing amplitude $R = \sqrt{3^2 + 4^2} = 5$

Solution:

We express $3 \sin x + 4 \cos x$ in single harmonic form $R \sin(x + \phi)$.

Comparing $3 \sin x + 4 \cos x$ with $a \sin x + b \cos x$:

$$a = 3$$

$$b = 4$$

The amplitude R is calculated as:

$$\begin{aligned} R &= \sqrt{a^2 + b^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} = 5 \end{aligned}$$

Let $3 = 5 \cos \phi$ and $4 = 5 \sin \phi$, where $\tan \phi = \frac{4}{3}$.

Substituting these into $f(x)$:

$$\begin{aligned} f(x) &= 5 \cos \phi \sin x + 5 \sin \phi \cos x \\ &= 5(\sin x \cos \phi + \cos x \sin \phi) \\ &= 5 \sin(x + \phi) \end{aligned}$$

Since $-1 \leq \sin(x + \phi) \leq 1$ for all real x :

$$\begin{aligned} -5(1) &\leq 5 \sin(x + \phi) \leq 5(1) \\ -5 &\leq f(x) \leq 5 \end{aligned}$$

Thus:

$$\text{Maximum value} = 5$$

$$\text{Minimum value} = -5$$

Final Answer

Maximum value = 5, Minimum value = -5.

Question 10

Find the maximum and minimum values, if any, of the following functions:

(i) $f(x) = x$ in $(0, 1)$

(ii) $f(x) = x$ in $[0, 1]$

TYPE 79e Finding Local Extrema by Inspection – Analyzing Extrema of Identity Function on Intervals

Given: Function $f(x) = x$ defined on (i) open interval $(0, 1)$, and (ii) closed interval $[0, 1]$.

To Find: The maximum and minimum values of $f(x)$ in each domain.

Concept / Formula Used: $[a, b]$ (a, b)

Solution:

Part (i): $f(x) = x$ on $(0, 1)$

The domain is the open interval $x \in (0, 1)$.

For any $x \in (0, 1)$, we have $0 < f(x) < 1$.

Suppose $f(x)$ attains a minimum value at some point $c \in (0, 1)$. Then $f(c) = c > 0$.

However, choosing $x_1 = \frac{c}{2} \in (0, 1)$ gives:

$$f(x_1) = \frac{c}{2} < c = f(c)$$

Thus, no element in $(0, 1)$ can serve as a minimum value.

Similarly, suppose $f(x)$ attains a maximum value at $d \in (0, 1)$. Then $f(d) = d < 1$.

Choosing $x_2 = \frac{d+1}{2} \in (0, 1)$ gives:

$$f(x_2) = \frac{d+1}{2} > d = f(d)$$

Thus, no element in $(0, 1)$ can serve as a maximum value.

Therefore, $f(x)$ has neither a maximum value nor a minimum value on $(0, 1)$.

Part (ii): $f(x) = x$ on $[0, 1]$

The domain is the closed interval $x \in [0, 1]$.

Since $0 \leq x \leq 1$ for all $x \in [0, 1]$:

$$0 \leq f(x) \leq 1$$

The boundary points are included in the domain:

$$f(0) = 0 \quad (\text{Minimum value at } x = 0)$$

$$f(1) = 1 \quad (\text{Maximum value at } x = 1)$$

Final Answer

- (i) No maximum value, no minimum value in $(0, 1)$.
 (ii) Maximum value = 1 at $x = 1$, Minimum value = 0 at $x = 0$.

Question 11

Prove that the function $f(x) = x^3 + x^2 + x + 1$ does not have maxima or minima.

TYPE 84 Proving Non-Existence of Extrema via Strictly Monotonic Derivative

To Prove: The function $f(x) = x^3 + x^2 + x + 1$ has no local or global maxima or minima on $\mathbb{R}$.

Concept / Formula Used: $x = c$ $f'(c) = 0$ $f'(x) > 0$ $\mathbb{R}$ $f(x)$

Solution:

The given function is:

$$f(x) = x^3 + x^2 + x + 1$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 + x^2 + x + 1) \\ &= 3x^2 + 2x + 1 \end{aligned}$$

To check for critical points, set $f'(x) = 0$:

$$3x^2 + 2x + 1 = 0$$

Calculating the discriminant $D = b^2 - 4ac$ for this derivative quadratic equation:

$$\begin{aligned} D &= 2^2 - 4(3)(1) \\ &= 4 - 12 \\ &= -8 < 0 \end{aligned}$$

Since the coefficient of x^2 is $a = 3 > 0$ and the discriminant $D < 0$, the quadratic derivative $3x^2 + 2x + 1$ is strictly positive for all real numbers $x \in \mathbb{R}$.

Alternatively, completing the square for $f'(x)$:

$$\begin{aligned} f'(x) &= 3 \left(x^2 + \frac{2}{3}x \right) + 1 \\ &= 3 \left(x^2 + \frac{2}{3}x + \frac{1}{9} \right) + 1 - 3 \left(\frac{1}{9} \right) \\ &= 3 \left(x + \frac{1}{3} \right)^2 + 1 - \frac{1}{3} \\ &= 3 \left(x + \frac{1}{3} \right)^2 + \frac{2}{3} \end{aligned}$$

Since $\left(x + \frac{1}{3} \right)^2 \geq 0$, we get:

$$f'(x) \geq \frac{2}{3} > 0 \quad \forall x \in \mathbb{R}$$

Because $f'(x) \neq 0$ for all $x \in \mathbb{R}$, there exist no critical points.

Hence Proved

Since $f'(x) > 0$ for all real x , $f(x)$ has no critical points and is strictly increasing on $\mathbb{R}$. Therefore, $f(x)$ does not have maxima or minima. Hence proved.

Question 12

Find the (absolute) maximum and the (absolute) minimum values of the following functions in the indicated intervals. Also find the points of (absolute) maxima and minima:

- (i) $f(x) = (x - 1)^2 + 3$ in $[-3, 1]$
- (ii) $f(x) = 2x^3 - 15x^2 + 36x + 1$ on the interval $[1, 5]$
- (iii) $f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ in $[0, 3]$
- (iv) $f(x) = x + \sin x$ in $[0, 2\pi]$
- (v) $f(x) = \sin x + \sqrt{3} \cos x$ in $[0, \pi]$
- (vi) $f(x) = 3 + |x + 1|$ in $[-2, 3]$

TYPE 85a Finding Absolute Extrema on a Closed Interval – Finding Absolute Extrema of Diverse Functions on Closed Interval

Given: Functions and continuous domains for subparts (i) through (vi).

To Find: Absolute maximum value, absolute minimum value, and corresponding points of absolute maxima and minima for each part.

Concept	/	Formula	Used:	$f(x)$	$[a, b]$	c	$\in$	(a, b)	$f'(c)$	$=$
0		$f'(c)$		$f(x)$	a, b					

Solution:

Part (i): $f(x) = (x - 1)^2 + 3$ on $[-3, 1]$

Differentiating $f(x)$:

$$f'(x) = 2(x - 1)$$

Setting $f'(x) = 0 \implies 2(x - 1) = 0 \implies x = 1$.

The evaluation points in $[-3, 1]$ are $x = -3$ and $x = 1$.

Evaluating $f(x)$:

$$f(-3) = (-3 - 1)^2 + 3 = (-4)^2 + 3 = 16 + 3 = 19$$

$$f(1) = (1 - 1)^2 + 3 = 0 + 3 = 3$$

Comparing values:

Absolute Maximum Value = 19 at point of maximum $x = -3$

Absolute Minimum Value = 3 at point of minimum $x = 1$

Part (ii): $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$

Differentiating $f(x)$:

$$f'(x) = 6x^2 - 30x + 36$$

$$= 6(x^2 - 5x + 6)$$

$$= 6(x - 2)(x - 3)$$

Setting $f'(x) = 0 \implies x = 2$ or $x = 3$. Both critical points lie in $[1, 5]$.

Evaluating $f(x)$ at critical points $x = 2, 3$ and boundary points $x = 1, 5$:

$$f(1) = 2(1)^3 - 15(1)^2 + 36(1) + 1 = 2 - 15 + 36 + 1 = 24$$

$$f(2) = 2(2)^3 - 15(2)^2 + 36(2) + 1 = 16 - 60 + 72 + 1 = 29$$

$$f(3) = 2(3)^3 - 15(3)^2 + 36(3) + 1 = 54 - 135 + 108 + 1 = 28$$

$$f(5) = 2(5)^3 - 15(5)^2 + 36(5) + 1 = 250 - 375 + 180 + 1 = 56$$

Comparing values:

$$\text{Absolute Maximum Value} = 56 \quad \text{at point of maximum } x = 5$$

$$\text{Absolute Minimum Value} = 24 \quad \text{at point of minimum } x = 1$$

Part (iii): $f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ on $[0, 3]$

Differentiating $f(x)$:

$$f'(x) = 12x^3 - 24x^2 + 24x - 48$$

$$= 12(x^3 - 2x^2 + 2x - 4)$$

$$= 12[x^2(x - 2) + 2(x - 2)]$$

$$= 12(x - 2)(x^2 + 2)$$

Setting $f'(x) = 0 \implies (x - 2)(x^2 + 2) = 0$.

Since $x^2 + 2 > 0$ for all real x , the only real critical point is $x = 2 \in [0, 3]$.

Evaluating $f(x)$ at $x = 0, 2, 3$:

$$f(0) = 3(0)^4 - 8(0)^3 + 12(0)^2 - 48(0) + 25 = 25$$

$$f(2) = 3(2)^4 - 8(2)^3 + 12(2)^2 - 48(2) + 25$$

$$= 3(16) - 8(8) + 12(4) - 96 + 25$$

$$= 48 - 64 + 48 - 96 + 25 = -39$$

$$f(3) = 3(3)^4 - 8(3)^3 + 12(3)^2 - 48(3) + 25$$

$$= 3(81) - 8(27) + 12(9) - 144 + 25$$

$$= 243 - 216 + 108 - 144 + 25 = 16$$

Comparing values:

$$\text{Absolute Maximum Value} = 25 \quad \text{at point of maximum } x = 0$$

$$\text{Absolute Minimum Value} = -39 \quad \text{at point of minimum } x = 2$$

Part (iv): $f(x) = x + \sin x$ on $[0, 2\pi]$

Differentiating $f(x)$:

$$f'(x) = 1 + \cos x$$

Setting $f'(x) = 0 \implies 1 + \cos x = 0 \implies \cos x = -1$.

In $[0, 2\pi]$, $\cos x = -1$ gives critical point $x = \pi$.

Evaluating $f(x)$ at $x = 0, \pi, 2\pi$:

$$f(0) = 0 + \sin 0 = 0$$

$$f(\pi) = \pi + \sin \pi = \pi + 0 = \pi$$

$$f(2\pi) = 2\pi + \sin 2\pi = 2\pi + 0 = 2\pi$$

Comparing values:

$$\text{Absolute Maximum Value} = 2\pi \quad \text{at point of maximum } x = 2\pi$$

$$\text{Absolute Minimum Value} = 0 \quad \text{at point of minimum } x = 0$$

Part (v): $f(x) = \sin x + \sqrt{3} \cos x$ on $[0, \pi]$

Differentiating $f(x)$:

$$f'(x) = \cos x - \sqrt{3} \sin x$$

Setting $f'(x) = 0$:

$$\cos x - \sqrt{3} \sin x = 0$$

$$\sqrt{3} \sin x = \cos x$$

$$\tan x = \frac{1}{\sqrt{3}}$$

In $[0, \pi]$, $\tan x = \frac{1}{\sqrt{3}}$ gives critical point $x = \frac{\pi}{6}$.

Evaluating $f(x)$ at $x = 0, \frac{\pi}{6}, \pi$:

$$f(0) = \sin 0 + \sqrt{3} \cos 0 = 0 + \sqrt{3}(1) = \sqrt{3}$$

$$f\left(\frac{\pi}{6}\right) = \sin \frac{\pi}{6} + \sqrt{3} \cos \frac{\pi}{6} = \frac{1}{2} + \sqrt{3} \left(\frac{\sqrt{3}}{2}\right) = \frac{1}{2} + \frac{3}{2} = 2$$

$$f(\pi) = \sin \pi + \sqrt{3} \cos \pi = 0 + \sqrt{3}(-1) = -\sqrt{3}$$

Comparing values:

$$\text{Absolute Maximum Value} = 2 \quad \text{at point of maximum } x = \frac{\pi}{6}$$

$$\text{Absolute Minimum Value} = -\sqrt{3} \quad \text{at point of minimum } x = \pi$$

Part (vi): $f(x) = 3 + |x + 1|$ on $[-2, 3]$

The function is continuous on $[-2, 3]$ but non-differentiable at $x = -1$, making $x = -1$ a critical point.

Evaluating $f(x)$ at boundary points $x = -2, 3$ and critical point $x = -1$:

$$f(-2) = 3 + |-2 + 1| = 3 + |-1| = 3 + 1 = 4$$

$$f(-1) = 3 + |-1 + 1| = 3 + |0| = 3$$

$$f(3) = 3 + |3 + 1| = 3 + |4| = 3 + 4 = 7$$

Comparing values:

$$\text{Absolute Maximum Value} = 7 \quad \text{at point of maximum } x = 3$$

$$\text{Absolute Minimum Value} = 3 \quad \text{at point of minimum } x = -1$$

Final Answer

- (i) Max = 19 at $x = -3$; Min = 3 at $x = 1$
- (ii) Max = 56 at $x = 5$; Min = 24 at $x = 1$
- (iii) Max = 25 at $x = 0$; Min = -39 at $x = 2$
- (iv) Max = 2π at $x = 2\pi$; Min = 0 at $x = 0$
- (v) Max = 2 at $x = \frac{\pi}{6}$; Min = $-\sqrt{3}$ at $x = \pi$
- (vi) Max = 7 at $x = 3$; Min = 3 at $x = -1$

Question 13

Find the maximum and minimum values of $x + \sin 2x$ on $[0, 2\pi]$. Also find points of maxima and minima.

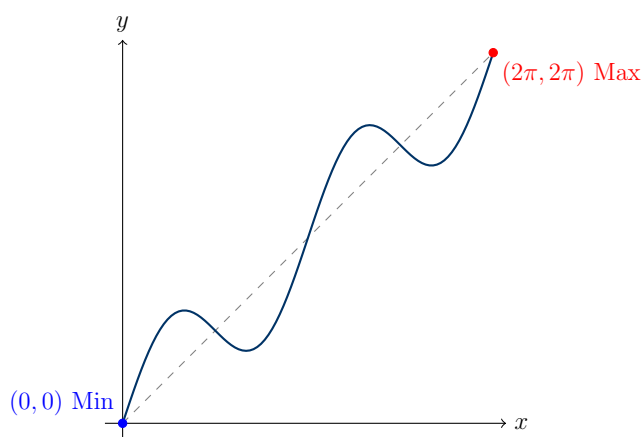
TYPE 85b Finding Absolute Extrema on a Closed Interval – Finding Absolute Extrema of Trigonometric-Linear Function on

★ **Likely Exam Question**

Given: Function $f(x) = x + \sin 2x$ defined on the closed interval $x \in [0, 2\pi]$.

To Find: Absolute maximum value, absolute minimum value, and corresponding points of maxima and minima.

Concept / Formula Used: $f(x)$ $[a, b]$ $f'(x) = 0$ $x = a, b$



Sketch of $f(x) = x + \sin 2x$ on $[0, 2\pi]$ highlighting boundary extrema

Solution:

The given function is:

$$f(x) = x + \sin 2x, \quad x \in [0, 2\pi]$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x + \sin 2x) \\ &= 1 + 2 \cos 2x \end{aligned}$$

To find the critical points, set $f'(x) = 0$:

$$\begin{aligned} 1 + 2 \cos 2x &= 0 \\ 2 \cos 2x &= -1 \\ \cos 2x &= -\frac{1}{2} \end{aligned}$$

Since $x \in [0, 2\pi]$, the transformed variable $2x \in [0, 4\pi]$.

The solutions for $\cos 2x = -\frac{1}{2}$ in $[0, 4\pi]$ are:

$$2x = \pi - \frac{\pi}{3}, \quad \pi + \frac{\pi}{3}, \quad 3\pi - \frac{\pi}{3}, \quad 3\pi + \frac{\pi}{3}$$

$$2x = \frac{2\pi}{3}, \quad \frac{4\pi}{3}, \quad \frac{8\pi}{3}, \quad \frac{10\pi}{3}$$

Dividing by 2, the critical points $x \in [0, 2\pi]$ are:

$$x = \frac{\pi}{3}, \quad \frac{2\pi}{3}, \quad \frac{4\pi}{3}, \quad \frac{5\pi}{3}$$

Evaluating $f(x)$ at all critical points and boundary points $x = 0, 2\pi$:

1. At $x = 0$:

$$f(0) = 0 + \sin 0 = 0$$

2. At $x = \frac{\pi}{3}$:

$$f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} + \sin\left(\frac{2\pi}{3}\right) = \frac{\pi}{3} + \frac{\sqrt{3}}{2} \approx 1.913$$

3. At $x = \frac{2\pi}{3}$:

$$f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + \sin\left(\frac{4\pi}{3}\right) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \approx 1.228$$

4. At $x = \frac{4\pi}{3}$:

$$f\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + \sin\left(\frac{8\pi}{3}\right) = \frac{4\pi}{3} + \frac{\sqrt{3}}{2} \approx 5.055$$

5. At $x = \frac{5\pi}{3}$:

$$f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} + \sin\left(\frac{10\pi}{3}\right) = \frac{5\pi}{3} - \frac{\sqrt{3}}{2} \approx 4.370$$

6. At $x = 2\pi$:

$$f(2\pi) = 2\pi + \sin(4\pi) = 2\pi + 0 = 2\pi \approx 6.283$$

Comparing all candidate values:

Absolute Maximum Value = 2π at point of maximum $x = 2\pi$

Absolute Minimum Value = 0 at point of minimum $x = 0$

Final Answer

Maximum value = 2π at point of maxima $x = 2\pi$.

Minimum value = 0 at point of minima $x = 0$.

Question 14

Find the absolute maximum and absolute minimum values of the function $f(x) = \cos^2 x + \sin x$, $x \in [0, \pi]$.

TYPE 85c Finding Absolute Extrema on a Closed Interval – Finding Absolute Extrema of Quadratic-Trigonometric Functions

Given: Function $f(x) = \cos^2 x + \sin x$ for $x \in [0, \pi]$.

To Find: The absolute maximum and absolute minimum values of $f(x)$.

Concept / Formula Used: $\cos^2 x = 1 - \sin^2 x$ $f(x)$ $[a, b]$ $f'(x) = 0$ a, b

Solution:

Using the identity $\cos^2 x = 1 - \sin^2 x$, we re-write $f(x)$:

$$f(x) = 1 - \sin^2 x + \sin x, \quad x \in [0, \pi]$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= -2 \sin x \cos x + \cos x \\ &= \cos x(1 - 2 \sin x) \end{aligned}$$

Setting $f'(x) = 0$ to determine critical points:

$$\cos x(1 - 2 \sin x) = 0$$

This yields two factor equations:

$$\text{Case 1: } \cos x = 0 \implies x = \frac{\pi}{2} \in [0, \pi]$$

$$\text{Case 2: } 1 - 2 \sin x = 0 \implies \sin x = \frac{1}{2} \implies x = \frac{\pi}{6}, \frac{5\pi}{6} \in [0, \pi]$$

Thus, the critical points in $[0, \pi]$ are $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$.

Evaluating $f(x)$ at critical points and domain boundary points $x = 0, \pi$:

1. At $x = 0$:

$$f(0) = \cos^2 0 + \sin 0 = 1^2 + 0 = 1$$

2. At $x = \frac{\pi}{6}$:

$$\begin{aligned} f\left(\frac{\pi}{6}\right) &= \cos^2\left(\frac{\pi}{6}\right) + \sin\left(\frac{\pi}{6}\right) \\ &= \left(\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2} \\ &= \frac{3}{4} + \frac{2}{4} = \frac{5}{4} \end{aligned}$$

3. At $x = \frac{\pi}{2}$:

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= \cos^2\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right) \\ &= 0^2 + 1 = 1 \end{aligned}$$

4. At $x = \frac{5\pi}{6}$:

$$\begin{aligned} f\left(\frac{5\pi}{6}\right) &= \cos^2\left(\frac{5\pi}{6}\right) + \sin\left(\frac{5\pi}{6}\right) \\ &= \left(-\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2} \\ &= \frac{3}{4} + \frac{2}{4} = \frac{5}{4} \end{aligned}$$

5. At $x = \pi$:

$$f(\pi) = \cos^2 \pi + \sin \pi = (-1)^2 + 0 = 1$$

Comparing all calculated values:

$$\text{Absolute Maximum Value} = \frac{5}{4} \quad \text{at } x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\text{Absolute Minimum Value} = 1 \quad \text{at } x = 0, \frac{\pi}{2}, \pi$$

Final Answer

Absolute maximum value = $\frac{5}{4}$.
Absolute minimum value = 1.

6.6 – Maxima and Minima Using Derivative Tests

Question 1

Find the turning points of the following functions and distinguish between them. Also find the local maximum and minimum values of the functions:

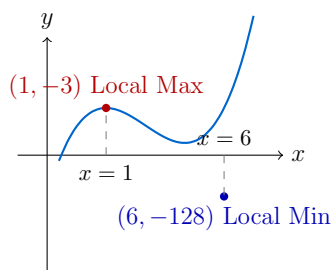
(i) $f(x) = 2x^3 - 21x^2 + 36x - 20$

(ii) $f(x) = x^3 - 3x^2 + 3x$

TYPE 88a Finding and Classifying Local Extrema via Derivative Tests – Finding and Classifying Local Extrema of Cubic Polynomials

Given: Functions $f(x) = 2x^3 - 21x^2 + 36x - 20$ and $f(x) = x^3 - 3x^2 + 3x$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x_0) < 0$ x_0 $f''(x_0) > 0$
0 x_0



Turning points and behavior of cubic polynomial $f(x) = 2x^3 - 21x^2 + 36x - 20$

Solution:

Part (i): Given function:

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(2x^3 - 21x^2 + 36x - 20) \\ &= 6x^2 - 42x + 36 \\ &= 6(x^2 - 7x + 6) \\ &= 6(x - 1)(x - 6) \end{aligned}$$

To find the turning points, set $f'(x) = 0$:

$$\begin{aligned} 6(x - 1)(x - 6) &= 0 \\ x &= 1 \quad \text{or} \quad x = 6 \end{aligned}$$

Differentiating $f'(x)$ to find the second derivative:

$$\begin{aligned} f''(x) &= \frac{d}{dx}(6x^2 - 42x + 36) \\ &= 12x - 42 \end{aligned}$$

Testing at critical point $x = 1$:

$$\begin{aligned} f''(1) &= 12(1) - 42 \\ &= -30 < 0 \end{aligned}$$

Since $f''(1) < 0$, $x = 1$ is a point of local maximum. Evaluating the function at $x = 1$:

$$\begin{aligned} f(1) &= 2(1)^3 - 21(1)^2 + 36(1) - 20 \\ &= 2 - 21 + 36 - 20 \\ &= -3 \end{aligned}$$

Turning point 1: $(1, -3)$ is a point of local maximum.

Testing at critical point $x = 6$:

$$\begin{aligned} f''(6) &= 12(6) - 42 \\ &= 72 - 42 \\ &= 30 > 0 \end{aligned}$$

Since $f''(6) > 0$, $x = 6$ is a point of local minimum. Evaluating the function at $x = 6$:

$$\begin{aligned} f(6) &= 2(6)^3 - 21(6)^2 + 36(6) - 20 \\ &= 2(216) - 21(36) + 216 - 20 \\ &= 432 - 756 + 216 - 20 \\ &= -128 \end{aligned}$$

Turning point 2: $(6, -128)$ is a point of local minimum.

Part (ii): Given function:

$$f(x) = x^3 - 3x^2 + 3x$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 - 3x^2 + 3x) \\ &= 3x^2 - 6x + 3 \\ &= 3(x^2 - 2x + 1) \\ &= 3(x - 1)^2 \end{aligned}$$

To find the turning points, set $f'(x) = 0$:

$$\begin{aligned} 3(x - 1)^2 &= 0 \\ x &= 1 \end{aligned}$$

Differentiating $f'(x)$ to find the second derivative:

$$\begin{aligned} f''(x) &= \frac{d}{dx}(3x^2 - 6x + 3) \\ &= 6x - 6 \end{aligned}$$

Testing at critical point $x = 1$:

$$\begin{aligned} f''(1) &= 6(1) - 6 \\ &= 0 \end{aligned}$$

Since $f''(1) = 0$, we test the sign of $f'(x)$ in the neighborhood of $x = 1$: For $x < 1$, $f'(x) = 3(x - 1)^2 > 0$. For $x > 1$, $f'(x) = 3(x - 1)^2 > 0$. Since $f'(x)$ does not change sign as x increases through 1, $x = 1$ is neither a local maximum nor a local minimum; it is a point of inflexion.

Evaluating the function at $x = 1$:

$$\begin{aligned} f(1) &= (1)^3 - 3(1)^2 + 3(1) \\ &= 1 - 3 + 3 \\ &= 1 \end{aligned}$$

Final Answer

(i) Turning points are $(1, -3)$ (Local Maxima) and $(6, -128)$ (Local Minima).

Local maximum value = -3 , Local minimum value = -128 .

(ii) Turning point is $(1, 1)$, which is a point of inflexion (neither local maximum nor local minimum).

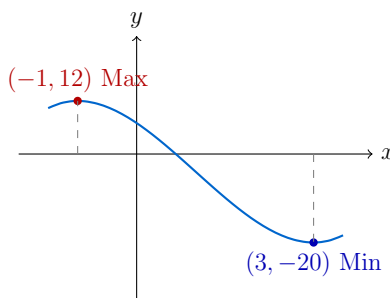
Question 2

Find the coordinates of stationary points on the curve $y = x^3 - 3x^2 - 9x + 7$, and distinguish between points of local maxima and minima.

TYPE 88a Finding and Classifying Local Extrema via Derivative Tests – Finding and Classifying Local Extrema of Cubic Polynomials

Given: Curve equation $y = x^3 - 3x^2 - 9x + 7$.

Concept / Formula Used: $\frac{dy}{dx} = 0$ $\frac{d^2y}{dx^2} < 0$ $\frac{d^2y}{dx^2} > 0$



Stationary points on curve $y = x^3 - 3x^2 - 9x + 7$

Solution:

Given equation of the curve:

$$y = x^3 - 3x^2 - 9x + 7$$

Differentiating y with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^3 - 3x^2 - 9x + 7) \\ &= 3x^2 - 6x - 9 \\ &= 3(x^2 - 2x - 3) \\ &= 3(x - 3)(x + 1)\end{aligned}$$

For stationary points, set $\frac{dy}{dx} = 0$:

$$\begin{aligned}3(x - 3)(x + 1) &= 0 \\ x &= 3 \quad \text{or} \quad x = -1\end{aligned}$$

Differentiating $\frac{dy}{dx}$ to get the second derivative:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}(3x^2 - 6x - 9) \\ &= 6x - 6\end{aligned}$$

At $x = -1$:

$$\begin{aligned}\frac{d^2y}{dx^2} &= 6(-1) - 6 \\ &= -12 < 0\end{aligned}$$

Since $\frac{d^2y}{dx^2} < 0$, $x = -1$ is a point of local maximum. Calculating the y -coordinate at $x = -1$:

$$\begin{aligned} y &= (-1)^3 - 3(-1)^2 - 9(-1) + 7 \\ &= -1 - 3 + 9 + 7 \\ &= 12 \end{aligned}$$

Stationary point $(-1, 12)$ is a point of local maximum.

At $x = 3$:

$$\begin{aligned} \frac{d^2y}{dx^2} &= 6(3) - 6 \\ &= 18 - 6 \\ &= 12 > 0 \end{aligned}$$

Since $\frac{d^2y}{dx^2} > 0$, $x = 3$ is a point of local minimum. Calculating the y -coordinate at $x = 3$:

$$\begin{aligned} y &= (3)^3 - 3(3)^2 - 9(3) + 7 \\ &= 27 - 27 - 27 + 7 \\ &= -20 \end{aligned}$$

Stationary point $(3, -20)$ is a point of local minimum.

Final Answer

Point of local maximum is $(-1, 12)$.

Point of local minimum is $(3, -20)$.

Question 3

Find the points of local maxima and minima (if any) of each of the following functions. Find also the local maximum and minimum values:

- (i) $f(x) = x^2$
- (ii) $f(x) = x^3 - 3x$
- (iii) $f(x) = x^3 - 12x$
- (iv) $f(x) = x^3 - 6x^2 + 9x + 15$
- (v) $f(x) = x^3 - 3x + 3$
- (vi) $f(x) = 2x^3 - 6x^2 + 6x + 5$
- (vii) $f(x) = \frac{x}{2} + \frac{2}{x}, \quad x > 0$
- (viii) $f(x) = \frac{1}{x^2 + 2}$
- (ix) $f(x) = 2 \sin x - x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$
- (x) $f(x) = \sin x + \cos x, \quad 0 < x < \frac{\pi}{2}$
- (xi) $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$

TYPE 88b Finding and Classifying Local Extrema via Derivative Tests – Finding Local Extrema of Polynomial, Rational, and

Given: Various functions $f(x)$ listed in subparts (i) to (xi).

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0 \implies$ $f''(x) < 0 \implies$

Solution:

(i) $f(x) = x^2$

$$f'(x) = 2x = 0 \implies x = 0$$

$$f''(x) = 2 > 0 \quad \text{for all } x$$

Point of local minimum at $x = 0$. Local minimum value $f(0) = 0$. No point of local maximum.

(ii) $f(x) = x^3 - 3x$

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \implies x = 1, -1$$

$$f''(x) = 6x$$

At $x = -1$: $f''(-1) = -6 < 0 \implies$ point of local maximum at $x = -1$. Local maximum value $f(-1) = (-1)^3 - 3(-1) = 2$. At $x = 1$: $f''(1) = 6 > 0 \implies$ point of local minimum at $x = 1$. Local minimum value $f(1) = 1^3 - 3(1) = -2$.

(iii) $f(x) = x^3 - 12x$

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 0 \implies x = 2, -2$$

$$f''(x) = 6x$$

At $x = -2$: $f''(-2) = -12 < 0 \implies$ point of local maximum at $x = -2$. Local maximum value $f(-2) = (-2)^3 - 12(-2) = 16$. At $x = 2$: $f''(2) = 12 > 0 \implies$ point of local minimum at $x = 2$. Local minimum value $f(2) = 2^3 - 12(2) = -16$.

(iv) $f(x) = x^3 - 6x^2 + 9x + 15$

$$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0 \implies x = 1, 3$$

$$f''(x) = 6x - 12$$

At $x = 1$: $f''(1) = -6 < 0 \implies$ point of local maximum at $x = 1$. Local maximum value $f(1) = 1 - 6 + 9 + 15 = 19$. At $x = 3$: $f''(3) = 6 > 0 \implies$ point of local minimum at $x = 3$. Local minimum value $f(3) = 27 - 54 + 27 + 15 = 15$.

(v) $f(x) = x^3 - 3x + 3$

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \implies x = 1, -1$$

$$f''(x) = 6x$$

At $x = -1$: $f''(-1) = -6 < 0 \implies$ point of local maximum at $x = -1$. Local maximum value $f(-1) = -1 + 3 + 3 = 5$. At $x = 1$: $f''(1) = 6 > 0 \implies$ point of local minimum at $x = 1$. Local minimum value $f(1) = 1 - 3 + 3 = 1$.

(vi) $f(x) = 2x^3 - 6x^2 + 6x + 5$

$$f'(x) = 6x^2 - 12x + 6 = 6(x - 1)^2 = 0 \implies x = 1$$

$$f''(x) = 12(x - 1)$$

At $x = 1$, $f''(1) = 0$. Also, $f'(x) \geq 0$ for all x , so $f'(x)$ does not change sign across $x = 1$. Hence, $x = 1$ is a point of inflexion. There are no points of local maximum or local minimum.

$$\text{(vii)} \quad f(x) = \frac{x}{2} + \frac{2}{x}, \quad x > 0$$

$$f'(x) = \frac{1}{2} - \frac{2}{x^2} = 0 \implies x^2 = 4 \implies x = 2 \quad (\text{since } x > 0)$$

$$f''(x) = \frac{4}{x^3}$$

At $x = 2$: $f''(2) = \frac{4}{8} = \frac{1}{2} > 0 \implies$ point of local minimum at $x = 2$. Local minimum value $f(2) = \frac{2}{2} + \frac{2}{2} = 2$. No point of local maximum.

$$\text{(viii)} \quad f(x) = \frac{1}{x^2 + 2}$$

$$f'(x) = -\frac{2x}{(x^2 + 2)^2} = 0 \implies x = 0$$

$$f''(x) = \frac{-2(x^2 + 2)^2 - (-2x) \cdot 2(x^2 + 2)(2x)}{(x^2 + 2)^4} = \frac{6x^2 - 4}{(x^2 + 2)^3}$$

At $x = 0$: $f''(0) = \frac{-4}{8} = -\frac{1}{2} < 0 \implies$ point of local maximum at $x = 0$. Local maximum value $f(0) = \frac{1}{0+2} = \frac{1}{2}$. No point of local minimum.

$$\text{(ix)} \quad f(x) = 2 \sin x - x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$f'(x) = 2 \cos x - 1 = 0 \implies \cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3} \quad \text{or} \quad x = -\frac{\pi}{3} \quad \text{in} \quad \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$f''(x) = -2 \sin x$$

At $x = \frac{\pi}{3}$: $f''\left(\frac{\pi}{3}\right) = -2 \sin \frac{\pi}{3} = -\sqrt{3} < 0 \implies$ point of local maximum at $x = \frac{\pi}{3}$. Local maximum value $f\left(\frac{\pi}{3}\right) = 2\left(\frac{\sqrt{3}}{2}\right) - \frac{\pi}{3} = \sqrt{3} - \frac{\pi}{3}$.

At $x = -\frac{\pi}{3}$: $f''\left(-\frac{\pi}{3}\right) = -2 \sin\left(-\frac{\pi}{3}\right) = \sqrt{3} > 0 \implies$ point of local minimum at $x = -\frac{\pi}{3}$. Local minimum value $f\left(-\frac{\pi}{3}\right) = 2\left(-\frac{\sqrt{3}}{2}\right) - \left(-\frac{\pi}{3}\right) = -\sqrt{3} + \frac{\pi}{3}$.

$$\text{(x)} \quad f(x) = \sin x + \cos x, \quad 0 < x < \frac{\pi}{2}$$

$$f'(x) = \cos x - \sin x = 0 \implies \tan x = 1 \implies x = \frac{\pi}{4}$$

$$f''(x) = -\sin x - \cos x$$

At $x = \frac{\pi}{4}$: $f''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2} < 0 \implies$ point of local maximum at $x = \frac{\pi}{4}$. Local maximum value $f\left(\frac{\pi}{4}\right) = \sin \frac{\pi}{4} + \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$. No point of local minimum.

$$(xi) f(x) = 3x^4 + 4x^3 - 12x^2 + 12$$

$$f'(x) = 12x^3 + 12x^2 - 24x = 12x(x^2 + x - 2) = 12x(x+2)(x-1) = 0$$

$$x = 0, 1, -2$$

$$f''(x) = 36x^2 + 24x - 24$$

At $x = 0$: $f''(0) = -24 < 0 \implies$ point of local maximum at $x = 0$. Local maximum value $f(0) = 12$.

At $x = 1$: $f''(1) = 36 + 24 - 24 = 36 > 0 \implies$ point of local minimum at $x = 1$. Local minimum value $f(1) = 3(1)^4 + 4(1)^3 - 12(1)^2 + 12 = 7$.

At $x = -2$: $f''(-2) = 36(4) + 24(-2) - 24 = 72 > 0 \implies$ point of local minimum at $x = -2$. Local minimum value $f(-2) = 3(16) + 4(-8) - 12(4) + 12 = -20$.

Final Answer

Subpart	Local Max at	Local Min at	Max Value	Min Value
(i)	None	$x = 0$	—	0
(ii)	$x = -1$	$x = 1$	2	-2
(iii)	$x = -2$	$x = 2$	16	-16
(iv)	$x = 1$	$x = 3$	19	15
(v)	$x = -1$	$x = 1$	5	1
(vi)	None	None	—	—
(vii)	None	$x = 2$	—	2
(viii)	$x = 0$	None	$\frac{1}{2}$	—
(ix)	$x = \frac{\pi}{3}$	$x = -\frac{\pi}{3}$	$\sqrt{3} - \frac{\pi}{3}$	$-\sqrt{3} + \frac{\pi}{3}$
(x)	$x = \frac{\pi}{4}$	None	$\sqrt{2}$	—
(xi)	$x = 0$	$x = 1, -2$	12	7, -20

Question 4

For what values of a and b , the function $f(x) = x^3 + ax^2 + bx - 3$ has local maximum value at $x = 0$ and local minimum value at $x = 1$?

TYPE 90a Finding Function Parameters from Given Extrema – Finding Polynomial Parameters from Given Local Extrema

★ Likely Exam Question

Given: Function $f(x) = x^3 + ax^2 + bx - 3$ with critical points at $x = 0$ (local max) and $x = 1$ (local min).

Concept / Formula Used: $f(x)$ $x = x_1$ $x = x_2$ $f'(x_1) = 0$ $f'(x_2) = 0$

Solution:

Given function:

$$f(x) = x^3 + ax^2 + bx - 3$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 + ax^2 + bx - 3) \\ &= 3x^2 + 2ax + b \end{aligned}$$

Since $f(x)$ has a local maximum at $x = 0$, $x = 0$ is a stationary point, so $f'(0) = 0$:

$$\begin{aligned} 3(0)^2 + 2a(0) + b &= 0 \\ b &= 0 \end{aligned}$$

Since $f(x)$ has a local minimum at $x = 1$, $x = 1$ is a stationary point, so $f'(1) = 0$:

$$3(1)^2 + 2a(1) + b = 0$$

Substituting $b = 0$:

$$\begin{aligned} 3 + 2a + 0 &= 0 \\ 2a &= -3 \\ a &= -\frac{3}{2} \end{aligned}$$

Verification using second derivative test:

$$\begin{aligned} f''(x) &= \frac{d}{dx}(3x^2 + 2ax + b) \\ &= 6x + 2a \\ &= 6x + 2\left(-\frac{3}{2}\right) \\ &= 6x - 3 \end{aligned}$$

At $x = 0$: $f''(0) = -3 < 0$, which confirms local maximum at $x = 0$.

At $x = 1$: $f''(1) = 6(1) - 3 = 3 > 0$, which confirms local minimum at $x = 1$.

Final Answer

$$a = -\frac{3}{2}, \quad b = 0$$

Question 5

Given $2f(x) = \log|x|^a - bx^2 + x$, if $f(x)$ has extreme values at $x = -1$ and $x = 2$, find the value of a and b .

TYPE 90b Finding Function Parameters from Given Extrema – Finding Function Parameters from Given Extreme Value Location

Given: Function relation $2f(x) = \log|x|^a - bx^2 + x$ with extreme values at $x = -1$ and $x = 2$.

Concept / Formula Used: $x = -1$ $x = 2$ $f'(-1) = 0$ $f'(2) = 0$ $\log|x|^a = a \log|x|$

Solution:

Given equation:

$$2f(x) = a \log|x| - bx^2 + x$$

Dividing by 2:

$$f(x) = \frac{a}{2} \log|x| - \frac{b}{2}x^2 + \frac{x}{2}$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{a}{2x} - \frac{b}{2}(2x) + \frac{1}{2} \\ &= \frac{a}{2x} - bx + \frac{1}{2} \end{aligned}$$

Since $f(x)$ has extreme values at $x = -1$ and $x = 2$, we have $f'(-1) = 0$ and $f'(2) = 0$.

Setting $f'(-1) = 0$:

$$\begin{aligned} \frac{a}{2(-1)} - b(-1) + \frac{1}{2} &= 0 \\ -\frac{a}{2} + b + \frac{1}{2} &= 0 \end{aligned}$$

Multiplying the entire equation by 2:

$$\begin{aligned} -a + 2b + 1 &= 0 \\ a - 2b &= 1 \quad \text{--- (Equation 1)} \end{aligned}$$

Setting $f'(2) = 0$:

$$\begin{aligned} \frac{a}{2(2)} - b(2) + \frac{1}{2} &= 0 \\ \frac{a}{4} - 2b + \frac{1}{2} &= 0 \end{aligned}$$

Multiplying the entire equation by 4:

$$\begin{aligned} a - 8b + 2 &= 0 \\ a - 8b &= -2 \quad \text{--- (Equation 2)} \end{aligned}$$

Subtracting Equation 2 from Equation 1 to solve the simultaneous system:

$$\begin{aligned} (a - 2b) - (a - 8b) &= 1 - (-2) \\ 6b &= 3 \\ b &= \frac{3}{6} \\ b &= \frac{1}{2} \end{aligned}$$

Substituting $b = \frac{1}{2}$ back into Equation 1:

$$\begin{aligned} a - 2\left(\frac{1}{2}\right) &= 1 \\ a - 1 &= 1 \\ a &= 2 \end{aligned}$$

Final Answer

$$a = 2, \quad b = \frac{1}{2}$$

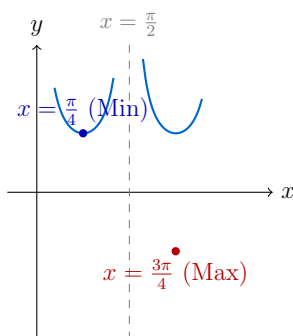
Question 6

Aditya hires a speed boat in an island to do water sports, which is moving along a curve $y = \frac{1}{\cos x \sin x}$, $x \in (0, \pi)$ in water. Find the number of stationary points.

TYPE 92 Finding the Number of Stationary Points of a Trigonometric Function

Given: Curve equation $y = \frac{1}{\cos x \sin x}$ for domain $x \in (0, \pi)$.

Concept / Formula Used: $2 \sin x \cos x = \sin 2x$ $\frac{dy}{dx} = 0$



Sketch of curve $y = 2 \csc 2x$ for $x \in (0, \pi)$ showing stationary points

Solution:

Given curve:

$$y = \frac{1}{\cos x \sin x}, \quad x \in (0, \pi)$$

Multiplying numerator and denominator by 2:

$$\begin{aligned} y &= \frac{2}{2 \sin x \cos x} \\ &= \frac{2}{\sin 2x} \\ &= 2 \csc 2x \end{aligned}$$

Differentiating y with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(2 \csc 2x) \\ &= 2(-\csc 2x \cot 2x) \cdot \frac{d}{dx}(2x) \\ &= -4 \csc 2x \cot 2x \end{aligned}$$

For stationary points, set $\frac{dy}{dx} = 0$:

$$-4 \csc 2x \cot 2x = 0$$

Since $\csc 2x \neq 0$ for any $x \in (0, \pi)$, we must have:

$$\cot 2x = 0$$

Given $x \in (0, \pi)$, the argument $2x \in (0, 2\pi)$. The values of $2x \in (0, 2\pi)$ for which $\cot 2x = 0$ are:

$$\begin{aligned} 2x = \frac{\pi}{2} &\implies x = \frac{\pi}{4} \\ 2x = \frac{3\pi}{2} &\implies x = \frac{3\pi}{4} \end{aligned}$$

Both $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ lie within the open interval $(0, \pi)$. Therefore, there are exactly 2 stationary points.

Final Answer

Number of stationary points = 2 (at $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$).

Question 7

Anu enjoys a roller coaster ride by first going downwards and then upwards to the maximum height. The relation between the distance travelled (cm) with respect to the time taken to complete the ride by Anu is given by the following equation:

$$y = 4x - \frac{1}{2}x^2$$

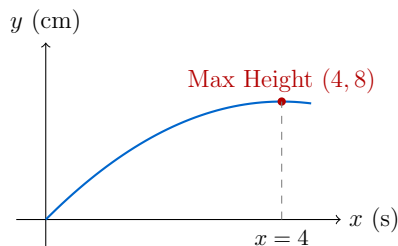
where x = time in seconds.

- What is the rate of change of displacement with respect to the time?
- How many seconds will it take her to go to its maximum height?

TYPE 93 Solving Kinematics Maxima Problems using First Derivative

Given: Displacement relation $y = 4x - \frac{1}{2}x^2$, where x is time in seconds.

Concept / Formula Used: $\frac{dy}{dx}$ $\frac{dy}{dx} = 0$ $\frac{d^2y}{dx^2} < 0$



Trajectory of displacement $y = 4x - \frac{1}{2}x^2$ showing maximum height

Solution:

Part (a): Rate of change of displacement with respect to time Displacement equation:

$$y = 4x - \frac{1}{2}x^2$$

Differentiating y with respect to time x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(4x - \frac{1}{2}x^2 \right) \\ &= 4 - \frac{1}{2}(2x) \\ &= 4 - x\end{aligned}$$

Hence, the rate of change of displacement with respect to time is $4 - x$.

Part (b): Time taken to reach maximum height At maximum height, the rate of change of displacement $\frac{dy}{dx}$ equals zero:

$$\begin{aligned}\frac{dy}{dx} &= 0 \\ 4 - x &= 0 \\ x &= 4 \text{ seconds}\end{aligned}$$

Verification of maximum using the second derivative test:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}(4 - x) \\ &= -1 < 0\end{aligned}$$

Since $\frac{d^2y}{dx^2} < 0$, y is indeed maximum at $x = 4$ seconds.

Final Answer

- (a) Rate of change of displacement $= 4 - x$
(b) Time taken to reach maximum height $= 4$ seconds

6.7 – Applications of Maxima and Minima

Question 1

Find two numbers whose sum is 24 and whose product is as large as possible.

TYPE 94 Maximizing Product of Two Numbers with Constant Sum

Given: Two numbers x and y such that $x + y = 24$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) < 0$

Solution:

Let the two numbers be x and y . We are given:

$$\begin{aligned}x + y &= 24 \\ y &= 24 - x\end{aligned}$$

Let P be the product of the two numbers:

$$P = xy$$

Substituting $y = 24 - x$ into the product equation:

$$\begin{aligned} P(x) &= x(24 - x) \\ P(x) &= 24x - x^2 \end{aligned}$$

Differentiating $P(x)$ with respect to x :

$$\begin{aligned} \frac{dP}{dx} &= \frac{d}{dx}(24x - x^2) \\ \frac{dP}{dx} &= 24 - 2x \end{aligned}$$

For critical points, we set $\frac{dP}{dx} = 0$:

$$\begin{aligned} 24 - 2x &= 0 \\ 2x &= 24 \\ x &= 12 \end{aligned}$$

Now, we find the second derivative to test for maximum:

$$\begin{aligned} \frac{d^2P}{dx^2} &= \frac{d}{dx}(24 - 2x) \\ \frac{d^2P}{dx^2} &= -2 \end{aligned}$$

Since $\frac{d^2P}{dx^2} = -2 < 0$, the product P is maximum at $x = 12$.

Now, back-substitute $x = 12$ to find y :

$$\begin{aligned} y &= 24 - 12 \\ y &= 12 \end{aligned}$$

Thus, the two numbers are 12 and 12.

Final Answer

The two numbers are 12 and 12.

Question 2

The sum of two positive numbers is 20. Find the numbers

- if the sum of their squares is minimum.
- if the product of the square of one and the cube of the other is maximum.

TYPE 95 Optimizing Sum of Squares or Products with Constant Sum

Given: Two positive numbers x and y such that $x + y = 20$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$ $f'(x) = 0$ $f''(x) < 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$\begin{aligned}x + y &= 20 \\ y &= 20 - x\end{aligned}$$

(i) If the sum of their squares is minimum:

Let S be the sum of the squares of the two numbers:

$$S = x^2 + y^2$$

Substituting $y = 20 - x$:

$$\begin{aligned}S(x) &= x^2 + (20 - x)^2 \\ S(x) &= x^2 + 400 - 40x + x^2 \\ S(x) &= 2x^2 - 40x + 400\end{aligned}$$

Differentiating $S(x)$ with respect to x :

$$\begin{aligned}\frac{dS}{dx} &= \frac{d}{dx}(2x^2 - 40x + 400) \\ \frac{dS}{dx} &= 4x - 40\end{aligned}$$

For critical points, set $\frac{dS}{dx} = 0$:

$$\begin{aligned}4x - 40 &= 0 \\ 4x &= 40 \\ x &= 10\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2S}{dx^2} &= \frac{d}{dx}(4x - 40) \\ \frac{d^2S}{dx^2} &= 4\end{aligned}$$

Since $\frac{d^2S}{dx^2} = 4 > 0$, the sum of squares S is minimum at $x = 10$.

Back-substituting $x = 10$ to find y :

$$\begin{aligned}y &= 20 - 10 \\ y &= 10\end{aligned}$$

Thus, the numbers are 10 and 10.

(ii) If the product of the square of one and the cube of the other is maximum:

Let P be the product of the square of x and the cube of y :

$$P = x^2y^3$$

Substituting $y = 20 - x$:

$$P(x) = x^2(20 - x)^3$$

Differentiating $P(x)$ with respect to x using the product rule:

$$\begin{aligned}
 \frac{dP}{dx} &= \frac{d}{dx} [x^2(20-x)^3] \\
 &= 2x(20-x)^3 + x^2 \cdot 3(20-x)^2 \cdot (-1) \\
 &= 2x(20-x)^3 - 3x^2(20-x)^2 \\
 &= x(20-x)^2 [2(20-x) - 3x] \\
 &= x(20-x)^2 (40-2x-3x) \\
 &= x(20-x)^2 (40-5x)
 \end{aligned}$$

For critical points, set $\frac{dP}{dx} = 0$:

$$x(20-x)^2(40-5x) = 0$$

Since x and y are positive numbers, $x \neq 0$ and $x \neq 20$. Thus:

$$\begin{aligned}
 40 - 5x &= 0 \\
 5x &= 40 \\
 x &= 8
 \end{aligned}$$

Now, let us find the second derivative to confirm maximum:

$$\frac{d^2P}{dx^2} = \frac{d}{dx} [(40x - 5x^2)(20-x)^2]$$

Using the product rule at $x = 8$:

$$\frac{d^2P}{dx^2} = (40 - 10x)(20-x)^2 + (40x - 5x^2) \cdot 2(20-x)(-1)$$

Substitute $x = 8$:

$$\begin{aligned}
 \left. \frac{d^2P}{dx^2} \right|_{x=8} &= (40 - 80)(12)^2 + (320 - 320) \cdot 2(12)(-1) \\
 &= -40 \cdot 144 + 0 \\
 &= -5760 < 0
 \end{aligned}$$

Since the second derivative is negative, P is maximum at $x = 8$.

Back-substituting $x = 8$ to find y :

$$\begin{aligned}
 y &= 20 - 8 \\
 y &= 12
 \end{aligned}$$

Thus, the numbers are 8 and 12.

Final Answer

- (i) The numbers are 10 and 10.
- (ii) The numbers are 8 and 12.

Question 3

- The product of two positive numbers is 16. Find the numbers
- (i) if their sum is least.

(ii) if the sum of one and the square of the other is least.

TYPE 96 Minimizing Sum of Two Numbers with Constant Product

Given: Two positive numbers x and y such that $xy = 16$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$xy = 16$$

$$y = \frac{16}{x}$$

(i) If their sum is least:

Let S be the sum of the two numbers:

$$S = x + y$$

Substituting $y = \frac{16}{x}$:

$$S(x) = x + \frac{16}{x}$$

Differentiating $S(x)$ with respect to x :

$$\begin{aligned}\frac{dS}{dx} &= \frac{d}{dx} \left(x + \frac{16}{x} \right) \\ \frac{dS}{dx} &= 1 - \frac{16}{x^2}\end{aligned}$$

For critical points, set $\frac{dS}{dx} = 0$:

$$\begin{aligned}1 - \frac{16}{x^2} &= 0 \\ x^2 &= 16 \\ x &= 4 \quad (\text{since } x > 0)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2S}{dx^2} &= \frac{d}{dx} \left(1 - \frac{16}{x^2} \right) \\ \frac{d^2S}{dx^2} &= \frac{32}{x^3}\end{aligned}$$

At $x = 4$:

$$\left. \frac{d^2S}{dx^2} \right|_{x=4} = \frac{32}{4^3} = \frac{32}{64} = \frac{1}{2} > 0$$

Since the second derivative is positive, the sum S is minimum at $x = 4$.

Back-substituting $x = 4$ to find y :

$$y = \frac{16}{4}$$

$$y = 4$$

Thus, the numbers are 4 and 4.

(ii) If the sum of one and the square of the other is least:

Let S be the sum of x and the square of y :

$$S = x + y^2$$

Substituting $y = \frac{16}{x}$:

$$S(x) = x + \left(\frac{16}{x}\right)^2$$

$$S(x) = x + \frac{256}{x^2}$$

Differentiating $S(x)$ with respect to x :

$$\frac{dS}{dx} = \frac{d}{dx} \left(x + \frac{256}{x^2} \right)$$

$$\frac{dS}{dx} = 1 - \frac{512}{x^3}$$

For critical points, set $\frac{dS}{dx} = 0$:

$$1 - \frac{512}{x^3} = 0$$

$$x^3 = 512$$

$$x = 8$$

Finding the second derivative:

$$\frac{d^2S}{dx^2} = \frac{d}{dx} \left(1 - \frac{512}{x^3} \right)$$

$$\frac{d^2S}{dx^2} = \frac{1536}{x^4}$$

At $x = 8$:

$$\left. \frac{d^2S}{dx^2} \right|_{x=8} = \frac{1536}{8^4} = \frac{1536}{4096} = \frac{3}{8} > 0$$

Since the second derivative is positive, the sum S is minimum at $x = 8$.

Back-substituting $x = 8$ to find y :

$$y = \frac{16}{8}$$

$$y = 2$$

Thus, the numbers are 8 and 2.

Final Answer

- (i) The numbers are 4 and 4.
- (ii) The numbers are 8 and 2.

Question 4

Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.

TYPE 97 Maximizing Product of Powers of Two Numbers with Constant Sum

Given: Two positive numbers x and y such that $x + y = 60$.

Concept / Formula Used: $f(y)$ $f'(y) = 0$ $f''(y) < 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$\begin{aligned}x + y &= 60 \\x &= 60 - y\end{aligned}$$

Let P be the product xy^3 :

$$P = xy^3$$

Substituting $x = 60 - y$:

$$\begin{aligned}P(y) &= (60 - y)y^3 \\P(y) &= 60y^3 - y^4\end{aligned}$$

Differentiating $P(y)$ with respect to y :

$$\begin{aligned}\frac{dP}{dy} &= \frac{d}{dy}(60y^3 - y^4) \\ \frac{dP}{dy} &= 180y^2 - 4y^3 \\ \frac{dP}{dy} &= 4y^2(45 - y)\end{aligned}$$

For critical points, set $\frac{dP}{dy} = 0$:

$$4y^2(45 - y) = 0$$

Since y is a positive number, $y \neq 0$. Thus:

$$\begin{aligned}45 - y &= 0 \\y &= 45\end{aligned}$$

Now, find the second derivative to confirm maximum:

$$\begin{aligned}\frac{d^2P}{dy^2} &= \frac{d}{dy}(180y^2 - 4y^3) \\ \frac{d^2P}{dy^2} &= 360y - 12y^2\end{aligned}$$

At $y = 45$:

$$\begin{aligned}\left. \frac{d^2P}{dy^2} \right|_{y=45} &= 360(45) - 12(45)^2 \\ &= 45(360 - 540) \\ &= 45(-180) \\ &= -8100 < 0\end{aligned}$$

Since the second derivative is negative, the product P is maximum at $y = 45$.

Back-substituting $y = 45$ to find x :

$$x = 60 - 45$$

$$x = 15$$

Thus, the two positive numbers are 15 and 45.

Final Answer

The numbers are $x = 15$ and $y = 45$.

Question 5

Find two positive numbers x and y such that their sum is 35 and the product x^2y^5 is maximum.

TYPE 97 Maximizing Product of Powers of Two Numbers with Constant Sum

Given: Two positive numbers x and y such that $x + y = 35$.

Concept / Formula Used: $f(y)$ $f'(y) = 0$ $f''(y) < 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$x + y = 35$$

$$x = 35 - y$$

Let P be the product x^2y^5 :

$$P = x^2y^5$$

Substituting $x = 35 - y$:

$$P(y) = (35 - y)^2y^5$$

Differentiating $P(y)$ with respect to y using the product rule:

$$\begin{aligned} \frac{dP}{dy} &= \frac{d}{dy} [(35 - y)^2y^5] \\ &= 2(35 - y)(-1)y^5 + (35 - y)^2 \cdot 5y^4 \\ &= -2y^5(35 - y) + 5y^4(35 - y)^2 \\ &= y^4(35 - y)[-2y + 5(35 - y)] \\ &= y^4(35 - y)(175 - 7y) \end{aligned}$$

For critical points, set $\frac{dP}{dy} = 0$:

$$y^4(35 - y)(175 - 7y) = 0$$

Since x and y are positive numbers, $y \neq 0$ and $y \neq 35$. Thus:

$$175 - 7y = 0$$

$$7y = 175$$

$$y = 25$$

Let us check the sign of $\frac{dP}{dy}$ around $y = 25$:

- For $y < 25$ (slightly), $175 - 7y > 0 \implies \frac{dP}{dy} > 0$.
- For $y > 25$ (slightly), $175 - 7y < 0 \implies \frac{dP}{dy} < 0$.

Since $\frac{dP}{dy}$ changes sign from positive to negative as y increases through 25, the product P is maximum at $y = 25$.

Back-substituting $y = 25$ to find x :

$$x = 35 - 25$$

$$x = 10$$

Thus, the two positive numbers are 10 and 25.

Final Answer

The numbers are $x = 10$ and $y = 25$.

Question 6

Sum of two numbers is 5. If the sum of the cubes of these numbers is least, then find the sum of the squares of these numbers.

TYPE 98 Minimizing Sum of Cubes with Constant Sum

Given: Two numbers x and y such that $x + y = 5$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$

Solution:

Let the two numbers be x and y . We are given:

$$x + y = 5$$

$$y = 5 - x$$

Let S be the sum of the cubes of the two numbers:

$$S = x^3 + y^3$$

Substituting $y = 5 - x$:

$$S(x) = x^3 + (5 - x)^3$$

Differentiating $S(x)$ with respect to x :

$$\begin{aligned} \frac{dS}{dx} &= \frac{d}{dx} [x^3 + (5 - x)^3] \\ &= 3x^2 + 3(5 - x)^2 \cdot (-1) \\ &= 3x^2 - 3(25 - 10x + x^2) \\ &= 3x^2 - 75 + 30x - 3x^2 \\ &= 30x - 75 \end{aligned}$$

For critical points, set $\frac{dS}{dx} = 0$:

$$30x - 75 = 0$$

$$30x = 75$$

$$x = \frac{75}{30}$$

$$x = \frac{5}{2}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2S}{dx^2} &= \frac{d}{dx}(30x - 75) \\ \frac{d^2S}{dx^2} &= 30\end{aligned}$$

Since $\frac{d^2S}{dx^2} = 30 > 0$, the sum of cubes S is minimum at $x = \frac{5}{2}$.

Back-substituting $x = \frac{5}{2}$ to find y :

$$\begin{aligned}y &= 5 - \frac{5}{2} \\ y &= \frac{5}{2}\end{aligned}$$

Now, we need to find the sum of the squares of these numbers, which is $x^2 + y^2$:

$$\begin{aligned}x^2 + y^2 &= \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2 \\ &= \frac{25}{4} + \frac{25}{4} \\ &= \frac{50}{4} \\ &= \frac{25}{2}\end{aligned}$$

Final Answer

The sum of the squares of the numbers is $\frac{25}{2}$ (or 12.5).

Question 7

Three numbers are given whose sum is 180 and the ratio of first two of them is 1 : 2. If the product of the numbers is greatest, find the numbers.

TYPE 99 Maximizing Product of Three Numbers with Constant Sum

Given: Three numbers x, y, z such that $x + y + z = 180$ and $x : y = 1 : 2$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) < 0$

Solution:

Let the three numbers be x, y , and z . We are given:

$$\begin{aligned}x + y + z &= 180 \quad \text{--- (1)} \\ \frac{x}{y} &= \frac{1}{2} \implies y = 2x \quad \text{--- (2)}\end{aligned}$$

Substituting $y = 2x$ into equation (1):

$$\begin{aligned}x + 2x + z &= 180 \\ 3x + z &= 180 \\ z &= 180 - 3x \quad \text{--- (3)}\end{aligned}$$

Let P be the product of the three numbers:

$$P = xyz$$

Substituting $y = 2x$ and $z = 180 - 3x$:

$$P(x) = x(2x)(180 - 3x)$$

$$P(x) = 2x^2(180 - 3x)$$

$$P(x) = 360x^2 - 6x^3$$

Differentiating $P(x)$ with respect to x :

$$\frac{dP}{dx} = \frac{d}{dx}(360x^2 - 6x^3)$$

$$\frac{dP}{dx} = 720x - 18x^2$$

$$\frac{dP}{dx} = 18x(40 - x)$$

For critical points, set $\frac{dP}{dx} = 0$:

$$18x(40 - x) = 0$$

Since the numbers must be positive for a non-zero maximum product, $x \neq 0$. Thus:

$$40 - x = 0$$

$$x = 40$$

Finding the second derivative:

$$\frac{d^2P}{dx^2} = \frac{d}{dx}(720x - 18x^2)$$

$$\frac{d^2P}{dx^2} = 720 - 36x$$

At $x = 40$:

$$\begin{aligned}\left.\frac{d^2P}{dx^2}\right|_{x=40} &= 720 - 36(40) \\ &= 720 - 1440 \\ &= -720 < 0\end{aligned}$$

Since the second derivative is negative, the product P is maximum at $x = 40$.

Now, back-substitute $x = 40$ to find y and z :

$$y = 2(40) = 80$$

$$z = 180 - 3(40) = 180 - 120 = 60$$

Thus, the three numbers are 40, 80, and 60.

Final Answer

The numbers are 40, 80, and 60.

Question 8

Find the maximum profit that a company can make, if the profit function is given by $p(x) = 41 + 72x - 18x^2$.

TYPE 100 Finding Maximum of Quadratic Profit Function

Given: Profit function $p(x) = 41 + 72x - 18x^2$.

Concept / Formula Used: $p(x)$ $p'(x) = 0$ $p''(x) < 0$

Solution:

The profit function is given by:

$$p(x) = 41 + 72x - 18x^2$$

Differentiating $p(x)$ with respect to x :

$$\begin{aligned} p'(x) &= \frac{d}{dx}(41 + 72x - 18x^2) \\ p'(x) &= 72 - 36x \end{aligned}$$

For critical points, set $p'(x) = 0$:

$$\begin{aligned} 72 - 36x &= 0 \\ 36x &= 72 \\ x &= 2 \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} p''(x) &= \frac{d}{dx}(72 - 36x) \\ p''(x) &= -36 \end{aligned}$$

Since $p''(x) = -36 < 0$, the profit is maximum at $x = 2$.

Now, substitute $x = 2$ into the profit function to find the maximum profit:

$$\begin{aligned} p(2) &= 41 + 72(2) - 18(2)^2 \\ &= 41 + 144 - 18(4) \\ &= 41 + 144 - 72 \\ &= 41 + 72 \\ &= 113 \end{aligned}$$

Final Answer

The maximum profit is 113.

Question 9

The cost (in Rs.) per article C , of manufacturing a certain article is given by the formula $C = 5 + \frac{48}{x} + 3x^2$, where x is the number of articles manufactured per hour. Find the minimum value of C .

TYPE 101 Finding Minimum of Rational-Quadratic Cost Function

Given: Cost function $C(x) = 5 + \frac{48}{x} + 3x^2$ for $x > 0$.

Concept / Formula Used: $C(x)$ $C'(x) = 0$ $C''(x) > 0$

Solution:

The cost function is:

$$C(x) = 5 + \frac{48}{x} + 3x^2$$

Differentiating $C(x)$ with respect to x :

$$\begin{aligned}\frac{dC}{dx} &= \frac{d}{dx} \left(5 + \frac{48}{x} + 3x^2 \right) \\ \frac{dC}{dx} &= -\frac{48}{x^2} + 6x\end{aligned}$$

For critical points, set $\frac{dC}{dx} = 0$:

$$\begin{aligned}-\frac{48}{x^2} + 6x &= 0 \\ 6x &= \frac{48}{x^2} \\ x^3 &= 8 \\ x &= 2\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2C}{dx^2} &= \frac{d}{dx} \left(-\frac{48}{x^2} + 6x \right) \\ \frac{d^2C}{dx^2} &= \frac{96}{x^3} + 6\end{aligned}$$

At $x = 2$:

$$\begin{aligned}\left. \frac{d^2C}{dx^2} \right|_{x=2} &= \frac{96}{2^3} + 6 \\ &= \frac{96}{8} + 6 \\ &= 12 + 6 \\ &= 18 > 0\end{aligned}$$

Since the second derivative is positive, the cost C is minimum at $x = 2$.

Now, substitute $x = 2$ into the cost function to find the minimum cost:

$$\begin{aligned}C(2) &= 5 + \frac{48}{2} + 3(2)^2 \\ &= 5 + 24 + 3(4) \\ &= 5 + 24 + 12 \\ &= 41\end{aligned}$$

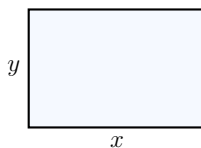
Final Answer

The minimum value of C is 41.

Question 10

Prove that of all rectangles with given area, the square has the smallest perimeter.

TYPE 102 Minimizing Perimeter of Rectangle with Constant Area



Rectangle with dimensions x and y

To Prove: The perimeter of a rectangle with given area A is minimum when it is a square ($x = y$).

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Let the sides of the rectangle be x and y . The area A is given and is constant:

$$A = xy$$

$$y = \frac{A}{x}$$

The perimeter P of the rectangle is:

$$P = 2(x + y)$$

Substituting $y = \frac{A}{x}$ into the perimeter equation:

$$P(x) = 2\left(x + \frac{A}{x}\right)$$

Differentiating $P(x)$ with respect to x :

$$\frac{dP}{dx} = 2\left(1 - \frac{A}{x^2}\right)$$

For critical points, set $\frac{dP}{dx} = 0$:

$$\begin{aligned} 2\left(1 - \frac{A}{x^2}\right) &= 0 \\ 1 - \frac{A}{x^2} &= 0 \\ x^2 &= A \\ x &= \sqrt{A} \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2P}{dx^2} &= \frac{d}{dx} \left[2\left(1 - \frac{A}{x^2}\right) \right] \\ \frac{d^2P}{dx^2} &= \frac{4A}{x^3} \end{aligned}$$

At $x = \sqrt{A}$:

$$\left. \frac{d^2 P}{dx^2} \right|_{x=\sqrt{A}} = \frac{4A}{(\sqrt{A})^3} = \frac{4}{\sqrt{A}} > 0$$

Since the second derivative is positive, the perimeter P is minimum when $x = \sqrt{A}$.

Now, back-substitute $x = \sqrt{A}$ to find y :

$$y = \frac{A}{\sqrt{A}}$$

$$y = \sqrt{A}$$

Since $x = y = \sqrt{A}$, the rectangle is a square.

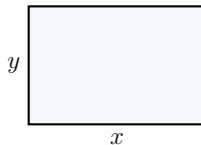
Hence Proved

LHS = RHS, hence proved that the rectangle with given area has the smallest perimeter when it is a square.

Question 11

Of all rectangles each of which has perimeter 24 cm, find the one having maximum area. Also find that area.

TYPE 103 Maximizing Area of Rectangle with Constant Perimeter



Rectangle with dimensions x and y

Given: Perimeter of the rectangle $P = 24$ cm.

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Let the sides of the rectangle be x and y . We are given:

$$2(x + y) = 24$$

$$x + y = 12$$

$$y = 12 - x$$

The area A of the rectangle is:

$$A = xy$$

Substituting $y = 12 - x$:

$$A(x) = x(12 - x)$$

$$A(x) = 12x - x^2$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx}(12x - x^2) \\ \frac{dA}{dx} &= 12 - 2x\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned}12 - 2x &= 0 \\ 2x &= 12 \\ x &= 6\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dx^2} &= \frac{d}{dx}(12 - 2x) \\ \frac{d^2A}{dx^2} &= -2\end{aligned}$$

Since $\frac{d^2A}{dx^2} = -2 < 0$, the area is maximum when $x = 6$ cm.

Back-substituting $x = 6$ to find y :

$$\begin{aligned}y &= 12 - 6 \\ y &= 6 \text{ cm}\end{aligned}$$

Thus, the rectangle is a square of side 6 cm.

The maximum area is:

$$\begin{aligned}A &= 6 \times 6 \\ &= 36 \text{ cm}^2\end{aligned}$$

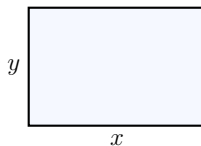
Final Answer

The rectangle has dimensions 6 cm $\times$ 6 cm and its maximum area is 36 cm².

Question 12

Prove that of all rectangles with given perimeter, the square has the largest area.

TYPE 103 Maximizing Area of Rectangle with Constant Perimeter



Rectangle with dimensions x and y

To Prove: The area of a rectangle with given perimeter P is maximum when it is a square ($x = y$).

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Let the sides of the rectangle be x and y . The perimeter P is given and is constant:

$$\begin{aligned} 2(x + y) &= P \\ x + y &= \frac{P}{2} \\ y &= \frac{P}{2} - x \end{aligned}$$

Let A be the area of the rectangle:

$$A = xy$$

Substituting $y = \frac{P}{2} - x$:

$$\begin{aligned} A(x) &= x \left(\frac{P}{2} - x \right) \\ A(x) &= \frac{P}{2}x - x^2 \end{aligned}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned} \frac{dA}{dx} &= \frac{d}{dx} \left(\frac{P}{2}x - x^2 \right) \\ \frac{dA}{dx} &= \frac{P}{2} - 2x \end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned} \frac{P}{2} - 2x &= 0 \\ 2x &= \frac{P}{2} \\ x &= \frac{P}{4} \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2A}{dx^2} &= \frac{d}{dx} \left(\frac{P}{2} - 2x \right) \\ \frac{d^2A}{dx^2} &= -2 \end{aligned}$$

Since $\frac{d^2A}{dx^2} = -2 < 0$, the area A is maximum when $x = \frac{P}{4}$.

Back-substituting $x = \frac{P}{4}$ to find y :

$$\begin{aligned} y &= \frac{P}{2} - \frac{P}{4} \\ y &= \frac{P}{4} \end{aligned}$$

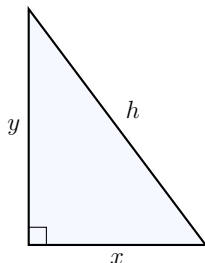
Since $x = y = \frac{P}{4}$, the rectangle is a square.

Hence Proved

LHS = RHS, hence proved that the rectangle with given perimeter has the largest area when it is a square.

Question 13

Prove that the perimeter of a right-angled triangle of given hypotenuse is maximum when the triangle is isosceles.

TYPE 104 Maximizing Perimeter of Right-Angled Triangle with Constant Hypotenuse

Right-angled triangle with hypotenuse h

To Prove: The perimeter of a right-angled triangle with given hypotenuse h is maximum when the other two sides are equal ($x = y$).

Concept / Formula Used: $x^2 + y^2 = h^2$ $P = x + y + h$

Solution:

Let the two legs of the right-angled triangle be x and y , and the hypotenuse be h (constant). By Pythagoras' theorem:

$$\begin{aligned}x^2 + y^2 &= h^2 \\y &= \sqrt{h^2 - x^2}\end{aligned}$$

The perimeter P of the triangle is:

$$P = x + y + h$$

Substituting $y = \sqrt{h^2 - x^2}$:

$$P(x) = x + \sqrt{h^2 - x^2} + h$$

Differentiating $P(x)$ with respect to x :

$$\begin{aligned}\frac{dP}{dx} &= \frac{d}{dx} \left(x + \sqrt{h^2 - x^2} + h \right) \\&= 1 + \frac{1}{2\sqrt{h^2 - x^2}} \cdot (-2x) \\&= 1 - \frac{x}{\sqrt{h^2 - x^2}}\end{aligned}$$

For critical points, set $\frac{dP}{dx} = 0$:

$$\begin{aligned}1 - \frac{x}{\sqrt{h^2 - x^2}} &= 0 \\ \frac{x}{\sqrt{h^2 - x^2}} &= x\end{aligned}$$

Squaring both sides:

$$\begin{aligned}h^2 - x^2 &= x^2 \\2x^2 &= h^2 \\x^2 &= \frac{h^2}{2} \\x &= \frac{h}{\sqrt{2}} \quad (\text{since } x > 0)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2P}{dx^2} &= \frac{d}{dx} \left(1 - \frac{x}{\sqrt{h^2 - x^2}} \right) \\&= - \frac{\sqrt{h^2 - x^2} \cdot 1 - x \cdot \frac{-2x}{2\sqrt{h^2 - x^2}}}{h^2 - x^2} \\&= - \frac{\sqrt{h^2 - x^2} + \frac{x^2}{\sqrt{h^2 - x^2}}}{h^2 - x^2} \\&= - \frac{h^2 - x^2 + x^2}{(h^2 - x^2)^{3/2}} \\&= - \frac{h^2}{(h^2 - x^2)^{3/2}}\end{aligned}$$

At $x = \frac{h}{\sqrt{2}}$:

$$\begin{aligned}\left. \frac{d^2P}{dx^2} \right|_{x=\frac{h}{\sqrt{2}}} &= - \frac{h^2}{\left(h^2 - \frac{h^2}{2}\right)^{3/2}} \\&= - \frac{h^2}{\left(\frac{h^2}{2}\right)^{3/2}} \\&= - \frac{2\sqrt{2}}{h} < 0\end{aligned}$$

Since the second derivative is negative, the perimeter is maximum when $x = \frac{h}{\sqrt{2}}$.

Back-substituting $x = \frac{h}{\sqrt{2}}$ to find y :

$$\begin{aligned}y &= \sqrt{h^2 - \left(\frac{h}{\sqrt{2}}\right)^2} \\&= \sqrt{h^2 - \frac{h^2}{2}} \\&= \frac{h}{\sqrt{2}}\end{aligned}$$

Since $x = y = \frac{h}{\sqrt{2}}$, the triangle is isosceles.

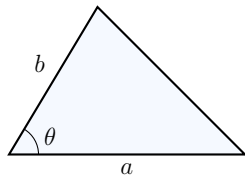
Hence Proved

LHS = RHS, hence proved that the perimeter is maximum when the triangle is isosceles.

Question 14

Two sides of a triangle have lengths a and b , and the angle between them is θ . What value of θ will maximize the area of the triangle? Also find the maximum area of the triangle.

TYPE 105 Maximizing Area of Triangle with Variable Angle



Triangle with sides a, b and angle θ

Given: Two sides of a triangle of lengths a and b with included angle θ .

Concept / Formula Used: $A = \frac{1}{2}ab \sin \theta$

Solution:

The area A of the triangle is given by:

$$A(\theta) = \frac{1}{2}ab \sin \theta$$

Differentiating $A(\theta)$ with respect to θ :

$$\begin{aligned}\frac{dA}{d\theta} &= \frac{d}{d\theta} \left(\frac{1}{2}ab \sin \theta \right) \\ \frac{dA}{d\theta} &= \frac{1}{2}ab \cos \theta\end{aligned}$$

For critical points, set $\frac{dA}{d\theta} = 0$:

$$\frac{1}{2}ab \cos \theta = 0$$

Since $a, b > 0$, we must have:

$$\begin{aligned}\cos \theta &= 0 \\ \theta &= \frac{\pi}{2} \quad (\text{since } 0 < \theta < \pi)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{d\theta^2} &= \frac{d}{d\theta} \left(\frac{1}{2}ab \cos \theta \right) \\ \frac{d^2A}{d\theta^2} &= -\frac{1}{2}ab \sin \theta\end{aligned}$$

At $\theta = \frac{\pi}{2}$:

$$\begin{aligned}\left. \frac{d^2A}{d\theta^2} \right|_{\theta=\frac{\pi}{2}} &= -\frac{1}{2}ab \sin \left(\frac{\pi}{2} \right) \\ &= -\frac{1}{2}ab < 0\end{aligned}$$

Since the second derivative is negative, the area is maximum when $\theta = \frac{\pi}{2}$ (or 90°).

The maximum area of the triangle is:

$$\begin{aligned} A_{\max} &= \frac{1}{2}ab \sin\left(\frac{\pi}{2}\right) \\ &= \frac{1}{2}ab \end{aligned}$$

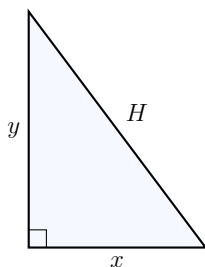
Final Answer

The area is maximum when $\theta = \frac{\pi}{2}$ and the maximum area is $\frac{1}{2}ab$.

Question 15

A right-angled triangle with constant area is given. Prove that the hypotenuse of the triangle is least when the triangle is isosceles.

TYPE 106 Minimizing Hypotenuse of Right-Angled Triangle with Constant Area



Right-angled triangle with legs x, y

To Prove: The hypotenuse of a right-angled triangle with constant area A is minimum when the legs are equal ($x = y$).

Concept / Formula Used: $A = \frac{1}{2}xy$ $H^2 = x^2 + y^2$

Solution:

Let the legs of the right-angled triangle be x and y . The area A is constant:

$$\begin{aligned} A &= \frac{1}{2}xy \\ y &= \frac{2A}{x} \end{aligned}$$

Let Z be the square of the hypotenuse:

$$Z = x^2 + y^2$$

Minimizing the hypotenuse is equivalent to minimizing its square Z . Substituting $y = \frac{2A}{x}$:

$$\begin{aligned} Z(x) &= x^2 + \left(\frac{2A}{x}\right)^2 \\ Z(x) &= x^2 + \frac{4A^2}{x^2} \end{aligned}$$

Differentiating $Z(x)$ with respect to x :

$$\begin{aligned}\frac{dZ}{dx} &= \frac{d}{dx} \left(x^2 + \frac{4A^2}{x^2} \right) \\ \frac{dZ}{dx} &= 2x - \frac{8A^2}{x^3}\end{aligned}$$

For critical points, set $\frac{dZ}{dx} = 0$:

$$\begin{aligned}2x - \frac{8A^2}{x^3} &= 0 \\ 2x &= \frac{8A^2}{x^3} \\ x^4 &= 4A^2 \\ x^2 &= 2A \\ x &= \sqrt{2A} \quad (\text{since } x > 0)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2Z}{dx^2} &= \frac{d}{dx} \left(2x - \frac{8A^2}{x^3} \right) \\ \frac{d^2Z}{dx^2} &= 2 + \frac{24A^2}{x^4}\end{aligned}$$

At $x = \sqrt{2A}$:

$$\begin{aligned}\left. \frac{d^2Z}{dx^2} \right|_{x=\sqrt{2A}} &= 2 + \frac{24A^2}{(\sqrt{2A})^4} \\ &= 2 + \frac{24A^2}{4A^2} \\ &= 2 + 6 \\ &= 8 > 0\end{aligned}$$

Since the second derivative is positive, the square of the hypotenuse Z (and thus the hypotenuse itself) is minimum when $x = \sqrt{2A}$.

Back-substituting $x = \sqrt{2A}$ to find y :

$$\begin{aligned}y &= \frac{2A}{\sqrt{2A}} \\ y &= \sqrt{2A}\end{aligned}$$

Since $x = y = \sqrt{2A}$, the triangle is isosceles.

Hence Proved

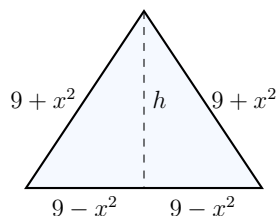
LHS = RHS, hence proved that the hypotenuse is least when the triangle is isosceles.

Question 16

The lengths of the sides of a triangle are $9 + x^2$, $9 + x^2$ and $18 - 2x^2$. Calculate:

- the area of the triangle in terms of x .
- the value of x for which this area is maximum.

TYPE 107 Maximizing Area of Isosceles Triangle with Variable Side Lengths



Isosceles triangle with sides $9 + x^2, 9 + x^2, 18 - 2x^2$

Given: An isosceles triangle with sides $a = 9 + x^2$, $b = 9 + x^2$, and base $c = 18 - 2x^2$.

Concept / Formula Used: $A = \frac{1}{2} \times \text{base} \times \text{height}$

Solution:

Let the base of the isosceles triangle be $c = 18 - 2x^2$. The altitude h drawn from the vertex to the base bisects the base. Therefore, half of the base is:

$$\begin{aligned}\frac{c}{2} &= \frac{18 - 2x^2}{2} \\ &= 9 - x^2\end{aligned}$$

Using Pythagoras' theorem on one of the right-angled halves of the triangle:

$$\begin{aligned}h^2 + \left(\frac{c}{2}\right)^2 &= a^2 \\ h^2 + (9 - x^2)^2 &= (9 + x^2)^2 \\ h^2 &= (9 + x^2)^2 - (9 - x^2)^2\end{aligned}$$

Using the algebraic identity $(A + B)^2 - (A - B)^2 = 4AB$:

$$\begin{aligned}h^2 &= 4(9)(x^2) \\ h^2 &= 36x^2 \\ h &= 6x \quad (\text{since } x > 0)\end{aligned}$$

(i) Area of the triangle in terms of x :

$$\begin{aligned}A &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2}(18 - 2x^2)(6x) \\ &= (9 - x^2)(6x) \\ &= 54x - 6x^3\end{aligned}$$

(ii) Value of x for which this area is maximum:

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx}(54x - 6x^3) \\ \frac{dA}{dx} &= 54 - 18x^2\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned} 54 - 18x^2 &= 0 \\ 18x^2 &= 54 \\ x^2 &= 3 \\ x &= \sqrt{3} \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2A}{dx^2} &= \frac{d}{dx}(54 - 18x^2) \\ \frac{d^2A}{dx^2} &= -36x \end{aligned}$$

At $x = \sqrt{3}$:

$$\left. \frac{d^2A}{dx^2} \right|_{x=\sqrt{3}} = -36\sqrt{3} < 0$$

Since the second derivative is negative, the area is maximum when $x = \sqrt{3}$.

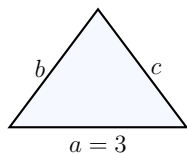
Final Answer

- (i) Area $A = 54x - 6x^3$.
- (ii) $x = \sqrt{3}$.

Question 17

The perimeter of a triangle is 8 cm. If one of the sides is 3 cm, what are the lengths of the other sides for maximum area of the triangle?

TYPE 108 Maximizing Area of Triangle with Constant Perimeter



Triangle with sides a, b, c

Given: Perimeter of the triangle $P = 8$ cm and one side $a = 3$ cm.

Concept / Formula Used: $A = \sqrt{s(s-a)(s-b)(s-c)}$ $s = \frac{P}{2}$

Solution:

Let the sides of the triangle be a, b , and c . We are given:

$$\begin{aligned} a &= 3 \text{ cm} \\ a + b + c &= 8 \\ 3 + b + c &= 8 \\ b + c &= 5 \\ c &= 5 - b \end{aligned}$$

The semi-perimeter s is:

$$s = \frac{a + b + c}{2} = \frac{8}{2} = 4 \text{ cm}$$

Using Heron's formula, the area A of the triangle is:

$$\begin{aligned} A &= \sqrt{s(s-a)(s-b)(s-c)} \\ A &= \sqrt{4(4-3)(4-b)(4-c)} \\ A &= \sqrt{4(1)(4-b)(4-(5-b))} \\ A &= \sqrt{4(4-b)(b-1)} \end{aligned}$$

To maximize A , we can maximize its square, $Z = A^2$:

$$\begin{aligned} Z(b) &= 4(4-b)(b-1) \\ Z(b) &= 4(4b-4-b^2+b) \\ Z(b) &= 4(-b^2+5b-4) \end{aligned}$$

Differentiating $Z(b)$ with respect to b :

$$\begin{aligned} \frac{dZ}{db} &= \frac{d}{db} [4(-b^2+5b-4)] \\ \frac{dZ}{db} &= 4(-2b+5) \end{aligned}$$

For critical points, set $\frac{dZ}{db} = 0$:

$$\begin{aligned} 4(-2b+5) &= 0 \\ -2b+5 &= 0 \\ 2b &= 5 \\ b &= 2.5 \text{ cm} \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2Z}{db^2} &= \frac{d}{db} [4(-2b+5)] \\ \frac{d^2Z}{db^2} &= -8 \end{aligned}$$

Since $\frac{d^2Z}{db^2} = -8 < 0$, the area is maximum when $b = 2.5$ cm.

Back-substituting $b = 2.5$ cm to find c :

$$\begin{aligned} c &= 5 - 2.5 \\ c &= 2.5 \text{ cm} \end{aligned}$$

Thus, the lengths of the other two sides are 2.5 cm and 2.5 cm.

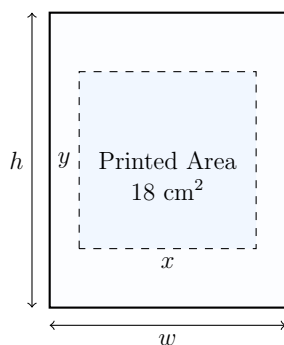
Final Answer

The other sides are 2.5 cm and 2.5 cm.

Question 18

A sheet of paper is to contain 18 cm^2 of printed matter. The margins at the top and bottom are 2 cm each, and at the sides 1 cm each. Find the dimensions of the sheet which require the least amount of paper.

TYPE 109 Minimizing Area of Rectangular Sheet with Margins



Sheet of paper with margins

Given: Printed area $A_p = 18 \text{ cm}^2$. Margins: top/bottom = 2 cm, sides = 1 cm.

Concept / Formula Used: $A = wh$ $w = x + 2$ $h = y + 4$

Solution:

Let the dimensions of the printed area be x (width) and y (height). We are given:

$$xy = 18$$

$$y = \frac{18}{x}$$

The total width of the sheet of paper is:

$$w = x + 2(1) = x + 2$$

The total height of the sheet of paper is:

$$h = y + 2(2) = y + 4$$

The total area A of the sheet of paper is:

$$A = wh$$

$$A = (x + 2)(y + 4)$$

Substituting $y = \frac{18}{x}$:

$$A(x) = (x + 2) \left(\frac{18}{x} + 4 \right)$$

$$= 18 + 4x + \frac{36}{x} + 8$$

$$= 26 + 4x + \frac{36}{x}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx} \left(26 + 4x + \frac{36}{x} \right) \\ \frac{dA}{dx} &= 4 - \frac{36}{x^2}\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned}4 - \frac{36}{x^2} &= 0 \\ 4 &= \frac{36}{x^2} \\ x^2 &= 9 \\ x &= 3 \text{ cm} \quad (\text{since } x > 0)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dx^2} &= \frac{d}{dx} \left(4 - \frac{36}{x^2} \right) \\ \frac{d^2A}{dx^2} &= \frac{72}{x^3}\end{aligned}$$

At $x = 3$:

$$\left. \frac{d^2A}{dx^2} \right|_{x=3} = \frac{72}{3^3} = \frac{72}{27} = \frac{8}{3} > 0$$

Since the second derivative is positive, the total area is minimum when $x = 3$ cm.

Now, find the dimensions of the sheet:

$$\begin{aligned}w &= x + 2 = 3 + 2 = 5 \text{ cm} \\ y &= \frac{18}{3} = 6 \text{ cm} \\ h &= y + 4 = 6 + 4 = 10 \text{ cm}\end{aligned}$$

Thus, the dimensions of the sheet of paper are 5 cm by 10 cm.

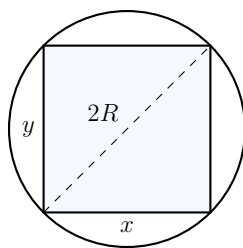
Final Answer

The dimensions of the sheet are 5 cm $\times$ 10 cm.

Question 19

Show that of all rectangles inscribed in a given fixed circle the square has the maximum area.

TYPE 110 Maximizing Area of Rectangle Inscribed in a Circle



Rectangle inscribed in a circle

To Prove: The area of an inscribed rectangle in a circle of fixed radius R is maximum when the rectangle is a square ($x = y$).

Concept / Formula Used: $x^2 + y^2 = (2R)^2 = 4R^2$ $A = xy$

Solution:

Let the rectangle inscribed in a circle of fixed radius R have length x and width y . The diagonal of the rectangle is the diameter of the circle, which is $2R$. By Pythagoras' theorem:

$$\begin{aligned}x^2 + y^2 &= (2R)^2 \\x^2 + y^2 &= 4R^2 \\y &= \sqrt{4R^2 - x^2}\end{aligned}$$

The area A of the rectangle is:

$$A = xy$$

To maximize A , we can maximize its square, $Z = A^2$:

$$\begin{aligned}Z(x) &= x^2 y^2 \\Z(x) &= x^2 (4R^2 - x^2) \\Z(x) &= 4R^2 x^2 - x^4\end{aligned}$$

Differentiating $Z(x)$ with respect to x :

$$\begin{aligned}\frac{dZ}{dx} &= \frac{d}{dx}(4R^2 x^2 - x^4) \\\frac{dZ}{dx} &= 8R^2 x - 4x^3 \\\frac{dZ}{dx} &= 4x(2R^2 - x^2)\end{aligned}$$

For critical points, set $\frac{dZ}{dx} = 0$:

$$4x(2R^2 - x^2) = 0$$

Since $x > 0$, we must have:

$$\begin{aligned}2R^2 - x^2 &= 0 \\x^2 &= 2R^2 \\x &= \sqrt{2}R\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2Z}{dx^2} &= \frac{d}{dx}(8R^2x - 4x^3) \\ \frac{d^2Z}{dx^2} &= 8R^2 - 12x^2\end{aligned}$$

At $x = \sqrt{2}R$:

$$\begin{aligned}\left.\frac{d^2Z}{dx^2}\right|_{x=\sqrt{2}R} &= 8R^2 - 12(2R^2) \\ &= 8R^2 - 24R^2 \\ &= -16R^2 < 0\end{aligned}$$

Since the second derivative is negative, the area is maximum when $x = \sqrt{2}R$.

Back-substituting $x = \sqrt{2}R$ to find y :

$$\begin{aligned}y &= \sqrt{4R^2 - (\sqrt{2}R)^2} \\ &= \sqrt{4R^2 - 2R^2} \\ &= \sqrt{2R^2} \\ &= \sqrt{2}R\end{aligned}$$

Since $x = y = \sqrt{2}R$, the rectangle is a square.

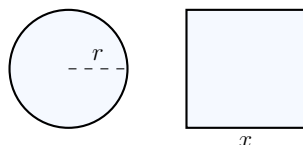
Hence Proved

LHS = RHS, hence proved that the inscribed rectangle of maximum area is a square.

Question 20

- The sum of perimeters of a circle and a square is k , where k is some constant. Prove that the sum of their areas is least when the side of the square is double the radius of the circle.
- Given the sum of perimeters of a circle and a square, show that the sum of areas is least when the diameter of the circle is equal to the side of the square.

TYPE 111 Minimizing Combined Area of Circle and Square



Circle and Square

To Prove:

- Sum of areas is least when $x = 2r$.
- Sum of areas is least when $2r = x$.

Concept / Formula Used: $P_c = 2\pi r$ $P_s = 4x$ $A_c = \pi r^2$ $A_s = x^2$

Solution:

Let the radius of the circle be r and the side of the square be x . We are given that the sum of their perimeters is a constant k :

$$\begin{aligned} 2\pi r + 4x &= k \\ 4x &= k - 2\pi r \\ x &= \frac{k - 2\pi r}{4} \end{aligned}$$

The sum of their areas A is:

$$A = \pi r^2 + x^2$$

Substituting $x = \frac{k-2\pi r}{4}$:

$$\begin{aligned} A(r) &= \pi r^2 + \left(\frac{k - 2\pi r}{4}\right)^2 \\ A(r) &= \pi r^2 + \frac{(k - 2\pi r)^2}{16} \end{aligned}$$

Differentiating $A(r)$ with respect to r :

$$\begin{aligned} \frac{dA}{dr} &= \frac{d}{dr} \left[\pi r^2 + \frac{(k - 2\pi r)^2}{16} \right] \\ &= 2\pi r + \frac{2(k - 2\pi r) \cdot (-2\pi)}{16} \\ &= 2\pi r - \frac{\pi(k - 2\pi r)}{4} \end{aligned}$$

For critical points, set $\frac{dA}{dr} = 0$:

$$2\pi r - \frac{\pi(k - 2\pi r)}{4} = 0$$

Dividing both sides by π :

$$\begin{aligned} 2r - \frac{k - 2\pi r}{4} &= 0 \\ 8r - (k - 2\pi r) &= 0 \\ 8r + 2\pi r &= k \\ 2r(4 + \pi) &= k \\ r &= \frac{k}{2(4 + \pi)} \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2A}{dr^2} &= \frac{d}{dr} \left[2\pi r - \frac{\pi(k - 2\pi r)}{4} \right] \\ &= 2\pi - \frac{\pi(-2\pi)}{4} \\ &= 2\pi + \frac{\pi^2}{2} \end{aligned}$$

Since $\frac{d^2A}{dr^2} = 2\pi + \frac{\pi^2}{2} > 0$, the sum of areas A is minimum (least) at this value of r .

Now, let us find the relation between x and r at this minimum. We know:

$$4x = k - 2\pi r$$

Substituting $k = 2r(4 + \pi)$:

$$4x = 2r(4 + \pi) - 2\pi r$$

$$4x = 8r + 2\pi r - 2\pi r$$

$$4x = 8r$$

$$x = 2r$$

(i) This proves that the side of the square (x) is double the radius of the circle ($2r$).

(ii) Since the diameter of the circle is $d = 2r$, the relation $x = 2r$ implies:

$$x = d$$

This proves that the sum of areas is least when the diameter of the circle is equal to the side of the square.

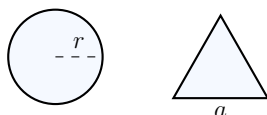
Hence Proved

LHS = RHS, hence proved for both parts (i) and (ii).

Question 21

A wire 10 metres long is cut into two parts. One part is bent into the shape of a circle and the other into the shape of an equilateral triangle. How should the wire be cut so that the combined area of the two figures is as small as possible?

TYPE 112 Minimizing Combined Area of Circle and Triangle



Circle and Equilateral Triangle

Given: Total length of wire $L = 10$ m.

Concept / Formula Used: $A_c = \pi r^2$ $A_t = \frac{\sqrt{3}}{4} a^2$

Solution:

Let the wire of length 10 m be cut into two pieces of length x and $10 - x$. Let the piece of length x be bent into a circle of radius r :

$$2\pi r = x$$

$$r = \frac{x}{2\pi}$$

Let the other piece of length $10 - x$ be bent into an equilateral triangle of side a :

$$3a = 10 - x$$

$$a = \frac{10 - x}{3}$$

The combined area A of the circle and the equilateral triangle is:

$$A = \pi r^2 + \frac{\sqrt{3}}{4}a^2$$

Substituting $r = \frac{x}{2\pi}$ and $a = \frac{10-x}{3}$:

$$A(x) = \pi \left(\frac{x}{2\pi} \right)^2 + \frac{\sqrt{3}}{4} \left(\frac{10-x}{3} \right)^2$$

$$A(x) = \frac{x^2}{4\pi} + \frac{\sqrt{3}}{36}(10-x)^2$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned} \frac{dA}{dx} &= \frac{d}{dx} \left[\frac{x^2}{4\pi} + \frac{\sqrt{3}}{36}(10-x)^2 \right] \\ &= \frac{2x}{4\pi} + \frac{\sqrt{3}}{36} \cdot 2(10-x) \cdot (-1) \\ &= \frac{x}{2\pi} - \frac{\sqrt{3}}{18}(10-x) \end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned} \frac{x}{2\pi} - \frac{\sqrt{3}}{18}(10-x) &= 0 \\ \frac{x}{2\pi} &= \frac{\sqrt{3}}{18}(10-x) \\ 18x &= 2\pi\sqrt{3}(10-x) \\ 9x &= \pi\sqrt{3}(10-x) \\ 9x &= 10\pi\sqrt{3} - \pi\sqrt{3}x \\ x(9 + \pi\sqrt{3}) &= 10\pi\sqrt{3} \\ x &= \frac{10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \text{ m} \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2A}{dx^2} &= \frac{d}{dx} \left[\frac{x}{2\pi} - \frac{\sqrt{3}}{18}(10-x) \right] \\ &= \frac{1}{2\pi} + \frac{\sqrt{3}}{18} \end{aligned}$$

Since $\frac{d^2A}{dx^2} = \frac{1}{2\pi} + \frac{\sqrt{3}}{18} > 0$, the combined area is minimum at this value of x .

The length of the other piece is:

$$\begin{aligned} 10 - x &= 10 - \frac{10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \\ &= \frac{10(9 + \pi\sqrt{3}) - 10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \\ &= \frac{90 + 10\pi\sqrt{3} - 10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \\ &= \frac{90}{9 + \pi\sqrt{3}} \text{ m} \end{aligned}$$

Thus, the wire should be cut into two pieces of lengths $\frac{10\pi\sqrt{3}}{9+\pi\sqrt{3}}$ m and $\frac{90}{9+\pi\sqrt{3}}$ m.

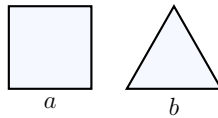
Final Answer

The lengths of the two pieces are $\frac{10\pi\sqrt{3}}{9+\pi\sqrt{3}}$ m and $\frac{90}{9+\pi\sqrt{3}}$ m.

Question 22

A wire of length 36 cm is cut into two pieces. One of the piece is turned into the form of a square and the other in the form of an equilateral triangle. Find the length of each piece so that the sum of the areas of the two figures be minimum.

TYPE 113 Minimizing Combined Area of Square and Triangle



Square and Equilateral Triangle

Given: Total length of wire $L = 36$ cm.

Concept / Formula Used: $A_s = a^2$ $A_t = \frac{\sqrt{3}}{4}b^2$

Solution:

Let the wire of length 36 cm be cut into two pieces of length x and $36 - x$. Let the piece of length x be bent into a square of side a :

$$\begin{aligned} 4a &= x \\ a &= \frac{x}{4} \end{aligned}$$

Let the other piece of length $36 - x$ be bent into an equilateral triangle of side b :

$$\begin{aligned} 3b &= 36 - x \\ b &= \frac{36 - x}{3} \end{aligned}$$

The combined area A of the square and the equilateral triangle is:

$$A = a^2 + \frac{\sqrt{3}}{4}b^2$$

Substituting $a = \frac{x}{4}$ and $b = \frac{36-x}{3}$:

$$\begin{aligned} A(x) &= \left(\frac{x}{4}\right)^2 + \frac{\sqrt{3}}{4} \left(\frac{36-x}{3}\right)^2 \\ A(x) &= \frac{x^2}{16} + \frac{\sqrt{3}}{36}(36-x)^2 \end{aligned}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx} \left[\frac{x^2}{16} + \frac{\sqrt{3}}{36}(36-x)^2 \right] \\ &= \frac{2x}{16} + \frac{\sqrt{3}}{36} \cdot 2(36-x) \cdot (-1) \\ &= \frac{x}{8} - \frac{\sqrt{3}}{18}(36-x)\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned}\frac{x}{8} - \frac{\sqrt{3}}{18}(36-x) &= 0 \\ \frac{x}{8} &= \frac{\sqrt{3}}{18}(36-x) \\ 18x &= 8\sqrt{3}(36-x) \\ 9x &= 4\sqrt{3}(36-x) \\ 9x &= 144\sqrt{3} - 4\sqrt{3}x \\ x(9+4\sqrt{3}) &= 144\sqrt{3} \\ x &= \frac{144\sqrt{3}}{9+4\sqrt{3}} \text{ cm}\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dx^2} &= \frac{d}{dx} \left[\frac{x}{8} - \frac{\sqrt{3}}{18}(36-x) \right] \\ &= \frac{1}{8} + \frac{\sqrt{3}}{18}\end{aligned}$$

Since $\frac{d^2A}{dx^2} = \frac{1}{8} + \frac{\sqrt{3}}{18} > 0$, the combined area is minimum at this value of x .

The length of the other piece is:

$$\begin{aligned}36-x &= 36 - \frac{144\sqrt{3}}{9+4\sqrt{3}} \\ &= \frac{36(9+4\sqrt{3}) - 144\sqrt{3}}{9+4\sqrt{3}} \\ &= \frac{324 + 144\sqrt{3} - 144\sqrt{3}}{9+4\sqrt{3}} \\ &= \frac{324}{9+4\sqrt{3}} \text{ cm}\end{aligned}$$

Thus, the lengths of the two pieces are $\frac{144\sqrt{3}}{9+4\sqrt{3}}$ cm and $\frac{324}{9+4\sqrt{3}}$ cm.

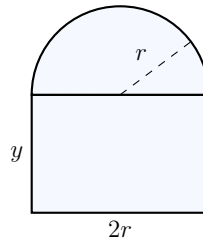
Final Answer

The lengths of the two pieces are $\frac{144\sqrt{3}}{9+4\sqrt{3}}$ cm and $\frac{324}{9+4\sqrt{3}}$ cm.

Question 23

A window is in the form of a rectangle above which there is a semicircle. If the perimeter of the window is p cm, show that the window will admit maximum possible light only when the radius of semicircle is $\frac{p}{\pi + 4}$ cm.

TYPE 114 Maximizing Area of Norman Window with Constant Perimeter



Window with rectangular base and semicircular top

To Prove: The area of the window is maximum when the radius of the semicircle is $r = \frac{p}{\pi + 4}$ cm.

Concept / Formula Used: $P = 2r + 2y + \pi r = p$ $A = 2ry + \frac{1}{2}\pi r^2$

Solution:

Let the radius of the semicircular top be r . Then the width of the rectangular base is $2r$. Let the height of the rectangular part be y . The perimeter P of the window consists of the bottom side ($2r$), two vertical sides ($2y$), and the semicircular arc (πr):

$$\begin{aligned} P &= 2r + 2y + \pi r \\ p &= (\pi + 2)r + 2y \\ 2y &= p - (\pi + 2)r \\ y &= \frac{p - (\pi + 2)r}{2} \end{aligned}$$

The area A of the window (which determines the amount of light admitted) is:

$$\begin{aligned} A &= \text{Area of rectangle} + \text{Area of semicircle} \\ A &= (2r)y + \frac{1}{2}\pi r^2 \end{aligned}$$

Substituting $y = \frac{p - (\pi + 2)r}{2}$:

$$\begin{aligned} A(r) &= 2r \left[\frac{p - (\pi + 2)r}{2} \right] + \frac{1}{2}\pi r^2 \\ &= r[p - (\pi + 2)r] + \frac{1}{2}\pi r^2 \\ &= pr - (\pi + 2)r^2 + \frac{1}{2}\pi r^2 \\ &= pr - \pi r^2 - 2r^2 + \frac{1}{2}\pi r^2 \\ &= pr - \left(\frac{\pi}{2} + 2 \right) r^2 \end{aligned}$$

Differentiating $A(r)$ with respect to r :

$$\begin{aligned}\frac{dA}{dr} &= \frac{d}{dr} \left[pr - \left(\frac{\pi}{2} + 2 \right) r^2 \right] \\ &= p - 2 \left(\frac{\pi}{2} + 2 \right) r \\ &= p - (\pi + 4)r\end{aligned}$$

For critical points, set $\frac{dA}{dr} = 0$:

$$\begin{aligned}p - (\pi + 4)r &= 0 \\ (\pi + 4)r &= p \\ r &= \frac{p}{\pi + 4} \text{ cm}\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dr^2} &= \frac{d}{dr} [p - (\pi + 4)r] \\ &= -(\pi + 4)\end{aligned}$$

Since $\frac{d^2A}{dr^2} = -(\pi + 4) < 0$, the area is maximum when $r = \frac{p}{\pi + 4}$ cm.

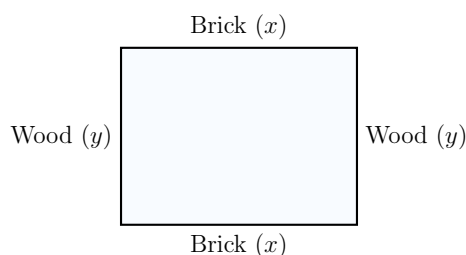
Hence Proved

LHS = RHS, hence proved that the window admits maximum light when the radius of the semicircle is $\frac{p}{\pi + 4}$ cm.

Question 24

A rectangular area of 9000 m^2 is to be surrounded by a fence with two opposite sides being made of bricks and the other two of wood. One metre of wooden fencing costs Rs. 250 while one metre of brick walling costs Rs. 100. What is the least amount of money that must be allotted for the construction of such a fence?

TYPE 116 Minimizing Cost of Fencing Rectangular Area



Fenced rectangular area with brick and wooden sides

Given: Area $A = 9000 \text{ m}^2$, cost of brick walling is Rs. 100/m, cost of wooden fencing is Rs. 250/m.

To Find: The least amount of money required for the construction of the fence.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$

Solution:

Let the length of the brick sides be x metres and the length of the wooden sides be y metres.

The area of the rectangular region is given as:

$$xy = 9000$$

Solving for y in terms of x :

$$y = \frac{9000}{x}$$

The cost of the two opposite brick sides of length x is:

$$\text{Cost}_{\text{brick}} = 2 \times 100 \times x = 200x$$

The cost of the two opposite wooden sides of length y is:

$$\text{Cost}_{\text{wood}} = 2 \times 250 \times y = 500y$$

The total cost C is the sum of these two costs:

$$C = 200x + 500y$$

Substitute $y = \frac{9000}{x}$ into the cost equation:

$$\begin{aligned} C(x) &= 200x + 500 \left(\frac{9000}{x} \right) \\ &= 200x + \frac{4500000}{x} \end{aligned}$$

Differentiating $C(x)$ with respect to x :

$$\frac{dC}{dx} = 200 - \frac{4500000}{x^2}$$

For the minimum cost, set $\frac{dC}{dx} = 0$:

$$\begin{aligned} 200 - \frac{4500000}{x^2} &= 0 \\ 200 &= \frac{4500000}{x^2} \\ x^2 &= \frac{4500000}{200} \\ x^2 &= 22500 \\ x &= 150 \end{aligned}$$

To verify if this is a minimum, find the second derivative:

$$\frac{d^2C}{dx^2} = \frac{9000000}{x^3}$$

At $x = 150$:

$$\begin{aligned} \frac{d^2C}{dx^2} &= \frac{9000000}{150^3} \\ &= \frac{9000000}{3375000} \\ &= \frac{8}{3} > 0 \end{aligned}$$

Since the second derivative is positive, the cost is minimum at $x = 150$.

Now, calculate the minimum cost:

$$\begin{aligned} C(150) &= 200(150) + \frac{4500000}{150} \\ &= 30000 + 30000 \\ &= 60000 \end{aligned}$$

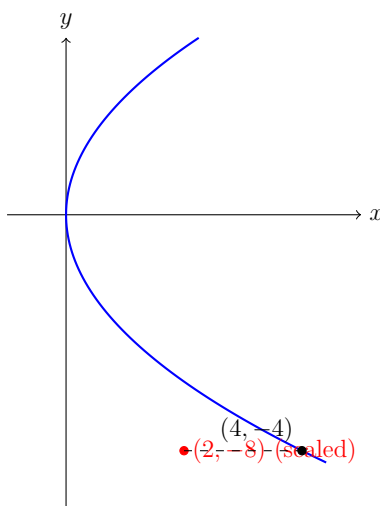
Final Answer

Rs. 60000

Question 25

- Find the point on the parabola $y^2 = 4x$ which is nearest to the point $(2, -8)$.
- Find the point on the curve $x^2 = 8y$ which is nearest to the point $(2, 4)$.

TYPE 117 Finding Point on Parabola Nearest to a Given Point



Parabola $y^2 = 4x$ and the nearest point to $(2, -8)$

Given: The curves (i) $y^2 = 4x$ and (ii) $x^2 = 8y$, and the respective points (i) $(2, -8)$ and (ii) $(2, 4)$.

To Find: The coordinates of the nearest point on each curve.

Concept	/	Formula	Used:	D	(x_1, y_1)	(x_2, y_2)	D	=
		$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	$Z = D^2$					

Solution:

- Let $Q(x, y)$ be any point on the parabola $y^2 = 4x$.

Expressing x in terms of y :

$$x = \frac{y^2}{4}$$

The distance D between $Q(x, y)$ and the point $P(2, -8)$ is:

$$D = \sqrt{(x - 2)^2 + (y + 8)^2}$$

Let $Z = D^2$:

$$Z = (x - 2)^2 + (y + 8)^2$$

Substitute $x = \frac{y^2}{4}$ into the equation for Z :

$$\begin{aligned} Z(y) &= \left(\frac{y^2}{4} - 2\right)^2 + (y + 8)^2 \\ &= \frac{y^4}{16} - y^2 + 4 + y^2 + 16y + 64 \\ &= \frac{y^4}{16} + 16y + 68 \end{aligned}$$

Differentiating $Z(y)$ with respect to y :

$$\frac{dZ}{dy} = \frac{4y^3}{16} + 16 = \frac{y^3}{4} + 16$$

For the minimum distance, set $\frac{dZ}{dy} = 0$:

$$\begin{aligned} \frac{y^3}{4} + 16 &= 0 \\ \frac{y^3}{4} &= -16 \\ y^3 &= -64 \\ y &= -4 \end{aligned}$$

Finding the second derivative:

$$\frac{d^2Z}{dy^2} = \frac{3y^2}{4}$$

At $y = -4$:

$$\frac{d^2Z}{dy^2} = \frac{3(-4)^2}{4} = 12 > 0$$

Since the second derivative is positive, the distance is minimum at $y = -4$.

Now, find the corresponding x -coordinate:

$$x = \frac{(-4)^2}{4} = 4$$

Thus, the nearest point is $(4, -4)$.

(ii) Let $Q(x, y)$ be any point on the curve $x^2 = 8y$.

Expressing y in terms of x :

$$y = \frac{x^2}{8}$$

The distance D between $Q(x, y)$ and the point $P(2, 4)$ is:

$$D = \sqrt{(x - 2)^2 + (y - 4)^2}$$

Let $Z = D^2$:

$$Z = (x - 2)^2 + (y - 4)^2$$

Substitute $y = \frac{x^2}{8}$ into the equation for Z :

$$\begin{aligned} Z(x) &= (x-2)^2 + \left(\frac{x^2}{8} - 4\right)^2 \\ &= x^2 - 4x + 4 + \frac{x^4}{64} - x^2 + 16 \\ &= \frac{x^4}{64} - 4x + 20 \end{aligned}$$

Differentiating $Z(x)$ with respect to x :

$$\frac{dZ}{dx} = \frac{4x^3}{64} - 4 = \frac{x^3}{16} - 4$$

For the minimum distance, set $\frac{dZ}{dx} = 0$:

$$\begin{aligned} \frac{x^3}{16} - 4 &= 0 \\ \frac{x^3}{16} &= 4 \\ x^3 &= 64 \\ x &= 4 \end{aligned}$$

Finding the second derivative:

$$\frac{d^2Z}{dx^2} = \frac{3x^2}{16}$$

At $x = 4$:

$$\frac{d^2Z}{dx^2} = \frac{3(4)^2}{16} = 3 > 0$$

Since the second derivative is positive, the distance is minimum at $x = 4$.

Now, find the corresponding y -coordinate:

$$y = \frac{4^2}{8} = 2$$

Thus, the nearest point is $(4, 2)$.

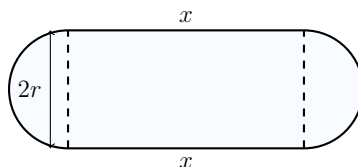
Final Answer

- (i) $(4, -4)$
- (ii) $(4, 2)$

Question 26

A running track of 440 m is to be laid out enclosing a football field. The football field is in the shape of a rectangle with semicircle at each end. If the area of the rectangular portion is to be maximum, then find the lengths of its sides. Also calculate the area of the football field.

TYPE 118 Maximizing Rectangular Area of Stadium with Semicircular Ends



Football field with rectangular portion and semicircular ends

Given: Total perimeter of the track $P = 440$ m.

To Find: The lengths of the sides of the rectangular portion for maximum area, and the total area of the football field.

Concept / Formula Used: $P = 2x + 2\pi r$ $A = 2xr$ $A_{\text{total}} = 2xr + \pi r^2$

Solution:

Let the length of the rectangular portion be x and the width be $2r$, where r is the radius of the semicircular ends.

The total perimeter of the track is given by:

$$P = 2x + 2\pi r = 440$$

Dividing by 2:

$$x + \pi r = 220$$

Solving for x :

$$x = 220 - \pi r$$

The area A of the rectangular portion is:

$$A = x(2r) = 2r(220 - \pi r) = 440r - 2\pi r^2$$

Differentiating A with respect to r :

$$\frac{dA}{dr} = 440 - 4\pi r$$

For maximum area, set $\frac{dA}{dr} = 0$:

$$\begin{aligned} 440 - 4\pi r &= 0 \\ 4\pi r &= 440 \\ r &= \frac{110}{\pi} \end{aligned}$$

Finding the second derivative:

$$\frac{d^2A}{dr^2} = -4\pi$$

Since $\frac{d^2A}{dr^2} < 0$, the area of the rectangular portion is maximum when $r = \frac{110}{\pi}$.

Now, find the dimensions of the rectangular portion: - Width ($2r$):

$$2r = 2 \left(\frac{110}{\pi} \right) = \frac{220}{\pi} \text{ m}$$

Using $\pi = \frac{22}{7}$:

$$2r = \frac{220}{\frac{22}{7}} = 70 \text{ m}$$

- Length (x):

$$x = 220 - \pi \left(\frac{110}{\pi} \right) = 110 \text{ m}$$

Now, calculate the total area of the football field:

$$\begin{aligned} A_{\text{total}} &= \text{Area of rectangle} + \text{Area of two semicircles} \\ &= 2xr + \pi r^2 \\ &= 2(110) \left(\frac{110}{\pi} \right) + \pi \left(\frac{110}{\pi} \right)^2 \\ &= \frac{24200}{\pi} + \frac{12100}{\pi} \\ &= \frac{36300}{\pi} \end{aligned}$$

Using $\pi = \frac{22}{7}$:

$$\begin{aligned} A_{\text{total}} &= 36300 \times \frac{7}{22} \\ &= 1650 \times 7 \\ &= 11550 \text{ m}^2 \end{aligned}$$

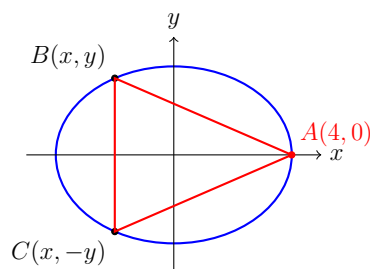
Final Answer

Lengths of sides: 110 m and 70 m; Total area: 11550 m²

Question 27

Find the maximum area of an isosceles triangle inscribed in the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with its vertex at one end of the major axis.

TYPE 119 Maximizing Area of Isosceles Triangle Inscribed in an Ellipse



Isosceles triangle inscribed in an ellipse

Given: Equation of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, vertex of the triangle at $(4, 0)$.

To Find: The maximum area of the inscribed isosceles triangle.

Concept / Formula Used: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a \cos \theta, b \sin \theta)$ $A = \frac{1}{2} \times \text{base} \times \text{height}$

Solution:

Let the equation of the ellipse be $\frac{x^2}{16} + \frac{y^2}{9} = 1$. Here, $a = 4$ and $b = 3$.

Let one vertex of the isosceles triangle be at $A(4, 0)$, which is one end of the major axis. Let the other two vertices be $B(4 \cos \theta, 3 \sin \theta)$ and $C(4 \cos \theta, -3 \sin \theta)$, where $\theta \in (0, \pi)$.

The base of the triangle is the length of BC :

$$BC = 3 \sin \theta - (-3 \sin \theta) = 6 \sin \theta$$

The height of the triangle is the distance from $A(4, 0)$ to the line $x = 4 \cos \theta$:

$$h = 4 - 4 \cos \theta = 4(1 - \cos \theta)$$

The area A of the triangle is:

$$\begin{aligned} A &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} (6 \sin \theta) [4(1 - \cos \theta)] \\ &= 12 \sin \theta (1 - \cos \theta) \end{aligned}$$

Differentiating A with respect to θ :

$$\begin{aligned} \frac{dA}{d\theta} &= 12 [\cos \theta (1 - \cos \theta) + \sin \theta (\sin \theta)] \\ &= 12 [\cos \theta - \cos^2 \theta + \sin^2 \theta] \\ &= 12 [\cos \theta - \cos^2 \theta + (1 - \cos^2 \theta)] \\ &= 12 [1 + \cos \theta - 2 \cos^2 \theta] \end{aligned}$$

Set $\frac{dA}{d\theta} = 0$ for critical points:

$$\begin{aligned} 1 + \cos \theta - 2 \cos^2 \theta &= 0 \\ 2 \cos^2 \theta - \cos \theta - 1 &= 0 \\ (2 \cos \theta + 1)(\cos \theta - 1) &= 0 \end{aligned}$$

This gives:

$$\cos \theta = 1 \quad \text{or} \quad \cos \theta = -\frac{1}{2}$$

Since $\theta \in (0, \pi)$, $\cos \theta = 1$ is not possible. Thus:

$$\cos \theta = -\frac{1}{2} \implies \theta = \frac{2\pi}{3}$$

To check for maximum, find the second derivative:

$$\frac{d^2 A}{d\theta^2} = 12(-\sin \theta + 4 \cos \theta \sin \theta) = 12 \sin \theta (4 \cos \theta - 1)$$

At $\theta = \frac{2\pi}{3}$:

$$\sin \left(\frac{2\pi}{3} \right) = \frac{\sqrt{3}}{2}, \quad \cos \left(\frac{2\pi}{3} \right) = -\frac{1}{2}$$

$$\begin{aligned}
 \frac{d^2 A}{d\theta^2} &= 12 \left(\frac{\sqrt{3}}{2} \right) \left(4 \left(-\frac{1}{2} \right) - 1 \right) \\
 &= 6\sqrt{3}(-3) \\
 &= -18\sqrt{3} < 0
 \end{aligned}$$

Since the second derivative is negative, the area is maximum at $\theta = \frac{2\pi}{3}$.

Calculate the maximum area:

$$\begin{aligned}
 A_{\max} &= 12 \sin \left(\frac{2\pi}{3} \right) \left(1 - \cos \left(\frac{2\pi}{3} \right) \right) \\
 &= 12 \left(\frac{\sqrt{3}}{2} \right) \left(1 - \left(-\frac{1}{2} \right) \right) \\
 &= 6\sqrt{3} \left(\frac{3}{2} \right) \\
 &= 9\sqrt{3} \text{ sq. units}
 \end{aligned}$$

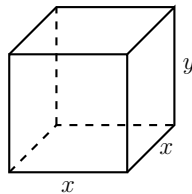
Final Answer

$9\sqrt{3}$ sq. units

Question 28

Show that the surface area of a closed cuboid with square base and given volume is minimum when it is a cube.

TYPE 120 Minimizing Surface Area of Closed Cuboid



Closed cuboid with square base

To Prove: The surface area of the closed cuboid is minimum when it is a cube (i.e., $x = y$).

Concept / Formula Used: $V = x^2y$ $S = 2x^2 + 4xy$

Solution:

Let the side of the square base of the cuboid be x and its height be y .

The volume V of the cuboid is given to be constant:

$$V = x^2y$$

Solving for y in terms of x :

$$y = \frac{V}{x^2}$$

The total surface area S of the closed cuboid is:

$$S = 2x^2 + 4xy$$

Substitute $y = \frac{V}{x^2}$ into the surface area equation:

$$\begin{aligned} S(x) &= 2x^2 + 4x \left(\frac{V}{x^2} \right) \\ &= 2x^2 + \frac{4V}{x} \end{aligned}$$

Differentiating $S(x)$ with respect to x :

$$\frac{dS}{dx} = 4x - \frac{4V}{x^2}$$

For the minimum surface area, set $\frac{dS}{dx} = 0$:

$$\begin{aligned} 4x - \frac{4V}{x^2} &= 0 \\ 4x^3 &= 4V \\ x^3 &= V \end{aligned}$$

Substitute $V = x^3$ back into the equation:

$$\begin{aligned} x^3 &= x^2y \\ x &= y \end{aligned}$$

To confirm that this is a minimum, find the second derivative:

$$\frac{d^2S}{dx^2} = 4 + \frac{8V}{x^3}$$

Since $x > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2S}{dx^2} > 0$$

Thus, the surface area is minimum when $x = y$, which means the cuboid is a cube.

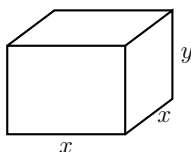
Hence Proved

The surface area is minimum when the cuboid is a cube.

Question 29

An open box with a square base is to be made of given iron sheet of area 27 sq. m. Show that the maximum volume of the box is 13.5 cu. m.

TYPE 121 Maximizing Volume of Open Box



Open box with square base

To Prove: The maximum volume of the box is 13.5 m³.

Concept / Formula Used: $S = x^2 + 4xy = 27$ $V = x^2y$

Solution:

Let the side of the square base of the open box be x and its height be y .

The surface area of the open box (which has no top) is given as:

$$S = x^2 + 4xy = 27$$

Solving for y in terms of x :

$$\begin{aligned} 4xy &= 27 - x^2 \\ y &= \frac{27 - x^2}{4x} \end{aligned}$$

The volume V of the box is:

$$\begin{aligned} V &= x^2y \\ &= x^2 \left(\frac{27 - x^2}{4x} \right) \\ &= \frac{x(27 - x^2)}{4} \\ &= \frac{27x - x^3}{4} \end{aligned}$$

Differentiating V with respect to x :

$$\frac{dV}{dx} = \frac{27 - 3x^2}{4}$$

For the maximum volume, set $\frac{dV}{dx} = 0$:

$$\begin{aligned} \frac{27 - 3x^2}{4} &= 0 \\ 27 - 3x^2 &= 0 \\ 3x^2 &= 27 \\ x^2 &= 9 \\ x &= 3 \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\frac{d^2V}{dx^2} = \frac{-6x}{4} = -1.5x$$

At $x = 3$:

$$\frac{d^2V}{dx^2} = -1.5(3) = -4.5 < 0$$

Since the second derivative is negative, the volume is maximum at $x = 3$.

Calculate the maximum volume:

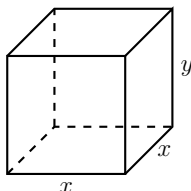
$$\begin{aligned} V_{\max} &= \frac{27(3) - 3^3}{4} \\ &= \frac{81 - 27}{4} \\ &= \frac{54}{4} \\ &= 13.5 \text{ m}^3 \end{aligned}$$

Hence Proved

The maximum volume of the box is 13.5 cu. m.

Question 30

The volume of a closed metal box with a square base is 4096 cm^3 . The cost of polishing the outer surface of the box is Rs. 4 per cm^2 . Find the dimensions of the box for the minimum cost of polishing it.

TYPE 122 Minimizing Polishing Cost of Closed Box

Closed box with square base

Given: Volume $V = 4096 \text{ cm}^3$, cost rate is Rs. 4/ cm^2 .

To Find: The dimensions of the box for minimum cost.

Concept / Formula Used: $V = x^2y$ $S = 2x^2 + 4xy$ $C = 4S$

Solution:

Let the side of the square base of the closed box be x cm and its height be y cm.

The volume of the box is:

$$V = x^2y = 4096 \implies y = \frac{4096}{x^2}$$

The total surface area S of the closed box is:

$$S = 2x^2 + 4xy$$

The cost of polishing C is:

$$C = 4S = 4(2x^2 + 4xy) = 8x^2 + 16xy$$

Substitute $y = \frac{4096}{x^2}$ into the cost equation:

$$\begin{aligned} C(x) &= 8x^2 + 16x \left(\frac{4096}{x^2} \right) \\ &= 8x^2 + \frac{65536}{x} \end{aligned}$$

Differentiating $C(x)$ with respect to x :

$$\frac{dC}{dx} = 16x - \frac{65536}{x^2}$$

For the minimum cost, set $\frac{dC}{dx} = 0$:

$$\begin{aligned} 16x - \frac{65536}{x^2} &= 0 \\ 16x^3 &= 65536 \\ x^3 &= 4096 \\ x &= 16 \text{ cm} \end{aligned}$$

Finding the second derivative:

$$\frac{d^2C}{dx^2} = 16 + \frac{131072}{x^3}$$

At $x = 16$:

$$\begin{aligned} \frac{d^2C}{dx^2} &= 16 + \frac{131072}{16^3} \\ &= 16 + \frac{131072}{4096} \\ &= 16 + 32 \\ &= 48 > 0 \end{aligned}$$

Since the second derivative is positive, the cost is minimum at $x = 16$.

Now, find the height y :

$$y = \frac{4096}{16^2} = \frac{4096}{256} = 16 \text{ cm}$$

Thus, the dimensions of the box for minimum cost are $16 \text{ cm} \times 16 \text{ cm} \times 16 \text{ cm}$.

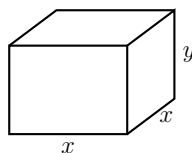
Final Answer

$16 \text{ cm} \times 16 \text{ cm} \times 16 \text{ cm}$

Question 31

An open tank with a square base and vertical sides is to be constructed from a metal sheet so as to hold a given quantity of water. Show that the cost of material will be least when the depth of the tank is half of its width.

TYPE 123 Minimizing Surface Area of Open Tank



Open tank with square base

To Prove: The cost of material (surface area) is least when the depth y is half of its width x (i.e., $y = \frac{x}{2}$).

Concept / Formula Used: $V = x^2y = \text{constant}$ $S = x^2 + 4xy$

Solution:

Let the side of the square base (width) of the open tank be x and its depth be y .

The volume V of water to be held is constant:

$$V = x^2y \implies y = \frac{V}{x^2}$$

The cost of material is proportional to the total surface area S of the open tank:

$$S = x^2 + 4xy$$

Substitute $y = \frac{V}{x^2}$ into the surface area equation:

$$\begin{aligned} S(x) &= x^2 + 4x \left(\frac{V}{x^2} \right) \\ &= x^2 + \frac{4V}{x} \end{aligned}$$

Differentiating $S(x)$ with respect to x :

$$\frac{dS}{dx} = 2x - \frac{4V}{x^2}$$

For the minimum surface area, set $\frac{dS}{dx} = 0$:

$$\begin{aligned} 2x - \frac{4V}{x^2} &= 0 \\ 2x^3 &= 4V \\ x^3 &= 2V \end{aligned}$$

Substitute $V = x^2y$ back into the equation:

$$\begin{aligned} x^3 &= 2(x^2y) \\ x &= 2y \\ y &= \frac{x}{2} \end{aligned}$$

To confirm that this is a minimum, find the second derivative:

$$\frac{d^2S}{dx^2} = 2 + \frac{8V}{x^3}$$

Since $x > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2S}{dx^2} > 0$$

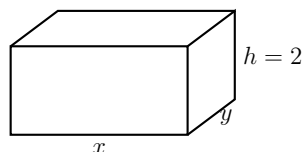
Thus, the cost of material is least when the depth of the tank is half of its width.

Hence Proved

The cost of material is least when the depth is half of its width.

Question 32

A tank with rectangular base and rectangular sides, open at the top, is to be constructed so that its depth is 2 m and volume is 8 m^3 . If building of tank costs Rs. 70 per square metre for the base and Rs. 45 per square metre for the sides, what is the cost of least expensive tank?

TYPE 124 Minimizing Construction Cost of Open Rectangular Tank*Open rectangular tank*

Given: Depth $h = 2 \text{ m}$, volume $V = 8 \text{ m}^3$, cost of base is Rs. $70/\text{m}^2$, cost of sides is Rs. $45/\text{m}^2$.

To Find: The cost of the least expensive tank.

Concept / Formula Used: $V = xyh$ $C = 70(xy) + 45(2xh + 2yh)$

Solution:

Let the length and width of the rectangular base be x and y respectively. The depth is given as $h = 2 \text{ m}$.

The volume of the tank is:

$$V = xyh = 2xy = 8 \implies xy = 4 \implies y = \frac{4}{x}$$

The cost of building the base is:

$$\text{Cost}_{\text{base}} = 70 \times xy = 70(4) = 280$$

The cost of building the four sides is:

$$\text{Cost}_{\text{sides}} = 45 \times (2xh + 2yh) = 45 \times (4x + 4y) = 180(x + y)$$

The total cost C is:

$$C = 280 + 180(x + y)$$

Substitute $y = \frac{4}{x}$ into the cost equation:

$$C(x) = 280 + 180 \left(x + \frac{4}{x} \right)$$

Differentiating $C(x)$ with respect to x :

$$\frac{dC}{dx} = 180 \left(1 - \frac{4}{x^2} \right)$$

For the minimum cost, set $\frac{dC}{dx} = 0$:

$$\begin{aligned} 180 \left(1 - \frac{4}{x^2} \right) &= 0 \\ 1 - \frac{4}{x^2} &= 0 \\ x^2 &= 4 \\ x &= 2 \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\frac{d^2C}{dx^2} = 180 \left(\frac{8}{x^3} \right)$$

At $x = 2$:

$$\frac{d^2C}{dx^2} = 180 \left(\frac{8}{8} \right) = 180 > 0$$

Since the second derivative is positive, the cost is minimum at $x = 2$.

Calculate the minimum cost:

$$\begin{aligned} C(2) &= 280 + 180 \left(2 + \frac{4}{2} \right) \\ &= 280 + 180(4) \\ &= 280 + 720 \\ &= 1000 \end{aligned}$$

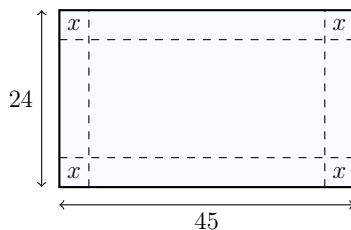
Final Answer

Rs. 1000

Question 33

A rectangular sheet of tin 45 cm by 24 cm is to be made into a box without top by cutting off squares from the corners and folding up the flaps. What should be the side of the square in order that the volume of the box is maximum?

TYPE 125 Maximizing Volume of Box from Rectangular Sheet



Tin sheet with cut-out corners

Given: Dimensions of the sheet are 45 cm $\times$ 24 cm.

To Find: The side of the square x to be cut off for maximum volume.

Concept / Formula Used: $V = \text{length} \times \text{width} \times \text{height} = (45 - 2x)(24 - 2x)x$

Solution:

Let the side of the square cut off from each corner be x cm.

The dimensions of the open box will be: - Length: $l = 45 - 2x$ - Width: $w = 24 - 2x$ - Height: $h = x$

For the dimensions to be positive, we must have:

$$24 - 2x > 0 \implies x < 12$$

The volume V of the box is:

$$\begin{aligned} V(x) &= x(45 - 2x)(24 - 2x) \\ &= x(1080 - 90x - 48x + 4x^2) \\ &= x(1080 - 138x + 4x^2) \\ &= 4x^3 - 138x^2 + 1080x \end{aligned}$$

Differentiating $V(x)$ with respect to x :

$$\frac{dV}{dx} = 12x^2 - 276x + 1080$$

For maximum volume, set $\frac{dV}{dx} = 0$:

$$\begin{aligned} 12x^2 - 276x + 1080 &= 0 \\ 12(x^2 - 23x + 90) &= 0 \\ x^2 - 23x + 90 &= 0 \\ (x - 18)(x - 5) &= 0 \end{aligned}$$

This gives $x = 18$ or $x = 5$. Since $x < 12$, we reject $x = 18$. Thus, $x = 5$.

Finding the second derivative:

$$\frac{d^2V}{dx^2} = 24x - 276$$

At $x = 5$:

$$\frac{d^2V}{dx^2} = 24(5) - 276 = 120 - 276 = -156 < 0$$

Since the second derivative is negative, the volume is maximum when $x = 5$ cm.

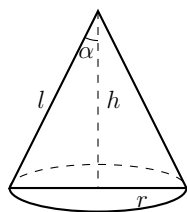
Final Answer

5 cm

Question 34

Show that the semi-vertical angle of a cone of the maximum volume and of given slant height is $\cos^{-1} \frac{1}{\sqrt{3}}$.

TYPE 126 Maximizing Volume of Cone with Constant Slant Height

Cone with semi-vertical angle α

To Prove: The semi-vertical angle α of the cone of maximum volume is $\cos^{-1} \frac{1}{\sqrt{3}}$.

Concept / Formula Used: $V = \frac{1}{3}\pi r^2 h$ $r = l \sin \alpha$ $h = l \cos \alpha$

Solution:

Let l be the given slant height of the cone, and α be its semi-vertical angle.

The radius r and height h of the cone are:

$$r = l \sin \alpha, \quad h = l \cos \alpha$$

The volume V of the cone is:

$$\begin{aligned} V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi (l \sin \alpha)^2 (l \cos \alpha) \\ &= \frac{1}{3}\pi l^3 \sin^2 \alpha \cos \alpha \end{aligned}$$

Using the identity $\sin^2 \alpha = 1 - \cos^2 \alpha$:

$$V = \frac{1}{3}\pi l^3 (1 - \cos^2 \alpha) \cos \alpha = \frac{1}{3}\pi l^3 (\cos \alpha - \cos^3 \alpha)$$

Differentiating V with respect to α :

$$\begin{aligned} \frac{dV}{d\alpha} &= \frac{1}{3}\pi l^3 (-\sin \alpha - 3 \cos^2 \alpha (-\sin \alpha)) \\ &= \frac{1}{3}\pi l^3 \sin \alpha (3 \cos^2 \alpha - 1) \end{aligned}$$

For maximum volume, set $\frac{dV}{d\alpha} = 0$: Since $\alpha \in (0, \pi/2)$, $\sin \alpha \neq 0$. Thus:

$$\begin{aligned} 3 \cos^2 \alpha - 1 &= 0 \\ \cos^2 \alpha &= \frac{1}{3} \\ \cos \alpha &= \frac{1}{\sqrt{3}} \\ \alpha &= \cos^{-1} \left(\frac{1}{\sqrt{3}} \right) \end{aligned}$$

To confirm that this is a maximum, find the second derivative:

$$\begin{aligned}\frac{d^2V}{d\alpha^2} &= \frac{1}{3}\pi l^3 \frac{d}{d\alpha} [\sin \alpha (3 \cos^2 \alpha - 1)] \\ &= \frac{1}{3}\pi l^3 [\cos \alpha (3 \cos^2 \alpha - 1) + \sin \alpha (-6 \cos \alpha \sin \alpha)] \\ &= \frac{1}{3}\pi l^3 [3 \cos^3 \alpha - \cos \alpha - 6 \cos \alpha \sin^2 \alpha]\end{aligned}$$

At $\cos \alpha = \frac{1}{\sqrt{3}}$ and $\sin^2 \alpha = 1 - \cos^2 \alpha = \frac{2}{3}$:

$$\begin{aligned}\frac{d^2V}{d\alpha^2} &= \frac{1}{3}\pi l^3 \left[3 \left(\frac{1}{3\sqrt{3}} \right) - \frac{1}{\sqrt{3}} - 6 \left(\frac{1}{\sqrt{3}} \right) \left(\frac{2}{3} \right) \right] \\ &= \frac{1}{3}\pi l^3 \left[\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} - \frac{4}{\sqrt{3}} \right] \\ &= -\frac{4\pi l^3}{3\sqrt{3}} < 0\end{aligned}$$

Since the second derivative is negative, the volume is maximum when $\alpha = \cos^{-1} \frac{1}{\sqrt{3}}$.

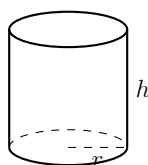
Hence Proved

The semi-vertical angle is $\cos^{-1} \frac{1}{\sqrt{3}}$.

Question 35

Show that a right circular cylinder of given volume, which is open at the top, has minimum total surface area if its height is equal to radius of its base.

TYPE 127 Minimizing Surface Area of Open Cylinder with Constant Volume



Open cylinder

To Prove: The total surface area of the open cylinder is minimum when $h = r$.

Concept / Formula Used: $V = \pi r^2 h = \text{constant}$ $S = \pi r^2 + 2\pi r h$

Solution:

Let r be the radius of the base and h be the height of the open cylinder.

The volume V is given to be constant:

$$V = \pi r^2 h \implies h = \frac{V}{\pi r^2}$$

The total surface area S of the cylinder (open at the top) is:

$$S = \pi r^2 + 2\pi r h$$

Substitute $h = \frac{V}{\pi r^2}$ into the surface area equation:

$$\begin{aligned} S(r) &= \pi r^2 + 2\pi r \left(\frac{V}{\pi r^2} \right) \\ &= \pi r^2 + \frac{2V}{r} \end{aligned}$$

Differentiating $S(r)$ with respect to r :

$$\frac{dS}{dr} = 2\pi r - \frac{2V}{r^2}$$

For the minimum surface area, set $\frac{dS}{dr} = 0$:

$$\begin{aligned} 2\pi r - \frac{2V}{r^2} &= 0 \\ 2\pi r^3 &= 2V \\ \pi r^3 &= V \end{aligned}$$

Substitute $V = \pi r^2 h$ back into the equation:

$$\begin{aligned} \pi r^3 &= \pi r^2 h \\ r &= h \end{aligned}$$

To confirm that this is a minimum, find the second derivative:

$$\frac{d^2 S}{dr^2} = 2\pi + \frac{4V}{r^3}$$

Since $r > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2 S}{dr^2} > 0$$

Thus, the total surface area is minimum when the height is equal to the radius of its base ($h = r$).

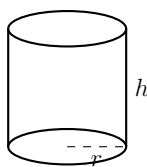
Hence Proved

The total surface area is minimum when $h = r$.

Question 36

- (i) Show that a closed cylindrical vessel of given volume has the least (total) surface area when its height is twice its radius.
- (ii) A closed right circular cylinder has a volume of 2156 cu. cm. What will be the radius of its base so that its total surface area is minimum? Find the height of the cylinder when its total surface area is minimum. Take $\pi = \frac{22}{7}$.

TYPE 128 Minimizing Surface Area of Closed Cylinder with Constant Volume



Closed cylinder

Given: For (ii), Volume $V = 2156 \text{ cm}^3$ and $\pi = \frac{22}{7}$.

To Find: For (ii), the radius r and height h for minimum surface area.

Concept / Formula Used: $V = \pi r^2 h$ $S = 2\pi r^2 + 2\pi r h$

Solution:

(i) Let r be the radius and h be the height of the closed cylinder.

The volume V is constant:

$$V = \pi r^2 h \implies h = \frac{V}{\pi r^2}$$

The total surface area S is:

$$S = 2\pi r^2 + 2\pi r h$$

Substitute $h = \frac{V}{\pi r^2}$ into the surface area equation:

$$\begin{aligned} S(r) &= 2\pi r^2 + 2\pi r \left(\frac{V}{\pi r^2} \right) \\ &= 2\pi r^2 + \frac{2V}{r} \end{aligned}$$

Differentiating $S(r)$ with respect to r :

$$\frac{dS}{dr} = 4\pi r - \frac{2V}{r^2}$$

For the minimum surface area, set $\frac{dS}{dr} = 0$:

$$\begin{aligned} 4\pi r - \frac{2V}{r^2} &= 0 \\ 4\pi r^3 &= 2V \\ 2\pi r^3 &= V \end{aligned}$$

Substitute $V = \pi r^2 h$ back into the equation:

$$\begin{aligned} 2\pi r^3 &= \pi r^2 h \\ 2r &= h \end{aligned}$$

Finding the second derivative:

$$\frac{d^2S}{dr^2} = 4\pi + \frac{4V}{r^3}$$

Since $r > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2S}{dr^2} > 0$$

Thus, the surface area is minimum when $h = 2r$.

(ii) From part (i), the surface area is minimum when:

$$2\pi r^3 = V$$

Substitute $V = 2156$ and $\pi = \frac{22}{7}$:

$$\begin{aligned} 2 \times \frac{22}{7} \times r^3 &= 2156 \\ \frac{44}{7} r^3 &= 2156 \\ r^3 &= 2156 \times \frac{7}{44} \\ r^3 &= 49 \times 7 \\ r^3 &= 343 \\ r &= 7 \text{ cm} \end{aligned}$$

Now, find the height h :

$$h = 2r = 2(7) = 14 \text{ cm}$$

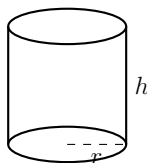
Final Answer

- (i) Proved that $h = 2r$
- (ii) Radius $r = 7$ cm, Height $h = 14$ cm

Question 37

Show that the radius of a closed right circular cylinder of given surface area and maximum volume is equal to half of its height.

TYPE 129 Maximizing Volume of Closed Cylinder with Constant Surface Area



Closed cylinder

To Prove: The radius r of the cylinder of maximum volume is equal to half of its height (i.e., $r = \frac{h}{2}$).

Concept / Formula Used: $S = 2\pi r^2 + 2\pi rh = \text{constant}$ $V = \pi r^2 h$

Solution:

Let r be the radius and h be the height of the closed cylinder.

The total surface area S is constant:

$$S = 2\pi r^2 + 2\pi rh \implies h = \frac{S - 2\pi r^2}{2\pi r}$$

The volume V of the cylinder is:

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi r^2 \left(\frac{S - 2\pi r^2}{2\pi r} \right) \\ &= \frac{r(S - 2\pi r^2)}{2} \\ &= \frac{Sr - 2\pi r^3}{2} \end{aligned}$$

Differentiating V with respect to r :

$$\frac{dV}{dr} = \frac{S - 6\pi r^2}{2}$$

For maximum volume, set $\frac{dV}{dr} = 0$:

$$\begin{aligned} \frac{S - 6\pi r^2}{2} &= 0 \\ S &= 6\pi r^2 \end{aligned}$$

Substitute $S = 2\pi r^2 + 2\pi rh$ back into the equation:

$$\begin{aligned} 2\pi r^2 + 2\pi rh &= 6\pi r^2 \\ 2\pi rh &= 4\pi r^2 \\ h &= 2r \\ r &= \frac{h}{2} \end{aligned}$$

To confirm that this is a maximum, find the second derivative:

$$\frac{d^2V}{dr^2} = \frac{-12\pi r}{2} = -6\pi r$$

Since $r > 0$, the second derivative is negative:

$$\frac{d^2V}{dr^2} < 0$$

Thus, the volume is maximum when the radius is equal to half of its height.

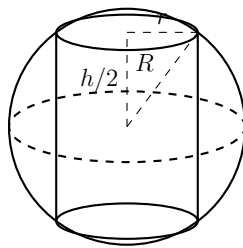
Hence Proved

The radius is equal to half of its height.

Question 38

Find the maximum volume of the cylinder which can be inscribed in a sphere of radius $3\sqrt{3}$ cm. (Leave the answer in terms of π).

TYPE 130 Maximizing Volume of Cylinder Inscribed in a Sphere



Cylinder inscribed in a sphere

Given: Radius of the sphere $R = 3\sqrt{3}$ cm.

To Find: The maximum volume of the inscribed cylinder.

Concept / Formula Used: $r^2 + \left(\frac{h}{2}\right)^2 = R^2$ $V = \pi r^2 h$

Solution:

Let $R = 3\sqrt{3}$ cm be the radius of the sphere. Let r and h be the radius and height of the inscribed cylinder.

From the right-angled triangle formed by the center of the sphere, the center of the cylinder's top face, and a point on the rim of the cylinder's top face:

$$r^2 + \left(\frac{h}{2}\right)^2 = R^2 \implies r^2 = R^2 - \frac{h^2}{4}$$

Substitute $R = 3\sqrt{3}$:

$$r^2 = (3\sqrt{3})^2 - \frac{h^2}{4} = 27 - \frac{h^2}{4}$$

The volume V of the cylinder is:

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi \left(27 - \frac{h^2}{4}\right) h \\ &= \pi \left(27h - \frac{h^3}{4}\right) \end{aligned}$$

Differentiating V with respect to h :

$$\frac{dV}{dh} = \pi \left(27 - \frac{3h^2}{4}\right)$$

For maximum volume, set $\frac{dV}{dh} = 0$:

$$\begin{aligned} 27 - \frac{3h^2}{4} &= 0 \\ \frac{3h^2}{4} &= 27 \\ h^2 &= 36 \\ h &= 6 \text{ cm} \quad (\text{since } h > 0) \end{aligned}$$

Finding the second derivative:

$$\frac{d^2V}{dh^2} = \pi \left(-\frac{6h}{4} \right) = -1.5\pi h$$

At $h = 6$:

$$\frac{d^2V}{dh^2} = -9\pi < 0$$

Since the second derivative is negative, the volume is maximum when $h = 6$ cm.

Calculate the maximum volume:

$$\begin{aligned} V_{\max} &= \pi \left(27(6) - \frac{6^3}{4} \right) \\ &= \pi \left(162 - \frac{216}{4} \right) \\ &= \pi(162 - 54) \\ &= 108\pi \text{ cm}^3 \end{aligned}$$

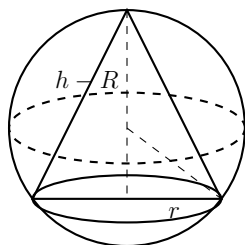
Final Answer

$108\pi \text{ cm}^3$

Question 39

A cone is inscribed in a sphere of radius 12 cm. If the volume of the cone is maximum, find its height.

TYPE 131 Maximizing Volume of Cone Inscribed in a Sphere



Cone inscribed in a sphere

Given: Radius of the sphere $R = 12$ cm.

To Find: The height h of the cone for maximum volume.

Concept / Formula Used: $r^2 + (h - R)^2 = R^2 \implies r^2 = 2Rh - h^2$ $V = \frac{1}{3}\pi r^2 h$

Solution:

Let $R = 12$ cm be the radius of the sphere. Let r and h be the radius and height of the inscribed cone.

From the geometry of the inscribed cone in the sphere:

$$r^2 + (h - R)^2 = R^2 \implies r^2 = 2Rh - h^2$$

Substitute $R = 12$:

$$r^2 = 24h - h^2$$

The volume V of the cone is:

$$\begin{aligned} V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi(24h - h^2)h \\ &= \frac{1}{3}\pi(24h^2 - h^3) \end{aligned}$$

Differentiating V with respect to h :

$$\begin{aligned} \frac{dV}{dh} &= \frac{1}{3}\pi(48h - 3h^2) \\ &= \pi(16h - h^2) \end{aligned}$$

For maximum volume, set $\frac{dV}{dh} = 0$:

$$\begin{aligned} \pi(16h - h^2) &= 0 \\ h(16 - h) &= 0 \end{aligned}$$

Since $h > 0$, we have:

$$h = 16 \text{ cm}$$

Finding the second derivative:

$$\frac{d^2V}{dh^2} = \pi(16 - 2h)$$

At $h = 16$:

$$\frac{d^2V}{dh^2} = \pi(16 - 32) = -16\pi < 0$$

Since the second derivative is negative, the volume is maximum when $h = 16$ cm.

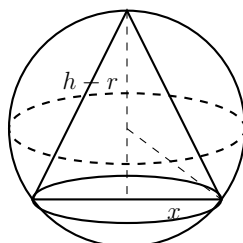
Final Answer

16 cm

Question 40

Show that the altitude of a right circular cone of maximum curved surface area which can be inscribed in a sphere of radius r is $\frac{4r}{3}$.

TYPE 132 Maximizing Curved Surface Area of Cone Inscribed in a Sphere



Cone inscribed in a sphere

To Prove: The altitude h of the cone of maximum curved surface area is $\frac{4r}{3}$.

Concept / Formula Used: $x^2 + (h - r)^2 = r^2 \implies x^2 = 2rh - h^2$ $l^2 = x^2 + h^2 = 2rh$
 $2rh$ $C = \pi x l$

Solution:

Let r be the radius of the sphere. Let x , h , and l be the base radius, height (altitude), and slant height of the inscribed cone respectively.

From the right-angled triangle in the sphere:

$$x^2 + (h - r)^2 = r^2 \implies x^2 = 2rh - h^2$$

The slant height l is given by:

$$l^2 = x^2 + h^2 = (2rh - h^2) + h^2 = 2rh$$

The curved surface area C of the cone is:

$$C = \pi x l$$

To maximize C , we can maximize $Z = C^2$:

$$\begin{aligned} Z &= \pi^2 x^2 l^2 \\ &= \pi^2 (2rh - h^2)(2rh) \\ &= 2\pi^2 r(2rh^2 - h^3) \end{aligned}$$

Differentiating Z with respect to h :

$$\frac{dZ}{dh} = 2\pi^2 r(4rh - 3h^2)$$

For maximum curved surface area, set $\frac{dZ}{dh} = 0$:

$$\begin{aligned} 4rh - 3h^2 &= 0 \\ h(4r - 3h) &= 0 \end{aligned}$$

Since $h > 0$, we have:

$$h = \frac{4r}{3}$$

Finding the second derivative:

$$\frac{d^2Z}{dh^2} = 2\pi^2 r(4r - 6h)$$

At $h = \frac{4r}{3}$:

$$\begin{aligned} \frac{d^2Z}{dh^2} &= 2\pi^2 r \left(4r - 6 \left(\frac{4r}{3} \right) \right) \\ &= 2\pi^2 r(4r - 8r) \\ &= -8\pi^2 r^2 < 0 \end{aligned}$$

Since the second derivative is negative, the curved surface area is maximum when the altitude is $h = \frac{4r}{3}$.

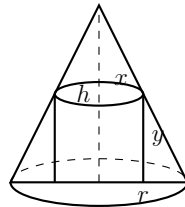
Hence Proved

The altitude of the cone is $\frac{4r}{3}$.

Question 41

Show that the height of a right circular cylinder of greatest volume which can be inscribed in a right circular cone of height h and radius r is one-third of the height of the cone and the greatest volume of the cylinder is $\frac{4}{9}$ times the volume of the cone.

TYPE 133 Maximizing Volume of Cylinder Inscribed in a Cone



Cylinder inscribed in a cone

To Prove: The height of the cylinder of greatest volume is $\frac{h}{3}$ and its volume is $\frac{4}{9}$ times the volume of the cone.

Concept / Formula Used: $\frac{h-y}{h} = \frac{x}{r} \implies y = h \left(1 - \frac{x}{r}\right)$ $V = \pi x^2 y$ $V_{\text{cone}} = \frac{1}{3} \pi r^2 h$

Solution:

Let r and h be the radius and height of the given cone. Let x and y be the radius and height of the inscribed cylinder.

Using similar triangles:

$$\frac{h-y}{h} = \frac{x}{r} \implies y = h \left(1 - \frac{x}{r}\right)$$

The volume V of the cylinder is

$$V = \pi x^2 y$$

Substituting $y = h \left(1 - \frac{x}{r}\right)$:

$$V = \pi x^2 \cdot h \left(1 - \frac{x}{r}\right) = \pi h \left(x^2 - \frac{x^3}{r}\right)$$

Differentiating with respect to x :

$$\frac{dV}{dx} = \pi h \left(2x - \frac{3x^2}{r}\right) = \pi h x \left(2 - \frac{3x}{r}\right)$$

Setting $\frac{dV}{dx} = 0$:

$$\pi h x \left(2 - \frac{3x}{r}\right) = 0$$

Since $x \neq 0$ (the cylinder has positive radius), we require

$$2 - \frac{3x}{r} = 0 \implies x = \frac{2r}{3}$$

Differentiating again to test for a maximum:

$$\frac{d^2V}{dx^2} = \pi h \left(2 - \frac{6x}{r}\right)$$

At $x = \frac{2r}{3}$:

$$\frac{d^2V}{dx^2} = \pi h \left(2 - \frac{6}{r} \cdot \frac{2r}{3} \right) = \pi h(2 - 4) = -2\pi h < 0$$

Since $\frac{d^2V}{dx^2} < 0$, the volume is maximum at $x = \frac{2r}{3}$.

Substituting back into $y = h \left(1 - \frac{x}{r} \right)$:

$$y = h \left(1 - \frac{2}{3} \right) = \frac{h}{3}$$

This proves the height of the cylinder of greatest volume is one-third the height of the cone.

Now computing the maximum volume:

$$\begin{aligned} V_{\max} &= \pi x^2 y = \pi \left(\frac{2r}{3} \right)^2 \cdot \frac{h}{3} \\ &= \pi \cdot \frac{4r^2}{9} \cdot \frac{h}{3} \\ &= \frac{4\pi r^2 h}{27} \end{aligned}$$

Comparing with the volume of the cone, $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$:

$$\begin{aligned} \frac{V_{\max}}{V_{\text{cone}}} &= \frac{\frac{4\pi r^2 h}{27}}{\frac{1}{3}\pi r^2 h} = \frac{4}{27} \times 3 = \frac{4}{9} \\ \therefore V_{\max} &= \frac{4}{9} V_{\text{cone}} \end{aligned}$$

Hence Proved

The height of the cylinder of greatest volume inscribed in the cone is $\frac{h}{3}$, and its greatest volume is $\frac{4}{9}$ times the volume of the cone.

Question 42

The sum of the surface areas of a rectangular parallelopiped with sides x , $2x$ and $\frac{x}{3}$ and a sphere is given to be constant. Prove that the sum of their volumes is minimum if x is equal to three times the radius of the sphere. Also find the minimum value of the sum of their volumes.

TYPE 115 Minimizing Combined Volume of Cuboid and Sphere

Given: Sides of the rectangular parallelopiped are x , $2x$, and $\frac{x}{3}$. Let the radius of the sphere be r . The sum of their surface areas is a constant, say S .

To Prove: The sum of their volumes V is minimum when $x = 3r$.

To Find: The minimum value of the sum of their volumes V .

Concept / Formula Used:

- Surface area of a rectangular parallelopiped: $S_{\text{cuboid}} = 2(ab + bc + ca)$
- Volume of a rectangular parallelopiped: $V_{\text{cuboid}} = abc$
- Surface area of a sphere: $S_{\text{sphere}} = 4\pi r^2$
- Volume of a sphere: $V_{\text{sphere}} = \frac{4}{3}\pi r^3$

Solution:

Let the sides of the rectangular parallelopiped be $a = x$, $b = 2x$, and $c = \frac{x}{3}$. Let the radius of the sphere be r .

The surface area of the rectangular parallelopiped is:

$$\begin{aligned}
 S_{\text{cuboid}} &= 2(ab + bc + ca) \\
 &= 2\left(x \cdot 2x + 2x \cdot \frac{x}{3} + \frac{x}{3} \cdot x\right) \\
 &= 2\left(2x^2 + \frac{2}{3}x^2 + \frac{1}{3}x^2\right) \\
 &= 2(3x^2) \\
 &= 6x^2
 \end{aligned}$$

The surface area of the sphere is:

$$S_{\text{sphere}} = 4\pi r^2$$

The sum of their surface areas is given to be constant, say S :

$$S = 6x^2 + 4\pi r^2 = \text{constant} \quad (1)$$

Differentiating equation (1) with respect to x :

$$\begin{aligned}
 \frac{d}{dx}(6x^2 + 4\pi r^2) &= 0 \\
 12x + 8\pi r \frac{dr}{dx} &= 0 \\
 8\pi r \frac{dr}{dx} &= -12x \\
 \frac{dr}{dx} &= -\frac{3x}{2\pi r}
 \end{aligned} \quad (2)$$

Now, let V be the sum of their volumes:

$$\begin{aligned}
 V &= V_{\text{cuboid}} + V_{\text{sphere}} \\
 &= x \cdot 2x \cdot \frac{x}{3} + \frac{4}{3}\pi r^3 \\
 &= \frac{2}{3}x^3 + \frac{4}{3}\pi r^3
 \end{aligned} \quad (3)$$

Differentiating V with respect to x :

$$\begin{aligned}
 \frac{dV}{dx} &= \frac{d}{dx}\left(\frac{2}{3}x^3 + \frac{4}{3}\pi r^3\right) \\
 &= 2x^2 + 4\pi r^2 \frac{dr}{dx}
 \end{aligned}$$

Substituting the value of $\frac{dr}{dx}$ from equation (2):

$$\begin{aligned}\frac{dV}{dx} &= 2x^2 + 4\pi r^2 \left(-\frac{3x}{2\pi r} \right) \\ &= 2x^2 - 6xr \\ &= 2x(x - 3r)\end{aligned}$$

For the volume to be minimum, we must have $\frac{dV}{dx} = 0$:

$$2x(x - 3r) = 0$$

Since $x > 0$ (as it represents a physical dimension), we have:

$$\begin{aligned}x - 3r &= 0 \\ x &= 3r\end{aligned}$$

To confirm that this point corresponds to a minimum, we find the second derivative $\frac{d^2V}{dx^2}$:

$$\begin{aligned}\frac{d^2V}{dx^2} &= \frac{d}{dx} (2x^2 - 6xr) \\ &= 4x - 6 \left(r + x \frac{dr}{dx} \right)\end{aligned}$$

At $x = 3r$, substituting $\frac{dr}{dx} = -\frac{3(3r)}{2\pi r} = -\frac{9}{2\pi}$:

$$\begin{aligned}\frac{d^2V}{dx^2} &= 4(3r) - 6 \left(r + 3r \left(-\frac{9}{2\pi} \right) \right) \\ &= 12r - 6r + \frac{81r}{\pi} \\ &= 6r + \frac{81r}{\pi} \\ &= r \left(6 + \frac{81}{\pi} \right)\end{aligned}$$

Since $r > 0$, we have $\frac{d^2V}{dx^2} > 0$. Thus, the sum of their volumes is minimum when $x = 3r$.

Now, we find the minimum value of the sum of their volumes by substituting $x = 3r$ into equation (3):

$$\begin{aligned}V_{\min} &= \frac{2}{3}(3r)^3 + \frac{4}{3}\pi r^3 \\ &= \frac{2}{3}(27r^3) + \frac{4}{3}\pi r^3 \\ &= 18r^3 + \frac{4}{3}\pi r^3 \\ &= \frac{2}{3}(27r^3 + 2\pi r^3) \\ &= \frac{2}{3}(27 + 2\pi)r^3\end{aligned}$$

Final Answer

The sum of the volumes is minimum when $x = 3r$ (i.e., x is equal to three times the radius of the sphere).

The minimum value of the sum of their volumes is:

$$V_{\min} = \frac{2}{3}(27 + 2\pi)r^3 \text{ cubic units}$$

where r is the radius of the sphere.

Question 43

A manufacturer plans to construct a cylindrical can to hold one cubic metre of oil. If the cost of constructing top and bottom of the can is twice the cost of constructing the side, what are the dimensions of the most economical can?

TYPE 134 Minimizing Cost of Cylindrical Can

Given: Volume of the cylindrical can $V = 1 \text{ m}^3$. Cost per unit area of the top and bottom is twice the cost per unit area of the curved side surface.

To Find: The radius r and height h of the cylinder that minimize the total cost of construction.

Concept / Formula Used: Volume of a cylinder $V = \pi r^2 h$, Curved surface area $S_{\text{side}} = 2\pi r h$, Area of top and bottom $S_{\text{ends}} = 2\pi r^2$.

Solution:

Let the radius of the cylindrical can be r metres and its height be h metres.

The volume V of the cylinder is given by:

$$\begin{aligned} V &= \pi r^2 h \\ 1 &= \pi r^2 h \end{aligned}$$

From this, we can express the height h in terms of r :

$$h = \frac{1}{\pi r^2} \quad (\text{i})$$

Let the cost of constructing the side surface per unit area be k (where $k > 0$). Then, the cost of constructing the top and bottom per unit area is $2k$.

The total cost of construction C is given by:

$$\begin{aligned} C &= (\text{Cost of top and bottom}) + (\text{Cost of side}) \\ C &= (2k) \cdot (2\pi r^2) + k \cdot (2\pi r h) \\ C &= 4k\pi r^2 + 2k\pi r h \\ C &= 2k(2\pi r^2 + \pi r h) \end{aligned}$$

To minimize the total cost C , we need to minimize the function:

$$f(r) = 2\pi r^2 + \pi r h$$

Substituting the value of h from equation (i) into $f(r)$:

$$f(r) = 2\pi r^2 + \pi r \left(\frac{1}{\pi r^2} \right)$$

$$f(r) = 2\pi r^2 + \frac{1}{r}$$

Now, we differentiate $f(r)$ with respect to r :

$$\frac{df}{dr} = \frac{d}{dr} (2\pi r^2 + r^{-1})$$

$$\frac{df}{dr} = 4\pi r - \frac{1}{r^2}$$

For a local extremum, we set $\frac{df}{dr} = 0$:

$$4\pi r - \frac{1}{r^2} = 0$$

$$4\pi r = \frac{1}{r^2}$$

$$r^3 = \frac{1}{4\pi}$$

$$r = \left(\frac{1}{4\pi} \right)^{1/3}$$

To verify whether this value of r gives a minimum, we find the second derivative $\frac{d^2f}{dr^2}$:

$$\frac{d^2f}{dr^2} = \frac{d}{dr} (4\pi r - r^{-2})$$

$$\frac{d^2f}{dr^2} = 4\pi + \frac{2}{r^3}$$

Since $r > 0$, we have:

$$\frac{d^2f}{dr^2} = 4\pi + \frac{2}{\frac{1}{4\pi}}$$

$$= 4\pi + 8\pi$$

$$= 12\pi > 0$$

Since the second derivative is positive, the cost is minimum when the radius is:

$$r = \left(\frac{1}{4\pi} \right)^{1/3} \text{ m}$$

Now, we substitute this value of r back into equation (i) to find the height h :

$$\begin{aligned}
 h &= \frac{1}{\pi r^2} \\
 h &= \frac{1}{\pi \left[\left(\frac{1}{4\pi} \right)^{1/3} \right]^2} \\
 h &= \frac{1}{\pi \left(\frac{1}{4\pi} \right)^{2/3}} \\
 h &= \frac{1}{\pi \cdot \frac{1}{(4\pi)^{2/3}}} \\
 h &= \frac{(4\pi)^{2/3}}{\pi} \\
 h &= \frac{4^{2/3} \cdot \pi^{2/3}}{\pi} \\
 h &= \frac{16^{1/3}}{\pi^{1/3}} \\
 h &= \left(\frac{16}{\pi} \right)^{1/3} \text{ m}
 \end{aligned}$$

Thus, the dimensions of the most economical can are:

$$\begin{aligned}
 \text{Radius } r &= \left(\frac{1}{4\pi} \right)^{1/3} \text{ metre} \\
 \text{Height } h &= \left(\frac{16}{\pi} \right)^{1/3} \text{ metre}
 \end{aligned}$$

Final Answer

$$\begin{aligned}
 \text{Radius } r &= \left(\frac{1}{4\pi} \right)^{1/3} \text{ m} \\
 \text{Height } h &= \left(\frac{16}{\pi} \right)^{1/3} \text{ m}
 \end{aligned}$$

Common Mistake: A common error is setting the cost function directly as the total surface area $2\pi r^2 + 2\pi r h$ without accounting for the different cost factors of the top/bottom and the side. Always write the cost function with the given cost ratios incorporated.